

Handbook of Biochemical Kinetics

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Table of Contents

Preface..... vii

**Abbreviations
& Symbols..... xi**

**Abbreviated
Binding Schemes xxi**

Source Words.....1

Wordfinder717

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Preface

The biotic world is doubtlessly the best known example of what Nobelist Murray Gell-Mann has termed “complex adaptive systems”—a name given to those systems possessing the innate capacity to learn and evolve by utilizing acquired information. Those familiar with living systems cannot but marvel at each cell’s ability to grow, to sense, to communicate, to cooperate, to move, to proliferate, to die and, even then, to yield opportunity to succeeding cells. If we dare speak of vitalism, especially as the new millennium is eager to dawn, we only do so to recognize that homeostatic mechanisms endow cells with such remarkable resilience that early investigators mistook homeostasis as a persuasive indication that life is self-determining and beyond the laws of chemistry and physics. A shared goal of modern molecular life scientists is to understand the mechanisms and interactions responsible for homeostasis. One approach for analyzing the mechanics of complex systems is to determine the chronology of discrete steps within the overall process—a pursuit called “kinetics.” This strategy allows an investigator to assess the structural and energetic determinants of transitions from one step to the next. By identifying voids in the time-line, one considers the possibility of other likely intermediates and ultimately identifies all elementary reactions of a mechanism.

Kinetics is an analytical approach deeply rooted in chemistry and physics, and biochemists have intuitively and inventively honed the tools of chemists and physicists for experiments on biological processes. Biochemical kinetics first began to flourish in enzymology—a field which has gainfully exploited advances in physical organic chemistry, structural chemistry, and spectroscopy in order to dissect the individual steps comprising enzyme mechanisms. No apology is offered, nor should any be required, for our strong emphasis on chemical kinetics and enzyme kinetics. Scientists working within these dis-

ciplines have enjoyed unparalleled success in dissecting complex multistage processes. Mechanisms are tools for assessing current knowledge and for designing better experiments. As working models, mechanisms offer the virtues of simplicity, precision, and generativity. “Simplicity” arises from the symbolic representation of the interactions among the minimal number of components needed to account for all observed properties of a system. “Precision” emerges by considering how rival models have nonisomorphic features (*i.e.*, testable differences) that distinguish one from another. “Generativity” results from the recombining of a model’s constituent elements to admit new findings, to predict new properties, and to stimulate additional rounds of experiment. For chemical and enzyme kineticists, the goal of this recursive enterprise is to determine a mechanism (a) that accounts for responses to changes in each component’s concentration, (b) that explains the detailed time-evolution of all chemical events, (c) that defines the concentration and structure of transient intermediates, (d) that makes sense of relevant changes in positional properties (*i.e.*, conformation, configuration, and/or physical location), and (e) that reconciles the thermodynamics of all reactions steps and transitions. In this respect, the rigor of chemical and enzyme kinetics teaches us all how best to invent new approaches that appropriately balance theory and experiment.

The inspiration for this HANDBOOK stemmed from our shared interest in teaching students about the logical and systematic investigation of enzyme catalysis and metabolic control. We began twenty-five years ago with the teaching of graduate-level courses (“Chemical Aspects of Biological Systems”; “Enzyme Kinetics and Mechanism”) at the University of California Santa Barbara as well as a course entitled “Enzyme Kinetics” at the Cornell University Medical College. More recently, we have

taught undergraduate students (“A Survey of Biochemistry and Molecular Biology”) as well as graduate students (“Advanced Metabolism”; “Physical Biochemistry and Structural Biology”; “Dynamic Processes in the Molecular Life Sciences”) here at the University of Florida. Our lectures have included material on the theory and practice of steady-state kinetics, rapid reaction kinetics, isotope-exchange kinetics, inhibitor design, equilibrium and kinetic isotope effects, protein oligomerization and polymerization kinetics, pulse-chase kinetics, transport kinetics, biomineralization kinetics, as well as ligand binding, cooperativity, and allostery. Because no existing text covered the bulk of these topics, we resorted to developing an extensive set of lecture notes—an activity that encouraged us to consider writing what we initially had envisioned as a short textbook on biochemical kinetics.

What also became clear was that, before and during any detailed consideration of a molecular process, teachers must always take pains to explain the associated terminology adequately. In 1789, the French chemist Antoine Lavoisier aptly declared: “Every branch of physical science must consist of the series of facts that are the objects of the science, the ideas that represent these facts, and the words by which these ideas are expressed. And, as ideas are preserved and communicated by means of words, it necessarily follows that we cannot improve the science without improving language or nomenclature.” We recognized that there was no published resource to help students come to grips with the far-ranging terminology of biochemical kinetics. Furthermore, as the distinction between scientific disciplines becomes blurred by what may be called “the interdisciplinary imperative,” students and practicing scientists from other disciplines will require a reliable sourcebook that explains terminology. Far too much time is wasted when students trace a finger over many pages of a textbook, only to find a partial definition for a sought-after term. Moreover, as more bioscientists come from countries not using English as a working language, there is an even greater need for a reliable and clearly written sourcebook of definitions.

A dictionary format became an appealing possibility for our HANDBOOK, but we also wished to treat many terms in greater depth than found in any dictionary. This led us to adopt the present word-list format which in many

respects resembles the “Micropaedia” section of the *ENCYCLOPAEDIA BRITANNICA*. One loses the seamless organization that can be realized in multichapter expositions that systematically develop a series of topics. We have accordingly attempted to mitigate this problem by including longer tracts on absorption and fluorescence spectroscopy, biomineralization, chemical kinetics, enzyme kinetics, Hill and Scatchard treatments, ligand binding cooperativity, kinetic isotope effects, and protein polymerization. Likewise, we have extensively inserted cross-references at appropriate locations within many entries. One intrinsic advantage of a mini-encyclopedia, however, is that in subsequent printings we should be able to make corrections and additions, or even deletions of an entire term, without upsetting the overall format. We also felt that readers should be encouraged to consult the most authoritative sources on particular topics. For this reason, we developed a collection of over 5000 literature references and, in many cases, our citations credit the original papers on a given topic. We have included the names of nearly 1000 enzymes, along with chemical reactions, EC numbers, and, in many instances, their biochemical and catalytic properties. The references cited are not intended to be comprehensive; rather, they serve to guide the reader to further interesting and helpful reading on subjects we have discussed. Where possible, at least one reference is included to provide information on assay protocols for the listed enzyme. We also urge readers to use the Wordfinder (included at the back of the Handbook) to take fullest advantage of the text and reference material. The nearly 8000 entries in the Wordfinder represent all listed source words as well as other subheadings, keywords, or synonyms. Each entry is immediately followed by the recommended source entries, and many source words are also cross-referenced to guide the reader to other related source words.

One of us (D.L.P.) has been a member of the *METHODS IN ENZYMOLOGY* family of editors for well over two decades. The volumes in this series on “Enzyme Kinetics and Mechanism” have become a standard for those interested in biological catalysis. As the form of this book began to emerge, we quickly recognized that *METHODS IN ENZYMOLOGY* could serve as an additional source for annotations on recommended theories and practices for kinetic studies on a wide range of topics. The Handbook contains nearly 6000 *METHODS IN ENZYMOLOGY* cita-

tions, and we have indicated the topic, volume, and beginning page for each at the foot of many of the source words. We trust that users of our Handbook will benefit from improved access to the first 280 volumes of *METHODS IN ENZYMOLOGY*.

For the derivations presented in this Handbook, we have assumed that the reader is familiar with the fundamentals of differential and integral calculus. To those who are loathe to engage in the rigor of mathematics, we say “Take heart!” The successful study of kinetics requires only that students work out a considerable number of problems which are both theoretical and numerical in character. We are reminded that Mithridates VI, the Grecian king of Pontus, is said to have acquired a tolerance to poison by taking gradually increasing doses. To aid those seeking their own intellectual mithridate (*i.e.*, acquired antidote), we provide scores of step-by-step derivations and practical advice on how to derive particular rate expressions. Likewise, we have included detailed protocols for H. J. Fromm’s systematic “theory-of-graphs” method as well as W. W. Cleland’s net reaction rate method. We are also greatly indebted to Dr. Charles Y. Huang for permitting us to include entire tracts from his chapter (which originally appeared in Volume 63 of *METHODS IN ENZYMOLOGY*) on the derivation of initial velocity and isotope exchange rate equations. We immediately recognized how daunting the task would be to attempt to surpass Dr. Huang’s truly outstanding treatment.

The success of our *HANDBOOK OF BIOCHEMICAL KINETICS* can only be judged by those using this manual for some period of time. We have come to recognize that we could not possibly represent all of the topics falling within the realm of biochemical kinetics—and certainly not within a first edition. We are also certain that, despite a determined effort to cover the terminology of chemical

and enzyme kinetics, we have still overlooked some important issues. We had also hoped to include additional kinetic techniques applied in pharmacokinetics, cell biology, electrophysiology, and metabolic control analysis. Time constraints also prevented our developing mathematical sections on Laplace transforms, vector algebra, distribution functions, and especially statistics. Eventually, we aspire to create a compact disk version of this Handbook, appropriately presented as hyperlinked text; that same CD should have room for selected problems/exercises along with step-by-step solutions, as well as down-loadable programs for kinetic simulation, algorithms for symbolic derivation of rate equations, molecular dynamics and related modeling techniques, and tried-and-true statistical methods. We also hope that our readers will not hesitate to advise us of shortcomings, missed terms, as well as techniques meriting definition, mention, or further explanation. We shall be forever grateful for such guidance.

We thank our students and colleagues for reading earlier drafts of our manuscript, and we are especially grateful to both Shirley Light and Dolores Wright of Academic Press for their insights, help, and thorough editing of the text. We also thank Academic Press for allowing us to incorporate the numerous annotations to *METHODS IN ENZYMOLOGY*.

Finally, as first- and second-generation disciples of Professor Herbert J. Fromm, we dedicate this book to him, in recognition of his germinal and indelible contributions to the field of enzyme kinetics and mechanism.

Daniel L. Purich

R. Donald Allison

January, 1999

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Abbreviations & Symbols

Roman Letters and Symbols

A	Molecule in the ground state Acceptor molecule (in fluorescence)	C or c	Molar concentration (M or moles/L)
A*	Molecule in the excited state	C	SI symbol for heat capacity (J K ⁻¹)
A	SI symbol for absorbance (unitless) SI symbol for Helmholtz energy (J) SI symbol for the pre-exponential term in Arrhenius equation (mol ⁻¹ m ³) ^{$n-1$} s ⁻¹	ΔC_p°	Constant pressure standard heat capacity per mole
ΔA	Change in absorbancy	$\hat{C}_i$ or $\hat{c}_i$	Weight concentration of the i th species
% A	Percent absorption of light (100 – % T)	c or C_i	Symbol for Curie (old unit of radioactivity)
[A]	Concentration of A	c_o	SI unit for speed of light in a vacuum; (value = 2.998×10^8 m s ⁻¹)
A, B, C, ...	Substrate A, B, C, ...	c	SI unit for speed of light in a medium (m s ⁻¹) Concentration
$A, B, C, \dots$	Coulombic contributions to the potential energy of interaction Moments of inertia of transition-state complex	D	SI symbol for debye (unitless) Donor molecule (in fluorescence)
A_{ij}	Amplitude of kinetic decay	D	SI symbol for translational diffusion constant (m ² s ⁻¹) Spectroscopic energy of dissociation of a diatomic molecule in the Morse equation
Å	Angstrom unit (10 ⁻¹⁰ m)	ΔD_o°	Difference in dissociation energies of products and reactants measured from zero-point energies
a_i	Thermodynamic activity of species i	Da	Dalton
a_o	Bohr radius	D_{rot}	Rotational diffusion constant
B	Generalized base	$d_{20,w}$	Density extrapolated to 20°C, water
B	$e^2\beta/2\epsilon kT$ or $[e^3/(\epsilon kT)^{2/3}](2\pi N_o/1000)^{1/2}$, a constant in the Debye-Hückel limiting law	d	Density Collision diameter
$B_{i,j}$	Second virial coefficient for the mutual interactions of species i and j	E	Free or uncomplexed enzyme General symbol for enzyme Effector molecule
Bq	Becquerel (unit of radioactivity = 1 disintegration per second)		
C	Coulomb		

Abbreviations & Symbols

E	Energy Initial kinetic energy of relative motion of reactants	ΔG°	Standard Gibbs free energy change per mole
$\mathbf{E}$	Electric vector of light	$\Delta^\ddagger G^\circ$	Standard Gibbs free energy of activation
E_a	Activation energy	g	Gram
[E _o] or E _o	Total enzyme concentration	g_e	Degeneracy of the lower state
Eq or eq	Equivalent Equilibrium	g_u	Degeneracy of the upper state
E°	Standard electromotive force	H	Henry (unit of self-inductance and mutual inductance)
E_s	Energy of molecule in excited singlet state	H	Enthalpy Hamiltonian
E_{tor}	Torsional potential energy	H_{local}	Magnetic field strength at nucleus of a molecule
e	Exponential function	ΔH°	Standard enthalpy change per mole
e	Charge on an electron (value = 1.602×10^{-19} coulombs) Equatorial position of a substituent on a molecule	$\Delta^\ddagger H^\circ$	Standard enthalpy of activation
e'	Pseudo-equatorial position of a substituent on a molecule	$\mathbf{H}$	Magnetic field
F	Modified enzyme form in ping pong mechanisms	$\mathbf{H}_{\text{res}}$	Magnetic field intensity at which resonance takes place
$\mathbf{F}$	Force	Hz	Hertz (unit of frequency cycles per second)
F	Free energy (<i>archaic</i>) Fluorescence Rotational-vibrational energy distribution function Momentum distribution function	h	Planck's constant = 6.626×10^{-34} J·sec or 6.626×10^{-27} erg·sec
$\mathcal{F}$	Faraday	$\hbar$	$h/2\pi = 1.055 \times 10^{-34}$ J·sec or 1.055×10^{-27} erg·sec
f	Fugacity Number of sites for acceptor on ligand (so-called "ligand valence") Oscillator strength Translational frictional coefficient	I	Nuclear spin quantum number Inhibitor
f	Fractional attainment of isotopic equilibrium	I	Intensity of radiation Ionic strength Light transmittance
F_{rel}	Fluorescence _{sample} /Fluorescence _{standard}	I_{50} or $I_{0.5}$	Inhibitor yielding 50% inhibition or 0.5 the uninhibited rate
G	Gibbs free energy Gravitational constant	$I(\lambda)$	Intensity of light at wavelength λ
		$I(\lambda)_f$	Intensity of emitted light at wavelength λ
		i	Square root of (-1)
		J	Joule

J	Nuclear magnetic resonance coupling coefficient Flux density (units = particles area ⁻¹ time ⁻¹)	k_d	Intrinsic dissociation constant (reciprocal of k_{ij})
j	Apparent order of a binding reaction	k_{iH}, k_{iD}, k_{iT}	A rate constant for isotopic isomers containing H, D, or T
K	Symbol for Kelvin	k_H/k_D	Kinetic isotope effect
K or K_{eq}	Macroscopic equilibrium constant	k_{ij}	Intrinsic association or binding constant (reciprocal of k_d) for interaction between sites on species i and j
K_a	Acid dissociation constant	KE	Kinetic energy
$K_A, K_B, \dots$	Dissociation constant for ligand A, B, C, ...	L	Liter
K_{ap}	Apparent equilibrium constant	L	Avogadro's number
K_D	Dissociation constant	L	Angular momentum
K_F	Formation constant (synonym of association constant) Dissociation constant for ligand F for an allosteric protein	$\mathcal{L}$	Allosteric constant equal to $[T_o]/[R_o]$
K_i	Macroscopic inhibition constant Macroscopic ionization constant	M	Molecular weight Molar
$K_{ia}, K_{ib}, \dots$	Dissociation constants in enzyme kinetics	$M^\ddagger$	Transition-state complex
K_R	Dissociation constant for ligand that binds to R-state of allosteric protein	M	Magnetization
K_S	Equilibrium constant for dissociation of ES complex	$\overline{M}_n$	Number average molecular weight
K_T	Dissociation constant for ligand that binds to T-state of an allosteric protein	M_r	Relative molecular mass
K_w	The constant equal to the product of $[H^+]$ (or, $[H_3O^+]$) and $[OH^-]$ in an aqueous solution	$\overline{M}_w$	Weight average molecular weight
$K_1, K_2, K_3, \dots$	Stepwise binding or dissociation constants for successive attachments of ligand to an oligomeric receptor	m	Meter
k or k_B	Boltzmann constant	m_e	Mass of electron at rest value = 9.1094×10^{-28} g
k	Rate constant Microscopic equilibrium constant	N	Newton (unit of force)
k_{cat}	Catalytic constant; turnover number	N_o	Avogadro's number = 6.0221×10^{23} mol ⁻¹
k_{cat}/K_m	Specificity constant	n	Refractive index Number of moles
		$n \rightarrow \pi^*$	Electronic transition
		$n \rightarrow \sigma^*$	Electronic transition
		n_H or n_{Hill}	Hill coefficient
		P	Generalized symbol for product
		P or p	Pressure
		pK_a	$-\log_{10} K_a$
		pO_2	Oxygen partial pressure

Abbreviations & Symbols

$(pO_2)_{0.5}$	Oxygen partial pressure at 0.5 saturation	T_1	Longitudinal relaxation time
Q	Coulomb (unit of electrostatic charge)	T_2	Transverse relaxation time
Q	Heat absorbed by a defined system	t	Time
Q_{CO_2}	Amount CO_2 released by tissue	$t_{1/2}$	Half-life
Q_{syn}	Synergism quotient	U	Internal energy
q	Quantum yield	V	Volume
q_o	Unquenched quantum yield	$\Delta V^\ddagger$	Volume of activation
R	Universal gas constant Electric resistance Gross rate of isotopic exchange	V_h	Hydrated volume
R_∞	Rydberg constant	V_m or V_{max}	Maximal velocity
$\bar{R}$	Fraction of allosteric protein in the R-state	$V_{m,f}$ or $V_{max,f}$	Maximal velocity in the forward direction
R_G	Radius of gyration	$V_{m,r}$ or $V_{max,r}$	Maximal velocity in the reverse direction
r	Radius Distance of separation	V/K	Ratio of V_{max} to K_m
$\mathbf{r}$	Polymer end-to-end vector	v	Speed Initial velocity of enzyme-catalyzed reaction Vibrational frequency
S	Svedberg unit (10^{-13} s)	$\dot{X}$	dX/dt
S_A	Partial molal entropy	$(\bar{X}_i)$	Equilibrium concentration of substance X_i
S_A'	Unitary part of the partial molal entropy	$\Delta(X_i)$	Difference between temporal and equilibrium concentration of X_i
ΔS°	Standard entropy change	z	Charge on a macromolecule or ion in units of e
$\Delta^\ddagger S^\circ$	Standard entropy of activation		
$\mathbf{S}$	Scattering vector		
S_1	Singlet state		
s	Second (unit of time)		
s	Sedimentation coefficient Equilibrium constant for helix growth		
$s_{20,w}$	Sedimentation coefficient corrected to 20°C, water		
$\hat{s}$	Unit vector along scattered radiation		
T	Temperature		
$\%T$	Percent transmission of light		
T_m	Melting temperature		

Greek Letters and Symbols

α	Degree of association Alpha particle Electric polarizability Reduced concentration ($[F]/K_F$) for allosteric protein
α_H	Hill coefficient
β	Reduced concentration ($[I]/K_I$) for allosteric protein
β_e	Bohr magneton

Γ	Surface concentration (mol m ⁻²) Parameter affecting relaxation amplitude	$\rho(r)$	Electron density
γ	Reduced concentration ($[A]/K_A$) for allosteric protein	Σ	Summation algorithm
δ	Phase shift Chemical shift in nuclear magnetic resonance	τ	Relaxation time (t) Lag time (s)
∂	Torque	Φ	Phi relation in enzyme kinetics Electrical potential
ε	Molar absorptivity Dielectric constant	ϕ	Quantum yield (unitless)
$\Delta\varepsilon$	Ellipticity in circular dichroism	χ	Mole fraction of component I
η	Solution viscosity	Ω	Solid angle
η_o	Solvent viscosity	$\Omega_{n,i}$	Statistical factor for ligand- i binding at n sites on a macromolecule
$[\eta]$	Intrinsic viscosity	ω	Angular momentum Circular frequency (Hz) Ionic strength (archaic)
θ_i	Fraction of ligand saturation of i th site	ω_o	Larmor frequency (Hz)
$[\theta]$	Molar ellipticity	ω	Angular velocity (rad s ⁻¹ or s ⁻¹)
κ	Transmission coefficient for transition state Inverse screening length	Mathematical Symbols α, β, γ Directional angles f' First derivative f'' Second derivative ∂ Partial derivative, Jacobian $\int$ Integral $\langle \rangle$ Average $\langle \rangle$ Overlap interval $\langle \rangle$ Expectation value integral $*$ Superscript designating radioactive substance Superscript designating excited state Subscript designating complex conjugate $\ddagger$ Superscript for transition state $(\partial x / \partial t)_y$ Partial differential of x with respect to time at constant y ∇ Vector differential or gradient, $\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$	
λ	Wavelength Kinetic decay time		
μ_i	Chemical potential of i th species per mole		
μ_i°	Standard chemical potential per mole		
μ_m	Magnetic moment		
μ	Ionic strength Electric dipole moment operator Reduced mass, $\mu = m_A m_B / (m_A + m_B)$		
ν	Frequency		
$\bar{\nu}$	Fractional saturation of ligand binding sites		
Ξ	Grand partition function in the Wyman treatment		
Π	Product algorithm Osmotic pressure		
ρ	Density (mass per unit volume)		

∇^2	Second derivative operator, $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
Δ	Constant time interval
()	Activity of a solute
(,)	Open interval
[]	Concentration of a solute
[,]	Closed interval
∞	Infinite dilution, typically as a subscript

Multiples/Submultiples

10^{12}	Tera (symbol = T)
10^9	Giga (symbol = G)
10^6	Mega (symbol = M)
10^3	Kilo (symbol = k)
10^{-1}	Deci (symbol = d)
10^{-2}	Centi (symbol = c)
10^{-3}	Milli (symbol = m)
10^{-6}	Micro (symbol = μ)
10^{-9}	Nano (symbol = n)
10^{-12}	Pico (symbol = p)
10^{-15}	Femto (symbol = f)
10^{-18}	Atto (symbol = a)
10^{-21}	Zepto (symbol = z)

Biochemical Abbreviations

A	Adenine Alanine or alanyl
aa	Amino acid
aaRS	Aminoacyl-tRNA
ACAT	Acyl-CoA:cholesterol acyltransferase
ACES	N-(2-Acetamido)-2-aminoethanesulfonic acid
ACh	Acetylcholine

ACP	Acyl carrier protein
ADA	Adenosine deaminase
Ade	Adenine
ADH	Alcohol dehydrogenase
ADP	Adenosine 5'-diphosphate
Ala	Alanine or alanyl
ALA	δ -Aminolevulinic acid or δ -aminolevulinate
AMP	Adenosine 5'-monophosphate
amu	Atomic mass unit (1.66×10^{-27} kg or 1.66×10^{-24} g)
Arg	Arginyl or arginyl
Asn	Asparagine or asparaginyl
Asp	Aspartic acid, aspartate, or aspartyl
Asx	Aspartate + asparagine or aspartyl + asparaginyl
ATCase	Aspartate transcarbamoylase
ATP	Adenosine 5'-triphosphate
B	Aspartate + asparagine (or aspartyl + asparaginyl)
BES	N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid
Bi	Two-substrate enzyme system
bp	Base pair
BPG	D-2,3-Bisphosphoglycerate
BPTI	Bovine pancreatic trypsin inhibitor
Bq	Becquerel
C	Cytosine Cysteine or cysteinyl
CaM	Calmodulin
cAMP	Cyclic AMP
CAP	Catabolite gene activating protein
cAPK	Protein kinase A (or cyclic AMP-stimulated protein kinase)
CAPS	3-(Cyclohexylamino)propanesulfonic acid

Cbz-	Benzyloxycarbonyl-	EDTA	Ethylenediaminetetraacetic acid or its conjugate base
cDNA	Complimentary strand DNA	EF	Elongation factor
CDP	Cytidine 5'-diphosphate	EGTA	Ethylene glycol bis(β -aminoethyl ether)- <i>N,N,N',N'</i> -tetraacetic acid or its conjugate base
CHES	3-(Cyclohexylamino)ethanesulfonic acid	EPPS	<i>N</i> -2-Hydroxyethylpiperazinepropane-sulfonic acid (also known as HEPPS)
Chl	Chlorophyll	EPR or ESR	Electron paramagnetic resonance or Electron spin resonance
CM	Carboxymethyl	F	Phenylalanine or phenylalanyl
cmc	Critical micelle concentration	FAD	Oxidized flavin adenine dinucleotide
CMP	Cytidine 5'-monophosphate	FADH $\cdot$	Radical form of reduced flavin adenine dinucleotide
CoA	Coenzyme A	FADH $_2$	Reduced flavin adenine dinucleotide
CoASH	Coenzyme A	FBP	Fructose 1,6-bisphosphate
CoQ	Coenzyme Q	Fd	Ferredoxin
CTP	Cytidine 5'-phosphate	fMet	<i>N</i> -Formylmethionine
Cys	Cysteine or cysteinyl	FMN	Flavin mononucleotide
D	Dalton Aspartic acid, aspartate, or aspartyl	F1P	Fructose 1-phosphate
d	Deoxy	F6P	Fructose 6-phosphate
dd	Dideoxy	G	Guanine Glycine or glycyI
DEAE	Diethylaminoethyl	GABA	γ -Aminobutyric acid
DFP	Diisopropyl fluorophosphate	Gal	Galactose
DG	<i>sn</i> -1,2-Diacylglycerol	GalNAc	<i>N</i> -Acetylglucosamine
DHAP	Dihydroxyacetone phosphate	GAP	Glyceraldehyde 3-phosphate
DHF	Dihydrofolate	GDP	Guanosine 5'-diphosphate
DHFR	Dihydrofolate reductase	Gla	4-Carboxyglutamic acid or 4-carboxyglutamyl
DMF	Dimethylformamide	Glc	Glucose
DMS	Dimethyl sulfate	Gln	Glutamine or glutamyl
DMSO	Dimethylsulfoxide	Glu	Glutamic acid, glutamate, or glutamyl
DNP	2,4-Dinitrophenyl	Gly	Glycine or glycyI
Dol	Dolichol	GMP	Guanosine 5'-monophosphate
L-DOPA	L-3,4-Dihydroxyphenylalanine		
DPN $^+$	see recommended abbreviation NAD $^+$		
DSC	Differential scanning calorimetry		
E	Glutamic acid, glutamate, or glutamyl		

Abbreviations & Symbols

G1P	Glucose 1-phosphate	IR	Infrared
G6P	Glucose 6-phosphate	IS	Insertion sequence
GSH	Glutathione (sometimes referred to as reduced glutathione)	ITP	Inosine 5'-triphosphate
GSSG	Glutathione disulfide (sometimes referred to as oxidized glutathione)	K	Lysine or lysyl
GTP	Guanosine 5'-triphosphate	kb	Kilobase pair
H	Histidine or histidyl	kD or kDa	Kilodalton
HA	Hemagglutinin	KF	Klenow factor
Hb	Hemoglobin	L	Leucine or leucyl
HbA	Adult hemoglobin	LCAT	Lecithin:cholesterol acyl transferase
HbCO	Carbon monoxide hemoglobin	LDH	Lactate dehydrogenase
HbO ₂	Oxyhemoglobin	LDL	Low density lipoprotein
HbS	Sickle cell hemoglobin	Leu	Leucine or leucyl
HbF	Fetal hemoglobin	Lys	Lysine or lysyl
HDL	High density lipoprotein	M	Methionine or methionyl
HEPES	<i>N</i> -2-Hydroxyethylpiperazine- <i>N'</i> -2-ethanesulfonic acid (also written as Hepes)	MALDI-MS	Matrix-assisted laser desorption ionization-mass spectroscopy
HEPPS	See EPPS	Man	Mannose
HGPRT	Hypoxanthine:guanine phosphoribosyltransferase	Mb	Myoglobin
His	Histidine or histidyl	MbCO	Carbon monoxide myoglobin
HMG-CoA	β -Hydroxymethylglutaryl-CoA	MbO ₂	Oxymyoglobin
hnRNA	Heterogenous nuclear RNA	MES	2-(<i>N</i> -Morpholino)ethanesulfonic acid
hsp	Heat shock protein	Met	Methionine or methionyl
Hyp	Hydroxyproline	MetHb	Methemoglobin (or, Fe(III)Hb)
I	Isoleucine or isoleucyl	MetMb	Metmyoglobin (or, Fe(III)Mb)
IDL	Intermediate density lipoprotein	MOPS	3-(<i>N</i> -Morpholino)propanesulfonic acid
IF	Initiation factor	MS	Mass spectroscopy/spectrometry
IgG	Immunoglobulin	N	Asparagine or asparaginyl
Ile	Isoleucine or isoleucyl	NAD ⁺	Oxidized nicotinamide adenine dinucleotide
IMP	Inosine 5'-monophosphate	NADH	Reduced nicotinamide adenine dinucleotide
IPTG	Isopropylthiogalactoside	NAG	<i>N</i> -Acetylglucosamine
		NAM	<i>N</i> -Acetylmuramic acid
		NANA	<i>N</i> -Acetylneuraminic acid

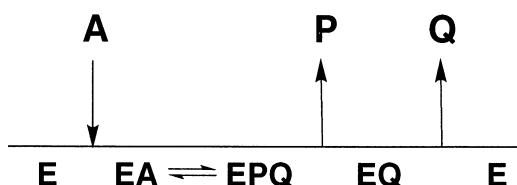
NMN	Nicotinamide mononucleotide	Q_{CO_2}	The amount (in microliters) of CO_2 given off (under standard conditions of pressure and temperature) per milligram of tissue per hour
NMR	Nuclear magnetic resonance		
NOESY	Nuclear Overhauser effect spectroscopy	QELS	Quasi-elastic laser light scattering
NTP	Nucleoside 5'-triphosphate	QH ₂	Ubiquinol
P	Proline or prolyl	Quad	Four substrate enzyme system
P (or p)	Phosphate	R	Arginine or arginyl
PEP	Phosphoenolpyruvate	r	Ribo
PFK	Phosphofructokinase	R5P	Ribose 5-phosphate
PG	Prostaglandin	RPC	Reverse phase chromatography
2PG	2-Phosphoglycerate	RT	Reverse transcriptase
3PG	3-Phosphoglycerate	RTK	Receptor tyrosine kinase
PGI	Phosphoglucoisomerase	Ru1,5P ₂	Ribulose 1,5-bisphosphate
PGM	Phosphoglucomutase	Ru5P	Ribulose 5-phosphate
Phe	Phenylalanine or phenylalanyl	S	Serine or seryl Svedberg constant
PIP ₂	Phosphatidylinositol 4,5-bisphosphate	SAM	S-Adenosylmethionine
PIPES	Piperazine- <i>N,N'</i> -bis(2-ethanesulfonic acid)	Ser	Serine or seryl
PK	Pyruvate kinase	T	Threonine or threonyl Thymine
PKA	Protein kinase A (or cyclic AMP-stimulated protein kinase)	TAPS	Tris(hydroxymethyl)methylamino-propanesulfonic acid
PKC	Protein kinase C	TCA	Tricarboxylic acid
PKU	Phenylketonuria	Ter	Three-substrate enzyme system
PLP	Pyridoxal 5-phosphate	TES	<i>N</i> -Tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid
Pol	DNA polymerase	THF	Tetrahydrofolate
PP _i	Inorganic pyrophosphate	Thr	Threonine or threonyl
Pro	Proline or prolyl	TIM or TPI	Triose-phosphate isomerase
PrP	Prion protein	TPP	Thiamin pyrophosphate (or thiamin diphosphate)
PRPP	5-Phosphoribosyl- α -pyrophosphate	Tris	Tris(hydroxymethyl)aminomethane
PS	Photosystem	TS	Thymidylate synthase Transition state
Q	Glutamine or glutamyl Ubiquinone (Coenzyme Q or CoQ)	TX	Transition state intermediate

Abbreviations & Symbols

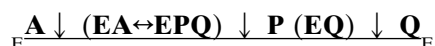
Tyr	Tyrosine or tyrosyl	Xaa	Unspecified amino acid or amino acyl residue
U	Uridine		
Uni	One-substrate enzyme system	XAFS	X-ray analysis for structure
V	Valine or valyl	Y	Tyrosine or tyrosyl
Val	Valine or valyl	YADH	Yeast alcohol dehydrogenase
X	Nonstandard or unknown amino acid or amino acyl	Z	Glutamate + glutamine or glutamyl + glutaminyl

Abbreviated Binding Schemes

The following diagrams indicate the binding interactions for enzyme kinetic mechanisms. To conserve space, the notation used in this Handbook is a compact version of the diagrams first introduced by Cleland¹. His diagram for the Ordered Uni Bi Mechanism is as follows:



Throughout this handbook, we have used the following single-line, compact notation:



This convention offers the advantage that it can be readily reproduced on virtually all word processors and typesetting devices without creating any special artwork. The enzyme surface is represented by the underline, in this case preceded by a subscript E to indicate the unbound (or free) enzyme before any substrate addition and after all products desorb. An arrow pointing to the line indicates binding, and the reader should understand that reversible binding is taken for granted. Moreover, unlike the original Cleland diagram, product release always is indicated by a downward arrow, because this systematic usage emphasizes the symmetry of certain mechanisms. The symbol “ $\text{A} \downarrow$ ” indicates that substrate A adds; the symbol “ $\text{A or B} \downarrow \downarrow$ ”, “ $\text{A or B or C} \downarrow \downarrow \downarrow$ ”, etc., indicates *random* addition of two or three substrates, respectively. Interconversions of enzyme-bound reactants can be reversible ($\leftrightarrow$) or irreversible ($\rightarrow$).

Taking the Ordered Uni Bi mechanism as an example, we can consider several additional possibilities:

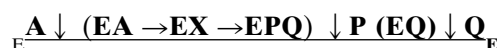
EA-to-EX and EX-to-EPQ reversible:



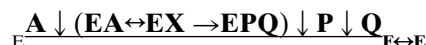
EA-to-EX reversible and EX-to-EPQ irreversible:



EA-to-EX irreversible and EX-to-EPQ irreversible:



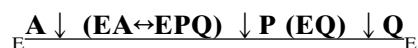
For the case of so-called iso mechanisms, the compact diagrams are as follows:



where $\text{F} \leftrightarrow \text{E}$ represents the reversible isomerization step.

Other examples of one-substrate and two-substrate kinetic mechanisms include:

Ordered Uni Bi Mechanism



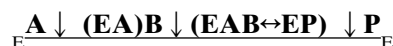
Random Uni Bi Mechanism



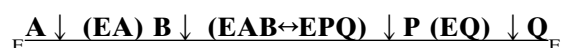
Random Bi Uni Mechanism



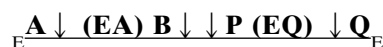
Ordered Bi Uni Mechanism



Ordered Bi Bi Mechanism

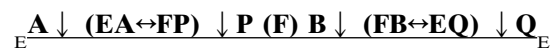


Ordered Bi Bi Theorell-Chance Mechanism

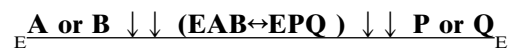


Abbreviated Binding Schemes

Ping Pong Bi Bi Mechanism



Random Bi Bi Mechanism



¹ W. W. Cleland (1963) *Biochem. Biophys. Acta.* **67**, 104.

Source Words

A

A, B, C, . . . / P, Q, R, . . .

Symbols for substrates and products, respectively, in multisubstrate enzyme-catalyzed reactions. In all ordered reaction mechanisms, A represents the first substrate to bind, B is the second, *etc.*, whereas P denotes the first product to be released, Q represents the second, *etc.* **See** *Cleland Nomenclature*

AB INITIO MOLECULAR-ORBITAL CALCULATIONS

A method of molecular-orbital calculations for determining bonding characteristics and other structural information about a wide variety of compounds and molecular configurations, including those that may not be directly observable (*e.g.*, transition state configurations with partial bonds). Although *ab initio* calculations are typically applied to systems with a small number of atoms, these computationally intensive calculations can be helpful in providing insights about the enzyme-catalyzed reactions. Related methods, known as semiempirical methods, use simplifying assumptions in the calculations and are determined more quickly than standard *ab initio* methods.

W. J. Hehre, L. Radom, P. von R. Schleyer & J. A. Pople (1986) *Ab Initio Molecular Orbital Theory*, Wiley, New York.

T. Clark (1985) *A Handbook of Computational Chemistry*, Wiley, New York.

W. G. Richards & D. L. Cooper (1983) *Ab Initio Molecular Orbital Calculations for Chemists*, 2nd ed., Oxford Press, Oxford.

W. Thiel (1988) *Tetrahedron* **44**, 7393.

ABM-1 & ABM-2 SEQUENCES IN ACTIN-BASED MOTORS

Consensus docking sites¹ for actin-based motility, defined by the oligoproline modules in *Listeria monocytogenes* ActA surface protein and human platelet vasodilator-stimulated phosphoprotein (VASP). Analysis of

known actin regulatory proteins led to the identification of distinct Actin-Based Motility (or Actin-Based-Motor) homology sequences:

ABM-1: (D/E)FPPPPX(D/E) [where X = P or T]

ABM-2: XPPPPP [where X = G, A, L, P, or S]

Actin-based motility involves a cascade of binding interactions designed to assemble actin regulatory proteins into functional locomotory units. *Listeria* ActA surface protein contains a series of nearly identical EFPPPP TDE-type oligoproline sequences for binding vasodilator-stimulated phosphoprotein (VASP). The latter, a tetrameric protein with 20-24 GPPPPP docking sites, binds numerous molecules of profilin, a 15 kDa regulatory protein known to promote actin filament assembly². Laine *et al.*³ recently demonstrated that proteolysis of the focal contact component vinculin unmasks an ActA homologue for actin-based *Shigella* motility. The ABM-1 sequence (PDFPPPPDL) is located at or near the C-terminus of the p90 proteolytic fragment of vinculin. Unmasking of this site serves as a molecular switch that initiates assembly of an actin-based motility complex containing VASP and profilin. Another focal adhesion protein zyxin⁴ contains several ABM-1 homology sequences that are also functionally active in reorganizing the actin cytoskeletal network.

¹D. L. Purich & F. S. Southwick (1997) *Biochem. Biophys. Res. Comm.* **231**, 686.

²F. S. Southwick & D. L. Purich (1996) *New Engl. J. Med.* **334**, 770.

³R. O. Laine, W. Zeile, F. Kang, D. L. Purich & F. S. Southwick (1997) *J. Cell Biol.* **138**, 1255.

⁴R. M. Golsteyn, M. C. Beckerle, T. Koay & E. Friedrich (1997) *J. Cell Sci.* **110**, 1893.

ABORTIVE COMPLEXES

Nonproductive reversible complexes of an enzyme with various substrates and/or products. The International Union of Biochemistry¹ distinguishes dead-end complex from abortive complex, and the latter term is regarded

as a synonym for “nonproductive complex”. Complexes that fail to undergo further reactions along the catalytic pathway are called dead-end complexes, and the reactions producing them are called dead-end reactions.

Some ambiguity exists in the literature regarding the usage of “abortive complex”. For example, the term has been used to describe the nonproductive complex formed between an enzyme and a competitive inhibitor² or to describe that inhibition of depolymerases resulting from shifted registration of the substrate within the enzyme’s set of subsites^{2,3}. Still others have used the term interchangeably with dead-end complexes⁴. The term abortive complexes formation is treated as a special case of dead-end complexation and is restricted to nonproductive complexes involving the binding of substrate(s) and/or product(s) to one or more enzyme forms. Thus, nonproductive complexes that culminate in substrate inhibition are abortive complexes. For a discussion concerning formation of EB and EP complexes in rapid-equilibrium ordered Bi Bi reactions, see the section on the Frieden Dilemma. Enzyme-substrate-product complexes that often form with multisubstrate enzymes are also abortive complexes.

Early product inhibition studies of *Aerobacter aerogenes* ribitol dehydrogenase⁵ demonstrated the formation of the E-NAD⁺-D-ribulose and E-NADH-D-ribitol complexes. In the lactate dehydrogenase reaction, the E-NADH-lactate and E-NAD⁺-pyruvate complexes are stable⁶, and determination of the K_d values indicates that the E-NAD⁺-pyruvate ternary complex is physiologically relevant⁷. Abortive complexes have been reported for a wide variety of enzymes. Isotope exchange at equilibrium is used to identify the E-NADH-malate abortive with bovine heart malate dehydrogenase⁷. Creatine kinase forms an E-MgADP-creatine complex⁸. Inhibition at high substrate-product concentrations may arise from factors other than abortive complexes; for example, the inhibition observed in an equilibrium exchange experiment may be related to high ionic strength of reaction solutions⁹.

Wong and Hanes¹⁰ pointed out that equilibrium exchange studies can be useful in detecting the presence of abortive species. Although abortive complexes can complicate exchange kinetic behavior, the Wedler-Boyer protocol¹¹ minimizes the influence of abortives on equilibrium exchange studies.

Different abortives may be formed with alternative products or substrates. Such procedures can be useful in helping to distinguish Theorell-Chance mechanisms from ordered systems with abortive complexes¹². In the case of lactate dehydrogenase, the E-pyruvate-NAD⁺ and E-lactate-NADH abortive complexes may play a regulatory roles in aerobic *versus* anaerobic metabolism.

Computer simulations¹³ also point to the regulatory potential of these non-productive complexes. **See** *Deadend Complexes; Inhibition; Nonproductive Complexes; Product Inhibition; Substrate Inhibition; Isotope Trapping; Isotope Exchange at Equilibrium; Enzyme Regulation*

¹International Union of Biochemistry (1982) *Eur. J. Biochem.* **28**, 281.

²M. Dixon & E. C. Webb (1979) *Enzymes*, 3rd ed., Academic Press, New York.

³J. D. Allen (1979) *Meth. Enzymol.* **64**, 248.

⁴H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

⁵H. J. Fromm & D. R. Nelson (1962) *J. Biol. Chem.* **231**, 215.

⁶H. J. Fromm (1963) *J. Biol. Chem.* **238**, 2938.

⁷H. Gutfreund, R. Cantwell, C. H. McMurray, R. S. Criddle & G. Hathaway (1968) *Biochem. J.* **106**, 683.

⁸E. Silverstein & G. Sulebele (1969) *Biochemistry* **8**, 2543.

⁹J. F. Morrison & W. W. Cleland (1966) *J. Biol. Chem.* **241**, 673.

¹⁰J. T.-F. Wong & C. S. Hanes (1964) *Nature* **203**, 492.

¹¹F. C. Wedler & P. D. Boyer (1972) *J. Biol. Chem.* **247**, 984.

¹²C. C. Wratten & W. W. Cleland (1965) *Biochemistry* **4**, 2442.

¹³D. L. Purich & H. J. Fromm (1972) *Curr. Topics in Cell. Reg.* **6**, 131.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Formation, **63**, 43, 419-424, 432-436; chymotrypsin, **63**, 205; isotope exchange, **64**, 32, 33, 39-45; isotope trapping, **64**, 58; limitation, **63**, 432-436; multiple, one-substrate system, **63**, 473, 474; pH effects, **63**, 205; practical aspects, **63**, 477-480; substrate inhibition, **63**, 500, 501; two-substrate system, **63**, 474-478; in product inhibition studies, **249**, 188-189, 193, 199-200, 205; identification of, **249**, 188-189, 202, 206, 208-209.

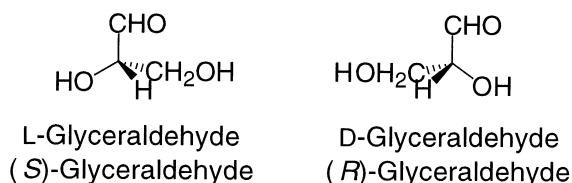
ABSCISSA

The x -coordinate axis for a graph of Cartesian coordinates $[x, y]$ or $[x, f(x)]$ or the x -value for any $[x, y]$ ordered pair. This corresponds to the *[Substrate Concentration]*-axis in v versus $[S]$ plots or the $1/[Substrate Concentration]$ -axis in so-called double-reciprocal or Lineweaver-Burk plots.

ABSOLUTE CONFIGURATION

A method for designating the stereoisomeric configuration of a chiral carbon atom within a molecular entity. The designation D was arbitrarily assigned to (+)-glyceraldehyde, and (–)-glyceraldehyde was assigned the label

L. Compounds that can be derived from L-glyceraldehyde without inversion reactions of the chiral center are likewise designated L- (and, their mirror images, D-). X-ray crystallographic studies later showed that D-glyceraldehyde had the configuration shown below¹. D- and L-Alanine are also depicted.



The DL-system is in common use with respect to amino acids and sugars, but the Cahn-Ingold-Prelog system (the RS-system) is more systematic and should be used.

The literature is replete with reports failing to specify the stereochemistry of certain reactions, and one must often infer the enantiomer. **See** *Configuration; Cahn-Ingold-Prelog System; Corn Rule; Diastereomers; Enantiomers; (R/S)-Convention*

¹J. M. Bijvoet, A. F. Peerdeman & A. J. van Bommel (1951) *Nature* **168**, 271.

²W. Klyne & J. Burkingham (1978) *Atlas of Stereochemistry*, 2nd ed., vol. **2**, Oxford Univ. Press, New York.

³J. Jacques, C. Gros, S. Bourcier, M. J. Brienne & J. Toullec (1977) *Absolute Configurations*, Georg Thieme Publ., Stuttgart.

ABSOLUTE TEMPERATURE

A temperature measured on an absolute temperature scale (*i.e.*, a scale in which zero degrees is equivalent to absolute zero). In the Kelvin scale, the degree unit is the kelvin, abbreviated as K; it does not have the superscript ° used to indicate degree as on the Celsius scale. K has the same magnitude as degree Celsius (°C).

ABSOLUTE UNCERTAINTY

The uncertainty in measured values expressed in units of the measurement. For example, a reaction velocity of 10.2 M/min is presumed to be valid to a tenth, and the absolute uncertainty is 0.1 M/min. **See** *Relative Uncertainty*

ABSOLUTE ZERO

The temperature at thermal energy of random motion of molecular entities of a system in thermal equilibrium is zero. This temperature is equal to -273.15°C or -459.67°F . Note that even at absolute zero, chemical bonds still retain zero point energy.

ABSORBANCE

A quantitative measure of photon absorption by a molecule, expressed as the $\log_{10}$ of the ratio of the radiant intensity I_o of light transmitted through a reference sample to the light I transmitted through the solution [*i.e.*, $A = \log(I_o/I)$]. Out-moded terms for absorbance such as optical density, extinction, and absorbandy should be abandoned.

The International Union of Pure and Applied Chemistry recommends that the definition should now be based on the ratio of the radiant power of incident radiation (P_o) to the radiant power of transmitted radiation (P). Thus, $A = \log(P_o/P) = \log T^{-1}$. In solution, P_o would refer to the radiant power of light transmitted through the reference sample. T is referred to as the transmittance. If natural logarithms are used, the quantity, symbolized by B , is referred to as the Napierian absorbance. Thus, $B = \ln(P_o/P)$. The definition assumes that light reflection and light scattering are negligible. If not, the appropriate term for $\log(P_o/P)$ is “attenuance.” **See** *Beer-Lambert Law; Absorption Coefficient; Absorption Spectroscopy*

ABSORBED DOSE

1. The quantity of absorbed energy absorbed per unit mass of a substance, object, or organism in an irradiated medium. Symbolized by D , the SI unit is the gray (Gy; joules per kilogram). The unit rad is also commonly used (1 rad = 0.01 Gy). 2. The amount of substance (*e.g.*, pharmaceutical) absorbed by an organism or cell.

ABSORPTION

1. The process of being taken up or becoming a part of another body. The unrelated, but often confused term “adsorption” describes a particular type of surface phenomenon. **See also** *Adsorption*. 2. The process of transfer of energy (*e.g.*, from an electromagnetic field) to a structure or entity (*e.g.*, a molecular entity). 3. The process of transport of a substance into a cell.

ABSORPTION COEFFICIENT

The log-base₁₀ attenuance (or absorbance) (*i.e.*, D or A) divided by the optical pathlength (l). This coefficient, symbolized by a , is thus equal to $l^{-1}\log(P_o/P)$. **See** *Beer-Lambert Law; Absorbance; Molar Absorption Coefficient; Absorption Spectroscopy*

ABSORPTION SPECTROSCOPY

Absorption spectroscopy is widely used to follow the course of enzyme-catalyzed reactions. Absorbance measurements should be made under conditions that permit use of the Beer-Lambert Law:

$$A = \log (I_0/I) = \epsilon cl$$

where A is the absorbance (a dimensionless parameter), I_0 and I are the incident radiant intensity and the transmitted radiant intensity, ϵ is the molar (base₁₀) absorption coefficient, c is the molar concentration of the absorbing species, and l is the absorption pathlength (*i.e.*, the distance through solution that light must pass). [Note: IUPAC¹ now favors $A = \log(P_0/P)$ where P_0 and P are the incident radiant power and the transmitted radiant power, respectively.]

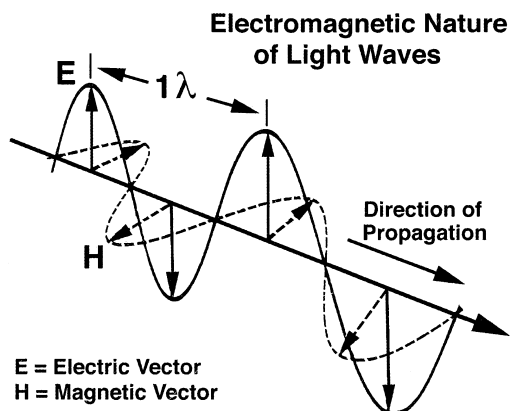


Figure 1. Perpendicular disposition of the electric vector E and magnetic vector H of a light wave traveling from its source in the direction of propagation shown by the arrow. Note that electromagnetic radiation interacts with molecules in two ways: (a) in absorption, the energy of a photon is absorbed by an electron (hence, the familiar term electronic absorbance spectrum) when the direction of the electric vector is aligned with the transition dipole of the molecule; (b) in light scattering, only the direction of propagation is changed, and very little, if any, energy is lost.

Practical Considerations. Typical absorption assay methods utilize ultraviolet (UV) or visible (vis) wavelengths. With most spectrophotometers, the measured absorbance should be less than 1.2 to obtain a strictly linear relationship (*i.e.*, to obey the Beer-Lambert Law). Nonlinear A versus c plots can result from micelle formation, sample turbidity, the presence of stray light (see below), bubble formation, stacking of aromatic chromophores, and even the presence of fine cotton strands from tissue used to clean the faces of cuvettes. One is well advised to confirm the linearity of absorbance with respect to product (or substrate) concentration under the exact assay conditions to be employed in

rate experiments. Prior centrifugation or filtration may be needed to reduce light scattering or turbidity.

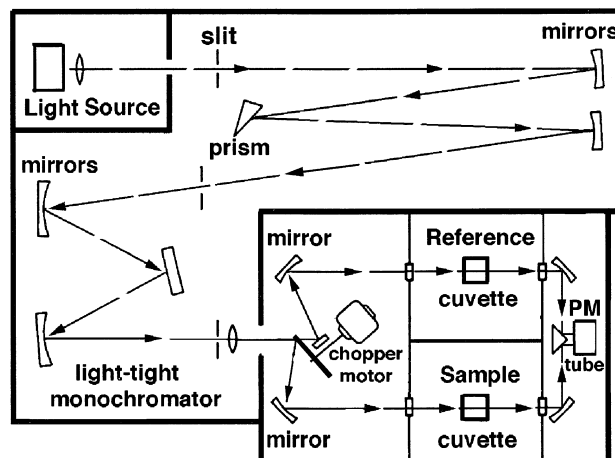


Figure 2. Design features of a double-beam UV/visible spectrophotometer. Note that rays of light pass through a set of slits as they enter the light-tight monochromator. Rotation of the prism determines the wavelength of dispersed light that passes on to the sample compartment. The chopper-motor rotates a beam-splitter that allows half of the in-coming light to travel to the reference compartment, while reflecting the other half of the in-coming light to the sample compartment.

Instruments with double monochromator configurations, or equivalent multi-pass configurations, can greatly reduce stray light (which is any radiation of wavelength other than that of the collimated light beam). Absorbance in the presence of stray light can be expressed as:

$$A = \log [(I_0 + I_s)/(I + I_s)]$$

where I_s is the stray light intensity. The larger the value of I_s (relative to I_0 or I), the greater the error in concentration or rate measurements².

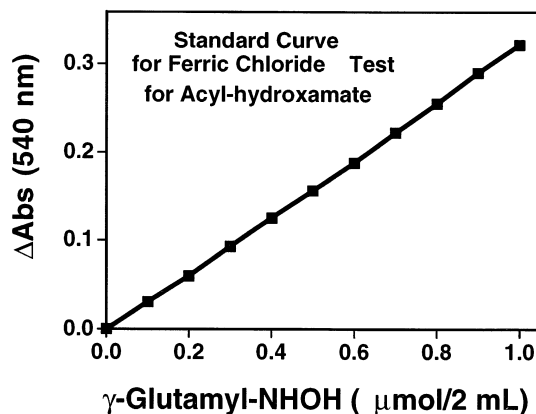


Figure 3. Example of the Beer-Lambert relationship.

The Beer-Lambert relationship is additive (*i.e.*, the absorption of light by one chemical species is unaffected

by the presence of other species, irrespective of whether those other species absorb light at the same wavelength). Thus, $A = \Sigma(\epsilon_i c_i l)$. The greater the difference in the molar absorption coefficients between the substrate(s) and the product(s), the larger the change in absorbance with time and the greater the ease in velocity determination. This statement assumes that the two chemical species do not interact with each other.

Note that the Beer-Lambert relationship does not require one to monitor a reaction at the wavelength maximum value ($\lambda_{\max}$) of either the substrate or the product. All other factors being equal, one should choose that wavelength yielding the greatest $\Delta\epsilon$ value. For example, AMP has a $\lambda_{\max}$ value at 259 nm at a pH value of 7, whereas IMP has a $\lambda_{\max}$ value of 248.5 nm. Yet, the AMP deaminase reaction is measured best at 265 nm, the wavelength affording the largest change in ϵ . Likewise, one can use a wavelength other than 340 nm to assay NADH or NADPH, but one should avoid unnecessary loss of signal-to-noise by working as close to the wavelength yielding maximal absorbance.

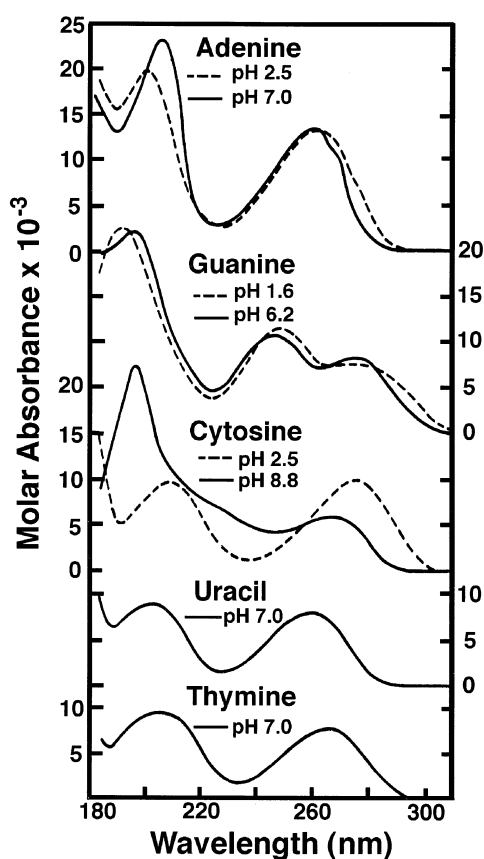


Figure 4. Ultraviolet spectrum of bases found in DNA and RNA.

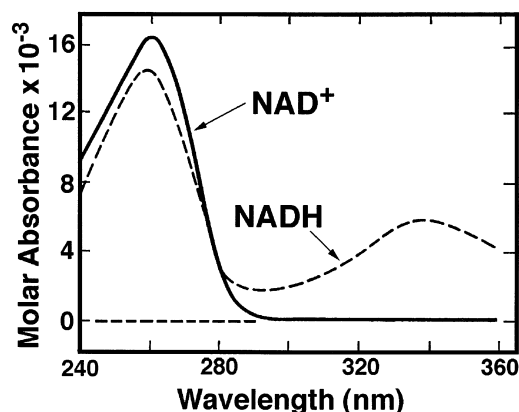


Figure 5. Ultraviolet spectrum of NAD^+ and NADH. Note that the absorption band centered at 340 nm serves as a valuable way to assay many dehydrogenases as well as other enzymes that form products that can be coupled to NAD^+ reduction or NADH oxidation.

Occasionally, one can increase the $\Delta\epsilon$ by utilizing alternative substrates. For example, 3-acetyl- NAD^+ or thio- NAD^+ can often be used with NAD^+ -dependent dehydrogenases. Note however that an alternative substrate may change the kinetic mechanism, as compared to that observed with the naturally occurring substrate. Alternative substrates are of particular value when the normal substrate(s) and product(s) do not efficiently absorb UV or visible light. For example, many *p*-nitroaniline or *p*-nitrophenyl derivatives have proved to be quite useful in enzyme assays because they exhibit intense absorption around 410 nm.

Ideally, other components in the reaction mixture should not absorb significantly at the monitored wavelength. In addition, colored impurities should be removed. For example, commercial imidazole, a commonly used buffer, contains a yellow impurity that can be easily removed upon recrystallization from ethyl acetate.

General Principles. Light absorption is quantized. The energy change associated with an electronic transition occurs within the ultraviolet (UV) or visible (vis) spectrum. Visible light absorption corresponds to low-energy electronic transitions, such as those observed with certain transition metal ions or with molecules having conjugated double bonds. The near ultraviolet (200 and 400 nm) corresponds to electronic transitions in molecular entities with smaller conjugated systems (e.g., 1,3,5-hexatriene, ATP, etc.). Isolated double bonds, such as those within peptide bonds, absorb light around 200-210 nm. Dioxygen, carbon dioxide, and water can absorb light of

wavelength less than 180 nm, and spectroscopy in the far-UV typically requires a vacuum, hence the term “vacuum UV” for low-wavelength light.

Sharp absorption bands are typically not observed in UV and visible absorption spectra of liquid samples. This is the consequence of the presence of the vibrational and rotational fine structure that become superimposed on the potential energy surfaces of the electronic transitions. Fine structure in UV/vis absorption spectra can be detected for samples in vapor phase or in nonpolar solvents.

The intensity of light absorption is governed by a number of factors that determine the transition probability (*i.e.*, the probability of interaction between the radiant energy and the electronic system). This probability is proportional to the square of the transition moment and is thus related to the electronic charge distribution in the molecular entity. Hence, absorption bands (with $\epsilon > 10,000 \text{ cm}^{-1} \text{ M}^{-1}$) suggest that the transition is accompanied by a large change in the transition moment. Intensity is also affected by the polarity of the excited state and the target area of the absorbing system. Transitions associated with UV-visible absorption spectroscopy consist of an electron in a filled molecular orbital being excited to the next higher energy orbital (an antibonding orbital). Although many exceptions are known, the relative transition energies roughly are: $\sigma \rightarrow \sigma^* > n \rightarrow \sigma^* > n \rightarrow \pi^* > \pi \rightarrow \pi^*$.

$\sigma \rightarrow \sigma^*$ Transitions. These transitions typically occur between 120 and 220 nm (*i.e.*, in the far-UV). The λ_{max} value for typical $\sigma \rightarrow \sigma^*$ transitions of carbon-carbon or carbon-hydrogen bonds is usually around 150 nm. For example, the λ_{max} value for ethane is 135 nm.

$n \rightarrow \sigma^*$ Transitions. These transitions typically occur at wavelengths greater than that needed for $\sigma \rightarrow \sigma^*$ transitions. Roughly, $n \rightarrow \sigma^*$ transitions involving $-\ddot{\text{O}}-$ occur with wavelengths at about 185 nm, with $-\ddot{\text{N}}-$ and $-\ddot{\text{S}}-$ at about 195 nm, and with carbonyls at about 190 nm. Examples of λ_{max} values (and ϵ values) of $n \rightarrow \sigma^*$ transitions include: water ($\lambda_{\text{max}} = 167 \text{ nm}$ with $\epsilon = 7000$), methanol ($\lambda_{\text{max}} = 183 \text{ nm}$ with $\epsilon = 500$), acetone ($\lambda_{\text{max}} = 188 \text{ nm}$ with $\epsilon = 1860$), and methyl iodide ($\lambda_{\text{max}} = 259 \text{ nm}$ with $\epsilon = 400$).

$n \rightarrow \pi^*$ Transitions. These are “forbidden” transitions according to symmetry rules, but molecular vibrations

allow these transitions to occur, albeit with low intensities. Nonbonding electrons of carbonyl groups will often have $n \rightarrow \pi^*$ transition λ_{max} values of around 300 nm. Some examples include acetone ($\lambda_{\text{max}} = 279 \text{ nm}$, $\epsilon = 15$), acetophenone (319 nm, $\epsilon = 50$), thiourea (a shoulder at 291 nm, $\epsilon = 71$), and acetic acid (204 nm, $\epsilon = 41$). The $n \rightarrow \pi^*$ transition can also be detected in optical rotatory dispersion measurements. Moreover, $n \rightarrow \pi^*$ transitions often exhibit a blue shift in polar solvents or environments.

$\pi \rightarrow \pi^*$ Transitions. Typically occurring at wavelengths in the near-UV, transitions of this type are the most commonly utilized spectral signals in kinetic and structural studies.

¹IUPAC (1988) *Pure and Appl. Chem.* **60**, 1055.

²R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

³C. R. Cantor & P. R. Schimmel (1980) *Biophysical Chemistry*, part II, pp. 344-408, Freeman, San Francisco.

⁴R. P. Bauman (1962) *Absorption Spectroscopy*, Wiley, New York.

⁵H. H. Jaffe & M. Orchin (1962) *Theory and Application of Ultraviolet Spectroscopy*, Wiley, New York.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Absorption Spectrophotometer: Application, **24**, 15-25; baseline compensation, **24**, 8-10; computerized, **24**, 19-25; light scattering, **24**, 13-15; monochromator, **24**, 4; photometer, **24**, 5-8; recorder, **24**, 8; sample compartment, **24**, 5; single-beam, **24**, 3-4; spectral characteristics, **24**, 10-12; split-beam, **24**, 3; stray light, **24**, 12-13.

Optical Spectroscopy: General principles and overview, **246**, 13; absorption and circular dichroism spectroscopy of nucleic acid duplexes and triplexes, **246**, 19; circular dichroism, **246**, 34; bioinorganic spectroscopy, **246**, 71; magnetic circular dichroism, **246**, 110; low-temperature spectroscopy, **246**, 131; rapid-scanning ultraviolet/visible spectroscopy applied in stopped-flow studies, **246**, 168; transient absorption spectroscopy in the study of processes and dynamics in biology, **246**, 201; hole burning spectroscopy and physics of proteins, **246**, 226; ultraviolet/visible spectroelectrochemistry of redox proteins, **246**, 701; diode array detection in liquid chromatography, **246**, 749.

Nanosecond Absorption Spectroscopy: Absorption apparatus, **226**, 131; apparatus, **226**, 152; detectors, **226**, 126; detector systems, **226**, 125; excitation source, **226**, 121; global analysis, **226**, 146, 155; heme proteins, **226**, 142; kinetic applications, **226**, 134; monochromators/spectrographs, **226**, 125; multiphoton effects, **226**, 141; nanosecond time-resolved recombination, **226**, 141; overview, **226**, 119, 147; probe source, **226**, 124; quantum yields, **226**, 139; rhodopsin, **226**, 158; sample holders, **226**, 133; singular value decomposition, **226**, 146, 155; spectral dynamics, **226**, 136; time delay generators, **226**, 130.

Time-Resolved Absorption Spectroscopy: Advantages, **232**, 389; applications, **232**, 387-388; detectors, **232**, 387, 392-393, 399; hemoglobin data analysis, **232**, 401-415; kinetic analyses, **232**, 390; photoselection effects, **232**, 390-391; kinetic intermediates and,

232, 389; spectrometer for, **232**, 392-401; performance characteristics, **232**, 389-390.

ABSORPTIVITY

A parameter in spectroscopy and photochemistry, equal to the absorbance ($1 - (P/P_0)$) divided by the optical pathlength (l) of the sample containing the absorbing agent. Thus, it equals $(1 - (P/P_0))/l$ where P_0 is the radiant power of light being transmitted through a reference sample, and P is the radiant power being transmitted through the solution. The Commission on Photochemistry does not recommend the use of this term. **See** *Absorbance*; *Absorption Coefficient*; *Beer-Lambert Law*; *Absorption Spectroscopy*

ABSTRACTION REACTION

Any chemical process in which one reactant removes an atom (neutral or charged) from the other reacting entity. An example is the generation of a free radical by the action of an initiator on another molecule. If abstraction takes place at a chiral carbon, racemization is almost always observed in nonenzymic processes. On the other hand, enzymes frequently abstract and reattach atoms or groups of atoms in a fashion that maintains stereochemistry.

ACCELERATION

In physics, the time rate of change of motional velocity resulting from changes in a body's speed and/or direction. In biochemistry, acceleration refers to an increased rate of a chemical reaction in the presence of an enzyme or other catalyst. **See** *Catalytic Rate Enhancement*; *Catalytic Proficiency*; *Efficiency Function*

ACCRETION

Solute or particulate accumulation onto an aggregated phase (or solid state) that grows together by the addition of material at the periphery. Both cohesive and adhesive forces are thought to be driving forces in accretion. Sea shells and kidney stones are also known to form as layers of crystallites and amorphous components by accretion of external substances.

ACCURACY

The closeness or proximity of a measured value to the true value for a quantity being measured. Unless the magnitude of a quantity is specified by a formal SI definition, one typically uses reference standards to establish

the accepted true value for a given quantity. **See also** *Precision*

ACETALDEHYDE DEHYDROGENASE (ACETYLATED)

This enzyme [EC 1.2.1.10] catalyzes the oxidation of acetaldehyde in the presence of NAD^+ and coenzyme A to form acetyl-CoA + NADH + H^+ . Other aldehyde substrates include glycolaldehyde, propanal, and butanal.

E. R. Stadtman & R. M. Burton (1955) *Meth. Enzymol.* **1**, 222 and 518.

F. B. Rudolph, D. L. Purich & H. J. Fromm (1968) *J. Biol. Chem.* **243**, 5539.

2-(ACETAMIDOMETHYLENE)SUCCINATE HYDROLASE

This enzyme [EC 3.5.1.29] catalyzes the hydrolysis of 2-(acetamidomethylene)-succinate to yield acetate, succinate semialdehyde, carbon dioxide, and ammonia.

R. W. Burg (1970) *Meth. Enzymol.* **18(A)**, 634.

ACETATE KINASE

This phosphotransferase [EC 2.7.2.1] catalyzes the thermodynamically favored phosphorylation of ADP to form ATP (*i.e.*, $K_{eq} = [\text{ATP}][\text{acetate}]/[\text{acetyl phosphate}][\text{ADP}] = 3000$). GDP is also an effective phosphoryl group acceptor. This enzyme is easily cold-denatured, and one must use glycerol to maintain full catalytic activity. Initial kinetic evidence, as well as borohydride reduction experiments, suggested the formation of an enzyme-bound acyl-phosphate intermediate, but later kinetic and stereochemical¹ data indicate that the kinetic mechanism is sequential and that there is direct in-line phosphoryl transfer. Incidental generation of a metaphosphate anion during catalysis may explain the formation of an enzyme-bound acyl-phosphate. Acetate kinase is ideally suited for the regeneration of ATP or GTP from ADP or GDP, respectively.

¹P. A. Frey (1992) *The Enzymes* **20**, 160.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Acetate assay with, **3**, 269; activation, **44**, 889; activity assay, **44**, 893, 894; alternative substrates, **87**, 11; bridge-to-nonbridge transfer, **87**, 19-20, 226, 232; chiral phosphoryl-ATP, **87**, 211, 258, 300; cold denaturation, **63**, 9; cysteine residues, **44**, 887-889; equilibrium constant, **63**, 5; exchange properties, **64**, 9, 39, **87**, 5, 18, 656; hydroxylaminolysis, **87**, 18; immobilization, **44**, 891, 892; inhibitor, **63**, 398; initial rate kinetics, **87**, 18; metal-ion bind-

ing, **63**, 275-278; metaphosphate, **87**, 12, 20; metaphosphate synthesis and, **6**, 262-263; nucleoside diphosphate kinase activity; **63**, 8; phosphorylation potential, **55**, 237; pH stability profile, **87**, 18; promoting microtubule assembly, **85**, 417-419; purine nucleoside diphosphate kinase activity, **63**, 8; regenerating GTP from GDP, **85**, 417; ribulose-5-phosphate 4-epimerase and, **5**, 253-254; *Veillonella alcalescens* acetate kinase [ATP formation assay, **71**, 312; hydroxamate assay, **71**, 311; properties, **71**, 315; stability to heat, **71**, 313; stimulation by succinate, **71**, 316; substrate specificity, **71**, 316]; xylulose-5-phosphate 3-epimerase and, **5**, 250-251; xylulose-5-phosphate phosphoketolase and, **5**, 26; purine inhibitor, **63**, 398; metal-ion binding, **63**, 275-278; phosphorothioates, **87**, 200, 205, 226, 232, 258; from *Bacillus stearothermophilus*; assay, **90**, 179; properties, **90**, 183; purification, **90**, 180; in acetyl phosphate and acetyl-CoA determination, **122**, 44; and hexokinase, in glucose 6-phosphate production, **136**, 52; dihydroxyacetone phosphate synthesis with, **136**, 277; glucose 6-phosphate synthesis with, **136**, 279; *sn*-glycerol 3-phosphate synthesis with, **136**, 276; in pyruvic acid phosphoroclastic system, **243**, 96, 99.

ACETATE KINASE (PYROPHOSPHATE)

This enzyme [EC 2.7.2.12] converts acetate and pyrophosphate to form acetyl phosphate and orthophosphate.

H. G. Wood, W. E. O'Brien & G. Michaels (1977) *Adv. Enzymol.* **45**, 8555.

ACETAZOLAMIDE

A diuretic agent (5-acetamido-1,3,4-thiadiazole-2-sulfonamide) that acts as a potent noncompetitive inhibitor (K_i 10^{-8} M) of carbonic anhydrase.

ACETOACETATE DECARBOXYLASE

This enzyme [EC 4.1.1.4] catalyzes the decarboxylation of acetoacetate to form acetone and carbon dioxide.

M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

D. S. Sigman & G. Mooser (1975) *Ann. Rev. Biochem.* **44**, 889.

I. Fridovich (1972) *The Enzymes*, 3rd ed., **6**, 255.

ACETOLACTATE SYNTHASE

This enzyme [EC 4.1.3.18] catalyzes the reversible carboxylation of 2-acetolactate with carbon dioxide to form two pyruvate ions. Thiamin pyrophosphate is a required cofactor.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 217.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

A. Schellenberger, G. Hübner & H. Neef (1997) *Meth. Enzymol.* **279**, 131.

J. S. Holt, S. B. Powles & J. A. M. Holtum (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 203.

R. Kluger (1992) *The Enzymes*, 3rd ed., **20**, 271.

J. H. Jackson (1988) *Meth. Enzymol.* **166**, 230.

ACETYLCHOLINESTERASE

This enzyme [EC 3.1.1.7], also known as true cholinesterase, choline esterase I, and cholinesterase, catalyzes the hydrolysis of acetylcholine to produce choline and acetate. The enzyme will also act on a number of acetate esters as well as catalyze some transacetylations.

D. M. Quinn & S. R. Feaster (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 455.

H. Okuda (1991) *A Study of Enzymes* **2**, 563.

T. L. Rosenberry (1975) *Adv. Enzymol.* **43**, 103.

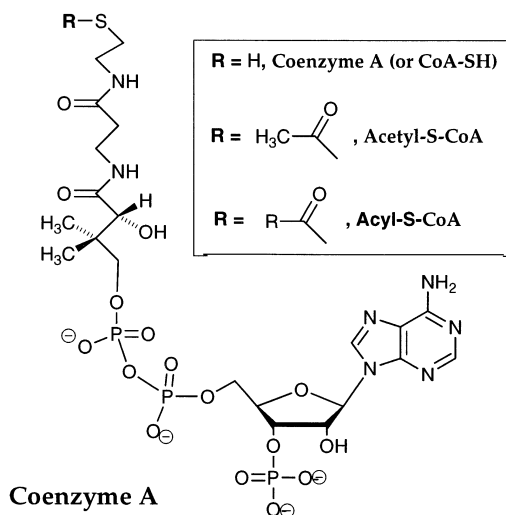
H. C. Froede & I. B. Wilson (1971) *The Enzymes*, 3rd ed., **5**, 87.

L. T. Potter (1971) *Meth. Enzymol.* **17(B)**, 778.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Acetylthiocholine as substrate, **251**, 101-102; assay by ESR, **251**, 102-105; inhibitors, **251**, 103; modification by symmetrical disulfide radical, **251**, 100; thioester substrate, **248**, 16; transition state and multisubstrate analogues, **249**, 305; enzyme receptor, similarity to collagen, **245**, 3.

ACETYL-CoA (or, ACETYL COENZYME A)



As the principal thiolester of intermediary metabolism, acetyl coenzyme A is involved in two-carbon biosynthetic and degradative steps. An essential component is the vitamin pantothenic acid, which provides the sulfur atom for the thiolester formation.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Assay, **1**, 611; **3**, 935-938; **63**, 33; separation by HPLC, **72**, 45; extraction from tissues, **13**, 439; formation of, **1**, 486, 518, 585; **5**, 466; free energy of hydrolysis, **1**, 694; substrate for the following enzymes [acetyl-coenzyme A acyl carrier protein transacylase, **14**, 50; acetyl-coenzyme A carboxylase, **14**, 3, 9; acetyl-coenzyme A synthetase, **13**, 375; *N*-acetyltransferase, **17B**, 805; aminoacetone

synthase, **17B**, 585; carnitine acetyltransferase, **13**, 387-389; **14**, 613; choline acetyltransferase, **17B**, 780, 788, 798; citrate synthase, **13**, 3, 4, 8, 9, 11, 12, 15-16, 19-20, 22, 25; **14**, 617; fatty acid synthase, **14**, 17, 22, 33, 40.

ACETYL-CoA C-ACETYLTRANSFERASE (or, THIOLASE)

This enzyme [EC 2.3.1.9], also known as thiolase, transfers an acetyl group from one acetyl-CoA molecule to another to form free coenzyme A and acetoacetyl-CoA.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

U. Gehring & F. Lynen (1972) *The Enzymes*, 3rd ed., **7**, 391.

ACETYL-CoA C-ACYLTRANSFERASE

This enzyme [EC 2.3.1.16], also known as 3-ketoacyl-CoA thiolase, transfers an acyl group from an acyl-CoA to acetyl-CoA to form free coenzyme A and 3-oxoacyl-CoA.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

ACETYL-CoA:ACP TRANSACYLASE

This enzyme [EC 2.3.1.38], also referred to as acetyl-CoA:[acyl-carrier protein] S-acetyltransferase, transfers an acetyl group from one acetyl-CoA to an acyl-carrier-protein (ACP) to form free coenzyme A and the acetyl-[acyl-carrier-protein]. **See also** *Fatty Acid Synthase*

S. J. Wakil & J. K. Stoops (1983) *The Enzymes*, 3rd ed., **16**, 3.

P. R. Vagelos (1973) *The Enzymes*, 3rd ed., **8**, 155.

A. W. Alberts, P. W. Majerus & P. R. Vagelos (1969) *Meth. Enzymol.* **14**, 50.

ACETYL-CoA CARBOXYLASE

This enzyme [EC 6.4.1.2] catalyzes the reaction of acetyl-CoA, bicarbonate, and ATP to form malonyl-CoA, orthophosphate, and ADP. The plant enzyme will also act on propionyl-CoA and butanoyl-CoA. The enzyme will also catalyze certain transcarboxylations and it requires biotin as a cofactor.

J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 495.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 3.

K.-H. Kim (1997) *Ann. Rev. Nutr.* **17**, 77.

S. B. Ohlrogge & J. G. Jaworski (1997) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 109.

R. W. Brownsey & R. M. Denton (1987) *The Enzymes*, 3rd ed., **18**, 123.

K. Bloch (1977) *Adv. Enzymol.* **45**, 1.

A. W. Alberts & P. R. Vagelos (1972) *The Enzymes*, 3rd ed., **6**, 37.

ACETYL-CoA SYNTHETASE

This enzyme [EC 6.2.1.1], also referred to as acetate-CoA ligase or acetate thiokinase, catalyzes the reaction of acetate, coenzyme A, and ATP to form acetyl-CoA, AMP, and pyrophosphate. The enzyme will also utilize propanoate and propenoate as substrates.

L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.

J. C. Londesborough & L. T. Webster, Jr. (1974) *The Enzymes*, 3rd ed., **10**, 469.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Adenosine 5'-O-(1-thiotriphosphate), **87**, 224, 230-231; bridge-nonbridge oxygens, **87**, 251-253; kinetics, **87**, 355; mechanism, **87**, 251, 355; NMR, **87**, 251-253; stereochemistry **87**, 206, 212, 224, 230-233, 251-253.

ACETYLENE MONOCARBOXYLATE HYDRATASE

This enzyme [EC 4.2.1.71] adds water to propynoate to form 3-hydroxypropenoate. The enzyme will also act on 3-butyrate to form acetoacetate.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

N-ACETYL GALACTOSAMINE-4-SULFATE SULFATASE

This enzyme [EC 3.1.6.12] acts on 4-sulfate groups of the N-acetylgalactosamine 4-sulfate moieties in chondroitin sulfate and dermatan sulfate.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

N-ACETYL GALACTOSAMINE-6-SULFATE SULFATASE

This enzyme [EC 3.1.6.4] acts on 6-sulfate groups of the N-acetylgalactosamine 6-sulfate moieties in chondroitin sulfate and the galactose 6-sulfate groups in keratan sulfate.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

N-ACETYL GALACTOSAMINIDE SIALYLTRANSFERASE

This enzyme [EC 2.4.99.3] catalyzes the reaction of a glycano-1,3-(N-acetylgalactosaminyl)-glycoprotein and CMP-N-acetylneuraminate to produce CMP and the glycano-(2,6- α -N-acetylneuraminyl)-(N-acetylgalactosaminyl)-glycoprotein.

T. A. Beyer, J. E. Sadler, J. I. Rearick, J. C. Paulson & R. L. Hill (1981) *Adv. Enzymol.* **52**, 23.

N-ACETYLGLUCOSAMINE KINASE

This enzyme [EC 2.7.1.59] catalyzes the phosphorylation by ATP of *N*-acetylglucosamine to generate ADP and *N*-acetylglucosamine 6-phosphate. The bacterial enzyme is also reported to act on glucose as well.

S. S. Barkulis (1966) *Meth. Enzymol.* **9**, 415.

N-ACETYLGLUCOSAMINE-6-PHOSPHATE 2-EPIMERASE

This enzyme catalyzes the epimerization at the 2-position of *N*-acetylglucosamine 6-phosphate. **See also** *N*-Acylglucosamine-6-phosphate 2-Epimerase

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

N-ACETYLGLUCOSAMINE-6-SULFATE SULFATASE

This enzyme [EC 3.1.6.14] catalyzes the hydrolysis of the 6-sulfate moieties of the *N*-acetylglucosamine 6-sulfate subunits of heparan sulfate and keratan sulfate. It has been suggested that this enzyme might be identical to *N*-sulfoglucosamine-6-sulfatase.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

α -N-ACETYLGLUCOSAMINIDASE

This enzyme [EC 3.2.1.50] catalyzes hydrolysis of terminal nonreducing *N*-acetylglucosamine residues in *N*-acetyl- α -glucosaminides.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

β -N-ACETYLGLUCOSAMINIDASE

This enzyme, reportedly catalyzing the hydrolysis of terminal, nonreducing *N*-acetyl- β -glucosamine moieties in chitobiose and higher analogs, is now a deleted EC entry [EC 3.2.1.30].

P. M. Dey & E. del Campillo (1984) *Adv. Enzymol.* **56**, 141.

N-ACETYLGLUTAMATE SYNTHASE

This enzyme [EC 2.3.1.1], also referred to as amino-acid *N*-acetyltransferase and acetyl-CoA:glutamate *N*-acetyltransferase, catalyzes the reaction of acetyl-CoA with glutamate to form coenzyme A and *N*-acetylglutamate. The enzyme will also acts on aspartate and, more slowly, with some other amino acids. The mammalian enzyme is activated by L-arginine. **See also** *Glutamate Acetyltransferase*

S. G. Powers-Lee (1985) *Meth. Enzymol.* **113**, 27.
T. Sonoda & M. Tatibana (1983) *J. Biol. Chem.* **258**, 9839.
H. J. Vogel & R. H. Vogel (1974) *Adv. Enzymol.* **40**, 65.

N-ACETYL- γ -GLUTAMYL-PHOSPHATE REDUCTASE

This enzyme [EC 1.2.1.38], also known as *N*-acetylglutamate semialdehyde dehydrogenase and NAGSA dehydrogenase, catalyzes the reaction of *N*-acetylglutamate 5-semialdehyde with NADP⁺ and phosphate to generate *N*-acetyl-5-glutamyl phosphate and NADPH.

H. J. Vogel & R. H. Vogel (1974) *Adv. Enzymol.* **40**, 65.

β -N-ACETYLHEXOSAMINIDASE

This enzyme [EC 3.2.1.52], also referred to as β -hexosaminidase and *N*-acetyl- β -glucosaminidase, catalyzes the hydrolysis of terminal nonreducing *N*-acetylhexosamine residues in *N*-acetyl- β -hexosaminides. *N*-Acetylglucosides and *N*-acetylgalactosides are substrates.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.
H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

O-ACETYLHOMOSERINE (THIOL)-LYASE

This enzyme [EC 4.2.99.10], also referred to as *O*-acetylhomoserine sulfhydrylase, catalyzes the reaction of *O*-acetylhomoserine with methanethiol to generate methionine and acetate. The enzyme can also act on other thiols or H₂S, producing homocysteine or thioethers. The enzyme isolated from baker's yeast will also catalyze the reaction exhibited by *O*-acetylserine (thiol)-lyase [EC 4.2.99.8], albeit more slowly.

S. Yamagata (1987) *Meth. Enzymol.* **143**, 465.
I. Shiiro & H. Ozaki (1987) *Meth. Enzymol.* **143**, 470.

N-ACETYLNEURAMINATE LYASE

This enzyme [EC 4.1.3.3], also known as *N*-acetylneuramate aldolase, will convert *N*-acetylneuramate to *N*-acetylmannosamine and pyruvate. The enzyme will also act on *N*-glycolylneuramate and on *O*-acetylated sialic acids, other than *O*⁴-acetylated derivatives.

K. N. Allen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 135.
W. A. Wood (1972) *The Enzymes*, 3rd ed., **7**, 281.

N²-ACETYLORNITHINE AMINOTRANSFERASE

This enzyme [EC 2.6.1.11] catalyzes the pyridoxal-phosphate-dependent reaction of 2-acetylornithine with α -

ketoglutarate to produce *N*-acetylglutamate 5-semialdehyde and glutamate.

- H. J. Vogel & R. H. Vogel (1974) *Adv. Enzymol.* **40**, 65.
 A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.
 H. J. Vogel & E. E. Jones (1970) *Meth. Enzymol.* **17(A)**, 260.

N²-ACETYLORNITHINE DEACETYLASE

This enzyme [EC 3.5.1.16], also known as acetylornithinase and *N*-acetylornithinase, catalyzes the reaction of water with *N*²-acetylornithine to produce acetate and ornithine. The enzyme also catalyzes the hydrolysis of *N*-acetylmethionine.

- H. J. Vogel & R. H. Vogel (1974) *Adv. Enzymol.* **40**, 65.

O-ACETYL SERINE (THIOL)-LYASE

This enzyme [EC 4.2.99.8], also known as cysteine synthase and *O*-acetylserine sulfhydrylase, catalyzes the pyridoxal-phosphate-dependent reaction of H₂S with *O*³-acetylserine to produce cysteine and acetate. Some alkyl thiols, cyanide, pyrazole, and some other heterocyclic compounds can also act as acceptors.

- T. Nagasawa & H. Yamada (1987) *Meth. Enzymol.* **143**, 474.
 A. E. Martell (1982) *Adv. Enzymol.* **53**, 163.
 L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

ACHIRAL

The absence of chirality. Achiral molecules have an internal plane or point of symmetry. A molecular configuration is said to be achiral when it is superimposable on its mirror image. **See** *Chirality*

ACID

A substance that liberates protons as a consequence of its dissolution or dissociation. **See** *Brønsted Theory*; *Lewis Acid*; *Lewis Base*

ACID-BASE EQUILIBRIUM CONSTANTS

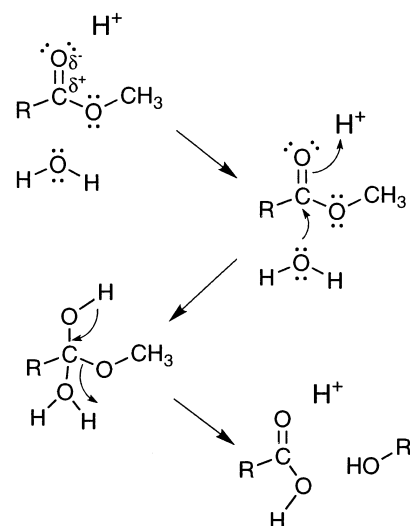
The following table contains *pK* values for a selected group of biochemically important substances. The associated thermodynamic parameters and the original literature references were presented by Edsall and Wyman¹.

pK_a Values for Selected Substances at 25°C

Substance	<i>pK_a</i>
Hydroxyproline (<i>pK₁</i>)	1.82
Proline (<i>pK₁</i>)	1.97
Aspartic acid (<i>pK₁</i>)	2.00
Threonine (<i>pK₁</i>)	2.10
Phosphoric acid (<i>pK₁</i>)	2.15
Serine (<i>pK₁</i>)	2.19
Chloroacetic acid	2.28
Glycine (<i>pK₁</i>)	2.35
Alanine (<i>pK₁</i>)	2.35
Formic acid	3.75
Lactic acid	3.86
Aspartic acid (<i>pK₂</i>)	3.91
Succinic acid (<i>pK₁</i>)	4.21
Acetic acid	4.76
Propionic acid	4.87
Succinic acid (<i>pK₂</i>)	5.63
Carbonic acid (apparent <i>pK₁</i>)	6.35
Glucose 1-phosphate (<i>pK₂</i>)	6.50
Phosphoric acid (<i>pK₂</i>)	7.20
Threonine (<i>pK₂</i>)	9.10
Serine (<i>pK₂</i>)	9.21
Ammonium ion	9.25
Ethanolammonium ion	9.50
Hydroxyproline (<i>pK₂</i>)	9.66
Glycine (<i>pK₂</i>)	9.78
Trimethylammonium ion	9.79
Alanine (<i>pK₂</i>)	9.87
Aspartic acid (<i>pK₃</i>)	10.00
Carbonic acid (<i>pK₂</i>)	10.33
Methylammonium ion	10.62
Proline (<i>pK₂</i>)	10.64
Dimethylammonium ion	10.77
Water (<i>pK_w</i>)	13.997

¹J. T. Edsall & J. Wyman (1958) *Biophysical Chemistry*, pp. 452, Academic Press, New York.

ACID CATALYSIS



Acid Catalysis of Ester Hydrolysis

Any process for which the rate of reaction is accelerated through the participation of an acid as a catalyst. **See** *General Acid Catalysis*; *Specific Acid Catalysis*

ACIDITY

1. The tendency for a Brønsted acid to act as a proton donor, expressed in terms of the compound's dissociation constant in water. 2. The term also refers to the tendency to form a Lewis adduct, as measured by a dissociation constant.

With reference to a solvent, this term is usually restricted to Brønsted acids. If the solvent is water, the pH value of the solution is a good measure of the proton-donating ability of the solvent, provided that the concentration of the solute is not too high. For concentrated solutions or for mixtures of solvents, the acidity of the solvent is best indicated by use of an acidity function. **See** *Degree of Dissociation; Henderson-Hasselbalch Equation; Acid-Base Equilibrium Constants; Brønsted Theory; Lewis Acid; Acidity Function; Leveling Effect*

ACIDITY FUNCTION

A thermodynamic measure of the proton-donating or proton-accepting ability of a solvent system (or closely related thermodynamic property such as the ability of a solvent system to form Lewis adducts)¹⁻³.

There are many types of acidity scales: depending on the nature of the indicator, of conjugate bases with a -1 charge, of acids that form stable carbocations, *etc.* Perhaps the best known acidity function is the Hammett acidity function, H_o , which is used for concentrated acidic solvents having a high dielectric constant^{4,5}. For any solvent (including a mixture of solvents in which the relative proportions are specified), H_o is defined to be $pK_{BHW+} - \log([BH^+]/[B])$. The value of H_o is measured using a weak indicator base whose pK value in water is known. Two common indicators are the aniline derivatives, *o*-nitroanilinium ion and 2,4-nitroanilinium ion, having pK_{BHW+} values of -0.29 and -4.53 , respectively, in water. The $[BH^+]/[B]$ ratio for one indicator is measured, usually spectrophotometrically, in the given solvent. Knowledge of a substance's pK in water (*i.e.*, pK_{BHW+}) allows the investigator to calculate H_o . Once H_o is known, pK_a values can be determined for any other acid-base pair.

The value of h_o is defined as $a_{H^+} f_I / f_{HI^+}$ where a_{H^+} is the chemical activity of the proton, and f_I and f_{HI^+} are the

activity coefficients of the indicator. H_o and h_o are related by $H_o = -\log h_o$. For dilute solutions, $H_o = \text{pH}$. This is the situation for most biochemical reactions. The subscript is a reference to the net charge on the base. Hence, H_- is the corresponding acidity function for mononegatively charged bases in equilibrium with neutral acids.

The Hammett acidity function only applies to acidic solvents having a high dielectric constant, further requiring that the f_I/f_{HI^+} ratio be independent of the nature of the indicator. Thus, the Hammett acidity function is applicable for uncharged indicator bases that are aniline derivatives. As mentioned above, other acidity functions have been suggested. No single formulation has been developed that satisfies different solvent systems or types of bases.

Bunnett and Olsen⁶⁻⁸ used a different approach and derived the equation:

$$\log([SH^+]/[S]) + H_o = \Phi(H_o + \log[H^+]) + pK_{SH^+}$$

in which S is a base that can be protonated by an acidic solvent. Plotting $\log([SH^+]/[S]) + H_o$ versus $H_o + \log[H^+]$ will result in a reasonably linear line having a slope of Φ . The method uses a linear free energy relationship to address the problems of defining basicity for weak organic bases in solutions of moderate concentrations of mineral acids. The value and sign of Φ is a measure of the response of the acid-base equilibria to changes in acid concentration. An analogous equation for kinetic data is

$$\log k_\psi + H_o = \Phi(H_o + \log[H^+]) + \log k_2^\circ$$

where k_ψ is the pseudo-first-order rate constant for a reaction in acidic solution and k_2° is the corresponding constant at infinite dilution in water. Here Φ is suggested to represent the response of the chemical reaction rate to changes in the acid concentration. Attempts have been made to apply this method to basic media. More-O'Ferrell⁹ suggested that the slope may be related semi-quantitatively to the structure of the transition state.

A related treatment of the acidity function considers changes in acidity of the solvent on an acid-base equilibrium¹⁰. With this procedure, an equilibrium is chosen as reference, having an equilibrium constant K'_o for the

reference reaction in a reference solvent and K' for the reference reaction in the particular medium under study. The reaction under study has the corresponding equilibria of K_o (in the reference solvent) and K (in the particular medium). Thus,

$$\log (K'/K_o') = m^* \log (K/K_o)$$

where the slope m^* corresponds to $1 - \Phi$ in the Bunnett-Olsen treatment. Bunnett¹¹ has also plotted $\log k_{\psi} + H_o$ vs. $\log a_{\text{water}}$ where a_{water} is the activity of water and has indicated that the slope of the line, w , suggests certain possibilities for the chemical reaction mechanism in moderately concentrated aqueous acids. For example, if w is between -2.5 and zero, then water does not participate in the formation of the transition state. If w is between 1.2 and 3.3 , water participates as a nucleophile in the rate-determining step. Long and Bakule¹² have criticized this approach. **See also** *Degree of Dissociation; Henderson-Hasselbalch Equation; Acid-Base Equilibrium Constants; Bunnett-Olsen Equations; Cox-Yeats Treatment*

¹C. H. Rochester (1970) *Acidity Functions*, Academic Press, New York.

²J. March (1985) *Advanced Organic Chemistry*, Wiley, New York.

³J. Hine (1962) *Physical Organic Chemistry*, McGraw-Hill, New York.

⁴L. Zucker & L. P. Hammett (1939) *J. Amer. Chem. Soc.* **61**, 2791.

⁵L. P. Hammett (1970) *Physical Organic Chemistry*. McGraw-Hill, New York.

⁶J. F. Bunnett & F. P. Olsen (1966) *Can. J. Chem.* **44**, 1899 and 1917.

⁷J. F. Bunnett, R. L. McDonald & F. P. Olsen (1974) *J. Amer. Chem. Soc.* **96**, 2855.

⁸V. Lucchini, G. Modena, G. Scorrano, R. A. Cox & Y. Yates (1982) *J. Am. Chem. Soc.* **104**, 1958.

⁹R. A. More O'Ferrall (1972) *J. Chem. Soc., Perkin Trans.* **2**, 976.

¹⁰A. Bagno, G. Scorrano & R. A. More O'Ferrall (1987) *Rev. Chem. Interim.* **7**, 313.

¹¹J. F. Bunnett (1961) *J. Amer. Chem. Soc.* **83**, 4956.

¹²F. A. Long & R. Bakule (1963) *J. Amer. Chem. Soc.* **85**, 2313.

ACID-LABILE SULFIDES

The bridging sulfur atoms in iron-sulfur proteins are often referred to as acid-labile sulfides, because treatment of such proteins with acids generates H_2S .

ACID PHOSPHATASE

This enzyme [EC 3.1.3.2], also referred to as acid phosphomonoesterase, phosphomonoesterase, and glycerophosphatase, catalyzes the hydrolysis of an orthophosphoric monoester to generate an alcohol and

orthophosphate. The enzyme, which has a wide specificity, will also catalyze transphosphorylations.

M. Cohn (1982) *Ann. Rev. Biophys. Bioeng.* **11**, 23.

V. P. Hollander (1971) *The Enzymes*, 3rd ed., **4**, 449.

ACONITASE

This [4Fe-4S] cluster-containing enzyme [EC 4.2.1.3], also known as citrate hydro-lyase and aconitate hydratase, will act on citrate to generate *cis*-aconitate ((*Z*)-prop-1-ene 1,2,3-tricarboxylate) and water. The enzyme will also catalyze the conversion of isocitrate into *cis*-aconitate.

V. E. Anderson (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 115.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

H. Lauble, M. C. Kennedy, H. Beinert & D. C. Stout (1992) *Biochemistry* **31**, 2735.

L. Zheng, M. C. Kennedy, H. Beinert & H. Zalkin (1992) *J. Biol. Chem.* **267**, 7895.

J. B. Howard & D. C. Rees (1991) *Adv. Protein Chem.* **42**, 199.

P. A. Srere (1975) *Adv. Enzymol.* **43**, 57.

J. P. Glusker (1971) *The Enzymes*, 3rd ed., **5**, 413.

ACONITATE DECARBOXYLASE

This enzyme [EC 4.1.1.6] catalyzes the conversion of *cis*-aconitate to itaconate (or, 2-methylenesuccinate) and carbon dioxide.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

R. Bentley (1962) *Meth. Enzymol.* **5**, 593.

ACONITATE Δ -ISOMERASE

This enzyme [EC 5.3.3.7] catalyzes the interconversion of *trans*-aconitate to *cis*-aconitate, the reaction reportedly to occur by an allelic rearrangement.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

ACROSIN

This enzyme [EC 3.4.21.10] catalyzes the hydrolysis of Arg-Xaa and Lys-Xaa peptide bonds. The enzyme belongs to the peptidase family S1 and is inhibited by naturally occurring trypsin inhibitors.

J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.

ACTIN ASSEMBLY ASSAYS

The 43 kDa actin monomer (often termed globular actin or G-actin) polymerizes to form filamentous (or F-) actin, a process that can be measured by a number of biochemical and biophysical techniques¹. Actin self-assembly obeys the kinetics of nucleated polymerization², and the stages of polymerization include nucleation, elongation, and polymer length redistribution. Cooper and Pollard¹ have presented the following table indicating the advantages and disadvantages of each of these techniques.

Table I

Methods to Measure Actin Polymerization

Method	Signal α [polymer]	Signal : noise ratio	Sensitivity to length	Shear rate	Native actin	Expense	Sample size (ml)
1. Capillary viscometry	$\pm$	High	Yes	High	Yes	Low	≥ 0.6
2. ΔOD_{232}	Yes	Moderate	No	0	Yes	High	≥ 0.6
3. Flow birefringence	$\pm$	High	Yes	Variable	Yes	High	≥ 1
4. Fluorescence							
a. NBD-NEM-actin	Yes	Moderate	No	0	No	High	≥ 0.5
b. Pyrene-actin	Yes	High	No	0	No	High	≥ 0.5
5. Light scattering	Yes	High	No	0	Yes	High	≥ 1
6. Electron microscopy	Yes	High	Yes	0	Yes	High	< 0.1
7. DNase inhibition	Yes	High	No	0	Yes	Moderate	< 0.1
8. Pelleting	Yes	High	Yes	Moderate	Yes	Moderate	0.2
9. Millipore filtration	Unknown	Moderate	Yes	High	Yes	Low	< 0.1

(1) *Capillary viscometry* is based on the higher viscosity of F-actin compared to G-actin, and one determines the time needed for a solution to pass through the orifice of a glass capillary viscometer. (2) *Difference spectroscopy* at 232 nm can be utilized to estimate the amount of F-actin in a solution, but one must exercise care to correct for light scattering contributions to the apparent optical density change (ΔOD_{232}). (3) *Flow birefringence* relies on the alignment of actin filaments with the direction of flow imparted by the rotation of one of two concentric cylinders containing the filaments in the annular space between the cylinders. (4) *Fluorescence* changes in an extrinsic chromophore (NBD or pyrene) covalently attached to actin monomers at cysteine-373 in the actin monomer. (5) *Light scattering* can be used to study polymerization because the filaments scatter light much more intensely than individual actin subunits. The 90° light scattering intensity is directly related to polymer weight concentration. (6) *Electron microscopy* can provide a

direct assessment of the length of suitably fixed and contrast-stained actin filaments. (7) *DNase inhibition* of actin polymerization takes advantage of the observation that DNase I preferentially binds to actin monomers with sufficient affinity to block any polymerization of DNase-actin complex. (8) *Pelleting* relies on the much higher sedimentation coefficient of actin filaments as compared to monomeric actin. (9) *Millipore filtration assays* allow one to rapidly separate monomeric and polymeric actin using filter disks with 0.45 μm pores.

Also see *Protein Polymerization; Self-Assembly Mechanisms*.

¹J. A. Cooper & T. D. Pollard (1982) *Meth. Enzymol.* **85**, 182.

²F. Oosawa & S. Asakura (1975) *Thermodynamics of Protein Polymerization*, Academic Press, New York.

ACTIN ASSEMBLY KINETICS

The self-assembly of actin filaments occurs through sequential head-to-tail polymerization, a process that passes through phases called nucleation, elongation, and establishment of the monomer-polymer equilibrium, polymer length redistribution, and treadmilling¹⁻³. The last two are properties of assembled filaments. Monomeric actin is called G-actin (or globular actin), and polymeric actin is termed F-actin (or filamentous actin). The most extensively studied actins are those obtained in high abundance from rabbit muscle and *Acanthamoeba castellanii*. Many features of actin polymerization are analogous to tubulin, and the reader may wish to consult that entry. **See** *Microtubule Assembly Kinetics*.

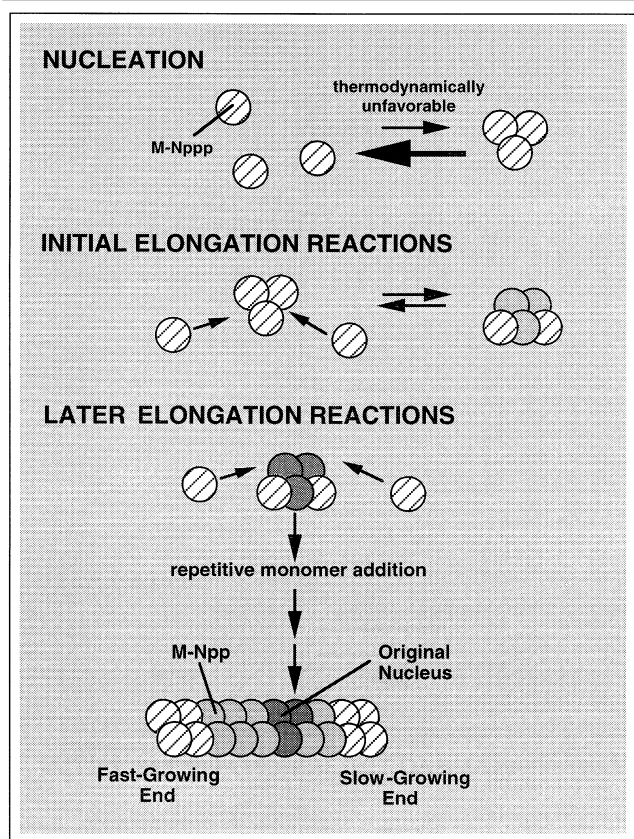


Figure 1. Nucleation and growth of actin filaments. Nucleation is shown here as a thermodynamically unfavored process, which in the presence of sufficient actin-ATP will undergo initial elongation to form small filament structures that subsequently elongate with rate constants that do not depend on filament length. Elongation proceeds until the monomeric actin (or G-actin) concentration equals the critical concentration for actin assembly.

Nucleation is the most thermodynamically unfavorable step (Fig. 1) in which three actin·ATP molecules must come together to form a polymerization nucleus. Only when the actin·ATP concentration is sufficiently high can these unstable nuclei persist at sufficient concentrations for polymerization to occur. Uncoordinated spontaneous nucleation must be suppressed in certain regions of the cytoplasm, and actin monomer-sequestering proteins may limit nucleation by decreasing the available concentration of actin·ATP monomers. The ADP·actin complex appears to be a potent inhibitor of nucleation. The nucleation process cannot persist for long if elongation is occurring. Indeed, if A_{total} is the total monomer concentration at the beginning of elongation, then the rate of spontaneous nucleation will be proportional to the third power of A_{total} :

$$\text{Rate}_{\text{Nucleation}} = k_N [A_{\text{total}}]^3$$

and a 20% drop in A_{total} should reduce the rate of nucleation by a factor of two:

$$\begin{aligned} \text{Rate}_{0.8}/\text{Rate}_{1.0} &= k_N [0.8A_{\text{total}}]^3 / k_N [A_{\text{total}}]^3 \\ &= (0.8)^3 [A_{\text{total}}]^3 / [A_{\text{total}}]^3 = 0.5 \end{aligned}$$

This effect will be even greater if there is any ADP-actin in the total pool of actin monomers, especially if ADP-actin inhibits nucleation.

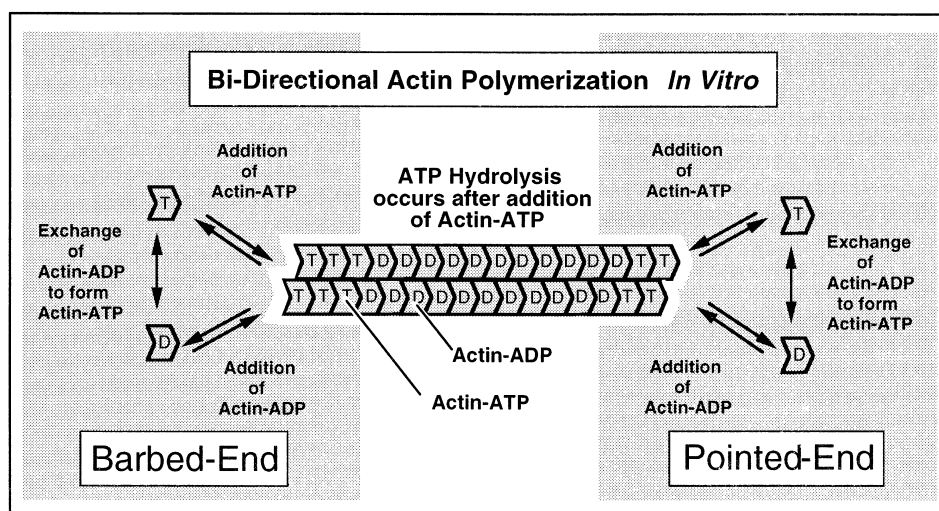


Figure 2. Scheme for the addition and loss of actin-ATP or actin-ADP from the barbed and pointed ends of an actin filament. The barbed end is the faster growing and more stable end of an actin filament. While the exchange of actin-ADP with ATP to yield actin-ATP and ADP is shown here as a spontaneous process, the actin regulatory protein profilin greatly accelerates the exchange process. Note also that hydrolysis is thought to occur after (and not coincident with) addition of actin-ATP at either end.

Elongation is the repetitive addition reactions of actin·ATP (Fig. 2). The actin-bound ATP is hydrolyzed during/after monomer addition to filaments, forming polymer-bound ADP and releasing orthophosphate. The kinetics of elongation conform to that predicted by the following rate law:

$$dC_p/dt = k_+N[A_{\text{total}}] - k_{-1}N$$

where C_p is the concentration of polymerized actin, k_+ is the bimolecular rate constant (equal to the sum of the on-rate constants for actin monomer addition to both ends of an actin filament, N is the number concentration of polymer ends that are capable of reacting with actin monomers, $[A_{\text{total}}]$ is the total monomer concentration, and k_{-1} is the rate constant for monomer dissociation. If N is constant during the elongation phase, the observed rate process will fit a simple first-order decay curve. Elongation continues until the elongation “on”-rate (i.e., $k_+[actin]$) is exactly balanced by the “off”-rate (k_-). This condition defines the critical actin concentration ($[actin]_{\text{critical}} = k_-/k_+$), a parameter which represents the concentration of actin that coexists with assembled filaments (Fig. 3, top).

Because the onset of monomer-polymer equilibrium can occur before the filaments achieve their own equilibrium concentration behavior, these filaments will undergo polymer length redistribution. This is a slow process *in vitro* that in many respects resembles crystallization (See *Ostwald Ripening*).

Treadmilling is the net opposite-end assembly/disassembly process first described by Wegner's study⁴ of actin polymer dynamics. He recognized that there was no reason to believe that the release of free energy of ATP hydrolysis need be identical for addition reactions occurring on opposite filament ends. The so-called (+)-end and (–)-end have different rate and equilibrium constants: $K^+ = k_{\text{off}}^+/k_{\text{on}}^+$ and $K^- = k_{\text{off}}^-/k_{\text{on}}^-$, such that $[actin]_{\text{critical}} = k_{\text{off}}^+/k_{\text{on}}^+ = (k_{\text{off}}^+ + k_{\text{off}}^-)/(k_{\text{on}}^+ + k_{\text{on}}^-)$. The macroscopic actin critical concentration obeys the following inequality:

$$k_{\text{off}}^+/k_{\text{on}}^+ < [actin]_{\text{critical}} < k_{\text{off}}^-/k_{\text{on}}^-$$

As the name implies, actin treadmilling does not involve any net increase in the amount of polymerized actin; instead, when ATP is present, monomers are spontane-

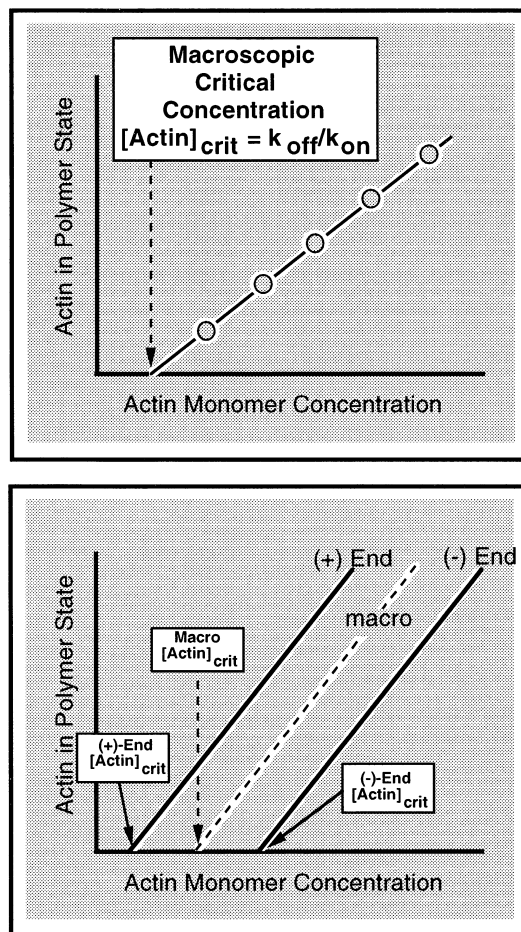


Figure 3. Critical concentration behavior of actin self-assembly. For the top diagram depicting the macroscopic critical concentration curve, one determines the total amount of polymerized actin by methods that measure the sum of addition and release processes occurring at both ends. Examples of such methods are sedimentation, light scattering, fluorescence assays with pyrene-labeled actin, and viscosity measurements. For the bottom curves, the polymerization behavior is typically determined by fluorescence assays conducted under conditions where one of the ends is blocked by the presence of molecules such as gelsolin (a barbed-end capping protein) or spectrin-band 4.1-actin (a complex prepared from erythrocyte membranes, such that only barbed-end growth occurs). Note further that the barbed end (or (+)-end) has a lower critical concentration than the pointed end (or (–)-end). This differential stabilization requires the occurrence of ATP hydrolysis to supply the free energy that drives subunit addition to the (+)-end at the expense of the subunit loss from the (–)-end.

ously released from the less stable (–)-ends, and monomers are taken up by the more stable (+)-ends (Fig. 3). This results in a flux of monomers that can be quantitatively assessed by the inclusion of [³H]ATP or [¹⁴C]ATP which exchanges with unlabeled actin in the monomeric pool and upon polymerization is trapped in filaments as actin-ADP.

Although investigations of the *in vitro* polymerization process have provided a theoretical underpinning for probing intracellular actin dynamics, there are obvious voids in our understanding. Actin-based motility in many cell types reaches a rate of 1 $\mu\text{m}/\text{second}$, corresponding to the steady-state addition/loss of 50 to 100 monomers per filament per second. This is much faster than any *in vitro* kinetic study, and the mechanism of assembly may differ substantially from the model studies. For example, although exchange of ATP for ADP on actin is not a rate-limiting reaction *in vitro*, this may not be true in the cell. Intracellular F-actin formation is also highly localized, occurring within discrete polymerization zones established by assembly of actin-based motor complexes from a battery of regulatory proteins. These include capping proteins, severing proteins, sequestering proteins, bundling and cross-linking proteins, actin-related proteins, as well as a growing list of actin-based motor complex components. Their interactions are controlled by binding interactions (**See Actin Filament Capping Protein, Actin Filament Severing Protein, Actin Filament Bundling/Cross-Linking Protein, and ABM-1 & ABM-2 Sequences in Actin-Based Motors**). In addition, actin assembly is controlled by extracellular cues transmitted to the cell's interior by focal adhesion and adherens junction proteins.

¹T. P. Stossel (1993) *Science* **260**, 1086.

²M. F. Carlier (1989) *Int. Rev. Cytol.* **115**, 139.

³T. D. Pollard & J. A. Cooper (1986) *Annual Rev. Biochem.* **55**, 987.

⁴A. Wegner (1982) *J. Mol. Biol.* **161**, 607.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Staining with fluorochrome-conjugated phalloidin, **194**, 729; binding activity of ponticulin, **196**, 58; crosslinking in agonist-stimulated cells, assay, **196**, 486; depolymerization [actin-binding protein effects, **215**, 74; myosin effects, **215**, 74; dilution effect on platelet cytoskeleton, **215**, 76]; expression in *Escherichia coli*, **196**, 368 [growth conditions for, **196**, 378; solubility, effect of bacterial lysis, **196**, 380; vector-related variability, **196**, 370]; fluorescein labeling, **196**, 50; actin-gelsolin complex [assay, **215**, 94; isolation from platelets, **215**, 89]; globular, purification from *Dictyostelium discoideum*, **196**, 89; isoforms, separation, **196**, 105; monomeric columns with, preparation, **196**, 310; nucleation activity [agonist-stimulated, assay of inhibitor, **196**, 495; measurement, **196**, 493]; platelet [characterization, **215**, 58; purification, **215**, 58, 66; recombination with actin-binding protein and α -actinin, **215**, 73]; polymerization in agonist-stimulated cells, assay, **196**, 486; preparation, **196**, 402; actin-profilin complex, isolation from cell extracts, **196**, 97; purification, **196**, 50, 74; separation from profilin:actin, **196**, 115; skeletal muscle, labeling with pyrene, **196**, 138; sliding over myosin-coated surfaces, assay, **196**, 399; stored, rejuvenation by recycling, **196**, 403.

Actin filaments: affinity column [preparation, **196**, 49; properties, **196**, 305]; assay [in cells, **196**, 486; in platelet lysates, **215**, 54]; associated proteins, identification, **215**, 50; attachment to col-

umn matrix, **196**, 309; chromatography, **196**, 312; crosslinking in cells, measurement, **196**, 491; fluorescent labeling, **196**, 403; imaging, **196**, 408; length measurement, **196**, 416; proteins binding, affinity chromatography, **196**, 303.

Actin-binding proteins: ABP-50, purification, **196**, 78; ABP-120, purification, **196**, 79; ABP-240 purification, **196**, 76; effect on actin depolymerization, **215**, 74; extraction, **196**, 311; isolation from *Dictyostelium discoideum*, **196**, 70; platelet-derived actin binding proteins [characterization, **215**, 58; purification, **215**, 58, 64; recombination with actin, **215**, 73]; 30-kDa *Dictyostelium discoideum* actin-crosslinking protein [assays, **196**, 91; preparation, **196**, 84]; actin-depolymerizing factor [assay, **196**, 132]; DNase assay, **196**, 136; platelet-derived α -actinin [characterization, **215**, 58; purification, **215**, 58, 70; recombination with actin, **215**, 73].

ACTIN ATPase REACTION (Apparent Irreversibility)

The pathway of ATP hydrolysis associated with rabbit muscle actin polymerization was investigated using an assay of intermediate ^{18}O -positional isotope exchange reactions. Under a variety of conditions that influence the rate and extent of F-actin self-assembly, ATP hydrolysis proceeded without any evidence of multiple reversals that are characteristic of any reversible phosphoanhydride-bond cleavage¹. These results are in harmony with published findings² indicating that GTP hydrolysis during tubulin polymerization also fails to display any evidence of intermediate exchange reactions.

¹M. F. Carlier, D. Pantaloni, J. A. Evans, P. K. Lambooy, E. D. Korn & M. R. Webb (1988) *FEBS Lett.* **235**, 211.

²J. M. Angelastro & D. L. Purich (1990) *Eur. J. Biochem.* **191**, 507.

ACTIN-BASED PATHOGEN MOTILITY

Actin-based motility involves the highly regulated assembly/disassembly of actin filaments in the cytoplasm nearest the peripheral membrane. The energetics and polarity of ATP-dependent filament assembly produce the expansive force and directionality for peripheral membrane protrusion during ameboid motion¹. Two bacterial systems bypass the complexities of signal transduction cascades as well as membrane geometry: *Listeria monocytogenes*, a gram-positive rod that causes severe meningitis and bacteremia, and *Shigella flexneri*, a gram-negative rod that is a leading cause of bacillary dysentery. *Listeria* produces proteins known as internalins to induce phagocytosis^{2,3}, and *Shigella* uses the surface proteins, Ipa B and C⁴. Although most bacteria are trapped and killed in phagolysosomes, ingested *Listeria* and *Shigella* readily escape by producing hemolysins that disrupt the phagolysosomal membrane⁵⁻⁷. After entering the host

cytoplasm, they grow readily, doubling every hour. These pathogens usurp host contractile proteins, allowing them to mimic the activated membrane state that stimulates localized actin filament assembly for their own motility. Then an initial halo of actin filaments surrounding the bacteria is remodeled over several hours, thereby confining actin assembly to one bacterial pole^{8,9}. They begin to locomote and to travel throughout the cytoplasm, eventually reaching the peripheral membrane, where growth of their actin-rich tails forces the membrane to protrude outwardly, forming filopods. Bacteria-laden filopods are ingested by adjacent cells, and the life-cycle begins anew. In this manner, both *Listeria* and *Shigella* evade humoral immunity and antibiotics that poorly penetrate host cells¹⁰.

Kinetics of Bacterial Motility. Examination of *Listeria*-infected host cells has provided many important clues about pathogen motility. Renal tubular epithelial PtK2 cells spread and flatten when grown on glass coverslips; after infection, extracellular bacteria are washed away, and addition of gentamicin (10 $\mu\text{g/mL}$) prevents extracellular bacterial growth. This antibiotic penetrates cells poorly and does not impair proliferation of intracellular bacteria^{8,11,12}. Within 3–4 hours postinfection¹³, *Listeria* begin moving at rates of 0.1 to 0.45 $\mu\text{m/sec}$. As they move, phase-dense tails form, and those bacteria moving more rapidly have longer tails. These tails contain numerous actin filaments, as indicated by staining with fluorescently conjugated phalloidin, a mushroom toxin that specifically binds to actin filaments. Microinjection of live, infected cells with rhodamine-conjugated actin monomers allows the kinetic analysis of actin assembly and disassembly in these tails. More rapid motility is always matched by more rapid actin assembly. By contrast, the rate of actin disassembly is constant ($t_{1/2} = 40\text{--}60\text{ sec}$) and does not depend on bacterial speed^{13,14}. Time-lapse video also revealed that the actin filament tails are not dragged along by the bacterium; rather, these tails remain fixed in the cytoplasm, and the progressive addition of actin subunits near the bacterium allows the pathogen to move forward. Labeled monomers are observed to add only at the junction between the bacterium and the actin tail¹³. Thus, new actin filament assembly takes place where the bacterium contacts the tail. These filaments become bundled by α -actinin, and they also become linked to the existing cy-

toskeletal network to form a stable platform. This prevents backward movement of the bacterium as actin filaments at the bacteria-tail junction lengthen; filament elongation propels each bacterium through the cytoplasm in the direction opposite to the tail. These experiments constitute the first clear demonstration that actin filaments can produce force and actively generate movement.

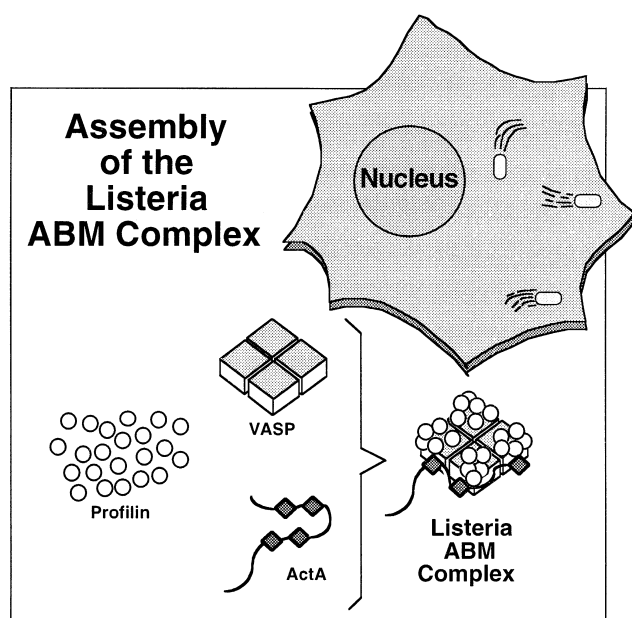
The site of motor assembly lies directly behind the moving bacterium in what has been termed a polymerization zone. *Listeria* assembles its actin-based motor by concentrating host cell proteins that stimulate actin assembly. Transposon mutagenesis^{15,16} demonstrated that a single 67 kDa protein, ActA, is needed for the formation of actin rocket tails as well as for the spread of bacteria from cell to cell. The ability to cause disease in mice was also markedly impaired (the LD_{50} increasing from 10^4 to 10^8 organisms) in ActA-deficient *Listeria*. ActA has a series oligoproline FEFPPPPTDE repeats. Microinjection¹⁷ of CFEFPPPPTDE, a synthetic peptide analogue of the second oligoproline repeat into *Listeria*-infected PtK2 cells competitively blocks *Listeria* actin-based motility. Time-lapse video phase microscopy revealed that within 30 sec after introducing the peptide, *Listeria* ceased all further motility. Over this same time-frame, the phase-dense tails also disappeared. At intracellular peptide concentration as low as 50 nM, *Listeria* actin-based motility was totally arrested¹⁷. Microinjection of a combinatorial library of peptides containing the very same amino acids, but in random order, was completely without effect at 60 μM . Similarly, two SH3 peptides failed to inhibit even at concentrations of 60 μM . These SH3 sequences contain intervening positively charged residues rather than negatively charged residues of the ActA peptide. Little or no inhibition was observable when 10 μM poly(L-proline) was microinjected. These experiments demonstrate the specificity of ActA peptide interactions in *Listeria* motility.

Chakrabarty *et al.*¹⁸ discovered that the host cell protein vasodilator-stimulated phosphoprotein (VASP) concentrates at the back of motile *Listeria* at the bacteria-tail junction. VASP docks onto the surface of *Listeria* by binding to the oligoproline sequences of ActA concentrated on the rearward pole of moving bacteria. In uninfected host cells, VASP is normally found in focal con-

tacts, regions that allow cells to adhere to the substratum and to process extracellular stimuli¹⁹. Focal contacts also contain high concentrations of actin filaments as well as other actin-binding proteins, including α -actinin, vinculin, zyxin, paxillin, and talin. Microinjection of the FEFPPPTDE peptide into uninfected cells dissociates VASP from these focal contacts, suggesting the existence of an ActA-like VASP binding protein in host cells¹⁸.

By binding profilin, a 15.5 kDa actin-regulatory protein, VASP acts as an adapter protein to link profilin to the surface of *Listeria*.¹⁹ Each 45 kDa VASP monomer contains a series of 5–6 oligoproline Gly-Pro-Pro-Pro-Pro repeats (abbreviated GP₅) that bind to profilin²⁰. The native VASP tetramer therefore contains 20–24 GP₅ sites for profilin binding. Immunofluorescence microscopy demonstrated that profilin concentrates at the back of *Listeria* in the very same region where VASP is found²¹. Microinjection of GP₅GP₅GP₅ blocks *Listeria* actin-based motility by competitively inhibiting profilin binding to VASP²⁰.

BASIC FEATURES OF THE ACTIN-BASED MOTOR COMPLEX. These experiments support the following model for *Listeria* actin-based motility. The *Listeria* surface protein ActA uses its oligoproline sequences (FEFPPPTDE) to bind the host cell protein VASP. VASP in turn deploys its own larger set of GPPPPP sequences to bind profilin.



This system allows one ActA molecule to concentrate up to 96 profilin molecules, and, *Listeria* may concentrate as much as 0.5–1 mM profilin on its surface²⁰. Profilin markedly potentiates growth of actin filaments²², and highly concentrated profilin in the polymerization zone would be expected to result in explosive actin filament assembly, thus generating the forces required for *Listeria* intracellular movement.

IDENTIFICATION OF HOST CELL ActA HOMOLOGUES. Sansonetti's group first demonstrated that the 120 kDa bacterial surface protein IcsA is the only protein required for the generation of the actin-based motor in *Shigella*²³. Comparisons of the derived amino acid sequences however demonstrate no homology between the ActA protein and IcsA, which contains no oligoproline sites. IcsA attracts a host cell protein as an ActA homologue (*i.e.*, the host cell protein must function in place of ActA by providing oligoproline sequences suitable for high-affinity binding of VASP). Zeile *et al.*¹² microinjected the *Listeria* ActA peptide into *Shigella*-infected cells to show that 80 nM FEFPPPTDE sequence also blocked *Shigella* actin-based motility. Furthermore, immunofluorescence experiments with anti-VASP antibody demonstrated that VASP became localized on the rearward pole of motile *Shigella*¹⁸. Microinjection of the GP₅ peptide into *Shigella*-infected cells also blocked profilin binding to VASP, and was effective in arresting *Shigella* motility at 10–12 μ M.

That motile *Shigella* concentrate an ActA homologue on their surface provided a way to identify a proteolytic vinculin fragment as the homologue. Laine *et al.*²⁴ used synthetic FEFPPPTDE as an immunogen to generate a polyclonal antibody (Ab). Microinjection of this antibody blocked *Shigella* motility in infected cells, and the Ab was localized to the bacterial tails. Western blots of

The *Listeria* Actin-Based Motility complex. The bacterial surface protein ActA contains four oligoproline repeats (designated as dark rectangles) that bind vasodilator-stimulated phosphoprotein VASP, a tetrameric actin-regulatory protein in the cytoplasm of the host cell. Each VASP tetramer contains 20–24 profilin-binding sites of the type GPPPPP, allowing high concentrations of profilin to bind onto the surface of *Listeria*. This creates a polymerization zone that enhances production and polymerization of ATP-actin monomers. Profilin probably ushers ATP-actin monomers to actin filaments growing at the bacterium-actin tail junction.

platelet extracts revealed a single, strongly cross-reactive polypeptide ($M_r = 90$ kDa; $pI = 6.5$) isolated by 2-dimensional IEF/SDS-PAGE. Microsequencing of five lysC protease fragments matched corresponding stretches in the head domain of vinculin, an actin regulatory protein localized in focal adhesion contacts²⁴. The carboxy-terminus of the 90 kDa head region contains a DFPPPPDLE sequence that is virtually identical to those in ActA. The anti-ActA Ab failed to react with full-length vinculin, binding preferentially to its 90 kDa fragment in cell extracts. This was confirmed using the p90 head domain (generated *in vitro* by thermolysin treatment) and full-length vinculin, indicating that this antigenic site was masked in the intact molecule. Thus, despite the harsh conditions of SDS-PAGE, refolding must occur upon processing the blot, and the ActA-like sequence is masked, rendering it incapable of attracting VASP until the head region is freed from the tail²⁴. Vinculin p90 accumulates in *Shigella*-infected PtK2 cells over the 1–3 hour period after initiating infection, but no p90 was observed in uninfected cells²⁴. Separate microinjections of intact vinculin and p90 head fragment into *Shigella*-infected cells demonstrated that 120 nM vinculin was completely without effect on the rate of *Shigella* locomotion, but 120 nM p90 dramatically accelerated the velocity of the *Shigella* intracellular movement, increasing mean velocities by a factor of three²⁴. These experiments suggest a variation on the *Listeria* model: (1) *Shigella* infection leads to vinculin proteolysis; (2) an ActA-like oligoproline sequence is unmasked in the vinculin head region; (3) IcsA binds the vinculin head domain, and the exposed ActA-like sequence binds VASP; (4) the latter in turn binds profilin onto the rearward pole of the bacterium; and (5) the high concentration of profilin within this polymerization zone stimulates actin filament assembly. Two phylogenetically unrelated bacteria have adopted similar subcellular mechanisms that permit them to spread from cell to cell. Both usurp host components to assemble actin-based motor complexes that are structurally and functionally related. *Listeria* uses ActA to attract VASP which then concentrates profilin. See ABM-1 and ABM-2 Sequences in Actin-Based Motors

¹T. P. Stossel (1994) *Sci. Am.* **271** (September), 54.

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³S. Dramsi, I. Biswas, E. Maguin, L. Braun, P. Mastroeni & P. Cossart (1995) *Mol. Microbiol.* **16**, 251.

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⁷S. Barzu, Z. Benjelloun-Touimi, A. Phalipon, P. Sansonetti & C. Parrot (1997) *Infect. Immun.* **65**, 1599.

⁸G. A. Dabiri, J. M. Sanger, D. A. Portnoy & F. S. Southwick (1990) *Proc. Natl. Acad. Sci. U. S. A.* **87**, 6068.

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¹⁰F. S. Southwick & D. L. Purich (1996) *New Engl. J. Med.* **334**, 770.

¹¹E. A. Havell (1986) *Infect. Immun.* **54**, 787.

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¹³J. M. Sanger, J. W. Sanger & F. S. Southwick (1992) *Infect. Immun.* **60**, 3609.

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¹⁶C. Kocks, E. Gouin, M. Tabouret, P. Berche, H. Ohayon & P. Cossart (1992) *Cell* **68**, 521.

¹⁷F. S. Southwick & D. L. Purich (1994) *Proc. Natl. Acad. Sci. U. S. A.* **91**, 5168.

¹⁸T. Chakraborty, F. Ebel & E. Domann (1995) *EMBO J.* **14**, 1314.

¹⁹M. Reinhard, M. Halbrugge, U. Scheer, C. Wiegand, B. M. Jockusch & U. Walter (1992) *EMBO J.* **11**, 2063.

²⁰F. Kang, R. O. Laine, M. R. Bubbs, F. S. Southwick & D. L. Purich (1997) *Biochemistry* **36**, 8384.

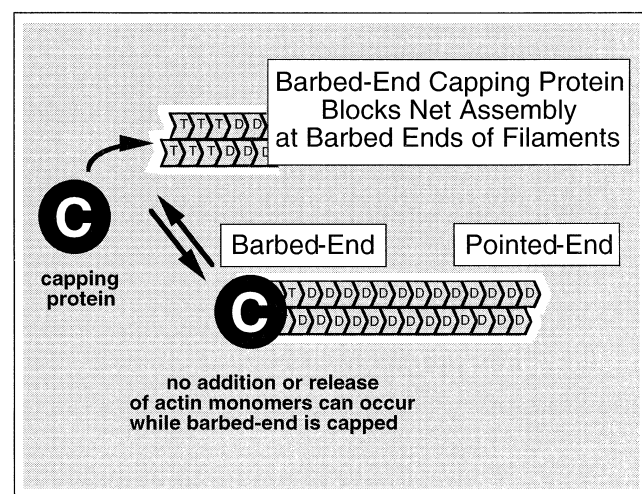
²¹J. A. Theriot, J. Rosenblatt, D. A. Portnoy, P. J. Goldschmidt-Clermont & T. J. Mitchison (1994) *Cell* **76**, 505.

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²³M. L. Bernardini, J. Mounier, H. d'Hauteville, M. Coquis-Rondon & P. J. Sansonetti (1989) *Proc. Natl. Acad. Sci. U. S. A.* **86**, 3867.

²⁴R. O. Laine, W. Zeile, F. Kang, D. L. Purich & F. S. Southwick (1997) *J. Cell Biol.* **138**, 1255.

ACTIN FILAMENT CAPPING PROTEIN



A cytoskeletal regulatory protein that binds to one end (usually the (+)-end or so-called barbed end) of

an actin filament, thereby blocking gain or loss of actin monomers. The affinity of a capping protein can be readily determined as the concentration of capping protein needed to inhibit the initial rate of dilution-induced disassembly of polymer containing pyrenyl-actin. The latter is a fluorescently tagged form of actin that gives a direct report of assembly indicator. Barbed-end capping proteins include Cap G, Cap Z, capping protein, as well as the actin capping/severing proteins below.

If a capping protein binds in a rapid equilibrium manner to the end of an actin filament, the capped filament (a) will no longer permit polymerization; (b) will thereby decrease the number of growing ends during elongation; and (c) should even retard subunit loss from the capped end if the sample is added to a actin-free solution containing a capping protein concentration sufficient to maintain the overall fraction of filaments that are capped. If $[N]$ equals the number concentration of uncapped filaments, and if $[N_{\text{capped}}]$ equals the number concentration of capped filaments, then in the presence of capping protein Y, the dissociation equilibrium constant for capping is given by

$$K_d = [Y_{\text{free}}][N]/[N_{\text{capped}}]$$

Because the total number concentration $[N_{\text{total}}]$ is the sum of $[N] + [N_{\text{capped}}]$, we can combine this relationship with the equilibrium equation to obtain:

$$[N] = [N_{\text{total}}] \{K_d / ([N_{\text{capped}}] + K_d)\}$$

We then see that the rate of elongation in the presence of a capping protein will be given by the following rate law:

$$d[A_{\text{poly}}]/dt = k_+[A_{\text{total}}][N_{\text{total}}] \{K_d / ([N_{\text{capped}}] + K_d)\} - k_{-1}N$$

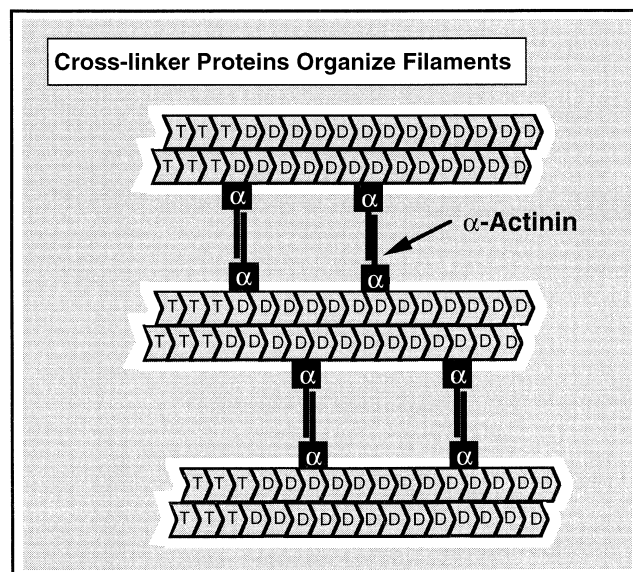
Because the concentration of capped filaments is a denominator term, we immediately recognize that the elongation rate will be retarded by capping.

Note that if a capping protein binds to monomeric actin, the capping protein will also be a monomer-sequestering agent. A good example of such behavior is profilin. **See also** *ABM-1 & ABM-2 Sequences in Actin-Based Motors; Actin-Based Bacterial Motility; Actin Assembly Kinetics*

T. P. Stossel (1993) *Science* **260**, 1086.

T. P. Stossel (1994) *Sci. Am.* **271**(September), 54.

ACTIN FILAMENT CROSS-LINKING PROTEIN/BUNDLING



A cytoskeletal regulatory protein that contains pairs or binding regions that attaching laterally to actin filament and resulting in cross-linking and/or bundling. These include α -actinin, ABP-30, ABP-120, ABP-240, spectrin, dematin, fascin, fimbrin, MARCKS, and dystrophin. Many of these proteins are antiparallel dimers of monomeric units that each contain a single actin filament binding site. **See also** *ABM-1 & ABM-2 Sequences in Actin-Based Motors; Actin-Based Bacterial Motility; Actin Assembly Kinetics*

T. P. Stossel (1993) *Science* **260**, 1086.

T. P. Stossel (1994) *Sci. Am.* **271**(September), 54.

ACTIN FILAMENT GROWTH (Polymerization Zone Model)

A model for localized, polar growth of actin filaments during lamellipod protrusion, filopod formation, or actin-based motility of intracellular pathogens such as *Listeria monocytogenes*, *Shigella flexneri*, or *vaccinia*. The polymerization zone is characterized by the presence of docking sequences that serve to concentrate or cluster filament growth-promoting factors, including but not limited to the actin regulatory component known as profilin. The cardinal feature of filament growth within the polymerization zone is that the concentration of polymerizable actin-ATP far exceeds that present in the bulk phase of the cytoplasm.

Actin filaments grow rapidly within cells, and the clearest evidence of this rapid growth is the ability of the cell's leading edge to move at rates of 0.5 to 1 micrometer per second. Likewise, actin-based motility of *Listeria* and *Shigella* can attain rates of nearly 0.5 micrometers per second. Because microfilaments contain about 360 actin monomers per micrometer of length, a motility rate of 0.5 to 1 micrometer per second corresponds to an apparent first-order rate constant (*i.e.*, $k_{\text{apparent}} = k_{\text{on}} [\text{Actin-ATP}]$) of about $180\text{--}360\text{ s}^{-1}$. The bimolecular rate constant k_{on} for actin-ATP addition to the barbed end has a nominal value of $2\text{--}3 \times 10^6\text{ M}^{-1}\text{ s}^{-1}$. Therefore, one can estimate that $60\text{--}180\text{ }\mu\text{M}$ actin-ATP (a value equal to $k_{\text{apparent}}/k_{\text{on}}$ or $180\text{--}360\text{ s}^{-1}/2\text{--}3 \times 10^6\text{ M}^{-1}\text{ s}^{-1} = 0.06\text{--}0.18\text{ mM}$) would be needed to sustain a filament growth rate of 0.5 to 1 $\mu\text{m/s}$. This local actin-ATP concentration range greatly exceeds the $0.25\text{--}0.3\text{ }\mu\text{M}$ concentration of G-actin thought to occur in most cells.

How then may one account for this discrepancy? We can first dispense with any trivial explanation suggesting that the value of k_{on} is unreasonably low, because bimolecular rate constants for protein addition to filaments typically have an upper bound of $10^7\text{ M}^{-1}\text{ s}^{-1}$. And although the total intracellular actin concentration often lies in the $200\text{--}300\text{ }\mu\text{M}$ concentration range, very little is present as uncomplexed actin-ATP monomer. In fact, cells typically have about 10–30% of total actin present as filaments, and the remainder is present as thymosin- $\beta 4$ -actin complex (60–70%), profilin-actin complex (10–20%), and substantially less than 1% as uncomplexed actin-ATP monomers. The most attractive idea is that regions of very active growth are apt to contain high local concentrations of actin-ATP that can readily mobilized to fulfill the needs of rapidly elongating filaments. Kang *et al.*¹ recently proposed one such model that relies on a hierarchy of oligoproline sequences (**See ABM-1 & ABM-2 Sequences In Actin-Based Motors**) to localize profilin (or the profilin-actin complex) at the sites of rapid filament assembly. As shown in Fig. 1, this cascade of binding interactions can result in the concentration of profilin and/or the profilin-actin complex within a polymerization zone defined by the activated cluster region. Host cells are also likely to utilize an analogous strategy for placing

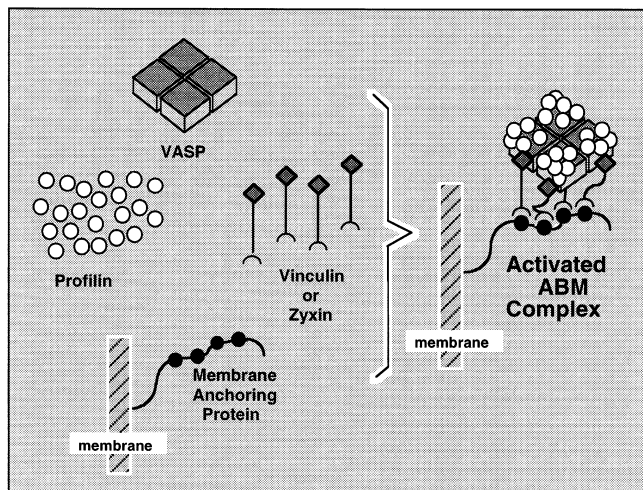


Figure 1. Localization of profilin and/or profilin-actin complex into an activated cluster. VASP, Vasodilator-stimulated phosphoprotein.

profilin and/or profilin-actin complex into an activated complex. This cluster creates a polymerization zone where the ability of profilin to facilitate ATP exchange with actin-ADP ensures that high levels of actin-ATP will be available for barbed-end growth of nearby actin filaments (see Fig. 2). Because actin-ADP is much less efficient in actin assembly, conversion to actin-ATP complex favors rapid elongation of filaments. Note also that depolymerization occurs exclusively at the pointed ends, and the much lower concentration of untethered profilin is likely to greatly limit the nucleotide exchange rate. The polymerization zone may also limit the ability of capping proteins (*e.g.*, Cap G, Cap Z, and gelsolin) to bind to the barbed end, an action that reduces the rate of filament assembly.

Other components, such as α -actinin and actin-related proteins (or ARPs), may also be recruited to the polymerization zone, whereas depolymerizing factors such as ADF and cofilin are apt to be selectively localized in the depolymerization zone. Note: Agents that disrupt the formation of activated ABM clusters will also suppress the rate of assembly in the polymerization zone. **See ABM-1 & ABM-2 Sequences In Actin-Based Motors**

¹F. Kang, R. O. Laine, M. R. Bubb, F. S. Southwick & D. L. Purich (1997) *Biochemistry* **36**, 8384.

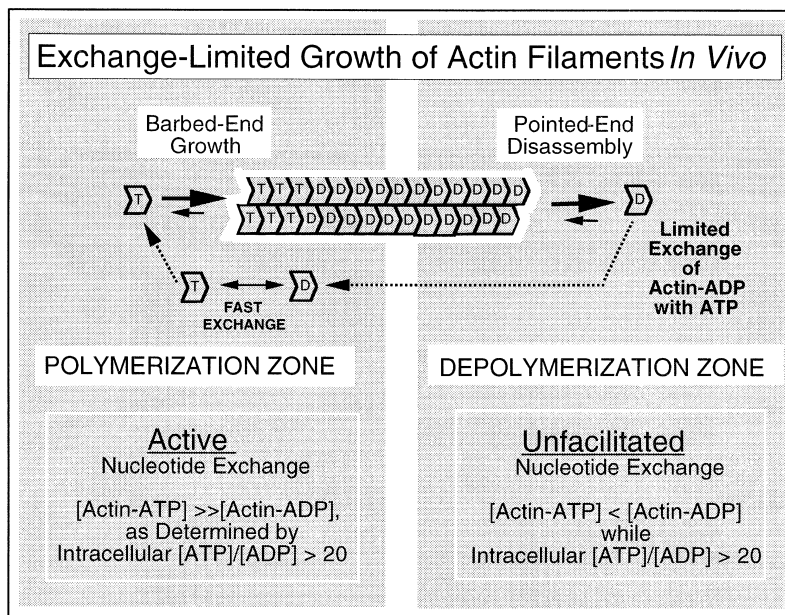


Figure 2. Characteristics of adenine nucleotide exchange that define actin polymerization and depolymerization zones within cells.

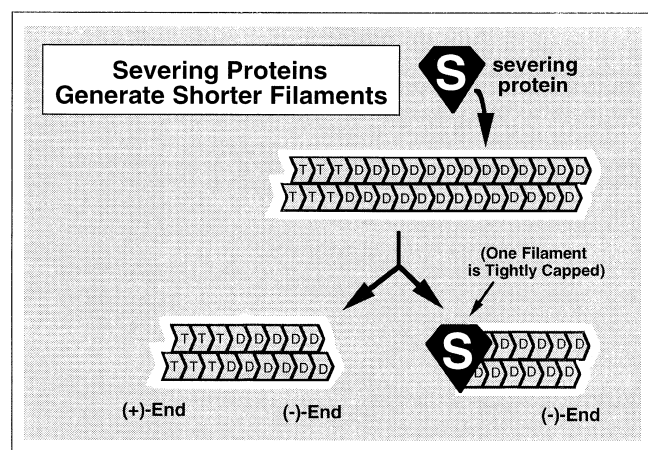
ACTIN FILAMENT LATERAL BINDING PROTEIN

A cytoskeletal regulatory protein that attaches laterally to actin filament without resulting in filament cross-linking and/or bundling. These include troponin, tropomyosin, calponin, tropomodulin, adducin, caldesmon, and hisactophilin. **See also** *ABM-1 & ABM-2 Sequences in Actin-Based Motors; Actin-Based Bacterial Motility; Actin Assembly Kinetics*

T. P. Stossel (1993) *Science* **260**, 1086.

T. P. Stossel (1994) *Sci. Am.* **271**(September), 54.

ACTIN FILAMENT SEVERING PROTEIN



A cytoskeletal regulatory protein that binds to and cleaves actin filaments. These proteins do not catalyze

filament cleavage; instead, they act stoichiometrically by remaining bound to one of the filament ends generated in the severing reaction. The prototypical severing protein is gelsolin, but other severing proteins include fragmin, villin, and ADF. Addition of a severing protein increases the number of polymer ends, and the rate of dilution-induced depolymerization rate will increase as more severing protein is added. Note also that many severing proteins form tightly bound caps on the so-called barbed end of one of the filaments produced by the severing process. A good example is gelsolin whose dissociation constant for the barbed end lies in the sub-picomolar range. **See also** *ABM-1 & ABM-2 Sequences in Actin-Based Motors; Actin-Based Bacterial Motility; Actin Assembly Kinetics*

T. P. Stossel (1993) *Science* **260**, 1086.

T. P. Stossel (1994) *Sci. Am.* **271**(September), 54.

ACTIN FILAMENTS (Mechanical Properties)

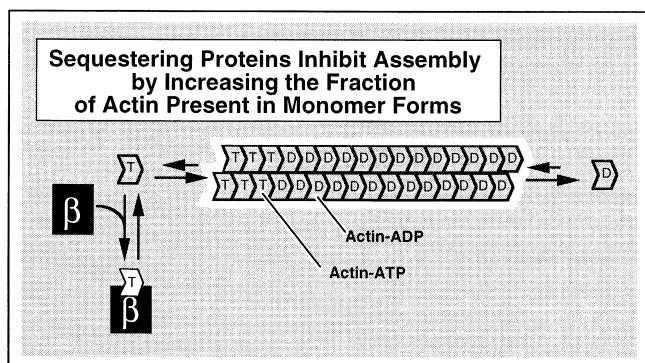
The mechanical properties of actin filament networks depend on the manner in which actin monomer is prepared and stored, as well as how they are polymerized conditions. Differences in mechanical properties are not the consequence of using two different types of forced oscillatory rheometers. Xu *et al.*¹ found that filaments assembled in EGTA and Mg^{2+} from fresh, gel-filtered ATP-actin monomer (1 mg/mL) have an elastic storage

modulus (G') of approximately 1 Pa at a deformation frequency of 0.1–1 Hz. G' is slightly higher when actin is polymerized in the presence of K^+ , Ca^{2+} , and Mg^{2+} . Actin monomer storage in the absence of frequent buffer changes to maintain ATP and reduced dithiothreitol can result in changes that increase the G' of filaments by a factor of 10 or more. The authors also stress that frozen storage can preserve the properties of monomeric actin, as long as one takes steps to prevent protein denaturation or aggregation due to freezing or thawing.

¹J. Xu, W. H. Schwarz, J. A. Kas, T. P. Stossel, P. A. Janmey & T. D. Pollard (1998) *Biophys. J.* **74**, 2731.

ACTIN MONOMER SEQUESTERING PROTEIN

A cytoskeletal regulatory protein that alters the distribution of monomeric and polymeric actin by sequestering monomers in a form that does not engage in polymer elongation reactions. These sequestering proteins do not directly affect the rate constants of actin polymerization/depolymerization or the magnitude of the critical concentration. The presence of a sequestering protein reduces the concentration of actin monomers that are polymerization-dependent, thereby altering the rate and extent of actin polymerization. Moreover, addition of sequestering protein to a solution containing actin polymer and monomer in an equilibrium (or steady-state) distribution results in net depolymerization, depending on the concentration and affinity of sequestering protein for actin monomers. Thymosin β_4 is the most abundant actin sequestering protein in nonmuscle cells, followed by profilin. Other sequestering proteins include actobindin, profilin, DNase I, ABP-50, and vitamin D binding protein.



The effect of monomer sequestration on actin assembly can be described as follows. Actin monomers assemble into filaments, forming

polymer-bound ADP and releasing orthophosphate. The kinetics of elongation conform to that predicted by the following rate law:

$$d[A_{\text{poly}}]/dt = k_+N[A_{\text{total}}] - k_-N$$

where $[A_{\text{poly}}]$ is the concentration of polymerized actin, k_+ is the bimolecular rate constant (equal to the sum of the on-rate constants for actin monomer addition to both ends of an actin filament), N is the number concentration of polymer ends that are capable of reacting with actin monomers, and $[A_{\text{total}}]$ is the total monomer concentration. Upon actin monomer sequestration by protein X, we must consider an additional form of actin, namely AX:

$$K_d = [A_{\text{free}}][X_{\text{free}}]/[AX]$$

where K_d is the dissociation constant of complex AX, and $[A_{\text{free}}]$, $[X_{\text{free}}]$, and $[AX]$ are the respective concentrations of free or uncomplexed actin monomer, free or uncomplexed sequestering protein, and actin·X complex. By conservation, we can write the following relationship:

$$[AX] = [A_{\text{total}}] - [A_{\text{free}}]$$

which upon substitution into the above expression for K_d yields

$$[A_{\text{free}}] = K_d [A_{\text{total}}]/([X_{\text{free}}] + K_d)$$

Sequestered monomers bound to the sequestering protein cannot bind to an actin filament, and only free actin can participate in elongation. Thus,

$$d[A_{\text{poly}}]/dt = k_+N[A_{\text{total}}] \{K_d/([X_{\text{free}}] + K_d)\} - k_-N$$

This rate law predicts that the rate of polymerization will be suppressed by the reduction in the free actin concentration as a consequence of AX formation.

In the absence of a sequestering protein, actin polymerization will proceed to equilibrium or it will reach a steady-state extent of polymerization, at which point $d[A_{\text{poly}}]/dt = 0$. $[A]_{\infty}$ is the critical concentration, equal to k_-/k_+ . In the presence of a sequestering protein, $[A_{\text{total}}] = [A_{\text{free}}] + [A_{\text{poly}}] + [AX]$, and after polymerization reaches equilibrium, the unpolymerized actin concentration $[A]_{\infty}'$ will equal $\{[AX] + [A_{\text{free}}]\}$. Hence,

$$[A]_{\infty}'/[A]_{\infty} = \{[AX] + [A_{\text{free}}]\}/[A_{\text{free}}]$$

and this relationship can be rearranged into the following equation:

$$[A]_{\infty}' = [A]_{\infty} \{1 + [X_{\text{free}}]/K_d\}$$

This equation clearly demonstrates that the observed critical concentration will be increased by the presence of sequestering protein. In this respect, addition of a sequestering protein should reduce *both* the rate and the extent of polymerization. **See also** *ABM-1 & ABM-2 Sequences in Actin-Based Motors; Actin-Based Bacterial Motility; Actin Assembly Kinetics*

T. P. Stossel (1993) *Science* **260**, 1086.

T. P. Stossel (1994) *Sci. Am.* **271**(September), 54.

ACTINOMETER

An instrument or suitable chemical system that allows one to determine the number of photons (or number of quanta) absorbed in a beam of electromagnetic radiation.

ACTION

The multiplicative product of work (unit = Joule) and time (unit = sec). The alternative term is *impulse*.

ACTION POTENTIAL

The transient change in the transmembrane potential upon excitation. An action potential cycle consists of a transient depolarization of the cell membrane of an excitable cell (such as a neuron) as a result of increased permeability of ions across the membrane, followed by repolarization, hyperpolarization, and finally a return to the resting potential. This cycle typically lasts 1–2 milliseconds and travels along the axon from the cell body (or, axon hillock) to the axonal terminus at a rate of 1–100 meters per second. **See** *Membrane Potential*

ACTION SPECTRUM

A plot of the photoresponse of a biological or chemical process (normalized with respect to the number of incident photons) as a function of wavelength or energy of radiation. Diverse biological phenomena have been investigated by determining action spectra¹. For example, action spectroscopy was used to demonstrate (a) that about 10 absorbed photons are sufficient to yield one molecule of dioxygen by photosynthesis; (b) that about 10¹⁰ photons are needed for hemolysis of red blood cells; (c) that only one photon is needed to dissociate carboxyhemoglobin into deoxyHb and CO. The method requires the use of a photomultiplier and amplifier suitable for measuring light intensity as well as a reference standard to calibrate the system. **See also** *Spectral Re-*

sponsivity; Spectral Effectiveness; Quantum Yield; Efficiency Spectrum; Excitation Spectrum

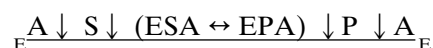
¹R. B. Setlow & E. C. Pollard (1962) *Molecular Biophysics*, pp. 267–351, Addison-Wesley, Reading, MA.

ACTIVATION

The increase in energy content of an atom, ion, or molecular entity or the process that makes an atom, ion, or molecular entity more active or reactive. In enzymology, activation often refers to processes that result in increased enzyme activity. For example, increasing temperature often can have a positive effect on enzyme activity (**See** *Arrhenius Equation*). Other examples of enzyme activation include: (1) proteolysis of zymogens; (2) alterations in ionic strength; (3) alterations due to pH changes; (4) activation in cooperative systems; (5) lipid or membrane interface activation; (6) metal ion effects; (7) autocatalysis; and (8) covalent modification.

Dixon and Webb¹ present an extensive consideration of activation mechanisms involving the reversible binding of an activator (less often termed an “agonist”) to the enzyme. Nonessential activation refers to enzyme-dependent processes that can convert substrate(s) to product(s) in the absence of the activator, albeit at a slower rate. Essential activators are molecular entities that are required by the enzyme in the catalysis of a reaction. In a sense, essential activators are similar to second (or third) substrates, albeit they are not converted to products. An example of an essential activator might be an enzyme that requires the binding of a metal ion for catalysis to proceed. Below are a few cases of essential activation.

Essential Activation in a Uni Uni Mechanism–Type I (Activator Binds First). Consider the following reaction scheme in which an activator (A) binds to the free enzyme (E) prior to the binding of the substrate (S):



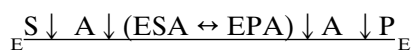
For the rapid equilibrium case, all binding steps are fast relative to the ESA-to-EPA interconversion step. Thus, $K_A = [\text{E}][\text{A}]/[\text{EA}]$, $K_S = [\text{EA}][\text{S}]/[\text{ESA}]$, and $K_P = [\text{EA}][\text{P}]/[\text{EPA}]$. The initial rate expression ($[\text{P}] = 0$) for this case is

$$v = V_{\max, f} [\text{A}][\text{S}] / \{K_S(K_A + [\text{A}]) + [\text{A}][\text{S}]\}$$

where $V_{\max,f} = k_5[E_{\text{total}}]$ where k_5 is the forward rate constant for the $\text{ESA} \rightarrow \text{EPA}$ interconversion. Note that, if $[A]$ is saturating (*i.e.*, $[A] \gg K_A$), then the equation reduces to the normal Michaelis-Menten expression. A double-reciprocal plot ($1/v$ versus $1/[A]$) at different, nonsaturating levels of the activator will yield a series of straight lines having a common intersection point on the vertical axis (at $1/v = 1/V_{\max,f}$).

The steady-state expression for this scheme (where $[P] \approx 0$ and the forward rate constants are k_1, k_3, k_5 , and k_7 for the $\text{E} + \text{A} \rightarrow \text{EA}$, the $\text{EA} + \text{S} \rightarrow \text{ESA}$, the $\text{ESA} \rightarrow \text{EPA}$, and the $\text{EPA} \rightarrow \text{EA} + \text{P}$ steps, respectively, and k_2, k_4, k_6 , and k_8 are the corresponding rate constants for the reverse reactions of those steps, respectively) is the same general expression as presented for the rapid equilibrium case except where $V_{\max,f} = k_5 k_7 [E_{\text{total}}] / (k_5 + k_6 + k_7)$, $K_A = k_2 / k_1$, and $K_S = (k_4(k_6 + k_7) + k_5 k_7) / (k_3(k_5 + k_6 + k_7))$.

Essential Activation in a Uni Uni Mechanism—Type II (Activator Binds Second). In this scheme, the essential activator can only bind to the enzyme-substrate or enzyme-product binary complexes:



Such a mechanism is a form of substrate-induced activation. If all of the binding steps are rapid relative to the ESA -to- EPA interconversion step, the initial-rate rapid-equilibrium equation for this scheme is

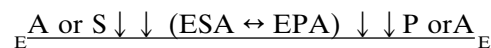
$$v = V_{\max,f} [A] [S] / \{K_A K_S + K_A [S] + [A] [S]\}$$

where $K_S = [E][S]/[ES]$, $K_A = [ES][A]/[ESA]$, and $V_{\max,f} = k_5[E_{\text{total}}]$ in which k_5 is the forward rate constant for the $\text{ESA} \rightarrow \text{EPA}$ conversion. Note the difference in this expression from the rapid equilibrium expression for the previous Type I scheme. In the earlier scheme, there is a $K_S[A]$ term in the denominator whereas there is a $K_A[S]$ term in this scheme. Hence, a double-reciprocal plot ($1/v$ versus $1/[S]$) will yield a series of straight lines intersecting at a common point not on the vertical axis. This intersection point will have coordinates of $1/[S] = -1/K_S$ and $1/v = 1/V_{\max,f}$. Hence, the rapid equilibrium Type II scheme should be readily distinguishable from the previous scheme.

The steady-state expression for this same scheme (in which the forward rate constants are k_1, k_3, k_5, k_7 , and

k_9 for the $\text{E} + \text{S} \rightarrow \text{ES}$, the $\text{ES} + \text{A} \rightarrow \text{ESA}$, the $\text{ESA} \rightarrow \text{EPA}$, the $\text{EPA} \rightarrow \text{EP} + \text{A}$ steps, and $\text{EP} \rightarrow \text{E} + \text{P}$ respectively, and k_2, k_4, k_6, k_8 , and k_{10} are the corresponding rate constants for the reverse reactions of those steps, respectively), in the absence of product is $v = V_{\max,f} [A] [S] / \{K_S K_A + K_S [A] + K_A [S] + [A] [S] + [A]^2 [S] / K_{IA}\}$ where $V_{\max,f} = k_3 k_5 k_7 k_9 [E_{\text{total}}] / \Phi$, $K_S = (k_2 k_4 k_6 k_8 + k_3 k_5 k_7 k_9) / (k_1 \Phi)$, $K_A = k_9 (k_4 (k_6 + k_7) + k_5 k_7) / \Phi$, $1/K_{IA} = k_3 k_8 (k_5 + k_6) / \Phi$, and $\Phi = k_4 k_6 k_8 + k_3 (k_9 (k_6 + k_7) + k_5 (k_7 + k_9))$. A double-reciprocal plot of such a steady-state mechanism, at varying, constant concentrations of the activator, will yield a series of straight lines. A replot of the slopes (slopes vs. $1/[A]$) will provide a value for K_A . However, a replot of the vertical intercepts (intercepts vs. $1/[A]$) will be nonlinear (curving up at elevated levels of the activator), depending on the relative magnitude of the squared term.

Essential Activation in a Uni Uni Mechanism—Type III (Activator and Substrate Bind Randomly).



where “A or P double arrows down” indicates random binding in two successive steps. The rapid equilibrium expression for this scheme, in which the sole rate-determining step is the ESA -to- EPA interconversion, is

$$v = V_{\max,f} [A] [S] / \{K_{iS} K_A + K_A [S] + K_S [A] + [A] [S]\}$$

where $V_{\max,f} = k_9[E_{\text{total}}]$, $K_{iS} = [E][S]/[ES]$, $K_A = [ES][A]/[ESA]$, $K_S = [EA][S]/[EAS]$, and k_9 is the forward rate constant for the $\text{ESA} \rightarrow \text{EPA}$ conversion. Double-reciprocal plots ($1/v$ versus $1/[S]$ or $1/[A]$) will yield a series of straight lines. In the $1/v$ versus $1/[S]$ plot, the intercept replot will provide a value for K_A and $V_{\max,f}$. The intercept replot of the $1/v$ versus $1/[A]$ data will provide a value for K_S . The slope replots will then yield K_{iS} . In the standard double-reciprocal primary plots (*i.e.*, $1/v$ vs. $1/[S]$), the lines will intersect at a common point having coordinates of $1/[S] = -1/K_{iS}$ and $1/v = (1/V_{\max,f})(1 - (K_S/K_{iS}))$. Hence, the point-of-intersection will be in the second quadrant if $K_S \gg K_{iS}$, in the third quadrant if $K_S \ll K_{iS}$, and on the horizontal axis if $K_S = K_{iS}$.

The steady-state expression for v for this scheme is rather complex, containing squared terms in $[S]$ and in $[A]$ in

the numerator as well as terms containing $[S]^2$, $[A]^2$, $[S]^2[A]$, $[S][A]^2$, $[A]^3$, $[S]^2[A]^2$, and $[S][A]^3$ in the denominator.

NONESSENTIAL ACTIVATION. In these cases, if the nonessential activator can bind to the free enzyme, the resulting equations are very similar in format to kinetic expressions for partial inhibition. Consider the following scheme for a Uni Uni mechanism in which a nonessential activator (A) can bind to the free enzyme. Thus, $E + S \rightleftharpoons ES \rightleftharpoons EP \rightleftharpoons E + P$ and $E + A \rightleftharpoons EA$ is followed by $EA + S \rightleftharpoons ESA \rightleftharpoons EPA \rightleftharpoons EA + P$. In addition, one can have $ES + A \rightleftharpoons ESA$ as well as $EP + A \rightleftharpoons EPA$. Considering the rapid equilibrium case; *i.e.*, where all binding steps are rapid relative to the interconversion steps ($ES \rightleftharpoons EP$ and $ESA \rightleftharpoons EPA$) and can be described by dissociation constants. Thus, $K_S = [E][S]/[ES]$, $K_A = [E][A]/[EA]$, $K_{AS} = [EA][S]/[ESA]$, $K_P = [E][P]/[EP]$, and $K_{AP} = [EA][P]/[EPA]$. In this scheme, the resulting rate expression will have the same general format irrespective of whether the activator can bind to the ES or EP complexes. Under initial rate conditions (*i.e.*, $[P] \approx 0$) and letting $k_{11} = \alpha k_3$ (where k_3 is the forward rate constant for the $ES \rightarrow EP$ conversion and k_{11} is the corresponding rate constant for the $ESA \rightarrow EPA$ conversion), and remembering that $\alpha > 1.0$ for an activator, then $v = V_{\max,f}[S](1 + (\alpha[A]K_S/(K_A K_{AS}))) \div \{K_S + [S] + ([A]K_S/K_A)(1 + [S]/K_{AS})\}$ where $V_{\max,f} = k_3[E_{\text{total}}]$. Note that, if $[A] = 0$, the equation reduces to the standard Michaelis-Menten equation. Plotting protocols are available that allow one to graphically determine the various kinetic parameters of this, and related, systems.

¹M. Dixon & E. C. Webb (1979) *Enzymes*, 3rd ed., Academic Press, New York.

ACTIVATION, CALMODULIN-DEPENDENT

Upon binding calcium ions, the small acidic protein known as calmodulin can activate enzymes by binding to a wide variety of proteins containing calmodulin-binding domains. Such proteins include cAMP phosphodiesterase, calmodulin-dependent nitric oxide synthase, calmodulin kinases, the plasma membrane calcium pump, calcineurin, and calmodulin-dependent inositol-(1,4,5)-trisphosphate 3-kinase. **See also** *Activation; Autoinhibition*

P. James, T. Vorherr & E. Carafoli (1995) *Trends in Biochem. Sci.* **20**, 38.

ACTIVATOR

A substance, agent, or factor (other than a catalyst) the presence of which will increase the rate of a catalyzed reaction. A substance that activates an enzyme-catalyzed reaction by binding to the enzyme is often referred to as an enzyme activator. **See** *Activation; Allosteric Effector; Linked Functions; London-Steck Plot*

ACTIVE SITE

The topologically defined region(s) on an enzyme responsible for the binding of substrate(s), coenzymes, metal ions, and protons that directly participate in the chemical transformation catalyzed by an enzyme, ribozyme, or catalytic antibody. Active sites need not be part of the same protein subunit, and covalently bound intermediates may interact with several regions on different subunits of a multisubunit enzyme complex. **See** *Lambda (Λ) Isomers of Metal Ion-Nucleotide Complexes; Lock and Key Model of Enzyme Action; Low-Barrier Hydrogen Bonds: Role in Catalysis; Yaga-Ozawa Plot; Yonetani-Theorell Plot; Induced-Fit Model; Allosteric Interaction*

ACTIVE-SITE TITRATION

The reaction of a chemically active substance with a group located in an enzyme's active site, such that the stoichiometry of active sites can be determined analytically. **See** *Acetylcholinesterase; Affinity Labeling; Diisopropylfluorophosphate*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Collagenase, **248**, 100–101, 502–503; DNA polymerase, **249**, 47; gelatinase, **248**, 97, 100–101, 474, 502–503; glycyl endopeptidase, **244**, 543–544; Kex2 protease, **244**, 158; matrix metalloproteinases, **248**, 502–503; myeloblastin, **244**, 63; peptidases, **248**, 85–101; stromelysin, **248**, 100–101, 460–461, 502–503; trypsin-like enzymes, **248**, 14.

ACTIVE TRANSPORT

Any process whereby a substance is transported from a region of low concentration to a region of higher concentration by coupling the otherwise unfavorable $\Delta G_{\text{diffusion}}$ with the highly favorable $\Delta G_{\text{hydrolysis}}$ for ATP conversion to ADP and orthophosphate. Other metabolites displaying high group transfer potential can replace ATP in the hydrolase-type reaction, as long as sufficient free energy is liberated. Contrary to generally held ideas that active transport requires a membrane, one can also have active transport in a single compartment, such as the

cytoplasm. The only condition that must be fulfilled is that the transport occurs against a concentration gradient. **See** *Membrane Transporters; Ion Pumps; $\Delta G_{\text{diffusion}}$*

ACTIVITY

1. With respect to an enzyme, the rate of substrate-to-product conversion catalyzed by an enzyme under a given set of conditions, either measured by the amount of substance (*e.g.*, micromoles) converted per unit time or by concentration change (*e.g.*, millimolarity) per unit time. **See** *Specific Activity; Turnover Number*. 2. Referring to the measure of a property of a biomolecule, pharmaceutical, procedure, *etc.*, with respect to the response that substance or procedure produces. 3. **See** *Optical Activity*. 4. The amount of radioactive substance (or number of atoms) that disintegrates per unit time. **See** *Specific Activity*. 5. A unitless thermodynamic parameter which is used in place of concentration to correct for nonideality of gases or of solutions. The absolute activity of a substance B, symbolized by λ_B , is related to the chemical potential of B (symbolized by μ_B) by the relationship $\mu_B = RT \ln \lambda_B$ where R is the universal gas constant and T is the absolute temperature. The ratio of the absolute activity of some substance B to some absolute activity for some reference state, $\lambda_B^\ominus$, is referred to as the relative activity (usually simply called “activity”). The relative activity is symbolized by a and is defined by the relationship $a_B = \lambda_B / \lambda_B^\ominus = e^{(\mu_B - \mu_B^\ominus) / RT}$. If the states are referenced with respect to Raoult’s Law, the relative activity is symbolized by a . However, if referenced with respect to Henry’s Law, the activities can have a molality, a concentration, or a mole fraction basis and are symbolized by a_m , a_c , and a_x , respectively. **See also** *Activity Coefficient; Chemical Potential; Fugacity; Turnover Number; Latent Activity; Biomineralization*

ACTIVITY COEFFICIENT

A unitless correction factor that relates the relative activity of a substance to the quantity of the substance in a mixture. Activity coefficients are frequently determined by emf (electromotive force) or freezing-point depression measurements. At infinite dilution, the activity coefficient equals 1.00. Activity coefficients for electrolytes can vary significantly depending upon the concentration of the electrolyte. Activity coefficients can exceed values of 1.00. For example, a 4.0 molal HCl solution has a coefficient of 1.76 and a 4.0 molal LiCl has a value of

1.46. The magnitude of the coefficient reflects the electric charge distribution of the ionic species. A 0.1 molal solution of $\text{Al}_2(\text{SO}_4)_3$ has an activity coefficient of only 0.035. It should also be noted that, in dilute solutions, activity coefficients of electrolytes decrease in magnitude with increasing concentration. A minimum is reached and the coefficient then increases with concentration. **See** *Activity; Debye-Hückel Law; Biomineralization*

ACTOMYOSIN ATPase

The ATPase enzyme activity of actomyosin has been assigned the classification number EC 3.6.1.32. Myosin catalyzes the hydrolysis of ATP to ADP and orthophosphate. In the absence of actin, myosin is a more feeble ATPase. **See also** *Myosin*

- A. Oplatka (1997) *Crit. Rev. Biochem. Mol. Biol.* **32**, 307.
- D. A. Schafer & J. A. Cooper (1995) *Ann. Rev. Cell Dev. Biol.* **11**, 497.
- P. J. McLaughlin & A. G. Weeds (1995) *Ann. Rev. Biophys. Biomol. Struct.* **24**, 643.
- J. Condeelis (1993) *Ann. Rev. Cell Biol.* **9**, 411.
- F. Solomon (1991) *Ann. Rev. Cell Biol.* **7**, 633.
- J. M. Squire (1990) *Molecular Mechanisms of Muscular Contraction*, CRC Press, Boca Raton.
- W. P. Jencks (1980) *Adv. Enzymol.* **51**, 75.

ACYL-ACYL-CARRIER-PROTEIN Δ^9 -DESATURASE

This enzyme catalyzes the NADPH- and dioxygen-dependent insertion of *cis* double bonds into the methylene region of fatty acyl structures covalently attached to the phosphopantetheine portion of an acyl carrier protein.

- B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.
- I. Nishida & N. Murata (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 541.

ACYLAL INTERMEDIATE

An enzymatic reaction intermediate formed by acylation of an acetal hydroxyl group. Such an intermediate is thought to occur in a number of reactions involving carbohydrates. The sucrose phosphorylase reaction is thought to proceed by way of an acyl-glucosyl intermediate.

ACYLAMINOACYL-PEPTIDASE

This enzyme [EC 3.4.19.1], also known as acylamino-acid releasing enzyme and *N*-acylpeptide hydrolase, catalyzes the hydrolysis of an acylaminoacyl-peptide to generate an acylamino acid and the free peptide. Catalysis is most

efficient when the P1 site is occupied by a seryl, alanyl, or methionyl residue. The enzyme is a poor catalyst if P1 is a glycyl, tyrosyl, aspartyl, asparaginyl, or prolyl residue. The EC number accounts for a group of similar enzymes that liberate *N*-acetyl or *N*-formyl amino acids from proteins and peptides.

W. M. Jones, A. Scaloni & J. M. Manning (1994) *Meth. Enzymol.* **244**, 227.

ACYL-CoA:CHOLESTEROL O-ACYLTRANSFERASE

This enzyme [EC 2.3.1.26], also known as sterol *O*-acyltransferase, sterol-ester synthase, and cholesterol acyltransferase, catalyzes the reaction of an acyl-coenzyme A derivative with cholesterol to produce coenzyme A and the cholesterol ester. The animal enzyme is highly specific for transfer of acyl groups having a single *cis* double bond at C9.

T.-Y. Chang, C. C. Y. Chang & D. Cheng (1997) *Ann. Rev. Biochem.* **66**, 613.

T.-Y. Chang & G. M. Doolittle (1983) *The Enzymes*, 3rd ed., **16**, 523.

ACYL-CoA DEHYDROGENASES

A number of enzymes given this name catalyze the formation of a 2,3-dehydroacyl-coenzyme A derivative from the acyl-CoA substrate. Acyl-CoA dehydrogenase (NADP⁺) [EC 1.3.1.8], also known as 2-enoyl-CoA reductase, utilizes NADP⁺. The liver enzyme acts on enoyl-CoA substrates having a carbon chain length of 4 to 16 carbons, with an optimum activity on 2-hexenoyl-CoA. In *E. coli*, both *cis*-specific and *trans*-specific enzymes exist (EC 1.3.1.37, *cis*-2-enoyl-CoA reductase (NADPH), and EC 1.3.1.38, *trans*-2-enoyl-CoA reductase (NADPH)). Medium-chain acyl-CoA dehydrogenase [EC 1.3.99.3] utilizes an electron-transferring flavoprotein (requiring FAD) instead of NADP⁺. **See also** *Butyryl-CoA Dehydrogenase*; *Long-Chain Acyl-CoA Dehydrogenase*

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed. **19**, 213.

ACYL-CoA DESATURASE

This enzyme [EC 1.14.99.5], also known as stearoyl-CoA desaturase, fatty acid desaturase, and Δ^9 -desaturase, will catalyze the reaction of stearoyl-CoA with a hydrogen donor and dioxygen to produce oleoyl-CoA, water, and

the oxidized factor. The rat liver enzyme is an enzyme system utilizing cytochrome *b*₅ and cytochrome *b*₅ reductase. **See also** *Desaturases*

I. Nishida & N. Murata (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 541.

ACYL-CoA:RETINOL O-ACYLTRANSFERASE

This enzyme [EC 2.3.1.76], also referred to as retinol fatty-acyltransferase, catalyzes the reaction of an acyl-CoA derivative with retinol to generate coenzyme A and the retinyl ester. The CoA derivative can be palmitoyl-CoA or other long-chain fatty-acyl derivatives of coenzyme A.

R. Blomhoff, M. H. Green & K. R. Norum (1992) *Ann. Rev. Nutr.* **12**, 37.

J. C. Saari & D. L. Bredberg (1990) *Meth. Enzymol.* **190**, 156.

A. C. Ross (1990) *Meth. Enzymol.* **189**, 442.

ACYL ENZYME INTERMEDIATES

An enzyme reaction intermediate (Enz—O—C(O)R or Enz—S—C(O)R), formed by a carboxyl group transfer (*e.g.*, from a peptide bond or ester) to a hydroxyl or thiol group of an active-site amino acyl residue of the enzyme. Such intermediates are formed in reactions catalyzed by serine proteases^{1–4}, transglutaminase⁵, and formylglycinamide ribonucleotide amidotransferase⁶. Acyl-enzyme intermediates often can be isolated at low temperatures, low pH, or a combination of both⁴. For acyl-seryl derivatives, deacylation at a pH value of 2 is about 10⁵-fold slower than at the optimal pH⁴. A primary isotope effect can frequently be observed with a ¹³C-labeled substrate. If an amide substrate is used, it is possible that a secondary isotope effect may be observed as well⁷. **See also** *Active Site Titration*; *Serpins (Inhibitory Mechanism)*

¹A. L. Fink & M. H. Geeves (1979) *Meth. Enzymol.* **63**, 336.

²P. Douzou (1977) *Cryoenzymology: An Introduction*, Academic Press, New York.

³A. L. Fink & G. A. Petsko (1981) *Adv. Enzymol.* **52**, 177.

⁴R. J. Coll, P. D. Compton & A. L. Fink (1982) *Meth. Enzymol.* **87**, 66.

⁵J. E. Folk (1982) *Meth. Enzymol.* **87**, 36.

⁶J. M. Buchanan (1982) *Meth. Enzymol.* **87**, 76.

⁷W. W. Cleland (1982) *Meth. Enzymol.* **87**, 625.

N-ACYLGLUCOSAMINE 2-EPIMERASE

This enzyme [EC 5.1.3.8], also known as *N*-acetylglucosamine 2-epimerase, catalyzes the epimerization of *N*-acylglucosamine to *N*-acylmannosamine. ATP is reported to be required in this reaction.

- M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.
E. Adams (1976) *Adv. Enzymol.* **44**, 69.
L. Glaser (1972) *The Enzymes*, 3rd ed., **6**, 355.

N-ACYLGLUCOSAMINE-6-PHOSPHATE 2-EPIMERASE

This enzyme [EC 5.1.3.9], also known as *N*-acetylglucosamine-6-phosphate 2-epimerase, catalyzes the interconversion of *N*-acylglucosamine 6-phosphate to *N*-acylmannosamine 6-phosphate.

- M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.
E. Adams (1976) *Adv. Enzymol.* **44**, 69.
L. Glaser (1972) *The Enzymes*, 3rd ed., **6**, 355.

1-ACYLGLYCEROL-3-PHOSPHATE ACYLTRANSFERASE

This enzyme catalyzes the reaction of an acyl-CoA derivative with 1-acyl-*sn*-glycerol 3-phosphate to generate coenzyme A and 1,2-diacyl-*sn*-glycerol 3-phosphate. The animal enzyme is reported to be specific for the transfer of unsaturated fatty acyl groups. Interestingly, the acyl-[acyl-carrier-protein] can also act as an acyl donor.

- R. M. Bell & R. A. Coleman (1983) *The Enzymes*, 3rd ed., **16**, 87.
J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

ACYLGLYCERONE-PHOSPHATE REDUCTASE

This enzyme [EC 1.1.1.101], also known as palmitoyldihydroxyacetone-phosphate reductase, catalyzes the reaction of 1-palmitoylglycerol 3-phosphate and NADP⁺ to produce palmitoylglycerone phosphate and NADPH. The enzyme can also utilize as substrates alkylglycerone 3-phosphate and alkylglycerol 3-phosphate.

- A. K. Hajra, S. C. Datta & M. K. Ghosh (1992) *Meth. Enzymol.* **209**, 402.

1-ACYLGLYCERO-3-PHOSPHOCHOLINE O-ACYLTRANSFERASE

This enzyme [EC 2.3.1.23], also called lysolecithin acyltransferase and lysophosphatidylcholine acyltransferase, catalyzes the reaction of an acyl-CoA derivative with 1-acyl-*sn*-glycero-3-phosphocholine to yield coenzyme A and 1,2-diacyl-*sn*-glycero-3-phosphocholine. The enzyme preferentially acts on unsaturated acyl-CoA derivatives, but 1-acyl-*sn*-glycero-3-phosphoinositol can also act as the acceptor.

- P. C. Choy, P. G. Tardi & J. J. Mukherjee (1992) *Meth. Enzymol.* **209**, 90.
J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

ACYL-LYSINE DEACYLASE

This enzyme [EC 3.5.1.17] catalyzes the hydrolysis of *N*⁶-acyllysine to generate a fatty acid anion and lysine.

- I. Chibata, T. Ishikawa & T. Tosa (1970) *Meth. Enzymol.* **19**, 756.

N-ACYLMANNOSAMINE KINASE

This enzyme [EC 2.7.1.60] catalyzes the ATP-dependent phosphorylation of *N*-acylmannosamine to yield ADP and *N*-acylmannosamine 6-phosphate. The enzyme can act on both the acetyl and the glycolyl derivatives.

- W. Kundig & S. Roseman (1966) *Meth. Enzymol.* **8**, 195.

N-ACYLNEURAMINATE CYTIDYLYLTRANSFERASE

This enzyme [EC 2.7.7.43], also referred to as CMP-*N*-acetylneuraminic acid synthetase and CMP-sialate synthase, catalyzes the conversion of CTP and *N*-acylneuramate to yield pyrophosphate and CMP-*N*-acylneuramate. The protein will act on both the *N*-acetyl and *N*-glycolyl derivatives.

- E. L. Kean & S. Roseman (1966) *Meth. Enzymol.* **8**, 208.

N-ACYLNEURAMINATE 9-PHOSPHATASE

This enzyme [EC 3.1.3.29] catalyzes the hydrolysis of *N*-acylneuramate 9-phosphate to produce *N*-acylneuramate and phosphate.

- G. W. Jourdian, A. Swanson, D. Watson & S. Roseman (1966) *Meth. Enzymol.* **8**, 205.

N-ACYLNEURAMINATE 9-PHOSPHATE SYNTHASE

This enzyme [EC 4.1.3.20] catalyzes the reaction of *N*-acylneuramate 9-phosphate with phosphate to produce *N*-acetylmannosamine 6-phosphate, phosphoenolpyruvate, and water. The protein will act on *N*-glycolyl and *N*-acetyl derivatives.

- D. Watson, G. W. Jourdian & S. Roseman (1966) *Meth. Enzymol.* **8**, 201.

ACYLPHOSPHATASE

This enzyme [EC 3.6.1.7] catalyzes the hydrolysis of an acyl phosphate to yield a fatty acid anion and orthophosphate.

- I. Harary (1963) *Meth. Enzymol.* **6**, 324.

ACYL-PHOSPHATE-HEXOSE PHOSPHOTRANSFERASE

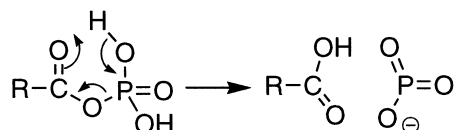
This enzyme [EC 2.7.1.61] catalyzes the reaction of an acyl phosphate with a hexose to produce an acid and hexose phosphate. If the sugar is D-glucose or D-mannose, phosphorylation is on O6. If the sugar is D-fructose, phosphorylation is on O1 or O6.

R. L. Andersin & M. Y. Karmel (1966) *Meth. Enzymol.* **9**, 392.

ACYL-PHOSPHATE INTERMEDIATE

An enzymatic reaction intermediate formed by phosphoryl transfer to a carboxyl group on an enzyme. Acyl-phosphates are structurally analogous to acid anhydrides ($R-CO-O-CO-R'$), and they are thermodynamically less stable than either of the two phosphoanhydride bonds in ATP. This is evident by the fact that the acetate kinase reaction ($ADP + \text{acetyl-phosphate} = ATP + \text{acetate}$) favors ATP formation with an equilibrium constant of about 3,000. Acetyl-phosphate can be chemically synthesized by reacting orthophosphate with acetic anhydride.

Acyl-phosphates can participate in so-called in-line phosphoryl transfer reactions, and depending on reaction conditions, acyl-phosphate compounds can form a metaphosphate anion:



Metaphosphate Formation from Acyl-P

Acyl-phosphates also display an unusual “U”-shaped pH profile for the rate of hydrolysis (see Fig. 1). This reflects the fact that acyl-phosphate compounds are susceptible to both acid- and base-catalyzed hydrolysis, and the shoulder at less acidic pH values reflects the slightly faster hydrolysis of the monoanion, compared to the dianion. Observation of a “U”-shape pH-rate profile for a previously uncharacterized phosphoryl-enzyme intermediate supports the inference that an acyl-phosphate has formed.

Acyl-phosphate compounds also undergo hydroxaminylation by hydroxylamine ($\text{HO}-\text{NH}_2$). Initial attack occurs by the oxygen atom of hydroxylamine, and the O-linked product reacts with a second molecule of hydroxylamine to yield the more stable N-acylated acid hydroxamate.

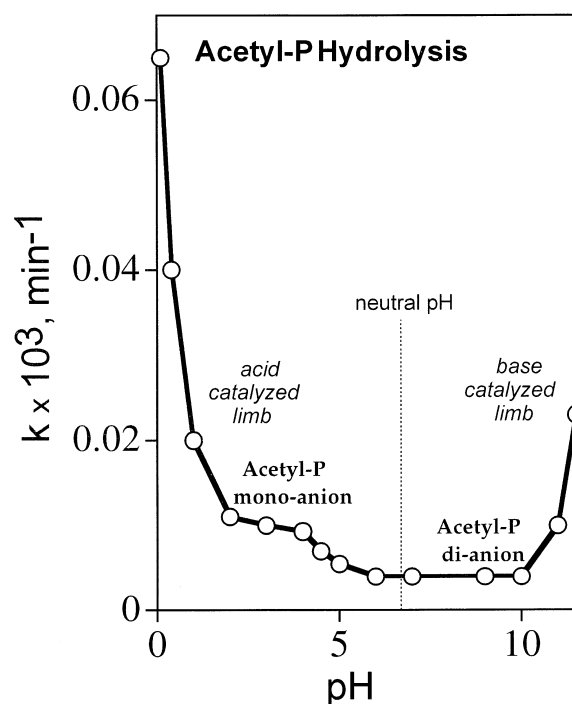
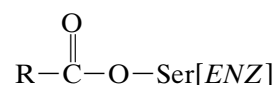


Figure 1. Acid- and base-catalyzed hydrolysis of acetyl-phosphate. From G. Di Sabato & Jencks (1961) *J. Am. Chem. Soc.* **83**, 4400; reproduced with permission of the authors and the American Chemical Society.

If an acyl-phosphate intermediate involves a side-chain carboxyl of glutamyl or aspartyl residue on the enzyme, the corresponding hydroxamate should undergo dinitrophenylation with Sanger's reagent, and Lössén rearrangement should decarboxylate the aspartate or glutamate. Likewise, an acyl-phosphate intermediate involving a side-chain carboxyl of glutamyl or aspartyl residue on the enzyme should undergo nucleophilic reduction by sodium [^3H]borohydride to form the corresponding ω -hydroxy- α -amino acids after acid hydrolysis of the protein. [Note: If an enzyme-catalyzed reaction involves a thiol-ester intermediate, these activated carboxyl compounds are also susceptible to the nucleophilic reactions cited above.] **See** *D-Alanine-D-Alanine Ligase; Mapping Substrate Interactions Using Substrate Data*

ACYL-SERINE INTERMEDIATE

An enzymatic reaction intermediate formed by carboxyl group transfer to a hydroxyl group of a serine residue within the active site.



Such an intermediate is known to be formed in reactions catalyzed by trypsin, chymotrypsin, thrombin, other enzymes of the blood-clotting cascade (except angiotensin-converting enzyme, which is an aspartic protease). An acyl-serine intermediate is also formed in the acetylcholinesterase reaction. The active site serine of this enzyme and the serine proteases can be alkylated by diisopropylfluorophosphate. **See also** *Active Site Titration*

ADAIR CONSTANTS

The thermodynamic association (or dissociation) constants used in the Adair equation for a ligand binding at sites in a multisite protein. The term “Adair constants” originally referred only to the four constants for the reversible binding of dioxygen to hemoglobin. **See** *Adair Equation*

ADAIR EQUATION

An equation presented by Gilbert Adair in 1924 to describe the binding of molecular oxygen to hemoglobin. Let the microscopic dissociation constant for the binding of O₂ to Hb be represented by k_1 , the microscopic dissociation constant for O₂ binding to the Hb(O₂) complex be represented by k_2 , and k_3 and k_4 are the corresponding microscopic dissociation constants for the third and fourth O₂, respectively. Thus, in terms of macroscopic dissociation constants, $k_1 = 4K_1$, $k_2 = (3/2)K_2$, $k_3 = (2/3)K_3$, and $k_4 = (1/4)K_4$, in which the coefficients are statistical factors derived from the number of available binding sites (and sites from which an already-bound ligand can leave). If Y_S represents the fractional saturation of the ligand binding (*i.e.*, fraction of occupied sites divided by total concentration of all sites), then

$$Y_S = \frac{\{([O_2]/k_1) + (3[O_2]^2/(k_1 k_2)) + (3[O_2]^3/(k_1 k_2 k_3)) + ([O_2]^4/(k_1 k_2 k_3 k_4))\}}{1 + (4[O_2]/k_1) + (6[O_2]^2/(k_1 k_2)) + (4[O_2]^3/(k_1 k_2 k_3)) + ([O_2]^4/(k_1 k_2 k_3 k_4))}$$

Analogous equations can be derived for the binding of other ligands to multisite proteins.

If the microscopic dissociation constants are not equal, then the Adair equation predicts a nonhyperbolic (*i.e.*, cooperative) curve. Values for k_1 and k_4 can be obtained by studying ligand binding at very low and very high ligand concentrations. Values for k_2 and k_3 can be ob-

tained from computer fitting protocols or by using a Hill plot.

The general Adair equation for the binding of a ligand X to a multisite protein where K_i represents the thermodynamic macroscopic association constant for the i th site and where n is the total number of sites is

$$nY_S = \left\{ \sum_{i=1}^n ([X]^i \prod_{j=1}^i (1/K_j)) \right\} \div \left\{ 1 + \sum_{i=1}^n ([X]^i \prod_{j=1}^i (1/K_j)) \right\}$$

Note that the Adair equation does not provide a reason for why identical sites would have different dissociation (or association) constants. **See** *Allosterism; Cooperativity; Koshland-Nemethy-Filmer Model*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Apparent fractional saturation, **232**, 615-618; global parameters, **232**, 625-626, 631-632; overview, **232**, 606-611; oxygen electrode data, treatment, **232**, 611-615; hemoglobin dimer-tetramer equilibrium in, **232**, 597-606; to oxygen equilibrium curve of hemoglobin; binding scheme, **232**, 560-563; importance of bottom and top data, **232**, 569-572; conversion of absorbance values to oxygen saturation values, **232**, 566-567; errors of best-fit Adair constant values, **232**, 572-574; extrapolation procedure, **232**, 566-567; nonlinear least-squares method, **232**, 565-566; statistical weighting, **232**, 567-569.

ADDITION REACTION

A general type of chemical reaction between two compounds, A and B, such that there is a net reduction in bond multiplicity (*e.g.*, addition of a compound across a carbon-carbon double bond such that the product has lost this π -bond). An example is the hydration of a double bond, such as that observed in the conversion of fumarate to malate by fumarase. Addition reactions can also occur with strained ring structures that, in some respects, resemble double bonds (*e.g.*, cyclopropyl derivatives or certain epoxides). A special case of a hydro-alkenyl addition is the conversion of 2,3-oxidosqualene to dammaradienol or in the conversion of squalene to lanosterol. Reactions in which new moieties are linked to adjacent atoms (as is the case in the hydration of fumarate) are often referred to as 1,2-addition reactions. If the atoms that contain newly linked moieties are not adjacent (as is often the case with conjugated reactants), then the reaction is often referred to as a 1, n -addition reaction in which n is the numbered atom distant from 1 (*e.g.*, 1,4-addition reaction). In general, addition reactions can take place via electrophilic addition, nucleophilic addition, free-radical addition, or via simultaneous or pericyclic addition.

ADDITIVITY PRINCIPLE

The principle that different structural domains, moieties, or features of a molecular substance contribute separately and additively to a property of a substance. In 1840, G. H. Hess introduced the Law of Constant Heat Summation, a relation that allows one to calculate the heat of a reaction from collected measurements of seemingly different reactions, as long as the summation of a series of reactions yields the same overall chemical reaction as the one of interest. Thermodynamic additivity requires that if two components, A and B, contribute independently to some process, then the total change in free energy (or enthalpy or entropy) is the sum of components, $\Delta G = \Delta G_A + \Delta G_B$. In view of its broad use in examining chemical and physical principles, Benson¹ has even offered the view that additivity is the “fourth” law of thermodynamics.

This principle has been utilized to assess intrinsic subsite binding energies for enzymes that have substrate binding subsites and exosites topologically distant from the reactive site (*e.g.*, polysaccharide depolymerases² and proteinases).

Dill³ recently discussed the merits and limitations of models that assume thermodynamic additivity and independence (of energy types, of neighbor interactions, of conformational freedom, of monomer contact pairing frequencies, *etc.*). He states that biological molecules may achieve stability in the face of thermal uncertainty, as polymers do, by compounding many small interactions; this summing can stump modelers because application of the additivity principle leads to accumulated error. Entropies and free energy may not be additive to describe weak interactions that are ensembles of states. He concludes that additivity principles appear to be few and limited in scope in biochemistry.

¹S. W. Benson (1976) *Thermochemical Kinetics: Methods for Estimation of Thermochemical Data and Rate Parameters*, 2nd ed., Wiley, New York.

²J. D. Allen (1979) *Meth. Enzymol.* **64**, 248.

³K. A. Dill (1997) *J. Biol. Chem.* **272**, 701.

ADDUCT

A distinct, chemical species that is formed by direct combination between two or more molecules such that no structural changes have occurred “within” the original interacting chemical species. The resulting interactions

between the species are often covalent or coordinate covalent. Examples include Lewis adducts and Meisenheimer adducts. The term “complex” usually refers to a looser association between interacting species. Intramolecular adducts are also possible. **See** *Complex*; *Lewis Adduct*; *Meisenheimer Adduct*

ADENINE DEAMINASE

This enzyme [EC 3.5.4.2], also called adenase, adenine aminohydrolase, and adenine aminase, catalyzes the hydrolysis of adenine to hypoxanthine and ammonia.

C. L. Zielke & C. H. Suelter (1971) *The Enzymes*, 3rd ed., **4**, 47.

ADENINE NUCLEOTIDE TRANSLOCASE

A nucleotide transporter (located in the outer mitochondrial membrane) that mediates one-for-one translocation/exchange of cytosolic ADP for mitochondrial ATP. This translocase is potentially inhibited by atractyloside and bongkregic acid.

M. Klingenberg, E. Winkler & S.-G. Huang (1995) *Meth. Enzymol.* **260**, 369.

M. Klingenberg (1991) *A Study of Enzymes* **2**, 367.

P. V. Vignais, M. R. Block, F. Boulay, G. Brandolin, P. Dalbon & G. J. M. Lauquin (1985) *Structure & Properties of Cell Membranes* **2**, 139.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Abundance, **260**, 370; atractyloside loading, **260**, 376-377, 379; exchange kinetics, **260**, 377-379; fluorescence probes, **260**, 384-385; inhibitors, **260**, 371-372; nucleotide binding, **260**, 370-371; purification; carboxyatractylate loading, **260**, 373; gel filtration, **260**, 374-375; hydroxyapatite chromatography, **260**, 374; solubilization, **260**, 373; quantitation, **260**, 379; reconstitution into phospholipid vesicles, **260**, 375-376; bovine heart protein, **260**, 376; yeast protein, **260**, 376-377; structure, **260**, 371; translocation intermediates in mitochondrial import, **260**, 270-271; and uncoupling protein, comparison, **260**, 370.

ADENINE PHOSPHORIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.7], also referred to as AMP pyrophosphorylase and transphosphoribosidase, catalyzes the reaction of AMP and pyrophosphate (or, diphosphate) to generate adenine and 5-phospho- α -ribose 1-diphosphate. In the reverse reaction, 5-amino-4-imidazolecarboxamide can replace adenine.

J. G. Flaks (1963) *Meth. Enzymol.* **6**, 136.

ADENOSINE DEAMINASE

This enzyme [EC 3.5.4.4], also known as adenosine aminohydrolase, catalyzes the hydrolysis of adenosine to yield inosine and ammonia.

- A. Radzicka & R. Wolfenden (1995) *Meth. Enzymol.* **249**, 284.
 D. K. Wilson & F. A. Quiocho (1994) *Nature Struct. Biol.* **1**, 691.
 J. J. Villafranca & T. Nowak (1992) *The Enzymes*, 3rd ed., **20**, 63.
 D. K. Wilson, F. B. Rudolph & F. A. Quiocho (1991) *Science* **252**, 1278.
 L. F. Thompson & J. E. Seegmiller (1980) *Adv. Enzymol.* **51**, 167.
 C. L. Zielke & C. H. Suelter (1971) *The Enzymes*, 3rd ed., **4**, 47.

ADENOSINE KINASE

This enzyme [EC 2.7.1.20] catalyzes the ATP-dependent phosphorylation of adenosine to generate AMP and ADP. 2-Aminoadenosine can also act as a substrate for this enzyme.

- P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.
 P. A. Frey (1989) *Adv. Enzymol.* **62**, 119.
 E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

S-ADENOSYLHOMOCYSTEINE LYASE

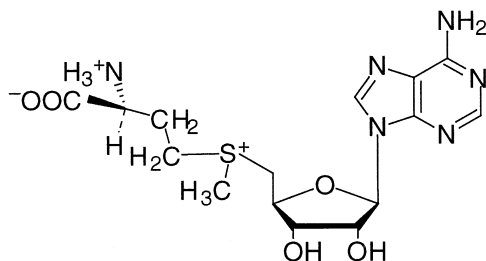
This enzyme [EC 3.3.1.1], also known as *S*-adenosylhomocysteinase, catalyzes the hydrolysis of *S*-adenosylhomocysteine to yield adenosine and homocysteine.

- A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed. **19**, 213.
 P. K. Chiang (1987) *Meth. Enzymol.* **143**, 377.
 A. Guranowski & H. Jakubowski (1987) *Meth. Enzymol.* **143**, 430.

S-ADENOSYLHOMOCYSTEINE NUCLEOSIDASE

This enzyme [EC 3.2.2.9] catalyzes the hydrolysis of *S*-adenosylhomocysteine to generate adenine and *S*-ribosylhomocysteine. The enzyme will also act on 5'-methylthioadenosine to give adenine and 5-methylthioribose. **See also** *Methylthioadenosine Nucleoside Hydrolase*

S-ADENOSYL-L-METHIONINE



A sulfonium derivative, abbreviated AdoMet or SAM, that is primarily generated by the action of methionine adenosyltransferase ($\text{ATP} + \text{L-methionine} + \text{H}_2\text{O} \rightarrow \text{SAM} + \text{P}_i + \text{pyrophosphate}$). SAM is a major methylat-

ing agent in the cell (**See Methyl Transfer Reaction**). SAM has six chiral centers: four in the ribose moiety, the α -carbon of methionine, and the sulfonium sulfur. The sulfur in the biologically active form of SAM has the *S*(+) configuration. This center readily undergoes epimerization; at a pH value of 7.5 and at 37°C, the half-life is about 12 hours¹. Even the solid halide salt of SAM will undergo 12% decomposition within a week at 3°C. Commercial SAM is subject to as much as 10% loss per day at 25°C; therefore, care must be exercised whenever studying SAM-dependent systems. This lability is not significant in biological systems, because the turnover time for SAM biosynthesis and consumption is short (*e.g.*, less than 25 minutes in human and rat liver). Protein binding of SAM may also improve its *in vivo* stability¹. Kinetic studies of SAM-dependent methyltransferases should use freshly prepared SAM. If solutions are to be stored, they should be kept for no longer than one month, preferably at -20°C under acidic conditions (typically below pH 2.5).

- ¹S. E. Wu, W. P. Huskey, R. T. Borchardt & R. L. Schowen (1984) *J. Amer. Chem. Soc.* **106**, 5762.

S-ADENOSYLMETHIONINE DECARBOXYLASE

This enzyme [EC 4.1.1.50] catalyzes the pyruvate-dependent conversion of *S*-adenosylmethionine to (5-deoxy-5-adenosyl)(3-aminopropyl)methylsulfonium salt and carbon dioxide.

- M. L. Hackert & A. E. Pegg (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 201.
 P. D. van Poelje & E. E. Snell (1990) *Ann. Rev. Biochem.* **59**, 29.
 M. H. Stipanuk (1986) *Ann. Rev. Nutr.* **6**, 179.
 C. W. Tabor & H. Tabor (1984) *Adv. Enzymol.* **56**, 251.
 F. Schlenk (1983) *Adv. Enzymol.* **54**, 195.
 E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

S-ADENOSYLMETHIONINE HYDROLASE

This enzyme [EC 3.3.1.2], also referred to as *S*-adenosylmethionine cleaving enzyme and methylmethionine-sulfonium-salt hydrolase, catalyzes the hydrolysis of *S*-adenosylmethionine to produce methylthioadenosine and homoserine. The enzyme will also convert methylmethionine sulfonium salt to dimethyl sulfide and homoserine.

- M. L. Gefter (1971) *Meth. Enzymol.* **17B**, 406.

ADENYLATE CYCLASE (or, ADENYLYL CYCLASE)

This enzyme [EC 4.6.1.1], also known as adenylyl cyclase and 3',5'-cyclic AMP synthetase, catalyzes the conversion of ATP to 3',5'-cyclic AMP and pyrophosphate. The enzyme requires pyruvate as a cofactor and will also utilize dATP as a substrate (thereby producing 3',5'-cyclic dAMP). In the presence of NAD(P)⁺-arginine ADP-ribosyltransferase, this enzyme is activated by covalent modification.

- R. Taussig & A. G. Gilman (1995) *J. Biol. Chem.* **270**, 1.
K. A. Mintzer & J. Field (1995) *Meth. Enzymol.* **255**, 468.
P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.
J. M. Stadel, A. De Lean & R. J. Lefkowitz (1982) *Adv. Enzymol.* **53**, 1.

ADENYLATE ISOPENTENYLTRANSFERASE

This enzyme [EC 2.5.1.27], also called 2-isopentenyl-diphosphate:AMP Δ^2 -isopentenyltransferase and cytokinin synthase, catalyzes the reaction of AMP with Δ^2 -isopentenyl diphosphate to yield pyrophosphate and N⁶-(Δ^2 -isopentenyl)adenosine 5'-monophosphate.

- R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.
D. S. Letham & L. M. S. Palmi (1983) *Ann. Rev. Plant Physiol.* **34**, 163.

ADENYLATE KINASE

This enzyme [EC 2.7.4.3], also known as myokinase, catalyzes the reversible reaction of MgATP with AMP to produce MgADP and ADP. Inorganic triphosphate can also act as substrate with this enzyme. **See** *Energy Charge; Metal Ions in Nucleotide-Dependent Reactions*

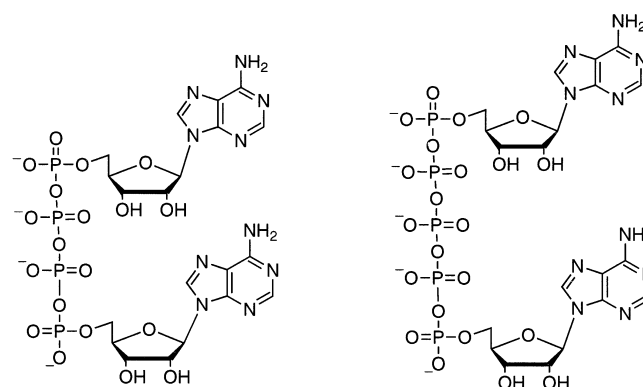
- M.-D. Tsai, R.-T. Jiang, T. Dahnke & Z. Shi (1995) *Meth. Enzymol.* **249**, 425.
L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.
J. J. Villafranca & T. Nowak (1992) *The Enzymes*, 3rd ed. **20**, 63.
M. Hamada, H. Takenaka, M. Sumida & S. A. Kuby (1991) *A Study of Enzymes* **2**, 403.
P. A. Frey (1989) *Adv. Enzymol.* **62**, 119.
A. S. Mildvan & D. C. Fry (1987) *Adv. Enzymol.* **59**, 241.
L. Noda (1973) *The Enzymes*, 3rd ed., **8**, 279.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

As frequent contaminant in enzyme preparations, **64**, 24; a contaminant in phosphotransferases, **63**, 7; coupled assay for, **63**, 6, 32; inhibition, **63**, 7, 398, 401, 483; isotope exchange properties, **64**, 8; mass action ratio, **63**, 18; mechanism, **63**, 18; substrate, metal-ion complex, **63**, 259; AMP phosphorylation, **238**, 34-35; bifurcation analysis, **240**, 809-812; kinetics, **240**, 809-810, 812; NMR spectroscopy, nonresonant effects, **239**, 77; in photoactivatable probe synthesis, **237**, 88-89, 93; substrates, **240**, 810; P¹,P⁵-di(adenosine-5')-pentaphosphate inhibition of adenylate kinase, **63**, 7, 401, 483; P¹,P⁴-di(adenosine-5')-tetraphosphate inhibition

of adenylate kinase, **63**, 7, 401, 483; phosphorus stereospecificity [toward adenosine 5'-monothiophosphate, **249**, 426-427 (enhancement with R97M mutant, **249**, 432-434; reversal with R44M mutant, **249**, 431-432; wild-type, confirmation, **249**, 429-431); toward adenosine 5-(1-thiotriphosphate), **249**, 433 (perturbation with T23A mutant, **249**, 437-440; relaxation with R128A mutant, **249**, 435-437); at AMP site, **249**, 426-427; at ATP site, **249**, 426-427; demonstration, **249**, 426; manipulation by site-directed mutagenesis, **249**, 425-443 (active site conformations and, **249**, 441-442; kinetic experiments, **249**, 428; methods, **249**, 428-429; microscopic rates and, **249**, 440-441; procedures, **249**, 429-433; results interpretation, **249**, 440-442; at P α of MgATP, **249**, 433-440; wild-type and site-directed mutant enzymes, comparison, **249**, 440-441].

ADENYLATE KINASE INHIBITORS



P¹,P⁴-di-(Adenosine-5')-P₄

P¹,P⁵-di-(Adenosine-5')-P₅

Because many cells maintain ATP, ADP, and AMP concentrations at or near the mass action ratio of the adenylate kinase reaction, the cellular content of this enzyme is often quite high. A consequence of such abundance is that, even after extensive purification, many proteins and enzymes contain traces of adenylate kinase activity. The presence of this kinase can confound the quantitative analysis of processes that either require ADP or are carried out in the presence of both ATP and AMP. Furthermore, the equilibrium of any reaction producing ADP may be altered if adenylate kinase activity is present. To minimize the effect of adenylate kinase, one can utilize the bisubstrate geometrical analogues Ap₄A and Ap₅A to occupy simultaneously both substrate binding pockets of this kinase^{1,2}. Typical inhibitory concentrations are 0.4 and 0.2 mM, respectively. Of course, as is the case for the use of any inhibitor, one must always determine whether Ap₄A or Ap₅A has a direct effect on a particular reaction under examination. For example, Powers *et al.*³ studied the effect of a series of α,ω -di(adenosine 5')-polyphosphates (*e.g.*, Ap_nA, where n =

2, 3, 4, 5 and 6) on carbamyl phosphate synthetase. Only AP_5A was found to be an effective inhibitor of the overall reaction as well as two partial reactions catalyzed by the enzyme (namely, HCO_3^- -dependent ATP hydrolysis and ATP synthesis from carbamyl-P and ADP).

¹D. L. Purich & H. J. Fromm (1972) *Biochim. Biophys. Acta* **276**, 563.

²G. E. Leinhard & I. I. Secemski (1973) *J. Biol. Chem.* **248**, 1121.

³S. G. Powers, O. W. Griffith & A. Meister (1977) *J. Biol. Chem.* **252**, 3558.

Selected entries from *Methods in Enzymology* [vol, page(s)]:
Adenylate kinase contamination in in phosphotransferases, **63**, 7; as contaminant enzyme preparations, **64**, 24; P^1, P^4 -di(adenosine-5')-tetraphosphate and P^1, P^5 -di(adenosine-5')-pentaphosphate inhibition of adenylate kinase, **63**, 7, 401, 483.

ADENYLOSUCCINATE LYASE

This enzyme [EC 4.3.2.2], also referred to as adenylosuccinase, catalyzes the conversion of N^6 -(1,2-dicarboxyethyl)AMP (or, adenylosuccinate) to yield fumarate and AMP. The enzyme will also convert 1-(5-phosphoribosyl)-4-(N -succinocarboxamide)-5-aminoimidazole to 5'-phosphoribosyl-5-amino-4-imidazolecarboxamide.

S. Ratner (1972) *The Enzymes*, 3rd ed., **7**, 167.

ADENYLOSUCCINATE SYNTHETASE

Adenylosuccinate synthetase¹⁻³ (AMPSase; EC 6.3.4.4) catalyzes the following reversible reaction:



Aside from the natural nucleophile, aspartate, the only other agent that can attack the 6-carbon of IMP in a highly specific manner is NH_2OH .

As the first committed step in the biosynthesis of AMP from IMP, AMPSase plays a central role in *de novo* purine nucleotide biosynthesis. A 6-phosphoryl-IMP intermediate appears to be formed during catalysis, and kinetic studies of *E. coli* AMPSase demonstrated that the substrates bind to the enzyme active sites randomly. With mammalian AMPSase, aspartate exhibits preferred binding to the E-GTP-IMP complex rather than to the free enzyme. Other kinetic data support the inference that Mg-aspartate complex formation occurs within the adenylosuccinate synthetase active site and that such a

complex may be an important factor in the activation of the protonated amino group of aspartate, enhancement of the enzyme's binding affinity, and its specificity for aspartate.

¹R. Honzatko & H. J. Fromm (1998) *Adv. Enzymol.* **73**, in press.

²B. F. Cooper & F. B. Rudolph (1995) *Meth. Enzymol.* **249**, 188.

³F. M. Raushel & J. J. Villafranca (1988) *Crit. Rev. Biochem.* **23**, 1.

Selected entries from *Methods in Enzymology* [vol, page(s)]:
Equilibrium isotope exchange study of kinetic mechanism, **249**, 466; site-directed mutagenesis of *Escherichia coli* enzyme, **249**, 93; positional isotope exchange studies, **249**, 423; product inhibition studies of three substrates:three products reactions, **249**, 207-208.

ADENYLYLSULFATE—AMMONIA ADENYLYLTRANSFERASE

This enzyme [EC 2.7.7.51] catalyzes reaction of adenylyl-sulfate and ammonia to form adenosine 5'-phosphoramidate and sulfate.

A. Schmidt & K. Jäger (1992) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **43**, 325.

H. Frankhauser, J. A. Schiff, L. J. Garber & T. Saidha (1987) *Meth. Enzymol.* **143**, 354.

ADENYLYLSULFATE KINASE

This enzyme [EC 2.7.1.25], also referred to as APS kinase, catalyzes the reaction of ATP and adenylylsulfate to generate ADP and 3'-phosphoadenylylsulfate.

T. S. Leyh (1993) *Crit. Rev. Biochem. Mol. Biol.* **28**, 515.

A. Schmidt & K. Jäger (1992) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **43**, 325.

J. A. Schiff & T. Saidha (1987) *Meth. Enzymol.* **143**, 329.

H. D. Peck, Jr. (1974) *The Enzymes*, 3rd ed., **10**, 651.

ADENYLYLSULFATE REDUCTASE

This enzyme [EC 1.8.99.2] catalyzes the reaction of AMP, sulfite, and some acceptors (for example, methyl viologen) to yield adenylylsulfate and the reduced acceptor. Both FAD and iron are cofactors.

J. Lampreia, A. S. Pereira & J. J. G. Moura (1994) *Meth. Enzymol.* **243**, 241.

C. Dahl, N. Speich & H. G. Trüper (1994) *Meth. Enzymol.* **243**, 331.

C. Dahl & H. G. Trüper (1994) *Meth. Enzymol.* **243**, 400.

Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

ADENYLYLSULFATE SULFOHYDROLASE

This enzyme [EC 3.6.2.1], also called adenylylsulfatase, catalyzes the hydrolysis of adenylylsulfate to generate AMP and sulfate.

A. Schmidt & K. Jäger (1992) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **43**, 325.

A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 1.

ADIABATIC PROCESS

1. Any process in which heat neither enters nor leaves the system. Thus, $q = 0$ or, for a differential process, $dq = 0$. If there is a reversible change in volume with such a system, $dU = -PdV$ where U is the internal energy, P is the pressure, and V is the volume. Any reversible adiabatic change is isentropic. (For an irreversible adiabatic change, $dU = -P_{\text{ex}}dV$ where P_{ex} is the external pressure.) An example of such a process is the adiabatic expansion of a gas, resulting in the cooling of the gas. The rapid release of vapor from an aerosol can is thus accompanied by a drop in temperature since the release is so fast there is little time for heat to be transferred. An adiabatic compression of a gas will result in the heating of the gas. An example can be provided by the use of a hand pump when inflating a basketball or bicycle tire. The act of pumping, by compressing the gas, is a near adiabatic process. Often, this increase in temperature is translated to the needle valve and can be felt by the individual doing the pumping. [Note that, for an ideal gas undergoing adiabatic expansion or compression, $PV^\gamma = K$ where K is a constant and γ is the ratio of the heat capacities, C_P/C_V (provided that the two heat capacities are constants). Recall that, for an isothermal, nonadiabatic process, $PV = \text{constant}$.] 2. In chemical reactions, one normally considers that the reaction path is continuous over a potential energy surface. Such a reaction is said to be an adiabatic reaction. If a reaction pathway moves from one potential energy surface to another, it is often termed a nonadiabatic (or diabatic) reaction. The use of the term in these instances is not the same as its use in thermodynamics. Nonadiabatic processes can occur when the movement of electrons is not rapid relative to the change of a nuclear position. Under these conditions, the Born-Oppenheimer approximation is not valid and one can have the crossing of potential energy surfaces. **See** *Isothermal Process*; *Heat Capacity*

ADP-RIBOSYLATION

Covalent modification of a protein by the linkage of an ADP-ribosyl moiety to the protein. The resulting product typically exhibits altered kinetic and/or regulatory properties. ADP-ribosyltransferases catalyze the trans-

fer of the ADP-ribosyl group of NAD^+ to a protein acceptor, producing the modified protein and free nicotinamide. The reaction scheme catalyzed by the cholera toxin A subunit and a rabbit muscle protein has been determined to be random¹. The amino acyl residues that have been reported to have been modified in these reactions include a cysteinyl residue (e.g., via pertussis toxin²), an arginyl residue (via cholera toxin A), an asparaginyl residue (via botulinum toxin), and a modified histidyl residue (via diphtheria toxin). In addition, the subunit structure appears to have a role in substrate specificity¹³.

The widespread occurrence of ADP-ribosyltransferases³⁻⁷ has generated considerable interest in assessing the role of ADP-ribosylation in cellular metabolism. Regulatory proteins associated with this modification event have also been identified. A small G-protein referred to as ARF (ADP-ribosylation factor) has been shown to enhance this covalent modification⁸⁻¹⁰. ARF actually represents a family of proteins, both cytosolic and membrane-bound. Interestingly, ARF has been shown to have a crucial role in vesicle transport from the Golgi^{11,12}. When ARF-GDP binds to a developing vesicle, a GDP|GTP exchange occurs, and the coatomer begins to form. After the vesicle pinches off, the GTP is hydrolyzed and the coatomer and ARF are released, allowing the vesicle to fuse with the target membrane.

¹J. S.-A. Larew, J. E. Peterson & D. J. Graves (1991) *J. Biol. Chem.* **266**, 52.

²A. G. Gilman (1987) *Ann. Rev. Biochem.* **56**, 615.

³M. D. Brightwell, C. E. Leech, M. K. O'Farrell, W. J. D. Whish & S. Shall (1975) *Biochem. J.* **147**, 119.

⁴J. Moss, S. J. Stanley & P. A. Watkins (1980) *J. Biol. Chem.* **255**, 5838.

⁵J. Moss & S. J. Stanley (1981) *Proc. Natl. Acad. Sci. U. S. A.* **78**, 4809.

⁶J. Moss & S. J. Stanley (1981) *J. Biol. Chem.* **256**, 7830.

⁷G. Soman, J. R. Mickelson, C. F. Louis & D. J. Graves (1984) *Biochem. Biophys. Res. Commun.* **120**, 973.

⁸R. A. Kahn & A. G. Gilman (1986) *J. Biol. Chem.* **261**, 7906.

⁹L. Monoco, J. J. Murtaugh, K. B. Newman, S.-C. Tsai, J. Moss & M. Vaughn (1990) *Proc. Natl. Acad. Sci. U. S. A.* **87**, 2206.

¹⁰M. Tsuchiya, S. R. Price, M. S. Nightingale, J. Moss & M. Vaughn (1989) *Biochemistry* **28**, 9668.

¹¹J. E. Rothman (1994) *Nature* **372**, 55.

¹²J. E. Rothman and L. Orci (1996) *Sci. Amer.* **274** (3), 70.

¹³D. J. Graves, B. L. Martin & J. H. Wang (1994) *Co- and Posttranslational Modification of Proteins*, Oxford Univ. Press, New York.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

By bacterial toxin, **235**, 617-632; by cholera toxin [catalysis, **237**, 45; assay, **235**, 642-647; G protein as subunit, **237**, 48]; by pertussis toxin, **237**, 24-26, 71, 132-133; G-protein subunits, **237**, 236-238; time-course of transducin ADP-ribosylation, **237**, 77-79, 91, 93-94.

ADRENODOXIN

A [2Fe-2S] ferredoxin participating in electron transfer reactions from NADPH to cytochrome P-450 within the adrenal gland.

ADSORPTION

The process of adhesion of molecular entities to surfaces of solids or liquids with which the entities are coming into contact. For example, adsorption chromatography utilizes different adsorption properties to separate molecules. **See also** *Absorption*; *Biom mineralization*; *Micellar Catalysis*; *Langmuir Isotherm*

AEQUORIN

A Ca^{2+} -binding protein isolated from jellyfish (*Aequorea* sp.) that is frequently used as a chemiluminescent calcium indicator.^{1,2} Aequorin contains a hydrophobic prosthetic group, coelenterazine. Investigators have been able to express the transfected gene recombinantly in a number of cells, and upon addition of the cofactor, one can measure intracellular calcium concentration. **See** *Fura-2*

¹E. R. Ridgway & C. C. Ashley (1967) *Biochem. Biophys. Res. Commun.* **29**, 229.

²A. Azzi & B. Chance (1969) *Biochim. Biophys. Acta* **189**, 141.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Calibration of calcium binding, **260**, 425-427; luminescence measurement, **260**, 424-425; mitochondria-targeted hybrid protein [calcium quantitation, **260**, 418, 422, 424-428; stable expression, **260**, 418-421; transient expression, **260**, 420; intracellular localization, **260**, 421-425; reconstitution with coelenterazine, **260**, 422-422]; structure, **260**, 418.

AFFINITY LABELING

The covalent modification of an enzyme, most frequently at a reactive amino acid residue located within the active site. Affinity labeling is usually achieved by employing a substrate or coenzyme analogue bearing a reactive group that undergoes chemical, or photochemical, reaction with a nearby nucleophilic or electrophilic group located at the active center. Classical examples of affinity labeling reagents include: (a) tosyl-L-phenylalanine chloromethyl ketone (abbreviated TPCK) which binds into the aromatic substrate binding pocket of chymotrypsin and subsequently undergoes a specific reaction with histidine-57 to inactivate the enzyme; (b) diisopropylphosphorfluoridate (abbreviated DIFP or DFP) which binds into the acetylcholine site of acetylcholinesterase and subsequently inactivates the enzyme by attaching to the

active site seryl residue with expulsion of fluoride ion; and (c) bromohydroxyacetone phosphate which inactivates triose-phosphate isomerase by esterification of the active site glutamate-165.

The basic kinetic scheme involves site-directed binding (hence, affinity) of a chemically or photochemically reactive substance I, followed by first-order conversion to a covalent adduct:



The rate law for covalent modification shows saturation kinetic behavior if the second step is rate-determining:

$$v_{\text{inactivation}} = -d[\text{E}]/dt = k[\text{E}][\text{I}]/(K_i + [\text{I}])$$

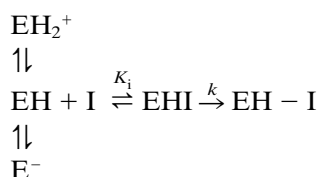
This equation will describe the loss of enzyme activity, if the affinity label is binding at the active site. Moreover, reversibly bound ligands capable of occupying the same site as the affinity label will competitively inhibit the rate of enzyme inactivation.

Baker¹ was among the earliest investigators to promote the study of active site-directed irreversible inhibitors as probes of enzyme active sites and as potential pharmacologic agents. He reasoned that while different isozymes may contain similar reactive groups in the active site, they may have unique sites outside the active site. His strategy was to locate reactive substituents in a way that promotes reactivity with residues lying outside the active site, thereby achieving what he termed "exo-alkylation". His goal was to inactivate certain isozymes that may be required by cancerous cells, but not their untransformed counterparts.

Plapp² has presented a valuable account of the use of affinity labeling to identify essential amino acid residues involved in enzymic catalysis, to map the general topological features of the active site, and to gain insights about the most likely mechanism of catalysis. He also discusses what one must consider in designing active site-directed reagents (a) to achieve specificity or, at least, selectivity and (b) to choose among the many options for the covalent chemistry (e.g., alkylation, acylation, or photoinsertion). Issues related to the evaluation of affinity labels (such as kinetics of inactivation, reactivity, substrate protection, and specificity) are also discussed.

Colman³ has also produced a definitive account of site-specific modification of enzymes, and her chapter is particularly instructive about the range and utility of reaction types that can be gainfully exploited in affinity labeling experiments.

As in the case of enzymic catalysis, one can have one protonation state of the enzyme, say EH, that reacts with the affinity label and two other states, say E⁻ and EH₂⁺, that do not.



In such a case, a Michaelis pH function becomes useful in describing the pH profile for inactivation. Starting with the conservation of enzyme equation:

$$[\text{E}_0] = [\text{EH}] + [\text{E}^-] + [\text{EH}_2^+]$$

or

$$[\text{E}_0]/[\text{EH}] = 1 + [\text{E}^-]/[\text{EH}] + [\text{EH}_2^+]/[\text{EH}]$$

or

$$[\text{E}_0]/[\text{EH}] = 1 + K_1/[\text{H}^+] + [\text{H}^+]/K_2$$

where $K_1 = [\text{H}^+][\text{E}^-]/[\text{EH}]$ and $K_2 = [\text{H}^+][\text{EH}]/[\text{EH}_2^+]$. If $v_{\text{inactivation}} = k[\text{EHI}]$, then one can rearrange the expression to get an equation that predicts a bell-shaped profile for the rate of inactivation as a function of pH. Likewise, the state of protonation of the affinity reagent can affect the kinetics of enzyme inactivation by affinity labels. Brocklehurst⁴ has given a useful account of two-protonic-state electrophiles as probes of enzyme catalytic mechanism.

¹B. R. Baker (1967) *Design of Active-Site Directed Irreversible Enzyme Inhibitors*, Wiley, New York.

²B. V. Plapp (1982) *Meth. Enzymol.* **87**, 469.

³R. F. Colman (1990) *The Enzymes*, 3rd. ed., **19**, 283.

⁴K. Brocklehurst (1982) *Meth. Enzymol.* **87**, 427.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

General Methodology: An overview, **46**, 3; application of affinity labeling for studying structure and function of enzymes, **87**, 469; transition state analogs as potential affinity labeling reagents, **46**, 15; mechanism-based irreversible enzyme inhibitors, **46**, 28; syncatalytic enzyme modification: characteristic features and differentiation from affinity labeling, **46**, 41; paracatalytic enzyme modification by oxidation of enzyme-substrate carbanion intermediates, **46**, 48; catalytic competence: a direct criterion for affinity labeling, **46**, 54; differential labeling: a general technique for selective modification of binding sites, **46**, 59; photoaffinity la-

beling, **46**, 69; design of exo affinity labeling reagents, **46**, 115; haloketones as affinity labeling reagents, **46**, 130; haloacetyl derivatives, **46**, 153; acetylenic irreversible enzyme inhibitors, **46**, 158; organic isothiocyanates as affinity labels, **46**, 164; photo-cross-linking of protein-nucleic acid complexes, **46**, 168; affinity labeling of multicomponent systems, **46**, 180.

Enzymes: Reaction of serine proteases with halomethyl ketones, **46**, 197; reaction of serine proteases with aza-amino acid and aza-peptide derivatives, **46**, 208; active site-directed inhibition with substrates producing carbonium ions: chymotrypsin, **46**, 216; peptide aldehydes: potent inhibitors of serine and cysteine proteases, **46**, 220; carboxypeptidases A and B, **46**, 225; *N*-substituted arginine chloromethyl ketones, **46**, 229; renin, **46**, 235; nucleotide and nucleic acid systems; adenosine derivatives for dehydrogenases and kinases, **46**, 240; alcohol dehydrogenases, **46**, 249; arylazido nucleotide analogs in a photoaffinity approach to receptor site labeling, **46**, 259; aromatic thioethers of purine nucleotides, **46**, 289; *N*⁶-*o*- and *p*-fluorobenzoyladenine 5'-triphosphates, **46**, 295; 6-chloropurine ribonucleoside 5'-phosphate, **46**, 299; carboxylic-phosphoric anhydrides isosteric with adenine nucleotides, **46**, 302; active-site labeling of thymidylate synthetase with 5-fluoro-2'-deoxyuridylate, **46**, 307; inert Co(III) complexes as reagents for nucleotide binding sites, **46**, 312; the active site of ribonucleoside diphosphate reductase, **46**, 321; adenosine deaminase, **46**, 327; direct photoaffinity labeling with cyclic nucleotides, **46**, 335; adenosine 3',5'-cyclic monophosphate binding sites, **46**, 339; 5-formyl-UTP for DNA-dependent RNA polymerase, **46**, 346; DNA-dependent RNA polymerase, **46**, 353; staphylococcal nuclease, **46**, 358.

Carbohydrate Systems: Carbohydrate binding sites, **46**, 362; glucosidases, **46**, 368; glycidol phosphates and 1,2-anhydrohexitol 6-phosphates, **46**, 381; β -galactosidase, **46**, 398; lysozyme, **46**, 403.

Amino Acid Systems: Glutamine binding sites, **46**, 414; labeling of the active site of L-aspartate β -decarboxylase with β -chloro-L-alanine, **46**, 427; active site of L-asparaginase: reaction with diazo-4-oxonorvaline, **46**, 432; labeling of serum prealbumin with *N*-bromoacetyl-L-thyroxine, **46**, 435; a pyridoxamine phosphate derivative, **46**, 441.

Steroid Systems: Labeling of steroid systems, **46**, 447; irreversible inhibitors of Δ^5 -3-ketosteroid isomerase: acetylenic and allenic 3-oxo-5,10-secosteroids, **46**, 461; labeling of Δ^5 -3-ketosteroid isomerase by photoexcited steroid ketones, **46**, 469.

Antibodies: Affinity labeling of antibody combining sites as illustrated by anti-dinitrophenyl antibodies, **46**, 479; *p*-azobenzene arsonate antibody, **46**, 492; affinity cross-linking of heavy and light chains, **46**, 501; bivalent affinity labeling haptens in the formation of model immune complexes, **46**, 505; DNP-based diazoketones and azides, **46**, 508; labeling of antilactose antibody, **46**, 516.

Other Proteins: The ouabain-binding site on (Na⁺/K⁺)-adenosine 5'-triphosphatase, **46**, 523; penicillin isocyanates for β -lactamase, **46**, 531; active site-directed addition of a small group to an enzyme: the ethylation of luciferin, **46**, 537; mandelate racemase, **46**, 541; dimethylpyrazole carboxamide and related derivatives, **46**, 548; labeling of catechol *O*-methyltransferase with *N*-haloacetyl derivatives, **46**, 554; affinity labeling of binding sites in proteins by sensitized photooxidation, **46**, 561; bromocolchicine as a label for tubulin, **46**, 567.

Receptors and Transport Systems: Affinity labeling of receptors, **46**, 572; nicotinic acetylcholine receptors, **46**, 582; β -adrenergic receptors, **46**, 591; opiate receptors, **46**, 601; amino acid transport proteins, **46**, 607; the biotin transport system, **46**, 613.

Nucleic Acids and Ribosomes: Affinity labeling of ribosomal functional sites, **46**, 621; photoaffinity labeling of 23S RNA in ribosomes, **46**, 637; photoaffinity labels for nucleic acids, **46**, 644; identification of GDP binding sites with 4-azidophenyl-GDP, **46**, 649; photoactivated GTP analogs, **46**, 656; a photoactivated analog of streptomycin, **46**, 660; haloacylated streptomycin and puromycin analogs, **46**, 662; chemically reactive oligonucleotides, **46**, 669; aromatic ketone derivatives of aminoacyl-tRNA as photoaffinity labels for ribosomes, **46**, 676; photoaffinity-probe-modified tRNA for the analysis of ribosomal binding sites, **46**, 683; analogs of chloramphenicol and their application to labeling ribosomes, **46**, 702; a homologous series of photoreactive peptidyl-tRNAs for probing the ribosomal peptidyltransferase center, **46**, 707; photoaffinity labeling of ribosomes with the unmodified ligands puromycin and initiation factor 3, **46**, 711.

AGARITINE γ -GLUTAMYLTRANSFERASE

This enzyme [EC 2.3.2.9] catalyzes the reaction of agaritine with an acceptor substrate to generate 4-hydroxymethylphenylhydrazine and the γ -glutamyl-acceptor. Examples of substrate acceptors include 4-hydroxyaniline, cyclohexylamine, 1-naphthylhydrazine, and similar compounds. The enzyme will also be the catalyst in the hydrolysis of agaritine. **See also** γ -Glutamyl Transpeptidase

B. Levenberg (1970) *Meth. Enzymol.* **17A**, 877.

AGMATINE DEIMINASE

This enzyme [EC 3.5.3.12], also known as agmatine iminohydrolase, catalyzes the hydrolysis of agmatine to produce *N*-carbamoylputrescine and ammonia. The plant enzyme also catalyzes the reactions of EC 2.1.3.3 (ornithine carbamoyltransferase), EC 2.1.3.6 (putrescine carbamoyltransferase) and EC 2.7.2.2 (carbamate kinase), thereby functioning as a putrescine synthase, converting agmatine and ornithine into putrescine and citrulline, respectively.

T. A. Smith (1985) *Ann. Rev. Plant Physiol.* **36**, 117.

C. W. Tabor & H. Tabor (1984) *Ann. Rev. Biochem.* **53**, 749.

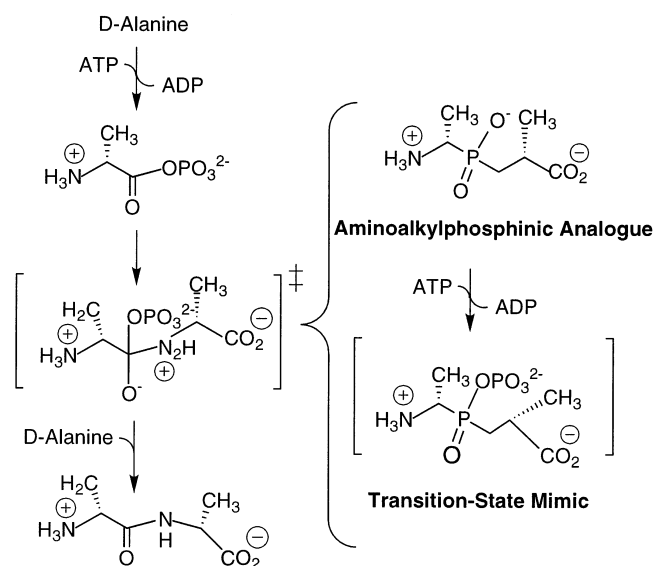
AGONIST

Any chemical substance (*i.e.*, hormone, drug, metabolite, *etc.*) that associates with a receptor binding site and produces a biological response through such binding. The occupancy assumption of agonist action states that the magnitude of the bioeffect is determined by the number of receptors occupied by agonist. Agonists may differ with respect to the affinity of their interactions with a

receptor; however, once occupied, the receptor is assumed to make the same contribution to the overall response in a manner that is independent of the bound agonist. If two different agonists are present at concentrations allowing each to occupy the same fraction of receptor sites, and if the response is different at this same level of receptor occupancy, then one uses the term efficacy to describe the biological effectiveness of each agonist-receptor complex. **See** *Receptor-Agonist Action*

D-ALANINE:D-ALANINE LIGASE

This enzyme [EC 6.3.2.4], also known as D-alanylalanine synthetase, catalyzes an essential reaction for growth of certain enterococci, and failure to synthesize this dipeptide prevents the cross-linking of the underlying peptidoglycan structure of the bacterial cell wall. The resulting aminoacyl-D-Ala-D-Ala strand is the target of vancomycin binding, such that this antibiotic arrests peptidoglycan cross-linking and blocks cell-wall synthesis.



Mechanism of D-Ala-D-Ala Synthesis and Inhibition

The kinetic reaction mechanism appears to be random, and for the reaction to proceed, all substrates must reside as a $E \cdot D\text{-Ala} \cdot D\text{-Ala} \cdot \text{MgATP}$ quaternary complex. Except for its activation of an α -carboxylate to form a peptide bond, the enzyme's mechanism appears to be completely analogous to that catalyzed by glutamine synthetase, which forms a γ -glutamyl-phosphate intermediate. There is strong evidence for the participation

of an acyl-phosphate intermediate which then suffers attack by the amino group of a second D-Ala molecule. Indeed, a phosphinate dipeptide analogue¹ is converted by enzymatic phosphorylation from low-affinity inhibitor to extremely tightly bound analogue of the ligase's reaction intermediates. [1(*S*)-Aminoethyl][2-carboxy-2(*R*)-methyl-1-ethyl]phosphinate is an ATP-dependent, slow-binding inhibitor of the D-Ala:D-Ala ligase from *Salmonella typhimurium*, and the enzyme-inhibitor complex (after ATP-dependent phosphorylation) has a half-life of 17 days at 37°C. The mechanism of inhibition is analogous to that of glutamine synthetase by methionine sulfoximine and phosphinothricin.

McDermott *et al.*² used rotational resonance, a then newly developed solid-state NMR method, for structural studies of an inhibited complex formed by reaction of D-alanine:D-alanine ligase, ATP, and the aminoalkyl dipeptide analogue. The measured NMR coupling properties indicate that the two species are bridged in a P—O—P linkage, with a P—P through-space distance of 2.7 ± 0.2 Å. This work unambiguously demonstrated that the inactivation mechanism involves phosphorylation of enzyme-bound inhibitor by ATP to form a phosphoryl-phosphinate adduct³.

Vancomycin-resistant enterococci produce a mutant enzyme⁴ (designated, VanA) that substitutes α -hydroxy acids in place of D-alanine. Because these so-called depsipeptides fail to bind vancomycin, but still function in the essential cell wall cross-linking reactions, these mutant bacteria are resistant to vancomycin antibiosis. Fan *et al.*⁵ determined the ligase structure (co-crystallized with an *S,R*-methylphosphinate analogue and ATP) by X-ray diffraction to a resolution of 2.3 Å. The authors suggest a catalytic mechanism in which a helix dipole and a hydrogen-bonded triad of tyrosine, serine, and glutamic acid facilitate the binding and deprotonation steps. A different triad exists in a D-alanine:D-lactate ligase (VanA) present in vancomycin-resistant enterococci. Another D-Ala:D-Ala-ligase-related enzyme has the capacity to make D-alanyl-D-serine⁶. Alignment of the deduced amino acid sequences with those of other related enzymes from gram-negative and gram-positive bacteria revealed the presence of four distinct sequence patterns in the putative substrate-binding sites, each correlating with specificity to a particular substrate.

¹K. Duncan & C. T. Walsh (1988) *Biochemistry* **27**, 3709.

²A. E. McDermott, F. Creuzet, R. G. Griffin, L. E. Zawadzke, Q. Z. Ye & C. T. Walsh (1990) *Biochemistry* **29**, 5767.

³C. T. Walsh (1993) *Science* **261**, 308 and *Science* **262**, 164.

⁴T. D. Bugg, S. Dutka-Malen, M. Arthur, P. Courvalin, & C. T. Walsh (1991) *Biochemistry* **30**, 2017.

⁵C. Fan, P. C. Moews, C. T. Walsh & J. R. Knox (1994) *Science* **266**, 439.

⁶S. Evers, B. Casadewall, M. Charles, S. Dutka-Malen, M. Gali-
mand & P. Courvalin (1996) *Mol. Evol.* **42**, 706.

ALANINE AMINOTRANSFERASE

This enzyme [EC 2.6.1.2], also known as glutamic-pyruvic transaminase and glutamic-alanine transaminase, catalyzes the pyridoxal-phosphate-dependent reaction of alanine with 2-ketoglutarate, resulting on the production of pyruvate and glutamate. 2-Aminobutanoate will also react, albeit slowly. There is another alanine aminotransferase [EC 2.6.1.12], better known as alanine-oxo-acid aminotransferase, which catalyzes the pyridoxal-phosphate-dependent reaction of alanine and a 2-keto acid to generate pyruvate and an amino acid. **See also** *Alanine:Glyoxylate Aminotransferase*

A. J. L. Cooper (1985) *Meth. Enzymol.* **113**, 69.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

D-ALANINE AMINOTRANSFERASE

This enzyme [EC 2.6.1.21], also known as D-aspartate aminotransferase, D-amino acid aminotransferase, and D-amino acid transaminase, catalyzes the reversible pyridoxal-phosphate-dependent reaction of D-alanine with α -ketoglutarate to yield pyruvate and D-glutamate. The enzyme will also utilize as substrates the D-stereoisomers of leucine, aspartate, glutamate, aminobutyrate, norvaline, and asparagine. **See** *D-Amino Acid Aminotransferase*

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

β -ALANINE AMINOTRANSFERASE

This enzyme [EC 2.6.1.18], also known as β -alanine-pyruvate aminotransferase, catalyzes the reversible pyridoxal-phosphate-dependent reaction of β -alanine with pyruvate to generate 3-oxopropanoate and alanine.

K. Yonaha, S. Toyama & K. Soda (1987) *Meth. Enzymol.* **143**, 500.

O. W. Griffith (1986) *Ann. Rev. Biochem.* **55**, 855.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

ALANINE DEHYDROGENASE

This enzyme [EC 1.4.1.1] catalyzes the reaction of alanine with water and NAD⁺ to produce pyruvate, ammonia, and NADH.

- A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
 N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.
 A. Yoshida & E. Freese (1970) *Meth. Enzymol.* **17A**, 176.

ALANINE:GLYOXYLATE AMINOTRANSFERASE

This enzyme [EC 2.6.1.44] catalyzes the reversible, pyridoxal-phosphate-dependent reaction of alanine with glyoxylate to generate pyruvate and glycine. One component of the animal enzyme can utilize 2-oxobutanoate as a substrate instead of glyoxylate. A second component of the enzyme can also catalyze the reaction of alanine with 3-hydroxypyruvate to produce pyruvate and serine.

See *Serine:Pyruvate Aminotransferase*

- A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

D-ALANINE HYDROXYMETHYLTRANSFERASE

This enzyme [EC 2.1.2.7], also called 2-methylserine hydroxymethyltransferase, catalyzes the reaction of D-alanine with water and 5,10-methylenetetrahydrofolate to produce tetrahydrofolate and 2-methylserine. The enzyme can also use 2-hydroxymethylserine as a substrate.

- E. W. Miles (1971) *Meth. Enzymol.* **17B**, 341.

ALANINE RACEMASE

This enzyme [EC 5.1.1.1] catalyzes the pyridoxal-phosphate-dependent interconversion of D-alanine with L-alanine.

- M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.
 R. A. John (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 173.
 M. A. Ator & P. R. Ortez de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 A. E. Martell (1982) *Adv. Enzymol.* **53**, 163.
 E. Adams (1976) *Adv. Enzymol.* **44**, 69.
 E. Adams (1972) *The Enzymes*, 3rd ed., **6**, 479.

β -ALANOPINE DEHYDROGENASE

This enzyme [EC 1.5.1.17] catalyzes the reversible reaction of 2,2'-iminodipropionate with water and NAD⁺ to produce alanine, pyruvate and NADH. In the reverse reaction, alanine can be replaced as a substrate by cysteine, serine, or threonine. Glycine acts very slowly as a substrate (that is, the reaction catalyzed by strombine dehydrogenase).

- J. Thompson & S. P. F. Miller (1991) *Adv. Enzymol.* **64**, 317.

D-ALANYL-D-ALANINE PEPTIDASE

This enzyme [EC 3.4.16.4], also known as serine-type D-alanyl-D-alanine carboxypeptidase, catalyzes the hydrolysis of D-alanyl-D-alanine to yield two D-alanine. This enzyme comprises a group of membrane-bound, bacterial enzymes of the peptidase family S11. They are distinct from the zinc D-alanyl-D-alanine carboxypeptidase [EC 3.4.17.14]. The enzyme also hydrolyzes the D-alanyl-D-alanine peptide bond in the polypeptide of the cell wall. In addition, the enzyme will also catalyze the transpeptidation of peptidyl-alanyl moieties that are N-acetyl-substituents of D-alanine. The protein is inhibited by β -lactam antibiotics, which acylate the active-site seryl residue.

- M. Jamin, J.-M. Wilkin & J.-M. Frère (1995) *Essays in Biochem.* **29**, 1.
 B. Granier, M. Jamin, M. Adam, M. Galleni, B. Lakaye, W. Zorzi, J. Grandchamps, J.-m. Wilkin, C. Fraipont, B. Joris, C. Duez, M. Nguyen-Distèche, J. Coyette, M. Leyh-Bouille, J. Dusart, L. Christiaens, J.-M. Frère & J.-M. Guysen (1994) *Meth. Enzymol.* **244**, 249.

β -ALANYL-CoA AMMONIA-LYASE

This enzyme [EC 4.3.1.6] catalyzes the conversion of β -alanyl-CoA to acrylyl-CoA and ammonia.

- P. R. Vagelos (1962) *Meth. Enzymol.* **5**, 587.

O-ALANYLPHOSPHATIDYLGlycerol SYNTHASE

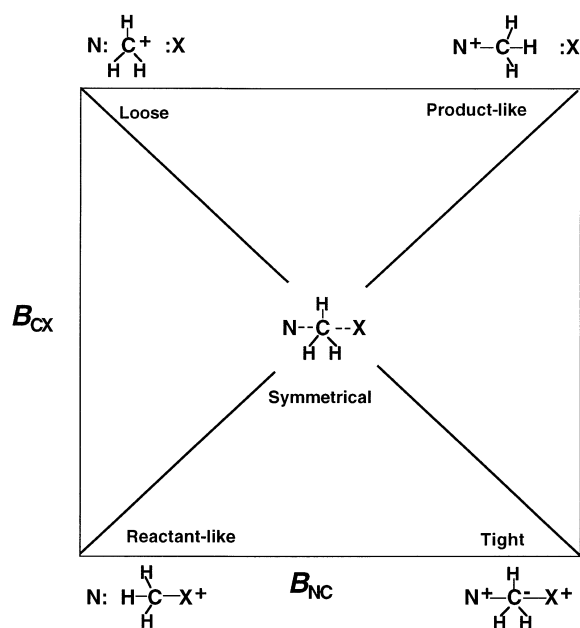
This enzyme [EC 2.3.2.11], also referred to as alanyl-tRNA:phosphatidylglycerol alanyltransferase, catalyzes the reaction of alanyl-tRNA and phosphatidylglycerol to generate tRNA and O³-alanyl-O¹-phosphatidylglycerol.

- R. L. Soffer (1974) *Adv. Enzymol.* **40**, 91.

ALBERY DIAGRAM

A graphical method for assessing the most likely transition-state geometry of a particular reaction. Albery¹ employed this diagram to evaluate the contributions of different components of a reaction to the formation of the transition state¹.

Consider the case of a simple S_N2 reaction in which the bond order for the breaking of the C—X bond in the reactant (where X is the leaving group), B_{CX} , varies from 1.0 to 0.0 and the bond order for the formation of the N—C bond (where N is the nucleophile), B_{NC} , varies from 0.0 to 1.0. The Albery diagram is a rectangular map (in this case, plotting B_{NC} on the horizontal axis, varying from 0.0 to 1.0, and B_{CX} on the vertical axis, going from



1.0 to 0.0) with the unreacted reactants in the lower left-hand corner (where $B_{CX} = 1.0$ and $B_{NC} = 0.0$) and the products in the upper right-hand corner ($B_{CX} = 0.0$ and $B_{NC} = 1.0$). The symmetric transition state lies at the center of the diagram. The lower right-hand corner represents tight transition-state structure in which the entering and leaving groups both have a large bond order associated with the reaction center. The upper left-hand corner represents a loose transition-state structure in which neither moiety exhibits any significant bonding interaction with the reaction center.

A perfectly symmetrical reaction would be represented by a diagonal running from the lower left-hand corner to the upper right-hand corner. The “looseness” and “tightness” of the leaving and entering group are represented by the other diagonal. Note that any transition state for a particular reaction can be represented by a point on this diagram.

The reader should note such a diagram is neither a free-energy diagram nor a potential-energy surface.

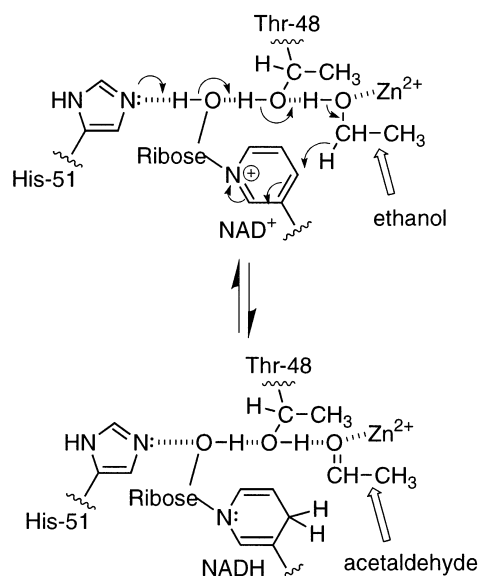
¹W. J. Albery (1993) *Adv. Phys. Org. Chem.* **29**, 139.

ALBUMIN

The abundance and ease of purification made bovine serum albumin (BSA) an early standard in protein chemistry, and BSA is widely used as protein standard in biuret, Lowry, and Bradford assays as well as a molecular weight

marker. BSA is also a nutrient in cell culture, an aid in antibody production, and a component that is frequently used for protein and enzyme stabilization. When added to a enzyme reaction mixture at about 1-2 mg/mL, BSA often stabilizes enzymes. BSA is also added to counter the effects of enzyme dilution and to prevent inactivation of the enzyme due to adsorption onto the walls of a reaction vessel. BSA binds a wide variety of substances, including substrates, cofactors, or other components of the reaction mixture. In such cases, BSA may inhibit enzyme activity, and one should exercise care in its use.

ALCOHOL DEHYDROGENASE



Hydride Transfer in the Alcohol Dehydrogenase Reaction

This zinc metalloenzyme [EC 1.1.1.1 and EC 1.1.1.2] catalyzes the reversible oxidation of a broad spectrum of alcohol substrates and reduction of aldehyde substrates, usually with NAD^+ as a coenzyme. The yeast and horse liver enzymes are probably the most extensively characterized oxidoreductases with respect to the reaction mechanism. Only one of two zinc ions is catalytically important, and the general mechanistic properties of the yeast and liver enzymes are similar, but not identical. Alcohol dehydrogenase can be regarded as a model enzyme system for the exploration of hydrogen kinetic isotope effects.

B. V. Plapp (1995) *Meth. Enzymol.* **249**, 91.

M. W. W. Adams & A. Kletzin (1996) *Adv. Protein Chem.* **48**, 101.

B. J. Bahnson & J. P. Klinman (1995) *Meth. Enzymol.* **249**, 373.

- W. W. Cleland (1995) *Meth. Enzymol.* **249**, 341.
T. A. Hansen & C. M. H. Hensgens (1994) *Meth. Enzymol.* **243**, 17.
N. J. Oppenheimer & A. L. Handlon (1992) *The Enzymes*, 3rd ed., **20**, 453.
C.-I. Brändén, H. Jörnvall, H. Eklund & B. Furugren (1975) *The Enzymes*, 3rd ed., **11**, 103.
K. Dalziel (1975) *The Enzymes*, 3rd ed., **11**, 1.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Abortive complex formation, **63**, 420; affinity chromatography, of NAD, **66**, 43-48; assay, of NAD, **66**, 41, 42, 80; binding, to agarose-bound nucleotides, **66**, 196-208; coupled assay, **63**, 34; cryoenzymology, **63**, 338; equilibrium perturbation, **64**, 120, 123; half-site reactivity, **64**, 184; inhibitor, **63**, 398, 408; isotope effect, **63**, 110; isotope exchange, **64**, 8, 28, 29; mechanism, **63**, 51; product inhibition, **63**, 436; reaction intermediates, stopped-flow absorbance studies, **61**, 319; substrate effect, **63**, 494; substrate inhibition, **63**, 507; affinity partitioning with dye ligands, **228**, 132-135; chromogenic substrates, **246**, 177-178, 189-190; cloning from bacteria, mutant complementation, **258**, 217-220; free sulfhydryl groups, determination by ESR, **251**, 97-98; horse liver ADH [active site, **249**, 94; isotope effects on oxidation of benzyl alcohol, **249**, 383-386; rapid scanning stopped-flow spectroscopy (alcohol oxidation, **246**, 184, 186-187, 189-190; aldehyde reduction, **246**, 184; cobalt substitution for zinc, **246**, 181-183; experimental design, **246**, 183-184; isobutyramide binding, **246**, 183; elucidation of mechanism, **246**, 191, 193; pyrazole inhibition, **246**, 183; structural assignment of intermediates, **246**, 190); reaction, hydrogen tunneling in, **249**, 390-393, 396-397; site-directed mutants (altered pH dependencies, **249**, 110-111; catalytic efficiency, **249**, 104-106; steady-state kinetic analysis, **249**, 101-104; transient kinetic analysis, **249**, 108-109); structure, **249**, 94]; human liver, site-directed mutants, acid-base catalysis, **249**, 117-118; liquid-liquid partition chromatography, **228**, 195, 197; mechanism, **246**, 178-179, 191, 193; metal ligands, **246**, 182-183; modification by symmetrical disulfide radical, **251**, 99; *Desulfovibrio gigas* NAD-dependent ADH, **243**, 17-18; NAD⁺ photoaffinity labeling, **237**, 72; secondary structure analysis, **246**, 514; catalytic efficiency, **249**, 104-107; transition state and multisubstrate analogs, **249**, 304; yeast ADH [reaction (hydrogen tunneling in, **249**, 383-390, 396-397; isotope effects, **249**, 383); site-directed mutants (altered pH dependencies, **249**, 110-111; catalytic efficiency, **249**, 104-106; steady-state kinetic analysis, **249**, 101-104.

ALCOHOL DEHYDROGENASE (QUINOLINE-DEPENDENT)

This enzyme [EC 1.1.99.8], also referred to as alcohol dehydrogenase (acceptor) and methanol dehydrogenase, catalyzes the oxidation-reduction reaction of a primary alcohol with an acceptor to generate an aldehyde and the reduced acceptor. The cofactor for this enzyme is pyrroloquinoline quinone (PQQ). A wide variety of primary alcohols can act as the substrate. **See also** *Alcohol Dehydrogenase*

- C. Anthony (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 155.
B. J. Bahnson & J. P. Klinman (1995) *Meth. Enzymol.* **249**, 373.
B. W. Groen & J. A. Duine (1990) *Meth. Enzymol.* **188**, 33.

ALCOHOL OXIDASE

This enzyme [EC 1.1.3.13], also known as methanol oxidase and AOX, catalyzes the reaction of a primary alco-

hol with dioxygen to generate an aldehyde and hydrogen peroxide. The enzyme utilizes FAD as the coenzyme and lower primary alcohols and unsaturated alcohols as the substrate. However, branched-chain and secondary alcohols are not acted upon by this enzyme.

- I. J. van der Klei, L. V. Bystrikh & W. Harder (1990) *Meth. Enzymol.* **188**, 420.

ALCOHOL SULFOTRANSFERASE

This enzyme [EC 2.8.2.2], also referred to as hydroxysteroid sulfotransferase, catalyzes the reaction of 3'-phosphoadenylylsulfate with an alcohol to produce adenosine 3',5'-bisphosphate and an alkyl sulfate. The alcohols that can act as substrates include aliphatic alcohols, ascorbate, chloramphenicol, ephedrine, hydroxysteroids, and other primary and secondary alcohols. However, phenolic steroids will not serve as substrates (such alcohols can be acted upon by steroid sulfotransferases).

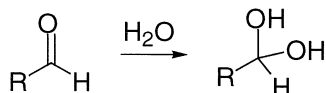
- J. D. Gregory (1962) *Meth. Enzymol.* **5**, 977.

ALDEHYDE DEHYDROGENASE

A number of enzymes have been referred to as aldehyde dehydrogenase. Aldehyde dehydrogenase (NAD⁺) [EC 1.2.1.3] catalyzes the reaction of an aldehyde with NAD⁺ and water to form a carboxylic acid and NADH. The enzyme is capable of acting on a wide variety of aldehydes. It will also oxidize D-glucuronolactone to form D-glucarate. Aldehyde dehydrogenase (NADP⁺) [EC 1.2.1.4] will also catalyze the oxidation of an aldehyde to an acid except in this case it uses NADP⁺ as the coenzyme. Aldehyde dehydrogenase (NAD(P)⁺) [EC 1.2.1.5] can use either NAD⁺ or NADP⁺ as the coenzyme. Aldehyde dehydrogenase (acceptor) [EC 1.2.99.3], better known as aldehyde dehydrogenase (pyrroloquinoline-quinone), catalyzes the oxidation of an aldehyde to an acid using PQQ. This enzyme can utilize a wide variety of aldehydes as substrates including straight-chain aldehydes containing as many as ten carbon atoms, aromatic aldehydes, glyoxylate, and glyceraldehyde.

- N. J. Oppenheimer & A. L. Handlon (1992) *The Enzymes*, 3rd ed., **20**, 453.
J. C. Murrell & W. Ashraf (1990) *Meth. Enzymol.* **188**, 26.
M. M. Attwood (1990) *Meth. Enzymol.* **188**, 314.
T.-K. Li (1977) *Adv. Enzymol.* **45**, 427.

ALDEHYDE HYDRATION



In aqueous solutions, aldehydes [$\text{RHC}=\text{O}$] undergo general acid-catalyzed addition of water to yield the hydrate [$\text{RHC}(\text{OH})_2$], and the equilibrium position lies in favor of the hydrate. Jencks¹ summarized the most likely mechanism for the hydration reaction.

¹W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, p. 211, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York.]

ALDEHYDE OXIDASE

This enzyme [EC 1.2.3.1] catalyzes the reaction of an aldehyde with water and dioxygen to produce a carboxylic acid and hydrogen peroxide. The enzyme uses both heme and molybdenum as cofactors. In addition, the enzyme can also catalyze the oxidation of quinoline and pyridine derivatives. In some systems this enzyme may be identical with xanthine oxidase.

P. E. Baugh, D. Collison, C. D. Garner & J. A. Joule (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 377.
J. G. Ferry (1992) *Crit. Rev. Biochem. Mol. Biol.* **27**, 473.
R. C. Bray (1980) *Adv. Enzymol.* **51**, 107.
R. C. Bray (1975) *The Enzymes*, 3rd ed., **12**, 299.
G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

ALDEHYDE OXIDOREDUCTASE

A family of enzymes that catalyze the molybdenum- or tungsten-cofactor dependent conversion of aldehydes to carboxylates, often using ferredoxin. **See also** *Aldose Reductase*

C. Kisker, H. Schindelin & D. C. Rees (1997) *Ann. Rev. Biochem.* **66**, 233.
M. W. W. Adams & A. Kletzin (1996) *Adv. Protein Chem.* **48**, 101.
J. J. G. Moura & B. A. S. Barata (1994) *Meth. Enzymol.* **243**, 24.
J. G. Ferry (1992) *Crit. Rev. Biochem. Mol. Biol.* **27**, 473.

ALDOLASE

Fructose-1,6-bisphosphate aldolase [EC 4.1.2.13] catalyzes the reversible cleavage of D-fructose 1,6-bisphosphate to dihydroxyacetone phosphate and D-glyceraldehyde 3-phosphate. Interestingly, the $\Delta G^\ominus$ value for this reaction is +23.9 kJ/mol, and under standard conditions the reaction is highly endergonic. Although appropriate for gluconeogenesis, it might appear that glycolysis would be hindered by this value. Nevertheless, in the

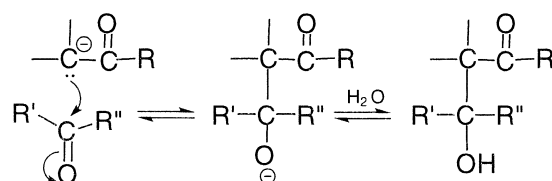
direction of glycolysis, under typical intracellular concentrations of reactants and products, the ΔG value is about -1.3 kJ/mol, and the cleavage reaction is thermodynamically favored.

Aldolases have been classified into two different groups: Class I aldolases are not inhibited by chelating agents such as EDTA and an intermediate can be trapped with borohydride treatment. This class of enzymes proceed via a mechanism involving covalent catalysis. Class II aldolases are inhibited by EDTA and a covalent intermediate with the enzyme is not formed. This class of enzymes, which require the presence of a metal ion such as Zn^{2+} or Fe^{2+} , proceed via a mechanism involving metal ion catalysis. Some organisms express both classes of aldolases. In Class II aldolases, a divalent cation (*e.g.*, Fe^{2+} or Zn^{2+}) stabilizes the enolate intermediate or forms a metal ion-sugar complex that acts as an electrophile. These aldolases occur in bacteria, fungi, and algae.

K. N. Allen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 135.
J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
J. A. Littlechild & H. C. Watson (1993) *Trends Biochem. Sci.* **18**, 36.
R. Kluger (1992) *The Enzymes*, 3rd ed., **20**, 271.
D. J. Hupe (1991) *A Study of Enzymes* **2**, 485.
D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.
B. L. Horecker, O. Tsolas & C. Y. Lai (1972) *The Enzymes*, 3rd ed., **7**, 213.

ALDOL CONDENSATION

The reversible reaction (usually base catalyzed) of two carbonyl compounds^{1,2} (*i.e.*, two aldehydes, two ketones, or one of each), in which at least one compound has an α -hydrogen, to produce a β -hydroxy carbonyl compound.



Originally, the term “aldol condensation” referred specifically to the reaction of an aldehyde (having an α -hydrogen) with an aldehyde/ketone to form a β -hydroxy aldehyde (the aldol). The reverse reaction is often referred to as a retrograde aldol reaction, a retro-aldol condensation (or reaction), or an aldol cleavage. March³ categorizes aldol condensations into five classes. The first is condensation between two identical aldehydes

(obviously, both aldehydes have to contain α -hydrogens). The reaction equilibrium lies to the right⁴. The second class is the condensation reaction between two identical ketones. In this case, the equilibrium is to the left. However, if the product is removed as the reaction progresses, then the synthesis can be successful. The third category is the condensation reaction between two different aldehydes, in which case a mixture of products will be obtained. If only one of the aldehydes has an α -hydrogen, then only two aldol products are possible. In many cases, the crossed product is the main product (the crossed aldol condensation is sometimes referred to as the Claisen-Schmidt reaction; see also category five). March's fourth category is the condensation between two different ketones, which is seldom attempted in the laboratory. The fifth and final category is the condensation between an aldehyde and a ketone. If there is no α -hydrogen on the aldehyde (e.g., an aromatic aldehyde), usually only a single product (the β -hydroxy ketone or the subsequent α,β -unsaturated ketone) is obtained since there is little competition with the ketone condensing with another ketone.

Beyond the classical glycolytic enzyme (**See Aldolase**), numerous enzymes catalyze aldol and aldol-like condensation/cleavage reactions. Among them are the following: (1) Sedoheptulose-1,7-bisphosphate aldolase, which converts sedoheptulose 1,7-bisphosphate to erythrose 4-phosphate and dihydroxyacetone phosphate. (2) Transaldolase, which catalyzes reactions with D-erythrose 4-phosphate and D-fructose 6-phosphate as substrates. As in the case of fructose-1,6-bisphosphate aldolase, this enzyme uses a ϵ -amino side-chain to form a Schiff base intermediate. In this case, however, the triose phosphate moiety is not released but is transferred to the other aldose (in this case, the aldotetrose). (3) Phospho-2-keto-3-deoxygluconate aldolase, an enzyme in the Entner-Doudoroff pathway, that catalyzes the cleavage 6-phospho-2-keto-3-deoxy-D-gluconate to form pyruvate and D-glyceraldehyde 3-phosphate. (4) 3-Deoxy-D-manno-octulosonate aldolase, which facilitates the reaction of pyruvate with D-arabinose to form 3-deoxy-D-manno-octulosonate, an important component of bacterial cell walls. (5) 2-Oxo-4-hydroxyglutarate aldolase, which converts 2-oxo-4-hydroxyglutarate to pyruvate and glyoxylate. (6) Deoxyribose-phosphate aldolase, which converts 2-deoxy-D-ribose 5-phosphate

to D-glyceraldehyde 3-phosphate and acetaldehyde. (7) δ -Aminolevulinate dehydratase (also called porphobilinogen synthase), which catalyzes the second step in protoporphyrin biosynthesis (two 5-aminolevulinates = porphobilinogen + 2H₂O). There are at least three distinct steps in this enzyme-catalyzed reaction: an aldol condensation, a dehydration, and a cyclic imine formation.

¹Z. G. Hajos (1979) *Carbon-Carbon Bond Formation*, vol. 1, p. 1, Marcel Dekker, New York.

²A. T. Nielsen and W.J. Houlihan (1968) *Org. React.* **16**, 1.

³J. March (1992) *Advanced Organic Chemistry*, 4th ed., p. 937, Wiley, New York.

ALDOSE 1-EPIMERASE

This enzyme [EC 5.1.3.3], also known as mutarotase, catalyzes the epimerization of the hemiacetal carbon atom of aldoses (thus, anomerization). Hence, α -D-glucose is reversibly converted to β -D-glucose. Other sugars can act as substrates (e.g., L-arabinose, D-xylose, D-galactose, maltose, and lactose).

M. E. Tanner (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 208.

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

ALDOSE REDUCTASE

This enzyme [EC 1.1.1.21], also known as aldehyde reductase and polyol dehydrogenase (NADP⁺), catalyzes the reaction of an alditol with NAD(P)⁺ to generate an aldose and NAD(P)H. The enzyme exhibits a broad specificity for the alditol.

J. G. Ferry (1992) *Crit. Rev. Biochem. Mol. Biol.* **27**, 473.

ALGINATE LYASE

This enzyme [EC 4.2.2.3], also referred to as poly(β -D-mannuronate) lyase and poly(mana) alginate lyase, catalyzes the eliminative cleavage of polysaccharides containing β -D-mannuronate residues to give oligosaccharides with 4-deoxy- α -L-erythro-hex-4-enopyranuronosyl groups at their ends.

J. Preiss (1966) *Meth. Enzymol.* **8**, 641.

ALKALINE PHOSPHATASE

This enzyme [EC 3.1.3.1], also known as alkaline phosphomonoesterase, phosphomonoesterase, and glycerophosphatase, catalyzes the hydrolysis of many orthophosphoric monoesters (the substrate specificity is quite wide) to generate an alcohol and orthophosphate. The

enzyme, which utilizes zinc and magnesium ions as cofactors, has a high pH optimum. In addition to this phosphatase activity, the enzyme can also catalyze certain transphosphorylations. In some systems, the enzyme can also act on diphosphate (pyrophosphate) (hence, a pyrophosphatase activity).

- A. C. Hengge (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 517.
 J. A. Gerlt (1992) *The Enzymes*, 3rd ed., **20**, 95.
 J. E. Coleman (1992) *Ann. Rev. Biophys. Biomol. Struct.* **21**, 441.
 B. L. Vallee & A. Galdes (1984) *Adv. Enzymol.* **56**, 283.
 H. W. Wyckoff, M. Handschumacher, K. Murthy & J. M. Sowadski (1983) *Adv. Enzymol.* **55**, 453.
 M. Cohn & G. H. Reed (1982) *Ann. Rev. Biochem.* **51**, 365.
 W. W. Cleland (1977) *Adv. Enzymol.* **45**, 373.
 T. W. Reid & I. B. Wilson (1971) *The Enzymes*, 3rd ed., **4**, 373.
 H. N. Fernley (1971) *The Enzymes*, 3rd ed., **4**, 417.

ALKANE HYDROXYLASE

This enzyme [EC 1.14.15.3], also known as alkane 1-monooxygenase, lauric acid ω -hydroxylase, ω -hydroxylase, and fatty acid ω -hydroxylase, catalyzes the reaction of octane with dioxygen and reduced rubredoxin to produce 1-octanol, water, and oxidized rubredoxin. The enzyme can also hydroxylate fatty acids in the ω -position. In some systems this enzyme is a heme-thiolated protein (P-450).

- B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

ALKENYLGLYCEROPHOSPHOCHOLINE HYDROLASE

This enzyme [EC 3.3.2.2], also known as lysoplasmalogenase, catalyzes the hydrolysis of 1-(1-alkenyl)-*sn*-glycero-3-phosphocholine to generate an aldehyde and *sn*-glycero-3-phosphocholine.

- M. S. Jurkowitz-Alexander & L. A. Horrocks (1991) *Meth. Enzymol.* **197**, 483.

ALKENYLGLYCEROPHOSPHOETHANOLAMINE HYDROLASE

This enzyme [EC 3.3.2.5], also known as lysoplasmalogenase, catalyzes the hydrolysis of 1-(1-alkenyl)-*sn*-glycero-3-phosphoethanolamine to yield an aldehyde and *sn*-glycero-3-phosphoethanolamine.

- M. S. Jurkowitz-Alexander & L. A. Horrocks (1991) *Meth. Enzymol.* **197**, 483.

1-ALKYL-2-ACETYLGLYCEROL CHOLINE PHOSPHOTRANSFERASE

This enzyme [EC 2.7.8.16] was previously thought to catalyze the reaction of CDP-choline with a 1-alkyl-2-acetyl-*sn*-glycerol to generate CMP and 1-alkyl-2-acetyl-*sn*-glycero-3-phosphocholine. It is now deleted as an EC entry.

- T.-c. Lee & F. Snyder (1992) *Meth. Enzymol.* **209**, 279.

1-ALKYL-2-ACETYLGLYCEROPHOSPHOCHOLINE ESTERASE

This enzyme [EC 3.1.1.47], also known as platelet-activating factor acetylhydrolase, 2-acetyl-1-alkyl-glycerophosphocholine esterase, and LDL-associated phospholipase A₂, catalyzes the hydrolysis of 2-acetyl-1-alkyl-*sn*-glycero-3-phosphocholine to yield 1-alkyl-*sn*-glycero-3-phosphocholine and acetate.

- D. M. Stafforini, T. M. McIntyre, G. A. Zimmerman & S. M. Prescott (1997) *J. Biol. Chem.* **272**, 17895.
 D. M. Stafforini, S. M. Prescott & T. M. McIntyre (1991) *Meth. Enzymol.* **197**, 411.

S-ALKYLCYSTEINE LYASE

This enzyme [EC 4.4.1.6] catalyzes the pyridoxal-phosphate-dependent hydrolysis of an *S*-alkyl-L-cysteine to generate an alkyl thiol, ammonia, and pyruvate. The reaction is an α,β -elimination. In yeast, the enzyme may be identical to cystathionine β -lyase. **See also** *Alliin Lyase*

- Y. Nishizuka (1971) *Meth. Enzymol.* **17B**, 470.

ALKYLDIHYDROXYACETONE PHOSPHATE SYNTHASE

This enzyme [EC 2.5.1.26], also known as alkylglycerone-phosphate synthase, catalyzes the reaction of 1-acylglycerone 3-phosphate with a long-chain alcohol to produce 1-alkylglycerone 3-phosphate and a long-chain acid anion. In this reaction, the ester-linked fatty acid of the substrate is removed and replaced with a long-chain alcohol in an ether linkage.

- A. Brown & F. Snyder (1992) *Meth. Enzymol.* **209**, 377.

ALKYLGLYCEROL KINASE

This enzyme [EC 2.7.1.93] catalyzes the reaction of ATP with 1-*O*-alkyl-*sn*-glycerol to generate ADP and 1-*O*-alkyl-*sn*-glycerol 3-phosphate.

- F. Snyder (1992) *Meth. Enzymol.* **209**, 211.

1-ALKYLGLYCEROPHOSPHOCHOLINE O-ACYLTRANSFERASE

This enzyme [EC 2.3.1.63] catalyzes the reaction of an acyl-CoA and 1-alkyl-*sn*-glycero-3-phosphocholine to yield coenzyme A and 1-alkyl-2-acyl-*sn*-glycero-3-phosphocholine. In some systems, this enzyme may be identical to 1-acylglycerophosphocholine acyltransferase [EC 2.3.1.23].

P. C. Choy & C. R. McMaster (1992) *Meth. Enzymol.* **209**, 86.

1-ALKYL-2-LYSOGLYCERO-3-PHOSPHOCHOLINE O-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.67], also referred to as 1-alkyl-glycerophosphocholine acetyltransferase, catalyzes the reaction of acetyl-CoA with 1-alkyl-*sn*-glycero-3-phosphocholine to produce coenzyme A and 1-alkyl-2-acetyl-*sn*-glycero-3-phosphocholine.

T.-c. Lee, D. S. Vallari & F. Snyder (1992) *Meth. Enzymol.* **209**, 396.

ALLANTOICASE

This enzyme [EC 3.5.3.4], also called allantoate amidinohydrolase, catalyzes the hydrolysis of allantoate to produce (–)-ureidoglycolate and urea. The enzyme can also catalyze the hydrolysis of (+)-ureidoglycolate to generate glyoxylate and urea.

N. E. Tolbert (1981) *Ann. Rev. Biochem.* **50**, 133.

ALLANTOINASE

This enzyme [EC 3.5.2.5] catalyzes the hydrolysis of allantoin to generate allantoate.

T. G. Cooper (1984) *Adv. Enzymol.* **56**, 91.

N. E. Tolbert (1981) *Ann. Rev. Biochem.* **50**, 133.

ALLENE-OXIDE CYCLASE

This enzyme [EC 5.3.99.6] catalyzes the conversion of (9*Z*)-(13*S*)-12,13-epoxyoctadeca-9,11,15-trienoate to (15*Z*)-12-oxophyto-10,15-dienoate. The allene oxides which are formed by the action of allene-oxide synthase [EC 4.2.1.92] are converted into cyclopentenone derivatives.

G. Sembdner & B. Parthier (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 569.

ALLENE-OXIDE SYNTHASE

This enzyme [EC 4.2.1.92], also called hydroperoxide dehydratase and hydroperoxide isomerase, catalyzes the

conversion of (9*Z*,11*E*,14*Z*)-(13*S*)-hydroperoxyoctadeca-(9,11,14)-trienoate to (9*Z*)-(13*S*)-12,13-epoxyoctadeca-9,11-dienoate and water. The enzyme uses a heme-thiolate cofactor and utilizes a number of unsaturated fatty-acid hydroperoxides as substrates, forming the corresponding allene oxides.

R. A. Creelman & J. E. Mullet (1997) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 355.

A. R. Brash & W. Song (1996) *Meth. Enzymol.* **272**, 250.

ALLIIN LYASE

This enzyme [EC 4.4.1.4], also known as alliinase and cysteine sulfoxide lyase, catalyzes the conversion of an *S*-alkyl-L-cysteine *S*-oxide to an alkyl sulfenate and 2-aminoacrylate. The enzyme requires pyridoxal phosphate.

A. E. Braunstein & E. V. Goryachenkova (1984) *Adv. Enzymol.* **56**, 1.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

ALLOPHANATE HYDROLASE

This enzyme [EC 3.5.1.54] catalyzes the hydrolysis of urea-1-carboxylate to yield 2 CO₂ and 2 ammonia. Interestingly, the enzyme isolated from yeast (but not the corresponding enzyme from green algae) will also catalyze the reaction of urea with ATP and CO₂ to produce ADP, orthophosphate, 2 ammonia, and 2 CO₂.

T. G. Cooper (1984) *Adv. Enzymol.* **56**, 91.

ALLOSE KINASE

This enzyme [EC 2.7.1.55], also referred to as allokinase, catalyzes the reaction of ATP with D-allose to yield ADP and D-allose 6-phosphate.

F. J. Simpson & L. N. Gibbins (1966) *Meth. Enzymol.* **9**, 412.

ALLOSTERY

Linked-function mechanisms for cooperative binding interaction of metabolites and/or drugs, based on the presence of two or more different conformational states of the protein or receptor. **See** *Adair Equation*; *Cooperative Ligand Binding*; *Hemoglobin*; *Hill Equation & Plot*; *Koshland-Némethy-Filmer Model*; *Monod-Wyman-Changeux Model*; *Negative Cooperativity*; *Positive Cooperativity*

ALLOTOPIC EFFECT

Interference in receptor-agonist interactions arising from the binding of an antagonist at a separate binding site.

While the antagonist site is topologically distinct from the agonist binding site, the two are linked by alterations in receptor conformation. The analogous behavior is commonly called allosterism for regulatory enzymes, and the prefix “allo” (Greek for “another”) emphasizes the involvement of spatially separate binding sites.

ALLYLIC SUBSTITUTION REACTION

A reaction of an allylic system ($\text{CH}_2=\text{CH}-\text{CH}_2-$) in which substitution occurs at the carbon adjacent to the double bond (e.g., at position one if the double bond is between positions 2 and 3). The incoming group may become attached to the same atom that contained the leaving group or it may become attached to the “relative position” 3, with the double bond shifting to the position between carbons one and two.

ALPHA (α)

1. First item in a series (for example, the first carbon linked to a carboxyl group). 2. Abbreviation for alpha particle. 3. Symbol for angle of optical rotation. 4. Symbol for degree of dissociation. 5. Symbol for electric polarizability of a molecule. 6. Often with a subscript number (*i*) the coefficient of $[\text{A}]^i$ in the numerator of the generalized rate expression. 7. Symbol for “is proportional to.” 8. Symbol for Napierian absorption coefficient. 9. In brackets, symbol for specific optical rotation.

ALPHA-ADDITION (α -ADDITION)

The reversal of α -elimination; an addition reaction between two or more reacting chemical species in which two new chemical bonds are formed on the same atom.

ALPHA CLEAVAGE (α -CLEAVAGE)

A homolytic cleavage of a bond associated with an atom or group that is bonded to a specified group. An example of an alpha-cleavage would be a Norrish type I cleavage.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

ALPHA EFFECT

Enhanced nucleophilicity of an attacking nucleophile attributable to one or more unshared pairs of electrons on an atom that lies immediately adjacent to the nucleophilic group¹⁻⁴. The increased nucleophilicity of hydrazines and hydroxylamines is due to an alpha effect, as is the increased nucleophilicity of HO_2^- compared to HO^- in solution. This effect on nucleophile reactivity can

be significant for substitution reactions at a carbonyl group, at an unsaturated carbon, or in reactions of a nucleophile with a carbocation. The effect is largest when the transition state displays considerable bond formation. Substitution reactions at a saturated carbon generally tend to exhibit little or no alpha effect.

As is also true for ambident anions, substances exhibiting alpha effects in their reactions consistently deviate from the anticipated structure-reactivity correlations known for simple nucleophiles.

One explanation for the alpha effect is ground-state destabilization: Repulsive electronic interactions between the alpha atom's lone-pair and the nucleophile occur in the ground-state, and such destabilization is expected to be relieved as a covalent bond is forming in the transition-state of a nucleophilic substitution reaction. Reduced solvation in molecules exhibiting the alpha effect may also play a role in the increased nucleophilicity⁵; for example, OH_2^- displays no effect in the gas phase, but a substantial effect is observed in solution⁶. Another factor may be the ability of the alpha lone-pair to stabilize any partially positive group formed in the transition state.

¹A. P. Grekov & V. Y. Veselov (1978) *Russ. Chem. Rev.* **47**, 631.

²N. J. Fina & J. O. Edwards (1973) *Int. J. Chem. Kinet.* **5**, 1.

³S. Wolfe, D. J. Mitchell, H. B. Schlegel, C. Minot & O. Eisenstein (1982) *Tet. Lett.* **23**, 615.

⁴S. Hoz & E. Buncl (1985) *Isr. J. Chem.* **26**, 313.

⁵D. Herschlag & W. P. Jencks (1990) *J. Amer. Chem. Soc.* **112**, 1951.

⁶C. H. DePuy, E. W. Della, J. Filley, J. J. Grabowski & V. M. Bierbaum (1983) *J. Amer. Chem. Soc.* **105**, 2481.

ALPHA-ELIMINATION (α -ELIMINATION)

A reaction involving the loss or elimination of two substituents attached to a common atom (usually carbon). α -Elimination is often observed in the synthesis of certain carbenes and nitrenes. **See also** Carbene

ALPHA EXPULSION

Any reaction characterized by the expulsion or loss of a group covalently bound to the carbon atom lying immediately adjacent (hence the designation “alpha”) to a chromophore. After photoexcitation, the group that is expelled either contains an odd number of electrons or is an anionic species.

ALPHA PARTICLE

A decay product (corresponding to the nucleus of a helium atom) formed in certain nuclear disintegration reactions. These are the least energetic particles formed during radioactive decay, and their motion can be stopped by a single thickness of paper.

ALTERNATIVE HYPOTHESIS

A statistical term for a hypothesis that a particular set of data is at odds with a given null hypothesis. **See** *Statistics (A Primer)*

ALTERNATIVE PRODUCT INHIBITION

Rate experiments that are typically carried out in the presence of different concentrations of an alternative product (or product analog) while using the normal substrates^{1,2}. This approach can be particularly useful when the normal product cannot be used because it is unstable, insoluble, or ineffective (the latter indicated by a very high K_i value). Moreover, the normal product may be consumed as an essential substrate in a coupled assay system² for the primary enzyme. Fromm and Zewe used the alternative product inhibition approach in their study of hexokinase³. Wratten and Cleland later applied this procedure to exclude the Theorell-Chance mechanism for liver alcohol dehydrogenase⁴. **See** *Abortive Complexes*

¹H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

²F. B. Rudolph (1979) *Meth. Enzymol.* **63**, 411.

³H. J. Fromm & V. Zewe (1962) *J. Biol. Chem.* **237**, 3027.

⁴C. C. Wratten & W. W. Cleland (1965) *Biochemistry* **4**, 2442.

ALTERNATIVE SUBSTRATES

Any compounds (other than the physiologically relevant substrate) that can serve as substrates for a particular enzyme. Alternative substrates compete with the natural substrate and with each other for access to the enzyme's active site. Thus, if one is utilizing an assay that measures the production of the true substrate, then the presence of the alternative substrate will result in competitive inhibition relative to the true substrate.

GENERAL PROPERTIES. Because enzymes rarely exhibit absolute substrate specificity, their use of alternative substrates affords the opportunity to investigate the

nature of enzymic catalysis under conditions where the reaction's kinetic and thermodynamic parameter may reveal more about rate-limiting steps or other mechanistic aspects. An excellent example of an enzyme with many alternative substrates is bovine brain hexokinase (Table I).

Table I

Substrate Specificity of Bovine Brain Hexokinase

Substrate	Michaelis Constant	Relative Rate
Glucose	0.05 mM	1.0
Fructose	1.6 mM	1.5
Mannose	0.07 mM	0.4
2-Deoxyglucose	0.27 mM	1.0
1,5-Anhydro-D-glucitol ^a	10 mM	1.0
Glucosamine	0.8 mM	0.6
Galactose	100 mM	0.02

^aNote: This cyclic ether analogue of glucose is conformationally analogous to the β -D-glucopyranose, except it lacks the C-1 hydroxyl group.

MULTISUBSTRATE SYSTEMS. Wong and Hanes^{1,2} were probably among the first to suggest that alternative substrates may be useful in mechanistic studies. Fromm's laboratory was the first to use and extend the theory of alternative substrate inhibition to address specific questions about multisubstrate enzyme kinetic mechanisms³⁻⁵. Huang demonstrated the advantages of a constant ratio approach⁶⁻⁸ when dealing with alternative substrate kinetics.

There are three basic methods for carrying out alternative substrate inhibition studies. In the first, the investigator seeks to observe numerical changes in the coefficients of the double-reciprocal form of the enzyme rate expression in the presence and absence of the alternative substrate. For some mechanisms, only certain coefficients will be altered². This method requires extremely accurate estimates of the magnitudes of the coefficients and should always be supplemented with other kinetic probes⁶.

In the second and most commonly used method, the investigator studies the alterations in the patterns of the initial rate data (usually graphically presented as double-reciprocal plots). In these studies of multisubstrate and multiproduct enzyme-catalyzed reactions, an investigator can measure the rate of the reaction by either by observing any increase in the concentration of a common

product (*i.e.*, a product formed by both the normal substrate and the alternative substrate) or by detecting an increase in the concentration of the product not formed from the alternative substrate. Each procedure results in its own characteristic set of rate expressions and will produce different patterns in the double-reciprocal plots ($1/v$ vs. $1/[A]$ at different, constant concentrations of the alternative substrate). The common-product approach, although very useful, will often produce nonlinearity in the double-reciprocal plots^{4,6,9}. Huang has suggested that a constant-ratio approach, in which the substrate and the alternative substrate are varied in a constant ratio, can further assist an investigator in distinguishing between mechanisms⁶⁻⁸.

Whenever using alternative substrate inhibition procedures, the investigator must demonstrate that initial rate conditions remain valid throughout the course of the experiment. This is particularly true of the other substrate(s) in multisubstrate enzymes. Because both the substrate under study and its analog are present in the reaction mixtures, the other cosubstrates will be depleted faster. This should always be a consideration in the design of the experiment.

ENERGETICS. While isozymes are apt to have different energies of activation, even under the same assay conditions, an enzyme acting on different substrates can in some circumstances exhibit the same energy of activation^{10,11}. Yeast sucrase, for example, has an energy of activation of $46 \text{ kJ} \cdot \text{mol}^{-1}$ (or $11.0 \text{ kcal} \cdot \text{mol}^{-1}$) for both sucrose and raffinose¹¹⁻¹³. The rate-determining step in the enzyme-catalyzed reaction may differ with alternative substrates, and this may be reflected in the observed energy of activation. Likewise, if the rate-determining step changes with protein modification, Arrhenius plots should reflect those changes. An example is the myosin ATPase which exhibits biphasicity in the Arrhenius plot with ITP as a substrate, but a typical linear Arrhenius plot with ATP as the substrate. Levy, Sharon, and Koshland¹⁴ suggested that this may be the result of the 6-amino group on ATP interacting with some functional moiety on the protein, thereby producing an enzyme-substrate complex insensitive to temperature change. **See** *Competitive Inhibitor; Abortive Complexes; Map-*

ping Substrate Interactions Using Substrate Data; Membrane Transport; Energy of Activation; Q_{10} ; Arrhenius Equation; van't Hoff Relationship

- ¹J. T. Wong & C. S. Hanes (1962) *Can. J. Biochem. Physiol.* **40**, 763.
- ²J. T. F. Wong (1975) *Kinetics of Enzyme Mechanisms*, Academic Press, New York.
- ³V. Zewe, H. J. Fromm & R. Fabino (1964) *J. Biol. Chem.* **239**, 1625.
- ⁴H. J. Fromm (1964) *Biochim. Biophys. Acta* **81**, 413.
- ⁵F. B. Rudolph & H. J. Fromm (1971) *Arch. Biochem. Biophys.* **147**, 515.
- ⁶C. Y. Huang (1979) *Meth. Enzymol.* **63**, 486.
- ⁷C. Y. Huang & S. Kaufman *J. Biol. Chem.* **248**, 4242.
- ⁸C. Y. Huang (1977) *Arch. Biochem. Biophys.* **184**, 488.
- ⁹I. G. Darvey (1976) *Mol. Cell. Biochem.* **11**, 3.
- ¹⁰M. Dixon & E. C. Webb, *Enzymes*, 3rd ed., Academic Press, New York (1979).
- ¹¹I. W. Sizer (1943) *Adv. Enzymol.* **3**, 35.
- ¹²I. W. Sizer (1938) *Enzymologia* **4**, 215.
- ¹³I. W. Sizer (1937) *J. Cellular and Comp. Physiol.* **10**, 61.
- ¹⁴H. M. Levy, N. Sharon & D. E. Koshland, Jr. (1959) *Biochim. Biophys. Acta* **33**, 288.

ALTRONATE DEHYDRATASE

This enzyme [EC 4.2.1.7] catalyzes the conversion of D-altronate to 2-dehydro-3-deoxy-D-gluconate and water.

- J. Robert-Baudouy, J. Jimeno-Abendano & F. Stoeber (1982) *Meth. Enzymol.* **90**, 288.
 W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

AMADORI REARRANGEMENT

A reaction in which *N*-glycosides of aldoses are converted into *N*-glycosides of the corresponding ketoses *via* acid or base catalysis. In tryptophan biosynthesis, the conversion of *N*-(5'-phosphoribosyl)anthranilate to enol-1-*o*-carboxyphenylamino-1-deoxyribulose phosphate involves an Amadori rearrangement. Similarly, in the biosynthesis of folic acid, riboflavin, and the dimethylbenzimidazole group of vitamin B₁₂, an Amadori rearrangement occurs at an early step. In histidine biosynthesis, the rearrangement occurs in the interconversion of *N*¹-5'-phosphoribosylformimino-5-aminoimidazole-4-carboxamide ribonucleotide to *N*¹-5'-phosphoribulosylformimino-5-aminoimidazole-4-carboxamide ribonucleotide. Amadori rearrangements can also occur during the glycation of proteins.

- ¹M. Amadori (1925) *Atti. Accad. Nazl. Lincei* **2**(6), 337.
- ²M. Amadori (1929) *Atti. Accad. Nazl. Lincei* **9**(6), 68 and 226.
- ³G. Hodge (1955) *Adv. Carbohydr. Chem.* **10**, 169.
- ⁴R. U. Lemieux (1964) in *Molecular Rearrangements* (P. de Mayo, ed.), p. 753, Wiley, New York.
- ⁵J. E. Hodge & B. E. Fisher (1963) *Methods in Carbohydr. Chem.* **2**, 99.

AMBIDENT NUCLEOPHILE

A term describing a chemical species containing two distinct, yet strongly interacting, reactive centers at which a covalent bond can be formed. These centers are positioned so that a reaction at one center either stops (or greatly retards) a reaction at the other center. The more basic group in the ambident anion preferentially reacts with the polarizable group under nucleophilic attack. Ambident anions consistently deviate from anticipated structure-activity correlations known for substances containing only a single reactive group.

Examples of such molecules include conjugated nucleophiles such as the enolate anion. Such nucleophiles have potentially two attacking atoms (in the case of the enolate anion, the oxygen or the α -carbon); reaction conditions affect which will be the more prevalent species. Other examples include the cyanide (CN^-) and the nitrite (NO_2^-) ions.

Molecules having two reactive centers that are noninteracting or only weakly interacting, such as the carboxylate anions in malonate or succinate, are not considered to be ambident. The term "bifunctional" should be used to describe these compounds. If a molecule contains more than two interacting centers, the terms "polydent" or "multident" should be used. Examples of such nucleophiles include anions derived from malonic esters, β -keto esters, β -diketones, as well as phenoxide ions.

ω -AMIDASE

This enzyme [EC 3.5.1.3] catalyzes the hydrolysis of a monoamide of a dicarboxylic acid to generate a dicarboxylate and ammonia. The enzyme can utilize as a substrate glutaramate, succinamate, and their corresponding 2-keto derivatives.

A. J. L. Cooper, T. E. Duffy & A. Meister (1985) *Meth. Enzymol.* **113**, 350.

A. J. L. Cooper & A. Meister (1976) *Crit. Rev. Biochem.* **4**, 281.

AMIDOPHOSPHORIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.14], also known as glutamine phosphoribosyl-pyrophosphate amidotransferase, catalyzes the reaction of glutamine with 5-phospho- α -D-ribose 1-diphosphate and water to produce 5-phospho- β -D-ribosylamine, diphosphate (or, pyrophosphate), and glutamate.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.

H. Zalkin (1985) *Meth. Enzymol.* **113**, 264.

W. D. L. Musick (1981) *Crit. Rev. Biochem.* **11**, 1.

W. N. Keley & J. B. Wyngaarden (1974) *Adv. Enzymol.* **41**, 1.

D. E. Koshland, Jr. & A. Levitzki (1974) *The Enzymes*, 3rd ed. **10**, 539.

AMINE OXIDASES

This group of enzymes catalyzes the oxidation of amines. Amine oxidase [EC 1.4.3.4], a flavin-containing enzyme (also known as monoamine oxidase, tyramine oxidase, tyraminase, or adrenalin oxidase) catalyzes the reaction of an organic amine (*i.e.*, $\text{R}-\text{CH}_2-\text{NH}_2$) with dioxygen and water to yield an aldehyde, ammonia, and hydrogen peroxide. The enzyme utilizes FAD as a cofactor and readily acts on primary amines. Secondary and tertiary amines can also serve as substrates.

Amine oxidase [EC 1.4.3.6] (also known as diamine oxidase, diamino oxyhydrase, and histaminase) catalyzes the reaction of an organic amine (*i.e.*, $\text{R}-\text{CH}_2-\text{NH}_2$) with dioxygen and water to yield an aldehyde, ammonia, and hydrogen peroxide. Cofactors are a copper ion and 6-hydroxy-DOPA.

C. Anthony (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 155.

A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.

C. Hartmann & W. S. McIntire (1997) *Meth. Enzymol.* **280**, 98.

J. P. Klinman (1996) *J. Biol. Chem.* **271**, 27189.

J. M. Janes & J. P. Klinman (1995) *Meth. Enzymol.* **258**, 20.

B. J. Bahnson & J. P. Klinman (1995) *Meth. Enzymol.* **249**, 373.

E. J. Brush & J. W. Kozarich (1992) *The Enzymes*, 3rd ed., **20**, 317.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

J. A. Duine, J. Frank & J. A. Jongejan (1987) *Adv. Enzymol.* **59**, 169.

V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

B. G. Malmström, L.-E. Andréasson & B. Reinhammer (1975) *The Enzymes*, 3rd ed., **12**, 507.

AMINE SULFOTRANSFERASE

This enzyme [EC 2.8.2.3], also known as arylamine sulfotransferase, catalyzes the reaction of 3'-phosphoadenylylsulfate with an arylamine to produce adenosine 3',5'-bisphosphate and an arylsulfamate. Primary and secondary amines which can act as substrates include aniline, 2-naphthylamine, cyclohexylamine, and octylamine.

S. G. Ramaswamy & W. B. Jakoby (1987) *Meth. Enzymol.* **143**, 201.

D-AMINO ACID AMINOTRANSFERASE

This enzyme catalyzes the reaction of a D-amino acid and an α -keto acid to produce a new D-amino acid and a new α -keto acid. This enzyme activity is often catalyzed by D-alanine aminotransferase [EC 2.6.1.21].

See D-Alanine Aminotransferase

R. A. John (1998) in *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 173.

H. Hayashi, H. Wada, T. Yoshimura, N. Esaki & K. Soda (1990) *Ann. Rev. Biochem.* **59**, 87.

C. T. Walsh (1984) *Ann. Rev. Biochem.* **53**, 493.

D-AMINO-ACID DEHYDROGENASE

This enzyme [EC 1.4.99.1] catalyzes the oxidation-reduction reaction of a D-amino acid with an acceptor and water to generate an α -keto acid, ammonia, and the reduced acceptor. The enzyme utilizes FAD as a cofactor. Most D-amino acids, with the noted exceptions of D-aspartate and D-glutamate, can be used as substrates.

K. Tsukada (1971) *Meth. Enzymol.* **17B**, 623.

L-AMINO-ACID DEHYDROGENASE

This enzyme [EC 1.4.1.5] catalyzes the oxidation-reduction reaction of an L-amino acid with NAD^+ and water to produce an α -keto acid, ammonia, and NADH. The amino acids used as substrates are the aliphatic amino acids.

N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

D-AMINO ACID OXIDASE

This FAD-dependent enzyme [EC 1.4.3.3] catalyzes the reaction of a D-amino acid with dioxygen and water to generate an α -keto acid, ammonia, and hydrogen peroxide. Substrate specificity is wide and includes glycine.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

K. Yagi (1991) *A Study of Enzymes* **2**, 271.

G. A. Hamilton (1985) *Adv. Enzymol.* **57**, 85.

H. J. Bright & D. J. T. Porter (1975) *The Enzymes*, 3rd ed., **12**, 421.

L-AMINO-ACID OXIDASE

This FAD-dependent enzyme [EC 1.4.3.2] catalyzes the reaction of an L-amino acid with dioxygen and water to generate an α -keto acid, ammonia, and hydrogen peroxide.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

H. J. Bright & D. J. T. Porter (1975) *The Enzymes*, 3rd ed., **12**, 421.

 ω -AMINO ACID : PYRUVATE AMINOTRANSFERASE

This enzyme catalyzes the reaction of an ω -amino acid with pyruvate to produce L-alanine and an aldehyde-containing carboxylic acid. **See** Lysine 6-Aminotransferase; Ornithine Aminotransferase

K. Yonaha, S. Toyama & K. Soda (1987) *Meth. Enzymol.* **143**, 500.

AMINO ACID RACEMASE

This enzyme [EC 5.1.1.10] catalyzes the interconversion of L- and D-amino acids, using pyridoxal phosphate.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

E. Adams (1972) *The Enzymes*, 3rd ed., **6**, 479.

AMINO ACID TURNOVER KINETICS

Wolfe¹ has presented an excellent description of the systematic application of stable and radioactive isotope tracers in determining the kinetics of leucine metabolism and other amino acids in living systems.

¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, p. 357, Wiley-Liss, New York.

AMINOACYLASE

This enzyme [EC 3.5.1.14] (also referred to as histozyme, hippuricase, benzamidase, dehydropeptidase II, aminoacylase I, and acylase I) catalyzes the hydrolysis of an N-acyl-L-amino acid to yield a fatty acid anion and an L-amino acid. The enzyme has a wide specificity for the amino acid derivative. It will also catalyze the hydrolysis of dehydropeptides.

J. P. Greenstein (1955) *Meth. Enzymol.* **2**, 109.

AMINOACYL-tRNA HYDROLASE

This enzyme [EC 3.1.1.29], also called peptidyl-tRNA hydrolase, catalyzes the hydrolysis of an N-substituted aminoacyl-tRNA to produce an N-substituted amino acid and tRNA.

P. Yot, D. Paulin & F. Chapeville (1971) *Meth. Enzymol.* **20**, 194.

AMINOACYL-tRNA SYNTHETASES

These enzymes [EC 6.1.1.x] catalyze the reaction of a specific amino acid with the corresponding tRNA and ATP to generate AMP, diphosphate (or, pyrophosphate) and the aminoacyl-tRNA derivative.

E. A. First (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 573.

- D. C. H. Yang (1996) *Curr. Top. Cell. Reg.* **34**, 101.
C. W. Carter, Jr. (1993) *Ann. Rev. Biochem.* **62**, 715.
D. D. Buechter & P. Schimmel (1993) *Crit. Rev. Biochem. Mol. Biol.* **28**, 309.
M. Mirande (1991) *Prog. Nucleic Acid Res. Mol. Biol.* **40**, 95.
P. R. Schimmel (1979) *Adv. Enzymol.* **49**, 187.
L. L. Kisselev & O. O. Favorova (1974) *Adv. Enzymol.* **40**, 141.
D. Söll & P. R. Schimmel (1974) *The Enzymes*, 3rd ed., **10**, 489.

2-AMINOADIPATE AMINOTRANSFERASE

This enzyme [EC 2.6.1.39] catalyzes the reversible reaction of 2-aminoadipate with 2-oxoglutarate (or, α -ketoglutarate) to generate 2-oxoadipate and glutamate. The enzyme requires pyridoxal phosphate.

H. P. Broquist (1971) *Meth. Enzymol.* **17B**, 119.

2-AMINOADIPATE 6-SEMIALDEHYDE DEHYDROGENASE

This enzyme [EC 1.2.1.31], also known as α -aminoadipate reductase, catalyzes the reaction of L-2-aminoadipate 6-semialdehyde with NADP⁺ and water to generate L-2-aminoadipate and NADPH.

V. W. Rodwell (1971) *Meth. Enzymol.* **17(B)**, 188.

4-AMINO BUTYRATE AMINOTRANSFERASE

This enzyme [EC 2.6.1.19] catalyzes the reversible reaction of γ -aminobutyrate with α -ketoglutarate to yield succinate semialdehyde and glutamate. A number of enzyme preparations have been reported to also use β -alanine, 5-aminopentanoate, and (R,S)-3-amino-2-methylpropanoate as substrates.

- R. A. John (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 173.
R. B. Silverman (1995) *Meth. Enzymol.* **249**, 240.
M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **9**, 213.
A. J. L. Cooper (1985) *Meth. Enzymol.* **113**, 80.
A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

α -AMINO- ϵ -CAPROLACTAM RACEMASE

This enzyme [EC 5.1.1.15], also referred to as 2-amino-hexano-6-lactam racemase, catalyzes the reversible interconversion of the L- and D-stereoisomers of 2-amino-hexano-6-lactam. The enzyme, which utilizes pyridoxal phosphate, will also catalyze the interconversion of 2-aminopentano-5-lactam and 2-amino-3-mercaptopentano-6-lactam. The enzyme exhibits a minor aminotransferase activity with certain α -amino acids.

- M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.
H. Hayashi, H. Wada, T. Yoshimura, N. Esaki & K. Soda (1990) *Ann. Rev. Biochem.* **59**, 87.

1-AMINOCYCLOPROPANE-1-CARBOXYLATE SYNTHASE

This enzyme [EC 4.4.1.14] catalyzes the conversion of S-adenosyl-L-methionine to 1-aminocyclopropane-1-carboxylate methylthioadenosine via an α,γ -elimination. The required cofactor for this enzyme is pyridoxal phosphate.

- H. Kende (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 283.
D. O. Adams & S. F. Yang (1987) *Meth. Enzymol.* **143**, 426.

2-AMINO-4-HYDROXY-6-HYDROXY-METHYLDIHYDROPTERIDINE PYROPHOSPHOKINASE

This enzyme [EC 2.7.6.3], also known as 6-hydroxymethyl-7,8-dihydropterin pyrophosphokinase and 7,8-dihydro-6-hydroxymethylpterin pyrophosphokinase, catalyzes the reaction of ATP with 2-amino-4-hydroxy-6-hydroxymethyl-7,8-dihydropteridine to generate AMP and 2-amino-7,8-dihydro-4-hydroxy-6-(diphospho-oxy-methyl)pteridine.

R. L. Switzer (1974) *The Enzymes*, 3rd ed., **10**, 607.

β -AMINOISOBUTYRATE AMINOTRANSFERASE

This enzyme [EC 2.6.1.40], also known as (R)-3-amino-2-methylpropionate-pyruvate aminotransferase, catalyzes the reversible reaction of (R)-3-amino-2-methylpropanoate with pyruvate to generate 2-methyl-3-oxopropanoate and alanine.

O. W. Griffith (1986) *Ann. Rev. Biochem.* **55**, 855.

5-AMINOLEVULINATE AMINOTRANSFERASE

This pyridoxal phosphate-dependent enzyme [EC 2.6.1.43] catalyzes the reversible reaction of 5-aminolevulinate with pyruvate to produce 4,5-dioxopentanoate and alanine.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

5-AMINOLEVULINATE DEHYDRATASE

This enzyme [EC 4.2.1.24], also known as porphobilinogen synthase, catalyzes the reaction of two molecules of

5-aminolevulinate to generate porphobilinogen and two water. This enzyme is a zinc-dependent system.

K. N. Allen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 135.

P. M. Jordan & P. N. B. Gibbs (1985) *Biochem. J.* **227**, 1015.

B. L. Vallee & A. Galdes (1984) *Adv. Enzymol.* **56**, 283.

S. Granick & S. I. Beale (1978) *Adv. Enzymol.* **46**, 33.

D. Shemin (1972) *The Enzymes*, 3rd ed., **7**, 323.

5-AMINOLEVULINATE SYNTHASE

This enzyme [EC 2.3.1.37] catalyzes the reaction of succinyl-CoA with glycine to yield 5-aminolevulinate, coenzyme A, and carbon dioxide. Pyridoxal phosphate is used as a cofactor in this reaction. In mammals, the enzyme isolated from erythrocytes is genetically distinct from that in other tissues.

P. M. Shoolingin-Jordan, J. E. LeLean & A. J. Lloyd (1997) *Meth. Enzymol.* **281**, 309.

S. Granick & S. I. Beale (1978) *Adv. Enzymol.* **46**, 33.

P. M. Jordan & D. Shemin (1972) *The Enzymes*, 3rd ed., **7**, 339.

AMINOMALONATE DECARBOXYLASE

This enzymatic activity is now included under aspartate β -decarboxylase.

E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

AMINOPEPTIDASE

Enzymes with this designation [EC 3.4.11.x] are α -aminoacylpeptide hydrolases. **See also** *specific aminopeptidase*

R. J. DeLange & E. L. Smith (1971) *The Enzymes*, 3rd ed., **3**, 81.

AMINOPEPTIDASE P

This enzyme [EC 3.4.11.9] (also known as Xaa-Pro aminopeptidase, X-Pro aminopeptidase, proline aminopeptidase, and aminoacylproline aminopeptidase) catalyzes the hydrolysis of a peptide bond at the *N*-terminus of a peptide provided that the *N*-terminal amino acyl residue is linked to a prolyl residue by that peptide bond. The enzyme will also act on dipeptides and tripeptides with that same restriction. Either manganese or cobalt is needed as a cofactor. This enzyme appears to be a membrane-bound system in both mammalian and bacterial cells. The protein belongs to the peptidase family M24B.

A. Yaron & A. Berger (1970) *Meth. Enzymol.* **19**, 521.

AMOUNT-OF-SUBSTANCE

A chemical term (symbolized by *n*, often with a subscript denoting the specific type of substance) denoting the

quantity of atoms, molecules, ions, radicals, or electrons present. The SI unit is the mole. The amount-of-substance divided by the total volume is the concentration.

See *Concentration*

AMP DEAMINASE

This enzyme [EC 3.5.4.6], also known as AMP amino-hydrolase and adenylic acid deaminase, catalyzes the hydrolysis of AMP to yield IMP and ammonia.

C. L. Zielke & C. H. Suelter (1971) *The Enzymes*, 3rd ed., **4**, 47.

AMPHIPHILIC

Referring to a chemical species having two distinct domains with opposing solubility properties (*e.g.*, a charged domain and an apolar or hydrophobic domain). Such molecules often readily form micelles when present in dilute aqueous solutions.

AMPHIPROTIC

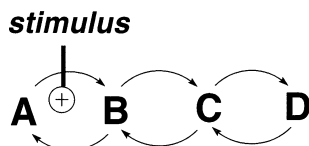
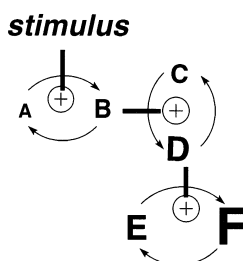
A term applied to solvents, such as water and ethanol, that are both protogenic (*i.e.*, release protons) and protophilic (*i.e.*, bind protons). **See** *Protogenic*

AMPHOTERIC

Referring to chemical species possessing the ability to exhibit either acidic or basic properties (*e.g.*, amino acids). This property actually depends on the medium. For example, sulfuric acid is an acid, when studied in water; however, it becomes amphoteric when studied in superacids.

AMPLIFICATION

The multiplicative enhancement of a signal, leading to an output of much greater magnitude through the relay action of one or more transducers. Concerned about how the dark-adapted eye senses single-photon events, Wald¹ was among the first to recognize that only certain cyclic reaction schemes display the capacity to amplify an initially weak signal. The diagram shown below illustrates (A) one-to-one stoichiometric processes that proceed without amplification and (B) a sequence of enzymatic activities arranged as cascades that produce robust amplification.

(A) Without amplification**(B) With amplification**

Hypothetical hierarchy of reactions: (A) those that fail to yield amplification and (B) those that can achieve any level of amplification, depending only on the number of cycles and the kinetic parameters of the active enzymes formed by successive conversions of proenzyme into active enzymes.

The classical example is blood clotting, where successive steps involving enzyme-catalyzed proteolysis converts an inactive (or weakly active) proenzyme into its highly active form. Although unknown at the time of Wald's classical report, kinase-type and nucleotidyltransferase-type reactions (**See Enzyme Cascade Kinetics**) are frequently the source of biological signal transduction and amplification.

¹G. Wald (1965) *Science* **150**, 1028.

AMPLITUDE

1. The maximum distance between the value of a periodic function (*i.e.*, a function with repeated values for $f(x)$ for all integer multiples of a constant displacement or increment along the independent variable axis) and the function's mean value. 2. A term used in classical mechanics to define the magnitude of the maximum displacement of a body experiencing an oscillatory motion. 3. A term used in relaxation kinetics to indicate the magnitude of displacement of a chemical reaction.

AMP NUCLEOSIDASE

This enzyme [EC 3.2.2.4] catalyzes the hydrolysis of AMP to yield adenine and D-ribose 5-phosphate.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

M. H. O'Leary (1989) *Ann. Rev. Biochem.* **58**, 377.

AMYLASES

Amylases are glycosidases that hydrolyze *O*-glucosyl bonds in glucans. α -Amylase [EC 3.2.1.1], also referred to as 1,4- α -D-glucan glucanohydrolase, catalyzes the endohydrolysis of a 1,4- α -D-glucosidic bond in a polysaccharide or oligosaccharide (there must be at least three glucosyl units in the oligosaccharide). Starch, glycogen, and related polysaccharides and oligosaccharides will serve as substrates. The endoglycosidic linkages are hydrolyzed in a random manner. The glucosyl group that has the newly generated anomeric carbon will be released with that anomeric carbon in the α -configuration. β -Amylase [EC 3.2.1.2], also known as saccharogen amylase and 1,4- α -D-glucan maltohydrolase, catalyzes the hydrolysis of 1,4- α -glucosidic linkages in polysaccharides near the nonreducing end of the polysaccharide chain such that successive maltose units are released from that nonreducing end. Starch, glycogen, and related polysaccharides and oligosaccharides can serve as substrates, all generating β -maltose by an inversion. **See Limit Dextran**

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.

E. H. Van Beers, H. A. Büller, R. J. Grand, A. W. C. Einerhand & J. Dekker (1995) *Crit. Rev. Biochem. Mol. Biol.* **30**, 197.

J. Lehmann & M. Schmidt-Schuchardt (1994) *Meth. Enzymol.* **247**, 265.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

I. S. Pretorius, M. G. Lambrechts & J. Marmur (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 53.

J. D. Allen (1980) *Meth. Enzymol.* **64**, 248.

J. A. Thoma, J. E. Spradlin & S. Dygert (1971) *The Enzymes*, 3rd ed., **5**, 115.

T. Takagi, H. Toda & T. Isemura (1971) *The Enzymes*, 3rd ed., **5**, 235.

AMYLOSUCRASE

This enzyme [EC 2.4.1.4], also referred to as sucrose-glucan glucosyltransferase, catalyzes the reaction of sucrose with a glucan (specifically, [(1,4)- α -D-glucosyl]_{*n*}) to yield D-fructose and a longer glucan (that is, [(1,4)- α -D-glucosyl]_{*n*+1}).

R. J. Hehre (1955) *Meth. Enzymol.* **1**, 178.

ANALOG COMPUTER SIMULATION

An approximate numerical integration corresponding to the time evolution of a chemical rate process, as achieved by an electronic circuit operating as a computer by manipulating continuous physical variables. The electronic analog computer uses voltages and/or currents in circuits

of resistors, capacitors, switches, and operational amplifiers to carry out analogous mathematical functions that simulate the time dependence of the process being modeled. Inherent limitations on the precision of circuitry components led kineticists to abandon analog computers and to favor the much greater reliability of digital computer methods. **See also** *Computer Simulation*; *KINSIM/FITSIM*

ANAPLEROTIC

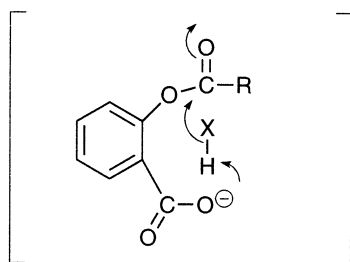
Referring to reactions, pathways, or processes that replenish or add to intermediates of a metabolic cycle (usually the tricarboxylic cycle). The process itself is referred to as anaplerosis.

ANATION

The replacement of one or more water ligands within a coordination complex (or coordination entity) by an anion.

ANCHIMERIC ASSISTANCE

The well-known facilitation^{1,2} of nucleophilic displacement reactions (particularly solvolytic reactions) resulting from a suitably positioned neighboring group. This functional group may help in directing the nucleophile to the electrophilic center or it may increase reactivity by stabilizing a reaction intermediate. Jencks³ provides the following example of the importance of intramolecular assistance: Hydrolysis of tetramethylsuccinilic acid at pH 5 has a half-time of 30 minutes, whereas the hydrolysis of acetanilide at the same pH has a half-time of around 300 years.



Anchimeric assistance may also explain slight changes in mechanism resulting from the presence of a neighboring group. One likely case is the acid-catalyzed hydrolysis of phenylglycosides. Raftery's group⁴ found that the secondary kinetic isotope effect (k_H/k_D) was 1.13 for acid hydrolysis of phenyl-4-*O*-(2-acetamido-2-deoxy- β -D-glucopyranosyl)- β -D-glucopyranoside, but k_H/k_D was later determined to be only 1.08 for phenyl-4-*O*-(2-acetamido-2-deoxy- β -D-glucopyranosyl)-2-acetamido-2-deoxy- β -D-glucopyranoside⁵. Such a result suggests that the kinetics for forming the likely oxocarbenium ion intermediate are influenced by the neighboring acetamido group. Interestingly, the value of k_H/k_D for lysozyme-catalyzed hydrolysis are 1.11 and 1.14, respectively, suggesting that substitution of the acetamido group for a hydrogen has little effect on the enzymic process. **See also** *Neighboring Group Mechanism*

copyranosyl)- β -D-glucopyranoside, but k_H/k_D was later determined to be only 1.08 for phenyl-4-*O*-(2-acetamido-2-deoxy- β -D-glucopyranosyl)-2-acetamido-2-deoxy- β -D-glucopyranoside⁵. Such a result suggests that the kinetics for forming the likely oxocarbenium ion intermediate are influenced by the neighboring acetamido group. Interestingly, the value of k_H/k_D for lysozyme-catalyzed hydrolysis are 1.11 and 1.14, respectively, suggesting that substitution of the acetamido group for a hydrogen has little effect on the enzymic process. **See also** *Neighboring Group Mechanism*

¹C. K. Ingold (1953) *Structure and Mechanism in Organic Chemistry*, p. 511 Cornell University Press, Ithaca.

²S. Winstein & E. Grunwald (1948) *J. Am. Chem. Soc.* **70**, 828.

³W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, p. 211, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York]

⁴F. W. Dahlquist, T. Rand-Meir & M. A. Raftery (1969) *Biochemistry* **8**, 4214.

⁵L. H. Mohr, L. E. H. Smith & M. A. Raftery (1973) *Arch. Biochem. Biophys.* **159**, 505.

ANCHOR PRINCIPLE

A concept suggesting that the relatively large size of coenzymes and other substrates is likely to assist in the proper positioning of the reaction center on the enzyme. This anchoring is attended by a corresponding loss of translational and rotational motion. The binding energy includes the energy required for achieving the proper orientation. The binding of well-anchored large substrates may initiate the conformational reorganization of the enzyme's active site. If these factors are enhanced in the transition state, so too will they favorably contribute to catalysis.

¹W. P. Jencks (1975) *Adv. Enzymol.* **43**, 219.

ANGIOTENSIN CONVERTING ENZYME

This zinc-dependent enzyme [EC 3.4.15.1] (also known as dipeptidyl carboxypeptidase I, dipeptidyl-dipeptidase A, kininase II, peptidase P, and carboxycathepsin) catalyzes the release of a C-terminal dipeptide at a neutral pH. The enzyme will also act on bradykinin. The presence of prolyl residues in angiotensin I and in bradykinin results in only single dipeptides being released due to the activity of this enzyme, a protein which belongs to the peptidase M2 family. The enzyme is a glycoprotein, generally membrane-bound, that is chloride ion-dependent.

- P. Corval, T. A. Williams & F. Soubrier (1995) *Meth. Enzymol.* **248**, 283.
 A. A. Patchett & E. H. Cordes (1985) *Adv. Enzymol.* **57**, 1.
 B. L. Vallee & A. Galdes (1984) *Adv. Enzymol.* **56**, 283.
 M. A. Ondetti & D. W. Cushman (1984) *Crit. Rev. Biochem.* **16**, 381.

ANGULAR MOMENTUM

A measure of the rotational momentum of a body. For circular motion, the angular momentum equals the product of a body's mass and its angular velocity.

ANION EFFECTS ON pH-RATE DATA

As noted by Dixon and Webb¹, the dependence of enzyme activity can be significantly altered by the presence of anions². For example, the presence of monovalent anions shifts the pH_{opt} of salivary α -amylase to a more alkaline pH value³. In addition, Cl^- and Br^- clearly can have an activating role with this enzyme as well. Another example is fumarase^{4,5}. This enzyme is activated by several divalent and trivalent anions. Monovalent anions have either little effect or are inhibitory. There is also a shift in the pH_{opt} to a more alkaline pH value whether the anion is activating or inhibiting. This behavior led a number of investigators to suggest that these anions alter the magnitude of the ionization constant of the enzyme-substrate complex, affecting the alkaline side of the pH-rate profile, with little or no effect on the dissociation constant governing the "acid" limb of the curve. Chloride ions behave differently: they show inhibitory effects on the "acid" limb of the curve and exhibit little influence on the other dissociation constant. Nevertheless, in both scenarios, there is a shift to a higher pH_{opt} . Anions can also effect k_{cat} without altering the magnitude of dissociation constants⁶.

Alberty⁷ analyzed the anion effect on pH-rate data. He first considered a one-substrate, one-product enzyme-catalyzed reaction in which all binding interactions were rapid equilibrium phenomena. He obtained rate expressions for effects on V_{max} and K_{m} , thereby demonstrating how an anion might alter a pH-rate profile. He also considered how anions may act as competitive inhibitors. The effect of anions on alcohol dehydrogenase has also been investigated⁸. Chloride ions appear to affect the on- and off-rate constants for NAD^+ and NADH binding. **See also** *pH Studies; Activation; Optimum pH*

¹M. Dixon & E. C. Webb (1979) *Enzymes*, 3rd ed., p. 395, Academic Press, New York.

²S. W. Cole (1904) *J. Physiol.* **30**, 202.

³K. Myrbäck (1926) *Hoppe-Seyler's Zeit. Physiol. Chem.* **159**, 1.

⁴P. J. G. Mann & B. Woolf (1930) *Biochem. J.* **24**, 427.

⁵V. Massey (1953) *Biochem. J.* **53**, 67.

⁶E. C. Webb & P. F. W. Morrow (1959) *Biochem. J.* **73**, 7.

⁷R. A. Alberty (1954) *J. Amer. Chem. Soc.* **76**, 2494.

⁸H. Theorell, A. P. Nygaard & R. Bonnichsen (1955) *Acta Chem. Scand.* **9**, 1148.

ANISOTROPIC

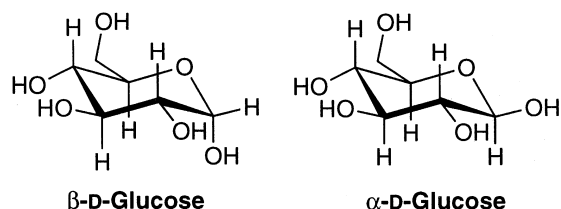
A term used with respect to a medium or a substance having one or more physical properties that have different values depending upon the direction with which that property is measured. **See also** *Isotropic*

ANNULATION (and ANNELATION)

A chemical transformation in which a ring structure is fused to a molecular entity by way of two newly formed bonds. Some investigators distinguish annulation (the formation of a ring from acyclic precursors) from annelation (the fusion of an additional ring to a ring already in existence).

ANOMER

One of the diastereomers of sugars (and sugar derivatives) differing only in the configuration about the internal hemiacetal or hemiketal carbon (*i.e.*, the so-called anomeric carbon)¹.



Anomers are a special class of epimers. The two anomeric forms are designated by α and β prefixes placed immediately before the configurational symbol (*e.g.*, α -D-glucose and β -L-mannose). The α anomeric configuration is defined as that configuration that has the same configuration as the reference carbon (in the cases above, C5) in the Fischer projection. The more dextrorotatory of the α/β -pair of D-sugar anomers is written as " α -D-," and the other is designated as " β -D-." The more levorotatory of the α/β -pair of L-sugar anomers is written as " α -L-," and its anomer is written as " β -L-." **See** *Epimers; Anomeric Specificity; Diastereomers*

¹J. F. Robyt (1998) *Essentials of Carbohydrate Chemistry*, p. 33, Springer, New York.

ANOMERIC EFFECT

An observed effect of certain substituents on the conformation and stability of glycosides. In the case of α - and β -glucosides, when the substituent attached to the anomeric position of D-glucose is a nonpolar alkyl group, such a substituent will favor the equatorial position (*i.e.*, β). However, if the substituent is polar, the α -position (*i.e.*, axial) is favored. Thus with polar moieties, α -glucosides exhibit greater stability than β -glucosides.

A. J. Kirby (1983) *The Anomeric Effect and Related Stereoelectronic Effects at Oxygen*, Springer-Verlag, New York.

ANTAGONIST

Any substance that blocks agonist binding and/or inhibits an agonist-induced biological response. Many antagonists are structural analogues of the corresponding agonists, and molecular mimicry permits antagonists to bind at a receptor binding site in a manner that blocks or alters the typical agonist-induced response. Antagonists can sometimes also interfere with downstream steps in signal-transduction cascades triggered by an agonist. Some antagonists show no obvious structural similarity to the agonist, and antagonist binding to the receptor is thought to convert the occupied receptor to a conformation which blocks agonist binding.

ANTARAFACIAL & SUPRAFACIAL MIGRATIONS

Intramolecular antarafacial and suprafacial reactions are associated with a sigmatropic rearrangement. In such reactions, a σ -bond, adjacent to one or more π -systems, migrates to a new position with the concomitant reorganization of the π -system. For example, if a hydrogen migrates along one face of the π -system, it is designated a suprafacial migration. If the hydrogen migrates across the π -system, it is designated an antarafacial migration. Orbital symmetry rules¹ can identify which migrations are forbidden and those that are allowed. For example, the rules predict that antarafacial thermal [1,3] sigmatropic rearrangements are allowed, whereas suprafacial migrations are forbidden. However, in photochemical reactions, the suprafacial pathway is allowed, but the antarafacial migration is forbidden. For [1,5] rearrangements, the thermal process is suprafacial whereas the photochemical path is antarafacial. In general, for [1,*j*] migrations, if *j* is of the form $4n + 1$, where *n* is an integer, suprafacial migrations are thermally allowed.

If *j* is of the form $4n - 1$, suprafacial migrations are photochemically allowed. For antarafacial migrations, the restrictions are reversed. **See also** *Antarafacial & Suprafacial Reactions*

¹R. B. Woodward & R. Hoffman (1970) *The Conservation of Orbital Symmetry*, Academic Press, New York.

ANTARAFACIAL & SUPRAFACIAL REACTIONS

Spatially distinct reactions occurring when a molecule or molecular fragment can simultaneously undergo two distinct changes in bonding at a common center or centers. If the reaction occurs at opposite faces of the portion of the molecule undergoing change, the reaction is designated antarafacial. If the changes occur at the same face of the molecule or molecular fragment, the reaction is designated suprafacial. The distinction between the two processes is obvious when considering reactions at a *p*-orbital or with a conjugated π -electron system^{1,2}. **See also** *Antarafacial & Suprafacial Migrations*

¹IUPAC (1979) *Pure Appl. Chem.* **51**,1725.

²T. L. Gilchrist & R. C. Storr, (1979) *Organic Reactions and Orbital Symmetry*, Cambridge Univ. Press, Cambridge.

ANTHOCYANIDIN SYNTHASE

This enzyme catalyzes the conversion of *cis*-leucoanthocyanidins to anthocyanidins.

A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.

ANTHRANILATE 1,2-DIOXYGENASE

This enzyme [EC 1.14.12.1], also known as anthranilate hydroxylase, decarboxylating, catalyzes the reaction of anthranilate with NAD(P)H, dioxygen, and two water molecules to produce catechol, carbon dioxide, NAD(P)⁺, and ammonia. The enzyme requires an iron ion as a cofactor.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

ANTHRANILATE HYDROXYLASE

This enzyme [EC 1.14.13.35], also referred to as anthranilate 3-monooxygenase (deaminating) and anthranilate 2,3-dioxygenase (a transferred entry, formerly EC 1.14.12.2), catalyzes the reaction of anthranilate with NADPH and dioxygen to generate 2,3-dihydroxybenzo-

ate, NADP^+ , and ammonia. The enzyme isolated from *Aspergillus niger* is an iron protein, whereas the yeast *Trichosporon cutaneum* enzyme utilizes FAD as a co-factor.

- B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

ANTHRANILATE PHOSPHORIBOSYL-TRANSFERASE

This enzyme [EC 2.4.2.18], also referred to as phosphoribosyl-anthranilate pyrophosphorylase, catalyzes the reaction of anthranilate with phosphoribosylpyrophosphate to produce *N*-5'-phosphoribosylanthranilate and pyrophosphate. In certain species, this enzyme is part of a multifunctional protein, together with one or more other components of the system for the biosynthesis of tryptophan (*i.e.*, indole-3-glycerol-phosphate synthase, anthranilate synthase, tryptophan synthase, and phosphoribosylanthranilate isomerase).

- R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.
 R. Bauerle, J. Hess & S. French (1987) *Meth. Enzymol.* **142**, 366.
 W. D. L. Musick (1981) *Crit. Rev. Biochem.* **11**, 1.

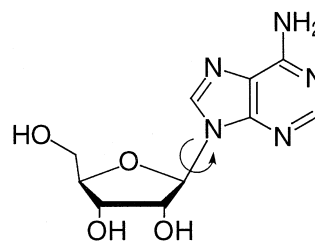
ANTHRANILATE SYNTHASE

This enzyme [EC 4.1.3.27] catalyzes the reaction of chorismate with glutamine to generate anthranilate, pyruvate, and glutamate. In certain species, this enzyme is part of a multifunctional protein together with one or more other components of the system for the biosynthesis of tryptophan (*i.e.*, indole-3-glycerol-phosphate synthase, anthranilate phosphoribosyltransferase, tryptophan synthase, and phosphoribosylanthranilate isomerase). The anthranilate synthase that is present in these complexes has been reported to be able to utilize either glutamine or ammonia as the nitrogen source. However, it has also been reported that when anthranilate synthase is separated from this complex, only ammonia can serve as a substrate.

- B. Bartel (1997) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 51.
 H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.
 R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.
 I. P. Crawford (1987) *Meth. Enzymol.* **142**, 300.
 H. Zalkin (1985) *Meth. Enzymol.* **113**, 287.
 D. E. Koshland, Jr., & A. Levitzki (1974) *The Enzymes*, 3rd ed., **10**, 539.
 J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.
 H. Zalkin (1973) *Adv. Enzymol.* **38**, 1.

ANTI

A prefix that designates stereochemical relationships in molecular entities by referring to opposite sides of a reference plane, as contrasted with the prefix “*syn*” which refers to the same side of the reference plane. For example, two substituents are said to be in an *anti* arrangement if they are attached to atoms that are joined by a single bond and if the dihedral angle formed by the substituents is greater than 90° .



For substituents attached to adjacent atoms of a carbon-carbon double bond, the terms *cis* and *trans* or *E* and *Z* are used instead of *anti* and *syn*. This is also true of oximes and related molecules as well as to cyclic structures. The terms *anti* and *syn* are also used in the context of chemical reactions or transformations, designating the relative orientation of substituent addition or elimination.

ANTIAROMATICITY

A property associated with compounds that are destabilized by a closed loop of electrons. Antiaromatic compounds are typically planar and contain $4n$ electrons, where n is a positive integer, in overlapping parallel *p* orbitals. These compounds also have a paramagnetic ring current. Thus, protons on the outside of the ring will exhibit an upfield NMR chemical shift. **See** *Aromatic*

- R. Breslow (1973) *Acc. Chem. Res.* **6**, 393.
 F. Sondheimer (1972) *Acc. Chem. Res.* **5**, 81.

ANTI-ARRHENIUS BEHAVIOR

Reaction behavior that fails to agree with the Arrhenius temperature dependence. Enzyme-catalyzed reactions have been reported to display anti-Arrhenius behavior. As the temperature rises, the temperature dependence of the reaction rate does exhibit a normal Arrhenius pattern; however, at elevated temperatures, the reaction velocity falls off as a result of denaturation.

A reaction for which the mechanism changes in a temperature-dependent manner may also exhibit anti-Arrhenius behavior. **See** *Arrhenius Equation; Arrhenius Plot, Nonlinear*

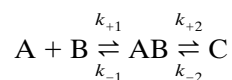
ANTIBODY-HAPTEN INTERACTIONS

Specific, rapid, and high-affinity recognition of antigens by antibodies is essential for the host's immune system to respond to invasion by foreign cells and pathogens. Pecht and Lancet¹ have presented a cogent analysis of antibody interactions with haptens, and the interested reader will find their examination of the kinetics and thermodynamics to be particularly lucid. Briefly, for the simple one-step binding scheme,



their examination of the literature indicated the following interesting information about the magnitudes for the forward and reverse rate constants: (a) k_{on} varies over a range of three orders of magnitude (from 0.005 to $5 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$); (b) k_{off} varies over a range of eight powers of ten (from $3 \times 10^{-5} \text{ s}^{-1}$ to $6 \times 10^3 \text{ s}^{-1}$); (c) rigorous treatment of the kinetics requires an additional isomerization step typical of most binding reactions. The latter is suggested, because the limiting rate of diffusion places an upper bound on the magnitude of the equilibrium constant for a one-step binding mechanism. Thus, in a one-step reaction mechanism, the value of k_{off} must be particularly important in models that attempt to explain wide variations in binding affinity that are achieved with different haptens as ligands.

A general two-step ligand-induced isomerization mechanism, as described by Eigen², can be written as:



His treatment puts the values for the first-step rate constants as

$$k_{+1} = (4\pi N/1000)\{(D_A + D_B)/R_{AB}\}$$

and

$$k_{-1} = 3(D_A + D_B)/R_{AB}^2$$

where N equals Avogadro's number, D_A and D_B are the diffusion constants, and R_{AB} is the so-called encounter distance. Pecht and Lancet¹ estimate the latter to be 3

to 15 Å, and for a hapten of molecular weight ~ 400 , they use a diffusivity D_B of about $5 \times 10^{-5} \text{ cm}^2/\text{s}$ and an R_{AB} value of 7.5 Å. Taking a typical diffusivity for an immunoglobulin ($D_A = 3 \times 10^{-9} \text{ cm}^2/\text{sec}$), the values of k_{+1} and k_{-1} turn out to be $3 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$ and $3 \times 10^9 \text{ s}^{-1}$, suggesting that the equilibrium constant K_1 for the weakly interacting, encounter-controlled first step is around 1 M. The second step involves the hapten-induced conformational change from species AB to form species C. For the encounter step, Pecht and Lancet¹ suggest that k_{-1} (as well as the $\Delta H_{-1}^\ddagger$ and $\Delta S_{-1}^\ddagger$) may be considered to be constant for virtually all hapten-immunoglobulin reactions. On the other hand, values of k_{+1} and $\Delta S_{+1}^\ddagger$ fall within a range that depends on the geometry of ligand approach to its receptor site. For the isomerization step, unimolecular rate constants range from 10 to 100 per second, and the slowness of the transition is suggestive of long-range conformational changes that involve more than a few residues contacting each other in the complex.

¹I. Pecht & D. Lancet (1977) in *Chemical Relaxation in Molecular Biology* (I. Pecht & R. Rigler, eds.) pp. 2-3, Springer-Verlag, Berlin.

²M. Eigen (1974) in *Quantum Statistical Mechanics in the Natural Sciences* (S. L. Minz & S. M. Wiedermayer, eds.) pp. 37-61, Plenum Press, New York.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Basic Principles of Antigen-Antibody Reactions: Proteins and polypeptides as antigens, **70**, 49; the experimental induction of antibodies to nucleic acids, **70**, 70; the preparation of antigenic hapten-carrier conjugates: a survey, **70**, 85; production of reagent antibodies, **70**, 104; preparation of Fab fragments from IgGs of different animal species, **70**, 142; use of carbodiimides in the preparation of immunizing conjugates, **70**, 151; use of glutaraldehyde as a coupling agent for proteins and peptides, **70**, 159; immunochemical analysis by antigen-antibody precipitation in gels, **70**, 166

Methods for the Detection of Antigens/Antibodies: Equilibrium and kinetic inhibition assays based upon fluorescence polarization, **70**, 3; fluorescence excitation transfer immunoassay (FETI), **70**, 28; indirect quenching fluorimmunoassay, **70**, 60; the homogeneous substrate-labeled fluorescent immunoassay, **70**, 79; fluorescence immunoassays using plane surface solid phases (FIAPS), **70**, 87.

ANTIBONDING ORBITAL

The combination of two atomic orbitals to form a bond between two elements actually produces two molecular orbitals having different energies. The molecular orbital with the lower energy is referred to as the bonding orbital; the other orbital tends to push atoms apart and is termed the antibonding orbital. **See** *Molecular Orbital; Bonding Orbital; Orbital*

ANTICOOPERATIVITY

Another descriptor used in place of “negative cooperativity” to describe ligand binding interactions. Anticooperativity results whenever partial ligand saturation of a cooperative system results in reduced binding, either as a consequence of changes in the apparent ligand binding constants or from the reduction in the concentration of sites available for ligand binding. **See** *Ligand Binding Cooperativity*; *Negative Cooperativity*

ANTIPOINT

A membrane protein or protein complex that is responsible for the simultaneous transport of two molecular entities or ions across the membrane, each entity traveling in opposite directions. **See** *Uniport*; *Symport*; *Membrane Transport*

APICOPHILICITY

The preference of a substituent to adopt an apical position in associative mechanisms for nucleophilic reactions on phosphorus. This property is discussed in detail by Benkovic and Shray¹, and the interested reader will also wish to consult Westheimer².

¹S. J. Benkovic & K. J. Shray (1978) in *Transition States of Biochemical Processes* (R. D. Gandour & R. L. Schowen, eds.) p. 496, Plenum Press, New York.

²F. H. Westheimer (1968) *Acc. Chem. Res.* **1**, 70.

APOACTIVATOR

Any regulatory protein that stimulates transcription from one or more genes in the presence of a coactivator molecule.

APROTIC

A term, usually referring to a solvent, describing a compound which act neither as a proton donor nor a proton acceptor. Examples of polar aprotic solvents include dimethylformamide, dimethylsulfoxide, acetone, acetonitrile, sulfur dioxide, and hexamethylphosphoramide. Examples of nonpolar aprotic solvents include benzene and carbon tetrachloride. Studies of reactions in protic and aprotic solvents have demonstrated the importance of solvation on reactants, leaving groups, and transition states. Degrees of nucleophilicity as well as acidity are different in aprotic solvents. For example, small, negatively charged nucleophiles react more readily in polar aprotic solvents. It should also be noted that extremely

strong acids or bases may still protonate or deprotonate compounds normally considered aprotic.

C. Reichardt (1988) *Solvents and Solvent Effects in Organic Chemistry*, 2nd ed., VCH, New York.

G. W. Klumpp (1982) *Reactivity in Organic Chemistry*, Wiley, New York.

T. W. Bentley & P. von R. Schleyer (1977) *Adv. Phys. Org. Chem.* **14**, 1.

F. G. Bordwell & D. L. Hughes (1981) *J. Org. Chem.* **46**, 3570.

A PRIORI PROBABILITY

A statistical method wherein the probability is assigned to an event or parameter prior to any testing.

APYRASE

This calcium ion-dependent enzyme [EC 3.6.1.5] (also known as ATP-diphosphatase, adenosine diphosphatase, and ATP-diphosphohydrolase) catalyzes the hydrolysis (with two H₂O) of ATP to generate AMP and two orthophosphate ions. The enzyme will also utilize ADP as a substrate as well as other nucleoside triphosphates and diphosphates.

P. S. Krishnan (1955) *Meth. Enzymol.* **2**, 591.

AQUACOBALAMIN REDUCTASE

Aquacobalamin reductase (NADH) [EC 1.6.99.8] catalyzes the reaction of two aquacob(III)alamin with NADH to form 2 cob(II)alamin and NAD⁺. Aquacobalamin reductase (NADPH) [EC 1.6.99.11] catalyzes the exact same reaction, albeit with NADPH. In addition, this enzyme can utilize hydroxycobalamin as a substrate, but not cyanocobalamin.

F. Watanabe & Y. Nakano (1997) *Meth. Enzymol.* **281**, 289.

AQUAMOLALITY

A concentration scale for solutes in aqueous solutions, equal to moles of solute/55.51 mol water. It is frequently used in studies of solvent isotope effects. As pointed out by Schowen and Schowen¹, the choice of standard states can change the sign for the free energy of transfer of a species from one solvent to another, even from HOH and DOD. The commonly used concentration scales are molarity, mole fraction, aquamolality, and molality. Free energies tend to be nearly the same on all but the molality scale, on which they are about 63 cal mol⁻¹ more positive at 298 K than on the first three scales. The interested reader should consult Table I of Schowen and Schowen¹

for a compilation of the relative stabilities of various molecules in isotopic waters at 298 K. **See** *Concentration*

¹K. B. Schowen & R. L. Schowen (1982) *Meth. Enzymol.* **87**, 551.

AQUATION

The process by which one or more water molecules are incorporated into another molecular entity until no other atoms, ions, and groups or moieties are further displaced. In the case of metal ions, aquation reflects the ability of nonbonded electrons on water oxygens to form coordinate covalent bonds with the metal ion. **See also** *Hydration*; *Water*

α -L-ARABINOFURANOSIDASE

This enzyme [EC 3.2.1.55], also called arabinosidase, catalyzes the hydrolysis of terminal nonreducing α -L-arabinofuranoside residues in α -L-arabinosides. Actual substrates include α -L-arabinofuranosides, α -L-arabinans containing (1,3)- and/or (1,5)-linkages, arabinoxylans, and arabinogalactans. It should be noted that some β -galactosidases and β -D-fucosidases will also hydrolyze α -L-arabinosides.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

D-ARABINOKINASE

This enzyme [EC 2.7.1.54] catalyzes the reaction of ATP with D-arabinose to generate ADP and D-arabinose 5-phosphate.

W. A. Volk (1966) *Meth. Enzymol.* **9**, 442.

ARABINONATE DEHYDRATASE

Arabinonate dehydratase [EC 4.2.1.5] catalyzes the conversion of D-arabinonate to 2-dehydro-3-deoxy-D-arabinonate and water whereas L-arabinonate dehydratase [EC 4.2.1.25] catalyzes the conversion of L-arabinonate to 2-dehydro-3-deoxy-L-arabinonate and water.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

ARABINOSE ISOMERASE

Arabinose isomerase [EC 5.3.1.3] catalyzes the interconversion of D-arabinose and D-ribulose. The enzyme will also utilize L-fucose and more slowly utilize L-galactose and D-altrose as substrates. L-Arabinose isomerase [EC 5.3.1.4] catalyzes the interconversion of L-arabinose and L-ribulose.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

I. A. Rose (1975) *Adv. Enzymol.* **43**, 491.

E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

ARABINOSE-5-PHOSPHATE ISOMERASE

This enzyme [EC 5.3.1.13] catalyzes the interconversion of D-arabinose 5-phosphate to D-ribulose 5-phosphate.

E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

ARACHIDONYL-CoA SYNTHETASE

This enzyme [EC 6.2.1.15], also known as arachidonate:CoA ligase, catalyzes the reaction of arachidonate with ATP and coenzyme A to generate arachidonyl-CoA, AMP, and pyrophosphate (or, diphosphate). The enzyme can also use 8,11,14-icosatrienoate as a substrate, but not the other long-chain fatty acids. It should be noted that this enzyme is not identical to long-chain acyl-CoA synthetase [EC 6.2.1.3].

M. Laposata (1990) *Meth. Enzymol.* **187**, 237.

ARGINASE

This enzyme [EC 3.5.3.1], also referred to as arginine amidinase and canavanase, catalyzes the hydrolysis of arginine resulting in the formation of urea and ornithine. The enzyme, which uses a manganese ion as a cofactor, will also utilize α -N-substituted L-arginines and canavanine as substrates.

J. E. Penner-Hahn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 439.

S. Ratner (1973) *Adv. Enzymol.* **39**, 1.

ARGININE DECARBOXYLASE

This enzyme [EC 4.1.1.19] uses pyridoxal phosphate as a cofactor in catalyzing the conversion of arginine to carbon dioxide and agmatine.

T. Hashimoto & Y. Yamada (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 257.

M. H. O'Leary (1980) *Meth. Enzymol.* **64**, 83.

E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

ARGININE DEIMINASE

This enzyme [EC 3.5.3.6], also known as arginine dihydrolase, catalyzes the hydrolysis of arginine (with canavanine as a substrate as well) to produce citrulline and ammonia.

J. Thompson & S. P. F. Miller (1991) *Adv. Enzymol.* **64**, 317.

ARGININE KINASE

This enzyme [EC 2.7.3.3] catalyzes the reaction of arginine with ATP to produce ADP and *N*-phosphoarginine.

J. F. Morrison (1973) *The Enzymes*, 3rd ed., **8**, 457.

ARGININE 2-MONOOXYGENASE

This enzyme [EC 1.13.12.1] catalyzes the reaction of arginine with dioxygen to generate 4-guanidobutanamide, carbon dioxide, and water. The enzyme can also utilize canavanine and homoarginine as substrates.

S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.

V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

ARGININE RACEMASE

This enzyme [EC 5.1.1.9] catalyzes the interconversion of L- and D-arginine. The enzyme requires pyridoxal phosphate as a cofactor.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

E. Adams (1972) *The Enzymes*, 3rd ed., **6**, 479.

ARGININOSUCCINATE LYASE

This enzyme [EC 4.3.2.1], also known as arginosuccinase, catalyzes the conversion of *N*-argininosuccinate to fumarate and arginine.

V. E. Anderson (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 115.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

L. S. Mullins & F. M. Raushel (1995) *Meth. Enzymol.* **249**, 398.

H. J. Vogel & R. H. Vogel (1974) *Adv. Enzymol.* **40**, 65–90.

S. Ratner (1973) *Adv. Enzymol.* **39**, 1.

S. Ratner (1972) *The Enzymes*, 3rd ed., **7**, 167.

ARGININOSUCCINATE SYNTHETASE

This enzyme [EC 6.3.4.5], also referred to as citrulline:aspartate ligase, catalyzes the reaction of citrulline with aspartate and ATP to produce argininosuccinate, AMP, and pyrophosphate (or, diphosphate).

F. M. Raushel & J. J. Villafranca (1988) *Crit. Rev. Biochem.* **23**, 1.

H. J. Vogel & R. H. Vogel (1974) *Adv. Enzymol.* **40**, 65.

S. Ratner (1973) *Adv. Enzymol.* **39**, 1.

ARISTOLOCHENE SYNTHASE

This enzyme [EC 2.5.1.40] catalyzes the conversion of *trans,trans*-farnesyl diphosphate (or, *trans,trans*-farnesyl pyrophosphate) to aristolochene and diphosphate (or, pyrophosphate). The enzyme-catalyzed reaction proceeds through an initial internal transfer of the farnesyl

group to the C10 position resulting in the monocyclic intermediate germacrene A. This is followed by further cyclization and an internal methyl transfer converting the intermediate into aristolochene.

R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.

AROMATIC

1. Referring to a substance that behaves chemically like benzene. 2. Referring to a molecule that exhibits an ability to sustain an induced ring current. With few exceptions, protons covalently linked to an aromatic ring exhibit a downfield chemical shift in an NMR experiment as compared to protons that are not associated with an aromatic ring. Because of resonance stabilization, aromatic compounds are especially stable, and they exhibit greater stability than that of any specific canonical form or so-called contributing form (e.g., any of the well known Kekulé or Dewar structures of benzene). Most commonly, aromatic molecules have cyclically delocalized π electrons, and the number of π electrons obeys Hückel's rule. In addition, aromatic rings will exhibit equal, or near equal, bond lengths, planarity, and the ability to undergo aromatic substitution reactions. **See** *Antiaromaticity*; *Hückel's Rule*

D. Lloyd (1989) *The Chemistry of Conjugated Cyclic Compounds*, Wiley, New York.

P. J. Garratt (1986) *Aromaticity*, Wiley, New York.

K. Jug & A. M. Koster (1991) *J. Phys. Org. Chem.* **4**, 163.

Z. Zhou & R. G. Parr (1989) *J. Amer. Chem. Soc.* **111**, 7371.

A. R. Katritzky, P. Barczynski, G. Musumarra, D. Pisano & M. Szafran (1989) *J. Amer. Chem. Soc.* **111**, 7.

AROMATIC-AMINO-ACID AMINOTRANSFERASE

This enzyme [EC 2.6.1.57] catalyzes the reversible reaction of an aromatic amino acid with α -ketoglutarate to generate an aromatic oxo acid and glutamate. Pyridoxal phosphate is a required cofactor. Methionine can also act as a weak substrate, substituting for the aromatic amino acid. Oxaloacetate substitutes for α -ketoglutarate.

J. DiRuggiero & F. T. Robb (1996) *Adv. Protein Chem.* **48**, 311.

C. Mavrides (1987) *Meth. Enzymol.* **142**, 253.

A. J. L. Cooper (1985) *Meth. Enzymol.* **113**, 73.

AROMATIC-AMINO-ACID DECARBOXYLASE

This enzyme [EC 4.1.1.28] (also referred to as DOPA decarboxylase, tryptophan decarboxylase, and hydroxy-

tryptophan decarboxylase) catalyzes the conversion of tryptophan to tryptamine and carbon dioxide. Pyridoxal phosphate is a required cofactor for this enzyme. The enzyme can also utilize 5-hydroxytryptophan and dihydroxyphenylalanine as substrates.

- T. Hashimoto & Y. Yamada (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 257.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 T. L. Sourkes (1987) *Meth. Enzymol.* **142**, 170.
 E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

AROMATIC-AMINO-ACID:GLYOXYLATE AMINOTRANSFERASE

This enzyme [EC 2.6.1.60] catalyzes the reversible reaction of an aromatic amino acid with glyoxylate to generate an aromatic oxo acid and glycine. Aromatic amino acids that can act as substrates include phenylalanine, kynurenine, tyrosine, and histidine. Pyruvate and hydroxypyruvate can act as glyoxylate substitutes as well.

- T. Takada & T. Noguchi (1987) *Meth. Enzymol.* **142**, 273.

ARRHENIUS CONSTANT

A dimensionless number equal to the activation energy of a reaction (E_a) divided by the product of the universal gas constant (R) and the absolute temperature, T ; thus, $E_a/(RT)$.

ARRHENIUS EQUATION & PLOT

A relationship¹ describing the temperature dependence of an individual rate constant:

$$(\partial \ln k / \partial T)_P = E_a / RT^2$$

where E_a is the energy of activation for the reaction having the rate constant k at constant pressure P , R is the universal gas constant, and T is the absolute temperature. The International Organization for Standardization (ISO), the International Union of Pure and Applied Chemistry (IUPAC), and the International Union of Pure and Applied Physics (IUPAP) have recommended that E_a or E , written in italic type², be the symbol for activation energy. Nevertheless, E_{act} , E^* , $E^\ddagger$, and $E^\#$ are often used (in the older literature, the symbol μ was also common).

If one assumes that E_a is not temperature-dependent, integration of the above differential equation yields:

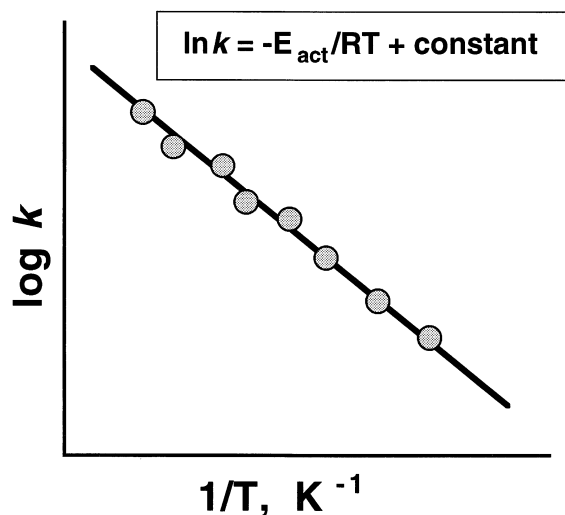
$$\ln k = -\frac{E_a}{RT} + \text{Constant}$$

Letting the constant of integration equal $\ln A$, the above equation becomes

$$k = Ae^{-E_a/RT}$$

This is the well-known Arrhenius equation in which A is referred to as the frequency factor or pre-exponential factor.

A plot of $\ln k$ against the reciprocal of the absolute temperature (an Arrhenius plot) will produce a straight line having a slope of $-E_a/R$. The frequency factor can be obtained from the vertical intercept, $\ln A$. The Arrhenius relationship has been demonstrated to be valid in a large number of cases (for example, colchicine-induced GTPase activity of tubulin³ or the binding of *N*-acetylphenylalanyl-tRNA^{Phe} to ribosomes⁴). In practice, the Arrhenius equation is only a good approximation of the temperature dependence of the rate constant, a point which will be addressed below.



Arrhenius recognized that for molecules to react they must attain a certain critical energy, E_a . On the basis of collision theory, the rate of reaction is equal to the number of collisions per unit time (the frequency factor) multiplied by the fraction of collisions that results in a reaction. This relationship was first developed from the kinetic theory of gases^{5,6}. For a bimolecular reaction, the bimolecular rate constant, k , can be expressed as

$$k = Ze^{-E_a/RT}$$

where $e^{-E_a/RT}$ is the Boltzmann factor representing the probability that a collision will be effective and Z is the number of collisions between reactants per second in 1 cm^3 .

Effect of Inhibitors on Arrhenius Plots. The presence of inhibitors can lead to nonlinear Arrhenius plots as well as an increase in E_a . For example, jack bean urease exhibits a linear Arrhenius plot in the absence of an inhibitor, but a curved plot in the presence of 34 mM sodium sulfite⁷.

Effect of Solvent on Arrhenius Plots. If water is a substrate, then the presence of an organic solvent, which may disrupt the structure and/or orientation of water, may alter the Arrhenius plot. For example, a linear plot is seen with fumarate hydratase in the presence of 10% methanol. However, the plot is biphasic in the presence of 10% ethanol⁸. **See** Boltzmann Distribution; Collision Theory; Temperature Dependency, Transition-State Theory; Energy of Activation; Q_{10}

¹S. Arrhenius (1889) *Zeit. für Physik. Chem.* **4**, 226.

²I. Mills (1988) *Quantities, Units, and Symbols in Physical Chemistry*, Blackwell Scientific Publ., Oxford.

³B. Perez-Ramirez & S. N. Timasheff (1994) *Biochemistry* **33**, 6262.

⁴S. Schilling-Bartetzko, A. Bartetzko & K. H. Nierhaus (1992) *J. Biol. Chem.* **267**, 4703.

⁵W. C. McC. Lewis (1918) *J. Chem. Soc.* **113**, 471.

⁶M. Trautz (1916) *Zeit. Anorg. Chem.* **96**, 1.

⁷G. B. Kistiakowsky & R. Lumry (1949) *J. Amer. Chem. Soc.* **71**, 2006.

⁸V. Massey (1953) *Biochem. J.* **53**, 72.

ARRHENIUS PLOTS (NONLINEAR)

Arrhenius plots of enzyme-catalyzed reactions occasionally exhibit curvilinearity¹ and appear to have two straight-line segments intersecting each other at an angle. While suggestive of a change in mechanism signaled by a break in the activation energy at some specific transition temperature, Kistiakowsky and Lumry² have pointed out that in many of those cases for which two intersecting lines are drawn, intersecting at a “transition temperature,” the data are better described by a smooth curve. Dixon and Webb³ offered the following summary of possible explanations for nonlinearity in Arrhenius plots. These include: (1) a change in the properties of the solvent with temperature; (2) the occurrence of two or more parallel or competing reactions (e.g., catalyzed by isozymes), each with different energies of activation;

(3) the presence of two successive reactions with different energies of activation; (4) the enzyme has two (or more) different forms, each with its own activation energy; (5) a reversible inactivation of the enzyme; or (6) a discontinuity affecting the forward reaction only.

Finally, yet another issue enters into the interpretation of nonlinear Arrhenius plots of enzyme-catalyzed reactions. As is seen in the examples above, one typically plots $\ln V_{\max}$ (or, $\ln k_{\text{cat}}$) versus the reciprocal absolute temperature. This protocol is certainly valid for rapid equilibrium enzymes whose rate-determining step does not change throughout the temperature range studied (and, in addition, remains rapid equilibrium throughout this range). However, for steady-state enzymes, other factors can influence the interpretation of the nonlinear data. For example, for an ordered two-substrate, two-product reaction, k_{cat} is equal to $k_5k_7/(k_5 + k_7)$ in which k_5 and k_7 are the off-rate constants for the two products. If these two rate constants have a different temperature dependency (e.g., $k_5 > k_7$ at one temperature but not at another temperature), then a nonlinear Arrhenius plot may result. **See** Arrhenius Equation; Q_{10} ; Transition-State Theory; van't Hoff Relationship

¹K. J. Laidler & B. F. Peterman (1979) *Meth. Enzymol.* **63**, 183.

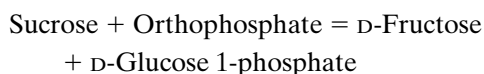
²G. B. Kistiakowsky & R. Lumry (1949) *J. Amer. Chem. Soc.* **71**, 2006.

³M. Dixon & E. C. Webb (1979) *Enzymes*, 3rd ed., Academic Press, New York.

ARSENOLYSIS

Arsenate-dependent “hydrolysis” of a covalent bond. Arsenate dianion (HAsO_4^{2-}) is a structural analogue of orthophosphate dianion (HPO_4^{2-}) and often acts as a phosphate analogue in many enzyme-catalyzed reaction utilizing phosphate as a substrate. (Note that the $\text{p}K_a$ values for arsenic acid (at 18°C) are 2.25, 6.77, and 11.60 whereas the $\text{p}K_a$ values for H_3PO_4 (at 25°C) are 2.15, 7.20, and 12.33.) The most significant fact is that arsenate esters (i.e., $\text{R}-\text{O}-\text{AsO}_3^{2-}$) are kinetically and thermodynamically unstable in aqueous solutions. When formed on and released from the enzyme's surface, they rapidly hydrolyze to form ROH and HAsO_4^{2-} . By contrast, phosphate esters are far more kinetically stable.

An example of an enzyme that exhibits arsenolysis is sucrose phosphorylase¹, an enzyme that catalyzes the reaction:



In the presence of arsenate (and absence of phosphate), the enzyme will exhibit a sucrase activity. The enzyme also catalyzes arsenolysis of D-glucose 1-phosphate. In the absence of the enzyme, there is very little hydrolysis (or arsenolysis) of D-glucose 1-phosphate. In the presence of the enzyme and arsenate, however, D-glucose 1-phosphate rapidly converts to D-glucose and orthophosphate. This observation supports the formation of a glucosyl-enzyme intermediate. When arsenate is present in significant amounts, the arsenate ester is formed and is almost immediately hydrolyzed to D-glucose and arsenate. Interestingly, maltose phosphorylase and cellobiose phosphorylase, which are both able to form D-glucose 1-phosphate as a product, do not facilitate arsenolysis. This is one of many lines of evidence indicating that the latter enzymes have different mechanisms than that of sucrose phosphorylase. Other enzyme reactions that undergo arsenolysis include glutamine synthetase, tubulin:tyrosine ligase (ADP-forming), and D-alanyl-D-alanine synthetase.

¹J. Mieyal & R. Abeles (1972) *The Enzymes*, 3rd ed., **7**, 515.

ARYLALKYLAMINE N-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.87], also known as serotonin acetyltransferase and serotonin acetylase, catalyzes the reaction of acetyl-CoA and arylalkylamine to generate coenzyme A and *N*-acetylarylalkylamine. The enzyme exhibits a rather narrow specificity toward other arylalkylamines. This enzyme is distinct from arylamine acetyltransferase.

M. A. Namboodisi, R. Dubbels & D. C. Klein (1987) *Meth. Enzymol.* **142**, 583.

ARYLAMINE N-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.5], also known as acetyl-CoA: arylamine *N*-acetyltransferase and arylamine acetylase, catalyzes the reaction of acetyl-CoA with an arylamine to produce coenzyme A and an *N*-acetylarylamine. This enzyme exhibits a low specificity with respect to the aromatic amine substrate. In fact, even serotonin can serve as a substrate. The enzyme has also been reported to catalyze acetyl-transfer reactions between arylamines without the use of coenzyme A.

ARYLFORMAMIDASE

This enzyme [EC 3.5.1.9], also called kynurenine formamidase and formylkynureninase, catalyzes the hydrolysis of *N*-formylkynurenine to yield formate and kynurenine. The enzyme will also use other aromatic formylamines as substrates.

E. Katz, D. Brown & M. J. M. Hitchcock (1987) *Meth. Enzymol.* **142**, 225.

ARYLSULFATASES

Arylsulfatase [EC 3.1.6.1], also known simply as sulfatase, catalyzes the hydrolysis of a phenol sulfate, thereby producing a phenol and sulfate. This enzyme classification represents a collection of enzymes with rather similar specificities. (1) Steryl-sulfatase [EC 3.1.6.2], also referred to as arylsulfatase C and steroid sulfatase, catalyzes the hydrolysis of 3- β -hydroxyandrost-5-en-17-one 3-sulfate to 3- β -hydroxyandrost-5-en-17-one and sulfate. The enzyme utilizes other steryl sulfates as substrates. (2) Cerebroside-sulfatase [EC 3.1.6.8], or arylsulfatase A, catalyzes the hydrolysis of a cerebroside 3-sulfate to yield a cerebroside and sulfate. The enzyme will also hydrolyze the galactose 3-sulfate bond present in a number of lipids. In addition, the enzyme will also hydrolyze ascorbate 2-sulfate and other phenol sulfates.

G. Lowe (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 627.

E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.

A. B. Roy (1987) *Meth. Enzymol.* **143**, 207.

Y.-T. Li & S.-C. Li (1983) *The Enzymes*, 3rd ed., **16**, 427.

A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 1.

R. G. Nicholls & A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 21.

ARYLSULFOTRANSFERASE

This enzyme [EC 2.8.2.1], also known as phenol sulfotransferase and sulfokinase, catalyzes the reaction of 3'-phosphoadenylylsulfate with a phenol to yield adenosine 3',5'-bisphosphate and an aryl sulfate. The enzyme can utilize a number of aromatic compounds as substrates.

G. Lowe (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 627.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

S. G. Ramaswamy & W. B. Jakoby (1987) *Meth. Enzymol.* **143**, 201.

ASCORBATE–CYTOCHROME *b*₅ REDUCTASE

This enzyme [EC 1.10.2.1] catalyzes the reaction of ascorbate with ferricytochrome *b*₅ to yield monodehydroascorbate and ferrocyclochrome *b*₅.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

ASCORBATE OXIDASE

This enzyme [EC 1.10.3.3], also called ascorbase, catalyzes the reaction of two ascorbate molecules with dioxygen to produce two dehydroascorbate molecules and two water molecules. The enzyme utilizes a copper ion as a cofactor.

- A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 E. T. Adman (1991) *Adv. Protein Chem.* **42**, 145.
 B. G. Malmström, L.-E. Andréasson & B. Reinhammer (1975) *The Enzymes*, 3rd ed., **12**, 507.

ASCORBATE PEROXIDASE

This enzyme [EC 1.11.1.11] catalyzes the reaction of ascorbate with hydrogen peroxide to produce dehydroascorbate and two water molecules.

- H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

ASPARAGINASE

This enzyme [EC 3.5.1.1] catalyzes the hydrolysis of asparagine to ammonia and aspartate.

- J. C. Wriston, Jr. (1985) *Meth. Enzymol.* **113**, 608.
 J. C. Wriston, Jr. & T. O. Yellin (1973) *Adv. Enzymol.* **39**, 185.
 J. C. Wriston, Jr. (1971) *The Enzymes*, 3rd ed., **4**, 101.

ASPARAGINE AMINOTRANSFERASE

This vitamin B₆-dependent enzyme [EC 2.6.1.14], also referred to as asparagine-oxo-acid aminotransferase, catalyzes the reversible reaction of asparagine and a 2-oxo acid to yield 2-oxosuccinamate and an amino acid.

- A. J. L. Cooper & A. Meister (1985) *Meth. Enzymol.* **113**, 602.

ASPARAGINE SYNTHETASE

Asparagine synthetase (glutamine-hydrolyzing) [EC 6.3.5.4] catalyzes the reaction of aspartate with ATP and glutamine to produce asparagine, AMP, diphosphate (or, pyrophosphate), and glutamate. The enzyme will also hydrolyze glutamine to glutamate and ammonia in the absence of the other substrates. Asparagine synthetase (ammonia-using) [EC 6.3.1.1], also known as aspartate—ammonia ligase, catalyzes the reaction of aspartate with ammonia and ATP to produce asparagine, AMP, and diphosphate (or, pyrophosphate). Asparagine synthetase (ammonia-using, ADP-forming) is an enzyme [EC 6.3.1.4] reported to catalyze the reaction of aspartate

with ammonia and ATP to produce asparagine, ADP, and orthophosphate.

- N. Richards & S. M. Schuster (1998) *Adv. Enzymol.* **72**, 145.
 H.-M. Lam, K. T. Coschigamo, I. C. Oliveira, R. Melo-Oliveira & G. M. Coruzzi (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 569.
 H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.
 A. Meister (1974) *The Enzymes*, 3rd ed., **10**, 561.
 J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

ASPARTATE AMINOTRANSFERASE

This enzyme [EC 2.6.1.1] (also known as transaminase A, glutamic:oxaloacetic transaminase, and glutamic:aspartic transaminase) catalyzes the reversible reaction of aspartate with α -ketoglutarate to produce oxaloacetate and glutamate. Pyridoxal phosphate is a required cofactor. The enzyme has a relatively broad specificity, and tyrosine, phenylalanine, and tryptophan can all serve as substrates.

- R. A. John (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 173.
 J. DiRuggiero & F. T. Robb (1996) *Adv. Protein Chem.* **48**, 311.
 H.-M. Lam, K. T. Coschigamo, I. C. Oliveira, R. Melo-Oliveira & G. M. Coruzzi (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 569.
 M. Martinez-Carrion, A. Artigues, A. Berezov, M. L. Bianconi, A. M. Reyes & A. Iriarte (1995) *Meth. Enzymol.* **259**, 590.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.
 C. Mavrides (1987) *Meth. Enzymol.* **142**, 253-267.
 T. Yagi, H. Kagamiyama, M. Nozaki & K. Soda (1985) *Meth. Enzymol.* **113**, 83.
 A. E. Martell (1982) *Adv. Enzymol.* **53**, 163.
 D. E. Metzler (1979) *Adv. Enzymol.* **50**, 1.
 A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

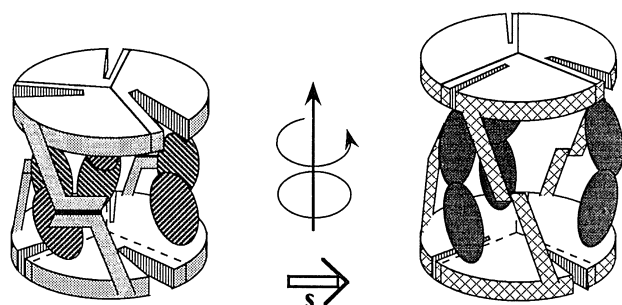
ASPARTATE AMMONIA-LYASE

This enzyme [EC 4.3.1.1], also known as aspartase and fumaric aminase, catalyzes the conversion of aspartate to fumarate and ammonia.

- V. E. Anderson (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 115.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 M. Tokushige (1985) *Meth. Enzymol.* **113**, 618.
 K. R. Hanson & E. A. Havir (1972) *The Enzymes*, 3rd ed., **7**, 75.

ASPARTATE CARBAMOYLTRANSFERASE

The prototypical allosteric regulatory protein isolated from *Escherichia coli* (molecular weight 310,000). The enzyme [EC 2.1.3.2], also known as aspartate transcarbamylase and carbamylaspartotranskinase, catalyzes the first committed step (*i.e.*, the transfer of a carbamoyl



T-STATE

R-STATE

Structural model for conformational changes in ATCase, based on the X-ray crystallographic investigations of Lipscomb and colleagues. Note that substrates are thought to enter through a channel, and allosteric effectors alter channel accessibility.

group from carbamoyl phosphate to the α -amino group of aspartate with the liberation of phosphate) in pyrimidine biosynthesis. This enzyme (ATCase) can be dissociated by mercurials into two catalytic trimers and three regulatory dimers. ATCase is feedback-inhibited by the pyrimidine nucleotide biosynthetic end-product CTP and is activated by ATP.

N. M. Allewell & V. J. LiCata (1995) *Meth. Enzymol.* **259**, 608.

W. W. Cleland (1995) *Meth. Enzymol.* **249**, 341.

W. N. Lipscomb (1994) *Adv. Enzymol.* **68**, 67.

E. Lolis & G. A. Petsko (1990) *Ann. Rev. Biochem.* **59**, 597.

E. R. Kantrowitz & W. N. Lipscomb (1988) *Science* **241**, 669.

J. S. Brabson, M. R. Maurizi & R. L. Switzer (1985) *Meth. Enzymol.* **113**, 627.

B. L. Vallee & A. Galles (1984) *Adv. Enzymol.* **56**, 283.

G. R. Jacobson & G. R. Stark (1973) *The Enzymes*, 3rd ed., **9**, 225.

Selected entries from *Methods in Enzymology* [vol, page(s)]: Activation, **64**, 177; concerted allosteric model, **64**, 173; isotope exchange properties, **64**, 10; negative cooperativity, **64**, 189; rapid relaxation measurement, **64**, 188, 189; sequential model, **64**, 173; site-directed mutagenesis, **202**, 717, 725; structure-function analysis, **202**, 694; supersecondary structures, genetic exchange within domains, **202**, 704; allosteric mechanism, **259**, 614, 627; binding site number, **259**, 615-616; concentration, **259**, 615; differential scanning calorimetry [assembly effects, **259**, 624-625; buffer sensitivity, **259**, 625; ligation effects, **259**, 625; mutation effects, **259**, 626]; electrostatic interactions, **259**, 626-628; *Escherichia coli* ATCase [cooperativity in (allosteric structures and model testing, **249**, 554-555; experimental evaluation, **249**, 548-554; heterotropic, **249**, 552-553; homotropic, **249**, 551-552; mutational analysis, **249**, 554; structural model, **249**, 549-551); inter-subunit ligand binding sites, **249**, 559-560; site-directed mutants, acid-base catalysis, **249**, 115-116; ligand binding [assays, **259**, 615-616; cooperativity, detection, **259**, 616-617; dissociation constants, **259**, 616; enthalpy, **259**, 618-620; entropy, **259**, 618-621; free energy determination, **259**, 623; nucleotide triphosphates, **259**, 619-620, 624; sites, interaction distances, **249**, 561-562; subunit interactions, **259**, 610, 622-624]; linkage analysis, **259**, 614, 621-624, 627; mechanism, **249**, 362-363 [contribution to protein structure knowledge, **259**, 612; equilibrium isotope exchange investigation, **249**, 470; molecular modeling, **259**, 612; thermodynamic definition by state functions, **259**, 611, 628]; modifier action, equilibrium isotope exchange investigation, **249**, 471-472; pK values, sensitivity to side-chain positions, **259**, 627-628; reac-

tion catalyzed, **259**, 609; regulation, **259**, 609-610; reversible denaturation, **259**, 626; site-directed mutants, equilibrium isotope exchange investigations, **249**, 474-478; structure, **259**, 609; subunit interactions, free energy, **259**, 613, 617-618, 620-621, 623.

ASPARTATE α -DECARBOXYLASE

This enzyme [EC 4.1.1.11] catalyzes the conversion of aspartate to β -alanine and carbon dioxide. Pyruvate is a cofactor for the *E. coli* enzyme.

M. L. Hackert & A. E. Pegg (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 201.

J. M. Williamson (1985) *Meth. Enzymol.* **113**, 589.

ASPARTATE β -DECARBOXYLASE

This enzyme [EC 4.1.1.12], also known as desulfonase, catalyzes the conversion of aspartate to alanine and carbon dioxide. Pyridoxal phosphate is a required cofactor. The enzyme will also catalyze the decarboxylation of aminomalonate as well as the desulfination of 3-sulfinoalanine to sulfite and alanine.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

ASPARTATE KINASE

Aspartate kinase [EC 2.7.2.4], also known as aspartokinase, catalyzes the reaction of aspartate with ATP to produce 4-phosphoaspartate and ADP. The enzyme isolated from *E. coli* is a multifunctional protein, also exhibiting the ability to catalyze the reaction of homoserine with NAD(P)⁺ to produce aspartate 4-semialdehyde and NAD(P)H (that is, the activity of homoserine dehydrogenase, EC 1.1.1.3).

I. Saint-Gerons, C. Parsot, M. M. Zakin, O. Bärzy & G. N. Cohen (1988) *Crit. Rev. Biochem.* **23**, S1.

G. N. Cohen (1985) *Meth. Enzymol.* **113**, 596.

P. Truffa-Bachi (1973) *The Enzymes*, 3rd ed., **8**, 509.

D-ASPARTATE OXIDASE

This enzyme [EC 1.4.3.1] catalyzes the reaction of D-aspartate with water and dioxygen to generate oxaloacetate, ammonia, and hydrogen peroxide. FAD is the cofactor for this enzyme.

G. A. Hamilton (1985) *Adv. Enzymol.* **57**, 85.

M. Dixon (1970) *Meth. Enzymol.* **17A**, 713.

ASPARTATE RACEMASE

This enzyme [EC 5.1.1.13] catalyzes the interconversion of L- and D-aspartate. The enzyme will also utilize alanine as a substrate, albeit at only half the reported rates of aspartate.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

ASPARTATE-SEMIALDEHYDE DEHYDROGENASE

This enzyme [EC 1.2.1.11] catalyzes the reaction of aspartate-4-semialdehyde with orthophosphate and NADP⁺ to produce 4-aspartyl phosphate and NADPH.

- I. Saint-Gerons, C. Parsot, M. M. Zakin, O. Bârzy & G. N. Cohen (1988) *Crit. Rev. Biochem.* **23**, S1.
 G. N. Cohen (1985) *Meth. Enzymol.* **113**, 600.
 G. D. Hegeman, G. N. Cohen & R. Morgan (1970) *Meth. Enzymol.* **17A**, 708.

ASPARTIC PROTEASES

The aspartic proteinases are a subset of peptide hydrolases [EC 3.4.23.x] that have a participating active-site carboxyl group and a pH optimum below 5. The microbial aspartic proteinases [formerly represented by EC 3.4.23.6] represents a large group of enzymes isolated from microorganisms with varying characteristics. For example, aspergillopepsin I [EC 3.4.23.18] (also known as awamori and trypsinogen kinase) is a peptidase, a member of peptidase family A1, isolated from several *Aspergillus* species (imperfect fungi). It catalyzes the hydrolysis of proteins with broad specificity. The enzyme generally favors hydrophobic residues in P₁ and P₁' sites. Its ability to accept a lysyl residue in P₁ site leads to trypsinogen activation. The enzyme will not clot milk. However, endothiapepsin [EC 3.4.23.22] (also known as *Endothia* aspartic proteinase) does clot milk. This enzyme, isolated from the ascomycete *Endothia parasitica* and a member of the peptidase family A1 as well, has a broad specificity similar to that of pepsin A. This enzyme prefers hydrophobic residues at P₁ and P₁'. **See also specific enzyme**

- T. D. Meek (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 327.
 N. D. Rawlings & A. J. Barrett (1995) *Meth. Enzymol.* **248**, 105.
 E. Lolis & G. A. Petsko (1990) *Ann. Rev. Biochem.* **59**, 597.

Selected entries from *Methods in Enzymology* [vol, page(s)]:
 Active site, structure, **241**, 214; catalytic mechanism, **241**, 223-224; crystal structure, **241**, 214, 216; comparative studies with HIV protease [catalytic properties, **241**, 205-224; evolutionary relationships, **241**, 196-197; screening for HIV-1 protease inhibitors, **241**, 318-321; structure, **241**, 254-257, 280; substrate specificity, **241**, 255, 283].

β-ASPARTYL-N-ACETYLGLUCOSAMINIDASE

This enzyme [EC 3.2.2.11], also referred to as 1-aspartamido-β-N-acetylglucosamine amidohydrolase, catalyzes the hydrolysis of 1-β-aspartyl-N-acetylglucosaminylamine to yield N-acetylglucosamine and asparagine.

- E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.
 E. H. Eylar & M. Murakami (1966) *Meth. Enzymol.* **8**, 597.

β-ASPARTYLDIPEPTIDASE

This enzyme [EC 3.4.19.5], now known as β-aspartyl-peptidase, is a mammalian cytosolic protein that catalyzes the hydrolysis of a β-linked aspartyl residue from the N-terminus of a polypeptide. Other isopeptide linkages (e.g., a γ-glutamyl linkage) are not hydrolyzed by this enzyme.

- E. E. Haley (1970) *Meth. Enzymol.* **19**, 730 and 737.

ASPARTYL-tRNA SYNTHETASE

This enzyme [EC 6.1.1.12], also known as aspartate:tRNA ligase, catalyzes the reaction of aspartate with ATP and tRNA^{Asp} to generate aspartyl-tRNA^{Asp}, AMP, and diphosphate (or, pyrophosphate). **See also Aminoacyl-tRNA Synthetases**

- E. A. First (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 573.
 J. Cavarelli, G. Eriani, B. Rees, M. Ruff, M. Boeglin, A. Mitschler, F. Martin, J. Gangloff, J.-C. Thierry & D. Moras (1994) *EMBO J.* **13**, 327.
 P. Schimmel (1987) *Ann. Rev. Biochem.* **56**, 125.

ASSOCIATION

The combining of two or more substances or molecular entities to yield a single substance or molecular entity, a process that involves either covalent or noncovalent bonding. Included in this definition is the formation of ion pairs from free ions, the noncovalent aggregation of monomers to form polymeric structures or complexes, as well as colligation. The opposite of association is dissociation.

ASSOCIATION CONSTANT

The equilibrium constant (often symbolized by K_a) describing the association of two or more compounds, ions, or atoms into a new compound or complex. For example, the association constant, or binding constant, for oxygen binding to myoglobin is [myoglobin-O₂]/[myoglobin][O₂], whereas the average association constant for hapten-antibody reaction is [hapten-bound antibody]/[free antibody][free hapten]. The reciprocal of the association constant is the dissociation constant. **See Dissociation Constant; Stability Constant**

ASTACIN

This enzyme [EC 3.4.24.21] is a zinc-dependent digestive endopeptidase from the cardia of the crayfish *Astacus*

fluviatilis. It catalyzes the hydrolysis of peptide bonds in substrates having five or more amino acids.

W. L. Mock (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 425.

W. Stöcker & R. Zwilling (1995) *Meth. Enzymol.* **248**, 305.

ASYMMETRIC INDUCTION

The process by which a stereochemically inactive center is converted to a specific stereoisomeric form. In most cases, the reacting center is prochiral. Such processes can occur with reactions involving an optically active reagent, solvent, or catalyst (e.g., an enzyme). The reaction produced by such a process is referred to as an enantioselective reaction¹⁻³. In principle, use of circularly polarized light in photochemical reactions of achiral reactants might also exhibit asymmetric induction. However, reported enantioselectivities in these cases^{4,5} have been very small.

¹J. D. Morrison (1985) *Asymmetric Synthesis*, vol. **5**, Academic Press, New York.

²K. Tomioka (1990) *Synthesis*, 541.

³G. Consiglio and R. Waymouth (1989) *Chem. Rev.* **89**, 257.

⁴O. Buchardt (1974) *Angew. Chem. Int. Ed. Engl.* **13**, 179.

⁵L. D. Barron (1986) *J. Amer. Chem. Soc.* **108**, 5539.

ASYMMETRY PARAMETER

A parameter, η , used for describing nonsymmetric fields in nuclear quadrupole resonance spectroscopy. It is defined as $\eta = (q_{xx} - q_{yy})/q_{zz}$ in which q_{xx} , q_{yy} and q_{zz} are the components of the field gradient q (which is the second derivative of the time-averaged electric potential) along the x -, y - and z -axes. By convention, q_{zz} refers to the largest field gradient, q_{yy} to the next largest, and q_{xx} to the smallest when all three values are different.

M. W. G. de Bolster (1997) *Pure Appl. Chem.* **69**, 1251.

ASYMPTOTE

A straight line whose perpendicular distance from a curve becomes progressively smaller as the distance from the origin at [0,0] becomes greater. For example, in a plot of velocity *versus* [Substrate Concentration] for an enzyme-catalyzed reaction, the asymptote reaches the maximal velocity when the enzyme molecules become saturated with substrate.

ATMOSPHERE

1. A unit of pressure, symbolized by atm, equal to 760 mmHg or 101,325 pascals (or, newtons per square meter) or 1.01325 bar. At sea level, atmospheric pressure

roughly fluctuates around a value of one atm. 2. The air surrounding the Earth. The composition of dry atmospheric air, in percent volume is: N₂, 78.084; O₂, 20.946; Ar, 0.934; CO₂, 0.033; and trace amounts of Ne, He, Kr, Xe, and CH₄. The density of atmospheric air at sea level is 1220 g/m³, while at an altitude of 10 km, the density is 425 g/m³.

ATOM PERCENT EXCESS

A measure of the amount of a stable isotopic label that exceeds its natural abundance in unlabeled tracee. This is most directly accomplished using an ion ratio mass spectrometer to measure the ratio of ion currents for isotopomers such as ¹²CO₂ at mass 44 and ¹³CO₂ at mass 45. From the difference between the ion current ratio for a sample (r_{sample}) and the ion current ratio for a reference gas ($r_{\text{reference}}$), the atom percent excess (APE) can be estimated. **See** *Tracer/Tracee Ratio; Compartmental Analysis; Isotope Exchange Kinetics*

ATOMIC MASS UNIT

A unit of mass (abbreviated by a.m.u. or amu) equal to 1/12-th the mass of the carbon atom of mass number 12 (¹²C). Thus, 1 a.m.u. = $m_u = (1/12)(^{12}\text{C mass}) = 1.6605402 \times 10^{-27}$ kg. Hence, carbon-12 has a mass of exactly 12.000000 a.m.u.

ATOMIC ORBITAL

The wavefunction of an electron associated with an atomic nucleus. The orbital is typically depicted as a three-dimensional electron density cloud. If an electron's azimuthal quantum number (l) is zero, then the atomic orbital is called an s orbital and the electron density graph is spherically symmetric¹. If l is one, there are three spatially distinct orbitals, all referred to as p orbitals, having a dumb-bell shape with a node in the center where the probability of finding the electron is extremely small. (Note: For relativistic considerations², the probability of an electron residing at the node cannot be zero.) Electrons having a quantum number l equal to two are associated with d orbitals.

¹P. S. C. Matthews (1986) *Quantum Chemistry of Atoms and Molecules*, Cambridge Univ. Press, Cambridge.

²R. E. Powell (1968) *J. Chem. Ed.* **45**, 558.

ATOMIZATION

The conversion of a molecule into its constituent atoms.

ATP

Adenosine 5'-triphosphate, the major energy-storing nucleotide within living organisms. The standard Gibbs free energy of hydrolysis is -30.5 kJ/mol.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

^{32}P -labeled [detection, **238**, 40, 48, 50-51; disposal, **238**, 53; half-life, **238**, 40; α -phosphate labeling specificity, **238**, 39-40; quality of preparations, **238**, 40]; regeneration systems, **238**, 34-35; stability, **238**, 352-354; tritiated [assay advantages, **238**, 38; detection, **238**, 39, 48, 50-51; disposal, **238**, 52-53; half-life, **238**, 38; stability of label, **238**, 39].

ATPase, Ca^{2+}

This class of enzymes [EC 3.6.1.38] includes calcium pumps that catalyze hydrolysis of ATP, producing ADP and orthophosphate; this hydrolysis is coupled to the transport of Ca^{2+} ions.

F. Wuytack & L. Raeymaekers (1992) *J. Bioenerg. Biomembr.* **24**, 285.

W. P. Jencks (1992) *Biochem. Soc. Trans.* **20**, 555.

G. Inesi, D. Lewis, D. Nikic, A. Hussain & M. E. Kirtley (1992) *Adv. Enzymol.* **65**, 185.

W. P. Jencks (1980) *Adv. Enzymol.* **51**, 75.

W. Hasselbach (1974) *The Enzymes*, 3rd ed., **10**, 431.

ATPase, F_1F_0

See F_1F_0 ATPase; *Binding Change Mechanism*; CF_1CF_0 ATPase

ATPase, H^+/K^+

This class of enzymes [EC 3.6.1.36] (also known as the hydrogen/potassium-exchanging ATPase, the potassium-transporting ATPase, proton pump, and the gastric H^+/K^+ ATPase) catalyzes the hydrolysis of ATP to ADP and orthophosphate, coupled with the exchange of H^+ and K^+ ions. The gastric mucosal enzyme has been the best characterized.

M. Besançon, J. M. Shin, F. Mercier, K. Munson, E. Rabon, S. Hershey & G. Sachs (1992) *Acta Physiol. Scand.* **146**, 77.

G. Sachs, M. Besançon, J. M. Shin, F. Mercier, K. Munson & S. Hershey (1992) *J. Bioenerg. Biomembr.* **24**, 301.

Q. Al-Awqati (1986) *Ann. Rev. Cell Biol.* **2**, 179.

C. Tanford (1983) *Ann. Rev. Biochem.* **52**, 379.

ATPase, Na^+/K^+

Nerve stimulation results in a net influx of sodium ions, and normal conditions are restored by the outward transport of sodium ions against an electrochemical gradient. While several earlier workers had identified ATPases in the sheath of giant squid axons, it was Skou^{1,2} who first connected the sodium, potassium ATPase [EC 3.6.1.37] with the ion flux of neurons. This discovery culminated

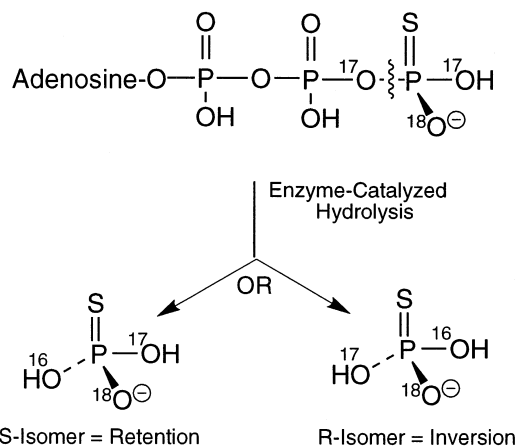
in his sharing the Nobel Prize in 1997, some forty years after his seminal work, with Boyer and Walker who were cited for their studies on the mechanism and structure of ATP synthase. Later studies by Kanazawa *et al.*³ measured the rates of formation and decomposition of the phosphorylated intermediate in the reaction catalyzed by sodium, potassium ATPase.

¹J. C. Skou (1957) *Biochim. Biophys. Acta* **23**, 394.

²J. C. Skou (1960) *Biochim. Biophys. Acta* **42**, 6.

³T. Kanazawa, M. Saito & Y. Tonomura (1970) *J. Biochem. (Tokyo)* **67**, 693.

ATPase (Stereochemistry of Phosphoryl Transfer)



Stereochemical Course of ATPase Reactions

[^{16}O , ^{17}O , ^{18}O , S]ATP γ S provides a powerful means for answering a fundamental question in ATPase catalysis, namely whether a particular ATPase is transiently phosphorylated during the catalytic reaction cycle. Formation and hydrolysis of an E-P intermediate requires two in-line nucleophilic substitution reactions, each suffering inversion of stereochemical configuration at the phosphorus atom of the phosphoryl group. This results in overall conservation of stereochemical configuration. However, when an ATPase catalyzes direct transfer to water, there will be an inversion of configuration.

M. R. Webb (1982) *Meth. Enzymol.* **87**, 301.

ATP CITRATE LYASE

This enzyme [EC 4.1.3.8], also known as ATP-citrate (*pro-S*)-lyase and citrate cleavage enzyme, catalyzes the reaction of citrate with ATP and coenzyme A to yield oxaloacetate, acetyl-CoA, ADP, and orthophosphate.

The multiprotein enzyme complex can be dissociated into smaller components of which two are identical with citryl-CoA lyase and citrate:CoA ligase.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

P. A. Srere (1975) *Adv. Enzymol.* **43**, 57.

L. B. Spector (1972) *The Enzymes*, 3rd ed., **7**, 357.

ATP:GLUCOSE-1-PHOSPHATE ADENYLYLTRANSFERASE

This enzyme [EC 2.7.7.27], also known as ADP–glucose synthase and ADP–glucose pyrophosphorylase, catalyzes the reaction of α -D-glucose 1-phosphate with ATP to produce ADP–glucose and pyrophosphate.

A. M. Smith, K. Denyer & C. Martin (1997) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 67.

J. Preiss (1973) *The Enzymes*, 3rd ed., **8**, 73.

ATP & GTP DEPLETION

Because ATP and GTP are required in many biosynthetic and mechanochemical reactions, the ability to manipulate their concentrations is particularly important in many kinetic studies. The three enzymes most commonly employed for this purpose are hexokinase, phosphofructokinase, and alkaline phosphatase. Each of these enzymes offers advantages and disadvantages, and one must consider the design requirements for any particular experiment before using one of these to deplete ATP or GTP.

Yeast hexokinase (1–2 international units per mL reaction solution) acting for 20 min in the presence of 10 mM D-glucose has proven to be quite effective in depleting ATP at levels not exceeding 1 mM. At pH 7 in the presence of 1–2 mM uncomplexed magnesium ion, the equilibrium constant for the hexokinase reaction is about 1500; thus, one can anticipate substantial conversion of 1 mM ATP, as indicated by the following equation:

$$\frac{[\text{ADP}]}{[\text{ATP}]} = 1500 \frac{[\text{D-Glucose}]}{[\text{D-Glucose 6-P}]}$$

Using the expression $\{x/(1\text{mM} - x) = 1500 (\sim 10\text{ mM})/x\}$, one can solve the quadratic equation or iteratively estimate the value of x that satisfies the equation. This would place x at about 0.99994, indicating that 99.99% of the ATP would be depleted if the hexokinase reaction equilibrium is reached.

As written, the above expression helps one to recognize that hexokinase can be used to maintain the $[\text{ADP}]/$

$[\text{ATP}]$ concentration ratio at any desired value, even in the presence of a slow ATPase activity by the enzyme under investigation. By knowing the equilibrium constant for the hexokinase reaction at the pH for a particular experiment, one can merely use D-glucose and D-glucose 6-phosphate in the ratio that will yield the desired $[\text{ADP}]/[\text{ATP}]$ in the presence of sufficient hexokinase to reach equilibrium.

GTP is also a substrate for yeast hexokinase, but phosphofructokinase (PFK) acts on GTP more efficiently than hexokinase. Typically, one uses 1–2 international units PFK per milliliter of reaction solution (for 20 min in the presence of 5 mM D-fructose 6-phosphate) to deplete GTP initially present at concentrations up to 1 mM.

Because alkaline phosphatase converts ATP or GTP to their respective nucleosides, use of alkaline phosphatase to deplete ATP or GTP should be reserved for those cases where this does not present a problem¹. Furthermore, alkaline phosphatase is potently inhibited by orthophosphate; so higher than would be anticipated amounts of enzyme are required for efficiently depleting ATP or GTP.

¹D. L. Purich & R. K. MacNeal (1978) *FEBS Lett.* **96**, 83.

ATP PHOSPHORIBOSYLTRANSFERASE

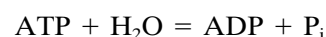
This enzyme [EC 2.4.2.17], also known as phosphoribosyl-ATP pyrophosphorylase, catalyzes the reaction of ATP with 5-phospho- α -D-ribose 1-diphosphate to generate 1-(5-phospho-D-ribosyl)-ATP and pyrophosphate.

W. D. L. Musick (1981) *Crit. Rev. Biochem.* **11**, 1.

R. G. Martin, M. A. Berberich, B. N. Ames, W. W. Davis, R. F. Goldberger & J. D. Yourno (1971) *Meth. Enzymol.* **17B**, 3.

ATP, THERMODYNAMICS OF HYDROLYSIS

Hydrolysis of the terminal (or, β - γ) phosphoanhydride bond of ATP is the primary thermodynamic driving force in metabolism. Written as a biochemical equation at a specified pH, ATP hydrolysis can be represented as:



The apparent equilibrium constant, $K'_{(\text{T,P,pH})}$ for this process is :

$$K'_{(\text{T,P,pH})} = [\text{ADP}] [\text{P}_i] / \{[\text{ATP}]c^\circ\}$$

where c° is the standard state concentration of 1 M. Biochemists can often advantageously describe biochemically relevant equations in terms of reactants at specified pH and concentrations of free metal ions that are bound by reactant species. The relevant equilibrium constant is written as K' . A reconceptualization of the thermodynamics of biochemical processes, based on equilibria written in terms of reactants at a specified pH and concentrations of free metal ions bound by reactant species^{1,2}.

In chemical thermodynamics, temperature and pressure are specified and a system is defined in terms of species, using the change in the Gibbs free energy:

$$dG = -SdT + VdP + \sum_{i=1}^N \mu_i dn_i$$

where G , S , T , V , and P are the Gibbs free energy, entropy, temperature, volume, and pressure. The terms μ_i , n_i , and N are the chemical potential of species i , the amount of species i , and the number of species in the system.

In the new biochemical thermodynamic approach, the various protonated species of a metabolite are treated as pseudoisomeric forms having the same transformed chemical potential μ_i at a specified pH, such that they are collected in the term n_i' (which is equal to $\sum n_i$).

$$(dG)_{\text{pH}} = -S' dT + VdP + \sum_{i=1}^{N'} \mu_i' dn_i'$$

¹R. A. Alberty (1994) *Biochim. Biophys. Acta* **1207**, 1.

²R. A. Alberty (1996) *Eur. J. Biochem.* **240**, 1. (A useful account of the IUPAC-IUBMB recommendations for nomenclature and tables in biochemical thermodynamics)

ATROPISOMERS

Isomers that lack internal planes of symmetry or points of symmetry (hence, are chiral) due to a prevention (or significant hindrance) to rotation about a single bond.

M. Oki (1983) *Top. Stereochem.* **14**, 1.

ATROXASE

This zinc-dependent nonhemorrhagic endopeptidase [EC 3.4.24.43] isolated from the venom of the rattlesnake *Crotalus atrox* is a peptidase family M12B member that catalyzes the hydrolysis of the His⁵—Leu⁶, Ser⁹—His¹⁰,

His¹⁰—Leu¹¹, Ala¹⁴—Leu¹⁵, and the Tyr¹⁶—Leu¹⁷ bonds in the insulin B chain. It will also act on fibrinogen.

J. B. Bjarnason & J. W. Fox (1995) *Meth. Enzymol.* **248**, 345.

ATTENUANCE FILTER

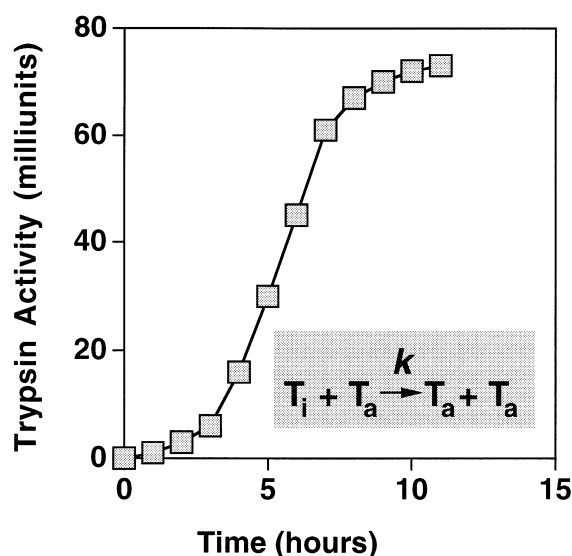
A substance or device (synonymous with neutral density filter) that reduces the radiant power of incident radiation by a constant factor over all wavelengths within the instrument's operating range. **See also** *Filter*

ATTO-

A prefix, symbolized by a, used in submultiples of units and corresponding to a value of 10^{-18} .

AUTOCATALYSIS

The self-activating process for converting an inactive catalyst, I, to its active form, A. Such processes typically display a discernible lag-phase followed by accelerated conversion of I to A. Zymogen activation can display autocatalysis, as exemplified by the conversion of pepsinogen to active pepsin at low pH. Many growth processes can also be described via an autocatalysis curve.



The time-course of an autocatalytic process. Note that during the early stages the amount of active catalyst is low, but changes dramatically as a function of time.

The process is a special case of a second-order autocatalytic reaction in which a product or reaction intermediate functions as a catalyst. In the early stages of the reaction, the rate-of-reaction of autocatalytic processes increases with the time from its initial value, as seen in the plot

above. If a reaction of A to P is autocatalytic such that the rate expression is

$$v = -\frac{d[A]}{dt} = k[A][P]$$

where t is time and k is a second-order rate constant, then the integrated rate expression is

$$\frac{1}{[A_0] + [P_0]} \ln \frac{[A_0][P]}{[P_0][A]} = kt$$

Chain reactions and explosions are also autocatalytic processes. **See also** *Bifurcation Theory; Prion Plaque Formation*

AUTOINHIBITION

A term used to describe a characteristic of certain enzymes that have intermolecular interactions which result in the inhibition of the enzyme's catalytic activity¹. For example, the calmodulin-binding domain of the human plasma-membrane Ca^{2+} pump overlaps with an autoinhibitory domain^{2,3}. In the absence of calmodulin and phosphorylation by protein kinase C, the Ca^{2+} pump is inhibited. Binding of calmodulin to the calmodulin-binding domain results in activation of the pump. A second such case is the flavin-containing heme protein nitric oxide synthase⁴ (NOS) which generates $\text{NO}\cdot$ in neurons from L-arginine, NADPH, and O_2 . The enzyme autoinactivates upon reacting with steady-state concentrations of $\text{NO}\cdot$ to form a ferrous-nitrosyl complex during aerobic catalysis. Other examples of autoinhibition include calmodulin-dependent cAMP phosphodiesterase⁵ and smooth-muscle myosin light-chain kinase⁶. In a sense, all enzymes that can undergo activation by intermolecular effectors display the phenomena of autoinhibition. Activation is simply a depression of an inhibitory or partially inhibitory aspect of the protein. However, the term autoinhibition remains quite useful in describing domains in such proteins as well as the conformational changes and binding equilibria associated with these phenomena. **See also** *Activation*

¹P. James, T. Vorherr & E. Carafoli (1995) *Trends in Biochem. Sci.* **20**, 38.

²M. Zurini, J. Krebs, J. T. Penniston & E. Carafoli (1984) *J. Biol. Chem.* **259**, 618.

³P. James, T. Vorherr, J. Krebs, A. Morelli, G. Castello, D. J. McCormick, J. T. Penniston, A. DeFlora & E. Carafoli (1989) *J. Biol. Chem.* **264**, 8289.

⁴H. M. Abu-Soud, J. Wang, D. L. Rousseau, J. M. Fukuto, L. J. Ignarro & D. J. Stuehr (1995) *J. Biol. Chem.* **270**, 22997.

⁵C. B. Klee (1980) in *Protein Phosphorylation and Bioregulation* (eds., G. Thomas, E. J. Podesta & J. Gardon), p. 61, Karger.

⁶A. M. Edelman, K. Takio, D. K. Blumental, R. S. Hansen, K. A. Walsh, K. Titani & E. G. Krebs (1985) *J. Biol. Chem.* **260**, 11275.

AUTONOMOUS CATALYTIC DOMAIN

A catalytic function or region of a multienzyme system that is associated with a domain that is separate and distinct from the domain(s) responsible for the other catalytic function(s).

Nomenclature Comm., IUB (1989) *Eur. J. Biochem.* **185**, 485.

AUTOPHAGY

Intracellular digestion occurring in living cells in which regions of the cytoplasm and/or cell components are digested. Many authors consider autophagy to include the normal processes occurring within the lysosome. Heterophagy then refers to intracellular digestion of exogenous materials brought into the cell by pinocytosis and phagocytosis. Some authors make a distinction between autophagy and autolysis, whereas others consider autolysis as a form of autophagy. In this context, autolysis is a process that occurs specifically in dying or dead cells. It is programmed and controlled cell destruction.

AUTOPHOSPHORYLATION

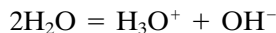
The self-phosphorylation process catalyzed by many protein kinases as part of the regulatory mechanism for their own activation. Because true autophosphorylation is a unimolecular reaction involving enzyme both as catalyst and phosphoryl acceptor, the fraction of autophosphorylated enzyme at any time after addition of ATP (or another phosphoryl donor) will be independent of the initial concentration of the enzyme. This criterion was first applied to the autophosphorylation of cardiac muscle cyclic AMP-stimulated protein kinase¹, now designated protein kinase A (PKA). At a fixed concentration of MgATP^{2-} , the fraction of autophosphorylated protein will follow the first-order rate laws, $[A]/[A_0] = e^{-kt}$, where k is a first-order rate constant.

¹J. A. Todhunter & D. L. Purich (1977) *Biochim. Biophys. Acta* **485**, 87.

AUTOPROTOLYSIS

Any process in which a molecule acts as an acid in a reaction with another identical molecule acting as a base. Autoprotolytic reactions can only occur with amphiprotic molecular entities.

The best known example of autoprotolysis is the ionization of water:



The equilibrium constant for this reaction is given by the expression

$$K_{\text{eq}} = [\text{H}_3\text{O}^+][\text{OH}^-]/[\text{H}_2\text{O}]^2$$

Because the activity (or concentration) values for H_3O^+ and OH^- are much smaller than the corresponding value for H_2O (*i.e.*, the solvent), the solvent concentration remains unchanged. (The density of pure water at 25°C is about 0.997 g/ml. Hence, the concentration of pure water at 25°C is (0.997 g/mL)(1000 mL/L)(18.0153 g/mol) = 55.3 M. This value is clearly larger than $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ and the value of $[\text{H}_2\text{O}]^2$ can be regarded as a constant.) Thus, the expression for the autolysis of water (or, the ionization of water) can be simplified to $K_{\text{eq}}[\text{H}_2\text{O}]^2 = K_{\text{w}} = [\text{H}_3\text{O}^+][\text{OH}^-]$. This value is equal to 10^{-14} M^2 at 25°C. **See** *Temperature Effects on pK_{w}*

Autoprotolysis constants exist for any amphiprotic solvent and can be determined from electric conductivity studies of the solvent and solutions. A few examples¹ include:

Solvent	Autoprotolysis Constant ^a
H_2O	$[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$
NH_3	$[\text{NH}_4^+][\text{NH}_2^-] = 10^{-33}$
H_2SO_4	$[\text{H}_3\text{SO}_4^+][\text{HSO}_4^-] = 2 \times 10^{-4}$
HCOOH	$[\text{HC(OH)}_2^+][\text{HCOO}^-] = 6 \times 10^{-7}$
CH_3COOH	$[\text{CH}_3\text{C(OH)}_2^+][\text{CH}_3\text{COO}^-] = 10^{-13}$
CH_3OH	$[\text{CH}_3\text{OH}_2^+][\text{CH}_3\text{O}^-] = 2 \times 10^{-17}$
$\text{CH}_3\text{CH}_2\text{OH}$	$[\text{CH}_3\text{CH}_2\text{OH}_2^+][\text{CH}_3\text{CH}_2\text{O}^-] = 3 \times 10^{-20}$

^aAll values are those measured at 25°C, except for NH_3 (−33°C).

¹L. Pauling (1970) *General Chemistry*, 3rd ed., Freeman, San Francisco (Dover republication, 1988).

AUTOPROTOLYSIS CONSTANT

The product of the molar concentrations (or, more accurately, the activities) of the species produced as a result of autoprotolysis. The autoprotolysis constant for water is K_{w} , equal to $[\text{H}_3\text{O}^+][\text{OH}^-]$, or 1.0×10^{-14} at 25°C. It is a temperature-dependent constant, increasing with

increasing temperature (at 37°C, $K_{\text{w}} = 2.51 \times 10^{-14}$). **See** *Autoprotolysis*

AVOGADRO'S NUMBER

1. The number of atoms, symbolized by either L or N_{A} , in 0.012 kg of pure carbon-12, equivalent to 6.0221367×10^{23} per mole. 2. The number of molecules in one gram-molecular weight (1 mole) of any compound.

AZEOTROPE

For a solution or mixture of two or more distinct liquid components, an azeotrope is that composition (typically measured in mole fractions or percent weight and referred to as the azeotropic solution) with which there is either a maximum point (a negative azeotrope) or a minimum point (a positive azeotrope) in a boiling point *versus* composition diagram at constant pressure.

AZIDO PHOTOAFFINITY REAGENTS

Any member of a class of substances that generate a highly reactive nitrene species upon absorption of photic energy.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

2-Azidoadenosine synthesis, **237**, 84; ^{32}P -labeled 2-azido-ADP-ribose, intra- and intermolecular transfer from G protein a subunit carboxy terminus, **237**, 95-99; 2-azido-AMP [nonradioactive, synthesis, **237**, 85, 93; ^{32}P -labeled, chemical synthesis, **237**, 82-86; in preparation of 2-azido- ^{32}P]NAD⁺, **237**, 82]; 4-azidoanilido-GTP [G protein labeling, **237**, 100-110; preparation, **237**, 101-103; structure, **237**, 101; tubulin labeling, **237**, 106-107]; P^3 -(4-azidoanilido)- P^1 -5'-guanosine triphosphate **195**, 282; 2-azido-ATP [^{32}P -labeled, enzymatic synthesis, **237**, 87-89, 92; in preparation of 2-azido- ^{32}P]NAD⁺, **237**, 82]; 3'-(*p*-azido-*m*-iodophenylacetyl)-2',5'-dideoxy-adenosine [adenylyl cyclase inactivation, **238**, 68-71; photocoupling, **238**, 67-68; radioactive labeling, **238**, 68; solubility, **238**, 68; synthesis, **238**, 65]; 3'-(*p*-azido-*m*-iodophenylbutyryl)-2',5'-dideoxy-adenosine [adenylyl cyclase inactivation, **238**, 68-71; photocoupling, **238**, 67-68; radioactive labeling, **238**, 68; solubility, **238**, 68; synthesis, **238**, 65]; 2-azido-NAD [as G-protein structure probe, **237**, 71; determination of kinetic constants, **237**, 91-92; nonradioactive, synthesis, **237**, 86-87; ^{32}P -labeled (enzyme cleavage analysis, **237**, 91; synthesis, **237**, 81-95)]; 5-azido-UDPGlucose [^{32}P -labeled, applications in glycobiology, **230**, 334-339; synthesis, **230**, 330-334]; 5-azido-UDPGlucuronic acid [^{32}P -labeled, applications in glycobiology, **230**, 334-339; synthesis, **230**, 330-334]; 4'-azidowarfarin, **206**, 57-60

AZURIN

A bacterial electron-transfer protein, dark blue in color, that contains a type 1 copper site.

B

BACTERIAL LEADER PEPTIDASE I

This enzyme [EC 3.4.21.89], also known as signal peptidase I and phage-procoat-leader peptidase, catalyzes the hydrolysis of N-terminal leader sequences from secreted and periplasmic protein precursors. It acts on a single bond Ala–Ala in the m-13 phage procoat protein and creates the signal (leader) peptide and coat protein. It is a member of the peptidase family S26 but is unaffected by inhibitors of most serine peptidases.

W. R. Tschantz & R. E. Dalbey (1994) *Meth. Enzymol.* **244**, 285.

BAIT-AND-SWITCH MECHANISM

A strategy for raising of catalytic antibodies for a desired reaction type that employs a hapten as “bait” to induce the production of antibodies that are most likely to possess certain side-chain residues that may be important in binding and/or catalysis. One then “switches” to the substrate.

R. A. Lerner & S. J. Benkovic (1988) *Bioessays* **9**, 107.
K. D. Janda, M. I. Weinhouse, D. M. Schloeder, R. A. Lerner & S. J. Benkovic (1990) *J. Amer. Chem. Soc.* **112**, 1274.

BAND-PASS FILTER

A substance or device which allows the transmission of electromagnetic radiation of a specified wavelength range and does not allow transmission of radiation outside that range. **See also** *Filter*; *Cut-off Filter*

BASAL RATE

The residual or unstimulated activity of (a) an enzyme reaction, (b) a series of reactions, or (c) the energy metabolism of an individual organism. Although one most frequently considers basal reaction rates in enzyme ki-

netics, any unfacilitated transport or leakage of ions and/or metabolites across a membrane barrier can also be considered a basal activity in studies of transporter kinetics.

BASAL METABOLIC RATE (or B.M.R.). An index of metabolic activity of an individual organism, usually measured by the rate of oxygen consumption while in a resting, nonsleeping state. The basal metabolic rate can also be determined as the rate of heat evolution in a state of resting and without recent consumption of foodstuffs. Because temperature control is of vital importance to mammals, the basal metabolic rate is roughly proportional to body surface area. A young healthy adult male (mass 70 kg) typically has a basal metabolic rate of 300–350 kJ/hour, corresponding to about 70–80 watts.

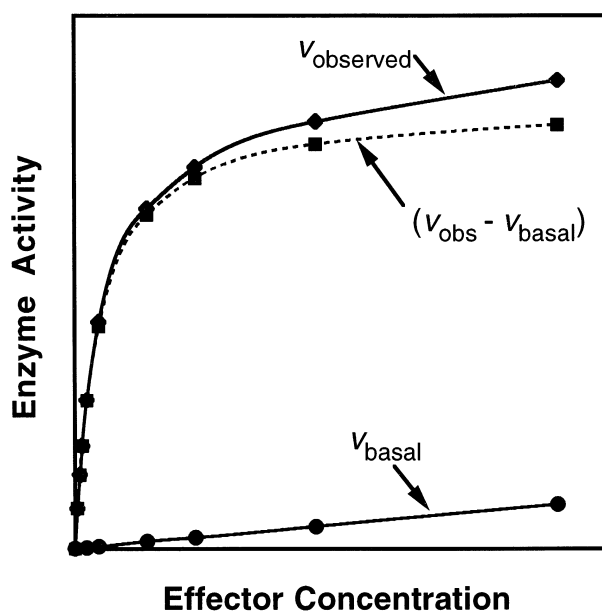
ENZYME KINETICS. Most enzyme-catalyzed reactions proceed at rates that are much faster than their nonenzymatic counterparts. Nonetheless, there are a number of conditions that account for basal rates for biochemical reactions in the absence, or apparent absence, of the enzyme under consideration:

(a) Some reactions proceed at significant uncatalyzed rates, including the hydration of carbon dioxide and the formation or decomposition of certain cyclic lactones. In these cases, one must determine the basal rate (in the absence of added enzyme) over the same substrate concentration range used for enzymatic rate measurements. The difference (*i.e.*, $v_{\text{enzymatic}} - v_{\text{nonenzymatic}}$) must then be used in any determination of enzyme rate behavior. Note that nonenzymatic processes can be catalyzed by certain buffers and/or metal ions. The judicious choice of components in an assay solution can often reduce the basal reaction rate.

(b) A substrate or other component in the reaction assay may be contaminated with the enzyme of interest or some other enzyme that affects the accurate determination of enzyme reaction rate. This can be a serious problem when carrying out investigations of posttranslational modification reactions involving protein substrates. Care must be exercised to ascertain whether the protein substrate has any endogenous enzymatic activity. This often can be accomplished merely by omitting the primary enzyme and determining whether any posttranslational modification has occurred. (One should also confirm that the apparent kinetics of posttranslational modification are unaffected by the denaturation or proteolysis of the protein substrate)

(c) A coupled enzyme assay may introduce trace amounts of the enzyme of interest as a contaminant in one or more of the auxiliary enzymes added to the reaction.

(d) A nucleotide-regenerating system may likewise introduce trace amounts of the enzyme of interest as a contaminant.

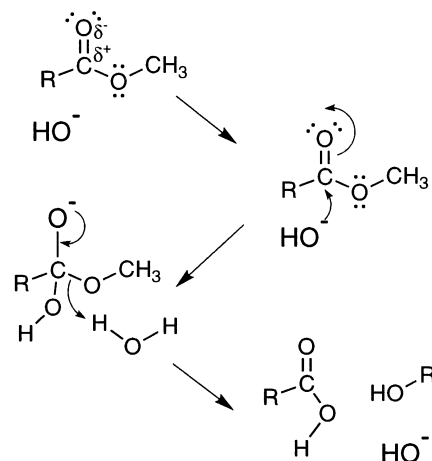


Response of an enzyme reaction in which the basal rate is proportional to the concentration of an effector. In this case, unless the basal rate is subtracted point-by-point from each of the observed velocities, the system will never achieve saturation.

ENZYME ACTIVATION. To characterize enzyme activation quantitatively, one must often correct for the

presence of any basal reaction rate or basal activity in the unstimulated state. Typically, one uses $(v_{\text{stimulated}} - v_{\text{basal}})$ or $(v_{\text{stimulated}} - v_{\text{basal}})/(v_{\text{maximally-stimulated}} - v_{\text{basal}})$ for this purpose in studies of enzyme activation.

BASE CATALYSIS



Base Catalysis of Ester Hydrolysis

Any chemical reaction that is accelerated through the catalytic participation of base. **See** Acidity; Brønsted Relation; General Acid Catalysis; General Base Catalysis

BASE PAIRING

The hydrogen-bond pairing interactions between pyrimidine and purine bases in DNA and RNA. **See** DNA Breathing

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Catalysts of proton exchange [concentration in base pair lifetime determination, **261**, 405-406; efficiency, **261**, 388; external proton acceptor, **261**, 389-391; intrinsic catalysis, **261**, 390-391; mechanism, **261**, 386; pK values, **261**, 387; selection, **261**, 404-405]; kinetics of opening and proton exchange [mechanism of proton exchange, **261**, 385-386, 388-389; pH dependence, **261**, 385; time scale of exchange, **261**, 391, 393]; proton exchange measurement by NMR [base pair lifetime determination, **261**, 404-406; B-DNA, **261**, 406-407; B'-DNA, **261**, 408, 410-411; cross-relaxation effects, **261**, 396-397; cytidine imino proton exchange, **261**, 411-413; data processing, **261**, 401, 403; dissociation constant determination, **261**, 403-404; imino proton chemical shift, **261**, 404; luzopeptin-DNA complex, **261**, 411; magnetization transfer from water protons, **261**, 393-395, 399; proton/deuterium exchange in real time, **261**, 391-393; relaxation rate increments, **261**, 395, 401; sample preparation, **261**, 391-392; solvent signal suppression, **261**, 398-399].

BASE STACKING

The diffusion-controlled bonding interactions between successive heterocyclic bases in oligo- and polynucleo-

tides that arise from extended, planar π -orbital interactions. The resulting structures usually exhibit a 3.4 Å plane-to-plane distance within the same chain. While the stabilization arising from base stacking is small (ΔG -1 kcal/mol), the additive effect of many such interactions can profoundly influence nucleic acid structure as well as interactions with enzymes that use nucleic acids as substrates. Generally speaking, purine-purine base stacking is more energetically favored.

BASICITY OF CARBONYL OXYGENS

As noted elsewhere in this Handbook, carbonyl groups are polarized: the carbon atom has a partial positive charge, whereas the carbonyl oxygen displays partial negative charge character. In fact, the carbonyl oxygen has weakly basic properties; hence, protonated carbonyls are usually strong acids, as evidenced by the conjugate acids of acetamide (pK_a 0), water (pK_a -1.7), methanol (pK_a -2), acetic acid (pK_a -6), ethylacetate (pK_a -6 to -7), and acetaldehyde (pK_a -8). The protonated form of a carbonyl oxygen is often called an oxonium salt, and such species do indeed factor into the mechanisms of acid-catalyzed reactions involving carbonyl-containing compounds.

BATHOCHROMISM

A shift (also known as a "red shift") in a substance's electronic absorption spectrum toward longer wavelengths, as a consequence of a substituent, solvent, environment, or other effect. The opposite of a bathochromic shift is referred to as a hypsochromic shift.

BECQUEREL

The SI unit of measure for radioactivity. One becquerel (Bq) is equal to one disintegration per second (dps), thereby corresponding to about 2.703×10^{-11} curie.

BEER-LAMBERT LAW

A function relating the absorbance of light relative to concentration and thickness of a sample. The absorbance of light is directly proportional to the concentration of the absorbing chromophore multiplied by the thickness of the substance, containing the chromophore, through which the light is being transmitted. Thus, $A = \epsilon bc$ where A is the absorbance, ϵ is the molar extinction coefficient (which varies with wavelength), b is the thickness of the

solution, and c is the concentration. **See** *Absorption Spectroscopy*

BENZENE 1,2-DIOXYGENASE

This enzyme [EC 1.14.12.3], also known as benzene hydroxylase, catalyzes the NADH-linked reaction of benzene and dioxygen to produce *cis*-1,2-dihydrobenzene-1,2-diol and NAD^+ . This multiprotein complex contains an iron-sulfur flavoprotein, an iron-sulfur oxygenase, and ferredoxin. **See also** *Phthalate Dioxygenase*

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

P. J. Geary, J. R. Mason & C. L. Joannou (1990) *Meth. Enzymol.* **188**, 52.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

BENZOATE-CoA LIGASE

This enzyme [EC 6.2.1.25] catalyzes the reaction of benzoate with ATP and coenzyme A to produce benzoyl-CoA, AMP, and pyrophosphate (or, diphosphate). The enzyme will also use 2-, 3- and 4-fluorobenzoate as substrate. The corresponding chlorobenzoates will all serve as substrates, albeit less effectively.

J. Gibson, J. F. Geissler & C. S. Harwood (1990) *Meth. Enzymol.* **188**, 154.

BENZOATE 1,2-DIOXYGENASE

Benzoate 1,2-dioxygenase [EC 1.14.12.10], also called benzoate hydroxylase, catalyzes the reaction of benzoate with dioxygen and NADH to generate catechol, carbon dioxide, and NAD^+ . This is a multiprotein system which contains a reductase which is an iron-sulfur flavoprotein (FAD) and an iron-sulfur oxygenase.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

BENZOYLFORMATE DECARBOXYLASE

This enzyme [EC 4.1.1.7] catalyzes the conversion of benzoylformate to benzaldehyde and carbon dioxide. Thiamin pyrophosphate is a cofactor.

M. H. O'Leary (1989) *Ann. Rev. Biochem.* **58**, 377.

BERBERINE BRIDGE ENZYME

This enzyme [EC 1.5.3.9], also called reticuline oxidase and tetrahydropyprotoberberine synthase, catalyzes the re-

action of (*S*)-reticuline with dioxygen to generate (*S*)-scoulerine and hydrogen peroxide. FAD is the cofactor for this enzyme. The reaction product is a precursor of protopine, protoberberine, and benzophenanthridine-alkaloid biosynthesis in plants.

T. Hashimoto & Y. Yamada (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 257.

BETA (β)

1. The second letter in the Greek alphabet; hence, used to denote the second item in a series (for example, the second methylene carbon from the carboxyl group of a fatty acid). 2. Symbol for the coefficient of $[B]^i$ in the denominator of a generalized rate expression. 3. Symbol for reciprocal temperature parameter, $\beta = 1/kT$. 4. Symbol for pressure coefficient, $\beta = (\partial p/\partial T)_v$. 5. Symbol for depth of penetration of light (Napierian). 6. β^- , Symbol for electron. 7. β^+ , Symbol for positron.

BETAINE-ALDEHYDE DEHYDROGENASE

This enzyme [EC 1.2.1.8] catalyzes the reaction of betaine aldehyde with NAD^+ and water to produce betaine and NADH.

D. Rhodes & A. D. Hanson (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 357.

BETAINE-HOMOCYSTEINE S-METHYLTRANSFERASE

This enzyme [EC 2.1.1.5] catalyzes the reaction of trimethylammonioacetate with homocysteine to yield dimethylglycine and methionine.

W. E. Skiba, M. S. Wells, J. H. Mangam & W. M. Awad, Jr. (1987) *Meth. Enzymol.* **143**, 384.

BIFUNCTIONAL CATALYSIS

A catalytic process (often involving proton transfer) in which both functional groups of a bifunctional chemical species participate in the rate-controlling step. In such systems, the catalytic coefficient is larger than would be expected were only one functional group present. Bifunctional catalysis is not the same as two different catalysts acting in concert.

BIFUNCTIONAL ENZYME

An enzyme possessing the capacity to catalyze two mechanistically different chemical reactions. Examples in-

clude the fructose-6-phosphate-2-kinase/fructose-2,6-bisphosphatase, glutathionylspermidine synthetase/aminidase, and the *Escherichia coli* glutamine synthetase cascade's UTase/UR enzyme reactions.

C. H. Lin, D. S. Kwon, J. M. Bollinger, Jr., & C. T. Walsh (1997) *Biochemistry* **36**, 14930.

BIFURCATION THEORY

A theoretical framework for considering how the behavior of dynamical systems change as some parameter of the system is altered. Poincaré first applied the term bifurcation for the "splitting" of asymptotic states of a dynamical system. A bifurcation is a period-doubling, -quadrupling, *etc.*, that precede the onset of chaos and represent the sudden appearance of a qualitatively different behavior as some parameter is varied. Bifurcations come in four basic varieties: flip bifurcations, fold bifurcations, pitchfork bifurcations, and transcritical bifurcations. In principle, bifurcation theory allows one to understand qualitative changes of a system change to, or from, an equilibrium, periodic, or chaotic state.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Biological applications, **240**, 781-782, 791; development, **240**, 784-790; enzyme kinetic applications [hysteresis, **240**, 782-783; autocatalysis, **240**, 799-803; substrate cycling, **240**, 803-806]; detection with Jacobian eigenvalues, **240**, 795; algorithms, **240**, 793-795 [AUTO computer program, **240**, 795-798, 815; DSTOOL computer program, **240**, 796-797; GEPASI computer program, **240**, 796; LOCBIF computer program, **240**, 796-797]; Newton's method, **240**, 792-794; numerical continuation, **240**, 792-797; model preparation, **240**, 797-798.

BILIVERDIN REDUCTASE

This enzyme [EC 1.3.1.24] catalyzes the reversible reaction of bilirubin with $NAD(P)^+$ to generate biliverdin and $NAD(P)H$.

B. F. Cooper & F. B. Rudolph (1995) *Meth. Enzymol.* **249**, 188.

BIMOLECULAR

Bimolecular reactions are elementary reactions involving two distinct entities that combine to form an activated complex. For reactions in solution, the solvent contributes to the reaction's molecularity only when it is a reactant of the system. Bimolecular reactions are usually second order, but it is important to stress that some second order reactions need not be bimolecular.

Bimolecular rate constants typically have units of $M^{-1} s^{-1}$. Strict compliance to SI units for bimolecular rate

constants requires expressing concentrations in the SI system as $\text{mol}\cdot\text{m}^{-3}$, but concentrations are more frequently given as $\text{mol}\cdot\text{L}^{-1}$ (or, M^{-1} or $\text{mol}\cdot\text{dm}^{-3}$). When gas-phase reactions are studied (and even in the case of gases in solution), “concentrations” are often reported in terms of partial pressures rather than molarity. Nevertheless, it is usually preferable to convert such units to true concentration units when able.

Bimolecular processes are very common in biological systems. The binding of a hormone to a receptor is a bimolecular reaction, as is substrate and inhibitor binding to an enzyme. The term “bimolecular mechanism” applies to those reactions having a rate-limiting step that is bimolecular. **See** *Chemical Kinetics; Molecular; Reaction Order; Elementary Reaction; Transition-State Theory*

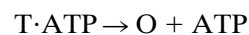
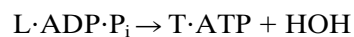
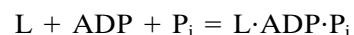
BINARY COMPLEX

A noncovalent complex between two molecules. Binary complex often refers to an enzyme-substrate complex, designated ES in single-substrate reactions or as EA or EB in certain multisubstrate enzyme-catalyzed reactions. **See** *Michaelis Complex*

BINDING CHANGE MECHANISM

The F_0F_1 -ATPase mechanism (a) for coupling chemiosmotic gradient free energy of proton translocation to the synthesis of ATP or (b) for coupling the free energy of ATP hydrolysis to create transmembrane proton gradients. While translocation of protons is the function of F_0 component, the formation or cleavage of ATP's β - γ phosphoanhydride bond is catalyzed by the $\alpha_3\beta_3$ F_1 component. When associated into a functional proton-translocating ATP synthase/hydrolase, F_0 and F_1 can mediate either of the processes mentioned above.

The binding change mechanism for the trimeric F_1 component resembles the action of a Wankle (or rotary) engine used in some automobiles. F_1 's “cylinders” are made up of three identical polypeptide chains, and each protomeric unit is in a different conformation (designated L for “Loose”, T for “Tight”, and O for “Open”). ADP and P_i bind to the L-site. Then, an energy-driven step occurs that successively converts $L\cdot\text{ADP}\cdot P_i$ to $T\cdot\text{ATP}$ and $T\cdot\text{ATP}$ to $O + \text{ATP}$:



These three conformational states are driven by the mechanical rotation of the $\alpha_3\beta_3$ F_1 trimer around an axis connecting F_0 to F_1 . Again by analogy to the rotary engine, the L-, T-, and O-states represent fueling, igniting, and discharging steps.

The concept of rotational catalysis¹ by ATP synthase is based on: (a) ^{32}P and ^{18}O exchange rate data attesting to strong cooperativity with sequential participation of several catalytic sites; (b) P_i and ATP ^{18}O -isotopomer distributions indicating that all catalytic sites exhibit identical catalysis; and (c) that catalysis is strongly influenced by the γ -subunit whose primary structure was not likely to account for spatially similar interactions with the β -subunits^{2,3}. The model was found to be compatible with the 2.8 Å resolution structure of bovine heart mitochondrial F_1 -ATPase⁴.

This radical mechanism was without precedent in biochemistry, and direct microscopical observation was reported in 1997 by Noji *et al.*⁵ Rotation of γ -subunit of thermophilic F_1 -ATPase was observed directly with an epifluorescence microscope. The enzyme was immobilized on a nickel-coated coverslip through a His-tag introduced to the N-termini of the β -subunit, and the enzyme was attached to a fluorescently labeled actin filament. The latter served like a hand of a clock for detecting rotatory motions in their observations. The authors describe the motor as follows: “a central rotor of radius approximately 1 nm, formed by its γ -subunit, turns in a stator barrel of radius approximately 5 nm formed by three α - and three β -subunits. F_1 -ATPase, together with the membrane-embedded proton-conducting unit F_0 , forms the H^+ -ATP synthase that reversibly couples transmembrane proton flow to ATP synthesis/hydrolysis in respiring and photosynthetic cells.” In the presence of ATP, the filament rotated for more than 100 revolutions in an anticlockwise direction when viewed from the “membrane” side. The rotary torque produced reached more than 40 pNewtons nm^{-1} under high load. As emphasized by Noji *et al.*⁵, cells employ many linear motors (or chemical-to-mechanical transducers), including dynein, kinesin, myosin, elongation factors, and RNA

polymerase, to move along and exert force on a filamentous structures. With the exception of bacterial flagellum's rotary motor, no other rotary motor had been demonstrated prior to this work on the F_0F_1 -ATPase.

The binding change mechanism represents an intellectual triumph in kinetic and mechanistic enzymology, and Paul Boyer was awarded the 1997 Nobel Prize in Chemistry for creating an entirely new way of analyzing energy transduction mechanisms. He shared the prize with John Walker, whose group determined the atomic level structure of F_1 , and Jens Skou who first identified the Na^+, K^+ -ATPase responsible for establishing the sodium/potassium gradients in neurons.

¹P. D. Boyer (1997) *Annu. Rev. Biochem.* **66**, 717.

²P. D. Boyer & W. E. Kohlbrenner (1981) in *Energy Coupling in Photosynthesis* (R. Selman & S. Selman-Reiner, eds.) pp. 231-240, Elsevier Biomedical, Amsterdam.

³P. D. Boyer (1983) in *Biochemistry of Metabolic Processes* (B. L. F. Lennon, F. W. Stratman & R. N. Zalten, eds.) pp. 465-477, Elsevier, Amsterdam.

⁴J. P. Abrahams, A. G. Leslie, R. Lutter & J. E. Walker (1994) *Nature* **370**, 621.

⁵H. Noji, R. Yasuda, M. Yoshida & K. Kinoshita, Jr. (1997) *Nature* **386**, 299.

BINDING INTERACTION

Any noncovalent association guided by electrostatics and complementarity, in which a smaller solute (known as a ligand) binds to a site on its larger molecular weight binding partner (frequently called a protein, enzyme, or receptor). Although kinetics is the major focus of this handbook, binding is a fundamental step in nearly all kinetic mechanisms. Detailed discussions of equilibrium ligand binding interactions lie beyond the scope of this book, and interested readers are encouraged to consult Winzor and Sawyer¹ or Wyman and Gill². **See** *Allosteric Interaction; Binding Isotherm; Biosensor; Cooperative Ligand Binding; Equilibrium Constant; Equilibrium Dialysis; Hummel-Dreyer Method; Linked Function Theory; Klotz Plot; Macroscopic Constant; Microscopic Constant; Molecular Crowding; Scatchard Plot*

¹D. J. Winzor & W. H. Sawyer (1995) *Quantitative Characterization of Ligand Binding*, p. 168, Wiley, New York.

²J. Wyman & S. J. Gill (1990) *Binding and Linkage: Functional Chemistry of Biological Macromolecules*, p. 330, University Science Books, Mill Valley, CA.

BINDING SITE

A topologically defined site on a macromolecular entity that interacts with another molecular entity (commonly

referred to as a ligand) *via* electrostatic interactions, hydrophobic interactions, π -orbital overlapping, hydrogen bonding, van der Waals forces, coordination, or even covalent bonding.

BINOMIAL THEOREM

A mathematical theorem defining the form of the polynomial corresponding to $(a + b)^n$, where n is a positive integer. The coefficients for each term of such a polynomial are given by Pascal's triangle. **See** *Pascal's Triangle*

BIOAVAILABILITY

The pharmacokinetic characteristics describing how the active drug in a particular formulation is absorbed and retained in the bloodstream or an organ. Typically determined parameters are (1) the circulating concentration of a drug at certain time intervals after its specified route of administration (*i.e.*, oral, intravenous, *etc.*), and (2) the duration of the drug's persistence in blood. Because differences in the pharmaceutical formulation of the same drug can greatly influence the chemical and physical determinants of bioavailability, any reliable study of bioavailability must be conducted with the formulated drug, and not the pure drug itself. Formulations exhibiting identical bioavailability are said to be bioequivalent.

BIOCATALYTIC ELECTRODE

Any molecule-selective electrode that incorporates an enzyme-catalyzed reaction to transduce the concentration of an analyte into an electrochemical signal. Such electrodes fall under the general category of biosensors, and they rely on the catalytic properties of a covalently-attached or matrix-enclosed enzyme for their high selectivity and responsiveness. Voltammetric sensors containing glucose oxidase (reaction catalyzed: glucose + $O_2 \rightarrow$ glucono-1,5-lactone + H_2O_2) have proved to be commercially successful; the hydrogen peroxide product can be oxidized to molecular oxygen, thereby generating several electrons (reaction: $H_2O_2 + 2OH^- \rightarrow H_2O + 2e^-$). **See also** *Biosensor*

BIOCHEMICAL NOMENCLATURE: RECOMMENDATIONS

The following are the exact literature references dealing with the recommended nomenclature for biochemical compounds and their associated enzyme-catalyzed reactions.

GENERAL:

Abbreviations and symbols for chemical names of special interest in biological chemistry

J. Biol. Chem. (1966) **241**, 527-533

Abbreviations and symbols: a compilation

Eur. J. Biochem. (1977) **74**, 1-6

Trivial names of miscellaneous compounds of importance in biochemistry

J. Biol. Chem. (1966) **241**, 2987-2994

Citation of bibliographic references in biochemical journals

J. Biol. Chem. (1973) **248**, 7279-7280

Fundamental stereochemistry

J. Org. Chem. (1970) **25**, 2849-2867

Final version may be found in *Pure Appl. Chem.* (1976) **45**, 11-30.

AMINO ACIDS, PEPTIDES & PROTEINS:

Recommended nomenclature and symbolism for amino acids and peptides

J. Biol. Chem. (1985) **260**, 14-42

Biochemistry (1975) **14**, 449-462

Abbreviations and symbols for the description of the conformation of polypeptide chains

J. Biol. Chem. (1970) **245**, 6489-6497

Nomenclature of iron-sulfur proteins

Eur. J. Biochem. (1979) **93**, 427-430

Corrections: *Eur. J. Biochem.* (1979) **102**, 315

Nomenclature of peptide hormones

J. Biol. Chem. (1975) **250**, 3215-3216

Nomenclature of human immunoglobulins

Eur. J. Biochem. (1974) **45**, 5-6

Recommended nomenclature of glycoproteins, glycopeptides, and peptidoglycans

J. Biol. Chem. (1987) **262**, 13-18

Recommended nomenclature of electron-transfer proteins

Eur. J. Biochem. (1991) **200**, 599-611

BIOETHERMODYNAMICS:

Recommendations for the measurement and presentation of biochemical equilibrium data

J. Biol. Chem. (1976) **251**, 6879-6885

Recommendations for the presentation of thermodynamic and related data in biology

Eur. J. Biochem. (1985) **153**, 429-434

CARBOHYDRATES:

Nomenclature of Carbohydrates (Recommendations 1996)

available at: <http://www.qmw.ac.uksim;ugca000/iupac/2carb>

Tentative rules for carbohydrate nomenclature. Part 1

J. Biol. Chem. (1972) **247**, 613-634

Corrections, *Eur. J. Biochem.* (1972) **25**, 4

Conformational nomenclature for five- and six-membered ring forms of monosaccharides and their derivatives

Eur. J. Biochem. (1980) **111**, 295-298

Nomenclature of unsaturated monosaccharides

Eur. J. Biochem. (1981) **119**, 1-3

Nomenclature of branched-chain monosaccharides

Eur. J. Biochem. (1981) **119**, 5-8

Abbreviated terminology of oligosaccharide chains

J. Biol. Chem. (1982) **257**, 3347-3351

Polysaccharide nomenclature

J. Biol. Chem. (1982) **257**, 3352-3354

Symbols for specifying the conformation of polysaccharide chains

Eur. J. Biochem. (1983) **131**, 5-7

CAROTENOIDS:

Tentative rules for the nomenclature of carotenoids

J. Biol. Chem. (1972) **247**, 2633-2643

Revision, *Biochemistry* (1975) **14**, 1803

Pure Appl. Chem. (1975) **41**, 407-431

CORRINOIDS:

Nomenclature of corrinoids

Biochemistry (1974) **13**, 1555-1560

CYCLITOLS:

The nomenclature of cyclitols. Recommendations

Eur. J. Biochem. (1975) **57**, 1-7

ENZYMES:

Enzyme Nomenclature. Recommendations

(1992) Academic Press, New York

Nomenclature of multiple forms of enzymes

J. Biol. Chem. (1977) **252**, 5939-5941

Catalytic activity

Units of enzyme activity

Eur. J. Biochem. (1979) **97**, 319-320

Symbolism and terminology in enzyme kinetics

Eur. J. Biochem. (1982) **128**, 281-291

FOLIC ACID:

Nomenclature and symbols for folic acid and related compounds

Eur. J. Biochem. **2**, 5-6 (1967)

LABELED COMPOUNDS:

Eur. J. Biochem. (1978) **86**, 9-25

Eur. J. Biochem. (1979) **102**, 315-316

Final version: *Pure Appl. Chem.* (1979) **51**, 353-380.

LIPIDS:

The nomenclature of lipids. Recommendations

Lipids (1977) **12**, 455-468

NUCLEOTIDES AND NUCLEIC ACIDS:

Abbreviations and symbols for nucleic acids, polynucleotides and their constituents

J. Biol. Chem. (1970) **245**, 5171-5176

Corrections, *J. Biol. Chem.* (1971) **246**, 4894

Abbreviations and symbols for the description of conformations of polynucleotide chains

Eur. J. Biochem. (1983) **131**, 9-15

Nomenclature for incompletely specified bases in nucleic acid sequences

J. Biol. Chem. (1986) **261**, 13-17

Nomenclature of junctions and branch points in nucleic acids

Eur. J. Biochem. (1995) **230**, 1-2

PHOSPHORUS:

Nomenclature of phosphorus-containing compounds of biochemical importance

Proc. Natl. Acad. Sci. U.S.A. (1977) **74**, 2222-2230

QUINONES:

Nomenclature of quinones with isoprenoid side chains

Eur. J. Biochem. (1975) **53**, 15-18

RETINOIDS:

Nomenclature of retinoids

J. Biol. Chem. (1983) **258**, 5329-5333

STEROIDS:

Nomenclature of steroids. Revised tentative rules

Biochemistry (1969) **8**, 2227-2242

Amendments (1971) and corrections

Biochemistry (1971) **10**, 4994-4995

Definitive rules for steroids

Pure Appl. Chem. (1972) **31**, 285-322.

TETRAPYRROLES:

The nomenclature of tetrapyrroles

Eur. J. Biochem. (1980) **108**, 1-30

TOCOPHEROLS:

Nomenclature of tocopherols and related compounds

Eur. J. Biochem. (1982) **123**, 473-475

VITAMIN B₆

Nomenclature for vitamin B₆ and related compounds

Eur. J. Biochem. (1973) **40**, 325-327

VITAMIN D

Nomenclature of vitamin D

Eur. J. Biochem. (1982) **124**, 223-227

BIOCHEMICAL SELF-ASSEMBLY

Caspar and Klug¹ distinguished two fundamental types of self-assembly processes: (a) *True self-assembly*—a series of reactions relying on the propensity of subunits to condense and form assembled structures strictly as a result of the information encoded in the architecture of the components. (b) *Template-directed assembly*—a process depending on the presence of a separate template that imparts structural constraints on the pathway for constructing the final assembled structure.

True self-assembly is observed in the formation of many oligomeric proteins. Indeed, Friedman and Beychok² reviewed efforts to define the subunit assembly and reconstitution pathways in multisubunit proteins, and all of the several dozen examples cited in their review represent true self-assembly. Polymeric species are also formed by true self-assembly, and the G-actin to F-actin transition is an excellent example³. By contrast, there are strong indications that ribosomal RNA species play a central role in specifying the pathway to and the structure of ribosome particles. And it is interesting to note that the assembly of the tobacco mosaic virus (TMV) appears to be a two-step hybrid mechanism: the coat protein subunits first combine to form 34-subunit disks by true self-assembly from monomeric and trimeric com-

ponents; then, the assembly of the virus particle is directed by single-stranded RNA at a specific site in the sequence of the polynucleotide.

The cardinal feature of template-free self-assembly is the ability of the protomers to engage in nucleation, a series of cooperative reactions leading to the formation of polymerization nuclei. These nuclei are thermodynamically unstable, and the high molecularity of nucleation hinders the process. Once formed, however, such nuclei can promote further assembly, because the rate constants for the subsequent protomer addition steps often favor elongation. These rate constants, of course, need not be identical because the first few steps beyond nucleation may be necessary to create the regular lattice for elongation reactions to become independent of polymer length. In this respect, the nuclei represent self-assembly intermediates with unfavorable equilibria leading to them and far more favorable equilibria beyond them. As noted earlier, pure tubulin contains all of the structural information required to form microtubules, and there has been considerable interest in characterizing the initiation process. **See** *Actin Assembly Kinetics*; *Hemoglobin S Polymerization*; *Microtubule Assembly Kinetics*

¹D. L. Caspar & A. Klug (1962) *Cold Spring Harbor Symp. Quant. Biol.* **27**, 1.

²F. K. Friedman & S. Beychok (1979) *Annu. Rev. Biochem.* **48**, 217.

³F. Oosawa & M. Kasai (1962) *J. Mol. Biol.* **4**, 10.

BIOCONJUGATE

Any biologically produced molecular entity or species having substituents derived from different origins or pathways. For example, glutathione reacts with many arene oxides, and the bioconjugate product is then processed for excretion. Bioconjugates are often more water-soluble, and they are typically more readily compartmentalized and/or metabolized.

BIOCONVERSION

The biological transformation of one molecular entity or substance to form another.

BIOLEACHING

Any biological process, usually involving microorganisms, that extracts substances, typically metals, from ores or soil.

BIOMASS

Any material(s) produced by the growth of a microorganism(s), a plant(s), or an animal(s).

BIOMIMETIC

Any process or procedure designed to imitate one or more natural chemical processes. A molecular entity may also be biomimetic by mimicking the structure and/or function of a biological substance.

BIOMINERALIZATION

Any process in which inorganic cations and inorganic/organic anions combine at concentrations exceeding their respective solubility product(s) to create crystalline or paracrystalline structures.

INTRODUCTORY REMARKS. The most common biominerals include calcium phosphate, calcium carbonate, and calcium oxalate. While these substances readily form regular crystals and paracrystals, living systems rely on biospecific binding interactions to achieve exquisite morphological control over the biomineralization process. The complex and varied crystal habits adopted by marine organisms demonstrate how stereospecific intervention can lead to an astonishing variety of shell forms through the directed deposition of the principal ingredient, calcium carbonate. In higher organisms, the diverse morphologies of endoskeleton, exoskeleton, and teeth are achieved (a) by modulating interactions with cell membranes, proteins, and other inorganic and organic electrolytes, and (b) by controlling concentration/supersaturation of mineral salts through the action of chelating agents, ion pumps, and energy metabolism. Biomineralization also represents an efficient method for removing excess cations, particularly calcium ion, from biological fluids. Within the urinary tract, crystallization of calcium oxalate and calcium phosphate occurs opportunistically and quite freely, and these events probably represent an efficient means for efficient excretion of low-solubility substances. Renal stone disease apparently arises when crystallites assemble into flow-obstructing stones¹⁻³. While a comprehensive treatment of biomineralization lies far beyond the bounds of this handbook, there is a set of physicochemical events that underlie all of the above-mentioned processes, especially as regards those solution and surface phenomena controlling nucleation and growth phases. This entry relies on examples using

calcium oxalate monohydrate, the major component in idiopathic renal biom mineralization.

SOLUTION BEHAVIOR. Biom mineralization is dominated by physical chemical considerations^{1,2}, and we begin with a discussion of real electrolyte solutions in which the concentration of a substance exceeds its thermodynamically defined solubility. In such a case, the presence of a coexisting crystal surface will lead to crystal growth.

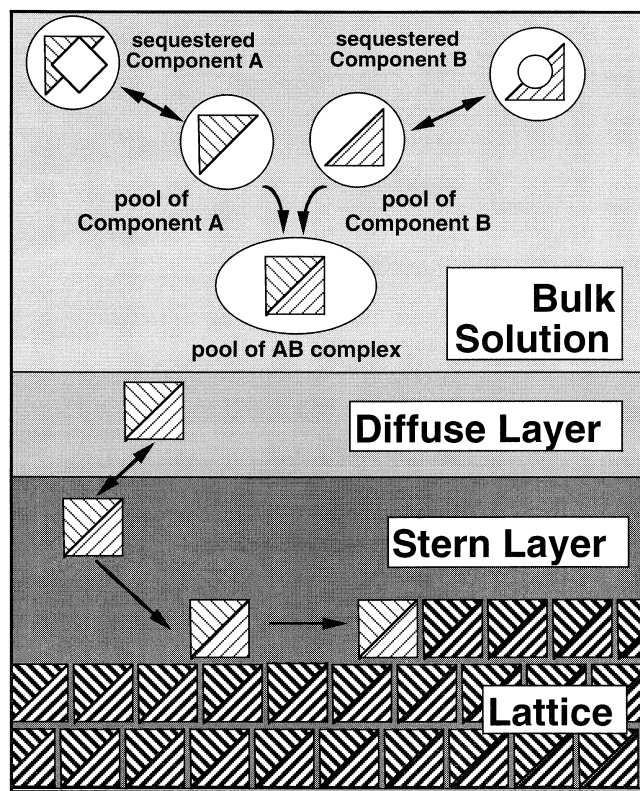


Figure 1. A model for solution and crystal phase behavior of uncomplexed and complexed components (shown here as light and dark triangles, respectively).

On the left-hand side of Fig. 1 are the various mineral-forming components in their complexed and uncomplexed forms. They can directly bind to the crystal growth sites or they can combine to form the crystallization monomers (half-filled squares). These processes can be blocked competitively by the presence of other substances that form nonproductive complexes, thereby depleting the concentrations of precursors through mass action. The diagram also shows a second phase that helps to explain the nature of oriented diffusion and subsequent adsorption of the monomers. This so-called “dou-

ble-layer region” includes the “diffuse layer” (or “Gouy-Chapman layer”) and the “Stern layer”. The double-layer model allows one to rationalize the forces that monomers experience within the gradient of the electric field created by the surface. This can result in attractive or repulsive diffusion of the monomer to or from the crystal surface. Upon approaching more closely to the surface, each monomer enters the Stern layer and encounters a combination of local electrostatic and chemical forces. Upon arriving at the crystal lattice surface, the monomer undergoes two-dimensional diffusion on the crystal lattice until it is incorporated into the growing crystal. These growth sites are depicted as discontinuities in the crystal surface, such as a step from one lattice layer to the next.

Once formed, crystals can interact with or adhere to other components. Adhesion can occur directly between crystals to form larger structures which may act as the nidus for further mineralization. Crystals can also interact with metabolites and/or tissues to form such structures. These interactions can and do alter the stereochemical course of crystal assembly, and molecular recognition probably accounts for the striations and intricate geometries adopted by biom mineralized structures. The role and composition of metabolites comprising the matrix may be useful to understanding the local and large-scale structure of biom minerals.

LIQUID-PHASE BEHAVIOR. The liquid phase contains dissolved substances and contacts the solid phase. For our purposes, the “liquid phase” is used synonymously with “aqueous phase”, and all processes discussed in this section take place in aqueous solutions. The dissolved monomers of the solid phase are formed in equilibrium with their uncomplexed components. Such components may be uncomplexed ions (which are charged atoms or molecules) free in solution or ionic complexes in equilibrium with dissociated ions. Concentrations of the uncomplexed ions, therefore, depend upon the concentrations of all chemical substances competing for binding interactions with them. Each complexation reaction is defined by either a solution equilibrium constant:

$$K_a = [\text{XY Complex}]/[\text{Component X}][\text{Component Y}]$$

or by a solubility product constant

$$K_{sp} = [\text{Component A}][\text{Component B}]$$

Accordingly, if a particular ion undergoes complexation with a number of other substances, then the free (or uncomplexed) concentration is defined as the total analytical concentration of the ion *minus* the total concentrations of all complexed forms of that ion. In other words, the total amount of an ion in solution is distributed into its ionic form (uncomplexed form) and the other complexed forms. In some cases, mineral salts are so weakly associated in solution that their uncomplexed ionic forms predominate; such is the case with sodium chloride. We use the term “speciation” to designate the detailed distribution of a substance’s ionic and complexed species in solution. Such an account must deal with the equilibria interconnecting such species, and the magnitude of their corresponding thermodynamic equilibrium constants quantitatively define each of their concentrations. To achieve a detailed description of a substance’s speciation, one must explicitly know all of the possible complexation reactions that the biomineral ions undergo, and one must have good values for the stability constant (synonymous with “formation constant” or “association constant” in this presentation) for each complex. The magnitude of a stability constant is expressed in reciprocal molar units (M^{-1}) and depends on the temperature and ionic strength of the solution. The reader may recall that ionic strength may be defined as one-half the sum of a series of terms, each corresponding to the concentration of a particular species multiplied by the square of its charge. **See** *Ionic Strength, Solubility Product*

So far, we have expressed the number of ions per unit volume by concentration. Physical chemists prefer to use “activity” to characterize the behavior of solutes in solution. As a solution becomes more concentrated and as the ionic strength increases, ions behave as if there were fewer of them present than would be indicated by their analytical concentrations. The activity of a component is related to its concentration by a proportionality constant known as an “activity coefficient”. Considerations regarding ionic activity become particularly important, when fluids are quite concentrated. An extension of this treatment is the use of the “activity product” to

represent the mathematical product of the activities for a set of ions.

Solubility is the measure of the amount of a solid that can be held in solution, and this is achieved when a solution remains in contact with the solid for a sufficient period to ensure equilibration (saturation). Solubility is typically expressed as an activity product which is generally called the “solubility product constant”. A more useful way of expressing the degree of saturation is provided by the “relative supersaturation,” which can be regarded as the activity product of a solid in solution divided by the solubility product constant. This ratio defines how much of a solid is in solution compared to how much should remain dissolved under stable, equilibrium conditions. Solutions can be made supersaturated with respect to a particular solid by combining solutions of the constituent ions of the solid, or by lowering the temperature of a saturated solution. Usually, the temperature dependence of saturation is not very high, so that lowering the temperature is a more gentle means of supersaturating a solution. Any amount of solute that exceeds the solubility product will over time reach equilibrium by precipitating from solution. Relative supersaturation is actually the driving force for nucleation and crystal growth.

THE DOUBLE LAYER. On a microscopic level, boundaries between solids and liquids (also referred to as “interfaces”) are complicated transition zones (see Fig. 2). The physical theories for interfacial behavior^{1,2} are different from those for regular solutions, and this is especially true for crystals in aqueous environments.

The surfaces of these crystals often possess characteristic “surface charge” which influences the composition of the solution near the surface. As noted earlier, the double-layer theory divides this region into two parts: (1) a layer near the surface that has an excess of ions whose electric charge is opposite to that of the surface itself (because opposite charges attract); and (2) an outer layer or region containing ions that are oppositely charged relative to that of the first (*i.e.*, the outer layer has the same charge as the surface). The double-layer also maintains overall electric neutrality. The innermost layer is called the Stern layer. Ions and molecules within this layer experience both electrostatic interaction with and

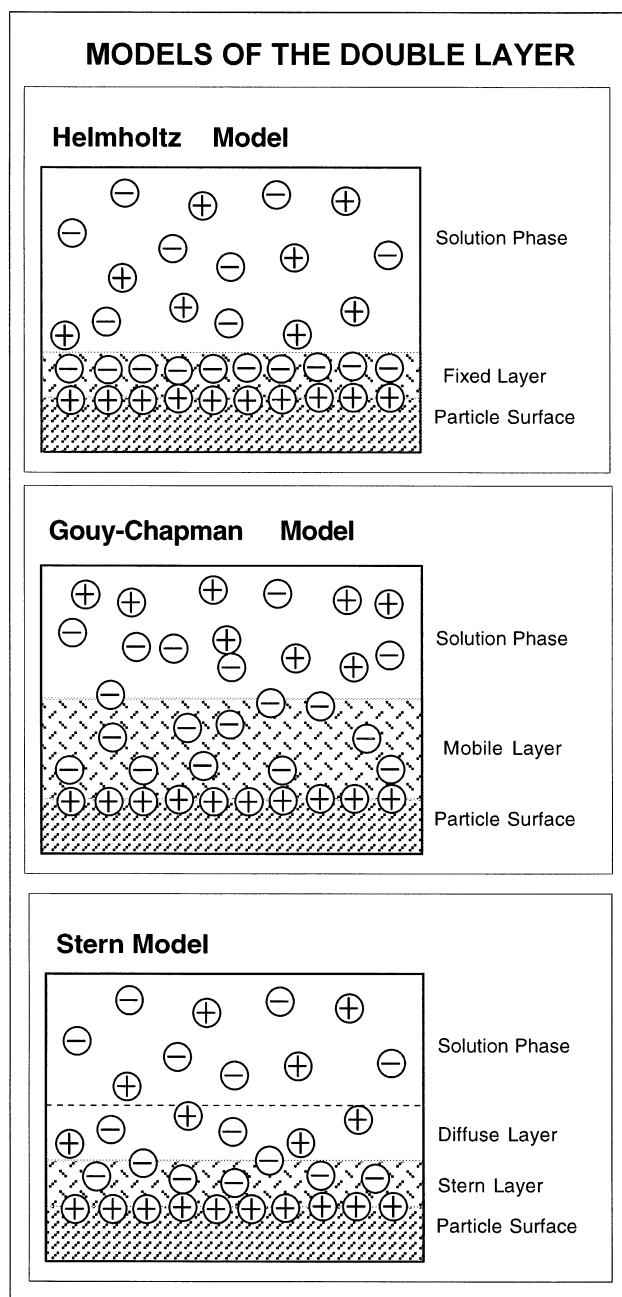


Figure 2. The ionic nature of the interactions at an interface of a electrically charged surface and an aqueous solution containing ionizable solutes. The oldest model, first proposed by Helmholtz, unsuccessfully treats the interface as though it were composed of rigidly held, oppositely charged ions, lined up next to each other in a manner akin to a parallel-plate capacitor. The Gouy-Chapman model was based on the idea that one must consider thermal motions that disturb ionic mobility and interactions at the surface. This model is characterized by the presence of a mobile phase of counterions immediately above a charged surface. The most realistic treatment, proposed by Stern in 1924, accounts for adsorption and exchange of counterions within a layer that extends from the interface for a distance equal to several ionic diameters. The electrical properties of double layers also allow one to treat colloidal interactions, in the same manner as mutual repulsion can explain the stability of so-called lyophobic sols. (Note: To avoid unnecessary duplication, only a positively charged surface is considered.)

chemical attachment to the surface. The composition of this layer gives rise to many of the distinctive behaviors of particles in solution, especially influencing aggregation. A very important parameter characterizing the Stern layer is the Debye-Hückel length, a parameter which expresses the thickness of this layer and strongly depends on solution ionic strength. The Gouy-Chapman layer is the outer or *diffuse* layer. Ions and molecules within this layer are not bound, yet they are influenced by the electrostatic properties of the Stern layer. While it is meaningful to consider the amounts of chemical species in the Stern layer, the Gouy-Chapman layer is best thought of as possessing only a total charge. *Oriented diffusion* is a term coined for the influence that solutes may experience in different ways by the charge structure of the double-layer region. This influence could amount to a facilitation or obstacle with regard to the participation of certain substances in surface processes.

Some ions and other molecular substances attach to surfaces under the influence of electrostatic and chemical forces. This process is known as *adsorption*, and those substances adhering to the surface are called “adsorbates.” Various theories treat adsorption as a hyperbolic binding curve (**See Langmuir Isotherm**) which relates the substance’s surface concentration to its bulk solution concentration. An isotherm describes the gross or macroscopically observable behavior of adsorbates, whereas the Stern theory is a way of describing adsorption at a molecular level, while still taking into account the specific properties of the adsorbates and the surfaces. Surface-adsorbed ions and molecules are usually not rigidly attached, and they often can migrate along the surface to other points of attraction where they may dwell for a time. This process is called “surface diffusion.” The interface may be pictured as an active region where molecules and ions approach and attach, detach and diffuse, or detach and return to the bulk solution. Some substances may anchor themselves more or less irreversibly to the surface, and this appears to be a particularly important viewpoint for larger biomolecules which may have multiple points of attachment, so-called *polypodal attachment*. At any instant, such molecules may release one or two points of attachment, but releasing all points of attachment simultaneously is less and less likely with every additional site of attachment. Naturally, some of the substances attached to the surface are those of which

the solid itself is composed, and various ionization states or complexes of these substances may still interact with the surface without leading to crystal growth. The distinction between incorporated and adsorbed species must be kept in mind. Incorporation into the solid leads to net crystal growth. Likewise, during dissolution, some of the exposed solid may leave, thereby returning to the solution. Other lattices may have just the right structure to permit other unrelated solid lattices to be formed. When a crystal of one composition gives rise to a crystal of another composition in this way, it is usually called *epitaxy* or *epitaxial growth*, but the term *oriented overgrowth* may also be seen.

For a solution supersaturated with solute, crystallization cannot take place in the absence of a higher-order process known as *nucleation*. In one sense, nucleation refers to any process affording a route for a substance from one physical phase to another, usually in the direction

of greater order. Examples are the slow formation of bubbles in carbonated beverages or the condensation of water vapor into droplets. Of greatest interest to biom mineralization is the spontaneous appearance of microscopic crystals from supersaturated solutions.

IONIC SPECIATION. Ions interact continually in aqueous solution. Ions are complexed with water molecules. Even when we say that a certain ion is uncomplexed, the fact is that the ion is still complexed, in this case with water molecules. Association constants (also known in the literature as stability or formation constants) allow one to quantitate the extent to which an ion is complexed with any particular substance in solution. They also allow comparisons of the relative affinity of different complexing agents for a particular chemical substance. Speciation is a chain of linked binding functions (see Fig. 3). Such diagrams show the relative concentrations of the various complexes in solution, and the reversible equilibria existing between these pools are shown by the arrows.

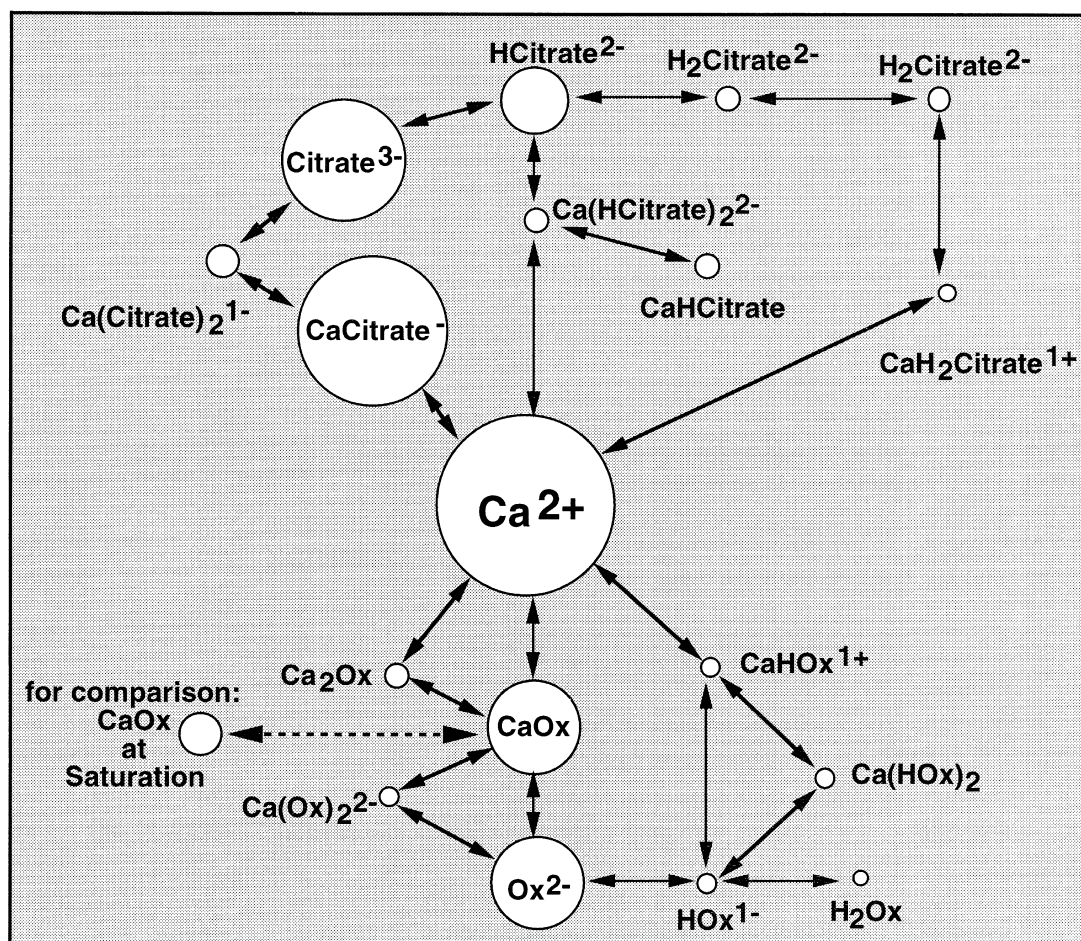


Figure 3. Graphical representation of ionic and complexed species as a "pool diagram."

The linkage between all the circles through various pathways is a graphical display of the fact that a change in the concentration of any complex or component on the diagram will be transmitted to all other circles on the diagram. For example, if citrate is reduced, the calcium which was bound as calcium citrate complex will be released, and this will increase the available calcium; increased available calcium will, in turn, compete with other cations for phosphate, sulfate, and so on. The net result will be an increased concentration of all other calcium complexes, a reduction of all other cation complexes, and an increase in the uncomplexed cations. Because of the number of complexes in the scheme, the effect of small changes in any one constituent is buffered. The key to the physical success of the scheme is that all the reactions in it are reversible; that is, though the complexes are stable relative to the component activities, if those activities decrease, there is nothing preventing the complexes from breaking down into their components. Irreversible reactions change the way this scheme works; such reactions could be the irreversible binding of a cation or anion, reactions of complexing agents with extremely high affinities, or, as we shall consider in the next section, the precipitation of insoluble salts.

SOLUBILITY PRODUCT. For ions that combine and form an insoluble phase, we use an alternative form of an equilibrium constant known as the solubility product. Consider the following reaction:



The ion solubility product (K_{sp}) is given by the expression,

$$K_{sp} = [\text{Anion}][\text{Cation}]/1$$

where “1” in the denominator designates the assigned activity of the solid phase, and is usually not written. The law of mass action requires that ions in solution will remain completely soluble so long as the product of their concentrations is less than or equal to their corresponding solubility product. Whenever the product of their concentrations exceeds the solubility product, the solution will be initially supersaturated, and if precipitation occurs, the system will reach thermodynamic equilibrium only when the product of their concentrations is once

again less than or equal to their corresponding solubility product.

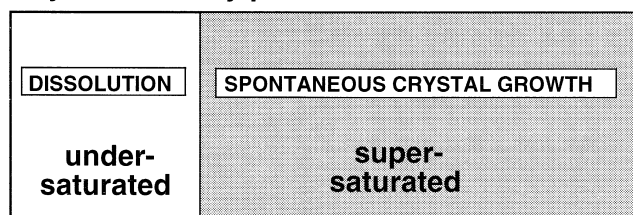
RELATIVE SUPERSATURATION. The free energy that drives nucleation and crystal growth is directly related to a substance’s relative supersaturation (RS), and the latter is a unitless parameter equal to $[X]_{\text{instantaneous}}/[X]_{\text{saturation}}$:

$$\Delta G = -RT \ln [X]_{\text{instantaneous}}/[X]_{\text{saturation}} = -RT \ln(RS)$$

The higher the relative supersaturation, the more likely nucleation becomes, and the faster crystal growth proceeds. Molecules or ions will remain dissolved provided that the conditions are energetically favorable. However, all molecules above a certain threshold solution activity will remain in, or become part of, the solid phase. Molecules are in equilibrium between the solid state and the dissolved state. The extent to which the equilibrium balance favors the dissolved phase indicates the degree of solubility.

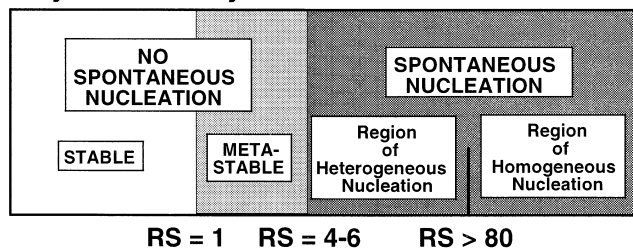
Once formed, the solid phase will grow until the activity of the complex from which the solid phase is formed reaches its equilibrium value.

Crystals already present:



RS=1

Crystals initially absent:



RS = 1 RS = 4-6 RS > 80

increasing relative supersaturation

Figure 4. Phase diagram for crystallization as a function of relative supersaturation.

These solid phases are connected to the components in Fig. 4, with which they are in reversible equilibrium. For example, if magnesium ion were added to a complex solution containing solid calcium oxalate monohydrate (COM), the magnesium would compete with calcium for an increased share of the oxalate; this would reduce the amount of the calcium oxalate complex, and finally a small amount of calcium oxalate solid would dissolve to restore the complex concentration to its equilibrium value. In urine, this picture must be extended to account for the molecular substances that coat crystals and reduce access of the solution to the surface; coated crystals do not redissolve readily.

SURFACE PHENOMENA. Assuming the presence of crystal surfaces in living systems, adsorption is the process most actively affecting their nature. Adsorbates alter the chemical, physical, and electrical properties of surfaces; by doing so, they affect crystal growth and aggregation. Whereas one might tend to identify a solid by its bulk composition, it is the surface composition that interacts with substances in the solution. Again using COM (calcium oxalate monohydrate) as our example, the bulk composition of this stone salt will be one calcium ion to one oxalate ion to one water molecule. The surface composition depends on how the crystal is “sliced” (*i.e.*, which of the different crystal faces are stable in solution). These different faces have different ratios of calcium to oxalate, and because these ratios need not be strictly one-to-one, different faces can actually have different charges. Each face of the crystal will have a specific surface charge. It is often impractical to try to view the charging of ions at such a microscopic level, but the overall charging of the crystal is experimentally determined in “Zeta potential” measurements using a microscope to follow the electrophoresis of crystals.

NUCLEATION. Nucleation creates a new phase that is organizationally more related to the crystal lattice than to the monomeric species that undergoes crystallization. This process permits solutions that are of high relative supersaturation to crystallize and thereby reach equilibrium between liquid and solid phases^{1,2,4}. Nucleation occurs when the local concentration of components that will comprise the solid phase exceeds a threshold level as a result of short-range concentration fluctuations in the bulk solution. In this respect, the kinetics of nucle-

ation reflect the frequency and duration of such fluctuations.

To understand nucleation, consider five different conditions that arise as the relative supersaturation is increased in the initial absence or presence of crystals. At RS values greater than one, the solution is supersaturated and will undergo spontaneous crystal growth until unit saturation occurs. One can identify so-called “stable” and “metastable” states which do not undergo spontaneous nucleation despite the fact that the metastable condition is somewhat supersaturated. At higher relative supersaturation (say RS 4–6 for the case of calcium oxalate monohydrate), one can observe spontaneous nucleation. This region of spontaneous nucleation can be further divided conceptually into two regions, based on the magnitude of the liquid-solid interfacial tension (σ). Heterogeneous nucleation allows other substances to reduce the metastable limit and is characterized by a σ value of approximately 30 erg/cm². On the other hand, homogeneous nucleation requires considerably higher relative supersaturation (hence, higher σ values) to allow local concentration fluctuations to be the primary route for forming nuclei. Homogeneous nucleation can occur slowly, even at the relative supersaturation values commonly associated with heterogeneous nucleation.

The rate at which the concentration of supersaturated components are depleted through crystal growth depends on the concentration of nuclei. The inhibitory action of various agents reflects a lowering of the nuclei concentration either through inhibition of nuclei formation or through promotion of nuclei dissolution, and promoters of nucleation must act to either increase the rate of nucleation or decrease the loss of nuclei from the solution. Inhibition or promotion of nucleation must be analyzed under conditions where the relative supersaturation remains the same. These definitions do not include mechanisms which could be explained as resulting merely from changes in the relative supersaturation, as would be the case for calcium ion and oxalate complexing agents.

The induction time τ for the precipitation process, where $\tau = gC^n$. C is the square root of the molar ion concentration product of the precipitating salt, and g and n are empirical constants. In this respect, n represents

the apparent molecularity for nucleation; for calcium oxalate at 25°C, these investigators obtained values for g and n of 1.03×10^{-7} S and -3.33 , respectively.

GROWTH & AGGREGATION. After nucleation occurs, crystal growth ensues until the RS approaches unity. During this growth, crystals can remain as single growing units or they can aggregate to form larger particles. In this regard, one should distinguish between crystal size distributions and particle size distributions; the former arise from growth of crystallites in the absence of aggregation, and the latter reflect a summation of individual crystal growth as well as that arising from aggregation.

Aggregation can be regarded as a higher-order process in which particles in aqueous media associate and form extended structures. Charged surfaces on particles can interact to bond them together, forming the larger structures or aggregates. Understanding the composition of the Stern layer is therefore an essential basis to a detailed account of the aggregation behavior of suspended particles. Aggregation is strongly dependent on the hydrodynamic forces in the medium, and it is divided broadly into two categories: (i) orthokinetic aggregation occurs in the presence of stirring, and (ii) perikinetic aggregation takes place in the effective absence of stirring. When particles aggregate so tightly that they merge together, they are said to be *sintered*. Aggregation might be effected also by substances that can cross-link surfaces by a sort of gluing phenomenon. Interactions leading to aggregation are: electrostatic attraction; van der Waals interactions; liquid bridging; capillarity; viscous binder effects; and solid bridging.

¹B. Finlayson (1978) *Kidney Internat.* **13**, 344.

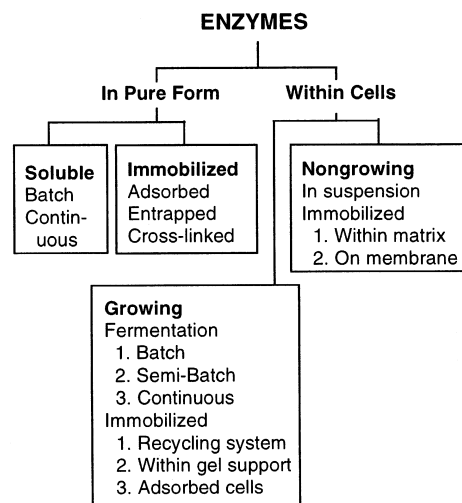
²C. M. Brown & D. L. Purich (1992) in *Disorders of Bone and Mineral Metabolism* (F. L. Coe & M. J. Flavin, eds.) pp. 613-624, Raven Press, New York.

³W. G. Robertson & M. Peacock (1985) in *Urolithiasis: Etiology-Diagnosis* (H.-J. Schneider, ed.) p. 183, Springer-Verlag, Berlin.

⁴J. A. Christiansen & A. E. Nielsen (1951) *Acta Chim. Scand.* **5**, 673.

BIOREACTOR

An open or closed reaction system or device designed to manipulate physical parameters (such as heat/mass transfer, pH, reactant concentration(s), *etc.*) as a means for regulating biocatalysis in order to maximize product



Range of bioreactor configurations gainfully employed in biotechnology and agri-business.

output and to minimize process costs. **See** *Immobilized Enzymes*

¹C. L. Cooney (1984) in *Biotechnology & Biological Frontiers* (P. H. Abelson, ed.) pp. 242-253, American Association for the Advancement of Science, Washington.

²R. A. Messing (1975) *Immobilized Enzymes for Industrial Reactors*, Academic Press, New York.

BIOSENSOR

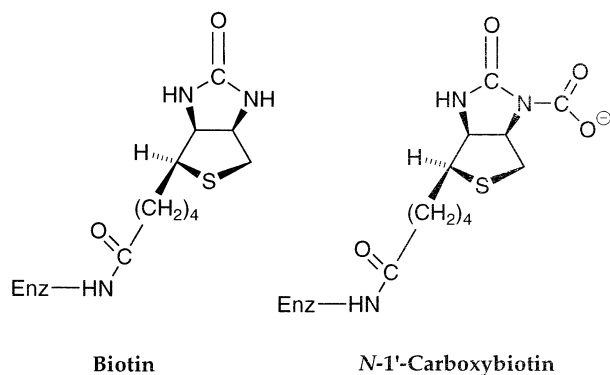
A highly sensitive detection system that relies on surface plasmon resonance (SPR) to obtain real time monitoring of biochemical interactions without requiring stable/radioactive isotopic labels or absorption/fluorescence reporter groups. The optical phenomenon known as SPR occurs when surface plasmon waves are excited at a metal/liquid interface. Gold is an excellent conductor as a result of its "gaseous cloud" of free electrons, and this cloud of ions can be a form of plasma. These free electrons in the detector's gold foil surface are remarkably sensitive to changes in the electrostatic interactions occurring in the liquid phase as a result of biospecific binding interactions. Moreover, these free electrons determine the refractive index near the foil's surface. In the BIACORE® probe, SPR is detected using the gold film at the sensor probe tip. "White" light traversing the probe's optical fiber undergoes total internal reflection at the surface interfaces, but SPR reduces the reflected light intensity at certain wavelengths. Binding interactions alter the refractive index close to the surface, such that the wavelength of the reflected light intensity minimum shifts. The BIACORE® probe allows the

instrument to sense the spectral output of the reflected light. Because nearly all proteins have very similar specific refractive indices, SPR is a mass detector that does not depend on the nature of the interacting species. The surface of a sensor probe is coated with a carboxymethylated-dextran matrix that permits covalent attachment of a detecting molecule. An intrinsic advantage of this methodology is that extremely small volumes of material are needed because all of the binding interactions occur in an ultrathin layer. **See also** *Biocatalytic Electrode*

Selected entries from *Methods in Enzymology* [vol, page(s)]:
 Theory, **240**, 323-324; instrumentation [BIACORE, **240**, 323-324; data output, **240**, 325]; data analysis [computer program, **240**, 330; linearity of models, **240**, 343-345; linear least-squares, **240**, 325, 343, 345-346, 348; nonlinear least-squares, **240**, 325-326, 343, 345-346, 348]; ligand immobilization, **240**, 329; ligate preparation, **240**, 329-330; phase data [association, **240**, 330-332; dissociation, **240**, 333-334]; rate constants [association, **240**, 340-342, 348-349; dissociation, **240**, 334-340, 348-349]; rate equations, **240**, 326-328; reproducibility in parameter determinations, **240**, 342-343; kinetic analysis of macromolecular interactions by surface plasmon resonance biosensors [theory, **295**, 269-270; experimental design, **295**, 271-277; data analysis, **295**, 277; global analysis of data, **295**, 279; CLAMP biosensor data analysis program, **295**, 279-285; testing reaction models experimentally, **295**, 285; establishing parameter confidence, **295**, 286; interpreting results, **295**, 290-293].

BIOTIN AND DERIVATIVES

Biotin¹ is a vitamin that was first recognized as an agent for preventing dermatitis and paralysis in rats fed large quantities of uncooked egg white (which contains abundant quantities of the protein avidin). The vitamin is covalently attached to an ϵ -amino group of an active site lysyl residue in biotin-dependent enzymes. Carboxyphosphate is thought to be formed prior to carboxyl group transfer to biotin. The N-1' position of biotin then acts as a carboxyl group carrier involved in carboxylation reactions.



Beyond its fundamental role in metabolism, the high affinity of biotin binding to avidin (K_d 10^{-15} M) and streptavidin has made biotin labeling especially valuable in cell biology¹. A protein of interest may be covalently labeled with suitably derivatized, commercially available biotin. For example, one can use *N*-hydroxysuccinimide or *p*-nitrophenyl derivatives of biotin to react with amino groups on the target protein.² The protein's location within cells can be detected by using streptavidin along with a fluorescently labeled anti-streptavidin antibody.

¹For a comprehensive discussion of the reactions and properties of vitamins and cofactors, readers are urged to consult *Biochemistry: The Chemical Reactions of Living Cells* (2nd ed.) by D. E. Metzler (2000) Academic Press, Orlando. This text and reference book contains a valuable compilation of literature citations.

²The Pierce Chemical Company (Rockford, Illinois) freely distributes *Avidin-Biotin Chemistry: A Handbook* to interested researchers.

Selected entries from *Methods in Enzymology* [vol, page(s)]:
 Isotopic dilution assay for biotin use of [¹⁴C]biotin, **62**, 279; use of [³H]biotin, **62**, 284; rapid fluorometric assay for avidin and biotin, **62**, 287; microassay of avidin, **62**, 290; radioimmunoassay for chicken avidin, **62**, 292; the synthesis and use of spin-labeled analogs of biotin in the study of avidin, **62**, 295; the avidin-biotin complex in affinity cytochemistry, **62**, 308; egg yolk biotin-binding protein: assay and purification, **62**, 316; antibodies that bind biotin and inhibit biotin-containing enzymes, **62**, 319; microbiological biosynthesis of biotin, **62**, 326; preparation of pimeloyl-coenzyme A, **62**, 339; 7,8-diaminopelargonic acid aminotransferase, **62**, 342; dethiobiotin synthetase, **62**, 348; synthesis and use of specifically tritiated dethiobiotin in the study of biotin biosynthesis by *Aspergillus niger*, **62**, 353; the biotin transport system in yeast, **62**, 371; enzymic reduction of D-biotin D-sulfoxide to D-biotin, **62**, 379; isolation and characterization of D-allobisnorbiotin, **62**, 385; analysis of microbial biotin proteins, **62**, 390; identification of biocytin, **62**, 398.

BIOTIN HOLOCARBOXYLASE SYNTHETASE

This enzyme [EC 6.3.4.10], also known as biotin—[propionyl-CoA-carboxylase] ligase and holocarboxylase synthetase, catalyzes the reaction of biotin with ATP and apo-[propanoyl-CoA:carbon-dioxide ligase (ADP-forming)] to produce AMP, pyrophosphate, and propanoyl-CoA:carbon-dioxide ligase (ADP-forming).

Y. Suzuki & K. Narisawa (1997) *Meth. Enzymol.* **279**, 386.
 K. Dakshinamurti & J. Chauhan (1988) *Ann. Rev. Nutr.* **8**, 211.
 N. H. Goss & H. G. Wood (1984) *Meth. Enzymol.* **107**, 261.

BIRADICAL

A molecule, molecular species, or entity containing two unpaired electrons on different atoms (e.g., $R-\dot{C}H-CH_2CH_2-R-\dot{C}H-R$); also referred to as diradicals. If

the unpaired electrons are widely separated, the entity behaves spectrally like two doublets. If the electrons are close enough for interaction, they are referred to as triplets. **See also** *Carbene; Free Radicals*

W. T. Borden (1982) *Diradicals*, Wiley, New York.

L. J. Johnston & J. C. Scaiano (1989) *Chem. Rev.* **89**, 521.

C. Doubleday, Jr., N. J. Turro & J. Wang (1989) *Acc. Chem. Res.* **22**, 199.

BISPHOSPHOGLYCERATE MUTASE

This enzyme [EC 5.4.2.4] (also known as bisphosphoglycerate synthase, diphosphoglycerate mutase, and glycerate phosphomutase) catalyzes the reversible interconversion of 3-phospho-D-glyceroyl phosphate to 2,3-bisphospho-D-glycerate. The “cofactor” for this enzyme is 3-phosphoglyceroyl phosphate. In the reaction scheme, the enzyme is phosphorylated by this “cofactor” to yield a phosphoenzyme and 3-phosphoglycerate. The latter compound is then rephosphorylated by the enzyme to yield 2,3-bisphosphoglycerate, but this reaction is slowed down by dissociation of 3-phosphoglycerate from the enzyme, which is therefore more active in the presence of added 3-phosphoglycerate.

L. A. Fothergill-Gilmore & H. C. Watson (1989) *Adv. Enzymol.* **62**, 227.

Z. B. Rose (1982) *Meth. Enzymol.* **87**, 42-51.

Z. B. Rose (1980) *Adv. Enzymol.* **51**, 211-253.

W. J. Ray, Jr., & E. J. Peck, Jr. (1972) *The Enzymes*, 3rd ed., **6**, 407.

BI TER REACTION

An enzyme-catalyzed reaction in which there are two substrates and three reaction products.

BI UNI REACTION

An enzyme-catalyzed reaction involving two substrates and one product. There are two basic Bi Uni mechanisms (not considering reactions containing abortive complexes or those categorized as Iso mechanisms). These mechanisms are the ordered Bi Uni scheme, in which the two substrates bind in a specific order, and the random Bi Uni mechanism, in which either substrate can bind first. Each of these mechanisms can be either rapid equilibrium or steady-state systems.

“BLACK BOX”

A term typically used in steady-state enzyme kinetics to acknowledge that there exist many kinetic models where

measurements of the changes in enzyme velocity as a function of substrate or effector concentration cannot provide any mechanistic information about the number or nature of isomerizations occurring within the so-called central complexes. The fact is that steady-state enzyme kinetics can provide useful tools for a very limited repertoire of mechanistic studies, and one must quickly proceed to experiments that employ other methods, such as isotope exchange, stopped-flow, rapid mix/quench, temperature-jump, rapid scan spectroscopy, kinetic isotope effects, transition-state inhibitor design, and NMR as well as any of the other environmentally sensitive spectroscopic methods.

A quote from W. P. Jencks' prefatory chapter¹ in the 1997 *Annual Review of Biochemistry* says it all rather succinctly. Referring to his twenty years investigating the ATP-dependent sarcoplasmic reticulum calcium pump, he remarked: “We studied this reaction cycle by using rapid mixing experiments to determine the individual steps of the reaction. I had previously decided that any conclusion that was reached from measurements of steady-state kinetics is wrong. This could be an overreaction. However, I think this is a good place to start, especially when one is examining complicated reactions.”

¹W. P. Jencks (1997) *Ann. Rev. Biochem.* **66**, 1.

BLEACHING

A term used in photochemistry and photomicroscopy to indicate the loss of absorption or emission intensity.

BLEOMYCIN

A glycopeptide molecule, often abbreviated BLM, that can serve as a metal chelating ligand. The bleomycins comprise a family of related antibiotics. The Fe(III) complex of bleomycin is an antitumor agent. Bleomycin's activity is associated with single-strand DNA cleavage through the generation of hydroxyl free radicals.

BLUE COPPER PROTEIN

An electron-transfer protein containing a type-1 copper site that manifests itself by a strong visible absorption spectrum. Another characteristic attributable to copper coordination by a cysteinyl sulfur is the EPR signal that displays an unusually small hyperfine coupling to the copper nucleus.

BOHR

An atomic unit of length used in quantum mechanical calculations of electronic wavefunctions. It is symbolized by a_0 and is equivalent to the Bohr radius, the radius of the smallest orbit of the least energetic electron in a Bohr hydrogen atom. The bohr is equal to $\alpha/(4\pi R_\infty)$, where α is the fine-structure constant, π is the ratio of the circumference of a circle to its diameter, and R_∞ is the Rydberg constant. The parameter α includes $\hbar$, as well as the electron's rest mass and elementary charge, and the permittivity of a vacuum. One bohr equals $5.29177249 \times 10^{-11}$ meter (or, about 0.529 angstroms).

BOHR EFFECT

Hemoglobin's linked function behavior of cooperative oxygen binding that results in reduced oxygen binding as the pH of a red cell or hemoglobin solution is lowered. Protonation of hemoglobin stabilizes the deoxygenated form, thereby reducing the ability of this heme protein to bind oxygen. In a reciprocal manner, oxygen binding must likewise alter the acid/base properties of hemoglobin. **See also** *Linked Functions*; *Adair Equation*; *Hemoglobin*

Selected entries from *Methods in Enzymology* [vol, page(s)]: Description, **231**, 97; **232**, 130; mechanism, **232**, 97-99; role of hemoglobin A histidyl residues, **232**, 130-133, 138; role of hemoglobin chloride-binding site, **231**, 225-226; ^1H NMR analysis, **232**, 97; role of hemoglobin side-chain carboxyl groups, **231**, 246-247; anion dependence, **259**, 524, 529, 534-535; characteristics, **259**, 16-18, 513; cryoelectrophoresis determination, **259**, 475-477; dissociation of individual proton-binding sites, **259**, 515, 531-534; electrostatic energy of proton binding, **259**, 526-527; energetics calculation, limitations of algorithms, **259**, 536-538; free energy [calculation from titration curves, **259**, 517-519; salt and pH dependence, **259**, 527]; measurement, **259**, 475-477, 520; microscopic proton-binding isotherms, **259**, 520-521, 535-536; proton binding site, average charge computation, **259**, 525-526; proton titration curves, **259**, 514-517, 523-524, 530-534.

BOLTZMANN DISTRIBUTION

A quantitative function¹ describing how large collections of electrons, atoms, or molecules are distributed in terms of their occupancy of quantum mechanically allowed energy states (*i.e.*, electronic, vibrational, rotational, and translational), based on energy differences defined by solution of the Schrödinger equation. At any given temperature, the Boltzmann distribution can be used to determine the average energy of the entire collection of molecules—a fact that helped determine an absolute

temperature scale. This distribution also helps one recognize how chemical reactivity is related to the fraction of molecules (within an ensemble) having sufficient energy to exceed some energy barrier to reactivity.

Consider a collection of particles with energies $\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots, \varepsilon_i$, each corresponding to a single quantum state, starting with the lowest lying state and proceeding to the i th state. The number of particles in each state can be written symbolically as $N_0, N_1, N_2, N_3, \dots, N_i$. The Boltzmann distribution function relating the relative occupancy of two states is written as:

$$\frac{N_i}{N_0} = \exp[-(\varepsilon_i - \varepsilon_0)/k_B T] = \exp(-\Delta\varepsilon_i)/k_B T$$

where k_B is the Boltzmann constant (where k_B equals R/N_0 , the gas constant R divided by Avogadro's number). If more than one state has the same energy, then the number of such states represents the degeneracy (symbolized by g). The number of particles in g degenerate states relative to the number in the lowest lying quantum state can be determined using the following expression:

$$\frac{N_i}{N_0} = \frac{g_i}{g_0} \exp(-\Delta\varepsilon_i)/k_B T$$

The exponential nature of this function results in very small values of (N_i/N_0) whenever $\Delta\varepsilon_i$ is much greater than $k_B T$.

The Boltzmann distribution helps us understand how intermediates can become trapped in energy wells between successive transition states in a multistep reaction mechanism. This behavior forms the basis for a special field of enzymology known as cryoenzymology. By appropriate choice of water-miscible solvents, enzymes can be studied at ultra-low temperatures where the rates of interconversion of enzyme species can be greatly retarded. **See** *Cryoenzymology*

¹Derivation of the Boltzmann distribution function is based on statistical mechanical considerations and requires use of Stirling's approximation and Lagrange's method of undetermined multipliers to arrive at the basic equation, $(N_i/N_0) = (g_i/g_0)\exp[-\beta \Delta\varepsilon_i]$. The exponential term β defines the temperature scale of the Boltzmann function and can be shown to equal $1/k_B T$. In classical mechanics, this distribution is defined by giving values for the coordinates and momenta for each particle in three-coordinate space and the lin-

ear moment, p_x , p_y , and p_z (each equal to the particles mass multiplied by the velocity component in the x , y , and z directions). The probability that a particle lies in the intervals x to $x + dx$, y to $y + dy$, and z to $z + dz$, and the probability lies correspondingly between p_x and $p_x + dp_x$, p_y and $p_y + dp_y$, and p_z and $p_z + dp_z$, can be written as: $P = e^{-W/k_B T} dx dy dz dp_x dp_y dp_z$, where W is the energy and k_B the Boltzmann constant, and T the absolute temperature. The probability of a particle having energy w above the ground state can likewise be written as: $N_i/N_0 = g e^{-W/k_B T}$, where g takes into account the number of possible states and any energetic degeneracy. Note that N_i is the number of molecules having energy W , and N_0 is the total number of molecules or particles. Further note that the ratio N_i/N_0 is the probability of occurrence of molecule N_i in a total ensemble N_0 . An analogous treatment can be obtained by defining the relative probabilities of various quantum states of a system in equilibrium.

BOLUS

A large quantity of substance delivered or added to an organism or system, all at once. A single administration of a large amount of a drug is often described as a "bolus dose".

BOLUS TRACER INJECTION

A bolus injection initially leads to an increase in the tracer/tracee ratio within a compartment, and this ratio then undergoes a first-order decline based on the rate constant for metabolite elimination¹.

¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, Wiley-Liss, New York.

BONDING ORBITAL

A molecular orbital having a lower energy level than the atomic orbitals from which the bonding orbital is formed. Such an orbital can contain two electrons, and their presence results in a strong bond when the overlap of the atomic orbitals is large. Two overlapping atomic orbitals combine to yield one low-energy bonding orbital (designated σ) and one high-energy antibonding orbital (designated σ^*). Two paired electrons are sufficient to fill the σ orbital, and any additional electrons must occupy the high energy σ^* orbital where, rather than stabilizing the bond, they lead to repulsion between the atoms.

BOND NUMBER

The number of electron-pair bonds shared by two atoms. In acetone, for example, the bond number for the carbon

and oxygen is two, whereas the bond number is one for carbon bonded to hydrogen.

BOND ORDER

An index of the degree of bonding between two atoms in a molecule. A normal single bond is given a bond order of one. In valence-bond theory, the bond order for a particular bond is calculated as the weighted sum of all the canonical forms of the molecule. For example, if benzene were completely represented by two Kekulé resonance forms, then one would predict that a carbon-carbon bond in benzene has a bond order of 1.5. However, the three so-called Dewar structures also contribute to the structure of benzene¹. The carbon-carbon bond order for benzene using valence-bond theory is thus determined to be 1.463. Hence, a carbon-carbon bond in benzene is not halfway between a single and a double bond, but actually somewhat less^{2,3}. Bond orders are calculated differently in molecular orbital theory^{1,2}. The order is determined from the weights of the atomic orbitals in each molecular orbital. With this method, the carbon-carbon bond order in benzene is 1.667.

¹A. Pullman & B. Pullman (1958) *Prog. Org. Chem.* **4**, 31.

²D. Clarkson, C. A. Coulson & T. H. Goodwin (1963) *Tetrahedron* **19**, 2153.

³W. C. Herndon & C. Parkanyi (1976) *J. Chem. Ed.* **53**, 689.

BOND VALENCE SUM ANALYSIS

An empirical method for correlating the oxidation state of a metal ion with the coordination geometry and the bond lengths¹. Bond valence sum analysis has been used in characterizing the structural features of vanadium-dependent haloperoxidases^{2,3}.

¹W. T. Liu & H. H. Thorp (1993) *Inorg. Chem.* **32**, 4102.

²A. Butler & M. J. Clague (1995) *Mechanistic Bioinorganic Chemistry* (H. H. Thorp & V. L. Pecoraro, eds.), pp. 329-349, ACS, Washington, DC.

³C. J. Carrano, M. Mohan & S. M. Holmes (1994) *Inorg. Chem.* **33**, 646.

BORN-OPPENHEIMER APPROXIMATION

An approximation stating that the motion of nuclei in ordinary molecular vibrations is slow relative to the motions of electrons. Thus, the nuclei can be held in fixed positions when doing calculations of electronic states. Such an assumption is useful in determining potential energy surfaces and is central in studying the quantum mechanical properties of molecules. **See also** *Adiabatic Photoreaction*; *Diabatic Photoreaction*

BORNYL PYROPHOSPHATE SYNTHASE

This enzyme [EC 5.5.1.8], also known as geranyl-diphosphate cyclase, catalyzes the conversion of geranyl diphosphate to (+)-bornyl diphosphate.

R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.

F. M. Raushel & J. J. Villafranca (1988) *Crit. Rev. Biochem.* **23**, 1.

BOROHYDRIDE REDUCTION

Sodium borohydride (NaBH_4) and its isotopic forms (NaB^2H_4 or NaB^3H_4) are impressively useful reducing agents¹⁻⁴. NaBH_4 exhibits much greater selectivity than lithium aluminum hydride (LiAlH_4), and normally reduces ketones and aldehydes to the corresponding alcohols without acting on carbon-carbon double bonds, nitriles, esters, nitro groups, *etc.* However, conditions can be altered (*e.g.*, the presence of transition metal salts) to increase the reducing power of NaBH_4 . For example, alkenes and alkynes can be reduced with NaBH_4 provided the unsaturated bonds are complexed with salts such as FeCl_2 or CoBr_2 . Amides are not reduced by NaBH_4 , unless other reagents are present. In general, NaBH_4 will reduce aldehydes and ketones to alcohols, acyl halides to alcohols (or to aldehydes) under some conditions, and act on Schiff bases, imines, and hydrazones. This nucleophilic reductant will also convert *N*-alkylimino esters (RC(OR')=NR'') to secondary amines.

In biochemistry, metal hydrides such as NaBH_4 have been widely used in synthesis. For example, NaBH_4 has been used in the preparation of alkyl cobalamins from cyanocobalamin^{5,6}, and in the synthesis of the chiral $[\beta, \gamma\text{-}^{17}\text{O}; \gamma\text{-}^{17}\text{O}, ^{18}\text{O}]\text{ATP}\gamma\text{S}$ from the 5'-aldehyde of adenosine^{7,8}. Lithium triethylborohydride (also known as "Su-

per Hydride") has been used in the preparation of chiral acetate and other chiral methyl groups^{9,10}.

NaBH_4 has greatly facilitated the chemical trapping of covalent intermediates that are formed in enzyme-catalyzed reactions. The metal hydrides have played a significant role in characterizing the mechanism of enzyme processes (Table I). Interestingly, borohydride treatment can occasionally be used to exclude certain mechanistic possibilities. Treating galactose-1-phosphate uridylyl-transferase¹¹ with NaB^3H_4 in the presence of substrate, for example, failed to result in radiolabel incorporation. This observation, when taken in the context of other experimental findings, indicated the very low probability of an acyl phosphate intermediate as a catalytic feature of this enzymic process.

¹A. Hajos (1979) *Complex Hydrides*, Elsevier, New York.

²H. O. House (1972) *Modern Synthetic Reactions*, 2nd ed., pp. 49-71, Benjamin, New York.

³O. H. Wheeler (1966) in *The Chemistry of the Carbonyl Group* (S. Patai, ed.), Part 1, pp. 507-566, Interscience, New York.

⁴H. C. Brown (1972) *Boranes in Organic Chemistry*, Cornell Univ. Press, Ithaca, New York.

⁵T. C. Stadtman (1971) *Science* **171**, 859.

⁶F. Wagner (1966) *Ann. Rev. Biochem.* **35**, Part I, 405.

⁷M. R. Webb & D. R. Trentham (1981) *J. Biol. Chem.* **256**, 4884.

⁸M. R. Webb (1982) *Meth. Enzymol.* **87**, 301.

⁹M. Kajiwara, S.-F. Lee, A. I. Scott, M. Akhtar, C. R. Jones & P. M. Jordan (1978) *J. Chem. Soc., Chem. Comm.*, p. 967.

¹⁰H. G. Floss (1982) *Meth. Enzymol.* **87**, 126.

¹¹J. C. Speck, P. T. Rowley & B. L. Horecker (1963) *J. Amer. Chem. Soc.* **85**, 1012.

¹²I. Fridovich (1972) *The Enzymes*, 3rd ed. **6**, 255.

¹³J. A. Todhunter & D. L. Purich (1975) *J. Biol. Chem.* **250**, 3505.

¹⁴R. Lane & E. Snell (1976) *Biochemistry* **15**, 4175, 4180.

¹⁵R. Wickner, C. Tabor & H. Tabor (1970) *J. Biol. Chem.* **245**, 2132.

¹⁶S. Shapiro & D. Dennis (1965) *Biochemistry* **4**, 2283.

¹⁷J. Butler, W. Alworth & M. Nugent (1974) *J. Amer. Chem. Soc.* **96**, 1617.

¹⁸J. A. Todhunter, K. B. Reichel & D. L. Purich (1976) *Arch. Biochem. Biophys.* **174**, 120.

¹⁹D. S. Hodgins & R. H. Abeles (1967) *J. Biol. Chem.* **242**, 5158.

²⁰P. A. Frey, L.-J. Wong, K.-F. Sheu & S.-L. Yang (1982) *Meth. Enzymol.* **87**, 20.

Table I

Borohydride Reduction as a Probe of Enzyme Reaction Intermediates

Enzyme Name	Reaction Type	Intermediate
Fructose-1,6-bisphosphate aldolase ¹²	Class I Aldolases	Imine
Acetoacetate decarboxylase ¹³	Decarboxylase	Imine
Glutamine synthetase ¹⁴	Synthetase	Acyl-phosphate
Histidine decarboxylase ¹⁵	Decarboxylase	Dehydroalanine
S-Adenosylmethionine decarboxylase ¹⁶	Decarboxylase	Dehydroalanine
Lactate racemase ¹⁷	Racemase	Lactyl thioester
5'-Dehydroquinase dehydrase ¹⁸	Dehydrase	Imine
Acetate kinase ¹⁹	Phosphotransferase	Acyl-phosphate (?)
D-Proline reductase ²⁰	Reductase	Dehydroalanine

BOSE-EINSTEIN CONDENSATE

A newly discovered, highly organized state of matter in which clusters of 20–30 component atoms are magnetically contained and adiabatically cooled to within $2-3 \times 10^{-9}$ K of absolute zero. At this point, the motions of the contained atoms are overcome by very weak cohesive forces of the Bose-Einstein condensate. While of no apparent relevance to biochemical kinetics, the Bose-Einstein condensate represents one of the most perfect forms of self-assembly, inasmuch as all atoms within the condensate share identical Schrödinger wave equations.

BRANCHED-CHAIN-AMINO-ACID AMINOTRANSFERASE

This enzyme [EC 2.6.1.42], also referred to as transaminase B, catalyzes the reversible reaction of leucine with α -ketoglutarate (or, 2-oxoglutarate) to produce 4-methyl-2-oxopentanoate and glutamate. The pyridoxal-phosphate-dependent enzyme will also utilize isoleucine and valine as substrates. However, this enzyme is distinct from that of valine:pyruvate aminotransferase [EC 2.6.1.66]. **See also** *Leucine Aminotransferase*

T. K. Korpela (1988) *Meth. Enzymol.* **166**, 269.

R. Kido (1988) *Meth. Enzymol.* **166**, 275.

A. J. L. Cooper (1985) *Meth. Enzymol.* **113**, 71.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

BRANCHED-CHAIN α -KETO ACID DEHYDROGENASE COMPLEX

This enzyme complex [EC 1.2.4.4], also known as 3-methyl-2-oxobutanoate dehydrogenase (lipoamide) and 2-oxoisovalerate dehydrogenase, catalyzes the reaction of 3-methyl-2-oxobutanoate with lipoamide to produce *S*-(2-methylpropanoyl) dihydrolipoamide and carbon dioxide. Thiamin pyrophosphate is a required cofactor. The complex also can utilize (*S*)-3-methyl-2-oxopentanoate and 4-methyl-2-oxopentanoate as substrates. The complex contains branched-chain α -keto acid decarboxylase, dihydrolipoyl acyltransferase, and dihydrolipoamide dehydrogenase [EC 1.8.1.4].

D. T. Chuang & R. P. Cox (1988) *Meth. Enzymol.* **166**, 135.

D. T. Chuang (1988) *Meth. Enzymol.* **166**, 146.

D. J. Danmer & S. C. Heffelfinger (1988) *Meth. Enzymol.* **166**, 298.

P. J. Randle, P. A. Patston & J. Espinal (1987) *The Enzymes*, 3rd ed., **18**, 97.

G. W. Goodwin, B. Zhang, R. Paxton & R. A. Harris (1988) *Meth. Enzymol.* **166**, 189.

BRANCHED-CHAIN α -KETO ACID DEHYDROGENASE KINASE

This enzyme phosphorylates branched-chain α -keto acid dehydrogenase using ATP.

J. Espanal, M. Beggs & P. J. Randle (1988) *Meth. Enzymol.* **166**, 166.

R. Paxton (1988) *Meth. Enzymol.* **166**, 313-320.

P. J. Randle, P. A. Patston & J. Espinal (1987) *The Enzymes*, 3rd ed., **18**, 97.

BRANCHING ENZYME

This enzyme [EC 2.4.1.18] (also known as 1,4- α -glucan branching enzyme, glycogen branching enzyme, amylo-(1,4 $\rightarrow$ 1,6)transglucosidase, and amylo-(1,4 $\rightarrow$ 1,6)transglycosylase) catalyzes the formation of 1,6-glucosidic linkages of glycogen by transferring a segment of a 1,4- α -glucan to a C6 hydroxyl of a similar glucan chain. The enzyme also converts amylose into amylopectin. The actual name given to the enzyme depends upon the product. If the product is glycogen, then glycogen branching enzyme should be used. If the product is amylopectin, then amylopectin branching enzyme is used. The latter has frequently been termed Q-enzyme.

O. Nelson & D. Pan (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 475.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

BRANCHPOINT

Any biochemical pathway intermediate that can proceed along more than one route in a network of metabolic reactions.

BREATHING

The continual loss and regain of local or global structure of a macromolecule. In terms of DNA structure, for example, breathing results from the breaking and remaking of hydrogen bonds that stabilize the double helix. The rate and extent of DNA breathing can often be assessed by tritium isotope exchange, because intact hydrogen bonds retain the isotope. In proteins, breathing also occurs, but lifetimes range over the nanosecond-to-hour time-scale.

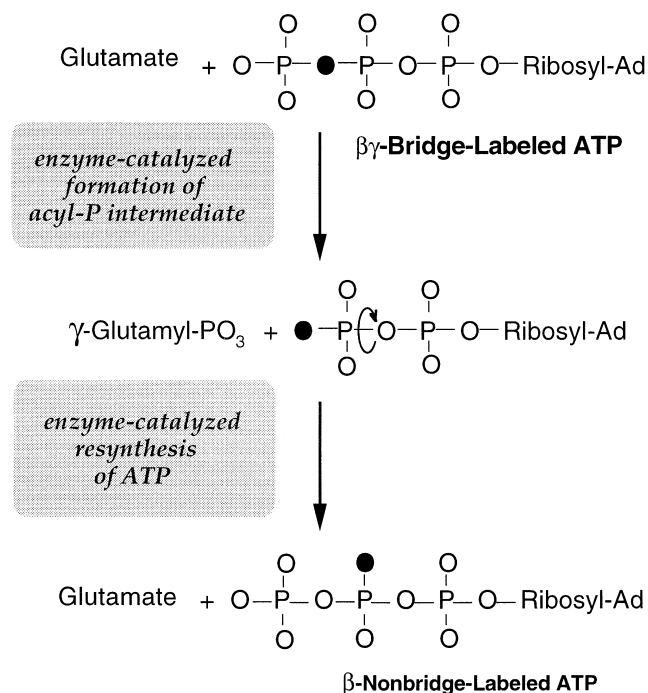
BRIDGED CARBOCATION

A carbocation in which a group or moiety bridges two or more potential carbenium centers (such that there are alternative Lewis structures having different carbenium centers).

BRIDGE-TO-NONBRIDGE OXYGEN SCRAMBLING

Midelfort and Rose¹ made a major advance in mechanistic enzymology through their development of an isotope

scrambling method for the detection of transient [Enz·ADP·P-X] formation from [^{18}O]ATP in ATP-coupled enzyme reactions. This ingenious method takes advantage of the torsional symmetry of the newly terminal phosphoryl oxygen atoms in the enzyme-bound ADP that is held near the newly formed acyl-phosphate intermediate of glutamine synthetase. ATP, specifically labeled with an ^{18}O atom in the β - γ bridge oxygen, was incubated with enzyme, and reversible cleavage of the $\text{P}_\beta\text{—O—P}_\gamma$ phosphoanhydride bond was detected by the appearance of ^{18}O in the β -nonbridge oxygens of the ATP pool. This requires formation and reversal of the phosphoryl transfer steps leading to formation of enzyme·ADP·acyl-phosphate, such that the resynthesized ATP is free to depart from the enzyme and to mix with the pool of [β - γ bridge-labeled ^{18}O]ATP. The Midelfort and Rose¹ experiments with sheep brain and *Escherichia coli* glutamine synthetases showed that cleavage of ATP to form enzyme-bound ADP and P—X requires glutamate.



Bridge-to-nonbridge positional isotope exchange catalyzed by glutamine synthetase. Note that ATP (labeled with oxygen-18 in the bridge position) is incubated with enzyme and glutamate. When formation of the acyl-phosphate occurs, the oxygen atoms on the terminal phosphoryl of ADP become torsio-symmetric. With rotation about the $\alpha\beta$ -single bond and subsequent resynthesis of ATP, the oxygen-18 atom scrambles into the nonbridge position, and this can be detected in subsequent analysis of the ATP.

The exchange catalyzed by the *E. coli* enzyme with glutamate occurs in the absence of ammonia and is partially

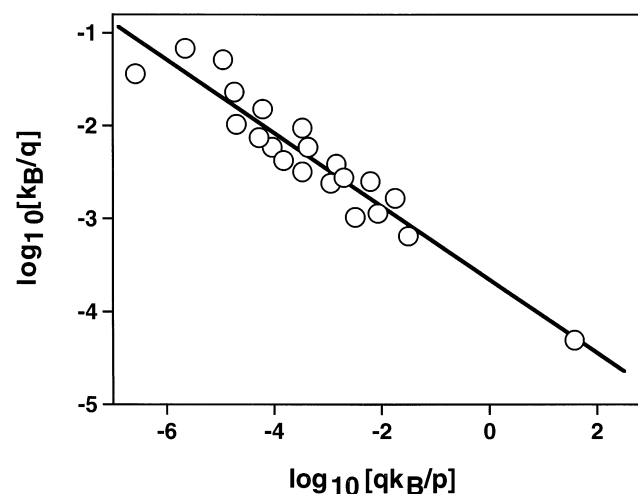
inhibited by added ammonium chloride, as would be expected if the exchange reflects the mechanistic pathway for glutamine synthesis. Their positional isotope exchange results provided compelling kinetic evidence for a two-step mechanism wherein phosphoryl transfer from ATP to glutamate to form the acyl-phosphate intermediate precedes reaction with ammonia.

¹C. F. Midelfort & I. A. Rose (1976) *J. Biol. Chem.* **251**, 5881.

BRIGGS-HALDANE EQUATION

The derivation of an enzyme initial-rate expression utilizing the steady-state assumption: *i.e.*, that the rate of change in the concentration of any enzyme form is much less than the rate of change of the product concentration or the substrate concentration. In such cases, the rate of change of the concentration of any enzyme form is set equal to zero. For a single-substrate, single-product, enzyme-catalyzed reaction, this equation (in the absence of significant product concentrations) is $v = V_{\max}[\text{S}]/(K_m + [\text{S}])$ where v is the initial rate, $V_{\max}$ is the maximum velocity for the forward reaction, $[\text{S}]$ is the initial substrate concentration, and K_m is the Michaelis constant. **See** *Enzyme Kinetics; Steady-State Assumption; Uni Uni Mechanism*

BRØNSTED PLOT



Brønsted plot for base catalysis of the mutarotation of glucose.

A graphical depiction of $\log(k_a/p)$ plotted as a function of $\log(qK_a/p)$, where k_a is the rate constant for general acid catalysis, K_a is the acid ionization constant, p is the number of equivalent protons on the acid, and q is the number of equivalent positions where a proton can be

accepted in the conjugate base. An analogous Brønsted plot is used for analyzing general base catalysis: $\log(k_b/q)$ is plotted as a function of $\log(pK_b/q)$.

BRØNSTED RELATION

A mathematical expression¹⁻⁶ relating the reaction rate constant to the strength of the acid or base participating in the reaction. This relationship, more commonly known as the Brønsted Catalysis Law, is an often used principle for analyzing systems undergoing general acid or general base catalysis. IUPAC⁷ recommends against the use of the term Brønsted Catalysis Law because the relationship has been demonstrated for a number of reactions (*e.g.*, certain proton transfer reactions) that are not catalyzed.

The Brønsted acid relation can be written as:

$$\frac{k_{HA}}{p} = G \left(\frac{qK_{HA}}{p} \right)^{\alpha}$$

where α and G are constants for a given reaction, solvent, and temperature (in which α is known as the Brønsted constant), k_{HA} is the rate constant for general acid catalysis (or the rate constant for an uncatalyzed reaction dependent on the concentration of HA [*i.e.*, the acid]), K_{HA} is the acid dissociation constant, and p and q are statistical factors. The number of equivalent protons in the acid HA is p , and the number of equivalent basic sites in the conjugate base A^- is q . Rate data can be plotted in the form $\log(k_{HA}/p) = \log G + \alpha \log(K_{HA}q/p)$. It should be noted that the chosen values for p and q should always be explicitly indicated by the authors. In addition, if one is applying transition-state theory, the rate constant should be corrected by using symmetry numbers for the transition state and the reactants. Thus, for general acid catalysis, k_{HA} is multiplied by $\sigma^\ddagger/(\sigma_{HA}\sigma_r)$. Unfortunately, establishing a good estimate for this particular correction factor is frequently challenging⁸⁻¹⁰.

The corresponding Brønsted base relation is given by:

$$\frac{k_A}{q} = G \left(\frac{qK_{HA}}{p} \right)^{-\beta}$$

where β is a Brønsted constant characteristic of the reaction, solvent, and temperature, and k_A is the rate

constant for general base catalysis (or, for uncatalyzed reactions, the rate constant dependent on the concentration of the base). Thus, $\log(k_A/q) = \log G - \beta \log(K_{HA}q/p)$.

For systems that undergo general acid and general base catalysis, one might expect that there would be a direct relationship between the observed rate constant and the strength of the acid or base. Brønsted's proposal^{1,2} in the mid-1920's was, in fact, the first linear free energy relationship. According to these relationships, a plot of the logarithm of the catalytic rate constant *versus* the logarithm of the dissociation constant for a number of acids (or bases) will yield a straight line if the system exhibits general acid (or general base) catalysis. The slope of this line provides the value for the Brønsted constant, α or β . Brønsted constants typically range between zero and one and are often considered to be an estimate of the extent of proton transfer in the transition state. A value of zero strongly suggests that the transition state resembles the reactants. On the other hand, the transition state will most likely resemble the products, when the Brønsted constant has a value near unity. Nevertheless, there are exceptions to this rule-of-thumb. In general, α will have a value of one when the rate constant is independent of the strength of the catalytic acid; the value is zero if the reaction is catalyzed by the solvent. Still others have suggested that Brønsted theory does not provide insight about the extent of proton transfer in the transition state¹¹⁻¹⁴.

There are a number of limitations on the Brønsted relationship. First of all, the relation holds only for similar types of acids (or bases). For example, carboxylic acids may have a different α values compared to sulfonic acids or phenols. Because charge, and likewise solvation, can greatly influence the reaction rate, deviations of net charge from one catalyst to another can also influence Brønsted plots. Another limitation on this relationship relates to temperature. Reaction rates and the corresponding dissociation constants for the acids must all be measured at the same temperature (and, most rigorously, in the same solvent). For some systems, this may prove infeasible. A third limitation is that the reaction must indeed be subject to general acid (or base) catalysis. For certain catalysts, deviations from a linear relationship may indicate other modes of action beyond general acid/

base catalysis (e.g., the occurrence of nucleophilic reactions). Other deviations of catalysts from a Brønsted line may be due to slow proton transfer processes and/or inefficient general acid/base catalysis (e.g., by carbon acids). Such deviations may arise as a consequence of resonance stabilization or structural reorganization of the resulting anion. Differential solvation of catalysts may also cause deviations from a Brønsted line.

As mentioned above, the Brønsted constant is an attempt to measure the relative stabilization of the transition state by different acids. However, values less than zero and greater than one have also been reported, possibly due to intermolecular effects on the transition state^{15,16}.

The linear part of the Brønsted relationship represents a segment of a curved line. The theory for the deviations from linearity was provided by Marcus^{11,12,17}.

Rowlett and Silverman¹⁸ used a Brønsted plot to examine the interaction of external buffers with human carbonic anhydrase II. The buffers act as proton acceptors in the removal of the proton generated by the enzyme-catalyzed reaction. The Brønsted plot displays a plateau at a value of about $10^9 \text{ M}^{-1}\text{s}^{-1}$ for the catalytic rate constant, which is close to the rate for proton transfer with small molecules. This plot indicated that the donor group on the enzyme has a $\text{p}K_{\text{a}}$ value of about 7.6. All of the data are “consistent with proton transfer directly between the active site and buffer or between a proton shuttle group on the enzyme of $\text{p}K_{\text{a}}$ near 7 and buffer”¹⁸. Using site-directed mutants of human carbonic anhydrase III having $\text{p}K_{\text{a}}$ values for zinc-bound water ranging between 5 and 9, Silverman *et al.*¹⁹ were able to demonstrate a Brønsted relationship and apply Marcus rate theory. The results suggest a “facile proton transfer” having a small intrinsic energy barrier of about 6.3 kJ/mol. The larger overall energy barrier of about 42 kJ/mol suggest that other processes, such as solvent reorganization, conformational changes, *etc.*, accompany the proton transfer.

¹J. N. Brønsted K. Pedersen (1924) *Zeit. Phys. Chem.* **A108**, 185.

²J. N. Brønsted (1928) *Chem. Rev.* **5**, 322.

³G. W. Klumpp (1982) *Reactivity in Organic Chemistry*, pp. 167-179, Wiley, New York.

⁴R. P. Bell (1978) in *Correlation Analysis in Chemistry: Recent Advances* (eds., N. B. Chapman & J. Shorter) pp. 55-84, Plenum Press, New York.

⁵J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, Wiley, New York.

⁶W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York].

⁷IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

⁸S. W. Benson (1958) *J. Amer. Chem. Soc.* **80**, 5151.

⁹E. Pollak & P. Pechukos (1978) *J. Amer. Chem. Soc.* **100**, 2984.

¹⁰D. R. Coulson, E. Pollak & P. Pechukos (1978) *J. Amer. Chem. Soc.* **100**, 2992.

¹¹R. A. Marcus (1968) *J. Phys. Chem.* **72**, 891.

¹²R. A. Marcus (1969) *J. Amer. Chem. Soc.* **91**, 7224.

¹³F. G. Bordwell & W. J. Boyle, Jr. (1971) *J. Amer. Chem. Soc.* **93**, 511.

¹⁴F. G. Bordwell & W. J. Boyle, Jr. (1972) *J. Amer. Chem. Soc.* **94**, 3901.

¹⁵A. J. Kresge, H. L. Chen, Y. Chiang, E. Murrill, M. A. Payne & D. S. Sagatys (1971) *J. Amer. Chem. Soc.* **93**, 413.

¹⁶A. J. Kresge (1970) *J. Amer. Chem. Soc.* **92**, 3210.

¹⁷J. Hine (1971) *J. Amer. Chem. Soc.* **93**, 3703.

¹⁸R. S. Rowlett & D. N. Silverman (1982) *J. Amer. Chem. Soc.* **104**, 6737.

¹⁹D. N. Silverman, C. Tu, X. Chen, S. M. Tanhauser, A. J. Kresge & P. J. Laipis (1993) *Biochemistry* **32**, 10757.

BRØNSTED THEORY

An acid/base theory stating (a) that an acid is any substance that tends to donate or release protons (also called hydrogen ions) to a base, and (b) that a base is a substance that accepts or removes protons from an acid. Examples of Brønsted acids include H_3O^+ , H_2O , CH_3COOH , and HSO_4^- . Examples of Brønsted bases include H_2O , OH^- , CH_3COO^- , and SO_4^{2-} .

BUBBLES

Those experimentalists who use spectrophotometry or spectrofluorimetry to measure rates of biochemical reactions should always be mindful that “bubble clearance” frequently displays first-order kinetics. This applies to bubbles adhering to the inside wall of the cuvette as well as bubbles released from solution itself. The presence of bubbles within a cuvette may introduce artifactual kinetic behavior resulting (a) from refractive index differences between the gas trapped in the bubbles and that of the test solution, and (b) from the high reflectance of the air/water interface surrounding some bubbles.

Slow warming of a cold viscous solution is a particularly effective way of generating bubbles. Should this prove to be the case, buffers can be degassed by centrifugation at 400-500 g for 10 min or by briefly warming the cold solution (obviously minus any protein constituents) to 60-70°C for 5 min. Centrifugation offers the advantage

that dust particles, another source of light scattering, can also frequently be removed from the buffer. Protein solutions often tend to froth, even upon careful mixing. This can lead to denaturation of any proteins passing into the thin bubble wall, a region where exposure to air and surface tension can wreak havoc on many proteins. Avoid shaking the cuvette after the addition of a protein or enzyme; repeated, delicate inversion of the cuvette, capped with a small strip of Parafilm, is usually sufficient to achieve adequate mixing without introducing bubbles.

BUFFER

Any substance that stabilizes the concentration of one of its dissociated species, most commonly a proton (or more properly a hydronium ion). The free concentration of a metal ion may also be buffered by use of a metal ion-chelator complex, such as Ca-EGTA or K·Mg-EDTA.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Buffer capacity, **63**, 4; choice, **63**, 19, 20, 285; metal ion chelation effects, **63**, 225, 226, 287, 298, 299; dielectric constant effect on pK, **63**, 226; dilution, **63**, 20; equilibrium constant effects, **63**, 18; heavy water, **63**, 226, 227; ionic strength effects, **63**, 226, 227; metal ion stability constant measurements, **63**, 298, 299; theory, **182**, 24; broad-range, overview, **182**, 30; broad-range, preparation, **104**, 410; metal ions and buffers [control of, **140**, 409; detection of catalytic metals, **186**, 125]; buffers: [solubilization of membrane proteins, choice, **104**, 310; **182**, 257; for cell permeabilization studies, ion selection, **192**, 297; for chemical cross-linking of histones, **170**, 553; for chromatofocusing proteins in different pH ranges, **182**, 384; for NMR studies, **104**, 412-414; for plant tissue homogenization, **182**, 179; for yeast extract preparation, **182**, 167; for preparation of extracts from higher eukaryotes, **182**, 196; for protein cleavage, **182**, 620; for reversed-phase HPLC, **193**, 404; volatile buffer removal, **182**, 29]; inhibitory effects: [of cacodylate, **9**, 669; of citrate, **9**, 620, 669; of 2,4,6-collidine, **9**, 722; of glycylglycine, **9**, 722; of imidazole, **9**, 722; of maleate, **9**, 722; of phosphate, **9**, 669; of Tris, **9**, 209, 345, 591, 722].

BUFFER CAPACITY

A measurement of the ability of a buffer system to limit the change in pH of a solution upon the addition of an increment of strong base¹⁻³. It is the reciprocal of the slope of the pH-neutralization curve². Consider the simple equilibrium, $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$ where $K_a' = [\text{H}^+][\text{A}^-]/[\text{HA}]$ in which K_a' is a practical dissociation constant determined under conditions of constant ionic strength. In such systems the practical $\text{p}K_a'$ is equal to the pH of a solution when there are equal concentrations of the two buffer species. Since the total concentration of the two

species remains constant (*i.e.*, $c = [\text{HA}] + [\text{A}^-]$) and charges have to be balanced (*i.e.*, the cation concentrations are equal to the anion concentrations: *e.g.*, $[\text{H}^+] + [\text{Na}^+] + \dots = [\text{OH}^-] + [\text{A}^-] + \dots$) then an expression for the buffer capacity, β , can be derived: $\beta = d[\text{B}]/d\text{pH} = (\log e)^{-1} [(K_a'c [\text{H}^+]/(K_a' + [\text{H}^+])^2) + [\text{H}^+] + K_w/[\text{H}^+]]$ where $[\text{B}]$ is the concentration of strong base added and K_w is the ion product of water ($= [\text{H}^+][\text{OH}^-]$). The second and third terms in the expression become important factors in strongly acidic or strongly basic solutions, respectively. Between pH values of about 3 and 11, these two terms are negligible. (However, this “rule-of-thumb” fails if the buffer concentration is very low.) Note that the buffer capacity is never zero since water will also contribute, albeit usually as a minor contributor.

See *Pseudo Buffers*

¹D. D. Van Slyke (1922) *J. Biol. Chem.* **52**, 525.

²D. D. Perrin & B. Dempsey (1974) *Buffers for pH and Metal Ion Control*, Chapman and Hall, London.

³R. G. Bates (1964) *Determination of pH, Theory and Practice*, Wiley, New York.

BUFFER EFFECTS ON ENZYMATIC PROTON TRANSFER

Beyond their control of bulk solvent pH, buffers can occasionally become an integral component in enzymatic catalysis. Carbonic anhydrase presents an instructive case where the catalytic efficiency is so great ($k_{\text{cat}} > 10^6 \text{ s}^{-1}$) that proton transfer becomes rate-limiting. The rate was found to depend on the concentration of the protonated form of buffers in the solution. Indeed, Silverman and Tu¹ adduced the first convincing evidence for the role of buffer in carbonic anhydrase catalysis through their observation of a imidazole buffer-dependent enhancement in equilibrium exchanges of oxygen isotope between carbon dioxide and water. The effect is strictly on k_{cat} , and $k_{\text{cat}}/V_{\text{max}}$ is unaffected because the latter is described by a rate law that doesn't include proton transfer steps.

¹D. N. Silverman & C. K. Tu (1975) *J. Amer. Chem. Soc.* **97**, 2263.

BUFFER SELECTION

A number of criteria are crucial in the selection of an appropriate buffer to be utilized in an enzyme-kinetic study. Included in these criteria are:

1. **Appropriate pK_a Value.** The pK_a value of the buffer should be within one pH unit of the pH value chosen for the study; preferably closer. For example, Tris should not be used as a buffer if the system is to be studied at pH 6.5. It should also be recalled that the pK_a of a buffer changes with temperature. For example, the dpK_a/dT value for Tris is -0.028 . If the assay solution is prepared and stored at one temperature, whereas the enzyme is assayed at another temperature, the buffer may very well not be appropriate under both temperature conditions.

2. **Stability.** The buffer should be stable under the conditions of the biochemical experiment or enzyme assay.

3. **Minimal Interactions with Enzyme.** The buffer should not be an effector of the enzyme system under study. It is not unusual to expect buffers containing carboxyl groups, phosphoryl groups, *etc.*, to bind to the enzyme and possibly to alter the enzyme activity. In the initial series of studies, an entire set of buffers should be examined. Buffers yielding maximal activity should be examined (under saturating substrate concentrations) at several buffer concentrations (at constant ionic strength). Buffer dilution should have minimal effect on enzyme activity. These studies are also useful in identifying potential classes of molecules that may act as potential activators or inhibitors. In addition, the investigator should supplement the study by looking at a number of different counterions.

4. **Minimal Interactions with Assay Components.** If a buffer binds to or reacts with a component of the system, it should be avoided. For example, borate buffers can complex with alcohols and carbohydrates and great care should be exercised when utilizing these compounds. Some arsenate-based buffers have oxidizing potential with respect to thiols. Buffers should also have a low binding capacity with respect to divalent cations.

5. **Effects with Respect to Assay System.** The buffer should not affect the physical procedure being utilized. For example, if proton NMR studies are being conducted, the buffer should not contain 1H . If a substance (*e.g.*, product, substrate, or protein) has to be isolated at a later step, it may be necessary to use a volatile

buffer. If a spectroscopic technique is being utilized (*e.g.*, UV absorption spectroscopy), the buffer (both protonated and deprotonated) should be transparent at the wavelength being utilized.

BUNDLING PROTEIN

Any of the cytoskeletal proteins that cross-link cytoskeletal filaments into colinear arrays, such as the action of α -actinin in promoting the formation of actin stress fibers. Bundling proteins typically contain pairs of binding sites for attachment to cytoskeletal filaments.

BUNNETT-OLSEN EQUATIONS

Certain expressions describing a solvent acidity function, where S is a base that is protonated by an aqueous mineral acid solution. The equations describe a linear free-energy relationship between $\log([SH^+]/[S]) + H_o$ and $H_o + \log[H^+]$, where H_o is Hammett's acidity function and where $H_o + \log[H^+]$ represents the activity function $\log(\gamma_S\gamma_{H^+}/\gamma_{SH^+})$ for the nitroaniline reference bases to build H_o . Thus, $\log([SH^+]/[S]) - \log[H^+] = (\phi - 1)(H_o + \log[H^+]) + pK_{SH^+}$ and $\log([SH^+]/[S]) + H_o = \phi(H_o + \log[H^+]) + pK_{SH^+}$. Here, ϕ represents the response of the $S + H^+ \rightleftharpoons SH^+$ equilibrium to changes in the acid concentration. A corresponding equation can be written for kinetic data: $\log k_\psi + H_o = \phi(H_o + \log[H^+]) + \log k_2^\circ$ where k_ψ is the pseudo-first-order rate constant for the reaction of a weak base in an acidic solution and k_2° is the second-order rate constant at infinite dilution in water. In this case, ϕ represents the response of the reaction rate to changing acid concentrations. **See also** *Acidity Function*

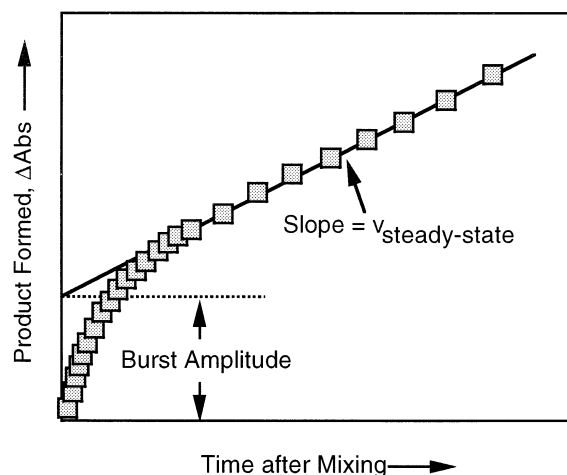
J. F. Burnett & F. P. Olsen (1966) *Can. J. Chem.* **44**, 1899.

J. F. Burnett, R. L. McDonald & F. P. Olsen (1974) *J. Amer. Chem. Soc.* **96**, 2855.

BURST KINETICS

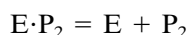
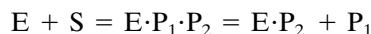
Bursts in product formation can occur when an enzyme is first combined with its substrate(s), depending on the nature of the kinetic mechanism, the relative magnitudes of the rate constants for each step, as well as the relative concentrations of active enzyme and substrate(s). This is especially apparent when one uses fast reaction kinetic techniques and when the chromophoric product is released in a fast step, which is then followed by a slower release of the second product. This is depicted below,

where product formation is indicated by the appearance of a chromophoric product.



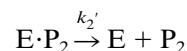
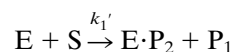
The classical example is chymotrypsin hydrolysis¹ of *p*-nitrophenyl ethylcarbonate or *p*-nitrophenyl acetate. In both cases, the first product released is *p*-nitrophenol, and the acyl-enzyme intermediate then breaks down at a slower rate. Because the first product absorbs visible light so strongly, product formation is readily detectable. In this diagram, the instantaneous velocity (*i.e.*, the line drawn tangent to the curve at any chosen moment) is highest at the time of mixing enzyme and substrate and then quickly falls off to a linear production rate characteristic of steady-state kinetic behavior. Because substrate depletion is relatively insignificant over the period shown in the figure, any change in reaction rate directly reflects the change in the concentration of free enzyme available to react with substrate. Moreover, extrapolation of the tangent line during the steady-state phase back to time zero gives the concentration change during the burst phase. This represents the concentration of active enzyme that was present at time zero.

A chemically realistic representation of the overall catalytic process can be written as follows:



In this case, P_1 is the chromophoric product that is released prior to the slower conversion of EP_2 to form P_2 ,

thereby completing the reaction cycle. For the purposes of understanding the kinetics of bursts, we can simplify the above scheme to allow for formation of EP_2 , followed by the slower release of P_2 :



As indicated by Fersht^{2,3}, the magnitude of the burst equals $[E_{active}]\{k_1'/(k_1' + k_2')\}^2$, which reduces to just the concentration of active enzyme $[E_{active}]$ if k_2' is much smaller than k_1' . Aside from providing valuable mechanistic information, burst kinetic experiments can be gainfully employed to estimate the fraction of active enzyme (*i.e.*, $[E_{active}]/[E_{total}]$) for an enzyme preparation.

¹B. S. Hartley & B. A. Kilby (1954) *Biochem. J.* **56**, 288.

²A. Fersht (1977) *Enzyme Structure and Mechanism*, pp. 108-109 & 175-176, Freeman, Reading.

³A. R. Fersht (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, pp. 650, W. H. Freeman, New York.

γ -BUTYROBETAINE HYDROXYLASE

This enzyme [EC 1.14.11.1], also known as γ -butyrobetaine, 2-ketoglutarate dioxygenase, catalyzes the reaction of 4-trimethylammoniobutanoate with δ -ketoglutarate (or, 2-oxoglutarate) and dioxygen to yield 3-hydroxy-4-trimethylammoniobutanoate, succinate, and carbon dioxide. Both iron ions and ascorbate are needed as cofactors.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

BUTYRYLCHOLINE ESTERASE

This enzyme [EC 3.1.1.8] (also known as cholinesterase, pseudocholinesterase, acylcholine acylhydrolase, non-specific cholinesterase, and benzoylcholinesterase) catalyzes the hydrolysis of an acylcholine to generate choline and a carboxylic acid anion. A variety of choline esters and a few other compounds can serve as substrates.

- D. M. Quinn & S. R. Feaster (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 455.
 H. Okuda (1991) *A Study of Enzymes*, **2**, 563.

BUTYRYL-CoA DEHYDROGENASE

This enzyme [EC 1.3.99.2], also referred to as short-chain acyl-CoA dehydrogenase and unsaturated acyl-CoA reductase, catalyzes the reaction of butanoyl-CoA and an electron-transferring flavoprotein to produce

2-butenoyl-CoA and the reduced electron-transferring flavoprotein. The enzyme, which uses FAD, forms a complex with the electron-transferring flavoprotein and electron-transferring flavoprotein dehydrogenase [EC 1.5.5.1] that will reduce ubiquinone and other acceptors.

- B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

C

CAFFEINE

A naturally occurring substance (1,3,7-trimethylxanthine) that competitively inhibits 3',5'-cyclic AMP phosphodiesterase and acts as a powerful central nervous system stimulant. The inhibitory effect can be reversed at high cAMP concentrations. A related substance, theophylline (also known as aminophylline) has a similar mode of action.

CAGE

An aggregate collection of molecules surrounding another molecular entity. A solvent cage forms by electrostatic interactions around various ions in solution.

CAGE COMPOUND

1. Clathrate. 2. Any molecular entity that is nonplanar and often polycyclic that encloses a cavity. 3. Photosensitive precursors that generate chemically reactive species upon irradiation with a laser or flash-lamp. **See** *Photoreactive Caged Compounds*

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

CAGED ATP

A chemically substituted ATP molecule (such as adenosine-5'-triphospho-1-(2-nitrophenyl)ethanol) that is not capable of participating in ATP-dependent hydrolase/phosphotransferase reactions until the protecting group is photochemically removed. Excitation with UV light at a wavelength of 360 nm immediately photolyzes caged ATP to produce ATP, thereby permitting the kinetic analysis of processes occurring on time-scales as short as 1-2 milliseconds.

CAHN-INGOLD-PRELOG SYSTEM

A self-consistent set of rules (also referred to as the *RS*-system) that is used to designate the three-dimensional configuration of chiral molecules¹. This system replaces the *DL* absolute configuration scheme. In publishing experimental results using chiral substrates, the *RS*-designation should always be provided. If the substrate under consideration is an amino acid or sugar, then the *DL* absolute configuration may be used. However, one should recall that there is no direct correlation between the two systems. For example, L-alanine (or, (+)-alanine) is equivalent to (*S*)-alanine whereas L-cysteine (or, (–)-cysteine) corresponds to (*R*)-cysteine. The individual rules for the Cahn-Ingold-Prelog system can be found in the references below or in any basic organic chemistry text. The important rules are: (1) List substituents directly attached to a chiral carbon in decreasing atomic number. (2) If two or more atoms are directly linked to the chiral carbon, then the atomic number of the atom(s) linked to those atoms determines the ranking. Thus, in (CH₃)₂CH–CHCl–CH₂SH, the –CH₂SH group takes precedence over the CH(CH₃)₂ group because sulfur has a higher atomic number than carbon. (3) Double bonds are counted as two single bonds, and triple bonds count as three single bonds. (4) When dealing with different isotopes of the same atom, the heavier isotope takes precedence over the lighter one. Thus, tritium takes precedence over hydrogen. (5) All atoms except hydrogen are considered to have a valence of four; thus, –NH⁺(CH₃)₂ takes precedence over –N(CH₃)₂.

IUPAC (1976) *Pure and Appl. Chem.* **45**, 13.

R. S. Cahn, C. K. Ingold & V. Prelog (1966) *Angew. Chem. Int. Ed. Engl.* **5**, 385.

V. Prelog & G. Helmchen (1982) *Angew. Chem. Int. Ed. Engl.* **21**, 567.

CALCINEURIN

A calmodulin-binding phosphoprotein phosphatase, found at highest abundance in brain tissue, that removes

phosphoryl groups from phosphoseryl and phosphothreonyl residues.

- B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.
 C. B. Klee, G. F. Draetta & M. J. Hubbard (1988) *Adv. Enzymol.* **61**, 149.
 C. B. Klee, M. H. Krinks, A. S. Manalan, P. Cohen & A. A. Stewart (1983) *Meth. Enzymol.* **102**, 227.

CALCIUM-45 (⁴⁵Ca)

A β -emitting (0.257 MeV) radioactive calcium nuclide with a decay half-life of 163 days. The following decay data indicate the time (days), followed by the fraction of original amount at the specified time: 0.0, 1.000; 5, 0.979; 10, 0.958; 15, 0.938; 20, 0.918; 25, 0.898; 30, 0.879; 35, 0.862; 40, 0.844; 45, 0.826; 50, 0.808; 100, 0.654; 150, 0.528; 200, 0.427; 250, 0.345; 300, 0.279; 350, 0.226; 400, 0.183; 450, 0.148; 500, 0.119; 550, 0.096; 600, 0.078.

Special precautions: Bone is the critical organ in terms of dose. The effective *biological* half-life in humans is nearly 20 years. Double gloving is recommended when working with this radionuclide.

CALCIUM/CALMODULIN-DEPENDENT PROTEIN KINASE

This enzyme [EC 2.7.1.123], also referred to as calcium/calmodulin-dependent protein kinase type II, and microtubule-associated protein MAP2 kinase, catalyzes the reaction of ATP with a protein to produce ADP and an *O*-phosphoprotein. The enzyme requires calcium ions and calmodulin. Proteins that can serve as substrates include vimentin, synapsin, glycogen synthase, the myosin light-chains, and the microtubule-associated tau protein. This enzyme is distinct from myosin light-chain kinase [EC 2.7.1.117], caldesmon kinase [EC 2.7.1.120], and tau-protein kinase [EC 2.7.1.135].

- P. I. Hanson & H. Schulman (1992) *Ann. Rev. Biochem.* **61**, 559.
 S. I. Walaas & P. Greengard (1987) *The Enzymes*, 3rd ed., **18**, 285.
 J. T. Stull, M. H. Nunnally & C. H. Michnauff (1986) *The Enzymes*, 3rd ed., **17**, 113.

CALCIUM ION

This Group IIA (or Group 2) element (atomic symbol, Ca; atomic number, 20; atomic weight, 40.078; electronic configuration = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$) loses both $4s^2$ electrons to form a divalent cation of 0.99 Å ionic radius. Ionic calcium combines readily with oxygen ligands (chiefly water, phosphates, polyphosphates, and carboxylates) to form stable metal ion complexes. ⁴¹Ca under-

goes nuclear disintegration by electron capture with a decay energy of 0.421 MeV and a half-life of 103,000 years. ⁴⁵Ca is a beta particle emitter with a decay energy of 0.257 MeV and a half-life of 162.7 days. The high degree of spatial and temporal precision occurring during calcium ion control of cellular processes suggests that there are efficient mechanisms for localized control release and sequestration of calcium ion. This is possible through the highly coordinated interplay of calcium ion channels, stores, and oscillations.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Chelation, **238**, 74, 76, 297; buffers [for analysis of exocytosis, **221**, 132; preparation, **219**, 186; modulation of cytosolic buffering capacity with quin2, **221**, 159]; fluorescence assay, **240**, 724-725, 740-742; fluorescence imaging, **225**, 531; **238**, 303-304, 322-325, 334-335; free intracellular levels after bacterial invasion, **236**, 482-489; free calcium in solutions for membrane fusion analysis, calculation and control, **221**, 149; homeostasis mechanisms, **238**, 80; hormonal elevation, **238**, 79; inositol phosphate effect on release, **238**, 207; determination of cytosolic levels [computer methods, **238**, 73-75; with fura-2, **238**, 73, 146; with indo-1, **238**, 298, 316-317; with quin-2, **238**, 297]; hormone effects, **238**, 79; ionomycin effects, **238**, 79; membrane depolarization effects, **238**, 80; NMR determination of calcium binding to proteins, **227**, 115; ⁴⁵Ca, **238**, 153-154; free ion determination, **238**, 139-140; spectrofluorometric assay, **236**, 485; calcium-binding proteins, **228**, 248-253; chelators, **236**, 473-475.

CALCIUM ION INDICATOR DYES

Tsien's development^{1,2} of calcium ion sensors in the form of fluorescent chelator/indicator molecules has allowed major advances regarding regulatory mechanisms based on calcium ion. It is now both feasible and routine to measure $[Ca^{2+}]$ directly in single living cells at high spatial resolution, using a digital imaging microscope and the highly fluorescent Ca^{2+} -sensitive dye known as fura-2.

Kao and Tsien³ studied the Ca^{2+} -binding kinetics of fura-2 and azo-1 by temperature-jump relaxation methods. In 140 mM KCl at 20°C, the respective association and dissociation rate constants for fura-2 were $6 \times 10^8 M^{-1}s^{-1}$ and $97 s^{-1}$; these kinetic properties were insensitive to hydrogen ion concentration over the pH range from 7.4 to 8.4. Azo-1 was studied in 140 mM KCl: At 10°C, azo-1 exhibited respective association and dissociation rate constants of $1.4 \times 10^8 M^{-1}s^{-1}$ and $780 s^{-1}$; at 20°C, these values were $3.99 \times 10^8 M^{-1}s^{-1}$ and $1200 s^{-1}$. Thus, fura-2 and azo-1 permit rapid monitoring changes in intracellular calcium ion concentration. **See Fura-2**

Tsien⁴ recently introduced a simple technique for loading cells with Ca^{2+} -selective chelators that are rendered tem-

porarily membrane permeable by masking their four carboxylates as esters which then hydrolyze once inside the cells and unmask the active chelator.

Ca^{2+} activation of processes at cell membranes (such as those involved in contraction, secretion, and neurotransmitter release) require *in situ* or *in vitro* Ca^{2+} concentrations that are typically 10–100 times higher than those measured in intact cells during stimulation. This discrepancy might be reconciled if the local cell membrane $[\text{Ca}^{2+}]$ is very different from that in the rest of the cell. Soluble Ca^{2+} indicators indicate spatially averaged cytoplasmic $[\text{Ca}^{2+}]$ and cannot resolve fluctuations in near-membrane Ca^{2+} signals. However, the new Ca^{2+} indicator FFP18 allows one to monitor near-membrane $[\text{Ca}^{2+}]$ selectively⁵. Images of the intracellular distribution of FFP18 show that >65% is located on or near the plasma membrane. Recorded $[\text{Ca}^{2+}]$ transients using FFP18 during membrane depolarization-induced Ca^{2+} influx show that near-membrane $[\text{Ca}^{2+}]$ rises faster and reaches micromolar levels earlier than cytoplasmic $[\text{Ca}^{2+}]$; the latter of which only reaches 0.1–0.2 μM . High-speed digital imaging revealed that the near-membrane $[\text{Ca}^{2+}]$ rises within 20 msec, peaks at 50–100 msec, and then subsides⁵.

See also *Calcium Ion; Aequorin; Fura-2*

¹R. Y. Tsien (1989) *Annu. Rev. Neurosci.* **12**, 227.

²R. W. Tsien & R. Y. Tsien (1990) *Annu. Rev. Cell. Biol.* **6**, 715.

³J. P. Kao & R. Y. Tsien (1988) *Biophys. J.* **53**, 635.

⁴R. Y. Tsien (1981) *Nature* **290**, 527.

⁵E. F. Etter, A. Minta, M. Poenie & F. S. Fay (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 5368.

CALMODULIN

A regulatory protein initially discovered as an activator of cyclic nucleotide phosphodiesterase (or 3',5'-cyclic nucleotide 5'-PDE, EC 3.1.4.17). Kinetic studies¹ on the activation of cyclic nucleotide PDE as a function of calmodulin and Ca^{2+} concentrations were analyzed by what is likely to be a general approach for characterizing activation mechanisms. The method accounts for the various interactions between phosphodiesterase and different calmodulin- Ca^{2+} complexes. All four Ca^{2+} must be bound to calmodulin for the protein to form an activated complex with PDE. The authors identified several regulatory advantages of engaging the four Ca^{2+} sites on calmodulin: (1) The activation of PDE as a function of Ca^{2+} is highly cooperative (if the PDE-calmodulin- $(\text{Ca}^{2+})_4$ complex is the dominant active species), and the

system behaves as an “on/off” switch for PDE activation. (2) At normal intracellular Ca^{2+} levels (<0.1 μM), PDE and calmodulin fail to complex, thereby allowing the distribution of calmodulin complexes with its various target enzymes to be reshuffled after resequestration of Ca^{2+} following each surge. (3) The affinity between the enzyme and the fully liganded calmodulin is 10^4 – 10^5 times that observed in the absence of Ca^{2+} . Thus, a very sharp increase in affinity is achievable through a 10- to 20-times increase in the affinity of Ca^{2+} for the enzyme-calmodulin complex in each of the four binding steps.

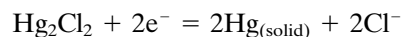
¹C. Y. Huang, V. Chau, P. B. Chock, J. H. Wang & R. K. Sharma (1981) *Proc. Natl. Acad. Sci. U.S.A.* **78**, 871.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Calmodulin purification and fluorescent labeling, **102**, 1; assay of calmodulin by Ca^{2+} -dependent phosphodiesterase, **102**, 39; myosin light chain phosphorylation in smooth muscle and nonmuscle cells as a probe of calmodulin function, **102**, 62; spectroscopic analyses of calmodulin and its interactions, **102**, 82; Ca^{2+} binding to calmodulin, **102**, 135; preparation of fluorescent labeled calmodulins, **102**, 148; techniques for measuring the interaction of drugs with calmodulin, **102**, 171; detection of calmodulin-binding polypeptides separated in SDS-polyacrylamide gels by a sensitive [²⁵I]calmodulin gel overlay assay, **102**, 204; use of calmodulin affinity chromatography for purification of specific calmodulin-dependent enzymes, **102**, 210; chemical approaches to the calmodulin system, **102**, 296; ¹³C chemical shift, **239**, 369; contamination in creatine phosphokinase, **238**, 76; removal of endogenous calmodulin, **238**, 77; heteronuclear relaxation studies, **239**, 564; hydrophobic affinity partitioning, **228**, 257; myosin light-chain kinase peptide complex, **239**, 664, 682, 708; regulation of adenylyl cyclase, **238**, 77; as retroviral protease substrate, **241**, 290; 3D ¹H-detected [¹³C-¹H] long-range correlation spectrum, **239**, 97.

CALOMEL ELECTRODE

A reference electrode composed of metallic mercury and solid mercurous chloride. The redox half-reaction is:



The potential of this electrode only depends on the chloride ion concentration which is strictly controlled by the saturated potassium chloride solution in the electrode. The E° value for the above reaction is 0.26808 volts (at 273.15 K and a pressure of 1 atm (or, 101325 Pa).

CALORIMETRY

The measurement of the amount of heat released or absorbed in a reaction; also included in this category are determinations of heat capacities, latent heats, and caloric values of fuels. **See also** *Differential Scanning Calorimetry*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Calorimetry: Accuracy, **240**, 98-99; algorithm for nonlinear least-

squares fit, **240**, 94; enthalpy change for ligand binding, **240**, 91-92, 96-99; entropy change for ligand binding, **240**, 93; free energy of ligand binding, **240**, 93-94; free ligand concentration calculation, **240**, 93; heat of binding, **240**, 92; heat capacity, analysis, **240**, 96-99; parameter estimation, **240**, 95; sequential analysis, **240**, 96-99; simultaneous analysis, **240**, 96-99; titration experiments, **240**, 91

Differential Scanning Calorimetry: Baseline estimation [effect on parameters, **240**, 542-543, 548-549; importance, **240**, 540; polynomial interpolation, **240**, 540-541, 549, 567; proportional method, **240**, 541-542, 547-548, 567]; changes in solvent accessible surface areas, **240**, 519-520, 528; cooperativity measurement error, **240**, 642; data simulation, **240**, 537-539; effect of ligand binding, **240**, 533; enthalpy determination, **240**, 504, 506-507, 513-514, 534-537; entropy determination, **240**, 506-507, 514-517, 535-536, 539; folding/unfolding partition function determination, **240**, 505, 508-509, 532, 535; F-statistics, **240**, 554-559, 562; gel-liquid crystalline phase transition, **240**, 571-572; Gibbs free energy determination, **240**, 503, 518, 521, 535-536; heat capacity determination, **240**, 506, 510-513, 531, 534, 546-547; industrial applications, **240**, 531, 568; α -lactalbumin molten globule state, **240**, 522, 528-530; noise effect on parameters, **240**, 539-540, 562, 565; two-dimensional, guanidinium hydrochloride thermal stability surfaces [accuracy, **240**, 566-567; methods, **240**, 537-543; multiple folding state fitting, **240**, 543-566].

CALPAIN

This enzyme [EC 3.4.22.17] is an intracellular, nonlysosomal member of the peptidase family C2. The enzyme catalyzes the calcium ion-dependent hydrolysis of peptide bonds with preference for Tyr-Xaa, Met-Xaa, or Arg-Xaa with a leucyl or valyl residue at the P₂ position. There are two main types of calpain. One has a high calcium sensitivity in the micromolar range and is called μ -calpain or calpain I. The other calpain has a low calcium sensitivity in the millimolar range and is called m-calpain or calpain II. Forms of calpain exhibiting intermediate calcium sensitivity also exist.

E. Shaw (1990) *Adv. Enzymol.* **63**, 271.

J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.

CAMPHOR 5-MONOOXYGENASE

This enzyme [EC 1.14.15.1], also known as camphor 5-*exo*-methylene hydroxylase, and cytochrome P450-cam, catalyzes the reaction of (+)-camphor with putidaredoxin and dioxygen to generate (+)-*exo*-5-hydroxycamphor, oxidized putidaredoxin, and water. A heme-thiolate acts as a cofactor. The enzyme can also utilize (–)-camphor as a substrate, and 1,2A-campholide will result in the formation of 5-*exo*-hydroxy-1,2-campholide.

V. Ullrich & W. Duppel (1975) *The Enzymes*, 3rd ed., **12**, 253.

cAMP PHOSPHODIESTERASE

This enzyme [EC 3.1.4.17] catalyzes the hydrolysis of 3',5'-cAMP or other 3',5'-cyclic nucleotides (cNMP) to

form AMP or NMP. The mammalian enzyme is activated by calcium ions in the presence of calmodulin.

E. Degerman, P. Belfrage & V. Manganiello (1997) *J. Biol. Chem.* **272**, 6823.

J. A. Gerlt (1992) *The Enzymes*, 3rd ed., **20**, 95.

cAMP-STIMULATED PROTEIN KINASE (PKA)

This protein kinase (known as protein kinase A or PK-A) has an R₂C₂ quaternary structure that binds 3',5'-cAMP at its dimeric regulatory (R) subunit with resultant release of two catalytic (C) subunits. The free energy of hydrolysis of the cyclic nucleotide activator is large (ΔG –13 kcal/mol) and allows the 3',5'-cAMP to be virtually irreversibly converted to AMP by the action of a specific phosphodiesterase. This protein kinase, originally discovered by the Nobelists Edwin Krebs and Edward Fischer, is now considered to be the prototype for over two thousand members of the protein kinase superfamily.

D. Bossemeyer, R. A. Engh, V. Kinzel, H. Ponstingl & R. Huber (1993) *EMBO J.* **12**, 849.

L. N. Johnson & D. Barford (1993) *Ann. Rev. Biophys. Biomol. Struct.* **22**, 199.

J. Larner (1990) *Adv. Enzymol.* **63**, 173.

S. I. Walaas & P. Greengard (1987) *The Enzymes*, 3rd ed., **18**, 285.

S. J. Beebe & J. D. Corbin (1986) *The Enzymes*, 3rd ed., **17**, 43.

D. A. Walsh & E. G. Krebs (1973) *The Enzymes*, 3rd ed., **8**, 555.

CANNIZZARO REACTION

The conproportionation reaction of aromatic aldehydes or aldehydes lacking α -hydrogens under alkaline conditions, yielding an alcohol and an acid.

S. Cannizaro (1853) *Ann.* **88**, 129.

T. A. Geissman (1944) *Org. React.* **2**, 94.

CARBAMOYL-PHOSPHATE SYNTHETASE I AND II

Carbamoyl-phosphate synthetase (ammonia) [EC 6.3.4.16], also known as carbamoyl-phosphate synthetase I, catalyzes the reaction of two molecules of ATP with carbon dioxide, ammonia, and water to produce two molecules of ADP, orthophosphate, and carbamoyl phosphate. Carbamoyl-phosphate synthetase (glutamine-hydrolyzing) [EC 6.3.5.5], also known as carbamoyl-phosphate synthetase II, catalyzes the reaction of two molecules of ATP with carbon dioxide, glutamine, and water to produce two molecules of ADP, orthophosphate, glutamate, and carbamoyl phosphate.

- J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 495.
 L. S. Mullins & F. M. Raushel (1995) *Meth. Enzymol.* **249**, 398.
 H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.
 A. Meister (1989) *Adv. Enzymol.* **62**, 315.
 D. S. Kaseman & A. Meister (1985) *Meth. Enzymol.* **113**, 305.
 I. A. Rose (1979) *Adv. Enzymol.* **50**, 361.
 D. E. Koshland, Jr., & A. Levitzki (1974) *The Enzymes*, 3rd ed., **10**, 539.
 S. Ratner (1973) *Adv. Enzymol.* **39**, 1.
 J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

N-CARBAMOYLPUTRESCINE AMIDOHYDROLASE

This enzyme [EC 3.5.1.53], also known as *N*-carbamoylputrescine amidase, catalyzes the hydrolysis of *N*-carbamoylputrescine to produce putrescine, carbon dioxide, and ammonia.

T. A. Smith (1985) *Ann. Rev. Plant Physiol.* **36**, 117.

CARBANION

An anion species in which an even number of electrons and a significant negative charge remain located on at least one carbon atom.

CARBENE

A highly reactive compound containing a neutral divalent carbon with two nonbonding electrons (*i.e.*, :CH₂ or a substitution derivative). The nonbonding electrons can have parallel spins (triplet state) or antiparallel spins (singlet state). The parent species, :CH₂, is also known as methylene. A number of carbene derivatives have been used as photoaffinity labels of proteins. Irradiation of 3'-*O*-(4-benzoyl)benzoyl-ATP will cause 70% inactivation of mitochondrial F₁-ATPase¹.

¹N. Williams & P. S. Coleman (1982) *J. Biol. Chem.* **257**, 2834.

CARBENIUM CENTER

The carbon atom (within a carbenium ion) on which the excess positive charge may be considered to be largely concentrated.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

CARBENIUM ION

A term for a positively-charged trivalent carbon atom (a carbocation). R₃C⁺ is a carbenium ion. The term "car-

benium ion" is intended to replace the term "carbonium ion," but the habits of practicing chemists persist. **See** *Carbocation; Carbonium Ion; Carbene*

CARBOCATION

A cation in which significant positive charge is located on at least one carbon atom, itself having an even number of electrons. Both carbenium ions and pentavalent species (such as the methanonium ion, CH₅⁺) are carbocations. However, radical cations and carbynium ions are not considered to be carbocations. **See** *Carbenium Ion; Bridged Carbocation; Carbonium Ion*

CARBOHYDRATE MODIFICATION REACTIONS

Because carbohydrates are so frequently used as substrates in kinetic studies of enzymes and metabolic pathways, we refer the reader to the following topics in Robyt's excellent account¹ of chemical reactions used to modify carbohydrates: formation of carbohydrate esters, pp. 77-81; sulfonic acid esters, pp. 81-83; ethers [methyl, p. 83; trityl, pp. 83-84; benzyl, pp. 84-85; trialkyl silyl, p. 85]; acetals and ketals, pp. 85-92; modifications at C-1 [reduction of aldehydes and ketones, pp. 92-93; reduction of thioacetals, p. 93; oxidation, pp. 93-94; chain elongation, pp. 94-98; chain length reduction, pp. 98-99; substitution at the reducing carbon atom, pp. 99-103; formation of glycosides, pp. 103-105; formation of glycosidic linkages between monosaccharide residues, 105-108]; modifications at C-2, pp. 108-113; modifications at C-3, pp. 113-120; modifications at C-4, pp. 121-124; modifications at C-5, pp. 125-128; modifications at C-6 in hexopyranoses, pp. 128-134.

¹J. F. Robyt (1998) *Essentials of Carbohydrate Chemistry*, Springer, New York.

CARBON-14 (¹⁴C)

A radioactive β -emitting (0.156 MeV) carbon nuclide. The decay half-life is 5715 years.

Special precautions: Bone concentrates carbonates and is the critical organ in terms of dose. The effective *biological* half-life in humans is around 10 days. Double gloving is recommended when working with this radionuclide.

CARBON DIOXIDE

Carbon dioxide is a symmetrical, linear triatomic molecule ($\text{O}=\text{C}=\text{O}$) with a zero dipole moment. The carbon-to-hydrogen bond distances are about $1.16\text{\AA}$, which is about $0.06\text{\AA}$ shorter than typical carbonyl double bonds. This shorter bond length was interpreted by Pauling¹ to indicate that greater resonance stabilization occurs with CO_2 than with aldehydes, ketones, or amides. When combined with water, carbonic acid (H_2CO_3) forms, and depending on the pH of the solution, carbonic acid loses one or two protons to form bicarbonate and carbonate, respectively. The various thermodynamic parameters² of these reactions are shown in Table I.

Table I

Thermodynamics of CO_2 Hydration and Ionization at 298 K

Reaction	pK	ΔG (kcal/mol)	ΔH (kcal/mol)	ΔS (e.u.)
$K_{\text{hydration}} = [\text{H}_2\text{CO}_3]/[\text{CO}_2]$	2.59	3.53	1.13	-8
$K_{\text{carbonic acid}} = [\text{H}^+][\text{HCO}_3^-]/[\text{H}_2\text{CO}_3]$	3.77	5.17	1.01	-14
$K_1 = [\text{H}^+][\text{HCO}_3^-]/\{[\text{CO}_2] + [\text{H}_2\text{CO}_3]\}$	6.35	8.67	2.24	-21.6
$K_2 = [\text{H}^+][\text{CO}_3^{2-}]/[\text{HCO}_3^-]$	10.3	14.1	3.60	-35.2

Edsall³ has presented a cogent analysis of the various rates of hydration and dehydration, including the ability of oxygen acids to catalyze these reactions. **See** *Oxygen, Oxides & Oxo Anions*

¹L. Pauling (1960) *The Nature of the Chemical Bond* (3rd ed.), pp. 267-269, Cornell Univ. Press, Ithaca.

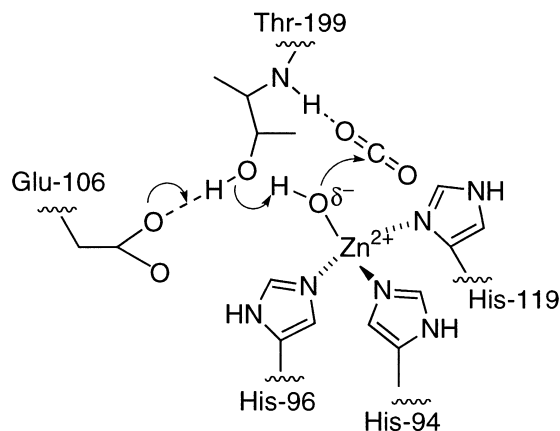
²J. T. Edsall & J. Wyman (1958) *Biophysical Chemistry*, vol. I, chap. 10, Academic Press, New York.

³J. T. Edsall (1969) in *CO₂: Chemical, Biochemical, and Physiological Aspects* (R. E. Forster, J. T. Edsall, A. B. Otis & F. J. W. Roughton, eds.), pp. 15-27, NASA Publication SP-188.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

As anesthetic, **77**, 115; binding equilibria analysis, **76**, 487-511; by chemical-chromatographic method, **76**, 487-496; breath tests, **77**, 3-6; carbonic anhydrase, **64**, 120; carbon isotope effect, **64**, 86-95; collection of ^{13}C -labeled carbon dioxide, **77**, 9; contamination, **64**, 93, 102; distillation, **64**, 91; effect on oxygen equilibrium curve for hemoglobin, **76**, 446, 468-470; equilibrium isotope effect, **64**, 109; exhalation, **77**, 3-9; pharmacokinetics, **77**, 8, 9; fixation [measurement in mutants, **69**, 16, 17, 20-23; photosynthetic mutants, **69**, 9; release of products, **69**, 608, 609]; formation, **77**, 3-6; controlled supply, **69**, 567-569; measurement, **69**, 563; mass spectrometer calibration, **69**, 554; measurement, **64**, 92; **69**, 563, 564; comparison of techniques, **77**, 6-8; reaction with α -amino and ϵ -amino groups, **76**, 488-489, 495, 504, 505; resistance to diffusion, **69**, 453; solubility, **64**, 93-94; in hemoglobin solutions, **76**, 494; standard curve, **69**, 564-566; nitrogen and water removal, **64**, 91.

CARBONIC ANHYDRASE



Carbonic Anhydrase-II Catalysis of H_2O Attack on CO_2

This enzyme [EC 4.2.1.1], also referred to as carbonate dehydratase, is a zinc-dependent enzyme that catalyzes the reaction of carbon dioxide with water to form carbonic acid (or, of bicarbonate and a proton). **See also** *Proton Transfer in Aqueous Solution; Manometric Assay Methods; Marcus Rate Theory*

J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 483.

D. N. Silverman (1995) *Meth. Enzymol.* **249**, 479.

W. S. Sly & P. Y. Hu (1995) *Ann. Rev. Biochem.* **64**, 375.

M. R. Badger & G. D. Price (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 369.

D. W. Christianson (1991) *Adv. Protein Chem.* **42**, 281.

B. L. Vallee & A. Galde (1984) *Adv. Enzymol.* **56**, 283.

D. N. Silverman & S. H. Vincent (1983) *Crit. Rev. Biochem.* **14**, 207.

D. N. Silverman (1982) *Meth. Enzymol.* **87**, 732.

Y. Pocker & S. Sarkonen (1978) *Adv. Enzymol.* **47**, 149.

S. Lindskog, L. E. Henderson, K. K. Kannan, A. Liljas, P. O. Nyman & B. Strandberg (1971) *The Enzymes*, 3rd ed., **5**, 587.

CARBONIUM ION

1. A carbocationic species in which there is at least one pentavalent carbon atom (e.g., CH_5^+). 2. Traditional name for chemical species that are now referred to as carbenium ions. Considerable confusion exists in the literature with this term for carbocations. The “-onium” suffix usually refers to a higher covalency when compared to the neutral atom; thus, CH_5^+ would be a true carbonium ion (in terms of the first definition). Additional ambiguity results when the term “ethyl carbonium ion” is used to describe both CH_3CH_2^+ and to $\text{R}-\text{CH}_2\text{CH}_2^+$. For these reasons, the terms carbocation or carbenium ion are now preferred.

CARBON MONOXIDE

The properties of this odorless and extremely poisonous diatomic gas are described under *Oxygen, Oxides & Oxo Anions*.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Hemoglobin binding, stepwise CO combination and dissociation rate constants, double mixing experiment, **232**, 432-436; infrared spectra, **232**, 140-151; interaction with partially liganded hemoglobin intermediates, double mixing experiments, **232**, 432-445; rebinding to hemoglobin [after photodissociation, **232**, 72-73, 416; after photolysis, **232**, 80-83, 416; carboxyhemoglobin Fe-C-O configuration, **232**, 280-281; optical spectroscopy, **232**, 62-63; structural relaxation dynamics studies, **232**, 345, 348].

CARBON MONOXIDE DEHYDROGENASE

This enzyme [EC 1.2.99.2], also known as acetyl-CoA synthase, catalyzes the reaction of carbon monoxide with water and an acceptor to produce carbon dioxide and the reduced acceptor. The cofactors of this enzyme include nickel and zinc ions as well as non-heme iron. Methyl viologen can act as the acceptor substrate. The enzyme is isolated from *Clostridium* sp. Interestingly, it also catalyzes an exchange reaction of carbon between C1 of acetyl-CoA and carbon monoxide. The protein participates in the synthesis of acetyl-CoA from carbon dioxide and hydrogen in the organisms.

B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.

P. E. Baugh, D. Collison, C. D. Garner & J. A. Joule (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 377.

N⁵-(CARBOXYETHYL)ORNITHINE SYNTHASE

This enzyme [EC 1.5.1.24], also referred to as N⁵-(1-L-carboxyethyl)-L-ornithine:NADP⁺ oxidoreductase, reversibly catalyzes the reaction of ornithine with pyruvate and NADPH to produce N⁵-(1-carboxyethyl)ornithine, NADP⁺, and water. Lysine can also serve as a substrate, acting on N⁶, albeit not as effectively as ornithine.

J. Thompson & J. A. Donkersloot (1992) *Ann. Rev. Biochem.* **61**, 517.

J. Thompson & S. P. F. Miller (1991) *Adv. Enzymol.* **64**, 317.

CARBOXYLESTERASES

These enzymes [EC 3.1.1.1] (also referred to as ali-esterase, B-esterase, monobutyrase, cocaine esterase, methylbutyrase, and procaine esterase) catalyze the hydrolysis of a carboxylic ester to yield a carboxylate anion and an alcohol. They exhibit a broad specificity, even acting on vitamin A esters.

K. Krisch (1971) *The Enzymes*, 3rd ed., **5**, 43.

CARBOXYPEPTIDASE A

This zinc-dependent enzyme [EC 3.4.17.1], a member of the peptidase family M14, catalyzes the hydrolysis of peptide bonds at the C-terminus of polypeptides. Little hydrolytic action occurs if the C-terminal amino acid is aspartate, glutamate, arginine, lysine, or proline. Carboxypeptidase A is formed from a precursor protein, procarboxypeptidase A.

J. E. Coleman (1992) *Ann. Rev. Biochem.* **61**, 897.

J. J. Villafranca & T. Nowak (1992) *The Enzymes*, 3rd ed., **20**, 63.

D. W. Christianson (1991) *Adv. Protein Chem.* **42**, 281.

J. A. Hartsuck & W. N. Lipscomb (1971) *The Enzymes*, 3rd ed., **3**, 1.

CARBOXYPEPTIDASE B

This zinc-dependent enzyme [EC 3.4.17.2], a member of the peptidase family M14, catalyzes the hydrolysis of peptide bonds located at the C-terminus of a polypeptide. The enzyme prefers C-terminal lysine or arginine residues.

J. E. Folk (1971) *The Enzymes*, 3rd ed., **3**, 57.

CARBOXYPEPTIDASE C

This enzyme [EC 3.4.16.5] (also known as serine-type carboxypeptidase I, cathepsin A, carboxypeptidase Y, and lysosomal protective protein) is a member of the peptidase family S10 and catalyzes the hydrolysis of the peptide bond, with broad specificity, located at the C-terminus of a polypeptide. The pH optimum ranges from 4.5 to 6.0. The enzyme is irreversibly inhibited by diisopropyl fluorophosphate and is sensitive to thiol-blocking reagents.

S. J. Remington & K. Breddan (1994) *Meth. Enzymol.* **244**, 231.

CARBOXYPEPTIDASE D

This enzyme [EC 3.4.16.6], also known as carboxypeptidase KEX1 and carboxypeptidase S1, is a member of the peptidase family S10 and catalyzes the hydrolysis of a peptide bond located at the C-terminus of a polypeptide with a preference for a C-terminal arginine or lysine residue. This broadly specific enzyme is inhibited by the diisopropyl fluorophosphate and thiol-blocking reagents. The pH optimum ranges from 4.5 to 6.0.

S. J. Remington & K. Breddan (1994) *Meth. Enzymol.* **244**, 231.

CARBOXYPEPTIDASE M

This enzyme [EC 3.4.17.12], which is a member of peptidase family M14, catalyzes the hydrolysis of a peptide

bond at the C-terminus of a polypeptide provided that the C-terminal amino acid is an arginine or a lysine residue. The protein is a membrane-bound enzyme which has a pH optimum at neutral pH. This enzyme is distinct from arginine carboxypeptidase (carboxypeptidase N) [EC 3.4.17.3] and from carboxypeptidase H [EC 3.4.17.10].

F. Tan, P. A. Deddish & R. A. Skidgel (1995) *Meth. Enzymol.* **248**, 663.

CARBOXYPEPTIDASE N

This enzyme [EC 3.4.17.3] (also referred to as lysine carboxypeptidase, arginine carboxypeptidase, kininase I, or anaphylatoxin inactivator) is a zinc-dependent member of peptidase family M14. The enzyme hydrolyzes the peptide bond at the C-terminus provided that the C-terminal amino acid is either arginine or lysine. The enzyme inactivates bradykinin and anaphylatoxins in blood plasma.

R. A. Skidgel (1995) *Meth. Enzymol.* **248**, 653.

CARBOXYPEPTIDASE T

This enzyme [EC 3.4.17.18], a member of peptidase family M14, catalyzes the hydrolysis of a peptide bond at the C-terminus of a polypeptide provided that the amino acid has a C-terminal amino acid that is either hydrophobic or positively charged.

V. M. Stepanov (1995) *Meth. Enzymol.* **248**, 675.

CARDIOLIPIN SYNTHASE

The following are recent reviews on the molecular and physical properties of this enzyme.

I. Shibuya & S. Hiraoka (1992) *Meth. Enzymol.* **209**, 321.
J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

CARNITINE

$(\text{CH}_3)_3\text{N}^+-\text{CH}_2-\text{CH}(\text{OH})-\text{CH}_2-\text{COO}^-$, an acyl-transfer metabolite, found in high abundance in skeletal muscle, liver, and yeast. *O*-Acylcarnitine, is a high-energy ester like its structurally related *O*-acetylcholine, $(\text{CH}_3)_3\text{N}^+-\text{CH}_2-\text{CH}_2-\text{O}-\text{C}(=\text{O})-\text{CH}_3$.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Radioisotopic assay of acetylcarnitine and acetyl-CoA, **123**, 259; short-chain acylcarnitines: identification and quantitation, **123**, 264; purification and assay of carnitine acyltransferases, **123**, 276; determination of the specific activity of long-chain acylcarnitine esters, **123**, 284; synthesis of carnitine precursors and related compounds, **123**, 290; synthesis of radioactive (–)-carnitine from γ -

aminobutyrate, **123**, 297; S-adenosylmethionine: ϵ -N-L-lysine methyltransferase, **123**, 303

CARNITINE O-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.7], also known as carnitine acetylase, catalyzes the reversible reaction of acetyl-CoA with carnitine to yield coenzyme A and *O*-acetylcarnitine. The enzyme can also use propanoyl-CoA and butanoyl-CoA as substrates.

L. L. Bieber & S. Farrell (1983) *The Enzymes*, 3rd ed., **16**, 627.
K. F. Tipton & H. B. F. Dixon (1979) *Meth. Enzymol.* **63**, 183.

CARNITINE O-OCTANOYLTRANSFERASE

This enzyme [EC 2.3.1.137] catalyzes the reversible reaction of octanoyl-CoA with carnitine to yield coenzyme A and *O*-octanoylcarnitine. The enzyme utilizes a range of acyl-CoA derivatives as substrates, with optimal activity reported with C₆ or C₈ acyl groups.

L. L. Bieber & S. Farrell (1983) *The Enzymes*, 3rd ed., **16**, 627.

CARNITINE O-PALMITOYLTRANSFERASE

This enzyme [EC 2.3.1.21] catalyzes the reversible reaction of palmitoyl-CoA with carnitine to yield coenzyme A and *O*-palmitoylcarnitine. The enzyme exhibits a broad specificity toward acyl-CoA substrates, covering a range from C₈ to C₁₈, but optimal activity is observed with palmitoyl-CoA.

L. L. Bieber & S. Farrell (1983) *The Enzymes*, 3rd ed., **16**, 627.

CARNOSINASE

This enzyme [EC 3.4.13.3] (also referred to as Xaa-His dipeptidase, X-His dipeptidase, aminoacylhistidine dipeptidase, and homocarnosinase), is a zinc-dependent dipeptidase that catalyzes the hydrolysis of Xaa-His dipeptides. Carnosine, homocarnosine, and anserine are preferred substrates for this mammalian cytosolic enzyme. Other aminoacylhistidine dipeptides are weaker substrates (including homoanserine). The enzyme is activated by thiols and inhibited by metal-chelating agents.

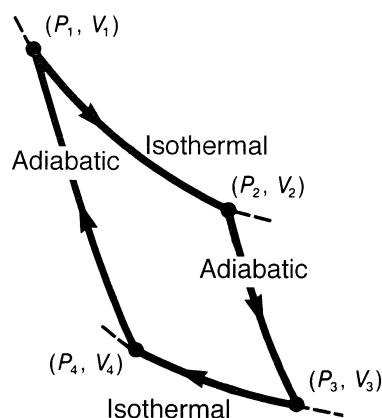
O. W. Griffith (1986) *Ann. Rev. Biochem.* **55**, 855.

CARNOSINE SYNTHETASE

This enzyme [EC 6.3.2.11] catalyzes the reaction of histidine with β -alanine and ATP to produce carnosine, AMP, and pyrophosphate.

O. W. Griffith (1986) *Ann. Rev. Biochem.* **55**, 855.
G. D. Kalyankar & A. Meister (1971) *Meth. Enzymol.* **17B**, 102.

CARNOT CYCLE



A hypothetical cycle for achieving reversible work, typically consisting of a sequence of operations: (1) isothermal expansion of an ideal gas at a temperature T_2 ; (2) adiabatic expansion from T_2 to T_1 ; (3) isothermal compression at temperature T_1 ; and (4) adiabatic compression from T_1 to T_2 . This cycle represents the action of an ideal heat engine, one exhibiting maximum thermal efficiency. Inferences drawn from thermodynamic consideration of Carnot cycles have advanced our understanding about the thermodynamics of chemical systems. **See** *Carnot's Theorem*; *Efficiency*; *Thermodynamics*

CARNOT'S THEOREM

A principle stating that for any engine working between the same two temperatures, maximum efficiency will occur by an engine working *reversibly* between those same two temperatures. Thus, all reversible engines have the same efficiency between the same temperatures and that efficiency is dependent only on those temperatures and not on the nature of the substance being acted upon. **See** *Efficiency*; *Thermodynamics*, *Laws of*; *Carnot Cycle*

β -CAROTENE (or, β -CAROTENOID) 15,15'-DIOXYGENASE

This enzyme [EC 1.13.11.21] catalyzes the reaction of β -carotene with dioxygen to produce two retinal molecules. Both iron ions and bile salts are required cofactors.

- L. Villard-Mackintosh & C. J. Bates (1993) *Meth. Enzymol.* **214**, 168.
M. R. Lakshman & C. Okoh (1993) *Meth. Enzymol.* **214**, 256.
J. A. Olson & M. R. Lakshman (1990) *Meth. Enzymol.* **189**, 425.
O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

CASBENE SYNTHASE

This enzyme [EC 4.6.1.7] catalyzes the conversion of geranylgeranyl diphosphate to the antifungal diterpene

casbene and pyrophosphate (or, diphosphate). The enzyme was first isolated from the castor bean.

R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.

CASPASES (or, CYSTEINYL ASPARTATE-SPECIFIC PROTEINASES)

Caspase-1 [EC 3.4.22.36] (also known as interleukin-1 β converting enzyme and interleukin-1 β convertase precursor) is a member of the peptidase family C14. It catalyzes the hydrolysis of the Asp¹¹⁶-Ala¹¹⁷ and Asp²⁷-Gly²⁸ in the precursor protein, resulting in the release of interleukin-1 β . The enzyme will also hydrolyze the small peptide, Ac-TyrValAlaAsp-NHMEC.

D. W. Nicholson & N. A. Thornberry (1997) *Trends Biochem. Sci.* **22**, 299.

CATALASE

This enzyme [EC 1.11.1.6] catalyzes the conversion of hydrogen peroxide to dioxygen and two water molecules. Both heme and manganese ions are used as cofactors. Several organic substances, *e.g.* ethanol, can act as the hydrogen donor. A manganese protein containing Mn(III) in the resting state is often called pseudocatalase.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

J. E. Penner-Hahn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 439.

D. C. Rees & D. Farrelly (1990) *The Enzymes*, 3rd ed., **19**, 37.

M. Ueda, S. Mozaffar & A. Tanaka (1990) *Meth. Enzymol.* **188**, 463.

G. R. Schonbaum & B. Chance (1976) *The Enzymes*, 3rd ed., **13**, 363.

CATALYST

A substance that accelerates a chemical reaction but does not become consumed, generated, or permanently changed by such reaction. Thus, a catalyst does not alter the overall stoichiometric expression for the reaction or the overall equilibrium constant. The enhanced reactivity produced by a catalyst is referred to as catalysis.

An important characteristic of a catalyst's action is that the mechanism of the reaction is altered in a manner that allows for a lower activation energy. In a number of nonenzymatic examples, a reactant or product can act as a catalyst as well, and the definition must be altered to include substances that appear in the overall rate expression with a power higher than the corresponding

stoichiometry number. One may also describe certain substances as catalysts even when, upon accelerating the reaction, they are destroyed by the process. In fact, nearly every catalyst for hydrogenation of organic molecules becomes progressively more and more poisoned with successive rounds of catalysis. In biology, we recognize that bacterial luciferases catalyze only a single turnover. Likewise, actin and tubulin respectively accelerate ATP and GTP hydrolysis in a single catalytic round associated with polymerization.

At very high concentrations, the enzyme can alter the equilibrium constant. If K_{eq} is calculated by determining the equilibrium concentrations of all free products and reactants, and if the products and reactants have different affinities for the free enzyme, then high $[E_{tot}]$ favors formation of significant amounts of EA and EP, and this may cause an apparent shift in K_{eq} . In such instances, the enzyme is now a stoichiometric participant in the reaction, and the true equilibrium constant has to take this into account.

CATALYST EFFECT ON EQUILIBRIUM CONSTANT

The cardinal feature of catalysis is that the equilibrium constant of a chemical reaction is unaffected by the presence of a catalyst. This is true as long as the concentration of the catalyst is insignificant relative to the concentration(s) of the least abundant reactant(s) or product(s). Thus for an uncatalyzed reaction ($A \rightleftharpoons B$) with rate constants k_+ and k_- (for the forward and reverse rates, respectively), the equilibrium constant $K_{eq} = k_+/k_-$. In the presence of catalyst, the rate constants are increased, say to xk_+ and yk_- , and the new equilibrium constant $K_{eq}' = xk_+/yk_- = (x/y)(k_+/k_-) = (x/y)K_{eq}$. Because the rate enhancement in the presence of enzyme will always be the same for forward and reverse reactions, $x = y$, such that K_{eq}' must still equal K_{eq} .

CATALYTIC ANTIBODIES

Antibodies possessing catalytic activity as a result of their selection using transition state mimics as immunogenic or antigenic agents. Antibody catalysis takes advantage of the molecular diversity of the immune system and current knowledge of mechanistic enzymology and physical organic chemistry¹, allowing production of novel catalysts for reactions previously lacking biological

catalysis. As pointed out by Schultz and Lerner¹, systematic mechanistic and structural investigations of catalytic antibodies should provide insights about enzymic catalysis and the evolution of catalytic function.

The concept of catalytic antibodies was suggested succinctly by Jencks². "If complementarity between the active site and the transition state contributes significantly to enzymatic catalysis, it should be possible to synthesize an enzyme by constructing such an active site. One way to do this is to prepare an antibody to a haptenic group which resembles the transition state of a given reaction. The combining sites of such antibodies should be complementary to the transition state and should cause an acceleration by forcing bound substrates to resemble the transition state."

Many of the 60 known reactions catalyzed by monoclonal antibodies involve kinetically favored reactions (*e.g.*, ester hydrolysis), but abzymes can also speed up kinetically disfavored reactions. Stewart and Benkovic³ applied transition-state theory to analyze the scope and limitations of antibody catalysis quantitatively. They found the observed rate accelerations can be predicted from the ratio of equilibrium binding constants of the reaction substrate and the transition-state analogue used to raise the antibody. This approach permitted them to rationalize product selectivity displayed in antibody catalysis of disfavored reactions, to predict the degree of rate acceleration that catalytic antibodies may ultimately afford, and to highlight some differences between the way that they and enzymes catalyze reactions.

The nature of antibody catalysis remains to be elucidated, and antibodies will not reach the efficiency of enzymes until they can emulate the conformational changes, acid/base, redox, and/or nucleophilic/electrophilic reactivities of catalytic residues along the entire reaction coordinate. It is worthy of note that Hollfelder *et al.*⁴ recently demonstrated that serum albumins catalyze the eliminative ring-opening of a benzoisoxazole at rates that are similar to those observed with catalytic antibodies⁵. They suggest that formal general base catalysis contributes only modestly to the efficiency of both systems, and they favor the view that the antibody catalysis may be enhanced in some cases by nonspecific medium effects.

Antibody Aldolases: An Illustrative Example. Catalytic antibodies can be prepared by reactive immunization^{6,7}, a method that selects antibody catalysts *in vivo* in terms of their ability to catalyze a particular chemical reaction. As pointed out by Barbas *et al.*⁸, the protocol differs from earlier methods relying on immunization with transition-state mimics that are as inert as possible so that resulting antibodies can react with target substrates that are in their native state. In the reactive immunization technique, a reactive antigen is designed so that a desired chemical reaction can take place within the antibody binding pocket during its induction. These efforts allowed Barbas *et al.*⁸ to prepare antibody aldolases that displayed typical Michaelis-Menten kinetics, and Michaelis constants for substrates in cross-aldol reactions ranged upward from 1 mM to 1 M. The acceptor aldehydes displayed K_m values in the 0.02–0.50 mM range. Some of the observed catalytic rate enhancements are shown in Fig. 1.

Barbas *et al.*⁸ offered the following mechanisms shown in Fig. 2 for the Class 1 aldolase reaction and Ab38C2 or Ab 33F12 antibody aldolase reactions, where R is 4-isobutyramidobenzyl or *n*-butyl. Both mechanisms involve a chemically reactive lysyl ϵ -amino group,

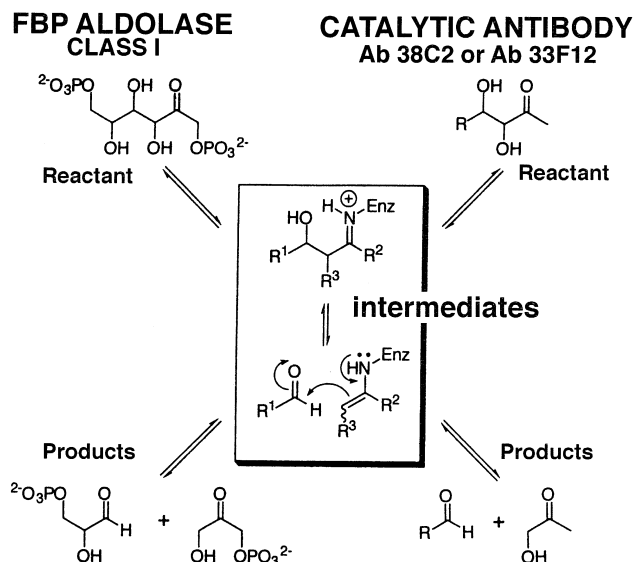


Figure 1. Kinetic parameters for the selection of antibody-catalyzed aldol and *retro*-aldol reactions, reflecting the biocatalyst's ability to accept substrates that differ clearly with respect to their molecular geometry. No background reaction was observed for the self-condensation of cyclopentanone. The indicated value for cyclopentanone addition to pentanal was estimated using the published k_{uncat} value of $2.28 \times 10^{-7} \text{ M}^{-1} \text{ s}^{-1}$ for the aldol addition of acetone to an aldehyde. Reproduced with permission of the authors and the American Association for the Advancement of Science.

Substrate	Product	$k_{\text{cat}} (\text{min}^{-1})$	$k_{\text{cat}}/k_{\text{uncat}}$
		0.086	3.6×10^6
		2.14	not determined
		0.0003	not determined
		1.02	4.5×10^6
		5.0	not determined

Figure 2. Mechanisms of aldolase catalysis. Reproduced with permission of the authors and the American Association for the Advancement of Science.

and Barbas *et al.*⁸ used binding studies with pyridoxal as well as aldolase substrates to demonstrate that Ab33F12 contains such a residue. From spectrophotometric studies of the pH dependence of the catalytic antibody's reaction with 3-methyl-2,4-pentanedione, the observed titration curve for enamine formation yielded a pK_a value of 5.5 to 6.0. This is in the range originally observed by Westheimer's group for the perturbed lysyl ϵ -amino group in the acetoacetate decarboxylase reaction, and Barbas *et al.*⁸ provided X-ray crystal structural evidence that the Fab' fragment has what they refer to as a promiscuous hydrophobic pocket surrounding a structurally unusual lysyl residue.

¹P. G. Schultz & R. A. Lerner (1995) *Science* **269**, 1835.

²W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, p. 288, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York].

³J. D. Stewart & S. J. Benkovic (1995) *Nature* **375**, 388.

⁴F. Hollfelder, A. J. Kirby & D. S. Tawfik (1996) *Nature* **383**, 60.

⁵S. N. Thorn, R. G. Daniels, M. T. Auditor & D. Hilvert (1995) *Nature* **373**, 228.

⁶J. Wagner, R. A. Lerner & C. F. Barbas III (1995) *Science* **270**, 1797.

⁷P. Wirsching, J. A. Ashley, C.-H. Lo, K. D. Janda & R. A. Lerner (1995) *Science* **270**, 1775.

⁸C. F. Barbas III, A. Heine, G. Zhong, T. Hoffmann, S. Gramatikova, R. Björnstedt, B. List, J. Anderson, E. A. Stura, I. A. Wilson & R. A. Lerner (1997) *Science* **278**, 2085.

Selected entries from *Methods in Enzymology* [vol, page(s)]: Design, **178**, 551; immunoassay, **178**, 542; production, **178**, 531; purification, **178**, 543; substrates and enzymatic assay, **178**, 544; derivatization with spectroscopic probe, **178**, 567; ester cleavage assays, **178**, 565; fluorescence quenching binding assay, **178**, 567; substrates, synthesis, **178**, 564; generation [for acyl-group transfer, **203**, 342; for β -elimination reactions, **203**, 330; for Claisen rearrangements, **203**, 339; for Diels-Alder reactions, **203**, 340; from existing antibodies, **203**, 347; for hydrolysis of esters, carbonates, and amide bonds, **203**, 333; for stereospecific hydrolysis of unactivated esters, **203**, 337; for thymine dimer cleavage, **203**, 333]; strategies, **203**, 329; thermodynamics of catalytic mechanism, compared to enzymes, **259**, 685.

CATALYTIC CONSTANT

A proportionality factor for the dependency of a particular reaction on the concentration of a particular catalyst. The rate of a reaction, v , is equal to the sum of the rates of the uncatalyzed reaction (having a proportionality constant (or, rate constant), k_o) and the rate of the catalyzed reaction. Thus,

$$v = (k_o + \sum k_i [C_i]^{n_i}) [A]^\alpha [B]^\beta \dots$$

where $[C_i]$ is the concentration of one of a set of i catalysts, k_i is the catalytic constant for catalyst i , n_i is the partial order of the reaction with respect to catalyst i

(usually $n_i = 1$), $[A]$, $[B]$, *etc.*, are the concentrations of the reactants, and α , β , *etc.*, are the partial order of the reaction with respect to each substrate. Thus, k_i is an $(\alpha + \beta + \dots + 1)$ th rate constant. **See** *Turnover Number*

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

CATALYTIC DOMAIN

That part of a polypeptide chain(s) possessing catalytic function. The catalytic domain may comprise more than one structural domain¹, and a multienzyme polypeptide contains at least two types of such domains. **See** *Multienzyme Polypeptide; Autonomous Catalytic Domain*

¹Nomenclature Committee, IUB (1989) *Eur. J. Biochem.* **185**, 485.

CATALYTIC ENHANCEMENT

The ratio of the rate of an enzyme- (or ribozyme-) catalyzed reaction to the rate of the reaction in the absence of catalysis. This ratio equals k_{cat}/k_{non} , where k_{cat} is the turnover number and k_{non} is the noncatalyzed rate constant. **See** *Catalytic Proficiency*

CATALYTIC PROFICIENCY

A quantitative measure¹ of an enzyme's ability to lower the activation barrier for the reaction of a substrate in solution. Catalytic proficiency (a unitless parameter) equals the enzyme-catalyzed reaction rate constant (expressed as k_{cat}/K_m) divided by the rate constant (k_{non}) for the noncatalyzed reference reaction.

In the case of orotic acid, nonenzymatic decarboxylation proceeds with a half-time ($t_{1/2}$) of about 2.45×10^{15} s near pH 7 at room temperature, as indicated by reactions in quartz tubes at elevated temperatures¹. Orotidine 5'-phosphate decarboxylase thus appears to be an extremely proficient enzyme which enhances the reaction rate by a factor of 10^{17} . They estimate the transition state form of the substrate has a dissociation constant that is less than 5×10^{-24} M.

Radzicka and Wolfenden¹ also report other catalytic proficiencies: staphylococcal nuclease, 5.9×10^{19} ; adenosine deaminase, 7.8×10^{16} ; AMP nucleosidase, 5.0×10^{16} ; cytidine deaminase, 9.1×10^{15} ; carboxypeptidase A, 2.2×10^{15} ; ketosterol isomerase, 1.8×10^{15} ; triose-phosphate isomerase, 5.6×10^{13} ; chorismate mutase, 4.2×10^{13} ; carbonic anhydrase, 9.2×10^8 ; prolylpeptidyl isomerase,

or cyclophilin, 5.3×10^8 . Estimated values of catalytic proficiencies represent upper limits on the dissociation of the enzyme-substrate complex in the transition state for cases where (a) the mechanisms of enzymatic and noncatalyzed reactions are not the same or (b) the enzyme reaction rate may be limited by a step other than bond-making or bond-breaking in substrates, such as the desorption of product(s)¹⁻².

Mandelate racemase, another pertinent example, catalyzes the kinetically and thermodynamically unfavorable α -carbon proton abstraction. Bearne and Wolfenden³ measured deuterium incorporation rates into the α -position of mandelate and the rate of (*R*)-mandelate racemization upon incubation at elevated temperatures. From an Arrhenius plot, they obtained a $\Delta G^\ddagger$ for racemization and deuterium exchange rate was estimated to be around 35 kcal/mol at 25°C under neutral conditions. The magnitude of the latter indicated mandelate racemase achieves the remarkable rate enhancement of 1.7×10^{15} , and a level of transition state affinity ($K_{\text{TX}} = 2 \times 10^{-19}$ M). These investigators also estimated the effective concentrations of the catalytic side chains in the native protein: for Lys-166, the effective concentration was 622 M; for His-297, they obtained a value 3×10^3 M; and for Glu-317, the value was 3×10^5 M. The authors state that their observations are consistent with the idea that general acid-general base catalysis is efficient mode of catalysis when enzyme's structure is optimally complementary with their substrates in the transition-state. **See Reference Reaction; Catalytic Enhancement**

¹A. Radzicka & R. Wolfenden (1995) *Science* **267**, 90.

²R. Wolfenden (1976) *Ann. Rev. Biophys. Bioeng.* **5**, 271.

³S. L. Bearne & R. Wolfenden (1997) *Biochemistry* **36**, 1646.

CATALYTIC RNA

Enzymatically active ribonucleic acid segments (some of which are known as ribozymes) with the capacity to catalyze RNA self-splicing¹ or peptide bond formation². The overall catalytic rate enhancement is around 10^{11} .

The hammerhead ribozyme was the first identified motif to catalyze sequence-specific self-cleavage in plant virus satellite RNA molecules (Fig. 1). This motif facilitates magnesium ion-dependent transesterification with a turnover rate of about 1 s^{-1} .

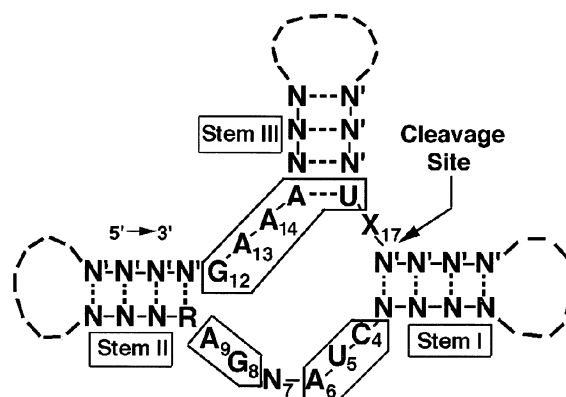


Figure 1. Consensus nucleotide sequence and stem-loop structure of hammerhead ribozyme.

In the microorganism *Tetrahymena*, group I intron splicing is accomplished in the catalytic cycle shown in Fig. 2. The intron first binds guanosine or one of its 5'-phosphorylated derivatives which is then used as a nucleophile in cleaving the 5' splice site.

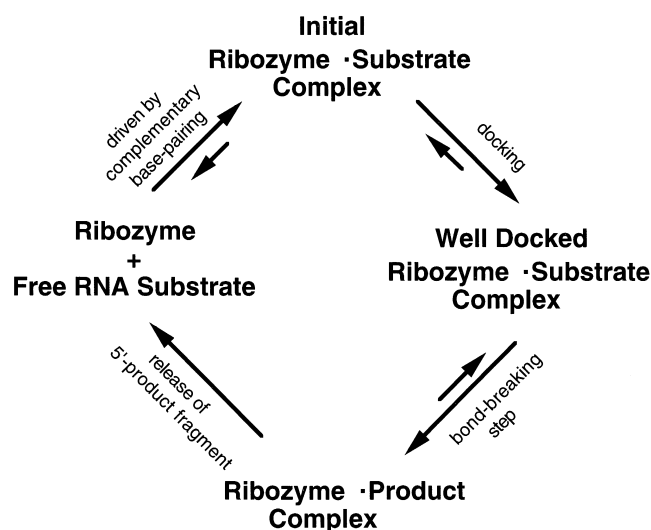


Figure 2. The catalytic reaction cycle of the *Tetrahymena* ribozyme (E), showing the binding and docking reactions (leading to the formation of E·S complex), followed by a bond cleavage breaking step (the rate constant for which is k_{chem}) and release of the 5'-fragment in the multiturnover steps (rate constant equals k_{mt}).

Ribozyme catalysis of phosphoryl transfer in appears to be promoted by destabilization of the ground state³. It also seems clear that magnesium ion complexation and hydrogen bonding interactions stabilize the negative charge that develops on the leaving group during entry of the nucleophile. This transesterification reaction is mechanistically analogous to that used in the mRNA spliceosome as well as in other DNA topoisomerase and

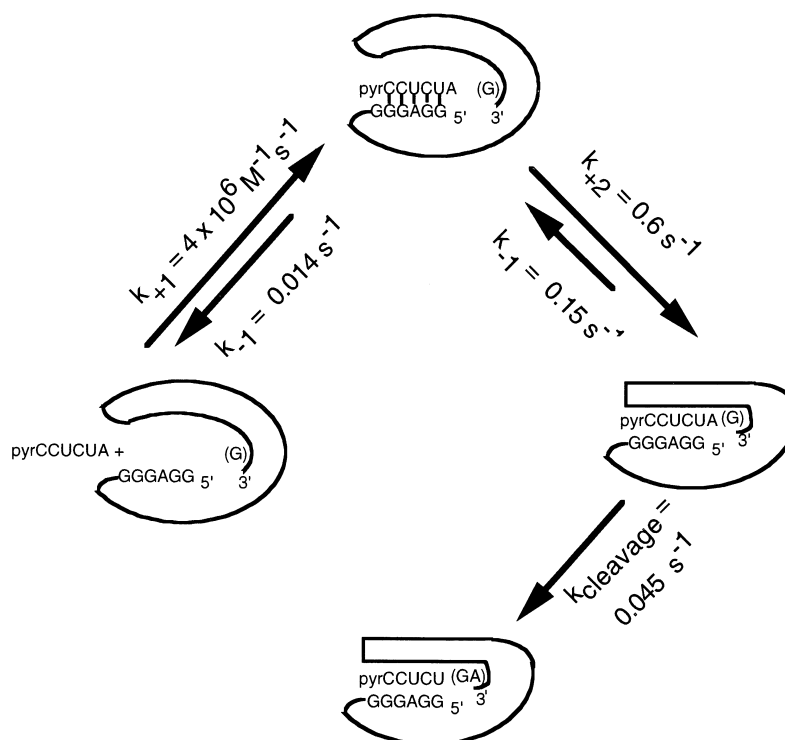


Figure 3. Minimal mechanism for pyrCCUCUA binding in the presence of the acceptor molecule pG. Vertical lines represent base-pairing interactions, and the boldface dots show those nucleotides that require 2'-OH groups needed for optimal binding. (From Turner *et al.*⁵ with permission of the authors and the publisher Springer-Verlag.)

transposition reactions. The true catalytic nature of the ribozyme was demonstrated by the discovery that the RNA component of RNase P catalytically processed tRNA precursors⁴.

As shown by Turner *et al.*⁵, fluorescence experiments using a 5'-pyrenylated oligonucleotides have aided the determination of rate constants and equilibrium constants that define (a) the initial base-pairing step in substrate binding, (b) the so-called docking step that reflects a substrate-induced conformational step, and (c) the bond cleavage step *per se*. The scheme shown in Fig. 3 represents a beautiful example of Koshland's induced-fit model at work in ribozyme action.

The self-splicing reaction of group II introns requires the folding of intronic RNA to form the catalytically active unit. The most likely chemical pathway during the splicing reaction is shown in Fig. 4. The reaction mechanism commences with proton abstraction from the ribose 2'-OH, and the 2'-alkoxide then attacks the phosphorus, forming a pentavalent intermediate. In the second step, rate-limiting breaking of the P—O bond may

be assisted by acid catalysis. Two magnesium ions are thought to play key roles in this process.

Containing only around 30 nucleotides in their catalytic core, the so-called hammerhead ribozymes are the smallest catalytic RNA's. They display Michaelis-Menten kinetics in their action on substrates: $k_{\text{cat}} 1\text{--}3 \times 10^{-2} \text{ s}^{-1}$;

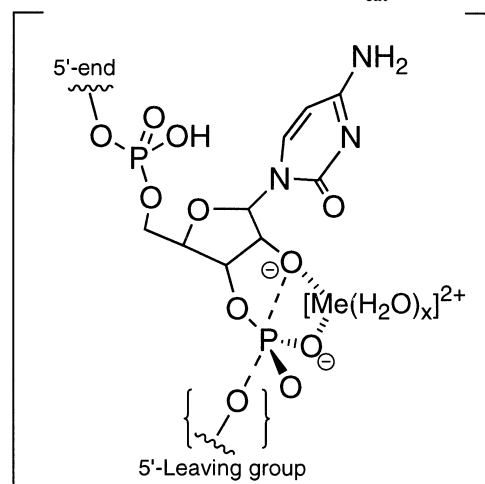


Figure 4. Transesterification reaction mechanism of group II introns. Nucleophilic attack leads to the pentavalent phosphorus intermediate which subsequently decomposes to yield the cleavage products.

K_m values 20–200 nM. Product release is fast, such that the rate-determining step is phosphodiester bond scission. Values for k_{cat}/K_m fall in the $1\text{--}4 \times 10^7 \text{ M}^{-1} \text{ s}^{-1}$ range.

In their lucid comparison of mechanisms of ribozyme and protein RNase action, Narlikar and Herschlag⁶ recently described the current state of RNA catalysis. Of particular relevance to catalysis is whether RNA catalysts share the same power exhibited by proteins to perturb the microenvironment of active-site relative to the aqueous solution. Proteins have a good mix of side-chain residues that can lower or raise the dielectric constant of regions within the active site. The authors argue that RNA is likely to be less effective in creating a low effective dielectric constant within the active site, and they suggest that the larger size of RNA catalysts may be related the need to be rigid to preferentially stabilize the transition state. They also believe that this bears directly on Jencks' observations about the role of intrinsic binding energy in enzymic catalysis (*i.e.*, the energy derived from binding interactions, while not strongly manifested in the ground-state enzyme-substrate complex, is profoundly engaged in transition state stabilization)⁷.

The bacterial ribosome is composed of three RNA components (5S, comprising 120 nucleotides; 16S, 1542 nucleotides; and 23S, 2904 nucleotides) along with some 55 protein components. Nitta *et al.*² demonstrated that the 23S ribosomal RNA (rRNA) catalyzes the peptide bond reaction without any requirement for ribosomal proteins. They also individually synthesized the six domains of the 23S rRNA, and after combining these components, peptide bond formation could be demonstrated. By omitting or including various combinations of the six domains, Nitta *et al.*² also showed that activity could be reconstituted by domain V only, when present at a concentration that was ten-times higher than that of intact 23S ribosomal RNA. This important finding demonstrates that RNA can catalyze peptide bond synthesis (*i.e.*, C–N ligation) in addition to the well-known phosphotransfer reactions of ribozymes.

¹A. J. Zaug & T. R. Cech (1982) *Nucl. Acids Res.* **10**, 2823.

²I. Nitta, Y. Kamada, H. Noda, T. Ueda & K. Watanabe (1998) *Science* **281**, 666.

³T. R. Cech & D. Herschlag (1996) *Catalytic RNA* (F. Eckstein & D. M. J. Lilley, eds.) p. 4, Springer-Verlag, Berlin.

⁴C. Guerrier-Takada, K. Gardiner, T. Marsh, N. Pace & S. Altman (1983) *Cell* **35**, 849.

⁵D. H. Turner, Y. Li, M. Fountain, L. Profenno & P. C. Bevilacqua (1996) *Catalytic RNA* (F. Eckstein & D. M. J. Lilley, eds.) p. 23, Springer-Verlag, Berlin.

⁶G. J. Narlikar & D. Herschlag (1997) *Ann. Rev. Biochem.* **66**, 19.

⁷W. P. Jencks (1975) *Adv. Enzymol.* **43**, 219.

CATASTROPHE THEORY

A formal mathematical approach for explaining the occurrence of sudden or abrupt changes in the behavior of a system. These abrupt changes occur at frequencies that often are dictated by one or more rate constants describing the interconversion of metastable and unstable states. The process is said to be stochastic if the frequency of transition is controlled by a random variable.

The self-assembly of microtubules is attended by abrupt switching between states of constant growth and shrinkage, and this process is termed "dynamic instability." The growth-to-shrinkage transitions occur catastrophically, and the reverse transition, known as rescue, returns the microtubules to a growth phase. These growth-to-shrinkage transitions had been viewed as random events occurring in proportion to the microtubule number concentration, thus following first-order kinetics. Odde *et al.*¹ recently applied a probabilistic analysis of data (based on direct microscopical observation of individual microtubules *in vitro*) in terms of the distribution of growth times. They observed that the slower growing and biologically inactive ends (termed (–)-ends) obeyed first-order catastrophe kinetics, but the faster growing and biologically active (+)-ends did not. The kinetic behavior at (+)-ends led them to infer that elongating (+)-ends of microtubules have an effective frequency of catastrophe that depends on the duration of growth prior to their experiencing a growth-to-shrinkage transition. While initially low, this frequency rises asymptotically to a limiting value. The authors conclude that the dynamic instability of microtubules is more complex than previous treatments would suggest.

¹D. J. Odde, L. Cassimeris & H. M. Buettner (1995) *Biophys. J.* **69**, 796.

CATECHOL 1,2-DIOXYGENASE

This enzyme [EC 1.13.11.1] catalyzes the reaction of catechol with dioxygen to produce *cis,cis*-muconate. Iron ions are used as a cofactor. Among its physiological roles, catechol 1,2-dioxygenase participates in the metabolism

of nitro-aromatic compounds by a strain of *Pseudomonas putida*.

- B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.
 K.-L. Ngai, E. L. Neidle & L. N. Ornston (1990) *Meth. Enzymol.* **188**, 122.
 O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

CATECHOL 2,3-DIOXYGENASE

This iron-dependent enzyme [EC 1.13.11.2], also called metapyrocatechase, catalyzes the reaction of catechol with dioxygen to produce 2-hydroxymuconate semialdehyde. The enzyme from *Alcaligenes* sp. strain O-1 reportedly catalyzes the reaction of 3-sulfocatechol with dioxygen and water to produce (2*E*,4*Z*)-2-hydroxymuconate and bisulfite.

- B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.
 S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.
 M. A. Ator & P. R. Ortez de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 I. A. Kataeva & L. A. Golovleva (1990) *Meth. Enzymol.* **188**, 115.
 O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

CATECHOL O-METHYLTRANSFERASE

This enzyme [EC 2.1.1.6] catalyzes the reaction of catechol with *S*-adenosylmethionine to produce *S*-adenosylhomocysteine and guaiacol (or, *o*-methoxyphenol).

- F. Takusagawa, M. Fujioka, A. Spies & R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 1.
 T. Lotta, J. Vidgren, C. Tilgmann, I. Ulmanen, K. Melén, I. Julkunen & J. Taskinen (1995) *Biochemistry* **34**, 4202.
 X. Cheng (1995) *Ann. Rev. Biophys. Biomol. Struct.* **24**, 293.
 J. Vidgren, L. A. Svenson & A. Liljas (1994) *Nature* **368**, 354.

CATECHOL OXIDASE

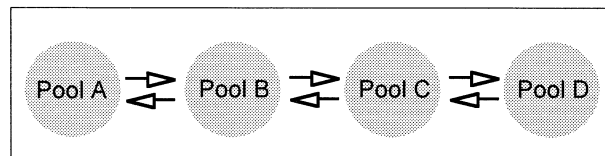
These copper ion-dependent enzymes [EC 1.10.3.1] (also referred to as diphenol oxidases, *O*-diphenolase, phenolases, polyphenol oxidases, or tyrosinases) catalyze the reaction of two catechol molecules with dioxygen to produce two 1,2-benzoquinone and two water. A variety of substituted catechols can act as substrates. Many of the enzymes listed under this classification also catalyze a monophenol monooxygenase activity [*i.e.*, EC 1.14.18.1]. **See also** *Monophenol Monooxygenase*; *Tyrosine Monooxygenase*

- N. H. Horowitz, M. Fling & G. Horn (1970) *Meth. Enzymol.* **17A**, 615.

CATENANE

A molecular entity consisting of two or more polymeric rings interlinked together but not covalently bonded to each other. Mitochondrial DNA are known to form catenated structures consisting of two interlinked nucleic acid circles (*i.e.*, a [2]catenane).

CATENARY MODEL FOR COMPARTMENT ANALYSIS



A linear series of metabolite compartments used in mathematical algorithms for modeling the kinetic behavior of metabolite and drug turnover.

CATHEPSIN B

This lysosomal enzyme [EC 3.4.22.1], also known as cathepsin B1, is a member of the peptidase family C1. The catalyzed reaction is the hydrolysis of peptide bonds with a broad specificity. The enzyme prefers the ArgArg—Xaa bond in small peptide substrates (thus distinguishing this enzyme from cathepsin L). The enzyme also exhibits a peptidyl-dipeptidase activity, releasing C-terminal dipeptides from larger polypeptides.

- H. Kirschke & B. Wiederanders (1994) *Meth. Enzymol.* **244**, 500.
 H. Scholze & E. Tannich (1994) *Meth. Enzymol.* **244**, 512.

CATHEPSIN D

This lysosomal endopeptidase [EC 3.4.23.5] is similar to pepsin A, except that the specificity is narrower and will not hydrolyze the Gln⁴—His⁵ peptide bond in the B chain of insulin. The enzyme is a member of the peptidase family A1.

- T. D. Meek (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 327.
 J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.

CATHEPSIN E

This enzyme [EC 3.4.23.34], also known as slow-moving proteinase and erythrocyte membrane aspartic proteinase, is similar to cathepsin D, albeit with a slightly broader specificity.

- T. D. Meek (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 327.
 T. Kageyama (1995) *Meth. Enzymol.* **248**, 120.

CATHEPSIN G

This endopeptidase [EC 3.4.21.20], a member of the peptidase family S1, has substrate specificity similar to that of chymotrypsin C.

J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.

CATHEPSIN H

This mammalian lysosomal endopeptidase [EC 3.4.22.16] is also known as aleurain, cathepsin B3, cathepsin BA, and benzoylarginine:naphthylamide hydrolase. A member of the peptidase family C1, the enzyme also acts with an aminopeptidase activity, preferring Arg–Xaa peptide bonds.

H. Kirschke & B. Wiederanders (1994) *Meth. Enzymol.* **244**, 500.

CATHEPSIN L

This peptidase family C1 enzyme [EC 3.4.22.15] is an lysosomal endopeptidase with specificity akin to papain. Cathepsin L displays a higher activity toward protein substrates than does cathepsin B.

H. Kirschke & B. Wiederanders (1994) *Meth. Enzymol.* **244**, 500.

CATHEPSIN S

This peptidase family C1 enzyme [EC 3.4.22.27] is a lysosomal cysteinyl-dependent endopeptidase with substrate specificity similar to cathepsin L.

H. Kirschke & B. Wiederanders (1994) *Meth. Enzymol.* **244**, 500.

CATHEPSIN T

This protease [EC 3.4.22.24] interconverts the three forms of tyrosine aminotransferase [EC 2.6.1.5].

H. C. Pitot & E. Gohda (1987) *Meth. Enzymol.* **142**, 279.

CATIONIC INTERMEDIATE ANALOGUES

Positively charged reaction intermediates generated in a wide variety of enzyme-catalyzed reactions. Analogues containing a protonated amino group in a position where a carbocation is generated during catalysis have proved to be excellent tightly bound enzyme inhibitors¹. For example, 1-amino sugars inhibit a number of glycosidases², and aza-substrate derivatives can serve as strong inhibitors.

¹A. Radzicka & R. Wolfenden (1995) *Meth. Enzymol.* **249**, 284.

²H. L. Levine, R. S. Brody & F. H. Westheimer (1980) *Biochemistry* **19**, 4993.

CATION π -INTERACTIONS

A relatively strong, noncovalent binding interaction of a cation to the π face of an aromatic structure. This interaction can be considered to result from an electrostatic attraction between a positive charge and the quadrupole moment of the aromatic ring. Dougherty¹ suggests that one can also view the side chains of phenylalanine, tyrosine, and tryptophan as candidates for these interactions.

¹D. A. Dougherty (1996) *Science* **271**, 163.

CDP-4-DEHYDRO-6-DEOXYGLUCOSE REDUCTASE

This enzyme [EC 1.17.1.1], also known as CDP-4-keto-6-deoxyglucose reductase, catalyzes the reversible reaction of CDP-4-dehydro-3,6-dideoxy-D-glucose with NAD(P)⁺ and water to produce CDP-4-dehydro-6-deoxy-D-glucose and NAD(P)H.

S. Matsushashi & J. L. Strominger (1966) *Meth. Enzymol.* **8**, 310.

CDP-DIACYLGLYCEROL:sn-GLYCEROL-3-PHOSPHATE 3-PHOSPHATIDYLTRANSFERASE

This enzyme [EC 2.7.8.5] (also referred to as phosphatidylglycerophosphate synthase, glycerophosphate phosphatidyltransferase, and 3-phosphatidyl-1'-glycerol-3'-phosphate synthase) catalyzes the reaction of CDP-diacylglycerol with glycerol 3-phosphate to produce CMP and 3-(3-phosphatidyl)glycerol 1-phosphate.

W. Dowhan (1992) *Meth. Enzymol.* **209**, 313.

R. A. Pieringer (1983) *The Enzymes*, 3rd ed., **16**, 255.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

T. S. Moore, Jr. (1982) *Ann. Rev. Plant Physiol.* **33**, 235.

CDP-GLUCOSE 4,6-DEHYDRATASE

This enzyme [EC 4.2.1.45] catalyzes the conversion of CDP-glucose to CDP-4-dehydro-6-deoxy-D-glucose and water. The enzyme uses NAD⁺ as a cofactor.

L. Glaser & H. Zarkowsky (1971) *The Enzymes*, 3rd ed., **5**, 465.

CELLOBIOS 2-EPIMERASE

This enzyme [EC 5.1.3.11] catalyzes the interconversion of cellobiose to D-glucosyl-D-mannose.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

L. Glaser (1972) *The Enzymes*, 3rd ed., **6**, 355.

CELLOBIOSE PHOSPHORYLASE

This enzyme [EC 2.4.1.20] catalyzes the reaction of cellobiose with orthophosphate to produce α -D-glucose 1-phosphate and D-glucose.

J. J. Mieyal & R. H. Abeles (1972) *The Enzymes*, 3rd ed., **7**, 515.

M. Cohn (1961) *The Enzymes*, 2nd ed., **5**, 179.

M. Doudoroff (1961) *The Enzymes*, 2nd ed., **5**, 229.

CELL PROLIFERATION KINETICS

When cells divide or undergo binary fission under batch culture conditions, the proliferation process often can be depicted as follows:

$$2^0 N_0 \rightarrow 2^1 N_0 \rightarrow 2^2 N_0 \rightarrow \dots \rightarrow 2^i N_0 \rightarrow \dots \rightarrow 2^n N_0$$

where $2^i N_0$ describes the algebraic product of 2^i (the number of doublings) and N_0 (the original number of cells). In the most ideal case, every cell will sustain its viability and proliferative potential. Even so, such a collection of otherwise identical cells will not divide synchronously for any substantial period, because access to nutrients as well as geometrical considerations will affect each living cell in slightly different ways. If the number of doublings ν equals the elapsed time t divided by the doubling time t_d (i.e., $\nu = t/t_d$), the factor f_t for the fold-increase in cell number can be written as:

$$f_t = N_t/N_0 = 2^\nu$$

This equation describes a geometric growth process which can be transformed into the following logarithmic expression:

$$\ln N_t - \ln N_0 = 0.693t/t_d$$

Note that this equation will only apply during the initially exponential growth phase. As the number of cells becomes much greater, the daughter cells compete for limiting quantities of one or more nutrients. This will reduce the efficiency of cell division, and the proliferation curve will reach a plateau that is often called the stationary growth phase.

CELL PROLIFERATION (Thymidine Incorporation Assay)

Measurements of the rate and extent of cell proliferation are essential for understanding the effects of cell growth factors and inhibitors. In the absence of aneuploidy, the DNA content is an invariant characteristic of each particular cell type, and typically the incorporation of tritiated thymidine into DNA is a useful measure of cell prolifera-

tion. Such measurements start with the incubation of the cells of interest with [3 H]thymidine, preferably ethanol-free because the presence of ethanol can alter the redox state of cells, and ethanol is also toxic to cells. One centrifuges the cells at low relative centrifugal field for 10–20 min, a procedure that works best with cells in suspension, although adherent cells can be trypsinized to release them from culture dishes without any confounding effect on the thymidine incorporation measurement. After resuspending cells in phosphate-buffered saline and subsequent recentrifugation to pellet the cells, one uses 10% (w/v) trichloroacetic acid to release low-molecular-weight nucleosides and nucleotides from the acid-precipitated DNA which can be collected on filter disks that are then further extracted with absolute ethanol. A nonradioactive assay can also be accomplished by replacing labeled thymidine with 5-bromo-2'-deoxyuridine (BrdU) which is subsequently detected with a BrdU-specific monoclonal antibody. This approach works best if cells are exposed simultaneously to [3 H]thymidine (or BrdU) and 5-fluorodeoxyuridine which blocks the thymidylate synthase reaction, thereby reducing any competition by endogenous thymidine in the DNA incorporation step.

CELLULASE

Also referred to as endo-1,4- β -glucanase and carboxymethyl cellulase, this enzyme [EC 3.2.1.4] catalyzes the endohydrolysis of 1,4- β -D-glucosidic linkages in cellulose. The enzyme also catalyzes the hydrolysis of 1,4-linkages in β -D-glucans also containing 1,3-linkages.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119-208.

P. Béguin & M. Lemaire (1996) *Crit. Rev. Biochem. Mol. Biol.* **31**, 201.

M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.

S. C. Fry (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 497.

D. R. Whitaker (1971) *The Enzymes*, 3rd ed., **5**, 273.

CELLULOSE POLYSULFATASE

This enzyme [EC 3.1.6.7] catalyzes the hydrolysis of the 2- and 3-sulfate groups of the polysulfates of cellulose and charonin.

A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 1.

CELLULOSE SYNTHASE

Also referred to as UDP-glucose- β -D-glucan glucosyltransferase and UDP-glucose-cellulose glucosyltransfer-

ase, cellulose synthase (UDP-forming) [EC 2.4.1.12] catalyzes the reaction of UDP-glucose with $[(1,4)\text{-}\beta\text{-D-glucosyl}]_n$ to produce UDP and $[(1,4)\text{-}\beta\text{-D-glucosyl}]_{(n+1)}$. This enzyme participates in the biosynthesis of cellulose. Cellulose synthase (GDP-forming) [EC 2.4.1.29] catalyzes the reaction of GDP-glucose with $[(1,4)\text{-}\beta\text{-D-glucosyl}]_n$ to produce GDP and $[(1,4)\text{-}\beta\text{-D-glucosyl}]_{(n+1)}$. This enzyme also participates in the biosynthesis of cellulose.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

CELSIUS TEMPERATURE

The temperature, symbolized by θ or t and measured in units of degree Celsius (symbolized by $^{\circ}\text{C}$), in which $\theta = T - 273.15$ (in which T is the temperature in kelvin) where the triple point of water is set at 273.16 K (or, 0.01°C). The conversion of Celsius temperature to Fahrenheit temperature ($^{\circ}\text{F}$) is given by $^{\circ}\text{F} = (9/5)^{\circ}\text{C} + 32$.

CENNAMO TREATMENT

A protocol¹⁻³ for assessing the behavior of enzymes having one or more isomerization steps. The method facilitates determination of K_{iip} , a kinetic parameter associated with the enzyme interconversion. **See** *Iso Mechanisms*

¹C. Cennamo (1968) *J. Theoret. Biol.* **21**, 260.

²C. Cennamo (1969) *J. Theoret. Biol.* **23**, 53.

³K. L. Rebholz & D. B. Northup (1995) *Meth. Enzymol.* **249**, 211.

CENTRAL ATOM

The centrally located atom in a coordination entity to which other atoms or group of atoms (generally referred to as ligands) binds.

¹M. W. G. de Bolster (1997) *Pure Appl. Chem.* **69**, 1251.

CERAMIDASE

This enzyme [EC 3.5.1.23], also known as acylsphingosine deacylase, catalyzes the hydrolysis of an *N*-acylsphingosine to produce a fatty acid anion and sphingosine.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

Y. Kishimoto (1983) *The Enzymes*, 3rd ed., **16**, 357.

CERAMIDE CHOLINEPHOSPHOTRANSFERASE

This enzyme [EC 2.7.8.3] catalyzes the reaction of CDP-choline with *N*-acylsphingosine to generate CMP and a sphingomyelin.

E. P. Kennedy (1962) *Meth. Enzymol.* **5**, 486.

CEREBROSIDE SULFATASE

This enzyme [EC 3.1.6.8], also known as arylsulfatase A, catalyzes the hydrolysis of a cerebroside 3-sulfate to produce a cerebroside and sulfate. The enzyme will also catalyze analogous reactions on the galactose 3-sulfate residues in a number of lipids as well as on ascorbate 2-sulfate and many phenol sulfates.

A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 1.

CERULOPLASMIN

A blue, copper-containing glycoprotein present in mammalian blood plasma and containing type 1, type 2, and type 3 copper centers. The type 2 and type 3 copper centers are close together, forming a trinuclear copper cluster. Ceruloplasmin has an important role in the transport and storage of copper ions. Thus, it participates in the metabolism of copper-containing enzymes.

CF₁CF_o-ATPase

H⁺-Transporting ATP synthase [EC 3.6.1.34] in plants, also referred to as chloroplast ATPase and CF₁CF_o-ATPase, catalyzes the hydrolysis of ATP to produce ADP and orthophosphate. When coupled with proton transport the reverse reaction results in the synthesis of ATP by this multisubunit complex. CF₁, isolated from the rest of the membrane-bound complex, retains the ATPase activity but not the proton-translocating activity.

B. J. Barkla & O. Pantoja (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 159.

M. R. Sussman (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 211.

L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.

B. Rubinstein & D. G. Luster (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 131.

R. Serrano (1989) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **40**, 61.

H. S. Penefsky (1974) *The Enzymes*, 3rd ed., **10**, 375.

cGMP-DEPENDENT PROTEIN KINASE

This general class of ATP-dependent protein kinases (abbreviated cGPK) catalyzes 3',5'-cyclic-GMP-stimulated phosphorylation of protein substrates. cGPK's are single-chain kinases containing two copies of the cyclic nucleotide-binding domain in their N-terminal regions. The nucleotide specificity of cAMP-protein kinases and cGMP-protein kinases can be traced to the conserved

region of β -barrel 7: a threonyl residue is invariant in cGPK, and an alanyl residue is invariant in most cAPK's. One preferred substrate is vasodilator-stimulated phosphoprotein (VASP), an adapter in actin-based motility mechanisms. (**See** *Actin-Based Pathogen Motility*) VASP's Ser-239 appears to be the preferred cGMP-dependent protein kinase phosphorylation site in intact cells and platelets.

- A. M. Edelman, D. K. Blumenthal & E. G. Krebs (1987) *Ann. Rev. Biochem.* **56**, 567.
 S. J. Beebe & J. D. Corbin (1986) *The Enzymes*, 3rd ed., **17**, 43.
 A. C. Nairn, H. C. Hemmings, Jr., & P. Greengard (1985) *Ann. Rev. Biochem.* **54**, 931.
 D. A. Flockhart & J. D. Corbin (1982) *Crit. Rev. Biochem.* **12**, 133.
 E. G. Krebs & J. A. Beavo (1979) *Ann. Rev. Biochem.* **48**, 923.

cGMP PHOSPHODIESTERASE

This membrane-bound homodimeric cyclic nucleotide phosphodiesterase (EC 3.1.4.17) catalyzes the conversion of 3',5'-cyclic-GMP to form 5'-GMP.

- A. Tar, T. D. Ting & Y.-K. Ho (1994) *Meth. Enzymol.* **238**, 3.
 N. D. Goldberg & M. K. Haddox (1977) *Ann. Rev. Biochem.* **46**, 823.

CHA METHOD

A method for deriving enzyme-rate expressions combining both rapid equilibrium and steady-state procedures first illustrated by Cha¹. With this method, demonstrated by Fromm² and Huang³, a different rate expression will be obtained depending on which steps are chosen to be in rapid equilibrium and which steps are not. **See** *Enzyme Kinetic Derivations; Turnover Number*

- ¹S. Cha (1988) *J. Biol. Chem.* **263**, 820.
²H. J. Fromm (1972) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.
³C. Y. Huang (1979) *Meth. Enzymol.* **63**, 54.

CHAIN PROPAGATION

The step that regenerates a reactive species, thereby permitting a chain reaction to proceed.

CHAIN REACTION

A reaction having a mechanism utilizing reactive intermediates (chain carriers) that participate in a cycle of steps, allowing reactive species to be regenerated after each cycle (*i.e.*, chain propagation steps). The chemical kinetic expression first given independently by Christiansen¹, Herzfeld², and Polanyi³, has the same general form of the empirical rate expression provided by Bodenstein

and Lind⁴. **See also** *Chain Transfer; Initiation; Termination*

- ¹J. A. Christiansen (1919) *K. Dan. Vidensk. Selsk., Mat.-Fys. Medd.* **1**, 14.
²K. F. Herzfeld (1919) *Ann. Phys.* **59**, 635.
³M. Polanyi (1920) *Zeit. Elektrochem.* **26**, 49.
⁴M. Bodenstein & S. C. Lind (1907) *Zeit. Phys. Chem.* **57**, 168.

CHAIN RULE

A procedure in calculus for changing variables in a differential function, such that:

$$dy/dx = dy/da + da/dx$$

as long as y and a are differentiable functions of a and x , respectively.

CHALCONE ISOMERASE

This enzyme [EC 5.5.1.6], also known as chalcone:flavonone isomerase, catalyzes the interconversion of a chalcone to a flavonone.

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 R. A. Dixon, P. M. Dey & C. J. Lamb (1983) *Adv. Enzymol.* **55**, 1.

CHALCONE SYNTHASE

This enzyme [EC 2.3.1.74] (also known as naringenin-chalcone synthase, flavonone synthase, and 6'-deoxy-chalcone synthase) catalyzes the reaction of three malonyl-CoA with 4-coumaroyl-CoA to produce four coenzyme A, three carbon dioxide, and naringenin-chalcone. If both NADH and a particular reductase is also present, the final product is 6'-deoxychalcone.

- R. A. Dixon, P. M. Dey & C. J. Lamb (1983) *Adv. Enzymol.* **55**, 1.

CHANGE IN MECHANISM

For any chemical or physical process, several different mechanisms may compete with each other, depending on the energy that is available to the system. A change in one or more reaction conditions can alter the degree to which competing pathways contribute to the kinetics of the overall process. The resulting shift from one pathway to another is referred to as a change in mechanism. An instructive, but as yet unproven, example might apply in the reaction catalyzed by hen egg white lysozyme. The enzyme is believed to operate via an oxocarbenium ion intermediate stabilized by neighboring carboxylate groups at the enzyme's active site. If another set of reaction conditions were to lead to the formation of an acylal intermedi-

ate (containing a covalent bond between one of the carboxyl groups and the C-1 position of the sugar), then we would say that a change in mechanism has occurred.

CHANNEL ACCESS

Control of enzymic activity arising from the modulated access of substrates to a channel leading to the active site. Such a scheme was suggested for aspartate carbamoyltransferase¹ which has its complement of active sites located on the interior surface of the complex comprised of catalytic and regulatory subunits. Nonetheless, isotope exchange studies of this enzyme suggest that this form of enzyme regulation does not apply in the case of aspartate transcarbamoylase^{2,3}.

¹S. G. Warren, B. F. Edwards, D. R. Evans, D. C. Wiley & W. N. Lipscomb (1973) *Proc. Natl. Acad. Sci. U.S.A.* **70**, 1117.

²F. C. Wedler & F. J. Gasser (1974) *Arch. Biochem. Biophys.* **163**, 69.

³F. C. Wedler & W. H. Shalongo (1982) *Meth. Enzymol.* **87**, 647.

CHAOTROPIC AGENTS

Agents which, upon addition to an aqueous solution, increase the solubility of nonpolar entities. These ions (*e.g.*, Li⁺, Mg²⁺, Ca²⁺, Ba²⁺, I⁻, ClO₄⁻, SCN⁻, and the guanidinium cation) are thought to promote some proteins to undergo denaturation by disrupting hydrogen-bonding interactions. Thus, in enzyme-catalyzed reactions, one should always be concerned about the chaotropic nature of various ionic species. The major intracellular monovalent cations (*e.g.*, potassium ion, arginine guanidinium ion, histidine imidazolium ion, and lysine ϵ -ammonium ion) are chaotropes. **See** *Hofmeister Series*; *Kosmotropes*; *Ion Interactions with Water*

CHAPERONES (Chaperonins)

Proteins that assist in the folding or translocation of other proteins¹. These specialized proteins possess the ability (a) to inhibit incorrect folding and/or aggregation of newly biosynthesized proteins during and following translation; (b) to target or direct denatured or oxidatively damaged proteins to lysosomes where they undergo degradation; and (c) to facilitate transmembrane translocation of targeted proteins.

Chaperonins frequently carry out their functions through ATP hydrolysis-dependent steps, and the kinetic properties of these reactions suggest that the equilibrium constants for the protein binding or releasing reactions are

modulated by thermodynamic coupling to the free energy of ATP hydrolysis. The heat shock cognate or Hsc70 protein is among the most highly conserved proteins in nature. The amino-terminal domain of this protein contains an ATP binding/hydrolysis function, and the carboxyl-terminus is thought to bear the polypeptide binding site protein. Indeed, the observed rate of the dissociation of denatured proteins from Hsc70 exceeds the rate of ATP hydrolysis has been taken as evidence that ATP binding, and not ATP hydrolysis itself, somehow facilitates the dissociation reaction^{2,3}. Ha and McKay⁴ recently provided evidence for an induced conformational change that follows ATP binding.

How molecular chaperones facilitate protein folding by rapidly binding non-native proteins of very different sequence and function is a question of central significance. Perrett *et al.*⁵ studied this process by determining the effect of ionic strength on the refolding of barnase (a bacterial RNase) on GroEL as well as on the thermal denaturation of the enzyme in the presence of chaperonins GroEL and SecB. Both chaperones bind barnase in its denatured state and lower the melting temperature (or, T_m) value. Refolding of barnase in the presence of GroEL is obviously multiphasic: (1) the slowest phase corresponds to the refolding of a singly bound molecule of barnase complexed to GroEL; and (2) the fastest phase involves association of barnase and GroEL. When Perrett *et al.*¹ used high GroEL-to-barnase ratios and an ionic strength less than 0.2 M, the fast phase matched the observed rate of binding. The association rate for barnase and GroEL was highly dependent on ionic strength; with a high ionic strength >0.6 M, most barnase molecules refolded free in solution, thereby escaping the binding interaction with GroEL. Their findings led them to suggest that ionic interaction between barnase and GroEL occurs initially and transiently; then hydrophobic binding appears to occur. Such a mechanism affords the opportunity for diffusion-controlled association, followed by slow dissociation of unfolded polypeptide.

While hydrogen exchange kinetics have been used to infer important features of protein folding pathways, Clarke *et al.*⁶ recently advanced the argument that hydrogen exchange does not represent a short cut for studying protein-folding pathways, because the technique does not allow one to distinguish paths involving intermedi-

ates from those that are unrelated side reactions. Studies of barnase and chymotrypsin inhibitor-2 indicate no obvious relationship between hydrogen exchange at equilibrium and their folding pathways⁷.

¹M. J. Gething & J. Sambrook (1992) *Nature* **355**, 33.

²D. R. Palleros, K. L. Reid, L. Shi, W. J. Welch & A. L. Fink (1993) *Nature* **365**, 664.

³K. Prasad, J. Heuser, E. Eisenberg & L. Greene (1994) *J. Biol. Chem.* **269**, 6931.

⁴J. H. Ha & D. B. McKay (1995) *Biochemistry* **34**, 11635.

⁵S. Perrett, R. Zahn, G. Stenberg & A. R. Fersht (1997) *J. Mol. Biol.* **269**, 892.

⁶J. Clarke, L. S. Itzhaki & A. R. Fersht (1997) *Trends Biochem. Sci.* **22**, 284.

⁷A. R. Fersht (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, pp. 650, W. H. Freeman, New York.

CHARGE DENSITY

The quantity (symbolized by ρ) of electrical charge per unit volume, equal to Q/V , with SI units of coulombs per cubic meter. More technically, this definition refers to the volume charge density. **See** *Surface Charge Density*

CHARGED INTERFACES

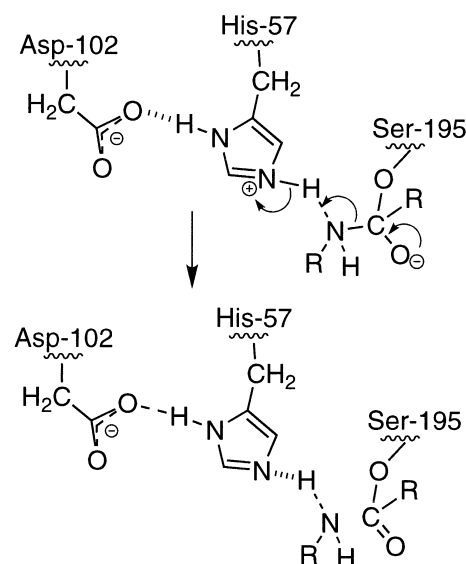
When particles or large molecules make contact with water or an aqueous solution, the polarity of the solvent promotes the formation of an electrically charged interface. The accumulation of charge can result from at least three mechanisms: (a) ionization of acid and/or base groups on the particle's surface; (b) the adsorption of anions, cations, ampholytes, and/or protons; and (c) dissolution of ion-pairs that are discrete subunits of the crystalline particle, such as calcium-oxalate and calcium-phosphate complexes that are building blocks of kidney stone and bone crystal, respectively. The electric charging of the surface also influences how other solutes, ions, and water molecules are attracted to that surface. These interactions and the random thermal motion of ionic and polar solvent molecules establishes a diffuse part of what is termed the electric double layer, with the surface being the other part of this double layer.

The Gouy-Chapman model describes the properties of the diffuse region of the double-layer. This intuitive model assumes that counterions are point charges that obey a Boltzmann distribution, with highest concentration nearest the oppositely charged flat surface. The polar solvent is assumed to have the same dielectric constant within the diffuse region. The effective surface

potential depends on the charge density of the surface and the ionic properties of the aqueous medium and its solutes. A more realistic case allows hydrated ions only to approach the surface at a finite distance, thereby defining what is called the Stern layer—so named to honor the model's originator. **See** *Biom mineralization*

D. J. Shaw (1980) *Introduction to Colloid and Surface Chemistry*, 3rd ed., Butterworth & Co., London.

CHARGE RELAY SYSTEM



Charge Relay in Chymotrypsin Catalysis

A term used to describe the catalytic action of a triad of side-chain residues in the so-called serine proteases and serine esterases. In the case of chymotrypsin, the β -carboxylate of Asp-102 tugs on the proton bound to the nitrogen at position-1 of the imidazole group of His-57; this is thought to increase the basicity of the other ring nitrogen (at position 3) which deprotonates the hydroxyl of Ser-195 as it attacks the carbonyl of the peptide bond. (One difficulty with this mechanistic proposal is that it remains unclear how His-57 is a sufficiently strong base to abstract a proton to generate an incipient alkoxide from the serine hydroxyl group.)

T. A. Steitz & R. G. Shulman (1982) *Ann. Rev. Biophys. Bioeng.* **11**, 419.

D. R. Corey & C. S. Craik (1992) *J. Am. Chem. Soc.* **114**, 1784.

CHARGE-TRANSFER COMPLEX

Any aggregate (containing two or more molecules, molecular entities, or functional groups) in which an electric charge is transferred from a donor to an acceptor.

M. W. G. de Bolster (1997) *Pure Appl. Chem.* **69**, 1251.

CHARGE-TRANSFER TRANSITION

An electronic transition in which a significant electric charge is transferred from an electron donor to an electron acceptor. A charge-transfer complex (also given the nonstandard appellation “electron donor-acceptor (EDA) complex”)^{1,2} is a ground-state complex exhibiting a charge-transfer absorption spectrum (in general, a spectrum that is not the same as the sum of the spectra of the individual molecular entities)³⁻⁶. Such complexes are usually colored.

Examples of such charge-transfer complexes include the complex formed between a metal ion and a π orbital of a double bond or an aromatic system^{7,8}, the complex formed between polynitro aromatics (such as picric acid)⁹ and other π -orbital-containing molecules, and complexes of I_2 and Br_2 with amines, ketones, aromatics, *etc.*¹⁰ Phenols and quinones also form charge-transfer complexes¹¹.

The term “charge-transfer” is not intended to imply that full transfer of charge is necessary for the complexation to occur.

¹IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

²Comm. on Photochem (1988) *Pure and Appl. Chem.* **60**, 1055.

³R. Foster (1969) *Organic Charge-Transfer Complexes*, Academic Press, New York.

⁴R. Foster (1976) *Chem. Berich.* **12**, 18.

⁵O. K. Poleshchuk & Y. K. Maksyutin (1976) *Russ. Chem. Rev.* **45**, 1077.

⁶D. V. Banthorpe (1970) *Chem. Rev.* **70**, 295.

⁷S. D. Ittel & J. A. Ibers (1976) *Advan. Organomet. Chem.* **14**, 33.

⁸F. R. Hartley (1973) *Chem. Rev.* **73**, 163.

⁹V. P. Parini (1962) *Russ. Chem. Rev.* **31**, 408.

¹⁰O. Hassel & C. Romming (1962) *Q. Rev. Chem. Soc.* **16**, 1.

¹¹R. Foster & M. I. Foreman (1974) in *The Chemistry of the Quinonoid Compounds* (ed., S. Patai), pt. 1, pp. 257-333, Wiley, New York.

CHELATION

The formation of two or more bonds between a single central atom (often a metal ion) and a chemical species, termed a multidentate ligand. The complex formed is referred to as a chelate. One naturally occurring example is the $MgATP^{2-}$ complex. The terms bidentate, tridentate, *etc.*, refer to the number of sites on the chelating ligand that are bond to the central atom. EDTA and EGTA have four carboxylic acid groups to form a tetradentate ligand, and many crown ethers are polydentate in terms of their ion binding character. **See also** *Sequestration*; *Cryptand*

CHELEX

A weak cation chelating resin [matrix: styrene divinylbenzene; functional group: $R-CH_2N(CH_2COO^-)_2$] that is highly selective for divalent cations and can be used to prepare metal-depleted biochemical solutions and cell culture media. The resin has the following properties: (a) operating pH range of 4–14; (b) chemically stable from pH 0 to 14; (c) minimal wet capacity of 0.4 mEq/mL; (d) nominal density of 0.65 g/mL; (e) wet bead size ranges of 75–150, 150–300, and 300–1200 μ m; and (f) approximate molecular weight exclusion of 3,500.

A typical batch procedure involves direct addition of one part Chelex to four parts buffer solution (weight/volume) with subsequent filtration to remove the resin. This treatment will readily remove iron, copper, zinc, calcium, and magnesium ions. Optimal chelation for iron and calcium ions occurs at pH 5–6.4; for copper, zinc, and magnesium ions, the optimal pH range is 7.4–8.0.

Proteins, especially those with higher isoelectric points, will bind to Chelex, and care must be exercised to minimize protein binding whenever Chelex is used to “demetallate” a solution containing protein. In such cases, one should operationally determine the least amount of Chelex and time of exposure needed to remove metal ions without significantly reducing the protein concentration. Chelex may even be used to study the effects of cations on cells in culture. Again, one must determine the most appropriate pH and resin/specimen to minimize loss of other essential nutrients and/or cofactors.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Affinity for calcium ion, **57**, 302, 307; arsenazo III purification, **56**, 318; assay of nucleoside phosphorylase, **51**, 539; for preparation of metal-free buffers, **54**, 476; removal of calcium ions, **57**, 301, 302; of paramagnetic contaminants, **54**, 154; for removing iron contamination, **233**, 85-86, 123; DNA preparation and, **29**, 376, 377, 380, 384, 404; methionyl tRNA and, **20**, 190; in deferration of laboratory media, **235**, 327; for iron contamination removal, **233**, 85-86, 123.

CHEMICAL BOND

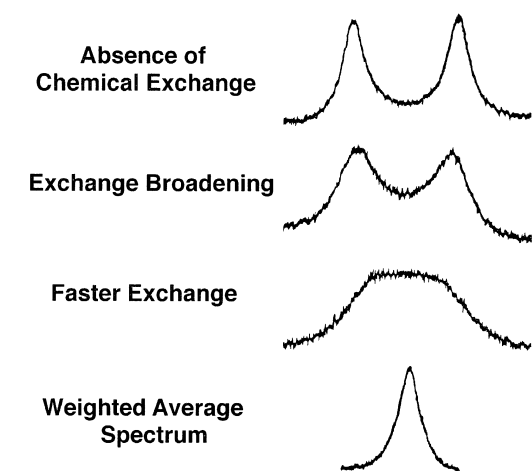
Pauling¹ offered the following definition of a chemical bond: “. . .there is a chemical bond between two atoms or groups of atoms. . . (if) forces acting between them are such as to lead to the formation of an aggregate with sufficient stability to make it convenient for the chemist to consider it as an independent molecular species.” Chemical bonds include ionic bonds, coordinate covalent

bonds, covalent bonds, and the fractional bonds of solid metals. Weak dipole-dipole interactions are not considered as forming a chemical bond.

¹L. Pauling (1960) *The Nature of the Chemical Bond*, 3rd ed., p. 10, Cornell University Press, Ithaca.

CHEMICAL EXCHANGE EFFECT ON NMR SPECTRUM

The electromagnetic environment surrounding a nucleus within molecule A may change if A undergoes a reversible chemical reaction to form molecule B, especially if the chemical transformation involves electrons that are sufficiently near the nucleus of interest. The same can be said of a nucleus within a ligand molecule undergoing reversible binding to another molecule, such as a protein; the nucleus will then experience two different electromagnetic fields in free and bound forms of the ligand, say A and B. At first thought, one might surmise that the NMR spectrum will always exhibit two resonances: one for A and another for B at chemical shifts δ_A and δ_B . The fact is that the observed NMR spectrum (see below) will depend on the time-scale of the chemical exchange between A and B relative to



Effect of chemical exchange on NMR resonance spectra.

the NMR relaxation rate. If exchange is very slow, two sharp resonances, or “lines” in the parlance of NMR spectroscopy, will be observed. At the other extreme where $A \leftrightarrow B$ exchange is very fast, the single sharp line observed in the spectrum will reflect a time-averaging of the electronic field effects of both A and B. At intermediate rates of exchange, one or two exchange-broadened lines will be observed, depending on the relative magnitude of rate constants for exchange and NMR relaxation.

CHEMICAL FLUX

The change in concentration of a molecular entity (being transformed) per unit time and usually symbolized by ϕ . This change in concentration (*i.e.*, dc_i/dt) occurs in one direction only and applies to the progress of a reaction step (or sequence of steps) in a complex scheme that may even involve a set of parallel reactions. The term chemical flux can also refer to the progress of a chemical reaction(s) in one direction while that system is at equilibrium (*i.e.*, via isotope exchange at equilibrium).

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

CHEMICAL KINETICS

GENERAL INTRODUCTION. When conducted with appropriate rigor and design, kinetic studies provide the opportunity to draw strong inferences about how reacting molecules combine with each other. One can also deduce how electrons and atoms within molecules rearrange geometrically and/or electronically as they approach the transition state. The validity of such inferences and deductions can also enable one to manipulate reaction conditions, thereby altering the outcome in terms of yields and types of products formed. Likewise, mechanistic information can be used to organize ideas about a large family of related reactions, and the inferential skills of all chemists ultimately stem from systematic studies of reaction kinetics.

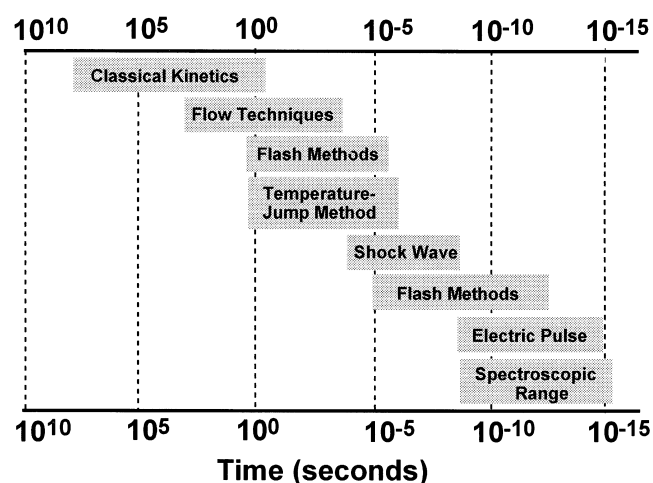


Figure 1. Techniques for determining the rates of chemical and physical processes occurring on the time-scales indicated as gray zones.

All kinetic studies begin with empirical rate equations accounting for the dependence of reaction velocity (v)

or reactant concentrations, with k as the temperature-dependent constant of proportionality:

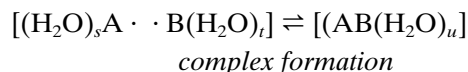
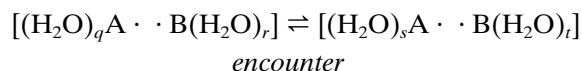
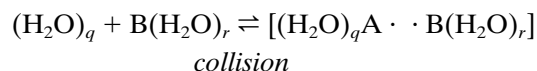
$$v = k \cdot f$$

where f can be regarded as a combination of reactant concentration(s) and a partition function having the properties of a well-defined mathematical system. The goal of kinetic mechanism studies is to identify all reaction schemes that reasonably account for the mathematical form of the empirical rate equation. One then develops other experiments to test which of the rival mechanistic schemes is self-consistent, allowing for a direct correspondence of observed rate behavior over a defined range of experimental conditions with that predicted by the rate equation. The ingenuity of the kineticist rests with his/her ability to create and to execute rate experiments that test the reasonableness of rival mechanisms. One approach is to choose a set of reaction conditions (taking into account such factors as pressure, temperature, solvent polarity, ionic strength, pH, *etc.*) that introduces unique and distinguishable features into the derived rate equations (or rate laws) for each rival mechanism. This amounts to a perturbation method wherein seemingly identical kinetic behavior under one set of conditions is altered or perturbed experimentally to divulge divergent behavior that is diagnostic with respect to a choice of reaction schemes.

Kineticists often apply a variety of mechanistic probes to avoid being duped into reaching an inappropriate conclusion based on limited information. Even so, one cannot test all possible sets of reaction conditions, and the “proof by exclusion” method allows one only to discard mechanisms that fail to be consistent with a given set of experimental results. Kinetic studies are therefore said to lack the ability to prove any reaction mechanism; they only have the power to disprove a particular mechanistic scheme. In mathematics, one is content with rigorous proofs achieved by exclusion or by *reductio ad absurdum* arguments. In fact, *all* science, not just kinetics, is based on the tenet that conclusions always remain open to question and can be overturned by reliable evidence to the contrary. Kineticists create mechanisms which are “logic boxes”—strong enough to persuade themselves and others of the chemical reasonableness of their findings. This hallmark of good science is hypothesis-driven

research, and mechanisms are hypotheses that unify seemingly diverse properties of nature.

SOLUTION BEHAVIOR. Virtually all biochemical reactions occur in water, and one should first consider some aspects of rate processes in aqueous solutions. From X-ray dispersion studies, one gets a sense that water displays considerable local order. (For further consideration, **See Water**). The hydrogen bonding capacity of water greatly influences the structure and reactivity of components present in an aqueous solution. In a condensed solution phase, ions and molecules move about less randomly than they would in the gas phase; this results from the fact that these ions and molecules are influenced by electronic and van der Waals interactions with other adjacent solvent and solute molecules. Translation diffusion is dependent both on the mean free path and electrostatic interactions, both of which are far more restricted in solutions. Any moving ion experiences drag exerted by neighboring ions and solvent. A variety of arguments suggest that while root-mean square velocities of molecules in solution and gas phases are somewhat comparable, collisions occur much more frequently in solution. In aqueous solutions, the local ordering of H₂O molecules creates a shell, or solvent cage, around solutes. Once two reactants do collide into each other, a solvent cage surrounds and holds them in an “encounter complex,” a transient and vaguely understood configuration of reactants and solvent that favors chemical reaction between encountering reactants. So while some ions and certainly protons can react at high rates, aqueous solute reactivity is said to be encounter-controlled, meaning that reactants must first succeed in forming an encounter complex. Accordingly, one may correctly anticipate that a more viscous medium retards the frequency of such encounters. This can be illustrated as follows with the water molecules in the solvent cages indicated in parentheses, along with subscripts to emphasize that the number of solvent molecules need not be the same for each solvent cage:



One's intuition also leads to the notion that the lifetime of an encounter complex must be of importance in promoting reactivity of ions and solutes.

Transition-state theory (*see below*) indicates that solvation can accelerate or retard the rates of a chemical reaction. In all cases, solvation lowers the potential energy of reactants. The single most important issue is the stability of the solvated ground state reactants in solution relative to that of the reactants in the transition state. Moreover, solvent interactions alter the equilibrium constants for many reactions, as might be inferred from the magnitude of $\Delta H_{\text{hydration}}$ for ions and even neutral solutes. Ionic solutes make substantial demands on the solvating properties of water, and this is especially true for ions of low ionic radius. Capable of acting both as an acid or a base, water is an especially good solvent for anions and cations. Furthermore, the extensive hydrogen network affects the ability of water to solvate molecules. During solvation, bonds between water molecules must break to create a cavity into which the solute may be placed. The ease with which such a cavity can form is greatly influenced by the surface tension of the solvent. Lower values for the latter permit cavity formation at lower energies.

Proton transfer in aqueous solutions can also be quite rapid (**See Water**). Protons also readily penetrate hydrogen-bonded complexes, and their transfer is sometimes aided by quantum mechanical tunneling. The ability of protons to move rapidly in hydrogen-bonded regions may also explain why proton binding and release are rarely rate-limiting in enzymic reactions. Contrary to intuition, electron transfer is not faster than proton transfer. Electrons do not exist as mobile charge carriers in water. Close proximity of electron donor and acceptor can overcome this limitation, and redox enzymes probably promote electron transfer by altering water structure or excluding water molecules altogether.

REACTION ORDER & MOLECULARITY. In the context of reaction rate equations, one can identify systems obeying the following mathematical forms:

$$\text{Zero Order} \quad v = k_0$$

$$\text{First Order} \quad v = k_1[A]$$

$$\text{Second Order} \quad v = k_2[A]^2 \text{ or } v = k[A][B]$$

$$\text{Third Order} \quad v = k_3[A]^3 \text{ or } v = k[A]^2[B] \\ \text{or } v = k[A][B][C]$$

where the respective units of k_0 , k_1 , k_2 , and k_3 are $\text{M}\cdot\text{s}^{-1}$, s^{-1} , $\text{M}^{-1}\cdot\text{s}^{-1}$, and $\text{M}^{-2}\cdot\text{s}^{-1}$. These expressions and the units of their rate constants all express reaction rate as a molarity change per unit time (*i.e.*, $-\text{d}[X]/\text{dt}$). Note also that, while we list a third-order reaction above, no such processes are known to occur for chemical reactions in the liquid phase.

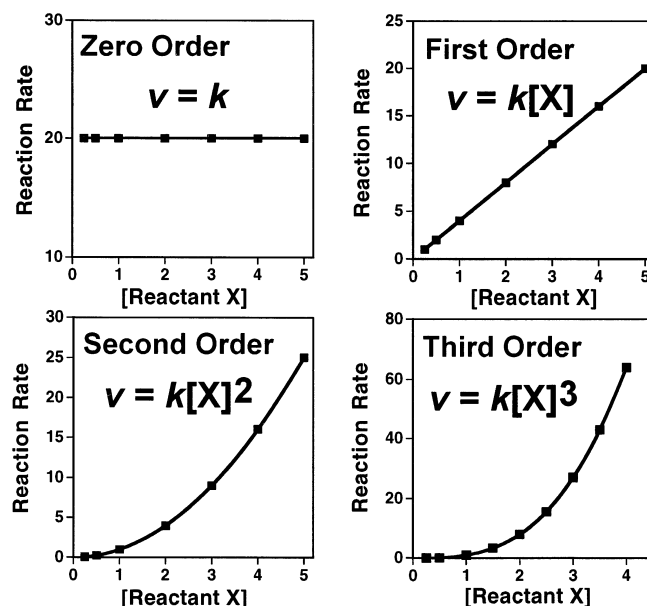


Fig. 2. Concentration dependency of zero-, first-, second-, and third-order chemical reactions.

The functional nature of some rate equations requires that rate constants occasionally can be of a mixed order, (*i.e.*, they can have nonintegral dimensions with respect to reactant molarity). Whether the reaction order is an integer or a fractional value, its value n is obtained from the slope of a plot of reaction rate *versus* $\log_{10} [\text{Reactant}]$.

Another term used to describe rate processes is “molecularity,” which can be defined as an integer indicating the molecular stoichiometry of an elementary reaction, which is a one-step reaction. Collision theory treats molecularity in terms of the number of molecules (or atoms, if one or more of the reacting entities are single atoms) involved in a simple collisional process that ultimately leads to product formation. Transition-state theory considers molecularity as the number of molecules (or entities) that are used to form the activated complex. For reactions in solution, solvent molecules are counted in the molecularity, only if they enter into the overall process and not when they merely exert an environmental or solvent effect.

Again, the molecularity of a reaction is always an integer and only applies to elementary reactions. Such is not always the case for the order of a reaction. The distinction between molecularity and order can also be stated as follows: molecularity is the theoretical description of an elementary process; reaction order refers to the entire empirically derived rate expression (which is a set of elementary reactions) for the complete reaction. Usually a bimolecular reaction is second order; however, the converse need not always be true. Thus, unimolecular, bimolecular, and termolecular reactions refer to elementary reactions involving one, two, or three entities that combine to form an activated complex.

The rate constant (sometimes called the specific reaction rate) is commonly designated by k . The SI unit of time is the "second" (symbolized by s). Thus, unimolecular rate constants are typically expressed in s^{-1} , and unimolecular processes are by definition concentration-independent reactions. A slight difficulty arises regarding SI units and bi- and termolecular rate constants. Concentrations in the SI system would be mol per cubic meter, but in chemistry concentrations are expressed in $\text{mol}\cdot\text{dm}^{-3}$ (or more commonly $\text{mol}\cdot\text{L}^{-1}$ or simply M^{-1}). Thus, a bimolecular rate constant typically has units of $M^{-1}s^{-1}$, whereas a termolecular rate constant is expressed with units of $M^{-2}s^{-1}$.

FIRST-ORDER PROCESSES. In a first-order reaction, the rate is directly proportional to the concentration of reactant, as indicated by the following rate law:

$$v = -d[A]/dt = k[A]$$

Rearranging this expression allows one to separate variables to form:

$$-d[A]/[A] = kdt$$

where the concentration of A at time t is $[A]_t$ and that originally present is $[A]_o$. Integration of the above equation yields:

$$[A]_t = [A]_o \exp(-kt)$$

The half-life of a first-order process (*i.e.*, the time $t_{1/2}$ required for $[A]_o$ to decay to $0.5[A]_o$) is given as:

$$\ln 2 = 0.693 = kt_{1/2} \quad \text{or} \quad \log 2 = (kt_{1/2})/(2.30)$$

These equations tell us that at successive time intervals equal to $n \times t_{1/2}$ where $n = 1, 2, 3, \text{etc.}$), the concentration

of A will fall to 0.50, 0.25, 0.125, *etc.*, of the original amount of A (Fig. 3).

First-order kinetic behavior is observed throughout Nature. One familiar example is radioactive decay:

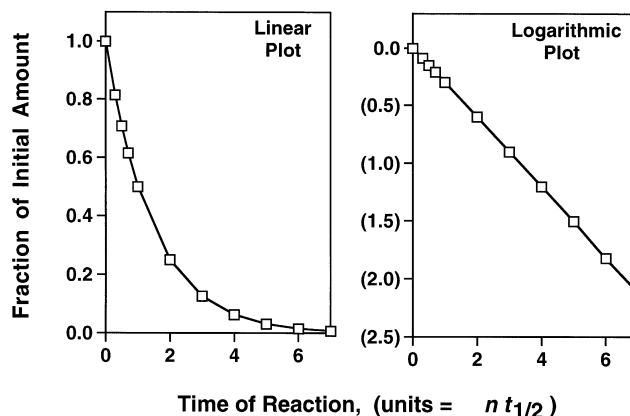
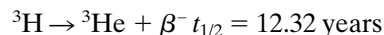
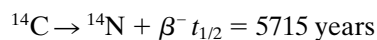
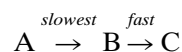


Fig. 3. Graphical representations of a first-order rate process.

First-order reaction kinetics is frequently observed in organic chemistry in the form of an S_N1 reaction, indicating it is a first-order nucleophilic substitution type. An example is the solvolysis of *tert*-butylbromide at alkaline pH to form *tert*-butanol and bromide ion. The reaction probably proceeds in two steps:

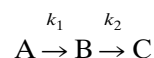


The first-order nature of this reaction reminds us that rate-limiting hydroxide attack on *tert*-butylbromide is not a feature of this S_N1 reaction.

Another instance where first-order kinetics applies is the conversion of reversible enzyme-substrate complex (ES) to regenerate free enzyme (E) plus product (P) as part of the Michaelis-Menten scheme:



Other reactions may exhibit the sequential first-order conversion of one reactant, followed by a second first-order process:



These are called series first-order reactions, and they are commonly analyzed using numerical integration, as shown in Fig. 4.

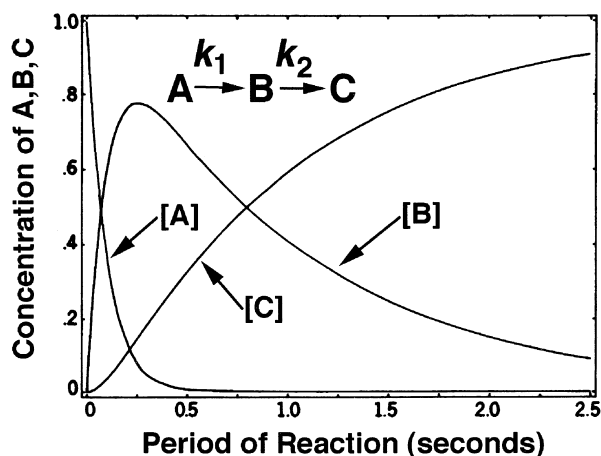
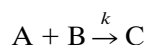


Fig. 4. Series first-order process $A \rightarrow B \rightarrow C$, beginning with rate constants $k_1 = 10 \text{ s}^{-1}$ and $k_2 = 1 \text{ s}^{-1}$. [Reactant A (concentration = 1 mM), intermediate species B (concentration = 0 mM), and product C (concentration = 1 mM)].

BIMOLECULAR RATE PROCESSES. Many chemical processes involve the rate-limiting reaction of two components:



where k has units equal to $\text{M}^{-1} \text{ s}^{-1}$.

1. General Rate Equation. The differential equation for a bimolecular rate process can be written as:

$$v = dx/dt = k[A][B] = k(a - x)(b - x)$$

where a equals the initial reactant A concentration; b equals the initial reactant B concentration; x equals the amount of product at time t ; $(a - x)$ equals the concentration of A at time t ; and $(b - x)$ equals the concentration of B at time t . After separating variables, we get:

$$dx/\{(a - x)(b - x)\} = kdt$$

After integration by the method of partial fractions, the result is:

$$\frac{\ln(a - x) - \ln(b - x)}{a - b} = kt + \text{Constant}$$

where at $t = 0$ and $x = 0$, the constant of integration

equals $\ln\{(a/b)/(a - b)\}$. This yields the integrated second-order rate law ($a \neq b$):

$$\frac{1}{(a - b)} \ln \frac{b(a - x)}{a(b - x)} = kt$$

2. Pseudo-First-Order Behavior. In some cases, one reactant may be present in great excess over the other, and the last two equations can be simplified to their pseudo-first-order form. In Fig. 5, we illustrate the time course of a bimolecular reaction between equivalent initial concentrations of A and B.

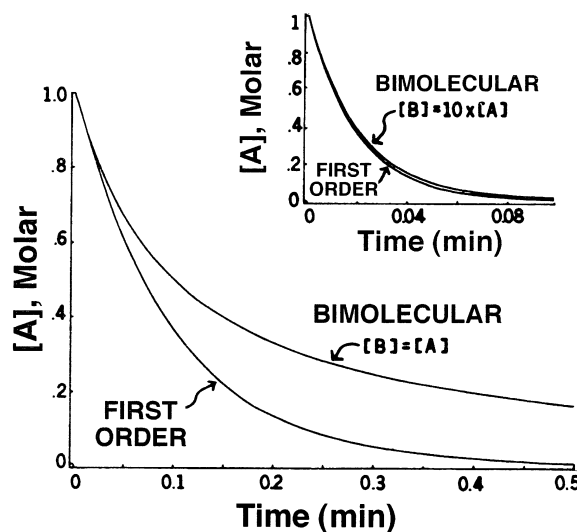
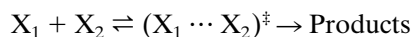


Fig. 5. Divergence in plots of the concentration of reactant A as a function of time for first-order and biomolecular rate processes (in the latter cases $[A] = [B]$). The inset shows what occurs in the pseudo-first-order condition (i.e., a condition where a ten-times molar excess of reactant B allows the conversion of reactant A in a nearly perfect first-order process).

The observant reader will notice that the rate of conversion of A is initially the same as the first-order process shown. However, as A and B decrease, the bimolecular reaction rate slows appreciably, and the decrease in A concentration tails off more slowly than the first-order process. As illustrated in the inset, however, when $[B_0] = 10 \times [A_0]$, the second-order and first-order reactions are virtually indistinguishable. In fact, plots of $\ln[A_0]$ versus time look virtually linear (i.e., pseudo-first-order) when $[B_0] \geq 3[A_0]$.

3. Effect of Electrostatic Interactions on the Bimolecular Rate Constant. The bimolecular rate equation presented above does not account for the effect of electrostatic interactions on reactivity of ionic molecules. Brønsted and Bjerrum, among others, recognized that the behavior

of ionic reactants will depend on ionic strength and not simply on the concentrations of each reacting component. These electrostatic effects must be factored into kinetic and thermodynamic expressions for transition-state behavior. The usual scheme, written here as:



where reactants X_1 and X_2 first form the transition-state species $(X_1 \cdots X_2)^\ddagger$, must be amended for ionic reactants of charge z_1 and z_2 .



In both of these schemes, the parentheses are used to indicate the chemical activities, and not simply the respective concentrations of each species. One can write an equilibrium relationship for the reactants and transition-state complex:

$$K^\ddagger = \frac{(X_1^{z_1} \cdots X_2^{z_2})^\ddagger}{(X_1^{z_1})(X_2^{z_2})} = \frac{[X_1^{z_1} \cdots X_2^{z_2}]^\ddagger}{[X_1^{z_1}][X_2^{z_2}]} \times \frac{\gamma_{12}^\ddagger}{\gamma_1 \gamma_2}$$

where the parentheses indicate activities, the brackets represent concentrations, and the Greek letter γ_i symbolizes the respective activity coefficients. As previously written, the bimolecular reaction rate is:

$$-\frac{dc_1}{dt} = k_{12}c_1c_2$$

From transition-state theory, the bimolecular rate constant k_{12} can be written as:

$$k_{12} = \frac{\nu c^\ddagger}{c_1c_2} = \frac{k_B T}{h} \times \frac{c^\ddagger}{c_1c_2} = \frac{k_B T}{h} \times K^\ddagger \times \frac{\gamma^\ddagger}{\gamma_1 \gamma_2}$$

This equation contains the activity coefficients γ_1 , γ_2 , and $\gamma^\ddagger$. Recall from the Debye-Hückel treatment of ionic interactions in dilute solutions that the magnitude of these coefficients shows the following dependence on ionic strength μ for a solution of electrolytes:

$$\log \gamma_+ = -0.5091 z_+^2 (\mu)^{1/2}$$

$$\log \gamma_- = -0.5091 z_-^2 (\mu)^{1/2}$$

Therefore, because the bimolecular rate constant deals with two components,

$$k_{12} = \log \left(\frac{k_B T}{h} K^\ddagger \right) + 1.0182 z_1 z_2 (\mu)^{1/2}$$

[Recall that $\mu = (1/2) \sum_{i=1}^n c_i z_i^2$, where c_i is the concentration of the i th ion, z_i is the charge, and the summation extends over all n ions within the solution.] The equation for k_{12} predicts that for two ionic reactants of the same charge the reaction rate will increase with ionic strength. This behavior is an immediate consequence of the reduced electrical repulsion of ions of like charge when present in a more concentrated electrolyte solution. Such behavior is often called ionic screening. If two reactants are of opposite ionic charge, their mutually attractive interactions will be favored at low ionic strength, but disfavored at higher ionic strength. Thus, for oppositely charged reactants, the observed reaction rate will decrease as the ionic strength increases.

Experimenters would do well to avoid any unnecessary changes in the ionic composition of reaction samples within a series of experiments. If possible, chose a standard set of reaction conditions, because one cannot readily correct data from one set of experimental conditions in any reliable manner that reveals the reactivity under a different set of conditions. Maintenance of ionic strength and solvent composition is desirable, and correction to constant ionic strength often effectively minimizes or eliminates electrostatic effects. Even so, remember that Debye-Hückel theory only applies to reasonably dilute electrolyte solutions. Another important fact is that ion effects and solvent effects on the activity coefficients of polar transition states may be more significant than more modest effects on reactants.

EXPERIMENTAL CONSIDERATIONS. Accurate analytical measurement of reactant depletion or product accumulation is of central importance in kinetics. Usually, the experimenter develops an assay based on product accumulation, especially when no product or product-like analyte is initially present. In this situation, $\Delta P = [P_t] - [P_{\text{initial}}]$, where $[P_{\text{initial}}]$ is most often negligible. This contrasts with the greater uncertainty in measuring a drop in substrate concentration ($\Delta S = [S_t] - [S_{\text{initial}}]$); here, the difference of two larger, nearly equal values challenges the precision or reproducibility of rate measurements. The analytical method must not introduce complications arising from lack of sensitivity or any confounding effects such as slow response of the measuring instrument. For example, one should avoid spectropho-

tometric assays where the molar absorption coefficient is unavoidably small; this situation can lead to great imprecision. Likewise, the long response time of some pH and ion-selective electrodes may overlap with processes having a similar time scale for reaction. **See** *Absorption Spectroscopy, Fluorescence Spectroscopy*

1. Progress Curve Analysis. Determination of the rate constants for a first-order or bimolecular rate process can be achieved by graphical analysis. In the first case, one plots $\ln\{([A_0] - [A_t])/([A_0] - [A_\infty])\}$ versus time, where $[A_0]$, $[A_t]$, and $[A_\infty]$ are the respective reactant concentrations at time zero, time t , and infinity. [Note: Many reactions are not irreversible, and one cannot assume that $[A_\infty]$ will be zero; likewise, there may be a baseline correction in the analytical method, and this will frequently show up in the value of $[A_\infty]$. The slope equals $-k$. For determining rate constants of a bimolecular process, a logarithmic plot is employed to obtain a linear graphical transformation of the reaction progress curve. In those cases where the so-called “infinity value” cannot be determined, one may apply the Guggenheim method. In this protocol, values of the physical property (e.g., pressure, absorbance, etc.) are measured at various times (i.e., property values, $\lambda_1, \lambda_2, \lambda_3, \dots$, are obtained at times $t_1, t_2, t_3, \dots$). Then, values of the same physical property ($\lambda'_1, \lambda'_2, \lambda'_3, \dots$) are obtained at times $t_1 + \Delta, t_2 + \Delta, t_3 + \Delta, \dots$, in which Δ is a constant time interval. Typically, for chemical reactions, Δ is a value between $t_{1/2}$ and $3t_{1/2}$. For λ_n and λ'_n , $\ln(\lambda'_n - \lambda_n) = -kt_n + \text{constant}$ where λ_n is the measurement of the physical property at t_n , λ'_n is the property measurement at $t_n + \Delta$, k is the first-order rate constant, and the constant is $\ln[(\lambda_\infty - \lambda_0)(1 - e^{-k\Delta})]$. Thus, a plot of $\ln(\lambda'_n - \lambda_n)$ versus time will yield a straight line having a slope of $-k$.

2. Initial Rate Assumption. The entire reaction progress curve, or at least a substantial portion of it, is typically required to accurately determine the rate constant for a first-order or second-order reaction. Nonetheless, one can frequently estimate the rate constant by measuring the velocity over a brief period (known as the initial rate phase) where only a small amount of reactant is consumed. This leads to a straight-line reaction progress curve (see Fig. 6) which is drawn as a tangent to the initial reaction velocity.

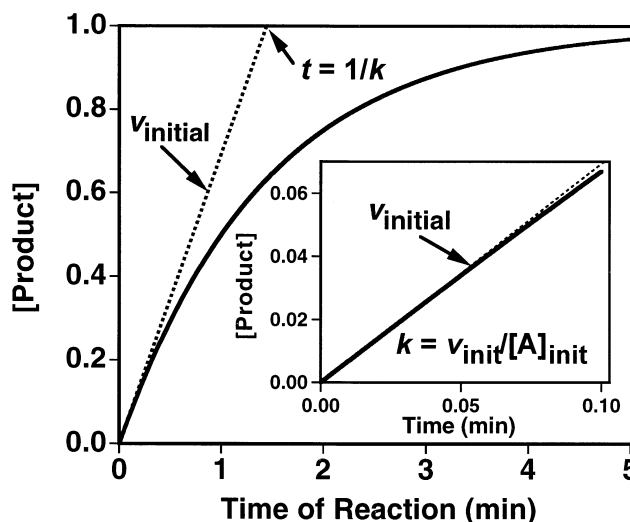


Fig. 6. Initial rate behavior of chemical reactions.

Then, k equals the observed reaction rate divided by the initial reactant concentration (i.e., $k = v_{\text{initial}}/[A]_{\text{initial}}$). This method is most useful when one has an assay method that is sufficiently sensitive to ensure that only a small fraction, say 3–5%, of the reactant is depleted during the rate measurements. Typically, this is satisfactorily achieved with a UV-visible spectrophotometer, a fluorescence spectrometer, or a radioactively labeled reactant. The initial rate method is extremely convenient, and the preponderance of enzyme rate data has been obtained by initial rate measurements. Finally, one should note that the initial rate method can yield erroneous results if the initial reactant concentration is in doubt. This is not true for the plots of $\ln\{([A_0] - [A_t])/([A_0] - [A_\infty])\}$ versus reaction time because one is considering the fraction of reactant A remaining.

PRINCIPLE OF DETAILED BALANCE. For chemical reactions at thermodynamic equilibrium, the likelihood of any particular molecular configuration strictly depends on the energy of that configuration. This is a requirement of microscopic reversibility, a principle rooted in classical equilibrium thermodynamics and equally well grounded in statistical mechanics and quantum mechanics. One may demonstrate that for chemical reactions at equilibrium, or rigorously at steady-state, the number of molecules passing from configurations “a” to “b” must be equivalently balanced by the number of molecules converting from configurations “b” to “a.” Another way of stating this principle is to say that should there exist a pathway from configuration “a” to configu-

ration “b,” a pathway must likewise exist for transition from “b” to “a.” Each step must be balanced by forward and reverse reaction steps. This Principle of Detailed Balancing can be most useful in efforts to construct kinetic mechanisms for chemical reactions and exchange processes.

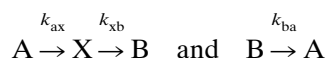
Consider the following reversible chemical reaction between reactant A and product B:



for which the rate of reaction at equilibrium can be written as:

$$\frac{d[\text{A}]}{dt} = -k_1[\text{A}] + k_{-1} = 0$$

Suppose for the sake of argument that there is an alternative path in which A passes through an intermediate form X on its way to B. This could be schematically drawn as follows:



One may wonder whether this cyclic scheme can provide a pathway to maintain A and B in equilibrium through the intermediate X. In fact, such a mechanism would violate the Principle of Detailed Balancing, because at equilibrium the rate of any particular reaction step must be exactly equal to (*i.e.*, balanced by) the rate of the exact reverse of that reaction step. In the case presented here, we are considering collections of molecules on a large or macroscopic scale. Were we to consider the same process on the molecular scale, the corresponding rule is known as the Principle of Microscopic Reversibility, a condition that can be rigorously proven on the basis of quantum mechanical expressions for transition probabilities. This is intuitively reasonable on the basis of transition state theory: molecules following a given path between reactant and product will naturally pass over the barrier of lowest potential energy. It also follows that one cannot propose any reaction mechanism that defies the principles of detailed balancing and microscopic reversibility.

TRANSITION STATE THEORY. Arrhenius and van't Hoff independently proposed that the rate constant for

a chemical reaction must depend on the absolute temperature T , thereby transforming the empirical relationship:

$$k = Ae^{-B/T}$$

(where A and B are suitably chosen constants) into a related form:

$$k = Ae^{-\Delta E/RT}$$

where R is the universal gas constant. This equation is widely known as the Arrhenius Law, and the pre-exponential term is known as the frequency factor (with the same units as k).

The constant A , in many respects, is analogous to the highest achievable magnitude of the rate constant k , and the $e^{-\Delta E/RT}$ term defines the fraction of molecules having sufficient energy to react. If $\Delta E \approx 0$, then $e^{-\Delta E/RT} \approx e^0$, which equals unity; thus, $k \approx A$. On the other hand, if ΔE is very large, such that $E \approx 20RT$, then $e^{-\Delta E/RT} \approx e^{-20}$, which is approximately 2×10^{-9} ; therefore, k is extremely small in this example. This behavior obeys the Boltzmann principle which states that the fraction of collisions with energy exceeding a threshold value of ΔE is $e^{-\Delta E/RT}$. This fraction, also known as the Boltzmann factor, is greater: (a) as the absolute temperature T becomes greater or (b) as ΔE becomes smaller.

The Arrhenius equation can be rearranged to yield the following expression:

$$\ln k = \ln A - (\Delta E/RT)$$

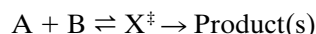
or

$$\log_{10} k = \log_{10} A - (\Delta E/2.30259RT)$$

If a process obeys the Arrhenius law, then a plot of $[\log_{10} k]$ versus T^{-1} should yield a straight line with a slope of $-(\Delta E/2.30259R)$. This indicates that ΔE represents a barrier to reactivity, perhaps related to organizational events that reflect the conversion of the reactant to an activated species.

Eyring and Polanyi extended the Arrhenius concept concerning the nature of barriers to chemical reactivity. They considered the concept of a transition state which can be defined as the highest energy species or configuration in a chemical reaction. Transition-state theory now forms the basis of uncatalyzed and catalyzed chemi-

cal reactions, whether homogeneous, heterogeneous, or interfacial processes (see below). The simplest schematic representation is:



where $X^\ddagger$ is the activated transition-state (TS) complex in the transition state. We deliberately treat the system as the simplest two-step process here (Fig. 7). [Another reasonable approach would be to involve an initial step forming AB before forming $X^\ddagger$.]

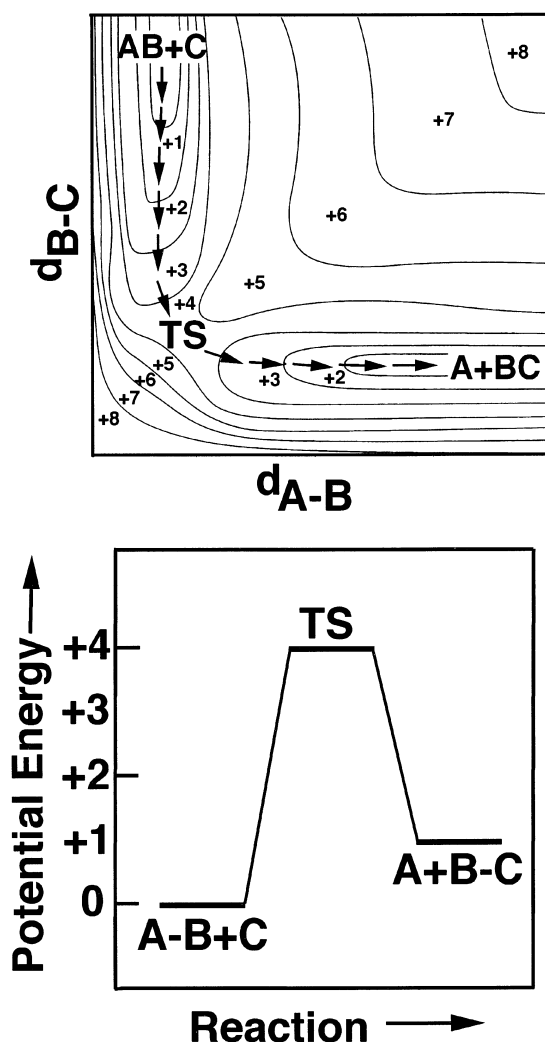


Fig. 7. Energetics of a bimolecular rate process. *Top:* Representation of the potential energy surface along coordinate axes corresponding to the interatomic distance of B-to-C and A-to-B, where incremental displacements along the potential energy axis are shown as a series of isoenergetic lines (each marked by arbitrarily chosen numbers to indicate increased energy of the transition-state (TS) intermediate relative to the reactants). *Bottom:* Typical reaction coordinate diagram for a bimolecular group transfer reaction.

Written as an equilibrium dissociation constant, $K^\ddagger$ equals $[A][B]/[X^\ddagger]$; and if we rearrange this expression (by substituting $\Delta G^\ddagger = -nRT \ln K^\ddagger$ in the form: $\exp(-\Delta G^\ddagger/nRT) = K^\ddagger$), we get:

$$[X^\ddagger] = [A][B]e^{-\Delta G^\ddagger/nRT}$$

This expression can be combined with the equation, $v = k_2[A][B] = k^\ddagger[X^\ddagger]$ where $k^\ddagger$ is a measure of the frequency with which product is formed from $X^\ddagger$. In this respect,

$$k_2 = k^\ddagger e^{-\Delta G^\ddagger/nRT}$$

This simplified approach is analogous to the more rigorous absolute rate treatment. The important conclusion is that the bimolecular rate constant is related to the magnitude of the barrier that must be surmounted to reach the transition state. Note that there is no activation barrier (*i.e.*, that $\Delta G^\ddagger = 0$) in cases where no chemical bond is broken prior to chemical reaction. One example is the combination of free radicals. (In other cases where electrons and hydrogen ions can undergo quantum mechanical tunneling, the width of the reaction barrier becomes more important than the height.)

The Eyring activated-complex (or transition-state) treatment relates the observed rate constant k to $K^\ddagger$ multiplied by the frequency factor $k_B T/h$, where k_B is the Boltzmann constant, T is the absolute temperature, and h is Planck's constant:

$$k = K^\ddagger(k_B T/h)$$

At 298 K, $k_B T/h \approx 6 \times 10^{12} \text{ s}^{-1}$, and when multiplied by $K^\ddagger$ which is an association equilibrium constant (with units of molarity⁻¹), the units M⁻¹s⁻¹ of k correspond to those of a bimolecular rate constant.

DIFFUSION-CONTROLLED PROCESSES. Even reactions having very small activation energies are not instantaneous; they still require their reactants to diffuse through the liquid before encountering each other. The rate of these so-called "diffusion-controlled reactions" depends solely on the time needed to form an encounter complex. The motions of molecules in solutions are greatly influenced by the potential fields exerted by their neighbors. Thus, the potential energy of individual molecules varies. There are also rapid collisions with molecules making up the solvent cage around a molecule as it proceeds to move by a series of discontinuous and

haphazard displacements. Two molecules in solution that become neighbors may even tend to collide with each other a number of times (often referred to as an encounter) before they either react or separate. Particles experiencing random or Brownian motion can be described by applying Fick's First Law and the Einstein diffusion relation, yielding an estimate of the upper limit of the bimolecular rate constant k :

$$k = 4\pi d_{AB}(D_A + D_B)L$$

where d_{AB} is the hard-sphere collision diameter discussed in collision theory, D_A and D_B are the diffusion coefficients for A and B, and L is Avogadro's number. This result was first obtained by Smoluchowski by assuming there is no intermolecular potential energy between the reacting molecules. With nominal values of d_{AB} at 5×10^{-8} cm, and $(D_A + D_B)$ at 10^{-5} cm²·sec⁻¹, and by adjusting for appropriate units, the above equation shows that k is equal to about 7×10^9 M⁻¹·sec⁻¹. The corresponding upper limit value for a bimolecular rate constant in the gas phase is about 10^{11} M⁻¹·sec⁻¹. Thus in solutions, bimolecular rate constants cannot exceed 10^9 – 10^{10} M⁻¹·sec⁻¹ because the rate of diffusion then becomes a limiting factor.

Recall that the diffusion coefficient of a molecule will decrease with increasing viscosity of the solvent. Thus, the rate of encounter complex formation will decrease in a viscous medium. Since viscosity is itself temperature dependent, such encounters in solution will have their own activation energy.

The above expression represents the maximum rate for an encounter in a solution. However, most reactions have an energy barrier and/or strict steric requirements. Typically, many collisions occur before a reaction ensues. One must also consider the duration of an encounter and the frequency of vibrational collision during an encounter. The duration of an encounter, τ_{AB} , between two molecules A and B can be estimated to be

$$\tau_{AB} = \lambda^2/6D$$

In this expression, $D = D_A + D_B$, the sum of the diffusion coefficients for A and B, and λ^2 is the mean square displacement needed to terminate an encounter. The

value of λ can be set equal to the diameter of a solvent molecule or set equal to $2d_{AB}$. The probability of a reaction occurring during an encounter is

$$P_r = \frac{\nu p e^{-LE/RT}}{\nu p e^{-LE/RT} + 1/\tau_{AB}}$$

where ν is the frequency of collision between A and B during an encounter and p is the steric factor. Thus, the collision theory bimolecular rate constant in solution becomes

$$k = (2/3)\pi d_{AB}^2 \nu p e^{-LE/RT} L$$

If the magnitude of τ_{AB}^{-1} is small relative to the magnitude of $\nu p e^{-LE/RT}$, then the reaction is effectively diffusion-controlled. **See Arrhenius Equation; Transition-State Theory; Fick's Laws**

CATALYSIS. Any condition promoting $X^\ddagger$ formation will tend to speed up the reaction rate, and catalysts are thought to accomplish rate enhancement chiefly by stabilizing the transition state. Shown in Fig. 8 is an enzyme-catalyzed process in which reactant S (more commonly called substrate in enzymology) combines with enzyme to form an enzyme-substrate complex. This complex leads to formation of the transition state complex $EX^\ddagger$ which may proceed to form enzyme-product complex. The catalytic reaction cycle is then completed by the release of product P, whereupon the uncombined enzyme returns to its original state.

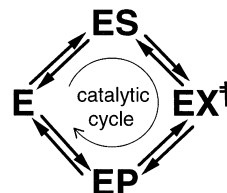


Fig. 8. Enzyme reaction cycle. In each catalytic round, the enzyme returns to the same chemical form.

In this case, there are three barriers to surmount: (1) the activated complex X_1^* which precedes ES formation; (2) the least stable configuration $X_2^\ddagger$, corresponding to the transition state; and (3) the activated complex X_3^* which precedes EP breakdown to form enzyme plus product (Fig. 9).

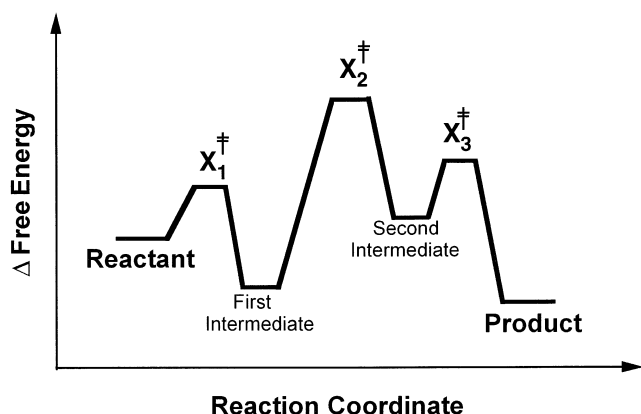


Fig. 9. Reaction coordinate diagram for a catalyzed reaction involving two intermediate species and three transition states. The highest lying species $X_2^\ddagger$ corresponds to the transition state (TS) of the reaction.

Catalysts are very effective at stabilizing the transition state, and enzymatic reactions often proceed at rates that are 10^7 – 10^{11} times faster than their uncatalyzed counterparts. Because a catalyst returns to the same chemical form after each catalytic round, the catalyst cannot alter the equilibrium constant of a reaction. Thus, an enzyme must accelerate the $S \rightarrow P$ interconversion by the equivalent factor representing the acceleration of $P \rightarrow S$.

Although enzyme-catalyzed reactions are described in many other entries in this Handbook, some mention of the time-evolution of an enzymatic process should be considered here. Shown in Fig. 10 is an representation of a typical reaction progress curve. A rapid rise in the concentration of reactant-bound species $\{ES + E^\ddagger + EP\}$ occurs in the pre-steady-state phase. Then, the total concentration of $ES + E^\ddagger + EP$ remains relatively stable during the steady-state phase, the period over which product formation is relatively constant. As substrate becomes depleted, the driving force for maintaining the concentrations of ES and $E^\ddagger$ weakens, and although not shown here, $[EP]$ becomes the dominant reactant-bound species as product accumulates.

In terms of enzyme catalysis, the following factors are likely to influence the magnitude of the rate enhancement in enzymatic processes: (a) proximity and orientation effects; (b) electrostatic complementarity of the enzyme's active site with respect to the reactant's stabilized transition state configuration; (c) enzyme-bound metal ions that serve as template, that alter pK 's of catalytic groups, that facilitate nucleophilic attack, and that have

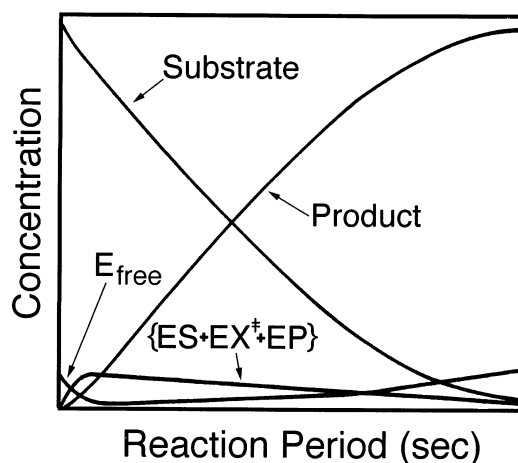


Fig. 10. Time-course of an enzyme-catalyzed reaction, illustrating time-dependent changes in all reactant concentrations.

special redox properties; (d) fine tuning of rate constants that optimize the throughput of reactants.

We use the term homogeneous catalysis for those cases occurring in a single phase, gaseous or liquid. Most enzymes are homogeneous catalysts. The term heterogeneous catalysis refers to catalysis of multiphase chemical reactions, and this form usually occurs at the interface of the different phases. Lipolysis, for instance, is catalyzed by lipases acting at the interface between aqueous and lipid phases. Another example, this time from bioorganic chemistry, is the so-called micellar catalysis of phosphate monoester hydrolysis in two-phase systems composed of aqueous and tertiary cetylammmonium micelles.

Chemical reactivity is also facilitated by what has been called acid-base catalysis. Because most nucleophilic, electrophilic, and redox reactions in water are strongly influenced by pH, the fundamental nature of acid-base catalysis becomes obvious. Indeed, systematic investigation of bioorganic and enzymic reactions requires an understanding of the salient features of acid-base catalysis. This field began in the early nineteenth century with Kirchoff's studies of the hydrolysis of starch into glucose, as well as Wilhemy's work on acid-catalyzed inversion of cane sugar. Research over the past fifty years has proven the overall significance of acid-base catalysis in biotic systems. **See** *Brønsted Theory; pH Kinetics*

RELAXATION KINETICS. The term "relaxation" describes the readjustment of a system from one state of

dynamic equilibrium to a new state in response to the abrupt addition or removal of energy to or from the system. The relaxation time refers to the period that must elapse for an exponentially decreasing variable to fall from its original value X_{initial} to X_{initial}/e (or approximately $0.368X_{\text{initial}}$). In 1866, J. C. Maxwell was the first to use the term “time of relaxation” to represent the period needed for the elastic force of a fluid to decay to X/e , where X is the initial value of the imposed force. Although Einstein was the first to treat the relaxation kinetics of a dissociating gas subjected to adiabatic compression and dilatation in a sound wave, many other chemical reactions in the solution phase also exhibit relaxation behavior.

1. BRIEF DESCRIPTION OF RELAXATION METHODS. Chemical relaxation theory and the ingenious methods for applying relaxation theory to rapid chemical reactions were developed largely through the work of Nobelists Eigen, Porter, and Norrish. In the temperature-jump method, energy is added to a system by the electrical discharge of a capacitor which results in Ohmic heating that arises as a consequence of the frictional drag on ions that are suddenly induced to move toward their respective electrodes. This technique (see below) is widely utilized to determine the kinetics of elementary reactions that make up a multistage chemical reaction mechanism.

In the pressure-jump technique, a system reaches equilibrium at an initially pressurized condition; then, the sudden rupture of a foil diaphragm causes rapid depressurization. The system responds to a loss of energy (*i.e.*, for a gas, any increase in volume will increase the mean free path between reactants, thereby lowering its energy; likewise, if conversion of reactant to product has a non-zero volume change, the chemical system will be perturbed by a sudden change in pressure). Pressure effects can also be induced by shock tubes that create a sudden compression wave that passes through a reacting medium. In pulsed radiolysis and flash photolysis, the sudden absorption of high energy radiation rapidly perturbs a system that was previously in dynamic equilibrium. In concentration correlation analysis, the chaotic movement of molecules leads to an increase or decrease in the statistically average number of molecules in a volume element, and the system relaxes in response to the pertur-

bation from thermodynamic equilibrium. Hammes has edited the definitive treatment on the investigation of elementary reaction steps and very fast reactions in solution.

2. BASIC KINETIC BEHAVIOR. Relaxation experiments are usually carried out with systems that are fairly close to equilibrium, and as such they can be treated by linear differential equations. If a perturbation leads to a larger displacement from equilibrium, the mathematics becomes intractable, and this is certainly the case if multiple step systems are substantially displaced from equilibrium (Fig. 11).

Chemical Relaxation

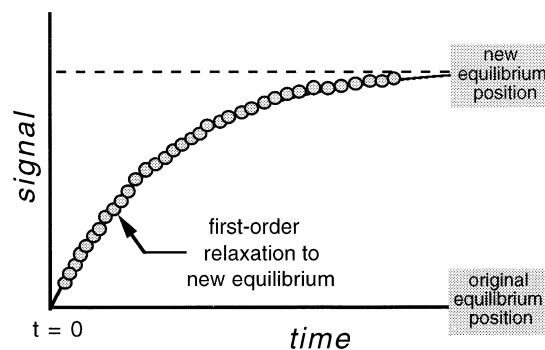


Fig. 11. Relaxation behavior of a chemical reaction where a sudden change from the original equilibrium position results in an increase in some signal that is proportional to the change in reactant concentration. Note that the process is first order.

Case 1. Unimolecular Isomerization: Consider a simple chemical reaction between species A and B at dynamic equilibrium:



If we let the quantity a equal the total concentration of A and B (*i.e.*, $a = [A] + [B]$), and if we designate the concentration of B at any time to be x , then we can write the following differential equation:

$$dx/dt = k_1(a - x) - k_{-1}x$$

which at equilibrium (designated below by the subscript e) becomes:

$$dx/dt = 0 = k_1(a - x_e) - k_{-1}x_e$$

and by solving for k_1a ,

$$k_1a = (k_1 + k_{-1})x_e$$

By analogy, we can express the displacement of x from its equilibrium concentration x_e as follows,

$$\Delta x = x - x_e$$

Thus, $x = \Delta x + x_e$, and the rate of change of Δx can be rewritten as

$$d(\Delta x)/dt = k_1 a - k_1 \Delta x - k_1 x_e - k_{-1} \Delta x - k_{-1} x_e$$

Now, substituting the previously expression $k_1 a = (k_1 + k_{-1})x_e$, we get:

$$d(\Delta x)/dt =$$

$$k_1 x_e + k_{-1} x_e - k_1 \Delta x - k_1 x_e - k_{-1} \Delta x - k_{-1} x_e$$

which upon canceling terms of opposite sign yields the final rate law:

$$d(\Delta x)/dt = -(k_1 + k_{-1})\Delta x$$

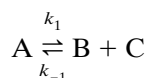
Integration of this differential equation yields the following equation:

$$\Delta x = \Delta x_0 \exp(-t/\tau)$$

where τ equals $(k_1 + k_{-1})$.

Case 2. Unimolecular Forward/Bimolecular Reverse:

Consider a simple chemical reaction between species A and B at dynamic equilibrium:



Again, we let a equal the total concentration of A and B (*i.e.*, $a = [A] + [B]$), and x is the concentration of B or C. We can write the following differential equation for the system at equilibrium:

$$dx/dt = 0 = k_1(a - x_e) - k_{-1}x_e^2$$

Using the expression $\Delta x = x - x_e$, we get

$$d(\Delta x)/dt = k_1(a - \Delta x - x_e) - k_{-1}(\Delta x + x_e)^2$$

Because $(\Delta x)^2$ is very small for a system near equilibrium, then

$$d(\Delta x)/dt = k_1 a - k_1 \Delta x - k_1 x_e - k_{-1} \Delta x - k_{-1} x_e^2$$

Now, by first substituting the previous expression $k_1 a = (k_1 x_e + k_{-1} x_e^2)$ and then canceling terms of opposite sign yields the final rate law:

$$d(\Delta x)/dt = -(k_1 + 2k_{-1}x_e)\Delta x$$

Integration yields the following equation:

$$\Delta x = \Delta x_0 e^{-t/\tau}$$

where τ equals $(k_1 + 2k_{-1}x_e)$. Note that $(A_e + B_e)$ make separate additive contributions from the forward and reverse reaction. It is possible to arrange to study the system at several different values of x_e , such that a plot of τ^{-1} versus x_e has a slope of $2k_{-1}$ and an intercept of k_1 .

3. IMPORTANT ONE-STEP REACTION SYSTEMS.

Without further derivation, we now summarize the relaxation kinetic parameters for the most frequently encountered one-step reactions.

Process	Relaxation Time
$\underset{k_{-1}}{\overset{k_1}{A}} \rightleftharpoons B$	$\tau^{-1} = k_1 + k_{-1}$
$A + B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} C$	$\tau^{-1} = k_1(A_e + B_e) + k_{-1}$
$2A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A_2$	$\tau^{-1} = 4k_1 A_e + k_{-1}$
$A + B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} C + D$	$\tau^{-1} = k_1(A_e + B_e) + k_{-1}(C_e + D_e)$
$A + C \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} C + B$	$\tau^{-1} = k_1 C_e + k_{-1} C_e$ [where C is a catalyst]
$A + B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} C$	$\tau^{-1} = k_1 B_e + k_{-1}$ [where B is buffered]

4. TEMPERATURE-JUMP METHOD. This rapid kinetic technique is frequently used to study the behavior of transient processes. The equilibrium position of most biochemical reactions is temperature-dependent, and a chemical equilibrium can be perturbed by suddenly increasing the reaction temperature. Moreover, for those reactions exhibiting little or no temperature dependence, one can link the formation of one reactant or product to another temperature-dependent process. For example, a reaction that produces or consumes a proton can be carried out in the presence of a buffer that is strongly temperature dependent. As noted earlier, Joule heating is accomplished by discharging a high-voltage capacitor through an aqueous electrolyte solution confined in the reaction cuvette. This compartment is like other cuvettes used in absorption or fluorescence spectrophotometry, insofar as it contains optically transparent windows. Typically, the optical path for the cuvette is around 1 cm. However, as shown below, the temperature-jump cell must also contain the high-voltage electrode and the

grounding electrode, suitably positioned to maximize the heating within a short period. Moreover, the electrodes must be free of any sharp edges that might permit unwanted sparking of the applied high voltage. For experiments with costly biochemical components, the reaction volume can be as little as 0.2 mL, and voltages of about 10 kV are required to obtain a 10°C temperature change in about 10 microseconds. If this results in a 5–10% displacement of a 1 mM reactant, then the corresponding 50–100 mM change in reactant concentration can easily be detected by absorption or fluorescence spectroscopy. If we consider an isomerization reaction with k_1 and k_{-1} each equal to 500 s^{-1} , then τ will be about 1 millisecond, a relaxation period that would be well separated from the 3–10 microsecond period needed for Joule heating.

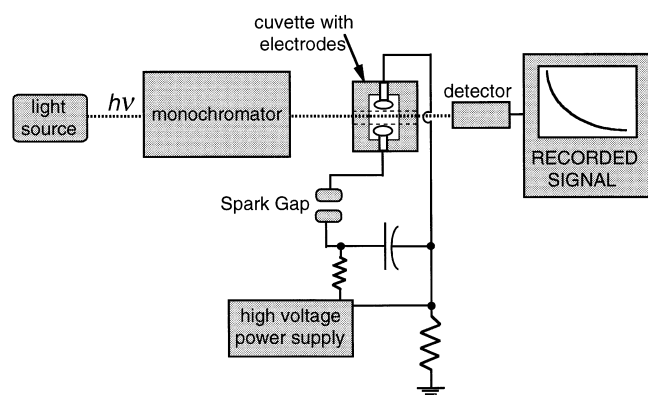


Fig. 12. Block diagram of a temperature-jump apparatus.

The schematic shown in Fig. 12 illustrates the general components that make up the temperature-jump apparatus. A spark-gap triggers the discharge of electrical energy stored within a capacitor, and discharge imparts a substantial electric field on the sample. As the principal charge carriers, small ions present in the reaction solution begin to migrate rapidly toward an oppositely charged pole; the acceleration of these ions must necessarily be balanced by frictional forces of the solvent, leading to the consequential release of energy as heat. Because the ions are present throughout the sample, this sudden heating occurs relatively uniformly throughout the sample. The speed with which the capacitor discharges tends to be the major factor limiting the time required for Joule heating. To assure rapid and uniform heating, large chemically inert, gold-plated electrodes are placed in the sample compartment and make direct and intimate contact with the reaction solution. The temperature change is experienced by the reaction system

as a change in rate constants, such that the system undergoes chemical relaxation (Fig. 13). Usually, absorbance or fluorescence signals are recorded on an oscilloscope or some other data acquisition device, and these can be analyzed to obtain values for the rate constants. Other optical techniques, such as polarimetry and light scattering, have been employed. The temperature-jump technique can also be combined with rapid mixing devices to increase the types of chemical processes that can be observed.

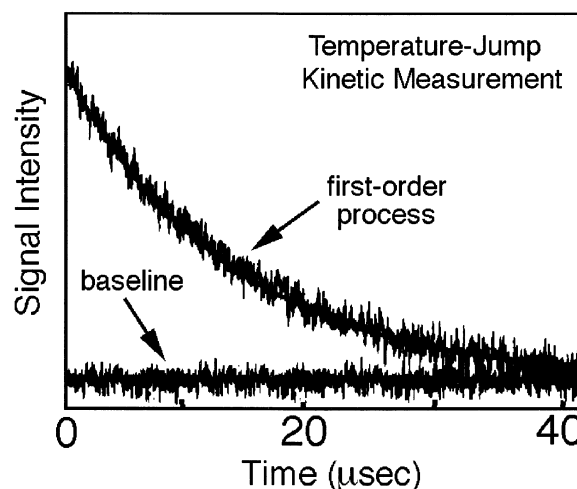


Fig. 13. Typical tracing of a kinetic process observed through the use of the temperature-jump method.

Temperature-jump devices provide satisfactory signal-to-noise characteristics (see Fig. 13). This is accomplished by using bright light sources such as xenon or mercury vapor sources as well as quartz halide lamps. The choice of photomultiplier depends on the method of detection, with fluorescence measurements requiring a higher quality PM tube than necessary for detecting changes in UV-visible absorbance.

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CHEMICAL POTENTIAL

The thermodynamic parameter μ_i which J. Willard Gibbs defined as the change in the free energy (now termed

the Gibbs free energy change or ΔG) with a change in the amount of substance in moles. For component B,

$$\mu_B = (\partial G / \partial n_B)_{T,P,n_j \neq B}$$

If component B is an ionic species,

$$\mu_B = RT \ln a_B + \mu_B^\ominus$$

where R and T are the universal gas constant and absolute temperature, respectively, and a_B is the activity of component B (i.e., $a_B = \gamma_B m_B$, with γ_B defined as the ion activity coefficient and m_B expressed as the molarity of B). Hence,

$$\mu_B = \mu_B(\text{ideal}) + \mu_B(\text{electric})$$

or

$$\mu_B = [RT \ln a_B + \mu_B^\ominus] + RT \ln \gamma_B$$

where $\ln \gamma_B = z_i^2 e_i^2 b / 8\pi \epsilon_0 k_B T$ in the well known Debye-Hückel electrostatic model. **See** *Chemical Thermodynamics, Debye-Hückel Model, Gibbs Free Energy*.

CHEMICAL RESCUE

The capacity of an acidic/basic solute to replace a functional catalytic group, usually located within or near an enzyme's active site, that has been removed by random mutation or site-directed mutagenesis¹. In their studies on aspartate aminotransferase, Toney and Kirsch¹ used site-directed mutagenesis to replace Lys-258, the endogenous general base, with an alanine residue. Catalytic activity was then restored to this catalytically inactive mutant by providing exogenous amines. Eleven amines were studied to generate a Brønsted correlation ($\beta = 0.4$) for the transamination of cysteine sulfinic acid. These investigators concluded that localized mutagenesis allows one to apply classical Brønsted analysis of transition-state structure to enzyme-catalyzed reactions. Another good example is the use of exogenous amines of a site-directed mutant of ribulose-1,5-bisphosphate carboxylase/oxygenase, also lacking a key lysyl residue². Harpel *et al.*² demonstrated that ethylamine enhances the carboxylation rate of K329A by about 80-fold and strengthens complexation of 2-carboxyarabinitol 1,5-bisphosphate. Rescue of K329A follows an apparent Brønsted relationship with a β of 1, suggesting complete protonation of amine in the rescued transition state. Rate saturation with respect to amine concentration and the different steric preferences for amines between K329A

and K329C suggest that the amines bind to the enzyme in the position voided by the mutation.

¹M. D. Toney & J. F. Kirsch (1989) *Science* **243**, 1485.

²M. R. Harpel & F. C. Hartman (1994) *Biochemistry* **33**, 5553.

CHEMICAL SHIFT

The displacement in the magnetic resonance frequency of a nucleus as a consequence of the electronic environment in which the nucleus resides. Because moving electrons generate their own magnetic fields, a nucleus surrounded by these electrons experiences an effective field, H_{eff} , which is defined by $(1 - \alpha)H_0$, where α is the so-called screening constant and H_0 is the applied magnetic field. A chemical shift is typically reported as a dimensionless displacement (units = parts per million, or simply ppm) from a reference standard. If the magnetic field is varied while the radio frequency ν is held constant, then the chemical shift (ppm) equals $\{[H_{\text{sample}} - H_{\text{reference}}]/H_{\text{reference}}\} \times 10^6$. If the radio frequency is varied at constant magnetic field, the chemical shift (ppm) equals $\{[\nu_{\text{sample}} - \nu_{\text{reference}}]/\nu_{\text{reference}}\} \times 10^6$. Tetramethylsilane is frequently used as a proton chemical shift reference compound in organic chemistry, because it yields a single sharp NMR signal in a region well removed from the resonance lines of most other protons.

CHEMISORPTION

Any adsorptive process or interaction involving significant electron sharing or transfer. Bond strengths typically fall in the 1–5 eV range per chemisorbed molecule, corresponding to about 100 to 500 kJ per mole. As opposed to physisorption, chemisorption is more specific, and the strength of binding much stronger. When chemisorption involves electron interactions with transition metals, the stronger interactions probably arise from the localized and highly directional d -orbital interactions.

CHEMOSELECTIVITY

The preferential chemical reaction of a reagent with one of two or more different functional groups. For example, sodium borohydride exhibits greater chemoselectivity as a reducing agent than does lithium aluminum hydride, because the latter reacts with a wider spectrum of substances.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

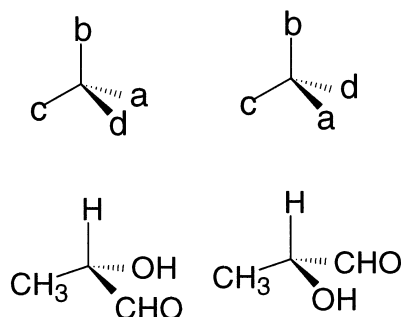
CHI (χ)

1. Symbol for the bond angle between the C2 (that is, α) carbon and the adjacent “side-chain” carbon of an amino acid in a peptide or protein. 2. Symbol for surface electric potential. 3. A parameter associated with a distribution in statistics, more commonly referred to as chi-squared distributions (χ^2 -distribution). Chi-square is a sum of terms in which each term is a quotient obtained by dividing the square of the difference between the observed and the theoretical value of a quantity by the theoretical value.

CHIRALITY

The condition of handedness or lack of superimposability of molecules differing only with respect to the stereochemical arrangement of identical substituents around a tetrahedral center. In 1893, Lord Kelvin succinctly defined chirality: “I call any geometrical figure, or any group of points, *chiral*, and say that it has *chirality*, if its image in a plane mirror, ideally realized, cannot be brought to coincide with itself.”

Because proteins are made up of L-amino acids, they exhibit chirality in their structures, lacking planes or points of symmetry. Proteins also can exhibit chirality in their interactions with other chiral molecules as well as prochiral centers in other molecules. This latter point is beautifully illustrated by fumarase’s catalysis of the dehydration of L-malate, a molecule containing two seemingly equivalent hydrogen atoms:



Nonsuperimposable Structures

The hydrogen in the *pro-R* position¹ is abstracted by the enzyme, and the *pro-S* hydrogen remains unaffected. In the reverse reaction, a single deuterium from D₂O is added

during the hydration of fumarate, and that deuterium always occupies the *pro-R* position. The stereospecificity of fumarase is completely attributable to its ability to bind substrates in a manner that permits a base on the enzyme’s surface to add or abstract only the *pro-R* hydrogen.

¹Note that the *pro-R* and *pro-S* hydrogens in L-malate can be readily discerned by mentally replacing either hydrogen by a deuterium atom and then applying the *RS* or Cahn-Ingold-Prelog system for defining *R* and *S* stereoisomers.

CHIRALITY ANALYSIS OF ENZYME REACTIONS

A method used to study the stereochemical course of an enzyme-catalyzed reaction involving the use of functional moieties (of the normal substrate) that are not typically chiral. Consider, for example, a prochiral methylene group (R—CH₂—R’). Because the methylene group is prochiral, an enzyme is usually able to distinguish between the two hydrogen atoms¹, but this is not true for an achiral methyl group (R—CH₃), unless specifically isotopically substituted. Nevertheless, the stereochemical course of an enzyme acting on an achiral center can often be analyzed by first converting the group into a chiral entity (e.g., for a methyl group into R—CHDT) and then treating with the enzyme, followed by chirality analysis of the substrate and/or product²⁻⁷. Synthesis of the stereospecifically pure enantiomer necessitates the use of a series of stereospecific reactions of which the stereochemical course has been well established.

If a chiral group is generated in the system to be studied, then chirality analysis of that group typically entails the removal of that moiety in a defined set of reactions such that the stereochemistry is not altered or the reactions have a known stereochemical course. For example, chirality analysis of chiral methyl groups typically involves conversion of the methyl groups into acetic acid.

Various applications of chirality analysis can be found in refs. 5, 6, and 7. **See** *Stereospecificity; Resolution*

¹A. G. Ogston (1948) *Nature* **162**, 963.

²J. W. Cornforth, J. W. Redmond, H. Eggerer, W. Buckel & C. Gutschow (1969) *Nature* **221**, 1212.

³J. Luthy, J. Retey & D. Arigoni (1969) *Nature* **221**, 1213.

⁴J. W. Cornforth, J. W. Redmond, H. Eggerer, W. Buckel & C. Gutschow (1970) *Eur. J. Biochem.* **14**, 1.

⁵J. W. Cornforth (1973) *Chem. Soc. Rev.* **2**, 1.

⁶H. G. Floss & M.-D. Tsai (1979) *Adv. Enzymol.* **50**, 243.

⁷H. G. Floss (1982) *Meth. Enzymol.* **87**, 126.

CHIRALITY PROBES

Stereochemical probes of the specificity of substrates, products, and effectors in enzyme-catalyzed reactions, receptor-ligand interactions, nucleic acid-ligand interactions, *etc.* Most chirality probe studies attempt to address the stereospecificity of the substrates or ligands or even allosteric effectors. However, upon use of specific kinetic probes, isotopic labeling of achiral centers, chromium- or cobalt-nucleotide complexes, *etc.*, other stereospecific characteristics can be identified, all of which will assist in the delineation of the kinetic mechanism as well as the active-site topology. A few examples of chirality probes include:

(1) Substrate/Product/Effector/Ligand Stereospecificity. Assessing the specificity for particular stereoisomers as substrates, products or effectors (surprisingly, effectors such as enzyme activators, rarely have their stereospecificities reported). In such studies, the investigators should clearly state the degree of chiral purity which is present and how it was determined.

(2) Anomeric Stereospecificity. For enzymes utilizing sugar or sugar derivatives as substrates or effectors, in addition to assessing issues of sugar specificity, it should be recalled that most sugars exist in two anomeric configurations. In many cases, the anomeric interconversion rate is slow and an investigator can assess the anomeric specificity for a given system¹. For example, D-glucose oxidase is specific for the β isomer². For systems with faster anomerization rates, both substrate analog and kinetic methods are available¹.

(3) Chiral Methyl or Methylene Groups. Upon following a specific set of reactions, utilizing isotopically labeled reagents, chiral methylene groups (*e.g.*, R-CHD—RO') or chiral methyl groups (*e.g.*, R—CHDT) can be substituted or incorporated into substrates. With such labeled compounds, stereochemical issues not apparent with the unlabeled substrate can be readily addressed³.

(4) Hydride Transfer in NAD⁺- and NADP⁺-Dependent Enzymes. The transfer of the hydride ion in redox reaction of NAD⁺- and NADP⁺-dependent enzymes can occur either to the *re*- or the *si*-face of the pyridine ring of the coenzyme^{4,5}. Such stereochemistry is crucial in the characterization of these enzymes. The same enzymes from different sources can express different stereospecificities. For example, *E. coli* NAD(P)⁺ transhydrogenase expressed one form of stereospecificity⁶ whereas the *Pseudomonas aeruginosa* enzyme catalyzes the identical reaction with the other NAD⁺ form⁷.

(5) Chromium(III) and Cobalt(III) Nucleotide Complexes. Coordination isomers of Cr(III)- and Co(III)-nucleotides can be separated and used to test for specificity. Since Cr(III) is diamagnetic, EPR and NMR experiments can supplement the chirality probes^{8,9}.

(6) Phosphorothioate Nucleotides Analogues. The stereochemical course of nucleotide-utilizing enzymes can be studied using a nucleotide analog in which a non-bridging oxygen at the α - or β -phosphorus of nucleoside 5'-triphosphate has been replaced by a sulfur atom. These phosphorothioate analogues have proven very useful in characterizing an enzyme-catalyzed reaction by observing whether inversion or retention has occurred¹⁰.

(7) Chiral Phosphates Utilizing Oxygen Isotopes as Nucleotide Analogues. Chiral phosphoryl groups in nucleotides can be achieved using various oxygen isotopes^{11,12}. Chiral [¹⁶O, ¹⁷O, ¹⁸O]phosphoric monoesters have the additional advantage in that they are chemically equivalent to the naturally occurring substrate and can also be used in a number of NMR probes¹³.

¹S. J. Benkovic (1979) *Meth. Enzymol.* **63**, 370.

²D. Keilin & E. F. Hartree (1952) *Biochem. J.* **50**, 341.

³H. G. Floss (1982) *Meth. Enzymol.* **87**, 126.

⁴K. You, L. J. Arnold, Jr., W. S. Allison & N. O. Kaplan (1978) *Trends in Biochem. Sci.* **3**, 265.

⁵K. You (1982) *Meth. Enzymol.* **87**, 101.

⁶J. B. Hoek, J. Rydström & B. Hojeberg (1974) *Biochem. Biophys. Acta* **333**, 237.

⁷D. D. Louie & N. O. Kaplan (1970) *J. Biol. Chem.* **245**, 5691.

⁸W. W. Cleland (1982) *Meth. Enzymol.* **87**, 159.

⁹J. J. Villafranca (1982) *Meth. Enzymol.* **87**, 180.

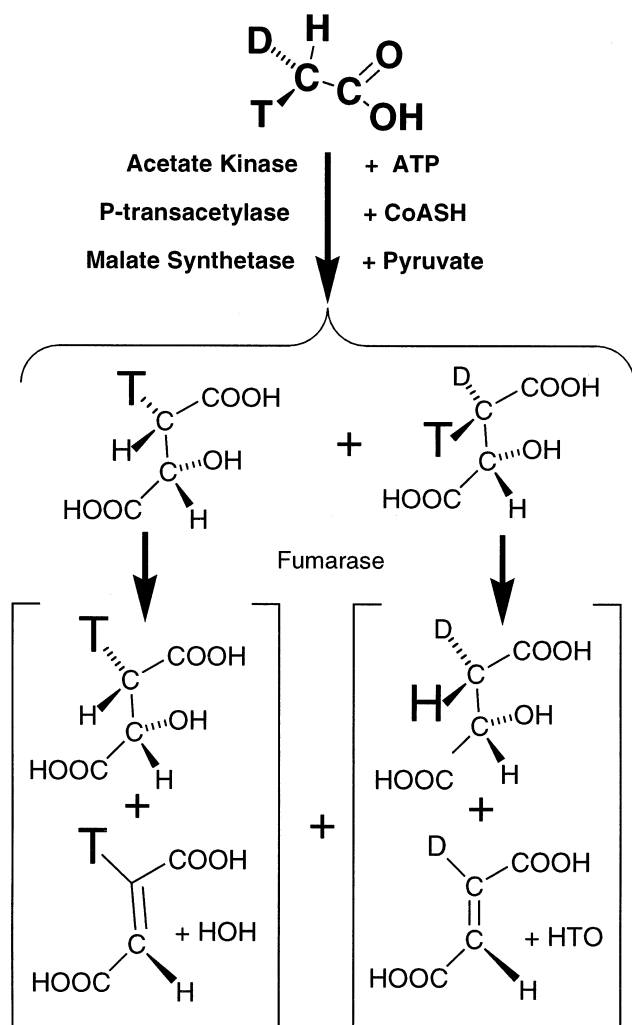
¹⁰F. Eckstein, P. J. Romaniuk & B. A. Connolly (1982) *Meth. Enzymol.* **87**, 197.

¹¹P. A. Frey, J. P. Richard, H. Ho, R. S. Brody, R. D. Sammons & K. Sheu (1982) *Meth. Enzymol.* **87**, 213.

¹²S. L. Buchwald, D. E. Hansen, A. Hassett & J. R. Knowles (1982) *Meth. Enzymol.* **87**, 279.

¹³M.-D. Tsai (1982) *Meth. Enzymol.* **87**, 235.

CHIRAL METHYL GROUPS



Stereochemically defined, isotopically prepared methyl groups used to determine the stereochemical course of methyltransferases^{1,2}. One typically utilizes [(*S*)-³H,²H,¹H]CH₃X or [(*R*)-³H,²H,¹H]CH₃X where X is the remaining part of the substrate. This process cannot be observed when all three hydrogen atoms are ¹H, because they are equivalent and stereochemically indistinguishable.

¹K. R. Hanson & I. A. Rose (1975) *Accts. Chem. Res.* **8**, 1.

²H. G. Floss (1982) *Meth. Enzymol.* **87**, 126.

CHI-SQUARE DISTRIBUTION

A statistical term referring to a monoparametric distribution used to obtain confidence intervals for the variance of a normally distributed random variable. The so-called chi-square (χ^2) test is a protocol for comparing the goodness of fit of observed and theoretical frequency distributions.

CHITINASE

This enzyme [EC 3.2.1.14] (also referred to as chitodextrinase, 1,4- β -poly-*N*-acetylglucosaminidase, and poly- β -glucosaminidase) catalyzes the hydrolysis of the 1,4- β -linkages of *N*-acetyl-D-glucosamine polymers of chitin. It should be noted that some chitinases will also display the activity observed with lysozyme [EC 3.2.1.17].

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

E. Cabib (1987) *Adv. Enzymol.* **59**, 59.

T. Imoto, L. N. Johnson, A. C. T. North, D. C. Phillips & J. A. Rupley (1972) *The Enzymes*, 3rd ed., **7**, 665.

CHITIN DEACETYLASE

This enzyme [EC 3.5.1.41] catalyzes the hydrolysis of chitin (or, more specifically, the *N*-acetamido groups of *N*-acetyl-D-glucosamine residues in chitin) to produce chitosan and acetate.

E. Cabib (1987) *Adv. Enzymol.* **59**, 59.

CHITIN SYNTHASE

This enzyme [EC 2.4.1.16], also known as chitin-UDP *N*-acetylglucosaminyltransferase, catalyzes the reaction of UDP-*N*-acetyl-D-glucosamine with [(1,4)-(N-acetyl- β -D-glucosaminyl)]_{*n*} to produce UDP and [(1,4)-(N-acetyl- β -D-glucosaminyl)]_(*n*+1). The enzyme ultimately converts UDP-*N*-acetyl-D-glucosamine into chitin and UDP.

E. Cabib (1987) *Adv. Enzymol.* **59**, 59.

CHLORAMPHENICOL ACETYLTRANSFERASE

This enzyme [EC 2.3.1.28] catalyzes the reaction of acetyl-CoA with chloramphenicol to produce coenzyme A and chloramphenicol 3-acetate.

W. V. Shaw & A. G. W. Leslie (1991) *Ann. Rev. Biophys. Biophys. Chem.* **20**, 363.

W. V. Shaw (1983) *Crit. Rev. Biochem.* **14**, 1.

CHLOROPEROXIDASE

This enzyme [EC 1.11.1.10], also called chloride peroxidase, catalyzes the reaction of hydrogen peroxide with two RH and two Cl⁻ to produce two R-Cl and two water molecules. A heme group is one of the cofactors. This enzyme can also catalyze bromination and iodination, but not fluorination.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

A. Butler (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 427.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

CHLOROPHYLLASE

This enzyme [EC 3.1.1.14] catalyzes the hydrolysis of chlorophyll to produce phytol and chlorophyllide. The enzyme has also been reported to catalyze chlorophyllide transfer reactions (for example, in converting chlorophyll to methylchlorophyllide).

S. Granick & S. I. Beale (1978) *Adv. Enzymol.* **46**, 33.

CHOLERA TOXIN

An ADP-ribosyltransferase that permanently activates the G_s regulatory protein in the adenylate cyclase pathway. Because ADP-ribosylated G_s-GTP complex cannot be converted to its metabolic inactive form, the adenylate cyclase remains in its activated state. The following are recent reviews on the molecular and physical properties of this ADP-ribosylating enzyme.

J. Moss, R. S. Hawn, S.-C. Tsai, C. F. Welsh, F.-J. S. Lee, S. R. Price & M. Vaughan (1994) *Meth. Enzymol.* **237**, 44.
J. Moss, S.-C. Tsai & M. Vaughan (1994) *Meth. Enzymol.* **235**, 640.
N. J. Oppenheimer & A. L. Handlon (1992) *The Enzymes*, 3rd ed., **20**, 453.
G. S. Kopf & M. J. Woolkalis (1991) *Meth. Enzymol.* **195**, 257.
J. Moss & M. Vaughan (1988) *Adv. Enzymol.* **61**, 303.

CHOLESTERYL ESTER HYDROLASE

This enzyme [EC 3.1.1.13] (also known as cholesterol esterase, sterol esterase, cholesterol ester synthase, and triterpenol esterase) catalyzes the hydrolysis of a sterol ester to produce a sterol and a fatty acid anion. This class represents a group of enzymes exhibiting broad specificity. They act on esters of sterols and long-chain fatty acids, and may also bring about the esterification of sterols. These enzymes are typically activated by bile salts. **See also** *Esterases*

D. P. Hajjar (1994) *Adv. Enzymol.* **69**, 45.
M. A. Wells & N. A. DiRenzo (1983) *The Enzymes*, 3rd ed., **16**, 113.

CHOLINE O-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.6], also known as choline acetylase, catalyzes the reaction of acetyl-CoA with choline to produce coenzyme A and *O*-acetylcholine. The enzyme can also utilize propionyl-CoA as a substrate, albeit as a weaker reactant.

S. H. Zeisel (1981) *Ann. Rev. Nutr.* **1**, 95.
H. G. Mautner (1976) *Crit. Rev. Biochem.* **4**, 341.
D. S. Sigman & G. Mooser (1975) *Ann. Rev. Biochem.* **44**, 889.
L. T. Potter (1971) *Meth. Enzymol.* **17(B)**, 778.
A. K. Prince (1971) *Meth. Enzymol.* **17(B)**, 788.
L. T. Potter & V. A. S. Glover (1971) *Meth. Enzymol.* **17(B)**, 798.

CHOLINE DEHYDROGENASE

This enzyme [EC 1.1.99.1] is PQQ-dependent and catalyzes the reaction of choline with the acceptor to produce betaine aldehyde and the reduced acceptor.

S. H. Zeisel (1981) *Ann. Rev. Nutr.* **1**, 95.
Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

CHOLINE KINASE

This enzyme [EC 2.7.1.32] catalyzes the phosphoryl transfer from ATP with choline to produce ADP and *O*-phosphocholine. Ethanolamine and its methyl and ethyl derivatives can also serve as substrates.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.
S. H. Zeisel & J. K. Blusztajn (1994) *Ann. Rev. Nutr.* **14**, 269.
K. Ishidate & Y. Nakazawa (1992) *Meth. Enzymol.* **209**, 121.
J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

CHOLINE OXIDASE

This enzyme [EC 1.1.3.17] catalyzes the reaction of choline with dioxygen to produce betaine aldehyde and hydrogen peroxide. FAD is a cofactor. The enzyme has also been reported to be able to oxidize betaine aldehyde to betaine.

J. H. Quastel (1955) *Meth. Enzymol.* **1**, 674.

CHOLINE-PHOSPHATE CYTIDYLYLTRANSFERASE

This enzyme [EC 2.7.7.15], also referred to as phosphorylcholine transferase, catalyzes the reaction of CTP with choline phosphate to produce pyrophosphate (or, diphosphate) and CDP-choline.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.
S. H. Zeisel & J. K. Blusztajn (1994) *Ann. Rev. Nutr.* **14**, 269.
Y. Kishimoto (1983) *The Enzymes*, 3rd ed., **16**, 357.

CHOLINEPHOSPHOTRANSFERASE

This enzyme [EC 2.7.8.2] (also known as CDP-choline:1,2-diacylglycerol cholinephosphotransferase, diacylglycerol cholinephosphotransferase, phosphorylcholine:glyceride transferase, alkylacylglycerol cholinephosphotransferase, and 1-alkyl-2-acetylgllycerol cholinephosphotransferase) catalyzes the reaction of CDP-choline with 1,2-diacylglycerol to produce CMP and a phosphatidylcholine. 1-Alkyl-2-acylglycerol derivatives can also serve as substrates.

S. H. Zeisel & J. K. Blusztajn (1994) *Ann. Rev. Nutr.* **14**, 269.
R. B. Cornell (1992) *Meth. Enzymol.* **209**, 267.
R. H. Hjelmstad & R. M. Bell (1992) *Meth. Enzymol.* **209**, 272.

CHOLINE SULFATASE

This enzyme [EC 3.1.6.6] catalyzes the hydrolysis of choline sulfate to produce choline and sulfate.

A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 1.

CHOLINE SULFOTRANSFERASE

This enzyme [EC 2.8.2.6] catalyzes the reaction of 3'-phosphoadenylylsulfate with choline to produce adenosine 3',5'-bisphosphate and choline sulfate.

J. D. Gregory (1962) *Meth. Enzymol.* **5**, 977.

CHOLYL-CoA SYNTHETASE

This enzyme [EC 6.2.1.7], also known as cholate:CoA ligase, catalyzes the reaction of ATP with cholate and coenzyme A to produce AMP, pyrophosphate (or, diphosphate) and cholyl-CoA.

W. H. Elliott (1962) *Meth. Enzymol.* **5**, 473.

CHONDROITIN 4-SULFOTRANSFERASE

This enzyme [EC 2.8.2.5] catalyzes the reaction of chondroitin with 3'-phosphoadenylylsulfate to produce adenosine 3',5'-bisphosphate and chondroitin 4'-sulfate. Specifically, the sulfation is at the 4-position of *N*-acetyl-galactosamine residues of chondroitin. This enzyme is distinct from that of chondroitin 6-sulfotransferase [EC 2.8.2.17].

S. Suzuki (1966) *Meth. Enzymol.* **8**, 496.

CHONDROSULFATASES

Chondro-4-sulfatase [EC 3.1.6.9] catalyzes the hydrolysis of 4-deoxy- β -D-gluc-4-enuronosyl-(1,4)-*N*-acetyl-D-galactosamine 4-sulfate to yield 4-deoxy- β -D-gluc-4-enuronosyl-(1,4)-*N*-acetyl-D-galactosamine and sulfate. The saturated analog will also act as a substrate but not higher oligosaccharides. Nor will 6-sulfate derivatives be hydrolyzed by this enzyme. Chondro-6-sulfatase [EC 3.1.6.10] catalyzes the hydrolysis of 4-deoxy- β -D-gluc-4-enuronosyl-(1,4)-*N*-acetyl-D-galactosamine 6-sulfate to yield 4-deoxy- β -D-gluc-4-enuronosyl-(1,4)-*N*-acetyl-D-galactosamine and sulfate. The saturated analog will also act as a substrate but not higher oligosaccharides. Nor will 4-sulfate derivatives be hydrolyzed by this enzyme.

A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 1.

CHORISMATE MUTASE

This enzyme [EC 5.4.99.5] catalyzes the interconversion of chorismate and prephenate.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

B. E. Davidson (1987) *Meth. Enzymol.* **142**, 432-439.

CHORISMATE SYNTHASE

This enzyme [EC 4.6.1.4], also known as 5-enolpyruvyl-shikimate-3-phosphate phospholyase, catalyzes the conversion of 5-*O*-(1-carboxyvinyl)-3-phosphoshikimate to chorismate and orthophosphate. It should be noted that shikimate is numbered so that the double bond is between C1 and C2. However, some of the early reports on this enzyme numbered shikimate in the reverse direction.

B. A. Palfrey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

F. H. Gaertner (1987) *Meth. Enzymol.* **142**, 362.

CHROMIUM-51 (⁵¹Cr)

A γ -emitting (0.320 MeV) radioactive chromium nuclide having a decay half-life is 27.70 days. The following decay data indicate the time (days), followed by the fraction of original amount at the specified time: 0.0, 1.000; 1, 0.975; 2, 0.951; 3, 0.928; 4, 0.905; 5, 0.882; 6, 0.861; 7, 0.839; 8, 0.819; 9, 0.798; 10, 0.779; 20, 0.606; 30, 0.472; 40, 0.368; 50, 0.286; 60, 0.223; 70, 0.174; 80, 0.135; 90, 0.105; 100, 0.082; 110, 0.064; 120, 0.050.

Special precautions: Lung is the critical organ in terms of inhaled dose. The effective *biological* half-life in humans is around 2 years. Store behind lead shielding. Follow institutional regulations regarding (a) type and location of dosimeters; and (b) whole body counting if significant uptake is suspected.

CHROMIUM-NUCLEOTIDE COMPLEXES

Exchange-inert complexes of Cr(III) with nucleotide ligands are very stable toward hydrolysis. Such complexes have proven to be extremely useful as chirality probes in that different coordination isomers can be prepared and separated¹. These nucleotide complexes have also proved useful as dead-end inhibitors of enzyme-catalyzed reactions. Because Cr(III) is paramagnetic, distances can be measured by measuring the effects of Cr(III) on the NMR signals of nearby atoms when the Cr(III)-nucleotide complex binds to the surface of a macromolecule². **See** *Exchange-Inert Complexes*

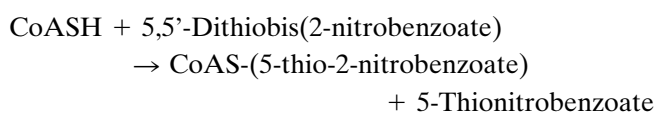
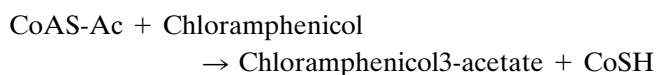
¹W. W. Cleland (1982) *Meth. Enzymol.* **87**, 159.

²J. J. Villafranca (1982) *Meth. Enzymol.* **87**, 180.

CHROMOGENIC SUBSTRATE

A synthetically derivatized substrate designed to undergo a change in absorption and/or fluorescence spectrum upon its enzymatic conversion to product. Chromogenic substrates provide valuable assays for enzymes that otherwise fail to produce a spectral change, especially phosphotransferases, amide bond synthases, isomerases, and hydrolases.

Probably the most familiar example is *p*-nitrophenylphosphate which upon treatment with alkaline phosphatase generates orthophosphate and *p*-nitrophenol. The latter is an intensely yellow substance with maximal absorbance at 410–412 nm. The continuous spectrophotometric rate assay of trypsin¹ uses the chromogenic substrate *N*- α -benzoyl-DL-arginine-*p*-nitroanilide and upon hydrolysis produces *p*-nitroaniline (λ_{max} 405 nm). β -*N*-Acetylglucosaminidase can be similarly assayed by using *p*-nitrophenyl-*N*-acetyl- β -D-glucosaminide^{2,3}. In other cases, the continuous spectral assay can be achieved by employing a chromogenic reagent that reacts with a reaction product. For instance, consider the chloramphenicol acetyltransferase reaction:



The reaction velocity is measured by the formation of thionitrobenzoate ion, a species that absorbs strongly at 412 nm.

The time course of protease reactions can be monitored continuously by using fluorogenic peptide substrates. One type contains a fluorescent substituent (termed a donor) and a second aromatic group (called an acceptor), itself incapable of fluorescing. Through Förster resonance energy transfer (or FRET), the photic energy from the donor's excited state is transferred to the acceptor, resulting in partial or complete quenching of fluorescence. Upon cleavage by the protease, the donor and acceptor are released from each other, and because the efficiency of FRET falls off as the inverse sixth power of the donor-to-acceptor distance, bond scission greatly

decreases the ability of the acceptor to quench the donor's fluorescence. In such instances, one observes an increase in the sample's fluorescence that can be used to evaluate the kinetics of the enzymatic reaction^{4,5}.

A severe limitation of chromogenic substrates is the presence of one or more reporter groups whose bulk and electrostatic properties can greatly alter the nature of the enzyme's interaction with its substrate. This would be especially significant when the chromogenic substituent contributes an atom near the reaction center (*i.e.*, the actual site of cleavage). In the example given above for alkaline phosphatase, the presence of a nitrophenyl group may even change important aspects of the bond making/breaking mechanism. A chromogenic substrate can also exhibit subsite interactions that are not at all representative of a protease's interactions with its natural substrate. To avoid such circumstances, the experimenter should consider resorting to the use of two substances: the chromogenic substrate and the second, a nonchromogenic substrate whose interactions with the enzyme are of primary interest. The latter can be treated as an alternative substrate inhibitor of the chromogenic substrate reaction. In this way, one can infer how the nonchromogenic substrate interacts with the substrate. This works best if the enzyme is noncooperative; otherwise, one runs the risk of site-site interactions that reflect site occupancy by both substrates. In the same way, an enzyme's interactions with nonsubstrates can be treated by using it as a competitive inhibitor relative to the chromogenic substrate. While these approaches require additional kinetic characterization, the experimenter is richly rewarded by a much better picture of the enzyme's interaction with the substrate, and not the chromogenic substituent. **See** *Absorption Spectroscopy; Fluorescence Spectroscopy*.

¹B. Kassell (1970) *Meth. Enzymol.* **19**, 844.

²O. A. Bessey, O. H. Lowry & M. J. Brock (1946) *J. Biol. Chem.* **164**, 321.

³S.-C. Li, S & Y.-T. Li (1970) *J. Biol. Chem.* **245**, 5153.

⁴R. M. Silverstein (1975) *Anal. Biochem.* **63**, 281.

⁵J. H. Hash (1975) *Meth. Enzymol.* **43**, 737.

CHROMOPHORE

That atom, group, or moiety within a molecular entity which is primarily responsible for the electronic transition resulting in the spectral band under consideration.

See *Absorption Spectroscopy*

CHYMASE

This enzyme [EC 3.4.21.39], also referred to as mast cell protease I and skeletal muscle (SK) protease, is an endopeptidase that has been isolated from mast cell granules. It belongs to the peptidase family S1 and catalyzes the hydrolysis of peptide bonds, preferring Phe-Xaa > Tyr-Xaa > Trp-Xaa > Leu-Xaa.

J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.

CHYMOPAPAIN

This enzyme [EC 3.4.22.6], also known as papaya proteinase II, is a member of the peptidase family C1. It is the major endopeptidase of papaya (*Carica papaya*) latex. It has a specificity similar to that of papain. In addition, there are a number of chromatographic forms of the enzyme.

D. K. Kunimitsu & K. T. Yasunobu (1970) *Meth. Enzymol.* **19**, 244.

D. M. Greenberg (1955) *Meth. Enzymol.* **2**, 54.

CHYMOSIN

This enzyme [EC 3.4.23.4], also known as rennin, is a member of the peptidase family A1, exhibiting a broad specificity similar to that of pepsin A. It is a neonatal gastric enzyme with high milk clotting (note that chymosin clots milk by catalyzing the hydrolysis of a single peptide bond in κ -casein [Phe¹⁰⁵-Met¹⁰⁶]) and weak general proteolytic activity. The protein is formed from the precursor, prochymosin. The enzyme has also been found among mammals with a postnatal uptake of immunoglobins.

D. R. Davies (1990) *Ann. Rev. Biophys. Biophys. Chem.* **19**, 189.

CHYMOTRYPSIN

This enzyme [EC 3.4.21.1], also known as α -chymotrypsin, is an endopeptidase belonging to the peptidase family S1. It catalyzes the hydrolysis of peptide bonds with the preference for Tyr-Xaa, Trp-Xaa, Phe-Xaa, and Leu-Xaa.

C. W. Wharton (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 345.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

E. T. Kaiser, D. S. Scott & S. E. Rokita (1985) *Ann. Rev. Biochem.* **54**, 565.

R. J. Coll, P. D. Compton & A. L. Fink (1982) *Meth. Enzymol.* **87**, 66.

J. S. Fruton (1982) *Adv. Enzymol.* **53**, 239.

D. M. Blow (1971) *The Enzymes*, 3rd ed., **3**, 185.

G. P. Hess (1971) *The Enzymes*, 3rd ed., **3**, 213.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Active site titration: [by 2-hydroxy-5-nitro- α -toluenesulfonic acid sultone, **19**, 6-14; by *p*-nitrophenyl ester substrates, **19**, 14-20; by rapidly reversible, covalently bound substrates, **19**, 14-20; by slowly reversible, covalently bound inhibitors, **19**, 6-14]; assay, **11**, 235; α -bromoacetyl aminoacyl derivatives, **11**, 679; intramolecular transformation of, **11**, 679; buried carboxyl groups in, **11**, 25, 618; derivative formed with 2-bromoacetamido-4-nitrophenol, difference spectra of, **11**, 869, 870; photolysis of, **11**, 679; diethylphosphonyl, reactivation with *N*-phenylnicotinohydroxamic acid, **11**, 701; dimerization, **26**, 197; **27**, 402, 438; infrared spectrum, **26**, 468; phenylmethane-sulfonyl, reactivation of, **11**, 709; reaction [with 2-bromoacetamido-4-nitrophenol, **11**, 868; with α -bromoacetanilide, **11**, 564; with α -bromoacetophenone, **11**, 564; with α -bromo-*N*-benzylacetamide, **11**, 564; with dansyl chloride, **11**, 858, 859; with diisopropyl fluorophosphate, **11**, 694-695; with phenylmethanesulfonyl fluoride, **11**, 708; with TPCK, **11**, 678]; specificity of, **11**, 222; stock solution preparation, **19**, 11; conversion to anhydrochymotrypsin, **11**, 709, 711; abortive complex, **63**, 205; activation energy, **63**, 243-245; **64**, 226; adsorption, onto glass, **61**, 77; bovine [action on β chain of oxidized insulin, **80**, 649; amino acid composition, **80**, 650; hydrolysis of low molecular weight synthetic substrates, **80**, 731; inhibition, **80**, 801; N-terminal sequence, **80**, 631]; substrate specificity, **80**, 603; burst, **64**, 199; chemical modification, **63**, 219; competitive labeling, **63**, 219; conformational change, **64**, 224, 225; cryoenzymology, **63**, 338; dielectric constant, effects, **63**, 212; hysteresis, **64**, 217, 220; infrared spectrum, **63**, 212; inhibition, **63**, 180; **80**, 771; inhibition by eglin, **80**, 812; inhibition by a 1-proteinase inhibitor, **80**, 763; interaction parameters, preferential, **61**, 48; kinetic parameters, **63**, 237, 238, 244; kinetics, calorimetric determination, **61**, 264; low-frequency dynamics, **61**, 443-445; low-temperature dynamics, **61**, 325, 326; microcalorimetric studies of ligand binding, **61**, 303; molecular form in normal serum, **74**, 273-275; negative cooperativity, **64**, 220; nitrogen isotope effect, **64**, 95; nuclear magnetic resonance properties, **63**, 212; partial specific volumes, **61**, 48; pK values, perturbed, **63**, 210, 220; progress curve analysis, **63**, 180; calorimetry of protonation, **61**, 265; radioimmunoassay, **74**, 273; reaction with a 2-antiplasmin, **80**, 404, 406; similarity to cathepsin G, **80**, 564; similarity to thrombin, **80**, 294, 295; spectrokinetic probe, **61**, 323; structure-function, **80**, 733; zymogen conformational change, **64**, 225; active-site titration, **248**, 87-89, 100-101; carboxy-terminal sequence analysis, **240**, 705, 708, 710, 712; active site residues, **244**, 23-24; members, **244**, 22-37; family [biological role, **244**, 23; catalytic serine codon, **244**, 30-31; catalytic unit, **244**, 23, 25; cleavage site specificity, **244**, 114; domains, **244**, 25, 29-30; evolution, **244**, 30-32; gene, **244**, 29-30; members, **244**, 22-23, 26-27; processing, **244**, 23; signal peptide, **244**, 23, 25; species distribution, **244**, 26-28, 31]; fluorimetric assay, **248**, 20-21; phosphonate inhibitors, **244**, 434; preparation, **235**, 566; site-directed mutants and acid-base catalysis, **249**, 114; structure, **240**, 703, 705; substrates [fluorogenic, **248**, 19; thioester, **248**, 10; transition state and multisubstrate analogs, **249**, 306; zymography in nondissociating gels with copolymerized substrates, **235**, 588-589.

CHYMOTRYPSIN C

This enzyme [EC 3.4.21.2], a member of the peptidase family S1, is an endopeptidase that exhibits a preference for hydrolysis of peptide bonds at Leu-Xaa, Tyr-Xaa, Phe-Xaa, Met-Xaa, Trp-Xaa, Gln-Xaa, and Asn-Xaa. The enzyme reacts more readily with Tos-

Leu—CH₂Cl than Tos—Phe—CH₂Cl in contrast to chymotrypsin (note: chymotrypsin A and chymotrypsin B were later found to be homologous and now are listed with EC 3.4.21.1).

J. E. Folk (1970) *Meth. Enzymol.* **19**, 109.

CHYMOTRYPSIN-LIKE SERINE PROTEASE SUPERFAMILY

Perona and Craik¹ recently described the serine proteases possessing a chymotrypsin-like fold. These enzymes have an almost identical fold made up of two β -barrels, with the catalytic Ser195, His57, and Asp102 residues at the interface of the two domains. To bring about accurate positioning of the nucleophilic Ser195, each enzyme contains five enzyme-substrate hydrogen bonds that serve to juxtapose the scissile peptide bond adjacent to the catalytic Ser—His couple. Many specificity-controlling structural determinants reside on surface loops, suggesting that rapid and varied evolutionary divergence can take place, while still conserving the overall tertiary fold.

¹J. J. Perona & C. S. Craik (1997) *J. Biol. Chem.* **272**, 29987.

trans-CINNAMATE 4-MONOOXYGENASE

This enzyme [EC 1.14.13.11], also known as cinnamate 4-hydroxylase, catalyzes the reaction of *trans*-cinnamate with NADPH and dioxygen to generate 4-hydroxycinnamate, NADP⁺, and water. The enzyme, which uses a heme-thiolate as a cofactor, can also replace NADPH with NADH (however, the reaction will proceed slower).

I. Raskin (1992) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **43**, 439.

R. A. Dixon, P. M. Dey & C. J. Lamb (1983) *Adv. Enzymol.* **55**, 1.

CIRCADIAN RHYTHM

Any biological process characterized by or occurring repeatedly with a 24-hour periodicity. Beyond their complex spatial structures, living organisms display remarkable time structures that integrate complex data sets comprised of daily and seasonal cues. These include the day-night cycle, length of day, the spectrum and intensity of solar illumination, lunar cycle and consequential changes in its luminosity, tidal rhythms, seasonal changes in the flow and content of rivers, *etc.* These changes are sensed as changes in photic and thermal energy which alter an organism's biochemical, physiological and behavioral characteristics. In *Drosophila*, the gene known as period (*PER*) is required for the circadian rhythms, and oscillations in levels of its mRNA and protein in

brain may strongly influence rhythmicity. A homologous mouse gene (*CLOCK*) encodes a novel transcription factor that is likely to control periodicity by changes in DNA binding, protein dimerization, and transcriptional activation, and *CLOCK* protein shares a motif with *PERIOD*, suggesting that the circadian clock mechanism is evolutionarily conserved.

On the basis of studies with *Drosophila* and *Neurospora* circadian clocks, such oscillatory behavior appears to involve cycling gene expression. Nishida¹ summarized recent progress using molecular biological approaches to elucidate the circadian clock mechanism. He also described the importance of transcription factors in providing clues about the nature of what may be a common mechanism of circadian clock in the divergent species.

Controlled chaos may also factor into the generation of rhythmic behavior in living systems. A recently proposed model² describes the central circadian oscillator as a chaotic attractor. Limit cycle mechanisms have been previously offered to explain circadian clocks and related phenomena, but they are limited to a single stable periodic behavior. In contrast, a chaotic attractor can generate rich dynamic behavior. Attractive features of such a model include versatility of period selection as well as use of control elements of the type already well known for metabolic circuitry.

¹N. Ishida (1995) *Neurosci. Res.* **23**, 231.

²A. L. Lloyd & D. Lloyd (1993) *Biosystems* **29**, 77.

CIRCE EFFECT

Utilization of strong attractive forces to lure a substrate into an active site and to force the substrate to undergo “extraordinary transformation of form and structure.”¹ Distortion of the substrate occurs as a consequence of favorable electrostatic interactions within and immediately surrounding the enzyme's active site. The Latin name *Circe* harkens back to the Greek mythological sorceress *Kirke* who transformed Odysseus's fellow sailors into swine, but Odysseus later succeeded in forcing her to return them to their previously human forms.

¹W. P. Jencks (1975) *Adv. Enzymol.* **43**, 219.

CIRCULAR DICHROISM SPECTROSCOPY

A spectroscopic technique that measures the differential absorption of left and right circularly polarized light as

a measure of electronic interactions arising from asymmetric (or chiral) as well as dissymmetric molecular groups. This method also allows one to detect conformational changes in biopolymers.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

General principles, **76**, 262-265; of hemoglobins, **76**, 262-275; globin effects, **76**, 271; information derived, **76**, 267-275 [in far UV, **76**, 267-270; in near UV, **76**, 270-272; in visible region, **76**, 272-275]; instrumentation, **76**, 266-267; of single crystals, **76**, 206-207; absorbance measurement, **240**, 637; aromatic residues, **232**, 248; hemoglobin, **232**, 248-257; near-UV region, **232**, 248, 252-257; secondary structure analysis, **232**, 248; Soret region, **232**, 257-260; visible region, **232**, 248-249, 260-263; monitoring of protein unfolding, **240**, 615; protein concentration needed for measurement, **240**, 632, 635; structural analysis correlated with X-ray crystallography, **232**, 247-248; in far-UV region, **232**, 7-8; in near-UV region, **232**, 7; ribonuclease A unfolding, **240**, 623-628; source code for two-state protein folding model, **240**, 619, 644-645; standard deviation determination, **240**, 621; staphylococcal nuclease A unfolding analysis, **240**, 628, 631-632; stationary CD, **232**, 249-263; time-resolved CD [apparatus for millisecond time range, **232**, 263; for nanosecond and microsecond time scale, **232**, 263-264; experimental kinetic data, **232**, 264-266]; high-pressure experiments, **259**, 369-370; hybrid oligonucleotide strand characterization, **246**, 31-33; instrumentation [calibration, **246**, 23-24, 41; design, **246**, 40-42; manufacturers, **246**, 41; stopped-flow, **246**, 61; light characteristics, **246**, 35-36]; low-density lipoprotein oxidation analysis, **233**, 428-430; multiple scanning, **246**, 43; nucleic acids [chromophores, **246**, 62; duplex stoichiometry, **246**, 4, 6, 13-15, 35; ionic strength effects, **246**, 63; ligand binding, **246**, 67-68; protein interactions, **246**, 6, 68-71; secondary structure, **246**, 4, 6, 13-15, 35, 62-67; temperature effects, **246**, 63-64]; photoacoustic detection, **246**, 60; proteins [baseline, **259**, 500; chromophores, **246**, 43-44; concentration for measurement, **240**, 632, 635; folding, **246**, 61; ligand interactions, **246**, 55-58; secondary structure analysis, **246**, 5, 13-15, 35; **259**, 499-500 (accuracy of determination, **246**, 513; a helix, **246**, 44-45; analysis methods, **246**, 49-52; β sheet, **246**, 45-46; β turns, **246**, 46-47; helix-coil transition, **246**, 47-49; membrane proteins, **246**, 58-60; random coil, **246**, 47; sources of error, **246**, 54; wavelengths for data collection, **246**, 53-54); structural analysis (correlation with X-ray crystallography, **232**, 247-248; in far-UV region, **232**, 7-8; in near-UV region, **232**, 7; tertiary structure analysis, **246**, 54-55; unfolding studies, **240**, 615; **259**, 498-500, 506-509; vibrational studies, **246**, 515-516; wavelength monitoring, **240**, 635]; reaction kinetics, **246**, 14; RNA [codon-anticodon interactions, **261**, 285, 287; duplex formation monitoring, **261**, 284; duplex stoichiometry, **246**, 24-28; ion concentration effects, **261**, 283-284; temperature effects, **261**, 283-284; wavelength sensitivity (backbone alterations, **261**, 283; base stacking, **261**, 283); sample cells, **246**, 42; sample requirements [buffers, **246**, 42; chirality, **246**, 14, 34; concentration, **246**, 42]; semiquinone intermediate in amine oxidase, **258**, 87; sensitivity, **258**, 274; theory, **258**, 274 [electric dipole transition moment, **246**, 39-40; magnetic dipole transition moment, **246**, 39-40; rotational strength, **246**, 39-40, 93-94]; transition metals [coordination geometry, **246**, 5, 16; spin state, **246**, 5]; units of measurement [ellipticity, **246**, 24, 36-37; molar circular dichroism, **246**, 24, 36]; vibrational circular dichroism; [instrumentation, **246**, 37; protein conformation determination, **246**, 52-53].

CITRAMALATE LYASE

The enzyme [EC 4.1.3.22] catalyzes the conversion of (3*S*)-citramalate to acetate and pyruvate. It should be

noted that the enzyme complex can be dissociated into subunits. Two of those components are citramalate CoA-transferase [EC 2.8.3.11] and citramalyl-CoA lyase [EC 4.1.3.25].

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

CITRAMALYL-CoA LYASE

This enzyme [EC 4.1.3.25] catalyzes the conversion of (3*S*)-citramalyl-CoA to acetyl-CoA and pyruvate. The (3*S*)-citramalyl thioacyl-carrier protein can also be utilized as a substrate. This enzyme has been reported to be a component of citramalate lyase [EC 4.1.3.22].

H. A. Lardy (1969) *Meth. Enzymol.* **13**, 314.

CITRATE LYASE

This enzyme [EC 4.1.3.6] (also known as citrate (*pro*-3*S*)-lyase, citrase, citratase, citritase, citridesmolate, and citrate aldolase) catalyzes the conversion of citrate to acetate and oxaloacetate. Citrate lyase can be dissociated into subunits, two components of which are identical with citrate CoA-transferase [EC 2.8.3.10] and citryl-CoA lyase [EC 4.1.3.34].

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

P. A. Srere (1975) *Adv. Enzymol.* **43**, 57.

L. B. Spector (1972) *The Enzymes*, 3rd ed., **7**, 357.

CITRATE SYNTHASE

Citrate (*si*)-synthase [EC 4.1.3.7] (also known as citrate condensing enzyme, citrogenase, and oxaloacetate transacetase) catalyzes the reaction of acetyl-CoA with oxaloacetate and water to produce citrate and coenzyme A (in which acetyl-CoA $\leftrightarrow$ (*pro*-3*S*)-CH₂COO⁻). Citrate (*re*)-synthase [EC 4.1.3.28] catalyzes the same reaction, albeit with the opposite stereochemistry (thus, acetyl-CoA $\leftrightarrow$ (*pro*-3*R*)-CH₂COO⁻).

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

S. J. Remington (1992) *Curr. Top. Cell. Reg.* **33**, 209.

R. F. Colman (1990) *The Enzymes*, 3rd ed., **19**, 283.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

G. Müller-Kraft & W. Babel (1990) *Meth. Enzymol.* **188**, 350.

P. A. Srere (1975) *Adv. Enzymol.* **43**, 57.

L. B. Spector (1972) *The Enzymes*, 3rd ed., **7**, 357.

CLAISEN CONDENSATION

A base-catalyzed reaction of esters (or thioesters) containing an α -hydrogen with another ester (in some cases

another molecule of the same ester) to produce a β -keto ester and an alcohol or thiol. A number of enzymes catalyze Claisen condensations, including thiolase, malate synthase, citrate synthase, and hydroxymethylglutaryl-CoA synthase.

CLATHRATE

A compound having sites or inclusion spaces for completely enclosing so-called guests, molecular entities that fit into the above-mentioned sites or spaces. A synonym for clathrate is cage compound. A clathrate compound can be formed with hydroquinone as the host and methanol, CO₂, argon, or SO₂ as the guest¹. Water can also serve as a host in clathrate compounds². **See** *Cage Compound*; *Guest*; *Host Molecule*

¹D. D. MacNicol, J. J. McKendrick & D. R. Wilson (1978) *Chem. Soc. Rev.* **7**, 65.

²G. H. Cady (1983) *J. Chem. Ed.* **60**, 915.

CLELAND NOMENCLATURE

A system for describing kinetic mechanisms for enzyme-catalyzed reactions^{1,2}. Reactants (*i.e.*, substrates) are symbolized by the letters A, B, C, D, *etc.*, whereas products are designated by P, Q, R, S, *etc.* Reaction schemes are also identified by the number of substrates and products utilized (*i.e.*, Uni (for one), Bi (two), Ter (three; occasionally Tri), Quad (four), Quin (five), *etc.*). Thus, a two-substrate, three-product enzyme-catalyzed reaction would be a Bi Ter system. In addition, reaction schemes are identified by the pattern of substrate addition to the enzyme's active site as well as the release of products. For a two-substrate, one-product scheme in which either substrate can bind to the free enzyme, the enzyme scheme is designated a random Bi Uni mechanism. If the substrates bind in a distinct order (note that, in such cases, A binds before B; for ordered multiproduct release, P is released prior to Q, *etc.*), the scheme would be ordered Bi Uni. If the binding scheme is different than the release of product, then that information should also be provided; for example, a two-substrate, two-product reaction in which the substrates bind to the enzyme in an ordered fashion whereas the products are released randomly would be designated ordered on, random off Bi Bi scheme. If one or more Theorell-Chance steps are present, that information is also given (*e.g.*, ordered Bi Bi-(Theorell-Chance)), with the prefixes included if there is more than one Theorell-Chance step.

The schemes provided as examples above are often referred to as being "sequential" in that all substrates must bind before any product is released. However, in other well-characterized examples, a product is released prior to the binding of one or more substrates. For example, in a two-substrate, two-product scheme, substrate A could bind to the free enzyme (E) to form an EA binary complex. A chemical "event" then occurs within the complex resulting in the formation of the first product (P) and a modified form of the enzyme (note: the initial free enzyme is designated by the letter E whereas the modified enzyme (or other isomerizational forms of the enzyme) are designated by the letters F, G, H, *etc.*). After P is released, the second substrate (B) binds to F to form the FB. Another chemical event occurs, the second product (Q) is formed and the original form of the enzyme (E) is regenerated. Such a mechanism is described as a ping pong Bi Bi scheme in Cleland's nomenclature. Ping pong schemes can also occur in systems having more than two substrates and/or products. For example, consider the three-substrate, three-product reaction scheme in which the first substrate binds to the free enzyme resulting in the formation of the first product followed by the ordered binding and release of the remaining substrates and products. Such a scheme is referred to as an ordered Uni Uni Bi Bi ping pong mechanism.

If a stable enzyme form isomerizes and these isomerization steps are a part of the reaction pathway, the term Iso is provided. For a Uni Uni mechanism in which an isomerization step occurs [*e.g.*, the reaction sequence: $E + A \rightleftharpoons EA$; $EA \rightleftharpoons FP$; $FP \rightleftharpoons F + P$; and $F \rightleftharpoons E$] is designated an Iso Uni Uni scheme. If more than one isomerization event occurs, then the prefixes di-, tri-, *etc.*, are utilized with Iso. Note that this aspect of the nomenclature does not refer to isomerization steps which may or may not occur between central complexes.

Complexes of enzyme, substrates, products, inhibitors, *etc.*, are often designated as being binary, ternary, quaternary, *etc.*, depending on the number of entities present in the complex. For example, EAB would be a ternary complex. Central complexes are those transient complexes that generate products (or substrates in the reverse reaction) or which isomerize to those forms which can generate products. Thus, in an ordered Bi Bi reaction scheme, the enzyme can exist in five forms: E, EA, EAB,

EPQ, and EQ. The central complexes are EAB and EPQ. The binary complexes EA and EQ are often referred to as Michaelis complexes in that they are generated by simple binding events; but no “chemistry” occurs until one or more other reactants bind to the active site. Note that central complexes can only participate in unimolecular events whereas Michaelis complexes can participate in both unimolecular and bimolecular events.

In Cleland nomenclature, the initial velocity and individual rate constants are designated by lower case italicized letters (*e.g.*, v , k , k_2 , *etc.*). Dissociation, Michaelis, and equilibrium constants utilize an upper case italicized K with the appropriate unitalicized lower case subscript. For example, the equilibrium constant would be symbolized by K_{eq} whereas the Michaelis constant for substrate B would be designated by K_b . Dissociation constants for a Michaelis complex contain a subscript i and a letter for the dissociating ligand (*e.g.*, for the EA binary complex, the dissociation constant would be K_{ia}). Maximum velocities are designated by a capital italicized V , usually with a subscript 1 or 2 depending on whether the forward or reverse reaction is referred to. (If the numerical subscript is not provided, the forward reaction is assumed. In most cases, the unitalicized subscript “max” is also provided.)

Occasionally rate expressions are described as 1/1, 2/1, *etc.*, functions, referring to the maximum power of the substrate concentration in the numerator (N) and denominator (D). For example, consider the case of the steady-state random Bi Uni mechanism. The reciprocal form of the rate expression (at constant [B]) has the general form of $1/v = (\phi_o + \phi_{DA}[A] + \phi_{DA2}[A]^2) / (\phi_{NA}[A] + \phi_{NA2}[A]^2)$ where the ϕ values are collections of rate constants. If both the numerator and denominator of this reciprocal form of the rate expression are divided by the substrate concentration raised to the highest power in which it appears (in this case, $[A]^2$), then the numerator has a term in $1/[A]^2$ (as well as $1/[A]$ and $1/[A]^0$) whereas the denominator has terms in $1/[A]$ and $1/[A]^0$. Thus, this rate expression is a 2/1 function. **See** *Multisubstrate Mechanisms*

¹W. W. Cleland (1963) *Biochim. Biophys. Acta* **67**, 104.

²W. W. Cleland (1963) *Biochim. Biophys. Acta* **67**, 173.

CLOSED SYSTEM

Any system (chemical, biological, mechanical, *etc.*) that does not permit the transfer of mass with its surrounding environment. Nonetheless, in closed systems, heat or energy can pass between the system and the surroundings. **See** *Isolated System*; *Open System*

CLOSTRIPAIN

This enzyme [EC 3.4.22.8], also referred to as clostridiopeptidase B, is an endopeptidase belonging to the peptidase family C11 that has been isolated from the bacterium *Clostridium histolyticum*. It catalyzes the hydrolysis of peptide bonds with a preference for the Arg—Xaa bond (including the Arg—Pro bond), but not the Lys—Xaa peptide bond. This calcium ion-activated enzyme is inactivated by EDTA.

E. Shaw (1990) *Adv. Enzymol.* **63**, 271.

K. Morihara (1974) *Adv. Enzymol.* **41**, 179.

W. M. Mitchell & W. F. Harrington (1971) *The Enzymes*, 3rd ed., **3**, 699.

C.M.C.

Abbreviation for critical micelle concentration. **See** *Micelle Formation*

CMP-N-ACETYLNEURAMINATE- β -GALACTOSIDE SIALYLTRANSFERASES

CMP-*N*-acetylneuraminate- β -galactoside α -2,6-sialyltransferase [EC 2.4.99.1], also referred to as β -galactosamide α -2,6-sialyltransferase and β -galactoside α -2,6-sialyltransferase, catalyzes the reaction of CMP-*N*-acetylneuraminate with β -D-galactosyl-1,4-acetyl- β -D-glucosamine to produce CMP and α -*N*-acetylneuraminyl-2,6- β -D-galactosyl-1,4-*N*-acetyl- β -D-glucosamine. Note that the terminal β -D-galactosyl residue of the oligosaccharide of glycoproteins, as well as lactose, can act as a substrate as well.

CMP-*N*-acetylneuraminate- β -galactoside α -2,3-sialyltransferase [EC 2.4.99.4], also referred to as β -galactoside α -2,3-sialyltransferase, catalyzes the reaction of CMP-*N*-acetylneuraminate with β -D-galactosyl-1,3-*N*-acetyl- α -D-galactosaminyl-R to produce CMP and α -*N*-acetylneuraminyl-2,3- β -D-galactosyl-1,3-*N*-acetyl- α -D-galactosaminyl-R. In the acceptor substrate, R can be H, a threonyl or seryl side chain in a glycoprotein, or a glycolipid. Lactose can also be utilized as a substrate. In some systems, this enzyme may be identical with CMP-

N-acetylneuraminate:monosialoganglioside sialyltransferase [EC 2.4.99.2].

D. M. Carlson & G. W. Jourdan (1966) *Meth. Enzymol.* **8**, 358.

COACTIVATOR

A molecule that functions in conjunction with a protein apo-activator. The classical example is cAMP, the coactivator of the CAP protein in bacteria.

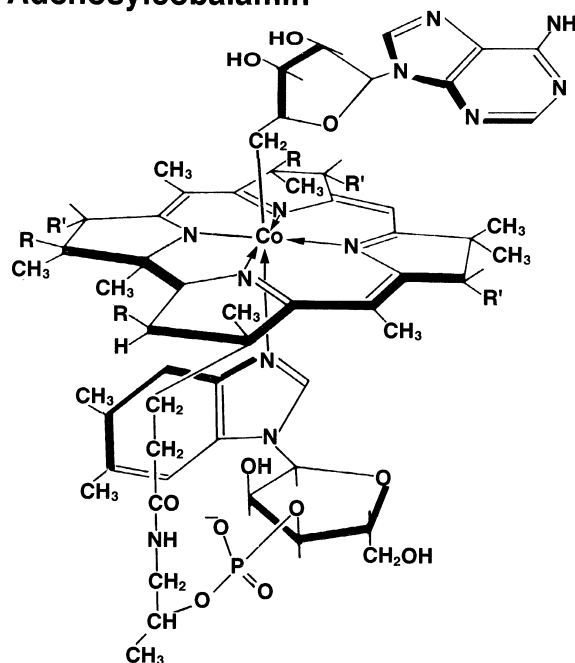
COBALAMIN ADENOSYLTRANSFERASE

This enzyme [EC 2.5.1.17], also referred to as cob(I)-alamin adenosyltransferase or aquacob(I)alamin adenosyltransferase, catalyzes the reaction of cob(I)alamin with ATP and water to produce adenosylcobalamin, orthophosphate, and pyrophosphate (or, diphosphate). A cofactor for this enzyme is manganese ion.

S. H. Mudd (1973) *The Enzymes*, 3rd ed., **8**, 121.

COBALAMINS AND COBAMIDES (B₁₂)

Adenosylcobalamin



Members of the vitamin B₁₂ or corrinoid group, containing a highly substituted tetrapyrrole ring which donates four nitrogen atoms to form coordinate covalent linkages with a centrally positioned hexacoordinate cobalt ion.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Spectroscopic analysis of vitamin B₁₂ derivatives, **67**, 5; preparation of cryptofluorescent analogs of cobalamin coenzymes, **67**, 12; polarography of compounds of the vitamin B₁₂ series in dimethylformamide, **67**, 20; comparison of serum vitamin B₁₂ (co-

balamin) determination by two isotope-dilution methods and by *Euglena* assay, **67**, 24; measurement of tissue vitamin B₁₂ by radioisotopic competitive inhibition assay and quantitation of tissue cobalamin fractions, **67**, 31; ribosome-associated vitamin B₁₂ adenosylating enzyme of *Lactobacillus leichmannii*, **67**, 41; immobilized derivatives of vitamin B₁₂ coenzyme and its analogs, **67**, 57; solubilization of the receptor for intrinsic factor-B₁₂ complex from guinea pig intestinal mucosa, **67**, 67; properties of proteins that bind vitamin B₁₂ in subcellular fractions of rat liver, **67**, 72; isolation and characterization of vitamin B₁₂-binding proteins from human fluids, **67**, 80; biosynthesis of transcobalamin II, **67**, 89; competitive binding radioassays for vitamin B₁₂ in biological fluids or solid tissues, **67**, 99; determination of cobalamins in biological material, **123**, 3; analysis of cobalamin coenzymes and other corrinoids by high-performance liquid chromatography, **123**, 14; isolation of native and proteolytically derived ileal receptor for intrinsic factor-cobalamin, **123**, 23; purification of B₁₂-binding proteins using a photodissociative affinity matrix, **123**, 28; solid-phase immunoassay for human transcobalamin II and detection of the secretory protein in cultured human cells, **123**, 36.

COBALT NUCLEOTIDES

Exchange-inert complexes of Co(III) with nucleotides that have proven to be extremely useful as chirality probes because the different coordination isomers are stable and can be prepared and separated¹. In addition, these nucleotides can be used as dead-end inhibitors of enzyme-catalyzed reactions and, since Co(III) is diamagnetic, a number of spectroscopic protocols can be utilized². **See** *Exchange-Inert Complexes*; *Chromium-Nucleotide Complexes*; *Metal Ion-Nucleotide Interactions*

¹W. W. Cleland (1982) *Meth. Enzymol.* **87**, 159.

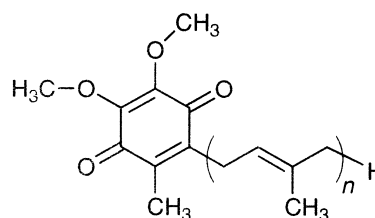
²J. J. Villafranca (1982) *Meth. Enzymol.* **87**, 180.

COCOONASES

Insect cocoonases are now classified under trypsin's EC number, [EC 3.4.21.4].

J. H. Law, P. E. Dunn & K. J. Kramer (1977) *Adv. Enzymol.* **45**, 389.

COENZYME Q (Ubiquinone)



Coenzyme Q (Ubiquinone)

A polyprenylated benzoquinone derivative that is a key component in mitochondrial electron transport, and the

reduced form is a hydroquinone. Coenzyme Q is not essential in humans.

CO-ION

An ion of like electric charge. Antonym: Counterion.

COLLAGENASES

(1) Microbial collagenase [EC 3.4.24.3] (also known as *Clostridium histolyticum* collagenase, clostridiopeptidase A, collagenase A, and collagenase I) is a member of the peptidase family M9. This enzyme catalyzes the hydrolysis of peptide bonds in native collagen in the triple helical region at Xaa—Gly bonds. (2) Widely distributed in vertebrates, the zinc-dependent interstitial collagenase [EC 3.4.24.7], also referred to as vertebrate collagenase and matrix metalloproteinase 1, is a member of the peptidase family M10B. It catalyzes the hydrolysis of a particular peptide bond in the native collagen triple helix. (3) Neutrophil collagenase [EC 3.4.24.34], also called matrix metalloproteinase 8, is a zinc-/calcium-dependent endopeptidase and is a member of the peptidase family M10B. (4) Hypodermin C [EC 3.4.21.49], also known as *Hypoderma* collagenase, is an endopeptidase that catalyzes the hydrolysis of peptide bonds in a number of proteins including native collagen. **See also** *Gelatinase*

M. Dioszegi, P. Cannon & H. E. Van Wart (1995) *Meth. Enzymol.* **248**, 413.

H. Tschesche (1995) *Meth. Enzymol.* **248**, 431.

J. D. Grubb (1994) *Meth. Enzymol.* **235**, 594.

B. L. Vallee & A. Galdes (1984) *Adv. Enzymol.* **56**, 283.

R. Kuttan & A. N. Radhakrishnan (1973) *Adv. Enzymol.* **37**, 273.

S. Seifter & E. Harper (1971) *The Enzymes*, 3rd ed., **3**, 649.

COLLISION CROSS-SECTION

A measure of the cross-sectional area swept through by a particle or chemical entity; this cross section influences collisions with another molecular or ionic entity. **See** *Collision Theory*

J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, Wiley, New York.

COLLISION THEORY

A gas-phase reactivity model that assumes molecules react as a result of the collision of reactant molecules. The basic idea is that the kinetic energy of the impacting molecules exceeds the activation energy required for reaction. Classical mechanics is used to estimate the fraction of the collisions with enough energy to allow re-

actant molecules to convert to product. A deficiency in the model is that steric or orientation factors are ignored, despite the fact that quantum mechanical considerations may require them. Furthermore, energy contributions from rotational and vibrational degrees of freedom may factor into the activation energy term in collision theory.

COLLOID SCIENCE

The systematic investigation of the chemical and physical properties of large molecules and particles having at least one dimension in the 1 nm to 1 mm range. Colloidal suspensions exhibit special properties not typically observed with smaller molecules and/or ions. In many cases, colloids are incapable of achieving a true state of dissolution, existing instead as dispersions. Their constituent macromolecules also develop an electric double layer at their junction with an aqueous electrolyte solution, and the distribution of ionic charge near their surface is not homogeneous. Colloids are often aggregates formed from both regular and irregular bonding patterns among their constituent atoms, molecules, or particles. The term “sol” describes an apparently uniform, or at least consistent, dispersion within a liquid or gaseous medium. The term “gel” refers to a disperse colloidal suspension of particles or macromolecules linked by chemical bonds or physisorption. In the sol state, colloids are characterized by substantial right angle light scattering (a phenomenon termed the Tyndall effect in honor of John Tyndall, although this property was probably first noted years earlier by Michael Faraday in his work on colloidal gold).

The nature of the forces maintaining a colloidal suspension can be largely regarded as electrostatic. In some instances, the component substance of a colloid can be brought to a state of dispersion that possesses such great stability that, however one may try, the original components can no longer be made to re-emerge from the dispersion. Such dispersions are said to be irreversible. Other colloidal suspensions (termed reversible) are readily returned to their undispersed condition.

Emulsions are colloidal suspensions produced by combining two liquids, and their dispersed state is usually stabilized by yet another component, termed an emulsifying agent. Foams are gaseous colloidal suspensions formed by combining two substances (of which one is typically a gas), and again the stability of a foam is aided

by a third and minor component (in this case, a foaming agent).

Doubtlessly, Nature's most astonishing colloidal suspension is protoplasm, a dispersion of biological macromolecules (chiefly proteins) as well as numerous hydrophilic, lipophilic, and amphipathic low-molecular-weight metabolites. The cytoskeleton, various vesicles and even glycogen particles form the underlying structure of protoplasm, which is constantly being remodeled locally through the involvement of ATP-dependent molecular motors as well as numerous regulatory proteins. The viscosity and consistency of protoplasm changes in gel-sol transitions that are associated with cellular motility.

COLUMBAMINE O-METHYLTRANSFERASE

This enzyme [EC 2.1.1.118] catalyzes the reaction of S-adenosylmethionine with columbamine to generate S-adenosylhomocysteine and palmatine. The product of this reaction is a protoberberine alkaloid that occurs in a wide variety of plants. The enzyme is distinct from that of 8-hydroxyquercetin 8-O-methyltransferase.

T. Hashimoto & Y. Yamada (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 257.

COMBINATORIAL LIBRARIES & MOLECULAR RECOGNITION

A method for simultaneous synthesis of a large number of peptides and/or ligands on solid supports for the discovery and/or development of high-affinity ligands that can be used as enzyme inhibitors or drugs. Initial work¹ centered on the identification of protease inhibitors using peptide libraries, but the sophistication of solution- and solid-phase chemistry has progressed to consider non-peptidic ligands that may exhibit even higher affinity complexation with target proteins. Lam² describes five general approaches using combinatorial peptide library: (a) biological libraries; (b) spatially addressable parallel solid-phase or solution-phase libraries; (c) synthetic library methods requiring deconvolution; (d) the 'one-bead/one-compound' library method; and (e) synthetic library methods using affinity chromatography selection. In some cases, synthetic combinatorial libraries allow the investigator to probe enzyme/receptor interactions with an extensive library of so-called "pharmacophore" motifs rather than a library of compounds³. Brenner *et al.*⁴ have described how the diversity of chemical synthe-

sis and the power of genetics can be linked to provide a powerful, versatile method for drug screening. They described a process of alternating parallel combinatorial synthesis for encoding individual members of a large combinatorial library, such that each is marked with unique nucleotide sequences. After the chemical entity is bound to a target enzyme/receptor, the genetic tag is then amplified by replication and used to enrich the bound ligand by serial hybridization to a subset of the library. The nature of the chemical structure bound to the receptor is readily decoded by sequencing the nucleotide tag.

The method is illustrated by the work of Bastos *et al.*⁵ who synthesized two sets of noncleavable peptide-inhibitor libraries to map the subsites for substrate/inhibitor interactions with human heart chymase. A chymotrypsin-like enzyme, chymase converts angiotensin I to angiotensin II. As is customary for describing protease subsites, P1, P2, *etc.*, are numbered toward the N-terminal direction from the scissile bond, whereas P'1, P'2, *etc.*, are numbered in sequence toward the C-terminal direction from the scissile bond. Their first library consisted of peptides with 3-fluorobenzylpyruvamides in the P1 position. The P'1 and P'2 positions were varied to contain each of the 20 natural amino acids and P'3 was always an arginine. The second library consisted of peptides with Phe ketoamides at P1, Gly in P'1, and benzyl-oxycarbonyl (Z)-Ile in P4. The P2 and P3 positions again were varied to contain each of the naturally occurring amino acids, except for Cys and Met. The peptides of both libraries are attached to a solid support (pins). The peptide interactions with enzyme were evaluated by immersing the "pins" in an enzyme solution and then determining the amount of absorbed enzyme. The libraries selected the best and worst inhibitors within each group of peptides and provided an approximate ranking of other peptides according to K_i . The authors found that Z-Ile-Glu-Pro-Phe-CO₂Me and (F)-Phe-CO-Glu-Asp-ArgOMe appeared to be the best inhibitors of chymase in this collection of peptide inhibitors. They then synthesized the peptides and found K_i values were 1 nM and 1 mM, respectively. The respective K_i values for chymotrypsin were 10 nM and 100 mM.

Feng *et al.*⁶ used combinatorial synthesis to discover several ligands containing nonpeptide binding elements to

the Src SH3 domain. Their encoded library used compounds of the form Cap—M₁—M₂—M₃—PLPPLP, in which the Cap and M_i's are composed of a diverse set of organic monomers. The PLPPLP portion provided a structural bias directing the nonpeptide fragment Cap—M₁—M₂—M₃ to the SH3 specificity pocket. Fifteen ligands were selected from > 1.1 million distinct compounds. The solution structures of the Src SH3 domain complexed with two ligands containing nonpeptide elements selected from the library were determined by multidimensional NMR spectroscopy. The nonpeptide moieties of the ligands interact with the specificity pocket of Src SH3 domain differently from peptides complexed with SH3 domains. Structural information about the ligands was used to design various homologues, and their affinities for the SH3 domain were measured. These investigators demonstrated the efficacy of cycles of protein structure-based combinatorial chemistry, followed by structure determination of the few highest-affinity ligands in defining the basis of molecular recognition by Src SH3 domain.

¹K. S. Lam, S. E. Salmon, E. M. Hersh, V. J. Hruby, W. M. Kazmier-ski & R. J. Knapp (1991) *Nature* **354**, 82 (publ. errata: *Nature* **358**, 434 (1992)).

²K. S. Lam (1997) *Anticancer Drug Des.* **12**, 145.

³N. F. Sepetov, V. Krchnak, M. Stankova, S. Wad, K. S. Lam & M. Lebl (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 5426.

⁴S. Brenner & R. A. Lerner (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 5381.

⁵M. Bastos, N. J. Maeji & R. H. Abeles (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 6738.

⁶S. Feng, T. M. Kapoor, F. Shirai, A. P. Combs & S. L. Schreiber (1996) *Chem. Biol.* **3**, 661.

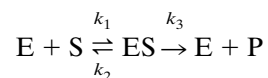
COMMITMENT FACTORS IN SOLVENT ISOTOPE EFFECTS

An approach to studying transition states in enzyme-catalyzed reactions using solvent isotope effects. In this treatment, very useful in isotope effect experiments, the relative rates of contributing steps in a multistep reaction are grouped into a fraction referred to as the commitment factor¹.

¹K. B. Schowen & R. L. Schowen (1982) *Meth. Enzymol.* **87**, 551.

COMMITMENT-TO-CATALYSIS

A unitless ratio of rate constants that serves as an index or measure of an enzyme's ability to promote catalysis as opposed to releasing its substrate before catalysis can occur. Consider the following scheme:



In this scheme, $V_{\max} = k_3[E_{\text{tot}}]$, $K_m = (k_2 + k_3)/k_1$, and the commitment to catalysis equals $k_3/(k_2 + k_3)$. (a) If k_3 is much less than k_2 , then K_m becomes the equilibrium or dissociation constant $K_{\text{ia}} (= k_2/k_1)$, indicating that E and S react rapidly and reversibly to form the ES complex. In this case, the commitment-to-catalysis is very low because ES has a much greater tendency to dissociate rather than to convert S to form P. (b) If k_3 equals k_2 , then $K_m = (k_2 + k_3)/k_1 = 2k_2/k_1 = 2K_{\text{ia}}$, and the commitment-to-catalysis equals 0.5. This means that the ES complex has an identical probability to dissociate to E + S or to convert to E plus P. (c) If k_3 is much greater than k_2 , the commitment-to-catalysis is high and approaches unity. Whenever ES is formed, product formation would then be highly favored. **See also** *Rapid Equilibrium Assumption; Kinetic Isotope Effect; Isotope Trapping*

COMPARTMENTAL ANALYSIS

A kinetic approach for determining the pools accessible to a tracee (*i.e.*, a metabolite, drug, or other substance) within a living organism. The basic strategy entails the following: (a) administration of a tracer, in the same chemical form as the tracee or as a metabolic precursor; (b) determination of changes in the specific activity (*i.e.*, [mol tracer]/[mol tracer + mol tracee]) in biological samples using radioactive or stable isotopes; (c) development of compartmental model(s) consistent with the time evolution of observed changes in specific activity, including rate constants and pool sizes for each tracee compartment; and (d) systematic experimentation to adduce sufficient information for discriminating rival models, along with due consideration of any statistical uncertainty in the data.

COMPARTMENTALIZATION

The division of a cell into different regions or domains containing different components and physiological functions. Compartmentalization is a means for more effective regulation arising from the partitioning of a cell into regions with different enzymatic and regulatory components. For example, fatty acid catabolism occurs within the mitochondria whereas fatty acid biosynthesis occurs in the cytoplasm. Hence, biomembranes often serve an

important role in the regulation of metabolic pathways by forming compartments as well as serving as a region of communication between different compartments. However, other types of compartmentalization can also occur. For example, proteins embedded in a biomembrane or proteins that are a part of a larger cytoplasmic aggregation of other proteins, can often be considered to be compartmentalized since communication to and from those proteins may be limited by nondiffusion processes.

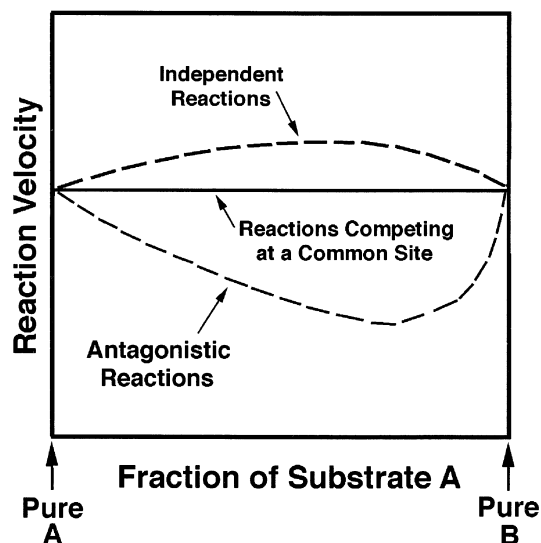
COMPENSATION EFFECT

Some reactions are characterized by straight-line plots of $T\Delta S^\ddagger$ versus $\Delta H^\ddagger$ having a slope of approximately one, where this linearity results from compensatory, or off-setting, changes of $\Delta H^\ddagger$ and $T\Delta S^\ddagger$. For this reason, the change in the Gibbs energy of activation, $\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$, is a better description of the variation in the reaction than either $\Delta H^\ddagger$ or $T\Delta S^\ddagger$ alone^{1,2}. **See also** *Isokinetic Relationship*

¹K. A. Connor (1991) *Chemical Kinetics*, VCH, New York.

²IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

COMPETITION PLOT



A plotting procedure¹ that serves to test whether two reactions occur at the same active site. One makes a plot of total rate versus p , where p varies from 0 to 1 and specifies the concentrations $(1 - p)A_0$ and pB_0 of two substrates in terms of reference concentrations A_0 and B_0 . If the two substrates react at the same active site,

the competition plot gives a horizontal straight line, *i.e.*, the total rate is independent of p . Independent reactions at two separate sites give a curve with a maximum. Separate reactions with cross-inhibition generate curves with either maxima or minima according to whether the Michaelis constants of the two substrates are smaller or larger than their inhibition constants in the other reactions. Ambiguous results can sometimes arise, but the authors present experimental strategies around such difficulties.

¹C. Chevillard, M. L. Cardenas & A. Cornish-Bowden (1993) *Biochem. J.* **289**, 599.

COMPETITIVE BINDING ASSAY

An assay protocol in which a macromolecule or binder (*i.e.*, the entity which is capable of binding a particular substance or ligand) competes for labeled and unlabeled ligand. Once binding has occurred, bound and free ligand are separated and quantitated. The bound and unbound ratios can be related to known standards. Examples of this procedure, also occasionally referred to as displacement analysis and saturation analysis, include enzyme-linked immunosorbent assays (ELISAs) and radioimmunoassays. Some investigators also use the term to describe binding assays involving an unlabeled ligand and a structurally similar ligand (*e.g.*, a competitive inhibitor) that binds to the same site on the macromolecule.

COMPETITIVE INHIBITION

Any reduction in catalytic activity by a substance binding to the same form of a enzyme (or receptor) to which a substrate (or ligand) binds. This binding is said to be competitive, because inhibitor I and substrate S cannot bind simultaneously (*i.e.*, their binding is mutually exclusive). Note: Substrate and inhibitor need not act at the same topological site on the enzyme in order to exhibit mutually exclusive binding.

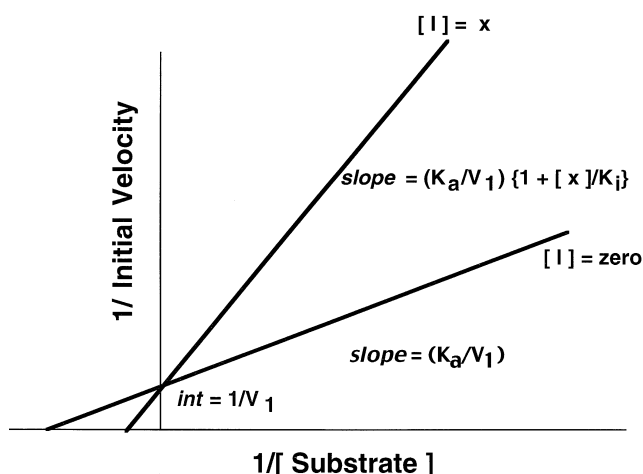
For a one-substrate enzyme-catalyzed reaction, the following scheme describes the action of a competitive inhibitor:



where $K_i = [E][I]/[EI]$, assuming rapid and reversible binding. The resulting rate law can be written in double-reciprocal form:

$$1/v = (1/V_{\max}) + (K_m/(V_{\max}[S]))\{1 + [I]/K_i\}$$

This equation has a $(1/v)$ axis intercept value of $(1/V_{\max})$ and a slope of $(K_m/V_{\max})\{1 + [I]/K_i\}$. Hence, all lines intersect at a common point on the $(1/v)$ axis. Plotting the slope values *versus* $[I]$ yields a secondary plot (or replot) whose slope is $K_m/(V_{\max}K_i)$ and horizontal intercept is $-(1/K_i)$.



Fromm¹ first demonstrated how competitive inhibitors can be employed to distinguish the order of substrate binding for multisubstrate enzyme mechanisms. Each competitive inhibitor, with respect to one substrate, displays distinctive pattern(s) relative to the other substrate(s)^{2,3}.

¹H. J. Fromm & V. Zewe (1962) *J. Biol. Chem.* **237**, 3027.

²H. J. Fromm (1979) *Meth. Enzymol.* **63**, 467.

³H. J. Fromm (1995) *Meth. Enzymol.* **249**, 123.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Theory, **63**, 468-477 [one-substrate, **63**, 468-474; two-substrate, **63**, 474-478; three-substrate, **63**, 477, 479]; practical aspects, **63**, 477-485; limitation, **63**, 485, 486; complications, **63**, 406-408; computer program, **63**, 114, 115, 126, 127, 137; constant ratio to substrate, **63**, 10; coupled enzyme assay, **63**, 40; cryoenzymology, **63**, 345; equation, **63**, 10, 468-470; metal-nucleotide complex, **63**, 268, 269, 292, 328, 329; pH effects, **63**, 214; product effects, **63**, 412, 413; statistical analysis, **63**, 110; substrate binding order, **63**, 467-486; tight-binding, **63**, 445.

COMPETITIVE INHIBITORS (Multisubstrate Enzyme Systems)

Fromm and Zewe¹ originally suggested that competitive inhibitors could aid an investigator in distinguishing be-

tween various enzyme kinetic mechanisms. The procedure^{2,3} utilizes a competitive inhibitor for one of the substrates in a multisubstrate enzyme-catalyzed reaction and determines whether that same inhibitor is competitive, noncompetitive, or uncompetitive with respect to the other substrate(s) of the enzyme. For example, if an investigator had a competitive inhibitor for one of the substrates of a non-ping pong bisubstrate enzyme and that inhibitor was uncompetitive with respect to the other substrate, the researcher could eliminate the random binding mechanism (Table I).

Table I
Use of Competitive Inhibitors to Distinguish Various Bi-Reactant Systems^a

Mechanism	1/v versus 1/[A]	1/v versus 1/[B]
Ordered Bi Uni	C	N
	U	C
Ordered Bi Bi'	C	N
	U	C
Ping Pong Bi Bi	C	U
	U	C
Random Bi Uni	C	N
	N	C
Random Bi Bi	C	N
	N	C

^aC = competitive inhibition pattern; N = noncompetitive inhibition pattern; and U = uncompetitive inhibition pattern.

The various ter-reactant enzyme systems can also be discriminated by the use of competitive inhibitors (see Table II). Fromm⁴ has also presented a cogent argument for exercising special care in choosing the concentrations of the nonvaried substrates in experiments on reactions involving three substrates.

While requiring the availability of competitive inhibitors for each of the substrates, Fromm's use of competitive inhibitors to distinguish multisubstrate enzyme kinetic pathways represents the most powerful initial rate method. **See Alternative Substrate Inhibition**

¹H. J. Fromm and V. Zewe (1962) *J. Biol. Chem.* **237**, 3027.

²H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, pp. 94-110, Springer-Verlag, New York.

³H. J. Fromm (1979) *Meth. Enzymol.* **63**, 467.

⁴H. J. Fromm (1967) *Biochim. Biophys. Acta* **139**, 221.

Table II. Use of Competitive Inhibitors to Distinguish Various Ter-Reactant Enzyme Systems^a

Mechanism	1/v versus 1/[A]	1/v versus 1/[B]	1/v versus 1/[C]
Ordered Ter Ter	C U U	N C U	N N ^b C
Random Ter Ter (rapid equilibrium)	C N N	N C N	N N C
Random AB, Ordered C (rapid equilibrium)	C N U	N C U	C ^c C ^d C
Ordered A, Random BC (rapid equilibrium)	C U U	N C N	N N C
Random AC, Ordered B (rapid equilibrium)	C N N	N C N	N N C
Hexa Uni Ping Pong	C U U	U C U	U U C
Ordered Bi Uni Uni Bi Ping Pong	C U U	N ^e C U	U U C
Ordered Uni Uni Bi Bi Ping Pong	C U U	U ^e C U	U N ^f C
Random Bi Uni Uni Bi Ping Pong	C N U	N C U	U U C
Random Uni Uni Bi Bi Ping Pong	C U U	U C N	U N C

^aC = competitive inhibition pattern; N = noncompetitive inhibition pattern; and U = uncompetitive inhibition pattern.

^bThis plot will be nonlinear if EAI binds C to form an EAIC complex.

^cThis plot will be noncompetitive if EIB binds C to form an EIBC complex.

^dThis plot will be noncompetitive if EIA binds C to form an EIAC complex.

^eThis plot will be nonlinear if EI binds B to form an EIB complex.

^fThis plot will be nonlinear if E'I binds C to form an E'IC complex.

COMPLEX REACTION

A reaction or set of reactions that consists of several elementary reactions. **See also** *Elementary Reaction*; *Primitive Change*; *Parallel Reactions*; *Series Reactions*

COMPONENT

1. A chemically distinct substance or species from which a phase or constituent of a system can be made. For example, for a system consisting of ice floating in water, there are three constituents: liquid water, ice, and water vapor. The number of components can exceed the number of phases. For example, D-glucose dissolved in water in a sealed ampule results is a system having one phase, albeit containing two components. **See also** *Constituent*
2. One of the coordinates or elements of a vector. 3. A vector term that is a part of a vector sum or vector product (*i.e.*, in which the resultant is a vector). 4. With

respect to any vectoral quantity, a component in any direction is the representation of the vector by the projection of that vector onto the line in the specified direction. For example, in the two-dimensional Cartesian coordinate system, for the vector going from the coordinate point (1,1) to point (2,2), the component of that vector along the *x*-axis runs from point (1,0) to point (2,0).
5. In statistics, a variable in a multivariate distribution.

COMPOSITE RATE CONSTANTS

A rate constant describing the reaction rate over a series of transition states, where if written as the sum of reciprocal equilibrium and rate constants, it describes the difference in free energy between the *n*th transition state and the original reactant¹. An excellent example is provided in Fig. 1 which is taken from Fisher *et al.*¹

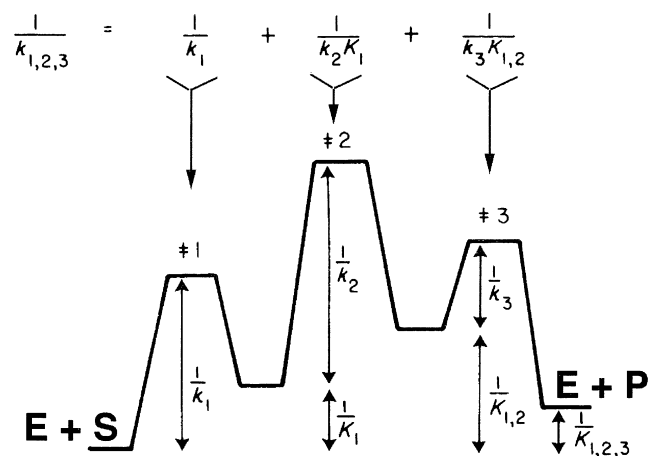


Figure 1. Free energy diagram of a three-step process, showing each term in the expression for the composite rate constant $k_{1,2,3}$ in the forward direction. Reproduced with permission of the authors and the American Chemical Society.

¹L. M. Fisher, W. J. Albery & J. R. Knowles (1986) *Biochemistry* **25**, 2529.

COMPOSITE REACTION

A chemical reaction whose rate expression includes rate constants from more than one elementary reaction. Examples of composite reactions include parallel reactions and stepwise reactions.

COMPRESSION

A potential factor affecting the catalytic power of enzyme-catalyzed reactions in which the enzyme forces the attacking moiety into the valence shell of the entity under "attack." Compression is thought to be manifested in the transition state complex. The results of kinetic isotope effect measurements have suggested that compression events factor in the catechol methyltransferase reaction¹.

¹F. Takusagawa, M. Fujioka, A. Spies & R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 1.

COMPROPORTIONATION

Reverse of disproportionation. **See** *Disproportionation*

COMPUTER ALGORITHMS & SOFTWARE

The widespread availability of digital computing has greatly facilitated data acquisition and analysis. No single text, and least of all a manual on the general applications of biochemical kinetics, can describe all the computing methods at the disposal of molecular life scientists. For this reason, readers are encouraged to consult specialty monographs dealing with a specific kinetic technique.

See also KINSIM/FITSIM, SAAM, Analog Computer, Meta-Model, etc.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

AGIRE, 249, 79-81, 225-226; **AMBER, 240**, 416, 428-429, 433; **243**, 562, 567-590; **261**, 5, 6, 28-29, 40, 56, 94, 292, 603; **AMBER/OPLS, 243**, 562; **AUTO, 240**, 795-798, 803, 805, 812, 815; **CAPP, 239**, 315; **CHARMM, 240**, 451; **243**, 562, 567, 569-571, 573, 578; **CHEOPS, 261**, 23; *Connolly Molecular Surface program, 243*, 604; *CURVEFIT function, 232*, 143, 151, 155, 159-160; *DelPhi program, 243*, 606; for determination of metal ion-nucleotide complex dissociation constants, **249**, 182, 184-188; *Dials&Windows program, 261*, 39, 43, 124; **DIANA, 239**, 304, 429; **240**, 416, 435-436; **DISMAN, 239**, 429; **DOCK, 241**, 355-358; **DSPP, 243**, 602; **ECEPP/2, 243**, 562; *EM algorithm, 240*, 178-180; **ENZFITTER, 248**, 68-70, 75-79, 98-99; **249**, 525; for estimation of cooperativity parameters, **249**, 525-526; **EZFIT, 249**, 525; **FELIX, 240**, 450; **FIRM, 240**, 427, 432-433, 435, 437; **FITSIM, 240**, 312-322; **FLOG, 241**, 359, 361-363; **FLUSYS, 248**, 20, 91; **GAMESS, 243**, 563; **GAUSSIAN, 243**, 563; **GRAFIT, 249**, 525; **GROMOS, 243**, 562, 606; **GROW, 241**, 358; **HABAS, 239**, 422; **HONDO, 243**, 563; **IGOR, 240**, 330; *Insight/Discover program, 243*, 561, 595; **ISOBI, 249**, 449, 455-457, 478; **ISOBI-HS, 249**, 456-459; **ISOCALC4, 249**, 349-350, 365-368; **ISOCALC5, 249**, 350, 366, 369-373; **ISO-COOP, 249**, 455-457, 471; **ISOMOD, 249**, 455-457, 471; **ISOTER, 249**, 449, 459-461, 464-465; **JMP, 249**, 525; **KALEIDOGRAPH, 249**, 525; **KINSIM, 240**, 312-322; **249**, 461; **LIGAND, 249**, 525; **LINSHA, 261**, 22-24; **MARDIGRAS, 239**, 420-421, 425, 427-428; **240**, 427-428; **261**, 10, 12-16, 74, 94, 601-603; *Marquardt algorithm, 246*, 696; **MATHEMATICA, 249**, 525; **MERLOT, 232**, 22; for molecular modeling, **243**, 560-561; *Monte Carlo methods (See Monte Carlo Methods)*; **MOPAC, 243**, 563; **MOPAC ESP, 243**, 564; for moving average interpolation, **235**, 32; *Nelder-Mead simplex algorithm, 240*, 188 *Newton algorithm, 246*, 696-697; for probit analysis, **235**, 31; for protein NMR spectral analysis, **239**, 289, 308; **PSEUROT, 261**, 255; **QUANTA, 243**, 561; **261**, 204; *Relaxation Analysis Program, 261*, 444-445; **SHAKE, 261**, 29, 37, 57; **SIMNOE, 261**, 78-79; **SPARKY, 261**, 8; **SPHINX, 261**, 22-24, 596; *Statistical Analysis System, 235*, 31, 39; **STEREO-SEARCH, 239**, 422; for structure-based computational chemistry calculations, **243**, 601-606; **ULTRAFIT, 249**, 525; **WESDYN, 241**, 186-187; **261**, 139; **XPLOR, 232**, 22; **239**, 304, 473; **261**, 205, 292, 603.

CONCAVE-DOWN

A term used to describe the behavior of a curvilinear function by specifying that, over a given interval, the first derivative increases, reaches a local or global maximum, and then decreases.

CONCAVE-UP

A term used to describe the behavior of a curvilinear function by specifying that, over a given interval, the first derivative decreases, reaches a local or global minimum, and then increases.

CONCENTRATION

The amount of substance, typically measured in moles, per unit volume of solution (commonly measured in

dm^{-3} [*i.e.*, liters, L]). Thus, amount-of-substance concentration, usually referred to simply as concentration and symbolized by c (often with a subscript denoting the species), can be described by $c_B = n_B/V$ where n_B is the amount-of-substance B and V is the total volume of the solution. Since the density of a solution will vary with temperature, the concentration c_B will be temperature-dependent. Hence, the temperature should always be reported. The SI unit is $\text{mol}\cdot\text{dm}^{-3}$ or molarity.

Concentration is the most common means for describing the composition of a solution in biochemistry. Enzyme kinetic expressions are typically expressed in these concentration units. Unless otherwise noted, this is the method used throughout this text. Nevertheless, other methods for describing compositions are utilized. For example, mole fractions are often used in Job plots. Gases in solution are commonly measured in terms of partial pressures. Below is a brief description of a few of these other conventions or methods.

Mole fraction, often symbolized by x or X followed by a subscript denoting the entity, represents the amount of a component divided by the total amount of all components. Thus, the mole fraction of component B of a solution, x_B , is equal to $n_B/\sum n_i$ where n_B is the amount of substance B and $\sum n_i$ is the total amount of all substances in solution. In biochemical systems, usually the solvent is disregarded in determining mole fractions. The mole fraction, a dimensionless number expressed in decimal fractions or percentages, is temperature-independent and is a useful description for solutions in theoretical studies and in physical biochemistry.

Molality, symbolized by m and usually followed by a subscript denoting the component, is also often used in physical biochemistry and is equal to the amount of substance per unit mass of solvent (*e.g.*, $m_B = n_B/(\text{mass}_{\text{solvent}})$). The SI unit for molality, another temperature-independent quantity, is $\text{mol}\cdot\text{kg}^{-1}$. A quantity related to molality, yet not as widely used in physical studies, is the volume molality, symbolized by m'' . The volume molality is equal to the amount of substance per unit volume of solvent (recall that concentration was equal to amount of substance per unit volume of solution).

As mentioned above, gases in solution are often measured in partial pressures. Recall that Dalton's Law of Partial Pressures states that the total pressure is the sum of the pressures partially exerted by all of the component, noninteracting gases. For one gaseous component in equilibrium with the same component dissolved in a liquid, the partial pressure of that gas in solution is the pressure that gas would exert in the atmosphere. It is symbolized by either p or P followed by the specific gas (*e.g.*, either P_{O_2} or p_{O_2}). For example, typical blood P_{O_2} levels range between 75 and 100 torr (or, 10.0 to 13.3 kPa since the pascal (Pa) is an SI unit).

In clinical studies, compositions are often described in terms of mass of substance per unit volume. For example, normal adult blood glucose levels typically range between 70 and 110 mg/dl. While these numbers have great utility in clinical studies, they are less useful in assessing the action of the metabolite with other proteins and systems. Thus, it is valuable to provide the concentration term as well in these reports.

Protein concentrations are often reported in terms of mass per unit volume (*e.g.*, mg/ml) or in units of activity per volume. Care should be exercised when using these forms of concentrations. To have valid steady-state kinetics, the total enzyme concentration must be much less than the substrate concentration. Hence, one should always be cognizant of the true concentration of active enzyme in terms of molarity.

If the molecular weight is known for each component of the solution, then mole fractions (in which the solvent is considered one of the components) and molality can be mathematically interconverted. For example, if a solution contains two components, A and B, in which x_A , x_B , m_A , m_B , M_A , and M_B represent the mole fractions, the molalities, and the molecular weights of A and B, respectively, then $x_B = m_B/[(1000/M_A) + m_B]$.

Similarly, concentration and mole fractions can be related. If ρ is the density of the solution then, for a solution containing two components A and B, $x_B = c_B/[(1000\rho - c_B M_B/M_A) + c_B]$. In dilute aqueous solutions, molarity is approximately equal to molality. **See** *Concentration Range Selection*

CONCENTRATION CORRELATION ANALYSIS

A chemical relaxation technique that measures the magnitude and time dependence of fluctuations in the concentrations of reactants¹⁻³. If a system is at thermodynamic equilibrium, individual reactant and product molecules within a volume element will undergo excursions from the homogeneous concentration behavior expected on the basis of exactly matching forward and reverse reaction rates. The magnitudes of such excursions, their frequency of occurrence, and the rates of their dissipation are rich sources of dynamic information on the underlying chemical and physical processes. The experimental techniques and theory used in concentration correlation analysis provide rate constants, molecular transport coefficients, and equilibrium constants. Magde⁴ has provided a particularly lucid description of concentration correlation analysis. **See** *Correlation Function*

¹Y. Chen (1973) *J. Chem. Phys.* **59**, 5810.

²Y. Chen & T. L. Hill (1973) *Biophys. J.* **13**, 1276.

³G. Feher & M. Weissman (1973) *Proc. Natl. Acad. Sci. U.S.A.* **70**, 870.

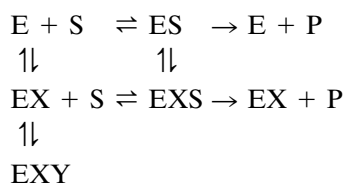
⁴D. Magde (1977) in *Chemical Relaxation in Molecular Biology* (I. Pecht & R. Rigler, eds.), pp. 2-3, Springer-Verlag, Berlin.

CONCENTRATION-RESPONSE CURVE

A plot of the biological response produced by a drug, typically rising from a basal value (frequently zero at zero drug concentration) to a maximal response as the drug concentration is increased to a point where the extent of drug complexation with its receptor is no longer concentration-dependent. Such curves are typically plotted as the fraction of maximal response *versus* the logarithm (base₁₀) of drug concentration.

CONCERTED FEEDBACK INHIBITION

The action of two (or more) feedback modifiers that must be present simultaneously for any inhibition to occur. An illustrative case is the ordered addition of two effectors (designated as X and Y) in the following reaction scheme:



This mechanism involves the ordered addition of inhibitors, such that X must bind before Y can. As a result, the following are essential properties: (a) the presence of only X along with substrate S has no effect of enzyme reaction rate, because X does not affect substrate binding or the rate of ES breakdown; (b) the presence of only Y along with S is without effect, because Y cannot bind in the absence of X; and (c) inhibition will take place only when X and Y are both present, thereby allowing inactive EXY complex to accumulate.

CONCERTED PROCESS

Two or more primitive chemical/physical changes occurring within the same elementary reaction step. Typically, the simultaneous progress of these changes is thermodynamically more favorable than a set of successive changes. **See** *Primitive Changes; Elementary Reactions; Synchronous*

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

CONDENSATION REACTION

A reaction occurring between two or more distinct molecular entities (or, between different sites within the same molecular entity), resulting in the formation of a main product with the concomitant release of a small molecule, most commonly water. In most chemical condensations, the reaction usually consists of an addition reaction, followed by an elimination reaction.

CONFIDENCE INTERVAL

A term used in statistics to indicate the range of an interval within which the true value for a parameter lies, given a specified confidence level.

CONFIDENCE LEVEL

A term used in statistics to indicate the reliability of a measured result. A confidence level of 90% indicates that there is a 10% chance that the result is not reliable.

CONFIGURATION, DETERMINATION OF

There are a number of methods for determining the configuration of an optically active molecular entity^{1,2}. Among the more common methods are:

1. Conversion of the particular compound of unknown configuration to a compound of known configuration, without altering the three-dimensional arrangement of substituents about the chiral center.

2. Conversion of the particular compound of unknown configuration to a compound of known configuration with the alternation of configuration about the chiral center, provided that the mechanism is known. For example, S_N2 mechanisms proceed with inversion at chiral carbons.

3. Use of biochemical agents such as enzymes. For example, if an enzyme has been reported to bind, or act on, only the *R*-isomers of a series of substrates then, if the enzyme acts on the unknown, it is likely, although not always, that the unknown also has the *R*-configuration.

4. In certain cases, the sign and extent of the optical rotation can aid in assigning the configuration. Often, with a homologous series of molecules, the optical rotation will change in the same direction as the particular entity increases in size. Hence, if the configurations of a number (not just one) are known, the unknown configuration of molecule that fits within that series may be determined.

5. The X-ray method of Bijvoet *et al.*³ can determine the configuration directly.

6. The use of optical rotary dispersion and circular dichroism also provide means for determining configurations.

7. Using techniques associated with a symmetric or stereoselective synthesis, a particular compound can be prepared, based on knowledge of the stereochemical course of the synthesis.

¹H. B. Kagan (1977) *Determination of Configuration by Chemical Means* (vol. 3 of Stereochemistry), Georg Thieme Publishers, Stuttgart.

²J. H. Brewster (1972) in *Elucidation of Organic Structures by Physical and Chemical Methods* (eds., K. W. Bentley & G. W. Kirby), vol. 4 of Techniques of Chemistry (ed., A. Weissberger), part 3, pp. 1-249, Wiley, New York.

³J. M. Bijvoet, A. F. Peerdeman & A. J. van Bommel (1951) *Nature* **168**, 271.

⁴H. B. Kagan (1977) *Determination of Configurations by Dipole Moments, CD, or ORD* (vol. 2 of Stereochemistry), Georg Thieme Publishers, Stuttgart.

⁵H. M. Smith (1983) *Chem. Rev.* **83**, 359.

CONFORMATIONAL CHANGES IN PROTEINS

Enzyme active sites and receptors rarely interact with ligands without some attendant change in conformation, and the ability to detect and quantify a conformational change lies at the heart of contemporary biochemical kinetics. **See** *Induced Fit Model*; *Fluorescence Spectroscopy*; *Linked Functions*; *Hemoglobin*; *Cooperativity*

Selected entries from *Methods in Enzymology* [vol., page(s)]:

2-Bromoacetamido-4-nitrophenol as a reporter group, **11**, 866; dansyl chloride (1-dimethylaminonaphthalene-5-sulfonyl chloride) as fluorescent reporter group, **11**, 857; difference spectroscopy, **11**, 748; environmentally sensitive groups attached to proteins, **11**, 856; fluorescence measurements, **11**, 776; hydrogen exchange, **11**, 734; immunological techniques, **11**, 917; proteolytic susceptibility, **11**, 905; microcalorimetry of protein conformational changes, **61**, 261, 287; detection of conformational changes in proteins by low temperature-rapid flow analysis, **61**, 318; investigation by [near-ultraviolet circular dichroism, **61**, 339; time-resolved fluorescence measurements, **61**, 378; low frequency vibrations and the dynamics of proteins and polypeptides, **61**, 425; carbon-13 nuclear magnetic resonance, **61**, 458]; enzymes, **61**, 259-335; **63**, 9; activation energetics, **64**, 225, 226; calcium binding, **64**, 387; calorimetry, **61**, 266; attending enzyme binding at interface, **64**, 388-392; fluorescence probes, **69**, 518, 519; hexokinase burst, **64**, 199; hysteresis, **64**, 194; isotope trapping, **64**, 58; lipase, **64**, 344, 348; lipid, **64**, 385, 388; low-temperature dynamics, **61**, 318-335; phospholipase A_2 , **64**, 348, 387, 388; ribonuclease, **64**, 220; specificity, reversal, **64**, 344; structural basis, **64**, 223-225; interactions, biological systems, **61**, 24, 25; spectroscopic studies, **61**, 337-549; studies with site-specific anti-cytochrome c antibodies, **74**, 257-262; temperature effects, **63**, 239, 348; surface pressure effects, **64**, 387.

CONFORMATIONAL EQUILIBRIA, MACROMOLECULAR

Any of the so-called *all-or-none* transitions between ordered structure and random coil for biological macromolecules. These highly cooperative processes can be defined by an equilibrium constant written as: $K = (\text{fraction of molecules in random coil})/(\text{fraction of molecules in ordered structure})$. Such transitions occur over very narrow temperature ranges, and the midpoint is usually given the name "the melting temperature" (T_m). The high degree of cooperativity in this two-state transition indicates that any break in local order results in loss of global order. This type of treatment can be used to characterize the conformational transitions of DNA as a function of temperature. They also apply to the heat-induced denaturation of proteins.

CONFORMATIONS

Three-dimensional arrangements in space of atoms and groups in a molecular entity that are interconvertible

by free rotations about chemical bonds^{1,2}. Specific conformations are often called conformers or rotamers. Typically, conformers tend to interconvert rapidly.

¹E. L. Eliel, N. L. Allinger, S. J. Angyal & G. A. Morrison (1965) *Conformational Analysis* Interscience, New York.

²J. Dale (1976) *Top. Stereochem.* **9**, 199.

CONSERVATION OF CHARGE, LAW OF

The condition satisfied whenever the total net charge of any closed system (*e.g.*, a chemical reaction) remains constant.

CONSERVATION OF ENERGY, LAW OF

The condition satisfied whenever the total energy of any closed system is constant. Thus, for a chemical reaction there is no loss or gain of energy in a closed system; however, energy may be transformed from one form into another. **See** *Conservation of Mass and Energy; Thermodynamics*

CONSTANT RATIO METHODS

1. A procedure used to assist in identifying sequential mechanisms when the double-reciprocal plots exhibit parallel lines¹⁻⁴. In some cases, bireactant mechanism can have various collections of rate constants that result in so-called parallel line kinetics, even though the mechanism is not ping pong. However, if the concentrations of A and B are kept in constant ratio with respect to each other, a sequential mechanism in a $1/v$ vs. $1/[A]$ plot would be nonlinear (since in the denominator the last term of the double-reciprocal form of the rate expression contains $[A]^2$: for example, for the steady-state ordered Bi Bi reaction scheme in which $[B] = \alpha[A]$, the double-reciprocal rate expression becomes $1/v = (1/V_{\max}) + (1/(V_{\max}[A]))(K_b/\alpha + K_a) + K_{ia}K_b/(\alpha V_{\max}[A]^2)$), whereas a ping pong mechanism would still be linear (that rate expression not containing a squared term). However, the nonlinearity is still dependent on the magnitude of this last term. The influence of this squared term can be increased by altering the ratio of A to B. Thus, in such studies it is advisable to conduct at least two sets of experiments: one in which $[A] \gg [B]$ and the other in which $[B] \gg [A]$. If substrate inhibition is a factor, however, the concentration of the

substrate has to be maintained below the concentration producing inhibition.

2. A similar procedure can be used in product inhibition studies for enzymes having three or more products. Different mechanisms can be distinguished if certain products are held in a constant ratio.

3. Constant ratio methods are also employed to maintain mass action ratios in measurements of isotope exchange at equilibrium.

4. An initial-rate method introduced by Fromm³ to discriminate between rival three-substrate enzyme kinetic mechanisms. **See** *Fromm Method for Ternary Systems*.

¹E. Garces & W. W. Cleland (1969) *Biochemistry* **8**, 633.

²W. W. Cleland (1970) *The Enzymes*, 3rd ed., **2**, 1.

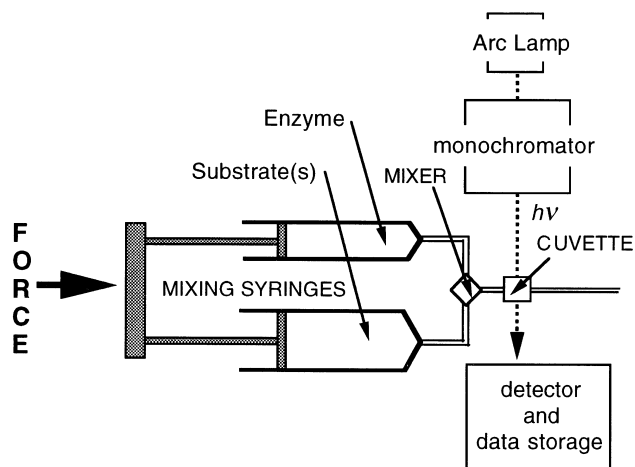
³H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

⁴D. L. Purich (1992) *Meth. Enzymol.* **87**, 3.

CONSTITUENT

A phase of a particular chemical or physical system. For ice floating in water, there are three constituents of the system: liquid water, ice, and air. However, there are only two components, water and air. **See also** *Component*

CONTINUOUS-FLOW DEVICE



A rapid mixing device, introduced nearly eighty years ago by Hartridge and Roughton, to mix two or more reactants (mixing time $\approx 10\text{-}50 \mu\text{sec}$) and to utilize a constant flow rate through a loop that then passes

through the observation port of a flow cuvette. The length of the loop and flow rate determine the “age” of the sample examined in the cuvette. The chief advantage of the method is that the sample is essentially time-invariant, and high quality spectral data can be obtained in the continuous-flow mode. The chief disadvantage is the consumption of high volumes of reactants. **See** *Stopped-Flow Technique*; *Rapid Mix-Quench Method*

CONTINUOUS *versus* STOPPED-TIME ASSAYS

The velocity of an enzyme-catalyzed reaction can be measured either by a continuous assay or by a stopped-time protocol. Whenever possible, the continuous measurement of a velocity (*e.g.*, the increase or decrease in absorbance *vs.* time) should be utilized. In stopped-time assays, the investigator must demonstrate that the reaction is completely terminated at the specified point in time and that products are readily and quantitatively separated from substrates. In addition, one must show that the system is under initial rate conditions. Thus, at least three or four different time points should be chosen. Stopped-time assays also require an assay blank (for $t = 0$). In this blank, typically the quenching conditions are applied prior to the initiation step. Whenever practicable, replicate kinetic analyses should be done, even with continuous assay protocols. **See** *Enzyme Assay Methods*; *Basal Rate*

CONTRADICTION

A lack of agreement or consistency in some experimental finding or theoretical argument. In the case of kinetic studies, one can take advantage of contradictions to eliminate rival schemes as long as one is confident of both the experimental results and the design on which they are based. Accordingly, one seeks to rule out potential explanations or rival mechanisms by seeking to find contradictions between theory and experiment.

In his influential philosophical essay “What is Life?,” Schrödinger¹ quotes Miguel de Unamuno: “*Si un hombre nunca se contradice, será porque nunca dice nada*” (Translated: “Should a man never contradict himself, the reason must be that he virtually never says anything at all.”) Philosophers have argued that valuable scientific publications must address points of contradiction con-

cerning earlier work and that they will raise new contradictions provoking continued inquiry.

¹E. Schrödinger (1956) *What is Life? and Other Scientific Essays*, Doubleday, Garden City, New York.

CONTRAST AGENT

Any paramagnetic (or ferromagnetic) metal complex or particle that decreases the relaxation times of nuclei detected in an image.

M. W. G. de Bolster (1997) *Pure Appl. Chem.* **69**, 1251.

CONTROL

A set of experimental conditions for which the outcome of measurement or observation is well defined on the basis of previous experience. A positive control describes a set of conditions known to result in a well characterized behavior, whereas a negative control confirms that a factor previously known to be without effect continues to be without effect.

For example, one may decide to alter the order of a series of rate assays with respect to substrate concentration (*i.e.*, in one case starting from low to high substrate concentration and then again from high to low substrate concentration). This would allow one to identify any effect of substrate concentration on the time-dependent inactivation of an enzyme. Without such a control, the experimenter might overlook an important effect of the substrate's effect on enzyme stability. Furthermore, depending upon the extent of enzyme inactivation and the order of rate assays with respect to substrate concentration, the observed rate saturation behavior could have the appearance of a cooperative system. Ideally, one can use a random number table to assign the order in which a set of measurements is obtained. In this case, the experimenter decides on a “randomization rule” (such as choosing every third entry from left to right in a table of random digits) and uses a pencil eraser to blindly locate a starting place in the table. After a random number is paired with each experimental condition, one can execute the experiment in an order determined by the magnitude of the random number in ascending or descending order. **See** *Statistics (A Primer)*

CONTROL COEFFICIENT

An index (symbolized by C_i^X) of the magnitude of change (or output) X exhibited by a system variable in response to a certain perturbation (or input):

$$C_i^X = \frac{\partial X}{\partial v_i} \cdot \frac{v_i}{X} = \frac{\partial \ln X_i}{\partial \ln v_i}$$

where X is the system variable, and v_i is steady-state rate of i th enzyme-catalyzed reaction. For metabolic pathways, a system variable would be a flux or a concentration of a given reactant, intermediate, or product.

COOPERATIVITY

Any process in which an earlier event affects the likelihood of occurrence of some subsequent event(s). For enzyme-catalyzed reactions, cooperativity is observed in the regulation of metabolic pathways by enzymes expressing non-hyperbolic rate-saturation behavior. The most widely accepted explanations for cooperativity relate equilibrium and kinetic models for symmetry-conserving and ligand-induced conformational models¹⁻³, but mnemonic models can account for some aspects of hysteretic behavior. **See** *Adair Equation*; *Allosterism*; *Hemoglobin*; *Hill Equation and Plot*; *Hysteresis*; *Koshland-Nemethy-Filmer-Model*; *Ligand Binding Analysis*; *Monod-Wyman-Changeux Model*; *Negative Cooperativity*; *Positive Cooperativity*; *Scatchard Equation and Plot*

¹K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

²K. E. Neet & G. R. Ainslie, Jr. (1980) *Meth. Enzymol.* **64**, 192.

³K. E. Neet (1995) *Meth. Enzymol.* **249**, 519.

Selected entries from *Methods in Enzymology* (vol, page(s)):

Adair model, **249**, 526-528; analysis [graphical representation in, **249**, 520, 522-525; mutational, **249**, 554]; definition, **249**, 519-520; apparent in [bisubstrate mechanism, **64**, 162, 163; concerted symmetrical model, **64**, 167, 168]; catalytic effect, **64**, 174-176; comparative aspects, **64**, 171-174; complex, **64**, 140, 141, 181-185; concerted Monod-Wyman-Changeux model [assumptions, **64**, 166, 166; **249**, 528-530, 533-534, 546-549, 553-554; binding model (exclusive, **64**, 166, 172; nonexclusive, **64**, 166, 172); catalytic effect, **64**, 174, 175; comparison with other model, **64**, 171-174; constant, **64**, 154; equilibrium expression, **64**, 166, 167; hysteresis, **64**, 214, 215; modifier action, **64**, 167; negative, **64**, 172-175]; concerted-symmetrical model, **64**, 165-168; constants, **64**, 153-159; crystal lattice as allosteric effector stabilizing quaternary conformation, **232**, 15; curvature, reciprocal plot, **64**, 142-144; data analysis, **64**, 150, 151; equilibrium model, **64**, 141, 151-153, 159-164, 185-189; estimation, **64**, 142-153; false, **63**, 6, 7; gamma coefficient, **64**, 142, 143, 149-151; heterotropic interaction, **64**, 140, 141; Hill model, **64**, 144-147, 150, 151; homotropic interaction, **64**, 140; hysteretic systems, **64**, 163, 164, 192-226; induced-fit model, **64**, 185-189; isotope exchange, **64**, 4, 17; kinetic aspects, **64**, 141, 159-164; kinetic mechanisms, **249**, 520-521, 539-546 [steady-state random enzyme, **249**, 539-541]; Koshland-Nemethy-Filmer model, **249**, 530-534, 548 Kurganov method, **64**, 148; least squares method, **64**, 156-159; lipase-colipase system, **64**, 379; metal ions, **63**, 266; multisite, **64**, 160-162; negative, **64**, 140, 181-183; **249**, 521, 540 [graphical represen-

tation, **249**, 523-524 chymotrypsin, **64**, 220; concerted model, **64**, 172, 173; double-reciprocal plot, **64**, 142-144; half-site reactivity, **64**, 183-185; Hill plot, **64**, 144-147, 150-153; hysteresis, **64**, 164, 195, 196, 205, 206, 210, 212, 214; maximal velocity determination, **64**, 145; mixed, **64**, 181-183; parameters describing allosteric enzymes [computer programs for, **249**, 525-526; estimation by curve fitting, **249**, 525-526]; product effect, **64**, 177; protomer-oligomer equilibrium, **64**, 180; Scatchard plot, **64**, 147, 148; sensitivity index, **64**, 302; sequential model, **64**, 170, 175; nomenclature, **64**, 159, 160; positive, **64**, 139, 140; **249**, 521, 540; protomer-oligomer equilibrium, **64**, 178-181; proton binding, **63**, 195-198, 232; PRPP:ATP phosphoribosyltransferase, **63**, 9; R_s , **64**, 147, 149, 150; R_i , **64**, 143, 144, 149-151; rate equation, **64**, 151-153; $S_{0.5}$, **64**, 146, 147, 150, 152; Scatchard plot, **64**, 145, 147, 148, 150; sequential model, **64**, 168-171; site-site interactions, **64**, 164-178; Sturgill-Biltonen method, **64**, 148; thermodynamic analysis, from graphs, **249**, 557; V system, **64**, 168.

COORDINATE-COVALENT BOND

A covalent bond formed when the two bonding electrons originate from or are said to be donated by the same atom.

COORDINATION

1. A covalent bond, also called a coordinate-covalent bond, in which the two electrons associated with that bond have come from only one of the two atoms or groups linked by that bond. While such bonds are often written as $A: \rightarrow B$, the nature of this bond is similar to that of a covalent bond with significant dipolar character.

2. The process of formation of a coordinate-covalent bond; the reverse of unimolecular heterolysis. 3. The number of ligands surrounding a central atom. **See also** *Dipolar Bond*

COORDINATION NUMBERS & GEOMETRIES

1. The number of atoms, groups, molecules, or ions surrounding a particular atom or ion in a crystal or complex. 2. The number of atoms, groups, or ions directly linked to a particular atom or ion in a crystal or complex.

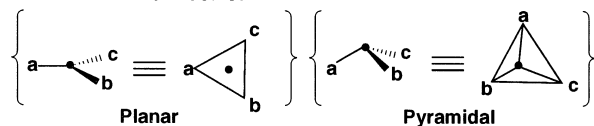
Pauling¹ offers the following distinction: the liganacy or coordination number of the central atom is the number of atoms bonded to that central atom, whereas its covalence is given by the number of covalent bonds formed by the central atom. The distinction becomes evident when double bonds join a central atom to a binding partner.

COORDINATION NUMBERS & GEOMETRIES

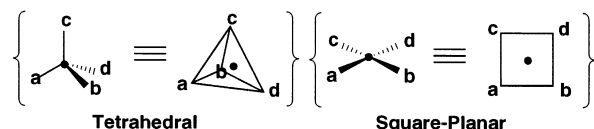
Coordination Number 2:



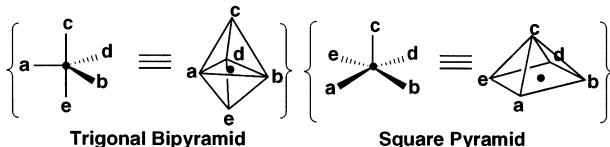
Coordination Number 3:



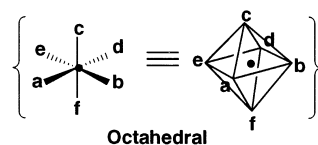
Coordination Number 4:



Coordination Number 5:



Coordination Number 6:



Note: The symbol • represents the central atom in the complex.

The following are descriptions and examples of complexes of metal ions having coordination numbers of two to six.

Coordination Number Two. Although rarely found in nature, complexes of Cu(I), Ag(I), Au(I) and Hg(I) adopt linear coordination geometries, such as that observed with $[\text{H}_3\text{N}-\text{Ag}-\text{NH}_3]$.

Coordination Number Three. These include planar complex ions such as boric acid $\text{B}(\text{OH})_3$ and out-of-plane pyramidal complexes (including SnCl_3^{1-}).

Coordination Number Four. These complex ions either have (a) a square planar geometry with the four ligands at each corner, such that the metal ion lies in-plane at the center, or (b) a tetrahedral complex where the centrally located metal ion has four ligands arranged as the hydrogen atoms in methane. Square planar complexes of

Cu(II), Ni(II), and Pt(II) which have eight *d*-orbital electrons, forming four electron pairs filling all *d* orbitals (but not the $d_{x^2-y^2}$ orbital) and hybridizing to create four dsp^2 orbitals extending in-plane from the central metal ion. A good example is the antineoplastic agent cisplatin, or $\text{cis}[\text{Pt}(\text{II})\text{Cl}_2(\text{NH}_3)_2]$ which has two chloride ions next to each (and hence *cis*) and two ammonia groups likewise situated next to each other. The *trans*-complex has alternating ligands, with no two alike next to each other. Examples of tetrahedral complexes are the rubredoxins which use four sulfhydryl groups to ligate the central iron atom. In this case, however, the bond angles are not all 109.5° , as observed in methane.

Coordination Number Five. Although much less frequently observed, two geometries have coordination numbers of five. The first has two apical ligands situated above and below a plane that contains three trigonally disposed ligands around it. [This pentavalent or trigonal bipyramid geometry is analogous to that achieved in phosphoryl transfer reactions, where the incoming nucleophile and the leaving group (*i.e.*, exiphile) are in apical positions, and the three other oxygen atoms are arranged trigonally and in-plane with respect to the phosphorus atom.] The second, so-called square-pyramid geometry resembles a five-sided Egyptian pyramid with a central metal ion and four ligands situated in the plane of its base, and the fifth ligand occupying an above-plane or apical position. Such a five-coordinate complex is that adopted by the high-spin Fe(II) atom in deoxyhemoglobin, except that the fifth ligand, a histidine imadazole nitrogen atom, pulls the iron out-of-plane by about $0.75 \text{ \AA}$.

Coordination Number Six. These are among the most frequently observed complexes. All six ligands are arranged in a hexacoordinate structure with octahedral (*i.e.*, eight-sided) geometry. A good example is O_2 bound as the sixth ligand in oxygenated hemoglobin. Likewise, the structure of cyanocobalamin (vitamin B_{12}) is octahedral, with four in-plane porphyrin ring nitrogens, and cyanide and a nitrogen supplied by the vitamin's dimethylbenzimidazole substituent. In living systems, the bound cyanide is hydrolyzed, allowing water to become the sixth ligand.

¹L. Pauling (1960) *The Nature of the Chemical Bond*, 3rd ed., p. 63, Cornell University Press, Ithaca.

COPROPORPHYRINOGEN OXIDASE

This enzyme [EC 1.3.3.3] (also referred to as coproporphyrinogenase, coproporphyrinogen-III oxidase, and coprogen oxidase) catalyzes the reaction of coproporphyrinogen-III with dioxygen to generate protoporphyrinogen-IX and two carbon dioxide molecules. Iron ions are required as cofactors.

T. Yoshinaga (1997) *Meth. Enzymol.* **281**, 355.

S. Granick & S. I. Beale (1978) *Adv. Enzymol.* **46**, 33.

CORNFORTH CHIRAL METHYL STEREOCHEMICAL METHOD

This procedure^{1,2} involves the enzymatic conversion of the chiral acetate (C(HDT)COO⁻), obtained from experiments involving chiral methyl groups², to labeled malate. The enzymes used in this procedure include acetate kinase, phosphotransacetylase, malate synthase, and fumarase.

¹J. W. Cornforth, J. W. Redmond, H. Eggerer, W. Buckel & C.

Gutschow (1970) *Eur. J. Biochem.* **14**, 1.

²H. G. Floss (1982) *Meth. Enzymol.* **87**, 126.

"CORN" RULE

The mnemonic for visualizing the correct three-dimensional representation of L-amino acids. As an individual sights along the H—C_α bond, with the hydrogen facing the person, the three remaining substituents should spell CO-R-N (for CO [*i.e.*, the carbonyl], R for the side chain, and N for the amino nitrogen) if the amino acid is an L-amino acid.

J. S. Richardson (1981) *Adv. Protein Chem.* **34**, 171.

CORPHIN

Corphin is the F-430 cofactor found in methyl-coenzyme M reductase, a nickel-containing enzyme that participates in the conversion of carbon dioxide to methane in methanogenic bacteria. The nickel ion in F-430 is coordinated by the tetrahydrocorphin ligand, which contains structural elements of both porphyrins and corrins.

M. W. G. de Bolster (1997) *Pure Appl. Chem.* **69**, 1251.

CORRELATION COEFFICIENT

A statistical parameter, often symbolized by *r*, that indicates the degree to which two variables have a linear relationship. The value of *r* will range between -1 and +1 in which a value of 1.00 indicates a perfect positive relationship between the two variables.

CORRELATION DIAGRAM

A depiction of how molecular orbitals are formed from atomic orbitals of separated atoms and/or how these molecular orbitals correlate with atomic orbitals of the atoms united as the nuclei come together to form a molecular species. Such diagrams are especially useful in rationalizing the symmetry properties of orbitals.

CORRELATION FUNCTION

A mathematical expression that defines how an ensemble average fluctuates with time. An example is the analysis of Brownian motion, where one may seek to understand the nature of the frictional force. If one considers τ as a time interval that is very small on the macroscopic scale, but large on the microscopic scale, then

$$\tau \gg \tau^*$$

where τ^* is the correlation time which will be on the same time-scale as the average period of fluctuations of the force $F(t)$. One can also say that τ^* is a direct measure of the relaxation time needed for the degrees of freedom responsible for the force to return to equilibrium after being suddenly disturbed. Using this formalism, one can ask for a certain time t_1 what the probability is that the force $F(t)$ assumes a value between $F(t_1)$ and $F(t_1) + dF(t_1)$. Likewise, one can evaluate the joint probability that at a time t_1 the force lies between $F(t_1)$ and $F(t_1) + dF(t_1)$ and that at t_2 the force lies between $F(t_2)$ and $F(t_2) + dF(t_2)$.

Correlation functions are powerful tools in statistical physics, and in the above example they permit one to examine the behavior of a fluctuating system from a reference time back to previous times. Such fluctuations can occur in the concentration of two (or more) interconverting chemical species in dynamic equilibrium, and the technique of concentration correlation analysis permits one to determine the forward and reverse rate constants for their interconversion. **See** *Concentration Correlation Analysis*

CORRELATION SPECTROSCOPY

A technique, also known as fluctuation correlation spectroscopy, that uses a correlation algorithm to detect similarity of a fluctuating signal occurring over a period of time. This property aids in the kinetic characterization of fluctuations in light scattering, or other electromagnetic

properties, from a small number of particles or molecules present in the small volume element that is sampled by an incident beam of light. Correlation spectroscopy allows an investigator to characterize aspects of the local nature of the system being studied.

Application of the correlation function shows how similar two signals are with respect to each other and indicates how long the two signals remain similar as one is shifted in a time domain relative to the other. Correlating a signal with itself is called autocorrelation. Different sorts of signals have distinctly different autocorrelation functions: (a) Random noise is said to be uncorrelated, meaning that noise is only instantaneously similar to itself. Upon applying a correlation function, any shift along the time axis fails to yield any correlation at all, and correlation function of random noise with itself is a single sharp spike at a time shift of zero. (b) Periodic signals go in and out of phase, as one is shifted with respect to the other. This means that such signals will show strong correlation at any shift allowing the peaks to coincide. The autocorrelation function of a periodic signal is also a periodic, with a time-period that is identical to that of the original signal. Signals that are of short duration can only remain self-similar for small time shifts, so their autocorrelation functions are short. **See** *Correlation Function*; *Concentration Correlation Analysis*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Acquisition of frequency-discriminated spectrum, **239**, 162-166, 170; sensitivity, **239**, 169-173; constant-time, **239**, 23-26; double-quantum filtered, **239**, 236; gradient pulse experiments, **239**, 185-189; protein structural information, **239**, 377-379; pulse sequence and coherence transfer pathway, **239**, 148-149; paramagnetic metalloprotein, **239**, 494-497; data recording, SWAT method, **239**, 166-169, 172; line shapes, effects of gradient pulses, **239**, 162-166; identification of protein amino acid resonances, **232**, 100; cyclosporin A, **239**, 240-241.

CORRELATION TIME

The average time (symbolized τ_c) between molecular collisions, taking into account rotational, dipolar, and electron spin lattice relaxation rates.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Anisotropy effects, **261**, 427-430; determination by dynamic laser light scattering (quasi-elastic light scattering), **261**, 432-433; determination for nucleic acids by NMR [accuracy, **261**, 432-433; algorithms, **261**, 11-13, 425, 430; carbon-13 relaxation, **261**, 11-12, 422-426, 431, 434-435; cross-relaxation rates, **261**, 419-422, 435; error sources, **261**, 430-432; phosphorus-31 relaxation, **261**, 426-427, 431; proton relaxation, **261**, 51, 418-422; relaxation matrix calculations, **261**, 12]; deuterium solvent viscosity effects, **261**, 433; effect

on NOE, **261**, 155, 413-414; length dependence in B-DNA, **261**, 432-433; rotational (**See** *Rotational Correlation Time*); spectral density function, **261**, 471-473; temperature effects, **261**, 431, 433-434.

CORTISONE β -REDUCTASE

This enzyme [EC 1.3.1.3] catalyzes the reversible reaction of 4,5- β -dihydrocortisone with NADP⁺ to produce cortisone and NADPH.

G. M. Tomkins (1962) *Meth. Enzymol.* **5**, 499.

COSMOGENIC DATING

Any geochronometric method for estimating the age of objects based upon the generation of radioactive isotopes by cosmic radiation, followed by isotopic incorporation into the biosphere/geosphere, and their subsequent first-order decay with release of radiation and/or accumulation of daughter isotopes. These methods take advantage of the lack of any dependence of the decay rate on temperature, pressure, pH, or other physical parameters. **See** *Radiocarbon Dating*

COULOMB

The SI unit for electric charge (symbolized by C) and equal to the amount of electricity transferred by a current of one ampere in one second.

COULOMB'S LAW

The relationship describing the interaction, in terms of mutual force (F), between two electrostatic point charges: $F = Q_1 Q_2 / (4 \pi \epsilon d^2)$ where Q_1 and Q_2 are the respective point charges, d is the distance separating them, and ϵ is the permittivity of the medium.

COULOMETRIC TITRATION

An analytical technique based upon the constant-current electrochemical generation of a redox-active titrant that can quantitatively react with a redox-active analyte. This technique takes advantage of the high precision with which the instrumentation controls electric current and time to produce standardized solutions of titrant.

4-COUMAROYL-CoA SYNTHETASE

This enzyme [EC 6.2.1.12] (also referred to as hydroxycinnamoyl-CoA ligase and 4-coumarate:CoA ligase) catalyzes the reaction of 4-coumarate with ATP and coenzyme A to yield 4-coumaroyl-CoA, AMP, and pyrophosphate (or, diphosphate).

R. A. Dixon, P. M. Dey & C. J. Lamb (1983) *Adv. Enzymol.* **55**, 1.

COUNTERION

An ion of opposite electric charge. Antonym: Co-ion.

COUNTING STATISTICS

While radioactive decay is itself a random process, the Gaussian distribution function fails to account for probability relationships describing rates of radioactive decay¹. Instead, appropriate statistical analysis of scintillation counting data relies on the use of the Poisson probability distribution function:

$$P = (\mu^{x_i}/x_i!) \exp(-\mu)$$

where P is the frequency of occurrence of a given count x_i , and μ is the mean for a large data set. If N is the number of counts over a certain time interval, then the relative standard deviation ($\sigma_{m,rel}$) can be written as follows:

$$\sigma_{m,rel} = \sigma_m/N = N^{-1/2}$$

because σ_m equals $N^{1/2}$. The standard deviation in the counting rate $r = dN/dt$, or simply N/t , one can show that the standard deviation in r is σ_r which is equal to $(r/t)^{1/2}$. Likewise, $\sigma_{r,rel}$ is given by the expression $(rt)^{-1/2}$. The confidence interval for a measurement can be defined as the limits around a measured quantity within which the true mean can be expected to fall with a stated probability. If the measured standard deviation closely approximates the true standard deviation, the confidence limit for the counting rate is the mean rate $\pm z\sigma$, and a tenfold decrease in the relative uncertainty necessitates approximately a hundred fold greater number of counts. A good rule of thumb is to collect 10,000 counts (above the accumulated background count) to achieve a precision of around one percent. **See Statistics (A Primer)**

¹G. Friedlander, J. W. Kennedy, E. S. Macias & J. M. Miller (1981) *Nuclear and Radiochemistry*, 3rd ed., chap. 9, Wiley, New York.

COUPLED ENZYME ASSAYS

A protocol for continuous enzyme assay that involves one or more auxiliary enzymes to convert a product of the primary reaction in a second or auxiliary reaction that produces a change in absorbance or fluorescence. As noted below, coupled enzyme assays, while convenient, are fraught with experimental limitations that must be overcome in order to obtain valid initial velocity data.

Primary enzyme reaction: $A \rightarrow B$

Auxiliary enzyme reaction: $B \rightarrow C$

McClure¹ was among the first to treat the kinetics of coupled enzyme assays, and others have discussed essential aspects of experimental design.^{2,3} Rudolph *et al.*³ provided an extensive list of coupled-enzyme assays for selected enzyme reaction products, and several examples are shown in Table I.

Among the practical considerations for utilizing coupled enzyme protocols are the following:

(1) The components of the coupling system should neither inhibit nor activate the primary enzyme. Moreover, care must be exercised to ascertain that the auxiliary enzyme(s) is not contaminated with other minor enzyme activities capable of influencing the primary enzymatic activity. The results from any coupled enzyme assay must exactly match the results obtained with other valid initial rate assays to ensure that the presence of the auxiliary system in no way affects the activity of the primary enzyme. This is typically accomplished by comparing data obtained from the coupled assay with stopped-time assay results to ensure that similar results are obtained.

(2) Sufficient coupling enzymes must be provided. Various treatments for determining the necessary concentration of auxiliary enzyme are available². Good rules-of-thumb are (a) that further addition of auxiliary enzyme should not change the rate of the primary enzyme reaction, (b) that doubling the concentration of the primary enzyme results in an exact doubling of observed reaction rate. These conditions should be confirmed at the highest and lowest concentrations of the primary reaction substrate(s) if the coupled assay system is to be used to generate a set of reaction velocity data as a function of those same substrate concentrations.

(3) Investigators should always determine that the assays are linear and that the lag times are reasonable (the amplitude change before the onset of linearity should rarely exceed 1–2% of total product formed if the reaction proceeds to equilibrium).

(4) When an investigator studies the effects of other substances (*e.g.*, competitive inhibitors, *etc.*) on the en-

Table I

Auxiliary Enzyme Assay Methods for Initial Rate Analysis

Product Measured	Species Monitored	Coupling Enzymes	Examples
ATP	NADH Disappearance	Hexokinase and Glucose-6-P Dehydrogenase	Adenylate Kinase, Creatine Kinase ^a
ADP	NADH Disappearance	Pyruvate Kinase and Lactate Dehydrogenase	Fructokinase, ^b γ -Glutamylcysteine Synthetase, ^c Mannokinase, ^d ATPase ^{e,f}
ADP	NADH Disappearance	Pyruvate Kinase, Lactate Dehydrogenase, and 3'-Nucleotidase	Adenosine-5'-phosphosulfate Kinase ^g
AMP	NADH Disappearance	Adenylate Kinase	Adenosine Phosphotransferase, ^h GMP Synthetase ⁱ
GDP	NADH Disappearance	Pyruvate Kinase and Lactate Dehydrogenase	Adenylosuccinate Synthetase ^j
GMP	NADH Disappearance	GMP Kinase, Pyruvate Kinase, and Lactate Dehydrogenase	Hypoxanthine-Guanine Phosphoribosyltransferase ^k
UDP	NADH Disappearance	Pyruvate Kinase and Lactate Dehydrogenase	Glycogen Synthetase ^l
Fructose 6-P	NADH Formation	Glucosephosphate Isomerase, Glucose-6-P Dehydrogenase	Fructokinase ^b
Galactose	NADH Formation	Galactose Dehydrogenase	Lactase ^m
D-Glucose	O ₂ Consumption	Glucose Oxidase and Catalase	Aldose-1-epimerase ⁿ
3-Phosphoglycerate	NADH Disappearance	Phosphoglyceromutase, Enolase, Pyruvate Kinase, and Lactate Dehydrogenase	Phosphoglycerate Kinase ^o
Pyruvate	NADH Disappearance	Lactate Dehydrogenase	Neuraminidase ^p
Acetyl-CoA	p-Nitroaniline Disappearance	Arylamine Acetyltransferase	Acetyl-CoA Synthetase, ATP-Citrate Lyase ^q
Fumarate	NADH Formation	Fumarase, Malate Dehydrogenase	Argininosuccinate Lyase ^r
Oxaloacetate	NADH Disappearance	Malate Dehydrogenase	Phosphoenolpyruvate Carboxy-kinase ^s
Aspartate	NADH Disappearance	Aspartate Aminotransferase, Malate Dehydrogenase	Asparaginase ^t

^aL. T. Oliver, *Biochem. J.* **61**, 116 (1955).

^bB. Sabater, J. Sebastián, and C. Asensio, *Biochim. Biophys. Acta* **284**, 414 (1972).

^cW. B. Rathbun and H. D. Gilbert, *Anal. Biochem.* **54**, 153 (1973).

^dB. Sabater, J. Sebastián, and C. Asensio, *Biochim. Biophys. Acta* **284**, 406 (1972).

^eB. C. Monk and G. M. Kellerman, *Anal. Biochem.* **73**, 187 (1976).

^fD. J. Horgan, R. K. Tume, and R. P. Newbold, *Anal. Biochem.* **48**, 147 (1972).

^gJ. N. Burnell and F. R. Whatley, *Anal. Biochem.* **68**, 281 (1975).

^hW. D. Park, M. E. Tischler, R. B. Dunlop, and R. R. Fisher, *Anal. Biochem.* **54**, 495 (1973).

ⁱT. Spector, R. L. Miller, J. A. Fyfe, and T. A. Krenitsky, *Biochim. Biophys. Acta* **370**, 585 (1974).

^jT. Spector and R. L. Miller, *Biochim. Biophys. Acta* **445**, 509 (1976).

^kA. Giacomello and C. Salerno, *Anal. Biochem.* **79**, 263 (1977).

^lJ. S. Passonneau and D. A. Rottenburg, *Anal. Biochem.* **51**, 528 (1973).

^mN. Asp and A. Dahlqvist, *Anal. Biochem.* **47**, 527 (1972).

ⁿM. K. Weibel, *Anal. Biochem.* **70**, 489 (1976).

^oM. Ali and Y. S. Brownstone, *Biochim. Biophys. Acta* **445**, 74 (1976).

^pD. N. Ziegler and H. D. Hutchinson, *Appl. Microbiol.* **23**, 1060 (1972).

^qG. Hoffman, L. Weiss, and O. H. Weiland, *Anal. Biochem.* **84**, 441 (1978).

^rJ. F. Sherwin and S. Natelson, *Clin. Chem.* **21**, 230 (1975).

^sR. J. Hansen, H. Hinz, and H. Holzer, *Anal. Biochem.* **74**, 576 (1976).

^tH. N. Jayaram, P. A. Cooney, S. Jayaram, and L. Rosenblum, *Anal. Biochem.* **59**, 327 (1974).

zyme activity, those other substances should not affect the coupling system. An activator may accelerate the primary enzyme reaction to a point where the auxiliary system becomes limiting, and the presence of inhibitors, metal ion chelators, and other metabolites may reduce the effectiveness of the auxiliary enzyme. In both cases, increasing the amount of auxiliary enzymes may be essential for maintaining a valid coupled enzyme assay.

Even under seemingly ideal conditions, the steady-state concentration of the first reaction product may exceed the inhibitory constant for the binding of that product to the primary enzyme. In such cases, the linearity of the coupled assay can be misleading, and the investigator must validate the coupled enzyme kinetic data by direct comparison with the results obtained by another technique such as a stopped-time radiometric assay. This

reference assay must be sufficiently sensitive to permit activity assays under conditions that avoid significant accumulation of product.

Easterby⁴ proposed a generalized theory of the transition time for sequential enzyme reactions where the steady-state production of product is preceded by a lag period or transition time during which the intermediates of the sequence are accumulating. He found that if a steady state is eventually reached, the magnitude of this lag may be calculated, even when the differentiation equations describing the process have no analytical solution. The calculation may be made for simple systems in which the enzymes obey Michaelis-Menten kinetics or for more complex pathways in which intermediates act as modifiers of the enzymes. The transition time associated with each intermediate in the sequence is given by the ratio of the appropriate steady-state intermediate concentration to the steady-state flux. The theory is also applicable to the transition between steady states produced by flux changes. Application of the theory to coupled enzyme assays makes it possible to define the minimum requirements for successful operation of a coupled assay. The theory can be extended to deal with sequences in which the enzyme concentration exceeds substrate concentration.

Yang and Schulz⁵ also formulated a treatment of coupled enzyme reaction kinetics that does not assume an irreversible first reaction. The validity of their theory is confirmed by a model system consisting of enoyl-CoA hydratase (EC 4.2.1.17) and 3-hydroxyacyl-CoA dehydrogenase (EC 1.1.1.35) with 2,4-decadienoyl coenzyme A as a substrate. Unlike the conventional theory, their approach was found to be indispensable for coupled enzyme systems characterized by a first reaction with a small equilibrium constant and/or wherein the coupling enzyme concentration is higher than that of the intermediate. Equations based on their theory can allow one to calculate steady-state velocities of coupled enzyme reactions and to predict the time course of coupled enzyme reactions during the pre-steady state.

Occasionally, one may also wish to use an auxiliary enzyme not as an assay system but strictly as a means for maintaining the steady-state concentration of a primary reactant in a multisubstrate reaction system. For instance, acetate kinase (and its substrate acetyl phosphate) or creatine kinase (and its substrate creatine phos-

phate) can be utilized to regenerate the concentration of ATP, a substrate of the primary enzyme. **See** *ATP/GTP Regenerating Systems*

¹W. R. McClure (1969) *Biochemistry* **8**, 2782.

²A. Storer & A. Cornish-Bowden (1974) *Biochem. J.* **141**, 205.

³F. B. Rudolph, B. W. Baugher & R. S. Beissner (1979) *Meth. Enzymol.* **63**, 22.

⁴J. S. Easterby (1981) *Biochem. J.* **199**, 155.

⁵S. Y. Yang & H. Schulz (1987) *Biochemistry* **26**, 5579.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Analysis, **63**, 22-30; assumptions, **63**, 23; dehydrogenase, **63**, 31, 32; fluorescence assay, **63**, 37; isozymes, **63**, 30; kinase, **63**, 5, 6; kinetic theory, **63**, 22-30; lag time, **63**, 23-29; levels, **63**, 23, 26, 28, 37-39; model, **63**, 22-30; pH effects, **63**, 225; practical aspects, **63**, 30-39; precautions, **63**, 39-41; progress curve, **63**, 177.

COVALENT BOND

A strongly stabilizing interaction between two atoms in which certain of their atomic orbitals overlap, thereby resulting in a region of high electron density. This sharing of two electrons in the orbital by the two bonded atoms describes a molecular orbital.

COX-YEATS EQUATION

A modification of the Bunnett-Olsen equation concerned with solvent acidity in which $\log([SH^+]/[S]) - \log[H^+] = m^*X + pK_{SH^+}$ where $[S]$ and $[SH^+]$ are the solvent and protonated solvent concentrations, and X is the activity function $\log[(\gamma_S\gamma_{H^+}/\gamma_{SH^+})]$ for an arbitrary reference base. In practice, $X = -(H_o + \log[H^+])$, called the excess acidity (where H_o is the Hammett acidity function, $m^* = 1 - \phi$, and ϕ represents the response of the $S + H^+ \rightleftharpoons SH^+$ equilibrium to changes in the acid concentration). **See** *Acidity Function; Bunnett-Olsen Equation*

R. A. Cox & K. Yates (1978) *J. Amer. Chem. Soc.* **100**, 3861.

R. A. Cox & K. Yates (1981) *Can. J. Chem.* **59**, 2116.

V. Lucchini, G. Modena, G. Scorrano, R. A. Cox & K. Yates (1982) *J. Amer. Chem. Soc.* **104**, 1958.

CRATIC FREE ENERGY

A term introduced by Gurney¹ to describe the contribution to the total free energy of a solution and to free energy changes arising as a result of mixing. The cratic free energy can be regarded as the entropy of mixing, and this contribution to the free energy of binding must be taken into account when attempting to relate the energetics of subsite binding in multisite enzymes with microscopic thermodynamic parameters as well as in the thermodynamics of micelle formation.

¹R. W. Gurney (1953) *Ionic Processes in Solution*, McGraw-Hill, New York.

CREATINE KINASE

This muscle phosphotransferase (EC 2.7.3.2) catalyzes the reversible rephosphorylation of ADP to form ATP (*i.e.*, $K_{eq} = [\text{ATP}][\text{Creatine}]/([\text{Creatine phosphate}][\text{ADP}]) = 30$). In resting muscle, creatine phosphate is synthesized at the expense of abundant stores of ATP; intracellular creatine phosphate stores often reach 50–60 mM. If ATP is suddenly depleted by muscle contraction, its product ADP is immediately converted back into ATP by the reverse of the creatine kinase reaction. Depending on the pH at which the enzyme is studied, the kinetic reaction can be either rapid equilibrium random or rapid equilibrium ordered. *N*-Ethylglycocyamine can also act as a substrate.

J. J. Villafranca & T. Nowak (1992) *The Enzymes*, 3rd ed., **20**, 63.

G. L. Kenyon & G. H. Reed (1983) *Adv. Enzymol.* **54**, 367.

D. L. Purich & R. D. Allison (1980) *Meth. Enzymol.* **64**, 3.

D. C. Watts (1973) *The Enzymes*, 3rd ed., **8**, 384-455.

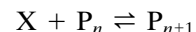
Selected entries from *Methods in Enzymology* [vol, page(s)]:

Abortive complex formation, **63**, 432; adenylate cyclase assay, **79**, 165; coupled enzyme assay, **63**, 32; inhibition, **63**, 463 [competitive, **63**, 292; tight-binding, **63**, 463]; mechanism, **64**, 7, 23, 32, 33, 39; metal ion activation, **63**, 258, 263, 264, 275-278, 292, 327-329; (2'-5')-oligoadenylate synthetase assay, **79**, 157, 158; pH effect, **63**, 280-283; creatine kinase isoenzymes [clinical significance, **74**, 198-199; hybridization, **74**, 201; normal plasma levels, **74**, 199; properties, **74**, 198; purification, **74**, 200-202]; radioimmunoassay, **74**, 198-209; ATP-regeneration system, **238**, 34-35; contamination with calmodulin, **238**, 76; use in photoactivatable probe synthesis, **237**, 88-89, 93.

CRITICAL CONCENTRATION

The threshold concentration of monomer that must be exceeded for any observable polymer formation in a self-assembling system. In the context of Oosawa's condensation-equilibrium model for protein polymerization, the cooperativity of nucleation and the intrinsic thermodynamic instability of nuclei contribute to the sudden onset of polymer formation as the monomer concentration reaches and exceeds the critical concentration. Condensation-equilibrium processes that exhibit critical concentration behavior *in vitro* include F-actin formation from G-actin, microtubule self-assembly from tubulin, and fibril formation from amyloid β protein. Critical concentration behavior will also occur in indefinite isodesmic polymerization reactions that involve a stable template. One example is the elongation of microtubules from centrosomes, basal bodies, or axonemes.

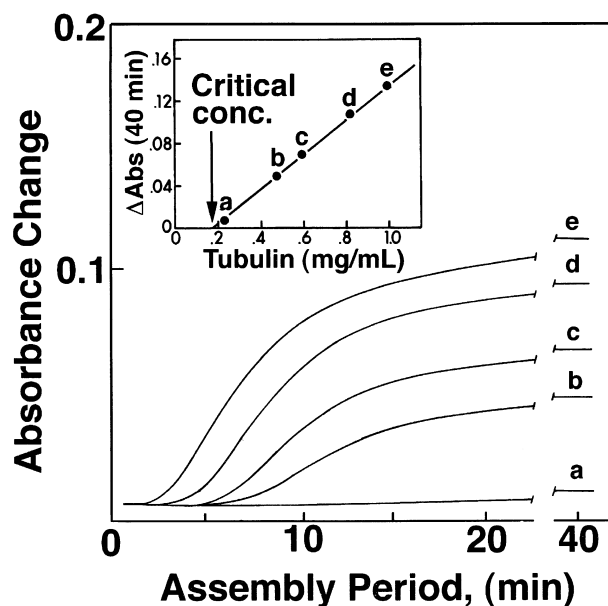
The condensation equilibrium model allows us to consider the equilibrium between unpolymerized protein X and its polymeric form P,



where n and $n + 1$ indicate the total number of assembled units of protein. In this process, the rate of monomer addition is governed by the rate constant k_{on} , and the rate of monomer release depends on the magnitude of k_{off} . If both rate constants are taken to be independent of the length of the polymer, then the kinetics for monomer addition/release from a large collection of polymers of varying lengths will be:

$$\frac{d[\text{X}]}{dt} = k_{on}[\text{X}] \sum_{i=1}^m [\text{P}_i] - k_{off} \sum_{i=1}^{m+1} [\text{P}_{i+1}]$$

The summations are made over all polymer lengths from one to a maximal length m and typically represents the polymer weight concentration; *i.e.*, the total amount of monomer present as polymeric species. Thus the two summations are alternate expressions of $[\text{P}]$, the approximate concentration of polymer ends. At equilibrium, $d\text{X}/dt = 0$, and the dissociation constant ($K_d = k_{off}/k_{on}$) reduces to $[\text{X}]$, which is the concentration of monomer that coexists with polymer at equilibrium. We chose to use the infinity sign as a subscript to distinguish this equilibrium concentration of monomer (*i.e.*, $[\text{M}]_\infty$) from that concentration $[\text{M}]$ that may be present at any other extent of reaction.



We can consider three situations: (a) if $[M]_{\infty} > [M]_{\infty}$, there will be a net increase in polymerization; (b) if $[M] < [M]_{\infty}$, there will be net depolymerization; and (c) when $[M] = [M]_{\infty}$, the monomer and polymer are in equilibrium. The last case does not imply a static condition; rather, monomer addition and loss can constantly occur, but the total polymer weight concentration will always remain unchanged. In head-to-tail polymerizations, like those of actin and tubulin assembly *in vitro*, each of the two polymer ends can interact with monomer; the critical concentration $[M]_{\infty}$ then equals $(k_{\text{off}}^{+} + k_{\text{off}}^{-}) / (k_{\text{on}}^{+} + k_{\text{on}}^{-})$, where the superscripts indicate the plus and minus ends of the assembled polymers. **See** *Actin Assembly Kinetics; Microtubule Assembly Kinetics; End-wise Depolymerization*

CRITICAL MICELLE CONCENTRATION

The threshold concentration above which a free amphiphile will self-assemble and spontaneously form micelles. Below this concentration, no micelles are detectable. Note that micelle formation is not equivalent to a phase separation¹. For most systems, the concentration range associated with the monomer-to-micelle transition is fairly narrow, and c.m.c. values can be obtained from graphical procedures. The symbol for c.m.c. is c_m . Whenever c.m.c. values are reported, the method for the determination should be clearly stated, because different physical techniques may be more or less sensitive to changes in the amphiphile's aggregation state. **See** *Micelle Formation*

¹C. Tanford (1973) *The Hydrophobic Effect*, Wiley-Interscience, New York.

CRITICAL MICELLE TEMPERATURE

The temperature, abbreviated c.m.t., at which a detergent/solvent system or a lipid/solvent system passes from a hydrated crystalline state to an isotropic micellar solution. For a number of lipids, the c.m.t. is below the freezing point of the solvent. The Krafft point, T_k , is the c.m.t. at the critical micelle concentration.

CROSS-CONJUGATION

A form of p orbital overlap in certain types of molecules in which two groups of moieties are not conjugated with each other yet both are conjugated with a third.

An example of a cross-conjugated system would be $C=C-C(=O)-C=C$.

N. F. Phelan & M. Orchin (1968) *J. Chem. Educ.* **45**, 633.

CROSSOVER THEOREM

A principle in metabolic regulation that allows one to identify the inhibited step within a metabolic pathway as that reaction for which the concentrations of reactants and products rise and fall, respectively, from their steady-state values when an inhibitor is introduced. In the context of the electron transfer chain, the crossover-point refers to that reaction step demarking the transition from more reduced to more oxidized respiratory enzymes.

CROSS-REACTION

The binding interaction between an antibody (or antiserum) and a substance that was not the original immunogen employed in the antibody's production. The interaction results from the presence of one or more binding determinants (called epitopes), and at least one of two cases holds true: (1) the substance has an epitope that mimicks the conformation and/or electrostatic properties of the original epitope; or (2) the original immunogen was contaminated with another substance giving rise to the cross-reactivity. Because each antibody has its own well-defined epitope binding site, the second case above is unlikely with monoclonal antibodies. Should a monoclonal Ab show cross-reactivity, the first case is the most reasonable explanation.

CROTONASE SUPERFAMILY

Members of this superfamily catalyze coenzyme A thiolester reactions requiring a stabilized oxyanion intermediate. Babbitt and Gerlt¹ point out that an enolate anion is generated by abstraction of the α -proton of the thiolester in reactions, resulting in β -addition or elimination of water (crotonase and carnitine racemase), 1,3-proton transfer (3,2-*trans*-enoyl-CoA isomerase), and carbon-carbon bond formation (naphthoate synthase, intramolecular addition of the enolate anion to a carboxyl group). An enolate anion is also generated by nucleophilic aromatic addition in the reaction catalyzed by 4-chlorobenzoyl-CoA dehalogenase. **See** *Enoyl-CoA Hydratase*

¹P. C. Babbitt & J. A. Gerlt (1997) *J. Biol. Chem.* **272**, 30591.

CRYOENZYMOLOGY

The application of very low temperatures to detect, to thermally trap, and to characterize intermediates in enzyme-catalyzed reactions¹⁻⁵. This is made possible by the fact that each individual, elementary step in a reaction pathway has its own activation energy. Lowering the temperature reduces the fraction of molecules that can react in certain steps, thereby permitting otherwise reactive species to accumulate.

Because cryosolvents must be used in studies of biochemical reactions in water, it is important to recall that the dielectric constant of a solution increases with decreasing temperature. Fink and Geeves¹ describe the following steps: (1) preliminary tests to identify possible cryosolvent(s); (2) determination of the effect of cosolvent on the catalytic properties; (3) determination of the effect of cosolvent on the structural properties; (4) determination of the effect of subzero temperature on the catalytic properties; (5) determination of the effect of subzero temperature on the structural properties; (6) detection of intermediates by initiating catalytic reaction at subzero temperature; (7) kinetic, thermodynamic, and spectral characterization of detected intermediates; (8) correlation of low-temperature findings with those under "normal" conditions; and (9) structural studies on trapped intermediates.

¹A. L. Fink & M. A. Geeves (1979) *Meth. Enzymol.* **63**, 336.

²A. L. Fink (1977) *Acc. Chem. Res.* **10**, 233.

³P. Douzou (1977) *Cryobiochemistry*, Academic Press, New York.

⁴P. Douzou (1977) *Adv. Enzymol.* **45**, 157.

⁵R. J. Coll, P. D. Compton & A. L. Fink (1982) *Meth. Enzymol.* **87**, 66.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Theory, **63**, 340-352; measurement, **63**, 365; cryosolvent [catalytic effect, **63**, 344-346; choice, **63**, 341-343; dielectric constant, **63**, 354; electrolyte solubility, **63**, 355, 356; enzyme stability, **63**, 344; pH measurements, **63**, 357, 358; preparation, **63**, 358-361; viscosity effects, **63**, 358]; intermediate detection, **63**, 349, 350; mixing techniques, **63**, 361, 362; rapid reaction techniques, **63**, 367-369; temperature control, **63**, 363-367; temperature effect on catalysis, **63**, 348, 349; temperature effect on enzyme structure, **63**, 348.

CRYPTANDS

Monocyclic or polycyclic molecules containing binding sites for one or more so-called guest molecules or ions. Examples include the crown ethers as well as numerous macrocycles containing other electronegative atoms besides oxygen. **See** *Inclusion Complexes; Host-Guest Interactions; Cyclodextrins*

J.-M. Lehn (1988) *Angew. Chem. Int., Ed. Engl.* **27**, 90.

P.G. Potvin & J.-M. Lehn (1987) *Prog. Macrocycl. Chem.* **3**, 167.

W. Kiggen & F. Vögtle (1987) *Prog. Macrocycl. Chem.* **3**, 309.

CRYPTATE EFFECT

An effect arising from the binding of a ligand to a cryptand: namely, high stability, high selectivity, slow exchange rates, and shielding of the substrate or guest from the environment. One simple example of a cryptate effect is the binding of NH_4^+ to a protonated polyamine cryptand. The $\text{p}K_a$ value of the NH_4^+ -cryptate is about six pH units higher than that of free NH_4^+ .

J.-M. Lehn (1988) *Angew. Chem. Int. Ed. Engl.* **27**, 89.

CRYPTIC CATALYSIS

A kinetic behavior first proposed by Wedler and Boyer¹ based on their studies of modifier effects on bacterial glutamine synthetase. They noted that particular effectors inhibited certain equilibrium exchanges, while other enzymic exchange reactions remained largely unaffected. They suggested that isotope-exchange-at-equilibrium experiments with the appropriate metabolic modifier may divulge characteristics of the catalytic process not typically observable in typical initial-rate studies (hence, use of the term cryptic catalysis). Nonetheless, before one attributes such behavior to cryptic catalysis, great care should be exercised whenever the effector can also act as an alternative substrate or when impurities are present (*e.g.*, the presence of GTP in the GDP stock solutions or *vice versa*). **See** *Isotope Exchange at Equilibrium*

¹F. C. Wedler & P. D. Boyer (1972) *J. Biol. Chem.* **247**, 993.

CRYPTIC STEREOCHEMISTRY

The stereochemical course of a reaction at an achiral center which can be only observed by utilizing isotopic substitution (*i.e.*, a procedure for converting the achiral group into a chiral moiety).

K. R. Hanson & I. A. Rose (1975) *Acc. Chem. Res.* **8**, 81.

H. G. Floss (1982) *Meth. Enzymol.* **87**, 126.

CRYSTAL FIELD SPLITTING

Loss of degeneracy of energy levels of molecular entities due to lower site symmetry created by a crystalline environment; this term is not synonymous with ligand field splitting¹. **See** *Ligand Field Splitting*

¹Comm. on Photochem. (1988), *Pure and Appl. Chem.* **60**, 1055.

Selected entries from *Methods in Enzymology* [vol, page(s)]:
 Degenerate perturbation treatment, **246**, 84; ligand position functions, **246**, 81-82; matrix elements, **246**, 81-83; octahedral field potential, **246**, 80; potential energy term, **246**, 78-79; re-source materials, **246**, 16; tetrahedral field, **246**, 81

CRYSTAL FIELD THEORY

A systematic approach for rationalizing the properties of coordination entities by assuming that ligands interact with the central metal atom solely by electrostatics, with ligands as negative point charges surrounding an electro-positive central atom. No covalent bonding is assumed to occur. Crystal field splitting refers to separation of a transition metal ion's five degenerate *d* orbitals into energetically different orbitals containing paired or unpaired electrons.

CTP:ETHANOLAMINEPHOSPHATE CYTIDYLYLTRANSFERASE

This enzyme [EC 2.7.7.14], also referred to as ethanolamine-phosphate cytidylyltransferase and phosphoryl-ethanolamine transferase, catalyzes the reaction of CTP with ethanolamine phosphate to produce CDP-ethanolamine and pyrophosphate.

L. M. Tijburg, P. S. Vermeulen & L. M. G. van Golde (1992) *Meth. Enzymol.* **209**, 258.

CTP SYNTHETASE

This enzyme [EC 6.3.4.2], also referred to as UTP:ammonia ligase, catalyzes the reaction of UTP with ATP and ammonia (or glutamine) to produce CTP, ATP, and orthophosphate (and glutamate if glutamine had been used as the nitrogen source).

H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.

H. Zalkin (1985) *Meth. Enzymol.* **113**, 282.

D. E. Koshland, Jr., & A. Levitzki (1974) *The Enzymes*, 3rd ed., **10**, 539.

J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

CUMULATIVE DISTRIBUTION FUNCTION

A distribution function for a random variable *X* defined as:

$$F(x) = P(X \leq x)$$

where *x* is any real number and *F(x)* is a function. The distribution function is obtained from the probability function by noting that:

$$F(x) = P(X \leq x) = \sum f(u)$$

where the sum is taken over all values of *u* for which *u* is less than or equal to *x*.

CUMULATIVE FEEDBACK INHIBITION

A proposal for the regulation of bacterial glutamine synthetase by end products containing nitrogen atoms derived from the amido group of glutamine^{1,2}. Such regulatory behavior would allow feedback inhibitors to throttle the enzyme's catalytic activity without completely arresting glutamine biosynthesis, unless all of the end products reached saturating concentrations. Each feedback inhibitor was believed to bind at its own topologically distinct regulatory site on the synthetase, and this incorrect model has become a standard entry in general biochemistry textbooks. Dahlquist & Purich³ demonstrated that no special sites are involved, and X-ray crystallography has verified their suggestion that many ligands bind directly at the active site⁴. Purich⁵ recently presented a critical analysis of the cumulative inhibition of glutamine synthetase.

¹C. A. Woolfolk & E. R. Stadtman (1964) *Biochem. Biophys. Res. Commun.* **17**, 313.

²C. A. Woolfolk & E. R. Stadtman (1967) *Arch. Biochem. Biophys.* **118**, 736.

³F. W. Dahlquist & D. L. Purich (1975) *Biochemistry* **14**, 1980.

⁴S. H. Liaw, C. Pan & D. S. Eisenberg (1993) *Proc Natl. Acad. Sci. U.S.A.* **90**, 4996.

⁵D. L. Purich (1998) *Adv. Enzymol.* **72**, 9.

CURIE

(1) A unit of measurement of radioactivity; one curie (Ci) is equal to 3.70×10^{10} disintegrations per second (dps) or 3.70×10^{10} becquerels (Bq). (2) Formerly, the amount of radioactivity equal to the amount of radon in equilibrium with one gram of radium.

CURVE FITTING

Any statistical procedure that defines a set of experimental data in terms of a mathematical function. The most common curve-fitting protocol is the least squares method (also known as analysis of covariance).

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Computer programs, **240**, 312; infrared S-H stretch bands for hemoglobin A, **232**, 159-160; determination of enzyme kinetic parameter, **240**, 314-319; kinetics program, in finite element analysis of hemoglobin-CO reaction, **232**, 523-524, 538-558; nonlinear least-squares method, **240**, 3-5, 10; to oxygen equilibrium curve, **232**, 559, 563; parameter estimation with Jacobians, **240**, 187-191.

β -CYANOALANINE SYNTHASE

This enzyme [EC 4.4.1.9] catalyzes the reaction of cysteine with cyanide to produce H_2S and 3-cyanoalanine.

A. E. Braunstein & E. V. Goryachenkova (1984) *Adv. Enzymol.* **56**, 1.

3',5'-CYCLIC AMP

This prototypical second messenger exerts excitatory control in cells principally by stimulating protein phosphorylation through the activation of kinase cascades.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Radioimmunoassay, **73**, 109; 79, 163, 164; binding assays, **65**, 859, 860; **237**, 394-397; protein binding assays, **238**, 90; ^3H -labeled cAMP [formation from ^3H adenine, **195**, 22; chromatographic separation, **195**, 27]; ^{32}P -labeled [synthesis, **195**, 35; purification, **195**, 40]; radio-immunoassay assays with tritiated adenine, **238**, 91-92; binding, GTP inhibition, **237**, 399-400; isotope dilution assays, **237**, 390-394; phospholipase C stimulation, **238**, 212-213; purification, **238**, 45-47, 49-52, 92-94; second messenger responses, **237**, 398-399.

3',5'-CYCLIC-AMP PHOSPHODIESTERASE

This enzyme [EC 3.1.4.17] catalyzes the hydrolysis of a nucleoside 3',5'-cyclic phosphate to produce a nucleoside 5'-phosphate. The enzyme can utilize 3',5'-cyclic AMP, 3',5'-cyclic dAMP, 3',5'-cyclic IMP, 3',5'-cyclic GMP, and 3',5'-cyclic CMP as substrates.

CYCLODEXTRIN

A host-type molecule (also called cycloamyloses and Schardinger dextrans) consisting of six (α -cyclodextrin), seven (β -cyclodextrin), or eight (γ -cyclodextrin) glucosyl residues interlinked in a large ring by glycosidic bonds. The structure formed is a hollow, truncated cone. The interior portion of the cone is less polar than the exterior and the cyclodextrins (CDs) can act as hosts to many different kinds of guests. Covalently modified cyclodextrins have served as models for enzyme catalysis¹⁻⁷. In addition, if cyclodextrins are stacked, they can act as channel-type complexes.

For additional information on cyclodextrins, reader should consult the indicated pages in Robyt⁸: properties of the α -, β -, γ -, δ -, and ϵ -cyclodextrins, p. 248; solubilities of α -, β -, and γ -cyclodextrins, p. 249; cyclic β -1 $\rightarrow$ 2-glucans, pp. 251-252; cyclic β -1 $\rightarrow$ 6- and β -1 $\rightarrow$ 3-glucans, pp. 252-254; cyclomaltodextrins, pp. 254-255; cycloalterotetraose, pp. 255-256; cyclonulodextrin, p. 256; synthesis of cyclodextrins, pp. 256-258; and macrocyclic maltodextrins, p. 258.

¹M. L. Bender & M. Komiyama (1978) *Cyclodextrin Chemistry*, Springer, New York.

²W. Saenger (1984) in *Inclusion Compounds* (J. L. Atwood, J. E. Davies & D. D. MacNicol, eds.), vol. **2**, pp. 231-259, Academic Press, New York.

³R. Bergeron (1984) in *Inclusion Compounds* (J. L. Atwood, J. E. Davies & D. D. MacNicol, eds.), vol. **3**, pp. 391-443, Academic Press, New York.

⁴I. Tabushi (1984) in *Inclusion Compounds* (J. L. Atwood, J. E. Davies & D. D. MacNicol, eds.), vol. **3**, pp. 445-471, Academic Press, New York.

⁵R. Breslow (1984) in *Inclusion Compounds* (J. L. Atwood, J. E. Davies & D. D. MacNicol, eds.), vol. **3**, pp. 473-508, Academic Press, New York.

⁶A. P. Croft & R. A. Bartsch (1983) *Tetrahedron* **39**, 1417.

⁷I. Tabushi & Y. Kuroda (1983) *Adv. Catal.* **32**, 417.

⁸J. F. Robyt (1998) *Essentials of Carbohydrate Chemistry*, Springer, New York.

CYCLODEXTRIN GLYCOSYLTRANSFERASE

Cyclodextrins are formed in the degradation of starch and dextran by the action of cyclodextrin/glucanoglycosyltransferases [EC 2.4.1.19]. These enzymes also catalyze so-called acceptor reactions in which the cyclodextrin ring is opened and an acceptor molecule (e.g., glucose) is added to the reducing end of the maltodextrin chain.

J. F. Robyt (1998) *Essentials of Carbohydrate Chemistry*, Springer, New York.

I. S. Pretorius, M. G. Lambrechts & J. Marmur (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 53.

M. Vihinen & P. Mäntsälä (1989) *Crit. Rev. Biochem. Mol. Biol.* **24**, 329.

CYCLOEUCALENOL-OBTUSIFOLIOL ISOMERASE

This enzyme [EC 5.5.1.9], also called cycloeucalenol cycloisomerase, catalyzes the interconversion of cycloeucalenol to obtusifoliol.

P. Benveniste (1986) *Ann. Rev. Plant Physiol.* **37**, 275.

CYCLOHEXADIENYL DEHYDROGENASE

This enzyme [EC 1.3.1.43], also referred to as arogenate dehydrogenase and pretyrosine dehydrogenase, catalyzes the reaction of arogenate with NAD^+ to produce tyrosine, NADH, and carbon dioxide. Both prephenate and D-prephenyllactate can act as alternative substrates.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

R. Fischer & R. Jensen (1987) *Meth. Enzymol.* **142**, 488.

CYCLOHEXANONE MONOOXYGENASE

This enzyme [EC 1.14.13.22] catalyzes the reaction of cyclohexanone with NADPH and dioxygen to produce 6-hexanolide, NADP^+ , and water. FAD is used as a co-

factor. A number of other cyclic ketones can serve as substrates.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

CYCLOPENTANONE 1,2-MONOOXYGENASE

This enzyme [EC 1.14.13.16] catalyzes the reaction of cyclopentanone with NADPH and dioxygen to produce 5-valerolactone, NADP⁺, and water [EC 1.14.13.16].

P. W. Trudgill (1990) *Meth. Enzymol.* **188**, 77.

CYSTATHIONINE β -LYASE

This pyridoxal-phosphate-dependent enzyme [EC 4.4.1.8], also referred to as β -cystathionase and cystine lyase, catalyzes the hydrolysis of cystathionine to yield homocysteine, pyruvate, and ammonia.

J. Giovanelli (1987) *Meth. Enzymol.* **143**, 443.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

CYSTATHIONINE γ -LYASE

This pyridoxal-phosphate-dependent enzyme [EC 4.4.1.1] (also referred to as homoserine deaminase, homoserine dehydratase, γ -cystathionase, cystine desulfhydrase, cysteine desulfhydrase, and cystathionase) catalyzes the hydrolysis of cystathionine to produce cysteine, ammonia, and α -ketobutanoate (or, 2-oxobutanoate).

T. Nagasawa, H. Kanzaki & H. Yamada (1987) *Meth. Enzymol.* **143**, 486.

A. E. Braunstein & E. V. Goryachenkova (1984) *Adv. Enzymol.* **56**, 1.

A. E. Martell (1982) *Adv. Enzymol.* **53**, 163.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

CYSTATHIONINE β -SYNTHASE

This pyridoxal-phosphate-dependent enzyme [EC 4.2.1.22] (also known as serine sulfhydrase, β -thionase, and methylcysteine synthase) catalyzes the reaction of homocysteine with serine to produce cystathionine and water.

J. P. Kraus (1987) *Meth. Enzymol.* **143**, 388.

A. E. Braunstein & E. V. Goryachenkova (1984) *Adv. Enzymol.* **56**, 1.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

CYSTEAMINE DIOXYGENASE

This enzyme [EC 1.13.11.19] catalyzes the reaction of cysteamine with dioxygen to produce hypotaurine. The protein uses iron ions as cofactors.

R. B. Richerson & D. M. Ziegler (1987) *Meth. Enzymol.* **143**, 410.

A. J. L. Cooper (1983) *Ann. Rev. Biochem.* **52**, 187.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

CYSTEINE S-CONJUGATE N-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.80] catalyzes the reaction of acetyl-CoA with an S-substituted L-cysteine to produce coenzyme A and an S-substituted N-acetyl-L-cysteine.

M. W. Duffel & W. B. Jakoby (1985) *Meth. Enzymol.* **113**, 516.

CYSTEINE S-CONJUGATE β -LYASE

This pyridoxal-phosphate-dependent enzyme [EC 4.4.1.13] catalyzes the conversion of an S-substituted cysteine (that is, RS-CH₂-CH(NH₃⁺)COO⁻) to RSH, ammonia, and pyruvate. **See also** S-Substituted Cysteine Sulfoxide Lyase

J. L. Stevens & W. B. Jakoby (1985) *Meth. Enzymol.* **113**, 510.

D-CYSTEINE DESULFHYDRASE

This enzyme [EC 4.4.1.15] catalyzes the hydrolysis of D-cysteine to produce pyruvate, ammonia, and H₂S.

A. Schmidt (1987) *Meth. Enzymol.* **143**, 449.

CYSTEINE DIOXYGENASE

This enzyme [EC 1.13.11.20] catalyzes the reaction of cysteine with dioxygen to yield 3-sulfinioalanine. Both iron ions and NAD(P)H are used as cofactors.

K. Yamaguchi & Y. Hosokawa (1987) *Meth. Enzymol.* **143**, 395.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

CYSTEINE LYASE

This pyridoxal-phosphate-dependent enzyme [EC 4.4.1.10] catalyzes the reaction cysteine with sulfite to produce cysteate and H₂S. The enzyme can also catalyze the reaction of two cysteines (thereby producing lantionine) as well as other alkyl thiols as substrates.

A. E. Braunstein & E. V. Goryachenkova (1984) *Adv. Enzymol.* **56**, 1.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

CYSTEINE SULFINATE DECARBOXYLASE

This pyridoxal-phosphate-dependent enzyme [EC 4.1.1.29], also referred to as sulfinioalanine decarboxylase, catalyzes the conversion of 3-sulfinio-L-alanine to

hypotaurine and carbon dioxide. The enzyme can also utilize L-cysteate as a substrate.

C. L. Weinstein & O. W. Griffith (1987) *Meth. Enzymol.* **143**, 404.
E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

CYTIDINE DEAMINASE

This enzyme [EC 3.5.4.5], also called cytidine aminohydrolase, catalyzes the hydrolysis of cytidine to produce uridine and ammonia.

A. Radzicka & R. Wolfenden (1995) *Meth. Enzymol.* **249**, 284.

CYTIDYLATE KINASE

This enzyme [EC 2.7.4.14], also called deoxycytidylate kinase, catalyzes the reaction of CMP with ATP to form CDP and ADP, as well as similar reactions with dCMP, UMP, and dUMP.

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

CYTOCHROME *c*

This heme-dependent peripheral membrane protein is a member of the electron-transport chain, and shuttles electrons between components of Complexes III and IV.

H. B. Gray & J. R. Winkler (1996) *Ann. Rev. Biochem.* **65**, 537.
R. J. Maier (1996) *Adv. Protein Chem.* **48**, 35.
G. R. Moore & G. W. Pettigrew (1990) *Cytochromes c. Evolutionary, Structural and Physicochemical Aspects*, Springer-Verlag, New York.
D. C. Rees & D. Farrelly (1990) *The Enzymes*, 3rd ed., **19**, 37.
A. A. DiSpirito (1990) *Meth. Enzymol.* **188**, 289.
T. E. Meyer & M. D. Kamen (1982) *Adv. Protein Chem.* **35**, 105.
R. E. Dickerson & R. Timkovich (1975) *The Enzymes*, 3rd ed., **11**, 397.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Acid-denatured, conformation, **232**, 3; acid-induced refolding, **232**, 5-6; properties, **234**, 342-343; synthesis, **234**, 339-342; as chemiluminescence catalyst for luminol HPLC hydroperoxide assay, **233**, 325-327; engineered metal-binding sites, in metal affinity partitioning, **228**, 178-179; amino acid composition, **243**, 159-160; amino acid sequence, **243**, 163-165; domain structure, **243**, 164-165; heme-binding sites, **243**, 163-165; heme ligands, **243**, 164-165; histidine residues, **243**, 163-165; properties, **243**, 158-162; redox properties, **243**, 160-161; spectral properties, **243**, 158-159; spin state, **243**, 161; HETCOR experiment with low-spin iron(III), **239**, 500; metal affinity partitioning, **228**, 173; reduction [NADPH oxidase assay with, **233**, 223-225; by superoxide, **233**, 155]; superfamily, **243**, 104-105.

CYTOCHROME *c*₃ HYDROGENASE

This enzyme [EC 1.12.2.1], also called hydrogenase, catalyzes the reaction of H₂ with two ferricytochrome *c*₃ to produce two H⁺ and two ferrocycytochrome *c*₃. The enzyme uses iron ions as well as a cofactor. Methylene blue

and other acceptors can be reduced with H₂ with this system. **See also** *Hydrogenase; Hydrogen Dehydrogenase*

M. A. Halcrow (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 359.

CYTOCHROME *c* OXIDASE

This enzyme [EC 1.9.3.1] (also referred to as cytochrome *aa*₃ and cytochrome oxidase) catalyzes the reaction of four ferrocycytochrome *c* with dioxygen to produce four ferricytochrome *c* and two water molecules. This protein also contains copper ions as cofactors.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.
A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.
H. B. Gray & J. R. Winkler (1996) *Ann. Rev. Biochem.* **65**, 537.
R. J. Maier (1996) *Adv. Protein Chem.* **48**, 35.
R. O. Poyton, B. Goehring, M. Droste, K. A. Sevarino, L. A. Allen & X.-J. Zhao (1995) *Meth. Enzymol.* **260**, 97.
R. A. Capaldi, M. F. Marusich & J.-W. Taanman (1995) *Meth. Enzymol.* **280**, 117.
S. M. Musser, M. H. B. Stowell & S. I. Chan (1995) *Adv. Enzymol.* **71**, 79.
B. L. Trumpower & R. B. Gennis (1994) *Ann. Rev. Biochem.* **63**, 675.
E. T. Adman (1991) *Adv. Protein Chem.* **42**, 145.
W. S. Caughey, W. J. Wallace, J. A. Volpe & S. Yoshikawa (1976) *The Enzymes*, 3rd ed., **13**, 299.

CYTOCHROME *c* PEROXIDASE

This enzyme [EC 1.11.1.5], having a heme derivative as a cofactor, catalyzes the reaction of two ferrocycytochrome *c* with hydrogen peroxide to produce two ferricytochrome *c* and two water molecules.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.
J. S. Zhou & B. M. Hoffman (1994) *Science* **265**, 1693.
H. Pelletier & J. Kraut (1992) *Science* **258**, 1748.
D. C. Rees & D. Farrelly (1990) *The Enzymes*, 3rd ed., **19**, 37.
T. E. Meyer & M. D. Kamen (1982) *Adv. Protein Chem.* **35**, 105.
T. L. Poulos & J. Kraut (1980) *J. Biol. Chem.* **255**, 10322.
T. Yonetani (1976) *The Enzymes*, 3rd ed., **13**, 345.

CYTOCHROME P-450

A superfamily of heme-dependent monooxygenases that utilize molecular oxygen and NADPH. These enzymes are localized in the endoplasmic reticulum, and are often used as a marker for microsomal fractions obtained upon homogenization of cells. Especially abundant in liver, cytochrome P-450 enzymes play a major role in detoxifi-

cation. They also catalyze the first step in ω -oxidation of medium- and long-chain fatty acids.

- H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.
 Y. Watanabe & J. T. Groves (1992) *The Enzymes*, 3rd ed., **20**, 405.
 F. P. Guengerich (1991) *J. Biol. Chem.* **266**, 10019.
 D. C. Rees & D. Farrelly (1990) *The Enzymes*, 3rd ed., **19**, 37.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 S. D. Black & M. J. Coon (1987) *Adv. Enzymol.* **60**, 35.
 P. Ortiz de Montellano, ed., (1986) *Cytochrome P-450: Structure, Mechanism, and Biochemistry*, Plenum, New York.
 M. J. Coon & D. R. Koop (1983) *The Enzymes*, 3rd ed., **16**, 645.
 I. C. Gunsalus & S. G. Sligar (1978) *Adv. Enzymol.* **47**, 1.
 V. Ullrich & W. Duppel (1975) *The Enzymes*, 3rd ed., **12**, 253.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Activity in ethanol oxidation, **233**, 118; in hydroxyethyl radical formation; analysis with reconstituted vesicles, **233**, 127; characterization, **233**, 123-125; monooxygenase activity, **231**, 574-575; reductase [hemoglobin-catalyzed reactions with, **231**, 573-574; oxygen concentration and, **231**, 579; pH dependence, **231**, 580; reaction mixtures, **231**, 578; oxygen content, measurement, **231**, 587-588; reductase concentration and, **231**, 580; time dependence, **231**, 578; preparation, **231**, 577].

CYTOCHROME P-450 REDUCTASE

This enzyme [EC 1.6.2.4] (also referred to as NADPH:ferrihemoprotein reductase, NADPH:cytochrome P450 reductase, TPNH₂ cytochrome *c* reductase, and ferrihemoprotein P450 reductase) catalyzes the reaction of NADPH with two ferricytochrome to produce NADP⁺ and two ferrocytochrome. The protein requires FMN and FAD. In addition, it also catalyzes the reduction of heme-thiolate-dependent monooxygenases (*e.g.*,

the unspecific monooxygenase [EC 1.14.14.1]) and it is a part of the microsomal hydroxylating system. This reductase also is capable of reducing cytochrome *b*₅ and cytochrome *c*.

- B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

β -(9-CYTOKININ)-ALANINE SYNTHASE

This enzyme [EC 4.2.99.13], also known as lupinic acid synthase, catalyzes the reaction of *O*-acetyl-L-serine with zeatin to produce lupinate and acetate. Zeatin is *N*⁶-(4-hydroxy-3-methyl-butyl-*trans*-2-enylamino)purine. A number of other *N*⁶-substituted purines can function as substrates for this enzyme.

- D. S. Letham & L. M. S. Palmi (1983) *Ann. Rev. Plant Physiol.* **34**, 163.

CYTOKININ 7 β -GLUCOSYLTRANSFERASE

This enzyme [EC 2.4.1.118], also known as UDP-glucose-zeatin 7-glucosyltransferase, catalyzes the reaction of UDP-glucose with an *N*⁶-alkylaminopurine to produce UDP and an *N*⁶-alkylaminopurine-7- β -D-glucoside. The enzyme can act on a number of *N*⁶-substituted adenine derivatives (*e.g.*, zeatin and *N*⁶-benzylaminopurine). However, it has been reported that *N*⁶-benzyladenine does not serve as a substrate. Depending on the substrate, a 9- β -D-glucoside may be the product formed.

- D. S. Letham & L. M. S. Palmi (1983) *Ann. Rev. Plant Physiol.* **34**, 163.

D

DAKIN-WEST REACTION

A chemical reaction in which the carboxyl group of an α -amino acid is replaced by an acyl group (*i.e.*, an acyl-decarboxylation). In this reaction, an α -amino acid is reacted with an anhydride in the presence of pyridine and the resulting product is an *N*-acylated ketone. In certain instances, *N*-substituted amino acids will also act as reactants.

H. D. Dakin & R. West (1928) *J. Biol. Chem.* **78**, 745.
G. L. Buchanan (1988) *Chem.Soc. Rev.* **17**, 91.

DARVEY PLOT

A graphical procedure for characterizing isomerization mechanisms^{1,2}. The protocol uses data from product inhibition, and $1/[v_{[P]=0} - v_{[P]_0}]$ is plotted *versus* $1/[P]$ at various constant concentrations of the substrate (where $v_{P=0}$ is the initial velocity in the absence of product and $v_{[P]_0}$ is the initial velocity in the presence of product). Secondary and ternary replots allows one to characterize the mechanism^{1,2}. This procedure requires very accurate estimation of initial rates.

¹I. G. Darvey (1972) *Biochem. J.* **128**, 383.

²K. L. Rebholz & D. B. Northrop (1995) *Meth. Enzymol.* **249**, 211.

DAVIES EQUATION

An equation for estimating how the activity coefficient of an ion in dilute solutions is influenced by ionic strength. **See** *Debye-Hückel treatment*

C. W. Davies (1962) *Ion Association*, Butterworth, London.

dCMP DEAMINASE

This enzyme [EC 3.5.4.12], also called deoxycytidylate deaminase, catalyzes the hydrolysis of dCMP to produce

dUMP and ammonia. A number of 5-substituted dCMPs can also serve as substrates.

P. Reichard (1988) *Ann. Rev. Biochem.* **57**, 349.

DEAD-END INHIBITION

Decreased enzymatic activity observed when complexation of substance with an enzyme (or a specific enzyme form in a kinetic mechanism) will not permit catalytic turnover to form product(s), unless that substance dissociates.

DEAD-END INHIBITION (CLELAND'S RULES)

Cleland^{1,2} described the following rules for reversible inhibition patterns observed in double-reciprocal plots of initial rate behavior.

Rule 1. Upon obtaining a double-reciprocal plot of $1/v$ vs. $1/[A]$ (where $[A]$ is the initial substrate concentration and v is the initial velocity) at varying concentrations of the inhibitor (I), if the vertical intercept varies with the concentration of the reversible inhibitor, then the inhibitor can bind to an enzyme form that does not bind the varied substrate. For example, for the simple Uni Uni mechanism ($E + A \rightleftharpoons EX \rightleftharpoons E + P$), a noncompetitive or uncompetitive inhibitor (both of which exhibit changes in the vertical intercept at varying concentrations of the inhibitor), I binds to EX, a form of the enzyme that does not bind free A. In such cases, saturation with the varied substrate will not completely reverse the inhibition.

If the inhibitor can bind more than once to the enzyme form that doesn't bind the varied substrate (*e.g.*, forming EXII in the example above), then a secondary replot (vertical intercepts vs. $[I]$) will be nonlinear, exhibiting

curvature that is concave up. If multiple binding does not occur, then the secondary plot will be linear.

Rule 2. For a plot of $1/v$ vs. $1/[A]$ at varying concentrations of reversible inhibitor I, if the slope of the lines varies with $[I]$, then (a) the inhibitor either binds to the same enzyme form to which the varied substrate binds or (b) the inhibitor binds to an enzyme form capable of altering the concentration of another enzyme form that reacts with the varied substrate. (Note that in case (b), there must be a reversible reaction between these two enzyme forms.) Example of these cases can be seen with the steady-state ordered Bi Bi reaction scheme having the reciprocal rate expression $1/v = (1/V_{\max}) + (K_a + (V_{\max} [A])) + (K_b/(V_{\max} [B])) + (K_{ia}K_b/(V_{\max}[A][B]))$, where $V_{\max}$ is the maximum velocity in the forward direction, K_a is the Michaelis constant for the substrate A, K_b is the Michaelis constant for the substrate B, and K_{ia} is the dissociation constant for A. A competitive inhibitor of A, in a $1/v$ vs. $1/[A]$ plot, will exhibit a change in the slope at different concentrations of I. Note also that the slope will vary with $[I]$ in the $1/v$ vs. $1/[B]$ plot. Here, the inhibitor binds to the free enzyme, which is an enzyme form reversibly connected to the enzyme form binding B (*i.e.*, EA).

Note that this rule requires reversible steps between the appropriate enzyme forms. Fromm³ has pointed out that, for product inhibition with saturating levels of one of the substrates, an effectively irreversible step may exist (*e.g.*, saturating B in the ordered Bi Bi example above: thus, $EA + B \rightleftharpoons EAB$ is essentially irreversible). In this case there would be no slope effect of the product P in the $1/v$ vs. $1/[A]$ plot (at saturating B).

Rule 3. Both Rules 1 and 2 are applied if the reversible inhibitor binds to more than one enzyme form. Secondary plots can be nonlinear. **See Abortive Complexes**

¹W. W. Cleland (1963) *Biochim. Biophys. Acta* **67**, 188.

²K. M. Plowman (1972) *Enzyme Kinetics*, McGraw-Hill, New York.

³H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

DEBRANCHING ENZYME

4 α -Glucanotransferase [EC 2.4.1.25] (also referred to as disproportionating enzyme, dextrin glycosyltransferase, oligo-1,4 $\rightarrow$ 1,4-glucantransferase, and amylomaltase) catalyzes the transfer of a segment of a (1,4)- α -D-glucan to

a new 4-position in an acceptor which may be glucose or a (1,4)- α -D-glucan. This enzymatic activity is also a portion of the mammalian and yeast glycogen debranching system (with an amylo-1,6-glucosidase activity). Amylo-1,6-glucosidase [EC 3.2.1.33], also referred to as dextrin 6- α -D-glucosidase, catalyzes the endohydrolysis of 1,6- α -D-glucoside linkages at points of branching in the polysaccharide chains of 1,4-linked α -D-glucose residues. As mentioned above, in mammals and yeast this enzyme is linked to a glycosyltransferase similar to 4 α -glucanotransferase [EC 2.4.1.25]. Together, these two activities constitute the glycogen debranching system.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

E. Y. C. Lee & W. J. Whelan (1971) *The Enzymes*, 3rd ed., **5**, 191.

DEBYE

The SI unit for the measurement of dipole moments (symbolized by D) equal to 10^{-18} esu-cm or 3.33564×10^{-30} coulomb-meter.

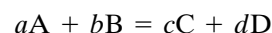
DEBYE-HÜCKEL TREATMENT

A mathematical approach that attempts to explain how the solution behavior of electrolytes depends on their ionic charge and the overall ionic strength of the solution. This treatment allows one to estimate the mean activity coefficients that relate the analytical concentration of an electrolyte to its solution activity. We also can come to understand how the activity coefficient behaves as a function of ionic strength.

Significance of Activity Coefficients. While we typically focus our attention on the analytical concentration of reactant(s) and product(s) for a given chemical process, the thermodynamic concept of equilibrium depends on the chemical potential of a species. This is shown by the following relationship

$$dG = \sum \mu_i dn_i$$

where μ_i is the chemical potential and dn_i is the change in moles of substance i added to (+ sign) or removed from (− sign) a solution. Let us consider making a solution by adding the reactants for the following equilibrium



We initially add reactants A and B in only minute amounts (say A_ϵ and B_ϵ) or by adding products C and

D, again only minute amounts (say $C\varepsilon$ and $D\varepsilon$). In such a case, dG_{forward} and dG_{reverse} can be written as follows

$$dG_{\text{forward}} = \mu_A A\varepsilon + \mu_B B\varepsilon$$

$$dG_{\text{reverse}} = \mu_C C\varepsilon + \mu_D D\varepsilon$$

Equilibrium is attained when dG_{forward} and dG_{reverse} are equal, and

$$\mu_A A\varepsilon + \mu_B B\varepsilon = \mu_C C\varepsilon + \mu_D D\varepsilon$$

We used small amounts of reactants and products to assure that we reached equilibrium, and we now cancel the subscripts in each term to arrive at

$$\mu_A A + \mu_B B = \mu_C C + \mu_D D$$

We can then write out the chemical potential μ_i as

$$\mu_i = \mu_i^\circ + RT \ln[A_i/A_i^\circ]$$

where μ_i° is the standard chemical potential of substance i , R is the universal gas constant in calories/mole or Joules/mole, T is the absolute temperature in units K, A_i is the activity (or the *effective concentration*) of the substance, and A_i° is its activity under standard conditions. In the explanation given above, we added infinitesimal amounts of reactants and products to the solution, but we need to relate activity A_i to concentration $[A]$, because in practice we most often work at higher concentrations. The activity coefficient γ_i is a proportionality factor which changes under different solution conditions, such that

$$A_i = \gamma_i[A]$$

To understand why γ_i depends on solution conditions, we must recognize that the behavior of solutes in solution depends upon the presence of other similar and/or dissimilar solutes, and electrolytes (*i.e.*, charged solutes) are especially affected by the presence of all ionic species in solution. Unless we can account for these effects on the value of α for each substance, we cannot know the effective concentration of a substance at any particular analytical concentration, and we cannot comprehend the thermodynamic properties of these substances in solution. The Debye-Hückel treatment offers us a means for estimating α_i .

The Debye Hückel Limiting Law. Although beyond the scope of this Handbook, the derivation of this quantitative relationship rests on the following simplifying assumptions: (a) Electrolytes are assumed to be completely

dissociated. The ions are taken to have a spherical charge distribution, such that they can be treated as if all their electric charge is localized as a point charge. (b) Interactions between charged solute species are dominated by electrostatic attraction/repulsion, as dictated by Coulomb's Law ($F = q_1 q_2 / Dr^2$), where F is the force of attraction/repulsion, q_1 and q_2 are the charges expressed in electrostatic units, r is the distance between the ions which are considered to be point charges, and D is the dielectric constant of the medium. (c) The long-range interactions are considered to occur in a uniform medium, and the dielectric constant and viscosity of pure water are used in this model. (d) No contribution is made by short-range forces such as ion-dipole interactions, ion pair formation, and van der Waals attractions. (e) Thermal motions of ions disrupt any oriented interionic attractions in a manner that can be defined by the Boltzmann distribution law.

For simple monovalent cations and anions, the Debye-Hückel limiting law can be written as follows:

$$-\log \gamma_i = Az_i^2(I)^{1/2}$$

where γ_i is the activity coefficient, z_i is the charge, I is the ionic strength ($I = (1/2)\sum c_i z_i^2$). The proportionality constant A is written as:

$$A = (1/2.303)(e^2/2Dk_B T)(8\pi e^2 N/1000Dk_B T)^{1/2}$$

where N is Avogadro's number, k_B is the Boltzmann constant, and T is the absolute temperature. The value of A for water is 0.512, such that

$$-\log \gamma_i = 0.512 z_i^2(I)^{1/2}$$

By introducing the mean activity coefficient (γ)

$$(\gamma)^{(z_M + z_N)} = (\gamma_M)^{z_N} (\gamma_N)^{z_M}$$

Taking the negative logarithm of each side, we obtain

$$-(z_M + z_N) \log \gamma = -z_N \log \gamma_M - z_M \log \gamma_N$$

Substituting the Debye-Hückel limiting law into each term on the right-hand side of the equation, and then rearranging, we get:

$$-\log \gamma = 0.512 z_M z_N I^{1/2}$$

where one uses the absolute values of z_M and z_N .

To illustrate how this equation is used, we can calculate the value of γ for a solution of 0.1 F HCl. [Note that

formality (F) is the number of gram formula weights per liter solution.] We distinguish formality and molarity: the first indicates explicitly how the solution was prepared, and the second specifies the concentration of actual species in solution. As to the value of ionic strength, we can write the following:

$$I = 1/2 \sum c_i z_i^2 = 0.5 \{(0.1)(1)^2 + (0.1)(-1)^2\} \\ = 0.5\{0.2\} = 0.1$$

$$-\log \gamma = 0.512 z_M z_N I^{1/2} = 0.512(1)(1)(0.1)^{1/2} = 0.16$$

Taking the antilogarithm of each side, we get γ equal to 0.69, and this value is to be compared to a measured activity of 0.80.

The Extended Debye-Hückel Equation. This exercise reminds us that the Debye-Hückel limiting law is not sufficiently accurate for most physicochemical studies. To estimate the calculated activity coefficient more accurately, one must consider the fact that ions are not point charges. To the contrary, ions are of finite size relative to the distance over which the ions interact electrostatically. This brings us to the extended Debye-Hückel equation:

$$-\log \gamma = 0.512 z_M z_N \{I^{1/2} [1 + 50.3 (DT)^{1/2} a I^{1/2}]\}$$

For pure water at 298 K, D equals 78.5, such that

$$-\log \gamma = 0.512 z_M z_N \{\mu^{1/2} [1 + 0.328 a I^{1/2}]\}$$

where the adjustable parameter a is the radius of the hydrated ion in angstrom units.

Dependence of Single-Ion Activity Coefficients on Ionic Strength

Ionic Species	Ionic Strength of Solution		
	$\mu = 0.01$	$\mu = 0.05$	$\mu = 0.10$
H ⁺	0.91	0.86	0.83
Na ⁺	0.90	0.82	0.76
Cl ⁻	0.90	0.81	0.76
Mg ²⁺	0.69	0.51	0.45
Ca ²⁺	0.69	0.49	0.41

DEFINITE INTEGRAL

An integral Y that can be evaluated between two defined (or definite) limits $x = a$ and $x = b$, and written as:

$$Y = \int_{x=a}^{x=b} f(x) dx$$

Definite integrals often can be graphically evaluated as so-called Riemann sums by partitioning the area under the curve into rectangles, the horizontal width of which

corresponds to intervals along the x -axis. Because the area of each partition can be determined and summed, one can obtain the overall area.

$$Y = \int_{x=a}^{x=b} f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{x_i + x_{i-1}}{2}\right) \cdot \Delta x_i$$

In this expression, we use so-called Δ -notation to indicate increments along the horizontal axis, such that the area of each partition is $f(x_i) \cdot (\Delta x_i)$, where $f(x_i)$ is the value of the function at the midpoint $(x_i + x_{i-1})/2$ of the interval $\Delta x_i = (x_i - x_{i-1})$.

DEGREE OF ACTIVATION

A dimensionless quantity, symbolized by ε_a , used to assess the extent of activation by a particular compound on the initial rate of an enzyme-catalyzed reaction. The degree of activation is equal to $(v_a - v_o)/v_o$ where v_a is the reaction rate in the presence of the activator and v_o is the initial rate in the absence of the activator. Whenever ε_a values are reported, the activator concentration and initial rate conditions have to be reported as well. The degree of activation, a useful parameter in the initial stages of an investigation, does not address issues related to the mechanism of activation. **See** *Activation; Basal Rate*

DEGREE OF DISSOCIATION

A dimensionless parameter (usually symbolized as α) indicating the fraction of monoprotic acid present as its dissociated form. This parameter can be defined quantitatively as follows:

$$\alpha = \frac{[\text{conjugate base}]}{[\text{total acid}]} = \frac{[\text{conjugate base}]}{[\text{acid HA}] + [\text{conjugate base}]}$$

If c is the initial or total concentration of an acid HA, then the concentrations of A⁻ and H⁺ are each equal to αc , where α lies in the interval $[0 < \alpha < 1]$. Likewise, the equilibrium concentration of HA must be $(1 - \alpha)c$. Thus,

$$K_{\text{dissociation}} = \frac{\alpha^2 c^2}{(1 - \alpha)c} = \frac{\alpha^2 c}{(1 - \alpha)}$$

We can consider several important cases. (1) While α can approach unity, the degree of dissociation can never be complete (*i.e.*, $\alpha \neq 1$); otherwise, the system would no longer be at equilibrium. This is a reasonable expectation for any equilibrium process, and in the case of an acid, there must be some HA present, however low its

concentration. (2) Under the condition where $\alpha \ll 1$, the above equation becomes:

$$\alpha^2 = \frac{K_{\text{dissociation}}}{c} \quad \text{or} \quad \alpha = \sqrt{\frac{K_{\text{dissociation}}}{c}}$$

Because $K_{\text{dissociation}}$ is an equilibrium dissociation constant, this expression reminds us that the degree of dissociation will depend on the total concentration of HA; α will get smaller as c is raised, and α will get larger as c is lowered.

In most cases, $K_{\text{dissociation}}$ is written as K_a or K_A , and one may take logarithms of the dissociation constant to obtain the familiar Henderson-Hasselbalch equation:

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]} = \text{p}K_a + \log \frac{\alpha}{1 - \alpha}$$

If $\text{p}K_a$ is known, this equation permits one to calculate α at any pH. **See** *Henderson-Hasselbalch Equation*

DEGREE OF INHIBITION

A dimensionless quantity used to assess the extent of inhibition by a particular compound at a specific concentration on the initial rate of an enzyme-catalyzed reaction. The degree of inhibition, symbolized by ε_i , is equal to $(v_o - v_i)/v_o$ where v_o is the reaction rate in the absence of the inhibitor and v_i is the rate in the presence of the inhibitor. Whenever ε_i values are reported, the inhibitor concentration and initial rate conditions have to be provided as well. The degree of inhibition is a useful parameter in the early stages of an investigation. However, it does not address the type of inhibition nor provide much information on the dissociation constant for the inhibitor. **See** *Inhibition (as well as specific type of inhibition)*

DEGREES OF FREEDOM

1. The number of independent variables required to specify the state of a mechanical or thermodynamic system. Degrees of freedom arise from the possible motions of molecules or particles in a system. (The term “generalized coordinates” is also used in physics to designate the minimal number of coordinates needed to specify the state of a mechanical system.) 2. The number of independent or unrestricted random variables constituting a statistic. **See** *Statistics (A Primer)*

The heat capacity of a substance depends on the extent to which thermal energy can be absorbed and stored in the form of molecular motions. The kinetic theory of gases was first used to define the contributions of degrees of freedom to the heat capacity. Naumann¹ concluded that a monoatomic ideal gas possesses no thermal energy other than that of translation. Elementary kinetic theory predicts that the contribution equals $3R/2$ for the constant volume heat capacity (C_V) or $5R/2$ for the constant pressure heat capacity (C_P). R is the universal gas constant. A diatomic molecule has five degrees of freedom: three translational motions (along x , y , and z axes) as well as separate rotational motions (one about each of the two mutually perpendicular axes). One must also consider the vibrational motion of the two atoms as they periodically move toward and away from one another, and this contribution to the heat capacity was first formulated by Einstein². Polyatomic molecules are slightly more complicated. For carbon dioxide, each of the three atoms has three degrees of freedom, yielding nine degrees of freedom for the $\text{O}=\text{C}=\text{O}$ molecule: three correspond to the motions of the molecule along the x , y , and z axes; two to rotations in the xz and xy planes; and four to vibrations that are most frequently called the two symmetric and asymmetric stretching modes and the two bending modes.

In chemical reaction kinetics, there may be thermodynamic contributions arising from any change(s) in the degrees of freedom of the atoms involved in bond-breaking and bond-making steps. Coupled vibratory motions can also be a factor: whenever the frequencies of two uncoupled vibrators are sufficiently close to each other, introduction of a physical connection between the vibrators will result in a frequency shift for the coupled system.

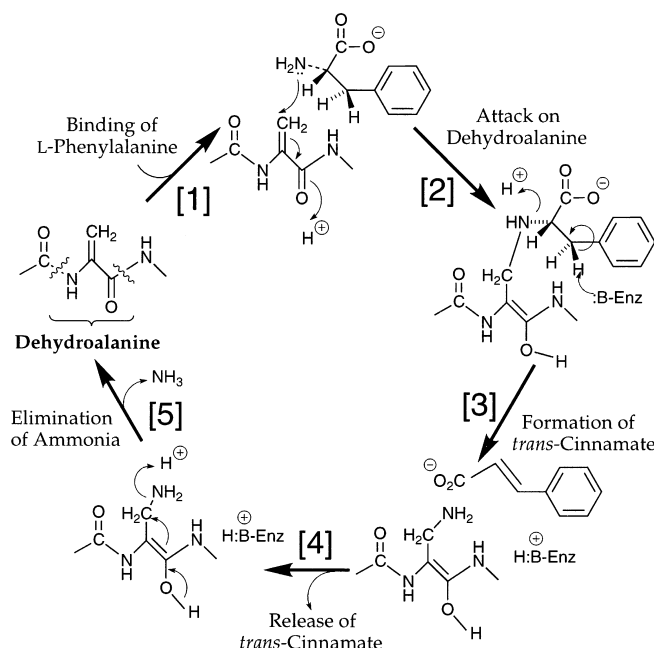
¹A. Naumann (1867) *Ann. Chem. Pharm.* **142**, 265.

²A. Einstein (1907) *Ann. Physik* **22**, 180.

DEHYDROALANINE

A specialized amino acid residue that serves as an essential electrophilic center in several enzymatic reactions, including those catalyzed by L-phenylalanine ammonia lyase (Reaction: L-phenylalanine $\rightarrow$ *trans*-cinnamate + NH_3) and L-histidine ammonia lyase (Reaction: L-histidine $\rightarrow$ urocanate + NH_3). The former facilitates the elimination of ammonia and the *pro-S* hydrogen of phenylalanine, and the initial step is nucleophilic attack of

phenylalanine's amino group on the methenyl group of a dehydroalanyl residue.



The dehydroalanyl is thought to be formed via expulsion of the elements of water from an active site serine of an inactive precursor form of these lyases. **See also** *Borohydride Reduction*

DEHYDROASCORBATE REDUCTASE

This enzyme [EC 1.6.5.4], also known as monodehydroascorbate reductase (NADH), catalyzes the reaction of NADH with two monodehydroascorbate molecules to produce NAD⁺ and two molecules of ascorbate.

E. Maellaro, B. Del Bello, L. Sugherini, M. Comporti & A. F. Casini (1997) *Meth. Enzymol.* **279**, 30.

7-DEHYDROCHOLESTEROL REDUCTASE

This enzyme [EC 1.3.1.21] catalyzes the reaction of cholesterol with NADP⁺ to yield cholesta-5,7-dien-3- β -ol and NADPH.

M. E. Dempsey (1969) *Meth. Enzymol.* **15**, 501.

2-DEHYDRO-3-DEOXY-D-GLUCONATE 5-DEHYDROGENASE

This enzyme [EC 1.1.1.127], also known as 2-keto-3-deoxygluconate dehydrogenase, catalyzes the reaction of 2-dehydro-3-deoxy-D-gluconate with NAD⁺ to produce (4S)-4,6-dihydroxy-2,5-dioxohexanoate and NADH. The enzyme isolated from *Pseudomonas* can utilize NADP⁺ as the coenzyme whereas the enzymes isolated

from *Erwinia chrysanthemi* and *E. coli* are more specific for NAD⁺.

J. Preiss (1966) *Meth. Enzymol.* **9**, 203.

2-DEHYDRO-3-DEOXY-D-GLUCONATE 6-DEHYDROGENASE

This enzyme [EC 1.1.1.126], also known as 2-keto-3-deoxy-D-gluconate dehydrogenase, catalyzes the reaction of 2-dehydro-3-deoxy-D-gluconate with NADP⁺ to produce (4S,5S)-4,5-dihydroxy-2,6-dioxohexanoate and NADPH.

J. Preiss (1966) *Meth. Enzymol.* **9**, 206.

2-DEHYDRO-3-DEOXYGLUCONOKINASE

This enzyme [EC 2.7.1.45], also known as 2-keto-3-deoxygluconokinase and 3-deoxy-2-oxo-D-gluconate kinase, catalyzes the reaction of ATP with 2-dehydro-3-deoxy-D-gluconate to produce ADP and 6-phospho-2-dehydro-3-deoxy-D-gluconate.

G. Ashwell (1962) *Meth. Enzymol.* **5**, 190.

DEHYDROGLUCONOKINASE

This enzyme [EC 2.7.1.13], also known as ketogluconokinase, catalyzes the reaction of 2-dehydro-D-gluconate with ATP to produce ADP and 6-phospho-2-dehydro-D-gluconate.

W. A. Wood & E. W. Frampton (1962) *Meth. Enzymol.* **5**, 291.

3-DEHYDROQUINATE DEHYDRATASE

This enzyme [EC 4.2.1.10] catalyzes the reaction of 3-dehydroquininate to produce 3-dehydroshikimate and water.

K. N. Allen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 135.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

S. Chaudhuri, K. Duncan & J. R. Coggins (1987) *Meth. Enzymol.* **142**, 320.

3-DEHYDROQUINATE SYNTHASE

This enzyme [EC 4.6.1.3] catalyzes the conversion of 3-deoxy-*arabino*-heptulosonate 7-phosphate to form 3-dehydroquininate and orthophosphate. The enzyme requires cobalt and the hydrogen atoms located on C7 of the substrate are retained on C2 of the product.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

3-DEHYDROSPHINGANINE REDUCTASE

This enzyme [EC 1.1.1.102] catalyzes the reversible reaction of sphinganine with NADP⁺ to produce 3-dehydro-sphinganine and NADPH.

A. H. Merrill, Jr., & E. Wang (1992) *Meth. Enzymol.* **209**, 427.

DEINHIBITION

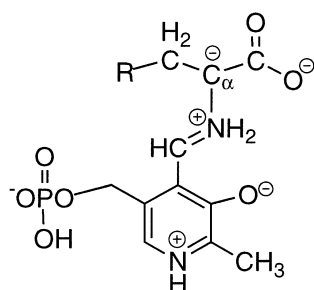
Any ligand-induced process that nullifies the inhibitory effect of an enzyme-bound inhibitor. In such cases, the ligand is said to deinhhibit the system. Such behavior is distinct from activation, because increased activity is not observed when the deinhibiting ligand interacts with the enzyme in the absence of the inhibitor.

An excellent example of deinhhibition is the reversal of glucose 6-phosphate inhibition of brain hexokinase (type I isozyme)¹. As the pacemaker of glycolysis in brain tissue, hexokinase exhibits unique regulatory properties in that physiological levels of phosphate relieve potent inhibition by the reaction product, glucose 6-phosphate. On the other hand, orthophosphate is without effect on brain hexokinase in the absence of glucose 6-phosphate. This behavior contributes in part to the action of orthophosphate in accelerating glycolysis, a process widely known as the Crabtree Effect.

¹D. L. Purich, F. B. Rudolph & H. J. Fromm (1973) *Advan. Enzymol.* **39**, 249.

DELOCALIZATION

A chemical concept that applies to π -bonding in conjugated systems. Electrons in conjugated π bonds are not restricted to just two atoms. Each bond of conjugated systems has a partial double bond character, reflected by the bond order for a given bond.



Delocalized α -Carbanion

With conjugated systems having a net charge, delocalization of the charge also occurs. The situation is analogous for unpaired electrons in conjugated radicals.

DELTA (Δ , δ)

Δ : 1. Symbol for application of heat in a reaction (e.g., $A \xrightarrow{\Delta} P$). 2. Symbol for change. 3. In organic structures, used to denote the presence of a double bond, usually with a superscript number to indicate position. 4. Symbol for mass excess.

δ : 1. Fourth letter in the Greek alphabet; hence, used to denote the fourth item in a series (for example, the fourth methylene carbon in a fatty acid). 2. Symbol for thickness. 3. Symbol for chemical shift in NMR.

DEOXYADENOSINE KINASE

This enzyme [EC 2.7.1.76] catalyzes the reaction of ATP with deoxyadenosine to produce ADP and dAMP. The enzyme will also utilize deoxyguanosine as a substrate. In some systems, this enzyme may be identical with deoxycytidine kinase.

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

DEOXYADENYLATE KINASE

This enzyme [EC 2.7.4.11] catalyzes the reaction of ATP with dAMP to produce ADP and dADP. The enzyme can also use AMP as a substrate (thus producing another ADP as a product; hence, an activity identical to that of adenylate kinase).

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

DEOXYCYTIDINE KINASE

This enzyme [EC 2.7.1.74] catalyzes the reaction of a nucleoside triphosphate with deoxycytidine to produce a nucleoside diphosphate and dCMP. This enzyme can use any nucleoside triphosphate (except dCTP) as the phosphate-donor substrate and it can phosphorylate cytosine arabinoside as well.

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

DEOXYCYTIDYLATE HYDROXYMETHYLTRANSFERASE

This enzyme [EC 3.6.1.12], also known as dCTP pyrophosphatase, catalyzes the hydrolysis of dCTP to produce dCMP and pyrophosphate (or, diphosphate). The enzyme can also utilize dCDP as a substrate, producing CMP and orthophosphate.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

3-DEOXY-D-manno-OCTULOSONATE ALDOLASE

This enzyme [EC 4.1.2.23] catalyzes the reversible conversion of 3-deoxy-D-manno-octulosonate to pyruvate and D-arabinose.

W. A. Wood (1972) *The Enzymes*, 3rd ed., **7**, 281.

3-DEOXYOCTULOSONATE 8-PHOSPHATE SYNTHASE

This enzyme [EC 4.1.2.16] (also known as phospho-2-dehydro-3-deoxyoctonate aldolase, phospho-2-keto-3-deoxyoctonate aldolase, and 3-deoxy-D-manno-octulosonic acid 8-phosphate synthetase) catalyzes the reaction of 2-dehydro-3-deoxy-D-octonate 8-phosphate and orthophosphate to produce phosphoenolpyruvate, D-arabinose 5-phosphate, and water.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

D. H. Levin & E. Racker (1966) *Meth. Enzymol.* **8**, 216.

(DEOXY)NUCLEOSIDE MONOPHOSPHATE KINASE

This enzyme [EC 2.7.4.13] catalyzes the reaction of ATP with a deoxynucleoside phosphate to produce ADP and a deoxynucleoside diphosphate. The enzyme can also utilize dATP as the phosphorylating substrate as well.

M. J. Bessman (1963) *Meth. Enzymol.* **6**, 166.

DEOXYRIBONUCLEASES

Enzymes catalyzing cleavage of DNA, including: endodeoxyribonucleases that generate 5'-phosphomonoesters [EC 3.1.21.x], endodeoxyribonucleases that produce products other than 5'-phosphomonoesters [EC 3.1.22.x], site-specific endodeoxyribonucleases acting on altered bases [EC 3.1.25.x], and exodeoxyribonucleases producing 5'-phosphomonoesters [EC 3.1.11.x]. A few examples are:

(1) Deoxyribonuclease I [EC 3.1.21.1], also known as pancreatic DNase and thymonuclease, catalyzes the endonucleolytic cleavage of DNA, preferring dsDNA, to a 5'-phosphodinucleotide and 5'-phosphooligonucleotide end products.

(2) Deoxyribonuclease IV (phage T₄-induced) [EC 3.1.21.2], also called endodeoxyribonuclease IV (phage T₄-induced) and endonuclease IV, catalyzes the endonucleolytic cleavage of DNA, preferring single-stranded, to 5'-phosphooligonucleotide end products.

(3) Type I site-specific deoxyribonuclease [EC 3.1.21.3], also referred to as type I restriction enzyme, catalyzes the endonucleolytic cleavage of DNA to give random, double-stranded fragments with terminal 5'-phosphates. ATP (or dATP) is simultaneously hydrolyzed to ADP and orthophosphate. This classification represents a large group of enzymes that have an absolute requirement for ATP (or dATP) and S-adenosylmethionine. They recognize specific short DNA sequences and hydrolyze bonds at sites remote from the recognition sequence.

(4) Type II site-specific deoxyribonuclease [EC 3.1.21.4], also referred to as type II restriction enzyme, catalyzes the endonucleolytic cleavage of DNA to give specific, double-stranded fragments with terminal 5'-phosphates. Magnesium ions are required as cofactors.

(5) Type III site-specific deoxyribonuclease [EC 3.1.21.5], also referred to as type III restriction enzyme, catalyzes the endonucleolytic cleavage of DNA to give specific, double-stranded fragments with terminal 5'-phosphates. This class of enzymes has an absolute requirement for ATP but does not hydrolyze the ATP. S-Adenosylmethionine stimulates the reaction, but is not absolutely required. These enzymes recognize specific short DNA sequences and hydrolyze bonds that are a short distance away from the recognition sequence.

(6) Deoxyribonuclease II [EC 3.1.22.1], also referred to as pancreatic DNase II, catalyzes the endonucleolytic hydrolysis of DNA (preferring double-stranded DNA) to produce 3'-phosphomononucleotide and 3'-phosphooligonucleotide end products.

(7) Deoxyribonuclease V [EC 3.1.22.3], also known as endodeoxyribonuclease V, catalyzes the endonucleolytic cleavage of DNA at apurinic or apyrimidinic sites to products with a 3'-phosphate.

(8) Crossover junction endoribonuclease [EC 3.1.22.4], also referred to as Holliday junction nuclease, catalyzes the endonucleolytic hydrolysis of a bond(s) in DNA at a junction such as a reciprocal single-stranded crossover between two homologous DNA duplexes (Holliday junction).

(9) Deoxyribonuclease X [EC 3.1.22.5] catalyzes the endonucleolytic cleavage of supercoiled plasma DNA to linear DNA duplexes. The enzyme exhibits a preference for supercoiled DNA.

(10) Deoxyribonuclease (pyrimidine dimer) [EC 3.1.25.1] catalyzes the endonucleolytic hydrolysis of a bond in DNA near pyrimidine dimers to generate products with 5'-phosphates. The enzyme acts on damaged strands of DNA, 5' from the damaged site.

(11) DNA-(apurinic or apyrimidinic site) lyase [EC 4.2.99.18, formerly EC 3.1.25.2] acts on the C—O—P bond 3' to the apurinic or apyrimidinic site in DNA. This bond is broken by a β -elimination reaction, leaving a 3'-terminal unsaturated sugar and a product with a terminal 5'-phosphate. Note that this "nicking" of the phosphodiester bond is a lyase-type reaction, not hydrolysis.

(12) Exodeoxyribonuclease I [EC 3.1.11.1] catalyzes the hydrolysis of bonds in DNA (preferring single-stranded DNA), acting progressively in a 3'- to 5'-direction, releasing 5'-phosphomononucleotides.

(13) Exodeoxyribonuclease III [EC 3.1.11.2] catalyzes the hydrolysis of bonds in DNA (preferring double-stranded DNA), acting progressively in a 3'- to 5'-direction, releasing 5'-phosphomononucleotides.

(14) Exodeoxyribonuclease (lambda-induced) [EC 3.1.11.3], also called lambda exonuclease, catalyzes the hydrolysis of double-stranded DNA, acting progressively in a 5'- to 3'-direction and releasing 5'-phosphomononucleotides. This enzyme does not attack single-strand breaks.

(15) Exodeoxyribonuclease (phage SP₃-induced) [EC 3.1.11.4] catalyzes exonucleolytic cleavage of DNA (preferring single-stranded) in the 5'- to 3'-direction to yield 5'-phosphodinucleotides.

(16) Exodeoxyribonuclease V [EC 3.1.11.5] catalyzes the exonucleolytic cleavage of double-stranded DNA in the presence of ATP, in either the 5'- to 3'- or the 3'- to 5'-direction to yield 5'-phosphooligonucleotides. Thus, this enzyme exhibits a DNA-dependent ATPase activity.

(17) Exodeoxyribonuclease VII [EC 3.1.11.6] catalyzes the exonucleolytic cleavage of DNA (preferring single-stranded) in either the 5'- to 3'- or the 3'- to 5'-direction to yield 5'-phosphomononucleotides.

N. H. Williams (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 543.

J. A. Gerlt (1992) *The Enzymes*, 3rd ed., **20**, 95.

Two entire volumes dedicated to DNases: *The Enzymes*, 3rd ed., **14** and **21**.

I. R. Lehman (1971) *The Enzymes*, 3rd ed., **4**, 251.

G. Bernardi (1971) *The Enzymes*, 3rd ed., **4**, 271.

M. Laskowski, Sr., (1971) *The Enzymes*, 3rd ed., **4**, 289.

2-DEOXYRIBOSE-5-PHOSPHATE ALDOLASE

This enzyme [EC 4.1.2.4], also referred to as phosphodeoxyriboaldolase and deoxyriboaldolase, catalyzes the reversible conversion of 2-deoxy-D-ribose 5-phosphate to D-glyceraldehyde 3-phosphate and acetaldehyde.

K. N. Allen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 135.

D. S. Feingold & P. A. Hoffee (1972) *The Enzymes*, 3rd ed., **7**, 303.

DEOXYTHYMIDINE KINASE

This enzyme [EC 2.7.1.21], better known as thymidine kinase, catalyzes the reaction of ATP with thymidine to produce ADP and thymidine 5'-phosphate. Deoxyuridine can serve as a substrate and dGTP can substitute for ATP. The deoxypyrimidine kinase complex induced by herpes simplex virus catalyzes this reaction as well as those of AMP:thymidine kinase, ADP:thymidine kinase, and dTMP kinase.

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

DEPHOSPHO-CoA KINASE

This enzyme [EC 2.7.1.24] catalyzes the reaction of ATP with dephospho-CoA to produce ADP and coenzyme A.

G. W. E. Plaut, C. M. Smith & W. L. Alworth (1974) *Ann. Rev. Biochem.* **43**, 899.

Y. Abiko (1970) *Meth. Enzymol.* **18A**, 358.

DEPOLARIZATION

1. Reduction, neutralization, or change in direction of polarity. 2. Transient reduction in the membrane potential. **See** *Action Potential*; *Hyperpolarization*

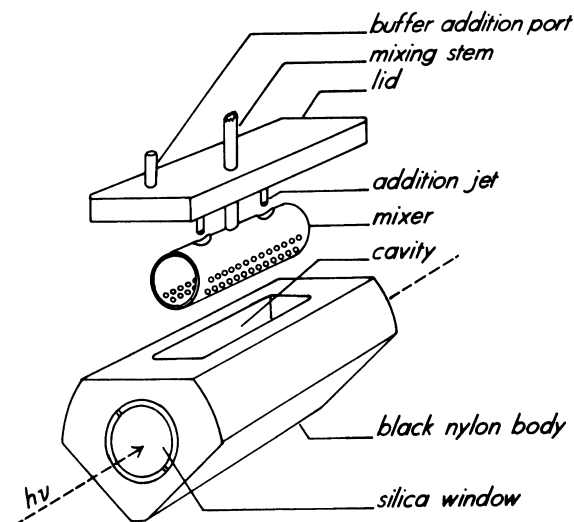
DEPOLYMERIZATION

The spontaneous or enzyme-catalyzed breakdown (or disassembly) of a biopolymer into its component subunits.

Biopolymers are typically formed from subunits through the formation of covalent and/or noncovalent bonds. Examples of the covalently linked biopolymers include proteins formed from amino acids, nucleic acids formed from ribo- or deoxyribonucleotides, and polysaccharides formed from monosaccharides. These are depolymerized by hydrolases which can add and release from the biopolymers during each hydrolytic cycle or remain associated with the polymer for several or many covalent bond-breaking cycles. Biopolymers formed by noncovalent forces include oligomeric proteins [such as hemoglobin ($\alpha_2\beta_2$), tubulin ($\alpha\beta$), and lactate dehydrogenase (M_4 , M_3H_1 , M_3H_1 , M_2H_2 , M_1H_3 , and H_4 tetramers formed from muscle (M) and heart (H) polypeptide chains) or polymeric structures [such as microtubules, filamentous actin, and intermediate filaments]. These depolymerize spontaneously in response to changes in pH, temperature or as the result of the presence/absence of other solutes (metal ions, metabolites, or other inorganic anions). **See also** Processivity; Dynamic Instability; Treadmilling; Polymerization; Depolymerization, End-Wise

DEPOLYMERIZATION, END-WISE

Linear supramolecular polymers (such as those formed from actin, tubulin, and tobacco mosaic virus coat protein) are formed as products of efficient entropy-driven self-assembly^{1,2}, and the dynamics of depolymerization can be quantitatively analyzed^{3,4} if the following conditions hold true: (a) the polymer undergoes stepwise release of subunits in a series first-order manner; (b) the off-rate constant for subunit release is independent of polymer length over the course of depolymerization; (c) the rate of polymerization is effectively zero under the conditions of the experiment; (d) turbidity or some other physically determinable parameter is a measure of the polymer weight concentration and is independent of changes in polymer length distribution; and (e) the concentrations of polymers of various lengths can be estimated by electron microscopy or some other suitably accurate method. For microtubules, these conditions can be met by isothermal dilution of the total tubulin concentration to below its critical concentration⁵, by cold-induced depolymerization⁵ after lowering the sample temperature to 5–6°C, or by calcium ion induced disassembly⁶. Rapid isothermal dilution can be accomplished within 1–3 seconds after dilution using the mixer shown in Fig. 1.



Rapid Dilution Cuvette

Figure 1. Mixing device for microtubule depolymerization kinetic studies requiring prompt dilution while minimizing shearing forces that may alter the polymer length distribution.

A typical fit of the theoretical depolymerization kinetics to the experimentally determined rate behavior for dilution-induced depolymerization is illustrated in Fig. 2

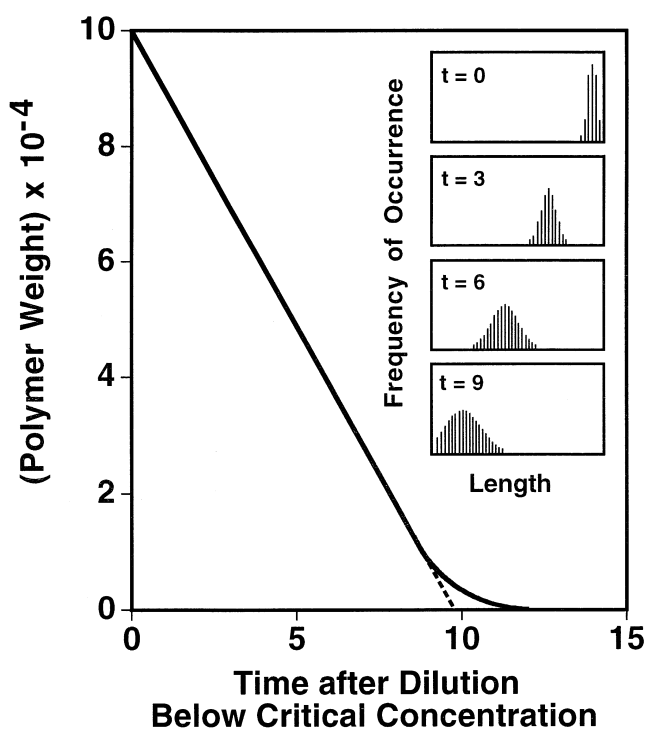


Figure 2. Progress curve for dilution-induced microtubule depolymerization. Inset: Polymer length distribution prior to dilution-induced disassembly. The data points are experimentally determined, and the solid line is based on the theoretical treatment^{3,4}.

for microtubules composed of tubulin and microtubule-associated proteins. The theoretical depolymerization curve (solid line) was obtained by taking into account the polymer length distribution for the microtubule sample immediately preceding depolymerization.

The generality of the end-wise depolymerization kinetic model is indicated by the comparison of the observed and predicted time-courses of cold-induced microtubule disassembly (Fig. 3). **See Self-Assembly Protein Polymerization**

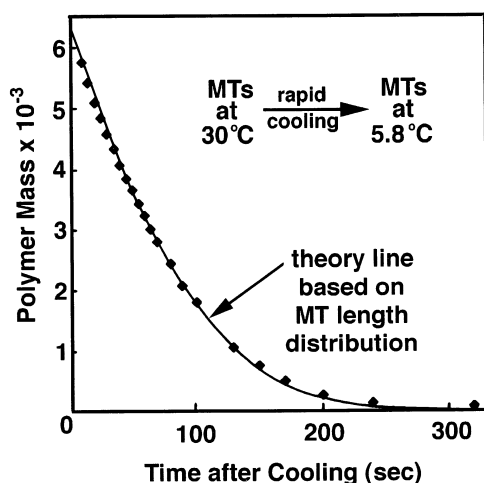


Figure 3. Progress curve for cold-induced depolymerization after chilling microtubules (assembled at 30°C using 2.1 mg/ml tubulin and 0.4 mg/ml microtubule-associated proteins). The data points are experimentally determined, and the solid line is based on the theoretical treatment^{3,4}.

¹M. A. Lauffer (1975) *Entropy-Driven Processes in Biology*, Springer-Verlag, Berlin.

²F. Oosawa and M. Kasai (1962) *J. Mol. Biol.* **4**, 10.

³D. Kristofferson, T. L. Karr & D. L. Purich (1980) *J. Biol. Chem.* **255**, 8567.

⁴D. L. Purich, T. L. Karr & D. Kristofferson (1980) *Meth. Enzymol.* **85**, 439.

⁵T. L. Karr, D. Kristofferson & D. L. Purich (1980) *J. Biol. Chem.* **255**, 8560.

⁶T. L. Karr, D. Kristofferson & D. L. Purich (1980) *J. Biol. Chem.* **255**, 11853.

DESATURASES, FATTY ACYL-CoA

There are several fatty acyl-CoA desaturases. Stearoyl-CoA desaturase [EC 1.14.99.5] (also referred to as acyl-CoA desaturase, fatty acid desaturase, and Δ^9 -desaturase) catalyzes the reaction of stearoyl-CoA with dioxygen and a hydrogen donor (AH_2) to produce oleoyl-CoA, two water molecules, and A. The enzyme requires iron as a cofactor. Linoleoyl-CoA desaturase [EC 1.14.99.25], also referred to as Δ^6 -desaturase, cata-

lyzes the reaction of linoleoyl-CoA with dioxygen and AH_2 to produce γ -linolenoyl-CoA, A, and two water molecules.

I. Nishida & N. Murata (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 541.

P. W. Holloway (1983) *The Enzymes*, 3rd ed., **16**, 63.

DESOLVATION

A potential factor for enhancing the effectiveness of catalyzed reactions. The relative importance of this factor will depend on the polarity of the substrate(s), transition state, and reaction product(s). The energy associated with desolvation of substrates must be compensated for by the binding interactions between the substrates and the enzyme.

DETHIOBIOTIN SYNTHASE

This enzyme [EC 6.3.3.3] catalyzes the reaction of ATP with 7,8-diaminononanoate and carbon dioxide to produce ADP, orthophosphate, and dethiobiotin. CTP can be used as a substrate as well although it is not as effective as ATP.

G. Schneider & Y. Lindqvist (1997) *Meth. Enzymol.* **279**, 376.

DEXTRANASE

This enzyme [EC 3.2.1.11], also known as α -1,6-glucan-6-glucanohydrolase, catalyzes the endohydrolysis of 1,6- α -D-glucosidic linkages in dextran.

J.-C. Janson & J. Porath (1966) *Meth. Enzymol.* **8**, 615.

DEXTRANSUCRASE

This enzyme [EC 2.4.1.5], also known as sucrose 6-glucosyltransferase, catalyzes the reaction of sucrose with [(1,6)- α -D-glucosyl]_n to produce fructose and [(1,6)- α -D-glucosyl]_(n+1).

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

DEUTERIUM

An isotope of hydrogen having a nucleus (referred to as the deuteron) consisting of one proton and one neutron. Deuterium is a stable isotope (symbolized by 2H or D) having an atomic weight of 2.0140 amu and a natural abundance of 0.015% relative to all hydrogen isotopes.

DEVIATION

The extent of variation within a series of replicate determinations of some experimental parameter. Let a_1 , a_2 , and a_3 represent several values of replicate measurements, and let a represent the arithmetic mean (or average). The deviations Δ_1 , Δ_2 , and Δ_3 are $(a_1 - a)$, $(a_2 - a)$, and $(a_3 - a)$, respectively. These quantities may be greater than, less than, or equal to zero, and the sign indicates the direction of deviation.

DEWAR STRUCTURES

Canonical forms of benzene that are calculated to contribute about 22% to the resonance stabilization of benzene. Such resonance structures have no separate physical reality or independent existence. For the case of benzene, the two Kekulé structures with alternating double bonds (*i.e.*, “cyclohexatriene” structures) contribute equally and predominantly to the resonance hybrid structure. A dotted circle is often used to indicate the resonance-stabilized bonding of benzene. Nonetheless, the most frequently appearing structures of benzene are the two Kekulé structures. **See** *Kekulé Structures*

A. Pullman & B. Pullman (1958) *Prog. Org. Chem.* **4**, 31.

DIACYLGLYCEROL O-ACYLTRANSFERASE

This enzyme [EC 2.3.1.20], also known as diglyceride acyltransferase, catalyzes the reaction of an acyl-CoA with 1,2-diacylglycerol to produce coenzyme A and a triacylglycerol. The acyl-CoA derivative can be palmitoyl-CoA or other long-chain acyl-CoA compounds.

R. A. Coleman (1992) *Meth. Enzymol.* **209**, 98.

R. M. Bell & R. A. Coleman (1983) *The Enzymes*, 3rd ed., **16**, 87.

1,2-DIACYLGLYCEROL 3- β -GALACTOSYLTRANSFERASE

This enzyme [EC 2.4.1.46], also referred to as UDP-galactose:1,2-diacylglycerol 3- β -galactosyltransferase, catalyzes the reaction of UDP-galactose with a 1,2-diacylglycerol to produce UDP and a 3- β -D-galactosyl-1,2-diacylglycerol.

J. Browse & C. Somerville (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 467.

DIACYLGLYCEROL KINASE

This enzyme [EC 2.7.1.107], also known as diglyceride kinase, catalyzes the reaction of ATP with a 1,2-diacyl-

glycerol to produce ADP and a 1,2-diacylglycerol 3-phosphate.

J. P. Walsh & R. M. Bell (1992) *Meth. Enzymol.* **209**, 153.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

R. A. Pieringer (1983) *The Enzymes*, 3rd ed., **16**, 255.

DIAMINE AMINOTRANSFERASE

This enzyme [EC 2.6.1.29] catalyzes the reaction of an α,ω -diamine with α -ketoglutarate (or, 2-oxoglutarate) to generate an ω -aminoaldehyde and glutamate.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

DIAMINOHEXANOATE DEHYDROGENASE

This enzyme [EC 1.4.1.11], better referred to as L-erythro-3,5-diaminohexanoate dehydrogenase, catalyzes the reaction of L-erythro-3,5-diaminohexanoate with NAD⁺ and water to produce (S)-5-amino-3-oxohexanoate, ammonia, and NADH.

N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

2,4-DIAMINOPENTANOATE DEHYDROGENASE

This enzyme [EC 1.4.1.12] catalyzes the reaction of 2,4-diaminopentanoate with NAD(P)⁺ and water to produce 2-amino-4-oxopentanoate, ammonia, and NAD(P)H. The enzyme can also utilize 2,5-diaminohexanoate as a substrate (although not as effectively as the substrate mentioned above) forming 2-amino-5-oxohexanoate, which then cyclizes nonenzymically to form 1-pyrroline-2-methyl-5-carboxylate.

N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

DIAMINOPIMELATE DECARBOXYLASE

This enzyme [EC 4.1.1.20] is a pyridoxal-phosphate-dependent protein that catalyzes the conversion of meso-2,6-diaminoheptanedioate to lysine and carbon dioxide.

E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

DIAMINOPIMELATE DEHYDROGENASE

This enzyme [EC 1.4.1.16] catalyzes the reaction of meso-2,6-diaminoheptanedioate with NADP⁺ and water to produce L-2-amino-6-oxoheptanedioate, ammonia, and NADPH.

N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

DIAMINOPIMELATE EPIMERASE

This enzyme [EC 5.1.1.7] catalyzes the interconversion of L,L-2,6-diaminoheptanedioate to *meso*-diaminoheptanedioate.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

E. Adams (1972) *The Enzymes*, 3rd ed., **6**, 479.

DIASTEREOMERS

Stereoisomers with more than one chiral center and which are not mirror images of each other; hence, stereoisomers that are not enantiomers of each other. For example, L-threonine and D-threonine are an enantiomeric pair whereas L-threonine and D-allothreonine are a diastereomeric pair (as is L-threonine and L-allothreonine). Diastereomers will have similar physical, chemical, and spectral properties but those properties will *not* be identical. If n is the number of chiral centers, then the maximum number of stereoisomers will be equal to 2^n . However, the actual number for a given set of isomers may be less than 2^n due to the presence of meso forms.

See *Enantiomer; Epimer; Meso Form*

DIATHERMIC WALL

A wall that separates systems but allows them to come into thermal equilibrium; necessary for isothermal processes. **See** *Adiabatic Wall*

DICHLOROMETHANE DEHALOGENASE

This enzyme [EC 4.5.1.3] catalyzes the hydrolysis of dichloromethane to generate formaldehyde and two HCl. Glutathione is required as a cofactor.

T. Leisinger & D. Kohlert-Staub (1990) *Meth. Enzymol.* **188**, 355.

DIELECTRIC CONSTANT

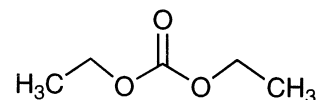
A measure of the ease of dissociation of ion pairs (or point charges) dissolved (or embedded) in a dielectric solvent. The force of ionic attraction or repulsion can be treated in accordance with Coulomb's inverse-square law, such that $F = q_1 q_2 / D r^2$, where the numerator is the product of the ionic charges q_1 and q_2 , r is the charge-to-charge distance, and D is the dielectric constant. D is a dimensionless constant that has a characteristic value for each dielectric substance or insulator. Suppose we are dealing with a pair of oppositely charged ions; the attraction will be correspondingly lessened in a medium where D is large. The following are dielectric constants

for several liquids at 20°C: hexane, 1.87; benzene, 2.28; diethyl ether, 4.33; chloroform, 5.05; acetone, 21.40; ethanol, 24.0; water, 80; and hydrogen cyanide, 116.0. Water has a high dielectric constant, whereas the value of D for benzene is much lower. In the former case, the electric field around each ion is effectively lowered because water is easily polarized; the induced dipoles of the water molecules are aligned so as to yield an overall dipole moment that greatly reduces the field strength. Benzene is not so easily polarized, and the interionic force of attraction of oppositely charged ions is much greater.

See also *Debye-Hückel Treatment*

The reader should also recall that the dielectric constant of the medium greatly influences chemical reactivity. This is especially true for nucleophilic substitution, where electric charge plays an essential role in all steps of the reaction.

DIETHYL PYROCARBONATE



A potent enzyme inhibitor (abbreviated DEP) that acts by ethoxyformylation of proteins, usually at histidine residues. DEP is an irreversible inhibitor of ribonuclease, and rinsing glassware with a 0.1% (weight/volume) DEP solution is recommended to render glassware nuclease-free. Aqueous solutions must be freshly prepared for maximal effectiveness, because DEP will hydrolyze in 6–12 hours at neutral pH.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Inhibitory properties, **68**, 212; inactivator, of ribonucleases, **65**, 681; lipase modification, **64**, 390; nuclease inactivation, **79**, 63; ribonuclease inactivation, **79**, 52, 112-113, 267.

DIFFERENTIAL SCANNING CALORIMETRY (DSC)

A thermochemical method that simultaneously measures differences in heat flow into a test substance and a reference substance (whose thermochemical properties are already well characterized) as both are subjected to programmed temperature ramping of the otherwise thermally isolated sample holder. The advantage of differential scanning calorimetry is a kinetic technique that allows one to record differences in heat absorption directly rather than measuring the total heat evolved/

absorbed by a sample of interest. Moreover, one can make measurements during the course of chemical reactions or as certain polymerization reactions proceed. Power-compensated DSC involves heating of the sample and reference substances by separate heaters, and one determines the power (usually in milliwatts) on each heater as the temperatures of sample and reference are kept equal. Heat flux DSC actually detects the differential heat flow from chromel discs lying beneath pans containing the sample and reference as the temperature is ramped. Instead of measuring the power of each heater (or furnace), heat flux DSC relies on high sensitivity thermocouples to generate the differential signal.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Aspartate transcarbamylase [assembly effects, **259**, 624-625; buffer sensitivity, **259**, 625; ligation effects, **259**, 625; mutation effects, **259**, 626]; baseline estimation [effect on parameters, **240**, 542-543, 548-549; importance of, **240**, 540; polynomial interpolation, **240**, 540-541, 549, 567; proportional method for, **240**, 541-542, 547-548, 567]; baseline subtraction and partial molar heat capacity, **259**, 151; changes in solvent accessible surface areas, **240**, 519-520, 528; characterization of membrane phase transition, **250**, 179-181; cooperativity measurement error, **240**, 642; data simulation, **240**, 537-539; enthalpy determination, **240**, 504, 506-507, 513-514, 534-537; entropy determination, **240**, 506-507, 514-517, 535-536, 539; excess heat capacity function [deconvolution analysis, **259**, 156-158; estimation, **259**, 152; statistical thermodynamic definition, **259**, 152-154]; folding/unfolding partition function determination, **240**, 505, 508-509, 532, 535; F-statistics, **240**, 554-559, 562; gel-liquid crystalline phase transition, **240**, 571-572; Gibbs free energy determination, **240**, 503, 518, 521, 535-536; heat capacity determination, **240**, 506, 510-513, 531, 534, 546-547; industrial application, **240**, 531, 568; α -lactalbumin molten globule state, **240**, 522, 528-530; ligand binding effects, **240**, 533; lipid-protein interaction analysis, **259**, 605-607; noise effect on parameters, **240**, 539-540, 562, 565; oligonucleotide thermodynamic parameters [entropy change, **259**, 237, 239; excess heat capacity, **259**, 234-235; free energy change, **259**, 155, 237, 239; van't Hoff enthalpy, **259**, 235-237]; protein measurement [enthalpy, **259**, 155-156, 168, 488, 595; entropy, **259**, 155-156, 168; heat capacity (absolute heat capacity, **259**, 145, 150-151, 154-155; change on unfolding, **259**, 165; conformational analysis, **259**, 163-165; curve-fitting, **259**, 164; hydration contribution, **259**, 158, 161-162, 164-165; noncovalent interactions, contribution, **259**, 158, 161, 163, 165; prediction, **259**, 167; primary heat capacity, **259**, 158-159; protonizable groups, contribution, **259**, 162-163; unfolded state, **259**, 162); transition temperature, **259**, 156]; protein unfolding, monitoring, **259**, 502, 601; RNA structure determination [data analysis, **259**, 289-290; sensitivity, **259**, 284]; two-dimensional, guanidinium hydrochloride thermal stability surfaces [accuracy, **240**, 566-567; methods, **240**, 537-543; multiple folding state fitting, **240**, 543-566].

DIFFERENTIATION

A procedure employed in calculus to describe the rate of change of a function with respect to one or more variables. The following are differentials for functions (where x is a variable and A is a constant) frequently encountered in chemistry.

$$dA/dx = 0$$

$$dx/dx = 1$$

$$dAx/dx = A \, dx/dx = A$$

$$d(e^x)/dx = e^x$$

$$d(e^{Ax})/dx = Ae^{Ax}$$

$$d(\ln x)/dx = 1/x$$

$$d(A \ln x)/dx = A/x$$

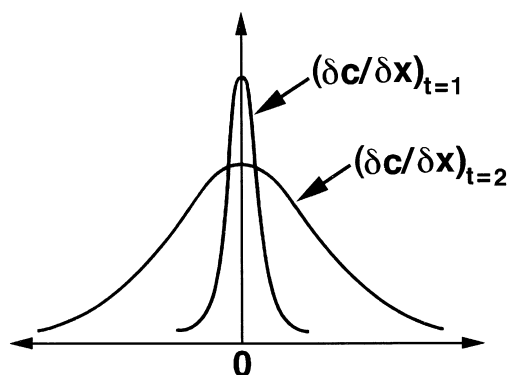
$$d[f(x) + g(x)]/dx = d[f(x)]/dx + d[g(x)]/dx$$

$$d[f(x)g(x)]/dx = g(x)d[f(x)]/dx + f(x)d[g(x)]/dx$$

$$d[f(x)/g(x)]/dx = \{g(x)d[f(x)]/dx - f(x)d[g(x)]/dx\}/g(x)^2 = [1/g(x)]df(x)/dx$$

DIFFUSION

The process by which random molecular motions result in a net flow of molecules from regions of high concentration to those of low concentration.



Concentration of a diffusible substance at times t_1 and t_2 . Note that the concentration gradient decreases over a period of time.

Diffusing molecules achieve a rate of flow J that is proportional to the concentration gradient $(\partial c/\partial x)$, characterized by a proportionality constant, D (also known as the diffusivity or diffusion coefficient):

$$J = -D \frac{\partial c}{\partial x}$$

The negative sign designates that the direction of flow occurs spontaneously from the region of higher concentration. The diffusion coefficient D represents the amount of solute that diffuses across a 1 cm^2 area per sec under the influence of a concentration gradient of 1 g/cm^2 . D is characteristic for each type of solute molecule, and its value takes into account both molecular mass and shape. **See also** *Transport Processes; Fick's Laws*

DIFFUSION COEFFICIENT/CONSTANT

A parameter indicating the mass of material that will diffuse across a unit area in unit time under a concentration gradient of unity. The diffusion coefficient, D , is the proportionality constant in Fick's Laws of Diffusion. **See Fick's Laws**

Diffusion Constants of Biomolecules

Particle	Diffusion Coefficient (cm ² /s) × 10 ⁷	Molecular Mass (Daltons)
Hemoglobin	6.3	67,000
Catalase	4.1	250,000
Urease	3.5	470,000
Tobacco Mosaic Virus	0.5	31,000,000

DIFFUSION CONTROL

Most treatments encountered in discussions of collision theory primarily are concerned with reactions in the gas phase. However, most of the reactions in chemistry and biochemistry occur in solutions. In solutions, the molecules are moving in the potential field of their neighbors rather than freely as in the gas phase. Thus, the potential energy varies and holes in the solvation shell permits displacement of the molecule from its original position. There are also rapid collisions with molecules that make up the solvent cage as the molecule makes its series of discontinuous displacements. Two molecules in solution that become neighbors will tend to collide a number of times (often referred to as an encounter) before they separate (or before they react).

Fick's laws describe the interactions or encounters between noninteracting particles experiencing random, Brownian motion. Collisions in solution are diffusion-controlled. As is discussed in most physical chemistry texts^{1,2}, by applying Fick's First Law and the Einstein diffusion relation, the upper limit of the bimolecular rate constant k would be equal to

$$k = 4\pi d_{AB}(D_A + D_B)L$$

where d_{AB} is the hard-sphere collision diameter discussed in collision theory, D_A and D_B are the diffusion coefficients for A and B, L is Avogadro's number, and there is an assumption that there is no intermolecular potential energy between the reacting molecules (this result was first obtained by Smoluchowski^{3,4}). Using typical values of d_{AB} (5×10^{-8} cm), $D_A + D_B$ (10^{-5} cm²·sec⁻¹), and adjusting for appropriate units, the above equation shows that k is equal to about 7×10^9

M⁻¹·sec⁻¹. The corresponding upper limit value for a bimolecular rate constant in the gas phase is about 10^{11} M⁻¹·sec⁻¹. Thus in solutions, bimolecular rate constants cannot exceed 10^9 - 10^{10} M⁻¹·sec⁻¹ since diffusion control takes over from collision control.

It should also be recalled that the diffusion coefficient of a molecule will decrease with increasing viscosity of the solvent. Thus, as might be expected with cytosols or lipid bilayers, a viscous medium will slow down the rate of encounters. Since viscosity is itself temperature dependent, such encounters in solution will have their own activation energy.

The above expression represents the maximum rate for an encounter in a solution. However, most reactions have an energy barrier and/or strict steric requirements. Typically, many collisions occur before a reaction ensues. One must also consider the duration of an encounter and the frequency of vibrational collision during an encounter.

The duration of an encounter, τ_{AB} , between two molecules A and B can be estimated¹ to be

$$\tau_{AB} = \frac{\bar{\lambda}^2}{6D}$$

where $D = D_A + D_B$, the sum of the diffusion coefficients for A and B, and $\bar{\lambda}^2$ is the mean square displacement needed to break off an encounter. The value of λ can be set equal to the diameter of a solvent molecule or set equal to $2d_{AB}$.

The probability of a reaction occurring during an encounter¹ is

$$P_r = \frac{\nu p \exp(-LE/RT)}{\nu p \exp(-LE/RT) + \tau_{AB}^{-1}}$$

where ν is the frequency of collision between A and B during an encounter and p is the steric factor. Thus, the collision theory bimolecular rate constant in solution becomes

$$k = \frac{2\pi d_{AB} \bar{\lambda}^2 \nu p L \exp(-LE/RT)}{3}$$

If τ_{AB}^{-1} is small relative to $\nu p \exp(-LE/RT)$, then the reaction is diffusion controlled. If not, then $k = 4\pi d_{AB} D L P_r$. This

is the liquid-phase equivalent of the equation for the bimolecular constant derived in the gas phase with the steric factor p included. Collision theory predicts that Arrhenius frequency factors will be slightly larger in solution than in the gas phase for bimolecular reactions. **See** Arrhenius Equation; Transition-State Theory; Fick's Laws

¹J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, Wiley, New York.

²W. J. Moore (1972) *Physical Chemistry*, Prentice-Hall, Englewood Cliffs, New Jersey.

³M. V. Smoluchowski (1916) *Phys. Zeit.* **17**, 557.

⁴M. V. Smoluchowski (1916) *Phys. Zeit.* **17**, 583.

DIFFUSION-CONTROLLED CRYSTAL GROWTH

For an aqueous suspension of crystals to grow, the solute must (a) make its way to the surface by diffusion, (b) undergo desolvation, and (c) insert itself into the lattice structure. The first step involves establishment of a stationary diffusional concentration field around each particle. The elementary step for diffusion has an activation energy ($\Delta G^\ddagger$), and a molecule or ion changes its position with a frequency of $(k_B T/h) \exp[-\Delta G^\ddagger/k_B T]$. Einstein's treatment of Brownian motion indicates that a displacement of λ will occur within a time τ if λ equals the square root of $2D\tau$. Thus, the rate constant for change of position equal to one ionic diameter d will be

$$\tau^{-1} = \frac{2D}{d^2} = (k_B T/h) \exp(-\Delta G^\ddagger/k_B T)$$

Nielsen¹ treats the solute concentration $c(\rho, t)$ in the space surrounding a particle as a function of distance (radius ρ from the center) and time t , and the diffusion process obeys the following equations for dc/dt and j (defined as the amount of solute diffusing in the direction of the solute concentration gradient per unit area of a cross section lying perpendicular to the gradient):

$$j = Ddc/d\rho$$

$$dc/dt = D[d^2c/d\rho^2 + (2/\rho)dc/d\rho]$$

The following boundary conditions apply: (a) $c(\rho, 0) = c_{\text{bulk}}$, the concentration at "infinite distance" or bulk concentration; (b) $c(r, t) = c_{\text{solubility}}$, the solute concentration at saturation; and (c) $c(\infty, t) = c_{\text{bulk}}$. The solution (assuming constant r and c_{bulk}) is

$$c = s + (c_{\text{bulk}} - s) \{1 - (r/\rho) (1 - \text{erf}[(\rho - r)/(2Dt)^{1/2}])\}$$

which, when t is infinity (*i.e.*, the system becomes stationary), reduces to

$$c = s + (c_{\text{bulk}} - s) \{1 - (r/\rho)\}$$

Nielsen² obtained the approximate solution for the time evolution of the radius:

$$r = [2Dv(c_{\text{bulk}} - s)t/q]^{1/2}$$

where q is a function of $v(c_{\text{bulk}} - s)$. This equation also predicts that diffusion-controlled crystal growth depends on solution viscosity which is one determinative factor in the magnitude of D . **See** Biomineralization

¹A. E. Nielsen (1964) *Kinetics of Precipitation*, pp. 135-138, Pergamon, Oxford.

²A. E. Nielsen (1961) *J. Phys. Chem.* **65**, 46.

DIFFUSION-LIMITED REACTION

A reaction whose rate is limited (or controlled) only by the speed with which reactants diffuse to each other. For a ligand binding to a protein, the bimolecular rate constant for diffusion-limited association is around $10^9 \text{ M}^{-1}\text{s}^{-1}$. The enzyme acetylcholinesterase has an apparent on-rate constant of $1.6 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$ with its natural cationic substrate acetylcholine, and the on-rate constant of about 6×10^8 with acetylselenoylcholine and about 2.5×10^8 with acetylthiocholine. On the other hand, neutral substrates such as phenylacetate and isoamylacetate have bimolecular rate constants of 6×10^5 and $1.6 \times 10^7 \text{ M}^{-1}\text{s}^{-1}$. This illustrates the degree to which electrostatic interactions can change a reaction from a diffusion-limited process to one that involves other types of protein-ligand interactions. **See** Chemical Kinetics; Diffusion-Limited Reaction

DIFFUSION OF LIGAND TO RECEPTOR

The random-motion process by which a ligand approaches and penetrates a tissue surface: first, by passing down a concentration gradient by three-dimensional Brownian motion in the bulk solution; then, by reaching the surface where the dimensionality of random excursions reduces it to a two-dimensional process; and finally, by docking at the receptor site¹⁻³. Once even weakly associated within the tissue surface, receptor electrostatics and mobility probably work together to guide the ligand to its binding site. Moreover, the kinetics of ligand diffusion to its receptor(s) is likely to greatly influence the time dependence of metabolite and drug action.

¹H. C. Berg & E. M. Purcell (1977) *Biophys. J.* **20**, 193.

²J. Crank (1975) *The Mathematics of Diffusion*, 2nd ed., Clarendon Press, Oxford.

³T. Kenakin (1993) *Pharmacologic Analysis of Drug-Receptor Interaction*, 2nd ed., pp. 20-24, Raven Press, New York.

DIFFUSION OF MOLECULES INTO A PORE

As illustrated by experiments on mediated uptake of proteins¹ through the nuclear pores of frog oocyte nuclei, molecular diffusion through large pores depends on the pore's cross-sectional area and the size and shape of the diffusing molecule¹⁻³. As the dimension of the molecule approaches that of the pore, movement into and through the pore will become hindered, because molecules must enter without hitting the rim of the pore. Assuming no bulk solvent flow and pure diffusion of a rigid sphere through a pore, the pore's effective cross-sectional area, $A_{\text{effective}}$, will be reduced by the radius of the entering molecule. For a sphere entering a cylindrical pore,

$$A_{\text{effective}} = A_o(r - a)2/K_1r^2$$

where A_o is the actual cross-sectional area, a is the sphere's radius, and K_1 is the drag coefficient; the latter of which is a complex function of a/r .

¹C. M. Feldherr, R. J. Cohen & J. A. Ogburn (1983) *J. Cell Biol.* **96**, 1486.

²W. L. Haberman & R. M. Sayre (1958) Motion of Rigid and Fluid Spheres in Stationary and Moving Liquids Inside Cylindrical Tubes, *David Taylor Model Basin Report No. 1143*, U. S. Navy, Washington, D. C.

³P. L. Paine & P. Scherr (1975) *Biophys. J.* **15**, 1087.

DIFFUSION OF OXYGEN WITHIN RED CELLS

As pointed by Schroeder and Holmquist¹, the rate of oxygen transport within an erythrocyte depends on several parameters: the partial pressure of O₂ at the inner surface of the red cell membrane; the solubility of molecular oxygen, which also depends on the hemoglobin concentration (typically, 34% w/v), the fractional saturation of heme sites with oxygen, the kinetics of oxygen binding and release steps, the self-diffusion constant of hemoglobin, and the shape and volume of the red cell itself. Using the Einstein relation, $t = x^2/2D$, Schroeder and Holmquist¹ estimated the approximate time needed for diffusion of an O₂ molecule to the red cell's interior by using values of 1 micrometer for the half-thickness x of a red cell and 7.3×10^{-6} cm²/s for the diffusivity of molecular oxygen. This puts the diffusion time at about 0.0007 s, which is about 100 times faster than the

0.058 s period required for hemoglobin ($D_{\text{Hb}} = 1.75 \times 10^{-7}$ cm²/s) to diffuse a similar distance at 37°C. Because half-life for oxygen dissociation from hemoglobin is 0.0078 s, each oxygen molecule must successively "hitch a ride" on 7–8 hemoglobin molecules on the way to the red cell's center. Note also that the transfer of oxygen does not involve what Scholander² termed a "bucket brigade" wherein diffusing hemoglobin molecules hand off oxygen molecules to each other. Theoretical and experimental grounds³ offer no support for direct bimolecular transfer of Hb-bound O₂ to another hemoglobin.

¹W. A. Schroeder & W. R. Holmquist (1968) in *Structural Chemistry and Biology* (A. Rich and N. Davidson, eds.), pp. 245-246, W. H. Freeman, San Francisco.

²P. F. Scholander (1960) *Science* **131**, 505.

³J. B. Wittenberg (1966) *J. Biol. Chem.* **241**, 104.

DIFFUSION TIME

The period of time t required for a molecule with a diffusion coefficient D (units of cm² s⁻¹) to diffuse a mean square distance x^2 , as given by the expression

$$t = \frac{x^2}{2D}$$

For example, a molecule with a diffusivity of 10^{-5} cm² s⁻¹ requires about 0.5 seconds to diffuse about 10 micrometers (or 10^{-1} cm), which corresponds to the dimension of many cells.

DIFFUSION WITHIN CELLULAR/ EXTRACELLULAR SPACES

Free diffusion of molecules in solution is characteristically a haphazard process with net directionality determined only by solute gradients and diffusion coefficients. Within cellular and extracellular spaces, however, diffusion can be strongly influenced by noncovalent interactions of solvent and solute molecules with membranes as well as the cellular and extracellular matrix¹⁻³. Channels and orifices can also alter the movement of solute and solvent molecules. These interactions can greatly alter the magnitude of the diffusion coefficient for a molecule from its isotropic value D in water to apparent diffusion coefficient D^* (which often can be directionally resolved into D_x^* , D_y^* , and D_z^*). The parameter λ , known as the tortuosity, equals $(D/D^*)^{1/2}$. In principle, λ has x , y , and z components that need not be equal if there is any anisotropy in the local electrical fields or porosity of the matrix.

Diffusional anisotropy can be measured by many procedures. Fluorescence microscopy¹ is an especially powerful technique that permits one to use video microscopy to follow the redistribution of fluorescently tagged molecules introduced into specific cellular or extracellular spaces. With the availability of microinjection methods, one can initiate such studies by placing a bolus dose of tagged molecules with considerable spatial precision. In NMR microscopy, one takes advantage of the fact that NMR signal intensities depend in part on translational self-diffusion, and any field gradient affecting transverse magnetization prior to or during signal acquisition will result in loss of signal strength^{4,6}. An example of the NMR approach is provided by the work of Inglis *et al.*⁴ on the diffusional anisotropy of water molecules in excised rat spinal cord. Anisotropic diffusion coefficients were measured in perfused-fixed rat cervical spinal cord in a 300 MHz imaging spectrometer by using apparent diffusion tensor (ADT) imaging. The method has an in-plane resolution of 10 μm , sufficient to discern diffusional characteristics in defined histologic regions (see the table below).

Anisotropic Diffusion Coefficients for Water Protons in Rat Cervical Spinal Cord⁴

Anatomical Region	D_{xx}	D_{yy}	D_{zz}
Dorsal funiculus	0.026 ^a	0.033	0.209
Ventrolateral funiculus	0.039	0.051	0.194
Ventral funiculus	0.027	0.037	0.187
Dorsolateral funiculus	0.049	0.045	0.195
Dorsal horn	0.144	0.163	0.183
Ventral horn	0.181	0.157	0.150
Gray commissure	0.178	0.097	0.124

^aAll anisotropic apparent diffusion coefficients are expressed in units of $10^{-3} \text{ mm}^2 \text{ s}^{-1}$. The z-axis runs parallel to the spinal cord.

¹C. Nicholson & E. Sykova (1998) *Trends Neuro. Sci.* **21**, 207.

²H. D. Lux & E. Neher (1973) *Exp. Brain Res.* **17**, 190.

³C. Nicholson & L. Tao (1993) *Biophys. J.* **65**, 2277.

⁴B. A. Inglis, L. Yang, E. D. Wirth, D. Plant & T. H. Mareci (1997) *Magn. Reson. Imag.* **15**, 441.

⁵H. C. Torrey (1955) *Phys. Rev.* **104**, 563.

⁶P. T. Callaghan (1991) *Principles of Nuclear Magnetic Resonance Microscopy*, pp. 157-169 and pp. 201-208, Clarendon Press, Oxford.

DIHYDROFOLATE REDUCTASE

This enzyme [EC 1.5.1.3], also called tetrahydrofolate dehydrogenase, catalyzes the reversible reaction of 7,8-dihydrofolate with NADPH to produce 5,6,7,8-tetrahydrofolate and NADP⁺. The enzyme isolated from mammals and some microorganisms can also slowly catalyze

the reduction of folate to 5,6,7,8-tetrahydrofolate. **See also** *Dihydrofolate Reductase-Thymidylate Synthase*

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

R. L. Blakley (1995) *Adv. Enzymol.* **70**, 23.

H. Eisenberg, M. Mevarech & G. Zaccai (1992) *Adv. Protein Chem.* **43**, 1.

K. A. Johnson & S. J. Benkovic (1990) *The Enzymes*, 3rd. ed., **19**, 159.

R. F. Colman (1990) *The Enzymes*, 3rd ed., **19**, 283.

J. Kraut & D. A. Matthews (1987) *Biological Macromolecules & Assemblies* (F. A. Jurnak & A. McPherson, eds.), pp. 1-71, Wiley, New York.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Cryoenzymology, **63**, 338; inhibitor, **63**, 396, 399; tight-binding inhibitor, **63**, 437, 462, 466; hysteresis, **240**, 318; deuteration for protein-ligand interaction study, **239**, 684-685; mechanism, **240**, 317-318; NADPH off-rate, **240**, 318; pre-steady-state burst, **240**, 318; unfolding, reaction coordinate diagram, **202**, 117; catalysis, transient kinetics, **249**, 23-27; *Escherichia coli* DHFR [hysteresis, structural basis, **249**, 545; kinetic mechanism, derivation, **249**, 25-27; ligand binding, association and dissociation rate constants, **249**, 24-27; site-directed mutants, altered pH dependencies, **249**, 111-112; transient kinetics, **249**, 23-27 (inhibitor binding, **249**, 23-27; site-directed mutant, **249**, 108-109)]; hysteresis, **240**, 318; mechanism, **240**, 317-318; mitochondrial import of fusion protein, **260**, 243-245, 248, 271-272; NADPH off-rate, **240**, 318; pre-steady-state burst, **240**, 318; reaction catalyzed, **249**, 23; transition state and multisubstrate analogs, **249**, 304

DIHYDROFOLATE SYNTHETASE

This enzyme [EC 6.3.2.12] catalyzes the reaction of ATP with dihydropteroate and glutamate to generate ADP, orthophosphate, and dihydrofolate.

G. M. Brown & J. M. Williamson (1982) *Adv. Biochem.* **53**, 345.

DIHYDROLIPOAMIDE DEHYDROGENASE

This enzyme [EC 1.8.1.4] (also known as lipoamide reductase (NADH), the E3 component of α -ketoacid dehydrogenase complexes, lipoyl dehydrogenase, dihydrolipoil dehydrogenase, and diaphorase) requires FAD and catalyzes the reaction of dihydrolipoamide with NAD⁺ to produce lipoamide and NADH. The protein is also a component of the pyruvate dehydrogenase complex and the α -ketoglutarate (or, 2-oxoglutarate) dehydrogenase complex. **See also** *Pyruvate Dehydrogenase*

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

M. S. Patel, N. N. Vettakkorumakankav & T.-C. Liu (1995) *Meth. Enzymol.* **252**, 186.

C. H. Williams, Jr. (1976) *The Enzymes*, 3rd ed., **13**, 89.

DIHYDRONEOPTERIN ALDOLASE

This enzyme [EC 4.1.2.25] catalyzes the conversion of 2-amino-4-hydroxy-6-(*D*-erythro-1,2,3-trihydroxypropyl)-7,8-dihydropteridine to 2-amino-4-hydroxy-6-hydroxymethyl-7,8-dihydropteridine and glycolaldehyde.

G. M. Brown & J. M. Williamson (1982) *Adv. Biochem.* **53**, 345.

DIHYDRONEOPTERIN TRIPHOSPHATE PYROPHOSPHOHYDROLASE

The following is a recent review on the molecular and physical properties of this enzyme.

G. M. Brown & J. M. Williamson (1982) *Adv. Biochem.* **53**, 345.

DIHYDROOROTASE

This enzyme [EC 3.5.2.3], also called carbamoylaspartic dehydrase, catalyzes the hydrolysis of (*S*)-dihydroorotate to yield *N*-carbamoyl-L-aspartate.

M. E. Jones (1980) *Ann. Rev. Biochem.* **49**, 253.

DIHYDROOROTATE DEHYDROGENASE

This enzyme [EC 1.3.99.11] catalyzes the reaction of (*S*)-dihydroorotate with an acceptor substrate to produce orotate and the reduced acceptor. Both iron and zinc ions are needed as cofactors. Acceptor substrates include 2,6-dichloroindophenol, 1,10-phenanthroline, and dioxygen (although dioxygen isn't as effective as the first two).

M. E. Jones (1980) *Ann. Rev. Biochem.* **49**, 253.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

DIHYDROOROTATE OXIDASE

This enzyme [EC 1.3.3.1] catalyzes the reaction of (*S*)-dihydroorotate with dioxygen to produce orotate and hydrogen peroxide. The enzyme requires FAD and FMN as cofactors. Ferricyanide can also serve as a substrate as well.

M. H. O'Leary (1989) *Ann. Rev. Biochem.* **58**, 377.

DIHYDROPTERIDINE REDUCTASE

This enzyme [EC 1.6.99.7] catalyzes the reaction of NAD(P)H with 6,7-dihydropteridine (that is, the quinoid form of dihydropteridine) to produce NAD(P)⁺ and 5,6,7,8-tetrahydropteridine. The enzyme is not identical with dihydrofolate reductase.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

S. Kaufman (1987) *Meth. Enzymol.* **142**, 97.

H. Hasegawa & N. Nakanishi (1987) *Meth. Enzymol.* **142**, 103.

F. A. Firgaira, R. G. H. Cotton, I. Jennings & D. M. Danks (1987) *Meth. Enzymol.* **142**, 116.

K. G. Scrimgeour & S. Cheema-Dhadli (1987) *Meth. Enzymol.* **142**, 127.

DIHYDROPTEROATE SYNTHASE

This enzyme [EC 2.5.1.15], also known as dihydropteroate pyrophosphorylase, catalyzes the reaction of 2-amino-4-hydroxy-6-hydroxymethyl-7,8-dihydropteridine diphosphate with 4-aminobenzoate to produce pyrophosphate and dihydropteroate.

G. M. Brown & J. M. Williamson (1982) *Adv. Enzymol.* **53**, 345.

DIHYDROPYRIMIDINASE

This enzyme [EC 3.5.2.2], also called hydantoinase, catalyzes the hydrolysis of 5,6-dihydrouracil to produce 3-ureidopropionate. The enzyme can also utilize dihydrothymine and hydantoin as substrates.

O. W. Griffith (1986) *Ann. Rev. Biochem.* **55**, 855.

DIHYDROURACIL DEHYDROGENASE

Dihydrouracil dehydrogenase (NAD⁺) [EC 1.3.1.1] catalyzes the reaction of 5,6-dihydrouracil with NAD⁺ to produce uracil and NADH. Dihydrouracil dehydrogenase (NADP⁺) [EC 1.3.1.2], also called dihydropyrimidine dehydrogenase (NADP⁺) and dihydrothymine dehydrogenase, catalyzes the reaction of 5,6-dihydrouracil with NADP⁺ to produce uracil and NADPH. The enzyme can also use dihydrothymine as a substrate.

O. W. Griffith (1986) *Ann. Rev. Biochem.* **55**, 855.

DIHYDROXYACETONE PHOSPHATE ACYLTRANSFERASE

This enzyme [EC 2.3.1.42], also known as glycerone-phosphate *O*-acyltransferase, catalyzes the reaction of an acyl-CoA with dihydroxyacetone phosphate (or, glycerone phosphate) to produce coenzyme A and an acyldihydroxyacetone phosphate (or, an acylglycerone phosphate). The acyl-CoA derivatives of palmitate, stearate, and oleate can all be utilized as substrates, with highest activity observed with palmitoyl-CoA.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.

K. O. Webber & A. K. Hajra (1992) *Meth. Enzymol.* **209**, 92.

R. M. Bell & R. A. Coleman (1983) *The Enzymes*, 3rd ed., **16**, 87.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

DIHYDROXYACETONE SYNTHASE

This enzyme catalyzes the reaction of formaldehyde with xylulose 5-phosphate to produce dihydroxyacetone and glyceraldehyde 3-phosphate.

L. V. Dystrikh, W. de Koning & W. Harder (1990) *Meth. Enzymol.* **188**, 435.

DIHYDROXY-ACID DEHYDRATASE

This enzyme [EC 4.2.1.9] catalyzes the conversion of 2,3-dihydroxy-3-methylbutanoate to 3-methyl-2-oxobutanoate and water.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

K. Kiritani & R. P. Wagner (1970) *Meth. Enzymol.* **17A**, 755.

2,3-DIHYDROXYBIPHENYL 1,2-DIOXYGENASE

This enzyme [EC 1.13.11.39], also referred to as biphenyl-2,3-diol 1,2-dioxygenase, catalyzes the reaction of biphenyl-2,3-diol with dioxygen to produce 2-hydroxy-6-oxo-6-phenylhexa-2,4-dienoate and water. The enzyme can also use 3-isopropylcatechol as a substrate, forming 7-methyl-2-hydroxy-6-oxoocta-2,4-dienoate as a product. This protein is not identical with catechol 2,3-dioxygenase.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

DIHYDROXYFUMARATE DECARBOXYLASE

This enzyme [EC 4.1.1.54] catalyzes the conversion of dihydroxyfumarate to tartronate semialdehyde and carbon dioxide.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

2,3-DIHYDROXYINDOLE 2,3-DIOXYGENASE

This enzyme [EC 1.13.11.23] catalyzes the reaction of 2,3-dihydroxyindole with dioxygen to produce anthranilate and carbon dioxide.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

3,4-DIHYDROXYPHENYLACETATE 2,3-DIOXYGENASE

This enzyme [EC 1.13.11.15], also called homoprotocatechuate 2,3-dioxygenase, is an iron-dependent enzyme

that catalyzes the reaction of 3,4-dihydroxyphenylacetate with dioxygen to produce 2-hydroxy-5-carboxymethylmuconate semialdehyde.

J. E. Penner-Hahn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 439.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

DIHYDROXYPTHALATE DECARBOXYLASE

This enzyme [EC 4.1.1.55] catalyzes the conversion of 4,5-dihydroxyphthalate to 3,4-dihydroxybenzoate and carbon dioxide.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

2,5-DIHYDROXYPYRIDINE 5,6-DIOXYGENASE

This enzyme [EC 1.13.11.9] is an iron-dependent protein that catalyzes the reaction of 2,5-dihydroxypyridine with dioxygen to produce maleamate and formate.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

2,3-DIHYDROXY-9,10-SECOANDROSTA- 1,3,5(10)-TRIENE-9,17-DIONE 4,5-DIOXYGENASE

This enzyme [EC 1.13.11.25], also referred to as steroid 4,5-dioxygenase and 3-alkylcatechol 2,3-dioxygenase, is an iron-dependent enzyme that catalyzes the reaction of 3,4-dihydroxy-9,10-secoandrost-1,3,5(10)-triene-9,17-dione with dioxygen to produce 3-hydroxy-5,9,17-tri-oxo-4,5:9,10-disecoandrost-1(10),2-dien-4-oate. The enzyme can also use 3-isopropylcatechol and 3-*tert*-butyl-5-methylcatechol as substrates.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

DIIODOPHENYLPYRUVATE REDUCTASE

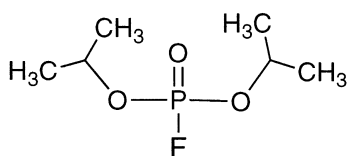
This enzyme [EC 1.1.1.96] catalyzes the reversible reaction of β -(3,5-diiodo-4-hydroxyphenyl)lactate with NAD⁺ to produce β -(3,5-diiodo-4-hydroxyphenyl)pyruvate and NADH. The substrates for this enzyme must contain an aromatic ring with a pyruvate side chain (or lactate), the most active substrates being halogenated derivatives. Potential substrates with hydroxyl or amino groups in the 3 or 5 position are inactive.

V. G. Zannoni (1970) *Meth. Enzymol.* **17A**, 665.

DIODOTYROSINE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.24] catalyzes the reversible reaction of 3,5-diiodotyrosine with α -ketoglutarate (or, 2-oxoglutarate) to generate 3,5-diiodo-4-hydroxyphenylpyruvate and glutamate. Also acting as substrates are the 3,5-dichloro, 3,5-dibromo, and the 3-iodo derivatives of tyrosine, as well as thyroxine and triiodothyronine.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

DIISOPROPYL FLUOROPHOSPHATE

A potent irreversible inhibitor (abbreviated DFP) of many serine proteinases and serine esterases (especially acetylcholinesterase). This substance is **EXTREMELY POISONOUS**, but the vapor state can be minimized by using dry, water-miscible solvents such as 2-propanol¹. Aqueous solutions become inactivated by hydrolysis, but solutions made with dry 2-propanol are stable at -20°C for many months.

Enzyme inactivation by DFP and related organophosphorus compounds (diethyl fluorophosphate, dipropyl fluorophosphate, isopropylmethyl fluorophosphate, diethyl-4-nitrophenylphosphate) can be treated as a bimolecular process:



where the enzyme engages in nucleophilic attack on the electrophilic phosphorus atom. Bimolecular rate constants¹ for inhibition of acetylcholinesterase, chymotrypsin, and trypsin are 4.6×10^4 , 1.5×10^4 , and $9.2 \times 10^2 \text{ M}^{-1} \text{ min}^{-1}$, respectively.

¹Cohen, R. A. Osterman & F. Berends (1967) *Meth. Enzymol.* **11**, 686.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Types of organophosphorus inhibitors, **11**, 686-688; toxicity hazards, **11**, 688; purity and analysis, **11**, 688; solutions of organophosphorus compounds, **11**, 689; estimation of specific radioactivity of organophosphorus compounds, **11**, 689-690; method for estimating phosphorus content, **11**, 691; reactions with enzymes, **11**, 691-701 [rate constants, **11**, 692; phosphorylation of chymotrypsin, **11**, 694-696; identification of phosphoryl and phosphonyl peptides, **11**, 697]; inhibition of serine peptidases, **244**, 7-8, 432; mechanism

of inactivation, **244**, 432-433; phosphorylserine bond formation, **244**, 433; stability inhibitor, **244**, 432; toxicity, **244**, 7-8, 432.

DILUTION EFFECTS

The reduction in concentration of reactants, enzymes, and solute molecules can provide important information about kinetic systems. For example, one can readily differentiate a first-order process from a second-order process by testing whether the period required to reduce a reactant concentration to 50% of its initial value depends on dilution. First-order processes and intramolecular processes should not exhibit any effect on rate by diluting a reactant. In terms of enzyme-catalyzed processes, the Michaelis-Menten equation requires that the initial reaction velocity depends strictly on the concentration of active catalyst. Dilution can also be used to induce dissociation of molecular complexes or to promote depolymerization of certain polymers (such as F-actin and microtubules).

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Dilution of enzyme samples, **63**, 10; lipolysis substrate effect, **64**, 361, 362; dilution jump kinetic assay, **74**, 14-19, 28; dilution method [for dissociation equilibria, **61**, 65-96; continuous dilution cuvette, **61**, 78-96; data analysis, **61**, 74, 75; equations, **61**, 70-74; errors, **61**, 76-78; experimental procedures, **61**, 69, 70; merits, **61**, 75, 76; theory, **61**, 68, 69

DIMERIZATION

A reaction between two identical molecular entities of chemical species to give a new species. Dimerization need not require covalent bond formation. Formation of a complex between two polypeptide chains or formation of an excimer are processes that do not require covalent bond formation. If the two polypeptide chains are identical, the complex is usually referred to as a homodimer; otherwise, the resulting complex is referred to as heterodimer.

DIMETHYLALLYLTRANSTRANSFERASE

This enzyme [EC 2.5.1.1] (also referred to as prenyltransferase and geranyl-diphosphate synthase) catalyzes the reaction of dimethylallyl diphosphate and isopentenyl diphosphate to produce geranyl diphosphate and pyrophosphate (or, diphosphate). The enzyme will not accept larger prenyl diphosphates as substrates.

H. Kleinig (1989) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **40**, 39.

DIMETHYLANILINE MONOOXYGENASE

This enzyme [EC 1.14.13.8], also referred to as microsomal flavin-containing monooxygenase and dimethyl-

aniline oxidase, catalyzes the reaction of *N,N*-dimethylaniline with NADPH and dioxygen to produce *N,N*-dimethylaniline *N*-oxide, NADP⁺, and water. The enzyme uses FAD and can utilize a number of dialkylarylamines as substrates.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

F. P. Guengerich (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 97.

DIMINISHED ISOTOPE EFFECT

The value corresponding to the observed isotope effect, expressed as the ratio: $[(k/k') - 1]$, as a consequence of other rate-contributing steps¹.

¹D. B. Northrop (1982) *Meth. Enzymol.* **87**, 607.

DIOL DEHYDRASE

The following are recent reviews on the molecular and physical properties of this enzyme.

J. A. Stubbe (1989) *Ann. Rev. Biochem.* **58**, 257.

B. M. Babior & J. S. Krouwer (1979) *Crit. Rev. Biochem.* **6**, 35.

R. H. Abeles (1971) *The Enzymes*, 3rd ed., **5**, 481.

DIPEPTIDASES

There are many dipeptidases [EC 3.4.13.x]. Cytosol non-specific dipeptidase [EC 3.4.13.18] (also referred to as peptidase A, glycylglycine dipeptidase, glycylleucine dipeptidase, and *N*²- β -alanylarginine dipeptidase) catalyzes the hydrolysis of dipeptides. Membrane dipeptidase [EC 3.1.13.19] (also known as microsomal dipeptidase, renal dipeptidase, and dehydropeptidase I) is a zinc-dependent enzyme (a member of the peptidase family M19) that also catalyzes the hydrolysis of dipeptides.

S. S. Tate (1985) *Meth. Enzymol.* **113**, 471.

R. J. DeLange & E. L. Smith (1971) *The Enzymes*, 3rd ed., **3**, 81.

DIPEPTIDYL-PEPTIDASE I

This enzyme [EC 3.4.14.1], also called cathepsin C and cathepsin J, catalyzes the hydrolysis of a peptide bond resulting in the release of an N-terminal dipeptide, XaaXbb–Xcc, except when Xaa is an arginyl or a lysyl residue, or Xbb or Xcc is a prolyl residue. This enzyme, a member of the peptidase family C1, is a Cl[−]-dependent lysosomal cysteine-type peptidase.

M. J. Mycek (1970) *Meth. Enzymol.* **19**, 285.

DIPEPTIDYL-PEPTIDASE IV

This enzyme [EC 3.4.14.5] (also referred to as dipeptidyl aminopeptidase IV, Xaa–Pro-dipeptidylaminopepti-

dase, Gly–Pro naphthylamidase, and postproline dipeptidyl aminopeptidase IV) catalyzes the hydrolysis of a peptide bond in a protein such that there is a release of an N-terminal dipeptide, XaaXbb–Xcc, preferentially when Xbb is a prolyl residue and provided Xcc is neither a prolyl nor a hydroxyprolyl residue. This protein, a member of the peptidase family S9B, is a membrane-bound serine-type peptidase isolated from mammalian tissue.

Y. Ikehara, S. Ogata & Y. Misumi (1994) *Meth. Enzymol.* **244**, 215.

DIPHOSPHOMEVALONATE DECARBOXYLASE

This enzyme [EC 4.1.1.33], also known as mevalonate pyrophosphate decarboxylase, catalyzes the reaction of ATP with (*R*)-5-diphosphomevalonate to produce ADP, orthophosphate, carbon dioxide, and isopentenyl diphosphate.

DIPHThERIA TOXIN

This bacterial ADPRtransferase inactivates host cell ribosomal elongation factor eEF-2 by NAD⁺-dependent covalent modification of diphthamide, an acceptor-modified histidyl residue, 2-[3-carboxyamido-3-(trimethylammonio)propyl]histidine.

L. Passador & W. Iglewski (1994) *Meth. Enzymol.* **235**, 617.

N. J. Oppenheimer & A. L. Handlon (1992) *The Enzymes*, 3rd ed., **20**, 453.

C.-Y. Lai (1986) *Adv. Enzymol.* **58**, 99.

DIPOLAR BOND

A bond formed by coordination of two neutral molecules or moieties. IUPAC now favors use of this term over various synonyms, such as coordinate covalent bond.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

DIPOLE-DIPOLE INTERACTION

An interaction, either intermolecular or intramolecular, between molecules, groups, or bonds having a permanent electric dipole moment. The distance and relative orientation between the two dipoles governs the strength of this interaction. **See also** *van der Waals Forces*

DIPOLE MOMENT

A vectoral entity (usually symbolized by μ) having a magnitude equal to Qr where Q represents the magnitude of either charge, r is the distance separating them.

The dipole moment is a fundamental property of a molecule (or any dipole unit) in which two opposite charges are separated by a distance^{1,2}. This entity is commonly measured in debye units (symbolized by D), equal to 3.33564×10^{-30} coulomb·meters, in SI units). Since the net dipole moment of a molecule is equal to the vectorial sum of the individual bond moments, the dipole moment provides valuable information on the structure and electrical properties of that molecule. The dipole moment can be determined by use of the Debye equation for total polarization. Examples of dipole moments (in the gas phase) are water (1.854 D), ammonia (1.471 D), nitromethane (3.46 D), imidazole (3.8 D), toluene (0.375 D), and pyrimidine (2.334 D). Even symmetrical molecules will have a small, but measurable dipole moment, due to centrifugal distortion effects. Methane^{3,4}, for example, has a value of about 5.4×10^{-6} D.

Dipole Moments (μ) of Selected Molecules

Molecule	μ	Molecule	μ
Oxygen, O ₂ (gas)	0	Ethyl acetate (gas)	1.76
Methane (gas)	0	Methyl bromide (gas)	1.80
Carbon dioxide (gas)	0	Water, H ₂ O (liquid)	1.85
Trimethylamine (gas)	0.62	Methyl chloride (gas)	1.86
Dimethylamine (gas)	1.02	Formic acid (in dioxane)	2.07
Hydrogen chloride (gas)	1.08	Pyridine (in benzene)	2.21
Methylamine (gas)	1.33	Acetaldehyde (gas)	2.68
Phenol (gas)	1.40	Acetone (gas)	2.84
Ammonia (gas)	1.46	Nitromethane (gas)	3.50
Methyl iodide (gas)	1.59	Acetamide (in benzene)	3.63
n-Propyl alcohol (gas)	1.66	Urea (in dioxane)	4.56

¹P. Debye (1945) *Polar Molecules*, Dover, New York.

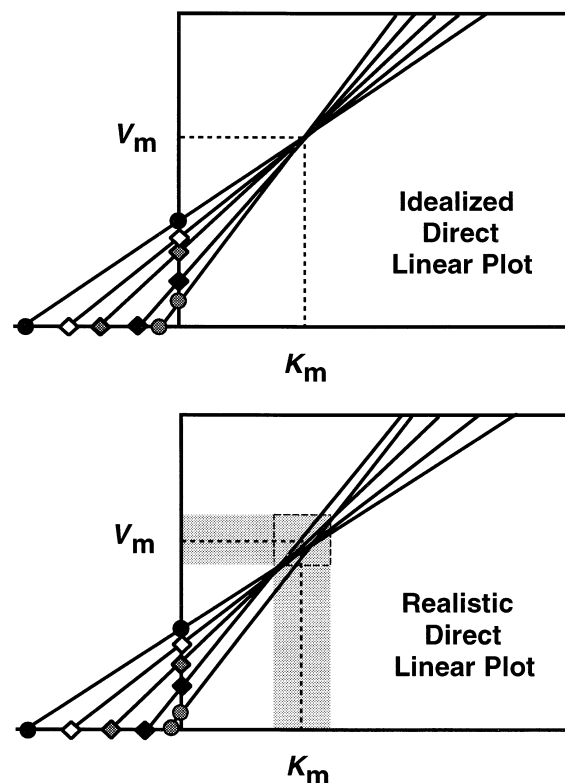
²O. Exner (1975) *Dipole Moments in Organic Chemistry*, Georg Thieme, Stuttgart.

³I. Ozier (1971) *Phys. Rev. Lett.* **27**, 1329.

⁴A. Rosenberg, I. Ozier & A. K. Kudian (1972) *J. Chem. Phys.* **57**, 568.

DIRECT-LINEAR PLOT

A graphical method for analyzing initial rate enzyme kinetic data that avoids some of the weighting and statistical problems associated with the other graphical procedures¹⁻³. The method offers the advantage that it is relatively insensitive to aberrant observations or data points. In a Cartesian-coordinate system, a substrate concentration is marked along the horizontal axis (*i.e.*, the *x* axis) to the left of the origin. The initial velocity (*v*) observed with that substrate concentration is placed on the vertical axis (*i.e.*, the *y* axis) above the origin. A line is then drawn connecting those two points. The process is repeated for every substrate concentration and initial velocity pair.



With perfect data, there will be a common intersection point at coordinates (K_a , V_{max}) in the first quadrant. However, for real experimental data, there will be a number of intersection points. The maximum number of intersection points will be $(n/2)(n - 1)$ where n is the number of lines. If different sets of lines have a common intersection point, then that point is treated with weighting. The estimate of the K_a value will be the median value for the *x*-coordinate of these intersection points. Correspondingly, the median value of the *y*-coordinate of these intersection points provides the value for V_{max} .

The method is not as useful with multisubstrate systems; however, Cornish-Bowden and Endrenyi⁴ have presented a robust regression method to treat data with more parameters.

¹R. Eisingal & A. Cornish-Bowden (1974) *Biochem. J.* **139**, 715.

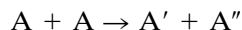
²A. Cornish-Bowden & R. Eisingal (1974) *Biochem. J.* **139**, 721.

³A. Cornish-Bowden & R. Eisingal (1978) *Biochim. Biophys. Acta* **523**, 268.

⁴A. Cornish-Bowden & L. Endrenyi (1986) *Biochem. J.* **234**, 21.

DISPROPORTIONATION

Any reaction in which one molecular entity reacts with an identical molecular entity, transferring an electron, atom or group from one reactant to the other:



The reverse of disproportionation is referred to as comproportionation. A special case of disproportionation (or dismutation) is radical disproportionation which serves as a termination process in the fate of free radicals. For example, $2 \text{CH}_3\text{CH}_2\cdot \rightarrow \text{CH}_3\text{CH}_3 + \text{CH}_2=\text{CH}_2$.

DISSOCIATION

The taking apart or separation of a single substance or molecular entity into two or more other substances or different molecular entities. Dissociation includes heterolytic and homolytic reactions, the release of free ions from ion pairs, the depolymerization process of a polymer consisting of smaller subunits, *etc.*

DISSOCIATION CONSTANT

The equilibrium constant (commonly symbolized K_d) describing the dissociation of a compound into two or more compounds, ions, or atoms. For the reaction $A + B \rightleftharpoons A-B$, the dissociation constant is $[A][B]/[A-B]$. The dissociation constant is the reciprocal of the association constant and the negative logarithm of the K_d is the pK_d . The dissociation constant for a Brønsted acid (commonly symbolized K_a) is $[H^+][A^-]/[HA]$, where HA is the undissociated acid and A^- represents the conjugate base. In aqueous solutions, water participates in this equilibrium. However, since the concentration of water is large and essentially constant, it has been incorporated into the constant K_a . The negative logarithm is referred to as the pK_a for the acid.

The dissociation constant for a base (symbolized K_b) equals $[B^+][OH^-]/[BOH]$, where BOH represents the undissociated base. As mentioned above, in aqueous solutions, the concentration of water is considered high and constant and is a portion of K_b . The “dissociation constant” (or, autoprotolysis constant) of water (symbolized K_w) equals $[H^+][OH^-] = 10^{-14}$ at 25°C. At 30°C, $K_w = 10^{-13.8}$ and at 37°C, $K_w = 10^{-13.6}$. The negative logarithm is pK_w . **See** Association Constant

DISSOCIATION CONSTANTS (Metal-Nucleotide Complexes)

O’Sullivan and Smithers¹ describe various protocols for determining dissociation constants (or, stability constants) of metal ion-nucleotide complexes. Morrison and Cleland^{2,3} have presented a kinetic method that has

proven to be quite useful. In this procedure the nucleotide competes for Mg^{2+} and another metal ion which binds to the nucleotide more tightly than Mg^{2+} . An enzyme is chosen that utilizes the Mg^{2+} -nucleotide complex as a substrate but which does not use the other metal ion-nucleotide complex. In addition, it is assumed that the metal ion under study (as well as Mg^{2+}) does not form an insoluble precipitate at the pH value of the studies and the concentration of the free metal ion. Upon determining the K_m value for MgATP, the K_d value for the MgATP complex, and the K_i value for the metal ion-nucleotide complex with the enzyme, all under the same set of experimental conditions, the K_d for the metal ion-nucleotide complex can be readily obtained. A computer program has been provided to assist in K_d determinations³.

¹W. J. O’Sullivan & G. W. Smithers (1979) *Meth. Enzymol.* **63**, 294.

²J. F. Morrison & W. W. Cleland (1980) *Biochemistry* **19**, 3127.

³W. W. Cleland (1995) *Meth. Enzymol.* **249**, 181.

DISSOCIATION CONSTANTS (Temperature Effects)

A temperature change can have an effect on the degree of ionization of various moieties on a protein as well as on substrates and effectors. Since proton dissociation constants (*i.e.*, K_a values) are thermodynamic parameters, a change in temperature can result in a significant alteration of the pH-activity curve¹.

Investigators have often used pH studies to aid in the identification of active-site residues. However, care must be exercised in this regard since observed pK values may refer to macroscopic parameters rather than microscopic values and pK values of groups can be greatly affected by the immediate environment. Because of these influences, it may be difficult to identify specific moieties responsible for binding and/or catalysis.

As was demonstrated in the discussion for the van’t Hoff relationship:

$$\Delta H = -2.303RT^2 dpK_a/dT$$

(Note that care should be exercised here. Changes in enthalpies are determined from equilibrium constants. The K_a values used in pH studies are dissociation constants and not association constants.) Plotting pK_a vs. the reciprocal of the absolute temperature will provide a value for ΔH . An early study using this procedure was

reported by Gutfreund² with trypsin. Gutfreund reported a pK_a value of 6.25 at 25°C and a ΔH value of 7000 cal·mol⁻¹ (29 kJ·mol⁻¹), concluding that the data agree with the participation of an imidazole moiety at the active site. In another early study (although only two temperatures were studied), Massey and Alberty³ concluded that the ΔH of ionization was less than 4500 cal·mol⁻¹ (19 kJ·mol⁻¹) in the inhibition studies of fumarate hydratase with thiocyanate, suggesting the participation of a carboxyl group. However, Knowles⁴ has pointed out that the factors that are sufficient to perturb a pK_a can also affect changes in enthalpy and/or entropy.

An investigator should also be concerned about the buffer since the pK_a of a buffer can also vary with temperature⁵⁻⁷. The pH of a buffer solution will vary with temperature depending on the variance of the activity coefficient terms and the pK_a of the buffer.

The temperature-dependent change in pK_a for some compounds can be significant. For instance, the ΔpK_a for Tris (tris(hydroxymethyl)aminomethane) between 4°C and 37°C is 0.924. It is therefore advisable to verify the pH of the solution throughout the temperature range studied when this buffer is used. Ellis and Morrison⁸ have discussed the practical pK_a value (written as pK_a^*), referring to the pK_a value under chosen experimental conditions. They present basic programs for calculating pK_a^* values at different temperatures and ionic strengths. **See Buffers; pH Effects; van't Hoff Relationship**

¹K. F. Tipton & H. B. F. Dixon (1979) *Meth. Enzymol.* **63**, 183.

²H. Gutfreund (1955) *Trans. Faraday Soc.* **51**, 441.

³V. Massey & R. A. Alberty (1954) *Biochim. Biophys. Acta* **13**, 354.

⁴J. R. Knowles (1976) *CRC Crit. Rev. Biochem.* **4**, 165.

⁵D. D. Perrin & B. Dempsey (1974) *Buffers and pH and Metal Ion Control*, Chapman and Hall, London.

⁶D. D. Perrin, B. Dempsey & E. P. Serjeant (1981) *pKa Prediction for Organic Acids and Bases*, Chapman and Hall, London.

⁷D. D. Perrin (1964) *Austral. J. Chem.* **17**, 484.

⁸K. J. Ellis & J. F. Morrison (1982) *Meth. Enzymol.* **87**, 405.

DISSOCIATION KINETICS

Measurement of the rates of dissociation of enzyme-substrate and protein-ligand complexes, usually promoted by dilution. The utility of dissociation kinetics is well illustrated in the report of Dunn and Raftery¹ who examined the kinetics of [³H]acetylcholine and [³H]suberylcholine dissociation from the membrane-bound

acetylcholine receptor. Each agonist binds to two high-affinity sites on the receptor, and the equilibrium dissociation constant is around 15 nM. Upon triggering dissociation of [³H]acetylcholine from the receptor complex by dilution, a monophasic process is observed with an apparent rate of 0.023 ± 0.010 s⁻¹. When μ M levels of acetylcholine, carbamylcholine, or suberyldicholine was included in the dilution buffer, this rate increased about 5-fold. This accelerating effect, which is not expected in simple independent binding processes, occurred even when the two high-affinity sites were initially saturated with the radiolabeled ligand. The authors inferred that there is likely to be an additional site (or subsite) for agonist with a K_d in the μ M range. Yet, over the 0 to 20 μ M range, no additional sites for [³H]acetylcholine were detectable. These findings led Dunn and Raftery¹ to propose that each high-affinity site has two mutually exclusive subsites, designated A and B. With [³H]acetylcholine initially occupying the A site, occupancy of site B by unlabeled ligand reduces the affinity for site A and accelerates the dissociation of the radiolabeled ligand. Their other dissociation kinetic measurements using a large bis-quaternary agonist, [³H]suberyldicholine, provided an additional clue about the properties of these proposed subsites. Its dissociation rate was similar to that observed with acetylcholine (0.028 ± 0.012 s⁻¹), but this rate was only weakly influenced by the presence of unlabeled ligands. They suggest that suberyldicholine may cross-link the two subsites or sterically occlude the second site.

¹S. M. Dunn & M. A. Raftery (1997) *Biochemistry* **36**, 3846.

DISTORTION, FREE ENERGY OF

Enzymes are thought to achieve their enormous rate accelerations by distorting the substrate into their most reactive electronic configuration and conformation (**See Entropy Trap Model**). Accordingly, there is great interest in estimating the free energy of distortion. One way to treat this problem was considered by Schowen¹ who defines $\Delta G_{\text{distortion}}$ as the total work needed (a) to convert the substrate from its standard-reaction state to the transition state, including that work for breaking away of any extra covalently attached groups, desolvation, *etc.*, and (b) to convert the enzyme from its substrate-free form into exactly the enzyme structure achieved in the transition state, again including protein conformational changes, desolvation of the active site, *etc.* This

“thought” experiment provides a way to estimate the energy changes from the substrate’s and enzyme’s ground states to their so-called poised structures (*i.e.*, their respective structures in the transition state). Schowen also defines the vertical binding energy as the net change in free energy that is realized when one combines the poised structure of the substrate with the poised structure of the enzyme.

¹R. L. Schowen (1978) in *Transition States of Biochemical Processes*, p. 88 (R. D. Gandour & R. L. Schowen, eds.), Plenum, New York.

DISTRIBUTION

The set of possible values of a random variable, most typically presented as a plot of frequency of occurrence on the vertical axis *versus* magnitude of the variable on the horizontal axis. **See Boltzmann Distribution Law**

DITHIONITE

A nonbiological reductant ($\text{S}_2\text{O}_4^{2-}$) that has proven to be of immense value in converting uncomplexed and porphyrin-bound Fe(III) to the +2 oxidation state with the concomitant formation of two molecules of SO_2 gas. Dithionite also reacts with heme-bound oxygen to produce deoxyhemoglobin, and treatment of intact red blood cells with dithionite can induce sickling in cells containing hemoglobin S. Dithionite also reduces NAD^+ to NADH.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Use in cytochrome assays, **69**, 181, 182; ferredoxin-nitrite reductase assay, **69**, 256; flavodoxins, **69**, 405, 406; interference, with analysis of nitrogenase, **69**, 755, 756; iron-sulfur center reduction, **69**, 241, 243, 246; oxygen reaction, **76**, 637-638; oxygen removal, **76**, 267, 293; oxyhemoglobin reaction, **76**, 637-639; preparation, **76**, 58-59, 791-792; reaction center isolation, **69**, 175; induction of sickling in hemoglobin S, **76**, 788; use in ligand-binding studies, **76**, 427; to reduce methemoglobin, **76**, 11-12.

DITHIOTHREITOL

A reagent introduced by W. W. Cleland (hence the alternative cognomen “Cleland’s Reagent”) to take advantage of the ability of 1,4-dithiothreitol to react with disulfide cross-linked protein (P-S-S-P) to form 2 P-SH plus the oxidized six-membered cyclic disulfide form of the reagent. This reagent and its partner, dithioerythritol, are frequently added (at 0.5 to 2.0 mM concentrations) to cell-free extracts to prevent adventitious oxidation of protein sulfhydryl groups during purification.

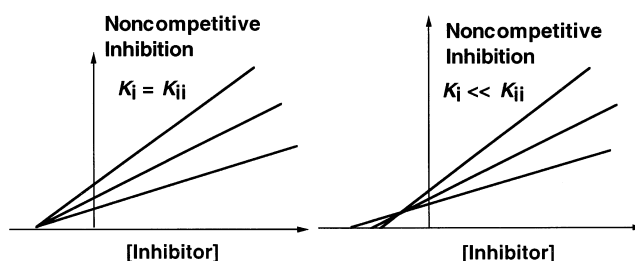
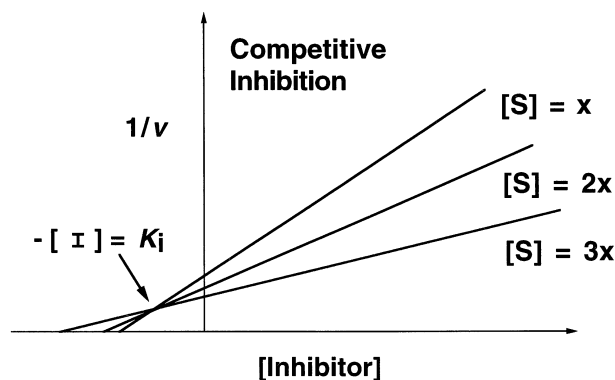
Selected entries from *Methods in Enzymology* [vol, page(s)]:

Absorption maxima of disulfide bond, **251**, 55; fatty acylation of

purified proteins, **250**, 462; oxidation potential, **251**, 20; protective effect against protease inactivation, **241**, 111-112; protein disulfide reduction, **233**, 398, 405; reaction with protein sulfhydryls, **251**, 361; reaction with superoxide, **251**, 84; redox potential, **252**, 179; reduction of disulfide bond, rate, **251**, 167, 171, 173; glutathione disulfide, **251**, 171; immunoglobulin, **251**, 171; measurement, **251**, 169-170; papain, **251**, 170-172; trypsinogen, **251**, 170-171; thiyl radical, **251**, 35.

DIXON PLOT

A plot of the reciprocal of the initial velocity as a function of inhibitor concentrations at different fixed concentrations of substrate¹. For a competitive inhibitor, the lines passing through the data will intersect at a common point in the second quadrant (whose coordinates are the ordered pair $(-K_{is}, 1/V_{\max})$. Here, K_{is} is the dissociation constant for the competitive inhibitor and $V_{\max}$ is the maximum velocity.



The intersecting line pattern observed for competitive inhibition (top plot) can be distinguished from the action of a noncompetitive inhibitor characterized by $K_i = K_{ii}$. However, if K_i is much less than K_{ii} , the observed pattern cannot be distinguished.

For a noncompetitive inhibitor², the intersection point occurs at coordinates $(-K_{is}, (1/V_{\max})(1 - [K_{is}/K_{ii}]))$, where K_{ii} is the dissociation constant of the inhibitor with respect to the enzyme-substrate complex. Thus, a noncompetitive inhibitor can have an intersection point in the second quadrant (when $K_{is} < K_{ii}$), on the horizontal axis (when $K_{is} = K_{ii}$), or in the third quadrant

(when $K_{is} > K_{ii}$). For this reason, a Dixon plot should *never* be used for the purpose of distinguishing modes of inhibitor action³. Standard double-reciprocal plots (*i.e.*, $1/v$ versus $1/[A]$ at varying concentrations of the inhibitor) offer a more reliable means.

¹M. Dixon (1953) *Biochem. J.* **55**, 170.

²D. L. Purich & H. J. Fromm (1972) *Biochem. Biophys. Acta*, **268**, 1.

³Note: Uncompetitive inhibitors display parallel lines in Dixon plots.

DIXON-WEBB PLOTS

A plot used in pH studies of enzyme activity: a plot of $\log V_{\max}$ vs. pH or a plot of $\log (V_{\max}/K_m)$ vs. pH. These plots are useful when the pK values of the enzyme are separated by at least 3.5 pH units. **See** *pH Studies*

M. Dixon & E. C. Webb (1979) *Enzymes*, 2nd ed., Longmans, Green, New York.

K. F. Tipton & H. B. F. Dixon (1979) *Meth. Enzymol.* **63**, 183.

DNA (ADENINE-N⁶)-METHYLTRANSFERASE

This enzyme [EC 2.1.1.72] catalyzes the reaction of *S*-adenosyl-L-methionine with an adenine residue in DNA to produce *S*-adenosyl-L-homocysteine and a 6-methylaminopurine residue in DNA.

F. Takusagawa, M. Fujioka, A. Spies & R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 1.

I. Ahmad & D. N. Rao (1996) *Crit. Rev. Biochem. Mol. Biol.* **31**, 361.

D. Landry, J. M. Barsomian, G. R. Feehery & G. G. Wilson (1992) *Meth. Enzymol.* **216**, 244.

DNA BREATHING

The continual loss and regain of local and global structure in DNA that results from the breaking and remaking of hydrogen bonds between A and T as well as G and C base pairs in double-stranded DNA. The ability of these base pairs within DNA to break and remake their hydrogen bonds is essential for DNA replication and transcription, and this so-called “breathing” process must also factor in the recognition, flexibility, and structure of DNA. Base-pair opening is most frequently studied by following the exchange of protons from imino groups with water using nuclear magnetic resonance measurements based on line broadening, longitudinal relaxation, and magnetization transfer from water. At room temperature, base-pair lifetimes occur on the 10–40 ms time-scale.

DNA (CYTOSINE-5)-METHYLTRANSFERASE

DNA (cytosine-5)-methyltransferase [EC 2.1.1.37] catalyzes the reaction of *S*-adenosyl-L-methionine with DNA to produce *S*-adenosyl-L-homocysteine and DNA containing a 5-methylcytosine residue. Site-specific DNA methyltransferase (cytosine-specific) [EC 2.1.1.73] catalyzes the reaction of *S*-adenosyl-L-methionine with DNA containing a cytosine to produce *S*-adenosyl-L-homocysteine and DNA containing a 5-methylcytosine.

I. Ahmad & D. N. Rao (1996) *Crit. Rev. Biochem. Mol. Biol.* **31**, 361.

X. Cheng (1995) *Ann. Rev. Biophys. Biomol. Struct.* **24**, 293.

S. Kumar, X. Cheng, S. Klimasauskas, S. Mi, J. Posfai, R. J. Roberts & G. G. Wilson (1994) *Nucleic Acid Res.* **22**, 1.

D. Landry, J. M. Barsomian, G. R. Feehery & G. G. Wilson (1992) *Meth. Enzymol.* **216**, 244.

S. Hattman (1981) *The Enzymes*, 3rd ed., **14**, 517.

S. J. Kerr & E. Borek (1973) *The Enzymes*, 3rd ed., **9**, 167.

DNA GLYCOSYLASES

DNA α -glucosyltransferase [EC 2.4.1.26] catalyzes the transfer of an α -D-glucosyl residue from UDPglucose to a hydroxymethylcytosine residue in DNA. DNA β -glucosyltransferase [EC 2.4.1.27] catalyzes the transfer of a β -D-glucosyl residue from UDPglucose to a hydroxymethylcytosine residue in DNA. Glucosyl-DNA β -glucosyltransferase [EC 2.4.1.28] catalyzes the transfer of a β -D-glucosyl residue from UDPglucose to a glucosylhydroxymethylcytosine residue in DNA.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

L. D. Samson (1992) *Essays in Biochem.* **27**, 69.

B. Weiss & L. Grossman (1987) *Adv. Enzymol.* **60**, 1.

E. C. Friedberg, T. Bonura, E. H. Radany & J. D. Love (1981) *The Enzymes*, 3rd ed., **14**, 251.

B. K. Duncan (1981) *The Enzymes*, 3rd ed., **14**, 565.

DNA GYRASES & DNA REVERSE GYRASE

This entry contains a set of enzymes for which the terms gyrase and reverse gyrase have been used. DNA topoisomerase (ATP-hydrolyzing) [EC 5.99.1.3] (also referred to as DNA topoisomerase II, DNA gyrase, and Type II DNA topoisomerase) catalyzes the ATP-dependent breakage, passage, and rejoining of double-stranded DNA. The enzyme can introduce negative superhelical turns into double-stranded circular DNA. One domain of the protein has a nicking-closing activity, whereas another unit catalyzes the supertwisting and hydrolysis of ATP. **See also** *DNA Topoisomerases*

R. A. Grayling, K. Sandman & J. N. Reeve (1996) *Adv. Protein Chem.* **48**, 437.

- D. B. Wigley (1995) *Ann. Rev. Biophys. Biomol. Struct.* **24**, 185.
 R. J. Reece & A. Maxwell (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 335.
 M. Gellert (1981) *The Enzymes*, 3rd ed., **14**, 345.

DNA LIGASES

DNA ligase (ATP) [EC 6.5.1.1] catalyzes the reaction of ATP with (deoxyribonucleotide)_n and (deoxyribonucleotide)_m to produce AMP, pyrophosphate, and (deoxyribonucleotide)_(n+m). This activity results in the formation of a phosphodiester at the site of a single-strand break in duplex DNA. RNA can also act as substrate to some extent.

DNA ligase (NAD⁺) [EC 6.5.1.2] (also referred to as polydeoxyribonucleotide synthase (NAD⁺), polynucleotide ligase (NAD⁺), DNA repair enzyme, and DNA joinase) catalyzes the reaction of NAD⁺ with (deoxyribonucleotide)_n and (deoxyribonucleotide)_m to produce AMP, nicotinamide nucleotide, and (deoxyribonucleotide)_(n+m). This forms a phosphodiester at the site of a single-strand break in duplex DNA. RNA can also act as substrate to some extent.

- A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
 B. F. Cooper & F. B. Rudolph (1995) *Meth. Enzymol.* **249**, 188.
 P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.
 M. J. Engler & C. C. Richardson (1982) *The Enzymes*, 3rd ed., **15**, 3.
 I. R. Lehman (1974) *The Enzymes*, 3rd ed., **10**, 237.

DNA METHYLTRANSFERASES

Methylation of certain bases within DNA molecules plays a major role in gene expression, as is evidenced by the transcriptional silencing of X chromosome genes through methylation. The following are recent reviews on the molecular and physical properties of these enzymes [EC 2.1.1.x]. **See also specific methyltransferase**

- F. Takusagawa, M. Fujioka, A. Spies & R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 1.
 I. Ahmad & D. N. Rao (1996) *Crit. Rev. Biochem. Mol. Biol.* **31**, 361.
 D. Landry, J. M. Barsomian, G. R. Feehery & G. G. Wilson (1992) *Meth. Enzymol.* **216**, 244.
 R. L. P. Adams (1990) *Biochem. J.* **265**, 309.
 R. L. P. Adams & R. H. Burdon (1982) *Crit. Rev. Biochem.* **13**, 349.
 S. Hattman (1981) *The Enzymes*, 3rd ed., **14**, 517.
 S. J. Kerr & E. Borek (1973) *The Enzymes*, 3rd ed., **9**, 167.

DNA OXIDATIVE DAMAGE AND REPAIR

Henle and Linn¹ recently reviewed the formation, prevention, and repair of DNA damage by iron/hydrogen peroxide (so-called Fenton oxidants). Berlett and Stadt-

man² also discussed the corresponding pathways by which oxidatively modified forms of proteins accumulate during aging, oxidative stress, and in some pathological conditions.

¹E. S. Henle & S. Linn (1997) *J. Biol. Chem.* **272**, 19095.

²B. S. Berlett & E. R. Stadtman (1997) *J. Biol. Chem.* **272**, 20313.

DNA PHOTOLYASE

This enzyme [EC 4.1.99.3] catalyzes the reactivation by light of irradiated DNA by converting cyclobutadipyrimidine residues in DNA to two pyrimidine residues in DNA. This is a flavoprotein containing a second chromophore group.

- B. A. Palfe & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
 S.-T. Kim, P. F. Heelis & A. Sancar (1995) *Meth. Enzymol.* **258**, 319.
 A. Sancar (1994) *Biochemistry* **33**, 2.
 E. J. Brush & J.W. Kozarich (1992) *The Enzymes*, 3rd ed., **20**, 317.
 B. M. Sutherland (1981) *The Enzymes*, 3rd ed., **14**, 482.

DNA POLYMERASES

DNA-directed DNA polymerases [EC 2.7.7.7], also called DNA nucleotidyltransferases (DNA-directed), are enzymes that catalyze the DNA template-directed extension of the 3'-end of a nucleic acid strand one nucleotide at a time. Thus, *n* deoxynucleoside triphosphates produce *n* pyrophosphate (or, diphosphate) ions and DNA_n. This enzyme cannot initiate the synthesis of a polymeric chain *de novo*; it requires a primer which may be DNA or RNA. RNA-directed DNA polymerases [EC 2.7.7.49], also referred to as reverse transcriptases, DNA nucleotidyltransferases (RNA-directed), and revertases, are enzymes that catalyze the RNA template-directed extension of the 3'-end of a nucleic acid strand one nucleotide at a time. Thus, *n* deoxynucleoside triphosphates produce *n* pyrophosphate (or, diphosphate) ions and DNA_n. As was the case above, this enzyme cannot initiate the synthesis of a polymeric chain *de novo*; it requires a primer which may be DNA or RNA.

- N. H. Williams (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 543.
 F. B. Perler, S. Kumar & H. Kong (1996) *Adv. Protein Chem.* **48**, 377.
 Z. Kelman & M. O'Donnell (1995) *Ann. Rev. Biochem.* **64**, 171.
 C. S. McHenry (1991) *J. Biol. Chem.* **266**, 19127.
 P. A. Frey (1989) *Adv. Enzymol.* **62**, 119.
 V. Mizrahi & S. J. Benkovic (1988) *Adv. Enzymol.* **61**, 437.
 B. L. Vallee & A. Galdes (1984) *Adv. Enzymol.* **56**, 283.

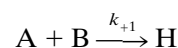
See also chapters in *The Enzymes*, 3rd ed., vols. **10**, **14**, **19**, and **20**.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Phosphate analogs for study of DNA polymerases, **262**, 189; mechanisms of inhibition of DNA polymerases by 2'-deoxyribonucleoside 5'-triphosphate analogs, **262**, 202; analyzing the fidelity of DNA polymerases, **262**, 217; gel fidelity assay measuring nucleotide misinsertion, exonucleolytic proofreading, and lesion bypass efficiencies, **262**, 232; kinetic analysis of nucleotide incorporation and misincorporation by Klenow fragment of *Escherichia coli* DNA polymerase I, **262**, 257; methods of analyzing processivity, **262**, 270; mutational analysis of bacteriophage f29 DNA polymerase, **262**, 283; rationale for mutagenesis of DNA polymerase active sites: DNA polymerase A, **262**, 294; mutational analysis of DNA polymerase substrate recognition and subunit interactions using herpes simplex virus as prototype, **262**, 303; use of genetic analyses to probe structure, function, and dynamics of bacteriophage T4 DNA polymerase, **262**, 323; limited proteolysis of DNA polymerases as a probe of functional domains, **262**, 331; assays for retroviral reverse transcriptase, **262**, 347; structure-function analysis of the 3'-5' exonuclease of DNA polymerases, **262**, 363.

DNA RENATURATION KINETICS

Upon mixing, complementary DNA sequences will tend to recombine and form a double helix in a process commonly called DNA renaturation. Treatment of DNA renaturation kinetics is greatly simplified by assuming: (a) that the rate behavior can be treated as a bimolecular process limited by the chains finding proper registration of their respective hydrogen bonding partners on complementary strands; (b) that, after initial chain recombination, there is a much more rapid zippering process for hydrogen bonding and formation of the double helix; (c) that a collection of complementary strands does not contain inhibitors (*i.e.*, components of slightly mismatched that retard the reassociation of genuinely complementary strands of DNA. When these conditions hold, the following scheme will describe how complementary sequences A and B will recombine:



By expressing the molar concentrations of each sequence A and B and helix H, we can obtain the following rate expression describes this process:

$$\frac{d[H_i]}{dt} = k_{+1}[A][B]$$

As noted by Bloomfield, Crothers, and Tinoco¹, however, there is value in re-expressing the molar concentrations of each strand in terms of the number of sequence repeats, where a total double-helix length of T base pairs, broken into fragments of length L , may contain n_i copies of repeat sequence- i . In this way, one can designate the concentration of strands having sequence- i as $[A]_i$, and $[B]_i$, which in turn can be defined with respect to their nucleotide content:

$$[A]_i = (n_i/2T)[A]_N \text{ and } [B]_i = (n_i/2T)[B]_N$$

where the factor "2" in the denominator changes base pairs into bases, and the subscript N indicates that concentration is expressed in terms of nucleotides. Under these conditions,

$$\frac{d[H_i]_N}{dt} = 2L \frac{d[H_i]}{dt} = k_{+1} 2L \left(\frac{n_i}{2T}\right)^2 = \frac{k_{+1}L}{2} \left(\frac{n_i}{T}\right)^2$$

This expression can also be written as follows:

$$\frac{d[H_i]_N}{dt} = \frac{k_N}{2} \left(\frac{n_i}{T}\right)^2 C_N$$

where C_N is the total single-strand nucleotide concentration, and k_N equals $k_{+1}n_i/T$.

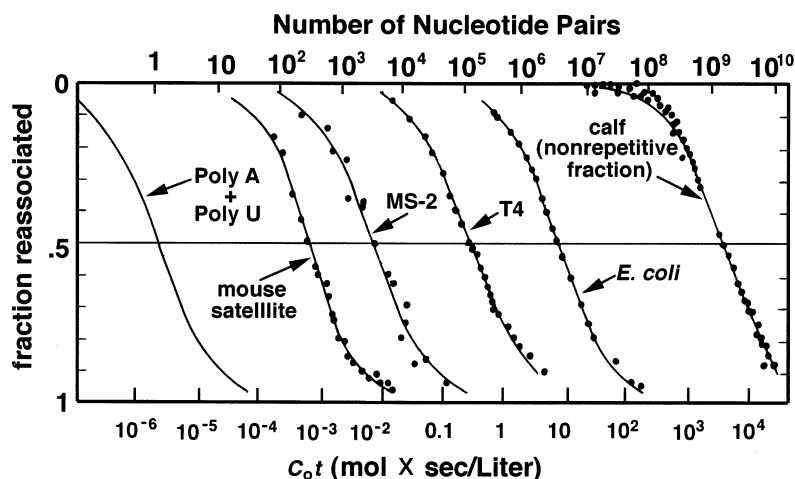


Figure 1. Progress curves for renaturation of double-stranded nucleic acid fragments (roughly 400 bases/fragment). From Britten and Kohne² with permission.

C_0t Curve Analysis. By making a few simplifying assumptions (including that L remains constant), Britten and Kohne² developed a straightforward kinetic approach to demonstrate the high frequency of occurrence of repeated sequences in DNA from the genomes of higher organisms. To analyze the time-course of DNA renaturation, they rearranged the integrated rate equation:

$$\left(\frac{1}{C_N} - \frac{1}{C_0} \right) = \frac{k_N t}{2}$$

(where C_0 is the total nucleotide concentration) to obtain the following final kinetic expression:

$$f = \frac{1}{1 + 0.5 k_N C_0 t}$$

Because the time-courses for renaturation are presented in terms of C_0t , and not time, the progress curves shown in Fig. 1 are frequently called C_0t curves. Here, the parameter f is the fraction of total nucleotides in the helical form (*i.e.*, $f = C_N/C_0$), and when $f = 0.5$, k_N can be evaluated readily as $C_0t/2$.

¹V. A. Bloomfield, D. M. Crothers & I. Tinoco, Jr. (1974) *Physical Chemistry of Nucleic Acids*, pp. 361-367 Harper & Row, New York.

²R. J. Britten & D. E. Kohne (1968) *Science* **161**, 529.

DNA TOPOISOMERASES

DNA topoisomerase [EC 5.99.1.2] (also referred to as DNA topoisomerase I, relaxing enzyme, untwisting enzyme, swivelase, type I DNA topoisomerase, nicking-closing enzyme, and ω -protein) catalyzes the ATP-independent breakage of single-stranded DNA, followed by the passage and rejoining of that DNA. The enzyme catalyzes the conversion of one topological isomer of DNA into another (*e.g.*, the relaxation of superhelical turns in DNA, the interconversion of simple and knotted rings of single-stranded DNA, and the intertwisting of single-stranded rings of complementary sequences. DNA topoisomerase (ATP-hydrolyzing) [EC 5.99.1.3] (also referred to as DNA topoisomerase II, DNA gyrase, and Type II DNA topoisomerase) catalyzes the ATP-dependent breakage, passage, and rejoining of double-stranded DNA. The enzyme can introduce negative superhelical turns into double-stranded circular DNA. One unit of the protein has a nicking-closing activity whereas

another unit catalyzes the supertwisting and hydrolysis of ATP.

J. C. Wang (1996) *Ann. Rev. Biochem.* **65**, 635.

R. A. Grayling, K. Sandman & J. N. Reeve (1996) *Adv. Protein Chem.* **48**, 437.

R. J. Reese & A. Maxwell (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 335.

A. Maxwell & M. Gellert (1986) *Adv. Prot. Chem.* **38**, 69.

S. A. Wasserman & N. R. Cozzarelli (1986) *Science* **232**, 951.

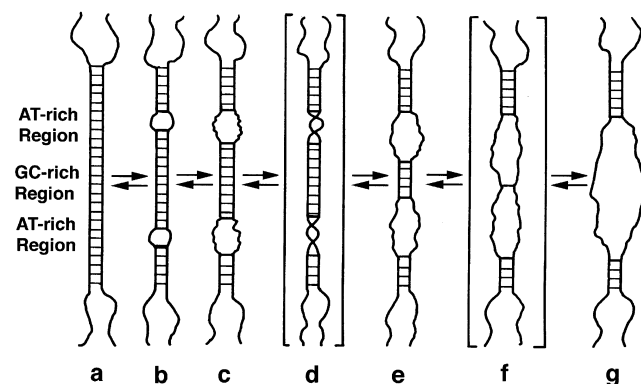
J. C. Wang (1981) *The Enzymes*, 3rd ed., **14**, 331.

M. Gellert (1981) *The Enzymes*, 3rd ed., **14**, 345.

H. A. Nash (1981) *The Enzymes*, 3rd ed., **14**, 471.

DNA UNWINDING (Kinetic Model For Small DNA)

Frictional resistance of the viscous medium is thought to be the chief energy barrier determining the unwinding rate¹. Longer polymers experience much greater friction, and the time for untwisting of DNA was thought to be proportional to at least the second power of length. Nonetheless, Cohen and Crothers² demonstrated that even relatively small DNA molecules (with masses of 65,000–2,300,000 Daltons) exhibit frictionally limited unwinding. The rate of unwinding is also influenced by the rate of breaking hydrogen bonds.



Diagrammatic representation of DNA denaturation. Upon perturbation, the first "melting" away of structure occurs within AT-rich regions, and further perturbation causes this local disorder to grow, such that GC-rich regions are the last to become disordered. The DNA unwinding reaction is limited by viscous resistance.

DNA unwinding in the above diagram³ occurs initially in the AT-rich regions of the DNA duplex (indicated as "a" above), and these local regions of disorder (shown in "c" and "d") require some twisting motions (shown in "d"). As these areas grow ("e"), one reaches the point of nearly complete loss of local order ("f"), followed by formation of a large loop ("g"). As would be expected,

the reverse rate constants are much lower for larger loops, because achieving proper registration of complementary bases pairs may lead to a higher activation energy.

Cohen and Crothers² also present an excellent account of their use of the Guggenheim method for extracting rate constants from incomplete segments of decay processes, in which either the starting- and/or end-points of the spectral decay are not well defined. **See also** *Chemical Kinetics (1. Progress Curve Analysis)*

¹D. M. Crothers (1969) *Acc. Chem. Res.* **2**, 225.

²R. J. Cohen & D. M. Crothers (1971) *J. Mol. Biol.* **61**, 525.

³We are indebted to our colleague, Professor Robert J. Cohen, for the original drawing on which this diagram is based.

DOCKING METHODS

Computer-based techniques for locating and analyzing ligand binding sites on macromolecules. These methods must take into account the steric and electrostatic properties of ligand and receptor as well as ligand-induced conformational changes.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

DOCK computer program, **241**, 355-358; applications to HIV protease, **241**, 355-358, 362-370; automated FLOG approach, **241**, 359, 361-365, 367-370; MINDEX flexibase, **241**, 362-370; ligand-receptor matching and evaluation, **241**, 361-362; ligand representation, **241**, 359-360; ligand selection, **241**, 366-370; receptor definition, **241**, 365-366; receptor representation, **241**, 360-361; search results, **241**, 362.

DOMAIN

A topologically separate or autonomous subregion of a protein, usually encoded by a contiguous sequence. Such regions are often stable to limited proteolysis, allowing them to be excised from the polypeptide chain without significant loss of structure or functional binding of ligands. Most domains in multidomain proteins form from continuous segments and adopt the same structural class.

Domains may be regarded as the basic units for the structure, function and evolution of proteins, but the definition of a domain remains fuzzy. They are most often treated as compact or connected areas that are apparent from a visual inspection of protein models. To avoid subjectivity and ambiguities of visual inspection, computer algorithms have been developed to localize domains. Rashin¹ offered an alternative interpretation: domains are stable globular fragments, generated in biochemical experiments that refold autonomously and retain specific functions. He proposed a method for localiz-

ing these globular fragments on the basis of surface area measurements. Islam *et al.*² used an algorithm based on interresidue contacts to identify domains in proteins. The results of applying their algorithm agreed with those derived by visual inspection for nearly four-fifths of the 284 nonredundant chains that they considered. They also examined the relationship between accessible surface area and molecular weight for domains and chains. Rigid domains, defined by Nichols *et al.*³ as a local region of tertiary structure common to two or more different protein conformations, can be identified numerically from atomic coordinates by finding sets of residues, such that the distance between any two residues within the set belonging to one conformation is the same as the distance between the two structurally equivalent residues within the set belonging to any other conformation. They described two algorithms that employ the difference-distance matrix to search directly for rigid domains from atomic coordinates.

¹A. A. Rashin (1981) *Nature* **291**, 85.

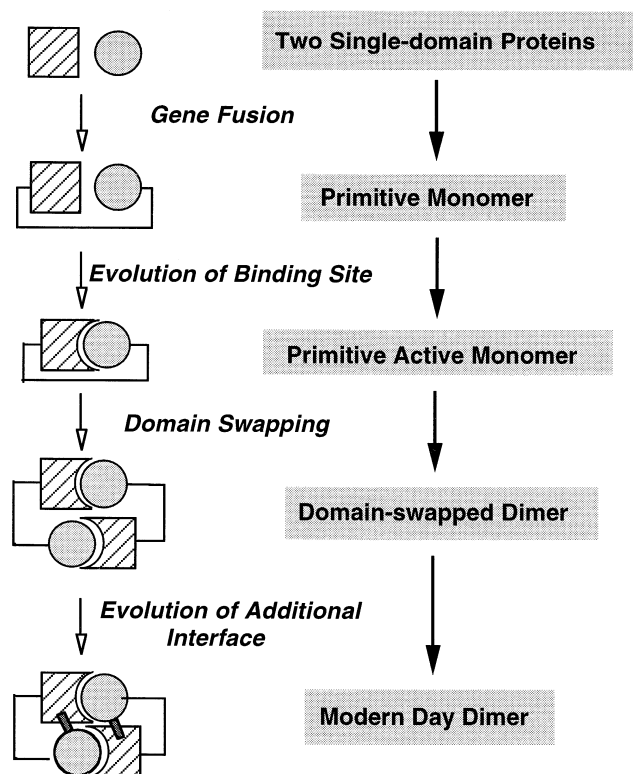
²S. A. Islam, J. Luo & M. J. Sternberg (1995) *Protein Engineering* **8**, 513.

³W. L. Nichols, G. D. Rose, L. F. Ten Eyck & B. H. Zimm (1995) *Proteins* **23**, 38.

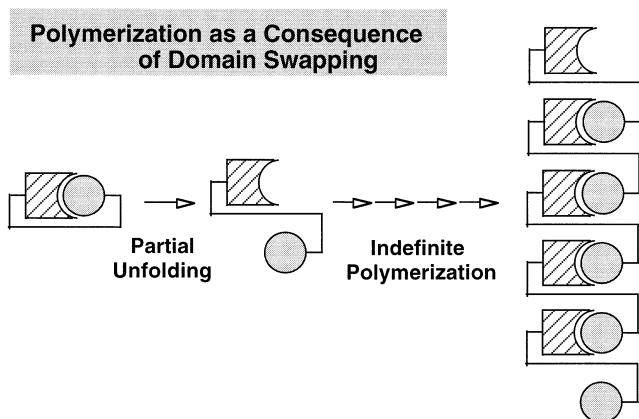
DOMAIN SWAPPING

A mechanism¹ for explaining how oligomeric and polymeric forms of a protein or polypeptide can arise by three-dimensional swapping of internal domains in monomers. The basic idea is that a protein's ternary structure is stabilized by the contacts between its domains, and under certain circumstances, the domains of several polypeptide chains may bind in a way that gives rise to nonphysiologic oligomers, polymers, and aggregates.

The following diagram, adapted from that first presented by Bennett *et al.*¹, describes a postulated pathway for evolution of a protein dimer from single-domain proteins. The scheme begins with the fusion of two single-domain polypeptides and proceeds through the evolution of interdomain contacts, and in the case of enzymes, development of an active site. These same interdomain contacts can also stabilize formation of a domain-swapped dimer which then undergoes further evolution into a present-day dimer.



As illustrated in the diagram below, domain swapping can also result in indefinite polymerization to form linear supramolecular structures. These may correspond to present-day polymers of proteins such as microtubules, or they may represent abnormal structures, like the straight and paired-helical filaments in the neurofibrillary tangles observed in the brain tissue of those afflicted with Alzheimer's disease.



The idea of domain swapping traces its origin to the work of Crestfield *et al.*² who noted that a mixture of His-12 alkylated RNase monomer and His-119 alkylated RNase monomer could recover catalytic activity through

the formation of a dimeric species. As pointed out by Bennett *et al.*¹, this is an excellent example of protein complementation³, wherein two different protein molecules, each lacking an essential protein motif, can combine to form a functional molecule.

¹M. J. Bennett, M. P. Schlunegger & D. Eisenberg (1995) *Protein Science* **4**, 2455.

²A. M. Crestfield, W. H. Stein & S. Moore (1963) *J. Biol. Chem.* **238**, 2421.

³I. Zabin & M. R. Villarejo (1975) *Annu. Rev. Biochem.* **44**, 295.

DONNAN EQUILIBRIUM

The equilibrium (also known as the Donnan effect) established across a semipermeable membrane or the equivalent of such a membrane (such as a solid ion-exchanger) across which one or more charged substances, often a protein, cannot diffuse. Diffusible anions and cations are distributed on the two sides of the membrane, such that the sum of concentrations (in dilute solutions) of diffusible and nondiffusible anions on either side of the membrane equals the sum of concentrations of diffusible and nondiffusible cations. Thus, the diffusible ions will be asymmetrically distributed across the membrane and a Donnan potential develops.

DOPAMINE β -MONOOXYGENASE

This enzyme [EC 1.14.17.1], also known as dopamine β -hydroxylase, is a copper-dependent system catalyzing the reaction of 3,4-dihydroxyphenethylamine with ascorbate and dioxygen to produce noradrenaline, dehydroascorbate, and water. The enzyme is stimulated by fumarate.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

F. P. Guengerich (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 97.

T. Ljones (1987) *Meth. Enzymol.* **142**, 596.

R. P. Frigon (1987) *Meth. Enzymol.* **142**, 603.

R. M. Weppelman (1987) *Meth. Enzymol.* **142**, 608.

S. M. Miller & J. P. Klinman (1982) *Meth. Enzymol.* **87**, 711.

V. Ullrich & W. Duppel (1975) *The Enzymes*, 3rd ed., **12**, 253.

DOSE

1. A measured quantity of a substance or solution given to an organism or system at one time. 2. The energy or amount of photons absorbed by an object or substance per unit area or unit volume in a given exposure time; the word is often preceded by the type of light (*e.g.*, UV dose). 3. Fluence; the energy or amount of photons received by an object or substance per unit area or unit volume during a given exposure time.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Median effective dose, **235**, 29, 31, 33; determination of median infectious dose, **235**, 29-39; median lethal dose, **235**, 29; moving average interpolation, **235**, 32-34, 36-37, 39; probit analysis, **235**, 31-36, 39; Reed-Muench method, **235**, 30, 33-35; staircase method, **235**, 33, 37-39; median response, **235**, 29; dose range for animal experiments, **235**, 29; 50% end-point determination, **235**, 29-30.

DOSE, LETHAL

The dose of a substrate which is likely to cause death in an organism. It is symbolized by LD and is usually supplemented with a subscript number indicating the percentage of test animals that died: the absolute lethal dose is LD₁₀₀, the median lethal dose is LD₅₀, and the minimal lethal dose is LD₀₅. These values will vary with the type of test animal and the route of administration; hence, that information should always be provided.

DOUBLE-RECIPROCAL PLOT

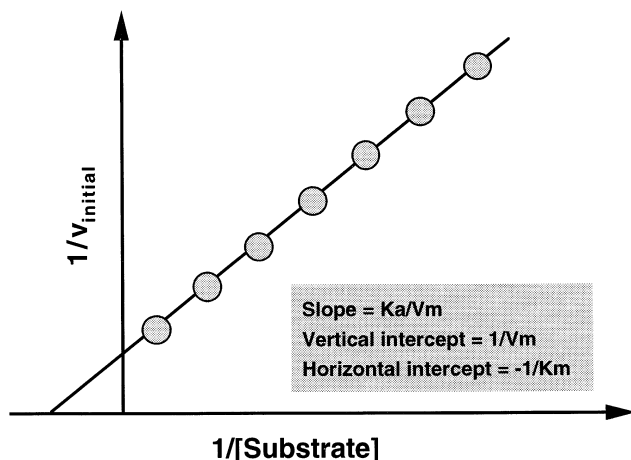
A linear reciprocal transformation of a function of the form $[f(\alpha) = \alpha/(1 + \alpha)]$, such that $1/f(\alpha)$ is plotted on the vertical axis and $1/\alpha$ is plotted on the horizontal axis. In the case of one-substrate enzyme kinetics, the hyperbolic function is:

$$v = V_{\max}[S]/(K_m + [S])$$

and the double-reciprocal form¹ is as follows:

$$1/v = (1/V_{\max}) + (K_m/(V_{\max}[S]))$$

Because any linear equation can be expressed in terms of its slope (m) and vertical intercept (b), such that $y = mx + b$, the double-reciprocal form of the Michaelis-Menten equation yields a slope of $K_m/V_{\max}$, a vertical-axis intercept of $1/V_{\max}$, and a horizontal-axis intercept of $-1/K_m$.



This graphical representation of initial rate data for enzyme-catalyzed reactions is also referred to as the Lineweaver-Burk plot² or a Woolf-Lineweaver-Burk plot^{1,2}. Double-reciprocal plots are also used for Bi-, Ter-, and higher reactant systems. Its broad usage and ease of interpretation have made the double-reciprocal plot the most common graphical representation of initial-rate data. However, investigators^{3,4} have identified problems with the double-reciprocal plot, particularly in terms of a greater variation in statistical weighting compared to other graphical representations. Nevertheless, Fromm has reviewed practical reasons for favoring double-reciprocal plots^{5,6}, and he has also demonstrated that data weighting and computer methods can eliminate or minimize the problems encountered by earlier workers⁷.

¹B. Woolf quoted in J. B. S. Haldane & K. G. Stern, *Allgemeine Chemie der Enzyme*, S. 119, Steinkopff, Dresden-Leipzig.

²H. Lineweaver & D. Burk (1934) *J. Amer. Chem. Soc.* **56**, 658.

³G. N. Wilkinson (1961) *Biochem. J.* **80**, 324.

⁴J. E. Dowd & D. S. Riggs (1965) *J. Biol. Chem.* **240**, 863.

⁵H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

⁶F. B. Rudolph & H. J. Fromm (1979) *Meth. Enzymol.* **63**, 138.

⁷D. Siano, J. W. Zyskind & H. J. Fromm (1975) *Arch. Biochem. Biophys.* **170**, 587.

DRUG ABSORPTION/ELIMINATION (First-Order Model)

The time evolution of plasma drug concentration can be treated by a simple model that assumes first-order absorption and first-order elimination of the drug. If $[D_o]$ is the effective concentration of the drug dose, appearance of drug in the plasma will obey the relationship:

$$[D] = [D_o]\exp(-k_a t)$$

where $[D]$ is the instantaneous plasma drug concentration at time t and k_a is the rate constant for absorption. Letting $[X]$ represent the amount of drug absorbed into the circulation,

$$d[X]/dt = k_a[D] - k_e[X]$$

where k_e is the first-order rate constant for elimination of the drug from plasma. Separating variables, we get

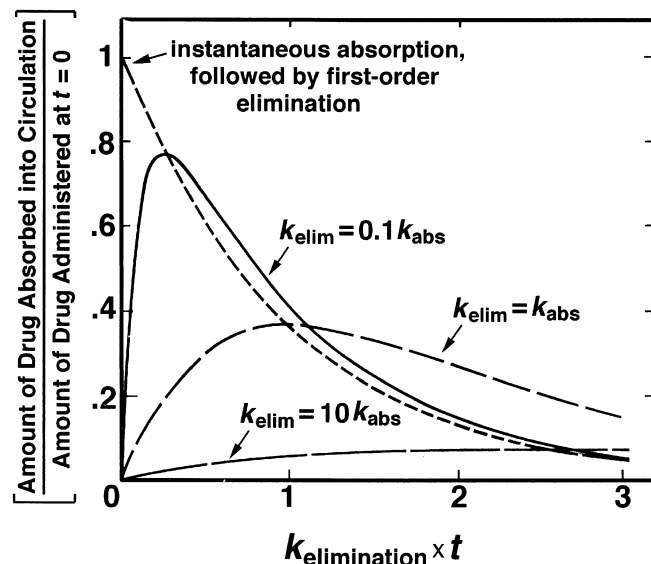
$$d[X]/dt + k_e[X] = k_a[D]$$

and substituting $[D_o]\exp(-k_a t)$ for $[D]$ then yields

$$d[X]/dt + k_e[X] = k_a[D_o]\exp(-k_a t)$$

Integration of this rate expression yields

$$[X] = \{k_a/(k_e - k_a)\}[D_o] \{\exp(-k_a t) - \exp(-k_e t)\}$$



Time-courses of drug absorption and elimination, showing how the relative magnitudes of k_{abs} and k_{elim} alter the amount of drug absorbed into the circulation relative to the amount of drug administered initially.

As shown above, the plasma drug concentration usually exhibits a biphasic rise and fall that are characteristic of all two-step series first-order processes. When k_e is zero, there will only be first-order absorption without any elimination, and when $k_a \gg k_e$, drug absorption will be virtually instantaneous, followed by first-order elimination.

DRUG ACTION (Operational Model)

An alternative model to the classical drug-receptor occupation treatment for drug action^{1,2}. The tissue response as a function of agonist concentration can be described by the equation:

$$E_a/E_m = [AR]/[R_t] = [A_{free}]/([A_{free}] + K_A)$$

where E_a is the response, E_m is the maximal response, $[A]$ is the agonist concentration, $[R_t]$ is the total concentration of receptor, K_A is the dissociation constant for the AR complex, and K_E is the value of $[AR]$ that evokes half-maximal tissue response.

¹J. W. Black & P. Leff (1983) *Proc. Roy. Soc. Lond. [Biol.]* **220**, 141.

²T. Kenakin (1993) *Pharmacologic Analysis of Drug-Receptor Interaction*, 2nd ed., pp. 20-24, Raven Press, New York.

DRUG RECEPTOR

A specific drug binding site that transduces site occupancy into a biological response. Although originally proposed as receptor sites located on the outer surface of a target cell membrane, drug receptors can reside at any location within a cell. They typically display a high degree of stereochemical recognition, such that nonspecific drug binding can be compensated for by using a pharmacologically inactive, stereoisomeric form of the active drug. Many specific drug receptors also can be classified into subtypes which are products of highly related genes, alternative mRNA splicing, unique post-translational covalent modification, or even characteristic subcellular localization.

DYAD SYMMETRY

A molecular structural property wherein a rotation of 180° leads to the same superimposable structure. Because proteins are made up of L-amino acid residues, neither a single subunit nor a multisubunit protein can possess a plane or point of symmetry. However, proteins do exhibit dyad symmetry. This structural property allows many DNA-binding proteins to make equivalent contacts to the two antiparallel chains of the nucleic acid.

DYNAMIC EQUILIBRIUM

The state of relentless interconversion of reactant(s) and product(s) at thermodynamic equilibrium. This fundamental concept that reactant(s) and product(s) are ceaselessly shuttling forth and back was adduced on the basis of kinetic observations, years before the field of chemical thermodynamics had undergone its modern transformation. Wilhemy¹ in 1850 used polarimetry to demonstrate that the rate of acid-catalyzed hydrolysis of sucrose was linearly dependent on the concentration of sugar. A few years later, Berthelot and St. Gilles^{2,3} reached the same conclusion in their studies of ethyl acetate hydrolysis. These observations led Guldberg and Waage^{4,5} to postulate that chemical equilibria are dynamic, realizing as they did that the reaction rate in each direction depended on the solution concentration (an intensive variable), and not the amount (an extensive variable) of reactant(s) and product(s).

- ¹L. Wilhelmy (1850) *Ann. Physik u. Chem., Poggendorff's* **81**, 413 and 499.
²M. Berthelot & P. De St. Gilles (1862) *Ann. Chim. Phys.* [3]**65**, 385.
³M. Berthelot (1862) *Ann. Chim. Phys.* [3]**66**, 110.
⁴C. M. Guldberg & P. Waage (1867) *Études sur les affinités chimiques*, Brøgger and Christie, Christiania (Oslo).
⁵C. M. Guldberg & P. Waage (1879) *J. prakt. Chem.* [2]**19**, 69.

DYNAMICS

The branch of physics concerned with matter in motion as well as the forces associated with those motions, as contrasted with its interchangeable usage for the kinetics of processes in chemistry and biochemistry.

DYNEIN

This family of ATP-dependent motors [EC 3.6.1.33] that forms ADP and orthophosphate during its energy-dependent translocation along the surface of microtubules. Unique “cargo” sites on dynein molecules allow for the specific transport of cellular organelles and other macromolecular components.

- S. Khan & M. Sheetz (1997) *Ann. Rev. Biochem.* **66**, 785.
 R. Vallee & E. L. F. Holzbauer (1994) *Ann. Rev. Cell Dev. Biol.* **10**, 339.
 K. A. Johnson (1992) *The Enzymes*, 3rd ed., **20**, 1.
 I. R. Gibbons (1988) *J. Biol. Chem.* **263**, 15837.

E

e

A special irrational number known in mathematics as the Euler number in honor of the prolific Swiss mathematician Leonhard Euler (1707-1783). This number is the least upper bound of the set of all numbers

$$\left(\frac{n+1}{n}\right)^n$$

where n is a positive integer. (A rational approximation of this number to ten significant figures is 2.718281828.)

The number e emerges from the exponential function, which is the sum of the exponential series, such that

$$\begin{aligned} e^x &= \exp x = \sum_{n=0}^{\infty} \frac{z^n}{n!} \\ &= 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} + \dots \end{aligned}$$

When z equals unity, we get the number e , which can be expressed in several equivalent ways:

$$\begin{aligned} e &= \sum_{n=1}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \dots \\ &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^n \end{aligned}$$

This number forms the basis of the natural (or Napierian) logarithmic scale. For each positive number x , there is unique number y , such $x = e^y$, and y is called the natural logarithm of x . We can therefore write

$$y = \log_e x = \ln x$$

With this definition, one should immediately recognize that for every number x , $e^{\ln x} = x$, where $x > 0$, such that

$\ln e^x = x$. This leads to the following properties of natural logarithms: (1) $\ln x \cdot y = \ln x + \ln y$; (2) $\ln (x/y) = \ln x - \ln y$; (3) $\ln x^c = c \ln x$; (4) $\ln x = \ln y$, if and only if $x = y$; and (5) $\ln x < \ln y$, if and only if $x < y$.

The number e is a special number that allows one to describe the exponential laws of growth and decay. If $f(t) > 0$ is the amount of a substance present at time t , then the rate of change of f at time t is proportional to $f(t)$ and can be written as: $f'(t) = k f(t)$ for some constant k and every time t in some interval. We integrate this differential equation as follows:

$$\int_{t=0}^{t=t} \frac{f'(t)}{f(t)} dt = \int_0^t k dt$$

Changing variables by allowing $u = f(t)$ and $du = f'(t) dt$ allows one rewrite the above integral as:

$$\int_{f(0)}^{f(t)} \frac{1}{u} du = \int_0^t k dt \text{ or } \ln u \Big|_{f(0)}^{f(t)} = kt \Big|_0^t$$

Therefore, $\ln f(t) - \ln f(0) = kt$; $\ln \{f(t)/f(0)\} = kt$; $f(t)/f(0) = e^{kt}$; and $f(t) = f(0)e^{kt}$. For growth, the exponent is positive [i.e., $f(t) = f(0)e^{kt}$]; for decay, the exponent is negative [i.e., $f(t) = f(0)e^{-kt}$].

The number e also appears in considerations of another form of growth, namely compound interest (i.e., the interest that accumulates when a principal P earns, and thereby grows from, successive payments at a fixed interest rate of i percent). The total principal after n periods (years in the case of many financial investments) can be determined by the following expression:

$$P_{\text{total}} = P\{1 + (i/100)\}^n$$

If the interest is paid m times per period n , the total principal will be greater:

$$P_{\text{total}} = P\{1 + (i/100nm)\}^{nm}$$

Furthermore, if the interest payments are instantaneous (*i.e.*, as n approaches infinity), growth in the total principal is governed by the following expression:

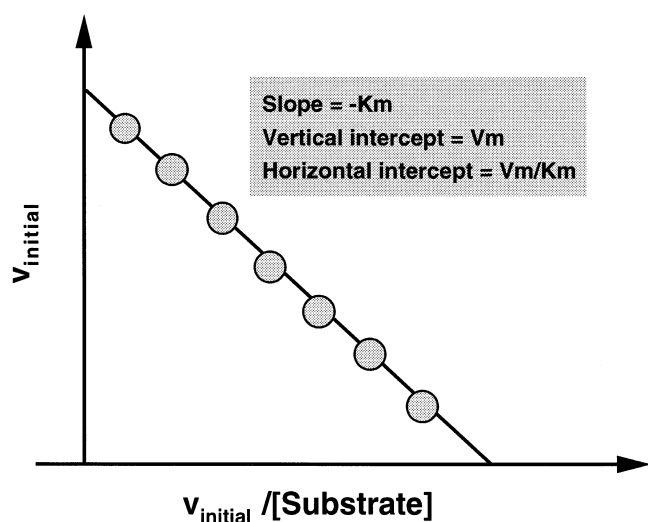
$$P_{\text{total}} = P \times \exp(i/100)$$

The principle of compound interest also applies to many biological processes, where an inherited incremental advantage of one organism allows it to gain greater and greater advantage over a competitor. For example, suppose a graduate student accomplishes 10% more per year than another student; then after five years, his/her total accomplishment will be $\{1 + (10/100)\}^5$ or 1.61 greater than the other student. (This moral is instructive!)

Another special property of e is that the derivative of e^x exactly equals e^x ; thus, this is a function that also happens to be its own derivative!

EADIE-HOFSTEE PLOT

A graphical procedure¹⁻⁴ for plotting enzyme initial rate data as v versus $v/[S]$ (also known as the Woolf-Augustinsson-Hofstee plot), where the initial velocity v is the so-called (y)-axis variable and v divided by the initial substrate concentration $[S]$ is the so-called (x)-axis variable, such that the vertical-axis intercept equals V_{max} , the slope equals $-K_m$, and the horizontal-axis intercept is V_{max}/K_m .



This transformation of the Michaelis-Menten equation for a one-substrate enzyme can be written as:

$$v = -K_m(v/[S]) + V_{\text{max}}$$

Others have plotted the data by changing the horizontal and vertical axes: thus, $v/[S]$ versus v (also known as the Eadie-Scatchard plot). In this case, the slope is $-1/K_m$, the vertical intercept is now V_{max}/K_m , and the horizontal intercept is V_{max} (thus, $v/[S] = -(v/K_m) + (V_{\text{max}}/K_m)$).

¹B. Woolf (1932), cited by J. B. S. Haldane & K. G. Stern, *Allgemeine Chemie der Enzyme*, p. 119, Steinkopff, Dresden-Leipzig.

²G. S. Eadie (1942) *J. Biol. Chem.* **146**, 85.

³K.-B. Augustinsson (1948) *Acta Physiol. Scand.* **15**, Suppl., 52.

⁴G. Scatchard (1949) *Ann. N. Y. Acad. Sci.* **51**, 660.

EC₅₀

The molarity of an effector or agonist that produces 50% of the maximal effect of that effector or agonist.

EDITING

Any process leading to the elimination or correction of errors in molecular structure, particularly as it applies to the sequence of monomeric units in nucleic acids and proteins as well as the synthesis of aminoacyl-tRNA's.

EDITING MECHANISMS

Intrinsic enzymatic activities of enzymes that minimize or prevent erroneous introduction of closely related structural analogues of metabolites into biopolymers. Hopfield¹ has discussed the general need and likely strategies for kinetic editing or proofreading. Fersht's group^{2,3} carried out an exhaustive analysis of aminoacyl-tRNA synthetase editing. Mischarged aminoacylated-tRNA's are recognized and hydrolyzed to prevent their inclusion in polypeptides by translation. DNA and RNA polymerases also use editing to excise incorrectly incorporated deoxyribotide and ribotide subunits. Yarus⁴ presented a cogent analysis showing that enzymic editing (or proofreading) requires (1) a branched pathway that diverts and recycles any incorrectly produced product; (2) constrained use of substrate specificity; (c) consumption of energy; and (d) a compromise between accuracy and yield. His report also treats the ways in which a branched Michaelis-Menten-type enzyme might produce useful proofreading properties.

¹J. J. Hopfield (1974) *Proc. Natl. Acad. Sci. U.S.A.* **71**, 4135.

²A. R. Fersht, R. S. Mulvey & G. L. E. Koch (1975) *Biochemistry* **14**, 13.

³A. R. Fersht (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, pp. 650, W. H. Freeman, New York.

⁴M. Yarus (1992) *Trends Biochem. Sci.* **17**, 130.

EDTA

Ethylenediamine-*N,N,N',N'*-tetraacetic acid, an important chelating agent and metal ion buffer: $(\text{HOOCCH}_2)_2\text{NCH}_2\text{CH}_2\text{N}(\text{CH}_2\text{COOH})_2$. The pK_a values of EDTA (at an ionic strength of 0.1 and a temperature of 20°C) are 2.0, 2.67, 6.16, and 10.26. Thus, at a pH range of 4 to 5 most of the EDTA in an aqueous solution is present as the divalent anion whereas between pH 7 to 9, the trivalent anion is the more prevalent species. However, the formation of a metal ion-EDTA complex will result in the displacement of the remaining protons from EDTA. Thus, $\text{M}^{n+} + \text{H}_2(\text{EDTA})^{2-} \rightleftharpoons \text{M}(\text{EDTA})^{(n-4)+} + 2\text{H}^+$ between pH 4 and 5 and $\text{M}^{n+} + \text{H}(\text{EDTA})^{3-} \rightleftharpoons \text{M}(\text{EDTA})^{(n-4)+} + \text{H}^+$ between pH 7 and 9. Hence, the tetravalent anion of EDTA is the species of importance in equilibrium calculations.

Metal ion complex formation is typically measured via stability, or formation, constants. Thus, for $\text{M}^{n+} + \text{EDTA}^{4-} \rightleftharpoons \text{M}(\text{EDTA})^{(n-4)+}$, $K_{\text{M}(\text{EDTA})} = [\text{M}(\text{EDTA})^{(n-4)+}] / ([\text{M}^{n+}][\text{EDTA}^{4-}])$. A number of values for $\log K_{\text{M}(\text{EDTA})}$ at 25°C and an ionic strength approaching zero are provided in the table below:

Formation Constants for Selected Metal-EDTA Complexes

Metal Ion	$\log K_{\text{M}(\text{EDTA})}$	Metal Ion	$\log K_{\text{M}(\text{EDTA})}$
Co(III)	36	Zn	16.4
V(III)	25.9	Co(II)	16.31
Fe(III)	24.23	Al	16.13
Cr(III)	23	Fe(II)	14.33
Hg(II)	21.80	Mn(II)	13.8
Cu(II)	18.7	V(II)	12.7
Ni	18.56	Ca	11.0
Pb(II)	18.3	Mg	8.64
V(V)	18.05	Ag	7.32
V(IV)	18.0	Na(I)	1.7

EFFECTIVE DOSE

The dose, symbolized by ED, producing a desired effect by a substance. The symbol is usually followed by a subscript number indicating the dose that has the desired effect on a certain percentage of the tested population. The median effective dose is ED_{50} .

EFFECTIVE MOLARITY

The ratio of a first-order rate constant for an intramolecular reaction (involving two functional groups or moieties within the same molecular entity) to the second-

order rate constant of the analogous intermolecular elementary reaction¹. Effective molarity has units of concentration. While the highest effective concentration in an aqueous solution was thought for many years to be 55.5 M (corresponding to that of pure water), Kirby² has tabulated much higher effective concentrations. These include values of 3,700 M for cyclic thiolester formation from 4-thiolbutyrate; 200 M for dithiolane ring formation with 1,4-dithioerythritol; and 190,000 M for cyclic anhydride formation from succinic acid. **See also** *Intramolecular Catalysis*.

¹IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

²A. J. Kirby (1980) *Adv. Phys. Org. Chem.* **17**, 183.

EFFECTOR

A general term applying to substances or agents that elevate or depress the efficient action of a catalyst, transporter, or receptor. Both activators and inhibitors are considered to be effectors of enzyme action and transport, and the analogous terms for receptors are agonists and antagonists. In many instances, the effector and modifier are used synonymously, but the Nomenclature Committee¹ of the International Union of Biochemistry suggests that effector be restricted to those substances producing a significant physiological effect. The term modifier applies to substances that are artificially added to *in vitro* systems. **See also** *Modifier; Activator; Inhibitor*

¹International Union of Biochemistry (1982) *Eur. J. Biochem.* **128**, 281.

EFFICIENCY FUNCTION

A quantitative expression developed by Albery and Knowles¹ to describe the effectiveness of a catalyst in accelerating a chemical reaction. The function, which depends on magnitude of the rate constants describing individual steps in the reaction, reaches a limiting value of unity when the reaction rate is controlled by diffusion. For the interconversion of dihydroxyacetone phosphate and glyceraldehyde 3-phosphate, the efficiency function equals 2.5×10^{-11} for a simple carboxylate catalyst in a nonenzymic process and 0.6 for the enzyme-catalyzed process. Albery and Knowles¹ suggest that evolution has produced a nearly perfect catalyst in the form of triose-phosphate isomerase. **See** *Reaction Coordinate Diagram*

¹W. J. Albery & J. R. Knowles (1977) *Biochemistry* **15**, 5631.

EF-HAND

A motif (found in more than 160 different calcium-binding proteins) consisting of two alpha helices, designated “E” and “F,” that are joined by a Ca^{2+} -binding loop. Functional EF-hands always occur in pairs. The EF-hand family includes calmodulin, troponin C, myosin regulatory light chain, parvalbumin, calbindins, α -actinin, calcineurin B subunit, calpain, and fimbrin. A striking feature of the EF-hand family is the ability to modulate the activity of a number of enzymes. EF-hands have been identified inferentially in numerous calcium ion binding proteins based upon similarity classifications of amino acid sequence and in some cases confirmed directly by crystal structures.

EIGENFUNCTION

Any solution to one or more differential equations for which there are only certain allowed parameters.

EIGEN-TAMM MECHANISM FOR METAL ION COMPLEXATION

A kinetic description of rapid metal ion complexation reactions. Desolvation of the primary coordination sphere of the metal ion is treated as the rate-limiting step in forming its inner-sphere coordination complex with a ligand.

M. Eigen & K. Tamm (1962) *Z. Elektrochem.* **66**, 107.

EINSTEIN

A unit used principally in photochemistry to designate the energy equivalent to an Avogadro's number of quanta (*i.e.*, 6.0221367×10^{23} quanta).

EINSTEIN-SUTHERLAND EQUATION

The quantitative expression ($D = k_B T/f$) describing the relationship between the diffusion constant D of a particle and its frictional coefficient f , where k_B and T are the Boltzmann constant and the absolute temperature, respectively. **See** *Diffusion*

ELASTASE

This enzyme catalyzes the hydrolysis of proteins, including elastin, with preferential cleavage at Ala–Xaa. The following are reviews on the molecular and physical properties of this enzyme [EC 3.4.21.36 (pancreatic) and EC 3.4.21.37 (leukocyte)].

- L. Rust, C. R. Messing & B. H. Iglewski (1994) *Meth. Enzymol.* **235**, 554.
 E. Lolis & G. A. Petsko (1990) *Ann. Rev. Biochem.* **59**, 597.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 L. Robert & W. Hornebeck, eds. (1989) *Elastin and Elastases*, CRC Press, Boca Raton.
 J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.
 E. T. Kaiser, D. S. Scott & S. E. Rokita (1985) *Ann. Rev. Biochem.* **54**, 565.
 T. A. Steitz & R. G. Schulman (1982) *Ann. Rev. Biophys. Bioeng.* **11**, 419.
 R. J. Coll, P. D. Compton & A. L. Fink (1982) *Meth. Enzymol.* **87**, 66.
 B. S. Hartley & D. M. Shotton (1971) *The Enzymes*, 3rd ed., **3**, 323.

ELASTICITY

A metabolic control parameter (symbolized by ε) that provides a measure of the direction and magnitude of a change in enzyme (or pathway) velocity as a function of a change in the concentration x of substance X acting as a substrate or effector¹.

$$\varepsilon \equiv (\partial \ln v / \partial \ln x) \equiv (x \partial v / v \partial x)$$

When ε equals zero, v shows no functional dependence on the concentration of X. If ε is positive, v increases as the concentration X increases; if ε is negative, v decreases as x increases.

For a lucid account of the kinetics of multi-enzyme systems, the reader should consult Cornish-Bowden¹ who defines such related parameters as flux control coefficients, summation relationships, and response coefficients.

¹A. Cornish-Bowden (1996) *Fundamentals of Enzyme Kinetics*, Portland Press, London.

ELASTICITY COEFFICIENT

The partial derivative of the velocity of a metabolic step or pathway with respect to the concentration of a substrate, product, and/or effector.

ELECTRIC CHARGE

The positive or negative charge (commonly symbolized by Q or q) on a molecule, radical, or particle resulting from the deficient or excess accumulation of electrons. Electric charge need not be an integer value. The SI unit “coulomb” (abbreviated C) equals the quantity of electricity transferred by an electric current of 1 ampere over the period of 1 second.

ELECTRIC POTENTIAL

A quantity (commonly symbolized by V) for the work needed to bring a unit positive charge to that point in space from an infinite distance. Thus, $V = dw/dQ$ where w is the work and Q is the electric charge. The SI unit for electric potential is the volt (V). The electric potential difference, also measured in volts and symbolized by U , ΔV , or $\Delta\Phi$, is equal to the difference in potential between two points (*i.e.*, $U = V_2 - V_1$) as measured by the work needed to transfer a unit positive charge from one point to the other. **See also** *Electromotive Force*

ELECTROACTIVE SPECIES

Any chemical compound that can be oxidized or reduced in a manner that generates a flow of electrons from a solution to an electrode.

ELECTROCATALYSIS

A term used in electrochemistry to describe facilitated electron transfer, resulting in an increased rate of half-cell reactions at electrode surfaces.

ELECTRODE KINETICS

The systematic representation of the time dependence of chemical reactions occurring at an electrode surface. These reactions can be analyzed as a succession of events: (a) diffusion of reactants within the bulk solution to the more condensed layer of solution near the electrode surface; (b) penetration of that layer to achieve adsorption on the electrode's surface; (c) electron transfer to (*i.e.*, reduction) or from (*i.e.*, oxidation) the adsorbed reactant(s); product desorption, penetration of the condensed layer, and diffusion into the bulk solution.

K. J. Laidler (1970) *J. Chem. Education* **47**, 600.

ELECTROFUGE

If the leaving group in a chemical reaction does not leave with the bonding electron pair, that leaving group is referred to as an electrofuge. The tendency of such a moiety to leave is referred to as electrofugicity. **See also** *Electrophile*; *Nucleofuge*

ELECTROMOTIVE FORCE

The amount of energy obtained from an electrical source per unit of electric charge. The SI unit for electromotive force (commonly symbolized by E) is the volt or V. The electromotive force is equal to $\int (F/Q) \cdot ds$ where F is

the force, Q is the electric charge, and s is the position vector. **See also** *Electric Potential*

ELECTRON

A fundamental elementary particle possessing a negative charge of $1.60217733 \times 10^{-9}$ coulomb (the elementary charge) and a rest mass, m_e , of $9.1093897 \times 10^{-31}$ kg (or, 0.51099906 MeV).

ELECTRON ACCEPTOR

1. A substance which will accept or interact with an electron. 2. A Lewis acid (Note, however, that IUPAC now discourages the use of this term in this manner.)

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

ELECTRON AFFINITY

The change in enthalpy due to the association of an additional electron to a particular atom or molecular entity. This physical quantity is usually measured in the gas phase.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

ELECTRON DENSITY

The electron probability distribution in a molecular entity. If $P(x,y,z)$ is the electron density (in Cartesian coordinates) at coordinates (x,y,z) , then $P(x,y,z) dx dy dz$ represents the probability of finding an electron in the volume element $dx dy dz$ at coordinates (x,y,z) . **See also** *Charge Density*

ELECTRON DENSITY FUNCTION

An approximate quantum mechanical expression^{1,2} that allows one to calculate the electrostatic surface potential around atoms, radicals, ions, and molecules by assuming that the ground-state electron density uniquely specifies the Hamiltonian of the system and thereby all the properties of the ground state. This approach greatly facilitates computational schemes for exact calculation of the ground-state energy and electron density of orbitals.

¹P. Hohenberg & W. Kohn (1964) *Phys. Rev. B* **136**, 864.

²W. Kohn & L. J. Sham (1965) *Phys. Rev. A* **140**, 1133.

ELECTRON DONOR

A substance or molecular entity that can transfer (or donate) an electron to another chemical species. Note:

IUPAC has discouraged the use of this term as a synonym for Lewis base.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

ELECTRONEGATIVITY

A measure of the ability of an atom within a molecule to attract bonding electrons toward itself^{1,2}. For a bond between two atoms of different electronegativities, the electron molecular orbital cloud is not symmetric, and the atom with the higher electronegativity attracts the larger proportion of the cloud. The most popular quantitative description was presented by Pauling³, who based his scale on bond dissociation energies (measured in kcal per mol).

¹K. D. Sen & C. K. Jorgensen (1987) *Electronegativity*, Springer, New York.

²S. S. Batsanov (1968) *Russ. Chem. Rev.* **37**, 332.

³L. Pauling (1960) *The Nature of the Chemical Bond*, 3rd ed., Cornell Univ. Press, Ithaca.

ELECTRONEUTRALITY PRINCIPLE

A conceptual framework, first proposed by Pauling^{1,2}, for writing the correct structural formulas for polyatomic molecules and ions. The electroneutrality principle requires that the electronic structure of stable molecules is such that the electric charge on each atom is close to zero (*i.e.*, electroneutrality), and varies between at most +1 and −1. Application of this principle requires knowledge of electronegativities as well as a general understanding of the relationship between ionic character of a bond and the electronegativity difference of the bonded atoms.

One classical example that applies the electroneutrality principle is the electronic structure of carbon monoxide, a diatomic molecule with a very small dipole moment of 0.110 debye. The only electronic structure that satisfies the octet rule for CO is :C≡O:, a structure that corresponds to C[−] and O⁺, if the shared electron pairs are equally divided by the two atoms. Pauling showed that the electronegativity difference of 1.0 would correspond to about 22% partial ionic character for each bond, and to charges of C^{0.34−} and O^{0.34+}. A second possible electronic structure, :C=Ö:, does not complete the octet for carbon. The partial ionic character of the bonds corresponds to C^{0.44+} and O^{0.44−}. If these two structures contribute equally to a hybrid molecular structure, the resultant electronic charge on each atom would be very small.

Thus, in agreement with the electroneutrality principle and the very small dipole moment, one can reasonably write the electron configuration for CO as the resonance hybrid: {C≡O: ↔ :C=Ö:}.

¹L. Pauling (1948) *J. Chem. Soc.*, 1461.

²L. Pauling (1948) *General Chemistry*, Freeman, San Francisco (also, the 1988 Dover republication of the third edition of Pauling's classic text (1970) *General Chemistry*, pp. 192-194).

ELECTRON-NUCLEAR DOUBLE RESONANCE (ENDOR)

A magnetic resonance spectroscopic technique used for detect hyperfine interactions between electrons and nuclear spins. In its continuous-wave mode, the ESR signal intensity is measured as radio frequency is applied. In pulsed mode, pulses of radio frequency energy are applied, and the ESR signal is detected as a spin-echo. In either case, enhanced EPR signal strength occurs when the radio frequency is in resonance with the coupled nuclei.

ELECTRON SINK

A term describing that region of a molecule having sufficient electrophilic character, such that it can capture or trap an electron during the course of a reaction.

ELECTRON SPIN RESONANCE

A condition arising when a chemical species containing an odd or unpaired electron is placed in a magnetic field (typically around 3000 gauss or 0.3 tesla), such that the components (corresponding to $+h/4\pi$ and $-h/4\pi$) of spin angular momentum of the electron are quantized along the direction of the applied magnetic field. In such cases, the difference in energy levels corresponds to that observed in the microwave region of the electromagnetic spectrum (*e.g.*, 10^{-23} joules is equivalent to a microwave frequency of about 0.3 cm^{-1}). When positioned in a magnetic field, a resonance effect is observed when the sample is irradiated by microwaves of energy equal to the aforementioned energy difference. The following band designations are used: L (1.1 gigahertz or GHz), S (3.0 GHz), X (9.5 GHz), K (22.0 GHz), and Q (35.0 GHz). Free radicals and other biological substances, such as nitric oxide (with its 15 electrons) and manganous ions, can be gainfully examined by ESR methods. Because the energy difference is relatively small, ESR spectra are frequently sensitive to the environment in which the unpaired electron resides. Synthetic moieties containing

free radicals can often be attached to macromolecules or membranes to use the ESR spectral properties to investigate mechanistic issues.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Theory, **74**, 156-157; **76**, 313; line shapes and g values, **76**, 315-316; hyperfine splitting, **76**, 315; axial symmetry, **76**, 313, 315, 319; rhombic symmetry, **76**, 313, 315, 320; optimizing conditions for observation, **76**, 328-329 [of Fe chromophores, **69**, 791; of Fe proteins, **69**, 775, 776, 778; of hemoglobin, **76**, 312-329; of hemoglobin S, **76**, 796-797; of iron-sulfur centers, **69**, 239, 240; of iron-sulfur proteins, **69**, 778; of nitrite reductase, **69**, 265, 267; of Mo-Fe proteins, **69**, 767-770; of nitrogenase Mo-Fe proteins, **69**, 761; of photoreduction centers, **69**, 240, 241]; low-temperature enzyme dynamics, **61**, 327; manganese environment, **69**, 355, 356, 358; high-spin ferric type, **76**, 314, 316-319; low-spin ferric type, **76**, 314, 319-324; metal-substituted, **76**, 91, 326-328; oxidation-reduction potentiometry, **69**, 242-245; spin trapping, **76**, 329; using circularly polarized microwaves, **76**, 318; ferredoxin assay, **69**, 233; immunoassay technique, **74**, 30, 140-151; ESR measurement of lipid bilayer changes, **79**, 468-473; ESR of plasma membrane changes, **79**, 468-473; ESR in determining stability constants, **63**, 311; acetylcholinesterase assay, **251**, 101-104; aconitase, **246**, 583; adenylsulfate reductase from sulfate-reducing bacteria, **243**, 248-249; aldehyde oxidoreductase from *Desulfovibrio gigas*, **243**, 30-32; amino-semiquinone of methylamine dehydrogenase, **258**, 184; anisotropy, **258**, 370-373; ascorbate free radicals *in vivo*, **233**, 149-154; bishistidine complexes, **231**, 470; chromanoxyl radicals, **234**, 364-365; contaminants, **246**, 543; continuous wave, **246**, 599-600; copper-dependent hydroxyl radical formation, **234**, 69-70; copper-DNA adduct-dependent hydroxyl radical formation, **234**, 70-73; desulfoferredoxin, **243**, 229-230; dissociation constant determination, **249**, 181; distance between spin labels, determination, **246**, 604; DNA [internal motion analysis, **261**, 500-505; radius measurement, **261**, 500; structure determination, **261**, 500]; DNA photolysis, **258**, 327-343; Fourier transform, **258**, 374-375; free radicals [identification, **246**, 548; production, **246**, 534; proteins, **246**, 547-550; spin trapping, **246**, 534-535]; galactose oxidase, **258**, 236-237, 254, 267-268; g factor magnitude, **246**, 538-539, 548; helix-forming peptides, **246**, 601-605; hemichromes, **231**, 453-454, 474-477; hexadecaheme cytochrome c, **243**, 161; hyperfine splitting, **246**, 539-540, 554 [dipole-dipole contribution, **246**, 541; isotropic contribution, **246**, 541; radical identification, **246**, 548-549; resolution, **246**, 562]; iron, **235**, 324; iron protein containing six-iron cluster, **243**, 237-240; length-dependent mean square oscillation amplitude, **261**, 501-503; ligand binding studies, **246**, 581, 583-584; ligand interactions at transition metal centers, **246**, 7; line broadening, **246**, 542-543; lineshape and motion time scale, **261**, 496-498; low-density lipoprotein oxidation studies, **233**, 428-430; low-spin sulfite reductase, **243**, 300-301; low-temperature, low-spin complexes, **231**, 467; lysine 2,3-aminomutase [isotopic substitution, **258**, 367-369, 377; liquid phase measurements, **258**, 366; π -radical properties, **258**, 370-373; rapid-mix/freeze-quench trapping of intermediates, **258**, 365; signal characterization, **258**, 367-369; slow substrate trapping of intermediates, **258**, 365-366; spectral properties, **258**, 363-365, 367; structure determination, **258**, 379]; magnetic field [frequency, **246**, 537, 544, 608; strength, **246**, 535, 537-538, 544, 608]; manganese binding affinity, determination for Klenow fragment, **262**, 148-150; membrane proteins, **246**, 606-608; metalloproteins [copper, **246**, 532-533; iron, **246**, 531-532, 551-552; metal coupling, **246**, 553-554; spin states, **246**, 551-552]; microwave [absorption by water, **246**, 542; frequency in determining g values, **246**, 549; power saturation, **246**, 544-546]; nickel-iron-selenium hydrogenase-ligand interaction studies, **243**, 89-93; nitroxide free radical, **246**, 530, 589, 592-593; nitroxides, **234**, 583-

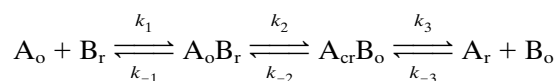
584; nucleic acid peroxy radicals, **251**, 52; origin of signal, **246**, 529-530, 536, 538-541, 554; peroxidation effects on membranes, **233**, 274-275, 280; photoexcited triplet states, **246**, 611; photosynthetic reaction centers, **246**, 533-534, 548-549; photosystem II tyrosyl radicals [apparatus, **258**, 312; D° , **258**, 312-314; deuteration effects, **258**, 314, 316; M° , **258**, 306, 315-316; Z° , **258**, 312-314]; powder pattern line shape, **246**, 541, 593-594; protein conformation studies, **246**, 5-6; pulsed saturation recovery EPR, **246**, 599-600; pulse-power Fourier transform spectroscopy, **246**, 608; pyruvate formate-lyase, **258**, 349-350; quantitative analysis, **246**, 536, 546-547, 553; redox titration, tetraheme cytochrome c₃, **243**, 131-133; resolution enhancement, **258**, 373-376; ribonucleotide reductase R2 protein reaction with excess Fe(II) and O₂, rapid freeze-quench studies, **258**, 286-303; rubrerythrin from *Desulfovibrio vulgaris*, **243**, 221-222; sample requirements, **246**, 543-544; saturation transfer EPR, **246**, 600-601; semiquinone intermediate in amine oxidase, **258**, 85-87; sensitivity, **246**, 2, 543-544; **251**, 52; **258**, 264-265; **261**, 500; signal-to-noise ratio optimization, **246**, 544-545; spectral simulation, **258**, 376-377; spin label relaximetry, **246**, 606; spin probes for DNA duplex [motion sources of DNA probes, **261**, 496, 503-504; placement, **261**, 505; types, **261**, 496]; spin trapping [alcohol derived radicals, **233**, 117-127; hydroxyl radicals, **233**, 71, 96-98; radical adducts *in vivo* (in biological fluids, **233**, 114-115; bioreduction of nitroxide moiety of radical adducts, **233**, 116-117; free radical formation *ex vivo*, **233**, 115-116; lipid peroxidation *ex vivo*, **233**, 116; sample handling, **233**, 113-114; in tissue extracts, **233**, 113-115]; temperature of experiment, **246**, 541-543; tensor averaging, **261**, 498-499; theory, **258**, 265; thiol assay with biradical method [accuracy, **251**, 198-199; advantages, **251**, 202-203; blood sample preparation, **251**, 94-95; calibration curves, **251**, 195-196; cell suspensions, **251**, 197-199; free sulfhydryl groups in proteins, **251**, 96-98; glutathione, **251**, 193-202; kinetic method, **251**, 93; limitations, **251**, 203; perfused heart, **251**, 200-202; sensitivity, **251**, 194, 197; spectra, **251**, 88-89, 192, 194-195, 197; static method, **251**, 93-94]; thyl radicals, **251**, 73-74, 110-112; two-dimensional spectroscopy, **246**, 609-610; tyrosine radical, **246**, 539, 548-550; uniform mode [estimation, **261**, 499-500; removal, **261**, 501]; unpaired spin restriction, **246**, 126; vitamin E phenoxyl radicals, **234**, 316-320, 351-353; VP-16 phenoxyl radicals [detection, **234**, 632-637; reduction reactions in aqueous solution, **234**, 638-640; reduction reactions in cell and nuclear homogenates, **234**, 641-642]; Zeeman energy of spin states, **246**, 538, 593; zero-field splitting, **246**, 550-551.

ELECTRON TRANSFER

The exchange of an electron between two molecular entities or between two sites within the same molecular entity. *See* Marcus Equation; Electrode Kinetics

ELECTRON TRANSFER REACTIONS

Rees and Farrelly¹ have presented an instructive general treatment of the following electron transfer reaction mechanism:



where the three successive equilibria are association of reactants, electron transfer, and product dissociation; each has an assigned formation constant K_1 , K_2 , and K_3 equal to k_i/k_{-i} . Treating the system in a manner akin

to a steady-state one-substrate/two-intermediate enzyme mechanism with $k_{-3}[A_r][B_o] = 0$, the reaction velocity v is given as follows:

$$v = k_{\text{obs}}[A_o][B_r] \\ = \{k_1 k_2 k_3 / (k_{-1} k_{-2} + k_{-1} k_3 + k_2 k_3)\} [A_o][B_r]$$

They describe three cases: (a) rate-limiting electron transfer, in which event, $v = k_{-2} K_1 [A_o][B_r]$; (b) rate-limiting substrate association, thereby yielding $v = k_1 [A_o][B_r]$; and pre-equilibration of the first two steps with rate-limiting product release, giving $v = k_3 K_1 K_2 [A_o][B_r]$. They note that k_{obs} never gives k_2 alone, and one would need to know which of the aforementioned cases applies in order to get a good estimate of k_2 . The reader is encouraged to consult their assessment of the occasions to which each case is apt to pertain. Rees and Farrelly¹ also give an account of how Marcus theory and quantum electronic theories might apply to electron transfer reactions.

¹D. C. Rees & D. Farrelly (1990) *The Enzymes*, 3rd. ed., **20**, 37.

ELECTRON TRANSPORT

The transfer of one or more electrons within and/or between molecules or other electroactive species. Unlike classical mechanics in which a system must surmount any potential energy barrier between two states, quantum mechanics allows certain electron transfer systems to tunnel or leak through the barrier without surmounting it. Tunneling can occur when the de Broglie wavelength of small particles (particularly electrons, hydrogen atoms, as well as protons and hydride ions) is comparable to the width of the potential energy barrier. In these cases, the Uncertainty Principle allows for the probability of the electron residing on either side of the barrier.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Electron-transport chain [components, **69**, 205, 206; sites of inhibition, **69**, 676, 677]; chloroplast [autoxidizable carriers, **69**, 416, 417; DBMIB, **69**, 422, 423; dichlorophenolindophenol and related carriers, **69**, 418; ferricyanide, **69**, 417, 418; isolated, **69**, 85, 86; NADP/ferredoxin, **69**, 416; substituted benzoquinones, **69**, 422; inhibitors, chemical modifiers, **69**, 503, 504; measurement, **69**, 653, 654; mutations, **69**, 16, 19; reverse, **69**, 630, 631; acid-base-driven, **69**, 639-641; luminescence, **69**, 639, 640; properties, **69**, 640, 641; reduction of coenzyme Q, **69**, 639; ATP-driven, **69**, 632-639; nitrate reductase sequence, **69**, 278, 279.

ELECTROPHILE

A reagent or reactant that is preferentially attracted to a region of high electron density within a particular molecular entity (or, in the case of intramolecular processes,

within the same molecular entity). Typically, an electrophile bonds by accepting both bonding electrons for the reaction partner (termed the nucleophile). The term electrophile is also used with respect to certain polar radicals exhibiting high reactivities with sites of high electron density. **See also** *Electrophilicity; Nucleophile*

ELECTROPHILIC CATALYSIS

Catalysis arising from the participation of a Lewis acid in catalyzed or uncatalyzed chemical reactions.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

ELECTROPHILICITY

1. The relative reactivity expressed by an electrophile. Electrophilicity is measured by the relative rate constants for a particular reaction of different electrophiles for a common substrate. 2. The property of being electrophilic. **See also** *Electrophile; Nucleophilicity*

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

ELECTROPHILIC SUBSTITUTION REACTION

A reaction in which an electrophile participates in heterolytic substitution of another molecular entity that supplies both of the bonding electrons. In the case of aromatic electrophilic substitution (AES), one electrophile (typically a proton) is substituted by another electron-deficient species. AES reactions include halogenation (which is often catalyzed by the presence of a Lewis acid salt such as ferric chloride or aluminum chloride), nitration, and so-called Friedel-Crafts acylation and alkylation reactions. On the basis of the extensive literature on AES reactions, one can readily rationalize how this process leads to the synthesis of many substituted aromatic compounds. This is accomplished by considering how the transition states structurally resemble the carbonium ion intermediates in an AES reaction.

ELECTROSTATIC BOND

A bond resulting from attractive electric interactions between chemical species that include, but are not limited to, ions. In this regard, polarized species such as carbonyl and amino groups can engage in electrostatic interactions that give rise to hydrogen bonds.

ELECTROSTATIC SURFACE POTENTIAL

A measure of the electric charge density and distribution along the surface of molecules. The surface potential is

approximately determined by the laws of classical electrostatics or more rigorously by quantum mechanics. The magnitude of the attractive or repulsive energy (U) between two point charges (q_1 and q_2) separated at a distance r is equal to (kq_1q_2/Dr) , where k is 9×10^9 joule-meter/coulomb² and D is the dielectric constant of the medium. To estimate the electrostatic surface potential, calculations are made for the attractive or repulsive interactions experienced by a probe proton passing above the molecule's surface at a fixed distance.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Computation, **240**, 648-649, 652-653; electrostatic potential difference effect on protein partitioning away from isoelectric point, **228**, 229; salt effects, **240**, 653-654; hemoglobin, **240**, 656-657; ion binding site determination, **240**, 648; static accessibility parameter algorithm, **240**, 648-649; Tanford-Kirkwood model, **240**, 648, 654.

ELECTROSTRICTION

1. The deformation of a dielectric substance or body in an applied electric field. 2. The orientation of solvent water molecules interacting with ions or charge surfaces of molecules. Proteins contain many ionic groups on their surface, and each group has the capacity to attract and deform nearby water molecules. These tightly bound water molecules experience an electrostrictive effect of about $-16 \text{ \AA}^3$ per charged group, and electrostriction slightly increases the density of bound water. Thus, a protein containing around 50 charged groups on its surface will increase the number of water molecules in its hydration shell by about 25, amounting to a density increase of 1–2%. Moreover, when a peptide bond is hydrolyzed, there is most frequently a decrease in the volume of the aqueous medium; this occurs when the newly formed carboxylate and protonated amino groups contract the surrounding water. Volume changes are directly determined by the use of a dilatometer which is comprised of the glass bulb of a pycnometer whose tapered glass cap is replaced by a tightly fitting cap with a vertical bore connected to a calibrated capillary.

Because the motions of bound water are restricted, electrostriction results in a considerable reduction in entropy. A consequence is that ionization in a solution reduces the entropy; by contrast, neutralization of two aquated counterions leads to a release of bound water and a rise in entropy.

ELECTROTAXIS

The directed movement or locomotion of an organism placed within an electric field.

ELECTROVALENT BOND

A less frequently used synonym for ionic bond.

ELEMENTARY CHARGE

The magnitude of electric charge on an electron or proton that gives rise to their mutually attractive interaction. The charge on the elementary particles is referred to as elementary charge, is symbolized by e , and has a value of $1.60217733 \times 10^{-19}$ coulombs.

ELEMENTARY REACTION

A reaction that takes place on a molecular scale in a single step following an individual collision or other elementary process and in which no stable intermediate need be postulated and no simpler reaction can be suggested.

ELIMINAND

An alternative term for the leaving group in an elimination reaction.

ELONGATION (With No Change in Polymer Number Concentration)

Although the self-assembly of polymeric structures can involve nucleation and elongation steps (*See Actin Assembly Kinetics; Microtubule Assembly Kinetics*), one can simplify the assembly process through what is known as “seeded” assembly. At an initial monomer concentration $[M]$, “seeded” assembly is induced by the addition of pre-assembled polymeric structures; consequently the polymer number concentration must remain constant. The rate of monomer incorporation into indefinite length polymers can be written as follows:

$$-d[M]/dt = k_+[M] - k_-$$

where k_+ corresponds to the product of the bimolecular rate constant for elongation and the concentration of elongating seeds, and k_- is likewise the product of the unimolecular rate constant for monomer dissociation and the concentration of elongating seeds. Multiplying the right-hand side of the equation by k_+/k_+ , we obtain

$$-d[M]/dt = k_+[M_0] - k_+k_-/k_+ = k_+\{[M_0] - [M_\infty]\}$$

Here, the ratio k_-/k_+ defines the critical monomer concentration $[M]$, the point below which no polymerization can occur. Integration of $-d[M]dt = k_+\{[M_0] - [M_\infty]\}$ yields the expression:

$$[M] = \{[M_0] - [M_\infty]\} \exp[-k_+ t]$$

which defines a single exponential decay from the initial monomer concentration $\{[M] - [M_\infty]\}$ to the equilibrium monomer concentration $[M_\infty]$.

If an experimenter wishes to study actin filament elongation, for example, there are three ways to induce the elongation process: (a) by adding short preformed filaments that have unblocked barbed and pointed ends (thereby yielding bidirectional elongation); (b) by introducing gelsolin-capped actin filaments (thereby permitting elongation strictly at the pointed end); or (c) by supplying red cell spectrin-4.1 complex (thereby inducing barbed-end elongation). In the case of tubulin polymerization, one may employ: (a) short pre-assembled microtubule seeds to induce bidirectional elongation; (b) flagellar axonemes to initiate morphologically distinguishable bidirectional growth; or (c) centrosomes to promote plus-end growth of microtubules. When bidirectional growth takes place, k_+ represents the sum of the on-rate constants for both ends.

EMBRYO

A subcritical aggregate having fewer subunit components than a nucleus. When this term is applied in the kinetics of precipitation, n refers to the number of subunits in a particle and n^* defines the number of subunits in a particle of critical size. This definition avoids confusion by distinguishing between subcritical ($n < n^*$ subunits), critical ($n = n^*$ subunits), and supercritical ($n > n^*$ subunits) particle sizes. If a nucleus is defined as containing n^* subunits, then an embryo contains $n < n^*$ subunits. Note that in this treatment, we are not using a phase-transition description to describe nucleation, and we are focusing on the smallest step in the process that leads to further aggregation.

EMISSION

1. Radiative deactivation of an excited state of a molecular entity. 2. The release of a nucleon (usually an electron, neutron, or photon) during radioactive decay.

EMISSION SPECTRUM

The emitted spectral radiant power (or exitance) or the emitted spectral photon irradiance (or exitance) plotted as a function of the frequency, wavenumber, or wavelength. The corrected emission spectrum has been cor-

rected for wavelength-dependent variations in the equipment response. **See also** *Action Spectrum; Excitation Spectrum*

ENANTIOSELECTIVE REACTION

A reaction in which an optically inactive compound (or achiral center of an optically active molecule) is selectively converted to a specific enantiomer (or chiral center). **See** *Asymmetric Induction*

ENCOUNTER COMPLEX

Any weakly attractive, short-lived complex that is typically formed as an intermediate in a reaction mechanism. When there are only two such molecular entities engaged in the formation of a particular encounter complex, that complex is often called an encounter pair.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

ENCOUNTER-CONTROLLED RATE

A reaction velocity equal to the rate of encounter of reacting molecular entities (also known as diffusion-controlled rate). For a bimolecular reaction in aqueous solutions at 25°C, the corresponding second-order rate constant for the encounter-controlled rate is typically about $10^{10} \text{ M}^{-1}\text{s}^{-1}$. **See** *Diffusion Control for Bimolecular Collisions in Solution*

ENDERGONIC PROCESS

A change or reaction for which the Gibbs free energy (ΔG) becomes more positive. (The origin of this compound term is derived from the Greek prefix *end-* which is a contraction of *endo* and the Greek root *ergon*, standing for work.)

ENDO- α (or β)-N-ACETYL GALACTOSAMINIDASE

A number of enzyme have been reported to catalyze the hydrolysis of glycosidic linkages in glycoproteins at *N*-acetylgalactosamine residues, exhibiting specificities at either the α - or the β -linkage. Enzymes that are reported to act on both α - and β -linkages have to be clearly characterized as to their specificity.

R. J. Staneloni & L. F. Leloir (1982) *Crit. Rev. Biochem.* **12**, 289.
H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

ENDO- β -N-ACETYLGLUCOSAMINIDASE

This family of enzymes [EC 3.2.1.96] catalyzes the endohydrolysis of the di-*N*-acetylchitobiosyl unit in high-man-

nose glycopeptides and [Man(GlcNAc)₂]Asn-containing glycoproteins, releasing the rest of the oligosaccharide intact, while leaving a single *N*-acetyl-D-glucosamine residue on the protein.

E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.
H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

ENDOENZYME

Any enzyme catalyzing a hydrolytic reaction that occurs within a polymeric substrate molecule. Those enzymes acting at the terminus of a polymeric substrate molecule are called exoenzymes.

ENDO- β -GALACTOSIDASE

A family of enzymes that catalyzes the endohydrolysis of β -galactosides. They include the endogalactosidase [EC 3.2.1.102] that hydrolyzes the 1,4- β -D-galactosidic linkages in blood groups A and B and the endohydrolase [EC 3.2.1.103] that acts on the 1,4- β -D-galactosidic linkages in keratan sulfate.

R. J. Staneloni & L. F. Leloir (1982) *Crit. Rev. Biochem.* **12**, 289.
H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

ENDONUCLEASE

The following reviews describe the molecular and physical properties of this broad class of enzymes that catalyze the endohydrolysis of deoxyribonucleic acids and ribonucleic acids. The class includes deoxyribonuclease II [EC 3.1.22.1], *Aspergillus* deoxyribonuclease K₁ [EC 3.1.22.2], deoxyribonuclease V [EC 3.1.22.3], crossover junction endoribonuclease [EC 3.1.22.4], and deoxyribonuclease X [EC 3.1.22.5]. **See also** *Deoxyribonucleases*; *Restriction Enzymes*; *Ribonucleases*

N. H. Williams (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 543.
V. E. Anderson (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 115.
A. S. Bhagwat (1992) *Meth. Enzymol.* **216**, 199.
I. R. Lehman (1981) *The Enzymes*, 3rd ed., **14**, 193.
E. C. Friedberg, T. Bonura, E. H. Radany & J. D. Love (1981) *The Enzymes*, 3rd ed., **14**, 251.
L. Grossman, A. Braun, R. Feldberg & I. Mahler (1975) *Ann. Rev. Biochem.* **44**, 19.

ENDONUCLEASE III

This enzyme (also known as DNA-(apurinic or apyrimidinic site) lyase, AP endonuclease class I, *E. coli* endonuclease III, Phage-T4 UV endonuclease, and *Micrococcus luteus* UV endonuclease) catalyzes the cleavage of the phosphodiester bond in a lyase-type reaction, not hydro-

lysis. This action leaves a 3'-terminal unsaturated sugar and a product with a terminal 5'-phosphate. This family of enzymes was previously listed under EC 3.1.25.2.

B. Demple & L. Harrison (1994) *Ann. Rev. Biochem.* **63**, 915.
E. C. Friedberg, T. Bonura, E. H. Radany & J. D. Love (1981) *The Enzymes*, 3rd ed., **14**, 251.

ENDONUCLEASE IV

This enzyme [EC 3.1.21.2], also known as endodeoxyribonuclease IV (phage T4-induced), catalyzes the endonucleolytic cleavage of DNA to 5'-phosphooligonucleotide end products. The enzyme exhibits a preference for single-stranded DNA.

B. Demple & L. Harrison (1994) *Ann. Rev. Biochem.* **63**, 915.
E. C. Friedberg, T. Bonura, E. H. Radany & J. D. Love (1981) *The Enzymes*, 3rd ed., **14**, 251.

ENDONUCLEASE V (PYRIMIDINE DIMER)

The enzyme [EC 3.1.25.1] acts on a damaged strand (at a position that is 5' from the damaged site) and catalyzes the endonucleolytic cleavage near pyrimidine dimers to products with 5'-phosphate.

A. Sancar & G. B. Sancar (1988) *Ann. Rev. Biochem.* **57**, 29.
T. Lindahl (1982) *Ann. Rev. Biochem.* **51**, 61.
E. C. Friedberg, T. Bonura, E. H. Radany & J. D. Love (1981) *The Enzymes*, 3rd ed., **14**, 251.

ENDONUCLEASE VII

This apurinic/apyrimidinic endonuclease from *E. coli* catalyzes the hydrolysis of single-stranded polydeoxyribonucleotides and DNA that contains depyrimidinic sites.

E. C. Friedberg, T. Bonura, E. H. Radany & J. D. Love (1981) *The Enzymes*, 3rd ed., **14**, 251.

ENDO-OLIGOPEPTIDASE A

This zinc-dependent enzyme [EC 3.4.24.15], also referred to as thimet oligopeptidase and soluble metalloendopeptidase, catalyzes the hydrolysis of peptide bonds with a preferential cleavage at positions with hydrophobic residues at P1, P2, and P3' and a small amino acid residue at P1'. Substrates for this enzyme contain five to fifteen amino acid residues.

E. Shaw (1990) *Adv. Enzymol.* **63**, 271.

ENDOPEPTIDASE Clp

This enzyme [EC 3.4.21.92], also referred to as endopeptidase Ti and caseinolytic protease, catalyzes the hydroly-

sis of peptide bonds in small peptides and proteins in the presence of ATP and magnesium ion. The protein usually used in assays of this enzyme is α -casein. If ATP is absent, only pentapeptides or smaller are hydrolyzed.

M. R. Maurizi, M. W. Thompson, S. K. Singh & S.-H. Kim (1994) *Meth. Enzymol.* **244**, 314.

ENDOPOLYGALACTURONIDASE

This cell wall hydrolase catalyzes the hydrolysis of α -(1 $\rightarrow$ 4)-galacturonide linkages.

R. L. Fischer & A. B. Bennett (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 675.

ENDOPOLYPHOSPHATASE

This enzyme [EC 3.6.1.10] (also known as polyphosphate depolymerase, metaphosphatase, and polyphosphatase) catalyzes the hydrolysis of polyphosphate to yield oligophosphate products containing four or five phosphate residues.

H. G. Wood & J. E. Clark (1988) *Ann. Rev. Biochem.* **57**, 235.

ENDOTHERMIC PROCESS

A change in a system or chemical reaction for which there is an absorption of heat (*i.e.*, the process requires heat to proceed). In such systems, ΔH is a positive value (where H is the enthalpy). **See also** *Enthalpy*; *Exothermic*; *Endogenic*; *Endoergic*

ENDOTHIAPEPSIN

This aspartic proteinase [EC 3.4.23.22], from the ascomycete *Endothia parasitica*, catalyzes the hydrolysis of proteins with broad specificity similar to that of pepsin A, with preferential action on substrates containing hydrophobic residues at P1 and P1'.

D. R. Davies (1990) *Ann. Rev. Biophys. Biophys. Chem.* **19**, 189.

END PRODUCT

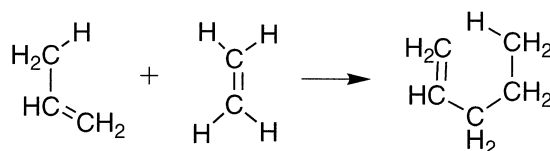
The final product(s) generated by a sequence of enzyme-catalyzed steps within a metabolic pathway. End products frequently act as potent inhibitors of the first committed step in a metabolic pathway leading to their synthesis. End products are also known to accelerate the rate of another metabolic pathway leading to a different set of end products. The highly coordinated biosynthesis of adenine and guanine nucleotides provides examples of negative and positive effects of end products of each pathway.

END-TO-END ANNEALING

The reattachment of two supramolecular polymers, such as that observed when a high concentration of sheared microtubules is permitted to incubate *in vitro*. End-to-end annealing of microtubules is an unlikely process in living cells, because (a) the viscosity of the cytoplasm is apt to reduce greatly the tumbling of these polymers and (b) the so-called minus ends of the microtubules are usually firmly attached to microtubule-organizing centers.

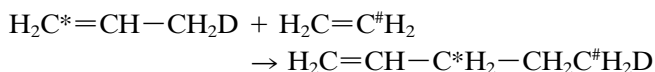
ENE REACTION

A reaction occurring between a molecular entity possessing both a double bond and an allylic hydrogen (known as the ene) and a molecule with a multiple bond (or enophile).



Ene Reaction of Propene with Ethene

In an ene reaction, there is a transfer of the allylic hydrogen and subsequent reorganization of the bonding. An example of an ene reaction is the addition of ethene (the enophile) to propene (the ene), thereby producing 1-pentene:



Superscript symbols are provided to facilitate an understanding of the reaction. The reverse of an ene reaction is often referred to as the retro-ene reaction.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

ENERGY BARRIER

1. In a reaction, the threshold energy that must be exceeded for molecules to become activated toward chemical reaction. 2. In conformational isomerizations, the energy needed to interconvert stable conformers. 3. More generally, the energy needed to transform, activate, or interconvert a particular molecular entity(ies).

ENERGY CHARGE (or, ADENYLATE ENERGY CHARGE)

An index of the phosphoanhydride (*i.e.*, P—O—P) bond content of the adenine nucleotides of a cell, based on a hypothetical model¹⁻³ that attempts to explain the metabolic basis for control of ATP utilization and regeneration. Later studies^{4,5} demonstrated that the energy charge model is overly simplistic and that its principles are unlikely to constitute a useful model for the control of energy metabolism within biological systems.

The energy charge (or adenylate energy charge) is estimated by substituting the measured ATP, ADP, and AMP concentrations into the following expression:

$$\text{Energy Charge} = \frac{[\text{ATP}] + 0.5[\text{ADP}]}{[\text{ATP}] + [\text{ADP}] + [\text{AMP}]}$$

The energy charge quotient has a value of unity (or, 1.00) when only ATP is present and a value of zero when only AMP is present. Thus, the adenine nucleotide system is said to be fully charged at EC = 1 and fully discharged at EC = 0. At intermediate values, the adenine nucleotides are interconverted by adenylate kinase, and their concentrations are constrained by the adenylate kinase mass action ratio:

$$K_{\text{eq}} = [\text{ADP}]^2/([\text{ATP}][\text{AMP}])$$

The adenylate energy charge model asserts that ATP-utilizing enzymes (called U-systems) and ATP-regenerating enzymes (called R-systems) constitute a homeostatic mechanism for buffering the energy charge in a range of 0.75–0.85. Accordingly, a drop in the cellular energy charge would be expected to reduce the velocity of ATP-utilizing reactions, while simultaneously stimulating ATP-replenishing processes.

The model has numerous short-comings^{4,5}. Chiefly, it fails to account for the fact that U- and R-enzymes frequently have other phosphoryl-acceptor and -donor substrates, and the model does not explain how fluctuations in these cosubstrate concentrations may alter the so-called responsiveness of enzymes. Moreover, the concentration of orthophosphate is wholly omitted from the model, despite the fact that key regulatory steps in ATP utilization and regeneration are exquisitely sensitive to changes in orthophosphate concentration. The energy-charge model also fails to consider magnesium ion com-

plexion by nucleotides or the effects of adenine nucleotide compartmentation in living cells. Moreover, many signal transduction processes involve U-systems (*e.g.*, ATP-dependent protein kinases) to stimulate enzymes in other so-called R-systems by downstream activation of ATP-regenerating enzymes. This would suggest that the initial hormone- or effector-stimulated kinase is itself unlikely to obey the predicted energy charge behavior of a U-system.

In view of these serious flaws, as well as other technical difficulties^{4,6}, the energy charge model is misleadingly simplistic in attempting to interpret the intracellular behavior of regulatory enzymes utilizing or replenishing ATP. In limited circumstances and with the understanding that the energy-charge hypothesis is flawed, the EC quotient may still be helpful in discussing the broad homeostatic features of ATP-utilizing and ATP-replenishing reaction.

¹D. E. Atkinson (1968) *Biochemistry* **11**, 4030.

²D. Atkinson (1977) *Trends Biochem. Sci.* **2**, N 198.

³D. E. Atkinson & A. G. Chapman (1979) *Meth. Enzymol.* **55**, 229.

⁴D. L. Purich & H. J. Fromm (1972) *J. Biol. Chem.* **247**, 249.

⁵D. L. Purich & H. J. Fromm (1973) *J. Biol. Chem.* **248**, 461.

⁶H. Fromm (1977) *Trends Biochem. Sci.* **2**, N 198.

ENERGY OF ACTIVATION

Any energy change (from the ground state to the transition state in a chemical reaction) that accounts (a) for the presence of a reaction barrier and (b) for the temperature dependence of a rate constant. This quantity is symbolized by E_a and may also be referred to as the Arrhenius energy of activation and activation energy.

An elementary reaction where radicals (*e.g.*, $A\cdot + B\cdot \rightarrow A-B$) recombine may not involve an energy barrier and would therefore have no energy of activation.

Occasionally, a negative activation energy is reported. For example, an E_a value of $-13.0 \text{ kJ}\cdot\text{mol}^{-1}$ was obtained for tRNA^{Phe} binding to the E-site of poly(U)programmed ribosomes containing unlabeled tRNA^{Phe} at its P-site¹. In such circumstances, one must be especially wary, because there are several reasons why an erroneously negative E_a value is obtained:

(a) The actual experimental design was flawed, and the rate parameter was not correctly measured. If the true

activation energy is a very small number, experimental inaccuracy or imprecision may be especially problematic. One should conduct an error analysis that appropriately evaluates the overall effect of imprecise measurement on the derived parameter.

(b) The observed process is not governed by a simple bimolecular reaction and involves at least one endothermic equilibration with a large $-\Delta H$ value. An example is the reaction of nitric oxide with molecular oxygen ($2 \cdot \text{NO} + \text{O}_2 \rightarrow 2 \cdot \text{NO}_2$) which was originally described by the third-order rate law: $(-d[\cdot \text{NO}]/dt = 2k_{\text{obs}} [\cdot \text{NO}]^2[\text{O}_2])^{2,3}$. The actual process is more likely to involve a fast equilibrium that is followed by a slower bimolecular step ($\cdot \text{NO} + \text{O}_2 \rightleftharpoons \text{NO}_3$; $\text{NO}_3 + \cdot \text{NO} \rightarrow 2 \cdot \text{NO}_2$)^{4,5}. The observed rate constant, k_{obs} , is the product of a bimolecular rate constant, k , for the second step and an equilibrium constant, K , for the formation of NO_3 . Thus, the observed activation energy, E_{obs} , is the sum of the E_a for the second step and ΔH for the equilibration step (*i.e.*, $E_{\text{obs}} = E_a + \Delta H$). Because the equilibration step has a particularly negative ΔH value, a negative activation energy is observed⁵.

(c) A much less likely explanation is that a process thought to be bimolecular is actually trimolecular. Application of transition-state theory to a trimolecular reaction is known to result in an abnormal activation energy⁶. **See Arrhenius Equation; Transition-State Theory; Q_{10}**

¹S. Schilling-Bartetzko, A. Bartetzko & K. H. Nierhaus (1992) *J. Biol. Chem.* **267**, 4704.

²M. Boderstein (1918) *Zeit. Elektrochem.* **24**, 183.

³M. Boderstein (1922) *Zeit. Phys. Chem.* **100**, 68.

⁴R. G. Pearson (1976) *Symmetry Rules for Chemical Reactions*, Wiley-Interscience, New York.

⁵J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, 3rd ed., Wiley, New York.

⁶H. Gershinowitz & H. Eyring (1935) *J. Amer. Chem. Soc.* **57**, 985.

ENERGY TRANSFER

1. The process by which a molecular entity absorbs light and a phenomenon originates from the excited state of another molecular entity. 2. The process by which an excited state of one molecular entity is deactivated to a lower-lying state upon the transfer of energy to a second molecular entity, thus raising that new entity to a higher energy state. If the two entities involved are different moieties of the same molecular entity, the process is referred to as an intramolecular energy transfer. **See**

also Förster Resonance Excitation Transfer (FRET); Radiative Energy Transfer; Spectral Overlap; Fluorescence Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

ENHANCER

Any DNA sequence that binds protein factors and thereby stimulates transcription. This stimulatory effect may occur over an appreciable distance from the enhancer's site. Enhancers act in either strand orientation, and they can be found either upstream or downstream with respect to the promoter.

ENOLASE

Enolase [EC 4.2.1.11] catalyzes the interconversion of 2-phosphoglycerate and phosphoenolpyruvate in a step-wise mechanism. Proton abstraction and at least one additional step (*e.g.*, hydroxide loss or product release) may limit the overall reaction rate.

R. Jaenicke, H. Schurig, N. Beaucamp & R. Ostendorp (1996) *Adv. Protein Chem.* **48**, 181.

J. J. Villafranca & T. Nowak (1992) *The Enzymes*, 3rd ed., **20**, 63.

L. Leboida & B. Stec (1991) *Biochemistry* **30**, 2817.

F. Wold (1971) *The Enzymes*, 3rd ed., **5**, 499.

ENOLASE SUPERFAMILY

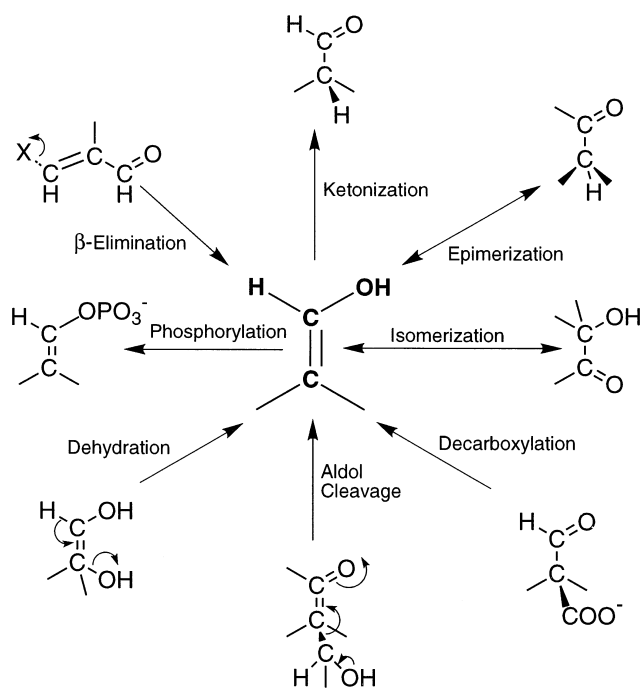
Members of this superfamily catalyze at least eleven different chemical reactions¹, including racemization, epimerization, and both *syn* and *anti* β -elimination reactions involving water, ammonia, or an intramolecular carboxylate group as a leaving group. Babbitt and Gerlt¹ point out that reactions catalyzed by members of the enolase superfamily are initiated by a common partial reaction, metal-assisted, general base-catalyzed abstraction of the α -proton of a carboxylate anion to generate a stabilized enolate anion intermediate. They further state that the fate of the intermediate (protonation in the case of racemization and epimerization reactions and vinylogous β -elimination in the others) must be determined by the different functional groups that "surround" the intermediate in the active site.

¹P. C. Babbitt & J. A. Gerlt (1997) *J. Biol. Chem.* **272**, 30591.

ENOLIZATION

Any process that produces an enol, including the formation of enolate anions. Enols (*i.e.*, entities containing the moiety $\text{HO}-\text{C}(\text{R}_1)=\text{C}(\text{R}_2\text{R}_3)$) appear as intermediates in a wide variety of enzyme-catalyzed reactions, and Rose¹ has presented the following diagram to describe

the various categories of stepwise processes involving enol intermediates.



Enzyme Reaction Types Involving Enol Intermediates

A number of methods¹ can assist in identifying and characterizing enol intermediates (as well as enamine and carbanion intermediates) in enzyme-catalyzed reactions. These include: (1) proton isotope exchange; (2) oxidation of the intermediate; (3) coupled elimination; (4) spectrophotometric methods; (5) use of transition-state inhibitors; (6) use of suicide inhibitors; (7) isolation of the enol; and (8) destructive analysis.

¹I. A. Rose (1982) *Meth. Enzymol.* **87**, 84.

5-EnolPYRUVYLSHIKIMATE-3-PHOSPHATE SYNTHASE

This enzyme [EC 2.5.1.19] catalyzes the reaction of phosphoenolpyruvate and 3-phosphoshikimate, yielding orthophosphate and 5-*O*-(1-carboxyvinyl)-3-phosphoshikimate.

J. S. Holt, S. B. Powles & J. A. M. Holtum (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 203.

K. A. Johnson (1992) *The Enzymes*, 3rd ed., **20**, 1.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

ENOYL-ACP REDUCTASE

This family of enzymes catalyzes the reaction of an acyl-[acyl-carrier-protein] with NAD⁺ [EC 1.3.1.9] or with

NADP⁺ [EC 1.3.1.10] to produce 2,3-dehydroacyl-[acyl-carrier-protein] and NADH or NADPH. **See also** *Fatty Acid Synthetase*

J. Browse & C. Somerville (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 467.

S. J. Wakil & J. K. Stoops (1983) *The Enzymes*, 3rd ed., **16**, 3.

ENOYL-CoA HYDRATASE or (CROTONASE)

This enzyme catalyzes the addition of water to *trans*-2(or 3)-enoyl-CoA to yield (3*S*)-3-hydroxyacyl-CoA. The enzyme will act on *cis* derivatives, generating the (3*R*)-3-hydroxyacyl-CoA product.

The following is a classical review on the molecular and physical properties of this enzyme [EC 4.2.1.17].

R. L. Hill & J. W. Teipel (1971) *The Enzymes*, 3rd ed., **5**, 539.

ENOILPYRUVATE

Although enolpyruvate has been long regarded as a pyruvate kinase reaction intermediate, direct chemical trapping and/or isolation remained elusive. Seeholzer *et al.*¹ recently developed a method that distinguishes enolpyruvate, based on its reaction with bromine in acid, followed by derivatization of the bromopyruvate with thionitrobenzoate. The thioether derivative (10 pmol) could be reliably quantitated, allowing their determination of the internal equilibria, including the E·ATP·enolpyruvate complex. Phosphoenolpyruvate also reacts with bromine to yield bromopyruvate, and to quantitate enolpyruvate specifically in a background of phosphoenolpyruvate, the authors took advantage of the much greater acid stability of phosphoenolpyruvate.

¹S. H. Seeholzer, A. Jaworowski & I. A. Rose (1991) *Biochemistry* **30**, 727.

ENTATIC STATE

A hypothetical conformational state¹ defined by a geometrically and electronically strained site within an enzyme thought to facilitate the conversion of an enzyme-substrate complex to the transition state. Vallee and Williams¹ defined the entatic state as an abnormal condition of localized strain transmitted by relief of compression or other steric clashes elsewhere in the enzyme. They suggested that catalytic rate enhancements arise from the heightened reactivity of catalytic group(s) that have experienced unimolecular activation. Williams²

presented additional ideas on how metalloenzymes may utilize such a mechanism to assist in the conversion of enzyme-bound substrate to the reaction's transition state.

¹B. L. Vallee & R. J. P. Williams (1968) *Proc. Natl. Acad. Sci. U.S.A.* **59**, 498.

²R. J. P. Williams (1971) *Cold Spring Harbor Symp. Quant. Biol.* **36**, 53.

ENTERING GROUP

The atom, ion, or group of a molecular entity that approaches and forms a covalent or coordinate covalent bond with the substrate during a reaction. Any attacking nucleophile is an entering group.

ENTEROPEPTIDASE

This enzyme [EC 3.4.21.9], also known as enterokinase, activates trypsinogen by catalyzing the cleavage of Lys⁶–Ile⁷ bond in trypsinogen.

J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.

ENTEROTOXIN

There are a large number of proteins that share this designation. Perhaps the best known is the heat-labile enterotoxin from *E. coli* that catalyzes the ADP-ribosylation of a number of proteins.

L. Passador & W. Igilewski (1994) *Meth. Enzymol.* **235**, 617.

C.-Y. Lai (1986) *Adv. Enzymol.* **58**, 99.

ENTHALPY

A thermodynamic quantity, symbolized by H , and equal to $U + PV$, where U is the thermal energy, P is pressure, and V is volume. If the pressure is constant and no other work is done other than $P \Delta V$ work, then the increase in enthalpy is equal to the heat absorbed at constant pressure ($\Delta H = q_P$). If $\Delta H < 0$, the process is regarded as exothermic; if $\Delta H > 0$, the process is regarded as endothermic. **See also** *Innate Thermodynamic Quantities*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Antibody-antigen interactions, **247**, 295-296; binding enthalpy [ligand concentration effect, **259**, 212; limitations on interpretation, **259**, 211-212; measurement by calorimetry, **259**, 196, 209-210; pH effect, **259**, 212; protein isomerization effects, **259**, 214-218; range and ligand interactions, **259**, 212; temperature effect, **259**, 212, 214-221; ternary complex formation, **259**, 213, 217-218]; calculation, **247**, 290-291; coupled equilibria, driving of processes, **259**, 708-709; differential scanning calorimetric determination, **240**, 504, 506-507, 513-514, 531, 534-537; enthalpy-entropy compensation in titration calorimetry, **247**, 300-302; errors in isothermal equilibria studies, **259**, 629; ionization enthalpies of buff-

ers, **240**, 518; motive enthalpy, temperature dependence, **259**, 641-642; protein analysis by differential scanning calorimetry, **259**, 155-156, 168; protonation effects, **240**, 517-518; spectroelectrochemical determination, **246**, 711; van't Hoff analysis [oligonucleotide (differential scanning calorimetry, **259**, 235-237; melting curves, **259**, 225-234); protein, **259**, 149-150, 196, 618-620]; van't Hoff enthalpy, **240**, 507-508, 528, 531.

ENTHALPY-DRIVEN REACTION

Any reaction or process for which the enthalpy change is large and negative, such that the ΔH term greatly dominates the $-T\Delta S$ term in the Gibbs equation ($\Delta G = \Delta H - T\Delta S$), and acts thereby as the thermodynamic driving force for that reaction or process.

ENTHALPY OF ACTIVATION

The standard enthalpy difference between reactant(s) of a reaction and the activated complex in the transition state at the same temperature and pressure. It is symbolized by $\Delta H^\ddagger$ and is equal to $(E_a - RT)$, where E_a is the energy of activation, R is the molar gas constant, and T is the absolute temperature (provided that all non-first-order rate constants are expressed in temperature-independent concentration units, such as molarity, and are measured at a fixed temperature and pressure). Formally, this quantity is the enthalpy of activation at constant pressure. **See** *Transition-State Theory (Thermodynamics)*; *Transition-State Theory*; *Gibbs Free Energy of Activation*; *Entropy of Activation*; *Volume of Activation*

ENTROPY

A fundamental thermodynamic state function (symbolized by S), and as such, not dependent on the path by which a particular state is reached. For a reversible process, the differential change in entropy, dS , is equal to the amount of energy absorbed by the system, dq , divided by the absolute temperature, T . Thus,

$$\Delta S = \int_A^B dq/T$$

The SI units for entropy is joules per kelvin. If irreversibility occurs at any point in the process under consideration, then $dS > dq/T$. The entropy of an isolated system undergoing an irreversible process always increases (in other words, for any isolated system $\Delta S \geq 0$).

In statistical mechanics¹, entropy of a system with respect to a particular state is related to the probability, W , of a system being in that state: *i.e.*, $S = k_B \ln W + b$ where b is a constant and k_B is the Boltzmann constant. Hence, entropy represents the degree of disorder within a sys-

tem. **See also** *Gibbs Free Energy; Enthalpy; Thermodynamics*

¹R. Dickerson (1969) *Molecular Thermodynamics*, p. 452, W. A. Benjamin, New York. (This is a remarkably lucid treatment of thermodynamics.)

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Amino acid side chain contribution, **240**, 516; approximate entropy [biological application, **240**, 71; calculation, **240**, 74-75; computer implementation, **240**, 78, 88; data points, **240**, 75-76, 88; data processing, **240**, 73; development, **240**, 71; family of statistics, **240**, 76; filters, **240**, 74-77, 88; implicit noise filtering, **240**, 79; length of compared runs, **240**, 74-75; model independence, **240**, 79; noise effects, **240**, 82, 88; outlier sensitivity, **240**, 78-79, 88; quantification of data regularity, **240**, 70, 74-75, 82; relationship to other approaches (feature recognition algorithms, **240**, 80-81; Kolmogorov-Sinai entropy, **240**, 71, 76, 81-83; moment statistics, **240**, 80; phase space plots, **240**, 83-84; power spectra, **240**, 83-84); relative consistency, **240**, 77; statistical validity, **240**, 79-80]; Kolmogorov-Sinai entropy [comparison with approximate entropy, **240**, 71, 76, 81-83; limitations, **240**, 81; noise effects, **240**, 81]; coupled equilibria, driving of processes, **259**, 708-709; differential scanning calorimetric determination, **240**, 506-507, 514, 516-517, 535-536, 539; errors in isothermal equilibria studies, **259**, 629, 639-640, 709; gas, **259**, 562-563; ligand binding-induced change, **240**, 93; maximum entropy method [applications (anisotropy decay, **240**, 301-310; distribution of protein conformational states, **240**, 269-273; frequency domain fluorescence, **240**, 273-286; time-correlated single-photon counting, **240**, 290-310); classical approach (automatic noise scaling, **240**, 267-268; formula, **240**, 266-267; inference about reconstruction, **240**, 267; positive-negative distributions, **240**, 268); computer implementation, **240**, 268-269, 290; derivation from Bayesian probability theory, **240**, 264-265; historic maximum entropy, **240**, 269-270; Laplace transform handling, **240**, 262, 287; in multidimensional NMR, **239**, 257-259, 265; properties, **240**, 310; roots, **240**, 263-264]; mixing, **259**, 709, 715; motive, crystalline solids, **259**, 639-641; peptide backbone contribution, **240**, 516-517; protein analysis by differential scanning calorimetry, **259**, 155-156, 168; range and ligand interactions, **259**, 212; single-particle translational partition function, **259**, 709; solvent reorganization, **259**, 561-564; spectroelectrochemical determination, **246**, 710-711; thermal [crystalline solids, **259**, 639-641; derivation of change, **259**, 631; water, **259**, 561-562].

ENTROPY AND ENTHALPY OF ACTIVATION (Enzymatic)

Determination of values for $\Delta H^\ddagger$ and $\Delta S^\ddagger$ allows one the opportunity to further understand the action of enzyme catalysis, provided that these quantities can be unambiguously evaluated. High catalytic efficiency is often attributed to $\Delta S^\ddagger$ effects. The magnitude and sign of $\Delta S^\ddagger$ can provide information on the nature of the activated complexes. Both positive and negative entropies of activation have been reported: as pointed out by Laidler^{1,2}, Fromm³, and Jencks⁴, many factors can affect $\Delta S^\ddagger$ values, including a reversible conformational change in the enzyme during or after substrate binding. In addition, whenever a substrate interacts with an enzyme, both

substrate and enzyme are “lost” to the solvent as separate solute species. Laidler^{2,5} referred to this contribution as the “unmixing” of the solute species with the solvent and affecting $\Delta S^\ddagger$, whereas Jencks⁴ attributed this to the entropy of dilution. Other factors affecting the entropy of activation include the orientation of the substrates in the transition state relative to the unassociated molecules, the relative hydrophobicity of both substrate and protein, the net change in number of hydrogen bonds between substrate and enzyme, the loss of translational and possibly rotational degrees of freedom upon the binding of the substrate^{6,7} (if the substrate has lost degrees of freedom, then the enzyme-substrate complex must have gained degrees of freedom⁸), the release of water molecules from the enzymes active site upon binding of the substrate, the relative orientation of solvent molecules in the transition state and in the solvent cage around the substrate molecule, the molecularity of the reaction³ (the net volume of activation is expected to be negative for a bimolecular reaction), and electrostatic effects.

The physical meaning for the entropy of activation is often overinterpreted, and care should be exercised in analyzing multistep processes that may not attain thermodynamic equilibrium during each catalytic reaction cycle. If there is a temperature-dependent change in the rate-determining step for a reaction, reliable interpretations of $\Delta S^\ddagger$ may be exceedingly difficult to achieve. This will also be true if there is a change in the heat capacity of the solvent or a temperature-dependent change in protein structure.

Hammes⁹ suggested that “energy compensation” has a role in enzyme-catalyzed reactions, such that there is synergism between bond cleavage and bond formation. **See** *Transition-State Theory; Transition-State Theory (Thermodynamics); Transition-State Theory in Solutions; Entropy of Activation; Volume of Activation*

¹K. J. Laidler & P. S. Bunting (1973) *The Chemical Kinetics of Enzyme Action*, 2nd ed., Clarendon Press, Oxford.

²K. J. Laidler (1955) *Discussion Faraday Soc.* **20**, 83.

³H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

⁴W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York].

⁵M. L. Barnard & K. J. Laidler (1952) *J. Amer. Chem. Soc.* **74**, 6099.

⁶P. D. Bartlett & R. R. Hiatt (1958) *J. Amer. Chem. Soc.* **80**, 1398.

⁷E. G. Foster, A. C. Pope & F. Daniels (1947) *J. Amer. Chem. Soc.* **69**, 1893.

⁸F. H. Westheimer (1962) *Adv. Enzymol.* **24**, 441.

⁹G. G. Hammes (1964) *Nature* **204**, 342.

ENTROPY OF ACTIVATION

The standard entropy difference between the reactant(s) of a reaction and the activated complex of the transition state, at the same temperature and pressure. Entropy of activation is symbolized by either $\Delta S^\ddagger$ or $\Delta^\ddagger S^\ominus$ and is equal to $(\Delta H^\ddagger - \Delta G^\ddagger)/T$ where $\Delta H^\ddagger$ is the enthalpy of activation, $\Delta G^\ddagger$ is the Gibbs free energy of activation, and T is the absolute temperature (provided that all rate constants other than first-order are expressed in temperature-independent concentration units such as molarity). Technically, this quantity is the entropy of activation at constant pressure, and from this value, the entropy of activation at constant volume can be deduced.

See *Transition-State Theory (Thermodynamics)*; *Gibbs Free Energy of Activation*; *Enthalpy of Activation*; *Volume of Activation*; *Entropy and Enthalpy of Activation (Enzymatic)*

ENTROPY TRAP MODEL

The concept that catalytic rate enhancement stems from an enzyme's ability to freeze the substrate into a particularly reactive configuration, which in the absence of an enzyme would only be present at vanishingly low concentration. In this model, reacting groups within an enzyme's active site are brought together in precisely the correct positions that maximize reactivity. The ability to achieve such high precision in orienting reactive groups occurs at a price, namely a very large loss of entropy: hence the name "entropy trap". Jencks¹ has estimated that rate enhancements of 10^8 to 10^{12} might be realistically accessible for two reactants at 0.01 M. Such precise binding requires restricted motion of reactant on the enzyme, a thermodynamically unfavored circumstance that most probably reduces the observed affinity of an enzyme for its substrate. Jencks has also emphasized that this reduced affinity stems from strain needed to force the substrate to bind at an active site, which is itself designed to be complementary to the transition state, and not the substrate's ground state.

¹W. P. Jencks (1975) *Adv. Enzymol.* **43**, 219.

ENZYME

Any biological substance acting as a catalyst. This expanded definition now includes proteins classically de-

fined as enzymes, other proteins that facilitate otherwise rate-limiting exchanges of macromolecule-bound effector molecules, catalytic antibodies designed in the laboratory, as well as ribozymes which are catalysts made up of ribonucleic acid.

The Enzyme Commission of the International Union of Biochemistry devised a scheme for enzyme classification based upon the types of reactions catalyzed. There are six main classes: the oxidoreductases (all enzymes that catalyze oxidation-reduction reactions, such as dehydrogenases, reductases, oxidases, transhydrogenases, peroxidases, desaturases, dismutases, and a few synthases); the transferases (those enzymes that catalyze the transfer of some moiety, such as a methyl or acetyl group, from one compound to another); the hydrolases which catalyze the hydrolytic cleavage of a bond; lyases, those enzymes that catalyze the cleavage of a bond by elimination, producing double bonds, or, conversely, adding groups to double bonds; the isomerases which catalyze configurational or structural changes within a molecule; and the ligases (and synthetases) that catalyze bond formation with the concomitant hydrolysis of a pyrophosphate bond in ATP or similar molecule. **See** *Enzyme Commission Nomenclature*

ENZYME CASCADE KINETICS

The rate behavior of linear and multicyclic systems of enzymes arranged such that each enzyme in a step or cycle is itself a substrate for an enzyme-catalyzed covalent modification reaction. The systematic investigation of the kinetics of these cascade systems promises to provide insights about the design features accounting for the signal amplification and information processing achieved by such hierarchies of enzymes with other enzymes serving as their substrates.

Covalent interconversion of enzymes is well established as a fundamental theme in metabolic regulation¹. The prototypic reversible interconverting systems include the sequence of phosphorylation/dephosphorylation steps in the activation of mammalian glycogen phosphorylase and pyruvate dehydrogenase² as well as the nucleotidylation/denucleotidylation using UTP and ATP in the bacterial glutamine synthetase cascade³ (see Fig. 1.).

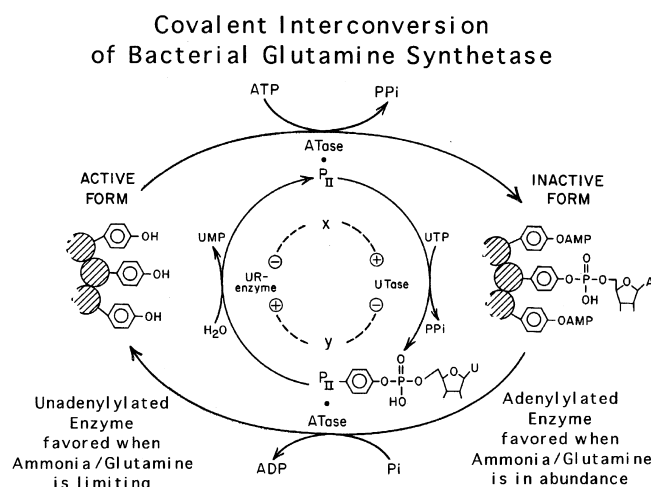


Figure 1. The *Escherichia coli* glutamine synthetase (GS) cascade, indicating the metabolic regulation of glutamine synthesis by multiple effector molecules that directly reflect the status of nitrogen metabolism in the bacterium. In this scheme, GS is the metabolically active form of glutamine synthetase, and GS-AMP is the metabolically inactive adenylylated enzyme. The same converting enzyme, designated ATase for adenylyltransferase, catalyzes adenylation and deadenylation to form GS-AMP and GS, respectively. As shown in the diagram, however, those metabolites serving as positive effectors for adenylation behave as negative effectors for deadenylation, and vice versa. Even more striking is the involvement of a bifunctional enzyme UR/UT which serves to release UMP from P_{II} -UMP or to uridylate tyrosinyl residues on P_{II} . The dotted lines from P_{II} and P_{II} -UMP indicate the ability of these forms of the regulatory protein to direct the ATase's action on glutamine synthetase.

Unidirectional activation of the intrinsic and extrinsic pathways of the blood clotting cascade exemplify a linear sequence of proteolysis steps that amplify an initial signal into a massive, well-controlled biological response. In this respect, enzyme cascades are not merely amplification systems; the multiplicity of enzymes involved in a cascade allows for a multiplicity of effector binding sites, thereby enhancing a cascade's responsiveness to changes in a regulatory input signal³. This is illustrated by the numerous calcium ion binding sites in the blood clotting cascade, and a small change in the concentration of this divalent cation elicits a major change in the rapidity with which fibrin is generated in the last step of this sequence of converting proteolytic zymogens into their catalytically active forms. A schematic representation of a generalized multicyclic cascade is shown in Fig. 2, where the increasing number of vertical arrows indicate an increase in catalytic activity of the enzymes.

Cascade of Enzyme-Catalyzed Interconversion Reactions

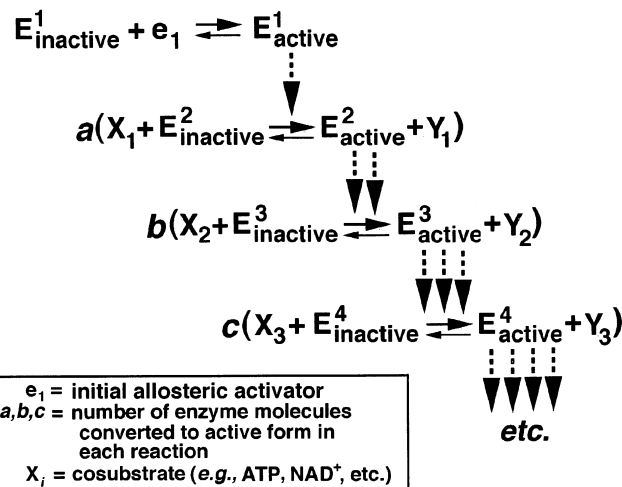


Figure 2. A multistep enzyme interconversion cascade illustrating the sequential modification and conversion of target enzymes from one state of activity to another. Although written here as a cascade of increasing catalytic activity, enzyme-catalyzed covalent modification can either activate or inhibit target enzymes, depending on the particular system under study.

Accordingly, the kinetic properties of cascades involving “converter” and “target” enzymes⁵ becomes of paramount significance in any consideration of covalently interconverting enzyme systems. Stadtman and Chock⁴⁻⁶ led the way in developing a steady-state treatment for analyzing such cascades. They have offered the following emerging concepts regarding the behavior of enzyme cascades:

1. Covalent interconversion of an enzyme is a dynamic process, and the specific activity of the target enzyme is determined by coupling of two opposing, dynamic cascades. This results in a steady-state distribution of active and inactive species of the interconvertible enzyme. The target enzyme's activity smoothly and continuously responds to feedback signals that alter the efficiency of the enzymes catalyzing the interconversion steps as well as the ability of the target enzyme to be an effective substrate for the interconverting enzymes.
2. A minimum of three enzymes are typically involved in each interconvertible enzyme cycle that can integrate many allosteric effectors into a single output, the specific activity of the target enzyme in the cascades.

3. Cyclic cascades can achieve great signal amplification, and interconvertible enzymes can respond to effectors at concentrations considerably below their respective dissociation constants.

4. Cyclic cascades can propagate a sigmoidal dose-response curve showing how an increasing concentration of allosteric effector manifests itself in activity changes of the target enzyme.

Future kinetic work on the time evolution of cascades, rather than steady-state rate behavior, is likely to reveal how time-dependent changes in the responsiveness are related to the kinetic properties of individual components. Presently, no multicyclic cascade has been rigorously experimentally analyzed, either by the steady-state approach or through the use of an integrated rate equation method. An effort in this direction is the recent work of Huang and Ferrell⁷ on the mitogen-activated protein kinase cascade. These investigators found that this cascade shows a sharp stimulus-response curve that may be fundamentally important in the function of those molecular switches controlling such high-level processes as mitogenesis.

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⁶P. B. Chock & E. R. Stadtman (1977) *Proc. Natl. Acad. Sci. U.S.A.* **74**, 2766.

⁷C.-Y. F. Huang & J. E. Ferrell (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 10078.

ENZYME COMMISSION NOMENCLATURE

The International Union of Biochemistry and Molecular Biology (IUBMB) system for rational naming and categorizing of enzymes into six major classes: [1] oxidoreductases which catalyze reactions involving the loss or gain of electrons by substrates; [2] transferases which catalyze the transfer of reactive groups from a donor substrate to an acceptor substrate; [3] hydrolases which catalyze the introduction of the elements of water into a substrate; [4] lyases which catalyze elimination of a group from a substrate with formation of a double bond or addition of a group to a double bond; [5] isomerases

which catalyze intramolecular rearrangements; and [6] ligases which catalyze the joining of molecules by the formation of covalent bonds.

The EC number contains four numerical elements each separated by points. For example, alcohol dehydrogenase is assigned the number EC 1.1.1.1. The first numerical element (the furthest one to the left) identifies which of the six main classes the enzyme belongs. For alcohol dehydrogenase, this class is the oxidoreductases. The second element identifies the subclass; for EC 1.1.x.x, this refers to oxidoreductases acting on the CH—OH group of donors. The third element identifies the sub-sub-class; for EC 1.1.1.x this refers to oxidoreductases that use NAD⁺ or NADP⁺ as the acceptor. The final sub-sub-sub-class is unique for that enzyme-catalyzed reaction.

1. X. X.X	OXIDOREDUCTASES (Acceptor = Oxidant; Donor = Reductant)
1. 1. X.X	Acting on the CH—OH group of donors.
1. 1. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .
1. 1. 2.X	Acceptor: a cytochrome.
1. 1. 3.X	Acceptor: oxygen.
1. 1. 4.X	Acceptor: a disulfide.
1. 1. 5.X	Acceptor: a quinone or similar compound.
1. 1.99.X	Acceptor: other.
1. 2. X.X	Acting on the aldehyde or oxo group of donors.
1. 2. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .
1. 2. 2.X	Acceptor: a cytochrome.
1. 2. 3.X	Acceptor: oxygen.
1. 2. 4.X	Acceptor: a disulfide.
1. 2. 7.X	Acceptor: an iron-sulfur protein.
1. 2.99.X	Acceptor: other.
1. 3. X.X	Acting on the CH—CH group of donors.
1. 3. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .
1. 3. 2.X	Acceptor: a cytochrome.
1. 3. 3.X	Acceptor: oxygen.
1. 3. 5.X	Acceptor: a quinone or related compound.
1. 3. 7.X	Acceptor: an iron-sulfur protein.
1. 3.99.X	Acceptor: other.
1. 4. X.X	Acting on the CH—NH ₂ group of donors.

1. 4. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .	1.11. X.X	Acting on a peroxide as acceptor (peroxidases).
1. 4. 2.X	Acceptor: a cytochrome.		
1. 4. 3.X	Acceptor: oxygen.	1.12. X.X	Acting on hydrogen as donor.
1. 4. 4.X	Acceptor: a disulfide.	1.12. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .
1. 4. 7.X	Acceptor: an iron-sulfur protein.	1.12. 2.X	Acceptor: a cytochrome.
1. 4.99.X	Acceptor: other.	1.12.99.X	Acceptor: other.
1. 5. X.X	Acting on the CH—NH group of donors.	1.13. X.X	Acting on single donors with incorporation of molecular oxygen (oxygenases).
1. 5. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .		
1. 5. 3.X	Acceptor: oxygen.		
1. 5. 4.X	Acceptor: a disulfide.	1.13.11.X	Acceptor: incorporation of two atoms of oxygen.
1. 5. 5.X	Acceptor: a quinone or similar compound.	1.13.12.X	Acceptor: incorporation of one atom of oxygen.
1. 5.99.X	Acceptor: other.		
1. 6. X.X	Acting on NADH or NADPH.	1.13.99.X	Miscellaneous (requires further characterization).
1. 6. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .		
1. 6. 2.X	Acceptor: a cytochrome.	1.14. X.X	Acting on paired donors with incorporation of molecular oxygen.
1. 6. 4.X	Acceptor: a disulfide.		
1. 6. 5.X	Acceptor: a quinone or similar compound.	1.14.11.X	Donor: 2-oxoglutarate as one donor, and incorporation of one atom each of oxygen into both donors.
1. 6. 6.X	Acceptor: a nitrogenous group.		
1. 6. 8.X	Acceptor: a flavin.		
1. 6.99.X	Acceptor: other.	1.14.12.X	Donor: NADH or NADPH, and incorporation of two atoms of oxygen into one donor.
1. 7. X.X	Acting on other nitrogenous compounds as donors.		
1. 7. 2.X	Acceptor: a cytochrome.	1.14.13.X	Donor: NADH or NADPH, and incorporation of one atom of oxygen.
1. 7. 3.X	Acceptor: oxygen.		
1. 7. 7.X	Acceptor: an iron-sulfur protein.		
1. 7.99.X	Acceptor: other.	1.14.14.X	Donor: reduced flavin or flavoprotein, and incorporation of one atom of oxygen.
1. 8. X.X	Acting on a sulfur group of donors.		
1. 8. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .		
1. 8. 2.X	Acceptor: a cytochrome.	1.14.15.X	Donor: a reduced iron-sulfur protein, and incorporation of one atom of oxygen.
1. 8. 3.X	Acceptor: oxygen.		
1. 8. 4.X	Acceptor: a disulfide.		
1. 8. 5.X	Acceptor: a quinone or similar compound.	1.14.16.X	Donor: reduced pteridine, and incorporation of one atom of oxygen.
1. 8. 7.X	Acceptor: an iron-sulfur protein.		
1. 8.99.X	Acceptor: other.	1.14.17.X	Donor: ascorbate, and incorporation of one atom of oxygen.
1. 9. X.X	Acting on a heme group of donors.		
1. 9. 3.X	Acceptor: oxygen.	1.14.18.X	Donor: another compound, and incorporation of one atom of oxygen.
1. 9. 6.X	Acceptor: a nitrogenous group.		
1. 9.99.X	Acceptor: other.		
1.10. X.X	Acting on diphenols and related substances as donors.	1.14.99.X	Miscellaneous (requires further characterization).
1.10. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .	1.15. X.X	Acting on superoxide radicals.
1.10. 2.X	Acceptor: a cytochrome.	1.16. X.X	Oxidizing metal ions.
1.10. 3.X	Acceptor: oxygen.	1.16. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .
1.10.99.X	Acceptor: other.	1.16. 3.X	Acceptor: oxygen.

1.17. X.X	Acting on —CH ₂ groups.	2. 7. 4.X	Phosphotransferases with a phosphate group as acceptor (nucleophile).
1.17. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .		
1.17. 3.X	Acceptor: oxygen.		
1.17. 4.X	Acceptor: a disulfide.	2. 7. 6.X	Diphosphotransferases.
1.17.99.X	Acceptor: other.	2. 7. 7.X	Nucleotidyltransferases.
1.18. X.X	Acting on reduced ferredoxin as donor.	2. 7. 8.X	Transferases for other substituted phosphate groups.
1.18. 1.X	Acceptor: NAD ⁺ or NADP ⁺ .		
1.18. 6.X	Acceptor: dinitrogen.	2. 7. 9.X	Phosphotransferases with paired acceptors.
1.18.99.X	Acceptor: H ⁺ .		
1.19. X.X	Acting on reduced flavodoxin as donor.	2. 8. X.X	Transferring sulfur-containing groups.
1.19. 6.X	Acceptor: dinitrogen.	2. 8. 1.X	Sulfurtransferases.
1.97. X.X	Other oxidoreductases.	2. 8. 2.X	Sulfotransferases.
		2. 8. 3.X	CoA-transferases.
2. X. X.X	TRANSFERASES.	2. 9. X.X	Transferring selenium-containing groups.
2. 1. X.X	Transferring one-carbon groups.		
2. 1. 1.X	Methyltransferases.		
2. 1. 2.X	Hydroxymethyl-, formyl- and related transferases.		
2. 1. 3.X	Carboxyl- and carbamoyltransferases.	3. X. X.X	HYDROLASES.
2. 1. 4.X	Amidinotransferases.	3. 1. X.X	Acting on ester bonds.
2. 2. X.X	Transferring aldehyde or ketone residues.	3. 1. 1.X	Carboxylic ester hydrolases.
2. 3. X.X	Acyltransferases.	3. 1. 2.X	Thiolester hydrolases.
2. 3. 1.X	Acyltransferases.	3. 1. 3.X	Phosphoric monoester hydrolases.
2. 3. 2.X	Aminoacyltransferases.	3. 1. 4.X	Phosphoric diester hydrolases.
2. 4. X.X	Glycosyltransferases.	3. 1. 5.X	Triphosphoric monoester hydrolases.
2. 4. 1.X	Hexosyltransferases.	3. 1. 6.X	Sulfuric ester hydrolases.
2. 4. 2.X	Pentosyltransferases.	3. 1. 7.X	Diphosphoric monoester hydrolases.
2. 4.99.X	Transferring other glycosyl groups.	3. 1. 8.X	Phosphoric triester hydrolases.
2. 5. X.X	Transferring alkyl or aryl groups, other than methyl groups.	3. 1.11.X	Exodeoxyribonucleases producing 5'-phosphomonoesters.
2. 6. X.X	Transferring nitrogenous groups.	3. 1.13.X	Exoribonucleases producing 5'-phosphomonoesters.
2. 6. 1.X	Transaminases (aminotransferases).	3. 1.14.X	Exoribonucleases producing other than 5'-phosphomonoesters.
2. 6. 3.X	Oximinotransferases.	3. 1.15.X	Exonucleases active with either ribo- or deoxyribonucleic acids and producing 5'-phosphomonoesters.
2. 6.99.X	Transferring other nitrogenous groups.		
2. 7. X.X	Transferring phosphorus-containing groups.	3. 1.16.X	Exonucleases active with either ribo- or deoxyribonucleic acids and producing other than 5'-phosphomonoesters.
2. 7. 1.X	Phosphotransferases with an alcohol group as acceptor (nucleophile).		
2. 7. 2.X	Phosphotransferases with a carboxyl group as acceptor (nucleophile).	3. 1.21.X	Endodeoxyribonucleases producing 5'-phosphomonoesters.
2. 7. 3.X	Phosphotransferases with a nitrogenous group as acceptor (nucleophile).	3. 1.22.X	Endodeoxyribonucleases producing other than 5'-phosphomonoesters.
		3. 1.25.X	Site-specific endodeoxyribonucleases specific for altered bases.

3. 1.26.X	Endoribonucleases producing 5'-phosphomonoesters.	3. 7. X.X	Acting on carbon-carbon bonds.
		3. 7. 1.X	In ketonic substances.
3. 1.27.X	Endoribonucleases producing other than 5'-phosphomonoesters.	3. 8. X.X	Acting on halide bonds.
		3. 8. 1.X	In C-halide compounds.
3. 1.30.X	Endonucleases active with either ribo- or deoxyribonucleic acids and producing 5'-phosphomonoesters.	3. 9. X.X	Acting on phosphorus-nitrogen bonds.
		3.10. X.X	Acting on sulfur-nitrogen bonds.
		3.11. X.X	Acting on carbon-phosphorus bonds.
3. 1.31.X	Endonucleases active with either ribo- or deoxyribonucleic acids and producing other than 5'-phosphomonoesters.	3.12. X.X	Acting on sulfur-sulfur bonds.
		4. X. X.X	LYASES.
3. 2. X.X	Glycosidases.	4. 1. X.X	Carbon-carbon lyases.
3. 2. 1.X	Hydrolysing <i>O</i> -glycosyl compounds.	4. 1. 1.X	Carboxy-lyases.
3. 2. 2.X	Hydrolysing <i>N</i> -glycosyl compounds.	4. 1. 2.X	Aldehyde-lyases.
3. 2. 3.X	Hydrolysing <i>S</i> -glycosyl compounds.	4. 1. 3.X	Oxo-acid-lyases.
3. 3. X.X	Acting on ether bonds.	4. 1.99.X	Other carbon-carbon lyases.
3. 3. 1.X	Thioether hydrolases.	4. 2. X.X	Carbon-oxygen lyases.
3. 3. 2.X	Ether hydrolases.	4. 2. 1.X	Hydro-lyases.
		4. 2. 2.X	Acting on polysaccharides.
3. 4. X.X	Acting on peptide bonds (peptide hydrolases).	4. 2.99.X	Other carbon-oxygen lyases.
		4. 3. X.X	Carbon-nitrogen lyases.
3. 4.11.X	Aminopeptidases.	4. 3. 1.X	Ammonia-lyases.
3. 4.13.X	Dipeptidases.	4. 3. 2.X	Amidine-lyases.
3. 4.14.X	Dipeptidyl-peptidases and tripeptidyl-peptidases.	4. 3. 3.X	Amine-lyases.
		4. 3.99.X	Other carbon-nitrogen-lyases.
3. 4.15.X	Peptidyl-dipeptidases.	4. 4. X.X	Carbon-sulfur lyases.
3. 4.16.X	Serine-type carboxypeptidases.	4. 5. X.X	Carbon-halide lyases.
3. 4.17.X	Metallo-carboxypeptidases.	4. 6. X.X	Phosphorus-oxygen lyases.
3. 4.18.X	Cysteine-type carboxypeptidases.	4.99. X.X	Other lyases.
3. 4.19.X	Omega peptidases.		
3. 4.21.X	Serine endopeptidases.	5. X. X.X	ISOMERASES.
3. 4.22.X	Cysteine endopeptidases.	5. 1. X.X	Racemases and epimerases.
3. 4.23.X	Aspartic endopeptidases.	5. 1. 1.X	Acting on amino acids and derivatives.
3. 4.24.X	Metalloendopeptidases.		
3. 4.99.X	Endopeptidases of unknown catalytic mechanism.	5. 1. 2.X	Acting on hydroxy acids and derivatives.
3. 5. X.X	Acting on carbon-nitrogen bonds, other than peptide bonds.	5. 1. 3.X	Acting on carbohydrates and derivatives.
		5. 1.99.X	Acting on other compounds.
3. 5. 1.X	In linear amides.	5. 2. X.X	<i>cis-trans</i> -Isomerases.
3. 5. 2.X	In cyclic amides.	5. 3. X.X	Intramolecular oxidoreductases.
3. 5. 3.X	In linear amidines.	5. 3. 1.X	Interconverting aldoses and ketoses.
3. 5. 4.X	In cyclic amidines.	5. 3. 2.X	Interconverting keto- and enol-groups.
3. 5. 5.X	In nitriles.		
3. 5.99.X	In other compounds.	5. 3. 3.X	Transposing C=C groups.
3. 6. X.X	Acting on acid anhydrides.	5. 3. 4.X	Transposing S—S bonds.
3. 6. 1.X	In phosphorus-containing anhydrides.	5. 3.99.X	Other intramolecular oxidoreductases.
3. 6. 2.X	In sulfonyl-containing anhydrides.		

- 5. 4. X.X Intramolecular transferases (mutases).
- 5. 4. 1.X Transferring acyl groups.
- 5. 4. 2.X Phosphotransferases (phosphomutases).
- 5. 4. 3.X Transferring amino groups.
- 5. 4.99.X Transferring other groups.
- 5. 5. X.X Intramolecular lyases.
- 5.99. X.X Other isomerases.

- 6. X. X.X **LIGASES.**
- 6. 1.X.X Forming carbon—oxygen bonds.
- 6. 1. 1.X Ligases forming aminoacyl-tRNA and related compounds.
- 6. 2. X.X Forming carbon—sulfur bonds.
- 6. 2. 1.X Acid—thiol ligases.
- 6. 3. X.X Forming carbon—nitrogen bonds.
- 6. 3. 1.X Acid—ammonia (or amine) ligases (amide synthases).
- 6. 3. 2.X Acid—amino-acid ligases (peptide synthases).
- 6. 3. 3.X Cyclo-ligases.
- 6. 3. 4.X Other carbon—nitrogen ligases.
- 6. 3. 5.X Carbon—nitrogen ligases with glutamine as amido-*N*-donor.
- 6. 4. X.X Forming carbon—carbon bonds.
- 6. 5. X X Forming phosphoric ester bonds.

ENZYME CONCENTRATION EFFECTS

In steady-state kinetic studies, the total concentration of the enzyme should be much less than the concentration of the substrate(s), product(s), and effector(s); typically, by at least a thousandfold. When this condition is not true, the steady-state condition will not be valid and other methods, such as global analysis, have to be utilized to analyze the kinetic data.

In standard kinetic studies, the initial velocity (v) should be directly proportional to the total enzyme concentration. Indeed, this plot should be one of the first items analyzed in the kinetic characterization of an enzyme. The enzyme concentration range in this plot should span from below to above the concentration to be used in the initial rate studies (and, ideally, the line should go through the origin). When curvature is observed in such plots, there are several potential explanations^{1,2}.

1. A toxic impurity present in the reaction mixture (but not in the stock enzyme solution) may result in upward curvature; at high concentrations of enzyme, a larger amount of enzyme remains active.
2. A dissociable activator (or coenzyme) present in the enzyme solution may result in upward curvature; as the enzyme concentration increases, so also does the concentration of the activator.
3. Upward curvature may occur when the enzyme undergoes self-association or aggregation into a more active form; such a process is favored at elevated concentrations.
4. If the enzyme under consideration is assayed via a coupling enzyme(s) system, downward curvature in the plot will be observed when the primary enzyme fails to limit the observed velocity. In such instances, it would be necessary to increase the amount of coupling enzymes such that they remain in excess under all conditions. This problem is often encountered in manometric assays, where the rate of diffusion of gas into the liquid begins to limit the enzyme reaction rate.
5. Elevated concentrations of enzyme can also affect the observed rate if those concentrations interfere with the physical methods of observing the reaction velocity (*e.g.*, increased light scattering at high $[E_{\text{total}}]$ in spectrophotometric assays; increased absorbance by the protein; *etc.*)
6. Nonlinearity can also be observed if increased amounts of the stock enzyme solution alter some other reaction parameter (*e.g.*, pH, ionic strength, *etc.*). These issues are easily assessed and remedied.
7. If a coenzyme or activator is present in the reaction mixture at relatively low concentrations, then addition of increasing concentrations of total enzyme to the reaction mixture may reduce the coenzyme or activator to suboptimal levels. In such cases, a downward-oriented curve may be seen in the v vs. $[E_{\text{total}}]$ plot.
8. Downward curvature can also be seen if steady-state and initial rate conditions are not observed (*e.g.*, the product concentration approaches inhibitory levels or substrate depletion occurs).

9. If a reversible inhibitor is present in the stock enzyme solution, then as more enzyme is added to the reaction, more inhibitor is also simultaneously introduced, and this can lead to progressively less activity in such v versus $[E_{\text{total}}]$ plots.

10. If the enzyme undergoes aggregation or a conformational change at elevated concentrations that results in reduced activity, the v versus $[E_{\text{total}}]$ plots will curve downward at elevated enzyme concentrations.

11. If the enzyme is unstable and steadily degrades in the stock solution, nonlinearity may be observed in plots of initial velocity v versus $[E_{\text{total}}]$. This can be corrected by using a standardized enzyme activity assay periodically during the course of the kinetic experiment to determine the amount of lost activity. The likelihood for enzyme instability arises whenever the kinetic experiment requires one to transfer the enzyme from its storage buffer to that used in the experiment.

¹M. Dixon & E. C. Webb (1979) *Enzymes*, 3rd ed., Academic Press, New York.

²D. L. Purich & R. D. Allison (1980) *Meth. Enzymol.* **63**, 3.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Dilution of enzyme samples, **63**, 10; lipolysis substrate effect, **64**, 361, 362; dilution jump kinetic assay, **74**, 14-19, 28; dilution method [for dissociation equilibria, **61**, 65-96; continuous dilution cuvette, **61**, 78-96; data analysis, **61**, 74, 75; equations, **61**, 70-74; errors, **61**, 76-78; experimental procedures, **61**, 69, 70; merits, **61**, 75, 76; theory, **61**, 68, 69.

ENZYME DEPLETION (Apparent Noncompetitive Inhibition)

Enzyme inhibition by an extremely tight-binding inhibitor¹. When the substrate(s), *regardless of the detailed mode of inhibition*, has (have) a negligible effect on the formation of enzyme-inhibitor (E-I) complex, the net result is depletion (*i.e.*, the removal of enzyme by the inhibitor from the reaction). The observed kinetic pattern is identical to the simple noncompetitive inhibition case; the substrate and the inhibitor do not affect each other's binding, because only V_{max} is changed due to reduced enzyme concentration, while K_m remains unaltered.

An example of enzyme depletion is the ribonuclease inhibitor isolated from human placenta by Blackburn, Wilson & Moore². This protein forms a 1:1 complex with bovine pancreatic RNase A and is a noncompetitive

inhibitor of the pancreatic enzyme, with a K_i of 3×10^{-10} M.

The rate equation to be used for kinetic analysis of enzyme depletion is that for *simple noncompetitive* inhibition. If the Henderson equation or similar types are not employed, keep in mind that the inhibitor concentration $[I]$ is the *free* inhibitor concentration. Determination of K_i may not be feasible if the rate assay is insensitive and requires an enzyme concentration much greater than K_i . Alternatively, K_i may be obtained by measuring the on-off rate constants of the E-I complex, provided the rate constants for any conformational change steps involved are also known.

¹The authors thank Dr. Charles Y. Huang for contributing this entry.

²P. Blackburn, G. Wilson & S. Moore (1977) *J. Biol. Chem.* **252**, 5904.

ENZYME ENERGETICS (The Case of Proline Racemase)

Although triosephosphate isomerase¹ is the enzyme that has been used in the most widely known systematic investigation of catalysis, every student interested in using kinetics to explore catalysis should also school herself or himself on the reaction mechanism and energetics of proline racemase. This bacterial enzyme efficiently interconverts L-proline to D-proline, and Rudnick and Abeles² showed that the enzyme is a homologous dimer containing a single active site made up of residues contributed from both subunits. They also demonstrated that catalysis proceeds by a two-base mechanism, and proton abstraction alternatively occurs from one or the other of the two faces of the proline ring. Two enzyme forms, designated E_1 and E_2 , each have one protonated base and one unprotonated base, between which there appears to be relatively little or no proton exchange (Fig. 1).

The research team headed by Alberly and Knowles presented a series of seven papers³⁻⁹ on the dynamics of the catalytic process. First, they investigated the reaction kinetics over a range of substrate concentration³. While the standard free energy changes for a series of intermediate steps and the associated transition states are fixed for a particular enzyme acting on a given substrate, the actual distribution of enzyme among substrate-bound and substrate-free species is sensitive to the degree of enzyme saturation by substrate. Accordingly, Fisher *et*

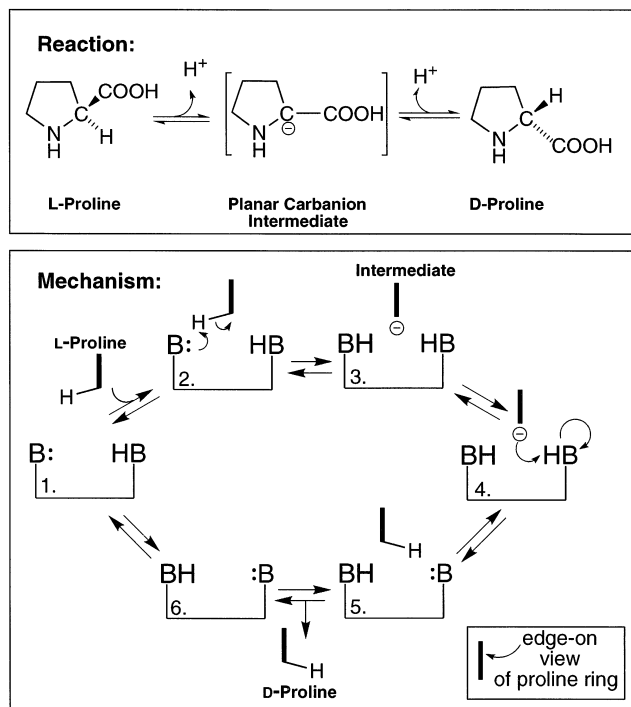
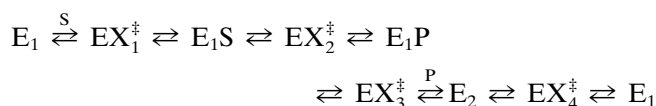


Figure 1. Reaction and mechanism of proline racemase.

*al.*³ considered three different substrate concentration regimes, which they termed: (a) *undersaturation*, a condition where the rate-limiting free energy difference is that between the lowest energy free enzyme form and the highest energy bound transition state; (b) *saturation*, a situation where the reaction proceeds from a bound intermediate over a bound transition state or from the free enzyme form over a free transition state; and (c) *oversaturation*, a condition where reactivity is determined by the rate of conversion from the lowest bound enzyme form over the highest free enzyme transition state.

Over these substrate concentration changes, kinetic measurements can be made to determine rate constants and other parameters that define the dynamic properties of segments along the reaction coordinate are shown in Fig. 2. The reaction catalyzed follows the scheme:



From left to right, these forms are represented in Fig. 1 as E_1 , 1, E_1S , 2, E_2P , 3, E_2 , 4, and back to E_1 . Note that different segments of the reaction cycle manifest

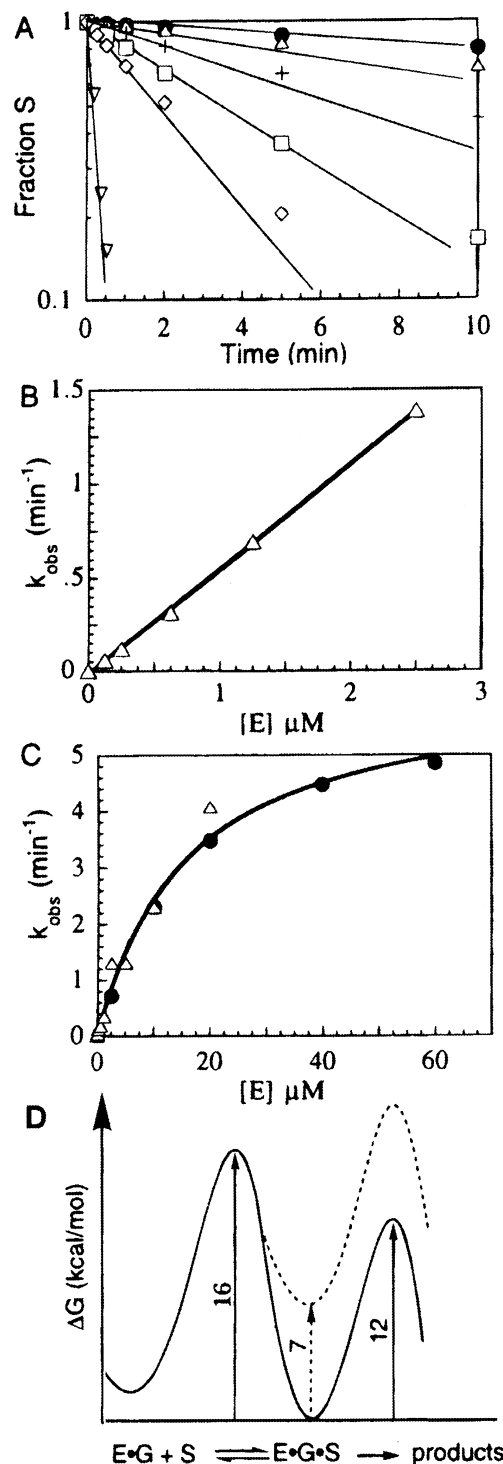


Figure 2. Free energy diagrams illustrating the kinetic regimes of undersaturation, saturation, and oversaturation. The states that determine the reaction rate observed under these conditions are marked as heavy bars. The changes for interversion of free enzyme forms E_1 and E_2 are shown by the dotted lines. Reproduced with permission of the authors and the American Chemical Society.

their kinetic properties under conditions of undersaturation, saturation and oversaturation. In the second paper, Fisher *et al.*⁴ used the tracer perturbation technique under oversaturation conditions. This clever enzyme kinetic technique, introduced by Britton and co-workers^{10,11}, permits one to measure the equilibrium distribution of enzyme-bound substrate(s), intermediate(s), and product(s). Radiolabeled substrate or product is initially permitted to react with enzyme for sufficient time to equilibrate. At thermodynamic equilibrium, all the different enzyme-bound species will be present at concentrations reflecting their stability relative to each other. One can then add a large excess of unlabeled substrate. Any unbound *or* newly released radiolabel will mix with this large unlabeled pool of substrate or product, where it will undergo substantial reduction in radiospecific activity. This dilution effectively reduces or eliminates any detectable recycling of released radiolabel. One can then sample the system as a function of time, and the rates and amounts of labeled substrate and labeled product depend on the kinetics of synthesis and/or interconversion of bound radiolabel to substrate and product, which upon release is unlikely to rebinding. One can also evaluate the “peak-switch” concentration c_p , a parameter that reveals the point of substrate saturation at which the rate-determining step shifts from substrate-to-product interconversion and then depends on the $[E_2 \leftrightarrow EX_2^{\ddagger} \leftrightarrow E_1]$ conversion. Fisher *et al.*⁵ then determined the rates, fractionation factors, and buffer catalysis in the oversaturated region to explore the interconversion of these two forms of free enzyme. Using equimolar substrate and product that was specifically deuterated at the 2-position for some or all of one enantiomer, competitive deuterium wash-out experiments were carried out after the reaction mixture reached chemical and isotopic equilibrium in the presence of enzyme. The authors indicate that their experimental results were most compatible with a stepwise reaction for the interconversion of the free enzyme forms, a process in which a proton is abstracted from a bound water molecule to give a reaction intermediate bearing a hydroxide ion bound to the diprotonated form of the enzyme. Other fractionation factors provided additional information on (1) the catalytic reaction's concertedness⁶; (2) the nature of essential catalytic groups⁷; and (3) the interconversion of the two monoprotonated enzyme forms E_1 and E_2 ⁸. These efforts culminated in the construction of what is probably the most

detailed free energy profiles for an enzyme catalyzed reaction^{9,12}. This profile is shown in Fig. 3, in which the transition state has an enzyme-proline carbanion sandwiched between the two catalytic thiols.

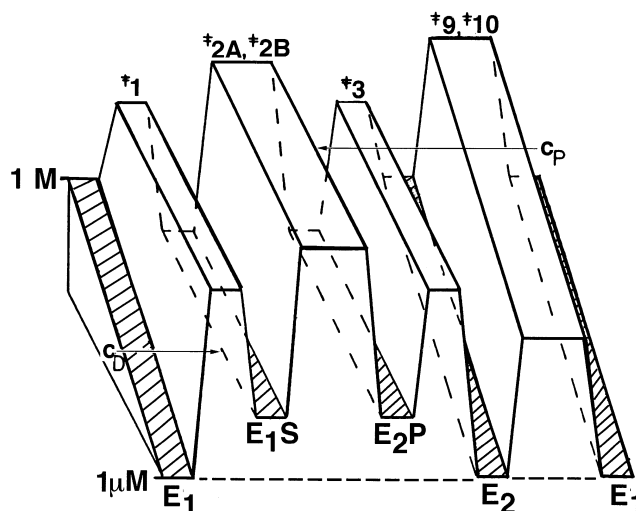


Figure 3. Computed three-dimensional free energy diagram for proline racemase: x-axis, the reaction coordinate; y-axis, free energy; and z-axis, substrate saturation. At the front of the diagram, $[S] = [P] = 1 \mu\text{M}$, where the enzyme is unsaturated; at the back of the diagram, $[S] = [P] = 1 \text{ M}$, where the enzyme is oversaturated. Reproduced from reference 9 with permission of the authors and the American Chemical Society.

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²G. Rudnick & R. H. Abeles (1975) *Biochemistry* **14**, 4515.

³L. M. Fisher, W. J. Albery & J. R. Knowles (1986) *Biochemistry* **25**, 2529.

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⁶J. G. Belasco, W. J. Albery & J. R. Knowles (1986) *Biochemistry* **25**, 2552.

⁷J. G. Belasco, T. W. Bruice, W. J. Albery & J. R. Knowles (1986) *Biochemistry* **25**, 2558.

⁸J. G. Belasco, T. W. Bruice, L. M. Fisher, W. J. Albery & J. R. Knowles (1986) *Biochemistry* **25**, 2564.

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¹⁰H. G. Britton & J. B. Clarke (1972) *Biochem. J.* **110**, 161.

¹¹H. G. Britton & J. B. Clarke (1972) *Biochem. J.* **130**, 397.

¹²W. J. Albery & J. R. Knowles (1986) *J. Theoret. Biol.* **124**, 173.

ENZYME EQUIVALENT WEIGHT

The weight (or more correctly, the mass) of a protein expressed in grams per mole of active sites. Not all oligomeric proteins, even some with identical subunits, have a number of active sites equal to the number of subunits. Enzyme normality (*i.e.*, the concentration of enzyme active sites) is typically determined by active site titration with an active-site-directed irreversible inhibitor. This is

illustrated by the use of ^{32}P -labeled diisopropylfluorophosphate¹ to determine the enzyme equivalent weight of acetylcholinesterase. Alternatively, one can use so-called “burst” kinetics using stoichiometric quantities of enzyme and substrate.

¹H. O. Michel & S. Krop (1951) *J. Biol. Chem.* **190**, 119.

ENZYME INHIBITION

In addition to the entries on specific types of reversible and irreversible inhibitors, one may also wish to consult the following selected entries from *Methods in Enzymology* [vol, page(s)].

Competitive, **249**, 123, 146, 190 [partial, **249**, 124; progress curve equations for, **249**, 176, 180; for three-substrate systems, **249**, 133, 136]; competitive-uncompetitive, **249**, 138; concave-up hyperbolic, **249**, 143; dead-end, **249**, 124 [for bireactant kinetic mechanism determination, **249**, 130-133; definition of kinetic constants, **249**, 220-221; effects on enzyme progress curves, nonlinear regression analysis, **249**, 71-72; inhibition constant evaluation, **249**, 134-135; kinetic analysis with, **249**, 123-143; one-substrate systems, **249**, 124-126; unireactant systems, theory, **249**, 124-130]; hyperbolic, **249**, 145 [identification, **249**, 162]; irreversible types, **249**, 240 (*See also Mechanism-Based Enzyme Inhibition*); kinetic models [with constant concentration of some components, **249**, 165-166; differential and conservation equation sets, **249**, 166-168; experimental data, processing, **249**, 168-169; introduction of appropriate kinetic mechanism, **249**, 163; introduction of simplifying assumptions, **249**, 163-166; parameters for, **249**, 166-170; progress curve equations, derivation, **249**, 168; with rapid equilibrium in some steps, **249**, 164-165; rate equations, derivation, **249**, 168; steady-state assumptions, **249**, 163-164; linear [identification, **249**, 162; uncompetitive, one-substrate systems, **249**, 128-129]; noncompetitive, **249**, 146, 190-191 [one-substrate systems, **249**, 126-128; progress curve equations for, **249**, 176, 180; pure, **249**, 146 (progress curve equations for, **249**, 175-178, 180); nonlinear, **249**, 162 [competitive, **249**, 138; noncompetitive, **249**, 138; one-substrate systems, **249**, 129-130]; parabolic, **249**, 138, 192-193 [one-substrate systems, **249**, 130]; parameters, **249**, 166 [determination, **249**, 160-170 (slow, tight-binding inhibition, **249**, 159-160); reliability, **249**, 169-170]; product inhibition (*See Product Inhibition*); slow, tight-binding [definition, **249**, 149; inhibition parameters, determination, **249**, 159-160; isomerization and, kinetic models, **249**, 160; kinetic models, **249**, 158-160]; slow-binding [definition, **249**, 149; inhibition parameters, **249**, 170-172; isomerization and, kinetic models, **249**, 155-158; kinetic models, **249**, 155-158, 170-172; rapid equilibrium in some steps, **249**, 164-165; time dependence, **249**, 149]; tight-binding [characteristics, **249**, 148-149; consequences, **249**, 148; definition, **249**, 148-149; hyperbolic, **249**, 173-174; identification, **249**, 148-149, 162; inhibition parameters, **249**, 173-178; kinetic models, **249**, 149-153, 173-178; progress curve equations for, **249**, 174-178; reaction conditions, **249**, 148]; uncompetitive, **249**, 146, 190-192 [progress curve equations for, **249**, 176, 180].

ENZYME-LINKED IMMUNOSORBENT ASSAYS (ELISA)

Any immunoassay that is based on the binding of soluble enzyme-linked antibody to an surface-immobilized anti-

gen, such that the amount of antibody-antigen complexation is quantitatively measured by the activity of the linked enzyme. The major advantage offered by ELISA methodologies is that there is no need for radioactivity, because the high catalytic power of the covalently linked enzyme gives a high signal-to-noise ratio in these measurements. Various strategies may be employed in immobilizing the antigen, but the technique requires that the binding of enzyme-linked antibody to antigen in no way reduces the enzyme's activity. In some competitive ELISA techniques, the enzyme-linked antibody is reacted with a soluble antigen that forms a complex and can no longer react with the surface-immobilized antigen.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Detection and quantification, **74**, 608-616; effect of antigen/antibody ratio, **74**, 613, 614; applications, **74**, 614-615; controls, **74**, 612; data analysis, **74**, 612-613; immunoglobulin G [conjugation to glucose oxidase, **74**, 610; heat aggregation, **74**, 609-610; heat denaturation, **74**, 610; immobilization, **74**, 610]; precision, **74**, 612, 614; principle, **74**, 608-609; procedure, **74**, 611-612 [adsorption of endogenous rheumatoid factor, **74**, 611; enzyme activity assay, **74**, 612; enzyme-immunoglobulin conjugate binding, **74**, 611-612; sample competitive binding, **74**, 611]; reference serum, preparation, **74**, 610-611; comparative results, **74**, 611, 614; sample storage, **74**, 611; sensitivity, **74**, 612; specificity, **74**, 613]; leukocyte interferon, **79**, 595; solid phase, **73**, 483, 523-550, 656, 666 [classification, **73**, 483-486; conjugate; (preparation, **73**, 147-166; purification, **73**, 163, 166-175; stability, **73**, 162, 163); enzyme selection, **73**, 148, 657-659; factors affecting, **73**, 158, 159, 506-511; kinetic analysis, **73**, 521, 522; magnetic method, **73**, 471-482, 663, 664 [for immunoglobulin E (enzyme conjugation, **73**, 475; immobilization of protein, **73**, 474, 475; reagents, **73**, 474; specific direct method, **73**, 477-479 indirect method, **73**, 479-481; total bridged avidin-biotin method, **73**, 481, 482 sandwich method, **73**, 477, 478; magnetic materials, **73**, 472-474)]; protein A as tracer, **73**, 186-191; conjugation procedure, **73**, 176-191; substrate choice, **73**, 657-659.

ENZYME MECHANISM

A systematic representation of the bond-breaking and bond-making chemical steps that explains: (a) the solvent and electrostatic factors affecting ligand binding and desorption steps; (b) the order of substrate binding and product release; (c) steric factors influencing an enzyme's chiral recognition and disposition of substrates, intermediates, products, and effectors; (d) the configurations of all atoms of the enzyme's active site and reactants during the entire reaction cycle; (e) the energetics and rates affecting the accumulation and interconversion of intermediates; (f) the structure and properties of the transition state; and (g) the activation or inhibition of enzyme catalysis by regulatory molecules.

In principle, no reaction mechanism can ever be completely understood until one achieves a complete quantum mechanical determination of the state-to-state differential scattering cross section. Enzyme catalysis involves far too many atoms for successful application of such a detailed quantum mechanical approach, even with the availability of supercomputers. This does not imply that quantum mechanics is without value in characterizing enzyme catalysis. Quantum mechanics allows one to comprehend aspects of chemical reactivity by considering the detailed properties of substrate(s), likely intermediates, and reaction products. One may also apply quantum mechanical to analyze the behavior of a limited number of atoms comprising a restricted region surrounding the reaction center. In any case, the inability to attain a complete quantum mechanical treatment in no way minimizes the quality of critical thinking required to conduct rigorous mechanistic studies on enzymes. In view of the celerity of enzyme processes, great ingenuity and experimental skill is required to investigate intermediates that accumulate at such low concentrations during the reaction cycle.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Acid-base catalysis [with site-directed mutants, **249**, 110-118; altered pH dependencies, **249**, 110]; commitment to [in determination of intrinsic isotope effects, **249**, 343, 347-349; in interfacial catalysis, **249**, 598-599; equilibrium isotope exchange in, **249**, 443-479; hydrogen tunneling in, **249**, 373-397]; interfacial [competitive inhibitors, kinetic characterization, **249**, 604-605; equilibrium parameters, **249**, 587-594; forward commitment to, **249**, 598-599; interpretation, **249**, 578-586 (constraining variables for high processivity, **249**, 582-586; kinetic variables at interface, **249**, 581-582); kinetic basis for, **249**, 567-614; kinetic parameters, **249**, 594-599; uses, **249**, 599-605; Michaelis-Menten formalism, **249**, 570, 578, 581-582; in scooting mode, **249**, 581, 583, 585-586, 600-603, 613]; site-directed mutagenesis studies, **249**, 91-119 [active site essential residues, **249**, 91-93; catalysis evolution, **249**, 92; catalytic efficiency, **249**, 92, 104-107; role of specific interactions or residues, **249**, 91-96; transition state theory, **249**, 284-288].

ENZYME PURITY

The availability of completely pure enzyme affords the opportunity (a) to investigate mechanistic and regulatory properties in the absence of contaminating enzyme activities, and (b) to determine those kinetic parameters requiring an accurate knowledge of the enzyme concentration. Nevertheless, most kinetic studies do not require pure enzyme. What is required is that there be no (or, at least, minimal) contaminating activities that interfere with the initial-rate assay. A contaminating protein may act on the substrate(s), product(s), or effector(s). Hence,

each of the central reaction constituents should be examined separately¹. One method for doing this is to incubate the substrate, product, or effector with the enzyme preparation (under the reaction conditions) in which a required constituent of the reaction (*e.g.*, one of the substrates in a multisubstrate reaction) is absent. The presence of the enzyme preparation should not result in a change in the concentration of a substrate, product, inhibitor, effector, or any other constituent (*e.g.*, buffer) in the absence of a necessary constituent for the reaction (at least, not in the time frame of the experiment).

In a number of cases there may be a contaminating enzyme present which acts on one or more of the substrates, products, or effectors of the system under study. It may be necessary to include in the reaction mixture a specific inhibitor for that contaminating activity. For example, adenylate kinase is often present in preparations of a number of phosphotransferases. It is often advantageous, in such instances, to include a specific inhibitor of adenylate kinase (*e.g.*, P^1, P^4 -di(adenosine-5')-tetraphosphate or P^1, P^5 -di(adenosine-5')-pentaphosphate)^{2,3}. If an inhibitor of the contaminating activity is added as an additional constituent of the reaction mixture, the investigator should demonstrate that the inhibitor is not an effector of the enzyme under study.

It should be recalled that, in some instances, the contaminating activity may be an intrinsic property of the enzyme. For example, sucrose phosphorylase exhibits a transglycosylation reaction⁴, γ -glutamyl transpeptidase has a glutaminase activity⁵, *E. coli* acetate kinase has a purine nucleoside-5'-phosphate kinase activity⁶, and hexokinase has a weak ATPase activity⁷. In order for a system to be thoroughly characterized, each of these other activities should be studied as well. However, it may not be easily demonstrated that the same polypeptide (or polypeptide complex) is responsible for both the central reaction and the side reaction(s); or, that both activities have a common active site. Protocols to address this issue include: (1) inactivation studies demonstrating parallel losses of activities; (2) successive proteolysis studies in which activities are lost at the same or different steps; and (3) site-directed mutagenesis studies identifying both activities with the same protein and loss of both or separate activities with the modification of particular amino acid residues.

Isozymes are also a common presence in enzyme preparations and they can often be detected via polyacrylamide gel electrophoresis. The detected presence of isozymes may result in the need for further purification steps and the kinetic characterization of each isozyme. It may be necessary to use nondenaturing electrophoretic procedures to separate the different isozymes. *See Isozymes; Enzyme Concentration*

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Selected entries from *Methods in Enzymology* [vol, page(s)]:
Why Purify Enzymes?, **182**, 1; strategies and considerations for protein purifications, **182**, 9; rethinking your purification procedure, **182**, 779.

General Methods for Handling Proteins and Enzymes: Buffers, **104**, 404; **182**, 24; preparation of high-purity laboratory water, **104**, 391; measuring enzyme activity, **182**, 38; quantitating protein, **182**, 50; protein assay for dilute solutions, **104**, 415; concentrating proteins and removing solutes, **182**, 68; maintaining protein stability, **182**, 83; vectors, hosts, and strategies for over-producing proteins in *Escherichia coli*, **182**, 93; overexpressing proteins in eukaryotes, **182**, 112.

Extracts and Subcellular Fractionation: Preparing extracts from prokaryotes, **182**, 147; preparing yeast extracts, **182**, 154; preparing extracts from plants, **182**, 174; preparing extracts from higher eukaryotes, **182**, 194; isolating subcellular organelles, **182**, 203; preparing membrane fractions, **182**, 225.

Solubilization Procedures; Detergents: An overview, **182**, 239; solubilization of functional membrane proteins, **104**, 305; solubilization of native membrane proteins, **182**, 253; solubilization of protein aggregates, **182**, 264; removal of detergents from membrane proteins, **104**, 318; **182**, 277.

Purification Procedures: Precipitation techniques, **182**, 285; protein precipitation with polyethylene glycol, **104**, 351; **182**, 301; ion exchange methods, **182**, 309; gel filtration, **182**, 317; hydroxyapatite columns, **182**, 329; hydrophobic chromatography, **182**, 339; chromatography on immobilized reactive dyes, **182**, 343; affinity chromatography, **104**, 3; **182**, 357, 371; immobilization of affinity ligands with organic sulfonyl chlorides, **104**, 56; hydrophobic chromatography, **104**, 69; affinity chromatography on immobilized dyes, **104**, 97; displacement chromatography of proteins, **104**, 113; chromatofocusing, **182**, 380; HPLC, **182**, 392; high-performance size exclusion chromatography, **104**, 154; high-performance ion-exchange chromatography, **104**, 170; reversed-phase HPLC, **104**, 190; high-performance liquid affinity chromatography, **104**, 212; care of HPLC columns, **104**, 133; optimal pH conditions for ion exchangers on macroporous supports, **104**, 223.

Electrophoretic Methods: Systems for polyacrylamide gel electrophoresis, **104**, 237; preparative isoelectric focusing, **104**, 256; gel

protein stains: silver stain, **104**, 441; gel protein stains: glycoproteins, **104**, 447; gel protein stains: phosphoproteins, **104**, 451; Western blots, **104**, 455; fluorography for the detection of radioactivity in gels, **104**, 460; affinity electrophoresis, **104**, 275; preparative isotachopheresis, **104**, 281; enzyme localization in gels, **104**, 416; one-dimensional gel electrophoresis, **182**, 425; two-dimensional polyacrylamide gel electrophoresis, **182**, 441; isoelectric focusing, **182**, 459; gel staining techniques, **182**, 477; elution of protein from gels, **182**, 488.

Specialized Purification Procedures: Purification of membrane proteins, **182**, 499; purification of integral membrane proteins, **104**, 329; reconstitution of membrane proteins, **104**, 340; purification of DNA-binding proteins by site-specific DNA affinity chromatography, **182**, 521; purification of glycoproteins, **182**, 529; purification of multienzyme complexes, **182**, 539.

Characterization of Purified Proteins: Assessing purity, **182**, 555; determining size, molecular weight, and presence of subunits, **182**, 566; amino acid analysis, **182**, 587; limited N-terminal sequence analysis, **182**, 602; peptide mapping, **182**, 613; analysis for protein modifications and nonprotein cofactors, **182**, 626; protein crystallization, **182**, 646.

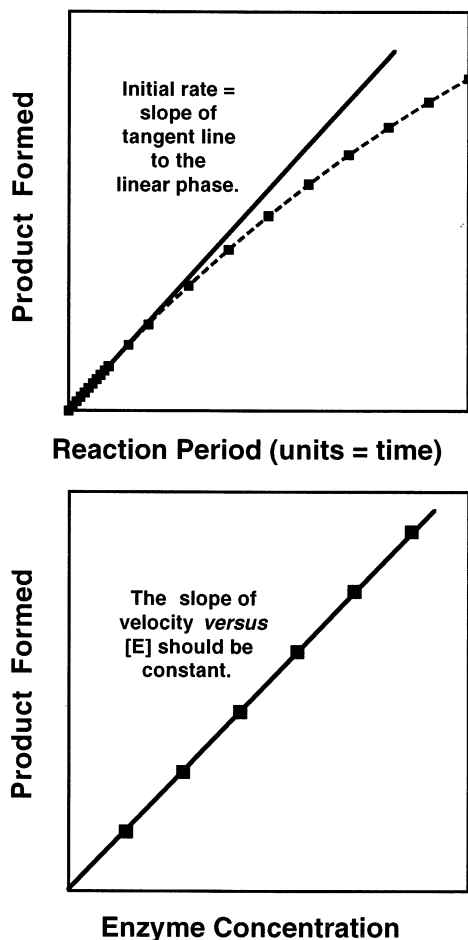
Immunological Procedures: Immunosorbent separations, **104**, 381; preparing polyclonal antibodies, **182**, 663; preparing monoclonal antibodies, **182**, 670; protein blotting and immunodetection, **182**, 679; immunoprecipitating proteins, **182**, 688; immunoassays, **182**, 700.

Additional Techniques: Radiolabeling of proteins, **182**, 721; using purified protein to clone its gene, **182**, 738; computer analysis of protein structure, **182**, 751.

ENZYME RATE EQUATIONS

(1. The Basics)

The time course and kinetics of an enzyme-catalyzed reaction can be evaluated by a variety of procedures, but the initial rate method is by far most widely applied. The initial reaction phase refers to a period of time over which initial substrate concentration remains constant and for which the accumulated product has no measurable effect. This phase is characterized by a linear accumulation of product with respect to the elapsed time after mixing enzyme and substrate. The slope of a line drawn tangent to the initial progress curve has a slope of $\Delta[P]/\Delta t$ or $\{[P]_t - [P]_{\text{initial}}\}/\Delta t$, where $[P]$ is the product concentration (initial and at time t), and this represents the reaction rate. Note that this fits with the accepted convention of expressing the rate of a chemical reaction as the change in molarity of a substrate or product, normalized with respect to the stoichiometry of the reaction. A doubling of the reaction time during the initial rate phase should result in a doubling of product formed.



The relationship between reaction velocity and enzyme concentration (in the absence of self-association of the enzyme) should also be adjusted such that reaction rate is linearly related to catalyst concentration, $[E_{\text{total}}]$. Initial rates typically fail to obtain if $[E_{\text{total}}] = 0.01 [A]_{\text{initial}}$ where $[A]_{\text{initial}}$ is the initial substrate concentration. As a general rule, the substrate concentration will not have changed more than 5–10% of its value over the initial rate phase of the reaction. This rule-of-thumb applies only to thermodynamically favorable reactions, and investigators are well advised to limit substrate consumption to well below 5%.

It was Henri who first proposed that enzyme catalysis depended on the formation of a transient complex of enzyme and substrate, followed by the breakdown (*i.e.*, chemical conversion) of bound substrate into product. Nonetheless, credit for derivation of the rate expression for the initial rate phase of one-substrate enzyme-catalyzed reactions is given to Michaelis and Menten. Both treatments gave the same general result:

$$v = V_{\text{max}}[S]/(K_m + [S])$$

where v is the initial reaction velocity, V_{max} is the maximal reaction velocity, K_m is the Michaelis constant, and $[S]$ is the initial substrate concentration.

DERIVATION OF THE MICHAELIS-MENTEN EQUATION. To obtain the Michaelis-Menten equation (that is, the rapid equilibrium expression) for a one-substrate enzyme-catalyzed reaction, one must make a series of assumptions: (a) that all experimental conditions (*e.g.*, pH, temperature, ionic strength) remain constant over the course of the experiment; (b) that the enzyme, substrate, and product are all stable under the assay conditions over the time course of the experiment; (c) that at least one binary central complex forms (*e.g.*, EX); (d) that E and S react rapidly and reversibly to form this complex; (e) that the binding of S to E can be described by an equilibrium dissociation constant, $K_s = k_2/k_1$, where k_1 is the forward bimolecular rate constant for S binding to E and k_2 is the unimolecular rate constant for the release of S from the central complex; (f) that the conversion of EX to E + P, having the unimolecular rate constant k_3 , is slow and irreversible (or, if reversible, the low concentration of [P] over the time course of the experiment precludes the reverse reaction); (g) that the total enzyme concentration $[E_{\text{tot}}]$ is the sum of uncomplexed enzyme concentration, [E], and the enzyme-substrate central complex (*i.e.*, conservation of matter requires that $[E_{\text{total}}] = [E] + [EX]$); (h) that the velocity, v , of the reaction, equal to $d[P]/dt$, equals [EX] multiplied by the unimolecular rate constant k_3 ; (i) that the concentration of substrate is essentially unchanged during the initial velocity period (*i.e.*, $\Delta[S] \ll [S]_{\text{initial}}$); and (j) that the total substrate concentration, $[S_{\text{total}}]$, far exceeds the total enzyme concentration, $[E_{\text{total}}]$, such that $[S_{\text{total}}] = [S]$, where [S] is the uncomplexed substrate concentration.

We can then begin the derivation by considering two independent equations:

$$K_s = [E][S]/[EX]$$

where K_s is the dissociation constant for S and

$$[E_{\text{total}}] = [E] + [EX]$$

This first equation can be rewritten as $[E] = K_s [EX]/[S]$ and substituted for [E] in the second equation

to yield

$$[E_{\text{total}}] = (K_s[EX]/[S]) + [EX]$$

Dividing both sides by $[EX]$, we get

$$[E_{\text{total}}]/[EX] = (K_s/[S]) + 1$$

Then, by multiplying the left side by k_3/k_3 (note that $v = k_3[EX]$ and defining the maximum velocity, V_{max} , as being equal to $k_3[E_{\text{total}}]$), we get:

$$v = \frac{V_{\text{max}}[S]}{K_s + [S]}$$

The above expression represents the equation of a hyperbola (*i.e.*, $f(x) = ax/(b + x)$) and a plot of the initial velocity as a function of $[S]$ will result in a rectangular hyperbola. The limiting value of the reaction rate at saturating substrate concentration is termed V_{max} . The K_m is obtained by determining the substrate concentration yielding one-half maximal velocity. At low $[S]$, the reaction rate is first-order with respect to substrate concentration, but at saturating substrate concentrations, the reaction is said to display zero-order kinetics. Another way for representing the Michaelis-Menten equation is by using the double-reciprocal (or Lineweaver-Burk) transformation:

$$1/v = (1/V_{\text{max}}) + (K_s/V_{\text{max}})(1/[S])$$

This method is widely used because it provides linear transformation of the hyperbolic function describing the rate saturation process. Double-reciprocal plots can be reasonably accurate if rate data can be obtained over a reasonable range of saturation, say from $0.3 V_{\text{max}}$ to $0.8 V_{\text{max}}$.

STEADY STATE TREATMENT. While the Michaelis-Menten model requires the rapid equilibrium formation of ES complex prior to catalysis, there are many enzymes which do not exhibit such rate behavior. Accordingly, Briggs and Haldane considered the case where the enzyme and substrate obey the steady state assumption, which states that during the course of a reaction there will be a period over which the concentrations of various enzyme species will appear to be time-invariant (*i.e.*, $d[EX]/dt \cong 0$). Such an assumption then provides that

$$d[EX]/dt = k_1[E][S] - k_2[EX] - k_3[EX] \cong 0$$

where we write a positive sign in front of all rates leading to EX and a minus sign in front of terms going away from EX. Transposing and collecting like terms gives

$$k_1[E][S] = k_2[EX] + k_3[EX] = (k_2 + k_3)[EX]$$

or

$$[E] = \{(k_2 + k_3)/k_1[S]\}[EX]$$

We can now substitute our new expression for $[E]$ into the same conservation equation ($[E_{\text{total}}] = [E] + [EX]$) as used earlier to get

$$[E_{\text{total}}]/[EX] = 1 + (k_2 + k_3)/(k_1[S])$$

and, by multiplying each enzyme species by k_3 , we arrive at

$$V_{\text{max}}/v = 1 + (k_2 + k_3)/(k_1[S])$$

(This means that we again use $k_3[EX]$ to represent v). The term $(k_2 + k_3)/k_1$ has the same units as the constant K_s obtained in the rapid equilibrium treatment. In honor of Leonor Michaelis, we use the designation K_m (or Michaelis constant) to represent this quotient, and the steady state rate law reduces to:

$$V_{\text{max}}/v = 1 + K_m/[S]$$

This result is identical in overall form to that obtained by Michaelis and Menten, and both equations may be used to deal with one-substrate enzymic processes. In fact, without detailed knowledge of the magnitudes k_1 , k_2 , and k_3 , one would be unable to discern a difference between the two equations.

[The reader should also readily recognize that the same equation is obtained if we use the conservation equation along with the following differential equation:

$$d[E]/dt = k_2[EX] + k_3[EX] - k_1[E][S] \cong 0$$

This fact reminds us that in making the steady state assumption, we are treating both E and EX as time-invariant forms.]

THE MICHAELIS-MENTEN EQUATION AS A LIMITING CASE OF THE STEADY STATE EQUATION. To achieve a rapid equilibrium between E and EX, $k_1[S]$ and k_2 must each be much greater than k_3 . [Note: the rate constant k_1 is a bimolecular rate constant with units of molarity⁻¹ seconds⁻¹, and we must use $k_1[S]$

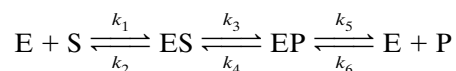
to compare its magnitude to either k_2 or k_3 .] Thus, returning to K_m , we see that when $k_2 \gg k_3$,

$$K_m = (k_2 + k_3)/k_1 \cong k_2/k_1 = K_s$$

Moreover, the rapid equilibrium assumption used in obtaining the Michaelis-Menten equation requires that

$$\begin{aligned}\text{Rate}_{\text{Substrate Binding}} &= k_1[E][S] \gg k_3[EX] \\ &= \text{Rate}_{\text{Product Formation}}\end{aligned}$$

EFFECT OF ADDITIONAL “CENTRAL COMPLEX” SPECIES ON THE GENERAL FORM OF THE STEADY STATE RATE EQUATION. Up to now, we have actually considered a chemically unrealistic model for enzyme catalysis in that we have assumed that a single enzyme-bound species, namely EX, accounts for the catalytic process. We now treat a more reasonable representation of the kinetic mechanism



by using an additional constant k_5 for the breakdown of EP, the enzyme-product complex, and the rate constants for the ES-to-EP interconversion. We will specify that $v = k_5[EP] - k_6[P]$. Again, if we take $[P] \cong 0$, then $v = k_5[EP]$ and the rate law can be derived by using the conservation equation (now containing EP as an additional species), along with two linear differential equations to solve for the three different enzyme forms:

$$[E_{\text{total}}] = [E] + [ES] + [EP]$$

$$d[ES]/dt = k_1[E][S] + k_4[EP] - k_2[ES] - k_3[ES] \cong 0$$

$$d[EP]/dt = k_3[ES] + k_6[E][P] - k_4[EP] - k_5[EP] \cong 0$$

This last differential equation is further simplified since we are considering the case in which $[P] \cong 0$: thus, $d[EP]/dt = k_3[ES] - (k_4 + k_5)[EP] \cong 0$. We need only to use the differential equations to obtain expressions for $[E]$ and $[ES]$ in terms of $[EP]$ (since the initial rate is directly proportional to $[EP]$: $v = k_5[EP]$) and then substitute into the first equation to obtain

$$\begin{aligned}[E_{\text{total}}] &= [E] + [ES] + [EP] \\ &= \left(\frac{k_2(k_4 + k_5) + k_3k_5}{k_1k_3[S]} + \frac{k_4 + k_5}{k_3} + 1 \right) [EP]\end{aligned}$$

which is substituted into the denominator of $v = k_5[EP][E_{\text{total}}]/[E_{\text{total}}]$. The $[EP]$ terms thus cancel and the equation rearranges into

$$v = \frac{k_1k_3k_5[S][E_{\text{total}}]}{k_2(k_4 + k_5) + k_3k_5 + k_1(k_3 + k_4 + k_5)[S]}$$

By dividing the numerator and denominator by $k_1(k_3 + k_4 + k_5)$, the coefficient of the substrate concentration term in the denominator, this equation can be converted to the same general form as before:

$$v = V_{\text{max}}[S]/([S] + K_m)$$

where $V_{\text{max}} = k_3k_5[E_{\text{total}}]/(k_3 + k_4 + k_5)$ and $K_m = \{k_2(k_4 + k_5) + k_3k_5\}/\{k_1(k_3 + k_4 + k_5)\}$. In this case, if $k_3 \ll k_5$ and $k_4 \ll k_5$, the rate limiting step becomes the breakdown of ES in the forward direction, and the rate expression reduces to the steady state derivation for the result we obtained earlier with the single EX intermediate.

It is worth repeating that in the absence of knowledge concerning the magnitudes of individual rate constants, the two-intermediate case is indistinguishable from the single intermediate example derived previously. Thus, initial rate steady-state kinetics alone tells us nothing about the rate processes occurring after substrate binding and before product release.

DERIVATION OF MORE COMPLICATED RATE EQUATIONS. So far, the rate equations that describe one-substrate enzyme systems have been fairly simple, and the usual algebraic manipulations of substitution and/or addition of simultaneous equations have permitted us to obtain the pertinent rate law. When the number of steps increases and especially when there are branched pathways involved, these manual methods become cumbersome, and more systematic procedures are required. The next two sections should allow the reader to develop a working knowledge of effective methods for obtaining multisubstrate enzyme rate expressions.

ENZYME RATE EQUATIONS (2. Derivation of Initial Velocity Equations*)

A rate equation for an enzymic reaction is a mathematical expression that depicts the process in terms of rate constants and reactant concentrations. It serves as a link between the experimentally observed kinetic behavior

and a plausible model or mechanism. The characteristics of the rate equation permit tests to be designed to verify the mechanism. Conversely, the experimental observations provide clues to what the mechanism may be, hence, what form the rate expression shall take.

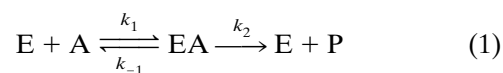
Derivation of rate equations is an integral part of the effective usage of kinetics as a tool. Novel mechanisms must be described by new equations, and familiar ones often need to be modified to account for minor deviations from the expected pattern. The mathematical manipulations involved in deriving initial velocity or isotope exchange-rate laws are in general quite straightforward, but can be tedious. It is the purpose of this entry, therefore, to present the currently available methods with emphasis on the more convenient ones.

The derivation of initial velocity equations invariably entails certain assumptions. In fact, these assumptions are often conditions that must be fulfilled for the equations to be valid. Initial velocity is defined as the reaction rate at the early phase of enzymic catalysis during which the formation of product is linear with respect to time. This linear phase is achieved when the enzyme and substrate intermediates reach a steady state or quasi-equilibrium. Other assumptions basic to the derivation of initial rate equations are as follows:

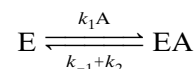
1. The enzyme and the substrate form a complex.
2. The substrate concentration is much greater than the enzyme concentration, so that the free substrate concentration is equivalent to the total concentration. This condition further requires that the amount of product formed is small, such that the reverse reaction or product inhibition is negligible.
3. During the reaction, constant pH, temperature, and ionic strength are maintained.

STEADY-STATE TREATMENT. During the steady state, the concentrations of various enzyme intermediates are essentially unchanged; that is, the rate of formation of a given intermediate is equal to its rate of disappearance. This assumption was first introduced to the derivation of enzyme kinetic equations by Briggs and Haldane¹.

To derive a rate equation, the first step is to write a reaction mechanism. The nomenclature used by Fromm² will be adopted here with the exception that rate constants in the forward and reverse directions will be denoted by positive and negative subscripts. For example, the simplest one substrate—one product reaction can be written as:



Since both the k_{-1} and k_2 steps (or branches) lead from EA to E, the two branches, as has been shown by Volkenstein and Goldstein³, can be combined into a single branch. This simplification procedure will be used whenever feasible.



The initial rate is given by

$$v = d[P]/dt = k_2[EA]$$

Applying the steady-state assumption, we have

$$d[EA]/dt = k_1[A][E] - (k_{-1} + k_2)[EA] = 0 \quad (2)$$

To obtain an expression for [EA], the enzyme conservation equation

$$\text{Total enzyme} = [E_o] = [E] + [EA] \quad (3)$$

is required. Substitution of $[E] = ([E_o] - [EA])$ into Eq. (2) yields

$$[ES] = \frac{[E_o][A]}{\{(k_{-1} + k_2)/k_1\} + [A]}$$

and

$$\begin{aligned} v = k_2[EA] &= \frac{k_2[E_o][A]}{\{(k_{-1} + k_2)/k_1\} + [A]} \\ &= \frac{V_1[A]}{K_m + [A]} \end{aligned} \quad (4)$$

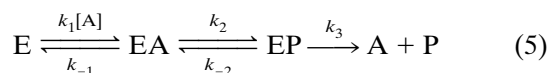
where V_1 is the maximum velocity in the forward direction and K_m is the Michaelis constant.

It should be noted that the validity of the steady-state method does not depend on the assumption $d[EA]/dt = 0$. Without setting Eq. (2) equal to zero, one can obtain the following expression from Eqs. (2) and (3):

$$[EA] = \frac{k_1[A][E_o] - d[EA]/dt}{k_1[A] + k_{-1} + k_2}$$

Wong⁴ has pointed out that the steady-state approximation only requires that $d[\text{EA}]/dt$ be small compared with $k_1[\text{A}][\text{E}_0]$. In the early phase of the reaction, if $[\text{A}] \gg [\text{E}_0]$, the rate of change of $[\text{EA}]$ due to diminishing $[\text{A}]$ will be relatively slow. It is clear that the validity of steady state is intimately tied to the condition of high substrate to enzyme ratio.

The Determinant Method. For a mechanism involving several enzyme-containing species, derivation of the rate equation can be done by solving the simultaneous algebraic equations by the determinant method. Consider the mechanism described by Eq. (1) with the addition of an EP intermediate.



The three simultaneous equations are given in the following form:

$$\begin{array}{l} \text{E} \quad \text{EA} \quad \text{EP} \\ d[\text{E}]/dt = \begin{vmatrix} -k_1[\text{A}] & k_{-1} & k_3 \\ k_1[\text{A}] & -(k_{-1} + k_2) & k_{-2} \\ 0 & k_2 & -(k_{-2} + k_3) \end{vmatrix} = 0 \\ d[\text{EA}]/dt = \begin{vmatrix} k_1[\text{A}] & -(k_{-1} + k_2) & k_{-2} \\ 0 & k_2 & -(k_{-2} + k_3) \end{vmatrix} = 0 \\ d[\text{EP}]/dt = \begin{vmatrix} 0 & k_2 & -(k_{-2} + k_3) \end{vmatrix} = 0 \end{array}$$

The determinant, or distribution term for $[\text{E}]$, for example, can be calculated from the coefficients listed above, after deleting the E column. For a mechanism of n intermediates, only $n - 1$ equations are needed. Thus, by leaving out the $d[\text{EP}]/dt$ row, we can write

$$[\text{E}] = \begin{vmatrix} k_{-1} & k_3 \\ -(k_{-1} + k_2) & k_{-2} \end{vmatrix} = k_{-1}k_{-2} + k_3(k_{-1} + k_2)$$

If the $d[\text{E}]/dt$ row is omitted instead, we have

$$\begin{aligned} [\text{E}] &= \begin{vmatrix} -(k_{-1} + k_2) & k_{-2} \\ k_2 & -(k_{-2} + k_3) \end{vmatrix} \\ &= k_{-1}(k_{-2} + k_3) + k_2(k_{-2} + k_3) - k_2k_{-2} \\ &= k_{-1}k_{-2} + k_3(k_{-1} + k_2) \end{aligned}$$

Note that deletion of different equations often leads to different amounts of algebraic manipulations. Application of the same operations to $[\text{EA}]$ and $[\text{EP}]$ yields

$$\begin{aligned} [\text{EA}] &= k_1(k_{-2} + k_3)[\text{A}] \\ [\text{EP}] &= k_1k_2[\text{A}] \end{aligned}$$

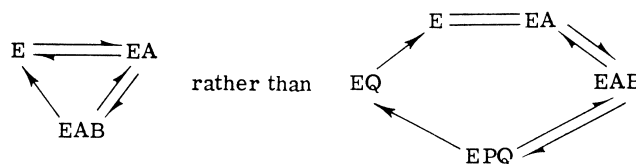
The rate equation is readily obtained as

$$\begin{aligned} \frac{v}{[\text{E}_0]} &= \frac{k_3[\text{EP}]}{[\text{E}] + [\text{EA}] + [\text{EP}]} \\ &= \frac{k_1k_2k_3[\text{A}]}{k_{-1}k_{-2} + k_3(k_{-1} + k_2) + k_1(k_{-2} + k_3)[\text{A}] + k_1k_2[\text{A}]} \end{aligned}$$

or

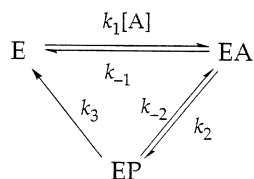
$$\begin{aligned} v &= \frac{k_2k_3[\text{E}_0][\text{A}]/(k_2 + k_{-2} + k_3)}{\{[k_{-1}k_{-2} + k_3(k_{-1} + k_2)]/[k_1(k_2 + k_{-2} + k_3)]\} + [\text{A}]} \\ &= \frac{V_1[\text{A}]}{K_m + [\text{A}]} \quad (6) \end{aligned}$$

Equation (6) is identical in form with Eq. (4). In fact, if $k_3 \gg k_2, k_{-2}$, Eq. (6) reduces to Eq. (4). Although Eq. (5) is a more realistic mechanism compared with Eq. (1), especially when the rapid-equilibrium treatment is applied to the reversible reaction, the information obtainable from initial-rate studies of such unireactant system remains nevertheless the same: V_1 and K_m . This serves to justify the simplification used by the kineticist; that is, the elimination of certain intermediates to maintain brevity of the rate equation (provided the mathematical form is unaltered). Thus, the *forward* reaction of an ordered Bi Bi mechanism is generally written as diagrammed below.



This use of the determinant method for complex enzyme mechanisms is time-consuming because of the stepwise expansion and the large number of positive and negative terms that must be canceled. It is quite useful, however, in computer-assisted derivation of rate equations⁵.

The King and Altman Method. King and Altman⁶ developed a systematic approach for deriving steady-state rate equations, which has contributed to the advance of enzyme kinetics. The first step of this method is to draw an *enclosed* geometric figure with each enzyme form as one of the corners. Equation (5), for instance, can be rewritten as:



The second step is to draw all the possible patterns that connect all the enzyme species without forming a loop. For a mechanism with n enzyme species, or a figure with n corners, each pattern should contain $n - 1$ lines. The number of valid patterns for any single-loop mechanism is equal to the number of enzyme forms. Thus, there are three patterns for the triangle shown above:



The determinant for a given enzyme species is obtained as the summation of the product of the rate constants and concentration factors associated with all the branches in the patterns *leading toward* this particular enzyme species. The same patterns are used for each species, albeit the direction in which they are read will vary. However, when an irreversible step is present, *e.g.*, the $EP \rightarrow E$ step, some patterns become invalid for certain enzyme forms.

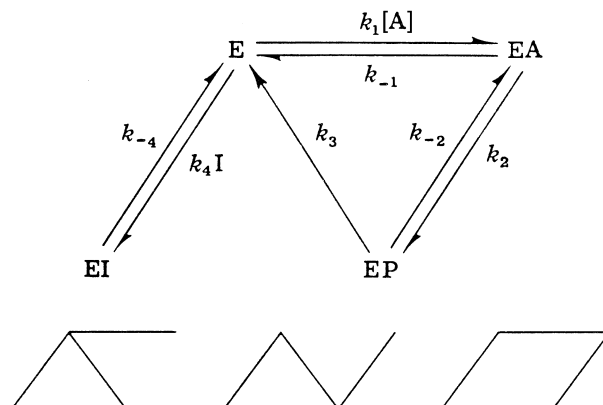
$$\begin{aligned}
 [E] &= \begin{array}{c} \xleftarrow{k_{-1}} \\ \swarrow k_3 \end{array} + \begin{array}{c} \swarrow k_3 \\ \searrow k_2 \end{array} + \begin{array}{c} \xleftarrow{k_{-1}} \\ \searrow k_{-2} \end{array} \\
 &= k_{-1}k_3 + k_2k_3 + k_{-1}k_{-2} \\
 [EA] &= \begin{array}{c} \xrightarrow{k_1[A]} \\ \swarrow k_3 \end{array} + 0 + \begin{array}{c} \xrightarrow{k_1[A]} \\ \searrow k_{-2} \end{array} \\
 &= k_1k_3[A] + k_{-1}k_{-2}[A] \\
 [EP] &= 0 + 0 + \begin{array}{c} \xrightarrow{k_1[A]} \\ \searrow k_2 \end{array} \\
 &= k_1k_2[A]
 \end{aligned}$$

The rate equation is obtained as

$$\frac{v}{[E_o]} = \frac{k_3[EP]}{[E] + [EA] + [EP]}$$

where $[E]$, $[EA]$, and $[EP]$ are the determinants for E, EA, and EP, respectively.

The presence of an enzyme intermediate(s) that is not part of a loop will not affect the number of King-Altman patterns. For instance, the addition of a competitive inhibitor, I, to the above system will result in the same number of patterns.



The additional $E \rightleftharpoons EI$ branch is present in *all* the diagrams. Thus, in calculating the number of valid King-Altman patterns, only the *closed loops* need be considered. The determinants of E, EA, EAB, and EI can be obtained by the method just described:

$$[E] = k_{-4}(k_{-1}k_3 + k_2k_3 + k_{-1}k_{-2})$$

$$[EA] = k_{-4}(k_1k_3[A] + k_{-1}k_{-2}[A])$$

$$[EP] = k_{-4}(k_1k_2[A])$$

$$[EI] = k_4[I](k_{-1}k_3 + k_2k_3 + k_{-1}k_{-2})$$

It is more convenient, however, to treat this case by first considering only the loop portion (ignoring the additional $E \rightleftharpoons EI$ step for the time being).

$$[E] = k_{-1}k_3 + k_2k_3 + k_{-1}k_{-2}$$

$$[EA] = k_1(k_{-2} + k_3)[A]$$

$$[EP] = k_1k_2[A]$$

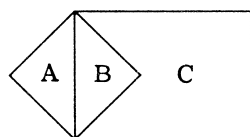
The determinant for EI is then obtained as

$$\begin{aligned}
 [EI] &= [E]k_4[I]/k_{-4} \\
 &= (k_{-1}k_3 + k_2k_3 + k_{-1}k_{-2})k_4[I]/k_{-4} \\
 &= (k_{-1}k_3 + k_2k_3 + k_{-1}k_{-2})[I]/K_i
 \end{aligned}$$

where $K_i = k_{-4}/k_4$.

The King-Altman method is most convenient for single-loop mechanisms. In practice, there is no need to write down the patterns. One can use an object; say, a paper clip, to block one branch of the loop, write down the appropriate term for each enzyme species, then repeat the process until every branch in the loop has been blocked once.

For more complex mechanisms having alternate pathways that form several closed loops, the precise number of valid King-Altman patterns must be calculated to avoid omission of terms. To illustrate the various situations that may occur, let us consider Scheme 1.



Scheme 1

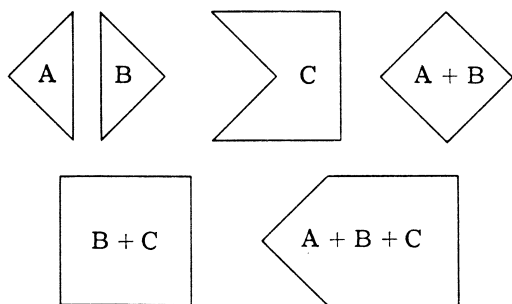
The total number of patterns with $n - 1$ lines is given by

$$\frac{m!}{(n-1)!(m-n+1)!}$$

where m = the number of lines in the complete geometric figure. In the above scheme, $m = 8$ and $n = 6$, and the total number of patterns with 5 lines is

$$\frac{8!}{5!3!} = \frac{(8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1)}{(5 \times 4 \times 3 \times 2 \times 1)(3 \times 2 \times 1)} = 56$$

This number, however, includes patterns that contain the following loops, which must be subtracted from the total:

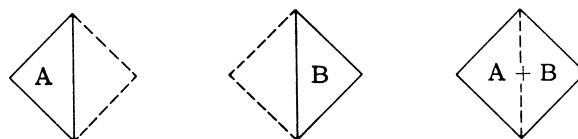


The number of patterns for a loop with r lines is given by

$$\frac{(m-r)!}{(n-1-r)!(m-n+1)!}$$

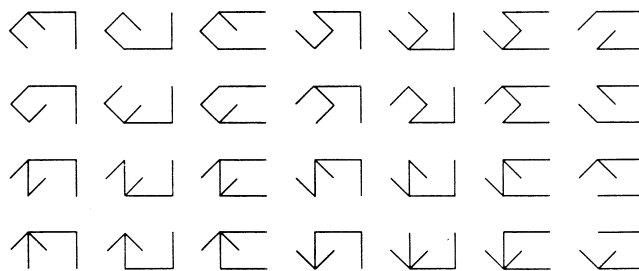
According to this equation, for loops A and B, $r = 3$, we have 10 patterns each; for loops A + B and B + C,

$r = 4$, we have 4 patterns each; and for loops C and A + B + C, $r = 5$ (note that $0! = 1$), we have 1 pattern each. The total number of loop-containing patterns to be subtracted is 30. One of the patterns, however, occurred three times in the above calculations, but should be discarded only once. This pattern involves both loop A and loop B (solid lines indicate the loop that gives rise to this pattern).



Thus, the total loop-containing patterns is 28, and the total number of valid patterns is $56 - 28 = 28$.

The 28 5-lined patterns are shown below.



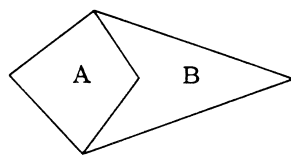
The conventional way of computing the valid King-Altman patterns is rather tedious. A set of formulas developed by the author allows the calculation of the desired number in a very short time. Each of these formulas is applicable to a particular geometric arrangement. For any figure consisting of three subfigures arrayed in sequence like the one shown in Scheme 1, the general formula for calculating the exact number of the valid King-Altman pattern, π , is

$$\pi = a \cdot b \cdot c - (l_{AB}^2 \cdot c + l_{BC}^2 \cdot a)$$

where a , b , and c = the number of lines in subfigures A, B, and C; l_{AB} and l_{BC} = the number of lines in the common boundaries between A and B, and B and C, respectively. For $a = 3$, $b = 3$, $c = 5$, $l_{AB} = 1$, and $l_{BC} = 2$ (Scheme 1), we have

$$\begin{aligned} \pi &= 3 \times 3 \times 5 - (1^2 \times 5 + 2^2 \times 3) \\ &= 45 - (5 + 12) \\ &= 28 \end{aligned}$$

In the case of two subfigures A and B sharing a common boundary as shown in Scheme 2



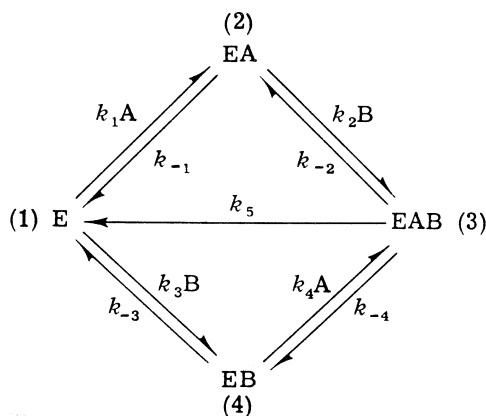
Scheme 2

the formula is given by

$$\begin{aligned}\pi &= a \cdot b - l_{AB}^2 \\ &= 4 \times 4 - 2^2 \\ &= 12\end{aligned}$$

Formulas for calculation of up to four subfigures in every possible geometric arrangement have been established.

The Method of Volkenstein and Goldstein. Volkenstein and Goldstein³ have applied the theory of graphs to the derivation of rate equations. Their approach has three main features: the use of an auxiliary “node,” the “compression” of a path into a point, and the addition of parallel branches. These can be best explained by an example (Scheme 3).

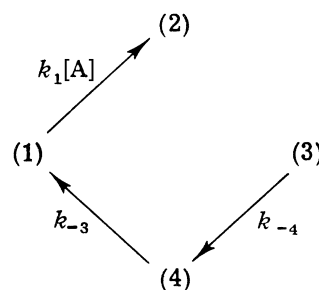


Scheme 3

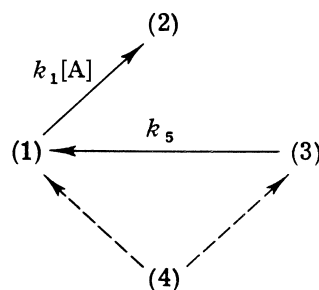
Each enzyme-containing species is assigned a number and referred to as a node.

Suppose we want to calculate the determinant for EA (node 2). First, we choose another node, say node 3, as the auxiliary node (a reference starting point). The choice of the auxiliary node is arbitrary; it will not affect the outcome of the derivation, but may affect the amount of work involved.

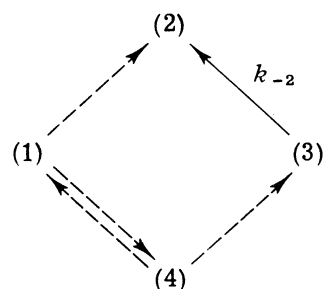
All the possible pathways (flow patterns) leading from (3) to (2) are then written (marked by solid branches).



Path 3412

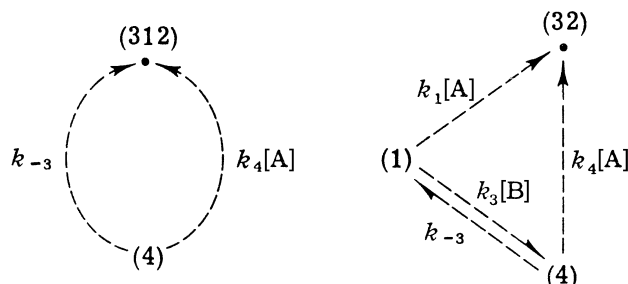


Path 312

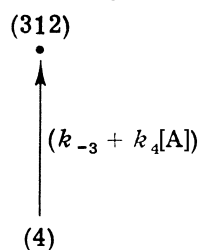


Path 32

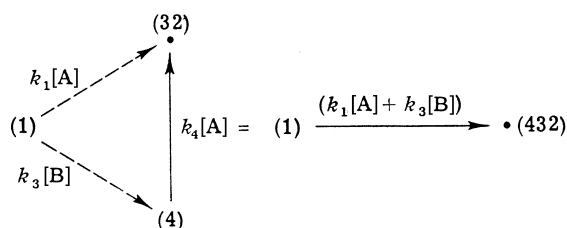
The nodes not included in the pathways retain the branches leading away from them (dashed branches). Since path 3412 flows through all the nodes, it is one of the terms of the determinant with a path value of $k_1 k_{-3} k_{-4} [A]$. Path 312 ($= k_1 k_5 [A]$) and path 32 ($= k_{-2}$) are now compressed into points.



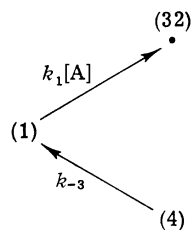
The parallel branches leading from (4) to the compressed point (312) can be added together to yield



and the expression for this part is $(P312)(k_{-3} + k_4[A]) = k_1k_5[A](k_{-3} + k_4[A])$. The part containing point (32) can be treated by selecting a secondary auxiliary node, say node (4), and repeating the procedure described at the onset.



and



The contribution from this part is $(P432)(k_1[A] + k_3[B]) = k_{-2}k_4[A](k_1[A] + k_3[B])$ and $(P32)(k_1k_{-3}[A]) = k_1k_{-2}k_{-3}[A]$. Adding the terms together, we obtain the determinant for EA

$$\begin{aligned} [EA] &= k_1k_{-3}k_{-4}[A] + k_1k_5[A](k_{-3} + k_4[A]) \\ &\quad + k_{-2}k_4[A](k_1[A] + k_3[B]) + k_1k_{-2}k_{-3}[A] \\ &= k_1k_{-3}(k_{-2} + k_{-4} + k_5)[A] + k_{-2}k_3k_4[A][B] \\ &\quad + k_1k_4(k_{-2} + k_5)[A]^2 \end{aligned}$$

The determinants for E, EB, and EAB can be obtained in a similar fashion. The complete rate equation is given by

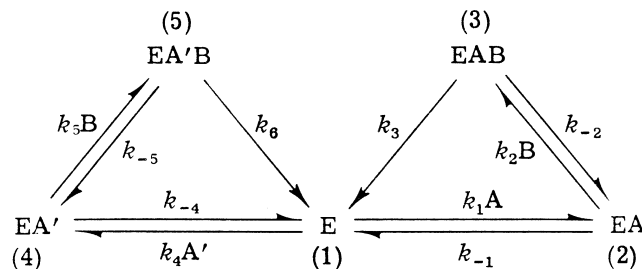
$$\frac{v}{[E_o]} = \frac{k_5[EAB]}{[E] + [EA] + [EB] + [EAB]}$$

Rate equations for more complex mechanisms can be derived by repeating the procedure described above as many times as necessary. The choice of the auxiliary

point becomes important for reaction schemes containing several loops. The process is analogous to deciding which row (equation) should be omitted from the matrix in the determinant method. In general, one should choose, by inspection of the geometric structure of the mechanism, a node such that, if one removes from the figure the auxiliary node and the node whose determinant is desired, the remaining nodes do not form a closed loop. In addition, one should select a node situated in a symmetrical position with respect to the desired node. For instance, node (4) is a better choice as an auxiliary node for the calculation of the determinant for node (2). Node (3) was chosen for the sole purpose of illustrating the use of secondary auxiliary nodes.

The Systematic Approach. The systematic approach for deriving rate equations was first devised by Fromm⁷ based on certain concepts advanced by Volkenstein and Goldstein³. Its underlying principles, however, are more akin to the graphic method of King and Altman⁶. The procedure to be described here⁸ is a modified method that includes the contributions from the aforementioned workers and from Wong and Hanes⁹.

Let us use as an example the ordered Bi Bi mechanism, in which an alternative substrate, A', for the first substrate, A, is present (Scheme 4).



Scheme 4

Each enzyme-containing species is assigned a node number as previously described in the Volkenstein and Goldstein method. For each node, a node value is written, which is simply the summation of all branch values (rate constant and concentration factor) leading away from the node⁷:

- (1) = $k_1[A] + k_4[A']$
- (2) = $k_{-1} + k_2[B]$
- (3) = $k_{-2} + k_3$
- (4) = $k_{-4} + k_5[B]$
- (5) = $k_{-5} + k_6$

The determinant of EA' and EA'B can be obtained by the same approach, and the complete rate equation is expressed as $v/[E_o]$. For the example given here, if the common product P is measured, we have

$$\frac{v}{[E_o]} = \frac{k_3[EAB] + k_6[EA'B]}{[E] + [EA] + [EAB] + [EA'] + [EA'B]}$$

The first rule for using the systematic approach, broadly stated, is as follows:

Rule 1: The determinant of a given enzyme-containing species is equal to the product of the node values of the other enzyme species, minus the reversible-step terms. When the nodes form one or more closed loops, apply the one-branch approach. Apply the consecutive-branch approach to any *remaining* loop-containing terms until all loops are eliminated.

It should be noted that unnecessary application of the one-branch approach may lead to needless algebraic manipulations. Furthermore, the branching technique often results in "redundant terms" that must be searched out and deleted. As an example, let us write the determinant for E in Scheme 4 by the one-branch method:

$$\begin{aligned} [E] &= (21)(3)(4)(5) + (31)(2)(4)(5) + (41)(2)(3)(5) \\ &\quad + (51)(2)(3)(4) \\ &= k_{-1}(k_{-2} + k_3)(k_{-4} + k_5[B])(k_{-5} + k_6) \\ &\quad + k_3(k_{-1} + k_2[B])(k_{-4} + k_5[B])(k_{-5} + k_6) \\ &\quad + k_{-4}(k_{-1} + k_2[B])(k_{-2} + k_3)(k_{-5} + k_6) \\ &\quad + k_6(k_{-1} + k_2[B])(k_{-2} + k_3)(k_{-4} + k_5[B]) \end{aligned}$$

Expansion of the above equation will generate many redundant terms; e.g., $k_{-1}k_3k_{-4}k_6$ can be found in all four terms above, but only one is needed. These redundant terms can be eliminated by using Rule 2.

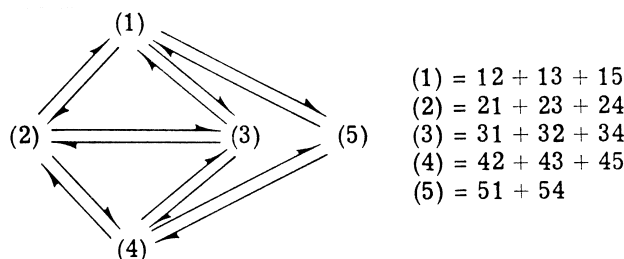
Rule 2: The nearest branch values cannot appear in *subsequent* node values. When the consecutive-branch approach is used, the *product* of the consecutive branches cannot appear in *subsequent* terms.

Consequently, the 21 term, k_{-1} , is crossed out from all subsequent terms; the 31 term, k_3 , from all subsequent node (3) terms, but not from the node (3) term preceding it; and the 41 term, k_{-4} , from the subsequent (4) term:

$$\begin{aligned} [E] &= k_{-1}(k_{-2} + k_3)(k_{-4} + k_5[B])(k_{-5} + k_6) \\ &\quad + k_3(k_{-1} + k_2[B])(k_{-4} + k_5[B])(k_{-5} + k_6) \\ &\quad + k_{-4}(k_{-1} + k_2[B])(k_{-2} + k_3)(k_{-5} + k_6) \\ &\quad + k_6(k_{-1} + k_2[B])(k_{-2} + k_3)(k_{-4} + k_5[B]) \end{aligned} \left. \vphantom{\begin{aligned} [E] &= k_{-1}(k_{-2} + k_3)(k_{-4} + k_5[B])(k_{-5} + k_6) \\ &\quad + k_3(k_{-1} + k_2[B])(k_{-4} + k_5[B])(k_{-5} + k_6) \\ &\quad + k_{-4}(k_{-1} + k_2[B])(k_{-2} + k_3)(k_{-5} + k_6) \\ &\quad + k_6(k_{-1} + k_2[B])(k_{-2} + k_3)(k_{-4} + k_5[B]) \end{aligned}} \right\} \text{Canceled}$$

After elimination of the redundant terms, the last two terms are canceled because they all contain the $k_2[B] \cdot k_{-2}$ reversible-step product. The remaining terms are identical with the expression obtained from $[E] = (2)(3)(4)(5)$, demonstrating the fact that unnecessary branching may lead to wasteful algebraic exercise.

There are situations where certain redundant terms are difficult to eliminate. The following example serves to illustrate a procedure useful for the complicated cases. Consider the hypothetical mechanism shown in Scheme 5.



Scheme 5

$$\begin{aligned} (1) &= 12 + 13 + 15 \\ (2) &= 21 + 23 + 24 \\ (3) &= 31 + 32 + 34 \\ (4) &= 42 + 43 + 45 \\ (5) &= 51 + 54 \end{aligned}$$

The numerical notation used for Scheme 5 has several advantages. (a) The tedium of writing down all the rate constants and concentration factors is avoided. The constants can be substituted into the final expression (after canceling all the loop terms, redundant terms, and reversible-step terms) to obtain the desired determinant. (b) When two or more steps are assigned the same rate constant, e.g., k_1 and k_{-1} , this method prevents the cancelation of $k_1 \cdot k_{-1}$ product formed from different steps. (c) The reversible-step terms can still be easily identified and eliminated by their numbers, e.g., $12 \cdot 21$, $35 \cdot 53$, etc.

Suppose we want to calculate the determinant for (5), we can write $D_5 = (1)(2)(3)(4)$. Applying the consecutive-

branch technique, we have

$$\begin{aligned}
 D_5 &= (15) (2) (3) (4) + (45) (1) (2) (3) \\
 &= 15 [21 (3) (4) + 31 (2) (4)] + 45 [24 (1) (3) \\
 &\quad + 34 (1) (2)] \\
 &= 15 [21 (31 + 32 + 34)(42 + 43 + 45) \\
 &\quad + 31 (\underline{21} + 23 + 24)(42 + 43 + 45)] \\
 &\quad + 45 [24 (12 + 13 + \underline{15})(\underline{31} + 32 + 34) \\
 &\quad + 34 (12 + 13 + \underline{15})(\underline{21} + 23 + \underline{24})]
 \end{aligned}$$

Note that within the same brackets, the redundant terms (underlined by solid lines) are readily removed, but the products of consecutive branch terms, $15 \cdot 21$ and $15 \cdot 31$ (underlined by dashed lines) within the second set of brackets cannot be canceled without further manipulation. One can either eliminate them during further expansion of the equation or repeat the consecutive-branch approach until *all* the consecutive branches end with a branch leading away from a *common* node. This “common-node consecutive branching” approach is always valid—even when all the loop-containing terms have been eliminated. Using this approach, we can write

$$\begin{aligned}
 D_5 &= 15 [21 (3) (4) + 31 (2) (4)] + 45 [24 (1) (3) \\
 &\quad + 34 (1) (2)] \\
 &= 15 \{21 (3) (4) + 31 [23 (4) + 43 \cdot 24]\} \\
 &\quad + 45 \{24 (1) (3) + 34 [23 (1) + 13 \cdot 21]\}
 \end{aligned}$$

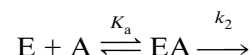
In the above operation, we chose node (2) as the common node; we could have chosen node (3) and obtained the same determinant. The principle involved here is analogous to the “auxiliary node” used by Volkenstein and Goldstein. It is not routinely used because the extra operations are not needed under most circumstances.

Comparison of Different Steady-State Methods. For relatively simple mechanisms, all the diagrammatic and systematic procedures illustrated in the foregoing sections are quite convenient. The King-Altman method is best suited for single-loop mechanisms, but becomes laborious for more complex cases with five or more enzyme forms because of the work involved in the calculation and drawing of valid patterns. With multiloop reaction schemes involving four to five enzyme species, the systematic approach requires the least effort, especially

when irreversible steps are present, since it does away with pattern drawing. When the number of enzyme forms reaches six or more in a mechanism with several alternate pathways, all the manual methods become tedious owing to the sheer number of terms involved.

THE RAPID-EQUILIBRIUM TREATMENT. The first rate equation for an enzyme-catalyzed reaction was derived by Henri and by Michaelis and Menten, based on the rapid-equilibrium concept. With this treatment it is assumed that there is a slow catalytic conversion step and the combination and dissociation of enzyme and substrate are relatively fast, such that they reach a state of quasi-equilibrium or rapid equilibrium.

The derivation of initial-velocity equations for any rapid equilibrium system is quite simple. When the equilibrium relationships among various enzyme-substrate complexes are defined, the rate equation can be written simply by inspection. Consider the one-substrate system

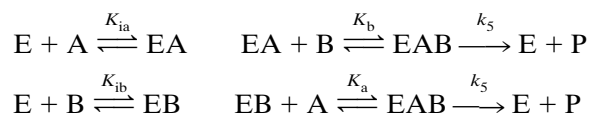


$$[EA] = \{[E][A]/K_a, v = k_2[EA]\}$$

We can write

$$\begin{aligned}
 \frac{v}{[E_o]} &= \frac{k_2[EA]}{[E] + [EA]} = \frac{k_2[E][A]/K_a}{[E] + [E][A]/K_a} \\
 &= \frac{k_2([A]/K_a)}{1 + [A]/K_a} = \frac{k_2[A]}{(K_a + [A])}
 \end{aligned}$$

It is clear that there is no need to write down $[E]$, and one can obtain the rate expression by replacing $[E]$ with 1. Thus, for the Random Bi Bi mechanism shown below



we can quickly write

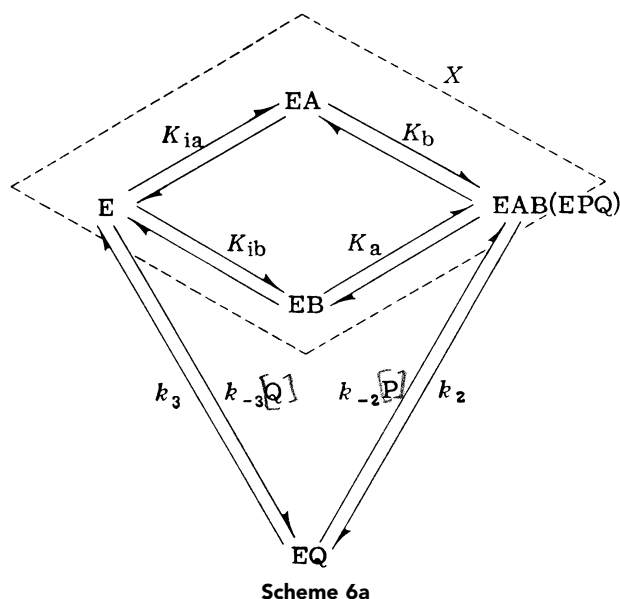
$$\frac{v}{[E_o]} = \frac{k_5[A][B]/(K_{ia}K_b)}{1 + [A]/K_{ia} + [B]/K_{ib} + [A][B]/K_{ia}K_b}$$

Note that although two equations describe the formation of EAB, only one of them is used (because $K_{ia}K_b = K_{ib}K_a$).

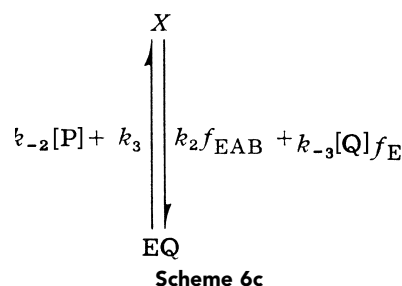
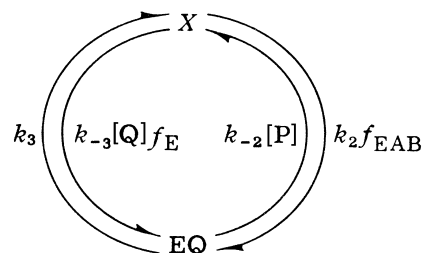
In using the equilibrium treatment, one should bear in mind that the rate laws so obtained are generally different in form from those derived by the steady-state assumption for the same mechanism.

THE COMBINED EQUILIBRIUM AND STEADY-STATE TREATMENT. There are a number of reasons why a rate equation should be derived by the combined equilibrium and steady-state approach. First, the experimentally observed kinetic patterns necessitate such a treatment. For example, several enzymic reactions have been proposed to proceed by the rapid-equilibrium random mechanism in one direction, but by the ordered pathway in the other. Second, steady-state treatment of complex mechanisms often results in equations that contain many higher-order terms. It is at times necessary to simplify the equation to bring it down to a manageable size and to reveal the basic kinetic properties of the mechanism.

The procedure to be described here was originally developed by Cha¹⁰. The basic principle of his approach is to treat the rapid-equilibrium segment as though it were a single enzyme species at steady state with the other species. Let us consider the hybrid Rapid-Equilibrium Random-Ordered Bi Bi system:



The area enclosed by dashed lines in Scheme 6a is the rapid-equilibrium random segment. The segment is called *X*, and Scheme 6a is reduced to a basic figure of *X* and EQ connected by two pathways (Scheme 6b):



Scheme 6b can be further condensed into one line (Scheme 6c) according to the principle of “addition of parallel branches.” The rate constants k_2 and $k_{-3}[Q]$ are multiplied by the fractional concentration of the enzyme species within the equilibrium segment that participates in that pathway, f_{EAB} and f_E .

$$f_{EAB} = \frac{[EAB]}{[E] + [EA] + [EB] + [EAB]}$$

$$= \frac{[A][B]/K_{ia}K_b}{1 + [A]/K_{ia} + [B]/K_{ib} + [A][B]/K_{ia}K_b}$$

$$f_E = \frac{1}{1 + [A]/K_{ia} + [B]/K_{ib} + [A][B]/K_{ia}K_b}$$

From Scheme 6c, we have

$$[EQ] = k_2 f_{EAB} [X] + k_{-3}[Q] f_E$$

$$[X] = k_{-2}[P] + k_3$$

Thus, we obtain the rate equation

$$\frac{v}{[E_o]} = \frac{k_2 f_{EAB} [X] - k_{-2}[P][EQ]}{[X] + [EQ]}$$

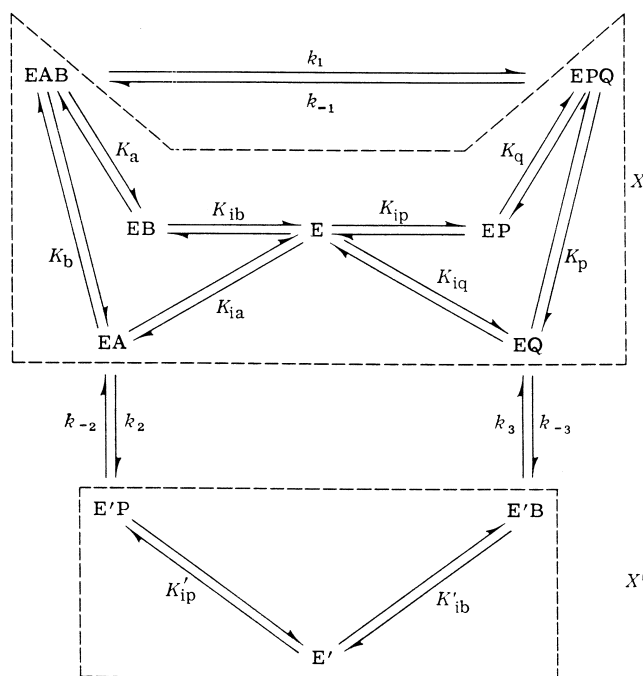
$$= \frac{k_2 k_3 f_{EAB} - k_{-2} k_{-3} [P][Q] f_E}{k_{-2}[P] + k_3 + k_2 f_{EAB} + k_{-3}[Q] f_E}$$

$$= \frac{k_2 k_3 ([A][B]/K_{ia}K_b - k_{-2} k_{-3} [P][Q])}{(k_{-2}[P] + k_3)(1 + [A]/K_{ia} + [B]/K_{ib} + [A][B]/K_{ia}K_b) + k_2([A][B]/K_{ia}K_b) + k_{-3}[Q]}$$

This equation reveals atypical product inhibition patterns for a random mechanism: P is noncompetitive with both A and B; Q is competitive with both A and B. Whenever abnormal product inhibition patterns are ob-

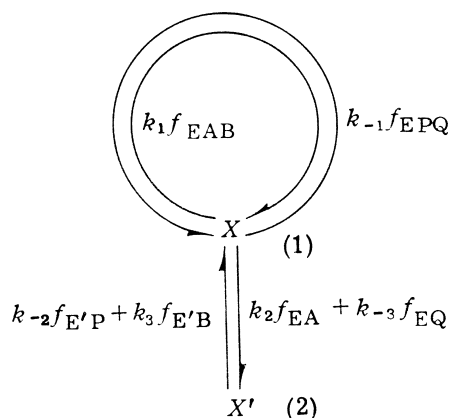
served, therefore, partial equilibrium treatment of the mechanism may be considered.

In the case of a rate-limiting step *within* a rapid equilibrium segment, it is necessary to include such a rate-limiting step in the velocity equation. Scheme 7a serves as an example for this type of mechanism:



Scheme 7a

The two rapid-equilibrium segments X and X' are again indicated by areas enclosed by dashed lines. Within X , there is a rate-limiting step involving the interconversion of EAB and EPQ. By treating X and X' as though they were two enzyme species, adding parallel branches together, and multiplying the rate constants with appropriate fractional enzyme concentrations, we arrive at Scheme 7b.



Scheme 7b

To obtain the determinants of X and X' , the internal rate-limiting step in X is not included since it leads both away from and toward X . This can be readily shown by applying the systematic method described in the derivation of steady-state equations. Recall that the node values for X and X' are the summation of branch values leading *away* from them, as follows:

$$(1) = k_2 f_{EA} + k_{-3} f_{EQ}$$

$$(2) = k_{-2} f_{E'P} + k_3 f_{E'B}$$

The determinants for X and X' are defined as the product of other node values excluding their own: that is, $X = (2) = k_{-2} f_{E'P} + k_3 f_{E'B}$ and $X' = (1) = k_2 f_{EA} + k_{-3} f_{EQ}$.

The fractional enzyme concentrations are obtained as their fractions in the appropriate equilibrium segments:

$$f_{EA} = \frac{[A]/K_{ia}}{1 + [A]/K_{ia} + [B]/K_{ib} + [A][B]/K_{ia}K_b + [P]/K_{ip} + [Q]/K_{iq} + [P][Q]/K_{ip}K_q}$$

$$f_{EQ} = \frac{[Q]/K_{iq}}{1 + [A]/K_{ia} + [B]/K_{ib} + [A][B]/K_{ia}K_b + [P]/K_{ip} + [Q]/K_{iq} + [P][Q]/K_{ip}K_q}$$

$$f_{EAB} = \frac{[A][B]/K_{ia}K_b}{1 + [A]/K_{ia} + [B]/K_{ib} + [A][B]/K_{ia}K_b + [P]/K_{ip} + [Q]/K_{iq} + [P][Q]/K_{ip}K_q}$$

$$f_{EPQ} = \frac{[P][Q]/K_{ip}K_q}{1 + [A]/K_{ia} + [B]/K_{ib} + [A][B]/K_{ia}K_b + [P]/K_{ip} + [Q]/K_{iq} + [P][Q]/K_{ip}K_q}$$

$$f_{E'B} = \frac{[B]/K_{ib}'}{1 + [B]/K_{ib}' + [P]/K_{ip}'}$$

$$f_{E'P} = \frac{[P]/K_{ip}'}{1 + [B]/K_{ib}' + [P]/K_{ip}'}$$

In writing an expression for the initial velocity for Scheme 7b, however, one must include the internal rate-limiting step. Thus, we have

$$v = (k_1 f_{EAB} - k_{-1} f_{EPQ})X + k_2 f_{EA}X - k_{-2} f_{E'P}X'$$

Note that in steady-state treatment, any of the pathways can be used to write the velocity expression. Either one of the two pathways linking X and X' (see Scheme 7a) will yield the same expression:

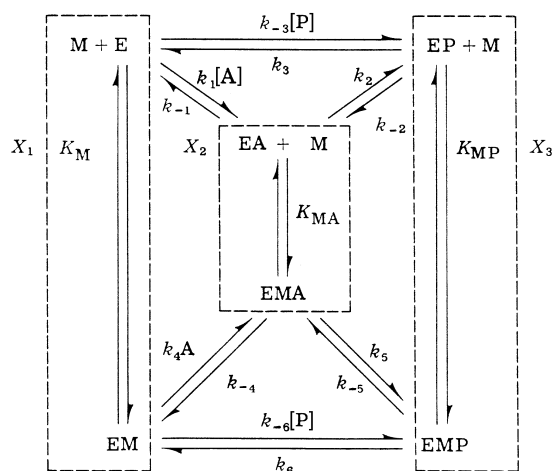
$$\begin{aligned}
 k_2 f_{EA} X - k_{-2} f_{EP} X' &= k_2 f_{EA} (k_{-2} f_{EP} + k_3 f_{EB}) \\
 &\quad - k_{-2} f_{EP} (k_2 f_{EA} + k_3 f_{EQ}) \\
 &= k_2 f_{EA} k_3 f_{EB} - k_{-2} f_{EP} k_{-3} f_{EQ}
 \end{aligned}$$

$$\begin{aligned}
 k_3 f_{EB} X' - k_{-3} f_{EQ} X &= k_3 f_{EB} (k_2 f_{EA} + k_{-3} f_{EQ}) \\
 &\quad - k_{-3} f_{EQ} (k_{-2} f_{EP} + k_3 f_{EB}) \\
 &= k_2 f_{EA} k_3 f_{EB} - k_{-2} f_{EP} k_{-3} f_{EQ}
 \end{aligned}$$

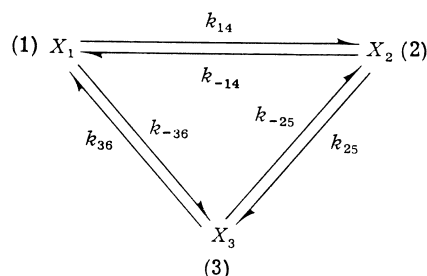
The complete rate equation is given by

$$\begin{aligned}
 \frac{\nu}{[E_o]} &= \frac{(k_1 f_{EAB} - k_{-1} f_{EPO})X + k_2 f_{EA} X - k_{-2} f_{EP} X'}{X + X'} \\
 &= \frac{(k_1 f_{EAB} - k_{-1} f_{EPO})(k_{-2} f_{EP} + k_3 f_{EB})}{k_{-2} f_{EP} + k_3 f_{EB} + k_2 f_{EA} + k_{-3} f_{EQ}}
 \end{aligned}$$

For mechanisms involving three or more rapid-equilibrium segments, once the segments are properly represented as “nodes” in a scheme, the rate equation can be obtained by the usual “systematic approach.” For example, consider the case of one substrate—one product reaction in which a modifier M is in rapid equilibrium with E, EA, and EP.



Treating X_1 , X_2 , and X_3 as nodes and adding parallel branches together, we obtain



where $k_{14} = k_1[A]f_E + k_4[A]f_{EM}$, $k_{-14} = k_{-1}f_{EA} + k_{-4}f_{EMA}$, $k_{25} = k_2f_{EA} + k_5f_{EMA}$, $k_{-25} = k_{-2}f_{EP} + k_{-5}f_{EMP}$, $k_{36} = k_3f_{EP} + k_6f_{EMP}$, and $k_{-36} = k_{-3}[P]f_E + k_{-6}[P]f_{EM}$. Note that $\nu = k_{36}X_3 - k_{-36}X_1$.

CONCLUDING REMARKS. In this entry, the derivation of initial-velocity equations under steady-state, rapid-equilibrium, and the hybrid rapid-equilibrium and steady-state conditions has been covered. Derivation of initial velocity equation for the quasi-equilibrium case is quite straightforward once the equilibrium relationships among various enzyme-containing species are defined. The combined rapid-equilibrium and steady-state treatment can be reduced to the steady-state method by treating the equilibrium segments as though they were enzyme intermediates.

Albeit several methods for deriving steady-state rate equations have been described, one has to be proficient at only one of them since, for a given reaction scheme, they all lead to the same equation. The method to be recommended here is the *systematic* approach. It requires the least amount of time and work, especially when irreversible steps are present in the reaction scheme. Furthermore, no pattern drawing is needed in this approach. The initial setup is done by systematic inspection of the reaction diagram, and the risk of omission of terms (due to omission of certain patterns) is minimized. It should be emphasized that the validity of a rate equation depends also on whether the reaction diagram is properly constructed; that is, whether the number and types of enzyme species involved, the pathways linking these species, the sequence of reaction steps, *etc.*, are carefully considered and appropriately arranged. While the procedures presented in this entry allow one to obtain correct algebraic expressions, the rate equations so obtained are only as valid as the assumptions made in constructing the reaction scheme. **See** *Fromm's Method For Deriving Enzyme Rate Equations; Net Rate Constant Method*

¹G. E. Briggs & J. B. S. Haldane (1925) *Biochem. J.* **19**, 338.

²H. J. Fromm (1979) *Meth. Enzymol.* **63**, 42.

³M. V. Volkenstein & B. N. Goldstein (1966) *Biochim. Biophys. Acta* **115**, 471.

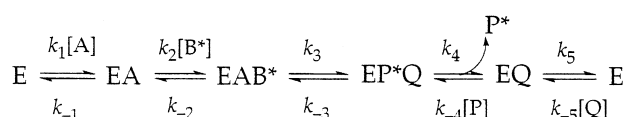
*This section, authored by Charles Y. Huang, originally appeared in *Methods in Enzymology*, volume 63, as the first part of Chapter [4], pp. 54-75 ("Derivation of Initial Velocity and Isotope Exchange Rate Equations"). All equations and references in this entry are numbered sequentially as they originally appeared.

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- ¹⁰S. Cha (1968) *J. Biol. Chem.* **243**, 820.

ENZYME RATE EQUATIONS (3. Derivation of Isotope Exchange Rate Equations*)

Isotope-exchange rate equations can be classified into two types: exchange at equilibrium and at steady state. The theory and technique of equilibrium exchange was pioneered by Boyer¹. Most applications of isotope-exchange methods to enzyme systems have been of this type. Under equilibrium conditions there are no net chemical changes; whereas under steady-state conditions there is net catalysis. Although the derivation of these two types of rate equations differs in the assumptions involved, an equation derived by the steady-state approach can be readily converted into one for equilibrium exchange. Therefore, procedures intended for steady-state exchange are equally applicable to the derivation of equilibrium-exchange rate equations.

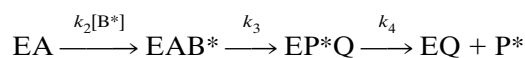
EQUATIONS FOR EXCHANGES AT EQUILIBRIUM. The derivation of equilibrium isotope-exchange equations for enzymic reactions was first formulated by Boyer¹. Others have subsequently contributed to its development. Yagil and Hoberman² and Flossdorf and Kula³ have devised generalized approaches that treat the flux of a label in a chemical reaction in a way analogous to the flow of charge in an electrical circuit. In this approach, (a) the equilibrium velocity of a reaction proceeding through n parallel steps is equal to the sum of the n individual rates; (b) when proceeding through n consecutive steps, the reciprocal of equilibrium velocity is equal to the sum of the n reciprocal. To demonstrate the use of this method, let us consider the B (R) P exchange in an Ordered Bi Bi mechanism (Scheme 1).



Scheme 1

The asterisks mark the labeled species. Since the system is at equilibrium and the exchange involves only three

steps, we can write a new scheme in the direction of isotopic flux:



The reverse steps are not shown because they are included in the equilibrium relationships to be substituted into the equation later.

Using the rule governing consecutive steps, we have

$$\frac{1}{v_{B \rightarrow P}^*} = \frac{1}{k_2[B^*][EA]} + \frac{1}{k_3[EAB^*]} + \frac{1}{k_4[EP^*Q]} \quad (1)$$

From the equilibrium relationships

$$\begin{aligned}
 [EAB^*] &= \frac{k_2[B^*][EA]}{k_{-2}} \\
 [EP^*Q] &= \frac{k_3[EAB^*]}{k_{-3}} = \frac{k_2k_3[B^*][EA]}{k_{-2}k_{-3}}
 \end{aligned}$$

we can substitute the expressions of $[EAB^*]$ and $[EP^*Q]$ into Eq. (1) to obtain

$$\frac{1}{v_{B \rightarrow P}^*} = \frac{k_3k_4 + k_{-2}(k_{-3} + k_4)}{k_2k_3k_4[B^*][EA]} \quad (2)$$

or

$$\frac{v_{B \rightarrow P}^*}{[E_o]} = \frac{k_2k_3k_4[B^*][EA]/[E_o]}{k_3k_4 + k_{-2}(k_{-3} + k_4)} \quad (3)$$

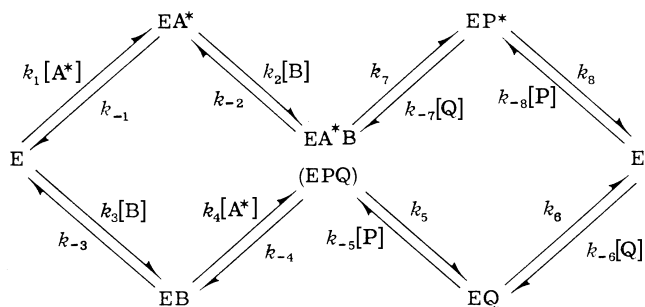
where

$$\frac{[EA]}{[E_o]} = \frac{[EA]}{[E] + [EA] + [EAB] + [EPQ] + [EQ]}$$

Hence,

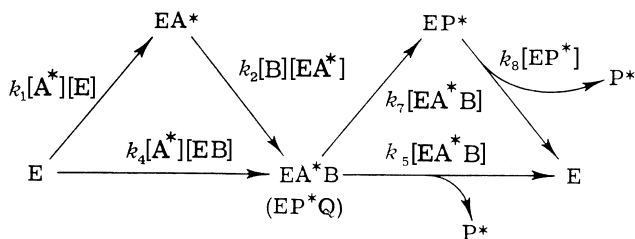
$$\begin{aligned}
 \frac{v_{B \rightarrow P}^*}{[E_o]} &= \frac{k_1k_2k_3k_4[A][B^*]}{k_{-1}[k_3k_4 + k_{-2}(k_{-3} + k_4)]} \\
 &\quad [1 + k_1[A]/k_{-1} + k_1k_2[A][B]/k_{-1}k_{-2} \\
 &\quad + k_{-5}[Q]/k_5 + k_{-4}k_{-5}[P][Q]/k_4k_5]
 \end{aligned}$$

The derivation of exchange-rate equations for mechanisms with branched pathways requires the rules governing consecutive and parallel steps. Consider as an example the $A \rightarrow P$ exchange in the Random Bi Bi mechanism (Scheme 2):



Scheme 2

Again, only the flux in one direction need be considered, and the steps not involved in the flux are ignored (Scheme 3).



Scheme 3

The first step is to obtain the expressions for the $E \rightarrow EA^* \rightarrow EA^*B$ and $EA^*B \rightarrow EP^* \rightarrow E$ pathways. Let us call them k_{12} and k_{78} :

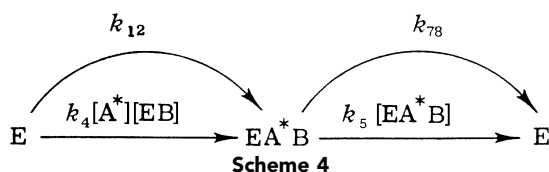
$$\frac{1}{k_{12}} = \frac{1}{k_1[A^*][E]} + \frac{1}{k_2[B][EA^*]};$$

$$k_{12} = \frac{k_1[A^*][E] \cdot k_2[B][EA^*]}{k_1[A^*][E] + k_2[B][EA^*]}$$

$$\frac{1}{k_{78}} = \frac{1}{k_7[EA^*B]} + \frac{1}{k_8[EP^*]};$$

$$k_{78} = \frac{k_7[EA^*B] \cdot k_8[EP^*]}{k_7[EA^*B] + k_8[EP^*]}$$

Scheme 3 can now be represented by Scheme 4.



Scheme 4

Since parallel steps are additive, we have



and finally

$$\frac{1}{v_{A \rightarrow P}^*} = \frac{1}{k_4[A^*][EB] + k_{12}} + \frac{1}{k_5[EA^*B] + k_{78}} \quad (4a)$$

The exchange-rate equation is obtained by substituting the expressions for k_{12} , k_{78} , and the equilibrium expressions for the enzyme intermediates into Eq. (4a).

From the following equilibrium relationships

$$[EA^*] = k_1[A^*][E]/k_{-1}$$

$$[EB] = k_3[B][E]/k_{-3}$$

$$[EA^*B] = k_4[A^*]k_3[B][E]/k_{-4}k_{-3}$$

$$[EP^*] = k_7[EA^*B]/k_{-7}[Q]$$

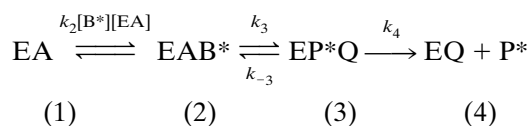
and

$$k_1[A]k_2[B] = k_3[A]k_4[B], \quad k_{-1}k_{-2} = k_{-3}k_{-4}$$

it can be shown that

$$\begin{aligned} \frac{1}{v_{A \rightarrow P}^*} &= \frac{1}{k_4[A^*][EB] + \{k_1[A^*][E] \cdot k_2[B][EA^*]\} / \{k_1[A^*][E] + k_2[B][EA^*]\}} \\ &= \frac{1}{k_5[EA^*B] + \{k_7[EA^*B] \cdot k_8[EP^*]\} / \{k_7[EA^*B] + k_8[EP^*]\}} \\ &= \frac{1}{[E]} \cdot \frac{\left\{ (k_{-1} + k_2[B])(k_{-7}[Q] + k_8)(k_{-4} + k_5) \right.}{\left\{ k_1[A]k_2[B] + k_4[A]k_3[B] \right\} \cdot \left\{ (k_{-1} + k_2[B])/k_{-3} \right\} \cdot \left\{ k_5(k_{-7}[Q] + k_8) + k_8k_7 \right\}} \end{aligned}$$

DERIVATION BY THE STEADY-STATE METHOD. Britton⁴ first derived isotope flux equations under steady-state rather than equilibrium conditions. To illustrate his procedure, we shall again use Scheme 1, the $B \rightarrow P$ exchange in Ordered Bi Bi mechanism, as an example, so that the results can be compared (Scheme 1a).



Scheme 1a

Note that (a) the reverse steps are needed for steady-state treatment; (b) the k_{-4} step is not shown because the initial rate of exchange is being measured; and (c) only the unlabeled enzyme concentration is included in the derivation because the concentration factors of labeled enzyme forms will cancel out during the derivation. Also, each enzyme form is assigned a number for reference purposes.

The procedure is to calculate the $B \rightarrow P$ or $1 \rightarrow 4$ isotope transfer in a stepwise manner using partition theory:

$$\text{Flux } 1 \rightarrow 2 = k_2[B^*][EA]$$

$$\text{Flux } 1 \rightarrow 3 = (1 \rightarrow 2) \cdot \frac{(2 \rightarrow 3)}{(2 \rightarrow 3) + (2 \rightarrow 1)}$$

$$= \frac{k_2[B^*][EA]k_3}{k_3 + k_{-2}}$$

$$\text{Flux } 1 \rightarrow 4 = (1 \rightarrow 3) \cdot \frac{(3 \rightarrow 4)}{(3 \rightarrow 4) + (3 \rightarrow 1)} \quad (5)$$

$$= \frac{k_2[B^*][EA]k_3}{k_3 + k_{-2}} \cdot \frac{k_4}{k_4 + \{k_{-2}k_{-3}/(k_3 + k_{-2})\}}$$

$$= \frac{k_2k_3k_4[B^*][EA]}{k_4(k_3 + k_{-2}) + k_{-2}k_{-3}}$$

$$= v_{B \rightarrow P}^*$$

Note that Eq. (5) is identical with Eq. (2). More recently, Cleland⁵ developed an approach that is more convenient. His procedure starts with the release of labeled product and works backwards as follows:

$$\text{Flux } 3 \rightarrow 4 = k_4$$

$$\text{Flux } 2 \rightarrow 4 = (2 \rightarrow 3) \cdot \frac{(3 \rightarrow 4)}{(3 \rightarrow 4) + (3 \rightarrow 2)}$$

$$= \frac{k_3k_4}{k_4 + k_{-3}}$$

$$\text{Flux } 1 \rightarrow 4 = (1 \rightarrow 2) \cdot \frac{(2 \rightarrow 4)}{(2 \rightarrow 4) + (2 \rightarrow 1)}$$

$$= \frac{k_2[B^*][EA]k_3k_4/(k_4 + k_{-3})}{\{k_3k_4/(k_4 + k_{-3})\} + k_{-2}}$$

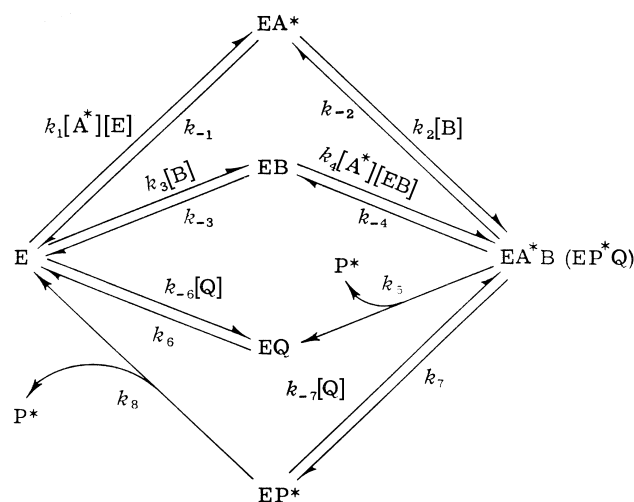
$$= \frac{k_2k_3k_4[B^*][EA]}{k_3k_4 + k_{-2}(k_4 + k_{-3})}$$

These procedures, however, are more suitable for deriving exchange-rate equations for mechanisms without branched pathways. For more complex mechanisms, the

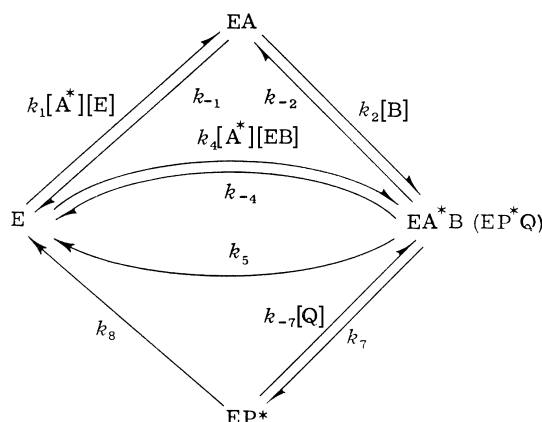
schematic method of Cleland⁶ based on the approach of King and Altman is quite convenient. This method can be adapted to the systematic approach and combined with the deletion of steps linking unlabeled species previously described to further reduce the amount of work involved⁷. Consider the $A \rightarrow P$ exchange in the Random Bi Bi mechanism (cf. Scheme 2; the figure has been redrawn in a folded geometric form) shown in Scheme 5.

The pathways connecting the unlabeled enzymes species, $E \rightleftharpoons EB$ and $E \rightleftharpoons EQ$, as has been shown in Scheme 3, can be eliminated; and the $EB \rightleftharpoons EAB$ and $EAB \rightarrow EQ$ steps can be directly linked to E (Scheme 5a).

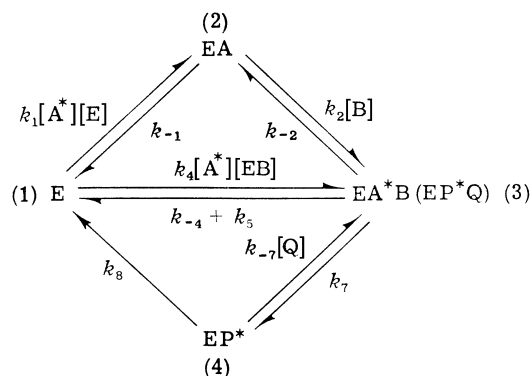
By combining the k_{-4} and k_5 branches, Scheme 5a can be further reduced to Scheme 5b.



Scheme 5



Scheme 5a



Scheme 5b

Now we apply the systematic method to Scheme 5b by assigning node numbers to the enzyme species and write down the node values:

$$(1) = k_1[A^*][E] + k_4[A^*][EB]$$

$$(2) = k_{-1} + k_2[B]$$

$$(3) = k_{-2} + k_{-4} + k_5 + k_7$$

$$(4) = k_{-7}[Q] + k_8$$

We then use Cleland's procedure to write the expression for the initial rate of formation of labeled P from A.

$$\begin{aligned} v^* &= d[P^*]/dt = k_5[EA^*B] + k_8[EP^*] \\ &= \frac{k_5 N_{EA^*B} + k_8 N_{EP^*}}{D} \end{aligned}$$

where N_{EA^*B} , N_{EP^*} , and D are the "determinants" of EA^*B , EP^* , and E in Scheme 5b. The N terms vary with mechanism, depending on the enzyme species from which the labeled product is released, but the D term is always obtained as the "determinant" of E .

Thus, we can write

$$\begin{aligned} N_{EA^*B} &= (1)(2)(4) = \{k_1[A^*][E] + k_4[A^*][EB]\} \\ &\quad (k_{-1} + k_2[B])(k_{-7}[Q] + k_8) \\ &= k_1[A^*][E](k_2[B])(k_{-7}[Q] + k_8) \\ &\quad + k_4[A^*][EB](k_{-1} + k_2[B])(k_{-7}[Q] + k_8) \end{aligned}$$

$$\begin{aligned} N_{EP^*} &= (3)(4)(1) = k_7\{k_1[A^*][E] \\ &\quad + k_4[A^*][EB]\}(k_{-1} + k_2[B]) \\ &= k_7 \cdot k_1[A^*][E] \cdot k_2[B] \\ &\quad + k_7 k_4[A^*][EB](k_{-1} + k_2[B]) \end{aligned}$$

$$\begin{aligned} D &= (2)(3)(4) = (k_{-1} + k_2[B]) \\ &\quad (k_{-2} + k_{-4} + k_5 + k_7)(k_{-7}[Q] + k_8) \\ &= (k_{-4} + k_5)(k_{-1} + k_2[B])(k_{-7}[Q] + k_8) \\ &\quad + k_{-1}k_{-2}(k_{-7}[Q] + k_8) + k_7k_8(k_{-1} + k_2[B]) \end{aligned}$$

The complete exchange equation is obtained as

$$\frac{v^*}{[E_o]} = \frac{(k_5 N_{EA^*B} + k_8 N_{EP^*})/D}{[E] + [EA] + [EB] + [EAB] + [EP] + [EQ]}$$

It should be noted that $[E]$, $[EA]$, $[EB]$, $[EAB]$, $[EP]$, and $[EQ]$ are now the *normal* determinants of E , EA , EB , EAB , EP , and EQ obtained from the steady-state treatment of the intact reaction scheme. If the equilibrium-exchange rate equation is desired, $[E]$, $[EA]$, $[EAB]$, $[EP]$, and $[EQ]$ should be obtained from the equilibrium relationships.

The following derivations demonstrate that an isotope exchange rate equation obtained by the steady-state treatment can be converted to one of exchange at equilibrium:

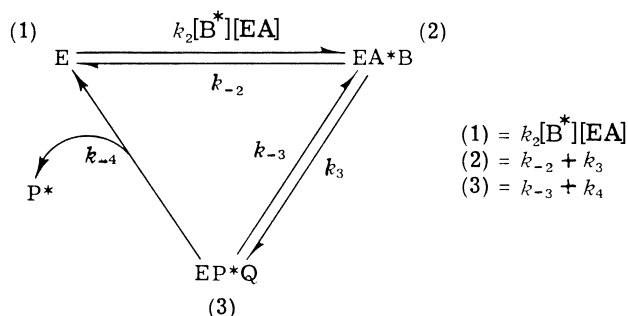
$$v^* = \frac{\left\{ \begin{aligned} &k_5\{k_1[A^*]k_2[B](k_{-7}[Q] + k_8)[E] + k_4[A^*] \\ &\cdot (k_{-1} + k_2[B])(k_{-7}[Q] + k_8)[EB]\} \\ &+ k_8\{k_7k_1[A^*]k_2[B][E] \\ &+ k_7k_4[A^*](k_{-1} + k_2[B])[EB]\} \end{aligned} \right\}}{\left\{ \begin{aligned} &(k_{-4} + k_5)(k_{-1} + k_2[B])(k_{-7}[Q] + k_8) \\ &+ k_{-1}k_{-2}(k_{-7}[Q] + k_8) + k_7k_8(k_{-1} + k_2[B]) \end{aligned} \right\}}$$

Substitution of $[EB] = k_3[B][E]/k_{-3}$ into the above expression yields

$$v^* = \frac{\begin{aligned} &k_5(k_{-7}[Q] + k_8) + k_8k_7\{k_1[A^*]k_2[B] \\ &+ k_4[A^*]k_3[B](k_{-1} + k_2[B])/k_{-3}\}[E] \end{aligned}}{(k_{-4} + k_5)(k_{-1} + k_2[B])(k_{-7}[Q] + k_8) + k_{-1}k_{-2}(k_{-7}[Q] + k_8) + k_7k_8(k_{-1} + k_2[B])}$$

which is identical with Eq. (4b) derived by the equilibrium treatment.

The systematic method is equally convenient for the derivation of rate equations for simple mechanisms. Scheme 1, for example, can be redrawn as an enclosed figure after deleting the pathways between unlabeled enzyme forms.



$$v_{B \rightarrow P}^* = \frac{k_4 N_{EP^*O}}{D}$$

$$N_{EP^*O} = (1)(2) = k_2[B^*][EA](k_{-2} + k_3) \\ = k_2 k_3 [B^*][EA]$$

$$D = (2)(3) = (k_{-2} + k_3)(k_{-3} + k_4) \\ = k_{-2}(k_{-3} + k_4) + k_3 k_4$$

Thus

$$v^* = \frac{k_2 k_3 k_4 [B^*][EA]}{k_{-2}(k_{-3} + k_4) + k_3 k_4}$$

¹P. D. Boyer (1959) *Arch. Biochem. Biophys.* **82**, 387.

²G. Yagil & H. D. Hoberman (1969) *Biochemistry* **8**, 352.

³J. Flossdorf & M. Kula (1972) *Eur. J. Biochem.* **30**, 325.

⁴H. G. Britton (1964) *J. Physiol. (London)* **170**, 1.

⁵W. W. Cleland (1975) *Biochemistry* **14**, 3220.

⁶W. W. Cleland (1967) *Ann. Rev. Biochem.* **36**, 77.

⁷C. Y. Huang (1978) *Fed. Proc., Fed. Am. Soc. Exp. Biol.* **37**, 1423.

*This entry, authored by Charles Y. Huang, originally appeared in *Methods in Enzymology*, volume 63, as the second part of Article [4], pp. 76–84 (“Derivation of Initial Velocity and Isotope Exchange Rate Equations”).

ENZYME STABILITY

In most kinetic investigations, one *assumes* the enzyme remains stable over the course of the measurement. When this is the case, corrective measures must be taken to obtain valid kinetic data. A useful test for any enzyme system is to plot enzyme activity *versus* time. This is readily accomplished by using a standardized assay (usually at optimal or saturating substrate concentrations) to measure the enzyme's specific activity periodically during the course of some experiment. This approach may fail to detect a reduction in activity characterized by lower affinity for substrate; however, use of a subsaturating substrate concentration in a time-course study will reveal this behavior.

Efforts should be made to stabilize an enzyme's activity. Certain agents (such as glycerol, ammonium ions, boric acid, polyethylene glycol, and even talcum powder or bentonite clay) have proven widely to be effective enzyme stabilizers. For multisubstrate enzymes, inclusion of one particular substrate with the enzyme (in the absence of other substrates or cofactors) often stabilizes an enzyme's catalytic activity. Such a substrate may also assist in “unlocking” the enzyme from a particularly inactive conformational form. In addition to substrates, other ligands and effectors (including reaction products,

effectors, and even reversible inhibitors) can often exert a stabilizing influence. Some investigators have successfully employed an inhibitor as a stabilizing agent in the concentrated stock solution of enzyme; when a small aliquot is added to the reaction mixture, the inhibitor's concentration is diluted, such that the inhibitory effect is undetectable¹.

Some interactions between macromolecules are relatively slow, and lags in the reaction are often minimized by preincubation. For multisubstrate enzymes acting on a protein substrate, one may need to incubate the enzyme and protein substrate together before initiating the reaction. An example is bovine brain tubulin:tyrosine ligase², and the observed lag is completely eliminated when the ligase and tubulin are incubated together prior to initiating the reaction.

Stabilization may also be achieved by altering reaction conditions (*e.g.*, pH, temperature, ionic strength, *etc.*). In other instances, loss of activity may attend the adsorption of the enzyme to the walls of the reaction container (*e.g.*, glass or plastic can strongly interact with proteins). Using a different container or simply including a “spectator” protein (*e.g.*, albumin) may increase the degree of stability.

The procedure adopted for isolating an enzyme can often influence enzyme stability. If the preparation procedure involves heat treatment or use of organic solvent, the enzyme may become “locked” into a less-active conformational form. The experimenter should consider using another preparative method.

Many enzymes (and protein substrates) require free thiol residues for proper conformation and/or activity. In this case, including a thiol reagent (such as 2-mercaptoethanol, dithiothreitol, dithioerythritol, or cysteine) may maintain activity.

Enzyme dilution often adversely influences enzyme stability. The presence of another protein (*e.g.*, 1–2 mg/mL serum albumin) may afford greater stability. When any protein is added to a system for stabilizing purposes, one should be certain that there is no effect on other substrates, products, or effectors in the assay mixture. Remember that some commercial preparations of bovine serum albumin, unless specifically “defatted” by extrac-

tion with an organic solvent, may contain fatty acids and other fat-soluble factors, Bovine serum albumin also binds ATP, and this can be a problem if the ATP concentration is varied in kinetic measurements conducted in the presence of BSA.

¹R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

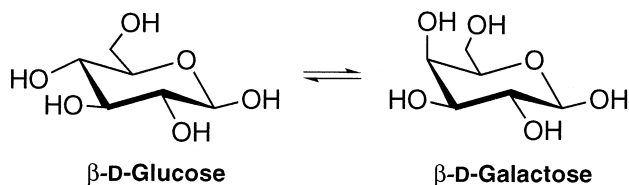
²N. L. Deans, R. D. Allison & D. L. Purich (1992) *Biochem. J.* **286**, 243.

ENZYME-SUBSTRATE COMPLEX

Any binary, ternary, or quaternary complex formed upon the association of a substrate with the free enzyme or other enzyme form; such a complex is often called the Michaelis complex.

EPIMERIZATION

The stereochemical isomerization or structural rearrangement resulting in the interconversion of epimers. For example, the conversion of β -D-glucose to β -D-galactose involves epimerization at the C-4 carbon atom, and the epimerization at the C-2 of β -D-glucose results in the synthesis of β -D-mannose.



EPITAXY

Growth of a crystal containing a different substance atop a pre-existing crystal. The occurrence of epitaxy suggests that the presence of the first crystal promotes a second, otherwise disfavored crystallization process. Epitaxy need not require the any intimate geometrical fit between the new and old solid phases, surface tension changes or weak adhesive forces can also be responsible. One theory of urinary stone disease posits that calcium oxalate monohydrate crystals do not spontaneously occur by homogeneous nucleation; rather, epitaxy is thought to take place at sites of mineralized calcium phosphate known as Randall's plaque.

EPOXIDE HYDROLASE

This enzyme [EC 3.3.2.3], also known as epoxide hydratase and arene-oxide hydratase, catalyzes the hydrolysis

of a variety of epoxides and arene oxides to yield the corresponding glycol.

R. N. Armstrong (1987) *Crit. Rev. Biochem.* **22**, 39.

trans-EPOXYSUCCINATE HYDROLASE

This enzyme [EC 3.3.2.4], also known as *trans*-epoxysuccinate hydratase and tartrate epoxidase, catalyzes the hydrolysis of *trans*-2,3-epoxysuccinate to yield *meso*-tartrate.

W. B. Jakoby & T. A. Fjellstedt (1972) *The Enzymes*, 3rd ed., **7**, 199.

EPSILON (ϵ)

1. The fifth letter in the Greek alphabet; hence, used to denote the fifth in a series (for example, the fifth methylene carbon in a fatty acid). 2. Symbol for molar absorption coefficient or extinction coefficient. 3. Symbol for permittivity (ϵ_0 refers to permittivity of a vacuum; ϵ_r refers to relative permittivity). 4. ϵ_a , Symbol for degree of activation (IUB (1982) *Eur. J. Biochem.* **128**, 281). 5. ϵ_i , Symbol for degree of inhibition. 6. Symbol for efficiency. 7. Symbol for linear strain. 8. Symbol for emittance.

EQUATIONS OF STATE

Thermodynamic expressions for the functional interdependence of a number of physical chemical parameters (usually, pressure, volume, temperature, and amount) of a particular substance. While equations of state have been developed for gases, liquids, and solids, the theories are most advanced for gaseous systems.

EQUI-EFFECTIVE DOSE/MOLARITY RATIO

The ratio (EDR) of two doses or the ratio (EMR) of two molarities of test and reference substances that are isoactive in their effectiveness as an agonist or antagonist.

EQUILIBRIUM

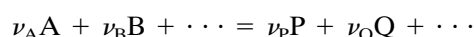
1. A condition of balance in a chemical system, at which no further change in reactant and product concentrations occurs. 2. If actions occurring within the reaction result in no net change in the reactant and product concentrations, the system is said to be in dynamic equilibrium. A chemical equilibrium is a dynamic equilibrium when the reaction rate in the forward direction is balanced by the rate in the reverse direction. The potential energy

of a system reaches a minimum when a system is at equilibrium. If the potential energy is at a maximum, the system's state is said to be unstable or metastable.

EQUILIBRIUM CONSTANT

A ratio equal to the product of the concentrations of all products at equilibrium divided by the product of the concentrations of all reactants. The equilibrium constant is most commonly symbolized by K_{eq} .

At equilibrium, there is no net change in the concentrations of any reactants or products, and the rate of the forward reaction equals the rate of the reverse reaction. For a general reaction,



then

$$K_{eq} = ([P]_{eq}^{\nu_P} [Q]_{eq}^{\nu_Q} \cdots) / ([A]_{eq}^{\nu_A} [B]_{eq}^{\nu_B} \cdots)$$

where $[A]_{eq}$, $[B]_{eq}$, etc., represent the concentrations of the substances at equilibrium. The negative logarithm of the equilibrium constant is pK_{eq} .

The equilibrium constant for a dissociation reaction is usually referred to as the dissociation constant (K_d), and the products of that dissociation are then written in the numerator. For reactions not at equilibrium, the value for the product of the concentrations of all products divided by the product of the concentrations of all reactants is frequently referred to as the mass action ratio.

See Association Constant; Dissociation Constant; Mass Action Ratio

EQUILIBRIUM CONSTANT DETERMINATIONS

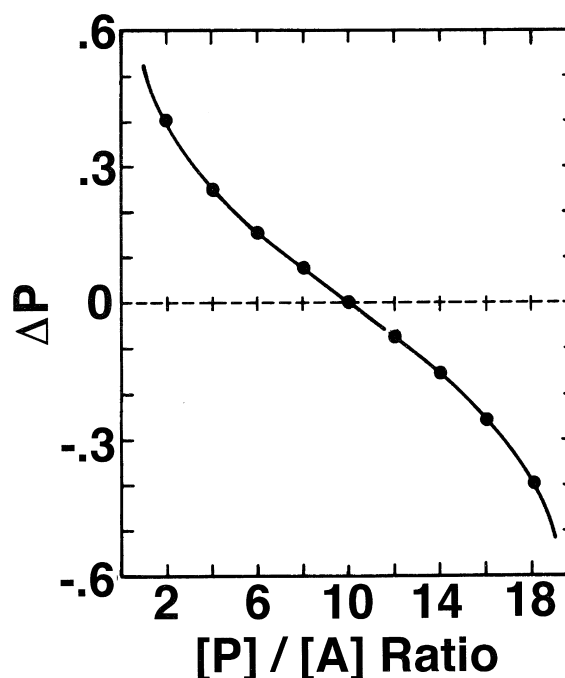
A number of methods can be utilized in determining the equilibrium position of chemical reactions:

1. Starting with substrates only (or products only), add sufficient enzyme to equilibrate the reaction and periodically sample the reaction mixture until substrate and product concentrations reach a time-invariant value. The difficulty with this method is that product inhibition can greatly retard the approach to equilibrium. In addition, if the enzyme is unstable, denaturation may occur before equilibration has in fact occurred. In such instances, a false K_{eq} will be obtained. Likewise, if a proton is con-

sumed or produced in the reaction, the pH of the reaction mixture may change. A good practice is to approach the equilibrium from both reaction directions; in such cases, the identical K_{eq} value must be obtained.

2. Applying the Haldane relation to obtain an equilibrium constant from initial rate kinetics. Because of the inherently greater uncertainty in determinations of rate parameters, this method often proves to be unreliable! **See** Haldane Relation

3. Another effective method of determining the equilibrium constant is to prepare a series of reaction samples in which all substrates and products are present at concentrations that would be expected to correspond roughly to the equilibrium constant. Once enzyme is added to mixtures having slightly different mass-action ratios, the investigator can plot these changes *versus* the initially chosen mass-action ratio. If the ratio is lower than the equilibrium constant, there will be net conversion to product (that is, measuring $D[P]$, possibly using labeled products and/or substrates). If the ratio is greater than the equilibrium constant, there will be net conversion to reactant. The mass action ratio that yields a zero deviation must exactly match the equilibrium constant.



For multisubstrate, multiproduct enzymes, one also has the opportunity to vary specific substrate:product pairs².

For example, consider a two-substrate, two-product reaction ($A + B = P + Q$). In a series of solutions, the concentrations of A and P can be set at several different absolute values such that the ratio is the same in all mixtures. The ratio of the other substrate:product pair ($[B]/[Q]$) can now be varied such that $[P][Q]/([A][B])$ varies about the estimated equilibrium constant. Once enzyme is added, deviations in one or more substrates or products are then measured after a sufficient incubation period.

4. The use of a doubly-labeled system is also ideal for demonstrating equilibration has occurred. For example, if ^{14}C -labeled substrate is mixed with ^3H -labeled product in the presence of enzyme, when equilibration has been reached, the $^3\text{H}:^{14}\text{C}$ ratio in substrate and product must be equal.

With each of these methods, the investigator should carefully note the amount of enzyme used. At very high enzyme concentrations, there is always the possibility a contaminating enzyme activity may alter the measurements. High enzyme concentrations can also lead to selective binding of a reaction component (e.g., metal ion), thereby displacing the K_{eq} . A good procedure is to determine the equilibrium constant at several enzyme concentrations. The value of K_{eq} should be independent of $[E_{\text{total}}]$. If this is not the case, the investigator has an opportunity to determine the basis for the perturbation.

The equilibrium constant of an enzyme-catalyzed reaction can depend greatly on reaction conditions. Because most substrates, products, and effectors are ionic species, the concentration and activity of each species is usually pH-dependent. This is particularly true for nucleotide-dependent enzymes which utilize substrates having $\text{p}K_{\text{a}}$ values near the pH value of the reaction. For example, both ATP^{4-} and HATP^{3-} may be the nucleotide substrate for a phosphotransferase, albeit with different K_{m} values. Thus, the equilibrium constant with ATP^{4-} may be significantly different than that of HATP^{3-} . In addition, most phosphotransferases do not utilize free nucleotides as the substrate but use the metal ion complexes. Both ATP^{4-} and HATP^{3-} have different stability constants for Mg^{2+} . If the buffer (or any other constituent of the reaction mixture) also binds the metal ion, the buffer (or that other constituent) can also alter the observed equilibrium constant^{1,2}.

Whenever reporting equilibrium constants, detailed information concerning the reaction conditions should always be indicated. Alberty³ has also presented an important review of biochemical thermodynamics in which he discusses the apparent equilibrium constant for biochemical reactions (K') in terms of sums of reactant species.

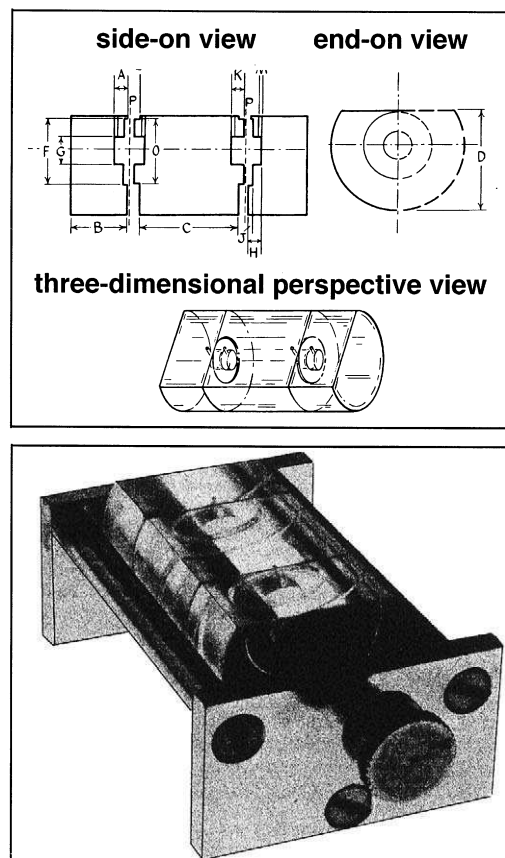
¹R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

²D. L. Purich & R. D. Allison (1980) *Meth. Enzymol.* **64**, 3.

³R. A. Alberty (1994) *Biochim. Biophys. Acta* **1207**, 1.

EQUILIBRIUM DIALYSIS

An equilibrium binding technique that allows one to measure the net accrual of macromolecule-bound ligand in one of two compartments separated by a semipermeable membrane.



Miniaturized dialysis cells for quantitative analysis of equilibrium ligand binding. Protein (20 μL) is placed in a compartment located on one side of a semipermeable membrane, and radiolabeled ligand is added to the second 20 μL -compartment. Equilibration is typically achieved within 90 min using these miniaturized cells. Equal volume aliquots are removed from both compartments, and the net accrual of radiolabel in the protein sample indicates the amount of bound ligand. (From P. T. Englund, J. A. Huberman, T. M. Jovin & A. Kornberg (1969) *J. Biol. Chem.* **244**, 3038, with permission of the authors and the American Society for Biochemistry & Molecular Biology.)

A now classical example of the power of equilibrium dialysis is the determination by Englund *et al.*¹ that DNA polymerase binds all four nucleoside 5'-triphosphate substrates at the same subsite in the enzyme's catalytic center. **See also** *Womack-Colowick Technique*

¹P. T. Englund, J. A. Huberman, T. M. Jovin & A. Kornberg (1969) *J. Biol. Chem.* **244**, 3038.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Theory, **76**, 561-562; determination, of association constant, **70**, 30; nonheme ligand studies, **76**, 535, 539-541; binding-parameter evaluation, **76**, 539-540; using bromthymol blue, **76**, 540-541; rapid rate of organophosphate-hemoglobin interaction, **76**, 559-577; binding assay, **76**, 567-570; buffers, **76**, 567; radioactive ligand synthesis, **76**, 562-567.

EQUILIBRIUM ENZYMES IN METABOLIC PATHWAYS

A commonly held belief is that, for an enzyme reaction within a metabolic pathway, a large excess of catalytic capacity relative to a pathway's metabolic flux ensures that a given step is at or near thermodynamic equilibrium. Brooks¹ recently treated the kinetic behavior of reaction schemes one might judge to be at equilibrium, and he showed that individual steps can remain far from equilibrium, even at a high ratio of an enzyme's flux to a pathway's steady-state flux. His calculations indicate that whether a reaction is near equilibrium depends on (a) the overall flux through the enzyme locus and (b) the kinetic parameters of the other enzymes in the pathway.

¹S. P. Brooks (1996) *Biochem. Cell Biol.* **74**, 411.

EQUILIBRIUM ISOTOPE EFFECT

A nonunity ratio (sometimes called a thermodynamic isotope effect) of the equilibrium constants ($K_{\text{light}}/K_{\text{heavy}}$) for two reactions differing only in the isotopic composition at one or more positions of their otherwise chemically identical substances^{1,2}. If the equilibrium isotope effect is attributable to a covalent bond making/breaking, then the effect is often referred to as a *primary* equilibrium isotope effect. If isotopic substitution at a position other than the scissile bond results in an equilibrium isotope effect, the term *secondary* equilibrium isotope effect is used.

Since potential energy surfaces of isotopic molecules are nearly identical, equilibrium isotope effects can only arise from the effect of isotopic mass on the nuclear motions of the reactants and products. Thus the ratio can be expressed in terms of partition functions for nuclear

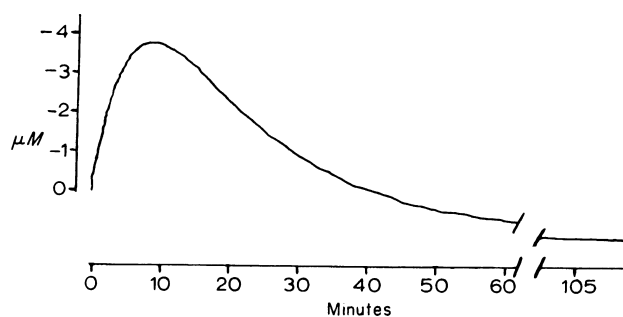
motion. In the case of light atoms at moderate temperatures (e.g., protium vs. deuterium), the equilibrium isotope effect is governed primarily by zero-point energy differences. **See** *Kinetic Isotope Effects*

¹IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

²M. Wolfsberg (1972) *Acc. Chem. Res.* **5**, 225.

EQUILIBRIUM PERTURBATION METHOD

The observed perturbation of a dynamic chemical process from its equilibrium position when isotopic substitution on a substrate or product changes the initial rate of conversion of substrate to product or the initial rate of conversion of product to substrate¹⁻³.



Equilibrium perturbation² with malic enzyme added to a pH 7.1 solution containing 0.419 mM malate-2-D, 0.079 mM NADPH, 20 mM potassium bicarbonate (3.8 mM carbon dioxide), 20 mM magnesium sulfate, 0.102 mM NADP⁺, and 3.83 mM pyruvate. NADPH initially disappears within the first 10 min, followed by its reappearance over the ensuing 60 min period.

The initially observed perturbation (or equilibrium isotope effect) will disappear as isotopic scrambling (or mixing) subsequently occurs. This is illustrated above for the malic enzyme; note the initial displacement of the equilibrium toward malate, followed by readjustment of the system to its final equilibrium position. The magnitude of the initial perturbation provides information on the occurrence of kinetic isotope effects and the nature of the rate-limiting step in an enzymatic process. **See also** *Kinetic Isotope Effect; Solvent Isotope Effect*

¹M. I. Schimerlik, J. E. Rife & W. W. Cleland (1975) *Biochemistry* **14**, 5347.

²W. W. Cleland (1980) *Meth. Enzymol.* **64**, 104.

³W. W. Cleland (1982) *Meth. Enzymol.* **87**, 641.

EQUILIBRIUM THERMODYNAMICS (Measurable Quantities)

Thermodynamic studies¹ of systems at equilibrium permit one to gain insights regarding mechanisms of chemical and biochemical reactions. Thermodynamic consider-

ations and inferences are founded on the experimenter's ability to evaluate accurately certain measurable quantities, mainly: heat capacities; latent heats, heats of mixing, heats of chemical reaction; equilibrium constants for reactions (including analogous electrochemical analogues of the chemical equilibrium constant); vapor pressures; solubilities; coefficients of compressibility and expansion; and volume changes attending a chemical reaction.

¹K. Denbigh (1968) *The Principles of Chemical Equilibrium*, 2nd ed., pp. 494, Cambridge University Press, London.

ERG

A unit of energy, work, or heat defined as the equivalent of a force of one dyne acting through a distance of one centimeter. A joule is equal to 10^7 ergs.

ERGODIC

A descriptor for a process that applies when every sizable sample represents the whole. The term stems from statistical mathematics and refers to any system which, after a sufficiently long time of wandering from one state to another, returns to states that are similar to those previously encountered. Assuming that system under study is ergodic is a standard practice in modern dynamics, statistical mechanics, and atomic theory. For example, chemical reaction dynamics frequently assumes that vibrational modes are well coupled (*i.e.*, ergodic) and redistribute energy rapidly with respect to the course of the reaction. Most large dynamical systems are assumed to exhibit ergodic dynamics, but in small systems, the free interchange of energy between degrees of freedom need not always occur.

ERGODIC PROCESS

A principle in statistical physics holding that, for each state within a system, there is an equal probability of occurrence; for this reason, a system can rapidly pass through all such states.

ERGOTHIONASE

This enzyme catalyzes the conversion of L-ergothioneine to 2-thiolurocanic acid and trimethylamine.

J. B. Wolff (1971) *Meth. Enzymol.* **17B**, 105.

ESTERASES

This is a broad class of enzymes that catalyze the hydrolysis of esters; some of these enzymes are quite specific.

See also *Cholesteryl Ester Hydrolase*; *specific esterase*

H. Okuda (1991) *A Study of Enzymes* **2**, 563.

K. Krisch (1971) *The Enzymes*, 3rd ed., **5**, 43.

ESTER HYDROLYSIS MECHANISMS

Esters can be hydrolyzed nonenzymatically in acidic and alkaline solutions. In both cases, a tetrahedral intermediate is indicated on the basis of the measured rates of hydrolysis and the rates of oxygen isotope exchange.

C. K. Ingold (1953) *Structure and Mechanism in Organic Chemistry*, Cornell University Press, Ithaca.

ETA (η)

1. Symbol for viscosity. 2. Symbol for the efficiency of a step. 3. The seventh letter of the Greek alphabet; hence, used to denote the seventh in a series (for example, the seventh methylene carbon in a fatty acid). 4. Symbol for overpotential.

ETHANOLAMINE AMMONIA-LYASE

This cobalamin-dependent enzyme [EC 4.3.1.7] catalyzes the interconversion of ethanolamine to form acetaldehyde and ammonia.

P. A. Frey & G. H. Reed (1993) *Adv. Enzymol.* **66**, 1.

J. A. Stubbe (1989) *Ann. Rev. Biochem.* **58**, 257.

B. M. Babior & J. S. Krouwer (1979) *Crit. Rev. Biochem.* **6**, 35.

T. C. Stadtman (1972) *The Enzymes*, 3rd ed., **6**, 539.

ETHANOLAMINE KINASE

This enzyme [EC 2.7.1.82] catalyzes the ATP-dependent phosphorylation of ethanolamine to yield *O*-phosphoethanolamine and ADP.

D. Rhodes & A. D. Hanson (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 357.

K. Ishidate & Y. Nakazawa (1992) *Meth. Enzymol.* **209**, 121.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

ETHANOLAMINE N-METHYLTRANSFERASE

This enzyme catalyzes the methylation of ethanolamine to produce *N*-methylethanolamine.

D. Rhodes & A. D. Hanson (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 357.

ETHANOLAMINEPHOSPHOTRANSFERASE

This enzyme [EC 2.7.8.1] converts CDP-ethanolamine and 1,2-diacylglycerol into CMP and phosphatidylethanolamine.

R. H. Hjelmstad & R. M. Bell (1992) *Meth. Enzymol.* **209**, 272.

ETHANOLAMINE—SERINE BASE-EXCHANGE ENZYME

This enzyme catalyzes the reaction of a phospholipid (for example, phosphatidylserine) with ethanolamine to produce phosphatidylethanolamine and the free base (*i.e.*, the amine-containing metabolite serine), thereby preserving the phosphodiester linkage. Ethanolamine can be replaced with serine, choline, monomethylethanolamine, and dimethylethanolamine.

J. N. Kanfer (1992) *Meth. Enzymol.* **209**, 341.

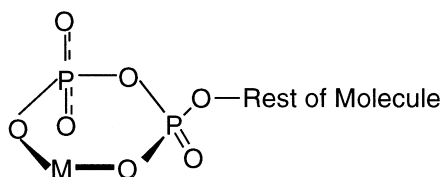
ETIOCHOLANOLONE SULFATASE

This enzyme catalyzes the hydrolysis of etiocholanolone sulfate to produce etiocholanolone and sulfate.

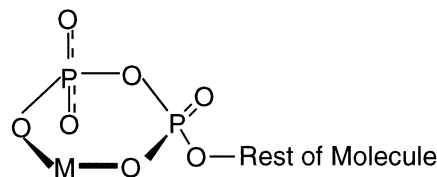
A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 1.

EXCHANGE-INERT METAL-NUCLEOTIDE COMPLEXES

A metal-nucleotide complex^{1,2} that exhibits low rates of ligand exchange as a result of substituting higher oxidation state metal ions with ionic radii nearly equal to the naturally bound metal ion. Such compounds can be prepared with chromium(III), cobalt(III), and rhodium(III) in place of magnesium or calcium ion. Because these exchange-inert complexes can be resolved into their various optically active isomers, they have proven to be powerful mechanistic probes, particularly for kinases, NTPases, and nucleotidyl transferases. In the case of Cr(III) coordination complexes with the two phosphates of ATP or ADP, the second phosphate becomes chiral, and the screw sense must be specified to describe the three-dimensional configuration of atoms.



Δ Isomer



Λ Isomer

In this case, the reference axis is drawn through the metal ion in a fashion perpendicular to the chelate ring. The skew line defining the helical sense (L for left-hand, and D for right-hand) is the bond from the chelate ring to the rest of the molecule. The L and D complexes are diastereomeric and can be separated from each other for direct stereochemical studies of their individual binding and inhibitory properties.

¹W. W. Cleland (1982) *Meth. Enzymol.* **87**, 159.

²J. J. Villafranca (1982) *Meth. Enzymol.* **87**, 180.

EXCITED STATE

A state of higher energy experienced by a molecular entity, relative to its ground state. In photic processes, the excited state can be achieved by absorption of photons, by heating, or by extreme pressurization.

EXERGONIC

An adjective describing a process, change, or reaction in which the change in Gibbs free energy (ΔG) is a negative value. If a compound has a negative change in the standard Gibbs free energy of formation, the substance is referred to as an exergonic compound.

EXOCYTOSIS

The release of intracellular constituents by fusion of an intracellular vesicle with the plasma membrane, thereby exposing the contents to the outer surface.

EXOENZYME

Any enzyme acting *at the terminus* of a polymeric substrate molecule. By contrast, the term endoenzyme refers to an enzyme catalyzing a hydrolytic reaction that occurs *within* a polymeric substrate molecule.

EXOERGIC

An adjective describing a process or reaction of nuclear systems which produces heat.

EXO-MALTOHEXAHYDROLASE

This enzyme [EC 3.2.1.98] removes successive maltohexaose residues from the nonreducing chain ends of 1,4- α -D-glucosidic linkages in amylaceous polysaccharides.

M. Vihinen & P. Mäntsälä (1989) *Crit. Rev. Biochem. Mol. Biol.* **24**, 329.

EXO-MALTOTETRAOHYDROLASE

This enzyme [EC 3.2.1.60] removes successive maltotetraose residues from the nonreducing chain ends by hydrolysis of 1,4- α -D-glucosidic linkages in amylaceous polysaccharides.

M. Vihinen & P. Mäntsälä (1989) *Crit. Rev. Biochem. Mol. Biol.* **24**, 329.

EXONUCLEASE

Any enzyme that catalyzes breakage of a phosphodiester linkage at one or the other end of a polynucleotide chain, resulting in the release of single nucleotides or small oligonucleotides. **See** *specific exonuclease*

B. Weiss (1981) *The Enzymes*, 3rd ed., **14**, 203.

K. M. T. Muskavitch & S. Linn (1981) *The Enzymes*, 3rd ed., **14**, 233.

L. Grossman, A. Braun, R. Feldberg & I. Mahler (1975) *Ann. Rev. Biochem.* **44**, 19.

EXONUCLEASE III

This enzyme [EC 3.1.11.2] degrades double-stranded DNA progressively in a 3'- to 5'-direction, thereby releasing 5'-phosphomononucleotides.

B. Demple & L. Harrison (1994) *Ann. Rev. Biochem.* **63**, 915.

B. Weiss (1981) *The Enzymes*, 3rd ed., **14**, 203.

E. C. Friedberg, T. Bonura, E. H. Radany & J. D. Love (1981) *The Enzymes*, 3rd ed., **14**, 251.

EXONUCLEASE V

This enzyme [EC 3.1.11.5] catalyzes 5'- to 3'- or 3'- to 5'-exonucleolytic cleavage (when ATP is present), yielding 5'-phosphooligonucleotides.

M. M. Cox & I. R. Lehman (1987) *Ann. Rev. Biochem.* **56**, 229.

K. M. T. Muskavitch & S. Linn (1981) *The Enzymes*, 3rd ed., **14**, 233.

EXOPOLY GALACTURONISIDASE

This enzyme [EC 3.2.1.82] catalyzes the hydrolysis of pectic acid from the nonreducing end, thereby releasing digalacturonate.

R. L. Fischer & A. B. Bennett (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 675.

EXOPOLYPHOSPHATASES

These enzymes [EC 3.6.1.11] catalyze the sequential hydrolysis of orthophosphate from polyphosphates.

H. G. Wood & J. E. Clark (1988) *Ann. Rev. Biochem.* **57**, 235.

EXO-SITE

A topologically distinct region on an enzyme that recognizes and binds specific molecules or structures and is far removed from the enzyme's active site. An example is the fibrinogen recognition site on thrombin. This site is spatially distinct from the site for cleavage of the scissile bond of fibrinogen.

J. W. Fenton (1981) *Ann. N.Y. Acad. Sci.* **370**, 468.

M. T. Stubbs & W. Bode (1995) *Trends in Biochem. Sci.* **20**, 23.

EXOTHERMIC

An adjective describing a process, change, or reaction that releases heat. If pressure is constant and no work other than $P\Delta V$ work is done, then $\Delta H = q_P$ where H is the enthalpy and q_P is the heat released at constant pressure. Thus, in such systems, exothermic systems will have negative ΔH values. Since volume changes in chemical and biochemical reactions in solution are small, typically $\Delta H = \Delta U$ in such systems in which U is the internal energy. A compound which has a negative standard enthalpy of formation ($\Delta_f H^\ominus$) is referred to as an exothermic compound. An example would be ethanol in which $\Delta_f H^\ominus$ (298.15) is -287 kJ/mol. **See also** *Endothermic; Exergonic; Exoergic*

EXOTOXIN A

Exotoxin A is an NAD^+ -dependent ADP-ribosylating protein from *Pseudomonas aeruginosa*.

L. Passador & W. Igilewski (1994) *Meth. Enzymol.* **235**, 617.

C.-T. Lai (1986) *Adv. Enzymol.* **58**, 99.

EXPERIMENTAL DESIGN OF INITIAL RATE ENZYME ASSAYS

The basic experimental design for enzyme kinetic assays is directed primarily by the focus of the particular study. If the objective of the assay is related to issues of clinical biochemistry or to enzyme purification, then design protocols are mainly concerned with optimizing each of the assay conditions (*e.g.*, substrate and effector concentrations, pH, temperature, ionic strength, metal ion levels, *etc.*). In such studies, the investigator is often addressing issues related to minimizing interfering agents, to increase the assay sensitivity such that low levels of enzymatic activity can be identified, and to provide an easy, straightforward, cheap, and reproducible protocol. However, the enzyme kineticist is typically working with sub-optimal conditions in addressing issues of enzyme-rate

behavior, mechanism, protein-ligand and protein-protein interactions, as well as regulation. It is through such detailed analyses that true optimal conditions can be identified.

1. Perhaps of first concern in determining the overall design of a particular assay is the actual method used for product identification (or for substrate depletion) per unit time. Many different methods have been utilized (*e.g.*, radiometric, spectrophotometric, fluorometric, pH-stat, polarimetric, *etc.*) No matter which method is used, the product has to be clearly identified (or substrate, if substrate depletion is being measured). With stopped-time assays, it may be necessary to separate product(s) from substrate(s) prior to determination of the amounts of the metabolite(s) present (as well as demonstration that product(s) and substrate(s) are truly separated). If so, the investigator should be able to demonstrate that the assay procedure clearly measures true initial rates (see below). Closely related to these issues are concerns about purity (**See** *Substrate Purity; Enzyme Purity; Water Purity, etc.*) and stability (**See** *Substrate Stability; Enzyme Stability, etc.*). If the components of the assay mixture are not stable over the time course of the experiment (or, if certain side reactions occur), then corrections have to be made in analyzing the rate behavior.

2. The assay protocol should measure true initial rates (**See** *Initial Rate Condition*). For most systems, this represents a time period in which less than ten percent of the substrate concentration has undergone conversion. However, if a reaction is not significantly favored thermodynamically or if product inhibition is particularly potent, then a much smaller percentage of substrate conversion may be needed such that true initial rate conditions are obtained. Addition of an auxiliary enzyme system may prove necessary to avoid product accumulation. **See** *Coupled Enzyme Assays*

3. Preincubation of the assay mixture is a crucial step in kinetic investigations. If the reaction is to be initiated by the addition of enzyme, then the reaction sample (minus the enzyme) should be incubated under the reaction conditions (particularly temperature) for several minutes. Whenever possible, the initiating agent (*e.g.*, the enzyme) should also undergo thermal equilibration; however, this may not be possible if the agent is relatively

unstable. In some systems, the substrate may be more unstable than the enzyme. In these cases, it may be better to initiate the reaction with the substrate. Nevertheless, whichever means is utilized to preincubate and to initiate the reaction, the volumes of both should be clearly stated in reports. In addition, the volume of the initiating agent should be the smaller of the two volumes, particularly if it is not possible to preincubate the initiator at the assay temperature.

4. In kinetic studies, it is best to eliminate any lag phases that may alter the initial velocity behavior of an enzyme. As mentioned above, preincubation is an important factor in this issue. Another is mixing. When the initiating agent is added to the reaction mixture, the entire solution should be thoroughly mixed such that true steady-state conditions can be attained (**See** *Mixing Time*). In some instances, the presence of a magnetic stir bar within the reaction vessel can be quite advantageous.

5. Investigators should always be concerned in steady-state investigations that the conditions for the steady-state (*e.g.*, the assumptions in the mathematical analysis of the system) are present. For example, the total enzyme concentration should be significantly less than the concentration of the substrate or any effector (*e.g.*, a non-competitive inhibitor). In most cases, if the concentration of the substrate(s) is 10^3 -fold or greater than that of the enzyme, the steady-state relationship should hold. If not, global analysis and analysis of the total progress curve may be required. A good check on this is to look at the initial velocity of the reaction at different enzyme concentrations (such a plot should always be made in the initial studies of the kinetic mechanism). This plot should be linear, with the line intersecting the origin, in the range of the concentration of the enzyme to be utilized. Nonlinearity can be due to a number of reasons (*e.g.*, polymerization of the enzyme at higher concentration may result in a more or less active enzyme form). In such cases, it may be necessary to study the kinetics of the system under more than one condition. (**See also** *Enzyme Purity*). In addition to the v vs. $[E_{\text{total}}]$ plot, a progress curve (*i.e.*, $[P]$ vs. time) should be made. With single point assays (*i.e.*, periodic sampling of a reaction mixture), the time points of the aliquots taken should be within the initial linear phase of this progress curve. When using continuous assay methods (*e.g.*, observing

the change in spectrophotometric absorbance with time), the initial velocity is obtained by measuring the slope of the progress curve. This slope is also measured in a time frame in which the progress curve is still linear (as mentioned above, typically less than 10% of substrate conversion). **See** *Continuous vs. Single-Point Assays; Progress Curve; Initial Rate Condition*

6. In single-point assays where the reaction is completely terminated upon addition of some quenching agent or condition, care should be exercised to demonstrate that the reaction has indeed been terminated and that the quenching process is fast relative to the reaction velocity. If product is being measured then the product should be stable under the quenching condition (or, at least stable enough under the time frame in which the product concentration is being measured). **See** *Quenching*

EXPONENTIAL

1. A term used to describe a function or expression containing powers or exponents (*e.g.*, y^x). 2. Referring to any function or equation containing or described by the exponential function, e^x or $(2.718281828 \dots)^x$. **See** *e*

EXPONENTIAL BREAKDOWN

1. The disintegration of a substance in a first-order manner. 2. A breakdown of the Swain-Schaad relationship in kinetic isotope effect studies¹, usually as a consequence of tunneling or kinetic complexity.

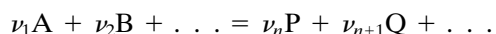
¹B. J. Bahnson & J. P. Klinman (1995) *Meth. Enzymol.* **249**, 373.

EXTENSIVE PROPERTY

Any property that depends on the concentration or amount of substance present. Examples of extensive properties include mass and internal energy.

EXTENT OF REACTION

An elementary description of the course of any chemical reaction by the amount of substance used up or generated in that reaction. Consider the general reaction:



where A, B, . . . , represent the reactants, P, Q, . . . , represent the products, and the corresponding stoichiometry numbers are represented by ν_i . Thus, the extent of reaction, symbolized by ξ , can be defined by $n_x = n_{x_0} + \nu_x \xi$, where n_x is the amount of substance X (*i.e.*, A or B or P, *etc.*), n_{x_0} is the amount of X at the beginning of the reaction, and ν_x is the corresponding stoichiometry number (with the convention that the number is positive for products and negative for reactants). It should be noted that the amount of any substance that has reacted, ν_i , can be represented as $dn_i = \nu_i d\xi$. The SI unit for the extent of reaction is the mole. **See** *Rate of Conversion; Rate of Reaction; Velocity*

EXTINCTION COEFFICIENT

An alternative term for the *molar absorption coefficient* or *molar absorptivity*. **See** *Absorption Spectroscopy*

EXTRINSIC FACTOR X ACTIVATING COMPLEX

This complex catalyzes the activation of factor IX or factor X as a part of the overall coagulation cascade.

L. Lorand & K. G. Mann, eds. (1993) *Meth. Enzymol.*, vols. **222** and **223**.

K. G. Mann, R. J. Jenny & S. Krishnaswamy (1988) *Ann. Rev. Biochem.* **57**, 915.

EYRING EQUATION

An equation for the bimolecular rate constant, k , obtained from transition-state theory. This constant is directly proportional to the equilibrium constant between reactants and the activated complex as well as the absolute temperature: $k = (RT/Lh)K^\ddagger$. **See** *Transition-State Theory*

E/Z CONVENTION

Because the *cis-trans* system for designating configurational isomers can frequently lead to confusion, the *E/Z* convention is the recommended IUPAC system. The two groups attached to each end of a double bond are given priority numbers in exactly the same way that applies to the *R/S* system for specifying stereochemistry. Higher priority is given to the substituent atom having the higher atomic number; in cases where the atomic number is the same, higher priority is given to the heavier isotopomer; and in those cases where there still remains

a possibility for ambiguity, one assigns priority based on the atomic number (then mass) at the first point of difference on two substituents. The so-called *Z*-isomer has the two groups of higher priority (attached to the two carbon atoms of the double bond) residing on the

same side of the molecule, whereas the *E*-isomer has the two groups of higher priority residing on the opposite side of the molecule. (*Z* represents the German word *zusammen* for “together,” whereas *E* represents the German word *entgegen* for “opposite.”)

F

F-430

A tetrapyrrole (related to porphyrins and corrins) containing a nickel ion. This cofactor, corphin, is a crucial component of methyl-coenzyme M reductase, a bacterial enzyme participating in the formation of methane.

FACILITATED TRANSPORT

Carrier-mediated passage of a molecular entity across a membrane (or other barrier). Facilitated transport follows saturation kinetics: *i.e.*, the rate of transport at elevated concentrations of the transportable substrate reaches a maximum that reflects the concentration of carriers/transporters. In this respect, the kinetics resemble the Michaelis-Menten behavior of enzyme-catalyzed reactions. Facilitated diffusion systems are often stereospecific, and they are subject to competitive inhibition. Facilitated transport systems are also distinguished from active transport systems which work against a concentration barrier and require a source of free energy. Simple diffusion often occurs in parallel to facilitated diffusion, and one must correct facilitated transport for the basal rate. This is usually evident when a plot of transport rate *versus* substrate concentration reaches a limiting non-zero rate at saturating substrate¹. While the term “passive transport” has been used synonymously with facilitated transport, others have suggested that this term may be confused with or mistaken for simple diffusion. **See** *Membrane Transport Kinetics*

¹W. D. Stein (1989) *Meth. Enzymol.* **171**, 23.

FARADAY'S CONSTANT

The amount of electricity required to reduce one equivalent of a monovalent ion. The Faraday, symbolized by

F or $\mathcal{F}$, is equal to 96,485.309 coulombs per mole. **See** *Nernst Equation*

FARADAY'S LAWS

1. The amount of an electrolyte decomposed by an electric current is proportional to the size of that current. 2. In a solution containing several electrolytes and through which a current is passing, the amounts of each substance decomposed is proportional to their chemical equivalents.

FARNESYL-DIPHOSPHATE FARNESYLTRANSFERASE

This enzyme [EC 2.5.1.21], also known as farnesyltransferase, presqualene di-diphosphate synthase, and squalene synthase, catalyzes the condensation of two molecules of farnesyl diphosphate to form presqualene diphosphate and diphosphate (or, pyrophosphate). The entire enzyme complex catalyzes the NADPH-dependent reduction of presqualene diphosphate to yield squalene.

E. D. Beytia & J. W. Porter (1976) *Ann. Rev. Biochem.* **45**, 113.

FARNESYLTRANSTRANSFERASE

This enzyme [EC 2.5.1.29], also known as geranylgeranyl-diphosphate synthase, catalyzes the condensation of *trans,trans*-farnesyl diphosphate and isopentenyl diphosphate to yield geranylgeranyl diphosphate and pyrophosphate.

H. Kleinig (1989) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **40**, 39.

FATTY ACID SYNTHASE

This enzyme complex [EC 2.3.1.85] catalyzes the conversion of acetyl-CoA, n moles malonyl-CoA, and $2n$ moles of NADPH to yield long-chain fatty acids, plus $(n+1)$

moles coenzyme A, n moles of carbon dioxide, and $2n$ moles of NADP^+ .

- S. B. Ohlrogge & J. G. Jaworski (1997) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 109.
 S. Smith (1994) *FASEB J.* **8**, 1248.
 L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.
 S.-I. Chang & G. G. Hammes (1990) *Acc. Chem. Res.* **23**, 363.
 S. J. Wakil & J. K. Stoops (1983) *The Enzymes*, 3rd ed., **16**, 3.
 K. Bloch (1977) *Adv. Enzymol.* **45**, 1.

FEEDBACK INHIBITION

A form of regulation in a metabolic pathway in which an end product (or even an intermediate in the pathway) binds to and inhibits an enzyme which catalyzes an earlier reaction in that same pathway. For example, consider the metabolic scheme $A \rightarrow B \rightarrow C \rightarrow D$ in which the steps are respectively catalyzed by the enzymes E_1 , E_2 , and E_3 . Feedback inhibition would be seen when elevated concentrations of D (or C) inhibited E_1 .

When more than one end product in branched pathways can bind to and inhibit a enzyme catalyzing an earlier step in the pathway, a number of possible forms of interaction may occur¹ (**See also** *Multiple Inhibition*). These include:

Synergistic Feedback Inhibition: Also referred to as cooperative feedback inhibition; in this scheme, binding of one inhibitor is not mutually exclusive of the other feedback inhibitor(s). For example, if one feedback inhibitor (at a particular concentration) resulted in a certain degree of inhibition, while, at another concentration, the other inhibitor exhibited a different degree of inhibition, synergistic feedback inhibition is present when the inhibition is greater than would be predicted when both inhibitors are present at those same concentrations. Various procedures are available to deal with such cases (*e.g.*, **See** *Yonetani-Theorell Plot*; *Yagi-Ozawa Plot*). The first step in the biosynthesis of the purine nucleotides, catalyzed by PRPP-glutamine amidotransferase, is reported to be an example of synergistic feedback inhibition^{2,3}.

Concerted Feedback Inhibition: Concerted feedback inhibition, also called multivalent feedback inhibition, is the limiting case of synergistic feedback inhibition. In this case neither end product alone will inhibit the target enzyme. However, when both are present, inhibition is observed.

Cumulative Feedback Inhibition: In cumulative feedback inhibition, the end products can inhibit the reaction of the target enzyme separately. Many textbooks erroneously indicate that the cumulative feedback inhibition of *E. coli* glutamine synthetase involves separate regulatory sites for each feedback inhibitor. **See** *Cumulative Feedback Inhibition*

Additive Feedback Inhibition: This term is restricted to cases in which the end product inhibitors bind to the target enzyme in a mutually exclusive manner and the degrees of inhibition are additive. Extreme care should be exercised when using this term; in most cases, the inhibition is really cumulative feedback inhibition.

¹I. H. Segel (1975) *Enzyme Kinetics*, Wiley, New York.

²C. T. Caskey, D. M. Ashton & J. B. Wyngaarden (1964) *J. Biol. Chem.* **239**, 2570.

³D. P. Nierlich & B. Magasanik (1965) *J. Biol. Chem.* **240**, 358.

FEED-FORWARD ACTIVATION

A regulatory effect observed in many biochemical pathways; an enzyme, which catalyzes a step (typically a late step) in a pathway, is activated by elevated levels of a precursor of a substrate for that enzyme. A possible example of feed-forward activation may be the action of elevated levels of fructose 1,6-bisphosphate on pyruvate kinase.

FeMo-COFACTOR

A crucial inorganic structure found in the FeMo protein of molybdenum-dependent nitrogenase. This cluster contains Fe, Mo, and S in a 7:1:9 ratio. There are two cuboidal subunits, Fe_4S_3 and MoFe_3S_3 , bridged by three S^{2-} ions and “anchored” to the protein by a histidyl residue linked to the Mo atom and by a cysteinyl residue linked to one of the iron ions of the Fe_4S_3 domain of the cluster. The Mo atom at the periphery of the molecule is six-coordinated. It has three sulfido ligands, the histidyl imidazole-nitrogen as a ligand, and two oxygen atoms from an (*R*)-homocitrate molecule.

FENTON REACTION

1. A ferrous ion-dependent reaction in which the highly reactive hydroxyl radical is generated from hydrogen peroxide: $\text{Fe}^{2+} + \text{H}_2\text{O}_2 \rightarrow \text{Fe}^{3+} + \text{HO}\cdot + \text{OH}^-$. This reaction possibly proceeds via an oxoiron(IV) intermediate. The addition of an additional reducing agent such as ascorbate leads to a cycle which can increase the

damage to biological molecules. **See also** *Haber-Weiss Reaction*; *Udenfriend's Reagent*

2. Originally, the Fenton reaction was specifically the oxidation of α -hydroxy acids to α -keto acids using Fenton's reagent, a fresh mixture of hydrogen peroxide and ferrous salts (usually ferrous sulfate)¹⁻³. The procedure will also convert 1,2-glycols to hydroxy aldehydes. Fenton's reagent can also be used to hydroxylate aromatic rings. In these reactions, biaryls are typically found as side products. **See also** *Udenfriend's Reagent*

¹H. J. H. Fenton (1893) *Proc. Chem. Soc.* **9**, 113.

²H. J. H. Fenton (1894) *J. Chem. Soc.* **65**, 899.

³C. Walling (1975) *Acc. Chem. Res.* **8**, 125.

FERREDOXIN

This redox protein participates in several enzyme-catalyzed reactions, including: glutamate synthase (ferredoxin) [EC 1.4.7.1]; ferredoxin-nitrite reductase [EC 1.7.7.1]; ferredoxin-nitrate reductase [EC 1.7.7.2], sulfite reductase (ferredoxin) [EC 1.8.7.1]; ferredoxin-NADP⁺ reductase [EC 1.18.1.2]; ferredoxin-NAD⁺ reductase [EC 1.18.1.3]; and NAD⁺-dinitrogen-reductase ADP-D-ribosyltransferase [EC 2.4.2.37]. **See also** *Resonance Raman Spectroscopy*

M. W. W. Adams & A. Kletzin (1996) *Adv. Protein Chem.* **48**, 101.

H. Cheng & J. L. Markley (1995) *Ann. Rev. Biophys. Biomol. Struct.* **24**, 209.

P. Schürmann (1995) *Meth. Enzymol.* **252**, 274.

J. J. G. Moura, A. L. Maredo & P. N. Palma (1994) *Meth. Enzymol.* **243**, 165.

H. Eisenberg, M. Mevarech & G. Zaccai (1992) *Adv. Protein Chem.* **43**, 1.

J. B. Howard & D. C. Rees (1991) *Adv. Protein Chem.* **42**, 199.

FERREDOXIN:NADP⁺ REDUCTASE

This FAD-enzyme [EC 1.18.1.2], also known as adrenodoxin reductase, catalyzes the reaction of reduced ferredoxin and NADP⁺, yielding oxidized ferredoxin and NADPH.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

FERROCHELATASE

This enzyme [EC 4.99.1.1], also known as protoheme ferro-lyase and heme synthetase, catalyzes the reaction of protoporphyrin with Fe(II) to yield protoheme and two protons.

V. M. Sellers & H. A. Dailey (1997) *Meth. Enzymol.* **281**, 378.

S. Granick & S. I. Beale (1978) *Adv. Enzymol.* **46**, 33.

FERROMAGNETIC

A large, positive magnetically susceptible substance that is capable of being magnetized by a weak magnetic field. If the material retains a high percentage of its magnetization after the applied magnetic field has been removed, that material is said to be "hard." If most of the magnetization is lost, the substance is designated "soft." Major ferromagnetic elements include iron, cobalt, and nickel. Above a certain temperature, known as the Curie point, ferromagnetic materials become paramagnetic. Below that temperature, an increasing applied magnetic field will result in increasing magnetization.

Ferrimagnetism is a special form of antiferromagnetism. Ferrites (an example of which is Fe₂O₃·XO where X is a divalent metal such as Co, Ni, Zn, or Mn) exhibit both ferromagnetism and ferrimagnetism. The presence of two ions in the solid of an antiferrimagnetic substance will result in antiparallel magnetic dipole moments of adjacent ions in which the difference of the moments is unequal (or in which the number of magnetic moments in one direction is greater than that in the opposite direction). It is possible, by the selection of an appropriate ion, to design ferrimagnetic substances with specific magnetizations.

FERROXIDASE

This copper-dependent enzyme [EC 1.16.3.1], also known as ceruloplasmin, converts 4 Fe(II) + 4 H⁺ + O₂ into 4 Fe(III) + 2 H₂O.

E. Frieden & H. S. Hsieh (1976) *Adv. Enzymol.* **44**, 187.

B. G. Malmström, L.-E. Andréasson & B. Reinhammer (1975) *The Enzymes*, 3rd ed., **12**, 507.

[2Fe–2S] CLUSTER

A designation for a two iron, two labile-sulfur cluster present in a number of iron-sulfur proteins. The [2Fe–2S] center consists of two iron atoms bridged by two sulfur atoms (the sulfido-bridges). The oxidation levels of the clusters are usually indicated by adding the charges on the iron and sulfide atoms (e.g., [2Fe–2S]²⁺).

[4Fe–4S] CLUSTER

A designation for a four iron, four labile-sulfur cluster present in a number of iron-sulfur proteins. Possible oxidation levels of this type of cluster include [4Fe–4S]³⁺, [4Fe–4S]²⁺, and [4Fe–4S]⁺. Note that the individual iron

atoms in a particular cluster do not have to have the same oxidation state. For example, there are two [4Fe-4S] clusters in ferredoxin and, in oxidized ferredoxin, there is one Fe(II) and three Fe(III) atoms in each cluster.

FICIN

The following articles describe some of the molecular and physical properties of this enzyme [EC 3.4.22.3] which catalyzes the hydrolysis of peptide bonds and exhibits a broad specificity (at Lys-, Ala-, Tyr-, Gly-, Asn-, Leu-, and Val-).

E. Shaw (1990) *Adv. Enzymol.* **63**, 271.

J. S. Fruton (1982) *Adv. Enzymol.* **53**, 239.

A. N. Glazer & E. L. Smith (1971) *The Enzymes*, 3rd ed., **3**, 502.

FICK'S LAWS OF DIFFUSION

1. The direction of diffusion or movement by solutes is always from a region of higher concentration to one of lower concentration. In addition, the diffusive flux, J_A , of a solute A across a plane at x is equal to $-D(\partial c_A/\partial x)$ where D is the diffusion coefficient and $\partial c_A/\partial x$ is the concentration gradient of A at x . 2. The increase of the concentration of A with time, $\partial c_A/\partial t$, is equal to $D(\partial^2 c_A/\partial x^2)$ where $\partial^2 c_A/\partial x^2$ is the change in the concentration gradient. **See also** *Diffusion; Transport Processes*

FIELD EFFECT

The polarization of a chemical bond in a molecule through space or through solvent molecules, by a remote pole or dipole, an effect that is dependent on the geometry of the molecule. Functional groups are often ranked as being either electron-withdrawing or electron-donating (with respect to hydrogen). Examples of electron-donating groups include O^- , COO^- , CH_3 , CH_2R , CHR_2 , CR_3 , D, as well as atoms of low electronegativity. Examples of electron-withdrawing groups include $COOH$, $COOR$, COR , the halogens, OH , SH , SR , OR , NH_3^+ , NR_3^+ , and $CH=CR_2$. In general, sp carbons have more electron-withdrawing effects than do sp^2 carbons which, in turn, have a greater effect than sp^3 carbons. Field effects, also known as direct effects, decrease with distance and orientation and can be influenced by the solvent. **See** *Inductive Effect*

J. D. Roberts & W. T. Moreland, Jr. (1953) *J. Amer. Chem. Soc.* **75**, 2167.

E. Ceppi, W. Eckhardt & C. A. Grob (1973) *Tet. Lett.* **1973**, 3627.

FIRST DERIVATIVE TEST

A procedure used in calculus to identify local maxima and minima in a continuous and differentiable function. The function at some point x in an interval (a,b) is a local maximum if the first derivative, $df(x)/dx$, is greater than zero over the interval to the left of x , is zero at x , and is less than zero over the interval to the right of x . Likewise, the function at some point x in an interval (a,b) is a local minimum if the first derivative, $df(x)/dx$, is less than zero over the interval to the left of x , is zero at x , and is greater than zero over the interval to the right of x .

FIRST-ORDER REACTION KINETICS

A reaction is considered first order if it agrees with the rate expression $v = -d[A]/dt = k[A]$, where v is the reaction rate, t represents time, and k is the first-order rate constant having units of reciprocal time. If $[A_0]$ represents the concentration of A at $t = 0$, and x the concentration of A at time t , then another method of expressing the rate law is $dx/dt = k([A_0] - x)$. Integration of the expression yields $\ln[A] = -kt + \ln[A_0] = \ln([A_0] - x)$ or $[A] = [A_0]e^{-kt}$. A plot of $\ln[A]$ vs. t will yield a straight line having a slope of $-k$. Alternatively, a value for k can be obtained from a determination of the reaction half-life, $t_{1/2}$. From the equation above, it is directly seen that the first-order rate constant is inversely proportional to $t_{1/2}$: i.e., $k = (\ln 2)/t_{1/2}$. As many authors have pointed out^{1,2}, true first-order rate constants are easily obtained from half-life measurements. Note that the half-life of a first-order reaction is independent of the initial reactant concentration. Thus, all that is necessary is to measure any quantity (such as spectric, radio-metric, volumetric, etc.) that changes with the concentration of A in such a way that the rate of change in the measured quantity has the same half-life.

There are many examples of first-order reactions: dissociation from a complex, decompositions, isomerizations, etc. The decomposition of gaseous nitrogen pentoxide ($2N_2O_5 \rightarrow 4NO_2 + O_2$) was determined to be first order ($-d[N_2O_5]/dt = k[N_2O_5]$) as is the release of product from an enzyme-product complex ($EP \rightarrow E + P$). In a single-substrate, enzyme-catalyzed reaction in which the substrate concentration is much less than the Michaelis constant (i.e., $[S] \ll K_m$) the reaction is said to be first-order since the Michaelis-Menten equation reduces to

$v = V_{\max} [S]/K_m$. A reaction of higher order is called pseudo-first-order if all but one of the reactants are high in concentration and do not change appreciably in concentration over the time course of the reaction. In such cases, these concentrations can be treated as constants.

See Order of Reaction; Half-Life; Second-Order Reaction; Zero-Order Reaction; Molecularity; Michaelis-Menten Equation; Chemical Kinetics

¹J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, Wiley, New York.

²W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York].

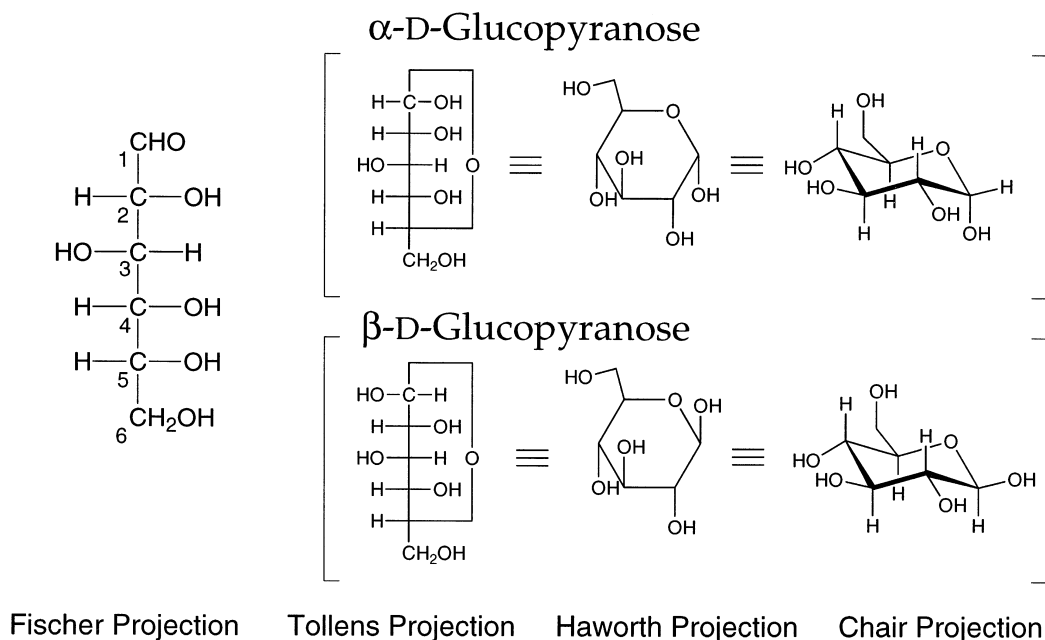
FISCHER PROJECTION

A method for depicting the geometry of one or more asymmetric tetrahedral centers, where each is repre-

In this way, the Fischer projection of central carbon of D-(+)-glyceraldehyde [also known as R-(+)-glyceraldehyde] has the hydroxyl group projecting to the right and the hydrogen atom pointed to the left. Likewise, the Fischer projection of D-glucose in the open chain form has the C1 aldehyde group at the top (pointing vertically) and the C6 hydroxymethyl group at the bottom (again vertically); the hydroxyl groups at C2, C3, C4, and C5 respectively point right, left, right, and right. **See** R/S Convention for Asymmetric Centers

FITSIM COMPUTER PROGRAM

Digital computer software that allows an investigator to determine the magnitude of rate constants for individual steps within a chemical reaction mechanism. **See** KINSIM/FITSIM



sented as the point of intersection of the vertical and horizontal lines of a cross. The horizontal lines that extend leftward and rightward represent bonds to those substituents extending out and toward the observer (*i.e.*, above the plane of the paper). The vertical lines that extend upward and downward represent bonds to two other substituents extending backward and behind the plane of the paper. For carbohydrates and alkanes, the convention is that the carbon chain is positioned vertically.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Data analysis flow chart, **240**, 314-315; data point number requirements, **240**, 314; determination of enzyme kinetic parameters: multisubstrate, **240**, 316-319; single substrate, **240**, 314-316; enzyme mechanism testing, **240**, 322; evaluation of binding processes, **240**, 319-321; file transfer protocol site, **240**, 312; instructions for use, **240**, 312-313.

FLAGELLAR ROTATION

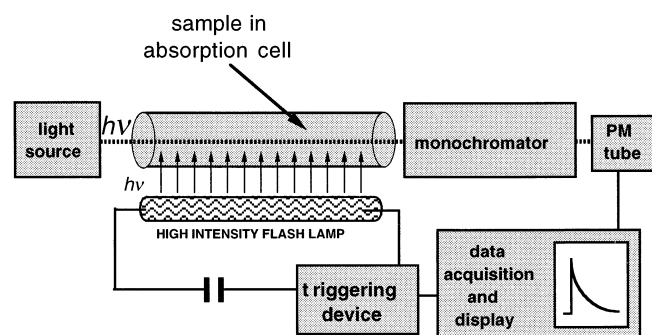
The mechanochemical rotatory motion of bacterial flagella, driven by electrochemical proton gradients across the peripheral membrane. Each complete turn requires

a coupling to the downhill transport of about a thousand protons. *Escherichia coli* can rotate their flagella at 10 Hertz (*i.e.*, cycles/s).

S. C. Schuster & S. Khan (1994) *Annu. Rev. Biophys. Biomol. Struct.* **23**, 509.

FLASH PHOTOLYSIS

A fast reaction technique that employs sudden photoactivation or photolysis to initiate or alter a chemical reaction system. This sudden perturbation creates a nonequilibrium situation, allowing one to determine the time course of the relaxation of a chemical reaction system back to equilibrium.



Flash Photolysis Apparatus

Light absorption is usually quite fast (time scale = 1–10 femtoseconds), and various physical measurements can be used to characterize the properties of intermediates that are formed along the reaction coordinate. This strategy was introduced by Porter who later shared the Nobel Prize in Chemistry with Eigen and Norrish for their germinal contributions to fast reaction kinetics. **See** *Chemical Kinetics*

G. Porter (1950) *Proc. Roy Soc.* **A200**, 284.

G. Porter (1960) *Zeit. Elektrochem.* **64**, 59.

G. Porter (1974) in *Investigation of Rates and Mechanisms of Reactions*, Part II, 3rd ed. (G. G. Hammes, ed.) vol. **VI**, pp. 367-462, Wiley-Interscience, New York.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Apparatus, **232**, 89; as central kinetic problem, **232**, 293; electron-transfer reactions, rate measurements, **232**, 89-91; for ligand binding cooperativity measurement, **231**, 147, 149; and ligand rebinding after photodissociation, **232**, 72-73; distribution of states and, **232**, 82; double distribution and, **232**, 82-83; kinetic simulations, **232**, 82; ligand for, **232**, 80-82; photolysis pulse length for, **232**, 79-80; comparison to pulse radiolysis, **233**, 3.

FLAVANONE 3 β -HYDROXYLASE

This enzyme [EC 1.14.11.9] uses iron and ascorbate as cofactors in a reaction that converts naringenin, 2-oxoglutarate and dioxygen into dihydrokaempferol, succinate, and carbon dioxide.

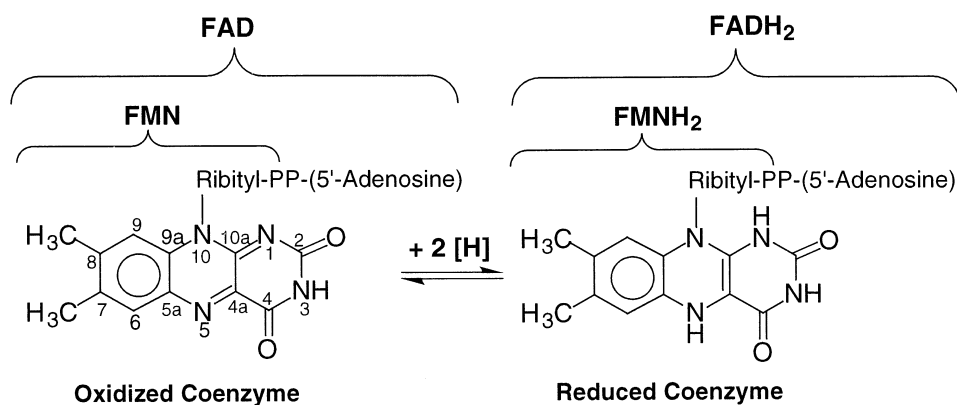
A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.

FLAVINS AND DERIVATIVES

This group includes the coenzyme forms of water-soluble vitamin B₂ or riboflavin. Synthesis occurs by initial cyclohydrolase action on the guanine ring of GTP and subsequent steps lead to the synthesis of the isoalloxazine ring structure (see structures below).

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Determination of FMN and FAD by fluorescence titration with apoflavodoxin, **66**, 217; purification of flavin-adenine dinucleotide and coenzyme A on *p*-acetoxymercurilaniline-agarose, **66**, 221; a convenient biosynthetic method for the preparation of radioactive flavin nucleotides using *Clostridium kluyveri*, **66**, 227; isolation, chemical synthesis, and properties of roseoflavin, **66**, 235; isolation, synthesis, and properties of 8-hydroxyflavins, **66**, 241; structure, properties and determination of covalently bound flavins, **66**, 253; a two-step chemical synthesis of lumiflavin, **66**, 265; syntheses of 5-deazaflavins, **66**, 267; preparation, characterization, and coenzymic properties of 5-carba-5-deaza and 1-



carba-1-deaza analogs of riboflavin, FMN, and FAD, **66**, 277; preparation of flavin 5'-phosphates using immobilized flavokinase, **66**, 287; flavin 1,*N*⁶-ethenoadenine dinucleotide, **66**, 290; flavin suicide inhibitor adducts, **66**, 294; GTP cyclohydrolase II from *Escherichia coli*, **66**, 303; riboflavin synthetase from *Eremothecium ashbyii* and a salvage pathway of the by-product in the enzyme reaction, **66**, 307; continuous fluorescence assay, partial purification, and properties of flavokinase from *Megasphaera elsdenii*, **66**, 323; riboflavin α -glucoside-synthesizing enzyme from pig liver, **66**, 327; isolation and identification of schizoflavins, **66**, 333; flavin affinity chromatography, **66**, 338; activity staining for flavoprotein oxidases, **66**, 344; temperature-difference spectra of flavins and flavoproteins, **66**, 350; fluorescence and optical characteristics of reduced flavins and flavoproteins, **66**, 360; time-resolved fluorescence on flavins and flavoproteins, **66**, 373; application of nuclear magnetic resonance and photochemically induced dynamic nuclear polarization to free and protein-bound flavins, **66**, 385; laser fluorescence techniques, **66**, 416.

FLAVONE SYNTHASE

This enzyme, a P-450 monooxygenase, catalyzes the NADPH-dependent conversion of flavanones to flavones.

A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.

FLAVONOL SYNTHASE

This enzyme catalyzes the insertion of a double bond in dihydroflavonols.

A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.

FLUCTUATIONS

Uncertain shifts or swings in the position, motion, or energy of particles or molecules. Fluctuations about some mean value of position, motion, or energy occur for individual particles/molecules, but these microscopic events are not typically sensed when averaged over a large population of particles/molecules. The statistical treatment of fluctuations in what is called statistical mechanics permits one to derive statistical analogues of entropy and free energy. Furthermore, the equilibrium state can be well understood on the basis of statistical mechanics. One can also use the concept of local fluctuations to account for molecular light scattering.

FLUORESCENCE

(1. General Aspects)

Fluorescence has become one of the most powerful tools in the molecular life sciences^{1,2}. This is understandable because most biological molecules (particularly macromolecules) absorb light over the 200–800 nm spectral range. Many biomolecules are intensely fluorescent, and

even nonfluorescent substances can be investigated with respect to the nature of their interactions by using other fluorescent molecules. Nonfluorescent molecules can be extrinsically labeled with a fluorescent reporter group that can be bound (a) noncovalently at one or more adsorption sites on the macromolecule or (b) covalently through linkages formed by reaction with nucleophiles or electrophiles present on the macromolecule of interest. Indeed, anyone familiar with fluorescence spectroscopy (and microscopy) has good reason to marvel at the variety of labeling procedures at one's disposal. Although only cottage industries a decade ago, specialty companies have grown rapidly in efforts to satisfy the burgeoning needs of spectroscopists and microscopists using fluorescent compounds.

Because fluorescence spectroscopy has become such a robust field of investigation, we cannot possibly cover all of the relevant techniques in this series of entries on fluorescence. For this reason, we will consider the basics of fluorescence, the design of a research fluorimeter, and illustrative cases of ligand binding measurements. Space constraints preclude any discussion of time-dependent fluorescence phenomena including the following topics: multiexponential decay processes, time-domain lifetime measurements, frequency-domain lifetime measurements, static and dynamic quenching, fluorescence anisotropy, energy transfer and distance determinations, as well as excited state reactions. For a more thorough treatment of these and other topics, the interested reader should consult reference 2 as well as the sets of *Methods in Enzymology* entries on fluorescence applications.

BASICS OF ABSORPTION AND FLUORESCENCE.

When light interacts with a substance, the incident photons typically collide elastically (in which case Rayleigh light scattering occurs) or they may be absorbed. In simplest terms, the quantum nature of light absorption reflects the fact that there are discrete energy differences (or steps) between successive electronic states, vibrational states, and even rotational states. Another basic principle is that an absorbed photon possessing more energy than that needed for an electronic transition will lead to the occupancy of higher vibrational and rotational energy states. In these cases, the excess energy is dissipated quickly as the molecule reaches the lowest-lying excited state. Furthermore, photon absorption is

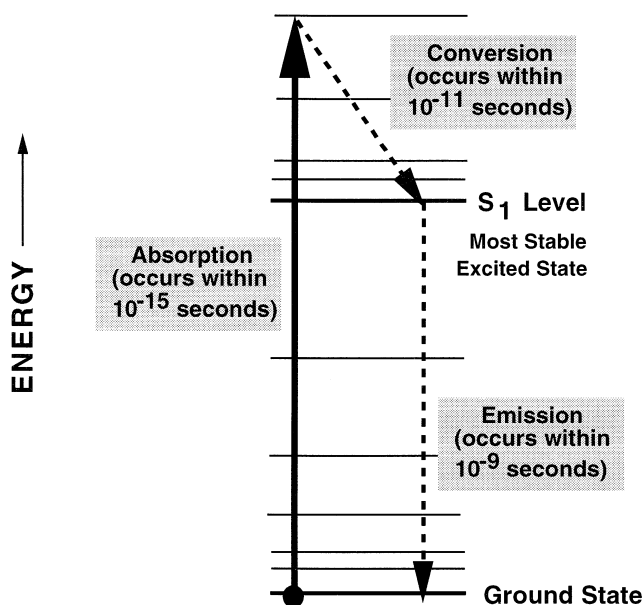


Figure 1. Diagrammatic representation of the electronic transitions taking place within a molecule after photon absorption (indicated here as a boldface upwardly pointed arrow). Excitation is followed by conversion to a less energetic excited state from which emission takes place.

virtually instantaneous (requiring only about 10^{-15} s), and this time is too short (by a factor of 100 to 1000) for changes to occur in the position of nuclei in the substance. (This so-called “vertical transition” is frequently regarded to be a consequence of the Franck-Condon Principle.) The lower probability of transitions to the lowest singlet state makes this transition slower (around 0.01 nanoseconds). Furthermore, the even lower probability of transition to the ground state renders the lifetime of the lowest lying excited state to be in the nanosecond or longer time range.

An absorbed photon’s energy can also be dissipated thermally (*i.e.*, as heat, when all of the excitation energy is converted into vibrational and rotational energy). Such transitions yield no photon in the ultraviolet or visible spectrum, and they are most commonly termed radiationless transitions. Another way in which a molecule can leave its excited state is for the molecule to lose only part of its energy as vibrational and/or rotational energy, thereby “dropping” down or relaxing to the lowest lying excited state (designated S_1 in Fig. 1 above). After this conversion, the S_1 singlet state may lose its energy in several ways: (a) by resonance energy transfer of its remaining energy to another molecule (termed an acceptor), and this molecule can then act as a quencher (by releasing its newly gotten energy thermally) or as a

fluorescer (by releasing a photon in the UV/visible range, though often at a longer wavelength than would have been produced by photon loss from the S_1 state of the donor molecule); (b) by colliding into another molecule which accepts some or all of the energy (a process called collisional quenching); (c) by fluorescence which is often called a “prompt” emission because it occurs in the nanosecond time range; or (d) by a so-called intersystem crossing to form a more stable triplet state which itself can lose the energy as phosphorescence (often termed “delayed” emission) or again by quenching.

Radiationless transitions have an associated rate constant ($k_{\text{radiationless}}$); intersystem crossings have a corresponding rate constant (k_{crossing}); and fluorescence is characterized by its own rate constant, designated here as $k_{\text{fluorescence}}$. [Each of these competing events are first-order relaxation processes. **See Chemical Kinetics: (9. Relaxation Kinetics)**] A photo-excited molecule rarely re-emits every photon by fluorescence, and the quantum yield is the ratio of the number of photons produced by fluorescence to the number of photons originally absorbed. Accordingly, the quantum yield ϕ is formally equal to $k_{\text{fluorescence}}/[k_{\text{fluorescence}} + k_{\text{crossing}} + k_{\text{radiationless}}]$. As we shall see later, the quantum yield of a substance can be greatly influenced by the fluorophore’s immediate environment. Likewise, shifts in the emission spectrum can also help one to detect changes in environment, and the physicochemical basis of such changes can often be inferred.

We should also note that there are other ways in which substances can luminesce (*i.e.*, produce light). One of the most common nonfluorescent mechanisms is called chemiluminescence. Chemical reactions can produce light when radicals (*i.e.*, atoms or molecules with reactive unpaired electrons) combine to form a covalent bond. Heating of molecules can also cause them to display chemiluminescence. And, of course, there are biological processes that are accompanied by chemiluminescence. The most familiar case is the luciferase reaction associated with fire flies and luminescent marine creatures. Peroxidase reactions also produce faint luminescence.

DESIGN FEATURES OF THE SPECTROPHOTOFUORIMETER. A sample’s fluorescence is almost

always dispersed radially, and in principle much greater sensitivity could be achieved by positioning a battery of photodetectors in a spherical arrangement to capture more of the emitted light from the sample. The design (Fig. 2) of the right-angle fluorimeter, however, provides a convenient means (a) for minimizing effects of stray light (*See Absorption Spectroscopy*); (b) for making polarization measurements; and (c) for sampling the same region within the cuvette. Collimating slits are mechanical devices that can be controlled manually or by servo circuitry to allow preset or adjustable apertures for light entering and leaving the sample compartment. Slits ensure that the experimenter can manipulate the intensity and purity of light reaching the sample and detector. A larger slit allows more light to pass, but this light may not be uniformly of the same wavelength. A smaller slit width helps to assure that only light within a certain wavelength range will reach the sample and detector, but this diminishes light intensity and likewise lowers instrument sensitivity. A good research fluorimeter allows one to control the intensity and wavelength of the exciting beam of light as well as the photomultiplier wavelength and slit widths of the emitted light.

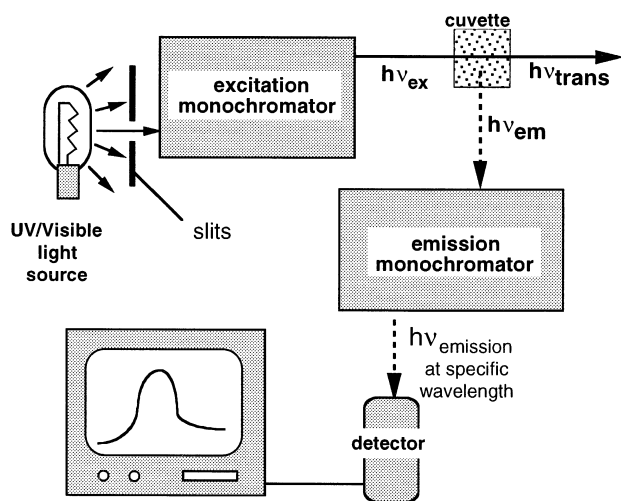


Figure 2. General design features of a fluorescence spectrophotometer.

Fluorimeters use a photomultiplier (or PM) tube that is positioned perpendicular to the incident light beam to detect fluorescence emission by a sample. The photomultiplier has a light-sensitive surface (the photocathode) which is maintained at a negative potential of about 950–1500 volts. So-called photoelectrons are released from photocathode upon light absorption, and they are

amplified by a succession of dynodes, culminating in a pulse of current at the anode. There are two basic ways to acquire fluorescence data, namely through continuous monitoring of the emission photocurrent or by single photon-counting. In the continuous or analog mode, one detects a time-averaged photocurrent, and this requires highly stable amplifiers and power supplies to avoid error. (Most modern instruments use an additional photodetector to monitor fluctuations in incident light intensity resulting from changes in the power output of the light source. Through the use of a partially silvered mirror to divert a fraction (say 5–10%) of the exciting light, this third detector contemporaneously samples the excitation beam and improves sensitivity by automatically correcting the emission photocurrent or photon count for fluctuations in incident light intensity.) In the photon-counting mode, single or individual pulses are registered upon arriving at the anode. This technique increases sensitivity, and any spurious thermally induced loss of electrons from the photocathode can be greatly reduced by using cryotechniques. (Modern photon-counting instruments have Peltier devices to keep the PM tube at -30° to -40°C .) One difficulty with single photon counting is called coincidence, which is the simultaneous (or nearly so) arrival of two pulses that are registered as one event. The probability of coincidence increases as the number of photons per second increases, and one should keep counting rates below 1 MHz to avoid this problem.

Unlike absorption spectroscopy, the measurement of fluorescence can yield unreliable data if the experimenter is unaware of intrinsic limitations of fluorescence. To appreciate these issues, we adopt the following “thought” experiment recommended by Laurence¹. We place a fluorescence cuvette containing an adequate volume of sample within the instrument located in a dark room, leaving the light-tight lid off the sample compartment. (In actual practice, this might require one to defeat any electronic interlocking mechanism used to protect the PM tube from exposure to bright light.) We can now view the cuvette from above, noting that this perpendicular geometry matches that of the pencil of emitted light that reaches the photomultiplier itself. Using the wavelength control on the excitation monochromator, we can scan the wavelength from longer to shorter wavelength. A sharp onset of fluorescence should be sensed by the viewer’s eye, and the wavelength yielding maximal out-

put of emitted light can be determined. If there is no large particle light scattering or turbidity in the sample, the color of the fluorescent light should not depend on the wavelength of the exciting light. Another reason why more than one color of fluorescence might occur is that the sample contains more than one fluorescent compound. The viewer should also take note of the intensity of light from the front face (*i.e.*, the surface closest to the exciting light beam) of the cuvette to the back face. The fluorescence should be of uniform intensity throughout the cuvette. If the fluorescence is brightest at the illuminated (or front) face, the solution is probably too concentrated, and insufficient light is able to reach the molecules nearer the back face. Because the emission beam is measured from the light near the center of a cuvette, extremely high concentrations of the sample will have the paradoxical effect of reducing the fluorescence detected by the photomultiplier tube. Diluting the sample should create a condition of uniform light intensity. Another reason why one might not observe uniform light intensity from the front to back face of the cuvette is the presence of a quenching agent. As noted earlier, a quenching agent can accept energy from the photo-excited sample molecule, and release of this energy by vibrational and rotational modes will reduce the intensity of fluorescence.

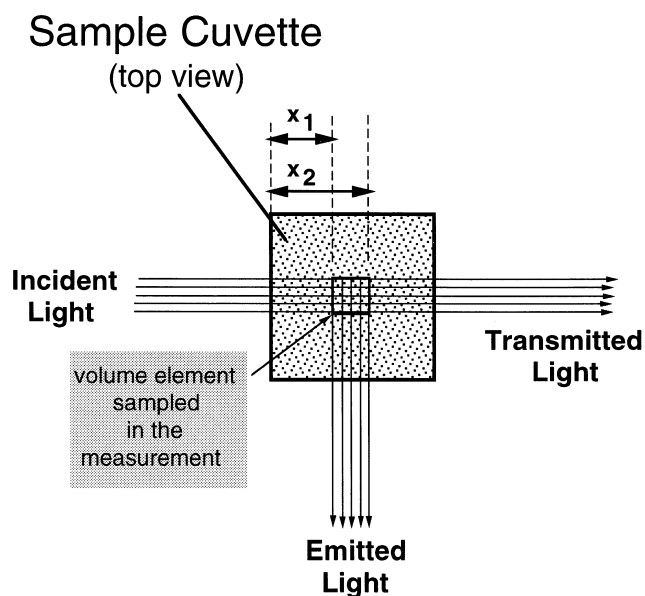


Figure 3. Illustration of the origin of inner filter effects arising by absorption of some exciting photons by one or more substances having an UV-visible absorbance spectrum in the wavelength region of light in the excitation beam.

Even if another type of molecule does not quench the fluorescence, it may still absorb light and reduce the number of photons available for exciting the fluorophore². Figure 3 reminds us that the volume element measured experimentally is located at the center of the sample cuvette. A fraction of incident light is observed over the distance x_1 , and the sample only absorbs the light traveling from x_1 to x_2 . Thus, the presence of one or more light-absorbing species in the cuvette reduces the photic energy available to excite the volume element that is actually measured. This latter phenomenon is called an “inner-filter” effect, because the presence of another absorbing molecule acts as though it is a filter present within the sample. If the absorbance values of the sample solution are determined at the excitation and emission wavelengths, it is possible to correct for inner-filter effects. A useful equation² is:

$$F_{\text{corrected}} = F_{\text{observed}} 10^{\left\{ \frac{(\text{Absorbance}_{\text{excitation}} - \text{Absorbance}_{\text{emission}})}{2} \right\}}$$

The measured absorbances are divided by two, because the midpoint of a 1 cm cuvette is located at a path length of 0.5. Figure 4 shows the observed and corrected fluorescence for a hypothetical compound having appreciable absorbance at the excitation and emission wavelengths.

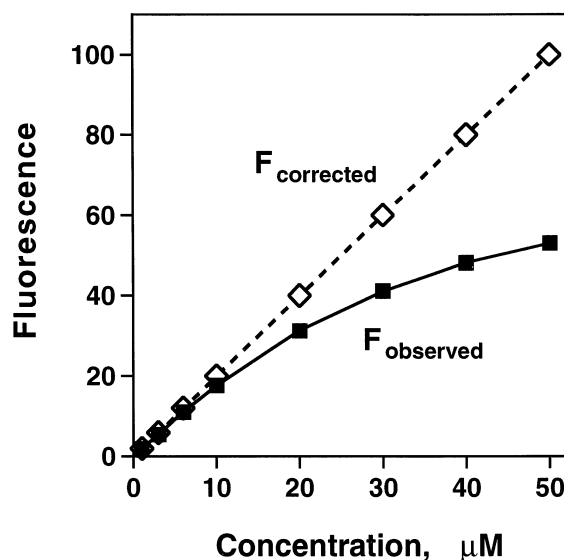


Figure 4. Uncorrected and corrected fluorescence signal as a function of the concentration of a hypothetical fluorophore having significant absorbance at the excitation wavelength (in this case $\epsilon_{\text{excitation}} = 10,000 \text{ M}^{-1}\text{cm}^{-1}$) as well as at the emission wavelength ($\epsilon_{\text{emission}} = 10,000 \text{ M}^{-1}\text{cm}^{-1}$).

Notice that the fluorescence intensity is directly proportional to the optical density only over a narrow range. Use of the above equation corrects for this apparent loss in fluorescence intensity.

INTRINSIC AND EXTRINSIC FLUORESCENCE.

Intrinsic fluorescence refers to the fluorescence of the macromolecule itself, and in the case of proteins this typically involves emission from tyrosinyl and tryptophanyl residues, with the latter dominating if excitation is carried out at 280 nm. The distance for tyrosine-to-tryptophan resonance energy transfer is approximately 14 Å, suggesting that this mode of tyrosine fluorescence quenching should occur efficiently in most proteins. Moreover, tyrosine fluorescence is quenched whenever nearby bases (such as carboxylate anions) accept the phenolic proton of tyrosine during the excited state lifetime. To examine tryptophan fluorescence only, one typically excites at 295 nm, where tyrosine weakly absorbs. [Note: While the phenolate ion of tyrosine absorbs around 293 nm, its high pK_a of 10–11 in proteins typically renders its concentration too low to be of practical concern.] The tryptophan emission is maximal at 340–350 nm, depending on the local environment around this intrinsic fluorophore.

Extrinsic fluorescence is used whenever the natural fluorescence of a macromolecule is inadequate for accurate fluorescence measurement. In this case, one can attach a fluorescent reporter group by using the reactive isocyanate or isothiocyanate derivatives of fluorescein or rhodamine, two intensely fluorescent molecules. One can covalently also label a protein's α - and ε -amino groups with dansyl chloride (*i.e.*, *N,N*-dimethylaminonaphthalenesulfonyl chloride). Another useful reagent is 8-anilino-1-naphthalenesulfonic acid (abbreviated ANS). This compound is bound noncovalently by hydrophobic interactions; in aqueous solutions, ANS is only very fluorescent, but upon binding within an apolar environment, the quantum yield of ANS becomes about 100 times greater.

We should note that there are other intrinsic fluorophores that can be used in binding studies. NADH and NADPH are both highly fluorescent coenzymes. So too are riboflavin and FAD, but flavoproteins are not typi-

cally fluorescent. With the exception of the Y base in tRNA, nucleic acids are nonfluorescent.

FLUORESCENCE MEASUREMENTS OF LIGAND BINDING. In principle, ligand binding may either enhance or quench the intrinsic or extrinsic fluorescence of its macromolecular receptor; or it may change the polarization of the fluorescence emission (see below).

The tandemly repeated Gly-Pro-Pro-Pro-Pro-Pro modules within platelet vasodilator stimulated phosphoprotein (VASP) are thought to serve as docking sites for the actin regulatory protein profilin³. Profilin is a ubiquitous protein involved in rearrangement of the actin cytoskeleton. An unusual property of profilin is its high binding affinity for poly(L-proline). The following fluorescence titration data illustrate how the binding of synthetic GPPPPP-triple repeat is attended by an enhancement in the intrinsic fluorescence of profilin² (Fig. 5). Note that the fluorescence enhancement reaches a limiting value that is indicative of saturation of available sites.

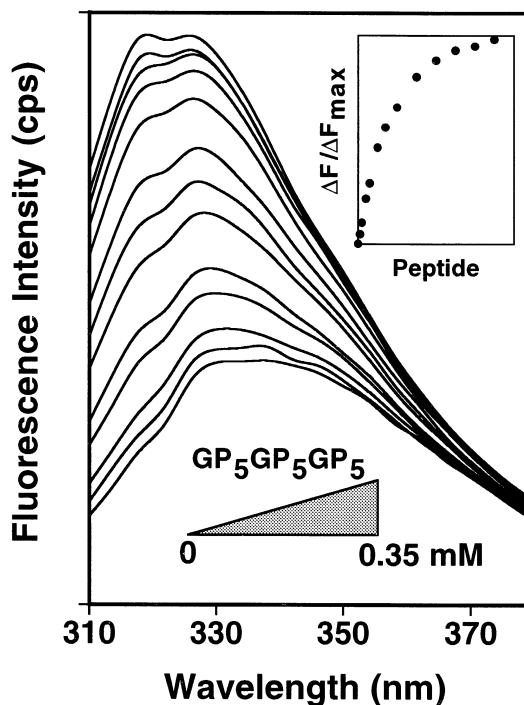


Figure 5. Enhancement of profilin fluorescence in the presence of the oligoproline peptide (GPPPPP)₃.

This is more evident in the replot of the extent of enhancement versus the concentration of the proline-rich peptide, and a dissociation constant of about 85 μ M was obtained. Fluorescence studies with [His133Ser]-profilin,

a site-directed mutant previously shown to be defective in binding poly(L-proline), exhibits little or no evidence of saturable $\text{GP}_5\text{GP}_5\text{GP}_5$ binding. On the basis of NMR experiments, the binding of oligoproline peptides to profilin appears to cover one or several tryptophan residues on the protein. Poly(L-proline) binds at the hydrophobic interface between profilin's N- and C-terminal helices and the upper face of its antiparallel β -sheet. As few as six contiguous prolines were deemed to be sufficient for binding profilin.

The binding of 8-anilino-1-naphthalenesulfonate (ANS) to ciliary dynein ATPase resulted in a marked increase in the dye's fluorescence intensity, accompanied by a blue shift in the observed fluorescence emission maximum⁴ (Fig. 6). While dynein has 37 ± 3 dye binding

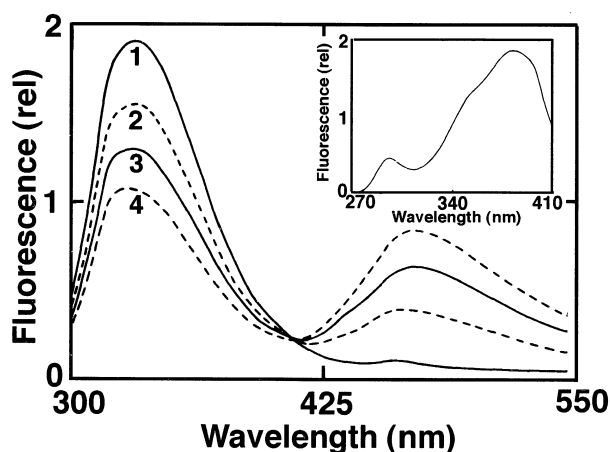


Figure 6. Enhanced ANS fluorescence attending binding of this extrinsic environmentally sensitive probe to sites on sea urchin sperm tail dynein.

sites, ANS binding does not alter enzyme activity. The fluorescence properties of the enzyme-dye complex were used as a conformational probe of dynein binding interactions with substrates and effectors. The fluorescence of the dynein-ANS complex was increased by a number of substrates, including ATP, GTP, and UTP. The transfer of excitation energy from dynein chromophores to adsorbed ANS was also investigated. Dynein appears to undergo a localized conformational change in its interaction with ATP. Native dynein was also found to be conformationally different from heat-activated or NEM-modified enzyme as evidenced by the emission and excitation spectra of the various enzyme-ANS complexes.

¹D. J. R. Laurence (1957) *Meth. Enzymol.* **4**, 174.

²J. R. Lakowicz (1983) *Principles of Fluorescence Spectroscopy*, pp. 496, Plenum Press, New York.

³F. Kang, R. O. Laine, M. R. Bubb, F. S. Southwick & D. L. Purich (1997) *Biochemistry* **36**, 8384.

⁴A. C. Saucier, S. Mariotti, S. A. Anderson & D. L. Purich (1985) *Biochemistry* **24**, 7581.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Energy level diagram, **246**, 336; excitation transfer, **246**, 688; expected intensity, **246**, 684; front-face, **232**, 231-246; hemoglobin adducts [cryogenic methods, **231**, 677-681; room-temperature methods, **231**, 674-676]; high-pressure experiments [hemoglobin subunit dissociation, **232**, 42-55; interpretation, **232**, 50-55]; inner filter effect, **246**, 698; instrumentation, **231**, 677-679; lipid-protein interactions, **259**, 604, 607; low-temperature spectroscopy, **246**, 160-163; measurement by ODMR, **246**, 611-612, 620, 625, 629; multilinear modeling, **246**, 684-685, 688, 700; photosynthetic complexes, **246**, 6; picosecond, **232**, 244-245; protein unfolding, **259**, 496-498, 506-509, 595, 598-603; sensitivity, **232**, 42; **246**, 2, 362; synchronous scanning, hemoglobin adducts, **231**, 674-676; time-resolved [See *Fluorescence Spectroscopy (Time-Resolved)*], time scale, **232**, 42.

FLUORESCENCE (2. Time-Resolved Methods)

Time-resolved fluorescence spectroscopy allows one to measure the rate constants of the processes directly influencing fluorescence lifetimes. This information permits one to infer details about the microenvironment of the fluorophore, thereby elucidating the nature of a macromolecule's internal motions.

One can employ linearly polarized light to excite selectively those fluorophores that are in a particular orientation. The difference between excitation and emitted light polarization changes whenever fluorophores rotate during the period of time between excitation and emission. The magnitude of depolarization can be measured, and one can therefore deduce the fluorophore's rotational relaxation kinetics. Extrinsic fluorescence probes are especially useful here, because the proper choice of their fluorescence lifetime will greatly improve the measurement of rotational relaxation rates. One can also determine the freedom of motion of the probe relative to the rotational diffusion properties of the macromolecule to which it is attached. When held rigidly by the macromolecule, the depolarization of a probe's fluorescence is dominated by the the motion of the macromolecule.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Applications, **246**, 335 [immunoassay, **246**, 343-344; nucleic acids, **246**, 344-345; photoreceptors, **246**, 341-343; protein conformation, **246**, 339-340; protein-membrane interactions, **246**, 340-341; two-dimensional imaging, **246**, 345]; energy level diagram, **246**, 336; excited state decay kinetics, **246**, 337-338; in-

terprobe distance determination, **246**, 320-321, 340; phase modulation method, **246**, 299, 345, 359 [data analysis, **246**, 361; detectors, **246**, 360; dynamic range, **246**, 360-361; excitation sources, **246**, 359-360; modulators, **246**, 360; sensitivity, **246**, 361]; pulse methods, **246**, 298-299, 345 [data deconvolution, **246**, 346-347; detectors, **246**, 352-356; global analysis, **246**, 349; lifetime analysis, **246**, 347-348; light sources, **246**, 351-352; scattering sample, **246**, 346; streak camera method, **246**, 356-357; time-correlated single-photon counting, **246**, 349-351; time shift parameter, **246**, 347; up-conversion technique, **246**, 357-359; wavelength dependence of detector response, **246**, 346-347]; resolution of fluorophore-binding extent, **246**, 299-300; rotational diffusion, **246**, 9.

FLUORESCENCE (3. Resonance Energy Transfer)

Fluorescence resonance energy transfer (or FRET) occurs if the absorbed energy of one fluorophore (termed the donor or D) is not emitted, but is instead transferred to a second molecule (termed an acceptor or A). The latter may act either as a fluorescer or as a quencher [See *Fluorescence (1) General Aspects*]. Changes in the conformation of a macromolecule can bring an acceptor molecule closer to or farther from a donor fluorophore, and the efficiency of dipole-dipole resonance energy transfer falls off as the inverse sixth power of the A-to-D distance. Thus, FRET serves as a molecular ruler in the 10–80 Å range. The most common ways for measuring FRET are as (a) a decrease in the emission intensity of the donor; (b) an increase in the emission intensity of the acceptor; or (c) a decreased lifetime for the donor.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Analysis of GTP-binding/GTPase cycle of G protein, **237**, 411-412; applications, **240**, 216-217, 247; **246**, 301-302 [diffusion rates, **246**, 303; distance of closest approach, **246**, 303; DNA (Holliday junctions, **246**, 325-326; hybridization, **246**, 324; structure, **246**, 322-324); dye development, **246**, 303, 328; reaction kinetics, **246**, 18, 302-303, 322]; computer programs for testing, **240**, 243-247; conformational distribution determination, **240**, 247-253; decay evaluation [donor fluorescence decay, **240**, 230-234, 249-250, 252; exponential approximation of exact theoretical decay, **240**, 222-229; linked systems, **240**, 234-237, 249-253; randomly distributed fluorophores, **240**, 237-243]; diffusion coefficient determination, **240**, 248, 250-251; diffusion-enhanced FRET, **246**, 326-328; distance measurement [accuracy, **246**, 330; effect of dye orientation, **246**, 305, 312-313; limitations, **246**, 300, 302, 305, 328, 330]; donor-acceptor pairs, as protein-protein interaction readout, **237**, 419-420; dye labeling [effect on macromolecular structure, **246**, 317; extrinsic fluorophores, **246**, 314-316; intrinsic fluorophores, **246**, 313-314; site-specific attachment, **246**, 316-317]; dye selection, **246**, 18, 314-316; effect of distance between pairs, **240**, 218-219, 251-252; effect on donor fluorescence lifetime, **246**, 302, 306; energy level diagram, **246**, 311; energy transfer equations, **246**, 306-307, 309-310; fluorescence decay functions, **240**, 222-223; fluorescence measurements, **237**, 420-422; Hamiltonian of donor-acceptor interaction,

246, 310; intramolecular fluorescence quenching mediated by, **241**, 54-56, 73-75; lanthanides, **246**, 330-334; measurement techniques, **237**, 420-422 [controls, **246**, 322; decrease in donor intensity, **246**, 317-318; donor lifetimes, **246**, 320-321, 340; donor photobleaching rates, **246**, 321-322; sensitized emission of acceptor, **246**, 318-320]; polarization measurements, **246**, 317-318; rate of inducing transitions, **246**, 310-311; theory, **246**, 10, 18, 306-313; time evolution for concentration of pairs, **240**, 239-240; transfer rate for dipole-dipole interaction, **240**, 216; uniformity of pair distribution, **240**, 237-239.

FLUORESCENCE CORRELATION SPECTROSCOPY (FCS)

A special case of fluctuation correlation, wherein the laser-induced fluorescence from a very small probe volume (defined by a laser beam focused by a water-immersion microscope objective to an open focal light cell) is autocorrelated in time¹⁻³. Random particle motion leads to fluctuations in the number of molecules within the volume element and likewise to fluctuations in the emitted light intensity. Fluctuation analysis offers a noninvasive determination of molecular dynamics in the single molecule range and can be carried out yielding chemical reaction constants or diffusion coefficients. Modest changes in the diffusion coefficient can be determined by measuring of the average decay time of the induced fluorescence light pulses. Fluorescence correlation spectroscopy was first carried out by Magde *et al.*¹, and there are numerous applications⁴. An important application is the determination of thermodynamic and association/dissociation rate constants for intermolecular reactions such as hybridization and renaturation processes between complementary DNA or RNA strands⁵ as well as antibody-antigen interactions.

¹D. Magde, E. L. Elson & W. W. Webb (1972) *Phys. Rev. Lett.* **29**, 705.

²E. L. Elson & D. Magde (1974) *Biopolymers* **13**, 1.

³D. Magde, E. L. Elson & W. W. Webb (1974) *Biopolymers* **13**, 29.

⁴M. Eigen & R. Rigler (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 5740.

⁵P. Schwille, F. Oehlenschläger & N. Walter (1996) *Biochemistry* **35**, 10182.

FLUORESCENCE DECAY

The process by which the energy of an excited atom, molecule, or molecular species is lost as a consequence of photon release as fluorescence. See *Fluorescence*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Analysis, **240**, 290-310; anisotropic, **240**, 301-310; effect of material diffusion, **240**, 219, 221; evaluation of donor fluorescence decay, **240**, 230-234; optimal length of time step, **240**, 224-229; exponential approximation of exact theoretical decay, **240**, 222-229; linked systems, **240**, 234-237; measurement techniques

[phase-modulation, **240**, 219-222, 273-286; pulse method, **240**, 219-220; randomly distributed fluorophores, **240**, 237-243; testing of accuracy and precision of computer programs, **240**, 243-247; time scale range of values, **240**, 725

FLUORESCENCE DEPOLARIZATION

A phenomenon that occurs when a collection of molecules is excited by plane-polarized light, such that only those molecules whose transition dipoles are appropriately aligned will absorb energy. If these excited molecules remain rigidly fixed during excitation and emission, all emitted photons will have their electric vectors lying in the same plane. However, if rotational diffusion occurs during the fluorescence lifetime, then such rotational mobility will reduce the fluorescence light intensity, as sampled by a second polarizer (called the analyzer).

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Additive properties of polarization, **246**, 286; angle-resolved, assessment of peroxidation effects on membranes, **233**, 274-275, 281-283, 287-288; binding isotherm construction, **246**, 287-288; effect of inner filter effects, **246**, 288; incoherent systems, **246**, 263-264; orientational averaging, **246**, 265, 269-270; Perrin equation, **246**, 284-285; polarization of emission, **246**, 284; rotational diffusion, **246**, 9, 260; time-resolved, assessment of peroxidation effects on membranes, **233**, 274, 283-285, 285-287.

FLUORESCENCE LIFETIME IMAGING MICROSCOPY

A microscopical technique that differentiates molecules on the basis of their respective fluorescence lifetimes.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Algorithms [amplitude image calculation, **240**, 733-736; Fourier transform, **240**, 734, 736; phase angle calculation, **240**, 733-736]; calcium quantitation, **240**, 740-742; chloride ion quantitation, **240**, 746; comparison to steady-state microscopy, **240**, 724; detectors and camera, **240**, 729-730, 732, 747; excitation source, **240**, 731, 747; factors affecting probe lifetime, **240**, 724-725; gain modulation signal, **240**, 730; image correction, **240**, 736-740; image processing, **240**, 748; instrumentation, **240**, 728-729, 731, 747-748; magnesium quantitation, **240**, 746; pH, **240**, 746.

FLUORESCENCE RECOVERY AFTER PHOTOBLEACHING

A spectroscopic technique (often abbreviated FRAP) that relies on highly localized photobleaching within specific cell regions to detect and/or measure (a) the lateral diffusibility of molecules within membranes, (b) the diffusion and replacement of molecules within assembled structures, such as the cytoskeleton, and (c) related diffusion and exchange of substances within the cell's aqueous phases. The recovery (or redistribution) phase that follows the photobleaching event provides valuable information

about the rates of diffusion and photolabel replacement into the photobleach target area, and one makes the implicit assumption that photobleaching does not alter the normal rates of diffusion and replacement that occur continuously within cells.

Wolf¹ has written a detailed chapter on how one goes about the task of designing, building, and using a FRAP instrument. After presenting an overview of the technique, he offers advice on: the choice of laser and microscope; methods for splitting, attenuating, and steering the laser microbeam; pertinent information on photomultipliers and output devices; and the operation of the instrument. Elson and Qian² have presented a cogent account of how one interprets FRAP data in terms of the underlying molecular interactions.

¹D. E. Wolfe (1989) *Meth. Cell Biol.* **30**, 271.

²E. L. Elson & H. Qian (1989) *Meth. Cell Biol.* **30**, 307.

FLUOROACETATE/FLUOROCITRATE

Potent metabolic inhibitors of the citric acid cycle. Fluoroacetate (F-CH₂COO⁻) must first be converted to fluoroacetyl-S-CoA (by acetyl-CoA synthetase) and thence to fluorocitrate (by citrate synthase) before it can act as a potent metabolic inhibitor of the aconitase reaction as well as citrate transport. Submicromolar concentrations of (-)-erythro-fluorocitrate can irreversibly inhibit citrate uptake by isolated brain mitochondria.

FLUX

1. A rate of transfer of entities, particles, fluids through a given point, surface, or pathway. For example, the different pathways for a particular enzyme-catalyzed reaction will have a different flux through each of those pathways. **See also** *Chemical Flux*. 2. A measure of the power associated with a particular quantity. **See** *Radiant Energy Flux*; *Radiant Power*. 3. A measure of the strength of a particular field of force (e.g., magnetic flux).

In recognition of the changing nature of virtually all phenomena observable in nature, Heraclitus (c. 500 B.C.) is attributed to be the first to assert¹: "All is flux, nothing stays still." In this respect, Heraclitus may be rightfully recognized as the first kineticist.

¹Plato (c. 360 B.C.) *Cratylus* and Diogenes Laertius (c. 250 A.D.) *Lives of the Philosophers*, book IX, sec. 8.

FLUX CONTROL IN METABOLIC PATHWAYS

The regulatory control of the rate of passage of metabolites through a biochemical pathway. This active area of research includes characterization of allosteric activation and inhibition, gene induction and repression, hysteretic interactions, and posttranslational modifications of all the proteins participating in a given metabolic pathway. For example in glycolysis, flux control includes regulation of glucose influx, the interactions of fructose 2,6-bisphosphate, feedforward activation, and feedback inhibition. Interestingly, studies of yeast fermentation rates, in which the rate-limiting enzymes of glycolysis are overexpressed, have not given any significantly positive results¹. Clearly, there are other aspects of flux control in this well-known pathway that need further elucidation. The recent reports that a subunit of the trehalose synthase complex participates in the control of the flux through yeast glycolysis suggests some unique aspects of regulatory controls in fungi and possibly other organisms². Interestingly, under heat-shock conditions, yeast glucose and trehalose concentrations increase whereas levels of hexose 6-phosphate and fructose 1,6-bisphosphate fall³. These observations hint at an interesting mechanism to restrict passage of metabolites through a pathway since trehalose 6-phosphate is an inhibitor of yeast hexokinase.

¹I. Schaaf, J. Heinisch & F. K. Zimmermann (1989) *Yeast* **5**, 285.

²J. M. Thevelein & S. Hohmann (1995) *Trends in Biochem. Sci.* **20**, 3.

³M. J. Neves & J. Francois (1992) *Biochem. J.* **288**, 859.

FLUX-GENERATING STEP

Any enzymic reaction that supplies substrate to a metabolic pathway. For all subsequent steps to maintain their steady-state concentrations, flux-generating reactions must exhibit zero-order kinetics.

FLUX RATIO

The ratio of two unidirectional fluxes of substances passing through some boundary or membrane.

FMN ADENYLYLTRANSFERASE

This enzyme [EC 2.7.7.2], also known as FAD synthase and FAD pyrophosphorylase, catalyzes the ATP-dependent transfer of an adenylyl group to FMN to yield FAD and diphosphate.

D. B. McCormick, M. Oka, D. M. Bowers-Komro, Y. Yanada & H. A. Hartman (1997) *Meth. Enzymol.* **280**, 407.

F₀F₁-ATPase (and Related Proton-Transporting ATPases)

This membrane-bound, multi-subunit enzyme [EC 3.6.1.34 and EC 3.6.1.35], also known as mitochondrial ATPase, chloroplast ATPase, coupling factors F₀ ("o" standing for oligomycin-sensitive component), F₁ and CF₁, catalyzes the hydrolysis/synthesis of ATP. In mitochondria, chloroplasts and bacteria ATP synthesis is coupled with transport of protons. When located within sealed vesicles, ATPase activity remains latent, but hydrolase activity can be observed upon addition of a proton ionophore whose shuttling of bound protons forth and back across the membrane collapses the transmembrane proton gradient. The phosphoryl transfer step proceeds as a single displacement reaction that is attended by inversion of stereochemical configuration at the phosphorus atom in the transferred phosphoryl group (*i.e.*, no phosphoenzyme intermediate is formed). **See Binding Change Mechanism**

P. D. Boyer (1997) *Ann. Rev. Biochem.* **66**, 717.

J. H. Hurley (1996) *Ann. Rev. Biophys. & Biomol. Struct.* **25**, 137.

R. H. P. Law, S. Manon, R. J. Devenish & P. Nagley (1995) *Meth. Enzymol.* **260**, 133.

J. E. Walker, I. R. Collinson, M. J. Van Raaij & M. J. Runswick (1995) *Meth. Enzymol.* **260**, 163.

J. P. Abrahams, A. G. W. Leslie, R. Lutter & J. E. Walker (1994) *Nature* **370**, 621.

H. S. Penefsky & R. L. Cross (1991) *Adv. Enzymol.* **64**, 173.

W. P. Jencks (1980) *Adv. Enzymol.* **51**, 75.

H. S. Penefsky (1974) *The Enzymes*, 3rd ed., **10**, 375.

FOLYLPOLY- γ -GLUTAMATE SYNTHASE

This enzyme [EC 6.3.2.17], which is also known as folylpoly- γ -glutamate synthetase, catalyzes the sequential ATP-dependent addition of glutamyl groups onto tetrahydrofolyl-[Glu]_n to yield tetrahydrofolyl-[Glu]_{n+1} plus ADP and orthophosphate.

I. Atkinson, T. Garrow, A. Brenner & B. Shane (1997) *Meth. Enzymol.* **281**, 134.

H. C. Imeson & E. A. Cossins (1997) *Meth. Enzymol.* **281**, 141.

FOOTPRINTING

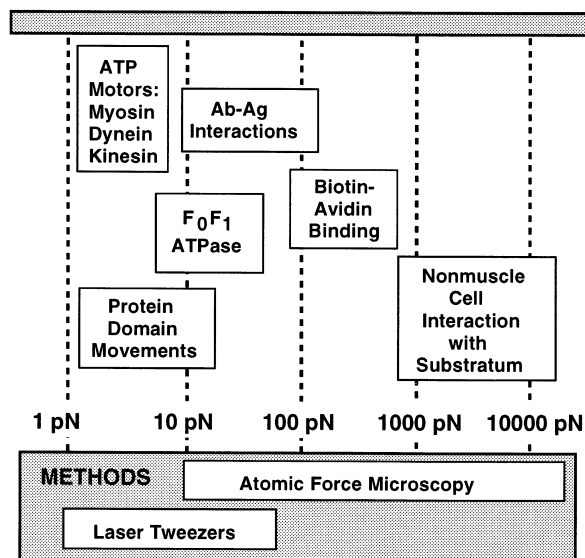
Any technique designed to characterize binding interactions by determining the accessibility of the backbone of macromolecules to cleavage or modification reactions. For nucleic acid interactions, footprinting was originally accomplished by changes in phosphodiester accessibility to DNase I, but numerous chemical and enzymatic methods continue to be elaborated.

FORCE

A vector quantity that changes the state of rest or motion of matter. The force, $\mathbf{F}$, is equal to the change in momentum, ρ , with respect to time (*i.e.*, $\mathbf{F} = d\rho/dt$). If the mass, m , of the entity is constant, the force is that which produces an acceleration, $\mathbf{a}$ (also a vector quantity), of that entity (*i.e.*, $\mathbf{F} = m\mathbf{a}$). The SI unit for force is the newton.

FORCE EFFECTS ON MOLECULAR MOTORS

Exertion of a mechanical force is essential for the functioning of cells and tissues. Various ATP/GTP hydrolysis-dependent motors play key roles in generating force and accomplishing useful work. Examples include the ATP-dependent polymerization of actin monomers, the GTP-dependent polymerization of tubulin dimers, the microtubule-associated dynein and kinesin motors, the ATP-dependent cross-bridging of myosin with actin filaments, various steps in mitosis and meiosis, ATP-dependent ciliary and flagellar dynein motors, the energy-driven bacterial flagellum rotor, and the F_0F_1 ATPase rotor. Within their natural context, cells also exhibit a balance of outwardly and inwardly directed forces, and Ingber¹ recently reviewed tensegrity, a term relating to the structural and organizational interconnectedness of the cytoskeleton with other cellular and extracellular elements. Tensegrity embodies the physical mechanisms that translate mechanical forces into a biochemical response and the integration of the resulting signals with those generated by growth factors and extracellular matrix.



The advent of laser tweezer and atomic force microscope technology has afforded the means for learning more about transduction of chemical and mechanical energy through the measurement of forces generated on a molecular scale. Khan and Scheetz² recently considered various mechanisms for force effects on enzymes, broadly defined to include certain slow NTPases such as the motors mentioned above and the assembly-induced ATP and GTP hydrolases (see Figure). Their review also provides several useful insights regarding the nature of force-producing steps in mechanochemical processes.

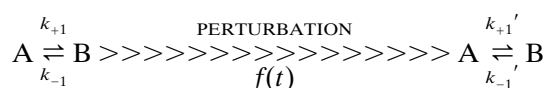
¹D. E. Ingber (1997) *Annu. Rev. Physiol.* **59**, 575.

²S. Khan & M. P. Sheetz (1997) *Annu. Rev. Biochem.* **66**, 785.

FORCING FUNCTION

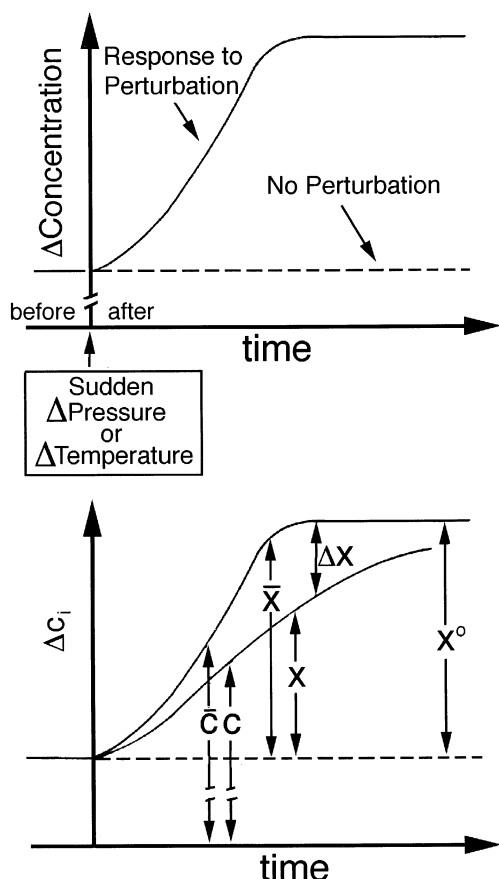
A mathematical function used in fast-reaction kinetics to describe how a perturbation of definable strength and duration leads to a change in the kinetic parameters from an initial condition or state to a final state preceding or overlapping with the ensuing chemical relaxation process under investigation.

A chemical system at equilibrium can be changed by a perturbing force applied in the form of a change in pressure, temperature, or electric field strength. The experimenter seeks to convert the chemical system from one set of rate constants before perturbation to another set of rate constants afterward:



The integrated form of the forcing function $f(t)$ describes the time evolution of the system's transition between these two states (shown in the following plots as the dashed lines; the adjustment of the chemical system in response to the altered rate constants is shown as the solid lines).

The plot shown below is realistic, because the forcing function and the relaxation process are overlapping on a temporal basis. These considerations can be formalized, as described for temporal changes of the concentration(s)^{1,2} of the reacting species or in terms of the advancements³ for each of the individual reaction step(s). Knowledge of the forcing function allows one to obtain corrected chemical relaxation data essential for appro-



A schematic representation of a forcing function (upper curve) and a relaxation process (lower curve). The system, originally at equilibrium, responds to changes in rate constants and equilibrium constants imposed by the forcing function. For a more complete account of forcing functions, see references 1 and 2 below. Based on a diagram from Ilgenfritz².

priate fitting to theoretical models for various potential reaction mechanisms.

¹M. Eigen & L. DeMaeyer (1973) in *Techniques of Chemistry*, vol. VI (Part 2), pp. 63-146, Wiley, New York.

²G. Ilgenfritz (1977) in *Chemical Relaxation in Molecular Biology* (I. Pecht & R. Rigler, eds.), pp. 2-3, Springer-Verlag, Berlin.

³P. R. Schimmel (1971) *J. Chem. Phys.* **54**, 4136.

FORMALDEHYDE DEHYDROGENASE

This enzyme [EC 1.2.1.46] converts formaldehyde and NAD^+ and water into formate and NADH. A similarly named enzyme [EC 1.2.1.1] converts formaldehyde and glutathione in the presence of NAD^+ into S-formylglutathione plus NADH. **See also** *Aldehyde Dehydrogenase*

M. M. Attwood (1990) *Meth. Enzymol.* **188**, 314.

K. T. Douglas (1987) *Adv. Enzymol.* **59**, 103.

FORMALITY

The number of gram formula weights of solute per liter of solution (designated by the unit Formal or F), as distinguished from molarity (M) which explicitly indicates the concentration of a molecular species. Though often confused and used interchangeably, formality defines how a particular solution is formulated (*i.e.*, prepared), whereas molarity defines what occurs after the solute is combined with its solvent. For example, a 0.01 F solution of lactic acid, when adjusted to a pH value corresponding to its $\text{p}K_a$, is actually 0.005 M lactic acid (the Brønsted acid) and 0.005 M lactate anion (the conjugate base). Most solutions are labeled “Molar” or M, whereas the experimenter should technically mark them as “Formal” or F. In electrochemistry, one frequently uses formality. For example, the standard hydrochloric acid solution in the glass pH electrode is 1 F. Likewise, “formal potential” is the potential of an electrode half-reaction if both reductant and oxidant concentrations are set at 1 F.

FORMAMIDOPYRIMIDINE GLYCOSIDASE

This enzyme catalyzes the release of formamidopyrimidines from methylated, alkali-treated DNA. The enzyme will also act on unmethylated formamidopyrimidine derivatives of both adenine and guanine. **See also** *Formamidomethyluracil-DNA Glycosidase*

B. Demple & L. Harrison (1994) *Ann. Rev. Biochem.* **63**, 915.

T. Lindahl (1982) *Ann. Rev. Biochem.* **51**, 61.

B. K. Duncan (1981) *The Enzymes*, 3rd ed., **14**, 565.

FORMATE DEHYDROGENASE

Enzymes that catalyze the oxidation/reduction reaction converting formate to carbon dioxide utilizing NAD^+ [EC 1.2.1.2], cytochrome b_1 [EC 1.2.2.1], cytochrome c -553 [EC 1.2.2.3], or NADP^+ [EC 1.2.1.43] as a redox cofactor.

T. C. Stadtman (1996) *Ann. Rev. Biochem.* **65**, 83.

W. W. Cleland (1995) *Meth. Enzymol.* **249**, 341.

D. R. Jollie & J. D. Lipscomb (1990) *Meth. Enzymol.* **188**, 331.

R. C. Bray (1980) *Adv. Enzymol.* **51**, 107.

T. C. Stadtman (1979) *Adv. Enzymol.* **48**, 1.

K. Dalziel (1975) *The Enzymes*, 3rd ed., **11**, 1.

FORMATION CONSTANT

An association constant (units = M^{-1}) for an equilibrium binding reaction of molecules, most frequently the binding interactions of a macromolecule ligand (which can be a proton, substrate, effector, anion, or cation such

as a metal ion). Concentrations of the products of an association or formation reactions appear in the numerator of formation constants. Macromolecules and ligands may undergo pH-dependent changes in charge, and many binding reactions are often also characterized by changes in entropy and/or enthalpy. Thus, formation constants frequently depend on such variables as pH, temperature, and pressure. **See also** *Ligand Binding*; *Metal Ion Complexation*; *Stability Constant*; *Biominer-alization*

FORMIMINOGLUTAMASE

This enzyme [EC 3.5.3.8] catalyzes the hydrolysis of formamide and L-glutamate from *N*-formimino-L-glutamate.

B. Magasanik, E. Kaminskas & J. Kimhi (1971) *Meth. Enzymol.* **17B**, 57.

FORMIMINOGLUTAMATE DEIMINASE

This enzyme [EC 3.5.3.13] catalyzes the hydrolytic deamination of *N*-formimino-L-glutamate to yield *N*-formyl-L-glutamate and ammonia.

R. B. Wickner & H. Tabor (1971) *Meth. Enzymol.* **17B**, 80.

5-FORMIMINOTETRAHYDROFOLATE CYCLODEAMINASE

This enzyme [EC 4.3.1.4] converts 5-formiminotetrahydrofolate to 5,10-methenyltetrahydrofolate and ammonia.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

FORMYLGLYCINAMIDE RIBONUCLEOTIDE AMIDOTRANSFERASE

This enzyme [EC 6.3.5.3], also known as phosphoribosyl-formylglycinamidine synthase, catalyzes the ATP-dependent synthesis of formylglycinamidine ribonucleotide (FGAM) from formylglycinamide ribonucleotide (FGAR) and glutamine, yielding ADP, orthophosphate, and glutamate as products¹⁻³. The enzyme is inhibited by 6-diazo-5-oxo-L-norleucine, as is true for many other glutamine-dependent enzymes. Buchanan³ has described the covalent chemistry of substrates and antimetabolites.

¹J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

²D. E. Koshland, Jr., & A. Levitzki (1974) *The Enzymes*, 3rd ed., **10**, 539.

³J. M. Buchanan (1982) *Meth. Enzymol.* **87**, 76.

FORMYLGLYCINAMIDINE SYNTHETASE

This enzyme converts 5'-phosphoribosylformylglycinamide in the presence of ATP, L-glutamine, and water into 5'-phosphoribosylformylglycinamidine, L-glutamate, ADP, and orthophosphate.

H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.

FORMYLMETHANOFURAN DEHYDROGENASE

This enzyme [EC 1.2.99.5] converts formylmethanofuran, in the presence of water and a suitable acceptor, into methanofuran, carbon dioxide, and reduced acceptor.

J. G. Ferry (1992) *Crit. Rev. Biochem. Mol. Biol.* **27**, 473.

FORMYLMETHANOFURAN:5,6,7,8-TETRAHYDROMETHANOPTERIN N-FORMYLTRANSFERASE

This enzyme [EC 2.3.1.101] catalyzes the reaction of *N*-formylmethanofuran with 5,6,7,8-tetrahydromethanopterin to yield methanofuran and 5-formyl-5,6,7,8-tetrahydromethanopterin.

J. G. Ferry (1992) *Crit. Rev. Biochem. Mol. Biol.* **27**, 473.

10-FORMYLTETRAHYDROFOLATE DEFORMYLASE

This enzyme [EC 3.5.1.10], also known as formyltetrahydrofolate amidohydrolase and formyl-FH₄ hydrolase, catalyzes the hydrolysis of 10-formyltetrahydrofolate to yield formate and tetrahydrofolate.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

10-FORMYLTETRAHYDROFOLATE DEHYDROGENASE

This enzyme [EC 1.5.1.6] catalyzes the conversion of 10-formyltetrahydrofolate in the presence of NADP⁺ and water to yield tetrahydrofolate, NADPH, and carbon dioxide.

V. Schirch (1997) *Meth. Enzymol.* **281**, 146.

FORMYLTETRAHYDROFOLATE HYDROLASE

This enzyme catalyzes the hydrolysis of *N*¹⁰-formyltetrahydrofolate to produce tetrahydrofolate and formate.

H. Zalkin (1997) *Meth. Enzymol.* **281**, 214.

10-FORMYLTETRAHYDROFOLATE SYNTHETASE

This enzyme [EC 6.3.4.3], whose recommended EC name is formate-tetrahydrofolate ligase, catalyzes the ATP-dependent conversion of formate and tetrahydrofolate to yield ADP, phosphate, and 10-formyltetrahydrofolate.

V. Schirch (1997) *Meth. Enzymol.* **281**, 146.

R. E. MacKenzie (1997) *Meth. Enzymol.* **281**, 171.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

FOSTER-NIEMANN PLOTS

A plot of an enzyme progress curve with respect to product inhibition. This plot is particularly useful with regard to enzymes with isomerization steps in which the product acts as a noncompetitive inhibitor. In the Foster-Niemann plot^{1,2}, $[P]/t$ is plotted as a function of $(1/t) \ln(1 - [P]/[P_\infty])$. As discussed by Orsi and Tipton³ in their review of progress curves, if product inhibition is competitive (as it would be in a standard Uni Uni mechanism), the Foster-Niemann plot would be linear (note that the integrated form of the Uni Uni mechanism, in the presence of product, is $[P]/t = [((V_{\max,f}/K_a) + (V_{\max,r}/K_p))/(K_a^{-1} - K_p^{-1})] + \{(1 + ([A_o]/K_p))/(K_a^{-1} - K_p^{-1}) + ([A_o]/(1 + K_{eq}))\}(1/t) \ln([P_\infty] - [P])/[P_\infty]$ where $V_{\max,f}$ and $V_{\max,r}$ are the maximum velocities in the forward and reverse reaction, respectively, K_a and K_p are the Michaelis constants for A and P, respectively, K_{eq} is the reaction equilibrium constant, $[P]$ is the concentration of product at time t , and $[P_\infty]$ is the final concentration of product.) However, the Foster-Niemann plot will be curved if either: (1) there is an error in zero time measurements⁴ (hence, it behooves the investigator to be precise as to the starting time and conditions of the reaction), or (2) product inhibition is noncompetitive (as is the case with isomerization mechanisms such as the Iso Uni Uni mechanism). As reviewed by Rebholz and Northrop², these plots are valuable in that progress curves using both high and low concentrations of substrate can be placed on a normalized scale, it is usually readily apparent if the lines are curved, and the goodness-of-fit of the data can be easily compared with other integrated expressions. **See** *Iso Mechanisms*

¹R. J. Foster & C. Niemann (1953) *Proc. Natl. Acad. Sci. U.S.A.* **39**, 999.

²K. L. Rebholz & D. B. Northrop (1995) *Meth. Enzymol.* **249**, 211.

³B. A. Orsi & K. F. Tipton (1979) *Meth. Enzymol.* **63**, 159.

⁴M. Taraszka (1962) Ph.D. Thesis, Univ. of Wisconsin, Madison.

FOURIER TRANSFORM

A mathematical procedure for expressing any single-valued periodic function as the infinite series summation of a succession of sine waveforms of higher order, such that $f(x)$, a function with a period of 2π , can be treated as a trigonometric series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

In addition to this capacity to deconvolute any periodic function, one may likewise reconstruct the original function.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Application in fluorescence, **240**, 734, 736, 757; convolution, **240**, 490-491; in NMR [discrete transform, **239**, 319-322; inverse transform, **239**, 208, 259; multinuclear multidimensional NMR, **239**, 71-73; shift theorem, **239**, 210; time-domain shape functions, **239**, 208-209]; FT infrared spectroscopy [iron-coordinated CO, in difference spectrum of photolyzed carbonmonoxymyoglobin, **232**, 186-187; for fatty acyl ester determination in small cell samples, **233**, 311-313; myoglobin conformational substrates, **232**, 186-187].

Applications in infrared spectroscopy: A and B bands, iron-coordinated CO, **232**, 186-187; application to allosteric mechanisms, **249**, 566; bacteriorhodopsin, **246**, 9, 380-381; caged compounds, **246**, 6, 520-521; DNA [base pair formation, **246**, 506; conformation, **246**, 506-507; denaturation thermodynamics, **246**, 506; ligand interactions, **246**, 6, 507; sample requirements, **246**, 506]; fatty acyl ester determination in small cell samples, **233**, 311-313; instrumentation [energy throughput, **246**, 503-504; interferometer, **246**, 504; multiplex advantage, **246**, 503-504; lipids [anesthetic perturbations, **246**, 509; disorder in membranes, **246**, 508-510; head groups, **246**, 510; membrane models, **246**, 507-508; phase transitions, **246**, 508-509; protein interactions, **246**, 510]; myoglobin conformational substrates, **232**, 186-187; photosystem II tyrosyl radicals, **258**, 316-318; polarization, protein order in membranes, **246**, 7; proteins [amide I band, **246**, 512-513; amide II mode, **246**, 512; derivative spectra, **246**, 513; proton exchange rates, **246**, 512; secondary structure, accuracy, **246**, 513-514; water interference in spectra, **246**, 138, 380, 387, 503, 513]; reaction-induced infrared difference spectroscopy [bacteriorhodopsin, **246**, 519-520; chromoproteins, **246**, 517-519]; trigger [electron transfer, **246**, 522; light, **246**, 517-522; requirements, **246**, 517]; sample requirements, **246**, 526; sensitivity, **246**, 380-381, 517; time-resolved spectroscopy [rapid-scan technique, **246**, 381, 523; step-scan method, **246**, 381, 523; stroboscopic method, **246**, 381, 523].

FOURIER TRANSFORM INFRARED/PHOTOACOUSTIC SPECTROSCOPY TO ASSESS SECONDARY STRUCTURE

Fourier transform infrared/photoacoustic spectroscopy (FT-IR/PAS) can be used to evaluate the secondary structure of proteins, as demonstrated by experiments on concanavalin A, hemoglobin, lysozyme, and trypsin, four proteins having different distributions of secondary

structures. Secondary structure assessed on the basis of FT-IR/PAS agreed well with that determined by X-ray diffraction, CD, and traditional FT-IR methods. This technique requires smaller amounts of sample, and the secondary structure can be evaluated with as little as 0.5 μg protein.

S. Luo, C. Y. Huang, J. F. McClelland & D. J. Graves (1994) *Anal. Biochem.* **216**, 67.

FRACTAL

A mathematically definable structure which exhibits the property of always appearing to have the same morphology, even when the observer “endlessly” enlarges portions of it. In general, fractals have three features: heterogeneity, self-similarity, and the absence of a well-defined scale of length. Fractals have become important concepts in modern nonlinear dynamics. **See** *Chaos Theory*

B. B. Mandelbrot (1983) *The Fractal Geometry of Nature*, Freeman, New York.

B. B. Mandelbrot (1984) *J. Stat. Phys.* **34**, 895.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Background literature, **210**, 638; in biological data, **210**, 648; definition, **210**, 665; identification in biochemical reactions, **210**, 651; related terminology, **210**, 655; for time scaling in biochemical networks, **210**, 636; noise scaling, **210**, 643; time history analysis, **210**, 643.

FRACTAL REACTION KINETICS

Reactions occurring at interfaces of two or more physical phases (namely those processes referred to as heterogeneous reactions) as well as reactions occurring in an under-stirred solution often do not obey the kinetic theory that applies to reactions taking place in well-stirred solutions. These diffusion-controlled reactions typically exhibit geometric constraints. A good example is crystal growth, a process occurring at the junction between liquid and crystal phases; crystal growth kinetics reflect a complex manifold of processes, including the diffusion of molecules from the bulk solution phase into the Gouy-Chapman layer (*i.e.*, bulk diffusion), diffusion of molecules into the Stern layer, two-dimensional surface transport, and capture by the crystal lattice. The converse (*i.e.*, dissolution) is likewise the culmination of a series of diffusion-limited steps. Fractals can be used to treat the kinetics at interfaces as well as percolation clusters¹, and such approaches allow one to deal with fractal orders for some elementary reactions, self-ordering and self-unmixing of reactants, as well as rate coefficients that exhibit temporal memory. Fractal network dynamics²

can also be applied to the special kinetic behavior of random structures that exhibit fractal geometry.

¹R. Kopelman (1988) *Science* **241**, 1620.

²R. Orbach (1986) *Science* **231**, 814.

FRACTIONATION FACTOR

A ratio defining the isotopic distribution of two isotopes equilibrated between two different chemical species. If X , followed by a subscript, represents the mole fraction of an isotope (denoted by that same subscript), then the fractionation factor, often symbolized by ϕ , with respect to chemical species A and B is $(X_1/X_2)_A/(X_1/X_2)_B$. Fractionation factors can also refer to different sites, A and B, within the same chemical species. As an example, the deuterium solvent fractionation factor, used in studying solvent isotope effects, is $\phi = (X_D/X_H)_{\text{solute}}/(X_D/X_H)_{\text{solvent}}$.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

FRAGMENTATION

1. The heterolytic cleavage of a molecular entity of the form $R_1-R_2-R_3-R_4-X \rightarrow R_1-R_2^+ + R_3-R_4 + :X^-$.
2. The conversion of a radical to a diamagnetic molecule and a smaller radical: *e.g.*, $R_1-R_2-X\cdot \rightarrow R_1-R_2 + X\cdot$.
3. The conversion of a radical ion to an ion and a separate free radical.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

FRANCK-CONDON PRINCIPLE

The most likely electronic transition will occur without changes in the positions of the nuclei (*e.g.*, little change in the distance between atoms) in the molecular entity and its environment. Such a state is known as a Franck-Condon state, and the transition is referred to as a vertical transition. In such transitions, the intensity of the vibronic transition is proportional to the square of the overlap interval between the vibrational wavefunctions of the two states. **See** *Fluorescence; Jablonski Diagram*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

FREE RADICAL

A molecular entity having an unpaired electron. Free radicals are usually short-lived and are highly reactive. Examples include methyl radical ($\cdot\text{CH}_3$), hydroxyl radical ($\cdot\text{OH}$), nitric oxide ($\text{NO}\cdot$), hydroperoxy radical ($\text{HOO}\cdot$), and the sodium atom (Na). Note that, since there is at least one unpaired electron, a free radical has

a net magnetic moment, is paramagnetic, and can yield an electron spin resonance spectrum.

The unpaired electron is usually depicted as a dot adjacent to the symbol of the atom having the highest spin density. Some investigators have suggested that the word “free” not be so widely used, favoring instead the restricted use of the term “free radical” for indicating those radicals not forming parts of radical pairs. Charged free radicals are more commonly called radical anions or radical cations. σ -Radicals are those molecular entities in which the unpaired electron occupies an orbital having considerable s character whereas π -radicals are those molecular entities with an unpaired electron in an orbital with p character. **See** *Radical*; *Biradical*; *Transient Chemical Species*; *Persistent*

FREQUENCY

1. The number of times that an event or a periodic function repeats during a unit value of an independent variable, usually time. 2. For any periodic motion, the frequency, symbolized by ν , is the number of repetitions (*e.g.*, number of revolutions or cycles or oscillations of an electromagnetic wave) of some process occurring in a unit period of time. The SI unit for frequency, reciprocal second, is often referred to as a hertz (Hz). However, the term hertz should only be used in reference to cycles per second and not for radial (circular) frequency or angular velocity symbolized by φ ($= 2\pi\nu$) and having SI units of $\text{rad}\cdot\text{s}^{-1}$.

FRIEDEN DILEMMA

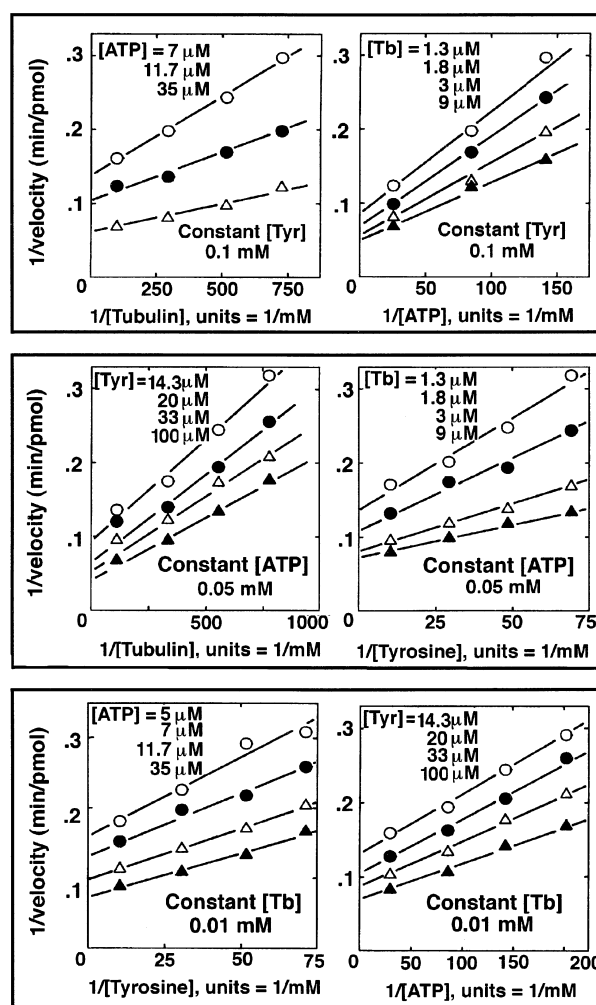
A potential limitation encountered when one seeks to characterize the kinetic binding order of certain rapid equilibrium enzyme-catalyzed reactions containing specific abortive complexes. Frieden¹ pointed out that initial rate kinetics alone were limited in the ability to distinguish a rapid equilibrium random Bi Bi mechanism from a rapid equilibrium ordered Bi Bi mechanism if the ordered mechanism could also form the EB and EP abortive complexes. Isotope exchange at equilibrium experiments² would also be ineffective. However, such a dilemma would be a problem only for those rapid equilibrium enzymes having k_{cat} values less than $30\text{--}50\text{ sec}^{-1}$.² For those rapid equilibrium systems in which k_{cat} is small, Frieden's dilemma necessitates the use of procedures other than standard initial rate kinetics.

¹C. Frieden (1976) *Biochem. Biophys. Res. Commun.* **68**, 914.

²D. L. Purich, R. D. Allison & J. A. Todhunter (1977) *Biochem. Biophys. Res. Commun.* **77**, 753.

FRIEDEN'S METHOD FOR TER-REACTANT ENZYMIC KINETICS

Frieden¹ presented a protocol for the kinetic analysis of enzymes having three substrates (A, B, and C). In this procedure, one substrate is held constant at a high concentration (*i.e.*, greater than its corresponding Michaelis constant), albeit nonsaturating. The system is then treated as a pseudo-two-substrate enzyme. Thus as an example, C would be held at a high, nonsaturating, and constant concentration while initial rates are measured at different concentrations of A (the concentrations of B being varied at different constant levels as well). The identical procedure is then followed with high, constant, nonsaturating levels of B and then of A.



Initial rate patterns for bovine brain tubulin:tyrosine ligase (Reaction: detyrosinated tubulin (Tb) + L-tyrosine + ATP = tyrosinated tubulin + ADP + orthophosphate). The results of each set of experiments are shown as a single frame containing the same data points plotted as a function of one or the other varied substrate. The nonvaried substrate concentration is shown in the lower right-hand corner of every double-reciprocal plot. (From ref. 2.)

Hence, a ter-reactant system will result in six double-reciprocal plots (or Eadie-Hofstee plots, *etc.*). Fromm³ has pointed out the importance of keeping the third substrate to nonsaturating conditions. Under saturating conditions, parallel lines might result, even though the enzyme is not ping pong in nature. If parallel lines are observed in ter-reactant (or higher) systems, it is strongly urged to repeat the kinetic study at a lower concentration of the unvaried substrate, in order to verify the parallel-lined behavior. The Frieden method is easily extended to quad- and higher systems. **See Fromm Method; Substrate Concentration Range**

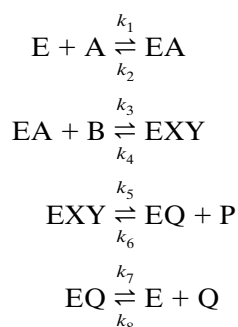
¹C. Frieden (1959) *J. Biol. Chem.* **234**, 2891.

²N. L. Deans, R. D. Allison, & D. L. Purich (1992) *Biochem. J.* **286**, 243.

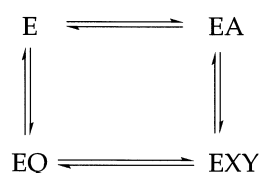
³H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

FROMM'S METHOD FOR DERIVING ENZYME RATE EQUATIONS

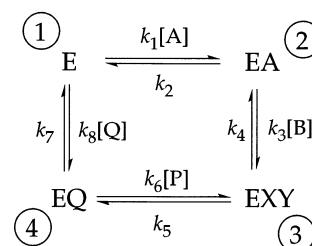
A useful procedure for deriving steady-state rate expressions for enzyme-catalyzed reactions^{1,2}. Although not as commonly used as the King and Altman method, it is far more convenient (and less error-prone) when attempting to obtain expressions for complicated reaction schemes. One of its values is that the approach is very systematic and straightforward. The systematic nature of the procedure can be illustrated by the derivation of the steady-state ordered Bi Bi reaction.



STEP ONE: Using just the different enzyme forms, write out the reaction scheme, such that a closed geometric pattern is generated. In this case (E) occurs at the start and end of the catalytic reaction cycle, and we get:



STEP TWO: Number each enzyme form with a circled number, and indicate each arrow with the corresponding unimolecular or bimolecular rate constant. For bimolecular rate constants, the corresponding reactant species (*i.e.*, substrate or product) should also be identified with the corresponding arrow. [To avoid errors by misplacing a minus sign, it is recommended that only positive numbers be used as subscripts for the rate constants (preferably odd numbers for forward reactions, even numbers for reverse reactions).] Thus, for our ordered Bi Bi scheme:



STEP THREE: Each circled number (*i.e.*, each enzyme form or enzyme-containing species) is characterized by one or more arrows leading away from that species. List these numbers in parentheses (the same numbers that were circled in step two) equal to a sum of those rate constants (including the substrate or product associated with bimolecular steps). Thus,

$$(1) = k_1 A + k_8 Q$$

$$(2) = k_2 + k_3 B$$

$$(3) = k_4 + k_5$$

$$(4) = k_6 P + k_7$$

Note that the letters represent concentrations (*i.e.*, [A], [B], [P], and [Q]).

STEP FOUR: For each enzyme form, write down all of the shortest one-step paths contributing to the formation of that enzyme form (including the rate constant and substrate or product for bimolecular steps, with each arrow). Thus, for our example:

$$\text{for 1 we have } 2 \xrightarrow{k_2} 1 \text{ and } 4 \xrightarrow{k_7} 1$$

$$\text{for 2 we have } 1 \xrightarrow{k_1 A} 2 \text{ and } 3 \xrightarrow{k_4} 2$$

$$\text{for 3 we have } 2 \xrightarrow{k_3 B} 3 \text{ and } 4 \xrightarrow{k_6 P} 3$$

$$\text{for 4 we have } 1 \xrightarrow{k_8 Q} 4 \text{ and } 3 \xrightarrow{k_5} 4$$

STEP FIVE: Adjacent to each one-step route (identified in the previous step) are written (in parentheses) the numbers of the enzyme species not appearing in that particular one-step path. These “collections” of one-step terms (*i.e.*, the “one-branch” approach of Fromm), are summed together to form the determinant for that enzyme species. Thus:

$$\begin{aligned}\textcircled{1} &= 2 \rightarrow 1(3)(4) + 4 \rightarrow 1(2)(3) = E \\ \textcircled{2} &= 1 \rightarrow 2(3)(4) + 3 \rightarrow 2(1)(4) = EA \\ \textcircled{3} &= 2 \rightarrow 3(1)(4) + 4 \rightarrow 3(1)(2) = EXY \\ \textcircled{4} &= 1 \rightarrow 4(2)(3) + 3 \rightarrow 4(1)(2) = EQ\end{aligned}$$

(One may also opt to use Huang's “consecutive-branch” approach.^{3,4})

STEP SIX: Substitute appropriate rate constants for the arrows (from step four) and the numbers in parentheses with the expressions obtained from step three. Thus,

$$\begin{aligned}[E] &= \textcircled{1} = k_2(k_4 + k_5)(k_6P + k_7) \\ &\quad + k_7(k_2 + k_3B)(k_4 + k_5) \\ [EA] &= \textcircled{2} = k_1A(k_4 + k_5)(k_6P + k_7) \\ &\quad + k_4(k_1A + k_8Q)(k_6P + k_7) \\ [EXY] &= \textcircled{3} = k_3B(k_1A + k_8Q)(k_6P + k_7) \\ &\quad + k_6P(k_1A + k_8Q)(k_2 + k_3B) \\ [EQ] &= \textcircled{4} = k_8Q(k_2 + k_3B)(k_4 + k_5) \\ &\quad + k_5(k_1A + k_8Q)(k_2 + k_3B)\end{aligned}$$

STEP SEVEN: Expand each term in step six.

$$\begin{aligned}[E] &= k_2k_4k_6P + k_2k_4k_7 + k_2k_5k_6P + k_2k_5k_7 \\ &\quad + k_2k_4k_7 + k_2k_5k_7 + k_3k_4k_7B + k_3k_5k_7B \\ [EA] &= k_1k_4k_6AP + k_1k_4k_7A + k_1k_5k_6AP \\ &\quad + k_1k_5k_7A + k_1k_4k_6AP + k_1k_4k_7A \\ &\quad + k_4k_6k_8PQ + k_4k_7k_8Q \\ [EXY] &= k_1k_3k_6ABP + k_1k_3k_7AB + k_3k_6k_8BPQ \\ &\quad + k_3k_7k_8BQ + k_1k_2k_6AP + k_1k_3k_6ABP \\ &\quad + k_2k_6k_8PQ + k_3k_6k_8BPQ \\ [EQ] &= k_2k_4k_8Q + k_2k_5k_8Q + k_3k_4k_8BQ \\ &\quad + k_3k_5k_8BQ + k_1k_2k_5A + k_1k_3k_5AB \\ &\quad + k_2k_5k_8Q + k_3k_5k_8BQ\end{aligned}$$

STEP EIGHT: Eliminate all redundant terms in the determinant for each enzyme form. For example, in the expanded expression for [E], the $k_2k_4k_7$ terms appear

twice. One of these is eliminated. The same procedure is followed for each enzyme form.

STEP NINE: Eliminate all “forbidden” terms appearing in any determinant. These are terms containing products of two rate constants associated with the same reversible step. For example, in the expression for [E], $k_3k_4k_7B$ and $k_2k_5k_6P$ are forbidden terms in that k_3k_4 and k_5k_6 , respectively, (*i.e.*, cyclic terms) are present.

STEP TEN: Eliminate any closed loop terms. There are no closed loop terms in the example given above (if there were it would either contain the product $k_1k_3k_5k_7$ or $k_2k_4k_6k_8$). Care should be exercised here. Although closed loop terms are rare, not considering them in enzyme rate derivations can lead to incorrect expressions and, thus, inaccurate and erroneous predictions of enzyme rate behavior. A good check is to write out all of the rate-constant products that would constitute a closed loop and then check each enzyme determinant to see if any are present. If so, they are eliminated. Huang's modification^{3,4} of Fromm's systematic approach also addresses this issue of closed loops.

After each of these elimination steps, the final determinants for the enzyme forms of the steady-state ordered Bi Bi reaction scheme are:

$$\begin{aligned}[E] &= k_2k_7(k_4 + k_5) + k_3k_5k_7[B] + k_2k_4k_6[P] \\ [EA] &= k_1k_7(k_4 + k_5)[A] + k_1k_4k_6[A][P] \\ &\quad + k_4k_6k_8[P][Q] \\ [EXY] &= k_1k_3k_7[A][B] + k_2k_6k_8[P][Q] \\ &\quad + k_1k_3k_6[A][B][P] + k_3k_6k_8[B][P][Q] \\ [EQ] &= k_1k_3k_5[A][B] + k_3k_5k_8[B][Q] \\ &\quad + k_2k_8(k_4 + k_5)[Q]\end{aligned}$$

Since $[E_{\text{total}}] = [E] + [EA] + [EXY] + [EQ]$, the relative proportion of each enzyme form during the steady-state conditions is given by the ratio of the determinant of the specific enzyme form to that of the total enzyme: *e.g.*, $[E]/[E_{\text{total}}]$.

Under the typical condition of initial rate studies in which [P] and [Q] are effectively zero (or, are sufficiently below their respective K_i values to be kinetically insignificant), the terms in the determinant containing [P] and [Q]

are eliminated (obviously, this would not be the case in product inhibition studies). The velocity of the enzyme catalyzed reaction is given by $d[Q]/dt = v = k_7[EQ] - k_8[E][Q]$ which reduces to $k_7[EQ]$ since $[Q] \approx 0$. Substituting the determinant for $[EQ]$ and multiplying by unity in the form of $[E_{\text{total}}]/[E_{\text{total}}] = [E_{\text{total}}]/\{[E] + [EA] + [EXY] + [EQ]\}$ in which the denominator is the sum of all the determinants obtained in the derivation, one obtains an expression for the steady-state rate of the reaction in terms of all the appropriate rate constants:

$$v = \frac{k_1 k_3 k_5 k_7 [A][B][E_{\text{total}}]}{\{k_2 k_7 (k_4 + k_5) + k_1 k_7 (k_4 + k_5) [A] + k_3 k_5 k_7 [B] + k_1 k_3 (k_5 + k_7) [A][B]\}}$$

If the standard form of the rate expression is to be used (*i.e.*, with Michaelis constants, maximum velocities, *etc.*) (**See Cleland Nomenclature**), then both the numerator and denominator are divided by the coefficient of the $[A][B]$ term appearing in the denominator (*i.e.*, divide by the coefficient of the term containing all of the substrates utilized in the reaction raised to the power equal to the stoichiometry number of each substrate). Thus, the ordered Bi Bi expression becomes

$$v = \frac{V_{\text{max}} [A][B]}{K_{\text{ia}} K_{\text{b}} + K_{\text{b}} [A] + K_{\text{a}} [B] + [A][B]}$$

in which V_{max} is the maximum forward reaction velocity equal to $k_5 k_7 [E_{\text{total}}]/(k_5 + k_7)$, K_{ia} is the dissociation constant for substrate A and is equal to k_2/k_1 , K_{a} is the Michaelis constant for A and is equal to $k_5 k_7 / \{k_1 (k_5 + k_7)\}$, and K_{b} is the Michaelis constant for B and is equal to $k_7 (k_4 + k_5) / \{k_3 (k_5 + k_7)\}$. **See Enzyme Rate Equations (2. Derivation of Initial Velocity Equations)**

¹H. J. Fromm (1970) *Biochem. Biophys. Res. Commun.* **40**, 692.

²H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

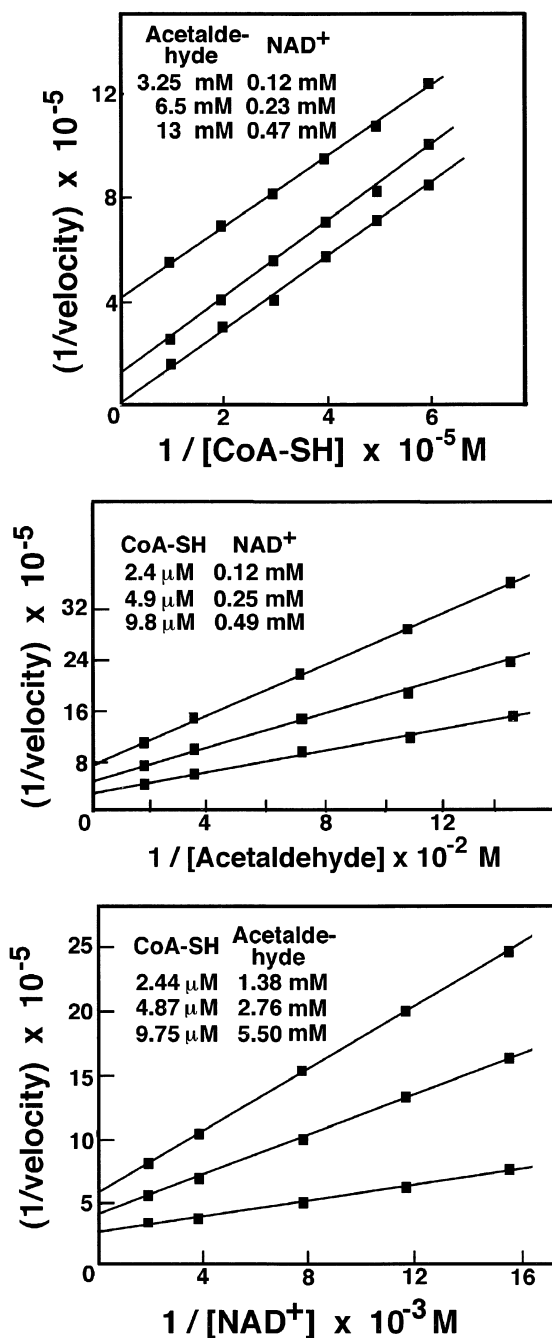
³C. Y. Huang (1978) *Fed. Proc., Fed. Am. Soc. Exp. Biol.* **37**, 1423.

⁴C. Y. Huang (1979) *Meth. Enzymol.* **63**, 54.

FROMM'S METHOD FOR TER-REACTANT ENZYMIC KINETICS

A procedure to simplify the experimental method in the kinetic analysis of three-substrate, enzyme-catalyzed reactions¹⁻⁴. In this method, the concentration of one substrate is varied while the other two substrates are kept in a constant ratio and in which the individual concentrations of these two substrates are in the neighborhood of their respective Michaelis constants. The experi-

ment is then repeated a number of times utilizing different absolute concentrations of the two substrates, albeit maintaining the same ratio. The same procedure is then used such that each substrate will be the varied reactant.



Initial rate patterns for *Escherichia coli* NAD⁺-dependent coenzyme A-linked aldehyde dehydrogenase (Reaction: NAD⁺ + CoA-SH + acetaldehyde = NADH + acetyl-S-CoA + H⁺). The results of each of three experiments are shown as a single double-reciprocal plot, and the nonvaried substrate concentrations for each curve are indicated above the data points.

With this protocol, ping pong reactions will yield at least one plot containing parallel lines. However, it is difficult with this procedure to obtain values for a number of the kinetic parameters (*e.g.*, Michaelis constants) for the reaction. An example of a system studied with this procedure is provided by *E. coli* CoA-linked aldehyde dehydrogenase². **See Frieden Protocol**

¹H. J. Fromm (1967) *Biochim. Biophys. Acta* **139**, 221.

²F. B. Rudolph, D. L. Purich & H. J. Fromm (1968) *J. Biol. Chem.* **243**, 5539.

³F. B. Rudolph & H. J. Fromm (1969) *J. Biol. Chem.* **244**, 3832.

⁴H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

FRONTIER ORBITALS

The highest-energy occupied molecular orbital (HOMO) and the lowest-energy unoccupied molecular orbital (LUMO) of a particular molecular entity. The HOMO can be completely or partially filled whereas the LUMO can be completely or partially vacant. The frontier orbital method allows one to interpret reaction behavior by studying HOMO and LUMO overlaps between reacting entities.

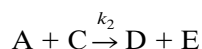
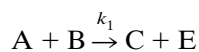
K. Fukai (1971) *Acc. Chem. Res.* **4**, 57.

I. Fleming (1976) *Frontier Orbitals and Organic Chemical Reactions*, Wiley, New York.

K. Fukai (1982) *Angew. Chem. Int. Ed. Engl.* **21**, 801.

FROST-SCHWEMER METHOD

A procedure for analyzing competitive, consecutive second-order reactions¹⁻³ of the type:



This method is of use in thiol-disulfide interchanges and when using two-protonic-state electrophiles⁴.

¹A. A. Frost & W. C. Schwemer (1952) *J. Amer. Chem. Soc.* **74**, 1268.

²W. C. Schwemer & A. A. Frost (1951) *J. Amer. Chem. Soc.* **73**, 4541.

³J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, 3rd ed., Wiley, New York.

⁴K. Brocklehurst (1982) *Meth. Enzymol.* **87**, 427.

1,2-β-FRUCTAN:1,2-β-FRUCTAN 1-β-FRUCTOSYLTRANSFERASE

This enzyme [EC 2.4.1.100] catalyzes fructosyl transfer between (1,2)-β-D-fructosyl_m and (1,2)-β-D-fructosyl_n to yield (1,2)-β-D-fructosyl_{m-1} plus (1,2)-β-D-fructosyl_{n+1}.

C. J. Pollock & A. J. Cairns (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 77.

FRUCTAN β-FRUCTOSIDASE

This enzyme [EC 3.2.1.80] catalyzes the hydrolysis of terminal nonreducing 2,1- and 2,6-linked β-D-fructofuranose residues in fructans.

C. J. Pollock & A. J. Cairns (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 77.

FRUCTOKINASE

This enzyme [EC 2.7.1.4] catalyzes the ATP-dependent phosphorylation of fructose to form fructose 6-phosphate and ADP.

W. W. Cleland (1977) *Adv. Enzymol.* **45**, 373.

FRUCTOSE-1,6-BISPHOSPHATASE

This hydrolase [EC 3.1.3.11] catalyzes conversion of D-fructose 1,6-bisphosphate to D-fructose 6-phosphate and orthophosphate.

H. C. Huppe & D. H. Turpin (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 577.

S. J. Pilkis & T. H. Claus (1991) *Ann. Rev. Nutr.* **11**, 465.

S. J. Pilkis, T. H. Claus, P. D. Kountz & M. R. El-Maghrabi (1987) *The Enzymes*, 3rd ed., **18**, 3.

E. Van Schaffingen (1987) *Adv. Enzymol.* **59**, 315.

S. Pontremoli & B. L. Horecker (1971) *The Enzymes*, 3rd ed., **4**, 611.

FRUCTOSE-2,6-BISPHOSPHATASE

This hydrolase [EC 3.1.3.46] catalyzes the conversion of D-fructose 2,6-bisphosphate to yield D-fructose 6-phosphate and orthophosphate. **See also 6-Phosphofructo-2-kinase**

S. J. Pilkis, T. H. Claus, I. J. Kurland & A. J. Lange (1995) *Ann. Rev. Biochem.* **64**, 799.

K. Uyeda (1991) *A Study of Enzymes* **2**, 445.

S. J. Pilkis, T. H. Claus, P. D. Kountz & M. R. El-Maghrabi (1987) *The Enzymes*, 3rd ed., **18**, 3.

E. Van Schaffingen (1987) *Adv. Enzymol.* **59**, 315.

D-FUCONATE DEHYDRATASE

This enzyme [EC 4.2.1.67] converts D-fuconate to 2-dehydro-3-deoxy-D-fuconate along with the release of water.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

FUCOSE-1-PHOSPHATE GUANYLYLTRANSFERASE

This enzyme [EC 2.7.7.30] catalyzes the conversion of GTP and L-fucose 1-phosphate to yield GDP-L-fucose and pyrophosphate (or, diphosphate).

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

α -L-FUCOSIDASE

This enzyme [EC 3.2.1.51] catalyzes the release of alcohols from α -L-fucosides to yield α -L-fucose.

S. C. Fry (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 497.

E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.

P. M. Dey & E. del Campillo (1984) *Adv. Enzymol.* **56**, 141.

H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

FUCOSYLTRANSFERASES

This class of enzymes includes galactoside 3(4)-L-fucosyltransferase, [EC 2.4.1.65]; glycoprotein 6- α -L-fucosyltransferase [EC 2.4.1.68], galactoside 2- α -L-fucosyltransferase [EC 2.4.1.69], and galactoside 3-fucosyltransferase [EC 2.4.1.152].

Y. Kishimoto (1983) *The Enzymes*, 3rd ed., **16**, 357.

K. A. Presper & E. C. Heath (1983) *The Enzymes*, 3rd ed., **16**, 449.

R. J. Staneloni & L. F. Leloir (1982) *Crit. Rev. Biochem.* **12**, 289.

T. A. Beyer, J. E. Sadler, J. I. Rearick, J. C. Paulson & R. L. Hill (1981) *Adv. Enzymol.* **52**, 23.

FUCULOSE-1-PHOSPHATE ALDOLASE

This enzyme [EC 4.1.2.17] catalyzes the aldol cleavage of L-fucose 1-phosphate to yield glycero phosphate and (S)-lactaldehyde.

D. S. Feingold & P. A. Hoffee (1972) *The Enzymes*, 3rd ed., **7**, 303.

FUMARASE

This hydratase [EC 4.2.1.2] catalyzes the reversible dehydration of (S)-malate to form fumarate and water.

I. A. Rose (1995) *Meth. Enzymol.* **249**, 315.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

D. J. T. Porter & H. J. Bright (1980) *J. Biol. Chem.* **255**, 4772.

W. W. Cleland (1977) *Adv. Enzymol.* **45**, 373.

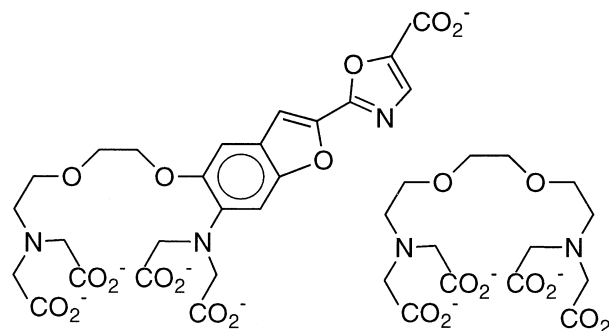
R. L. Hill & J. W. Teipel (1971) *The Enzymes*, 3rd ed., **5**, 539.

FUMARATE REDUCTASE

This oxidoreductase catalyzes the reaction of succinate and NAD⁺ to yield fumarate, NADH, and a proton.

K. M. Noll (1995) *Meth. Enzymol.* **251**, 470.

FURA-2



Structures of FURA-2 (left) and EGTA (right)

A calcium ion indicator dye (based on the structure of the chelator EGTA) that exhibits a strong fluorescence at 385 nm and can be used to measure changes in intracellular Ca²⁺ concentration. The approximate dissociation constant for the Ca²⁺-Fura-2 complex is 0.1 μ M, depending on cellular ion composition and pH. An esterified derivative of Fura-2 readily crosses the peripheral membrane of many cells and, after hydrolysis, the release of Fura-2 permits calcium ion measurements within cells.

See *Calcium Ion Indicator Dyes; Metal Ion Complexation*

R. Y. Tsien (1988) *Trends in Neurosciences* **11**, 419.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Cellular calcium determination, **238**, 73, 146, 298; calibration, **260**, 342-343; cell loading, **238**, 299; ratio fluorescence application, **238**, 298; Fura-2/AM cell loading for assay of free intracellular Ca²⁺, **236**, 484; in assay of agonist-promoted calcium levels inside sperm, **225**, 150.

FURIN

This calcium-activated enzyme [EC 3.4.21.75] catalyzes the hydrolysis of peptide bonds in protein precursors that results in the release of mature proteins from their proproteins by hydrolysis of ArgXaaYaaArg—Zaa bonds, where Xaa can be any amino acid and Yaa is an arginyl or a lysyl residue. Albumin, complement component C3, and von Willebrand factor are thus released from their respective precursors. Furin is a member of the peptidase family S8.

C. W. Wharton (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 345.

K. Nakayama (1994) *Meth. Enzymol.* **244**, 167.

G

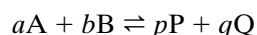
ΔG° (Relation to K_{eq})

The Gibbs equation relates the change in a reaction's Gibbs standard free energy (ΔG°) to its equilibrium constant:

$$\Delta G^\circ = -RT(\ln K_{eq})$$

where R is the universal gas constant, T is the absolute temperature, and K_{eq} is the equilibrium constant under standard conditions. This fundamental equation allows one to interpret the thermodynamic properties of reactions and to apply the Additivity Principle to assess the equilibrium constants of other reactions. Because this equation is of such importance, two brief derivations of the relationship are presented here.

BASIC DERIVATION. Consider a chemical reaction of the following general description:



where the lower case letters (a , b , c , and d) are stoichiometric coefficients for each reactant. Now, the concentration of A at a particular time can be greater than, less than, or equal to its equilibrium concentration; the same is true for B , P , and Q . The free energy of each reactant can be written as:

$$\text{Free energy of } a \text{ moles of } A = aG_A = \Delta G_A^\circ + aRT \ln A$$

$$\text{Free energy of } b \text{ moles of } B = bG_B = \Delta G_B^\circ + bRT \ln B$$

$$\text{Free energy of } p \text{ moles of } P = pG_P = \Delta G_P^\circ + pRT \ln P$$

$$\text{Free energy of } q \text{ moles of } Q = qG_Q = \Delta G_Q^\circ + qRT \ln Q$$

The free energy change for the reaction can now be written as shown in the following equations:

$$\begin{aligned} \Delta G &= G_{\text{products}} - G_{\text{reactants}} \\ &= pG_P + qG_Q - aG_A - bG_B \\ &= (\Delta G_P^\circ + pRT \ln P) + (\Delta G_Q^\circ + qRT \ln Q) \\ &\quad - (\Delta G_A^\circ + aRT \ln A) - (\Delta G_B^\circ + bRT \ln B) \end{aligned}$$

or

$$\begin{aligned} \Delta G &= [\Delta G_P^\circ + \Delta G_Q^\circ - (\Delta G_A^\circ + \Delta G_B^\circ)] \\ &\quad + RT \ln(P^p Q^q / [A^a B^b]) \end{aligned}$$

When the reaction system attains equilibrium, ΔG will equal zero, and

$$\Delta G^\circ + RT \ln(P^p Q^q / [A^a B^b]) = 0$$

or

$$\Delta G^\circ = -RT \ln(P^p Q^q / [A^a B^b]) = -RT \ln K_{eq}$$

If the stoichiometric coefficients are each equal to one, then we arrive at the expression

$$\Delta G^\circ = -RT \ln(PQ/(AB)) = -RT \ln K_{eq}$$

Note also that the standard free energy change (ΔG°) has units of joules per mole, whereas the free energy change (ΔG) has units of joules.

A SLIGHTLY MORE RIGOROUS APPROACH. In the above description, the system was treated as though the reactants were ideal gases. A better approach is to account for the free energy of the system as a function of added reactants, and for solutions, the most technically correct action is to use partial molal quantities. Because free energy, $G(T, P, n_1, n_2, \dots)$, is a state function depending on temperature, pressure, and the moles of

each reactant, one can write the total differential of the free energy, and this will include terms of the form:

$$\mu_i = (\partial G / \partial n_i)_{P, T, n_j (j \neq i)}$$

Thus, the total differential of the free energy function is

$$dG = \mu_A dn_A + \mu_B dn_B + \mu_P dn_P + \mu_Q dn_Q$$

During the course of a chemical reaction, the number of moles of a particular reactant will change in a manner consistent with the stoichiometric relationships defined by the stoichiometric coefficients (a , b , p , and q):

$$dn_B/dn_A = b/a \text{ or } dn_B = (b/a)dn_A$$

$$dn_P/dn_A = -p/a \text{ or } dn_P = (-p/a)dn_A$$

$$dn_Q/dn_A = -q/a \text{ or } dn_Q = (-q/a)dn_A$$

The chemical potential of each component can likewise be written as

$$\mu_A = \mu_A^\circ + RT(A)$$

$$\mu_B = \mu_B^\circ + RT(B)$$

$$\mu_P = \mu_P^\circ + RT(P)$$

$$\mu_Q = \mu_Q^\circ + RT(Q)$$

where the parentheses denote chemical activity not molar concentration. At thermodynamic equilibrium, $dG = 0$, and the sum of the chemical potentials of the substrates equals the sum of the chemical potentials of the products:

$$\mu_A dn_A + \mu_B dn_B = \mu_P dn_P + \mu_Q dn_Q$$

Substituting each of these terms by the quantities in the seven previous equations, we get:

$$(p\mu_P + q\mu_Q - a\mu_A - b\mu_B) = -RT \ln(P^p Q^q / [A^a B^b])$$

or

$$\Delta G^\circ = -RT \ln K_{eq}$$

See Activity Coefficients; Additivity Principle; Biochemical Thermodynamics; Chemical Potential; Equilibrium Constants; Hess's Law; Innate Thermodynamic Quantities; Molecular Crowding; Thermodynamics, Laws of; Thermodynamic Cycle; Thermodynamic Equations of State

$\Delta\Delta G$ AS AN INDEX OF COOPERATIVITY

The linkage of binding energy at multiple ligand sites is commonly referred to as cooperativity, and the free en-

ergy of interaction between such sites (*i.e.*, $\Delta\Delta G$) serves as a useful index of ligand binding cooperativity. A stringent definition of cooperativity can be stated in terms of the free energy changes associated with the relative magnitudes of the respective ligand binding constants. If the change in the free energy change is zero, then binding is strictly noncooperative. If the ligand affinity for one site is raised by binding to the other site, then $\Delta\Delta G$ will be positive, and this is termed positive cooperativity. By contrast, if the ligand affinity for one site is lowered by binding to the other site, then $\Delta\Delta G$ will be negative, and this is termed negative cooperativity. This thermodynamic approach is more rigorous than the use of the Hill coefficient.

GALACTARATE DEHYDRATASE

This enzyme [EC 4.2.1.42] catalyzes the conversion of D-galactarate to 5-dehydro-4-deoxy-D-glucarate and water.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

GALACTITOL 2-DEHYDROGENASE

This enzyme [EC 1.1.1.16] catalyzes the reaction of galactitol with NAD^+ to produce D-tagatose and NADH. The enzyme will also catalyze the conversion of other alditols containing an L-threo configuration adjacent to a primary alcohol group.

D. R. D. Shaw (1962) *Meth. Enzymol.* **5**, 323.

GALACTOKINASE

This enzyme [EC 2.7.1.6] catalyzes the reaction of ATP with D-galactose to produce ADP and D-galactose 1-phosphate. D-Galactosamine can also act as the substrate.

M. R. Heinrich & S. M. Howard (1966) *Meth. Enzymol.* **9**, 407.

D. B. Wilson & D. S. Hogness (1966) *Meth. Enzymol.* **8**, 229.

E. S. Maxwell, K. Kurahashi & H. M. Kalckar (1962) *Meth. Enzymol.* **5**, 174.

L. F. Leloir & R. E. Trucco (1955) *Meth. Enzymol.* **1**, 290.

GALACTONATE DEHYDRATASE

This enzyme [EC 4.2.1.6] catalyzes the conversion of D-galactonate to yield 2-dehydro-3-deoxy-D-galactonate and water.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

GALACTOSE 1-DEHYDROGENASE

This enzyme [EC 1.1.1.48] catalyzes the reaction of D-galactose with NAD^+ to produce D-galactono-1,4-lactone and NADH.

K. Wallenfels & G. Kurz (1966) *Meth. Enzymol.* **9**, 112.
M. Doudoroff (1962) *Meth. Enzymol.* **5**, 339.

GALACTOSE OXIDASE

This copper-dependent enzyme [EC 1.1.3.9] catalyzes the reaction of D-galactose with dioxygen to produce D-galacto-hexodialdose and hydrogen peroxide.

A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.
N. Ito, P. F. Knowles & S. E. V. Phillips (1995) *Meth. Enzymol.* **258**, 235.
J. W. Whittaker (1995) *Meth. Enzymol.* **258**, 262.
E. T. Adman (1991) *Adv. Protein Chem.* **42**, 145.
B. G. Malmström, L.-E. Andréasson & B. Reinhammer (1975) *The Enzymes*, 3rd ed., **12**, 507.

GALACTOSE-1-PHOSPHATE URIDYLTRANSFERASE

This enzyme [EC 2.7.7.10], also known as UTP:hexose-1-phosphate uridylyltransferase, catalyzes the reaction of UTP with α -D-galactose 1-phosphate to produce UDP-galactose and pyrophosphate (or, diphosphate). α -D-Glucose 1-phosphate can also function as a substrate, albeit not as effectively.

L. S. Mullins & F. M. Raushel (1995) *Meth. Enzymol.* **249**, 398.
P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.
P. A. Frey (1989) *Adv. Enzymol.* **62**, 119.
P. A. Frey, L.-J. Wong, K.-F. Sheu & S.-L. Yang (1982) *Meth. Enzymol.* **87**, 20.

α -GALACTOSIDASE

This enzyme [EC 3.2.1.22] catalyzes the hydrolysis of melibiose to yield galactose and glucose. It will also act on terminal, nonreducing α -D-galactose residues in α -D-galactosides. In addition, it will hydrolyze α -D-fucosides.

E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.
P. M. Dey & E. del Campillo (1984) *Adv. Enzymol.* **56**, 141.
R. O. Brady (1983) *The Enzymes*, 3rd ed., **16**, 409.
H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

β -GALACTOSIDASE

This enzyme [EC 3.2.1.23], also known as LacZ, catalyzes the hydrolysis of terminal (nonreducing) β -D-galactose residues in β -D-galactosides.

M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.
G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.
M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed. **19**, 213.
E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.
H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

GALACTOSIDE FUCOSYLTRANSFERASES

Galactoside 2-L-fucosyltransferase [EC 2.4.1.69], also known as blood group H α -2-fucosyltransferase, catalyzes the reaction of GDP-L-fucose with β -D-galactosyl-R to produce GDP and α -L-fucosyl-1,2- β -D-galactosyl-R. Lactose can also act as the acceptor substrate.

Galactoside 3(4)-L-fucosyltransferase [EC 2.4.1.65], also known as blood group Lewis α -4-fucosyltransferase, catalyzes the reaction of GDP-L-fucose with 1,3- β -D-galactosyl-N-acetyl-D-glucosaminyl-R to produce GDP and 1,3- β -D-galactosyl-(α -1,4-L-fucosyl)-N-acetyl-D-glucosaminyl-R. It can also act on the corresponding 1,4-galactosyl derivative, forming 1,3-L-fucosyl links.

Galactoside 3-fucosyltransferase [EC 2.4.1.152], also known as Lewis-negative α -3-fucosyltransferase, catalyzes the reaction of GDP-L-fucose with 1,4- β -D-galactosyl-N-acetyl-D-glucosaminyl-R to produce GDP and 1,4- β -D-galactosyl-(α -1,3-L-fucosyl)-N-acetyl-D-glucosaminyl-R. Unlike the galactoside 3(4)-L-fucosyltransferase [EC 2.4.1.65] above, this enzyme exhibits no action on the corresponding 1,3-galactosyl derivative.

T. A. Beyer, J. E. Sadler, J. I. Rearick, J. C. Paulson & R. L. Hill (1981) *Adv. Enzymol.* **52**, 23.

GALACTOSIDE SIALYLTRANSFERASES

The following articles describe some of the molecular and physical properties of this enzyme.

T. A. Beyer, J. E. Sadler, J. I. Rearick, J. C. Paulson & R. L. Hill (1981) *Adv. Enzymol.* **52**, 23.
A. P. Grollman (1966) *Meth. Enzymol.* **8**, 351.

GAMMA (Γ or γ)

Γ 1. Symbol for surface concentration (SI units = moles per square meter). 2. The symbol for the limit of the second derivative of the reciprocal of the initial rate of an enzyme-catalyzed reaction vs. the reciprocal of the substrate concentration as the substrate concentration approaches infinity (thus, a measure of the degree of curvature in a double-reciprocal plot). Hence, $\Gamma = \lim_{[A] \rightarrow \infty} (\delta^2(\nu^{-1})/\delta([A]^{-1})^2)$. This function provides a means for estimating the degree of cooperativity in an enzyme-catalyzed reaction which does not have a linear double-reciprocal plot¹. If Γ is positive, then the system exhibits positive cooperative; if negative, then negative cooperativity. This measurement of cooperativity is not com-

monly used since it is difficult to measure changes in slopes at high substrate concentrations¹. **See** *Cooperativity; Hill Plot*. 3. Symbol for the gamma function, an important mathematical relationship that appears in a number of physical systems: for example, in immobilized enzyme kinetics². It is defined by $\Gamma(1 + z) = \int_0^\infty e^{-x} x^z dx$ and has the property $\Gamma(z + 1) = z \Gamma(z)$. The gamma function also satisfies the relationship $\pi/(\sin(\pi z)) = \Gamma(z)\Gamma(1 - z)$ for all z whose absolute value is between zero and unity. Hence, $\Gamma(1/2)$ equals $\pi^{1/2}$.

γ 1. Symbol for activity coefficient. (γ_m on a molality basis, γ_c on a molarity basis, γ_x on a mole fraction basis). 2. Symbol for surface tension. 3. Symbol for photon. 4. Symbol for 10^{-4} gauss. 5. Symbol for shear strain. 6. Symbol for mass density. 7. Symbol for ratio of heat capacities, $\gamma = C_p/C_v$. 8. The third letter in the Greek alphabet; hence, used to denote the third item in a series (for example, the third methylene carbon in a fatty acid). $\gamma_{\pm}$, Symbol for mean ionic activity coefficient.

¹K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

²K. J. Laidler & P. S. Bunting (1980) *Meth. Enzymol.* **64**, 227.

GAUSS

A unit, symbolized by G, for magnetic flux density (or, magnetic induction) equal to a maxwell per square centimeter. One gauss is equivalent to 10^{-4} tesla. If a conductor, one centimeter long, moves through a magnetic field at a velocity of one centimeter per second at right angles to the magnetic flux, then one gauss is the magnetic flux density that will induce an electromotive force of 10^{-8} volt (or, one abvolt).

GDP-MANNOSE 4,6-DEHYDRATASE

The following is a review on the molecular and physical properties of this enzyme [EC 4.2.1.47] which catalyzes the dehydration of GDP-mannose to yield GDP-4-dehydro-6-deoxy-D-mannose and water.

L. Glaser & H. Zarkowsky (1971) *The Enzymes*, 3rd ed., **5**, 465.

GDP-MANNOSE DEHYDROGENASE

This enzyme [EC 1.1.1.132] catalyzes the following reaction: GDP-mannose + 2 NAD⁺ + H₂O to yield GDP-D-mannuronate + 2 NADH + 2H⁺.

S. Shankar, R. W. Ye, D. Schlichtman & A. M. Chakrabarty (1995) *Adv. Enzymol.* **70**, 221.

GDP-MANNOSE:DOLICHYL-PHOSPHATE MANNOSYLTRANSFERASE

This enzyme [EC 2.4.1.83] which catalyzes the following reaction: GDP-mannose + dolichyl phosphate to yield GDP + dolichyl D-mannosyl phosphate.

K. A. Presper & E. C. Heath (1983) *The Enzymes*, 3rd ed., **16**, 449.

GDP-MANNOSYLTRANSFERASES

This set of enzymes represent transferases that utilize GDP-mannose:

Glucomannan 4- β -mannosyltransferase [EC 2.4.1.32] catalyzes the reaction of GDP-mannose with [glucomannan]_n to produce GDP and [glucomannan]_(n + 1).

Heteroglycan α -mannosyltransferase [EC 2.4.1.48] catalyzes the reaction of GDP-mannose with a heteroglycan to produce GDP and a 1,2(or 1,3)- α -D-mannosylheteroglycan. In this reaction, in which the heteroglycan primer contains mannose, galactose, and xylose, 1,2- and 1,3-mannosyl bonds are formed.

Undecaprenyl-phosphate mannosyltransferase [EC 2.4.1.54] catalyzes the phosphatidylglycerol-dependent reaction of GDP-mannose with undecaprenyl phosphate to produce GDP and D-mannosyl-1-phosphoundecaprenol.

Phosphatidyl-*myo*-inositol α -mannosyltransferase [EC 2.4.1.57] catalyzes the transfer of one or more α -D-mannose units from GDP-mannose to positions 2, 6, and others in 1-phosphatidyl-*myo*-inositol.

GDP-mannose:dolichyl-phosphate mannosyltransferase [EC 2.4.1.83]: **See** *GDP-Mannose:Dolichyl-Phosphate Mannosyltransferase*

tRNA-queuosine β -mannosyltransferase [EC 2.4.1.110] catalyzes the reaction of GDP-mannose with tRNA^{Asp}-queuosine to produce GDP and tRNA^{Asp}-O^{5'}- β -D-mannosylqueuosine.

Glycolipid 2- α -mannosyltransferase [EC 2.4.1.131] catalyzes the transfer of an α -D-mannosyl residue from GDP-mannose into a lipid-linked oligosaccharide, forming an α -1,2-D-mannosyl-D-mannose linkage. The two 1,2-linked mannosyl residues in the mammalian lipid-linked

oligosaccharide of the structure $\text{Glc}_3\text{Man}_9\text{GlcNAc}_2$ are produced by this enzyme.

Glycolipid 3- α -mannosyltransferase [EC 2.4.1.132], also known as mannosyltransferase II, catalyzes the transfer of an α -D-mannosyl residue from GDP-mannose into a lipid-linked oligosaccharide, forming an α -1,3-D-mannosyl-D-mannose linkage. The 1,3-linked mannosyl residue in the mammalian lipid-linked oligosaccharide of the structure $\text{Glc}_3\text{Man}_9\text{GlcNAc}_2$ is produced by this enzyme.

Chitobiosyldiphosphodolichol α -mannosyltransferase [EC 2.4.1.142] catalyzes the reaction of GDP-mannose with chitobiosyldiphosphodolichol to produce GDP and α -D-mannosylchitobiosyldiphosphodolichol.

GEAR ALGORITHM

A predictor-corrector algorithm for automatic computer-assisted integration of stiff ordinary differential equations. This procedure carries the name of its originator¹.

Numerical integration of rate equations can be challenging whenever rate constants of widely disparate magnitudes are involved. For example, if a two-reaction process has a very large rate constant for the first reaction, followed by a very small rate constant for the second reaction, then many numerical integration procedures require one to make very small steps to achieve sufficient accuracy for the first decay process, and this necessitates using many of the same small step time intervals for the second reaction. The simulation becomes extraordinarily slow if one is dealing with a series of reactions with very slow and very fast processes nested together. This condition is often referred to as “stiffness” or “stiffness instability”. In the Gear Algorithm, the step size is chosen automatically so that once a fast reaction is simulated the step size can change to make the numerical integration of much slower steps far more tractable. **See** *Stiffness or Stiffness Instability*

¹C. W. Gear (1971) *Numerical Initial-Value Problems in Ordinary Differential Equations*, Prentice-Hall, Englewood Cliffs, NJ.

GELSOLIN

A calcium-regulated actin regulatory protein that binds to and severs actin filaments, thereby controlling the gelsol transitions of the actin cytoskeleton. After severing

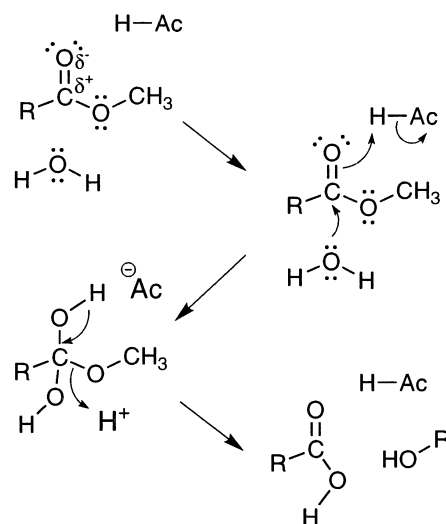
an actin filament, gelsolin remains tightly bound on the (+)- or barbed-end of one fragmented filament.

P. J. McLaughlin & A. G. Weeds (1995) *Ann. Rev. Biophys. Biomol. Struct.* **24**, 643.

T. P. Stossel, C. Chaponnier, R. M. Ezzell, J. H. Harding, P. A. Janney, D. J. Kwiatkowski, S. E. Lind, D. B. Smith, F. S. Southwick, H. L. Yin & K. S. Zaner (1985) *Ann. Rev. Cell Biol.* **1**, 353.

GENERAL ACID CATALYSIS

Catalysis of a reaction by a Brønsted acid, including the protonated solvent S, forming the lyonium ion, SH^+ . The rate of a general acid-catalyzed reaction is increased by an increase in the lyonium ion concentration and/or by an increase in the concentration of other acids, even when $[\text{SH}^+]$ is held constant.



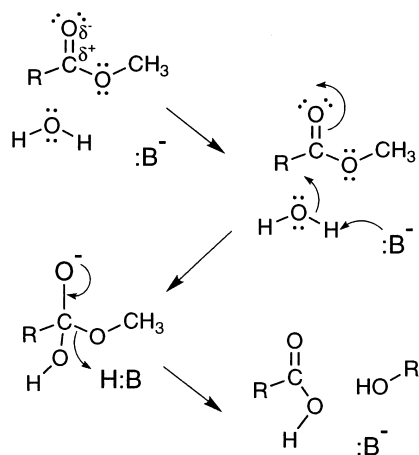
General Acid Catalysis of Ester Hydrolysis

The rate of a general acid-catalyzed reaction is equal to $\Sigma_{\text{HA}} k_{\text{HA}} [\text{HA}]$ multiplied by some function of the substrate concentration(s) where $[\text{HA}]$ is the general acid concentration and k_{HA} is the corresponding catalytic rate constant. Experimentally, general acid catalysis can be distinguished from specific acid catalysis by analysis of the effect of buffer concentration on the overall reaction rate. **See also** *Specific Acid Catalysis; Catalysis; General Base Catalysis*

GENERAL BASE CATALYSIS

Catalysis of a reaction by a Brønsted base, including the deprotonated solvent SH or SH^- , forming the lyate ion, S^- or S , respectively. The rate of such a catalyzed reaction is equal to $\Sigma_{\text{B}} k_{\text{B}} [\text{B}]$ multiplied by some function of the

substrate concentration(s), where $[B]$ is the general base concentration and k_B is the corresponding catalytic rate constant.



**General Base Catalysis
of Ester Hydrolysis**

Experimentally, general base catalysis can be distinguished from specific base catalysis by analysis of the effect of buffer concentration on the overall reaction rate. **See also** *Specific Base Catalysis; Catalysis; General Acid Catalysis*

GENETIC CODE

The following table indicates the standard codons that serve as the basis of the genetic code. Note: (a) Stop codons have no amino acids assigned to them; (b) Met* indicates the AUG start codon.

5'-BASE	MIDDLE-BASE				3'-BASE
	U	C	A	G	
U	UUU = Phe	UCU = Ser	UAU = Tyr	UGU = Cys	U
U	UUC = Phe	UCC = Ser	UAC = Tyr	UGC = Cys	C
U	UUA = Leu	UCA = Ser	UAA = Stop	UGA = Stop	A
U	UUG = Leu	UCG = Ser	UAG = Stop	UGG = Trp	G
C	CUU = Leu	CCU = Pro	CAU = His	CGU = Arg	U
C	CUC = Leu	CCC = Pro	CAC = His	CGC = Arg	C
C	CUA = Leu	CCA = Pro	CAA = Gln	CGA = Arg	A
C	CUG = Leu	CCG = Pro	CAG = Gln	CGG = Arg	G
A	AUU = Ile	ACU = Thr	AAU = Asn	AGU = Ser	U
A	AUC = Ile	ACC = Thr	AAC = Asn	AGC = Ser	C
A	AUA = Ile	ACA = Thr	AAA = Lys	AGA = Arg	A
A	AUG = Met*	ACG = Thr	AAG = Lys	AGG = Arg	G
G	GUU = Val	GCU = Ala	GAU = Asp	GGU = Gly	U
G	GUC = Val	GCC = Ala	GAC = Asp	GGC = Gly	C
G	GUA = Val	GCA = Ala	GAA = Glu	GGA = Gly	A
G	GUG = Val	GCG = Ala	GAG = Glu	GGG = Gly	G

GENTAMICIN 2"-NUCLEOTIDYLTRANSFERASE

This enzyme [EC 2.7.7.46] catalyzes the reaction of a nucleoside triphosphate with gentamicin to produce a 2"-nucleotidylgentamicin and pyrophosphate (or, diphosphate). The nucleoside triphosphate can be ATP, dATP, CTP, ITP, or GTP. Kanamycin, tobramycin, and sisomicin can also act as the acceptor substrate. The nucleotidyl residue is transferred to the 2-hydroxyl group of the 3-amino-3-deoxy-D-glucose moiety in the acceptor substrate.

P. A. Frey (1989) *Adv. Enzymol.* **62**, 119.

GENTISATE 1,2-DIOXYGENASE

This iron-dependent enzyme [EC 1.13.11.4] catalyzes the following reaction: 2,5-dihydroxybenzoate + O₂ = maleylpyruvate.

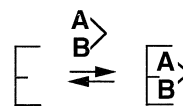
M. R. Harpel & J. D. Lipscomb (1990) *Meth. Enzymol.* **188**, 101.

GEOMETRICAL ANALOGUE

Sequential Binding of Substrates A and B:



Binding of A-B Geometric Analogue:



A multisubstrate inhibitor that achieves high-affinity because it interacts with binding determinants in more than a single substrate binding pocket on the enzyme. These inhibitors work with enzymes that have kinetic mechanisms that simultaneously require two or more of the substrates on the enzyme for catalysis. By binding to two ligand binding sites at the same time, the binding energy for ligand complexation is much greater than observed with competitive inhibitors acting only at a single substrate binding site. **See** *Adenylate Kinase Inhibitors*

GEOMETRIC MEAN, RULE OF THE

In the study of isotope effects, the isotope partitioning factor at a particular site is independent of the isotopic composition at any other site¹⁻³. Thus, an isotope effect for a diisotopically substituted species should be the product of the kinetic isotope effects observed for each of the monoisotopically substituted compounds.

¹V. Gold (1969) *Adv. Phys. Org. Chem.* **7**, 259.

²R. A. More O'Ferrall & A. J. Kresge (1980) *J. Chem. Soc., Perkin Trans.* **2**, 1840.

³K. B. Schowen & R. L. Schowen (1982) *Meth. Enzymol.* **87**, 551.

GERANOYL-CoA CARBOXYLASE

This biotin-dependent enzyme [EC 6.4.1.5] catalyzes the following reaction: $\text{ATP} + \text{geranoyl-CoA} + \text{HCO}_3^- = \text{ADP} + \text{P}_i + 3\text{-(4-methylpent-3-en-1-yl)pent-2-enedioyl-CoA}$.

H. G. Wood & R. E. Barden (1977) *Ann. Rev. Biochem.* **46**, 385.

GERANYLTRANSTRANSFERASE

This enzyme [EC 2.5.1.10], also known as farnesyl-diphosphate synthase, catalyzes the reaction of geranyl diphosphate with isopentenyl diphosphate to produce *trans,trans*-farnesyl diphosphate and pyrophosphate (or, diphosphate). Some forms of this enzyme will also utilize dimethylallyl diphosphate as a substrate. However, the enzyme will not accept larger prenyl diphosphates as an efficient substitute for geranyl diphosphate.

R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.

H. Kleinig (1989) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **40**, 39.

F. M. Raushel & J. J. Villafranca (1988) *Crit. Rev. Biochem.* **23**, 1.

GIBBERELLIN 2 β -HYDROXYLASE

This enzyme [EC 1.14.11.13], also known as gibberellin 2 β -dioxygenase, catalyzes the following reaction: $\text{gibberellin-1} + 2\text{-oxoglutarate} + \text{O}_2 = 2\text{-}\beta\text{-hydroxygibberellin-1} + \text{succinate} + \text{CO}_2$. The enzyme acts on other gibberellins as well.

A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.

GIBBERELLIN 3 β -HYDROXYLASE

This enzyme [EC 1.14.11.15], also referred to as gibberellin 3 β -dioxygenase, uses iron ions and ascorbate and catalyzes the following reaction: $\text{gibberellin-20} + 2\text{-oxoglutarate} + \text{dioxygen} = \text{gibberellin-1} + \text{succinate} + \text{carbon dioxide}$. Other related enzymes (including gibberellin 12 α -hydroxylase, gibberellin 13-hydroxylase, gibberellin 7-oxidase, and gibberellin 20-oxidase) are also reviewed in the articles below.

P. Hedden & Y. Kamiya (1997) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 431.

A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.

GIBBS FREE ENERGY

A fundamental thermodynamic parameter (typically symbolized by G) used to describe the thermodynamic driving forces governing a closed system(s) at constant temperature and pressure. The Gibbs free energy is a state function that equates the enthalpy, H , minus the absolute temperature, T , multiplied by the entropy, S : thus, $G = H - TS$. The Helmholtz energy, A , and the Gibbs energy are related by the expression $G = A + PV$, where P and V are pressure and volume. In a closed system at constant temperature and pressure, wherein only PV work is permitted, the system at equilibrium will have a minimum value for the Gibbs free energy.

Because most biological systems are operative under conditions of constant pressure, changes in G are quite useful to describe driving forces for a wide variety of processes. The sign of the change in Gibbs free energy, ΔG , is directly associated with the direction in which a particular chemical reaction can proceed. If $\Delta G < 0$ for a given set of conditions of a particular reaction, then the reaction will proceed spontaneously in the indicated direction until equilibrium is reached. Conversely, if ΔG is positive, then energy will be needed to shift the reaction further from its equilibrium condition. **See** *Helmholtz Energy; Endergonic; Exergonic; Enthalpy; Entropy; Thermodynamics; Biochemical Thermodynamics*

GIBBS FREE ENERGY DIAGRAM

A graphical depiction (frequently called a reaction coordinate diagram) of the relative Gibbs free energies (or, more commonly, the standard Gibbs free energies) of

the reactants, reaction intermediates, transition state(s), and products plotted in the same sequence in which they appear in the chemical reaction(s) under study. The horizontal coordinate expresses the ordered sequence, but is otherwise undefined in terms of time or any other parameter. The points representing the relative free energies are usually connected by a smooth curve (often referred to as the free energy profile). However, it should be pointed out that experimental observations typically can only provide information for the maxima and minima of these profiles and not on points in between.

The highest point on a Gibbs free energy diagram often, but not always, corresponds to the transition state of the rate-limiting step for the overall reaction(s). The concentration of a substrate not involved at that step may determine another step to be rate-limiting if the concentration of that substrate is sufficiently low¹. **See also** *Potential Energy Profile; Enzyme Energetics*

¹IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

GIBBS FREE ENERGY OF ACTIVATION

The standard Gibbs free energy difference between the reactant(s) of a reaction and the transition state. It is commonly symbolized by either $\Delta G^\ddagger$ or $\Delta^\ddagger G^\ominus$ and is equal to $-RT \ln K^\ddagger$ where $K^\ddagger$ represents the equilibrium constant between the reactant(s) and the transition state, R is the molar gas constant, and T is the absolute temperature. (It is more proper to designate $\Delta G^\ddagger$ as equal to $-RT \ln[K^\ddagger(c^\ominus)^{n-1}]$ where $c^\ominus$ is the standard-state concentration and n is the molecularity of the reaction. Thus, $K^\ddagger = \{\exp[-\Delta G^\ddagger/RT](c^\ominus)^{1-n}\}$. However, standard states are usually unit concentrations, the unit being used the same as that of the rate constant.) We should also recall the fact that $K^\ddagger$ differs from a true equilibrium constant due to the altered number of degrees of freedom in the transition state. The Gibbs free energy of activation can be determined from the bimolecular rate constant and the Eyring equation: $\Delta G^\ddagger = RT(\ln[k_B/h] - \ln[k/T])$ where k_B is Boltzmann's constant, h is Planck's constant, and k is the bimolecular rate constant. **See** *Transition-State Theory (Thermodynamics)*

GIBBS-HELMHOLTZ EQUATION

An important thermodynamic relationship for the temperature dependence of the Gibbs free energy, G :

$$(\partial(G/T)/\partial T)_P = -H/T^2$$

where T is the absolute temperature, P is the pressure, and H is the enthalpy. **See** *Gibbs Free Energy*

GLOBAL ANALYSIS

A systematic algorithm for data analysis that permits the experimenter to recover estimates of lifetimes (τ) from a multiexponential time-dependent process $F(t)$. The process is described by the generalized sum-of-exponentials expression, involving n steps:

$$F(t) = \sum a_i \exp[-t/\tau_i]$$

where the summation is taken from ($i = 1$) to ($i = n$), and a_i is the amplitude associated with each of the individual steps. This algorithm *simultaneously* analyzes data from multiple experiments. This is accomplished applying an internally consistent set of fitting parameters to nonlinear least squares software adapted to perform model-dependent summations of these nonlinear least-squares equations.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

In kinetic analysis of complex reactions, **210**, 382; fluorescence decay rate distributions, **210**, 357; implementation in Laplace deconvolution noniterative method, **210**, 293; in multiexponential decays, **210**, 296; partial global analysis by simulated annealing methods, **210**, 365; spectral resolution, **210**, 299.

GLUCAN ENDO-1,3(4)- β -GLUCANASE

This enzyme [EC 3.2.1.6] (also known as endo-1,3(4)- β -glucanase, glucan endo- β 1,3- β 1,4-glucosidase, endo-1,4- β -glucanase, endo-1,3- β -glucanase, and laminarinase) catalyzes the endohydrolysis of 1,3- or 1,4-linkages in β -D-glucans when the glucose residue, whose reducing group is involved in the linkage to be hydrolyzed, is itself substituted at C3. The enzyme will also utilize laminarin, lichenin, and cereal D-glucans as substrates. This glucanase is different from that of glucan endo-1,3- β -glucosidase [EC 3.2.1.39].

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

GLUCAN ENDO-1,2- β -GLUCOSIDASE

This enzyme [EC 3.2.1.71], also known as endo-1,2- β -glucanase, catalyzes the random hydrolysis of 1,2-glucosidic linkages in 1,2- β -D-glucans.

E. T. Reese & M. Mandels (1966) *Meth. Enzymol.* **8**, 607.

GLUCAN ENDO-1,3- α -GLUCOSIDASE

This enzyme [EC 3.2.1.59], also known as endo-1,3- α -glucanase, catalyzes the endohydrolysis of 1,3- α -D-glucosidic linkages in pseudonigeran (the products of which are nigerose and glucose), isolichenin, and nigeran.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

GLUCAN ENDO-1,3- β -GLUCOSIDASE

This enzyme [EC 3.2.1.39] (also known as (1 $\rightarrow$ 3)- β -glucan endohydrolase, endo-1,3- β -glucanase, and laminarinase) catalyzes the hydrolysis of 1,3- β -D-glucosidic linkages in 1,3- β -D-glucans. The enzyme exhibits very limited action on mixed-link (1,3/1,4)- β -D-glucans. However, it catalyzes the hydrolysis of laminarin, paramylon, and pachyman. The enzyme is not identical with endo-1,3(4)- β -glucanase [EC 3.2.1.6].

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

E. T. Reese & M. Mandels (1966) *Meth. Enzymol.* **8**, 607.

GLUCAN ENDO-1,6- β -GLUCOSIDASE

This enzyme [EC 3.2.1.75], also known as endo-1,6- β -glucanase, catalyzes the random hydrolysis of 1,6-linkages in 1,6- β -D-glucans. The enzyme will also act on lutean, pustulan, and 1,6-oligo- β -D-glucosides.

E. T. Reese & M. Mandels (1966) *Meth. Enzymol.* **8**, 607.

GLUCAN 1,3- β -GLUCOSIDASE

This enzyme [EC 3.2.1.58], also known as exo-1,3- β -glucanase and exo-1,3- β -glucosidase, catalyzes the successive hydrolysis of β -D-glucose units from the non-reducing ends of 1,3- β -D-glucans, resulting in the release of glucose. The enzyme will also act on oligosaccharides. However, laminaribiose is a weak substrate.

E. T. Reese & M. Mandels (1966) *Meth. Enzymol.* **8**, 607.

1,3- β -GLUCAN SYNTHASE

This enzyme [EC 2.4.1.34] (also known as 1,3- β -D-glucan-UDP glucosyltransferase, UDP-glucose-1,3- β -D-glucan glucosyltransferase, and callose synthetase) catalyzes the reaction of UDP-glucose with [(1,3)- β -D-glucosyl]_n to produce UDP and [(1,3)- β -D-glucosyl]_(n+1).

D. S. Feingold (1966) *Meth. Enzymol.* **8**, 404.

GLUCARATE DEHYDRATASE

This enzyme [EC 4.2.1.40] catalyzes the following reaction: D-glucarate = 5-dehydro-4-deoxy-D-glucarate + H₂O.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

GLUCOAMYLASE

This enzyme [EC 3.2.1.3] catalyzes the hydrolytic release of β -D-glucose from terminal 1,4-linked α -D-glucose residues successively from nonreducing ends of polysaccharide chains.

E. H. Van Beers, H. A. Büller, R. J. Grand, A. W. C. Einerhand & J. Dekker (1995) *Crit. Rev. Biochem. Mol. Biol.* **30**, 197.
G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

GLUCOCEREBROSIDASE

This enzyme [EC 3.2.1.45] catalyzes the hydrolysis of D-glucosyl-N-acylsphingosine to yield D-glucose and N-acylsphingosine.

G. A. Grabowski, S. Gatt & M. Horowitz (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 385.

E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.

R. O. Brady (1983) *The Enzymes*, 3rd ed., **16**, 409.

GLUCOKINASE

This enzyme [EC 2.7.1.2] catalyzes the ATP-dependent transphosphorylation of D-glucose to yield D-glucose 6-phosphate + ADP. The Michaelis constant for D-glucose is nearly 100 times greater than the corresponding value for hexokinase. Although originally thought to be specific for glucose, this enzyme also phosphorylates D-fructose, which binds some 30 to 50 times more weakly. **See Hexokinase**

J. H. Hurley (1996) *Ann. Rev. Biophys. & Biomol. Struct.* **25**, 137.

A. Cornish-Bowden & M. L. Cárdenas (1991) *Trends Biochem. Sci.* **16**, 281.

D. L. Purich, H. J. Fromm & F. B. Rudolph (1973) *Adv. Enzymol.* **39**, 249.

S. P. Colowick (1973) *The Enzymes*, 3rd ed., **9**, 1.

GLUCONATE DEHYDRATASE

This enzyme [EC 4.2.1.39] catalyzes the reaction: D-glucuronate $\rightleftharpoons$ 2-dehydro-3-deoxy-D-glucarate + H₂O.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

GLUCOSAMINATE AMMONIA-LYASE

This enzyme [EC 4.3.1.9] catalyzes the following reaction involving pyridoxal phosphate as a cofactor: D-glucosamine = 2-dehydro-3-deoxy-D-glucuronate + NH₃.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

GLUCOSAMINE N-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.3] catalyzes the reaction of acetyl-CoA with Δ -glucosamine to yield coenzyme A and N-acetyl- Δ -glucosamine.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

GLUCOSAMINE-6-PHOSPHATE ISOMERASE

This enzyme [EC 5.3.1.10] catalyzes the following isomerization and deamination reaction: D-glucosamine 6-phosphate + H₂O $\rightleftharpoons$ D-fructose 6-phosphate + NH₃.

- I. A. Rose (1975) *Adv. Enzymol.* **43**, 491.
J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.
E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

GLUCOSAMINE-6-PHOSPHATE SYNTHASE

This enzyme [EC 2.6.1.16], also referred to as glutamine-fructose-6-phosphate aminotransferase (isomerizing), catalyzes the reaction of L-glutamine and D-fructose 6-phosphate to yield L-glutamate and D-glucosamine 6-phosphate.

- H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.
H. Zalkin (1985) *Meth. Enzymol.* **113**, 278.
J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

GLUCOSE OXIDASE

This enzyme [EC 1.1.3.4] catalyzes the following reaction: β -D-glucose + O₂ $\rightleftharpoons$ D-glucono-1,5-lactone + hydrogen peroxide.

- H. J. Bright & D. J. T. Porter (1975) *The Enzymes*, 3rd ed., **12**, 421.

GLUCOSE-6-PHOSPHATASE

This enzyme [EC 3.1.3.9] catalyzes the hydrolysis of D-glucose 6-phosphate to yield D-glucose and orthophosphate. Some glucose phosphatases also catalyze transphosphorylation reactions from carbamoyl phosphate, hexose phosphates, pyrophosphate, phosphoenolpyruvate and nucleoside di- and triphosphates, using D-glucose, D-mannose, 3-methyl-D-glucose, or 2-deoxy-D-glucose as phosphoryl acceptors. **See** *Isotope Exchange (Reactions Away from Equilibrium)*

- K. A. Sukalski & R. C. Nordlie (1989) *Adv. Enzymol.* **62**, 93.
R. C. Nordlie (1982) *Meth. Enzymol.* **87**, 319.
R. C. Nordlie (1971) *The Enzymes*, 3rd ed., **4**, 543.

GLUCOSE-6-PHOSPHATE DEHYDROGENASE

This enzyme [EC 1.1.1.49] catalyzes the NADP⁺-dependent reduction of D-glucose 6-phosphate that yields 6-phospho-D-gluconolactone, NADPH, and a proton.

- L. V. Kletsova, M. Y. Kiriukhin, A. Y. Chistoserdov & Y. D. Tsyganov (1990) *Meth. Enzymol.* **188**, 335.
A. P. Sokolov & Y. A. Trotsenko (1990) *Meth. Enzymol.* **188**, 339.
H. R. Levy (1979) *Adv. Enzymol.* **48**, 97.

GLUCOSE-1-PHOSPHATE URIDYLYLTRANSFERASE

This enzyme [EC 2.7.7.9] catalyzes the UTP-dependent uridylylation of α -D-glucose 1-phosphate that yields UDP-glucose and pyrophosphate.

- L. S. Mullins & F. M. Raushel (1995) *Meth. Enzymol.* **249**, 398.
P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.
P. A. Frey (1989) *Adv. Enzymol.* **62**, 119.
R. L. Turnquist & R. G. Hansen (1973) *The Enzymes*, 3rd ed., **8**, 51.

GLUCOSE TRACER KINETICS

Wolfe¹ provided an excellent account of the systematic application of stable and radioactive isotope tracers in determining the kinetics of glucose production, glycolysis, and gluconeogenesis in living systems.

- ¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, pp. 283-316, Wiley-Liss, New York.

GLUCURONATE-1-PHOSPHATE URIDYLYLTRANSFERASE

This enzyme [EC 2.7.7.44] catalyzes the reaction of UTP with 1-phospho- α -D-glucuronate to produce UDP-D-glucuronate and pyrophosphate. CTP can also serve as the nucleotide substrate, albeit not as effectively.

- F. A. Loewus & M. W. Loewus (1983) *Ann. Rev. Plant Physiol.* **34**, 137.

GLUCURONATE-2-SULFATE SULFATASE

This enzyme [EC 3.1.6.18], also known as glucuronate-2-sulfatase and chondro-2-sulfatase, catalyzes the hydrolysis of the 2-sulfate groups of the 2-O-sulfo-D-glucuronate residues of chondroitin sulfate, heparin, and heparitin sulfate. The enzyme does not act on iduronate 2-sulfate residues.

- H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

β -GLUCURONIDASE

This enzyme [EC 3.2.1.31] catalyzes the hydrolysis of a β -D-glucuronoside to yield an alcohol and D-glucuronate.

- H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.
P. M. Dey & E. del Campillo (1984) *Adv. Enzymol.* **56**, 141.

GLUCURONOKINASE

This enzyme [EC 2.7.1.43] catalyzes the reaction of ATP with D-glucuronate to produce ADP and 1-phospho- α -D-glucuronate.

- F. A. Loewus & M. W. Loewus (1983) *Ann. Rev. Plant Physiol.* **34**, 137.

GLUCURONOLACTONE REDUCTASE

This enzyme [EC 1.1.1.20] catalyzes the reaction of L-gulonono-1,4-lactone with NADP⁺ to produce D-glucurono-3,6-lactone and NADPH.

Y. Mano & N. Shimazono (1970) *Meth. Enzymol.* **18A**, 55.

GLUCURONOSYLTRANSFERASES

This set of enzymes [EC 2.4.1.17], also known as UDP-glucuronosyltransferases, catalyzes the reaction of UDP-glucuronate with an acceptor to produce UDP and the β -D-glucuronoside of the acceptor. This family of proteins accepts a wide range of substrates, including phenols, alcohols, amines, and fatty acids.

A. Markovitz & A. Dorfman (1962) *Meth. Enzymol.* **5**, 155.

GLUTACONYL-CoA DECARBOXYLASE

This enzyme [EC 4.1.1.70] catalyzes the conversion of pent-2-enoyl-CoA to but-2-enoyl-CoA and carbon dioxide. The enzyme isolated from *Acidaminococcus fermentans* is a biotinyl-protein, requires sodium ions, and acts as a sodium pump.

J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 495.

GLUTAMATE N-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.35], also known as ornithine acetyltransferase, and ornithine transacetylase, catalyzes the reversible reaction of N²-acetyl-L-ornithine with L-glutamate to produce L-ornithine and N-acetyl-L-glutamate. This protein also exhibits a low hydrolysis activity (about 1% of that of the transferase activity) of N²-acetyl-L-ornithine to yield acetate and L-ornithine. This enzyme is not identical with N-acetylglutamate synthase [EC 2.3.1.1].

G. Dénes (1970) *Meth. Enzymol.* **17A**, 273.

D-GLUTAMATE: D-AMINO ACID AMINOTRANSFERASE

This enzyme catalyzes the transamination of a wide spectrum of α -amino acids and α -keto (or 2-oxo) acids, demonstrating absolute specificity for their D-isomers. The most likely physiologic role is to provide D-amino acids for peptidoglycan synthesis in bacterial cell wall formation.

W. M. Jones, T. S. Soper, H. Ueno & J. M. Manning (1985) *Meth. Enzymol.* **113**, 108.

GLUTAMATE DECARBOXYLASE

This pyridoxal-phosphate-dependent enzyme [EC 4.1.1.15] catalyzes the conversion of L-glutamate to 4-aminobutanoate and carbon dioxide. The mammalian brain enzyme also acts on L-cysteate, 3-sulfinol-L-alanine, and L-aspartate.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

M. H. O'Leary (1989) *Ann. Rev. Biochem.* **58**, 377.

J.-Y. Wu, L. Denner, C.-T. Lin & G. Song (1985) *Meth. Enzymol.* **113**, 3.

M. L. Fonda (1985) *Meth. Enzymol.* **113**, 11.

A. E. Martell (1982) *Adv. Enzymol.* **53**, 163.

E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

GLUTAMATE DEHYDROGENASE

Glutamate dehydrogenase [EC 1.4.1.2] catalyzes the reaction of L-glutamate with NAD⁺ and water to produce α -ketoglutarate (or, 2-oxoglutarate), ammonia, and NADH.

Glutamate dehydrogenase (NADP⁺) [EC 1.4.1.4] catalyzes the reaction of L-glutamate with NADP⁺ and water to produce α -ketoglutarate (or, 2-oxoglutarate), ammonia, and NADPH.

Glutamate dehydrogenase (NAD(P)⁺) [EC 1.4.1.3] catalyzes the reaction of L-glutamate with NAD(P)⁺ and water to produce α -ketoglutarate (or, 2-oxoglutarate), ammonia, and NAD(P)H.

J. DiRuggiero & F. T. Robb (1996) *Adv. Protein Chem.* **48**, 311.

H.-M. Lam, K. T. Coschigamo, I. C. Oliveira, R. Melo-Oliveira & G. M. Coruzzi (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 569.

N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

R. F. Colman (1991) *A Study of Enzymes* **2**, 173.

R. F. Colman (1990) *The Enzymes*, 3rd ed., **19**, 283.

H. F. Fisher (1973) *Adv. Enzymol.* **39**, 369.

GLUTAMATE FORMIMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.1.2.5], also known as glutamate formyltransferase, catalyzes the reaction of 5-formiminotetrahydrofolate with L-glutamate to produce tetrahydrofolate and N-formimino-L-glutamate. The enzyme will additionally catalyze the transfer of the formyl moiety from 5-formyltetrahydrofolate to L-glutamate. This protein occurs in eukaryotes as a bifunctional enzyme also having a formiminotetrahydrofolate cyclodeaminase activity [EC 4.3.1.4].

V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

GLUTAMATE FORMYLTRANSFERASE

This enzyme designation was formerly classified as EC 2.1.2.6. It is now classified under glutamate formimino-transferase [EC 2.1.2.5].

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

GLUTAMATE RACEMASE

This pyridoxal-phosphate-dependent enzyme [EC 5.1.1.3] catalyzes the interconversion of L-glutamate and D-glutamate.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

E. Adams (1972) *The Enzymes*, 3rd ed., **6**, 479.

GLUTAMATE γ -SEMIALDEHYDE DEHYDROGENASE

This enzyme [EC 1.2.1.41], also known as γ -glutamyl-phosphate reductase and glutamate-5-semialdehyde dehydrogenase, catalyzes the reaction of L-glutamate 5-semialdehyde with orthophosphate and NADP⁺ to produce L- γ -glutamyl 5-phosphate and NADPH.

E. Adams & L. Frank (1980) *Ann. Rev. Biochem.* **49**, 1005.

GLUTAMATE SYNTHASE

Glutamate synthase (NADPH) [EC 1.4.1.13], an iron-sulfur flavoprotein, catalyzes the reaction of L-glutamine with α -ketoglutarate (or, 2-oxoglutarate) and NADPH to produce NADP⁺ and two glutamate molecules. Ammonia can act as the nitrogen donor substrate instead of L-glutamine, albeit weaker.

Glutamate synthase (NADH) [EC 1.4.1.14], which uses FMN, catalyzes the reaction of L-glutamine with α -ketoglutarate (or, 2-oxoglutarate) and NADH to produce NAD⁺ and two glutamate molecules.

Glutamate synthase (ferredoxin) [EC 1.4.7.1], also known as ferredoxin-dependent glutamate synthase, is an iron-sulfur flavoprotein, that catalyzes the reaction of L-glutamine with α -ketoglutarate (or, 2-oxoglutarate) and two reduced ferredoxin molecules to produce two

oxidized ferredoxin molecules and two glutamate molecules.

H.-M. Lam, K. T. Coschigamo, I. C. Oliveira, R. Melo-Oliveira & G. M. Coruzzi (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 569.

L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.

H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.

R. F. Colman (1990) *The Enzymes*, 3rd ed., **19**, 283.

A. Meister (1985) *Meth. Enzymol.* **113**, 327.

J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

GLUTAMINASE

This enzyme [EC 3.5.1.2], also known as L-glutamine amidohydrolase, catalyzes the hydrolysis of L-glutamine to produce L-glutamate and ammonia.

N. P. Curthoys & M. Watford (1995) *Ann. Rev. Nutr.* **15**, 133.

E. Kvamme, I. AA. Torgner & G. Svenneby (1985) *Meth. Enzymol.* **113**, 241.

H. G. Windmueller (1982) *Adv. Enzymol.* **53**, 201.

S. C. Hartman (1971) *The Enzymes*, 3rd ed., **4**, 79.

GLUTAMINE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.15] (also known as glutamine:pyruvate aminotransferase, glutaminase II, glutamine:oxo-acid transaminase, and glutamine transaminase L) catalyzes the reaction of L-glutamine with pyruvate to produce 2-oxoglutarate and L-alanine. The enzyme will also catalyze the transfer of an amino group from L-methionine and glyoxylate can substitute for pyruvate.

A. J. L. Cooper & A. Meister (1985) *Meth. Enzymol.* **113**, 338.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

GLUTAMINE SYNTHETASE

This enzyme [EC 6.3.1.2], officially known as glutamate:ammonia ligase, catalyzes the reaction of L-glutamate with ammonia and MgATP to produce L-glutamine, MgADP and orthophosphate. The enzyme will also act on 4-methylene-L-glutamate (however, glutamine synthetase activity is not identical with 4-methyleneglutamine synthetase [EC 6.3.1.7]) and β -glutamate. **See** *Arsenolysis; Bridge-to-Nonbridge Oxygen Scrambling; Isotope Exchange at Equilibrium; Wedler-Boyer Technique; Cumulative Inhibition; Unconsumed Substrate; Cryptic Catalysis; Borohydride Reduction; Enzyme Cascade Kinetics*

D. L. Purich (1998) *Adv. Enzymol.* **72**, 9.

J. DiRuggiero & F. T. Robb (1996) *Adv. Protein Chem.* **48**, 311.

H.-M. Lam, K. T. Coschigamo, I. C. Oliveira, R. Melo-Oliveira & G. M. Coruzzi (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 569.

S.-H. Liaw & D. Eisenberg (1994) *Biochemistry* **33**, 675.

J. J. Villafranca & T. Nowak (1992) *The Enzymes*, 3rd ed., **20**, 63.

P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.

A. Oaks & B. Hirel (1985) *Ann. Rev. Plant Physiol.* **36**, 345.

I. A. Rose (1979) *Adv. Enzymol.* **50**, 361.

A. Meister (1974) *The Enzymes*, 3rd ed., **10**, 699.

E. R. Stadtman & A. Ginsburg (1974) *The Enzymes*, 3rd ed., **10**, 755.

GLUTAMINE SYNTHETASE ADENYLYLTRANSFERASE

This enzyme [EC 2.7.7.42], also known as glutamate:ammonia-ligase adenylyltransferase, plays a key role in controlling bacterial nitrogen metabolism by controlling the state of adenylylation of glutamine synthetase.

E. R. Stadtman (1973) *The Enzymes*, 3rd ed., **8**, 1.

E. R. Stadtman & A. Ginsburg (1974) *The Enzymes*, 3rd ed., **10**, 755.

S. G. Rhee, P. B. Chock & E. R. Stadtman (1989) *Adv. Enzymol.* **62**, 37.

GLUTAMINYL-tRNA SYNTHETASE

This enzyme [EC 6.1.1.18], also known as glutamine:tRNA ligase, catalyzes the reaction of ATP with L-glutamine and tRNA^{Gln} to produce L-glutaminyl-tRNA^{Gln}, AMP, and pyrophosphate (or, diphosphate).

E. A. First (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 573.

J. J. Perona, M. A. Rould & T. A. Steitz (1993) *Biochemistry* **32**, 8758.

P. Hoben & D. Söll (1985) *Meth. Enzymol.* **113**, 55.

γ-GLUTAMYL CYCLOTRANSFERASE

This enzyme [EC 2.3.2.4] catalyzes the conversion of a (5-L-glutamyl)-L-amino acid to 5-oxoproline and an L-amino acid.

A. Meister (1985) *Meth. Enzymol.* **113**, 438.

P. Van Der Werf & A. Meister (1975) *Adv. Enzymol.* **43**, 519.

M. Orlowski & A. Meister (1971) *The Enzymes*, 3rd ed., **4**, 123.

γ-GLUTAMYL-CYSTEINE SYNTHETASE

This enzyme [EC 6.3.2.2], also known as glutamate:cysteine ligase, catalyzes the reaction of ATP with L-glutamate and L-cysteine to produce ADP, orthophosphate, and γ-L-glutamyl-L-cysteine. L-Aminohexanoate can act as a substrate in place of glutamate. Certain thiol-containing molecules, including glutathione, will inhibit the mammalian enzyme.

R. C. Fahey & A. R. Sundquist (1991) *Adv. Enzymol.* **64**, 1-53.

G. F. Seelig & A. Meister (1985) *Meth. Enzymol.* **113**, 379 and 390.

A. Meister (1974) *The Enzymes*, 3rd ed., **10**, 671.

GLUTAMYL ENDOPEPTIDASES

This is a set of enzymes that act on peptide bonds containing a glutamyl residue. They include:

Glutamyl endopeptidase [EC 3.4.21.19] (also known as staphylococcal serine proteinase, V8 proteinase, protease V8, and endoproteinase Glu-C), a member of the peptidase family S2B, catalyzes the hydrolysis of Asp-Xaa and Glu-Xaa peptide bonds. In appropriate buffers, the specificity of the bond cleavage is restricted to Glu-Xaa. Peptide bonds involving bulky side chains of hydrophobic aminoacyl residues are hydrolyzed at a lower rate.

Glutamyl endopeptidase II [EC 3.4.21.82], also known as glutamic acid-specific protease, catalyzes the hydrolysis of peptide bonds, exhibiting a preference for Glu-Xaa bonds much more than for Asp-Xaa bonds. The enzyme has a preference for prolyl or leucyl residues at P2 and phenylalanyl at P3. Hydrolysis of Glu-Pro and Asp-Pro bonds is slow. This endopeptidase is a member of the peptidase family S2A.

J. J. Birktoft & K. Breddam (1994) *Meth. Enzymol.* **244**, 114.

γ-GLUTAMYL KINASE

This enzyme [EC 2.7.2.11], also known as glutamate 5-kinase, catalyzes the reaction of ATP with L-glutamate to produce ADP and L-glutamate 5-phosphate, which is rapidly converted in aqueous solutions to 5-oxoproline and orthophosphate.

E. Adams & L. Frank (1980) *Ann. Rev. Biochem.* **49**, 1005.

γ-GLUTAMYL TRANSPEPTIDASE

This enzyme [EC 2.3.2.2], also known as γ-glutamyl-transferase, catalyzes the reaction of a γ-glutamyl compound (such as glutathione, γ-glutamyl amino acid, glutamine, glutathione disulfide, leukotriene C₄, etc.) with an acceptor substrate to produce the γ-glutamylated acceptor derivative and the de-γ-glutamylated substrate. Thus, when the acceptor substrate is water, the enzyme catalyzes a glutathionase activity (that is hydrolysis of glutathione to produce L-glutamate and L-cysteinylglycine) and a glutaminase activity (L-glutamine + water → L-glutamate and ammonia). When the acceptor substrate is an amino acid (particularly L-cysteine, L-methionine, L-glutamine, and L-alanine), the product is the corresponding isopeptide derivative

(that is, γ -glutamyl-L-cystine, γ -glutamyl-L-methionine, γ -glutamyl-L-glutamine, and γ -glutamyl-L-alanine, respectively). Dipeptides and even glutathione can also serve as acceptor substrates. Elevated levels of amino acids can result in substrate inhibition of all the activities. The plant and yeast enzymes have a different range of specificities than that of the mammalian enzyme mentioned above.

R. D. Allison (1985) *Meth. Enzymol.* **113**, 419.

S. S. Tate & A. Meister (1985) *Meth. Enzymol.* **113**, 400.

S. C. Hartman (1971) *The Enzymes*, 3rd ed., **4**, 79.

GLUTAMYL-tRNA^{Gln} AMIDOTRANSFERASE

This enzyme catalyzes the ATP-dependent reaction of glutamyl-tRNA^{Gln} with glutamine to produce glutamyl-tRNA^{Gln}, ADP, orthophosphate, and glutamate.

H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.

H. Zalkin (1985) *Meth. Enzymol.* **113**, 303.

J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

GLUTAMYL-tRNA SYNTHETASE

This enzyme [EC 6.1.1.17], also known as glutamate:tRNA ligase, catalyzes the reaction of L-glutamate with tRNA^{Glu} and ATP to produce L-glutamyl-tRNA^{Glu}, AMP, and pyrophosphate.

P. Schimmel (1987) *Ann. Rev. Biochem.* **56**, 125.

J. Lapointe, S. Levasseur & D. Kern (1985) *Meth. Enzymol.* **113**, 42.

M. Proulx & J. Lapointe (1985) *Meth. Enzymol.* **113**, 50.

GLUTATHIONE

A naturally occurring intracellular redox metabolite (γ -L-glutamyl-L-cysteinylglycine; symbolized GSH) that is reversibly oxidized to glutathione disulfide [*i.e.*, $2\text{G-SH} - 2\text{H}^+ = \text{G-S-S-G}$; $E^\circ(\text{pH } 7) = -0.25\text{ V}$]. Cellular glutathione concentrations are typically at 1–5 mM concentrations, and glutathione plays major roles in cellular metabolism: for example, in the detoxification of certain arenes, in the mercapturic acid pathway, in the reduction of the Fe(III) in methemoglobin, in leukotriene and prostaglandin biosynthesis, in L-cystine translocation (as well as that of certain other amino acids), in the maintenance of thiol groups, in the destruction of hydrogen peroxide, other peroxides, and free radicals, in disulfide exchange reactions, *etc.*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Biosynthesis, **77**, 59–63 assay, **77**, 373–382; depletion *in vitro*, **77**, 59–63 [by acrylamide, **77**, 54; by acrylic esters, **77**, 54; by acrylonitrile, **77**, 54; by aspirin, **77**, 57; by *tert*-butyl hydroperoxide, **77**, 56,

57; depletion in cells [by buthionine sulfoximine, **77**, 62, 63; by cumene hydroperoxide, **77**, 56, 57; by diamide, **77**, 55, 56; by diazenecarboxylate derivatives, **77**, 55; by diethyl maleate, **77**, 52–54, 58; by diisopropylidene acetone, **77**, 54, 58; by ethylmorphine, **77**, 57; by formaldehyde, **77**, 57; by glycylglycine, **77**, 57; by products of cytochrome P-450, **77**, 54; by tetrathionate, **77**, 56; by transferase substrates, **77**, 54; by α,β -unsaturated carbonyls, **77**, 52–54; *in vivo*, **77**, 50–59]; distribution in tissue samples, **77**, 376; factors affecting levels, **77**, 51; S-substituted derivatives, **77**, 237 [synthesis, **77**, 137; turnover in kidney cells, **77**, 144, 145; turnover rate, **77**, 63; glutathione disulfide [assay, **77**, 373–382; N-ethylmaleimide, **77**, 375; with glutathione reductase, **77**, 381, 382; o-phthalaldehyde, **77**, 377; sample preparation, **77**, 375]; distribution in tissue samples, **77**, 376; efflux, **77**, 20; formation, **77**, 20].

GLUTATHIONE DEHYDROGENASE (ASCORBATE)

This enzyme [EC 1.8.5.1], also known as glutathione:dehydroascorbate oxidoreductase, catalyzes the reaction of two molecules of glutathione with dehydroascorbate to produce glutathione disulfide and ascorbate.

W. W. Wells, D. P. Xu & M. P. Washburn (1995) *Meth. Enzymol.* **252**, 30.

GLUTATHIONE PEROXIDASE

This selenium-dependent enzyme [EC 1.11.1.9] catalyzes the reaction of two molecules of glutathione with hydrogen peroxide to produce glutathione disulfide and two water molecules. Hydrogen peroxide can be replaced by steroid and lipid hydroperoxides, albeit not as effectively (nevertheless, this enzyme is not identical with phospholipid-hydroperoxide glutathione peroxidase [EC 1.11.1.12]). However, the hydroperoxy products formed by the action of lipoxygenase [EC 1.13.11.12] are not substrates.

F. Ursini, M. Maiorino, R. Brigelius-Flohé, K. D. Aumann, A. Roveri, D. Schomburg & L. Flohé (1995) *Meth. Enzymol.* **252**, 38.

R. C. Fahey & A. R. Sundquist (1991) *Adv. Enzymol.* **64**, 1.

K. T. Douglas (1987) *Adv. Enzymol.* **59**, 103.

B. Mannervik (1985) *Meth. Enzymol.* **113**, 490.

T. C. Stadtman (1979) *Adv. Enzymol.* **48**, 1.

GLUTATHIONE REDUCTASE

This FAD-dependent enzyme [EC 1.6.4.2] catalyzes the reaction of NADPH with glutathione disulfide to produce NADP⁺ and two glutathione molecules. The enzyme activity is dependent on a redox-active disulfide group in each of the active sites.

R. C. Fahey & A. R. Sundquist (1991) *Adv. Enzymol.* **64**, 1.

P. A. Karplus & G. E. Schulz (1989) *J. Mol. Biol.* **210**, 163.

K. T. Douglas (1987) *Adv. Enzymol.* **59**, 103.

I. Carlberg & B. Mannervik (1985) *Meth. Enzymol.* **113**, 484.

G. Schulz & E. F. Pai (1983) *J. Biol. Chem.* **258**, 1752.

C. H. Williams, Jr. (1976) *The Enzymes*, 3rd ed., **13**, 89.

GLUTATHIONE SYNTHETASE

This enzyme [EC 6.3.2.3] catalyzes the reaction of ATP with γ -L-glutamyl-L-cysteine with glycine to produce ADP, orthophosphate, and glutathione.

R. C. Fahey & A. R. Sundquist (1991) *Adv. Enzymol.* **64**, 1.

A. Meister (1985) *Meth. Enzymol.* **113**, 393.

A. Meister (1974) *The Enzymes*, 3rd ed., **10**, 671.

GLUTATHIONE S-TRANSFERASE

This set of enzymes [EC 2.5.1.18] (also known as glutathione *S*-alkyltransferase, glutathione *S*-aryltransferase, *S*-(hydroxyalkyl)glutathione lyase, and glutathione *S*-alkyltransferase) catalyzes the reaction of an RX with glutathione to produce HX and an *S*-substituted (with R) glutathione (thus, R-S-glutathione). R may be an aliphatic, aromatic, or heterocyclic group whereas X may be a sulfate, nitrate, or halide. In addition, some members of this set of enzymes, which typically have a broad specificity, will also catalyze the *S*-addition of glutathione to aliphatic epoxides and arene oxides. Others in this classification will catalyze the reduction of polyol nitrate by glutathione to polyol and nitrite as well as certain isomerization reactions and disulfide interchanges.

K. A. Marrs (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 127.

D. J. Meyer & B. Ketterer (1995) *Meth. Enzymol.* **252**, 53.

R. N. Armstrong (1994) *Adv. Enzymol.* **69**, 1.

T. H. Rushmore & C. B. Pickett (1993) *J. Biol. Chem.* **268**, 11475.

R. C. Fahey & A. R. Sundquist (1991) *Adv. Enzymol.* **64**, 1.

B. Mannervik & U. H. Danielson (1988) *Crit. Rev. Biochem.* **23**, 283.

K. T. Douglas (1987) *Adv. Enzymol.* **59**, 103.

R. N. Armstrong (1987) *Crit. Rev. Biochem.* **22**, 39.

B. Mannervik (1985) *Adv. Enzymol.* **57**, 357.

W. B. Jakoby & T. A. Fjellstedt (1972) *The Enzymes*, 3rd ed., **7**, 199.

GLYCERALDEHYDE-3-PHOSPHATE DEHYDROGENASE

Glyceraldehyde-3-phosphate dehydrogenase (phosphorylating) [EC 1.2.1.12], catalyzes the reaction of D-glyceraldehyde 3-phosphate with orthophosphate and NAD⁺ to produce 3-phospho-D-glyceroyl phosphate and NADH. The enzyme utilizes an active site thiol group to form a thiohemiacetal intermediate which upon oxidation is converted to a covalently bound thiolester. Based on his work with this enzyme, the metabolist Ephraim Racker was among the first to propose that enzyme catalysis can take advantage of the transient formation of covalently bound intermediates. Others, including Otto Warburg himself, initially surmised *incorrectly* that formation of covalently bound intermediates

would unnecessarily increase the number of bond-making/breaking steps and would decrease catalytic efficiency. D-Glyceraldehyde and several other aldehydes will also act as weaker substrates. A number of thiols can replace orthophosphate, thereby releasing the corresponding thiolester.

Glyceraldehyde-3-phosphate dehydrogenase (NADP⁺) (nonphosphorylating) [EC 1.2.1.9], also referred to as triose-phosphate dehydrogenase, catalyzes the reaction of D-glyceraldehyde 3-phosphate with NADP⁺ and water to produce 3-phospho-D-glycerate and NADPH.

Glyceraldehyde-3-phosphate dehydrogenase (NADP⁺) (phosphorylating) [EC 1.2.1.13], also known as triose-phosphate dehydrogenase (NADP⁺), catalyzes the reaction of D-glyceraldehyde 3-phosphate with orthophosphate and NADP⁺ to produce 3-phospho-D-glyceroyl phosphate and NADPH.

M. W. W. Adams & A. Kletzin (1996) *Adv. Protein Chem.* **48**, 101.

R. Jaenicke, H. Schurig, N. Beaucamp & R. Ostendorp (1996) *Adv. Protein Chem.* **48**, 181.

H. F. Fisher (1988) *Adv. Enzymol.* **61**, 1.

K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

J. I. Harris & M. Waters (1976) *The Enzymes*, 3rd ed., **13**, 1.

GLYCERATE DEHYDROGENASE

This enzyme [EC 1.1.1.29], also known as hydroxypyruvate reductase and hydroxypyruvate dehydrogenase, catalyzes the reaction of (*R*)-glycerate with NAD⁺ to produce hydroxypyruvate and NADH.

P. M. Goodwin (1990) *Meth. Enzymol.* **188**, 361.

J. W. Thorner & H. Paulus (1973) *The Enzymes*, 3rd ed., **8**, 487.

GLYCERATE KINASE

This enzyme [EC 2.7.1.31] catalyzes the reaction of ATP with (*R*)-glycerate to produce ADP and 3-phospho(*R*)-glycerate.

P. M. Goodwin (1990) *Meth. Enzymol.* **188**, 361.

J. W. Thorner & H. Paulus (1973) *The Enzymes*, 3rd ed., **8**, 487.

S. Black (1962) *Meth. Enzymol.* **5**, 352.

GLYCEROL

A substance (1,2,3-propanetriol; HOCH₂CH(OH)CH₂OH) that is a component and metabolite of many lipids and is commonly used in enzyme kinetic studies, serving as a cryoprotectant and stabilizing agent for proteins¹. Glycerol has also been utilized as a cryosolvent in a number of cryoenzymology studies: *e.g.*, catalase².

The freezing points of 10%, 30%, 50%, and 66.7% aqueous solutions (w/w) of glycerol are -1.6°C , -9.5°C , -23.0°C and -46.5°C , respectively. The melting point of pure glycerol is $+17.8^{\circ}\text{C}$. Because the vicinal hydroxyl groups form complexes with borates and boronates, borate-based buffers should be avoided if glycerol is present in the buffer. Likewise, glycerol interferes with periodate oxidation. The high viscosity of glycerol (at 30°C it is about 785 times that of pure water; at 25°C , this factor exceeds 1000) can influence NMR experiments³.

¹C. S. Milner & N. N. Dalton (1953) *Glycerol*, Reinhold, New York.

²G. K. Strother & E. Ackerman (1961) *Biochim. Biophys. Acta* **47**, 317.

³M.-D. Tsai (1982) *Meth. Enzymol.* **87**, 235.

GLYCEROL DEHYDRATASE

This cobalamin-dependent enzyme [EC 4.2.1.30] catalyzes the conversion of glycerol to 3-hydroxypropanal and water.

B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.

GLYCEROL KINASE

This enzyme [EC 2.7.1.30], also known as glycerokinase and ATP:glycerol 3-phosphotransferase, catalyzes the reaction of ATP with glycerol to produce ADP and glycerol 3-phosphate. Both glycerone (or, dihydroxyacetone) and L-glyceraldehyde can serve as substrates. The nucleoside triphosphate can be substituted by UTP and, in the case of the yeast enzyme, ITP and GTP.

J. H. Hurley (1996) *Ann. Rev. Biophys. & Biomol. Struct.* **25**, 137.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

E. C. C. Lin (1977) *Ann. Rev. Biochem.* **46**, 765.

J. W. Thorner & H. Paulus (1973) *The Enzymes*, 3rd ed., **8**, 487.

GLYCEROL-3-PHOSPHATE O-ACYLTRANSFERASE

This enzyme [EC 2.3.1.15] catalyzes the reaction of an acyl-CoA with *sn*-glycerol 3-phosphate to produce coenzyme A and a 1-acyl-*sn*-glycerol 3-phosphate. The acyl-CoA derivatives contain an acyl group with a chain length of at least ten carbon atoms. In addition, the acyl-CoA can be substituted by an acyl-[acyl-carrier protein] derivative.

I. Nishida & N. Murata (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 541.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.

M. A. Scheideler & R. M. Bell (1992) *Meth. Enzymol.* **209**, 55.

D. Haldar & A. Vancura (1992) *Meth. Enzymol.* **209**, 64.

R. M. Bell & R. A. Coleman (1983) *The Enzymes*, 3rd ed., **16**, 87.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

R. A. Pieringer (1983) *The Enzymes*, 3rd ed., **16**, 255.

GLYCEROL-3-PHOSPHATE DEHYDROGENASE

Glycerol-3-phosphate dehydrogenase (NAD^+) [EC 1.1.1.8] catalyzes the reaction of *sn*-glycerol 3-phosphate with NAD^+ to produce glycerone phosphate (or, dihydroxyacetone phosphate) and NADH. 1,2-Propanediol phosphate and glycerone sulfate can also act as substrates, having a weaker affinity.

Glycerol-3-phosphate dehydrogenase (NAD(P)^+) [EC 1.1.1.94], also known as NAD(P)H -dependent dihydroxyacetone-phosphate reductase, catalyzes the reaction of *sn*-glycerol 3-phosphate with NAD(P)^+ to produce glycerone phosphate (or, dihydroxyacetone phosphate) and NAD(P)H .

Glycerol-3-phosphate dehydrogenase [EC 1.1.99.5] is a flavoprotein that catalyzes the reaction of *sn*-glycerol 3-phosphate with an acceptor substrate to produce glycerone phosphate (dihydroxyacetone phosphate) and the reduced acceptor.

B. F. Cooper & F. B. Rudolph (1995) *Meth. Enzymol.* **249**, 188.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

GLYCERONE KINASE

This enzyme [EC 2.7.1.29], also known as triokinase and dihydroxyacetone kinase, catalyzes the reaction of ATP with glycerone (or, dihydroxyacetone) to produce ADP and glycerone phosphate (or, dihydroxyacetone phosphate). **See also** *Triokinase*

L. V. Bystrykh, W. de Koning & W. Harder (1990) *Meth. Enzymol.* **188**, 445.

K. H. Hofmann & W. Babel (1990) *Meth. Enzymol.* **188**, 451.

GLYCINE ACETYLTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.3.1.29], also known as glycine C-acetyltransferase and 2-amino-3-ketobutyrate:coenzyme A ligase, catalyzes the reaction of acetyl-CoA with glycine to produce coenzyme A and 2-amino-3-oxobutanoate.

D. McGilvray & J. G. Morris (1971) *Meth. Enzymol.* **17B**, 585.

I. Lieberman & H. A. Barker (1955) *Meth. Enzymol.* **1**, 616.

GLYCINE ACYLTRANSFERASES

This classification is for those enzymes that transfer acyl moieties to glycine. They include:

Glycine *N*-acyltransferase [EC 2.3.1.13] catalyzes the reaction of an acyl-CoA derivative with glycine to produce coenzyme A and an *N*-acylglycine. The acyl-CoA derivative can be one of a number of aliphatic and aromatic acids. However, neither phenylacetyl-CoA nor indole-3-acetyl-CoA can act as substrates.

Glycine *N*-benzoyltransferase [EC 2.3.1.71] catalyzes the reaction of benzoyl-CoA with glycine to produce coenzyme A and *N*-benzoylglycine. This enzyme is not identical with glycine *N*-acyltransferase or glutamine *N*-acyltransferase [EC 2.3.1.68].

H. Chantrenne (1955) *Meth. Enzymol.* **2**, 346.

GLYCINE AMIDINOTRANSFERASE

This enzyme [EC 2.1.4.1], also known as L-arginine:glycine amidinotransferase, catalyzes the reaction of L-arginine with glycine to produce L-ornithine and guanidinoacetate. Canavanine can serve as the substrate instead of arginine.

J. B. Walker (1979) *Adv. Enzymol.* **50**, 177.

J. B. Walker (1973) *The Enzymes*, 3rd ed., **9**, 497.

GLYCINE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.4] catalyzes the reaction of glycine with α -ketoglutarate (or, 2-oxoglutarate) to produce glyoxylate and L-glutamate. **See also** *Glycine:Oxaloacetate Aminotransferase*; *Glyoxylate Aminotransferase*

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

GLYCINE DECARBOXYLASE

This pyridoxal-phosphate-dependent enzyme [EC 1.4.4.2], also known as glycine dehydrogenase (decarboxylating) and the glycine cleavage system P-protein, catalyzes the reaction of glycine with a lipoylprotein to produce *S*-aminomethyldihydrolipoylprotein and carbon dioxide. Lipoamide can also act as acceptor substrate. This enzyme activity is a component, with aminomethyltransferase [EC 2.1.2.10], of glycine synthase. **See also** *Glycine Synthase*

D. J. Oliver (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 323.

E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

GLYCINE FORMIMINOTRANSFERASE

This enzyme [EC 2.1.2.4] catalyzes the reaction of 5-formiminotetrahydrofolate with glycine to produce tetrahydrofolate and *N*-formiminoglycine.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

GLYCINE METHYLTRANSFERASE

This enzyme [EC 2.1.1.20] catalyzes the reaction of *S*-adenosyl-L-methionine with glycine to produce *S*-adenosyl-L-homocysteine and sarcosine.

F. Takusagawa, M. Fujioka, A. Spies & R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 1.

M. H. Stipanuk (1986) *Ann. Rev. Nutr.* **6**, 179.

GLYCINE:OXALOACETATE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.35] catalyzes the reaction of glycine with oxaloacetate to produce glyoxylate and L-aspartate. **See also** *Glycine Aminotransferase*; *Glyoxylate Aminotransferase*

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

GLYCINE REDUCTASE

This enzyme catalyzes the reaction of glycine with orthophosphate to produce acetyl phosphate and ammonium ion. The oxidized form of the enzyme is subsequently reduced to its dithiol form in a NADH-linked step.

T. C. Stadtman (1996) *Ann. Rev. Biochem.* **65**, 83.

P. D. van Poelje & E. E. Snell (1990) *Ann. Rev. Biochem.* **59**, 29.

T. C. Stadtman (1990) *Ann. Rev. Biochem.* **59**, 111.

T. C. Stadtman (1980) *Ann. Rev. Biochem.* **49**, 93.

T. C. Stadtman (1979) *Adv. Enzymol.* **48**, 1.

GLYCINE SYNTHASE

This enzyme [EC 1.4.4.2], also known as glycine cleavage complex, consists of the components glycine dehydrogenase (decarboxylating) [EC 1.4.4.2], which catalyzes the pyridoxal-phosphate-dependent reaction of glycine with a lipoylprotein to produce *S*-aminomethyldihydrolipoylprotein and carbon dioxide, an aminomethyltransferase [EC 2.1.2.10], also known as glycine-cleavage system T-protein, which catalyzes the transfer of the group to tetrahydrofolate to generate (6*R*)-5,10-methylenetetrahydrofolate, ammonia, and dihydrolipoylprotein, and the L-protein which is a flavoenzyme that utilizes NAD⁺ to oxidize the reduced lipoate.

- V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.
 D. J. Oliver (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 323.
 J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.
 E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

GLYCOGEN PHOSPHORYLASE

This enzyme [EC 2.4.1.1], also called phosphorylase, catalyzes the reaction of [(1,4)- α -D-glucosyl] $_n$ with orthophosphate to produce [(1,4)- α -D-glucosyl] $_{n-1}$ and α -D-glucose 1-phosphate. The name to be used with this enzyme is dependent on the naturally occurring substrate: for example, glycogen phosphorylase, starch phosphorylase, maltodextrin phosphorylase, *etc.*

- N. B. Madsen (1991) *A Study of Enzymes* **2**, 139.
 N. D. Madsen (1986) *The Enzymes*, 3rd ed., **17**, 365.
 D. J. Graves & J. H. Wang (1972) *The Enzymes*, 3rd ed., **7**, 435.

GLYCOGEN SYNTHASE

Glycogen synthase [EC 2.4.1.11], also known as starch synthase and UDP-glucose-glycogen glucosyltransferase, catalyzes the reaction of UDP-glucose with [(1,4)- α -D-glucosyl] $_n$ to produce UDP and [(1,4)- α -D-glucosyl] $_{(n+1)}$. The name to be used with this enzyme is dependent on the nature of the product that is being formed: for example, glycogen synthase and starch synthase. Glycogen synthase from mammalian tissues is a multiprotein complex containing a catalytic subunit and the protein glycogenin. The enzyme requires glucosylated glycogenin as a primer (the reaction product of glycogenin glucosyltransferase [EC 2.4.1.186]). Glycogen synthase [EC 2.4.1.21], also known as starch (bacterial glycogen) synthase and ADP-glucose:starch glucosyltransferase, catalyzes the reaction of ADP-glucose with [(1,4)- α -D-glucosyl] $_n$ to produce ADP and [(1,4)- α -D-glucosyl] $_{(n+1)}$. The name to be used with this enzyme is dependent on the nature of the product that is being formed: for example, bacterial glycogen synthase and starch synthase. This enzyme is similar to the glycogen synthase [EC 2.4.1.11] above except that the nucleotide substrate is ADP-glucose.

- N. B. Madsen (1991) *A Study of Enzymes* **2**, 139.
 J. Larner (1990) *Adv. Enzymol.* **63**, 173.
 R. F. Colman (1990) *The Enzymes*, 3rd ed., **19**, 283.
 P. Cohen (1986) *The Enzymes*, 3rd ed., **17**, 461.
 P. J. Roach (1986) *The Enzymes*, 3rd ed., **17**, 499.
 W. Stalmans & H. G. Hers (1973) *The Enzymes*, 3rd ed., **9**, 309.

GLYCOLATE OXIDASE

This FMN-dependent enzyme [EC 1.1.3.15], also known as (*S*)-2-hydroxy-acid oxidase, catalyzes the reaction of a (*S*)-2-hydroxy acid with dioxygen to produce a 2-oxo acid and hydrogen peroxide. The enzyme exists as two major isoenzymes. The A form of the protein preferentially oxidizes short-chain aliphatic hydroxy acids. The B form preferentially oxidizes long-chain and aromatic hydroxy acids. The rat isoenzyme B form also acts as an L-amino-acid oxidase.

- B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
 L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.
 G. A. Hamilton (1985) *Adv. Enzymol.* **57**, 85.

GLYCOLYTIC OSCILLATION

The periodic and regular changes in glycolytic pathway kinetics, as first observed in yeast¹. In their recent consideration of three control models for glycolytic oscillations in yeast, Teusink *et al.*² applied metabolic control analysis and used operational definitions to quantify the control properties of enzymes with regard to glycolytic oscillations. Control of both frequency and amplitudes of the metabolite fluctuations were found to be distributed among the enzymes. They found little evidence supporting the role of an oscillophore (*i.e.*, the enzyme primarily held responsible for the generation of the oscillation of the system).

- ¹A. Ghosh & B. Chance (1964) *Biochem. Biophys. Res. Commun.* **16**, 174.
²B. Teusink, B. M. Bakker & H. V. Westerhoff (1996) *Biochim. Biophys. Acta* **1275**, 204.

GLYCOSIDASES

The following are recent reviews on the molecular and physical properties of this class of enzymes. Among the glycosidases are: oligoxyloglucan β -glycosidase [EC 3.2.1.120]; DNA-deoxyinosine glycosidase, also known as DNA(hypoxanthine) glycohydrolase [EC 3.2.2.15]; DNA-3-methyladenine glycosidase I, also known as DNA glycosidase I [EC 3.2.2.20]; DNA-3-methyladenine glycosidase II, also known as DNA glycosidase II [EC 3.2.2.21]; rRNA *N*-glycosidase [EC 3.2.2.22]; peptide-*N*⁴-(*N*-acetyl- β -glucosaminyl)asparagine amidase; and glycopeptide *N*-glycosidase [EC 3.5.1.52]. **See specific enzyme**

- G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.
 C. T. Walsh (1984) *Ann. Rev. Biochem.* **53**, 493.
 H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

GLYCYL ENDOPEPTIDASE

This enzyme [EC 3.4.22.25] catalyzes the hydrolysis of peptide bonds with a preference for Gly–Xaa in proteins and small molecule substrates. The enzyme, a member of the peptidase family C1, is isolated from the papaya plant, *Carica papaya*. It is not inhibited by chicken cystatin, unlike most other homologs of papain.

D. J. Buttle (1994) *Meth. Enzymol.* **244**, 539.

GLYCYL-tRNA SYNTHETASE

This enzyme [EC 6.1.1.14], also known as glycine:tRNA ligase, catalyzes the reaction of glycine with tRNA^{Gly} and ATP to produce glycyl-tRNA^{Gly}, AMP, and pyrophosphate (or, diphosphate).

P. Schimmel (1987) *Ann. Rev. Biochem.* **56**, 125.

GLYOXALASE I

This enzyme [EC 4.4.1.5], also known as lactoylglutathione lyase and methylglyoxalase, catalyzes the conversion of (*R*)-*S*-lactoylglutathione to produce glutathione and methylglyoxal. 3-Phosphoglycerol-glutathione will also serve as a substrate.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

K. T. Douglas (1987) *Adv. Enzymol.* **59**, 103.

A. S. Mildvan & D. C. Fry (1987) *Adv. Enzymol.* **59**, 241.

GLYOXALASE II

This enzyme [EC 3.1.2.6], also known as *S*-hydroxyacylglutathione hydrolase, catalyzes the hydrolysis of an (*S*)-(2-hydroxyacyl)glutathione to yield glutathione and a 2-hydroxy acid anion. *S*-Acetoacetylglutathione will act as a substrate as well, but not as efficiently.

K. T. Douglas (1987) *Adv. Enzymol.* **59**, 103.

GLYOXYLATE AMINOTRANSFERASE

This pyridoxal 5-phosphate-dependent enzyme [EC 2.6.1.4] catalyzes the transamination of glyoxylate from L-glutamate to produce glycine and α -ketoglutarate.

N. E. Tolbert (1981) *Ann. Rev. Biochem.* **50**, 133.

GLYOXYLATE REDUCTASE

Glyoxylate reductase [EC 1.1.1.26] catalyzes the reversible reaction of glycolate with NAD⁺ to produce glyoxylate and NADH. The enzyme will also catalyze the NADH-dependent interconversion of hydroxypyruvate to D-glycerate. Glyoxylate reductase (NADPH) [EC

1.1.1.79] catalyzes the reversible reaction of glycolate with NADP⁺ to produce glyoxylate and NADPH (as well as the hydroxypyruvate to D-glycerate conversion). The enzyme can use NAD⁺ as a substrate, although not as effectively as NADP⁺.

L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.

GMP SYNTHETASE

GMP synthetase [EC 6.3.4.1], also known as xanthosine-5'-phosphate:ammonia ligase, catalyzes the reaction of ATP with xanthosine 5'-phosphate and ammonia to produce GMP, AMP, and pyrophosphate (or, diphosphate). GMP synthetase (glutamine-utilizing) [EC 6.3.5.2] catalyzes the reaction of ATP with xanthosine 5'-phosphate, L-glutamine, and water to produce GMP, AMP, L-glutamate, and pyrophosphate.

H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.

H. Zalkin (1985) *Meth. Enzymol.* **113**, 273.

D. E. Koshland, Jr., & A. Levitzki (1974) *The Enzymes*, 3rd ed., **10**, 539.

J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

GOLDMAN EQUATION

An equation (also referred to as the constant field equation, the Goldman-Hodgkin-Katz equation, and the GHK equation) which relates the membrane potential ($\Delta\psi$) to the individual permeabilities of the ions (and their concentrations) on both sides of the membrane. Thus,

$$\Delta\psi = \frac{RT}{F} \ln \left[\frac{\sum_+ P_i [M_i^+]_{\text{out}} + \sum_- P_j [X_j^-]_{\text{in}}}{\sum_+ P_i [M_i^+]_{\text{in}} + \sum_- P_j [X_j^-]_{\text{out}}} \right]$$

where R is the universal gas constant, T is the absolute temperature, F is the Faraday constant, the sums ($\sum$) are taken over all cations ($\sum_+$) and anions ($\sum_-$) having significant permeabilities, P refers to the relative membrane permeability of these ions, and $[M^+]$ and $[X^-]$ are their concentrations.

If any one ion has a significantly greater permeability than any of the other ions, then the Goldman equation reduces to the Nernst equation for membrane potential.

If the permeability of one ion is altered, while all concentrations and other permeabilities remain unchanged, the Goldman equation permits one to predict the changes in the membrane potential. For example, consider the

case in which the permeability for a cation (the cation that is in higher concentration inside the cell) has increased. In such a case, that cation flows out of the cell and more positive charges now exist on the outer surface of the membrane and more negative charges on the inner surface. As can be seen from the Goldman equation, the membrane potential becomes more negative: the membrane is hyperpolarized.

GOODNESS-OF-FIT CRITERIA

Mannervik¹ has provided criteria for assessing the fitness of a particular kinetic model to the experimental data (utilizing regression analysis and statistical methods) and to aid in the discrimination between different models. He offers the following criteria: (1) a good model is expected to give convergence in the regression analysis; (2) a good model should give meaningful parameter values with low standard deviations; (3) a good model should give residuals showing random distribution about the zero level and lacking correlation with any of the dependent or independent variables; (4) a good model should give a low residual sum of squares that is compatible with the experimental variance. **See also** *Global Analysis; Statistics (A Primer)*

¹B. Mannervik (1982) *Meth. Enzymol.* **87**, 370.

GOOD'S BUFFERS

A collection of buffers¹⁻³ used in biochemical studies exhibiting minimal buffer salt interaction with proteins and/or metal ions. These buffers span the pH ranges normally used in biological experiments. While they tend not to bind metal ions strongly, this should be verified when metal ion-requiring substrates are involved⁴. The list below includes commercially available Good's buffers, along with their reported pK_a values (at 25°C, unless otherwise noted) and their corresponding dpK_a/dT values when known⁵. Those marked with an asterisk indicate the buffers which were originally reported by Good *et al.*¹

2-(N-Morpholino)ethanesulfonic acid* (abbreviated MES)	$pK_a = 6.1$, $dpK_a/dT = -0.011$
Bis(2-hydroxyethyl)iminotris-(hydroxymethyl)methane (BIS-TRIS)	$pK_a = 6.5$
N-(2-Acetamido)-2-iminodiacetic acid* (ADA)	$pK_a = 6.6$, $dpK_a/dT = -0.011$
Piperazine-N,N'-bis(2-ethanesulfonic acid* (PIPES)	$pK_a = 6.8$, $dpK_a/dT = -0.0085$

N-(2-Acetamido)-2-aminoethanesulfonic acid* (ACES*)	$pK_a = 6.8$, $dpK_a/dT = -0.020$
1,3-Bis[tris(hydroxymethyl)-methylamino]propane (BIS-TRIS PROPANE)	$pK_{a,1} = 6.8$, $pK_{a,2} = 9.0$
3-(N-Morpholino)-2-hydroxypropanesulfonic acid (MOPSO)	$pK_a = 6.9$
N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid* (BES)	$pK_a = 7.1$, $dpK_a/dT = -0.016$
3-(N-Morpholino)propanesulfonic acid (MOPS)	$pK_a = 7.2$
N-Tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid* (TES)	$pK_a = 7.45$, $dpK_a/dT = -0.020$
N-2-Hydroxyethylpiperazine-N'-2-ethanesulfonic acid* (HEPES)	$pK_a = 7.5$, $dpK_a/dT = -0.014$
3-[N,N-Bis(2-hydroxyethyl)-amino]-2-hydroxypropanesulfonic acid (DIPSO)	$pK_a = 7.6$
3-[N-Tris(hydroxymethyl)-methylamino]-2-hydroxypropanesulfonic acid (TAPSO)	$pK_a = 7.6$
N-(2-Hydroxyethyl)piperazine-N'-(2-hydroxypropanesulfonic acid) (HEPPSO)	$pK_a = 7.8$
Piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) (POPSO)	$pK_a = 7.8$
N-(2-Hydroxyethyl)piperazine-N'-(3-propanesulfonic acid) (EPPS, also called HEPPS)	$pK_a = 8.0$
N-Tris(hydroxymethyl)methylglycine* (Tricine)	$pK_a = 8.1$, $dpK_a/dT = -0.021$
N,N-Bis(2-hydroxyethyl)glycine* (Bicine)	$pK_a = 8.3$, $dpK_a/dT = -0.018$
Tris(hydroxymethyl)methylamino-propanesulfonic acid (TAPS)	$pK_a = 8.4$
3-[(1,1-Dimethyl-2-hydroxyethyl)-amino]-2-hydroxypropanesulfonic acid (AMPSO)	$pK_a = 9.0$
3-(N-Cyclohexylamino)-ethanesulfonic acid (CHES)	$pK_a = 9.3$ (9.55 at 20°C)
3-(Cyclohexylamino)-2-hydroxy-1-propanesulfonic acid (CAPSO)	$pK_a = 9.6$
3-(Cyclohexylamino)-1-propanesulfonic acid (CAPS)	$pK_a = 10.4$

¹N. E. Good, G. D. Winget, W. Winter, T. N. Connolly, S. Izawa & R. M. M. Singh (1966) *Biochemistry* **5**, 467.

²N. E. Good & S. Izawa (1972) *Meth. Enzymol.* **24B**, 53.

³W. J. Ferguson & N. E. Good (1980) *Anal. Biochem.* **104**, 300.

⁴J. F. Morrison (1979) *Meth. Enzymol.* **63**, 257.

⁵D. D. Perrin & B. Dempsey (1974) *Buffers for pH and Metal Ion Control*, Chapman and Hall, London.

G PROTEINS

A superfamily of GTPases (often referred to as GTP- or G-regulatory proteins) whose cardinal feature is a 200-residue domain comprising a central six-stranded β -pleated sheet surrounded by α -helices^{1,2}. Five polypeptide loops constituting the guanine nucleotide binding site are highly conserved among members of this protein superfamily. Examples of G proteins are (a) transducin,

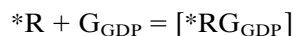
which in its bound-GTP state stimulates cyclic GMP phosphodiesterase, (b) elongation factor Tu which in its EF-Tu·GTP complex moves an aminoacyl-tRNA into the ribosome-mRNA complex A site for aminoacylation of a growing polypeptide, and (c) Ras which as Ras·GTP activates its downstream tyrosine kinase targets. Cleavage of the terminal phosphoanhydride bond of GTP allows each of these regulatory proteins to transit from its stimulatory conformational state to a quiescent state.

G proteins exhibit the ability to modulate both metabolic activation or inhibition based on the phosphorylation state of the bound guanine nucleotide. In this model, the activated receptor *R plays a catalytic role by activating more than one G-protein. This is evident in the following cycle of reactions:

1. Agonist-A Starts Process by Activating Receptor-R:



2. Activated Receptor *R Binds to GDP form of G Protein:



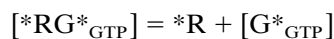
3. GDP Dissociates from [*RG_{GDP}] Complex:



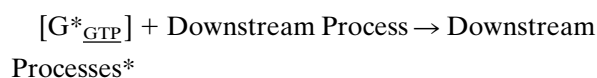
4. GDP-free Form [*RG_{___}] Binds GTP and Activates G Protein:



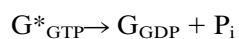
5. Activated Receptor *R Departs, Leaving Behind Activated G^{*}_{GTP}:



6. G^{*}_{GTP} Stimulates “Downstream” Metabolic Processes*:



7. G^{*}_{GTP} Undergoes GTP Hydrolysis to Deactivated G_{GDP} Form:



8. Activated Metabolic Response* is Terminated:
Downstream Process* → Downstream Process

A G-protein has two states, each characterized by the presence of bound GTP or GDP, and their interconversion is accomplished by an intrinsic GTPase as well as the participation of nucleotide-exchange factors. Bound GTP is converted to bound GDP by the GTPase, whereas bound GDP is converted to bound GTP by release of GDP and binding/exchange of GTP. The rate of interconversion between active and inactive states is limited by GTPase hydrolysis, and GTPase-activating proteins (GAPs) can accelerate hydrolysis, as can G protein association with a conformational state of its effector. In this respect, certain G proteins are “clocks”, others are “switches” or “adapters”, while still others are “sensors”.

These GTPases operate by in-line or S_N2 mechanisms, where the phosphoryl undergoes inversion of configuration as it is directly transferred to water. The kinetics of the GTPase and the guanine nucleotide exchange activities determine the time dependence of activated or inhibited states of the metabolic process under their direct control. G_{1α1} has a turnover number for GTP hydrolysis of around 3 min⁻¹; Ras has a *k*_{cat} of 0.3 min⁻¹; and EF-Tu has a value of 0.003 min⁻¹. Tetracoordinate AlF₄⁻ ion activates G_α subunits by mimicking GTP's γ-phosphoryl group as it undergoes hydrolysis and assumes a pentavalent geometry. GTPγS binding sustains the activated state of G proteins.

¹S. R. Sprang (1997) *Ann. Rev. Biochem.* **66**, 639.

²D. E. Coleman, A. M. Berghuis, E. Lee, M. E. Linder, A. G. Gilman & D. R. Sprang (1994) *Science* **265**, 1405.

Entire MIE Volumes Dedicated to This Topic: Vol. **237** (R. Iyengar, ed.); vol. **195** (R. A. Johnson & J. D. Corbin, eds.).

GRAPHICAL METHODS

See Double-Reciprocal Plot; Hanes Plot; Direct Linear Plot; Dixon Plot; Dixon-Webb Plot; Eadie-Hofstee Plot; Substrate Concentration Range; Frieden Protocol; Fromm Protocol; Point-of-Convergence Method; Dalziel Phi Relationships; Scatchard Plots; Hill Plots

GRAPH THEORY

A branch of mathematics concerned with the study and applications of graphs and their uses in kinetic behavior of linked processes. Graph theory, especially when applied to the dynamic nature of networks, has provided

useful protocols for deriving enzyme rate expressions. **See** *Fromm's Method for Deriving Enzyme Rate Equations*

GREEN FLUORESCENT PROTEIN

An intensely fluorescent protein¹ isolated from the jellyfish *Aequorea victoria* and other *Aequorea* species. This protein produces its fluorescence from an intrinsic chromophore generated by a series of steps involving residues 65–67 of the polypeptide chain. As a result, GFP has broad application, especially when expressed as a fused or chimeric protein containing a second polypeptide region whose intracellular location is of interest².

The steps leading to the formation of the intrinsic chromophore have recently been investigated³ kinetically with S65T-GFP. The process of chromophore formation is an ordered sequence of three distinct steps: (1) slow protein folding ($k_f = 2.44 \times 10^{-3} \text{ s}^{-1}$) that precedes chromophore modification; (2) an intermediate step occurs that includes, but may not be necessarily limited to, cyclization of the tripeptide chromophore motif ($k_c = 3.8 \times 10^{-3} \text{ s}^{-1}$); and (3) rate-limiting oxidation of the cyclized chromophore ($k_{ox} = 1.51 \times 10^{-4} \text{ s}^{-1}$). Reid and Flynn³ also reasoned that because chromophore forms *de novo* from purified denatured protein and is a first-order process, GFP chromophore formation is likely to be an autocatalytic process.

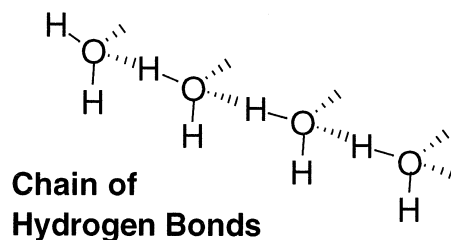
¹M. Chalfie, Y. Tu, G. Euskirchen, W. W. Ward & D. C. Prasher (1994) *Science* **263**, 802.

²S. R. Kain, M. Adams, A. Kondepudi, T. T. Yang, W. W. Ward & P. Kitts (1995) *Biotechniques* **19**, 650.

³B. G. Reid & G. C. Flynn (1997) *Biochemistry* **36**, 6786.

GROTTHUSS CHAIN

Extended chains of hydrogen-bonded water molecules proposed by Christian von Grotthuss in the early 19th century to explain electrolyte conductance in water¹.



While these extended chains failed to explain the ionic mobilities of most inorganic ions, they remain a viable

model for proton and hydroxide ion mobility. **See** *Grotthuss Mechanism; Water*

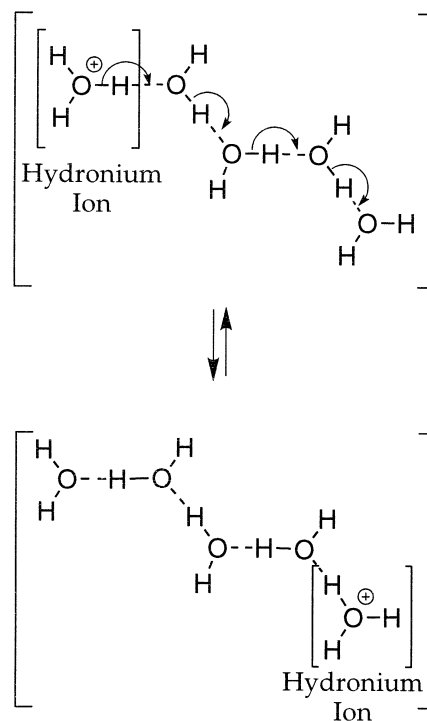
¹The molecular structure of water was not known in Grotthuss' time, and he proposed that water molecules, designated as $-$ and $+$, were arranged between the positive (p) and negative (n) electrodes in the following manner: $p - + - + - + - n$. (J. R. Partington (1964) *A History of Chemistry*, vol. 4, pp. 26-27, Macmillan, London.)

GROTTHUSS-DRAPER LAW

The so-called first law of photochemistry stating that only the radiation absorbed by a molecular entity or substance is effective in producing a photochemical change.

GROTTHUSS MECHANISM

A mechanism that explains the high *apparent* ionic mobilities¹ of protons and hydroxide ions in terms of hydrogen bond-making/-breaking steps along extended chains of water molecules. Solvation of a proton can be represented simply² as H_3O^+ , and this hydronium ion strongly interacts with other molecules in a hydrogen-bond network.



Shown in the figure above is a hypothetical scheme that illustrates how an organized series of protonation-deprotonation steps can have the net effect of rapidly "transferring" a proton from a hydronium ion (*i.e.*,

H_3O^+) in one region of an aqueous solution to produce a hydronium ion at a distant site. Note that the proton released locally from the initial H_3O^+ remains in its vicinity, and is not the same as the proton forming the hydronium ion at the distant site. For this reason, the ionic mobility appears to be much greater than would be expected on the basis of diffusion alone. Facilitated proton transfer along rigidly and accurately positioned hydrogen bonds could be of fundamental importance in enzyme catalysis. **See** *Water*

¹Ionic mobility refers to the velocity of an ion moving toward an oppositely charged electrode when a 1-volt potential is applied across a 1-centimeter electrochemical cell.

²A strongly hydrated molecular cluster, such as $[\text{H} \cdot (\text{OH}_2)_4]^+$, is probably a more realistic representation (M. Eigen (1964) *Angew. Chem. (Int. Eng. Edn.)* **3**, 1).

GROUND STATE

The state of a molecular entity (or system) with the lowest Gibbs free energy. The context of the study usually makes clear the type of ground state under consideration (e.g., electronic, vibrational, rotational, etc.). **See also** *Excited State; Fluorescence*

GROUP TRANSFER REACTIONS

Enzyme-catalyzed reactions in which a functional group or moiety is transferred from one molecular entity to another.

GTP CYCLOHYDROLASE I

This enzyme [EC 3.5.4.16] catalyzes the reaction of GTP with two water molecules to produce formate and 2-amino-4-hydroxy-6-(*erythro*-1,2,3-trihydroxypropyl)-dihydropteridine triphosphate. The reaction involves hydrolysis of two C–N bonds and isomerization of the pentose unit. The recyclization step may be nonenzymatic.

E. R. Werner, H. Wachter & G. Werner-Felmayer (1997) *Meth. Enzymol.* **281**, 53.

C. A. Nichol, G. K. Smith & D. S. Duch (1985) *Ann. Rev. Biochem.* **54**, 729.

G. M. Brown & J. M. Williamson (1982) *Adv. Biochem.* **53**, 345.

GTP CYCLOHYDROLASE II

This enzyme [EC 3.5.4.25] catalyzes the reaction of GTP with three water molecules to produce formate, 2,5-diamino-6-hydroxy-4-(5-phosphoribosylamino)pyrimidine, and pyrophosphate (or, diphosphate). In this reaction, two C–N bonds are hydrolyzed, releasing formate, with

the simultaneous hydrolysis of the terminal pyrophosphate.

A. Bacher, G. Richter, H. Ritz, S. Eberhardt, M. Fischer & C. Krieger (1997) *Meth. Enzymol.* **280**, 382.

G. M. Brown & J. M. Williamson (1982) *Adv. Biochem.* **53**, 345.

GTP DEPLETION or REGENERATION

See *ATP and GTP Depletion; Nucleoside 5'-Triphosphate Regeneration*

GUANINE DEAMINASE

This enzyme [EC 3.5.4.3], also known as guanine aminohydrolase and guanase, catalyzes the hydrolysis of guanine to produce xanthine and ammonia.

C. L. Zielke & C. H. Suelter (1971) *The Enzymes*, 3rd ed., **4**, 47.

GUANOSINE DEAMINASE

This enzyme [EC 3.5.4.15], also known as guanosine aminohydrolase, catalyzes the hydrolysis of guanosine to yield xanthosine and ammonia.

C. L. Zielke & C. H. Suelter (1971) *The Enzymes*, 3rd ed., **4**, 47.

GUANYLATE CYCLASE

This enzyme [EC 4.6.1.2], also known as guanylyl cyclase and guanyl cyclase, catalyzes the conversion of GTP to 3',5'-cyclic GMP and pyrophosphate (or, diphosphate). Both ITP and dGTP can act as substrates.

Entire volume of MIE that covers this topic: R. A. Johnson & J. D. Corbin, eds. (1991) *Meth. Enzymol.*, vol. **195**.

GUANYLATE KINASE

This enzyme [EC 2.7.4.8] catalyzes the reaction of ATP with GMP to produce ADP and GDP. dGMP can substitute for GMP and dATP can substitute for ATP.

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

GUEST

An inorganic or organic ion that binds to a host molecule.

See *Host Molecule; Inclusion Complexes*

GUGGENHEIM METHOD (FIRST-ORDER RATE CONSTANT)

A systematic procedure^{1,2} for determining the value of a first-order rate constant whenever the initial reactant concentration(s) or the final extent of reaction is unknown.

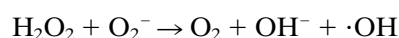
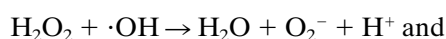
¹E. A. Guggenheim (1926) *Philos. Mag.* **2**, 538.

²J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, 3rd ed., Wiley, New York.

H

HABER-WEISS REACTION

A chemical reaction cycle involving hydrogen peroxide, hydroxide radical, molecular oxygen, and hydroxide ion:



The second reaction achieved notoriety as a possible source of hydroxyl radicals, but the reaction proceeds extremely slowly. Iron(III) complexes may catalyze this reaction; in this case, Fe(III) would first be reduced by the superoxide, followed by oxidation by hydrogen peroxide. **See also** Fenton Reaction

HALDANE RELATION

A mathematical equation¹ indicating how the equilibrium constant of an enzyme-catalyzed reaction (or half-reaction in the case of so-called ping pong reaction mechanisms) is related to the various kinetic parameters for the reaction mechanism. In the Briggs-Haldane steady-state treatment of a Uni Uni reaction mechanism, the Haldane relation can be written as follows:

$$K_{\text{eq}} = \frac{V_{\text{max,forward}} K_{\text{product}}}{V_{\text{max,reverse}} K_{\text{substrate}}} = \frac{k_{\text{cat,forward}}}{K_{\text{substrate}}} \times \frac{K_{\text{product}}}{k_{\text{cat,reverse}}}$$

where $V_{\text{max,forward}}$ (equal to $k_{\text{cat,forward}}[\text{Enzyme}_{\text{total}}]$) is the maximal reaction velocity in the forward direction, K_{product} is the Michaelis constant for the reverse reaction, $V_{\text{max,reverse}}$ (equal to $k_{\text{cat,reverse}}[\text{Enzyme}_{\text{total}}]$) is the maximal reaction velocity in the reverse direction, and

$K_{\text{substrate}}$ is the Michaelis constant for the forward direction.

Rigorous adherence of enzymes to the Haldane relation is well illustrated by the case of wild-type and Glu¹⁶⁵-to-Asp triose-phosphate isomerases². These enzymes differ only with respect to a single methylene in the side-chain carboxyl group of residue 165. The steady-state parameters for the wild-type enzyme are: $k_{\text{cat,forward}}$, 430 s⁻¹; K_{m} , 0.97 mM for dihydroxyacetone phosphate; $k_{\text{cat,reverse}}$, 4300 s⁻¹; and K_{m} , 0.47 mM for D-glyceraldehyde 3-phosphate. Substitution of these values into the above equation gives K_{eq} equal to 21 (for the reaction written in the direction of dihydroxyacetone phosphate formation), a value that is in excellent agreement with independent determinations of the overall equilibrium constant ($K_{\text{eq}} = 22$)³. The steady-state parameters² for the Glu¹⁶⁵-to-Asp enzyme are: $k_{\text{cat,forward}}$, 1.8 s⁻¹; K_{m} , 1.8 mM for dihydroxyacetone phosphate; $k_{\text{cat,reverse}}$, 2.8 s⁻¹; and K_{m} , 0.13 mM for D-glyceraldehyde 3-phosphate. Substitution of these values into the above equation again gives a value of 21.5 for K_{eq} . While there is every reason to expect that the wild-type and Glu¹⁶⁵-to-Asp forms of triose-phosphate isomerase operate by virtually identical catalytic mechanisms, strict adherence to the Haldane relation must occur, even for enzymes using different catalytic pathways as long as one is dealing with the same overall reaction.

The use of Haldane relationships to verify the magnitude of the equilibrium constant or, conversely, to determine (or verify) one of the kinetic parameters requires that all constants be measured under the same experimental conditions (e.g., temperature, pH, buffer species, ionic strength, free metal ion concentrations, *etc.*) If not, the Haldane relationship has no meaning. In addition, kinetic data are often limited in precision, unlike equilibrium measurements. For multisubstrate reactions, there are at least two different Haldane relationships. Thus,

there are a number of independent opportunities to compare the magnitude of derived equilibrium constant to that of the observed equilibrium constant. Such a procedure was used for nucleoside diphosphate kinase⁴ and for galactose-1-phosphate uridylyltransferase⁵.

Fromm⁶ and Cleland⁷ provide valuable discussions of the utility of Haldane relations in excluding certain kinetic reaction mechanisms based on a numerical evaluation of the constants on each side of the equal sign in the Haldane relation. If the equality is maintained, the candidate mechanism is consistent with the observed rate parameter data. Obviously, one must be concerned about the quality of experimentally derived estimates of rate parameters, because chemists have frequently observed that thermodynamic data (such as equilibrium constants) are often more accurate and precise than kinetically derived parameters. **See** *Haldane Relations for Multisubstrate Enzymes*

¹J. B. S. Haldane (1930) *Enzymes*, Longmans, Green and Co., London. (republished in 1965 by M.I.T. Press, Cambridge, MA).

²P. A. Raines & J. R. Knowles (1986) *Ann. N. Y. Acad. Sci.* **471**, 266.

³I. A. Rose & Z. B. Rose (1969) *Comprehensive Biochemistry* **17**, 126.

⁴Graces & W. W. Cleland (1969) *Biochemistry* **8**, 633.

⁵P. A. Frey, L.-J. Wong, K.-F. Sheu & S.-L. Yang (1982) *Meth. Enzymol.* **87**, 20.

⁶H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, Berlin.

⁷W. W. Cleland (1982) *Meth. Enzymol.* **87**, 366.

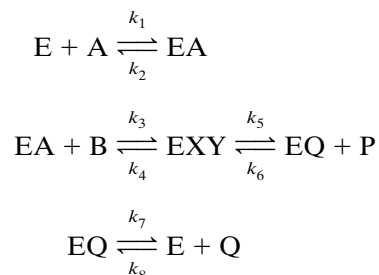
HALDANE RELATIONS FOR MULTISUBSTRATE ENZYMES

Alberty¹ first proposed the use of Haldane relations to distinguish among the ordered Bi Bi, the ordered Bi Bi Theorell-Chance, and the rapid equilibrium random Bi Bi mechanisms. Nordlie and Fromm² used Haldane relationships to rule out certain mechanisms for ribitol dehydrogenase.

Cleland³ pointed out that most Haldane relations have the general form $K_{eq} = \{(V_{max,f})^n K_{(p)} K_{(q)} K_{(r)} \dots\} / \{(V_{max,r})^n K_{(a)} K_{(b)} K_{(c)} \dots\}$ where $V_{max,f}$ and $V_{max,r}$ are the maximum velocities in the forward and reverse directions, respectively, n is an integer (ranging between -1 , 0 , 1 , 2 , $\dots$), $K_{(a)}$, $K_{(b)}$, $K_{(c)}$, *etc.*, are either the Michaelis constants or the dissociation constants for the substrates (A, B, C, *etc.*) and $K_{(p)}$, $K_{(q)}$, $K_{(r)}$, *etc.*, are either the Michaelis constants or dissociation constants for the products (P, Q, R, *etc.*). Cleland⁴ distinguished two types

of Haldane relations: thermodynamic and kinetic. Thermodynamic Haldane relations contain only the equilibrium constants associated with the individual steps of the reaction mechanism. An example of the thermodynamic type can be given for the ping pong Bi Bi reaction scheme ($E + A \rightleftharpoons EX \rightleftharpoons F + P$ followed by $F + B \rightleftharpoons FB \rightleftharpoons E + Q$). One Haldane is $K_{eq} = K_{ip}K_{iq}/(K_{ia}K_{ib})$ where K_{ip} , K_{iq} , K_{ia} , and K_{ib} refer to the dissociation constants of the products and reactants, respectively.

For a number of enzyme kinetic mechanisms, a thermodynamic Haldane cannot explicitly be defined in terms of K_{ix} parameters. An example is the ordered Bi Bi reaction:



In order to derive a thermodynamic Haldane relation for this mechanism, the true dissociation constants for B (*i.e.*, k_4/k_3) and for P (*i.e.*, k_5/k_6) are needed. From the definitions for the various kinetic parameters (**See** *Ordered Bi Bi Mechanism*) it is readily seen that $k_4/k_3 = K_{ib}K_aV_{max,r}/(K_{ia}V_{max,f})$ and $k_5/k_6 = K_{ip}K_qV_{max,f}/(K_{iq}V_{max,r})$. Thus, a thermodynamic Haldane relationship for the ordered Bi Bi mechanism can be written as $K_{eq} = V_{max,f}^2 K_{ip}K_q/(V_{max,r}^2 K_aK_{ib})$, where K_q and K_a are Michaelis constants.

Kinetic Haldane relations⁴ use a ratio of apparent rate constants in the forward and reverse directions, if the substrate concentrations are very low. For an ordered Bi Bi reaction, the apparent rate constant for the second step is $V_{max,f}/K_b$ (where K_b is the Michaelis constant for B) and, in the reverse reaction, $V_{max,r}/K_p$. Each of these is multiplied by the reciprocal of the dissociation constant of A and Q, respectively. The “forward” product is then divided by the “reverse” product. Hence, the kinetic Haldane relationship for the ordered Bi Bi reaction is $K_{eq} = (K_{ia}^{-1}V_{max,f}/K_b)/(K_{iq}^{-1}V_{max,r}/K_p) = V_{max,f}K_pK_{iq}/(V_{max,r}K_{ia}K_b)$. For completely random mechanisms, thermodynamic and kinetic Haldane relationships are equivalent.

In ping pong reactions, Haldane relations can also be written for the individual “half-reactions.” In such cases, Haldane expressions assist in analyzing isotope exchange studies involving these partial reactions.

Haldane relationships can also be useful in characterizing isozymes or the same enzyme isolated from a different source. Reactions catalyzed by isozymes must have identical equilibrium constants, but the magnitudes of their kinetic parameters are usually different (*e.g.*, the case of yeast and mammalian brain hexokinase⁵). Note that the Haldane relationship for the ordered Bi Bi mechanism is $K_{eq} = V_{max,f}K_p K_{iq}/(V_{max,r}K_{ia}K_b)$. This same Haldane is also valid for the ordered Bi Bi Theorell-Chance mechanism and the rapid equilibrium random Bi Bi mechanism. The reverse reaction of the yeast enzyme is easily studied: an observation not true for the brain enzyme, even though both enzymes catalyze the exact same reaction. A crucial difference between the two enzymes is the dissociation constant (K_{iq}) for Q (in this case, glucose 6-phosphate). For the yeast enzyme, this value is about 5 mM whereas for the brain enzyme the value is 1 μ M. Hence, in order for K_{eq} to remain constant (and assuming K_p , K_{ia} , and K_b are all approximately the same for both enzymes) the $V_{max,f}/V_{max,r}$ ratio for the brain enzyme must be considerably larger than the corresponding ratio for the yeast enzyme. In fact, the differences between the two ratios is more than a thousandfold. Hence, the Haldane relationship helps to explain how one enzyme appears to be more kinetically reversible than another catalyzing the same reaction.

¹R. A. Alberty (1953) *J. Amer. Chem. Soc.* **75**, 1928.

²R. C. Nordlie & H. J. Fromm (1959) *J. Biol. Chem.* **234**, 2523.

³W. W. Cleland (1963) *Biochim. Biophys. Acta* **67**, 104.

⁴W. W. Cleland (1982) *Meth. Enzymol.* **87**, 366.

⁵D. L. Purich & H. J. Fromm (1972) *Curr. Top Cell. Reg.* **6**, 131.

HALF-LIFE

The period of time required for the decay of one-half of a substance via a first-order process. In a first-order process

$$[A] = [A_0]e^{-kt}$$

where $[A]$ is the concentration of reactant at time t and $[A_0]$ is the concentration at time zero. When $[A] = 0.5[A_0]$, then $2 = e^{kt_{0.5}}$ (or, $\ln 2 = kt_{0.5}$). Hence, $t_{0.5} = (0.693147)/k$, where k is the first-order rate constant for the reaction.

The cardinal feature of all first-order processes is that the period required for reducing the reactant by a factor of 2 is independent of the amount of substance present.

CHEMICAL PROCESSES. In many chemical reactions and in *all* exchange reactions, the rate of decrease in the concentration of a reactant is directly proportional to the concentration of that reactant (*i.e.*, $v = k[A]$). For a first-order process, the relaxation period (corresponding to the period of time required to reach e^{-1} , or about 0.368, of the original amount) is given as:

$$t = 1/k = t_{0.5}/\ln 2$$

Many bimolecular rate processes can behave as though they are first-order processes, especially if one reactant is present at much greater concentration than the other. Such processes are called pseudo-first-order reactions.

RADIOACTIVE DECAY. Many atomic nuclei have unstable neutron-to-proton ratios and undergo spontaneous first-order decay through the emission of α , β^- , or β^+ particles or gamma rays.

The half-lives of radioisotopes typically used in biochemical experiments are:

Radionuclide	$t_{0.5}^*$	k^*
³ H	12.32 y	$1.783 \times 10^{-9} \text{ s}^{-1}$
¹¹ C	20.3 m	$5.69 \times 10^{-4} \text{ s}^{-1}$
¹⁴ C	$5.715 \times 10^3 \text{ y}$	$3.843 \times 10^{-12} \text{ s}^{-1}$
¹³ N	9.97 m	$1.16 \times 10^{-3} \text{ s}^{-1}$
²² Na	2.605 y	$8.432 \times 10^{-9} \text{ s}^{-1}$
³² P	14.28 d	$5.618 \times 10^{-7} \text{ s}^{-1}$
³³ P	25.3 d	$3.17 \times 10^{-7} \text{ s}^{-1}$
³⁵ S	87.2 d	$9.20 \times 10^{-8} \text{ s}^{-1}$
³⁶ Cl	$3.01 \times 10^5 \text{ y}$	$7.30 \times 10^{-14} \text{ s}^{-1}$
⁴⁰ K	$1.26 \times 10^9 \text{ y}$	$1.74 \times 10^{-17} \text{ s}^{-1}$
⁴² K	12.36 h	$1.558 \times 10^{-5} \text{ s}^{-1}$
⁴³ K	22.3 h	$8.63 \times 10^{-6} \text{ s}^{-1}$
⁴⁵ Ca	162.7 d	$4.931 \times 10^{-8} \text{ s}^{-1}$
⁴⁷ Ca	4.536 d	$1.769 \times 10^{-6} \text{ s}^{-1}$
⁵¹ Cr	27.70 d	$2.896 \times 10^{-7} \text{ s}^{-1}$
⁵² Fe	8.28 h	$2.33 \times 10^{-5} \text{ s}^{-1}$
⁵⁵ Fe	2.73 y	$8.05 \times 10^{-9} \text{ s}^{-1}$
⁵⁹ Fe	44.51 d	$1.802 \times 10^{-7} \text{ s}^{-1}$
⁵⁷ Co	271.8 d	$2.952 \times 10^{-8} \text{ s}^{-1}$
⁵⁸ Co	70.88 d	$1.132 \times 10^{-7} \text{ s}^{-1}$
⁶⁰ Co	5.271 y	$4.167 \times 10^{-9} \text{ s}^{-1}$
¹²³ I	13.2 h	$1.46 \times 10^{-5} \text{ s}^{-1}$
¹²⁵ I	59.4 d	$1.35 \times 10^{-7} \text{ s}^{-1}$
¹³¹ I	8.040 d	$9.978 \times 10^{-7} \text{ s}^{-1}$

*Abbreviations are y = years, d = days, h = hours, m = minutes, and s = seconds.

For a radioisotope decay, the rate is given as

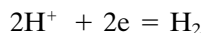
$$v = kN$$

where k is the first-order rate constant and N is the number of atoms ($N = (\text{number of moles})(N_A) = (\text{number of moles})(6.0221367 \times 10^{23} \text{ atoms} \cdot \text{mol}^{-1})$). [In converting years to seconds, it is useful to recall that there are about 3.15×10^7 seconds/year, or roughly $\pi \times 10^7$ seconds/year].

HALF-REACTIONS

1. A balanced hypothetical chemical equation indicating the transfer of electrons between two different oxidation states of the same element of chemical species. 2. One segment of a ping-pong (or double-displacement) enzyme mechanism.

REDOX HALF-REACTIONS. Electron transfer reactions involve oxidation (or loss of electrons) of one component and reduction (or gain of electrons) by a second component. Therefore, a complete redox reaction can be treated as the sum of two half-reactions such that the stoichiometry and electric charge is balanced across a chemical equilibrium. For each such half-reaction, there is an associated standard potential E° . The hydrogen ion-hydrogen gas couple is:



where by convention the charge on the electron is understood but not written. Because one cannot establish the absolute change in the electromotive force (emf) for any half-reaction, the E° value for the above reaction is arbitrarily set at 0.0000 volts. The standard potential for any reaction can then be defined as the emf change [(+) or (–) as well as magnitude] of an electrode consisting of the half-reaction measure against the hydrogen ion-hydrogen gas half-reaction under standard conditions:

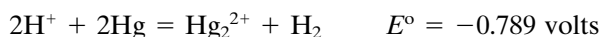
Reduction:



Oxidation:



Overall Reaction:



Likewise, any two half-reactions may be summed to establish the standard potential for an overall reaction; in so doing, care must be exercised to reverse the sign of

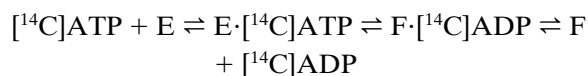
the emf associated with a half-reaction written in the opposite direction.

PING PONG HALF-REACTIONS. Many enzymes operate by double-displacement mechanisms involving covalent enzyme-substrate intermediates as shown in the following scheme:



E and F represent free enzyme and the modified enzyme, respectively, and the free energy change overall is the sum of the changes in free energy for the individual half-reactions. Examples of ping-pong enzymes are nucleoside-5'-diphosphate kinase (NDPK) and the amino acid-keto acid aminotransferases (such as glutamate:oxaloacetate transaminase), and F's in these respective cases are phosphoryl-enzyme and enzyme-bound pyridoxamine compounds. Early investigators were persuaded that the formation of a covalent intermediate would unnecessarily add more bond making/breaking steps, thereby lowering an enzyme's catalytic effectiveness. We now recognize that enzymes frequently experience no difficulty in bond making/breaking; instead, product release often is rate-determining. The formation of covalent intermediates actually facilitates catalysis in several ways: (a) by conserving the group-transfer potential of a reactive grouping of atoms (example: a phosphoryl group) to promote subsequent transfer with the second substrate acting as an acceptor molecule; (b) by using a single binding site that recognizes A or Q (in the case of E) and B or P (in the case of F) in the scheme shown above; (c) by repetitive use of the same catalytic components for acid/base and electrostatic interactions in each half-reaction. The reader should note that each such half-reaction may be accompanied by changes in stereochemistry of the transferred group, if one can detect such inversions. For example, by using [γ -(S) or (R)- ^{18}O , ^{17}O , ^{16}O]ATP one may determine the stereochemistry of phosphoryl transfer reactions.

Because each half-reaction can behave like an independent chemical process, ping pong enzymes catalyze exchange reactions. For example, yeast NDPK catalyzes an $\text{ADP} \leftrightarrow \text{ATP}$ exchange reaction:



Therefore, by mixing enzyme, unlabeled ADP, and $[^{14}\text{C}]\text{ATP}$ together, one can observe the synthesis of labeled ADP; this reaction can also be run in reverse by starting with enzyme, unlabeled ATP, and $[^{14}\text{C}]\text{ADP}$. (This is called a partial exchange reaction, because the total amount of ADP and ATP remains constant; only the isotopic forms change with time until they reach isotopic equilibrium.) **See** *Isotope Exchange; Substrate Synergism; Ping-Pong Mechanism; Covalent Intermediates; Phosphoryl Transfer Stereochemistry*

HALF-OF-THE-SITES REACTIVITY

A type of enzyme behavior¹⁻³ observed for some oligomeric proteins exhibiting cooperative behavior. For example, a negative, allosteric effector or covalent modifier (or even a substrate for an enzyme exhibiting negative homotropic cooperativity) may bind to one subunit, thereby inducing a conformational change that prevents the effector or covalent modifier from binding to the other subunit; hence the term, half-site reactivity. Lazdunski has also noted that half-site reactivity need not always exhibit cooperative curves in initial-rate kinetics^{2,3}.

¹M. Lazdunski, C. Petitclerc, D. Chappelet & C. Lazdunski (1971) *Eur. J. Biochem.* **20**, 124.

²M. Lazdunski (1972) *Curr. Top. Cell. Regul.* **6**, 267.

³K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

HAMMERHEAD RIBOZYME

The smallest known catalytic RNA motif (containing around 30 nucleotides) first detected in certain small satellite and viroid RNAs. The descriptor “hammerhead” refers to the motif’s shape when represented as a two-dimensional structure. Three Watson-Crick base-paired helices (designated I, II, and III) are linked by conserved sequences within the cleavage site located between helix I and III. When these two helices close with a hairpin loop, the resulting structure is the intramolecular (or *in cis*)-cleaving ribozyme. Closing only one of the helices leads to the intermolecular (or *in trans*)-cleaving form. Michaelis-Menten parameters were first established with the *in cis*-cleaving form¹, and pre-steady state kinetic studies have permitted the determination of individual rate constants for the Uni Bi random mechanism

of the *in trans*-cleaving hammerhead ribozyme^{2,3}. **See** *Catalytic RNA*

¹J. Haseloff & W. L. Gerlach (1988) *Nature* **334**, 585.

²M. J. Fedor & O. C. Uhlenbeck (1992) *Biochemistry* **31**, 12042.

³K. J. Hertel, D. Herschlag & O. C. Uhlenbeck (1994) *Biochemistry* **33**, 3374.

HAMMETT EQUATION

An equation used to assess the effect of structure on reactivity: $\log(k_X/k_H) = \sigma_X \rho$ where k_H is the corresponding rate constant when the position in the reactant under study contains a hydrogen as a substituent at that position, k_X is the rate constant when the substituent is X, σ_X is a constant characteristic of the group X, and ρ (the rho-value) is a constant for a given reaction under a given set of conditions. The Hammett equation is a linear free-energy relationship that has been applied to many reactions and functional groups. **See** ρ -Value; Linear Free-Energy Relationship; σ -Constant; Taft Equation; Extended Hammett Equation; Yukama-Tsuno Equation

L. P. Hammett (1970) *Physical Organic Chemistry*, 2nd ed., McGraw-Hill, New York.

C. D. Johnson (1973) *The Hammett Equation*, Cambridge Univ. Press, Cambridge.

J. Shorter (1982) *Correlation Analysis of Organic Reactivity*, Wiley, New York.

HAMMOND PRINCIPLE/POSTULATE

The postulate stating that if a chemical mechanism involves two consecutive steps (or states) of nearly the same energy, then interconversion between such states will require only a minor reorganization of the molecular structure. Hence, for a highly exergonic reaction, the transition state more closely resembles the reactant(s) or the immediately preceding intermediate, rather than the product(s) or immediately succeeding intermediate. This same idea is expressed in Leffler’s assumption, which is more commonly found in textbooks. In fact, the principle is sometimes referred to by the acronym “Bemahapothle” in recognition of the principle’s major proponents (*i.e.*, Bell, Marcus, Hammond, Polanyi, Thornton, and Leffler).

Note that any factor that stabilizes a reaction intermediate will also stabilize the transition state leading to that intermediate. **See also** *Leffler’s Assumption; Reacting Bond Rules*

J. E. Leffler (1953) *Science* **117**, 340.

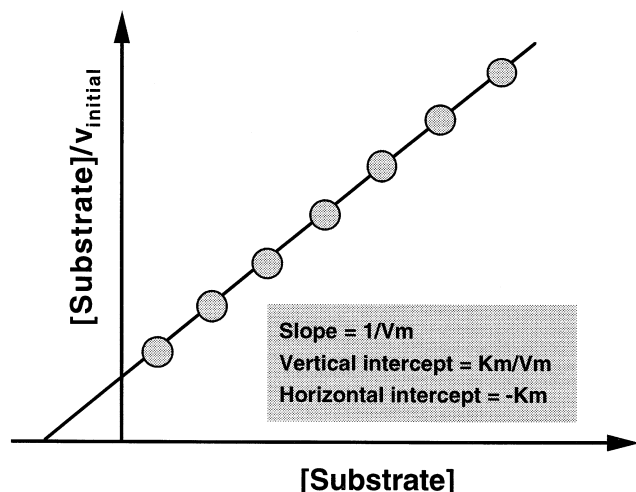
G. S. Hammond (1955) *J. Amer. Chem. Soc.* **77**, 334.

D. Fărcasiu (1975) *J. Chem. Educ.* **52**, 76.

A. Williams (1984) *Acc. Chem. Res.* **17**, 425.

HANES PLOT

A linear graphical method for analyzing the initial rate kinetics of enzyme-catalyzed reactions. In the Hanes plot, $[A]/v$ is plotted as a function of $[A]$, where v is the initial rate and $[A]$ is the substrate concentration¹⁻³.



Also referred to as the Hanes-Hultin plot and the Hanes-Woolf (or, Woolf-Hanes) plot, the method is based on a transformation of the Michaelis-Menten equation (*i.e.*, the expression for the Uni Uni mechanism): $[A]/v = (K_a/V_{\max}) + ([A]/V_{\max})$ where $V_{\max}$ is the maximum forward velocity and K_a is the Michaelis constant for A. In the Hanes plot, the slope of the line is numerically equal to $V_{\max}$, the vertical intercept is equivalent to $K_a/V_{\max}$, and the horizontal intercept to $-K_a$. **See Double-Reciprocal Plot**

¹B. Woolf (1932) quoted in *Allgemeine Chemie der Enzyme* (J. B. S. Haldane & K. G. Stern), p. 119, Steinkopff, Dresden-Leipzig.

²C. S. Hanes (1932) *Biochem. J.* **26**, 1406.

³E. Hultin (1967) *Acta Chem. Scand.* **21**, 1575.

HANSCH CONSTANT

A measure of the capability of a solute for hydrophobic interactions, based on the partition coefficient P for the distribution of the solute between 1-octanol and water.

C. Hansch & A. J. Leo (1979) *Substituent Constants for Correlation Analysis in Chemistry and Biology*, Wiley, New York.

HANSCH EQUATION

A quantitative relationship describing how the hydrophobicity of a parent compound is the result of additive contributions of substituents:

$$\pi = \log(P/P_o)$$

where P is the ratio of the solubility of a substituted molecule in n -octanol relative to its solubility in water, and P_o is the corresponding ratio for the unsubstituted parent compound. This relationship probably reflects the differences in the substituent's surface area, because the amount of water released in making a hydrophobic interaction is related to the water-accessible surface area.

C. Hansch & A. J. Leo (1979) *Substituent Constants for Correlation Analysis in Chemistry and Biology*, Wiley, New York.

HANSCH'S CORRELATION ANALYSIS

A procedure that utilizes quantitative structure-activity relationships to assist in the characterization and design of enzyme alternative substrates, inhibitors, and effectors. In this procedure, the inhibition (or activity) results of a series of structurally related compounds are analyzed to determine the coefficients in an equation of the form:

$$\log (1/C) = \sum a_n \pi_n + \sum b_n F_n + \sum c_n R_n + \sum d_n MR_n + \sum e_n E_{sn} + \dots$$

where C is the molar concentration of the inhibitor (or effector, *etc.*) producing 50% inhibition of enzyme (or, some other rate-related constant), π is a hydrophobic constant (often obtained from partition coefficients of organic compounds serving as references for hydrophobic interactions of enzymes) in which the subscript n represents different values for π at different positions (*e.g.*, *ortho* and *meta* positions on an aromatic ring), F and R are the Swain and Lupton factors for electrostatic interactions, MR is the molar refractivity which models nonspecific binding in nonhydrophobic space, and E_s is Taft's steric parameter. Regression analysis of a set of closely related molecules permits one to determine the magnitude of the coefficients. The final expression can be quite useful in the design of other effectors as well as in the characterization of the relative contributions to binding of hydrophobic interactions, polarizability and dispersion forces, electronic effects, steric interactions, *etc.*

C. Hansch (1975) *Adv. Pharmacol. Chemother.* **13**, 45.

C. Silipo & C. Hansch (1976) *J. Med. Chem.* **19**, 62.

M. Yoshimoto & C. Hansch (1976) *J. Med. Chem.* **19**, 71.

HARMONIC BEHAVIOR

A regularly periodic motion characteristic of a pendulum, spring, or many chemical bonds. **See Hooke's Law**

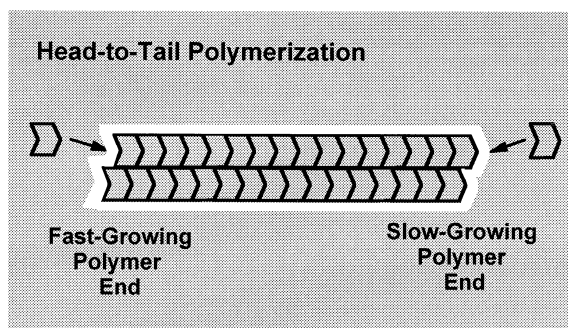
HARTREE-FOCK MOLECULAR-ORBITAL CALCULATIONS

A method of molecular-orbital calculations, also referred to as the self-consistent field method (SCF), to characterize the bonding in unsaturated and aromatic molecules while neglecting electron-electron repulsion. The method has been extended to all valence electrons.

M. J. S. Dewar (1969) *The Molecular Orbital Theory of Organic Chemistry*, McGraw Hill, New York.

H. H. Jaffe (1969) *Acc. Chem. Res.* **2**, 136.

HEAD-TO-TAIL POLYMERIZATION



Addition reactions that are characteristic of head-to-tail addition of subunits to an elongating polymer. Note that the rates of subunit addition to each of the two ends usually differ as a consequence of differences in the geometry of the associated transition states.

A term used in describing the geometric disposition of subunits during indefinite polymerization of protein subunits into supramolecular assemblies. This is illustrated by the ATP hydrolysis-dependent polymerization of actin onto a trimeric nucleus (marked here with gray shading). Because most protein subunits or protomers exhibit some degree of polarity, one would anticipate that such molecules can form linear oligomeric and polymeric structures reflecting their ability to bind to each other in a so-called “head-to-tail” arrangement. Tubulin and tobacco mosaic virus coat protein also undergo head-to-tail polymerization. **See** *Actin Assembly Kinetics*; *Microtubule Assembly Kinetics*

HEAT CAPACITY

A parameter (measured at constant volume; C_v) equal to dq_v/dT where q_v is the heat absorbed at constant volume and T is the temperature. Heat capacity is also equal to $(\partial U/\partial T)_v$ where U is the internal energy. The heat capacity measured at constant pressure (C_p) of a system is equal to dq_p/dT where q_p is the heat absorbed

at constant pressure. This value of C_p is also equal to $(\partial H/\partial T)_p$ where H is the enthalpy at constant pressure. In general, the magnitude of C_p is typically larger than that of C_v although the numbers are very close for liquids and solids. An exception is the case of water between 1°C and 4°C.

Thus, $q_v = \int_{T_1}^{T_2} C_v dT$ at constant volume. If C_v is independent of temperature, then $q_v = C_v(T_2 - T_1)$. This is an expression for ΔU . Similarly, $q_p = \int_{T_1}^{T_2} C_p dT$. If C_p is independent of temperature, $q_p = C_p(T_2 - T_1)$, an expression for ΔH . It should also be recalled that heat capacities are not state functions.

$C_p - C_v$ equals $[P + (\partial U/\partial V)_T](\partial V/\partial T)_p$. The $(\partial U/\partial V)_T$ term is often referred to as the internal pressure and is large for liquids and solids (**See** *Internal Pressure*). Since ideal gases do not have internal pressure, $C_p - C_v = nR$ for ideal gases. The ratio of the heat capacities, C_p/C_v , is commonly symbolized by γ .

The SI unit for heat capacity is $J \cdot K^{-1}$. Molar heat capacities (C_m) are expressed as the ratio of heat supplied per unit amount of substance resulting in a change in temperature and have SI units of $J \cdot K^{-1} \cdot mol^{-1}$ (at either constant volume or pressure). Specific heat capacities (C_v or C_p) are expressed as the ratio of heat supplied per unit mass resulting in a change in temperature (at constant volume or pressure, respectively) and have SI units of $J \cdot K^{-1} \cdot kg^{-1}$. Debye's theory of specific heat capacities applies quantum theory in the evaluation of certain heat capacities.

It should also be recalled that $dS = (C_p/T)dT$ at constant pressure and $dS = (C_v/T)dT$ at constant volume where S is the entropy.

HEAVY ATOM ISOTOPE EFFECT

An isotope effect (on either kinetic or equilibrium process) resulting from substitution by isotopes other than those of hydrogen. **See** *Kinetic Isotope Effects*

HELIX-COIL TRANSITIONS OF NUCLEIC ACIDS

As discussed by Pörschke¹, one can distinguish several basic features of helix-coil transitions in polynucleotides. Formation of a helix from separated complementary

strands most typically obeys second-order rate behavior^{2,3}, and forming a helix nucleus is considered to be the rate-determining step. With much longer chains, the so-called “zippering-up” phase that follows nucleation can be rate contributing. In such cases, one does not observe simple second-order kinetics. Whenever a mismatch occurs within a sequence of hydrogen-bonding base-pair partners, the annealing of the region to attain correct matching probably requires loss of helical structure, followed by a readjustment of interactions between partners. Another feature of helix-coil transitions is the dissociation behavior of long double helices, wherein the complexity of the available pathways for these transitions is reflected in the number of overlapping relaxation events. Spatz and Crothers⁴ offered the following scheme for processes taking place during DNA denaturation: With the occurrence of a perturbation, there is attendant rupture of structure in AT-rich regions. Further perturbation causes the loss of base pairing in the regions adjacent to these loops; the helical winding of the double helix leads to coiling of the polynucleotide within adjacent regions of coiling. More extensive unwinding follows, resulting in the formation of larger loop structures.

See DNA Unwinding Kinetics

¹D. Pörschke (1977) in *Chemical Relaxation in Molecular Biology* (I. Pecht & R. Rigler, eds.), pp. 190-218, Springer-Verlag, Berlin.

²P. D. Ross & J. M. Sturtevant (1960) *Proc. Natl. Acad. Sci. U.S.A.* **46**, 1360.

³J. G. Wetmur & N. Davidson (1968) *J. Mol. Biol.* **31**, 349.

⁴H. C. Spatz & D. M. Crothers (1969) *J. Mol. Biol.* **42**, 191.

HEME OXYGENASE

This enzyme [EC 1.14.99.3] catalyzes the reaction of heme with dioxygen and three AH₂ to produce biliverdin, carbon monoxide, Fe²⁺, three A, and three water molecules. This enzyme uses NAD(P)H as a cofactor and requires the presence of NADPH:ferrihemoprotein reductase [EC 1.6.2.4].

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.

HEMOGLOBIN

Within the $\alpha_2\beta_2$ hemoglobin tetramer, the α and β subunits (made up of 141 and 146 amino acid residues, respectively) can be thought of as two $\alpha\beta$ -dimeric units that are symmetrically positioned about a central cavity¹.

Hemoglobin binds oxygen cooperatively, but a simple all-or-nothing Hill-type binding model does not fit the observed saturation behavior; indeed, the observed Hill coefficient of 2.8 indicates that the system behaves as though the protein had 2.8 subunits that were infinitely cooperative. Because we know that four O₂ molecules are bound at saturation, a better model was proposed by Adair who could fit the entire binding equation by assuming that the intrinsic dissociation constants changed as a function of oxygen binding site occupancy.

A well-established fact is that deoxygenated and oxygenated forms differ with respect to each other by a slight 15° rotation and 1 Å translation between two $\alpha\beta$ dimers. The deoxygenated form is stabilized by protons as well as 2,3-bisphosphoglycerate, and these effector molecules respectively ensure higher than normal oxygen release during hypoxia and adaptation to high altitude. Perutz² first proposed that “the oxygenation of Hb is accompanied by structural changes in the subunits, triggered by shifts of the iron atoms relative to the porphyrin and, in the β -subunits, also by the steric effects of oxygen itself”. Under physiological conditions, the oxygen saturation curve and the kinetics of hemoglobin’s ligand binding reactions appear to fit the two-state Monod-Wyman-Changeux cooperativity model³. The relative abundance of the two states depend on the magnitude of the allosteric constant $\mathcal{L}$ in the Monod-Wyman-Changeux model (i.e., $\mathcal{L} = [T_0]/[R_0]$).

¹M. F. Perutz, A. J. Wilkinson, M. Paoli & G. G. Dodson (1998) *Annu. Rev. Biophys. Biomol. Struct.* **27**, 1.

²M. F. Perutz (1972) *Nature* **237**, 495.

³E. R. Henry, C. M. Jones, J. Hofrichter & W. R. Eaton. (1997) *Biochemistry* **36**, 6511.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Equilibrium Ligand Binding to Hemoglobin: Spectrophotometric procedures, **76**, 417; measurement of oxygen binding by means of a thin-layer optical cell, **76**, 427; measurement of accurate oxygen equilibrium curves by an automatic oxygenation apparatus, **76**, 438; thin-layer methods for determination of O₂ binding curves of hemoglobin solutions and blood, **76**, 449; analysis of ligand binding equilibria, **76**, 470; measurement of CO₂ equilibria: the chemical-chromatographic methods, **76**, 487; measurement of CO₂ binding: The ¹³C NMR method, **76**, 496; continuous determination of the oxygen dissociation curve of whole blood, **76**, 511; measurement of the Bohr effect: dependence on pH of oxygen affinity of hemoglobin, **76**, 523; measurement of binding of nonheme ligands to hemoglobins, **76**, 533; use of NMR spectroscopy in studies in ion binding to hemoglobin, **76**, 552; rapid-rate equilibrium analysis of the interactions between organic phosphates and hemoglobins, **76**, 559; measurement of the oxidation-reduction equilibria of hemoglobin and myoglobin, **76**, 577; photochemistry of hemoproteins, **76**, 582; measurement and analysis of ligand-linked subunit dissociation equilibria in human

hemoglobins, **76**, 596; ligand binding and conformational changes measured by time-resolved absorption spectroscopy, **232**, 387; hemoglobin-liganded intermediates, **232**, 445; hemoglobin-oxygen equilibrium binding: rapid-scanning spectrophotometry and singular value decomposition, **232**, 460; oxygen equilibrium curve of concentrated hemoglobin, **232**, 486; Adair fitting to oxygen equilibrium curves of hemoglobin, **232**, 559; weighted nonlinear regression analysis of highly cooperative oxygen equilibrium curves, **232**, 576; dimer-tetramer equilibrium in Adair fitting, **232**, 597; effects of wavelength on fitting Adair constants for binding of oxygen to human hemoglobin, **232**, 606; Adair equation: rederiving oxygenation parameters, **232**, 632; linkage thermodynamics, **232**, 655.

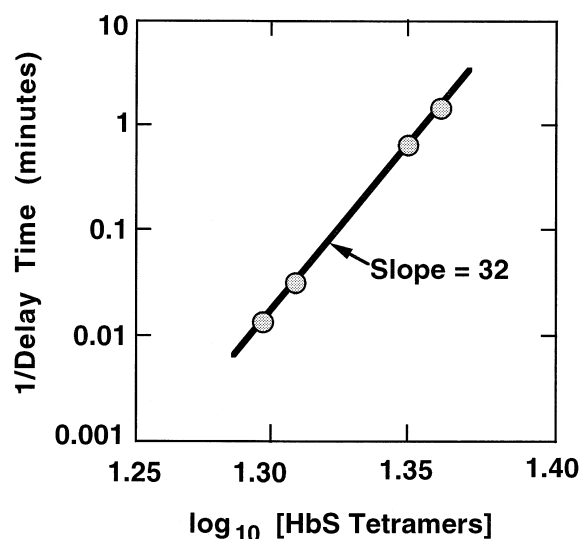
Kinetics of Ligand Reaction with Hemoglobin: Stopped-flow, rapid mixing measurements of ligand binding to hemoglobin and red cells, **76**, 631; numerical analysis of kinetic ligand binding data, **76**, 652; flash photolysis of hemoglobin, **76**, 667; temperature jump of hemoglobin, **76**, 681; assignment of rate constants for O₂ and CO binding to α and β subunits within R- and T-state human hemoglobin, **232**, 363; femtosecond measurements of geminate recombination in heme proteins, **232**, 416; double mixing methods for kinetic studies of ligand binding in partially liganded intermediates of hemoglobin, **232**, 430; simulation of hemoglobin kinetics using finite element numerical methods, **232**, 517.

HEMOGLOBIN-S POLYMERIZATION

Deoxygenated sickle cell hemoglobin (deoxyHbS), the β -Glu-6-Val point mutant form¹ of adult hemoglobin, appears to obey the following empirical rate law for nucleation of polymerization:

$$\text{Rate} = k[\text{deoxyHbS}]^n$$

The exponent n has been taken to be the number of protomers (in this case tetramers) in the smallest stable nucleus on which further self-assembly of fibers takes place². If nucleation is the rate-determining step in HbS polymerization, then one can estimate the rate of nucleation by determining the delay time t_d observed in lag-phase kinetic measurements of polymerization. A value of $n = 32$ was obtained from the slope of the plot shown below. One may question whether the apparent 32nd power dependence of polymerization rate on the concentration of HbS is physically realistic, especially when one recognizes that each tetramer within an assembled HbS fiber must make numerous contacts even in forming the so-called double strands. Moreover, the validity of the above kinetic analysis is valid only if implicit assumptions justifying the use of the above empirical rate law are satisfied. Among these assumptions are: that concentration and chemical activity are equivalent; that solute-solvent interactions are ideal; and that nucleation is the rate-determining step.



Plot of the reciprocal of the delay time for HbS fiber formation versus the logarithm of HbS concentration. From Hofrichter et al.² with permission of the authors and publisher.

This extraordinary result¹ suggests that polymerization is extremely sensitive to changes in HbS concentration, such that successful therapies for preventing sickling might be achievable by lowering the effective concentration of HbS. Modest changes in concentration should minimize the propensity for gelation long enough for the deoxygenated hemoglobin S molecules to rebound O₂ at the higher partial pressures of oxygen found in the lung. One should note that such high-order nucleation phenomena suggest that ligands such as carbon monoxide should paradoxically protect individuals from sickling, because even low concentrations of HbS(CO)₄ should interfere with the nucleation event. The late Eraldo Antonini and associates³ used microspectrophotometry with single red blood cells by taking advantage of the photosensitivity of the carbon monoxide complex with Hb; this method allows ligand release under strong illumination, followed by reforming of the complex when the light is switched off. Briefly, they found that the time necessary for complete sickling, as evidenced by full deformation of the cell, ranged in all examined cells from 3 to 5 seconds. Interestingly, the extent of photodissociation was only about 60%, suggesting that carboxy HbS tetramers were not effectively inhibiting the nucleation of polymers, if nucleation remains rate-limiting *in vivo*. By lowering the intensity of the photo-dissociating light beam after sickling was observed, they discovered that the deformation could be maintained by a light intensity several times lower than required to induce

sickling. Furthermore, the sickling process behaved as though it was fully reversible, an inference supported by the finding that recombination of hemoglobin and carbon monoxide after switching off the illuminating source was promptly attended by a return of the erythrocytes to their typical biconcave shape. The lack of any delay in these red cell sickling transitions leads one to question the physiologic relevance of the findings reported by Hofrichter and co-workers².

¹R. E. Dickerson & I. Geis (1983) *Hemoglobin: Structure, Function, Evolution, and Pathology*, pp. 125-145, Benjamin-Cummings, Menlo Park.

²J. Hofrichter, P. Ross & W. Eaton (1974) *Proc. Natl. Acad. Sci. U.S.A.* **71**, 4864.

³E. Antonini, M. Brunori, B. Giardina, P. A. Benedetti, G. Bianchini & S. Grassi (1978) in *Biophysical Discussions: Fast Biochemical Reactions in Solutions, Membranes and Cells* (V. A. Parsegian, ed.), pp. 187-189, Rockefeller University Press, New York.

HENDERSON-HASSELBALCH EQUATION

An equation^{1,2} relating the pH value of a solution and the concentrations of the basic and acidic species of an acid/base equilibria:

$$\text{pH} = \text{p}K_a' + \log \left(\frac{[\text{basic species}]}{[\text{acidic species}]} \right)$$

Note that K_a' is not the true dissociation constant for the weak acid (or weak base). For the reaction $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$, $K_a = (\text{H}^+)(\text{A}^-)/(\text{HA})$ where (H^+) , (A^-) , and (HA) , are the corresponding activities. However, K_a' is equal to $K_a f_{\text{HA}}/f_{\text{A}^-}$ where f_{HA} and f_{A^-} are the activity coefficients for HA and A^- , respectively. Under most practical conditions, particularly under conditions of low constant ionic strength, $K_a \cong K_a'$.

One should always be careful when using the Henderson-Hasselbalch equation: $\text{pH} = \text{p}K_a' + \log ([\text{A}^-]/[\text{HA}])$. The value of $[\text{HA}]$ is not always moles of the weak acid per unit volume, c_{HA} , that had been added to the moles of the conjugate base per unit volume, c_{A^-} . There are five species for this acid/base equilibria: HA, A^- , H^+ , OH^- , and Na^+ (or, whatever cation portion of the salt of the conjugate base is present). These species are all interrelated by five expressions:

$$(\text{H}^+)(\text{A}^-)/(\text{HA}) = K_a$$

$$(\text{H}^+)(\text{OH}^-) = K_w$$

$$[\text{HA}] + [\text{A}^-] = c_{\text{HA}} + c_{\text{A}^-}$$

$$[\text{Na}^+] = c_{\text{A}^-}$$

$$[\text{H}^+] + [\text{Na}^+] = [\text{OH}^-] + [\text{A}^-]$$

where parentheses represent activities and brackets represent concentrations. Thus, $[\text{A}^-] = c_{\text{A}^-} + ((\text{H}^+)/f_{\text{H}^+}) - K_w f_{\text{OH}^-}/(\text{H}^+)$ and $[\text{HA}] = c_{\text{HA}} - ((\text{H}^+)/f_{\text{H}^+}) + K_w f_{\text{OH}^-}/(\text{H}^+)$. Hence, the equation relating pH to $\text{p}K_a'$ now is $\text{pH} = \text{p}K_a' + \log [(c_{\text{A}^-} + (\text{H}^+)/f_{\text{H}^+} - K_w f_{\text{OH}^-}/(\text{H}^+))/(c_{\text{HA}} - (\text{H}^+)/f_{\text{H}^+} + K_w f_{\text{OH}^-}/(\text{H}^+))]$. This equation reduces to the standard Henderson-Hasselbalch equation if $[\text{H}^+]$ and $[\text{OH}^-]$ are much smaller than c_{HA} and/or c_{A^-} . If the $[\text{H}^+]$ or $[\text{OH}^-]$ values are less than 0.01 of the concentrations of the acidic and basic species, the error in neglecting the $[\text{H}^+]$ and $[\text{OH}^-]$ values is less than 0.009 pH unit. Perrin and Dempsey³ provide a simple table on when best to use the simple Henderson-Hasselbalch equation:

c_{HA} or c_{A^-}	pH Range
5 mM	4.3–9.7
10 mM	4.0–10
50 mM	3.3–10.7
100 mM	3.0–11

Outside this range, the full expression should be used.

¹L. J. Henderson (1908) *Amer. J. Physiol.* **21**, 169.

²K. Hasselbalch (1916) *Biochem. Zeit.* **78**, 112.

³D. D. Perrin & B. Dempsey (1974) *Buffers for pH and Metal Ion Control*, Chapman and Hall, London.

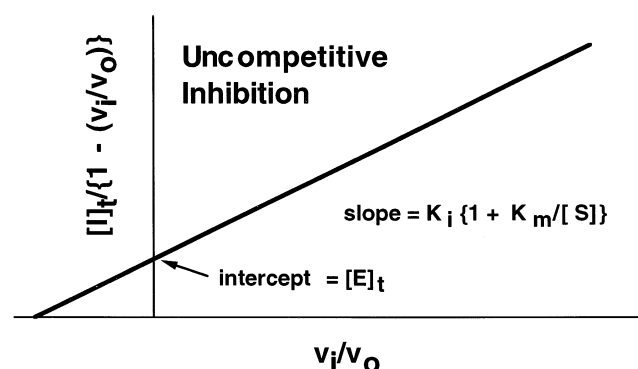
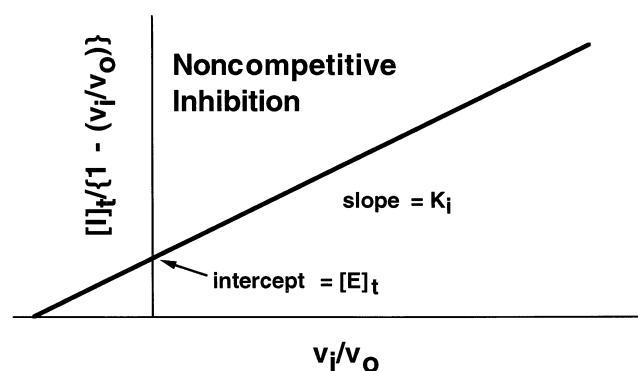
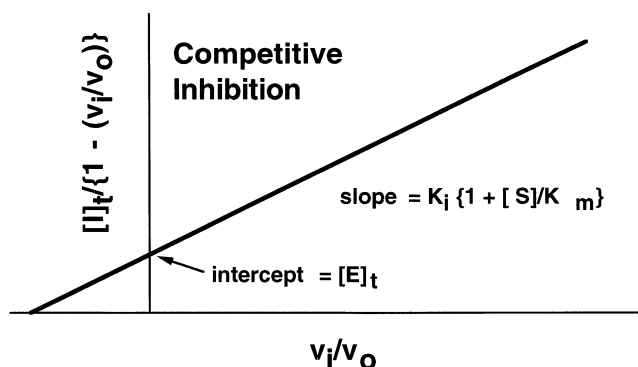
HENDERSON PLOT

A graphical protocol for the analysis of tight-binding inhibitors of enzyme-catalyzed reactions (see figure)^{1,2}.

This linearization of the tight-binding scheme allows the investigator the opportunity to calculate values for $[\text{E}_{\text{total}}]$ and K_i , the dissociation constant for the inhibitor. In the Henderson plot, $[\text{I}_{\text{total}}]/(1 - v/v_o)$ is plotted as a function of v_o/v where v_o is the steady-state velocity of the reaction in the absence of the inhibitor. The slope of the line is the apparent dissociation constant for the inhibitor. Secondary plots (from repeating the inhibition experiment at different substrate concentrations) will yield the K_i value. The vertical intercept is equal to $[\text{E}_{\text{total}}]$. Hence, repeating the experiment at a different concentration of enzyme will produce a parallel line.

¹P. J. F. Henderson (1972) *Biochem. J.* **127**, 321.

²J. W. Williams & J. F. Morrison (1979) *Meth. Enzymol.* **63**, 437.



HENRY

The SI unit (symbolized by H) for both self-inductance and mutual inductance. One henry represents the inductance produced when an induced electromotive force of one volt is generated as the inducing current is altered at a rate of one ampere per second. (Hence, one henry is equivalent to one volt-second per ampere or one weber per ampere). **See** *Magnetic Susceptibility; Permeability*

HERTZ

The SI unit for frequency (symbolized by Hz) equal to reciprocal seconds. The term hertz should only be used

with respect to cycles per second and not for radial (circular) frequency or angular velocity.

HESS' LAW

The amount of heat generated by a reaction is the same whether the reaction takes place in one step or in several steps. Hence, ΔH values (and, thus, ΔG values) are additive. This law, also known as the Law of Constant Heat Summation, was the earliest example of the Additivity Principle, which also states that ΔG values are additive. In biochemical processes, especially when dealing with multiple steps as in protein folding, application of the Additivity Principle can give spurious results if the accuracy and precision of the thermodynamic parameters is insufficient.

HETEROGENEOUS CATALYSIS

Catalysis of a reaction wherein one or more reactants is (are) in one phase and the catalytic process takes place at an interface between the phases. This phenomenon is also termed contact catalysis and surface catalysis. **See** *Homogeneous Catalysis*

HETEROGENEOUS REACTION

A reaction occurring at an interface of two phases. Some heterogeneous processes also display homogeneous aspects; for example, a particular reaction may occur in only one of the system's phases (*e.g.*, the rapid dissolution of a gas into a liquid followed by a particular reaction). **See** *Fractal Reaction Kinetics*

HETEROTROPIC EFFECT

Cooperative ligand binding in which a bound ligand molecule influences how a different type of ligand binds to its respective sites on a multisubunit macromolecule¹⁻³. The influence of the ligand can result in either weakened binding (hence, negative heterotropic cooperativity) or in enhanced binding (hence, positive heterotropic cooperativity). An example of negative heterotropic cooperativity is the action of 2,3-bisphosphoglycerate or hydrogen ion to reduce the extent of oxygen binding to hemoglobin A.

In the context of the Monod-Wyman-Changeux concerted-transition model for allosteric effects, one usually considers the effects of specific site occupancy on the behavior of other binding sites. Thus, a more correct

operational definition is that heterotropic effects represent the interactions on cooperative ligand binding exerted by molecules of unlike action. Applying this definition, we see that 2,3-bisphosphoglycerate or hydrogen ion act heterotropically with respect to oxygenation; yet, in terms of their mutual interactions with each other, 2,3-bisphosphoglycerate or hydrogen ion can be regarded as exerting homotropic effects. It is also true that while oxygen and carbon monoxide compete with respect to each other for the same sites (*i.e.*, the four heme groups on hemoglobin), both act on hemoglobin in the same manner—namely by increasing the fraction of macromolecules in the R-state. Thus, O₂ and CO can be said to act homotropically with respect to each other. **See** *Monod-Wyman-Changeux Model; Hemoglobin; Allosterism*

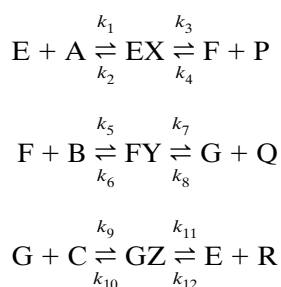
¹J. Monod, J. Wyman & J.-P. Changeaux (1965) *J. Mol. Biol.* **12**, 88.

²K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

³K. E. Neet (1995) *Meth. Enzymol.* **249**, 519.

HEXA UNI PING PONG ENZYME MECHANISM

A ping pong enzyme-catalyzed reaction mechanism in which three substrates react to form three products¹⁻³:



where F and G are modified forms of the enzyme E, the substrates are represented by A, B, and C, and the products are P, Q, and R.

The steady-state determinants for each enzyme form (in the absence of abortive complexes) are:

$$\begin{aligned} [E] &= k_5k_7k_9k_{11}(k_2 + k_3)[B][C] \\ &\quad + k_2k_4k_9k_{11}(k_6 + k_7)[C][P] \\ &\quad + k_2k_4k_6k_8(k_{10} + k_{11})[P][Q] \\ [EX] &= k_1k_5k_7k_9k_{11}[A][B][C] \\ &\quad + k_1k_4k_9k_{11}(k_6 + k_7)[A][C][P] \\ &\quad + k_1k_4k_6k_8(k_{10} + k_{11})[A][P][Q] \\ &\quad + k_4k_6kk_{10}k_{12}[P][Q][R] \end{aligned}$$

$$\begin{aligned} [F] &= k_1k_3k_9k_{11}(k_6 + k_7)[A][C] \\ &\quad + k_1k_3k_6k_8(k_{10} + k_{11})[A][Q] \\ &\quad + k_6k_8k_{10}k_{12}(k_2 + k_3)[Q][R] \end{aligned}$$

$$\begin{aligned} [FY] &= k_1k_3k_5k_9k_{11}[A][B][C] \\ &\quad + k_1k_3k_5k_8(k_{10} + k_{11})[A][B][Q] \\ &\quad + k_5k_8k_{10}k_{12}(k_2 + k_3)[B][Q][R] \\ &\quad + k_2k_4k_8k_{10}k_{12}[P][Q][R] \end{aligned}$$

$$\begin{aligned} [G] &= k_1k_3k_5k_7(k_{10} + k_{11})[A][B] \\ &\quad + k_5k_7k_{10}k_{12}(k_2 + k_3)[B][R] \\ &\quad + k_2k_4k_{10}k_{12}(k_6 + k_7)[P][R] \end{aligned}$$

$$\begin{aligned} [GZ] &= k_1k_3k_5k_7k_9[A][B][C] \\ &\quad + k_5k_7k_9k_{12}(k_2 + k_3)[B][C][R] \\ &\quad + k_2k_4k_9k_{12}(k_6 + k_7)[C][P][R] \\ &\quad + k_2k_4k_6k_8k_{12}[P][Q][R] \end{aligned}$$

Under initial rate conditions, where the product concentrations are effectively zero (*i.e.*, [P] ≈ 0, [Q] ≈ 0, and [R] ≈ 0), the steady-state expression for this reaction mechanism is

$$v = \frac{V_{\max}[A][B][C]}{\{K_c[A][B] + K_b[A][C] + K_a[B][C] + [A][B][C]\}}$$

where $V_{\max}$ is the maximum velocity in the forward direction (equal to $k_3k_7k_{11}[E_{\text{total}}]/(k_3k_7 + k_3k_{11} + k_7k_{11})$), K_a is the Michaelis constant for A (equal to $k_7k_{11}(k_2 + k_3)/\{k_1(k_3k_7 + k_3k_{11} + k_7k_{11})\}$), K_b is the Michaelis constant for B (equal to $k_3k_{11}(k_6 + k_7)/\{k_5(k_3k_7 + k_3k_{11} + k_7k_{11})\}$), and K_c is the Michaelis constant for C (equal to $k_3k_7(k_{10} + k_{11})/\{k_9(k_3k_7 + k_3k_{11} + k_7k_{11})\}$). This expression predicts that double-reciprocal plots (1/v vs. the reciprocal concentration of one of the substrates in which one of the other substrate concentration is varied and the third substrate is kept at a constant, nonsaturating level) will all exhibit parallel lines. Secondary replots of the data will provide values for the kinetic constants.

Since “half-reactions” are present in this reaction mechanism (actually, “third-reactions”), partial exchange experiments are also available for investigation. **See** *Multisubstrate Mechanisms; Isotope Exchange at Equilibrium*

¹Y. Milner, G. Michaels & H. G. Wood (1978) *J. Biol. Chem.* **253**, 878.

²N. H. Goss & H. G. Wood (1982) *Meth. Enzymol.* **87**, 51.

³C. S. Tsai, M. W. Burgett & L. J. Reed (1973) *J. Biol. Chem.* **248**, 8348.

HEXOKINASE

This phosphotransferase [EC 2.7.1.1] catalyzes phosphorylation of D-glucose as well as many other aldo- and keto-hexoses (with the notable exception of D-galactose)¹. MgATP²⁻ is the active phosphoryl donor, and the overall reaction is highly favorable (*i.e.*, $K_{eq} = [\text{ADP}][\text{D-Glc-6-P}]/[\text{D-Glc}][\text{ATP}] = 490$ at pH 6.5; $K_{eq} = 1500$ at pH 7). The kinetic reaction for the yeast enzyme involves random sequential binding of hexose and MgATP²⁻ to form a ternary complex. This was first demonstrated by the use of competitive inhibitors and isotope exchange at equilibrium. The rate of nucleotide exchange at equilibrium is about 1.5 times faster than the sugar $\leftrightarrow$ sugar-phosphate exchange, suggesting that the dissociation of sugar or sugar-phosphate (or both) must be rate-limiting. Proteolyzed forms of the yeast enzyme appear to favor glucose binding. The mammalian enzyme also operates by a random ternary complex mechanism. Contrary to earlier reports asserting that glucose 6-phosphate is an allosteric inhibitor of the mammalian erythrocyte and brain enzymes, the potency of this sugar phosphate's action appears to arise through its tight binding ($K_i = 12 \mu\text{M}$) at the γ -phosphoryl subsite of the ATP binding site. Based on the stereochemistry² of phosphoryl transfer, the hexokinase reaction is best described as a single in-line or S_N2 reaction involving facilitated attack of the 6-hydroxy group on MgATP²⁻. **See** ATP and GTP Depletion; Deinhibition; Induced Fit Hypothesis

¹D. L. Purich, F. B. Rudolph & H. J. Fromm (1973) *Adv. Enzymol.* **39**, 249.

²G. Lowe, P. M. Cullis, R. L. Jarvest, B. V. L. Potter & B. S. Sproat (1981) *Phil. Trans. R. Soc. London B* **293**, 75.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Activation, **63**, 258, 264, 292; assay, **63**, 32, 36, 37, 40; calorimetry, reactions catalyzed, **61**, 263, 268; competitive inhibition, **63**, 292, 476, 480; conformational change, **64**, 225, 226; cooperativity, **64**, 218, 221; hysteresis, **64**, 197, 215, 216, 218, 221, 222; inhibition, **63**, 5, 348; isomerization, **64**, 197; isotope exchange, **64**, 4, 7, 8, 25, 29-32, 37, 38; mechanism, **63**, 51; mnemonic mechanism, **64**, 221; nucleotide site, **64**, 180; product inhibition, **63**, 433, 434, 436; slow-binding inhibitor, **63**, 450; specificity, **63**, 378, 379; steady state assumption, **63**, 476; alternative substrate inhibition, **63**, 145, 494.

HEXOSAMINIDASES A AND B

This enzyme [EC 3.2.1.52] (also known as β -N-acetylhexosaminidase, β -hexosaminidase, hexosaminidase, and N-acetyl- β -glucosaminidase) catalyzes the hydrolysis of terminal nonreducing N-acetyl-D-hexosamine residues in

N-acetyl- β -D-hexosaminides. The enzyme can also utilize N-acetylglucosides and N-acetylgalactosides as substrates. The associated human genetic disorders are known as Sandhoff's disease and Tay-Sachs disease.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.

R. O. Brady (1983) *The Enzymes*, 3rd ed., **16**, 409.

HEXOSE-1-PHOSPHATE GUANYLYLTRANSFERASE

This enzyme [EC 2.7.7.29], also referred to as GDP-hexose pyrophosphorylase, catalyzes the reaction of GTP with α -D-hexose 1-phosphate to produce GDP-hexose and pyrophosphate (or, diphosphate).

R. G. Hansen, H. Verachtert, P. Rodriguez & S. T. Bass (1966) *Meth. Enzymol.* **8**, 269.

HIDDEN RETURN

A term used to designate any condition wherein a covalent molecule, R₁-R₂, is reformed without direct evidence of the prior formation of a tight ion pair (*i.e.*, R₁⁺R₂⁻).

HILDEBRAND PARAMETER

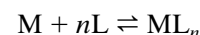
A measure of the cohesion of a solvent (*i.e.*, the energy required to create a cavity). This is an extension of earlier work by Thomson and others earlier in the twentieth century to treat solvent-solute interactions.

M. Chastrette, M. Rajzmann, M. Cannon & K. Purcell (1985) *J. Amer. Chem. Soc.* **107**, 1.

HILL COEFFICIENT

An all-or-none treatment of cooperative ligand binding, wherein a macromolecule with n sites for ligand exists in two states: (a) the uncomplexed form M; and (b) the fully ligated state ML_{*n*}.

To derive the Hill equation, we begin with the following equilibrium reaction:



Likewise, the following equilibrium constant can be written (in the form of a dissociation constant):

$$K = \frac{[\text{M}][\text{L}]^n}{[\text{ML}_n]} \quad \text{or} \quad [\text{M}] = K \frac{[\text{ML}_n]}{[\text{L}]^n}$$

Using the formalism of a saturation function, we get

$$Y_L = \frac{[ML_n]}{([M] + [ML_n])}$$

Y_L can vary over the interval from 0 to 1. Substituting for $[M]$ in terms of $[ML_n]$ and rearranging yields:

$$Y_L = \frac{1}{\left(\frac{K}{[L]^n}\right) + 1} \quad \text{or} \quad Y_L = \frac{[L]^n}{K + [L]^n}$$

The Hill plot is obtained by first rearranging this expression to obtain

$$\frac{Y_L}{1 - Y_L} = \frac{[L]^n}{K}$$

then, by taking the decadic logarithm of both sides, we get

$$\log\left(\frac{Y_L}{1 - Y_L}\right) = n \log[L] - \log K$$

As shown in Fig. 1, the linear region of a plot of $\log\{Y_L/(1 - Y_L)\}$ versus $\log[L]$ yields a slope of n . When one obtains experimental data from a Hill Plot, the symbol for “ n ” is typically designated as n_H .

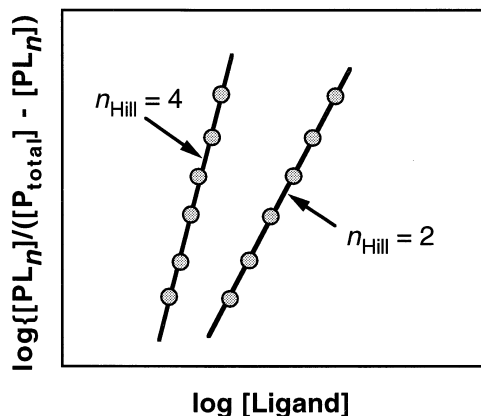


Fig. 1. Ligand binding behavior of a hypothetical protein exhibiting a Hill coefficient of 2 or 4.

Where n_H equals the number of sites determined by some other biophysical procedure, we say that the system shows “infinite cooperativity.” No such behavior has been rigorously demonstrated for an enzyme or receptor. In the case of hemoglobin oxygenation (Fig. 2) under physiologic conditions, the Hill coefficient has a value of about 2.8. Of course, from X-ray structural information, we know that hemoglobin has four sites. Thus, we

can say that hemoglobin behaves as though it has 2.8 infinitely cooperative sites, or we can state that the four subunits of human hemoglobin fail to show complete cooperativity.

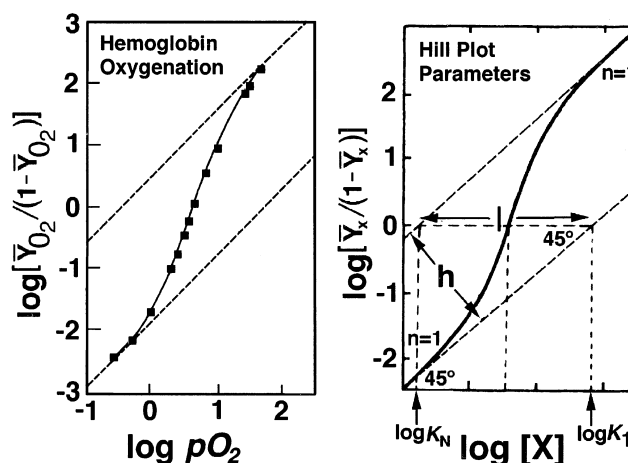
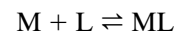


Fig. 2. Hill plot for oxygenation of human hemoglobin A as a function of the partial pressure (pO_2) of molecular oxygen. The diagram at the right shows that the Hill coefficient will reach a limiting value of one at both extremes of ligand concentration. For this reason this cooperativity index is best measured at ligand concentrations near half-maximal saturation.

Can the same enzyme yield a different Hill Plot under a different set of conditions? The answer is a most definite yes! The strength of interprotomer bonds determines the cooperativity of ligand binding, and agents that disrupt the quaternary structure of a protein can likewise change the nature of a Hill plot. The presence of an inhibitor or activator, treatment with a mercurial or some other alkylating agent, a change in pH or ionic strength, and changes in the temperature are all known to affect the cooperativity of ligand binding processes.

What would be the corresponding behavior of an enzyme obeying the Michaelis-Menten model? For the answer, let us restate the Michaelis-Menten model in the form of a saturation function:



Likewise, the following equilibrium constant can be written (again in the form of a dissociation constant):

$$K = \frac{[M][L]}{[ML]} \quad \text{or} \quad [M] = K \frac{[ML]}{[L]}$$

Using the formalism of a saturation function, we get

$$Y_L = [ML]/([M] + [ML])$$

Because there is only one site for ligand binding, no “ n ” appears in either the numerator or denominator. Nevertheless, Y_L can still vary over the interval from 0 to 1. Substituting for $[M]$ in terms of $[ML]$ yields:

$$Y_L = [ML]/\{(K[ML]/[L]) + [ML]\}$$

or

$$Y_L = 1/\{(K/[L]) + 1\}$$

and multiplying the right-hand side of the equation by $[L]/[L]$ gives the saturation function as

$$Y_L = [L]/\{K + [L]\}$$

This is rearranged to obtain:

$$Y_L/(1 - Y_L) = [L]/K$$

then, by taking the base 10 logarithm of both sides, we get

$$\log\{Y_L/(1 - Y_L)\} = \log[L] - \log K$$

The linear region of a plot of $\log\{Y_L/(1 - Y_L)\}$ versus $\log[L]$ yields a slope of one (see Fig. 3). Such ligand saturation behavior has been observed for myoglobin. Moreover, many oligomeric enzymes yield unit slopes, indicating that their binding sites operate independently (*i.e.*, there are no site-site interactions evident in their ligand binding).

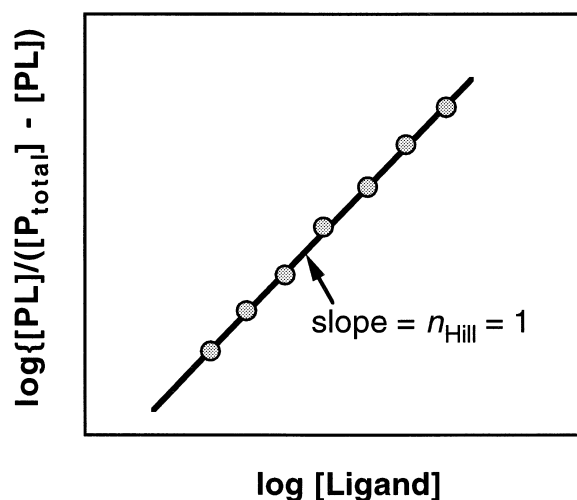


Fig. 3. Ligand binding behavior of a hypothetical protein exhibiting a unit Hill coefficient of ($n_H = 1$). Such behavior would be expected (a) for the case of a protein with only a single binding site or (b) for the case of an oligomeric protein containing n independent and noninteracting binding sites.

Should one use Hill plots to examine the initial velocity behavior of enzymes? Because infinite cooperativity is

assumed to be the basis of the Hill treatment, only rapidly equilibrating systems are suitable for Hill analysis. In fact, no enzyme system displaying steady-state kinetic behavior will ever satisfy this requirement. Even the Michaelis-Menten equation only has an initial, rapid-equilibrium segment (namely substrate binding to free enzyme and substrate release from the EX complex); product release is considered to be a slow step. For this reason, one must avoid the use of kinetic data in any application of the Hill equation.

HiPIP

An abbreviation for high-potential iron-sulfur protein, which is now regarded as a ferredoxin. In its role as a bacterial electron transfer component, this $[4Fe-4S]$ cluster protein can undergo interconversion to the $[4Fe-4S]^{2+}$ and $[4Fe-4S]^{3+}$ states.

HISTAMINE N-METHYLTRANSFERASE

This enzyme [EC 2.1.1.8] catalyzes the reaction of *S*-adenosyl-L-methionine with histamine to produce *S*-adenosyl-L-homocysteine and *N*^τ-methylhistamine.

J. Axelrod (1971) *Meth. Enzymol.* **17B**, 766.

HISTIDINE AMMONIA-LYASE

This enzyme [EC 4.3.1.3] (also known as histidinase, histidase, and histidine α -deaminase) catalyzes the conversion of histidine to urocanate and ammonia. **See Dehydroalanine**

V. E. Anderson (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 115.

K. R. Hanson & E. A. Havir (1972) *The Enzymes*, 3rd ed., **7**, 75.

HISTIDINE DECARBOXYLASE

This enzyme [EC 4.1.1.22] catalyzes the conversion of histidine to histamine and carbon dioxide. The enzyme requires either pyridoxal phosphate or pyruvate as a co-factor.

M. L. Hackert & A. E. Pegg (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 201.

M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.

H. Hayashi, H. Wada, T. Yoshimura, N. Esaki & K. Soda (1990) *Ann. Rev. Biochem.* **59**, 87.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

HISTIDINOL DEHYDROGENASE

This enzyme [EC 1.1.1.23] catalyzes the reaction of L-histidinol with two NAD^+ to produce L-histidine and two NADH. L-Histidinol will also serve as a substrate for this protein. The *Neurospora crassa* enzyme will also catalyze the reactions of phosphoribosyl-AMP cyclohydrolase [EC 3.5.4.19] and phosphoribosyl-ATP pyrophosphatase [EC 3.6.1.31].

R. G. Martin, M. A. Berberich, B. N. Ames, W. W. Davis, R. F. Goldberger & J. D. Yourno (1971) *Meth. Enzymol.* **17B**, 3.

HISTIDINOL-1-PHOSPHATE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.9], also called imidazolylacetol phosphate aminotransferase and imidazole acetol phosphate transaminase, catalyzes the reversible reaction of histidinol phosphate with α -ketoglutarate (or, 2-oxoglutarate) to produce 3-(imidazol-4-yl)-2-oxopropyl phosphate and glutamate.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

R. G. Martin, M. A. Berberich, B. N. Ames, W. W. Davis, R. F. Goldberger & J. D. Yourno (1971) *Meth. Enzymol.* **17B**, 3.

HISTIDINOL-PHOSPHATASE

This enzyme [EC 3.1.3.15] catalyzes the hydrolysis of histidinol phosphate to generate histidinol and orthophosphate.

R. G. Martin, M. A. Berberich, B. N. Ames, W. W. Davis, R. F. Goldberger & J. D. Yourno (1971) *Meth. Enzymol.* **17B**, 3.

HISTIDYL-tRNA SYNTHETASE

This enzyme [EC 6.1.1.21], also referred to as histidine:tRNA ligase, catalyzes the reaction of histidine with tRNA^{His} and ATP to produce histidyl-tRNA^{His}, AMP, and pyrophosphate. **See also** *Aminoacyl-tRNA Synthetases*

P. Schimmel (1987) *Ann. Rev. Biochem.* **56**, 125.

HISTONE KINASE

This enzyme [EC 2.7.1.70], also referred to as protamine kinase, catalyzes the reaction of ATP with protamine to produce ADP and an *O*-phosphoserine residue in protamine. The enzyme will also phosphorylate histones and it requires cAMP.

A. M. Edelman, D. K. Blumenthal & E. G. Krebs (1987) *Ann. Rev. Biochem.* **56**, 567.

HOFMEISTER SERIES

An empirically derived relationship¹ that describes the systematic effects of different neutral salts on the solubility of proteins. Collins and Washabaugh² indicate that the order of ionic species eluting from a Sephadex G-10 column corresponds to the known order of effectiveness ions in the Hofmeister series on protein solubility:



The order of elution indicates that ion size and charge density strongly influences the apparent molecular size of hydrated anions, hence the effect on the order of elution of these ions in gel permeation experiments. Those species eluting before chloride ion are termed polar kosmotropes to designate their ability to enhance water structure in their vicinity and to stabilize proteins. Those ions eluting after chloride ion are termed chaotropes, and they reduce local water structure and destabilize proteins. Chloride ion is regarded as having little effect on water structure or protein stability over a concentration range from 0.1–0.7 M². **See** *Chaotropic Agents; Kosmotropes*

¹F. Hofmeister (1888) *Naunyn-Schmiedeberg's Archiv für Experimentelle Pathologie und Pharmacologie (Leipzig)* **24**, 247.

²K. D. Collins & M. W. Washabaugh (1985) *Q. Rev. Biophys.* **18**, 323.

HOLO-[ACYL-CARRIER-PROTEIN] SYNTHASE

This enzyme [EC 2.7.8.7] catalyzes the reaction of coenzyme A with the apo-[acyl-carrier protein] to generate adenosine 3',5'-bisphosphate and the holo-[acyl-carrier protein].

R. H. Lambalot & C. T. Walsh (1997) *Meth. Enzymol.* **279**, 254.

HOMEOSTASIS

The self-regulated physiological/biochemical condition achieved through the action of built-in feedback loops that sense any departure from the system's set-point and then readjust input and/or output parameters to automatically return the system to that set-point. The term *homeostasis*, coined by American physiologist Walter Cannon to describe the resultant constancy of a living organism's internal state, has largely replaced the use of *milieu intérieur* (*i.e.*, internal environment) first proposed by the nineteenth century French physiologist Claude Bernard. The intellectual triumph of Bernard's recogni-

tion of this regulatory phenomenon is underscored by the fact that at just about the same time chemists were just realizing the occurrence of the dynamic equilibria in chemical reactions. Beyond the role of various organs in maintaining the internal environment, homeostasis is manifested at the subcellular level through feedback control of metabolic reactions and signal transduction processes.

Living organisms are self-stabilized by steady-state flows of metabolites, inorganic ions, and energy. As shown by Onsager and Prigogine, processes at steady state naturally resist change by outside forces. (*See Nonequilibrium Thermodynamics (A Primer)*) Homeostatic mechanisms endow cells with such remarkable resilience that early investigators mistook homeostasis as a persuasive indication that life is self-determining and beyond the laws of chemistry and physics. This tenet, known as vitalism, was based on Bergson's idea that living systems were maintained by a vital force (*élan vital*). To counter continued emphasis on vitalism, Bernard advanced the cause of the molecular life sciences by the following representative commentary¹: "But if we consider it, we shall soon see that the spontaneity of living bodies is simply the appearance and the result of a certain mechanism in completely determined environments; so that it will be easy, after all, to prove that the behavior of living bodies, as well as the behavior of inorganic bodies, is dominated by a necessary determinism linking them with conditions of a purely physico-chemical order."

¹C. Bernard (1865) *An Introduction to the Study of Experimental Medicine* (translated in 1927 by H. C. Greene) Dover edition, 1957, p. 61, Dover Publications, New York.

HOMOACONITATE HYDRATASE

This enzyme [EC 4.2.1.36], also referred to as homoaconitase, catalyzes the reversible conversion of 2-hydroxybutane 1,2,4-tricarboxylate to but-1-ene 1,2,4-tricarboxylate and water.

H. P. Broquist (1971) *Meth. Enzymol.* **17B**, 114.

HOMOCITRATE SYNTHASE

This enzyme [EC 4.1.3.21] catalyzes the reaction of 2-hydroxybutane 1,2,4-tricarboxylate with coenzyme A to produce acetyl-CoA, water, and α -ketoglutarate (or, 2-oxoglutarate).

M. J. P. Higgins, J. A. Kornblatt & H. Rudney (1972) *The Enzymes*, 3rd ed., **7**, 407.

H. P. Broquist (1971) *Meth. Enzymol.* **17B**, 113.

HOMOCYSTEINE DESULFHYDRASE

This pyridoxal-phosphate-dependent enzyme [EC 4.4.1.2] catalyzes the hydrolysis of homocysteine to produce ammonia, H₂S, and α -ketobutanoate (or, 2-oxobutanoate).

C. V. Smythe (1955) *Meth. Enzymol.* **2**, 315.

HOMOCYSTEINE S-METHYLTRANSFERASE

This enzyme [EC 2.1.1.10] catalyzes the reaction of S-adenosylmethionine with homocysteine to produce S-adenosylhomocysteine and methionine. With the bacterial enzyme, S-methylmethionine is a better substrate than S-adenosylmethionine.

S. K. Shapiro (1971) *Meth. Enzymol.* **17B**, 400.

HOMOGENEOUS CATALYSIS

Catalysis occurring in a single phase. Most enzymes are homogeneous catalysts; however if an enzyme embedded in a solid matrix acts on soluble substrates, aspects of heterogeneous catalysis become important. *See Heterogeneous Catalysis*

HOMOGENEOUS REACTION

A reaction that occurs wholly within one phase. It should be pointed out that certain homogeneous systems may still exhibit nonhomogeneous effects. For example, a reaction in a solution may be influenced by the walls of the container. If such phenomena occur, then corrections would have to be made. In general, this is not a problem for most enzyme-catalyzed processes.

HOMOGENTISATE 1,2-DIOXYGENASE

This enzyme [EC 1.13.11.5], an iron-dependent system (also called homogentisicase and homogentisate oxygenase), catalyzes the reaction of homogentisate with dioxygen to produce 4-maleylacetoacetate.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

K. Adachi & Y. Takeda (1970) *Meth. Enzymol.* **17A**, 638.

HOMOISOCITRATE DEHYDROGENASE

This enzyme [EC 1.1.1.155] catalyzes the reaction of (–)-1-hydroxy-1,2,4-butane tricarboxylate with NAD^+ to produce α -ketoadipate (or, 2-oxoadipate), carbon dioxide, and NADH.

H. P. Broquist (1971) *Meth. Enzymol.* **17B**, 118.

HOMOPHILIC INTERACTION

Any set of geometrically and energetically equivalent binding interactions between two identical polypeptide chains or proteins. In a homophilic interaction, one polypeptide chain serves as a ligand for a receptor site on the other polypeptide chain, and *vice versa*.

HOMOSERINE O-ACETYLTRANSFERASE

This enzyme [EC 2.3.1.31] catalyzes the reaction of acetyl-CoA with homoserine to produce coenzyme A and O-acetylhomoserine.

S. Nagai & D. Kerr (1971) *Meth. Enzymol.* **17B**, 442.

HOMOSERINE DEHYDROGENASE

This enzyme [EC 1.1.1.3] catalyzes the reaction of homoserine with NAD(P)^+ to produce aspartate 4-semialdehyde and NAD(P)H . NAD^+ is the better of the two coenzymes with the yeast enzyme whereas NADP^+ is preferred by the *Neurospora* enzyme. **See also** Aspartate Kinase

I. Saint-Gerons, C. Parsot, M. M. Zakin, O. Bârzy & G. N. Cohen (1988) *Crit. Rev. Biochem.* **23**, S1.

P. Truffa-Bachi (1973) *The Enzymes*, 3rd ed., **8**, 509.

HOMOSERINE O-SUCCINYLTRANSFERASE

This enzyme [EC 2.3.1.46], also referred to as homoserine O-transsuccinylase, catalyzes the reaction of succinyl-CoA with homoserine to produce coenzyme A and O-succinylhomoserine.

I. Saint-Gerons, C. Parsot, M. M. Zakin, O. Bârzy & G. N. Cohen (1988) *Crit. Rev. Biochem.* **23**, S1.

HOMOTROPIC INTERACTION

A term used in describing the interactions among dissimilar ligands in cooperative saturation. If an enzyme binds two ligands, say X and Y, then the effect of the binding of molecule X on the binding of other molecules of X is termed a homotropic interaction. Likewise, the effect of the binding of molecule Y on the binding of other molecules of Y is termed a homotropic interaction. Ef-

fects arising from the influence of ligand X binding on ligand Y binding, and *vice versa*, are termed heterotropic interactions.

In the context of the Monod-Wyman-Changeux concerted-transition model for allosteric effects, one usually considers the effects of specific site occupancy on the behavior of other binding sites. Thus, the more correct operational definition states that homotropic effects represent the interactions on cooperative ligand binding exerted by molecules of *like action*. Applying this definition, we see that while oxygen and carbon monoxide compete with respect to each other for the same sites (*i.e.*, the four heme groups on hemoglobin), both act on hemoglobin in the same manner—namely by increasing the fraction of macromolecules in the R-state. Thus, O_2 and CO act homotropically with respect to each other. It is also true that 2,3-bisphosphoglycerate or hydrogen ion act heterotropically with respect to oxygenation; yet, in terms of their mutual interactions with each other, 2,3-bisphosphoglycerate or hydrogen ion can be regarded as exerting homotropic effects. **See** Monod-Wyman-Changeux Model; Allosterism; Hemoglobin

HOOKE'S SPRING LAW

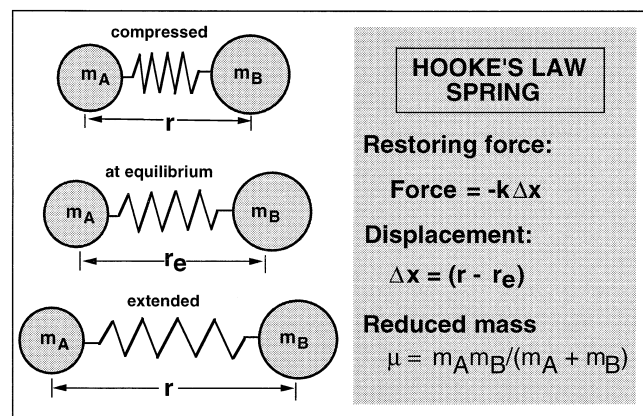


Fig. 1. Illustration of the forces produced during vibration of a Hooke's law spring.

The vibrational stretching mode of a chemical bond lies parallel to the direction of atom movement during bond scission. For this reason, by using Hooke's Law to estimate approximately the energy of bond stretching, one can infer aspects about the energetics that underlie kinetic isotope effects. Hooke's Law is a classical mechanical treatment of a massless spring joining two bodies of mass m_A and m_B at an equilibrium distance (r_e). Dis-

placement of either body from this equilibrium spring length is taken as a mechanical analogue of a vibrating chemical bond joining two atoms or groups of atoms within a molecule (Fig. 1). Any such displacement is accompanied by an oppositely directed restoring force,

$$f = -k(r - r_e)$$

where $(r - r_e)$ is the displacement, and k is the linear proportionality constant. By applying Newton's Law ($f = ma = d^2r_A/dt^2 = d^2r_B/dt^2$) for mechanical force, one may describe the classical motion of the system as follows:

$$m_A \cdot d^2r_A/dt^2 = m_B \cdot d^2r_B/dt^2 = -k(r - r_e)$$

where r_A and r_B are distances of m_A and m_B from the molecule's center of gravity. Recalling that the center of mass is related to these displacements by $r_A = [m_B/(m_A + m_B)]r$ and $r_B = [m_A/(m_A + m_B)]r$, we can substitute either of these expressions into the above equation to obtain:

$$[m_A/(m_A + m_B)] \cdot d^2(r - r_e)/dt^2 = -k(r - r_e)$$

where $[m_A/(m_A + m_B)]$ equals the reduced mass (symbolized by μ_{AB}). Because r_e is itself a constant, we can rewrite the above equation as:

$$\mu_{AB} = d^2(\Delta x)/dt^2 = -k(\Delta x)$$

This second-order differential equation has a harmonic function as its solution:

$$\Delta x = A_{\max} \sin [(k/\mu_{AB})^{1/2}t]$$

where $A_{\max}$ is the maximal amplitude (or displacement). Displacement is described by a sinusoidal function with respect to time—repeatedly achieving the identical displacement at any two times t_x and t_y differing by an integral multiple of the t_{period} , defined as the period of harmonicity (i.e., $A_x = A_y$ whenever $(t_x - t_y) = n \cdot t_{\text{period}}$, where $n = 1, 2, 3, \dots$).

Differences in the potential energy U of a Hooke's Law spring can be calculated as the product of the restoring force ($-f$) and the distance dx over which the force acts. Thus, by substituting $-kx$ in place of f , we get

$$dU = -kx \, dx$$

Integration yields a parabolic function

$$U = (1/2)kx^2$$

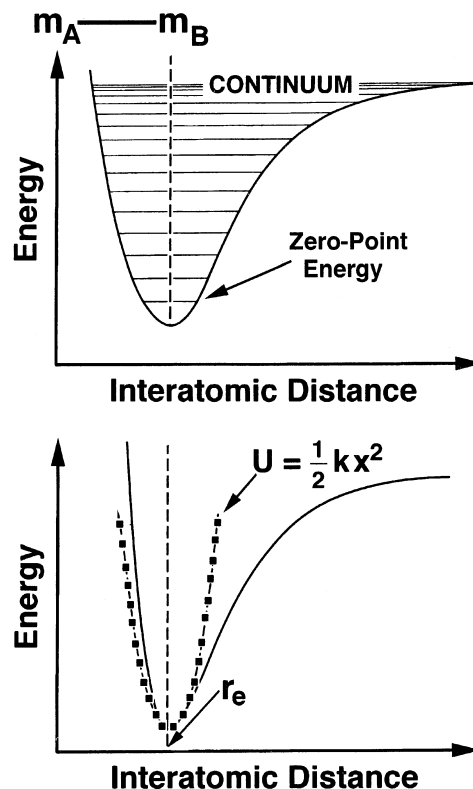


Fig. 2. Top: Energy well representation for a molecule m_A — m_B showing the various quantized states from the lowest-lying energy state (i.e., the zero-point energy state) to the classical mechanical continuum. Bottom: Same energy well with parabolic curve representing the potential energy of a Hooke's law spring. (Note that this approximation only correlates closely at the lowest energy level range. Because kinetic isotope effects are interpreted on the basis of changes in the zero-point energy, the Hooke's spring law approximation is useful in estimating the energy difference between two isotopomers.)

with non-negative energy values (hence, potential energy). This parabola is shown in Fig. 2 where there is good fit near the equilibrium position of the potential energy well for a chemical bond. This should not be surprising, because this classical mechanical model for a chemical bond assumed a linear restoring force; only the quantum mechanical treatment can fully account for the entire potential energy function, including the quantization of vibrational energies.

HORN-BÖRNIG PLOT

A graphical procedure used to determine, in cooperative systems, values for L (the ratio of the T to R state in the absence of any binding ligand in the Monod-Wyman-Changeux model) and n (the stoichiometry of binding) in exclusive binding systems ($c = 0$ where $c = K_R/K_T$; i.e., the ratio of the intrinsic dissociation constants for

the two states [K_R and K_T are the dissociation constants of the ligand with respect to the R and T state, respectively]; thus, in this case, the T state has no significant affinity for the ligand)^{1,2}.

This plot is based on the equation

$$\log[(\alpha V_{\max,f}/v) - \alpha - 1] = (1 - n)\log(1 + \alpha) + \log L'$$

where α is the ratio of the ligand concentration to the dissociation constant of the ligand with respect to the R state, $V_{\max,f}$ is the maximum velocity of the reaction in the forward direction, v is the initial velocity at the ligand concentration, and L' is the apparent allosteric constant equal to $(L(1 + \beta)^n/(1 + \gamma)^n)$ where β is the concentration of an allosteric inhibitor divided by the dissociation constant for that inhibitor (to the T state) and γ is the concentration of an allosteric activator divided by the dissociation constant for that activator (to the R state); hence, if β and γ equal zero, then $L = L'$; or, if $\beta = \gamma$, then $L = L'$). Thus, for exclusive binding allosteric systems, a plot of $\log((\alpha V_{\max,f}/v) - \alpha - 1)$ vs. $\log(1 + \alpha)$ will yield a straight line. The slope will be $1 - n$ and the vertical intercept will be $\log L'$. Values for L can be determined from data obtained in the absence of effectors. Thus, knowing the concentration of the effector can yield values for the dissociation constant for that effector. Note that this plot requires an accurate value for α . Hence, the dissociation constant of the ligand for the R state has to be known. Such a value can be obtained if an activator is available that completely changes the kinetics from nonhyperbolic to hyperbolic.

If the ligand can bind to either the R or T state (*i.e.*, $c = 0$), the Horn-Börnig plot will yield nonlinear plots.

The chief limitation of this plotting method is that, unlike the Adair Equation, the plot requires that the system obey all of the conditions required by the MWC model. This assumption will often prove to be incorrect!

¹A. Horn & H. Börnig (1969) *FEBS Lett.* **3**, 325.

²K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

HORSERADISH PEROXIDASE

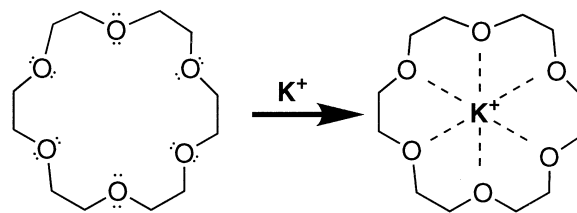
This enzyme [EC 1.11.1.7] catalyzes the reaction of hydrogen peroxide with a donor to produce two water and the oxidized donor.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

O. Ryan, M. R. Smyth & C. Fágáin (1994) *Essays in Biochem.* **28**, 129.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

HOST-GUEST INTERACTIONS



Binding of Potassium Ion to 18-Crown-6 Host Molecule

Structural recognition and binding interactions involving a naturally occurring or synthetic “host” (or template) and a smaller ligand known as a “guest”^{1,2}. Examples of host molecules include the cyclodextrins and the so-called crown ethers. Host-guest interactions have served as useful systems for modeling enzymic catalysis, resolving optical isomers of complicated organic molecules, making analogues for micellar catalysis, and for understanding the selective cation transport properties of valinomycin-like antibiotics. For his studies on host-guest interactions, Professor Donald Cram received the Nobel Prize in Chemistry in 1989.

¹D. W. Griffiths & M. L. Bender (1973) *Advances in Catalysis* **23**, 209.

²D. J. Cram (1976) *Techniques of Chemistry*, vol. **10**, pp. 815-873, Wiley, New York.

HOST MOLECULE

A molecule that contains one or more binding sites that can accommodate inorganic or organic ions referred to as guests. The binding site could even be a cavity within a crystal structure. Although enzymes clearly qualify as examples of host molecules, the term is usually restricted to structures such as crown ethers, macrocycles, and cyclodextrins. Nevertheless, these hosts do serve as models for molecular recognition. **See also** *Crown Ethers; Macrocycles; Inclusion Complexes*

HÜCKEL MOLECULAR-ORBITAL CALCULATIONS

A computational algorithm (also referred to as a HMO calculation) for characterizing delocalized chemical bonding patterns for unsaturated and aromatic mole-

cules assumes that σ orbitals can be treated as local bonds^{1,2}. The calculations involve only π electrons and either neglect or average out electron-electron repulsions. An extension of the Hückel method (often referred to as EHMO) has been used to treat other types of molecules^{1,3}. **See also** *Hartree-Fock Molecular-Orbital Calculations*; *Ab Initio Molecular Orbital Calculations*

¹K. Yates (1978) *Hückel Molecular Orbital Theory*, Academic Press, New York.

²C. A. Coulson, B. O'Leary & R. B. Mallion (1978) *Hückel Theory for Organic Chemists*, Academic Press, New York.

³R. Hoffman (1963) *J. Chem. Phys.* **39**, 1397.

HÜCKEL'S RULE

A rule for helping to predict if a monocyclic planar system of delocalized π electrons will exhibit aromatic character. The ring system will be aromatic if the number of delocalized π electrons equals $4n + 2$ where n is a positive integer or zero. For example, the ring system in phenylalanine obeys Hückel's rule. **See** *Aromaticity*

HUGHES-INGOLD THEORY FOR SOLVENT EFFECTS ON REACTIVITY

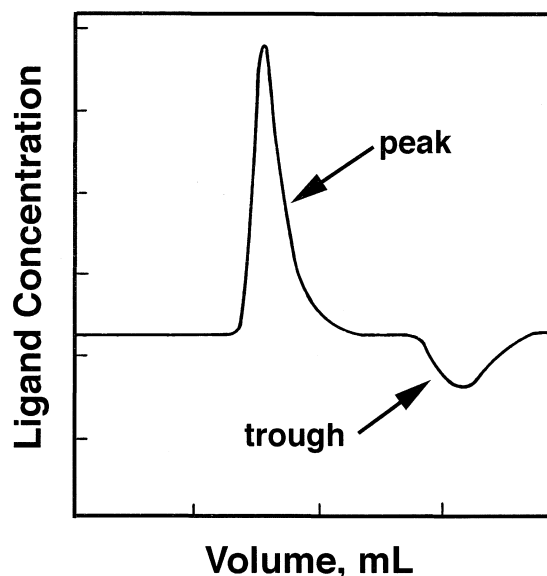
A principle applied in physical organic chemistry to account for the effect of solvation on the rates of nucleophilic reactions. This theory states that an increase in the ion-solvating power of the medium will tend to speed up the formation and concentration of charges, thereby inhibiting their breakdown or diffusion. This approach is predicated on the idea that a more polar solvent can potentially stabilize ionic intermediates and alter chemical reactivity.

Consider the attack of a nucleophile on the electrophilic carbon atom of a carbonyl group. Negative charge becomes concentrated on carbon and oxygen atoms of the carbonyl group as the reactants approach the transition state. The Hughes-Ingold theory suggests that the reaction rate will be increased by any increase in the ion-solvating power of the medium.

To date, no single concept provides a quantitative means for estimating the magnitude of the contribution of reaction medium to rate processes. Nevertheless, the Hughes-Ingold principle offers some insight regarding how electrostatic interactions in enzyme active sites might finely tune the properties of nucleophilic steps in enzymic catalysis.

HUMMEL-DREYER TECHNIQUE

A gel filtration method¹ for measuring a ligand's equilibrium binding interactions with a macromolecule. In this technique, a gel filtration column is pre-equilibrated with a ligand (that can be measured by UV absorbance or by radioactivity), and then one adds a small sample of protein solution (itself containing ligand at the same total concentration as used to equilibrate the column). If the protein forms a complex with the ligand, then the protein peak will contain an excess of ligand depending on the protein concentration, the ligand concentration, and the strength of the equilibrium binding interaction. This method has proved very useful in determining dissociation constants and stoichiometries of ligand binding interactions^{2,3}.



Ligand binding to a macromolecule can be quantitatively analyzed by passage of the macromolecule through a gel filtration column that has been previously equilibrated at a chosen ligand concentration. Upon elution, the macromolecule with bound ligand will be excluded from the gel and will emerge as a peak followed by a trough which represents the depletion of ligand from the less mobile low-molecular-weight ligand. These experiments are typically carried out at various concentrations of ligand, and the area beneath the peak to the baseline should equal the area above the trough and up to the baseline. This baseline is constructed by drawing the best line through the horizontal segments of the elution profile.

¹J. P. Hummel & W. J. Dreyer (1962) *Biochim. Biophys. Acta* **63**, 530.

²G. K. Ackers (1973) *Meth. Enzymology* **27**, 444.

³F. Sun, W. Kuo & R. A. Nash (1984) *J. Chromatog.* **288**, 377.

HYALURONATE LYASE

This enzyme [EC 4.2.2.1], also known as hyaluronidase, catalyzes the conversion of hyaluronate to *n* 3-(4-deoxy-

β -D-gluc-4-enuronosyl)-*N*-acetyl-D-glucosamine. The enzyme is also capable of using chondroitin as a substrate.

A. Linker (1966) *Meth. Enzymol.* **8**, 650.

HYALURONIDASES

A number of enzymes have been referred to as hyaluronidases. Hyaluronate 4-glycanohydrolase [EC 3.2.1.35], also known as hyaluronoglucosaminidase and hyaluronidase, catalyzes the random hydrolysis of 1,4-linkages between *N*-acetyl- β -D-glucosamine and D-glucuronate residues in hyaluronate. It will also catalyze the hydrolysis of 1,4- β -D-glycosidic linkages between *N*-acetylgalactosamine or *N*-acetylgalactosamine sulfate and glucuronate in chondroitin, chondroitin 4- and 6-sulfates, and dermatan. Hyaluronate 3-glycanohydrolase [EC 3.2.1.36], also known as hyaluronoglucuronidase and hyaluronidase, catalyzes the random hydrolysis of 1,3-linkages between β -D-glucuronate and *N*-acetyl-D-glucosamine residues in hyaluronate. **See** *Hyaluronate Lyase*

W. L. Hynes & J. J. Ferretti (1994) *Meth. Enzymol.* **235**, 606.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

K. Meyer (1971) *The Enzymes*, 3rd ed., **5**, 307.

HYBRIDIZATION

1. A quantum mechanical phenomenon wherein certain orbitals (such as s, p, and d) combine into new orbitals (referred to as hybrid orbitals). For example, the electrons within the four C—H covalent bonds of methane occupy four hybrid orbitals (designated sp^3) that are formed from unhybridized s and p orbitals, originally present in the isolated carbon atom, upon combination of one carbon atom with four hydrogen atoms. Hybridization of these orbitals in carbon creates directed covalent bonds of equivalent energy and geometry. The sp^3 hybridization arises from only one of several approximate solutions to the Schrödinger equation, and it is worth noting that the four C—H bonds of methane are not always equivalent in their reactivity. 2. The formation of DNA–DNA and DNA–RNA duplexes from individual nucleic acid chains. Typically, a gel electropherogram containing a series of DNA bands is denatured under alkaline conditions and transferred to a nitrocellulose membrane to make a replica of the original electropherogram. The nitrocellulose-bound DNA species are brought into contact with a solution containing ^{32}P -labeled, single-stranded DNA (or RNA) molecules of interest, and the blot is incubated and subsequently extensively washed under conditions that form and maintain

the structural integrity of DNA- ^{32}P DNA (or DNA- ^{32}P RNA) duplexes. Radioautography reveals the location(s) of the radiolabeled nucleic acid.

HYDRATION ATMOSPHERE

The short-range, semioordered water structure that surrounds ionic solutes and polar solutes as a consequence of dipolar and Coulombic interactions.

Anions and cations form strong coordination bonds with water molecules, and the polarizability of water factors into these interactions. Anions prefer to interact with the hydrogen atoms of a water molecule, whereas cations are attracted to the nonbonding electron pairs on the water oxygen. Because the nonbonded electron pairs tend to be localized and directional, cations make stronger bonds with water molecules than do anions of comparable ionic radius or electrical charge. This tightly bound water is said to be electrostricted, a phenomenon associated with the ability of ions to attract and deform nearby water molecules, such that there is a slight increase in the density of bound water (**See** *Electrostriction*). Proceeding outwardly from this inner hydration sphere, there is an outer coordination shell containing a larger number of semioordered water molecules clustered together in extended hydrogen-bonded chains. The motions of water molecules in this cluster zone are more restricted than free water molecules in the bulk solvent; however, they are also less restricted in their motions than water molecules in the first hydration sphere. In this respect, water molecules form a two-zone hydration atmosphere that can profoundly affect the kinetics of ligand exchange reactions of metal ions in an aqueous environment (**See** *Eigen-Tamm Mechanism*). Note also that the solvated ion tends to create a centrosymmetric ordering of water molecules, and the degree of order is highest at short distances from that solute ion. This is to be contrasted with the hydration atmosphere surrounding an electrically charged membrane vesicle or a biomineral surface, where the solute ions and water molecules form planes or layers with predictable properties. **See** *Biomineralization* (3. *The Double Layer*)

HYDRATION NUMBER

The hydration number represents the number of water molecules directly associated with the solute ion. Some approximate hydration numbers for biologically significant metal ions are: Li^+ , 3.7; Na^+ and K^+ , 4; Ca^{2+} , 8; and Mg^{2+} , 12.

HYDRIDE

1. A negatively charged hydrogen ion (H^-). The hydride ion typically does not exist free in solution although it is an important species in a significant number of enzyme-catalyzed reactions. **See** *Hydride Ion Transfer*. 2. A molecular entity containing hydrogen in which one or more hydrogens has (have) a formal negative charge; for example, lithium aluminum hydride (LiAlH_4). 3. Compounds containing only hydrogen and one other element; examples of hydrides, based on this definition, are AsH_3 , NH_3 , H_2O , HCl , CH_4 , H_2S , and PH_3 .

HYDRIDE ION TRANSFER

The transfer of a hydride ion (H^-) in a reaction. Such transfers occur in many enzyme-catalyzed reactions; perhaps the most common being in pyridine nucleotide-linked dehydrogenases in which hydride transfer is stereospecific (*i.e.*, to the *re*- or *si*-face of the coenzyme)¹.

A number of methods have been utilized to probe hydride transfer reactions. These include the efforts to follow the course of the reaction in deuterated or tritiated water². Perhaps one of the most effective methods is to study isotope effects³⁻⁶.

¹K.-S. You (1982) *Meth. Enzymol.* **87**, 101.

²I. A. Rose (1982) *Meth. Enzymol.* **87**, 84.

³K. B. Schowen & R. L. Schowen (1982) *Meth. Enzymol.* **87**, 551.

⁴D. B. Northrop (1982) *Meth. Enzymol.* **87**, 607.

⁵W. W. Cleland (1982) *Meth. Enzymol.* **87**, 625.

⁶W. W. Cleland (1995) *Meth. Enzymol.* **249**, 341.

HYDROCARBON MONOOXYGENASE

This enzyme system, also called ω -hydroxylase, catalyzes the NADH- and rubredoxin-dependent hydroxylation of terminal methyl groups of alkanes and fatty acids.

S. W. May & A. G. Katopodis (1990) *Meth. Enzymol.* **188**, 3.

HYDROGENASE

This enzyme [EC 1.18.99.1], also known as hydrogenlyase, catalyzes the reaction of H_2 with two oxidized ferredoxin to produce two H^+ and two reduced ferredoxin. This enzyme is an iron-sulfur protein and requires nickel ions. It can use molecular hydrogen to reduce a variety of substances. **See also** *Hydrogen Dehydrogenase*; *Cytochrome c_3 Hydrogenase*

M. A. Halcrow (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 359.

R. J. Maier (1996) *Adv. Protein Chem.* **48**, 35.

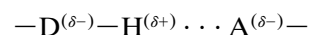
M. W. W. Adams & A. Kletzin (1996) *Adv. Protein Chem.* **48**, 101.

R. Cammark, V. M. Fernandez & E. C. Hatchikian (1994) *Meth. Enzymol.* **243**, 43.

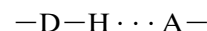
G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

HYDROGEN BONDS

The primarily electrostatic bond formed by interaction of a hydrogen atom with a highly electronegative element (*e.g.*, oxygen, nitrogen, or fluorine) and a second electronegative atom (*e.g.*, oxygen, nitrogen, or fluorine).¹⁻³ The electrostatic bond is formed between a hydrogen donor (D-H) and an acceptor (A):



or simply as



Hydrogen bonds (usually depicted with dotted or dashed lines) can be either intermolecular or intramolecular. In cases where oxygen or nitrogen acts as the electronegative element in the association, the oxygen or nitrogen atom can be singly or multiply bonded, albeit with different energies. The hydrogen bond is strongest (*i.e.*, most stable) when the three atoms of the bond are colinear (typically, within 15°). The distance between the donor and acceptor electronegative atoms is reduced from the calculated van der Waals radii as a result of the hydrogen bond interaction. Although they typically play only a compensatory role in stabilizing protein structure, hydrogen bonds can make significant contributions in both complexation and chemical reactions.

Jencks⁴ has pointed out that it is conceptually useful to consider the hydrogen bond as an intermediate stage of proton transfer from an acid to a base. A sufficiently strong base will hold the proton closer, thereby stretching and weakening the proton's association with the acid. Complete transfer to the base breaks the hydrogen bond. Hence, a hydrogen bond strength increases with the acidity of the donor and the basicity of the acceptor.

As pointed out by Fersht^{5,6}, hydrogen bonds can also play significant roles in molecular recognition. An individual hydrogen bond can contribute $0.5\text{--}1.8 \text{ kcal mol}^{-1}$ (or, $2\text{--}7.5 \text{ kJ mol}^{-1}$) to the binding energy and this may increase specificity by a factor of 2–20. This was determined by assessing the effects of deleting a hydrogen bond to an uncharged hydrogen bond donor/acceptor pair. A larger magnitude (around $3\text{--}6 \text{ kcal mol}^{-1}$ or

12.5–25 kJ mol⁻¹) is typically observed if one deletes a hydrogen bond for a charged donor/acceptor. **See** *Water; Low-Barrier Hydrogen Bonds*

¹L. Pauling (1960) *The Nature of the Chemical Bond*, Cornell Univ. Press, Ithaca, New York.

²M. D. Joesten & L. J. Schaad (1974) *Hydrogen Bonding*, Marcel Dekker, New York.

³G. C. Pimentel & A. L. McClellan (1960) *The Hydrogen Bond*, Freeman, San Francisco.

⁴W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, pp. 338–350, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York]

⁵A. R. Fersht (1987) *Trends Biochem. Sci.* **12**, 304.

⁶A. R. Fersht (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, pp. 650, W. H. Freeman, New York.

HYDROGEN DEHYDROGENASE

This enzyme [EC 1.12.1.2], also called hydrogenase on occasions, catalyzes the reaction of H₂ with NAD⁺ to produce H⁺ and NADH. This is a flavoprotein that requires iron and nickel. **See also** *Hydrogenase; Cytochrome c₃ Hydrogenase*

M. A. Halcrow (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 359.

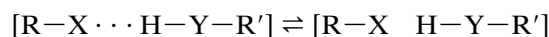
HYDROGEN EQUIVALENT

1. The number of replaceable hydrogen atoms per molecule of a particular substance. 2. The number of hydrogen atoms that will react with a molecule of a particular substance.

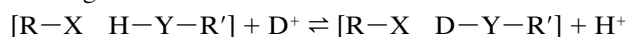
HYDROGEN EXCHANGE IN PROTEINS

The protons participating in hydrogen bonding in proteins are only released if their associated hydrogen bonds are broken and if there is direct access to additional hydrogens from the solvent. This is illustrated in the following scheme:

Dissociation:



Exchange:



Therefore, those hydrogen atoms that are present in stable H-bonds will resist exchange with the solvent. H-bonds lying on the surface of many proteins readily undergo exchange. In addition to deuterium, one can also use tritium-labeled water to follow the kinetics of protein hydrogen exchange.

This technique was developed decades ago in the laboratory of Karl Linderstrøm-Lang. In his technique¹, the protein of interest is dissolved in D₂O where the protein is kept for long periods over which all protium atoms (*i.e.*, ¹H) attached to N and O are replaced by D (or, ²H). After freeze drying and lyophilization to remove all solvent molecules, the protein is redissolved in H₂O and the time course of D $\rightleftharpoons$ H exchange is measured. Denatured proteins and random coiled proteins often exchange the hydrogens rapidly, whereas folded proteins typically show a spectrum of exchange rates.

This technique has been taken to new heights through the use of 2-D nuclear magnetic resonance^{2–4} to assign many, if not all, exchangeable protons in terms of the residues to which they are attached. As a result, one can assess individual rate constants defining the kinetics of exchange.

While hydrogen exchange kinetics have been used to infer important features of protein folding pathways, Clarke *et al.*⁵ argue that hydrogen exchange does not represent a short cut for studying protein-folding pathways because it is impossible to distinguish paths involving intermediates from side reactions. Their studies of barnase and chymotrypsin inhibitor-2 indicate no obvious relationship between hydrogen exchange at equilibrium and their folding pathways.

¹K. Linderstrøm-Lang (1955) *Chem. Soc. (London) Spec. Publ.*, No. 2, p. 1.

²Y. Bai, T. R. Sosnick, L. Mayne & S. W. Englander (1995) *Science* **269**, 192.

³M. Jamin & R. L. Baldwin (1996) *Nature Struct. Biol.* **3**, 613.

⁴S. W. Englander, L. Mayne, Y. Bai & T. R. Sosnick (1997) *Protein Sci.* **6**, 1101.

⁵J. Clarke, L. S. Itzhaki & A. R. Fersht (1997) *Trends Biochem. Sci.* **22**, 284.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Chemistry, **232**, 31, 39–40; functional labeling method, **232**, 29, 32–35, 40–41; kinetic labeling method, **232**, 28–29; local unfolding model, **232**, 39–41; NMR analysis, **232**, 28; isotope dilution assays for cAMP, cGMP, and inositol 1,4,5-trisphosphate, **237**, 390–394; hydrogen-deuterium [amide I infrared band frequencies and, **232**, 169; with NMR analysis, **232**, 28, 29]; hydrogen-tritium [functional labeling method, **232**, 29, 32–35; ³H into normal water, nickel-iron hydrogenase assay, **243**, 57–59; kinetic labeling method, **232**, 28–29; tritium loss curve, **232**, 38]; amide groups, **259**, 345–348, 350; free energy change, **259**, 352; hemoglobin [cooperative free energy change loss, **259**, 355–356; tritium exchange, **259**, 353–355]; infrared spectroscopy, **259**, 406; locating energetic changes in proteins, **259**, 354–355; nuclear magnetic resonance, **259**, 349–350, 352; nucleotide bases, **259**, 346–347; in proteins, dependence on [denaturant concentration, **259**, 348–350; pH, **259**, 345–346, 352; protein unfolding, **259**, 344, 347; thermal stability, **259**, 350–352]; structural physics, **259**, 347–348; structural protection factor, **259**, 346.

HYDROGEN SULFIDE:FERRIC ION OXIDOREDUCTASE

This glutathione-dependent enzyme catalyzes the reaction of hydrogen sulfide with Fe^{3+} to produce sulfite and Fe^{2+} .

I. Suzuki (1994) *Meth. Enzymol.* **243**, 455.

HYDROGEN TUNNELING

The passage of one or more protons or hydride ions through a potential-energy barrier without requiring these reactants to possess sufficient energy to surmount the barrier. Hydrogen tunneling has been identified in a number of enzyme-catalyzed reactions¹, typically through isotope effect studies. Some evidence suggests that tunneling may be greater in the enzyme-catalyzed reaction as compared to the same reaction occurring in the absence of the enzyme. A number of factors have been suggested to contribute to this increase in the probability of tunneling. One is the internal thermodynamics of enzymic catalysis. When comparing different substrates and products for an enzyme that can act on a number of substrates, as the ΔH° value nears zero, the degree of tunneling increases¹⁻⁴. In addition, the degree of tunneling is dependent on the ratio of the height and width of the reaction barrier⁵. Thus, compression of the reaction coordinate (*e.g.*, protein breathing motions bringing substrates close together thus increasing the probability of tunneling) could explain the increase seen in tunneling in enzyme-catalyzed reactions. Hence, both thermodynamic and geometric constraints seen in enzymes can potentially influence the occurrence of proton tunneling.

Large kinetic isotope effects are usually indicative of proton tunneling. However, if proton transfer is not rate-limiting or rate-contributing, proton tunneling may still be present, but its occurrence would be masked.

In studying both primary and secondary isotope effects, if the secondary effect is significantly reduced on the replacement of the primary H for D (*i.e.*, by the primary effect), then tunneling occurs. Note that this method assumes that the observed primary and secondary isotope effects occur in the same step of the reaction. A number of methods have been developed to characterize tunneling phenomena¹.

¹B. J. Bahnson & J. P. Klinman (1995) *Meth. Enzymol.* **249**, 373.

²J. R. De La Vega (1982) *Acc. Chem. Res.* **15**, 185.

³Y. Cha, C. J. Murraro & J. P. Klinman (1989) *Science* **243**, 1325.

⁴J. Rucker, Y. Cha, T. Jonsson, K. L. Grant & J. P. Klinman (1992) *Biochemistry* **31**, 11489.

⁵R. P. Bell (1980) *The Tunnel Effect in Chemistry*, Chapman and Hall, New York.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

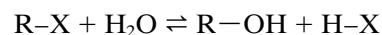
In enzyme catalysis, **249**, 373-397; [bovine serum amine oxidase, **249**, 393-394; coupled motion and, **249**, 386-388; demonstration, **249**, 374-386 (breakdown of rule of geometric mean in, **249**, 375-376, 388-389; by competitive comparison of k_H , k_D , and k_T , **249**, 382-383; exponential breakdown in, **249**, 376-378); factors affecting, **249**, 395-397; internal thermodynamics and, **249**, 396-397; monoamine oxidase B, **249**, 395; protein structure and, **249**, 397; verification, **249**, 386-395].

HYDROLASES

A major class of enzymes that catalyze hydrolytic cleavage reactions. Examples include esterases, phosphatases, sulfatases, nucleases, glycosidases, peptidases, proteinases, and amidases.

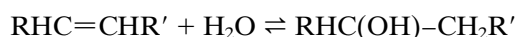
HYDROLYSIS

Any process of bond cleavage that is accompanied by the incorporation of the elements of a water molecule into the product(s):



Bonds typically hydrolyzed include carboxylic and phosphoric esters, amides, acetals, amidines, as well as metal ion complexes. (When a nucleophilic substitution reaction uses the solvent as the nucleophile, the reaction is often referred to as solvolysis.)

In the broader scope of this definition, we can also include the hydration of aldehydes or alkenes, where the C=O or C=C double bond is broken by incorporating the elements of water:



Jencks¹ considered the properties and mechanisms of hydrolysis and hydration. The interested reader may wish to consult his monograph (exact pages in parentheses) for information on the following: acetylacetone (p. 550); aldehydes (pp. 175-176; 211-215; 497-498); acylcyanides (p. 549); amides (pp. 523-526); amidines (pp. 471-477); anhydrides (pp. 508-517); anilides (pp. 224, 523-525; 553); aspirin (pp. 580-583); ATP (pp. 112-114); esters (pp. 399; 460; 483-487; 508-517; 570; 577-582);

ethyltrifluoroacetate (p. 515); formamidines (pp. 192, 234, 545–548; 553); hydration of carbonyl compounds (pp. 22, 33, 175–176, 198, 211–215, 497–498, 548–550, 608); imido esters (pp. 215–216, 487–489, 526–531, 536–542); ketones (pp. 548–550); *p*-nitrophenyl acetate (pp. 405, 408, 597–599); phenyl acetate (p. 314); phosphate esters (pp. 97, 169, 233, 305, 497); starch (p. 357); thiazolines (pp. 539–542, 552); and thiol esters (pp. 517–523, 534, 597).

¹W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York]

HYDRON

The H^+ ion in its natural abundance, typically used when one wishes not to distinguish between isotopes. The term “proton” thus refers specifically to $^1H^+$, “deuteron” to $^2H^+$, and “triton” to $^3H^+$.

HYDROPHILIC

Referring to the tendency or capacity of a molecular entity or one of its substituents to interact significantly with water or other polar solvents or polar groups.

HYDROPHOBIC EFFECT

The tendency^{1–4} of apolar side chains of amino acids (or lipids) to reside in the interior nonaqueous environment of a protein (or membrane/micelle/vesicle). This process is accompanied by the release of water molecules from these apolar side-chain moieties. The effect is thermodynamically driven by the increased disorder (*i.e.*, $\Delta S > 0$) of the system, thereby overcoming the unfavorable enthalpy change (*i.e.*, $\Delta H < 0$) for water release from the apolar groups.

Cantor and Schimmel⁵ provide a lucid description of the thermodynamics of the hydrophobic effect, and they stress the importance of considering both the unitary and cratic contributions to the partial molal entropy of solute-solvent interactions. Briefly, the partial molal entropy (S_A) is the sum of the *unitary* contribution (S_A') which takes into account the characteristics of solute A and its interactions with water) and the *cratic* term ($-R \ln \chi_{cA}$, where R is the universal gas constant and χ_A is the mole fraction of component A) which is a statistical term resulting from the mixing of component A with solvent molecules. The unitary change in entropy S_A'

for transfer of a hydrocarbon side chain from water to the nonaqueous interior of a protein is positive. Hence, its contribution ($-T\Delta S$) to the Gibbs free energy favors transfer of a side chain to the protein's interior. Note that in the presence of denaturing agent (such as 8 M urea), a side chain exhibits a reduced tendency to reside in the apolar interior⁶. About one-third to one-half of all amino acid residues in a protein have apolar side chains, thereby accounting for the cumulatively unfavorable effect of 8 M urea on protein folding.

Fersht and Serrano⁷ recently described the systematic application of site-directed mutagenesis, together with calorimetry, to evaluate how specific side chains play roles in stabilizing protein structure. Many self-assembly processes are entropy driven, and Lauffer⁸ proposed that water is released from apolar patches on the unpolymerized subunits as these patches interact with each other in the assembled state. In the case of microtubules, the warm-induced polymerization and cold-induced depolymerization are hallmarks of an entropy-driven process. Finally, although the common practices of enzyme purification are storage and processing at ice-water temperatures, some enzymes are cold-inactivated, and the experienced investigator will test the activity and structural properties of a newly characterized enzyme after storage at several different temperatures.

¹W. Kauzmann (1959) *Adv. Protein Chem.* **14**, 1.

²C. Tanford (1962) *J. Amer. Chem. Soc.* **84**, 4240.

³C. Tanford (1980) *The Hydrophobic Effect: Formation of Micelles and Biological Membranes*, 2nd ed., Wiley, New York.

⁴F. M. Richards in *Protein Folding* (T. E. Creighton, ed.) pp. 1–58, Freeman, San Francisco.

⁵C. R. Cantor & P. R. Schimmel (1980) *Biophysical Chemistry*, vol. 1, pp. 281–288, Freeman, San Francisco.

⁶P. L. Whitley & C. Tanford (1962) *J. Biol. Chem.* **237**, 1735.

⁷A. R. Fersht & L. Serrano (1993) *Curr. Opin. Struct. Biol.* **3**, 75.

⁸M. A. Lauffer (1975) *Entropy-Driven Processes in Biology*, Springer-Verlag, Berlin.

D-2-HYDROXY-ACID DEHYDROGENASE

This enzyme [EC 1.1.99.6] catalyzes the reaction of (*R*)-lactate with an acceptor to produce pyruvate and a reduced acceptor. The enzyme, which requires FAD and a zinc ion, will utilize a number of (*R*)-2-hydroxy acids as substrates.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

(S)-2-HYDROXY-ACID OXIDASE

This enzyme [EC 1.1.3.15] (also referred to as glycolate oxidase, hydroxy-acid oxidase A, and hydroxy-acid oxidase B) catalyzes the reaction of an (*S*)-2-hydroxy acid with dioxygen to produce a 2-oxo acid and hydrogen peroxide. FMN is the cofactor for this enzyme. This oxidase exists as two major isoenzymes. The A form preferentially oxidizes short-chain aliphatic hydroxy acids whereas the B form preferentially oxidizes long-chain and aromatic hydroxy acids.

N. E. Tolbert (1981) *Ann. Rev. Biochem.* **50**, 133.

3-HYDROXYACYL-CoA DEHYDROGENASE

This enzyme [EC 1.1.1.35] catalyzes the reaction of an (*S*)-3-hydroxyacyl-CoA with NAD⁺ to produce a 3-oxo-acyl-CoA and NADH. The enzyme will also utilize *S*-3-hydroxyacyl-*N*-acetylthioethanolamine and *S*-3-hydroxyacylhydrolipoate as substrates. The enzyme isolated from some sources can also utilize NADP⁺ as the coenzyme, albeit as a weaker substrate. In addition, there is a broad specificity with respect to the acyl chain length (note that there is a long-chain-length 3-hydroxyacyl-CoA dehydrogenase [EC 1.1.1.211]).

F. Lynen & O. Wieland (1955) *Meth. Enzymol.* **1**, 566.

 β -HYDROXYACYL THIOESTER (or, ACP) DEHYDRASE

This enzyme, officially known as 3-hydroxypalmitoyl-[acyl-carrier protein] dehydratase [EC 4.2.1.61], is the fatty-acid synthase component that catalyzes the conversion of (3*R*)-3-hydroxypalmitoyl-[acyl-carrier protein] to form 2-hexadecenoyl-[acyl-carrier protein] and water. This enzyme displays specificity toward 3-hydroxyacyl-[acyl-carrier protein] derivatives (with chain lengths from C₁₂ to C₁₆), with highest activity on the palmitoyl derivative. **See also** *Fatty Acid Synthetase*

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

S. J. Wakil & J. K. Stoops (1983) *The Enzymes*, 3rd ed., **16**, 3.

K. Bloch (1971) *The Enzymes*, 3rd ed., **5**, 441.

3-HYDROXYANTHRANILATE 3,4-DIOXYGENASE

This iron-dependent enzyme [EC 1.13.11.6] catalyzes the reaction of 3-hydroxyanthranilate and dioxygen to yield 2-amino-3-carboxymuconate semialdehyde. This product will spontaneously rearrange to quinolinic acid.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

3-HYDROXYANTHRANILATE OXIDASE

This enzyme [EC 1.10.3.5] catalyzes the reaction of 3-hydroxyanthranilate with dioxygen to produce 6-imino-5-oxocyclohexa-1,3-dienecarboxylate and hydrogen peroxide.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

3-HYDROXYASPARTATE ALDOLASE

This enzyme [EC 4.1.3.14] reversibly converts *erythro*-3-hydroxy-L-aspartate to glycine and glyoxylate.

R. G. Gibbs & J. G. Morris (1970) *Meth. Enzymol.* **17A**, 981.

***erythro*- β -HYDROXYASPARTATE DEHYDRATASE**

This pyridoxal-phosphate-dependent enzyme [EC 4.2.1.38] catalyzes the reaction of *erythro*-3-hydroxy-L-aspartate with water to produce oxaloacetate, ammonia, and water.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

R. G. Gibbs & J. G. Morris (1970) *Meth. Enzymol.* **17A**, 981.

4-HYDROXYBENZOATE DECARBOXYLASE

This enzyme [EC 4.1.1.61] catalyzes the conversion of 4-hydroxybenzoate to phenol and carbon dioxide.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

3-HYDROXYBENZOATE 4-MONOOXYGENASE

This enzyme [EC 1.14.13.23], also called 3-hydroxybenzoate 4-hydroxylase, catalyzes the reaction of 3-hydroxybenzoate with NADPH and dioxygen to produce 3,4-dihydroxybenzoate, NADP⁺, and water. FAD is used as a cofactor and the enzyme can also use analogs of 3-hydroxybenzoate substituted in the 2, 4, 5, and 6 positions as substrates.

B. Entsch (1990) *Meth. Enzymol.* **188**, 138.

V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

3-HYDROXYBENZOATE 6-MONOOXYGENASE

This enzyme [EC 1.14.13.24], also known as 3-hydroxybenzoate 6-hydroxylase, catalyzes the reaction of 3-hydroxybenzoate with NADH and dioxygen to produce 2,5-dihydroxybenzoate, NAD⁺, and water. NADPH can also be used as a substrate, although not as effectively

as NADH. FAD is used as a cofactor and the enzyme can utilize a number of analogs of 3-hydroxybenzoate substituted in the 2, 4, 5, and 6 positions as substrates.

B. Entsch (1990) *Meth. Enzymol.* **188**, 138.

V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

4-HYDROXYBENZOATE 3-MONOOXYGENASE

This enzyme [EC 1.14.13.2], also known as *p*-hydroxybenzoate hydroxylase, catalyzes the reaction of 4-hydroxybenzoate, NADPH, and dioxygen to produce protocatechuate, NADP⁺, and water. FAD is used as a cofactor by this enzyme. The enzyme from *Pseudomonas* is very specific for the coenzyme substrate whereas 4-hydroxybenzoate 3-monooxygenase (NAD(P)H) [EC 1.14.13.33] can utilize NADH and NADPH equally well. The enzyme isolated from *Corynebacterium cyclohexanicum* is highly specific for 4-hydroxybenzoate.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.

B. Entsch (1990) *Meth. Enzymol.* **188**, 138.

V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

3-HYDROXYBUTYRATE DEHYDROGENASE

This enzyme [EC 1.1.1.30] catalyzes the reversible reaction of (*R*)-3-hydroxybutanoate with NAD⁺ to produce acetoacetate and NADH. Other 3-hydroxymonocarboxylic acids can act as substrates as well.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

J. D. McGarry & D. W. Foster (1980) *Ann. Rev. Biochem.* **49**, 395.

4-HYDROXYBUTYRATE DEHYDROGENASE

This enzyme [EC 1.1.1.61] catalyzes the reaction of 4-hydroxybutanoate with NAD⁺ to produce succinate semialdehyde and NADH.

J. K. Hardman (1962) *Meth. Enzymol.* **5**, 778.

3-HYDROXYBUTYRYL-CoA EPIMERASE

This enzyme [EC 5.1.2.3] catalyzes the interconversion of (*S*)-3-hydroxybutanoyl-CoA and (*R*)-3-hydroxybutanoyl-CoA.

J. R. Stern (1962) *Meth. Enzymol.* **5**, 557.

HYDROXYINDOLE O-METHYLTRANSFERASE

This enzyme [EC 2.1.1.4], also referred to as acetylserotonin methyltransferase, catalyzes the reaction of *S*-adenosylmethionine with *N*-acetylserotonin to produce *S*-adenosylhomocysteine and *N*-acetyl-5-methoxytryptamine.

D. Sugden, V. Ceña & D. C. Klein (1987) *Meth. Enzymol.* **142**, 590.

4-HYDROXY-2-KETOGLUTARATE ALDOLASE

This enzyme [EC 4.1.3.16], also known as 2-keto-4-hydroxyglutarate aldolase and 2-oxo-4-hydroxyglutarate aldolase, catalyzes the reversible conversion of 4-hydroxy-2-oxoglutarate to pyruvate and glyoxylate. Interestingly, the enzyme is reported to be able to act on both stereoisomers.

K. N. Allen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 135.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

W. A. Wood (1972) *The Enzymes*, 3rd ed., **7**, 281.

HYDROXYLAMINE REDUCTASE

This enzyme [EC 1.6.6.11] catalyzes the reaction of hydroxylamine with NADH to produce ammonia, water, and NAD⁺. A number of hydroxamates can also act as substrates.

M. Zucker & A. Nason (1955) *Meth. Enzymol.* **2**, 416.

4-HYDROXYMANDELONITRILE LYASE

This enzyme [EC 4.1.2.11], also known as hydroxymandelonitrile lyase or hydroxynitrile lyase, catalyzes the conversion of (*S*)-4-hydroxymandelonitrile to cyanide and 4-hydroxybenzaldehyde. Aliphatic hydroxynitriles do not serve as substrates for this enzyme, unlike that of (*S*)-hydroxynitrilase [EC 4.1.2.39].

M. K. Seely & E. E. Conn (1971) *Meth. Enzymol.* **17B**, 239.

β-HYDROXY-β-METHYLGLUTARYL-CoA LYASE

This enzyme [EC 4.1.3.4] catalyzes the conversion of (*S*)-3-hydroxy-3-methylglutaryl-CoA to acetyl-CoA and acetoacetate.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

K. M. Gibson (1988) *Meth. Enzymol.* **166**, 219.

M. J. P. Higgins, J. A. Kornblatt & H. Rudney (1972) *The Enzymes*, 3rd ed., **7**, 407.

β -HYDROXY- β -METHYLGLUTARYL-CoA REDUCTASE

β -Hydroxy- β -methylglutaryl-CoA reductase (NADPH) [EC 1.1.1.34] catalyzes the reaction of (*S*)-3-hydroxy-3-methylglutaryl-CoA and two NADPH to produce (*R*)-mevalonate, coenzyme A, and two NADP⁺. The enzyme is inactivated by β -hydroxy- β -methylglutaryl-CoA reductase (NADPH) kinase [EC 2.7.1.109] and reactivated by [hydroxymethylglutaryl-CoA reductase (NADPH)]-phosphatase [EC 3.1.3.47]. β -Hydroxy- β -methylglutaryl-CoA reductase (NADH) [EC 1.1.1.88] catalyzes the reaction of 3-hydroxy-3-methylglutaryl-CoA and two NADH to produce mevalonate, coenzyme A, and two NAD⁺.

D. M. Gibson & R. A. Parker (1987) *The Enzymes*, 3rd ed., **18**, 179.
 T.-Y. Chang (1983) *The Enzymes*, 3rd ed., **16**, 491.
 Z. H. Beg & H. B. Brewer, Jr. (1981) *Curr. Top. Cell. Reg.* **20**, 139.
 V. W. Rodwell, D. J. McNamara & D. J. Shapiro (1973) *Adv. Enzymol.* **38**, 373.

 β -HYDROXY- β -METHYLGLUTARYL-CoA REDUCTASE (NADPH) KINASE

This enzyme [EC 2.7.1.109], also known simply as reductase kinase, catalyzes the reaction of ATP with the enzyme 3-hydroxy-3-methylglutaryl-CoA reductase (NADPH) producing ADP and the phosphorylated form of the reductase. This phosphorylation inactivates the reductase. Histones can substitute for the reductase as substrates.

A. M. Edelman, D. K. Blumenthal & E. G. Krebs (1987) *Ann. Rev. Biochem.* **56**, 567.

 β -HYDROXY- β -METHYLGLUTARYL-CoA SYNTHASE

This enzyme [EC 4.1.3.5] catalyzes the reversible reaction of acetyl-CoA with acetoacetyl-CoA and water to generate (*S*)-3-hydroxy-3-methylglutaryl-CoA and coenzyme A.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.
 M. J. P. Higgins, J. A. Kornblatt & H. Rudney (1972) *The Enzymes*, 3rd ed., **7**, 407.

3-HYDROXY-2-METHYLPYRIDINE-4,5-DICARBOXYLATE 4-DECARBOXYLASE

This enzyme [EC 4.1.1.51] catalyzes the conversion of 3-hydroxy-2-methylpyridine-4,5-dicarboxylate to 3-hy-

droxy-2-methylpyridine-5-carboxylate and carbon dioxide.

R. W. Burg (1970) *Meth. Enzymol.* **18A**, 634.

5-HYDROXYMETHYLPYRIMIDINE KINASE

This enzyme [EC 2.7.1.49] catalyzes the reaction of ATP with 4-amino-2-methyl-5-hydroxymethylpyrimidine to produce ADP and 4-amino-2-methyl-5-phosphomethylpyrimidine. In addition, ATP can be replaced as the substrate by CTP, UTP, and GTP.

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

6-HYDROXYNICOTINATE REDUCTASE

This enzyme [EC 1.3.7.1], also referred to as 6-oxotetrahydronicotinate dehydrogenase, catalyzes the reaction of 1,4,5,6-tetrahydro-6-oxonicotinate with oxidized ferredoxin to produce 6-hydroxynicotinate and the reduced ferredoxin.

L. Tsai & E. R. Stadtman (1971) *Meth. Enzymol.* **18B**, 233.

2-HYDROXY-3-OXOPROPIONATE REDUCTASE

This enzyme [EC 1.1.1.60], also referred to as tartronate semialdehyde reductase, catalyzes the reversible reaction of (*R*)-glycerate with NAD(P)⁺ to produce 2-hydroxy-3-oxopropanoate and NAD(P)H.

H. L. Kornberg & A. M. Gotto (1966) *Meth. Enzymol.* **9**, 240.

4-HYDROXYPHENYLACETATE 3-MONOOXYGENASE

This enzyme [EC 1.14.13.3], also called *p*-hydroxyphenylacetate 3-hydroxylase, catalyzes the reaction of 4-hydroxyphenylacetate with NADH and dioxygen to produce 3,4-dihydroxyphenylacetate, NAD⁺, and water. FAD is used as a cofactor.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

4-HYDROXYPHENYLPYRUVATE HYDROXYLASE

This enzyme [EC 1.13.11.27], also known as 4-hydroxyphenylpyruvate dioxygenase, catalyzes the reaction of 4-hydroxyphenylpyruvate with dioxygen to generate homogentisate and carbon dioxide. Iron ions are needed as cofactors.

A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.

- S. Lindstedt & B. Odelhög (1987) *Meth. Enzymol.* **142**, 139 and 143.
J. H. Fellman (1987) *Meth. Enzymol.* **142**, 148.
O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

4-HYDROXYPROLINE EPIMERASE

This enzyme [EC 5.1.1.8] catalyzes the interconversion of *trans*-4-hydroxy-L-proline to *cis*-4-hydroxy-D-proline. It also interconverts *trans*-4-hydroxy-D-proline and *cis*-4-hydroxy-L-proline.

- M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.
E. Adams & L. Frank (1980) *Ann. Rev. Biochem.* **49**, 1005.
E. Adams (1976) *Adv. Enzymol.* **44**, 69.
R. Kuttan & A. N. Radhakrishnan (1973) *Adv. Enzymol.* **37**, 273.
E. Adams (1972) *The Enzymes*, 3rd ed., **6**, 479.

β -HYDROXYPROPIONATE DEHYDROGENASE

This enzyme [EC 1.1.1.59] catalyzes the reaction of 3-hydroxypropanoate with NAD⁺ to produce 3-oxopropionate and NADH.

- M. J. Coon & W. G. Robinson (1962) *Meth. Enzymol.* **5**, 451.

15-HYDROXYPROSTAGLANDIN DEHYDROGENASE

15-Hydroxyprostaglandin dehydrogenase (NAD⁺) [EC 1.1.1.141] catalyzes the reaction of (5Z,13E)-(15S)-11 α , 15-dihydroxy-9-oxoprost-13-enoate with NAD⁺ to produce (5Z,13E)-11 α -hydroxy-9,15-dioxoprost-13-enoate and NADH. This enzyme will also use prostaglandins E₂, F_{2 α} and B₁ as substrates. However, prostaglandin D₂ will not serve as a substrate. In this case, 15-hydroxyprostaglandin D dehydrogenase (NADP⁺) [EC 1.1.1.196] is used. 15-Hydroxyprostaglandin dehydrogenase (NADP⁺) [EC 1.1.1.197] catalyzes the reaction of (5Z,13E)-(15S)-11 α ,15-dihydroxy-9-oxoprost-13-enoate with NADP⁺ to produce (5Z,13E)-11 α -hydroxy-9,15-dioxoprost-13-enoate and NADPH. This enzyme will also use prostaglandins E₂, F_{2 α} and B₁ as substrates. However, prostaglandin D₂ will not serve as a substrate. In some systems, this enzyme may be identical with prostaglandin E₂ 9-reductase [EC 1.1.1.189].

- C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

HYDROXYPYRUVATE REDUCTASE

This enzyme [EC 1.1.1.81], also referred to as D-glycerate dehydrogenase, catalyzes the reversible reaction of D-

glycerate with NAD(P)⁺ to produce hydroxypyruvate and NAD(P)H. **See also** *Glycerate Dehydrogenase*

- L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.
C. Krema & M. E. Lidstrom (1990) *Meth. Enzymol.* **188**, 373.

3 α -HYDROXYSTEROID DEHYDROGENASE

This enzyme [EC 1.1.1.50], also referred to as hydroxy-prostaglandin dehydrogenase, catalyzes the reaction of androsterone with NAD(P)⁺ to produce 5 α -androstane-3,17-dione and NAD(P)H. Other 3 α -hydroxysteroids can act as substrates as well as 9-, 11- and 15-hydroxy-prostaglandins. The stereochemistry is B-specific with respect to the pyridine coenzymes.

- O. Berséus, H. Danielsson & K. Einarsson (1969) *Meth. Enzymol.* **15**, 551.
S. S. Koide (1969) *Meth. Enzymol.* **15**, 651.

3(or 17) β -HYDROXYSTEROID DEHYDROGENASE

This enzyme [EC 1.1.1.51] catalyzes the reaction of testosterone with NAD(P)⁺ to produce androst-4-ene-3,17-dione and NAD(P)H. The enzyme can also use other 3 β - or 17 β -hydroxysteroids as substrates.

- J. Jarabak (1969) *Meth. Enzymol.* **15**, 746.

11 β -HYDROXYSTEROID DEHYDROGENASE

This enzyme [EC 1.1.1.146] catalyzes the reaction of an 11 β -hydroxysteroid with NADP⁺ to produce an 11-oxosteroid and NADPH.

- R. Benediktsson & C. R. W. Edwards (1996) *Essays in Biochem.* **31**, 23.

16 α -HYDROXYSTEROID DEHYDROGENASE

This enzyme [EC 1.1.1.147] catalyzes the reaction of a 16 α -hydroxysteroid with NAD(P)⁺ to produce a 16-oxosteroid and NAD(P)H.

- H. Breuer & R. Knuppen (1969) *Meth. Enzymol.* **15**, 691.

20-HYDROXYSTEROID DEHYDROGENASE

This enzyme [EC 1.1.1.53] catalyzes the reaction of androstane-3 α ,17 β -diol with NAD⁺ to produce 17 β -hydroxyandrostane-3-one and NADH. The 3 α -hydroxyl group or the 20 β -hydroxyl group of pregnane and androstane steroids can act as the substrate.

- W. G. Wiest (1969) *Meth. Enzymol.* **15**, 638.

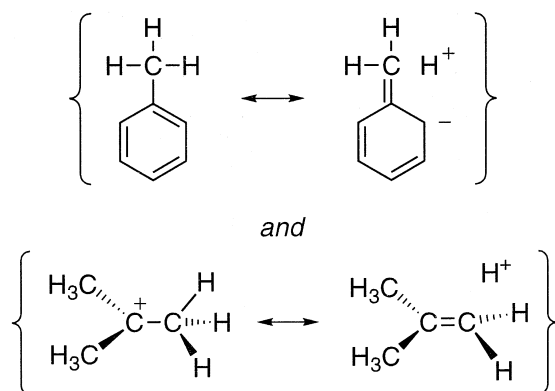
21-HYDROXYSTEROID DEHYDROGENASE

21-Hydroxysteroid dehydrogenase (NAD^+) [EC 1.1.1.150] catalyzes the reaction of pregnan-21-ol with NAD^+ to produce pregnan-21-al and NADH. Other 21-hydroxycorticosteroids can also serve as substrates. 21-Hydroxysteroid dehydrogenase (NADP^+) [EC 1.1.1.151] catalyzes the reaction of pregnan-21-ol with NADP^+ to produce pregnan-21-al and NADPH. Other 21-hydroxycorticosteroids can also serve as substrates.

C. Monder & C. S. Furfine (1969) *Meth. Enzymol.* **15**, 667.

HYPERCONJUGATION

Delocalization of electrons in one or more σ -bonds that interact with the electrons in a π -electron network.



Hyperconjugation is less effective than conjugation in providing a mechanism for delocalizing electrons. This effect can often be detected spectrally by a slight shift to higher wavelength of UV light absorption, and the magnitude of the spectral shift can be predicted by the Woodward's rules.

N. Muller & R. S. Mulliken (1958) *J. Amer. Chem. Soc.* **80**, 3489.
 M. J. S. Dewar (1962) *Hyperconjugation*, Ronald Press, New York.
 T. Sorensen (1967) *J. Amer. Chem. Soc.* **89**, 3782.
 E. M. Arnett & J. W. Larsen (1969) *J. Amer. Chem. Soc.* **91**, 34.
 L. Radom (1982) *Prog. Theoret. Org. Chem.* **3**, 1.

HYPERCYCLE

A model for biological self-organization based on autocatalytic processes during early stages of biological evolution. The concept was introduced by Eigen¹ to characterize a functional entity which integrates information stored in individual self-replicating elements. These self-replicating elements compete, much like species do in Nature, such that a single species eventually is selected, attended by the disappearance of less efficient competi-

tors. To suppress competition, Eigen and Schuster^{2,3} introduced specific second-order coupling terms [that are proportional to the product of two population numbers (or concentration in chemical processes)] into their kinetic equations that form a closed catalytic cycle.

¹M. Eigen (1971) *Quart. Rev. Biophys.* **4**, 149.

²M. Eigen & P. Schuster (1978) *Naturwissenschaften* **65**, 341.

³M. Eigen & P. Schuster (1979) *The Hypercycle- A Principle of Natural Self-Organization*, Springer-Verlag, Berlin.

HYPERPOLARIZATION

With respect to membranes, an increase in the polarization of a membrane. **See** *Depolarization*; *Action Potential*

HYPERREACTIVITY

Having an increased or elevated reactivity. This term has been used in reference to the relative activity of amino acyl residues at the active sites of enzyme. The immediate environment (*i.e.*, the microenvironment) may allow simple reagents to react faster with the amino acid than would normally be expected. Thus, in labeling of proteins with active site-directed reagents, an investigator should always consider the basis of increased reactivity: Is it due to facilitation of the reaction by increased affinity (*i.e.*, affinity labeling), or is it due to increased activity of the amino acyl side chain (*e.g.*, perhaps increased nucleophilicity due to the microenvironment).

Procedures¹⁻³ for distinguishing between the two means of facilitation include: (1) Use of a reagent that does not bind as an affinity label. If facilitation is due to hyperreactivity, this reagent should also be more reactive. (2) Use of transition-state analysis. A favorable change in the entropy of activation ($\Delta S^\ddagger$) would imply facilitation via affinity labeling whereas a more favorable change in the enthalpy of activation ($\Delta H^\ddagger$) implies hyperreactivity. However, a certain caution should always be exercised since other factors, *e.g.* differential solvation effects, can result in a certain degree of compensation between $\Delta H^\ddagger$ and $\Delta S^\ddagger$.

¹E. P. Lennette & B.V. Plapp (1979) *Biochemistry* **18**, 3933.

²E. P. Lennette & B.V. Plapp (1979) *Biochemistry* **18**, 3938.

³B. V. Plapp (1982) *Meth. Enzymol.* **87**, 469.

HYPERTONIC

1. Referring to a larger osmotic pressure (or ionic strength) of a particular solution relative to some stan-

dard solution (*e.g.*, blood plasma). In a hypertonic solution, cells would tend to shrink. 2. Referring to states that have a larger degree of tension.

HYPOTAUROCYAMINE KINASE

This enzyme [EC 2.7.3.6] catalyzes the reaction of ATP with hypotaurocyamine to produce ADP and *N*^w-phosphohypotaurocyamine. Taurocyamine will also act as a substrate, but not as efficiently.

N. v. Thoai, L.-A. Pradel & R. Kassab (1970) *Meth. Enzymol.* **17A**, 1002.

HYPOTHESIS

A trial idea, assumption, or conjecture regarding the behavior of a system. Such unproven assumptions help to establish organized thought and to develop experimental design. **See** *Statistics*; *Alternative Hypothesis*; *Null Hypothesis*

HYPOTHESIS TESTING

A statistical protocol for testing the likely validity of one hypothesis about the behavior of an experimental parameter against another, thereby affording a rational basis for accepting or rejecting a particular set of assumptions. **See** *Statistics*

HYPOXANTHINE (GUANINE) PHOSPHORIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.8] (also known as hypoxanthine phosphoribosyltransferase, IMP pyrophosphorylase, transphosphoribosidase, and guanine phosphoribosyltransferase) catalyzes the purine salvage reaction of hypoxanthine with 5-phospho- α -D-ribose 1-diphosphate to

produce IMP and pyrophosphate (or, diphosphate). The enzyme can also use guanine and 6-mercaptopurine as substrates.

W. N. Keley & J. B. Wyngaarden (1974) *Adv. Enzymol.* **41**, 1.

HYPSOCHROMIC SHIFT

An effect observed in the spectrum of a chemical species in which a substituent, solvent, change in environment, or other effect causes the electronic absorption spectrum to shift to shorter wavelengths. The opposite effect is referred to as a bathochromic shift. The hypsochromic shift is also known as the blue shift.

HYSTERESIS

Slow transitions produced by enzyme isomerizations. This behavior can lead to a type of cooperativity that is generally associated with ligand-induced conformational changes^{1,2}. A number of enzymes are also known to undergo slow oligomerization reactions, and these enzymes may display unusual kinetic properties. If this is observed, it is advisable to determine the time course of enzyme activation or inactivation following enzyme dilution. **See** *Cooperativity*; *Bifurcation Theory*; *Lag Time*

¹K. E. Neet & G. R. Ainslie, Jr., (1980) *Meth. Enzymol.* **64**, 192.

²K. E. Neet (1995) *Meth. Enzymol.* **249**, 519.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Activation energetics, **64**, 225, 226; analysis, **64**, 196-199; artifact, **64**, 196-200; classification, **64**, 217-220 [cooperativity (monomeric, **64**, 221; oligomeric, **64**, 222); incidental to function, **64**, 217-220; physiological damping mechanism, **64**, 220, 221]; computer program, **64**, 203-205; deadend branch, **64**, 206-208; graphical analysis, **64**, 200-203; isomerization, **64**, 194; isotope trapping, **64**, 58; model, **64**, 205-213; oligomerization, **64**, 214-217; oxygen-18 exchange, **64**, 82; physiological role, **64**, 195, 196; statistical method, **64**, 203-205; structural basis, **64**, 223-225.

I₅₀ (or I_{0.5})

The concentration of an inhibitor that reduces the velocity of an enzyme-catalyzed reaction or other metabolic process to 50% (or 0.5) the activity observed in the inhibitor's absence. For reversible inhibitors, the magnitude of I₅₀ or I_{0.5} is directly related to the enzyme-inhibitor dissociation constant(s) defining competitive, noncompetitive, and uncompetitive inhibition. However, for very tight-binding inhibitors, the magnitude of I_{0.5} can be substantially different than the dissociation constant. Likewise, if a metabolic inhibitor acts on more than one step (as effectors often do in metabolic pathways) or if one is dealing with cooperative inhibitor binding (as observed in allosteric control), then I_{0.5} may again yield an inaccurate estimate of the microscopic inhibition constants.

ICEBERG EFFECT

A negative change in entropy associated with the decreased freedom of motion of water molecules in the vicinity of a nonpolar group. As pointed out by Jencks¹, one should not be too literal when interpreting this behavior, because the system is dynamic, and not frozen. The term flickering cluster implies some reduction in libration and rotation without complete loss.

¹W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York].

IDURONATE-2-SULFATASE

This enzyme [EC 3.1.6.13], also known as iduronate-2-sulfate sulfatase and chondroitinsulfatase, catalyzes the

hydrolysis of the 2-sulfate groups of the L-iduronate 2-sulfate units of dermatan sulfate, heparan sulfate, and heparin.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

α-L-IDURONIDASE

This enzyme [EC 3.2.1.76] catalyzes the hydrolysis of α-L-iduronosidic linkages in desulfated dermatan.

H. Kresse & J. Glössl (1987) *Adv. Enzymol.* **60**, 217.

IMBALANCE

A condition arising when reaction parameters for different bond-breaking or bond-making processes have developed synchronously as the transition state is approached. Imbalance is common in elimination and addition reactions as well as in proton transfer reactions.

See also *Synchronous*; *Synchronization*

C. F. Bernasconi (1992) *Adv. Phys. Org. Chem.* **27**, 119.

IMIDAZOLEACETATE HYDROXYLASE

This enzyme [EC 1.14.13.5], also known as imidazoleacetate 4-monooxygenase, catalyzes the reaction of 4-imidazole acetate with NADH and dioxygen to produce 5-hydroxy-4-imidazole acetate, NAD⁺, and water. FAD is the cofactor for this enzyme.

S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.

IMIDAZOLEGLYCEROL-PHOSPHATE DEHYDRATASE

This enzyme [EC 4.2.1.19] catalyzes the conversion of D-erythro-1-(imidazol-4-yl)glycerol 3-phosphate to 3-(imidazol-4-yl)-2-oxopropyl phosphate and water.

R. G. Martin, M. A. Berberich, B. N. Ames, W. W. Davis, R. F. Goldberger & J. D. Yournon (1971) *Meth. Enzymol.* **17B**, 3.

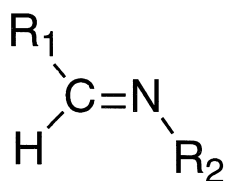
IMIDAZOLONEPROPIONASE

This enzyme [EC 3.5.2.7], also referred to as imidazolone-5-propionate hydrolase, catalyzes the hydrolysis of 4-imidazolone-5-propanoate to generate *N*-formimino-L-glutamate.

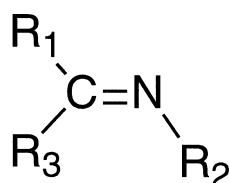
B. Magasanik, E. Kaminskas & J. Kimhi (1971) *Meth. Enzymol.* **17B**, 55.

IMINE INTERMEDIATE

An enzymatic reaction intermediate formed by nucleophilic attack by an amine on a carbonyl group of an aldehyde (forming an aldimine) or a ketone (forming a ketimine), followed by elimination of water.



Aldimine



Ketimine

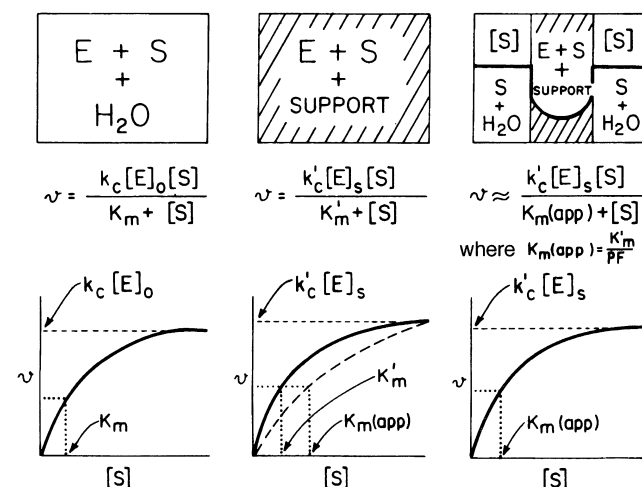
Such intermediates are known to form between substrate and enzyme in the aldolase reaction and between pyridoxal 5-phosphate and the amino group of enzyme or substrate in aminotransferase reactions. In the latter case, aldimine formation accounts for the high affinity of coenzyme binding to apotransaminases.

IMMOBILIZED ENZYME

Biological catalysts in the form of enzymes, cells, organelles, or synzymes that are tethered to a fixed bed, polymer, or other insoluble carrier or entrapped by a semi-impermeable membrane^{1,2}. Immobilization often confers added stability, permits reuse of the biocatalyst, and allows the development of flow reactors. The mode of immobilization may produce distinct populations of biocatalyst, each exhibiting different activities within the same sample. The study of immobilized enzymes can also provide insights into the chemical basis of enzyme latency, a well-known phenomenon characterized by the limited availability of active enzyme as a consequence of immobilization and/or encapsulization.

The specialized kinetic properties³⁻⁹ of matrix-bound enzymes are thought to result from at least four factors: (a) altered enzyme conformation and stability attending immobilization; (b) altered solvent and electrostatic en-

vironment for the immobilized enzyme; (c) changes in the effective substrate concentration accessible to the enzyme within the matrix; and (d) altered diffusional control on substrate binding.



Effect of immobilization on the dynamic properties of enzymes. Examples of three different experimentally imposed conditions include: (a) enzyme and substrate are present in an aqueous solution; (b) enzyme and substrate are immobilized on or within a solid support; and (c) immobilized enzyme acts on a substrate present within an aqueous solution. The rate laws and curves that are presented below each diagram indicate the expected rate behavior. (From K. J. Laidler & P. S. Bunting (1982) *Meth. Enzymol.* **64**, 229.)

Although the kinetics of immobilized enzymes is especially relevant to industrial process chemistry, there is growing recognition that enzymes in crowded solutions may likewise experience diffusional control of the catalytic properties.

Use of immobilized catalysts requires thorough characterization of physical and chemical properties, and one cannot overemphasize the importance of describing a system adequately to ensure reproducibility in the hands of other experimenters. Buchholz and Klein¹⁰ offered the following recommendations: (1) Provide a general description of the immobilized enzyme system, including the reaction scheme, type of carrier, exact source of the biocatalyst(s), known side reactions, and method of immobilization. (2) Describe biocatalyst(s) preparation in sufficient detail for others to replicate the preparation. Dry-weight yield of the preparation should be given as well as the amount of activity remaining in the supernatant after immobilization. The method for determining catalytic activity should always be provided as well. (3) Define physical and chemical properties, including the

shape of the biocatalyst particles, their size distribution, typical swelling behavior, as well as the methods used for making these determinations. In column configurations, dimensions should be provided along with compression behavior and flow rates when pertinent. In stirred systems, effectiveness of suspension as well as any abrasion should also be assessed. It is worth noting that abrasion can also occur with fluidized beds. (4) Determine kinetic parameters that describe the system: (a) Compare the dependence of initial rate on substrate concentration for both immobilized and free biocatalyst(s). The time elapsed between the preparation of the immobilized system and the kinetic experiments should be noted as well as pertinent storage conditions. (b) Assess effects of pH as well as buffer type and concentration on the kinetics and stability of the system. (c) If temperature effects are studied, then those conditions should be thoroughly described. (d) When using intact cells as biocatalyst, it is advisable to describe effects of growth conditions on the immobilized system. (e) Determine the diffusion limitations of the system (including the effects of particle size or enzyme load). (f) Present information concerning the degree of conversion for different values of residence time. (g) Describe details concerning the storage stability of the immobilized system should be as actual data values rather than as extrapolated parameters. (h) Study the operational stability (*i.e.*, residual initial rates after different periods of operation).

Immobilized enzymes often behave differently than their soluble counterparts with respect to their susceptibility to inhibition¹¹.

¹K. Mosbach, ed. (1977) *Meth. Enzymol.* vol. **44**, (1987) vols. **135** and **136**, and (1988) vol. **137**.

²M.-D. Legoy & D. Thomas, eds. (1995) *Ann. New York Acad. Sci.*, vol. **750**.

³K. J. Laidler & P. S. Bunting (1973) *The Chemical Kinetics of Enzyme Action*, Oxford Univ. Press, London.

⁴L. Goldstein (1976) *Meth. Enzymol.* **44**, 397.

⁵J. M. Engasser & C. Horvath (1976) in *Applied Biochemistry and Bioengineering* (L. B. Wingard, Jr., E. Katchalski-Katzir & L. Goldstein, eds.), Vol. **1**, p. 127, Academic Press, New York.

⁶K. J. Laidler & P. S. Bunting (1980) *Meth. Enzymol.* **64**, 227.

⁷D. Gabel & V. Kasche (1976) *Meth. Enzymol.* **44**, 526.

⁸S. K. Duggal & K. Buchholz (1982) *Eur. J. Appl. Microbiol. Biotech.* **16**, 81.

⁹D. S. Clark & J. E. Bailey (1984) *Biotechnol. Bioeng.* **16**, 231.

¹⁰K. Buchholz & J. Klein (1987) *Meth. Enzymol.* **135**, 3.

¹¹V. V. Mozhaev & K. Martinek (1982) *Enzyme Microb. Technol.* **4**, 299.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Functional groups on enzymes suitable for binding to matrices,

44, 11; a survey of enzyme coupling techniques, **135**, 30; immobilization of enzymes to agar, agarose, and Sephadex supports, **44**, 19; enzymes immobilized to cellulose, **44**, 46; immobilization of enzymes to various acrylic copolymers, **44**, 53; immobilization of enzymes on hydroxylalkyl methacrylate gels, **44**, 66; enzymes covalently bound to polyacrylic and polymethacrylic copolymers, **44**, 84; immobilization of enzymes on neutral and ionic carriers, **44**, 107; immobilization of enzymes on nylon, **44**, 118; covalent coupling methods for inorganic support materials, **44**, 134; tresyl chloride-activated supports for enzyme immobilization, **135**, 65; immobilization of proteins and ligands using chlorocarbonates, **135**, 84; 1,1'-carbonyldiimidazole-mediated immobilization of enzymes and affinity ligands, **135**, 102; covalent immobilization of proteins by techniques that permit subsequent release, **135**, 130; immobilization of glycoenzymes through carbohydrate chains, **135**, 141; immobilization of enzymes and microorganisms by radiation polymerization, **135**, 146; use of monoclonal antibodies for preparation of highly active immobilized enzymes, **135**, 160.

IMP CYCLOHYDROLASE

This enzyme [EC 3.5.4.10] catalyzes the hydrolysis of the purine ring of IMP to produce 5-formamido-1-(5-phosphoribosyl)imidazole-4-carboxamide.

J. G. Flaks & L. N. Lukens (1963) *Meth. Enzymol.* **6**, 52.

IMP DEHYDROGENASE

This oxidoreductase [EC 1.1.1.205], also known as inosine-5'-monophosphate dehydrogenase, catalyzes the reaction of IMP with NAD⁺ and water to produce xanthosine 5'-phosphate and NADH.

B. F. Cooper & F. B. Rudolph (1995) *Meth. Enzymol.* **249**, 188.

INDEFINITE POLYMERIZATION

Any polymerization reaction in which the product of each elongation step can itself also undergo further polymerization. When the same types of bonds and/or conformational states that are present in the reactant(s) are generated within product(s) during elongation, the process is referred to as isodesmic polymerization. Such is the case for the indefinite polymerization of actin, tubulin, hemoglobin S, and tobacco mosaic virus coat protein. **See** *Nucleation; Protein Polymerization; Actin Assembly Kinetics; Microtubule Assembly Kinetics; Microtubule Assembly Kinetics*

INDOLEAMINE 2,3-DIOXYGENASE

This enzyme [EC 1.13.11.42], also referred to as indoleamine-pyrrole 2,3-dioxygenase, catalyzes the reaction of tryptophan with dioxygen to form *N*-formylkynurenine. Heme participates as a cofactor. Many substituted and unsubstituted indoleamines, including melatonin, act as

substrates. In fact, among other roles, this enzyme participates in the degradation of melatonin.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

S. Kaufman (1993) *Ann. Rev. Nutr.* **13**, 261.

S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.

R. Yoshida & O. Hayaishi (1987) *Meth. Enzymol.* **142**, 188.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

INDOLEGLYCEROL PHOSPHATE SYNTHASE

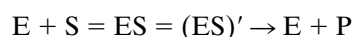
This enzyme [EC 4.1.1.48] catalyzes the conversion of 1-(2-carboxyphenylamino)-1-deoxy-D-ribulose 5-phosphate to generate 1-(indol-3-yl)glycerol 3-phosphate, carbon dioxide, and water. In some organisms, this enzyme is part of a multifunctional protein together with one or more components of the system for the biosynthesis of tryptophan (*i.e.*, anthranilate synthase, anthranilate phosphoribosyltransferase, tryptophan synthase, and phosphoribosylanthranilate isomerase).

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

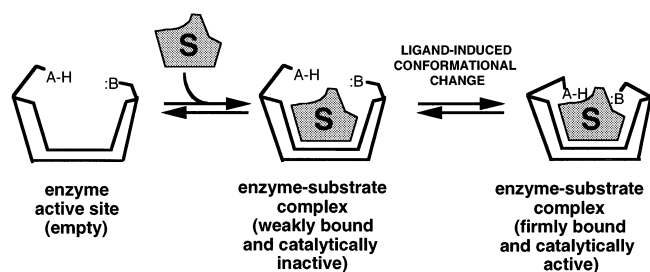
K. Kirschner, H. Szadkowski, T. S. Jardetzky & V. Hager (1987) *Meth. Enzymol.* **142**, 386.

INDUCED FIT MODEL

A conceptual framework¹⁻³ for interpreting the action of substrate binding in reorienting otherwise inactive groups on the enzyme into their catalytically active configuration. This implies an essential isomerization step:



Yeast hexokinase provides good examples of induced fit in two ways: first, the enzyme undergoes a dramatic conformational change upon binding glucose; and second, the binding of certain pentoses reorganizes the catalytic site so that water can replace glucose's 6-hydroxymethyl group with a resulting 50,000 times increase in ATPase activity.



Jencks⁴ has offered the view that the induced fit model provides a means for understanding the specificity and control of catalysis in terms of a substrate-induced conformational change, while not involving binding forces in reducing the reaction's activation energy. He indicates that induced fit has two consequences: (a) reduction in the observed binding constant relative to that which would have occurred were there perfect initial fit, and (b) the substrate that best utilizes specific binding forces to effect a conformational change should have the greatest maximal velocity. **See** *Cooperativity*; *Koshland-Nemethy-Filmer Model*

¹D. E. Koshland, Jr. (1958) *Proc. Natl. Acad. Sci. U.S.A.* **44**, 98.

²D. E. Koshland, Jr. (1960) *Adv. Enzymol.* **22**, 45.

³D. E. Koshland, Jr. & K. E. Neet (1968) *Annu. Rev. Biochem.* **37**, 359.

⁴W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, pp. 289-291, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York].

INDUCED SUBSTRATE INHIBITION

A term first introduced by Cleland^{1,2} to indicate that for ordered substrate binding mechanisms, addition of an inhibitor mimicking the first substrate may still permit binding of the second substrate. Hence, as long as the addition of the first substrate is not of the rapid equilibrium type, the presence of the inhibitor will "induce" substrate inhibition by the second substrate. An example of induced substrate inhibition is provided in the thymidylate synthase reaction³ where the second substrate methylene tetrahydrofolate becomes an inhibitor, but only in the presence of the inhibitor bromodeoxyuridine 5'-monophosphate.

¹W. W. Cleland (1990) *The Enzymes*, 3rd ed., **19**, 119.

²W. W. Cleland (1979) *Meth. Enzymol.* **63**, 500.

³P. V. Danenberg & K. D. Danenberg (1978) *Biochemistry* **17**, 4018.

INDUCED TRANSPORT ANALYSIS

An experimental protocol designed to detect and characterize isoenzyme mechanisms (that is, enzyme mechanisms requiring one or more isomerization steps of the free enzyme before the catalytic cycle is completed)¹⁻⁵. In iso mechanisms, the substrate and product bind to two different conformational forms of the enzyme. The induced transport approach requires preincubation of both labeled substrate and labeled product with enzyme, allowing the system to reach equilibrium; a large amount of unlabeled substrate is then added. If an iso step is

present, the labeled product will not compete with the unlabeled substrate, and the enzyme will catalyze the reverse synthesis of labeled substrate. Hence, there will be a transient decrease in labeled product concentration.

¹H. G. Britton (1966) *Arch. Biochem. Biophys.* **117**, 167.

²H. G. Britton (1973) *Biochem. J.* **133**, 255.

³H. G. Britton (1965) *Nature* **205**, 1323.

⁴H. G. Britton & J. B. Clarke (1968) *Biochem. J.* **110**, 161.

⁵K. L. Rebholz & D. B. Northrop (1995) *Meth. Enzymol.* **249**, 211.

INDUCTION

1. Cellular: A process by which the amount of a protein is increased through the activation of gene expression. 2. Electromagnetic: A change in the intensity or direction of a magnetic field resulting in an electromotive force in any conductor within the field: $V = -d\Phi/dt$ where V is the induced potential difference, Φ is the magnetic flux, and t is time. 3. Electrostatic: The process by which an electron charge is produced on a conductor due to the influence of an electric field.

INDUCTION PERIOD

Any slow phase (of a chemical reaction or physical process) that is followed by a faster phase. Such behavior is observed for systems prior to attaining the steady-state condition. **See also** *Series Reactions*

INDUCTIVE EFFECT

The polarization of a chemical bond caused by the polarization of an adjacent or nearby bond. The inductive effect depends only on the nature of the bonds and is transmitted through a chain of atoms by electrostatic induction. **See also** *Field Effect*

INDUCTOMERIC EFFECT

A molecular polarizability effect occurring by the inductive mechanism of electron displacement¹⁻³. Substituent polarizability is a factor governing reactivity.

¹IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

²C. K. Ingold (1953) *Structure and Mechanism in Organic Chemistry*, Cornell Univ. Press, Ithaca, New York.

³R. W. Taft & R. D. Topsom (1987) *Progr. Phys. Org. Chem.* **61**, 1.

INFLECTION, POINT OF

A point on a curve at which the concavity of the curve changes (*e.g.*, up to down or *vice versa*). The second derivative of the equation of the curve will be zero at this point. A number of curves found in enzyme kinetic

studies have inflection points: *e.g.*, cooperative systems, pH-rate studies, *etc.*

INHIBITION

The process by which the presence of a substance, known as the inhibitor, decreases reaction rate. **See** *Degree of Inhibition; Specific Type of Inhibition*

INHIBITOR

A substance, agent, or factor that lowers the rate of a reaction. An inhibitor that acts directly by binding to an enzyme is often referred to as an enzyme inhibitor. **See also** *Degree of Inhibition; Effector; Modifier; Inactivator; Competitive Inhibitor; Noncompetitive Inhibitor; Uncompetitive Inhibitor; Partial Inhibitor; Irreversible Inhibitor*

INHIBITORY INDEX

A value used to assess the effectiveness of an inhibitor acting on an enzyme or enzyme system derived from two different sources. It is a ratio equal to the concentration of the inhibitor X causing 50% inhibition of the host enzyme divided by the concentration of the inhibitor X causing 50% inhibition of the enzyme during some pathophysiologic condition or disease.

C. Silipo & C. Hansch (1976) *J. Med. Chem.* **19**, 62.

INITIAL RATE CONDITION

One of the basic assumptions in kinetic studies of an enzyme-catalyzed reaction is that true initial rates are being measured. In such cases, a plot of the product concentration *versus* time must yield a straight line. (This behavior is only observed when the substrate is at or near its initial (or, $t = 0$) concentration. As time increases, product accumulation and substrate depletion will result in a curvature of this progress curve; hence, the reaction velocity at these later times would be correspondingly lower.)

An oft-suggested check on the initial rate condition is that, over the duration of the assay, the initial substrate concentration (*i.e.*, $[S]$) should not decrease by more than 5% (and ideally, significantly less). However, the validity of this suggestion depends on a number of other factors. If the reaction equilibrium constant K_{eq} favors substrate(s), rather than product(s), then 10% conversion may amount to a significant fraction of that attainable

at equilibrium, and the system would clearly not satisfy initial rate conditions. Initial rate conditions are apt to apply when the substrate concentration is less than 5–10% the $\Delta[S]$ value, where $\Delta[S]$ is $[S]_{\text{initial}} - [S]_{\text{equilibrium}}$. If the reaction rate is not linear over this change in substrate, the experimenter is well advised to adopt a more sensitive rate assay.

The initial-rate phase may quickly end if one or more products accumulate to concentrations approaching the respective inhibition constants. In these cases, one may seek to minimize this problem by including an auxiliary enzyme (a) to remove product(s) or (b) to regenerate the substrate concentration. Ideally, the auxiliary enzyme should have a lower K_m value for the product of the primary reaction than the corresponding K_i value for the primary enzyme. **See** *Chemical Kinetics; ATP/GTP regeneration*

R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

INITIAL RATE ENZYME DATA, REPORTING OF

The initial rate assumption is one of the most powerful and widely used assumptions in the kinetic characterization of enzyme action. The proper choice of reaction conditions that satisfy the initial rate assumption is itself a challenge, but once conditions are established for initial rate measurements, the kinetic treatment of an enzyme's rate behavior becomes much more tractable¹. In reporting initial rate data, investigators would be well advised to provide the following information:

1. Provide a reference or direct experimental proof that the conditions chosen do provide initial rate measurements. At a minimum, the percentage of substrate consumed during the course of an initial rate determination should be specified. One should also show that under initial rate conditions a doubling of enzyme concentration should exactly produce a doubling in the observed initial rate. Likewise, if an auxiliary enzyme assay is used to monitor the primary enzyme's activity the observed rate at low or high substrate concentration should not depend on the concentration of additional enzyme(s), substrate(s), or factors used for the coupled assay.
2. Describe all assay conditions (*e.g.*, concentrations of substrates, products, inhibitors, and/or activators; enzyme concentration; temperature; pH and buffer composition;

other solutes; as well as the exact reaction volume and duration of assay, *etc.*). The specific activity and source of each enzyme should be explicitly indicated. Occasionally, the dilution of an enzyme stock solution to initiate the assay will result in a slow conformational change that effects catalytic activity. If this is the case, a complete description of assay conditions will also require an indication of the enzyme concentration in the stock solution and in the final assay itself.

3. All velocity data should be expressed as the molarity change per unit time, and not as micromoles per unit time. This is a requirement for all chemical kinetic studies, because the molarity change per unit time is an intensive variable.
4. Indicate the number of replicate kinetic runs as well as measures of the statistical variation in the rate data and in any derived kinetic parameters (*e.g.*, K_m , V_{max} , V_{max}/K_m , K_i , *etc.*).
5. Specify any unusual circumstances needed to obtain initial rate conditions. For example, the presence of saturating levels of an activator may be required to generate a linear double-reciprocal plot for a cooperative enzyme. Also indicate if any stabilizer is needed to reduce loss of enzyme activity and/or proteolysis.
6. Include adequate descriptions of any graphical treatment, computer modeling technique (*e.g.*, KINSIM, FITSIM, least squares analysis, ENZEO, *etc.*), global analysis method, or any other data fitting technique.

¹R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

INITIAL RATE PHASE

The period in a reaction curve that (a) is characterized by the full effect of initial substrate concentration and (b) for which any change in product concentration is minimal and has no measurable effect on the rate. There should be a linear relationship between product formation and time during the initial rate phase, and a doubling of the reaction time will result in a doubling of product formed. The relationship between reaction velocity and enzyme concentration (in the absence of self-association of the enzyme) should also be linear. Initial rates typically fail to be obtained when $[E_{\text{total}}]$ greater than or equal to $[\text{substrate}]_{\text{initial}}/100$. As a general rule, the substrate concentration will not have changed more than 5–10% from its initial value over the initial rate phase of the reaction. This oft-stated rule-of-thumb applies to reactions that are thermodynamically favorable, and investigators are well advised to work well below that 5–10% substrate depletion.

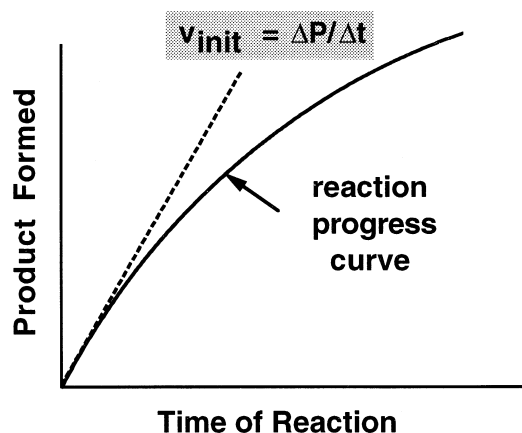


Figure 1. Plot of the change in product concentration as a function of time of reaction. The initial rate phase corresponds to the early linear region, and a tangent to this early region has a slope corresponding to the initial reaction velocity. (For a detailed description of how one obtains rate constants using the initial rate assumption, See *Chemical Kinetics*.)

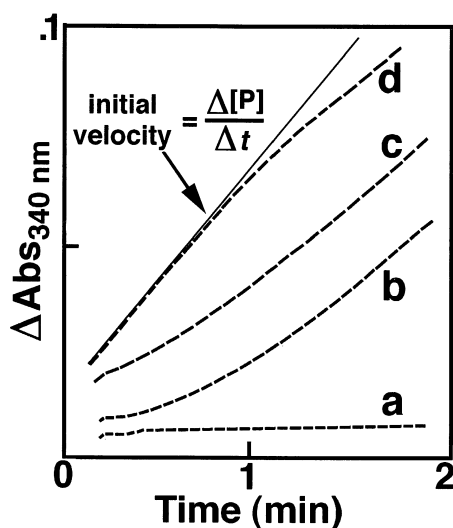


Figure 2. Illustration of the importance of the choice of reaction conditions on the determination of initial velocity. Shown are four conditions applied to examine the rate behavior of *Escherichia coli* NAD⁺-dependent coenzyme A-linked aldehyde dehydrogenase (Reaction: NAD⁺ + CoA-SH + Acetaldehyde = NADH + Acetyl-S-CoA + H⁺). All assay mixtures contained enzyme, 0.5 mM NAD⁺, 8 μM CoA-SH, 16 mM acetaldehyde, and 22.5 mM Tris buffer at pH 8.1. (a) Time-course observed when enzyme was added to the standard assay; (b) time-course observed when enzyme was added to standard assay augmented with 10 mM 2-mercaptoethanol; (c) time-course observed when enzyme was first preincubated for 15 min with 8 μM CoA-SH, 16 mM acetaldehyde, 10 mM 2-mercaptoethanol, and 22.5 mM Tris buffer at pH 8.1, and the reaction was initiated by addition of NAD⁺; (d) time-course observed when enzyme was preincubated with 10 mM 2-mercaptoethanol for 15 min and then added to standard assay augmented with 10 mM 2-mercaptoethanol. The data are most compatible with the idea that the enzyme has an active-site thiol group that must be reduced to express full catalytic activity during assay.

The development of an initial rate enzyme assay can also aid the discovery of special properties of the enzyme under investigation. Shown in Fig. 2, for example, are four conditions applied to examine the rate behavior of *Escherichia coli* NAD⁺-dependent Coenzyme A-linked aldehyde dehydrogenase (Reaction: NAD⁺ + CoA-SH + Acetaldehyde = NADH + Acetyl-S-CoA + H⁺). All assay mixtures contained enzyme, 0.4 mM NAD⁺, 8 μM CoA-SH, 16 mM acetaldehyde, and 22.5 mM Tris buffer (pH 8.1). The experiments demonstrated that the enzyme most probably contains an essential active-site thiol group.

When one or more product(s) exhibits a very small K_i value, the investigator must develop a more sensitive assay, such that true initial rates are actually obtained¹.

Occasionally, one can maintain initial rate conditions by using a coupled reaction system to regenerate one of the limiting substrates. For example, to regenerate ATP in a phosphotransferase reaction, one can use: creatine phosphate and creatine kinase; acetylphosphate and acetate kinase; or phosphoenolpyruvate and pyruvate kinase.

¹R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

INNATE THERMODYNAMIC QUANTITIES

Structural and molecular biologists often study the temperature dependence of the equilibrium position of a reaction or process. The Gibbs free energy undoubtedly provides the correct thermodynamic criterion of equilibrium. An understanding of this parameter can be achieved from either a macroscopic level (classical thermodynamics) or a molecular level (statistical thermodynamics). Ultimately, one seeks to understand the factors influencing ΔG° for a specific reaction.

Innate Thermodynamic Quantities*. Certain components of the total change in ΔG° are *innate*, because such parameters have nonzero values, even when extrapolated to 0 K. Other components change with temperature (e.g., at $T = 0$ K, $TS = 0$). Because $A = U - TS$ and $G = H - TS^{1-3}$, then $(H_o^\circ = G_o^\circ) \cong (A_o^\circ = U_o^\circ)$ at absolute zero. Except for entropy, the residual values of these quantities are the same at absolute zero, and they describe the innate thermodynamic behavior of the system.

Thermodynamic state functions change with temperature¹⁻³; this will be true if values of the heat capacity of any component is nonzero (which is almost always true). Whenever the heat capacity is not a constant, the various thermodynamic state functions will show nonlinear dependencies on temperature.

Reaction enthalpies are frequently measured at or near room temperature, such that

$$\Delta H_{298}^{\circ} = \Delta H^{\circ}(T_o) + \int_0^{298} \Delta C_p^{\circ} dT$$

where this last term represents the thermal agitation energy (heat capacity integrals): a variation of the Kirchhoff equation. With reactions involving small molecules only, one can derive $\Delta H_{\text{reaction}}^{\circ}$ in terms of inherent bond energy ($\Delta H(T_o)$ or ΔH_o° in some texts) by ignoring differences between this parameter and ΔH_{298}° . Cottrell⁴ calculated that the effect on chemical bond energy is only 1–2 kcal mol⁻¹, compared to the 80–100 kcal mol⁻¹ total. This is *not* true for systems involving biological macromolecules, and Chun⁵⁻⁹ has shown that the heat capacity integrals may be 40–50% of the magnitude of experimentally measured values of ΔH_{298}° .

Planck-Benzinger Thermal Work Function. In 1971, Benzinger¹⁰ proposed a thermal work function to account for both Boltzmann statistical energy effects and quantum-mechanical bond energies. While the latter are not usually significantly altered in reactions involving small molecules, Benzinger's conjecture was that the large-scale and long-range changes of conformation upon protein folding and/or assembly might generate significant energy differences arising from cumulative alteration of their numerous covalent bond structures.

$$\Delta G - \Delta H(T_o) = - \left\{ T \int_0^T (\Delta C_p/T) dT - \int_0^T \Delta C_p dT \right\} \\ = \Delta W$$

ΔW represents the thermal components of any intra-/intermolecular bonding term in a system (*i.e.*, energy other than the inherent difference of the 0 K portion of the interaction energy). Thus, ΔW completely expresses the thermal energy difference of the process involved. Application of the thermal work function permits one to separate so-called 0 K energy differences from those

associated with the heat capacity integrals. This offers a fuller understanding of reaction energies⁵⁻⁹. From this expression, it is possible to determine the temperature-invariant enthalpy at equilibrium conditions,

$$\Delta H^{\circ}(T_o) = \Delta W^{\circ}(T_s)_{\text{max}} + \Delta G^{\circ}(T_s)_{\text{min}}$$

At $\langle T \rangle_s$, the temperature-invariant enthalpy is a primary source of the chemical bond energy essential for any reaction to proceed in an interacting system. The magnitude of $\Delta H^{\circ}(T_o)$ is an indication of the type of macromolecular interaction taking place. This relationship has been designated as the Planck-Benzinger thermal work function. An analogous treatment developed by Giauque¹¹ for chemical systems has not been fruitfully applied to biological systems.

Application to Macromolecular Interactions. Chun⁹ describes how one can analyze the thermodynamics of a particular biological system as well as the thermal transition taking place. Briefly, it is necessary to extrapolate thermodynamic parameters over a broad temperature range. Enthalpy, entropy, and heat capacity terms are evaluated as partial derivatives of the Gibbs free energy function defined by Helmholtz-Kelvin's expression, assuming that the heat capacities integral is a continuous function.

$$\partial \Delta G(T) / \partial T = -\Delta S(T)$$

$$\{ \partial \Delta G(T) / T \} / \partial T = -\Delta H(T) / T^2$$

$$\partial \Delta H(T) / \partial T = \Delta C_p(T)$$

$$\partial \Delta S(T) / \partial T = C_p(T) / T$$

In this method, values of various thermodynamic properties are mathematically derived from the available data, based on thermodynamic principles. The method requires very accurate experimental data for the Gibbs free energy change showing the direction of the chemical change toward the minimum Gibbs function. Equilibrium sedimentation, calorimetric measurements, or spectroscopic data are useful in determinations of the temperature-invariant enthalpy. As to the question of whether the heat capacity integral is a continuous function, no phase change occurs over the accessible temperature range.

For self-associating protein systems, third-order polynomial functions provided a good fit over the accessible range⁵⁻⁹. The data on ΔG° must show the direction of the chemical change, toward the minimum in the Gibbs function. If this proves true, the equation can be applied in the standard or nonstandard state. For protein unfolding or DNA unwinding, nonlinear models are needed⁶⁻⁹. Consistent with Occam's razor, the simplest description is used to describe the system, and complexity is increased only if warranted by the experimental results.

The range of temperature over which any given biochemical system can be studied is quite limited: above some point, the system is destroyed; however, at too low a temperature, biological activity ceases. In favorable cases, data can be obtained from 283 to 333 K. Extrapolation to 0 K is a difficult, though manageable, problem.

This formalism may also be valuable in addressing protein unfolding, DNA unwinding, cold denaturation, and effects of site-direct mutagenesis on lysozyme unfolding⁹.

¹G. N. Lewis & M. Randall (1961) *Thermodynamics*, 2nd. ed. (revised by K. S. Pitzer & L. Brewer), McGraw-Hill, New York.

²J. W. Gibbs (1878) *Amer. J. Sci.* **16**, 441.

³M. Planck (1927) *Treatise on Thermodynamics*, 7th ed. (translated by A. Ogg, 3rd ed.), pp. 179-214 and 272-289, Longmans, Green and Co., London.

⁴T. L. Cottrell (1950) *The Strengths of Chemical Bonds*, pp. 21-70, Academic Press, London.

⁵P. W. Chun (1988) *Internat. J. Quantum Chem.* **15**, 247.

⁶P. W. Chun (1994) *J. Phys. Chem.* **86**, 6851.

⁷P. W. Chun (1995) *J. Biol. Chem.* **270**, 13925.

⁸P. W. Chun (1996) *J. Phys. Chem.* **100**, 7283.

⁹P. W. Chun (1997) *Meth. Enzymol.* **295**, 227.

¹⁰T. H. Benzinger (1971) *Nature* **229**, 100.

¹¹W. F. Giaque (1930) *J. Am. Chem. Soc.* **52**, 4808 and 4816.

*The authors thank their colleague, Professor Paul W. Chun, who is the leading proponent of the Planck-Benzinger thermal work function (P. W. Chun (1997) *J. Phys. Chem.* **101**, 7885; (1998) *Meth. Enzymol.* **295**, 227). Those interested in additional information may wish to contact Professor Chun by at: pwchun@biochem.med.ufl.edu.

INNER COORDINATION SPHERE

The region of a metal ion complex where ligands make direct binding interactions with the central metal ion. When the ligands do not completely neutralize the positive ionic charge of the central ion, other ions or electron-rich substances will become loosely associated with the complex through so-called outer coordination sphere interactions.

INNER FILTER EFFECT

1. An apparent decrease in the emission quantum yield and/or distortion of the band shape due to the reabsorption of emitted radiation. If such an effect is not corrected or compensated for, results of an emission experiment may prove to be incorrect. This is especially true in fluorescence quenching experiments conducted to evaluate the stoichiometry and affinity of ligand binding. 2. In a light irradiation experiment, the absorption of incident radiation by a species or molecular entity other than the intended primary absorber. **See** *Fluorescence*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

INNER-SPHERE ELECTRON TRANSFER

1. Electron transfer between two metal centers sharing a ligand or atom in their respective coordination shells. **See** *Metal Ion Coordination (Basic Chemical Behavior)* 2. Electron transfer between two centers in which the interaction between the donor and acceptor centers in the transition state is significant (*i.e.*, greater than 20 kJ/mol). **See also** *Outer Sphere Electron Transfer*

INOSAMINE-PHOSPHATE AMIDINOTRANSFERASE

This enzyme [EC 2.1.4.2] catalyzes the reaction of arginine with 1-amino-1-deoxy-*scyllo*-inositol 4-phosphate to produce ornithine and 1-guanidino-1-deoxy-*scyllo*-inositol 4-phosphate. Other substrates include 1D-1-guanidino-3-amino-1,3-dideoxy-*scyllo*-inositol 6-phosphate, streptamine phosphate and 2-deoxystreptamine phosphate. Canavanine can substitute for arginine.

J. B. Walker (1973) *The Enzymes*, 3rd ed., **9**, 497.

INOSINE KINASE

This enzyme [EC 2.7.1.73] catalyzes the ATP-dependent phosphorylation of inosine to generate IMP and ADP.

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.

INOSINE NUCLEOSIDASE

This enzyme [EC 3.2.2.2], also called inosinase, catalyzes the hydrolysis of inosine to generate hypoxanthine and D-ribose.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

myo-INOSITOL 2-DEHYDROGENASE

This enzyme [EC 1.1.1.18] catalyzes the reaction of *myo*-inositol with NAD⁺ to produce 2,4,6/3,5-pentahydroxycyclohexanone and NADH.

J. Larner (1962) *Meth. Enzymol.* **5**, 326.

myo-INOSITOL 1-KINASE

This enzyme [EC 2.7.1.64] catalyzes the reaction of ATP with *myo*-inositol to produce ADP and L-*myo*-inositol 1-phosphate.

F. A. Loewus & M. W. Loewus (1983) *Ann. Rev. Plant Physiol.* **34**, 137.

myo-INOSITOL 1-MONOPHOSPHATASE

This enzyme [EC 3.1.3.25], also referred to as *myo*-inositol-1(or 4)-monophosphatase and *myo*-inositol-1-phosphatase, catalyzes the hydrolysis of *myo*-inositol 1-monophosphate to generate *myo*-inositol and orthophosphate. Both enantiomers of *myo*-inositol 1-phosphate and *myo*-inositol 4-phosphate can act as substrates. However, the enzyme does not hydrolyze inositol bisphosphates, trisphosphates, or tetrakisphosphates.

P. W. Majerus (1992) *Ann. Rev. Biochem.* **61**, 225.
V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

myo-INOSITOL OXYGENASE

This enzyme [EC 1.13.99.1] catalyzes the reaction of *myo*-inositol with dioxygen to produce D-glucuronate and water. The enzyme requires iron for activity.

F. A. Loewus & M. W. Loewus (1983) *Ann. Rev. Plant Physiol.* **34**, 137.

myo-INOSITOL-1-PHOSPHATE SYNTHASE

This enzyme [EC 5.5.1.4] catalyzes the conversion of D-glucose 6-phosphate to 1L-*myo*-inositol 1-phosphate. NAD⁺ is required as a cofactor.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

INOSITOL POLYPHOSPHATE 1-PHOSPHATASE

Also known as inositol-1,4-bisphosphate 1-phosphatase [EC 3.1.3.57], this enzyme catalyzes the hydrolysis of D-*myo*-inositol 1,4-bisphosphate to generate D-*myo*-inositol 4-phosphate and orthophosphate. This enzyme can also act on inositol 1,3,4-trisphosphate (forming inositol 3,4-bisphosphate). Inositol-1,4,5-trisphosphate 1-phos-

phatase [EC 3.1.3.61] catalyzes the hydrolysis of D-*myo*-inositol 1,4,5-trisphosphate to produce D-*myo*-inositol 4,5-bisphosphate and orthophosphate.

P. W. Majerus (1992) *Ann. Rev. Biochem.* **61**, 225.
V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

INOSITOL POLYPHOSPHATE 3-PHOSPHATASE

Inositol 1,3,4,5-tetrakisphosphate 3-phosphatase [EC 3.1.3.62] catalyzes the hydrolysis of D-*myo*-inositol 1,3,4,5-tetrakisphosphate to generate D-*myo*-inositol 1,4,5-trisphosphate and orthophosphate. Inositol 1,3-bisphosphate 3-phosphatase [EC 3.1.3.65] catalyzes the hydrolysis of D-*myo*-inositol 1,3-bisphosphate to produce D-*myo*-inositol 1-monophosphate and orthophosphate. This enzyme is magnesium-ion-independent and is different from inositol-1,4-bisphosphate 1-phosphatase [EC 3.1.3.57], inositol-1,4,5-trisphosphate 1-phosphatase [EC 3.1.3.61], and inositol 3,4-bisphosphate 4-phosphatase [EC 3.1.3.66].

P. W. Majerus (1992) *Ann. Rev. Biochem.* **61**, 225.
V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

INOSITOL POLYPHOSPHATE 4-PHOSPHATASE

Inositol-3,4-bisphosphate 4-phosphatase [EC 3.1.3.66] catalyzes the hydrolysis of D-*myo*-inositol 3,4-bisphosphate to produce D-*myo*-inositol 3-monophosphate and orthophosphate. This enzyme is magnesium-ion-independent and is different from inositol-1,4-bisphosphate 1-phosphatase [EC 3.1.3.57], inositol-1,4,5-trisphosphate 1-phosphatase [EC 3.1.3.61], and inositol-1,3-bisphosphate 3-phosphatase [EC 3.1.3.65].

P. W. Majerus (1992) *Ann. Rev. Biochem.* **61**, 225.
V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

INOSITOL POLYPHOSPHATE 5-PHOSPHATASE

Inositol-1,4,5-trisphosphate 5-phosphatase [EC 3.1.3.56], also known as inositol trisphosphate phosphomonoesterase and inositol polyphosphate 5-phosphatase, catalyzes the hydrolysis of D-*myo*-inositol 1,4,5-trisphosphate to produce D-*myo*-inositol 1,4-bisphosphate and orthophosphate. The type I enzyme (but not the type II enzyme) will also hydrolyze inositol 1,3,4,5-tetrakisphosphate at the 5-position. However, neither of the two

types of the enzyme will utilize inositol 1,4-bisphosphate or inositol 1,3,4-trisphosphate as substrates.

P. W. Majerus (1992) *Ann. Rev. Biochem.* **61**, 225.

V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

INOSITOL-1,3,4,6-TETRAPHOSPHATE 5-KINASE

This enzyme [EC 2.7.1.140], also referred to as 1D-*myo*-inositol-tetrakisphosphate 5-kinase, catalyzes the reaction of ATP with 1D-*myo*-inositol 1,3,4,6-tetrakisphosphate to produce ADP and 1D-*myo*-inositol 1,3,4,5,6-pentakisphosphate.

V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

INOSITOL-3,4,5,6-TETRAPHOSPHATE 1-KINASE

This enzyme [EC 2.7.1.134], also referred to as 1D-*myo*-inositol-tetrakisphosphate 1-kinase, catalyzes the reaction of ATP with 1D-*myo*-inositol 3,4,5,6-tetrakisphosphate to produce ADP and 1D-*myo*-inositol 1,3,4,5,6-pentakisphosphate.

V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

INOSITOL-1,3,4-TRISPHOSPHATE 6-KINASE

This enzyme [EC 2.7.1.133], also referred to as 1D-*myo*-inositol-trisphosphate 6-kinase, catalyzes the reaction of ATP with 1D-*myo*-inositol 1,3,4-trisphosphate to produce ADP and 1D-*myo*-inositol 1,3,4,6-tetrakisphosphate.

V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

INOSITOL-1,4,5-TRISPHOSPHATE 3-KINASE

This enzyme [EC 2.7.1.127], also referred to as 1D-*myo*-inositol-trisphosphate 3-kinase, catalyzes the reaction of ATP with 1D-*myo*-inositol 1,4,5-trisphosphate to produce ADP and 1D-*myo*-inositol 1,3,4,5-tetrakisphosphate. The enzyme requires calcium ions as a cofactor.

P. W. Majerus (1992) *Ann. Rev. Biochem.* **61**, 225.

V. S. Bansal & P. W. Majerus (1990) *Ann. Rev. Cell Biol.* **6**, 41.

INSTANTANEOUS PROCESS

Any reaction or change that is immediate and occurs at the present, without any perceptible delay. In fact, the time-evolution or progress of many apparently instantaneous processes can be determined by using ultrafast

methods for detecting and/or quantifying a change in the system. The descriptor “instantaneous” should not be confused with “spontaneous”; the latter describes a process that is energetically poised to progress without the intervention of any external force or treatment. Although a spontaneous chemical process may be thermodynamically favorable, kinetic constraints may prevent the system from progressing to its equilibrium position. If the energy of activation is appreciable, then attainment of the transition state is less likely, and an otherwise spontaneous process may fail to progress.

INSULYSIN

This zinc-dependent enzyme [EC 3.4.24.56], a member of the peptidase family M16, catalyzes the degradation of insulin, glucagon, and other polypeptides.

A. B. Becker & R. A. Roth (1995) *Meth. Enzymol.* **248**, 693.

INTENSITY

1. In reference to a spectral feature, the scalar or magnitude of that feature. 2. Chromatic purity. 3. The magnitude of a particular force or energy per unit (e.g., surface, charge, mass, time, volume, *etc.*). 4. Synonym for photon irradiance. 5. Synonym for fluence rate. 6. Synonym for irradiance; illuminance. 7. Synonym for radiant power. 8. Symbolized by *I*, synonym for radiant intensity. 9. **See Magnetic Field Strength**. 10. **See Electric Field Strength**. 11. With respect to sound, symbolized by *I*, the rate of energy transfer per unit area normal to the direction of propagation. Thus, $I = (1/2)\rho_0 c a^2 f^2$ where ρ_0 is the mean density of the medium, *a* is the amplitude, *f* is the frequency, and *c* is the velocity of the sound.

INTERACTIVE

An adjective describing reciprocal action of two or more bodies, particles, or molecules upon each other. Such two-way action reflects the ability of interacting species to transfer energy. This descriptor is often used to describe the ability of ligand acting at one binding site on an enzyme or receptor to influence ligand binding at other sites.

INTERCEPT

1. The point at which a line, curve, or figure intersects a specified coordinate axis; this term also applies to the coordinates of that point. For example, in the double-reciprocal plot of the initial rates of the Uni Uni mecha-

nism ($1/v = 1/V_{\max} + (K_m/V_{\max})(1/[S])$) where v is the initial velocity, $V_{\max}$ is the maximum forward velocity, K_m is the Michaelis constant, and $[S]$ is the substrate concentration) the vertical intercept (*i.e.*, the ordinate intercept) is $1/V_{\max}$ whereas the horizontal intercept (*i.e.*, the abscissa intercept) is $-1/K_m$. 2. A point of intersection for two lines, curves, or figures.

INTERFACE ACTIVATION IN LIPOLYSIS

The hydrolysis of lipids rarely occurs in a single homogeneous phase, and the behavior of lipases at membrane-solvent and micelle-solvent interfaces has been discussed in detail by Verger¹ and Jain *et al.*² **See** *Micellar Catalysis*

¹R. Verger (1980) *Meth. Enzymol.* **64**, 340.

²M. K. Jain, M. H. Gelb, J. Rogers & O. G. Berg (1995) *Meth. Enzymol.* **249**, 570.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Adsorption effect, **64**, 345, 346; anchoring site, **64**, 390; bile salt interaction, **64**, 345; chain length effect, **64**, 344; charge density, **64**, 386; energy, **64**, 385; enzyme aggregation, **64**, 387, 388; enzyme inactivation, **64**, 347, 368, 372-375; free energy of adsorption, **64**, 365; hydration state, **64**, 385-386; insoluble lipid effect, **64**, 345-353; lipolytic enzymes, **64**, 341-345, 384-392; Michaelis constant, **64**, 356-358; orientation effect, **64**, 357, 385; insoluble lipid penetration, **64**, 345, 346; size effect, **64**, 356-358, 362; substrate concentration, **64**, 384, 385; surface dilution model, **64**, 355; surface pressure, **64**, 386, 387; organization and dynamics of phospholipids in aqueous dispersions, **249**, 570; structural features of phospholipase A₂, **249**, 572; neutral diluents, substrates, and transition-state mimics, **249**, 575; key considerations for interpreting interfacial catalysis, **249**, 578; kinetic variables at interface, **249**, 581; constraining variables for high processivity, **249**, 582; kinetic formalism for interfacial catalysis in scooting mode, **249**, 586; additional features of hydrolysis of vesicles in scooting mode, **249**, 586; determination of interfacial equilibrium parameters, **249**, 587; interfacial equilibrium constant determination by protection method, **249**, 588; table of kinetic and equilibrium constants, **249**, 591; binding of phospholipase A₂ to interface, **249**, 592; integrated Michaelis-Menten equation for interfacial lipolysis, **249**, 595; full activity by enzyme monomer, **249**, 600; competitive substrate specificity, **249**, 601; covalent modifiers of enzyme activity, **249**, 603; hydrolysis of zwitterionic vesicles, **249**, 605; phospholipid-detergent mixed micelles, **249**, 607; kinetic problems on monolayer interfaces, **249**, 610.

INTERMEDIACY

The state or condition of being intermediate, as in the middle place in a process. In most chemical reactions, intermediacy is characterized by a transition state barrier to reactivity.

INTERMEDIATE

1. A substance formed in the course of a chemical reaction that then proceeds further through the reaction. 2. A substance formed as a product of an enzyme-catalyzed

reaction and is precursor to other compounds in a metabolic pathway.

INTERMEDIATES, COVALENT ENZYME-SUBSTRATE

Many enzyme-catalyzed reactions proceed via the formation of enzyme-substrate covalent intermediates. Formation of an enzyme-bound covalent compound need not require its participation in enzyme catalysis. Establishing the catalytic competence of covalent intermediates is a major objective of enzyme chemistry. This competence has to be evaluated by a number of different methods¹, including compatibility with initial rate data, isotope exchange behavior, results of thermal or chemical trapping, spectrophotometric detection, stereochemical course of the reaction to observe inversion of configuration and/or isotope scrambling, the behavior of analogs of the putative covalent intermediate, and site-directed mutagenesis studies.

¹D. L. Purich (1982) *Meth. Enzymol.* **87**, 3.

INTERMOLECULAR

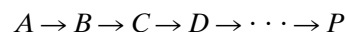
An adjective indicating an interaction between two or more distinct molecular entities. **See also** *Intramolecular*

INTERMOLECULAR ISOTOPE EFFECT

An isotope effect (either kinetic or equilibrium) resulting from a comparison made between isotopically different reactant molecular entities. **See** *Intramolecular Isotope Effect*; *Kinetic Isotope Effect*; *Equilibrium Isotope Effect* IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

INTERNAL EFFECTORS

Those metabolic intermediates whose concentrations are determined by the flux through the pathway in which they are formed and transformed. Consider the hypothetical pathway,



The internal intermediates are B , C , D , $\dots$, and A and P are the respective initial substrate and final product. Within cells, the enzymes forming internal intermediates are rarely saturated with respect to the concentration of their individual substrates which are internal effectors of the pathway dynamics.

B. Crabtree & E. A. Newsholme (1987) *Trends Biochem. Sci.* **12**, 4.

INTERNAL ENERGY

The change of internal energy (symbolized ΔU) in a closed system is equal to the sum(s) of the transfer-of-heat (q) on the system plus the performance of work (w) on the system. Thus, $\Delta U = q + w$; or, in differential form, $dU = dq + dw$. The internal energy, U , can only be defined in terms of a difference in energy between two different states. In some instances, a convenient standard state may be defined to be zero (*e.g.*, at absolute zero, *etc.*). The internal energy of a system is the sum of all the potential and kinetic energies of all the molecular entities within the system. This includes all energies associated with conformational changes of molecular entities as well as chemical reactions. If electromagnetic fields are present, then the internal energy includes electromagnetic energy. The energy function U is of importance in representing the first law of thermodynamics. The SI unit for ΔU is the joule.

INTERNAL ENTROPY

The internal freedom of motion of substrate molecules, intermediates, or reactive enzyme groups occurring during the binding and reaction of an enzyme-catalyzed reaction. When the bond hybridization changes during the course of a reaction (*e.g.*, sp^3 to sp^2 and then back to sp^3 in the triose-phosphate isomerase reaction), loss of internal rotatory motion (in this example, about the C–C bond during the formation of the ene-diolate intermediate) leads to a change in the internal entropy of this reaction. Jencks¹ has discussed how restricted motions during catalysis are apt to influence reactivity.

¹W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, pp. 363–366, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York].

INTERNAL EQUILIBRIA

Equilibria involving the productively bound substrates and the products formed during an enzyme-catalyzed reaction. These equilibria can be treated in terms of internal equilibrium constants (K_{int}) between these enzyme-bound species.

Burbaum *et al.*¹ considered how kinetic/thermodynamic features of present-day enzyme-catalyzed reactions suggest that enzyme evolution tends to maximize catalytic effectiveness. They analyzed Uni Uni enzymes in terms of reaction energetics. Catalytically optimized enzymes

were proposed to have “internal” equilibrium constants that depend on how close to equilibrium they maintain their reactions *in vivo*. For enzymes operating near equilibrium, the optimum values of K_{int} are near unity. For those operating far away from equilibrium, catalytic efficiency is less sensitive to the magnitude of K_{int} . Accordingly, they suggested that the internal thermodynamics of enzyme-catalyzed reactions dynamically match the concentrations of substrate- and product-containing complexes to the steady state *in vivo*. Burbaum & Knowles² also determined K_{int} for enolase (which interconverts 2-phosphoglycerate and phosphoenolpyruvate) and creatine kinase (a bisubstrate enzyme forming MgATP^{2-} and creatine from creatine phosphate and ADP). K_{int} values were determined by rapidly quenching equilibrium mixtures of enzyme and radiolabeled substrate and product, and these investigators chose conditions where virtually all of the marker substrate and product were bound. They also showed how published K_{int} values agree with their theory for kinetically optimized enzymes.

¹J. J. Burbaum, R. T. Raines, W. J. Albery & J. R. Knowles (1989) *Biochemistry* **28**, 9293.

²J. J. Burbaum & J. R. Knowles (1989) *Biochemistry* **28**, 9306.

INTERNAL PRESSURE

The change in internal energy per change in volume at constant temperature (expressed in differential terms as $(\partial U/\partial V)_T$, where U is the internal energy, V is the volume, and T is the temperature). For gases, this value (usually symbolized by P_i) is small. However, for liquids and solids, the internal pressure is large. The magnitude of the internal pressure provides an estimate of the magnitude of the cohesive forces within the liquid or solid state (or even the gaseous state for nonideal gases). The internal pressure is also obtained from one of the thermodynamic equations of state: namely, $(\partial U/\partial V)_T = T(\partial P/\partial T)_V - P$ where P is the pressure. For an ideal gas, the internal pressure is zero, because there are, by definition, no cohesive or intermolecular forces that apply to an ideal gas. In general, dipolar liquids have larger values than nonpolar liquids. For example, at 1 atm of external pressure and a temperature of 298 K, diethyl ether and benzene have P_i values of 2.40×10^8 Pa (or 2.37×10^3 atm) and 3.69×10^8 Pa (or 3.64×10^3 atm), respectively. By virtue of its strong hydrogen bond interactions, water has a corresponding P_i value of 2.0×10^9 Pa (or 2.0×10^4 atm).

INTERNET

The Internet has become a powerful resource for molecular life scientists, and the only limitation appears to be that many URL's are not maintained as stable sites for long-term use. Nonetheless, the reader may find the following addresses as a useful initial point of departure in efforts to locate valuable software for analyzing protein and nucleic acid structure.

BLAST (NCBI Sequence Similarity Search of Nucleotide and Protein Databases)

<http://www.ncbi.nlm.nih.gov/BLAST/>

Biochemist On-Line

<http://www.biochemist.com/>

Brookhaven Protein Data Base (Structural):

<http://www.pdb.bnl.gov/>

dbEST (Database of Expressed Sequence Tags)

<http://www.ncbi.nlm.nih.gov/dbEST/index.html>

Biochemical Pathways (Searchable Index)

[http://expasy.hcuge.ch/cgi-bin
/search-biochem-index](http://expasy.hcuge.ch/cgi-bin/search-biochem-index)

ENZYME (Enzyme Nomenclature Database)

<http://expasy.hcuge.ch/sprot/enzyme.html>

ExPASy Tools (A Menu of Sequence Analysis Tools)

<http://expasy.hcuge.ch/www/tools.html>

GenBank (Text and Similarity Searching)

[http://www.ncbi.nlm.nih.gov/Web/Search
/index.html](http://www.ncbi.nlm.nih.gov/Web/Search/index.html)

MMDB (The Molecular Modelling Database)

<http://www.ncbi.nlm.nih.gov/Structure/>

MULTIIDENT (Proteins Identification Using pI, MW, Amino Acid Composition, Sequence Tag and Peptide Mass Fingerprinting Data)

<http://expasy.hcuge.ch/sprot/multiident.html>

ORF Finder (Open Reading Frame Finder)

<http://www.ncbi.nlm.nih.gov/gorf/gorf.html>

Pedro's BioMolecular Research Tools

[http://www.public.iastate.edu/~pedro
/research_tools.html](http://www.public.iastate.edu/~pedro/research_tools.html)

Pfam (Comprehensive Collection of Protein Domain Families)

<http://www.sanger.ac.uk/Pfam/>

Principles of Protein Structure Using the Internet

<http://www.cryst.bbk.ac.uk/PPS2/top.html>

PROSITE (Dictionary of Protein Sites and Patterns)

<http://expasy.hcuge.ch/sprot/prosite.html>

PubMed (Med-Line Search Engine) :

[http://www4.ncbi.nlm.nih.gov/htbin-post
/Entrez](http://www4.ncbi.nlm.nih.gov/htbin-post/Entrez)

Structural Classification of Proteins

<http://scop.mrc-lmb.cam.ac.uk/scop/>

SWISS-3DIMAGE (Database of Annotated 3D Images)

<http://expasy.hcuge.ch/sw3d/sw3d-top.html>

Swiss-Model (An Automated Knowledge-Based Protein Modelling Server)

[http://expasy.hcuge.ch/swissmod
/SWISS-MODEL.html](http://expasy.hcuge.ch/swissmod/SWISS-MODEL.html)

SWISS-PROT (Annotated Protein Sequence Database)

<http://expasy.hcuge.ch/sprot/sprot-top.html>

Translate (Convert DNA or RNA Sequence to Protein Sequence)

<http://expasy.hcuge.ch/www/dna.html>

INTERPOLATION

Any mathematical algorithm or technique for estimating the magnitude of $f(x)$ for a continuous function of the variable x or for evaluating a parameter lying within an interval for which the first and last values of the interval are known.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Graphical techniques, **210**, 306; polynomial methods, **210**, 307; with rational functions, **210**, 311; with spline functions, **210**, 312; with trigonometric functions, **210**, 312

INTERSYSTEM CROSSING

A term in photochemistry and photophysics describing an isoenergetic radiationless transition between two electronic states having different multiplicities. Such a process often results in the formation of a vibrationally excited molecular entity, at the lower electronic state, which then usually deactivates to its lowest vibrational energy level. **See also** *Internal Conversion*; *Fluorescence*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

INTRAMOLECULAR CATALYSIS

Acceleration of a chemical reaction (at a particular site on a molecular entity or complex) by one or more func-

tional groups located at other site(s) within the same molecular entity. Intramolecular catalysis can often be expressed by comparison to the rate of the reaction when the functional group(s) is(are) removed from the same molecular entity or by determination of the effective molarity of the catalytic group.

INTRAMOLECULAR ISOTOPE EFFECT

An isotope effect (either kinetic or equilibrium) resulting from reactions in which the different isotopes occupy chemically equivalent alternative reactive sites within the same molecular entity. In such cases, isotopically distinct products are formed. **See** *Intermolecular Isotope Effect*; *Kinetic Isotope Effect*; *Equilibrium Isotope Effect* IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

INTRAMOLECULAR KINETIC ISOTOPE EFFECT

A kinetic isotope effect observed by a single reactant, having isotopic atoms at equivalent reactive positions, which reacts to produce isotopomeric products with a nonstatistical distribution. The pathway favored will be the one having lower force constants for the displacement of the isotopic nuclei in the transition state.

INTRINSIC BARRIER

The Gibbs energy of activation ($\Delta G^\ddagger$) when the thermodynamic driving force has been eliminated (*i.e.*, in the limiting case where $\Delta G^\ddagger = 0$). According to the Marcus equation, this barrier is directly related to the reorganization energy, λ , by the expression $\Delta G^\ddagger = \lambda/4$. **See** *Marcus Equation*; *Reorganization Energy*

R. D. Cannon (1980) *Electron Transfer Reactions*, Butterworths, London.

C. J. Schlesener, C. Amatore & J. K. Kochi (1986) *J. Phys. Chem.* **90**, 3747.

INTRINSIC BINDING ENERGY

The standard free energy change (ΔG_{int}) corresponding to the energy released when a protein binds to a compound (or a substituent group of that compound) without destabilization or energy losses attributable to translational, rotational, and internal entropy. As pointed out by Jencks¹, the intrinsic binding energy will never be directly observable, because binding of one molecule to another is almost always attended by a net loss of translational and rotational entropy. In other words, the observed binding energy of protein-ligand bond formation typically reflects

losses of translational and rotational degrees of freedom, often amounting to -40 entropy units. The intrinsic binding energy will also be underestimated by the net effect of the mutual freezing out of rotations and vibrations in the interacting components. Likewise, a protein or enzyme that destabilizes its ligand by distortion (*i.e.*, transition-state stabilization as opposed to ground-state stabilization) will also have an intrinsic binding energy that is better estimated through the binding of a so-called transition state analogue. Furthermore, Jencks¹ has stressed that the free energy of binding is affected by induced-fit interactions as well as nonproductive substrate complexation; these interactions are likely to exercise control over specificity without contributing significantly to catalytic rate enhancement.

¹W. P. Jencks (1975) *Adv. Enzymol.* **43**, 358.

INTRINSIC KINETIC ISOTOPE EFFECT

An effect of isotopic substitution within a reactant or substrate on a specific step in an enzyme-catalyzed reaction. The magnitude of an intrinsic isotope effect may not equal the magnitude of an isotope effect on collective rate parameters such as V_{max} or $V_{\text{max}}/K_{\text{m}}$, unless the isotopically sensitive step is the rate-limiting or rate-contributing step. **See** *Kinetic Isotope Effect*

INULINASE

This enzyme [EC 3.2.1.7], also referred to as 2,1- β -D-fructan fructanohydrolase and inulase, catalyzes the endohydrolysis of 2,1- β -D-fructosidic linkages in inulin.

G. Avigad & S. Bauer (1966) *Meth. Enzymol.* **8**, 621.

INULOSUCRASE

This enzyme [EC 2.4.1.9], also referred to as sucrose 1-fructosyltransferase, catalyzes the reaction of sucrose with [(2,1)- β -D-fructosyl]_{*n*} to produce D-glucose and [(2,1)- β -D-fructosyl]_{*(n+1)*}. Thus, this enzyme participates in the conversion of sucrose into inulin and glucose. Other fructosyl-containing sugars can serve as substrates as well instead of sucrose.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

INVERSE ISOTOPE EFFECT

A kinetic isotope effect in which the ratio $k_{\text{light}}/k_{\text{heavy}}$ is less than one. In such cases, there is an increase in the corresponding force constants on passing from the reactant to the transition state. **See** *Kinetic Isotope Effect*

INVERTASE

This enzyme [EC 3.2.1.26], also known as β -D-fructofuranosidase and saccharase, catalyzes the hydrolysis of terminal nonreducing β -D-fructofuranoside residues in β -D-fructofuranosides. Hence, the enzyme will also catalyze the hydrolysis of sucrose to glucose and fructose. In addition, the protein also exhibits a fructotransferase activity. **See also** *Sucrase*

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

P. M. Dey & E. del Campillo (1984) *Adv. Enzymol.* **56**, 141.

J. O. Lampen (1971) *The Enzymes*, 3rd ed., **5**, 291.

INVERTED/INVERSE MICELLE

A structure formed by the reversible association of amphiphiles in apolar solvents. In inverted micelles, the polar portion of the amphiphile is concentrated in the interior of the macrostructure. Such association usually occurs with aggregation and is not typically characterized by a definite nucleation stage. Thus, inverted micelles (also referred to as inverse or reverse micelles) often fail to exhibit critical micelle concentration behavior.

See *Micelle*

IODIDE PEROXIDASE

This enzyme [EC 1.11.1.8], also known as iodotyrosine deiodase, iodinase, and thyroid peroxidase, catalyzes the reaction of iodide with hydrogen peroxide to produce iodine and two water. The cofactor for this enzyme is heme.

M. Morrison (1970) *Meth. Enzymol.* **17A**, 653 and 658.

IODINE-125 (¹²⁵I)

A radioactive iodine nuclide that has three principal radiation emissions γ (0.035 MeV), $K\alpha$ X-ray (0.027 MeV), and $K\beta$ X-ray (0.031 MeV). The decay half-life is 60.14 days. The following decay data indicate the time (in days) followed by the fraction of original amount at the specified time: 0, 1.000; 2, 0.977; 4, 0.955; 6, 0.933; 8, 0.912; 10, 0.891; 12, 0.871; 14, 0.851; 16, 0.831; 18, 0.812; 20, 0.794; 40, 0.630; 60, 0.500; 80, 0.397; 100, 0.315; 120, 0.250; 140, 0.198; 160, 0.157; 180, 0.125; 200, 0.099; 220, 0.079; 240, 0.063.

Special precautions: Store this radioisotope behind 3 mm thick lead shielding. The effective *biological* half-life in humans is around 140 days. The thyroid gland is the critical organ in terms of dose, and one would be well

advised to adhere to institutional regulations regarding thyroid monitoring for localized uptake.

IODINE-131 (¹³¹I)

A radioactive iodine nuclide that has three principal radiation emissions: β (0.606 MeV), γ (0.364 MeV), and γ (0.637 MeV). The decay half-life is 8.040 days. The following decay data indicate the time followed by the fraction of original amount at the specified time: 0, 1.000; 6 hours, 0.979; 12, 0.958; 18, 0.938; 24, 0.918; 30, 0.898; 36, 0.879; 42, 0.860; 48, 0.842; 54, 0.824; 60, 0.807; 66, 0.789; 3 days, 0.773; 6, 0.597; 9, 0.461; 12, 0.356; 15, 0.275; 18, 0.213; 21, 0.164; 24, 0.127; 27, 0.098; 30, 0.076; 33, 0.059; 36, 0.045.

Special precautions: The thyroid gland is the critical organ in terms of dose. The effective *biological* half-life in humans is around 140 days. Store this radioisotope behind lead shielding. Follow institutional regulations regarding (a) type and location of dosimeters; and (b) monitoring thyroid for localized uptake.

ODOACETATE

An alkylating agent (ICH_2COO^-) that acts as a potent irreversible inhibitor of enzymes containing reactive thiol, ϵ -amino, and/or imidazole side-chain groups within their active sites. The carboxymethylation reaction with enzymes is typically a bimolecular process that takes place without rate-saturation behavior¹.

¹F. R. N. Gurd (1972) *Meth. Enzymol.* **25**, 424.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Principle of carboxymethylation, **25**, 424; precautions, **25**, 425; reaction products, **25**, 425-426; analysis of reaction time-course, **25**, 427-428; typical reaction conditions, **25**, 429-438; in affinity labeling, **87**, 469.

ION ATMOSPHERE

A term used in electrostatic descriptions of ions to denote the continuous electric charge density $[\rho(r)]$ surrounding an ionic species. On average, an ion will be surrounded by a spherically symmetrical distribution of counterions that form its ion atmosphere. **See** *Hydration Atmosphere*

ION CURRENT

1. The flux of ions across a membrane barrier, usually in the downward direction with respect to ion concentra-

tion. 2. The rate of chemical ionization in a mass spectrometer.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Analysis, software for, **207**, 717; barrier models, **207**, 818; closed and open time estimation, **207**, 755; data acquisition, **207**, 747; modal behavior analysis, **207**, 757; multiple channel problem, **207**, 756; single-channel [extraction of kinetic information, **207**, 765; measurement in tissue slices, **207**, 220]; synaptic, resolution improvement in patch clamp recording, **207**, 216; whole-cell recording in calcium channel, **207**, 181; fluctuation analysis, **207**, 192.

ION CYCLOTRON RESONANCE MASS SPECTROMETRY

An analytical mass spectrometric technique that traps an ionic species in a fluctuating magnetic field, such that the ion attains an angular or, in this case, a cyclotron frequency. This frequency ω_c equals the charge-to-mass ratio (z/m) multiplied by the product of the charge on a electron (e) and the magnetic field strength B . The ion cyclotron phenomenon is accomplished by using alternating current power source to create a fluctuating magnetic field. Coherent circular motion of resonating ions establishes a so-called image current that is used to determine the charge-to-mass ratio.

IONIC BOND

An electrostatic bond between two elements in which one or more electrons from one element have been donated to empty orbitals of the other element, thus imparting full charges to both elements; the bond between ions having opposite charges. The stable interaction of electropositive ions (*i.e.*, cations) and electronegative ions (*i.e.*, anions) resulting from a balancing of long-range ionic attraction and nearby repulsive forces. The bond length corresponds to the internuclear distance r between two charges (q_1 and q_2) in the following expression which includes terms for attractive potential ($U_{\text{attraction}} = -q_1q_2/r$) and the repulsive potential ($U_{\text{repulsion}} = be^{-r/a}$) where a and b are constants:

$$U = -q_1q_2/r + be^{-r/a}$$

This expression gives reasonable values for the equilibrium ionic bond length r and dipole moment μ , but the calculated curve for U versus r does not exactly match the experimental curve.

IONIC RADIUS

The calculated dimension (usually expressed in nm which is the preferred SI unit) of ions, based on X-

ray crystallographic studies of ionic crystals. Pauling¹ describes attempts to obtain self-consistent estimations based on measurements with a variety of ionic crystals. The following tabulated values are based on his calculations.

Table I

Ionic Radii of Metal Ions of Specified Coordination Number (CN)

Ion	CN ^a					
	2	4	6	8	10	12
Li ⁺	—	0.73Å	0.90Å	1.06Å	—	—
Na ⁺	—	1.13Å	1.16Å	1.32Å	—	1.53Å
Mg ²⁺	—	0.71Å	0.86Å	1.03Å	—	—
Al ³⁺	—	0.53Å	0.68Å	—	—	—
K ⁺	—	1.51Å	1.52Å	1.65Å	1.73Å	—
Ca ²⁺	—	—	1.14Å	—	—	—
V ²⁺	—	—	0.93Å	—	—	—
V ³⁺	—	—	0.81Å	—	—	—
Cr ²⁺	—	0.72Å (HS)	0.79Å (LS) 0.89Å (HS)	—	—	—
Cr ³⁺	—	0.76Å	—	—	—	—
Mn ²⁺	—	0.80Å (HS)	0.81Å (LS) 0.97Å (HS)	—	—	—
Mn ³⁺	—	—	0.72Å (LS) 0.79Å (HS)	—	—	—
Fe ²⁺	—	0.77 (HS)	0.75 (LS) 0.92 (HS)	—	—	—
Fe ³⁺	—	0.63Å (HS)	0.69Å (LS) 0.79Å (HS)	—	—	—
Co ²⁺	—	0.72Å (HS)	0.79Å (LS) 0.89Å (HS)	—	—	—
Co ³⁺	—	—	0.69Å (LS) 0.75Å (HS)	—	—	—
Ni ²⁺	—	0.69Å 0.63Å sq	0.83Å	—	—	—
Ni ³⁺	—	—	0.70Å (LS) 0.74Å (HS)	—	—	—
Cu ⁺	0.60Å	0.74Å	0.91Å	—	—	—
Cu ²⁺	—	0.71Å	0.87Å	—	—	—
Zn ²⁺	—	0.74Å	0.88Å	1.04Å	—	—
Rb ⁺	—	—	1.66Å	1.75Å	1.80Å	1.86Å
Mo ³⁺	—	—	0.83Å	—	—	—
W ⁴⁺	—	—	0.80Å	—	—	—

^aLS = low-spin state; HS = high-spin state; sq = square planar

¹L. Pauling (1960) *The Nature of the Chemical Bond*, 3rd ed., p. 514, Cornell University Press, Ithaca.

IONIC STRENGTH

An ionic concentration function (symbolized by I) designed to account for electrostatic interactions among charged substances in solutions, such that ions of higher charge are given greater weight as indicated by the following equation

$$I = 0.5 \sum m_i z_i^2$$

where m_i is the molal concentration (*i.e.*, the number of moles of the i -th component per 1000 g of solvent) and z_i is the number corresponding to the ionic charge (*i.e.*, $z = -1$ or $+1$ for monovalent anions and monovalent cations, respectively; $z = -2$ or $+2$ for divalent anions and divalent cations; *etc.*). It should be noted that the ionic strength for a solution is the summation for *all* ions present in the solution.

For dilute solutions, the molar concentration, c , can be used in place of molality. Thus, $I \cong (1/(2\rho_0)) \sum c_i z_i^2$ where ρ_0 is the density of the solvent. Hence, for aqueous solutions, $I \cong 0.5 \sum c_i z_i^2$. The following examples emphasize the importance of net charge in determining the magnitude of the ionic charge parameter: (a) for 10 mM NaCl, $I = 0.5[0.01 \times (-1)^2 + 0.01 \times (+1)^2] = 0.01$; (b) for 10 mM Na₂SO₄, $I = 0.5[0.01 \times (-2)^2 + 0.02 \times (+1)^2] = 0.03$; for 10 mM MgSO₄, $I = 0.5[0.01 \times (-2)^2 + 0.01 \times (+2)^2] = 0.04$; and (d) for 10 mM Na₃ATP, $I = 0.5[0.01 \times (-3)^2 + 0.03 \times (+1)^2] = 0.06$.

Rates of reactions involving ionic species, solubilities, activity coefficients, and a wide variety of other phenomena can be affected by the ionic strength. **See also** *Davies Equation; Debye-Hückel Treatment; Isotonic Buffers*

IONIZATION

Any process that generate one or more ions by (a) a unimolecular heterolysis of a neutral molecular entity into two or more ions or (b) a heterolytic substitution reaction involving neutral molecules. **See also** *Ionization Energy; Dissociation*

IONIZATION ENERGY

The minimum amount of energy or work (*i.e.*, enthalpy) required in the gaseous phase to remove an electron from

an isolated molecular entity (typically in its vibrational ground state). Adiabatic ionization energy is that energy needed to produce a new molecular entity that is also in its vibrational ground state. If the product is not in this ground state and possesses the energy dictated by the Franck-Condon principle, the energy needed is referred to as the vertical ionization energy.

IONOPHORE

A compound that binds to an ion in a manner which greatly facilitates the bound ion's permeability across a membrane. Naturally occurring ionophores include both mobile carriers (*e.g.*, valinomycin and nigericin) and channel formers (*e.g.*, gramicidin A).

B. C. Pressman (1976) *Ann. Rev. Biochem.* **45**, 501.

M. Dobler (1981) *Ionophores and Their Structures*, Wiley, New York.

ION PAIR

1. An intimate pair of oppositely charged ions, mutually attracted to each other by noncovalent, electrostatic interactions. Ion pairs often behave experimentally as a single unit. According to Bjerrum, an ion pair has atomic centers that are held closer together than a distance corresponding to the expression: $q = (8.36 \times 10^6 (z_+ z_- / \epsilon_r T))$, where q is expressed in picometers, z_+ and z_- are the charges of the ions, and ϵ_r is the relative permittivity of the medium. Ion pairs satisfying this requirement are often referred to as Bjerrum ion pairs. **See** *Tight Ion Pairs; Loose Ion Pairs; Solvent-Shared Ion Pair; Solvent-Separated Ion Pair*

2. The pair of oppositely charged ions produced in a single ionization reaction.

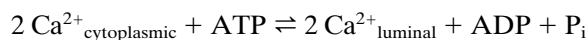
3. Occasionally, the monocation and electron generated in a redox half-reaction are also collectively referred to as an ion pair.

ION PUMP

A protein that brings about the active transport of ions by coupling the vectorial movement of an ion to some source of Gibbs free energy, often directly from ATP hydrolysis. These pumps exhibit tight ion binding in one state (*i.e.*, on the low ion concentration side of a membrane) and weak binding in a second state (*i.e.*, the high ion concentration side). This change in specificity (*i.e.*,

affinity) is driven by an ATP hydrolysis-dependent conformational change.

Jencks¹ has considered the mechanism of action of the sarcoplasmic reticulum calcium pump, which catalyzes the following reaction:



where there is stoichiometric coupling between the ATPase reaction and the active transport of an ion. One can write a reaction cycle involving two principal states E_1 and E_2 :

Sarcoplasmic Reticulum ATPase: The Ion Pump Cycle

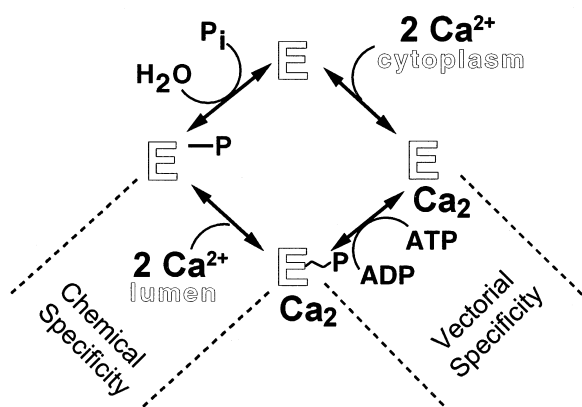


Figure 1. Reaction cycle showing how ATP hydrolysis and transient formation of a phosphoryl-enzyme intermediate are linked to the vectorial transport of calcium in the sarcoplasmic reticulum.

By analogy to the model proposed by Makinose², the cyclic transformations can also be depicted as follows:

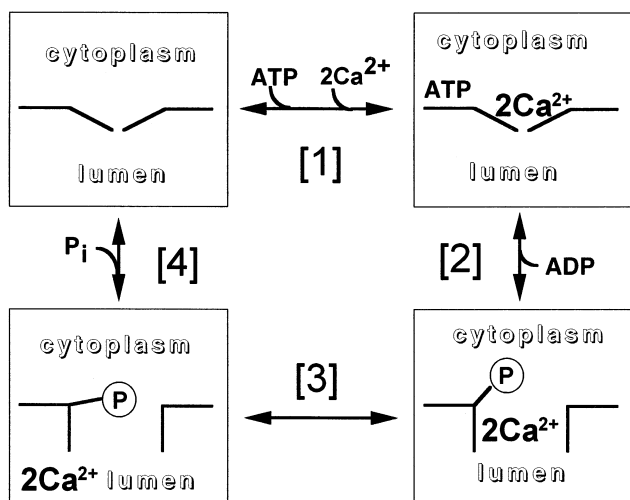


Figure 2. Schematic representation of the steps occurring during calcium ion pumping by the calcium-dependent ATPase.

In both schemes, the specificities of the pump for catalysis change in the two enzyme states. Jencks¹ points out that coupling is determined (a) by the chemical specificity achieved in catalyzing phosphoryl transfer to and from the enzyme (wherein $E \cdot \text{Ca}_2$ reversibly binds ATP, and E reacts reversibly with orthophosphate), and (b) by the vectorial specificity for ion binding and dissociation (wherein E reversibly binds/dissociates cytoplasmic calcium ion, and $E-P$ reversibly binds/dissociates luminal calcium). There must be a single conformation change during the reaction cycle: between E_1 and E_2 in the free enzyme and from $E_1 \sim P \cdot \text{Ca}_2$ to $E_2-P-\text{Ca}_2$ after enzyme phosphorylation.

Another indication that conformational changes are likely to account for changes in ion affinity was provided by Keillor and Jencks³ who recently examined the kinetic properties of the sheep kidney Na^+, K^+ -ATPase. This enzyme was preincubated with 120 mM sodium and 3 mM magnesium and then allowed to react with ATP (concentration range = 0.01–2.00 mM) to form a covalent phosphoenzyme intermediate ($E-P$). These investigators observed that the first-order rate constant for phosphorylation increases hyperbolically with ATP concentration to a maximum value of $4.6 \times 10^2 \text{ s}^{-1}$ and $K_{0.5} = 75 \mu\text{M}$. If phosphoryl-transfer is assumed to be rate-limiting, then the approach to equilibrium that yields 50% $E-P$ in the presence of ADP obeys the following rate law: $k_{\text{observed}} = k_f + k_r + 9.2 \times 10^2 \text{ s}^{-1}$. However, for $E-P$ formation from $E \cdot \text{Na}_3$ with 1.0 mM ATP and 2.0 mM ADP, $k_{\text{observed}} = 4.2 \times 10^2 \text{ s}^{-1}$, indicating that phosphoryl transfer from bound ATP to the enzyme is not rate-limiting for $E-P$ formation from $E \cdot \text{Na}_3$. Their results suggest that there is likely to be a rate-limiting conformational change of the $E \cdot \text{Na}_3 \cdot \text{ATP}$ intermediate, followed by rapid phosphoryl transfer, with $k_{\text{cat}} = 3000 \text{ s}^{-1}$. **See Membrane Transport; Binding Change Mechanism**

¹W. P. Jencks (1980) *Adv. Enzymol.* **51**, 75.

²M. Makinose (1973) *FEBS Lett.* **37**, 140.

³J. W. Keillor & W. P. Jencks (1996) *Biochemistry* **35**, 2750.

IRON-SULFUR CLUSTER

A structural unit within iron-sulfur proteins, comprising two or more iron atoms and bridging sulfur atoms (also termed sulfide ligands or sulfido-bridges). These bridging sulfur atoms are acid-labile, generating H_2S in the pres-

ence of strong acid. The designation of a cluster is based on the iron and sulfide content, and is given in square brackets (*e.g.*, [2Fe–2S], [4Fe–4S], and [3Fe–4S]). The possible oxidation levels are indicated by the net charge excluding the ligands (*e.g.*, [4Fe–4S]²⁺ cluster and [4Fe–4S]¹⁺ cluster). Note that the oxidized and reduced states of all such clusters differ by one formal charge, and this difference is independent of the number of iron atoms present. Individual iron atoms within a cluster can attain different oxidation states. **See** *Iron-Sulfur Proteins*

IRON-SULFUR PROTEINS

Iron-sulfur proteins contain nonheme iron that is liganded with sulfur atoms from cysteinyl residues and also usually contain inorganic sulfur (or, acid-labile sulfur) bridging two iron atoms. There are three major categories: (1) the rubredoxins, a group that lacks acid-labile sulfur, (2) simple iron-sulfur proteins containing only iron-sulfur clusters, and (3) complex iron-sulfur proteins with an additional redox center(s), such as a flavin, a heme, or another metal, often molybdenum. The clusters mainly function in electron transfer, but they can have other functions, including catalysis of hydratase/dehydratase reactions, maintenance of protein structure, and regulation of enzyme activity.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

H. Cheng & J. L. Markley (1995) *Ann. Rev. Biophys. Biomol. Struct.* **24**, 209.

T. E. Meyer (1994) *Meth. Enzymol.* **243**, 435.

J. B. Howard & D. C. Rees (1991) *Adv. Protein Chem.* **42**, 199.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Detection, **69**, 238-249; electron paramagnetic resonance spectroscopy, **69**, 239-245 [dark chemical reduction with reducing agents, **69**, 241, 242; oxidation-reduction potentiometry, **69**, 242-245; photoreduction, **69**, 240, 241]; quantitative identification, **69**, 781-784 [materials, **69**, 781; methods, **69**, 781-783, 791, 792 (precautions, **69**, 789-791; variations, **69**, 784-789)]; membrane-bound [chromatophore fragments, from photosynthetic bacteria, **69**, 247-249; oxygen-evolving organisms, **69**, 245-248]; occurrence, **69**, 238-249, 779, 780; redox nature, core extrusion, **69**, 785; reduction, **69**, 218, 221; types, **69**, 779, 780.

IRREVERSIBLE PATHS, REACTIONS, AND PROCESSES

If one or all of the intermediate states of a path, process, or reaction are not equilibrium states, then the path is regarded as irreversible. **See** *Nonequilibrium Thermodynamics (A Primer)*

IRREVERSIBLE POLYMERIZATION

Cohen¹ presented a method for determining the rate constants for irreversible polymerization where the different rate constants apply to the initial step and the propagation reactions. Data are obtained in the usual manner at moderate concentrations of primer. The approach is equally valid for template-directed and template-independent polymerization. Cohen applied this method to obtain rate constants for polyadenylate [(poly(A)) polymerase (EC 2.7.7.19)]. The report also contains useful information about simulating the time course of irreversible polymerization.

¹R. J. Cohen (1991) *J. Theoret. Biol.* **150**, 529.

ISOAMYLASE

This enzyme [EC 3.2.1.68], also called debranching enzyme, catalyzes the hydrolysis of α -(1,6)-D-glucosidic branch linkages in glycogen, amylopectin, and their β -limit dextrins. This particular enzyme is distinguished from α -dextrin endo-(1,6)- α -glucosidase [EC 3.2.1.41] by the inability of isoamylase to act on pullulan, and by the limited action on α -limit dextrins. Nevertheless, the action on glycogen is complete in contrast to the limited action by α -dextrin enzyme. Isoamylase will hydrolyze only 1,6-linkages at a branch point.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

I. S. Pretorius, M. G. Lambrechts & J. Marmur (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 53.

M. Vihinen & P. Mäntsälä (1989) *Crit. Rev. Biochem. Mol. Biol.* **24**, 329.

E. Y. C. Lee & W. J. Whelan (1971) *The Enzymes*, 3rd ed., **5**, 191.

ISOBUTYRYL-CoA MUTASE

This enzyme [EC 5.4.99.13] catalyzes the interconversion of 2-methylpropanoyl-CoA and butanoyl-CoA. The enzyme requires a cobalt ion for activity.

B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.

ISOCHORISMATE SYNTHASE

This enzyme [EC 5.4.99.6], also known as isochorismate synthase and isochorismate hydroxymutase, catalyzes the interconversion of chorismate and isochorismate.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

ISOCITRATE DEHYDROGENASE

Isocitrate dehydrogenase (NAD⁺) [EC 1.1.1.41], also known as β -ketoglutaric:isocitric carboxylase, catalyzes

the reaction of isocitrate (specifically, (1*R*,2*S*)-1-hydroxypropane 1,2,3-tricarboxylate, formerly termed *threo*-D(*S*)-isocitrate) with NAD^+ to produce α -ketoglutarate (or, 2-oxoglutarate), carbon dioxide, and NADH. The enzyme is reported not to decarboxylate added oxalosuccinate. Isocitrate dehydrogenase (NADP^+) [EC 1.1.1.42], also known as oxalosuccinate decarboxylase, catalyzes the reaction of isocitrate with NADP^+ to produce α -ketoglutarate, carbon dioxide, and NADPH. The enzyme is reported to be able to decarboxylate added oxalosuccinate.

- A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 L. N. Johnson & D. Barford (1993) *Ann. Rev. Biophys. Biomol. Struct.* **22**, 199.
 R. F. Colman (1990) *The Enzymes*, 3rd ed., **19**, 283.
 D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.
 W. W. Cleland (1980) *Meth. Enzymol.* **64**, 104.
 W. W. Cleland (1977) *Adv. Enzymol.* **45**, 373.
 K. Dalziel (1975) *The Enzymes*, 3rd ed., **11**, 1.

ISOCITRATE LYASE

This enzyme [EC 4.1.3.1] (also referred to as isocitrase, isocitritase, and isocitratase) catalyzes the conversion of isocitrate (specifically, (1*R*,2*S*)-1-hydroxypropane 1,2,3-tricarboxylate) to succinate and glyoxylate.

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.
 L. B. Spector (1972) *The Enzymes*, 3rd ed., **7**, 357.

ISODESMIC REACTION

A reaction in which the types of bonds formed in the product(s) are the same types of bonds broken in the reactant(s). **See** *Indefinite Polymerization*

ISOELECTRONIC

Referring to two or more molecular entities having the same number and connectivity of atoms (although not necessarily the same elements) as well as the same number of valence electrons. Thus, CO , N_2 , and NO^+ are isoelectronic, as are ketene ($\text{CH}_2=\text{C}=\text{O}$) and diazomethane ($\text{CH}_2=\text{N}=\text{N}$). **See** *Electroneutrality Principle*

ISOENTROPIC

A term referring to a reaction series in which each of the individual reactions has the same entropy of activation.

ISOEQUILIBRIUM RELATIONSHIP

A linear relationship between the standard enthalpies and entropies of a series of structurally related molecular entities undergoing the same reaction: thus, $\Delta H^\circ - \beta \Delta S^\circ = \text{constant}$ or $\Delta \Delta H^\circ = \beta \Delta \Delta S^\circ$. When $\beta \geq 0$, this relationship is referred to as an isoequilibrium relationship. When the absolute temperature equals the factor β (often referred to as the isoequilibrium temperature), then all substituent effects on the reaction disappear (*i.e.*, $\Delta \Delta G^\circ = 0$). In other words, a reaction studied at $T = \beta$ will exhibit no substituent effects. This would suggest that, when one studies substituent effects on a reaction rate, the reaction should be studied at more than one temperature. Note also that the ρ factor in the Hammett equation changes sign at the isoequilibrium temperature. **See** *Isokinetic Relationship*

- J. E. Leffler (1955) *J. Org. Chem.* **20**, 1202.

ISOERGONIC COOPERATIVITY

A novel form of allosteric protein-ligand binding behavior in which subsequent binding steps exhibit very different enthalpic properties, with only minor differences between their successive binding constants. One can detect isoergonic cooperativity by determinations of the change in the constant-pressure heat capacity (ΔC_p), but such behavior cannot be observed in experiments solely designed to determine the number of ligand molecules bound to a macromolecule.

- H. F. Fisher & J. Tally (1998) *Meth. Enzymol.* **295**, 331.

ISOKINETIC RELATIONSHIP

A condition said to exist for a series of structurally related molecular entities undergoing the same general reaction, wherein the enthalpies and entropies of activation can be related by the expression $\Delta H^\ddagger - \beta \Delta S^\ddagger = \text{constant}$ (*i.e.*, $\Delta \Delta H^\ddagger = \beta \Delta \Delta S^\ddagger$), where the factor β is independent of temperature. When the absolute temperature equals this factor (*i.e.*, $T = \beta$), all reactants of the series react at the same rate, and the temperature is referred to as the isokinetic temperature. If errors in determinations of $\Delta H^\ddagger$ result in compensating errors in $\Delta S^\ddagger$, values for β may be spurious. Thus, one must employ appropriate methods in establishing any isokinetic relationship. **See also** *Isoequilibrium Relationship*

- J. E. Leffler (1955) *J. Org. Chem.* **20**, 1202.

ISOLATED SYSTEM

Any system (chemical, biological, mechanical, *etc.*) that does not interact with its surroundings, such that matter and/or heat are not transferred into or out of the system. A change in environmental conditions does not result in a change in the system conditions.

An example of such a system is one in which the internal energy, U , and the volume, V , are constant. If these are the only two constraints on the system then, at thermodynamic equilibrium, the entropy, S , is at a maximum. On the other hand, if entropy and volume are constant for the isolated system then, at thermodynamic equilibrium, the internal energy is at a minimum. **See also** *Closed System*; *Open System*

ISOLEUCYL-tRNA SYNTHETASE

This enzyme [EC 6.1.1.5], also referred to as isoleucine:tRNA ligase, catalyzes the reaction of isoleucine with tRNA^{Ile} and ATP to produce isoleucyl-tRNA^{Ile}, AMP, and pyrophosphate.

W. Freist (1988) *Angew. Chem. Int. Ed. Engl.* **27**, 773.

P. Schimmel (1987) *Ann. Rev. Biochem.* **56**, 125.

D. Söll & P. R. Schimmel (1974) *The Enzymes*, 3rd ed., **10**, 489.

ISOMALTASE

This enzyme [EC 3.2.1.10] (also referred to as oligo-1,6-glucosidase, sucrase-isomaltase, and limit dextrinase) catalyzes the hydrolysis of 1,6- α -D-glucosidic linkages in isomaltose and dextrin products generated from starch and glycogen *via* α -amylase. **See also** *Sucrase*

E. H. Van Beers, H. A. Büller, R. J. Grand, A. W. C. Einerhand & J. Dekker (1995) *Crit. Rev. Biochem. Mol. Biol.* **30**, 197.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

M. Vihinen & P. Mäntsälä (1989) *Crit. Rev. Biochem. Mol. Biol.* **24**, 329.

ISO MECHANISMS

Enzyme mechanisms for which the enzyme, upon release of product(s), assumes a different form (*i.e.*, a different conformational structure, or one requiring rehydration, deprotonation; *etc.*) than the enzyme form originally present at the beginning of the catalytic cycle. Such mechanisms require reisomerization of the free enzyme back to the original form before another round of catalysis can occur. For example, the Iso Uni Uni mechanism is given by the following scheme: $E + A \rightleftharpoons EA \rightleftharpoons FP \rightleftharpoons F + P$ followed by $F \rightleftharpoons E$. A number of tools are

available to aid in the identification and characterization of Iso mechanisms^{1,2}. These include product inhibition studies, dead-end inhibition, isotope effect studies, and induced transport experiments. Other observations that may reflect Iso mechanisms. These include: biphasicity observed in Arrhenius plots as well as plots of pressure effects and unusual viscosity effects on kinetic parameters.

¹F. B. Rudolph (1979) *Meth. Enzymol.* **63**, 411.

²K. L. Rebholz & D. B. Northrop (1995) *Meth. Enzymol.* **249**, 211.

ISOMER

One of a number of molecular entities having the same atomic composition, but different line formulas and/or different stereochemical configurations. A distinction is usually made between structural isomers (molecular entities have the same atomic composition albeit different line formul^s), stereoisomers (entities having different arrangements, usually noninterconvertable and nonsuperimposable, in space), and *cis-trans* isomers (having different positions for substituents with respect to double bonds or sides of a ring structure).

ISOMERASES

A major class of enzymes that catalyze changes within one molecule. Examples include racemases, epimerases, mutases, and tautomerase^s.

ISOMERIZATION

A chemical reaction in which the principal product is isomeric with respect to the principal reactant. Isomerization can occur in the absence of molecular rearrangements (for example, between conformational isomers).

ISOPENTENYL-DIPHOSPHATE Δ -ISOMERASE

This enzyme [EC 5.3.3.2] catalyzes the interconversion of isopentenyl pyrophosphate and dimethylallyl pyrophosphate.

R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.

P. W. Holloway (1972) *The Enzymes*, 3rd ed., **6**, 565.

ISOPEPTIDE BOND

A substituted amide linkage between two amino acids in which the carbonyl group and/or the nitrogen is not

directly bonded to an α -carbon of one of the amino acids. Perhaps the most common example of an isopeptide bond in biological systems is the peptide linkage between the glutamyl residue and the cysteinyl residue in glutathione.

ISOPRENE RULE

A rule proposed by Ruzicka¹ that monoterpenes and sesquiterpenes are derived by electrophilic additions and rearrangements from a common intermediate.

¹L. Ruzicka (1953) *Experientia* **9**, 357.

3-ISOPROPYLMALATE DEHYDRATASE

This enzyme [EC 4.2.1.33], also referred to as isopropylmalate hydro-lyase, catalyzes the conversion of 3-isopropylmalate to 2-isopropylmaleate and water.

G. B. Kohlhaw (1988) *Meth. Enzymol.* **166**, 423.

3-ISOPROPYLMALATE DEHYDROGENASE

This enzyme [EC 1.1.1.85] catalyzes the reaction of 3-carboxy-2-hydroxy-4-methylpentanoate and NAD^+ to produce 3-carboxy-4-methyl-2-oxopentanoate and NADH. The product then decarboxylates, generating 4-methyl-2-oxopentanoate.

L. L. Searles & J. M. Calvo (1988) *Meth. Enzymol.* **166**, 225.

G. B. Kohlhaw (1988) *Meth. Enzymol.* **166**, 429.

2-ISOPROPYLMALATE SYNTHASE

This enzyme [EC 4.1.3.12] catalyzes the reaction of 3-carboxy-3-hydroxy-4-methylpentanoate with coenzyme A to produce acetyl-CoA, 3-methyl-2-oxobutanoate, and water. This enzyme has a potassium-ion requirement.

G. B. Kohlhaw (1988) *Meth. Enzymol.* **166**, 414.

M. J. P. Higgins, J. A. Kornblatt & H. Rudney (1972) *The Enzymes*, 3rd ed., **7**, 407.

ISOPULLULANASE

This enzyme [EC 3.2.1.57] catalyzes the hydrolysis of pullulan to produce isopanose (6- α -maltosylglucose). Panose (4- α -isomaltosylglucose) is a substrate and is hydrolyzed to isomaltose and glucose.

M. Vihinen & P. Mäntsälä (1989) *Crit. Rev. Biochem. Mol. Biol.* **24**, 329.

ISOSBESTIC POINT

A wavelength in the absorption spectra (measured as a wavelength, frequency, or wavenumber) in which absorption coefficients are equal. At such a point, the ab-

sorption of a sample does not change. For example the isosbestic point in the conversion of thionicotinamide adenine dinucleotide (thionicotinamide- NAD^+), an alternative substrate for many dehydrogenases, to the reduced form of the coenzyme, is 342 nm¹. Hence, the absorption of the sample at this wavelength will not change over the course of the reaction. (An obsolete synonym is isoabsorption point.) **See Beer-Lambert Law; Absorption Spectroscopy**

¹F. B. Rudolph & H. J. Fromm (1970) *Biochemistry* **9**, 4660.

ISOSTERIC AMINES AS CARBOCATION ANALOGUES

Positively charged amines that are structural analogues and are isosteric with putative carbocation intermediates in enzymic reactions. These compounds have proved their value in efforts to characterize enzyme mechanisms that proceed by the transient formation of carbocation intermediates.

ISOSTERY

A physiological mechanism of metabolic regulation via competitive inhibition by structural analogues of the substrate.

ISOTHERMAL PROCESS

Any process which occurs at constant temperature (thus, $dT = 0$). Such processes typically have a heat source or thermostat to control the temperature of the system. **See Adiabatic Process**

ISOTONIC

1. Referring or relating to or demonstrating equal tension. 2. Often referring to an equivalence of osmotic pressure (*i.e.*, isosmotic). Thus, "isotonic" often refers to solutions having identical ionic strength.

ISOTONIC BUFFERS

Buffer solutions that are isosmotic with respect to some standard, typically chosen such that suspended cells will neither shrink nor expand. Sodium chloride solutions (0.90% weight/volume or 0.155 M) at 37°C is often used to represent physiological conditions. These buffer systems are also important in studies of intact cells and membranal organelles; likewise, many pharmaceutical formulations must be prepared as isotonic solutions. Most enzyme-catalyzed reactions are affected by ionic

strength, and kinetic studies are typically conducted in solutions at constant ionic strength, approximately in the physiologic range. Care must be exercised to achieve an appropriate balance of sodium and potassium ions (*i.e.*, cytosolic = 140 mM K⁺ and 12 mM Na⁺; extracellular fluid or blood plasma = 4 mM K⁺ and 145 mM Na⁺).

Sörensen¹ introduced the concept of pH and was among the first to consider the effects of pH on enzyme-catalyzed reactions. That same publication describes a phosphate buffer system, now widely known as a Sörensen buffer. In its original formulation these buffer solutions were hypotonic, varying slightly in osmotic properties throughout the pH range. Curie and Sciarrone² described a modified Sörensen buffer system that is isotonic throughout the pH range. These authors provided a table for preparing isotonic phosphate buffers having a tonicity of 0.92%. For example, 50 mL 0.0667 M Na₂HPO₄ (*i.e.*, 9.470 g/L) is added to 50 mL 0.0667 M NaH₂PO₄ (*i.e.*, 9.208 g/L of NaH₂PO₄ · H₂O) together with 0.480 g of NaCl (for every 100 mL solution), and deionized water, yields a buffer of tonicity 0.92 and pH 6.72 at 25°C (6.71 at 37°C).

Krebs-Ringer physiologic buffer is prepared by mixing 100 mL solution A (0.9% w/v NaCl), 4 mL solution B (1.15% w/v KCl), 3 mL solution C (1.22% w/v CaCl₂), 1 mL solution D (2.11% w/v KH₂PO₄), and 1 mL solution E (3.8% w/v MgSO₄ · 7H₂O). Krebs-Ringer-bicarbonate buffer is prepared in the same manner, except that 21 mL solution F (1.3% w/v NaHCO₃) is added. Krebs-improved-Ringer buffer solution is prepared by mixing 80 mL solution A, 4 mL solution B, 3 mL solution C, 1 mL solution D, 1 mL solution E, 21 mL solution F, 4 mL 0.16 M sodium pyruvate, 7 mL 0.1 M sodium fumarate, 4 mL 0.16 M sodium glutamate, and 5 mL 0.3 M glucose.

Palitzsch³ described buffers utilizing sodium tetraborate and boric acid. Several authors^{2,4} have presented tables for modified Palitzsch buffers which are isotonic (tonicity of 0.90 at 37°C). For example, 30 ml of 0.05 M sodium tetraborate (Na₂B₄O₇ · 10H₂O dissolved in deionized water) and 70 ml of 0.2 M boric acid together with 240 mg NaCl for every 100 ml will yield a solution having a pH of 8.12 at 25°C (and 8.11 at 37°C). However, care should always be exercised when using borate-based buffers

since borate can form complexes with alcohols, polyols, and especially carbohydrates and ribonucleotides.

¹S. P. L. Sörensen (1909) *Biochem. Zeit.* **21**, 131.

²A. P. Curie & B. J. Sciarrone (1969) *J. Pharm. Sci.* **58**, 990.

³S. Palitzsch (1915) *Biochem. Zeit.* **70**, 333.

⁴D. Perrin & B. Dempsey (1974) *Buffers for pH and Metal Ion Control*, Chapman and Hall, London.

ISOTOPE DILUTION

An analytical chemical technique that utilizes radioactive (or stable) isotopes for the quantitative analysis of the amount of substance. In the absence of a kinetic isotope effect, isotopic isomers react identically with respect to their unlabeled counterparts. The method offers the advantage that specific activity (or gram-atom excess in the case of stable isotopes) is an intensive variable. Therefore, one only needs to recover sufficient labeled metabolite to determine amount of substance and disintegrations per minute (or, gram-atom excess) to reach an accurate determination of specific activity. The technique is feasible so long as one can accurately determine the initial and final specific activities.

We can define the specific activity of a substance A as,

$$\text{Specific Activity} = SA_A = \text{mol } A^*/(\text{mol } A + \text{mol } A^*)$$

When mol A* is much less than mol A (*i.e.*, if A* is truly a tracer species), then

$$SA_A = \text{mol } A^*/\text{mol } A \propto \text{dpm } A^*/\text{mol } A$$

Because specific activity is always an intensive variable, one need not recover all of the sample to determine the ratio mol A*/mol A. To evaluate SA_A, one need only obtain an amount that provides an accurate measurement of A* and total A. The former is easily achieved by liquid scintillation counting, and spectrophotometry or some other enzymatic assay usually allows accurate determination of mol A in a sample.

To illustrate how specific activity can be used to analyze the amount-of-substance, suppose that we combine isotopically labeled sample (SA_A as written above) with an unlabeled sample containing an unknown amount of the substance (call it A_x). Then, the new specific activity will be lower, such that

$$SA_A' = \text{mol } A^*/(\text{mol } A + \text{mol } A^* + \text{mol } A_x)$$